# Supplement to "Communication, Coherence and Collective Agents: An Active-Inference Framework"

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This supplement accompanies the main text and has four parts. The first collects, section by section, the secondary results, examples and extended discussion that the main text cites and does not state in full. The second contains extensions that address open problems posed in the literature on collective active inference; the main text summarizes them and records what remains open. The proofs of all statements, of the main text and of this supplement, follow, and the last part records the numerical checks. Section, equation, figure and statement numbers that carry the prefix S refer to this document; all other numbers refer to the main text.

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# Part I
# Additional results

## S1 Additional material for Section 1: the introduction

This section holds the extended discussion of Section 1 that the main text summarizes.

### S1.1 Status of hypothesis H1

Hypothesis H1 in the form stated in Section 1.2 is assembled in Theorems 4.23 and 5.20 for linear-Gaussian agents with fixed or exogenously scheduled attention. Because roles belong to channels, a member's sensing of the shared world is a sensory channel of the collective, not a leak (Remark S4.1). Part (a) is a theorem (Theorems 4.3 and S4.6 and Proposition S4.3); the Markov length is the decay length in $h$ of the mutual information between a site and everything farther than $h$ hops from it, given the sites in between. A family with correlation length zero has arbitrarily long Markov length, and nearest-neighbor passing has a diverging correlation length with Markov length zero (Proposition 4.6), so an earlier formulation of H1 that we had adopted, that every region whose shell is thicker than $\xi$ is an approximate blanket, is false in both directions (Remark S4.7). Blanket existence therefore selects nothing (Proposition S4.10). Part (b) is a definition; homogeneous lattices have bounded contrast, so the baseline is finite (Theorem 4.13), and the world-conditional precision is diffusive for reversible dynamics with nonnegative weights and row sums below one and for nonnegative normal dynamics with isotropic noise under a condition on row and column sums, but not in general (Theorem 4.9 and Example S4.12). Because the comparison medium keeps the declared geometry, structure that the declared geometry already contains is assumed, not found, and a finite closed population is a collective of its own comparison medium only if, for uniform leak, it is more connected than that medium, never under the exchangeable clause with the mean coupling, on a torus or on a vertex- and edge-transitive declared graph (Remark S4.16 and Proposition S4.17); collectives of collectives are judged level by level, each in the coarse medium of its level against that medium's own comparison medium (Definitions 4.19 and 4.21). In part (c), contrast above one, together with the dispersion of the members' total leaks, bounds the memory of the collective average in continuous time with the exterior held fixed, for a reversible diffusion whose drift is the block on $C$ of a diffusive precision; for a general drift a bound of the same kind holds in a drift-based contrast when the row and column dispersions are small, and a closure theorem in the contrast of Definition 4.10 is then open (Theorem 5.4 and Open problem S5.4); the deletion nulls, with any acts replaced by exogenous forcing, make the interaction order parameter, a mutual information between members' paths given the shared exogenous drive, vanish, and for scalar errors without records they bound every contrast by one (Proposition 5.16); and under a stated deliberation protocol the joint policy is a group posterior when the members agree on the redundancy of their inputs and some belief carries the agreed redundancies, and lies within explicit distances of one otherwise (Theorems 5.10 and 5.11). Agency does not follow from contrast (Proposition S5.12 and Example S5.14), and differing from partner randomization and surviving exact fusion are requirements of the definition of a meta-agent (Definition 5.19), not consequences of these theorems.

### S1.2 Status of hypothesis H2

Hypothesis H2 holds in stated scopes (Section 3). Overvaluation is proved in the scalar track model, for a band of positive error covariances, and in static common-source worlds that are scalar, aligned or read through invertible noiseless codes; in the symmetric scalar common-source world with channel noise the believed-to-true ratios of message values, for a receiver that retains its raw data, diverge like $(1-\phi)^{-2}$ as the shared fraction $\phi$ of precision tends to one; and in scalar dynamic worlds common process noise makes level-1 reading of belief messages between steady-state Kalman filters with independent sensors overconfident (Propositions 3.4 to S3.6). Undervaluation is proved for complementary errors in the track model and by closed-form examples (Proposition S3.4 and Examples S3.8 and S3.9); elsewhere it is found by numerical search in two random ensembles, which gives no frequency independent of the ensemble, and exact conditions are open (Numerical

observation S3.10 and Open problem S3.11). Social innovations, the differences between messages and the receiver's prediction, are blind to any error that all agents share (Lemma 3.6), and closed groups of scalar receivers that fuse linearly with fixed gains, homogeneous grounding and reversible weights are echo chambers, while exact inference forms no weak or self-sealed one (Theorem 3.10 and Proposition 3.11).

### S1.3 Status of hypothesis H3

The components of coherence named in hypothesis H3 (Section 1.2) are developed in Section 6. In scalar level-0 networks with exact codes, information about a distant source decays at twice the rate of influence, exactly on the infinite ring and, under row-sum contraction with a noise floor, as an upper bound otherwise (Proposition S6.3). In a static two-group model at infinite policy precision, with at least $\kappa$ in-group senders available, a calibrated receiver withdraws from every cross-group edge whose mistranslation cost exceeds the sender's information advantage, while the reverse inequality for one cross-group sender guarantees only that the receiver keeps at least one cross-group sender, not necessarily that one; ties aside, the attended graph splits along the groups exactly when every cross edge satisfies the withdrawal inequality. Receivers whose likelihood ignores mistranslation cannot withdraw (Theorem 6.4 and Proposition 6.3). Consensus with a given content exists exactly when the holonomies of the fundamental cycles fix it, and true translations between invertible encodings of one world are flat, so inconsistency among such encodings is learning error, which, for full-rank codes and learned translations that converge, with the true translations as limits on edges attended infinitely often with grounding, is confined to cycles with an edge attended only finitely often (Theorem 6.5 and Propositions S6.5 and S7.1). Coherence statistics miss amplification of belief messages around cycles, a calibration failure distinct from holonomy and from double counting (Theorem 6.8 and Proposition S6.10), and the grounding tension: in a scalar level-0 network with exact codes, nonnegative time-invariant weights and homogeneous grounding every agent's stationary mean-square error grows at least linearly with the correlation length (Theorem 6.12).

In the two-group model of Section 7, two groups whose private frames differ by a rotation, calibration turns translation error into message noise (Lemma 7.3). For the frozen-calibration mean-field map, the declared start reaches the split branch above a threshold $\gamma_S$ of the policy precision, a single fixed point on that branch attracts every state above $\gamma_T$, and only merged states exist below $\gamma_M$ (Theorem 7.6). At infinite policy precision with exact in-group translations the groups of the split are exactly the collectives at every level of an admissible window, which is $(7.954, 8.733]$ at the base point, with significance 1.098; over 162 grid points the significance lies between 1.009 and 1.415 (Theorem 7.8 and Numerical observation S7.6). With in-group angle noise of variance 0.01 the simulated contrasts are about 2 percent higher and the admissible window is not computed (Numerical observation 7.10). The split differs from the deletion nulls by proof, and from the frozen-environment null only when the initial states that frozen messages carry are deterministic or conditioned on (Proposition 7.9(a)); it differs from partner randomization by computation (Numerical observation S7.9). When receivers know their translation errors, no state attends along the split (Proposition 7.12).

### S1.4 Status of hypothesis H4

Hypothesis H4 in the form stated in Section 1.2 is Theorem 8.20. Light cones need only locality: under any local rule, attention selected by expected free energy included, agents more than $2t$ hops apart are world-conditionally independent after $t$ rounds from a world-conditionally independent

start independent of the noises (Theorem 8.5). Every length, time and volume needs (M), which the agents' own attention can violate (Example S8.1). Under row-sum contraction, which with the moment hypotheses of Lemma 8.2(a) gives (M), the world-conditional covariance clusters exponentially, and so it does under the spectral and reversible-chain contractions (Theorems 8.9 and S8.11 and Proposition S8.10). On $\mathbb{Z}^D$ with nearest-neighbor passing, self weight $c$, message weight $a$ and leak $\varepsilon = 1-c-2Da$, the covariance summed over hyperplanes decays at the rate $m$ with $\cosh m = 1+\varepsilon/(2a)$ in every dimension (Theorem 8.10), and $\xi_2^2/\tau \to a$ as $\varepsilon \to 0$ (Proposition S8.15). Contrast bounds extent through $\xi_{2,d}^2 = \varsigma_d^2/(2\varepsilon^{\Pi})$, with $\varsigma_d^2$ the second moment of the precision couplings along coordinate $d$ and $\varepsilon^{\Pi}$ the uniform precision leak, sharp up to the factor $\pi/\sqrt{12}$ (Theorem 4.14 and Corollary 4.16). None of these bounds says that collectives of the bounded size form.

### S1.5 Assumptions that become results

Several of the standing assumptions A5 to A11 of Section 1.2 become results: A7 the closure identity, by which a collective whose boundary hears its interior is never exactly closed, its reduced equation keeping a memory, initial-state or colored-noise term for some initial law (Theorem 5.3); A10 the identity time under turnover, which records written by members can extend and records written only by non-members whose writes do not depend on the members' states do not (Proposition 5.14); A11 the grounding tension (Theorem 6.12 and Remark S7.11); and A9 the multiplex populations of Section 10, whose blankets and ultrametric classes nest within a modality while classes of different modalities can cross (Proposition S4.26). Under A8 the size bound holds for every metric on a space of roles (Theorem 4.14), and A5 is why agency has to be proved under stated hypotheses. No theorem takes A6 as a hypothesis: it motivates the fixed or exogenously scheduled attention of the proofs, and its one quantitative reading is the number of consensus rounds a world step needs (Proposition 5.7). Beyond the standing assumptions, the theorems rest on hypotheses named in their statements, among them the conditions (Lin), (Loc), (B), (NN) and (M) of Assumption 8.1, the mean-field reduction of Assumption 7.4 and, for Theorem 11.13, per-channel noise and a stable joint process, $\rho(A) < 1$. The extension behind the fifth idea gives three of them further forms: A7 for aggregates of agents with different timescales and for the elimination of internal states (Propositions S13.10 and S14.13), A10 for records written by members, which, when they are read and do not decay, lengthen the identity time by a factor bracketed explicitly (Corollary 12.5), and A9 for one budget of attention shared across memberships (Proposition S14.18).

### S1.6 Status of hypothesis H5

Hypothesis H5 in the form stated in Section 1.2 is proved in linear-Gaussian models, with the exceptions named here; apart from one extension of the code game of Section 6 by a record and the ligand and sedimentation statistics of a cell (Propositions S12.21 and S14.8 and Example S14.9), discrete state spaces, nonlinear dynamics and renormalizing generative models appear only in remarks, conjectures and open problems. Part (a) is Theorem 11.13, an exact conditional theorem for a stable joint process with per-channel noise and a detectable coarse filter, with the criterion for an abstract physics in Theorem 11.8 and the meta-agent's inference in Proposition 11.12; its predictive law is that of the current exterior given the coarse blanket path, with the one-round latency of the joint process. Error bounds exist for components, a leak quadratic in a violation of the blanket conditions, a memory of an approximately closed exterior bounded through its drift-based contrast, and habit tilts bounded by the members' disagreement (Propositions 11.5 and 11.9 and Theorem 11.13), but for the predictive law under approximate closure only the conditional bound

of Theorem S18.7 and a finite-horizon bound that accumulates over the horizon (Proposition S39.6), so no approximate form of the theorem is stated (Open problem S11.13). Every property holds at every number of consensus rounds and none uses contrast, so part (a) describes the structure of a meta-agent and is not a criterion for which sets of agents are meta-agents (Remark 11.14). Part (b) collects Theorems 12.2 and 12.7, Propositions 12.3, 12.4 and 12.8, and Lemma 12.6 and Propositions S12.2, S12.6 and S12.7, proved for linear recursions, Gaussian trees, a static scalar world, drift with the exterior held fixed, scalar level-0 networks and linear conformity games; the floor is proved nondecreasing in the read weight and to grow by a factor below the memory factor, approaching it in a persistent world with weak grounding. Formation by conformity, the candidate mechanism of formation that does not rest on misreading, is a conjecture (Conjecture S12.25). Part (c) collects Theorem 14.3 and Proposition 14.4 and Propositions S14.2, S14.6, S14.12 and S14.13, proved for linear sensors, with the non-reversible case asymptotic and long memory proved for the limiting aggregate; the cell, the firm and the person are worked examples whose readings are interpretations (Examples S14.9, S14.14 and S14.16 and Remark S14.20). Part (d) collects Theorems 13.4 to 13.6, 13.8 and 13.10, Lemma 13.3, and Proposition 13.7 and Example S13.4 and Propositions S13.8 and S13.10, under the moment bound with a diffusive precision and positive leaks or the setting a statement names (Assumption 13.1); the depth bound for decimated media in two and more dimensions rests on an inequality proved in one dimension and checked for a single decimation step in two, and the first-order rate of the error of the memoryless reduction is a numerical observation (Numerical observation S13.11).

### S1.7 Further tables of main results

The tables of this subsection continue Table 1 of the main text, with the columns and the conventions on status explained in Section 1.3.

Table S1: Main results of Sections 6 to 10.

| Result | Where | Status | Model or hypotheses |
|---|---|---|---|
| Information about a source decays at twice the mass | Proposition S6.3 | proposition | scalar level-0 networks, exact codes; exact rate on the ring |
| Calibrated receivers withdraw from mistranslated edges | Theorem 6.4 and Proposition 6.3 | theorem; proposition | static two-group model, $\gamma = \infty$, orthogonal mistranslation |
| Holonomy criterion; true translations are flat | Theorem 6.5 and Proposition S6.5 | theorem; proposition | connected graphs; part (a) from Dennis (2026c) |
| Cycle amplification needs an edge ratio above one | Theorem 6.8 and Proposition S6.10 | theorem; proposition | excluded by a common prior or level-1 reading of posted priors |
| Code domains of length of order $(J/h)^2$ on rings | Theorem 6.11 and Conjecture S6.16 | theorem; conjecture | Gaussian pulls; higher dimensions conjectured |
| Grounding tension: error at least $Q\xi/2$ | Theorem 6.12 | theorem | scalar level-0 networks with exact codes, time-invariant $W$ and homogeneous sensing gain under (Lin), (Loc), (B), (NN); $\xi^2$ on the ring |
| Grounded learning flattens the attended subgraph | Proposition S7.1 | proposition | full-rank codes, convergent learning |

| Result | Where | Status | Model or hypotheses |
|---|---|---|---|
| Thresholds $\gamma_M$, $\gamma_S$, $\gamma_T$ of formation | Theorem 7.6 | theorem about the map | frozen-calibration mean-field map; stochastic step open |
| The groups of the split are the collectives | Theorem 7.8 and Numerical observation S7.6 | theorem; numerical observation | $\gamma = \infty$, exact in-group translations; openness given nondegenerate roots |
| Online calibration: the split's attention pattern in 0 to 7 of 8 replicates per cell | Numerical observation 7.11 | numerical observation | symmetric start; reruns of a cell differ by up to three replicates; not a contrast criterion; reachability open |
| Known translation errors remove the mechanism | Proposition 7.12 | proposition | translation-aware exact fusion, start exchangeable given the world path; full null heuristic |
| Drifting frames; formation beyond the model | Conjectures S7.12 and 7.13 | conjecture | exact-fusion null; populations with private frames |
| Light cone: independence beyond $2t$ hops | Theorem 8.5 | theorem | any local rule, conditionally independent start |
| The agents' own attention can violate (M) | Example S8.1 | example | two level-1 receivers, sender-specific codes |
| Clustering; Ornstein–Zernike rate $\cosh m = 1 + \varepsilon/(2a)$ | Theorems 8.9 and 8.10 | theorem | row-sum contraction; nearest-neighbor passing on $\mathbb{Z}^D$ |
| Exact criterion for records to lengthen memory | Propositions 8.12 and 8.13 | proposition | record model; size bound needs a diffusive precision |
| Condensation beyond trees; shared rank falls with scale | Conjectures S8.23 and S8.26 | conjecture | graphs with loops; relays with diverse codes |
| Hypothesis H4 in final form | Theorem 8.20 | theorem | (Lin), (Loc), (B), (M) and a named contraction; upper bounds apart from the exact lengths and the record criterion |
| A common rank-one code: growth at most the world's rate | Proposition 9.1 | proposition | level-1 receivers, isotropic world; symmetric listening or no sensing |
| A Walrasian price counts the common prior $N$ times | Propositions 10.5 and S10.16 | proposition | linear-Gaussian market; Hayek dimension of prices |

Table S2: Main results of Sections 11 and 12, the first part of the extension behind hypothesis H5.

| Result | Where | Status | Model or hypotheses |
|---|---|---|---|
| Exact coarse path blankets; the information-form statistic suffices; leaks quadratic in violations | Theorem 11.3 and Propositions 11.4 and 11.5 | theorem; proposition | per-channel noise; the exterior reads only the coarse act; no sensor reads one member's act |
| Exact abstract physics if and only if the kernel is invariant and the coarse exterior reads the members' channels only through the coarse act; minimal closed order $d^*$; memory bounded through drift-based contrast otherwise | Theorem 11.8 and Proposition 11.9 | theorem; proposition | declared linear coarse exterior, coarse act as input |

| Result | Where | Status | Model or hypotheses |
|---|---|---|---|
| Consensus on contributions gives the steady-state coarse filter at every number of rounds | Proposition 11.12 | proposition | memoryless readers, exact abstract physics, detectable coarse filter, sum-preserving consensus; the predictive mean from the centralized start with stationary initial covariance |
| Hypothesis H5a: a meta-agent with an abstract environment | Theorem 11.13 | theorem | exact and conditional; partial error bounds only; selects nothing |
| The two-level member model is exact if and only if the collective coordinate is closed | Proposition S12.2 | proposition | linear recursions without input |
| Likelihoods up, leave-one-out priors down; predictive coding exact in the means | Theorem 12.2 and Proposition S12.6 | theorem; proposition | Gaussian trees |
| Literal top-down reading: overconfidence below two once, without bound under rebuilt records | Propositions 12.3 and S12.7 | proposition | scalar trees (one reading); static scalar world, every rule but exact counting |
| A member-written record lengthens collective memory by a factor of at least $1+\beta/\omega$ | Proposition 12.4 and Corollary 12.5 | proposition; corollary | global record without decay, exterior fixed; separation gain not the product of factors in general |
| The shared-error floor rises with the read weight, by a factor below $1+\beta/\omega$, unseen in social and top-down innovations | Theorem 12.7 and Lemma 12.6 | theorem; lemma | scalar level-0 network, closed collective, exact codes |
| Linear conformity selects no undeclared partition | Proposition 12.8 | proposition | potential game; contrast on the Gibbs precision |
| Precision on a self-written record of a grounded coordinate raises the shared-error floor; the total error rose where local noise was a small share and fell by 1 to 4 percent, before rising, in three instances | Proposition S12.20 and Numerical observation S12.19 | proposition; numerical observation | setting of Theorem 12.7 with claimed-variance gains, $\beta+\omega\le 1$; computed instances |
| Formation by conformity | Conjecture S12.25 | conjecture | codes on a ring with records |

Table S3: Main results of Sections 13 and 14, the second part of the extension behind hypothesis H5.

| Result | Where | Status | Model or hypotheses |
|---|---|---|---|
| Exact flows: decimation on $\mathbb{Z}$, axis rate and $\xi_2$, the nested medium | Theorems 13.4 to 13.6 | theorem | diffusive precision, uniform leak |
| Block averages leave the diffusive class | Example S13.4 | example | nearest-neighbor field on $\mathbb{Z}$, every leak |
| Depth of a hierarchy bounded through $\xi_2$ | Proposition 13.7 | proposition | uniform leak on $\mathbb{Z}^D$; decimation in $D\ge 2$ conditional |
| Nested collectives at every level if and only if $\mathfrak{q} < b/(2b^2-1)$ and the leak exceeds a threshold | Proposition S13.8 | proposition | nested medium, no declared geometry, a level declared per level |

| Result | Where | Status | Model or hypotheses |
|---|---|---|---|
| Static error and integrated memory of block coordinates are one matrix, bounded entrywise by products of the blocks' equitability defects over the smallest eliminated rate | Lemma 13.3, Theorem 13.8, and Proposition 13.9 | lemma; theorem; proposition | reversible diffusion |
| Eliminating internal states drops an exact memory kernel; first-order rate of the error | Proposition S13.10 and Numerical observation S13.11 | proposition; numerical observation | linear diffusion with a particular partition; rate in two random instances |
| What coarse-graining preserves | Theorem 13.10 | theorem | block averaging under Assumption 13.1; exact for invariant block spans, otherwise, for uniform leak, up to an error bounded by the dispersion of the cut against the blocks' least internal stiffness $\varkappa_{\mathrm{in}}^{\min}$, which high contrast alone does not make small |
| Optimal linear sensors read leading canonical variates, in reversible exteriors under a budget fewer and slower ones as the horizon grows | Theorem 14.3 and Propositions S14.2 and 14.4 | theorem; proposition | information criterion; rank limit and Shannon budget; budget clause for distinct moduli; non-reversible case asymptotic |
| Selection by coupling, predictability and relevance; a cell senses its ligand, not gravity | Proposition S14.6 and Example S14.9 | proposition; example | attention objective at a horizon; the cell a worked model |
| Aggregates closed for interchangeable types; memory and long memory otherwise | Propositions S14.12 and S14.13 | proposition | linear populations; long memory in the limit, for a power-law density of persistences at one |
| A bound on the effective Hayek dimension falls with the horizon; a firm reads the futures price at its lag | Theorem 14.7 and Example S14.14 | theorem; example | quadratic losses; effective dimension in reversible exteriors; exogenous price; sufficiency of the futures price is the known separation result, recovered; readings for firms are interpretations |
| Selection beyond linear-Gaussian exteriors | Conjecture S14.22 | conjecture | stationary exteriors, finite predictive information |

## S1.8 What is not established

In the two-group model the formation chain is a theorem only at infinite policy precision with exact in-group translations, at the levels of a window that is nonempty by computation, where the declared start is itself the split; at finite policy precision the theorems are reachability of the split branch above $\gamma_S$ and, above $\gamma_T$, a single fixed point on that branch that attracts every state, both about the frozen-calibration mean-field map, and the model yields no formation result in the sense of Definition 7.1. The step from the stochastic model to the map, self-consistent calibration along trajectories, online calibration, which from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell, with reruns of a cell differing by up to three replicates (Numerical observation 7.11), and populations with anisotropic access, heterogeneous senders or geometry are open (Open problems S7.13 to S7.15); openness at finite policy precision is a sketch (Remark S7.8); and the stability classes of the mean-field fixed

points rest on a sign that is computed, not proved. From a start law permutation-invariant given the world path, such as the declared start with identical initial beliefs, the model's collectives fail the translation-aware form of the exact-fusion persistence clause (Proposition 7.12); they are therefore attributed to misreading of joint structure and are not meta-agents, a verdict proved for the translation-aware null and heuristic for full cross-covariance-aware exact fusion (Remark S5.28). Persistence with drifting frames and formation beyond the model are conjectures (Conjectures S7.12 and 7.13), and no formation result applies to cultures, markets or polities (Section 10.6).

Contrast is defined only for diffusive world-conditional precisions. Non-reciprocal attention and noisy records can produce non-diffusive ones, for which contrast and the size bound are open (Open problem S4.28), with the mutual-information selector used with the tests of Proposition 5.18 as the available substitute; receivers that fuse exactly raise the same question (Open problem S4.29). The bridge to correlation criteria beyond two equal blocks and the sharp constant of the size bound are open, and a Markov-length bound $\xi_M \le B(q/q_{\min}, K, D)(1+\xi)$ on $\mathbb{Z}^D$, with conditioning entering only through the noise anisotropy, is a conjecture (Open problems S4.31 and S4.32 and Conjecture S4.8).

Closure of the collective average in discrete time beyond Lemma 11.11 and under strong non-reversibility, a minimal stochastic memory, a uniform agency bound, policies that mix epistemic and performative acts, positivity of the interaction order parameter under selected attention, its estimation under an unrecorded shared drive and the agency of abstract collectives are open (Open problems S5.2, S5.4, S5.9, S5.21, S5.26, S5.27 and S10.30), and so is the existence of a meta-agent: no population analyzed here is shown to meet every clause of Definition 5.19 (Open problem S5.29); synchronous deliberation of three or more members and long identity from conformity are conjectures (Conjectures S5.13 and S5.22).

Code domains in two or more dimensions, coarsening and the bundling of isoglosses are conjectures (Conjectures S6.16 to S6.18), the two-dimensional exponent 4/3 resting on a declared heuristic identification of domain size with a correlation length of the random-field Ising model. Amplification under noisy circulation is not characterized, and under exact priors it is only computed (Open problem S6.14 and Numerical observation S6.12). Condensation of coherent information, reach to arbitrary distance above a branching threshold and confinement below it, is a theorem on trees and a conjecture on graphs with loops (Theorem S8.19 and Conjecture S8.23); the decline of shared rank with scale and the absence of amplification under frozen or periodic attention of level-1 receivers with one code and one channel noise in an isotropic world are conjectures (Conjectures S8.2 and S8.26); the pointwise Ornstein–Zernike form for $D \ge 2$ rests on a restated theorem and a heuristic step (Remark S8.13); and whether attention selected by expected free energy drives $a$ and $\varepsilon$ toward collectives as large as the size bound allows is open (Open problem S8.28). The scope conditions of misreading, its ratio in dynamic worlds and exact fusion in the minimal model are open (Open problems S3.7, S3.11 and S9.5); under Gaussian code uncertainty the second moment of the understanding error is finite only when the belief dimension exceeds the code rank $r_c$ by at least two, and its closed form there is open (Theorem S22.1 and Open problem S2.9); and fragmentation under a preference to be predicted is an untested conjecture (Conjecture S2.11).

For the fifth idea, hypothesis H5a has an error term only in the stable linear-Gaussian case with positive definite record noise and no computation noise of the members, under the condition $c_- > 0$ of Theorem S18.7 (Corollary S18.8, stated without input, for a stationary fine filter and the stationary filter of the moment-matched declared model); beyond that case the only error bound known for the meta-agent's predictive law when its coarse exterior is only approximately closed is the finite-horizon bound of Proposition S39.6, which needs neither stability nor $c_- > 0$ and accumulates over the horizon, and none is known for the memory of the coarse internal state under level-0 protocols beyond the exact terms proved for them (Proposition S11.10 and Open problems S11.13

and S18.13); mutually modeling meta-agents, a sharp constant for the leak of an approximate coarse blanket, and discrete and nonlinear meta-agents are open (Open problems S11.14 to S11.16). The population law of the two-group split under full exact fusion is open, only one receiver's deviation having been computed (Open problem S12.27 and Numerical observation S12.23); contrast with record nodes was defined only at the reversible record noise in the instances computed with outsiders, and no comparison medium is declared for populations with record nodes (Open problems S4.18 and S12.28); formation by conformity to a self-constituted prior is a conjecture (Conjecture S12.25); and for top-down precision on a self-written record only a rise of the shared-error floor is proved, the total error having fallen by 1 to 4 percent in three computed instances where local noise is a larger share (Proposition S12.20 and Numerical observation S12.19). Contrast for the non-diffusive laws that block averaging produces, closure in discrete time for non-reversible updates, the decay of decimated kernels in two and more dimensions, the baselines of nested media embedded in lattices beyond one instance, and error bounds for nonlinear and discrete renormalization are open (Open problem S13.22); the depth bound for decimated media in two and more dimensions rests on an inequality proved in one dimension and checked for one step in two (Proposition 13.7), the first-order rate of the error of the memoryless reduction is only observed (Numerical observation S13.11), and that Laplacian renormalization returns every region of high grounding-free ratio is refuted in general (Proposition S20.10): it holds for two equitable blocks under a gap condition and, with the constant threshold $b/(b-\mathfrak{q})$, for the level blocks below the top of the nested medium with $\mathfrak{q} < b$ (Proposition S13.19 and Theorem S20.9), and which regions it returns is otherwise an open problem (Open problem S13.21). Which sensor an agent that filters its readings should use is open, and selection beyond linear-Gaussian exteriors is a conjecture (Open problem S14.21 and Conjecture S14.22).

Throughout, the scope is that of the proofs: linear-Gaussian dynamics, mostly with fixed or exogenously scheduled attention; literal receivers in the network results; stationarity wherever a length or time is defined; and, for meta-agents and hierarchies, coarse states, partitions and comparison geometries declared before the analysis (Assumption A12). The results behind H1 to H4 that are stated for other models include the code game of Section 6, whose states are discrete, the deliberation model of Theorems 5.10 and 5.11, whose policy sets are finite, the mean-field map of Theorem 7.6, whose attention is endogenous, and the Séguy curves, which in the leaky regime read the signs of Gaussian fields (Proposition S10.13). Readings of social-science findings, and of the worked examples of firms, people and cells, are labeled as interpretations or conjectures (Section 10 and Remarks S11.12 and S14.20).

### S1.9 Relation to the author's gauge-theoretic program

The framework is distinct from the author's gauge-theoretic manuscripts (Dennis, 2026c,d) and does not take them as a foundation. Identity encodings recover standard active inference, and frames are optional private encodings: inference is frame covariant, encodings enter only through effective codes, and frames are identifiable only modulo the stabilizer of the public code (Propositions 2.11 to S2.13). Frames matter only through communication, where learned translations can carry holonomy around cycles (Definition 2.10 and Theorem 6.5). Results imported from that program are cited where they are used: the update subspace of a rank-limited message and the measure of meaning lost in reception (Section 2.3), the holonomy criterion on strongly connected graphs, a frustration corollary and the tangent learning model of translations (Theorem 6.5 and Proposition 6.6 and Section 6.3), from Dennis (2026c); the distinctions among a small memory residual, statistical sufficiency and a slow retained spectrum in closure, and the lumpability form of the block-elimination identity (Section 5.1 and Remark S5.1), from Dennis (2026d); and, from the author's formation document

(Dennis, 2026b), the criterion of its formation standard by which a selected structure differs from a null of type D and its characterization of the Gaussian location family under common-message dynamics (Definition 5.17 and Remark S5.7). The extension behind hypothesis H5 imports no result from that program; the program's top-down feedback, in which, as secondary descriptions of those documents state it, a child's prior is set from its parent's belief with a blend factor, is compared with literal top-down reading in Remark S12.11.

The minimal model of the author's report of November 7, 2026 (Dennis, 2026a) is the linear-Gaussian special case of the framework with private linear encodings, rank-limited messages, attention selected by expected free energy and law-header receivers, which read at level 1. That report obstructed, without refuting, the hypothesis that communicating agents condense into interior agents surrounded by a blanket of agents, and Section 9 places its model and results in the framework. The light cone for linear updates, exponential clustering, the Ornstein–Zernike rate on the ring and the double counting of the common innovation by header-reading receivers were first obtained in that report and are proved here in full (Theorems 8.5, 8.9 and 8.10 and Proposition S3.6).

## S2 Additional results for Section 2: the framework

This section collects statements, examples and remarks of Section 2 that the main text cites without stating in full. Proofs are in Section S26.

### S2.1 Two receivers of one message

Because the lift $O_{ij}$ of (5) uses the receiver's prior, two receivers read one message differently.

**Proposition S2.1** (Two receivers, one message)**.** *Receivers $i$ and $k$ have priors $\mathcal{N}(m_i, P_i)$ and $\mathcal{N}(m_k, P_k)$ and read one message $y$ with the same likelihood $\mathcal{N}(\hat{H}\eta, \hat{R})$. Write $q_i^+$ and $q_k^+$ for their posteriors, $S_i, S_k$ and $\mathcal{K}_i, \mathcal{K}_k$ for their innovation covariances and gains, and $\Lambda_k^+ = (P_k^+)^{-1}$.*

*(a)* $m_i^+ - m_k^+ = (I - \mathcal{K}_i\hat{H})(m_i - m_k) + (\mathcal{K}_i - \mathcal{K}_k)(y - \hat{H}m_k)$.

*(b)* $\mathcal{K}_i = \mathcal{K}_k$ *if and only if* $P_i\hat{H}^\top = P_k\hat{H}^\top$.

*(c) If* $P_i = P_k$, *then* $D_{\mathrm{KL}}(q_i^+\|q_k^+) = D_{\mathrm{KL}}(q_i\|q_k) - \frac{1}{2}(m_i - m_k)^\top\hat{H}^\top S_i^{-1}\hat{H}(m_i - m_k)$ *for every* $y$.

*(d) For any priors and any common likelihood with* $D_{\mathrm{KL}}(q_i\|q_k) < \infty$,

$$\mathbb{E}_{q_i(y)} D_{\mathrm{KL}}(q_i^+\|q_k^+) = D_{\mathrm{KL}}(q_i\|q_k) - D_{\mathrm{KL}}(q_i(y)\|q_k(y)),$$

*where $q_i(y)$ and $q_k(y)$ are the two receivers' predictive laws of the message.*

*(e) The decoded contents are* $\hat{y}_a^\circ = y - \hat{R}S_a^{-1}(y - \hat{H}m_a)$, $a \in \{i, k\}$. *If $\hat{H}$ has full row rank, then as $\hat{R} \to 0$, $\hat{H}m_i^+$ and $\hat{H}m_k^+$ tend to $y$, and in the limit* $m_i^+ - m_k^+ \in \ker\hat{H}$.

Receivers whose priors give the coded coordinates the same covariance with the whole state use the same gain, and receivers with equal prior covariances lose exactly the coded part of their disagreement, measured in the metric of the predicted message. In general the expected disagreement falls by the divergence between the receivers' predictions of the message, so receivers who predict a message alike are not brought closer on average. They disagree about what was said only through the channel noise, and about what it means through their prior lifts. Proof in Section S26; checked in Check S47.3. A message can even create disagreement where the means agreed.

**Example S2.2** (A message that creates disagreement)**.** Let $K = 2$, $\hat{H} = (1, 0)$, $\hat{R} = \sigma^2$, $m_i = m_k = 0$, and let the priors have unit variances and correlations $+\varrho$ and $-\varrho$ with $0 < \varrho < 1$. The receivers agree in their means and differ only in how they associate the uncoded coordinate with the coded one, and $D_{\mathrm{KL}}(q_i \| q_k) = 2\varrho^2/(1-\varrho^2)$. After a message $y$, $m_i^+ - m_k^+ = (0, 2\varrho y/(1+\sigma^2))$, the posterior covariances are mirror images with equal determinants, and

$$D_{\mathrm{KL}}(q_i^+ \| q_k^+) = \frac{2\varrho^2}{1-\varrho^2}\Big(\frac{\sigma^2}{1+\sigma^2} + \frac{y^2}{(1+\sigma^2)^2}\Big). \tag{S1}$$

The message leaves the receivers further apart than before if and only if $|y| > \sqrt{1+\sigma^2}$, that is, whenever it deviates from their common prediction $y \sim \mathcal{N}(0, 1+\sigma^2)$ by more than one predictive standard deviation, for every $\varrho$. At $\varrho = 0.8$ and $\sigma^2 = 0.5$ the prior divergence is 3.556, the divergence after $y = 0$ is 1.185 and the coefficient of $y^2$ is 1.580, so every message with $|y| > 1.225$ separates the receivers. If $y$ has second moment $v$ under its true law, the expected divergence is $D_{\mathrm{KL}}(q_i \| q_k) + 2\varrho^2(v - 1 - \sigma^2)/((1-\varrho^2)(1+\sigma^2)^2)$.

A stream of messages more variable than both receivers expect drives them apart on average. Read socially, and as an interpretation only, communities that agree on every coded quantity but read it into an uncoded one with opposite associations are polarized by news. Computation in Section S26; checked in Check S47.3.

## S2.2 Recursive and ungrounded learning of codes

The setting and notation are those of Section 2.5: messages $y_t = H\phi_t + \nu_t$, the receiver's regressor $\hat{\phi}_t$, its second moment $M_2$ and the orthogonal projector $\mathcal{P}_M$ onto $\mathrm{col}(M_2)$.

**Proposition S2.3** (A recursive learner keeps its prior where the data are silent)**.** *Let* $\mathrm{vec}H \sim \mathcal{N}(\mathrm{vec}H_0, P_0^H)$ *a priori, let* $R_w \succ 0$ *be a working noise and* $X_t = \hat{\phi}_t^\top \otimes I_{r_c}$*, and run, from* $\hat{h}_0 = \mathrm{vec}H_0$*,*

$$\begin{aligned} \hat{h}_{t+1} &= \hat{h}_t + P_t^H X_t^\top (X_t P_t^H X_t^\top + R_w)^{-1}(y_t - X_t \hat{h}_t), \\ P_{t+1}^H &= P_t^H - P_t^H X_t^\top (X_t P_t^H X_t^\top + R_w)^{-1} X_t P_t^H. \end{aligned} \tag{S2}$$

*(a)* $(\hat{h}_T, P_T^H)$ *are the posterior mean and covariance of the Gaussian linear regression model; under exact joint attention with an exactly observing receiver and* $R_w = R_c$ *they are the exact posterior of the code.*

*(b) If* $P_0^H$ *is a multiple of the identity, the posterior of* $H(I - \mathcal{P}_M)$ *equals its prior for every* $T$*.*

*(c) Under the hypotheses of (b) and of Proposition 2.7,* $\hat{h}_T \to \mathrm{vec}(H\mathcal{P}_M + H_0(I - \mathcal{P}_M))$ *almost surely.*

*(d) If, in addition, the rounds are independent and identically distributed and* $\mathrm{Cov}(\phi_t \mid \mathcal{Z}_t)$ *does not depend on* $\mathcal{Z}_t$*, then* $T^{1/2}((\mathcal{P}_M \otimes I_{r_c})\hat{h}_T - \mathrm{vec}(H\mathcal{P}_M))$ *converges in law to* $\mathcal{N}(0, M_2^+ \otimes R_{\mathrm{res}})$*, with* $R_{\mathrm{res}} = H\,\mathrm{Cov}(\phi_t \mid \mathcal{Z}_t)H^\top + R_c$*, whatever the working noise.*

Learning a code from data confined to a subspace keeps the prior on the rest, which is prior completion in code space. For dependent regressors least squares is strongly consistent under the eigenvalue conditions of Lai and Wei (1982). Proof in Section S26; checked in Check S47.4.

**Proposition S2.4** (Ungrounded learning measures how differently agents are informed)**.** *Keep the hypotheses of Proposition 2.7(a) except content calibration, and let* $C_{\phi\hat{\phi}} = \mathbb{E}[\phi_t \hat{\phi}_t^\top]$*.*

(a) *The minimum-norm least-squares estimate converges to* $HC_{\phi\hat{\phi}}M_2^+$*, the regression of messages on the receiver's beliefs. It equals the grounded limit* $H\mathcal{P}_M$ *if and only if* $H\mathbb{E}[(\phi_t - \hat{\phi}_t)\hat{\phi}_t^\top] = 0$*, which content calibration guarantees.*

(b) *Let both agents observe a state* $\eta \sim \mathcal{N}(0, P_0)$*, drawn afresh each round, with sensing noises of covariances* $R_i^s$*,* $R_j^s$ *and cross-covariance* $R_{ji}^s = \mathrm{Cov}(v_j, v_i)$*; let the sender send its posterior mean* $\phi = \mathcal{K}_j^s(\eta + v_j)$ *and the receiver regress on its own,* $\hat{\phi} = \mathcal{K}_i^s(\eta + v_i)$*, with* $\mathcal{K}_a^s = P_0(P_0 + R_a^s)^{-1}$ *invertible. The limit is* $H\mathcal{K}_j^s(P_0 + R_{ji}^s)P_0^{-1}$*, which does not depend on the receiver's noise. For scalars it is* $Hk_j^s(1 + R_{ji}^s/P_0)$ *with* $k_j^s = P_0/(P_0 + R_j^s)$*, and the bias is* $H(R_{ji}^s - R_j^s)/(P_0 + R_j^s)$*.*

(c) *Let each round be grounded with probability* $p_g \in (0, 1]$*, independently of everything else, with an exact public referent of second moment* $M_g \succ 0$ *as content and regressor, and ungrounded otherwise, with* $C_{\phi\hat{\phi}}$ *and* $M_2$ *those of the ungrounded rounds. Pooled least squares converges to* $H[p_gM_g + (1 - p_g)C_{\phi\hat{\phi}}][p_gM_g + (1 - p_g)M_2]^{-1}$*, and least squares on the grounded rounds alone converges to* $H$*.*

With independent noises the receiver learns the code attenuated by the sender's shrinkage, $H\mathcal{K}_j^s$, which is also the slope of $\mathbb{E}[y_t \mid \eta]$: the ungrounded learner gets the level-0 likelihood right and the code wrong, which matters when the code is used to attribute beliefs to the sender or to score being understood. With identical data the bias vanishes. The pooled bias is $(1 - p_g)H(C_{\phi\hat{\phi}} - M_2)[p_gM_g + (1 - p_g)M_2]^{-1}$, so it vanishes linearly as $p_g \to 1$, and a receiver that can tell grounded rounds from the others is consistent for every $p_g > 0$. Proof in Section S26; checked in Check S47.4.

### S2.3 Records and performative acts

The record model and its notation are those of Definition 2.9.

**Proposition S2.5** (Latency, delay and optimal persistence of a record)**.** *Consider one record with write weight* $\omega$ *and decay* $\delta$*,* $0 < \omega + \delta \leq 1$*.*

(a) *A write at round* $s$ *reaches a reader at rounds* $t \geq s + 1$ *with weight proportional to* $(1 - \omega - \delta)^{t-s-1}$*: the latency is one round, the normalized mean delay is* $1/(\omega + \delta)$ *and the decay time is* $-1/\ln(1 - \omega - \delta)$*.*

(b) *If readers attend the record independently with probability* $p_r$ *per round, the delay to the first reading is geometric with mean* $1/p_r$*, whatever the writer does next.*

(c) *A record read by several agents is a hub: two readers that never attend each other are two hops apart through it.*

(d) *Let* $\eta_{t+1} = F\eta_t + w_t$ *be a scalar world with* $0 < F < 1$ *and stationary variance* $P_0$*, let the written values be* $u_t = \eta_t + v_t^w$ *with noise variance* $R^w$ *and* $\mathrm{snr} = P_0/R^w$*, and let the record average them,* $z(t + 1) = (1 - \omega)z(t) + \omega u_t$*. Then*

$$\mathrm{Corr}^2(z_t, \eta_t) = \frac{F^2(1 - (1 - \omega)^2)}{1 - (1 - \omega)^2F^2 + \big(1 - (1 - \omega)F\big)^2/\mathrm{snr}}, \tag{S3}$$

*which is maximized at* $1 - \omega^* = (A - \sqrt{A^2 - 4F^2})/(2F) \in (0, F)$*, with* $A = 1 + F^2 + \mathrm{snr}(1 - F^2)$*. Equivalently* $\omega^* = 1 - F(1 - k^*)$*, with* $k^*$ *the steady-state Kalman gain for observing* $\eta$ *through the writes; at* $\omega^*$ *the record is proportional to the Kalman one-step predictor and* $\mathrm{Corr}^2(z_t, \eta_t) = 1 - P^-/P_0$*, with* $P^-$ *the steady-state predictive variance.*

Stigmergy decouples influence from co-presence, and a widely read record changes the graph on which distances, and hence correlation lengths, are measured. The optimal write weight is the leak $1 - F(1 - k^*)$ of the optimal predictor (Kalman, 1960): a record that forgets faster passes on write noise, and one that forgets more slowly keeps stale content. Proof in Section S26; checked in Check S47.5.

A performative act changes the world rather than the receiver's information about it, and reading it as a message is an error of a specific kind.

**Proposition S2.6** (Performative acts are interventions)**.** *Let receiver $i$ hold $\mathcal{N}(m_i, P_i)$ about $\eta_t$, and let $j$ announce an act $u$ that enters $\eta_{t+1} = F\eta_t + B_j^\eta u + w_t$ and is chosen independently of $\eta_t$ given $i$'s information.*

*(a) Read correctly, the act changes $i$'s prediction to $Fm_i + B_j^\eta u$ with covariance $FP_iF^\top + Q$ and carries no information about $\eta_t$. Read as a level-0 message $u = \hat{H}\eta_t + \nu$, it shifts $i$'s belief about $\eta_t$ by $\mathcal{K}(u - \hat{H}m_i)$, lowers its covariance by $\mathcal{K}\hat{H}P_i$ and omits $B_j^\eta u$, so the predicted mean is off by $F\mathcal{K}(u - \hat{H}m_i) - B_j^\eta u$.*

*(b) An act set from beliefs, $u = H_j^{\mathrm{a}} m_j$, as a price set from a valuation, is both a message with code $H_j^{\mathrm{a}}$ and an intervention, and the correct receiver uses both terms.*

*(c) An informative message changes $i$'s policy posterior only through its belief about the external states. A performative act changes the predicted outcomes $q_i(o \mid \pi)$ through the world's dynamics, and so changes the pragmatic value of $i$'s policies whenever it moves outcomes that enter $i$'s preferences, even when $i$'s beliefs about the act-independent world are unchanged.*

Performative acts coordinate by changing what the others' actions lead to, informative messages by changing what the others believe; mistaking one for the other is the confusion of intervention with observation (Pearl, 2009). What a receiver can learn about the sender's beliefs and about the world from acts that are both, such as prices set from valuations, is open. Proof in Section S26.

### S2.4 Expected free energy of communication

Attention, addressing and content are policies, selected by their expected free energy; attention in particular sets the couplings of the networks studied in the later sections. A one-round policy $\pi = (\mathcal{J}, \mathcal{A}, c, u)$ has one outcome slot per attended source, one response slot per addressee (an echo or a reply) and, for performative acts, the affected preference-relevant outcomes. The hidden states are $(\eta, \alpha)$, where $\alpha$ collects the parameters of the agent's partner models.

**Proposition S2.7** (Expected free energy of attention, addressing and content)**.** *Let the preference factorize over slots. Then*

$$G(\pi) = -I_q\big((\eta, \alpha); o^\pi\big) - \mathbb{E}_{q(o^\pi \mid \pi)} \ln \tilde{p}(o^\pi) = D_{\mathrm{KL}}\big(q(o^\pi \mid \pi) \| \tilde{p}(o^\pi)\big) + \mathbb{E}_{q((\eta,\alpha)\mid\pi)} \mathbb{H}\big[p(o^\pi \mid \eta, \alpha, \pi)\big],$$

*the equality requiring that the information term use the exact posterior of the agent's own model. For linear-Gaussian slots:*

*(a) Attention. With known code and noise models, a preference $\mathcal{N}(0, \Psi_{\mathrm{al}})$ over each attended message's innovation, an atom $p^\varnothing$ at the empty slot and no preference over world channels,*

$$G_i(\mathcal{J}) = -I_i(\eta; y_{\mathcal{J}}) + \sum_{j \in \mathcal{J}} \Big[\tfrac{1}{2} \operatorname{tr}\big(\Psi_{\mathrm{al}}^{-1} S_{ij}\big) + g_{\mathrm{al}}\Big] + \text{const},$$

*where $I_i$ is the mutual information and $S_{ij}$ the predictive covariance of $y_{ij}$ under $i$'s model of the messages at its interpretation level: $S_{ij} = \hat{H}_{ij} P_i \hat{H}_{ij}^\top + \hat{R}_{ij}$ at level 0, $S_{ij} = \hat{H}_{ij} P_i \hat{H}_{ij}^\top + \hat{R}^1_{ij}$ at level 1 and $S_{ij} = \mathrm{Cov}(y_{ij} \mid \mathcal{Z}_i)$ under exact fusion; and $g_{\mathrm{al}} = \frac{1}{2} \ln\det(2\pi\Psi_{\mathrm{al}}) + \ln(p^{\varnothing}/(1-p^{\varnothing}))$.*

(b) *Addressing. If sender $j$ predicts the response of addressee $i$ as $\mathcal{N}(\bar{o}_{ji}, S^{\mathrm{tx}}_{ji})$ under its partner model, with conditional noise $R^{\mathrm{tx}}_{ji}$, and prefers $\mathcal{N}(0, \Psi_{\mathrm{tx}})$, then*

$$G^{\mathrm{tx}}_j(\mathcal{A}, c, u) - G^{\mathrm{tx}}_j(\textit{silence}) = \sum_{i \in \mathcal{A}} \Big[ \tfrac{1}{2} \bar{o}_{ji}^\top \Psi_{\mathrm{tx}}^{-1} \bar{o}_{ji} + \tfrac{1}{2} \operatorname{tr}(\Psi_{\mathrm{tx}}^{-1} S^{\mathrm{tx}}_{ji}) - \tfrac{1}{2} \ln\det\big(S^{\mathrm{tx}}_{ji} (R^{\mathrm{tx}}_{ji})^{-1}\big) + g_{\mathrm{tx}} \Big],$$

*with $g_{\mathrm{tx}} = \frac{1}{2} \ln\det(2\pi\Psi_{\mathrm{tx}}) + \ln(p^{\varnothing}/(1-p^{\varnothing})) + g_{\mathrm{use}}$ and $g_{\mathrm{use}} \geq 0$ a cost of use per addressee.*

(c) *When two slots of one policy depend on a common hidden variable and are conditionally independent given it, as listening to $i$ and addressing $i$ both inform the model of $i$, their joint information is at most the sum of their informations.*

The mutual information in (a) is the believed value of attention, $\mathcal{V}^0_i(\mathcal{J})$ at level 0 and $\mathcal{V}^1_i(\mathcal{J})$ at level 1 in the notation of Definition 3.1, which Proposition S3.1 gives in closed form; under exact fusion it is the true value $\mathcal{V}^*_i(\mathcal{J})$. For one attended message the map $S \mapsto \frac{1}{2} \operatorname{tr}(\Psi_{\mathrm{al}}^{-1} S) - \frac{1}{2} \ln\det S$ is strictly convex with minimum at $S = \Psi_{\mathrm{al}}$, so the preference to predict attended messages balances messages the receiver cannot yet predict against messages it can. At levels 0 and 1 this term contains no sender mean; under exact fusion $S_{ij}$ falls as the sender's error correlates with the receiver's, so the preference pulls toward correlated senders. Proof in Section S26.

For world channels, over which (a) places no preference, the epistemic term is information about the present state, so it carries no horizon: for linear sensors under a binding budget on the information they carry, it takes the same value at every sensor that spends the budget, whatever the sensor reads (Remark S14.3). Selection by predictive information needs a horizon in the epistemic term or a pragmatic term, and the horizon-$h$ attention objective of Definition 14.5 has both, with a contingent act; Theorem 14.6 gives the sensors it selects, and Remark S14.4 computes the literal two-step expected free energy of sensing and then acting.

The author's preference to predict and to be predicted, which echoes the generalized synchrony of Friston and Frith (2015a), has two admissible formalizations as prior preferences of a sender $j$ over an outcome it observes after speaking to $i$. The receiver's reading of a message is its noiseless lift $\hat{m}_{ij} = m_i + O_{ij}(y_{ij} - \hat{H}_{ij} m_i)$, the belief it attributes to the sender, which satisfies $\hat{H}_{ij} \hat{m}_{ij} = y_{ij}$. Being understood is a preference $\mathcal{N}(0, \Psi_{\mathrm{u}})$ over the understanding error $\psi_{ij} = H_j \hat{m}_{ij} - u_j$, the discrepancy between the reading re-encoded with the true code and what was sent, which the sender receives as an echo or infers from the receiver's behavior. Being predicted is a preference $\mathcal{N}(0, \Psi_{\mathrm{p}})$ over the receiver's prediction error $\varphi_{ij} = y_{ij} - \hat{H}_{ij} m_i$, observed as a reply.

**Proposition S2.8** (Anatomy of the two errors)**.** *With $u_j = H_j m_j + h_j$ and $y_{ij} = u_j + \nu_{ij}$,*

$$\psi_{ij} = \nu_{ij} - (\hat{H}_{ij} - H_j)\hat{m}_{ij}, \qquad \varphi_{ij} = H_j(m_j - m_i) + h_j - (\hat{H}_{ij} - H_j) m_i + \nu_{ij}. \tag{S4}$$

*At correct code models, which identity encodings with a shared public code provide without any learning, $\psi_{ij} = \nu_{ij}$: being understood is satisfied by every content, honest or not, and is blind to agreement, whereas $\varphi_{ij}$ contains the coded disagreement $H_j(m_j - m_i)$. If the reading were the posterior mean $m_i^+$ instead of the lift, the understanding error at correct codes would be $\hat{R} S^{-1}(H_j m_i - u_j) + (I - \hat{R} S^{-1})\nu_{ij}$, which is free of beliefs only as the channel becomes noiseless.*

At correct code models the map from sent to read content is the identity and the echo is pure noise, so $\mathbb{E}[\psi_{ij} \psi_{ij}^\top] = R_c$ for all beliefs and every content: being understood reduces to being

heard. With uncertain codes the excess $-(\hat{H}_{ij} - H_j)\hat{m}_{ij}$ applies the code error to a reading that itself depends on the code model. Under Gaussian code error its second moment is finite only for $K \geq r_c + 2$ (Theorem S22.1), and where it is finite its closed form is open (Open problem S2.9); in the computed instances the term $\sigma_H^2 \mathbb{E}\|\hat{m}_{ij}\|^2 I$ with the reading at the true code carried between 69 and 98 percent of the diagonal excess of the second moment, truncated in the instance with $K = r_c + 1$ where the full moment is infinite, a share that depends on the instance (Check S47.6 and Remark S43.4). Being predicted keeps its content: the reply is the coded disagreement plus noise. Proof in Section S26; checked in Check S47.6.

**Open problem S2.9** (Understanding risk under code uncertainty)**.** Let $\mathrm{vec}(\hat{H}_{ij} - H_j) \sim \mathcal{N}(0, \sigma_H^2 I)$. Under the hypotheses of Theorem S22.1 the moment $\mathbb{E}\|\psi_{ij}\|^p$ of the understanding error is finite if and only if $p < K - r_c + 1$, so the second moment exists only for $K \geq r_c + 2$. For $K \geq r_c + 2$, find $\mathbb{E}[\psi_{ij}\psi_{ij}^\top]$ in closed form, including the terms that come from the dependence of the reading $\hat{m}_{ij}$ on the code model; Proposition S22.2 gives it to order $\sigma_H^2$. For $K \leq r_c + 1$, find instead the moment truncated to the event $\|\hat{H}_{ij} - H_j\|_F \leq \rho$ with $\rho$ below the smallest singular value of $H_j$, or the second moment when the reading uses the regularized gain $P_i\hat{H}_{ij}^\top(\hat{H}_{ij}P_i\hat{H}_{ij}^\top + \hat{R})^{-1}$ with a fixed $\hat{R} \succ 0$ in place of $O_{ij}$; both are finite when $m_i$ and $u_j$ have finite second moments.

**Proposition S2.10** (Bounded-confidence addressing and pandering)**.** *Let sender $j$'s partner model be $m_i \sim \mathcal{N}(\bar{m}_{ji}, \bar{P}_{ji})$ and let $\mathrm{vec}(\hat{H}_{ij} - H_j) \sim \mathcal{N}(0, \sigma_H^2 I)$, independent of each other and of $\nu_{ij}$, with content $u_j = H_j m_j + h_j$, and write $S_{ji} = H_j \bar{P}_{ji} H_j^\top + \sigma_H^2(\|\bar{m}_{ji}\|^2 + \mathrm{tr}\,\bar{P}_{ji})I + R_c$.*

*(a)* $\mathbb{E}[\varphi_{ij}\varphi_{ij}^\top] = \big(H_j(m_j - \bar{m}_{ji}) + h_j\big)\big(H_j(m_j - \bar{m}_{ji}) + h_j\big)^\top + S_{ji}$.

*(b) Let $I_{ji}^{\mathrm{tx}} := \mathbb{H}[\varphi_{ij}] - \frac{1}{2}\ln\det(2\pi e R_c)$ be the information that the reply to honest content carries about $(m_i, \hat{H}_{ij})$ under $j$'s model, with $\mathbb{H}[\varphi_{ij}]$ the entropy of the predictive law of the reply. It does not depend on $m_j$, and $I_{ji}^{\mathrm{tx}} \leq \frac{1}{2}\ln\det(S_{ji}R_c^{-1})$ with equality when that law is Gaussian, as it is at $\sigma_H^2 = 0$. At $\gamma \to \infty$ and with honest content, being predicted makes $j$ address $i$ if and only if*

$$\tfrac{1}{2}\big\|H_j(m_j - \bar{m}_{ji})\big\|^2_{\Psi_{\mathrm{p}}^{-1}} < \Xi_{ji}^{\mathrm{ex}} := -\tfrac{1}{2}\,\mathrm{tr}\big(\Psi_{\mathrm{p}}^{-1}S_{ji}\big) + I_{ji}^{\mathrm{tx}} - g_{\mathrm{tx}},$$
$$\Xi_{ji}^{\mathrm{ex}} \leq \Xi_{ji} := -\tfrac{1}{2}\,\mathrm{tr}\big(\Psi_{\mathrm{p}}^{-1}S_{ji}\big) + \tfrac{1}{2}\ln\det\big(S_{ji}R_c^{-1}\big) - g_{\mathrm{tx}}.$$

*At $\sigma_H^2 = 0$ the two thresholds coincide and $j$ addresses $i$ if and only if the left side is below $\Xi_{ji}$. For $\sigma_H^2 > 0$ the reply is a mixture of Gaussians over the partner's belief, the bound $\Xi_{ji}$ is the threshold of the Gaussian with the moments of (a), and the inequality with $\Xi_{ji}$ is necessary for addressing and can fail to be sufficient (Check S47.6). In both cases the log-odds of addressing $i$ against silence at finite $\gamma$ fall linearly, with slope $\gamma/2$, in the squared believed coded disagreement $\|H_j(m_j - \bar{m}_{ji})\|^2_{\Psi_{\mathrm{p}}^{-1}}$. Under being understood with known codes the value of addressing does not depend on beliefs.*

*(c) With an honesty habit $E(h_j) \propto \exp(-\frac{1}{2}h_j^\top \Psi_{\mathrm{h}}^{-1} h_j)$, the most probable content under being predicted is*

$$u_j^* = (I - \Gamma)H_j m_j + \Gamma H_j \bar{m}_{ji}, \qquad \Gamma = \big(\gamma\Psi_{\mathrm{p}}^{-1} + \Psi_{\mathrm{h}}^{-1}\big)^{-1}\gamma\Psi_{\mathrm{p}}^{-1}.$$

*Under being understood at known codes the expected free energy does not depend on $h_j$, and the habit selects honest content.*

(d) *If $j$ first hears $i$ and updates its partner model to $(\bar{m}^+_{ji}, \bar{P}^+_{ji})$ while its own belief $m_j$ is held fixed, then under $j$'s partner model $\mathbb{E}[(m_j - \bar{m}^+_{ji})(m_j - \bar{m}^+_{ji})^\top + \bar{P}^+_{ji}] = (m_j - \bar{m}_{ji})(m_j - \bar{m}_{ji})^\top + \bar{P}_{ji}$ and $\mathbb{E}\|\bar{m}^+_{ji}\|^2 + \operatorname{tr}\bar{P}^+_{ji} = \|\bar{m}_{ji}\|^2 + \operatorname{tr}\bar{P}_{ji}$, so the partner-model update alone leaves the expected being-predicted risk of addressing $i$ unchanged. If the message is $y_{ji} = H_i m_i + \nu_{ji}$ with known code $H_i$ and $\nu_{ji} \sim \mathcal{N}(0, R_c)$, the partner model is updated with the gain $L^{\mathrm{p}}_{ji} = \bar{P}_{ji} H_i^\top (S^{\mathrm{p}}_{ji})^{-1}$, $S^{\mathrm{p}}_{ji} = H_i \bar{P}_{ji} H_i^\top + R_c$, and $j$ also updates its own belief from the same message, $m^+_j = m_j + L_j(y_{ji} - H_i m_j)$ with any gain $L_j$, then with $d_{ji} = m_j - \bar{m}_{ji}$,*

$$\mathbb{E}[(m^+_j - \bar{m}^+_{ji})(m^+_j - \bar{m}^+_{ji})^\top + \bar{P}^+_{ji}] = (I - L_j H_i) d_{ji} d_{ji}^\top (I - L_j H_i)^\top + L_j S^{\mathrm{p}}_{ji} L_j^\top - L_j H_i \bar{P}_{ji} - \bar{P}_{ji} H_i^\top L_j^\top + \bar{P}_{ji},$$

*so by (a) the expected being-predicted risk of addressing $i$ with honest content changes by $\frac{1}{2}\operatorname{tr}(\Psi_{\mathrm{p}}^{-1} H_j \Delta_{ji} H_j^\top)$, with $\Delta_{ji}$ the right side minus $d_{ji} d_{ji}^\top + \bar{P}_{ji}$; this change vanishes at $L_j = 0$, and more generally whenever $H_j L_j = 0$, and for $H_j \neq 0$ it is nonzero for every gain outside a proper algebraic subset of the gains. For scalar beliefs with unit codes the left side changes by $-L_j(2 - L_j) d^2_{ji} + L_j(L_j - 2L^{\mathrm{p}}_{ji})(\bar{P}_{ji} + R_c)$, which is negative when $0 < L_j < 2L^{\mathrm{p}}_{ji}$ and positive when $d_{ji} = 0$ and $L_j > 2L^{\mathrm{p}}_{ji}$.*

Senders who want to be predicted address the partners they believe already agree with them, the rule of bounded-confidence opinion models (Deffuant et al., 2000; Hegselmann and Krause, 2002) applied on the sender's side, and they shade their content toward what partners expect. Senders who want to be understood choose partners for a shared code, not for agreement. The partner-model update produces no risk-based reciprocity between listening to and addressing the same partner. The listener's own belief update changes the risk of addressing that partner in the next round by the amount in (d), which is nonzero for generic gains: in the scalar case it lowers the risk, by reducing the believed coded disagreement, whenever the listener weights the message positively and less than twice as heavily in its own belief as in its partner model, and it can raise the risk otherwise. Whether a horizon-two expected free energy, which can value listening before addressing within one policy, produces reciprocity is open. The two readings of the alignment preference therefore select different communication graphs from one population. Proof in Section S26; checked in Check S47.6.

**Conjecture S2.11** (Bounded-confidence fragmentation)**.** *In populations of being-predicted senders whose partner models are learned from replies, the addressing graph converges to a union of belief clusters whose number decreases as the thresholds $\Xi^{\mathrm{ex}}_{ji}$ grow, as in bounded-confidence dynamics (Hegselmann and Krause, 2002), while populations of being-understood senders with shared codes keep a connected addressing graph.*

The conjecture is untested. Since $\Xi^{\mathrm{ex}}_{ji}$, like its Gaussian bound $\Xi_{ji}$, depends on the partner covariance $\bar{P}_{ji}$, which contact reduces, the correspondence with bounded confidence is exact only for frozen partner models.

## S2.5 Frames

Frames $U_i$, the public code $T_c$ and the effective codes $H_j = T_c U_j$ are as in Sections 2.2 and 2.7.

**Proposition S2.12** (Inference is frame covariant)**.** *(a) The Kalman filter run in internal coordinates returns $(U_i m_i, U_i P_i U_i^\top)$ at every step, where $(m_i, P_i)$ is the world-coordinate filter, so the decoded belief $U_i^{-1}\mu_i$ does not depend on the frame.*

(b) *Under a redescription $U_i \mapsto GU_i$ together with $i$'s own internal code $T_c^{(i)} \mapsto T_c^{(i)}G^{-1}$, every message, every belief in world coordinates, every information quantity and every expected free energy is unchanged; an agent-specific internal code therefore makes the frame unobservable.*

(c) *Under a change of world coordinates $\eta \mapsto \mathcal{T}\eta$, beliefs transform covariantly ($m_i \mapsto \mathcal{T}m_i$, $P_i \mapsto \mathcal{T}P_i\mathcal{T}^\top$) and codes as $H \mapsto H\mathcal{T}^{-1}$; messages, mutual informations, divergences between beliefs, Mahalanobis risks and expected free energies over sensory outcomes are invariant.*

(d) *Under a change of public coordinates $y \mapsto By$, $B \in \mathrm{GL}(r_c)$, with $T_c \mapsto BT_c$, $R_c \mapsto BR_cB^\top$ and every $\hat{H}_{ij} \mapsto B\hat{H}_{ij}$, decoded beliefs and informations are invariant. Expected free energy over messages is not: with the preference transformed as a density, each attended message shifts it by $\ln|\det B|$, so comparisons between attention sets with different numbers of messages depend on the public coordinates.*

Frames therefore have observable effects only through communication. With a shared public code, the frame-dependent quantities are the effective codes, the receivers' models of them and the translations between internal coordinates, and they are physical: senders with different frames mean different things by one signal. Proof in Section S26; checked in Check S47.7.

**Proposition S2.13** (Frames are identifiable only modulo the stabilizer of the code)**.** *Messages in modality $c$ depend on $U_j$ only through $T_cU_j$: encodings $U_j$ and $U_j'$ with $T_c(U_j' - U_j) = 0$ give the same conditional law of messages given the sender's belief, and when that belief has a positive-definite second moment, encodings with $T_c(U_j' - U_j) \neq 0$ give different conditional laws. The set $\{X \in \mathbb{R}^{K\times K} : T_cX = 0\}$ has dimension $K(K - r_c)$. With several modalities the equivalence is modulo $\{X : T_cX = 0$ for every $c \in \mathcal{M}\}$, of dimension $K(K - \mathrm{rank}[T_c]_{c\in\mathcal{M}})$, where $[T_c]_{c\in\mathcal{M}}$ stacks the codes. Mismatches in this set cost nothing in reception.*

The count agrees with that for $\mathrm{GL}(K)$ dictionaries in Dennis (2026a), and pooling modalities identifies more of a sender's frame, as in Dennis (2026c). Proof in Section S26; checked in Check S47.7. Outside the stabilizer a misidentified code has a cost, which Proposition S26.1 in Section S26 gives in closed form. In particular, for a receiver with prior mean $m_i = 0$ and error measured in the prior Mahalanobis norm $\|\cdot\|_{P_i^{-1}}$, if a misidentified frame rotates a rank-one code at fixed $P_i$-norm by a $P_i$-angle $\vartheta$, listening is worse than silence exactly when $\vartheta > 60°$, whatever the channel noise (Proposition S26.1(c)); with a nonzero prior mean or another weight the threshold angle depends on the instance.

## S2.6 Level-0 network

The remark concerns the scalar level-0 network of Definition 2.12.

*Remark* S2.14 (Scalar level-0 gains are convex weights). A scalar literal receiver with claimed prior variance $P_i$, sensing noise $R_i^s$ and assigned message noises $\hat{R}_{ij}$ has claimed posterior variance $P_i^+ = \left(P_i^{-1} + (H_i^s)^2/R_i^s + \sum_j \hat{H}_{ij}^2/\hat{R}_{ij}\right)^{-1}$ and gains $k_i^s = P_i^+(H_i^s)^2/R_i^s$ and $k_{ij} = P_i^+\hat{H}_{ij}^2/\hat{R}_{ij}$, so that $k_i^s + \sum_j k_{ij} = 1 - P_i^+/P_i < 1$. With exact code models and $F \geq 0$, every attention schedule therefore gives $W \geq 0$ entrywise with row sums $F(1 - k_i^s)$, which yields the row-sum contraction of Section 8 whenever $F < 1$ (Proposition 8.3(a)). Column sums are not controlled: a broadcaster's column sum grows with its audience.

## S3 Additional results for Section 3: misreading joint structure

This section collects statements, examples and remarks of Section 3 that the main text cites without stating in full. Proofs are in Section S27.

### S3.1 Believed values

**Proposition S3.1** (Believed values)**.** *Let the receiver's prior be $\mathcal{N}(m_i, P_i)$, and let $\Gamma^0_{ij} = \hat{H}_{ij}^\top \hat{R}_{ij}^{-1} \hat{H}_{ij}$ and $\Gamma^1_{ij} = \hat{H}_{ij}^\top (\hat{R}^1_{ij})^{-1} \hat{H}_{ij}$ be the precisions that levels 0 and 1 assign to a message, each defined when the noise it inverts is positive definite.*

*(a)* $\mathcal{V}_i^0(\mathcal{J}) = \frac{1}{2} \ln\det(I + P_i \sum_{j\in\mathcal{J}} \Gamma^0_{ij})$ *and* $\mathcal{V}_i^1(\mathcal{J}) = \frac{1}{2} \ln\det(I + P_i \sum_{j\in\mathcal{J}} \Gamma^1_{ij})$*, and* $\mathcal{V}_i^0 \geq \mathcal{V}_i^1$ *when* $\hat{R}_{ij} = R_c$*.*

*(b) Both believed values depend on a sender only through the vertex quantities $(\hat{H}_{ij}, \hat{R}_{ij}, \hat{P}_{ij})$ and on the receiver only through $P_i$; no cross-covariance of the two agents' errors enters.*

*(c) Both are monotone and submodular in $\mathcal{J}$, so greedy attention under the capacity $\kappa$ attains at least $1 - 1/e$ of the largest believed value over sets of at most $\kappa$ senders.*

Part (b) is the source of the trouble: a literal receiver scores a sender by its code, noise and posted confidence, never by what the two of them share. Part (c) is the classical guarantee for greedy maximization of monotone submodular functions (Nemhauser et al., 1978), and it gives literal attention a guarantee that the true value does not support. The proof, which includes the greedy bound, is in Section S27, and Check S47.9 tests submodularity numerically.

**Example S3.2** (The true value is not submodular)**.** Let a scalar $\eta$ be read through unit codes without channel noise. Receiver $i$ has an estimate with error variance 4, and senders $j$ and $k$ have estimates with error variances 1 and error correlation $-0.99$; all errors are independent of $\eta$, and the receiver's is independent of the senders'. The best combination of the two messages has error precision $2/(1 - 0.99) = 200$, so $\mathcal{V}_i^*(\{j\}) = \mathcal{V}_i^*(\{k\}) = \frac{1}{2}\ln 5 = 0.805$ while $\mathcal{V}_i^*(\{j,k\}) = \frac{1}{2}\ln 801 = 3.343 > 1.609$. Complementary errors cancel, and two individually poor senders are worth more than the sum of their values.

**Lemma S3.3** (A belief message is an attenuated observation)**.** *Let the world have the prior $\mathcal{N}(\bar{\eta}, P_0)$, let sender $j$ hold the posterior mean $m_j$ of $\eta$ given linear-Gaussian data, with posterior covariance $P_j$, and let $y_{ij} = H_j m_j + \nu$. Then*

$$y_{ij} - H_j\bar{\eta} = B_j(\eta - \bar{\eta}) + \tilde{\nu}_j, \qquad B_j = H_j(I - P_j P_0^{-1}), \tag{S5}$$

*where $\tilde{\nu}_j$ is Gaussian and independent of $\eta$, with covariance $H_j(P_j - P_j P_0^{-1} P_j)H_j^\top + R_c$.*

Proof in Section S27. The code that describes the message is the attenuated code $B_j$, and its noise includes the sender's estimation noise. Level 0 uses $H_j$ and $R_c$ and omits both; level 1 enlarges the noise but keeps the unattenuated code. When the receiver has information of its own, the overlap of the two agents' information must be made explicit, which the static common-source world of Definition 3.2 does.

## S3.2 Overvaluation

Double counting is the familiar form of misreading, and we first locate it exactly. The track model of multi-sensor fusion isolates error correlation: the receiver's and sender's estimates are $m_i = \eta + e_i$ and $m_j = \eta + e_j$ with jointly Gaussian errors independent of $\eta$, as under a flat prior, and the receiver's belief is $\mathcal{N}(m_i, \sigma_i^2)$.

**Proposition S3.4** (The overvaluation band)**.** *In the scalar track model let the errors have variances $\sigma_i^2, \sigma_j^2$ and covariance $\sigma_{ij}$, $|\sigma_{ij}| < \sigma_i\sigma_j$, let the code be $g \neq 0$, the channel noise $R_c \geq 0$ and the level-1 header $\sigma_j^2$. With $S = g^2(\sigma_i^2 + \sigma_j^2 - 2\sigma_{ij}) + R_c$ and $R^{\text{ex}} = g^2(\sigma_j^2 - \sigma_{ij}^2/\sigma_i^2) + R_c$, the true value is $\mathcal{V}_i^* = \frac{1}{2}\ln(S/R^{\text{ex}})$, $\mathcal{V}_i^1$ is its value at $\sigma_{ij} = 0$, $\mathcal{V}_i^0 = \frac{1}{2}\ln(1 + g^2\sigma_i^2/R_c) \geq \mathcal{V}_i^1$, and*

$$\mathcal{V}_i^* < \mathcal{V}_i^1 \iff 0 < \sigma_{ij} < \sigma_{\text{b}} := \frac{2\sigma_i^2(g^2\sigma_j^2 + R_c)}{g^2(\sigma_i^2 + \sigma_j^2) + R_c}, \tag{S6}$$

*while $\mathcal{V}_i^* > \mathcal{V}_i^1$ for $\sigma_{ij} < 0$ and for $\sigma_{ij} > \sigma_{\text{b}}$. For equal variances $\sigma_{\text{b}} \geq \sigma_i\sigma_j$, so every positive correlation is overvalued; at $R_c = 0$, $\sigma_{\text{b}}$ is the harmonic mean of the variances, below $\sigma_i\sigma_j$ when $\sigma_i \neq \sigma_j$. If $\sigma_i < \sigma_j$, then $\sigma_{ij} = \sigma_i^2$ gives $\mathcal{V}_i^* = 0$. For $R_c > 0$, as $\sigma_{ij} \to -\sigma_i\sigma_j$, $\mathcal{V}_i^* \to \frac{1}{2}\ln(1 + g^2(\sigma_i + \sigma_j)^2/R_c) > \mathcal{V}_i^0$, and as $\sigma_{ij} \to \sigma_i\sigma_j$ the limit of $\mathcal{V}_i^*$ exceeds $\mathcal{V}_i^0$ exactly when $\sigma_j > 2\sigma_i$.*

Proof in Section S27. Overvaluation is confined to a band of positive correlations. A sender whose error is the receiver's plus independent noise is worthless, yet a literal receiver counts it as independent evidence. Outside the band the exact receiver gains more than level 1 believes. The sender's error contains the multiple $\beta e_i$, $\beta = \sigma_{ij}/\sigma_i^2$, of the receiver's own error, which the receiver removes with its own estimate, leaving the observation $(1 - \beta)\eta$ plus noise independent of its information; below the band, where the errors are complementary ($\beta < 0$), and above it, this residual observation is worth more than the message read at level 1. The band is checked on random draws in Check S47.10. In the common-source world the prior correlates every error with $\eta$, and the scope is set by codes, noise and the distribution of information (Proposition 3.4).

**Proposition S3.5** (Level-1 overconfidence)**.** *In the static common-source world let $H_j$ be invertible and $R_c = 0$. A level-1 receiver adopts the precision $\Lambda_i + \Lambda_j$, which counts $\Lambda_{ij}^{\text{sh}}$ twice, and the mean $\hat{P}_i^{(1)}(\Lambda_i m_i + \Lambda_j m_j)$ with $\hat{P}_i^{(1)} = (\Lambda_i + \Lambda_j)^{-1}$. Its actual mean-square-error matrix is*

$$\hat{P}_i^{(1)}(\Lambda_i + \Lambda_j + 2\Lambda_{ij}^{\text{sh}})\hat{P}_i^{(1)}, \tag{S7}$$

*which exceeds the claimed covariance $\hat{P}_i^{(1)}$ by $2\hat{P}_i^{(1)}\Lambda_{ij}^{\text{sh}}\hat{P}_i^{(1)}$ and is at least the exact posterior covariance $(\Lambda_i + \Lambda_{j\setminus i})^{-1}$. In the symmetric scalar configuration the claimed variance is the actual one divided by $1 + \phi$.*

Proof in Section S27. Nothing in the receiver's model reveals the excess, and repeating the exchange compounds it. In a scalar static world where each member senses once and the members form a connected undirected graph, let every member add its neighbors' natural parameters, with their posted precisions, to its own round after round; this is literal consensus, the update of this level-1 receiver iterated over the graph. Every member's mean then converges to a pool weighted by eigenvector centrality, the believed precision diverges, and on a regular graph a proper common prior is counted $n$ times, once for each member (Proposition S5.5). The matrix (S7) is checked in Check S47.9. The same overvaluation arises across levels when a member takes as its prior its collective's estimate, which already contains the member's own report: in a scalar tree one such literal reading leaves the overconfidence factor below two (Proposition 12.3), while a record rebuilt

every round from its readers' posteriors has an overconfidence factor that grows without bound in the number of rounds under every aggregation rule of Proposition S12.7 other than the exact one, which keeps each contribution once.

**Proposition S3.6** (Dynamic worlds)**.** *(a) In the scalar world $\eta_{t+1} = F\eta_t + w_t$ with $Q > 0$, let agents $i$ and $j$ run steady-state Kalman filters on independent sensors $s_k = \eta + v_k$, $\mathrm{Var}(v_k) = R_k^s > 0$, with gains $k_k^s$. Their filtered errors have the stationary cross-covariance*

$$\Sigma_{ij} = \frac{(1-k_i^s)(1-k_j^s)Q}{1-(1-k_i^s)(1-k_j^s)F^2} > 0, \tag{S8}$$

*so a level-1 receiver of $j$'s filtered mean, $y_{ij} = H_j m_j + \nu$ with $H_j \neq 0$ and $j$'s claimed filtered variance as header, is overconfident after reading it, and level 1 is inexact for these belief messages whatever $F$.*

*(b) Let senders post predictive means with law headers, $y_{ij} = H_j F m_j + \nu$, let codes be known, and let $\Sigma_{kl} = \mathrm{Cov}(e_k, e_l)$ be the actual covariances of the filtered errors $e_k = m_k - \eta_t$ and $P_k$ their claimed covariances. A receiver that reads the header, which makes it a level-1 receiver, claims the innovation covariance $S_{ij}^{\mathrm{cl}} = H_j(FP_iF^\top + FP_jF^\top + 2Q)H_j^\top + R_c$; the actual one is $S_{ij}^{\mathrm{act}} = H_jF(\Sigma_{ii} + \Sigma_{jj} - \Sigma_{ij} - \Sigma_{ji})F^\top H_j^\top + R_c$, and*

$$S_{ij}^{\mathrm{cl}} - S_{ij}^{\mathrm{act}} = 2H_jQH_j^\top + H_jF\big[(P_i - \Sigma_{ii}) + (P_j - \Sigma_{jj}) + \Sigma_{ij} + \Sigma_{ji}\big]F^\top H_j^\top. \tag{S9}$$

Proof in Section S27. The world's innovation is a source every agent shares. For two tracks of one target, Bar-Shalom and Campo (1986) found that common process noise makes the fused uncertainty area about 70 percent of a single sensor's, against 50 percent if the dependence is ignored. Part (b), from Dennis (2026a), is the other face of that neglect: the common innovation cancels in the difference of two predictions but is counted twice in the claimed innovation covariance, so the correlation that makes a fused covariance optimistic makes the predicted innovation pessimistic. In reciprocal pairs that sense and listen to each other, the same report found claimed covariances of 0.35 to 0.65 times the actual error and claimed innovation covariances 3.0 to 13.4 times the actual ones. Part (a) is checked in Check S47.12.

**Open problem S3.7** (Dynamic ratios)**.** Find in closed form the believed-to-true ratio for belief messages in a dynamic world with common process noise when the exact receiver fuses given $(m_i, y_{ij})$; only the cross-covariance (S8) is known.

### S3.3 Undervaluation

**Example S3.8** (Common-information cancellation)**.** In the static common-source world let $K = 2$, $\mathcal{U}_i \subseteq \mathcal{U}_j$, $\Lambda_{ij}^{\mathrm{sh}} = \mathrm{diag}(\alpha_1, \alpha_2)$, $\Lambda_{j\setminus i} = \mathrm{diag}(0, \Gamma_{\mathrm{pr}})$, $H_j = (1,1)/\sqrt{2}$ and $R_c = 0$. Then

$$\mathcal{V}_i^* = \tfrac{1}{2}\ln\Big(1 + \frac{\Gamma_{\mathrm{pr}}}{\alpha_2}\Big), \qquad \mathcal{V}_i^1 = \tfrac{1}{2}\ln\Big(1 + \frac{\alpha_1^{-1} + \alpha_2^{-1}}{\alpha_1^{-1} + (\alpha_2 + \Gamma_{\mathrm{pr}})^{-1}}\Big), \tag{S10}$$

and $\mathcal{V}_i^* - \mathcal{V}_i^1 \to \tfrac{1}{2}\ln((\alpha_1 + \Gamma_{\mathrm{pr}})/\alpha_1) > 0$ as $\alpha_2 \to 0$. At $\alpha_1 = \Gamma_{\mathrm{pr}} = 1$ and $\alpha_2 = 0.01$, $\mathcal{V}_i^* = 2.308$ and $\mathcal{V}_i^1 = 1.973$; with $R_c = 0.1$ instead, $\mathcal{V}_i^0 = 3.113 > \mathcal{V}_i^* = 2.216 > \mathcal{V}_i^1 = 1.926$.

Derivation in Section S27, numbers in Check S47.10. The code mixes a coordinate on which the sender's information is entirely shared with one on which it is private. The exact receiver subtracts

the shared part, which it knows, and reads the private part free of the noise that level 1 attributes to the shared part. Shared information is redundant, which lowers a message's value, and predictable, which raises the value of the rest; in scalar and aligned worlds only redundancy acts. Since $\mathcal{U}_i \subseteq \mathcal{U}_j$ here, part (e) of Proposition 3.4 fails at level 1.

**Example S3.9** (Sender-side completion)**.** Let $K = 2$, $\bar{\eta} = 0$ and $P_0 = \left(\begin{smallmatrix} 1 & \psi \\ \psi & 1 \end{smallmatrix}\right)$ with $0 < |\psi| < 1$. The sender observes $\eta_1$ with noise precision $\Gamma_1$, the receiver observes $\eta_2$ with noise precision $\Gamma_2$, no source is shared, and $H_j = (0, 1)$, so the message codes $\eta_2$. As $\Gamma_1, \Gamma_2 \to \infty$,

$$\mathcal{V}_i^0 \to 0, \qquad \mathcal{V}_i^1 \to 0, \qquad \mathcal{V}_i^* \to \tfrac{1}{2}\ln(1 + \psi^2(1 - \psi^2)/R_c). \tag{S11}$$

At $\psi = 0.8$, $R_c = 0.01$ and $\Gamma_1 = \Gamma_2 = 10^4$, $\mathcal{V}_i^* = 1.587$, $\mathcal{V}_i^1 = 1.35 \times 10^{-4}$ and $\mathcal{V}_i^0 = 4.97 \times 10^{-3}$. With the full-rank code $H_j = I_2$ and $R_c = \sigma_c^2 I_2$ the limits are $\mathcal{V}_i^* \to \frac{1}{2}\ln(1 + (1 + \psi^2)(1 - \psi^2)/\sigma_c^2)$ and $\mathcal{V}_i^0 \to \frac{1}{2}\ln(1 + (1 - \psi^2)/\sigma_c^2)$, which are 0.966 and 0.763 at $\psi = 0.8$, $\sigma_c^2 = 0.1$.

Derivation in Section S27, numbers in Check S47.10. In the limit the coded content is the sender's prior completion $\psi\eta_1$ of the coordinate the receiver already knows. It carries information about $\eta_1$, which a literal receiver attributes to $\eta_2$ and discards, and the ratio of true to believed value is unbounded. The receiver map $O$ completes unsent directions from the receiver's prior (Proposition 2.3); here the completion happened at the sender, and only a receiver that models the sender's prior can read it.

**Numerical observation S3.10** (Undervaluation outside the proved cases)**.** In the two ensembles of random static common-source worlds specified at the head of Section S47.2, with the code and noise laws of Check S47.10 and one seed of the generator, $\mathcal{V}_i^* > \mathcal{V}_i^1$ in 221 of 5000 instances with invertible codes and channel noise (largest excess 0.70 nats) and in 454 of 4000 instances with rank-limited codes (largest excess 0.95 nats); $\mathcal{V}_i^* > \mathcal{V}_i^0$ in 2 of a second sample of 5000 instances with invertible noisy codes and in 17 of the 4000 rank-limited instances (largest excess 0.21 nats, among the rank-limited instances). Among 5000 instances with $\mathcal{U}_i \subseteq \mathcal{U}_j$, $\mathcal{V}_i^* > \mathcal{V}_i^1$ in 386 and $\mathcal{V}_i^* > \mathcal{V}_i^0$ in none, as part (e) of Proposition 3.4 requires. The counts and largest excesses are properties of these ensembles and this seed; the searches establish that undervaluation occurs at both levels in both code regimes and give no frequency that is independent of the ensemble.

**Open problem S3.11** (Scope of each direction)**.** Find necessary and sufficient conditions on codes, noise and the distribution of information for $\mathcal{V}_i^* \le \mathcal{V}_i^1$ and for $\mathcal{V}_i^* \le \mathcal{V}_i^0$ with rank-limited or noisy full-rank codes. Invariance of the coded subspace under the shared-information operator $P_j \Lambda_{ij}^{\mathrm{sh}}$ is not sufficient, since every invertible code has it and invertible noisy codes can be undervalued (Numerical observation S3.10).

### S3.4 Echo chambers

**Proposition S3.12** (Closed pure-listening sets)**.** *Let $C$, with $n = |C|$, be closed under attention and purely listening from round $t_0$ on, $k_i^s = 0$ for $i \in C$, with errors $e_C$ and means $m_C$.*

*(a) In a static world $I(\eta; m_C(t)) \le I(\eta; m_C(t_0))$, and in a stationary scalar world with $0 \le F < 1$, $I(\eta_t; m_C(t)) \le F^{2(t-t_0)} I(\eta_{t_0}; m_C(t_0))$.*

*(b) Let $F = 1$, $Q \ge 0$ and $W_{CC} = M$ constant and row-stochastic with stationary law $\varpi$, and let every eigenvalue of $M$ other than a simple eigenvalue 1 have modulus below one, as for irreducible aperiodic $M$. The common error $\bar{e} = \varpi^\top e_C$ is a random walk, $\bar{e}(t+1) = \bar{e}(t) + \varpi^\top \zeta_C(t) - w_t$, while the disagreement $(I - \mathbf{1}\varpi^\top)e_C$ has second moments bounded uniformly in $t$.*

(c) *Let the world be static, let every member attend $n_{\text{in}}$ members with assigned noise equal to the channel noise $R_c$, and let all members start with one claimed precision, so that $\hat{\Lambda}(t+1) = \hat{\Lambda}(t) + n_{\text{in}}/R_c$ and $W_{CC}(t) = c_t I + (1-c_t)M$ with $c_t = \hat{\Lambda}(t)/\hat{\Lambda}(t+1)$ and $M$ the normalized attention matrix. If $M$ is doubly stochastic and normal with 1 a simple eigenvalue, every member's error converges in mean square to one random limit $\bar{e}_\infty$ with $\operatorname{Var}(\bar{e}_\infty) \geq n^{-2}\mathbf{1}^\top \operatorname{Cov}(e_C(t_0))\mathbf{1}$, while the claimed variance $1/\hat{\Lambda}(t)$ decays like $R_c/(n_{\text{in}}t)$ and the believed information $\frac{1}{2}\ln(P_0\hat{\Lambda}(t))$ diverges like $\frac{1}{2}\ln t$.*

Proof in Section S27. A closed group of listeners thus turns channel noise into confidence: its members converge on a shared error while their believed information diverges and, by (a), their true information can only fall. In the computed instances of Check S47.14, five agents believe they hold 5.65 nats while their true information falls from 0.896 to 0.789 nats. Consensus on a random limit for weights that are not doubly stochastic and normal, such as stars, is only computed there, not proved. Sensing does not remove the effect when receivers read law headers.

**Example S3.13** (A complete group with law headers)**.** Take $n$ identical agents in a closed complete group in a scalar world with $F = 0.995$, $Q = 0.01$ ($P_0 = 1.0025$), sensing noise $R_i^s = 1$ and channel noise $R_c = 0.2$, each reading the others with the assigned noise $R_c + \hat{P}^-$, where $\hat{P}^-$ is the common claimed predicted variance at its stationary value; reading this header makes them level-1 receivers. An isolated Kalman filter has predicted error 0.1000. The stationary actual error is 0.159, 0.302 and 0.661 at $n = 2$, 8 and 64, while the error of a correlation-aware centralized fuser falls from 0.073 to 0.018 (Check S47.15). The group is a weak echo chamber for every $n \geq 2$ by Theorem 3.10, whose hypotheses hold because the sensing gain is common and the weights are symmetric: the floor of Proposition 3.8 is 0.116 at $n = 2$ and increases with $n$, so it exceeds the isolated error for every $n \geq 2$. As $n \to \infty$ the claimed predicted variance tends to $Q$ and the sensing gain to zero, so $\liminf_n \mathbb{E}e_i^2 \geq P_0$ and the ratio of actual to claimed variance has limit inferior at least $P_0/Q = 1/(1-F^2) \approx 100$; it is 52 at $n = 64$. Derivations are in Section S27.

*Remark* S3.14 (Echo chambers as equilibria of attention). In the earlier report (Dennis, 2026a), receivers that learned message noise from innovations or tracked agreement formed, at attention capacity two, disjoint triangles, reciprocal pairs and chains in which nobody sensed the shared part of the world, as strict equilibria on open parameter sets. Every listener ended at 1.10 to 1.26 times the prior variance while believing it gained 0.22 to 0.47 nats per round, and none of these structures was an equilibrium under exact fusion.

### S3.5 Exact fusion

**Proposition S3.15** (Fusion with correlated errors)**.** *Let receiver $i$'s prior error $e$ have covariance $P$, assigned correctly, and let a message $y = \hat{H}\eta + \tilde{\nu}$ have error covariance $R^{\text{a}} = \operatorname{Cov}(\tilde{\nu})$ and cross-covariance $X = \operatorname{Cov}(\tilde{\nu}, e)$. For a gain $\mathcal{K}$ the posterior error $e^+ = (I - \mathcal{K}\hat{H})e + \mathcal{K}\tilde{\nu}$ has covariance*

$$\operatorname{Cov}(e^+) = (I - \mathcal{K}\hat{H})P(I - \mathcal{K}\hat{H})^\top + \mathcal{K}R^{\text{a}}\mathcal{K}^\top + (I - \mathcal{K}\hat{H})X^\top\mathcal{K}^\top + \mathcal{K}X(I - \mathcal{K}\hat{H})^\top. \tag{S12}$$

(a) *If the innovation covariance $S^* = \hat{H}P\hat{H}^\top + R^{\text{a}} - \hat{H}X^\top - X\hat{H}^\top$ is invertible, the gain that minimizes $\operatorname{Cov}(e^+)$ in the Loewner order is $\mathcal{K}^* = (P\hat{H}^\top - X^\top)(S^*)^{-1}$.*

(b) *The gain $\mathcal{K}_0 = P\hat{H}^\top(\hat{H}P\hat{H}^\top + R^{\text{a}})^{-1}$, which ignores $X$, claims the covariance $(I - \mathcal{K}_0\hat{H})P$, and the actual covariance exceeds the claim by $(I - \mathcal{K}_0\hat{H})X^\top\mathcal{K}_0^\top + \mathcal{K}_0X(I - \mathcal{K}_0\hat{H})^\top$. For scalars with $R^{\text{a}} > 0$ the excess is $2\mathcal{K}_0(1 - \mathcal{K}_0\hat{H})X$, positive exactly when $\hat{H}X > 0$, that is,*

*when the decoded message error $\tilde{\nu}/\hat{H}$ and the receiver's error are positively correlated, and the receiver is then overconfident.*

*(c) An echo of the receiver's own belief, $y = \hat{H}m_i + \nu'$ with $\hat{H} \neq 0$ and $\nu'$ independent of everything else, has true value $I(\eta; y \mid m_i) = 0$, while level 0 believes it carries $\frac{1}{2}\ln\det(I+\hat{R}^{-1}\hat{H}P\hat{H}^\top) > 0$.*

Proof in Section S27. Exact inference is calibrated by construction, and this is enough to exclude echo chambers (Proposition 3.11).

The demands of exact fusion are informational. The receiver must know which of its sources the sender used and, in a dynamic world, the cross-covariance of the two agents' errors, which past messages create. On a tree, shared information can be subtracted link by link, as the channel filters of decentralized fusion do (Grime and Durrant-Whyte, 1994) and as part (c) of Lemma 3.3 does for one link. On a graph with cycles evidence returns along several paths, and exact fusion needs provenance, a record of where each piece of content came from; storage for links and records under which fusion is exact is derived for one shared linear-Gaussian state with known sensing models in Section S21. Gaussian belief propagation gives correct means when it converges (Weiss and Freeman, 2001) and converges on walk-summable models (Malioutov et al., 2006). Covariance intersection (Julier and Uhlmann, 1997) fuses estimates whose cross-covariance is unknown, but in the computations of the earlier report it left the claimed innovation covariance inflated by factors of 2.6 to 9.5 (Dennis, 2026a). For abstract agents, citations, audit trails and sourcing rules supply provenance; this reading is interpretive.

*Remark* S3.16 (Exact fusion along a process). In the report of Dennis (2026a), exact fusion in a dynamic world is cross-covariance-aware best linear unbiased fusion of the current message with the current estimate. It is Bayes under the conditional message likelihood of track-to-track fusion, its claimed covariance equals the true mean-square error along the process started from a common prior, and no update raises an agent's true error; but it is not the Bayes filter of all past data, and it needs the joint error covariance of all agents. It also couples agents in its own way: through the ambiguity term of the expected free energy a receiver can prefer the sender whose error is most correlated with its own.

### S3.6 Circulation and attribution

Literal reading has one further failure that is neither over- nor undervaluation. Content that circulates around a cycle of level-0 receivers, each completing the directions a message does not carry with its own prior regression, can return amplified. The relevant object is the product of the receivers' gain-code maps around the cycle (Definition 6.7): with noiseless codes of rank below $K$ a story that circulates repeatedly grows geometrically exactly when this product has spectral radius above one, and with any codes amplification requires an edge on which the sender's coded prior variance exceeds the receiver's predictive variance of the message (Theorem 6.8). A common prior covariance excludes it, and so does assigned noise that covers the sender's coded uncertainty, which level-1 receivers provide when the posted covariance is the sender's prior (Proposition S6.10). Amplification needs no correlated errors, so it is not double counting, and it occurs at identity encodings with correct codes, so it is not a holonomy; it is a calibration failure for belief messages under heterogeneous prior structure, treated in Section 6.4.

The exact-fusion receiver is also the reference against which collectives are attributed. The exact-fusion null of Definition 5.17 replaces every receiver's literal likelihood, fusion and attention by exact fusion, with provenance for records, and a collective must survive it: the same partition must be selected, with its blocks at contrast at least the declared level. A structure that vanishes under this null is attributed to misreading joint structure, not to communication as action. In the

two-group model of Section 7, clauses (T1) to (T3) and (T5) of a formation result (Definition 7.1) are theorems only at infinite policy precision $\gamma$ with noiseless angle observations, where the declared start is itself the split (Theorem 7.6(e), Theorem 7.8), and the contrast clause (T3) only at the levels of the window of Theorem 7.8(c), which is nonempty at every computed parameter point (Numerical observation S7.6). For finite $\gamma$ above the threshold $\gamma_S$ of (73), reachability of the split branch is a theorem about the frozen-calibration mean-field map (Theorem 7.6(a),(b)), with the stochastic step from the full model to that map open (Open problem 7.14) and the contrast of the reached state computed at the base point only (Numerical observation S7.7). The persistence clause (T4) fails for the translation-aware null, so the model yields no formation result in the sense of Definition 7.1, and what forms there is a calibration-level effect of misreading in this sense. Receivers that know their translation errors pay no mistranslation cost, so the split fails the translation-aware form of the exact-fusion persistence clause, as Proposition 7.12 proves, while for full cross-covariance-aware exact fusion this conclusion is heuristic; at large policy precision the split's partition is predictable from the codes alone (Section 7.6). Online calibration from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell, with reruns of a single cell differing by up to three replicates, so the counts show little beyond the pattern being reached in some replicates and missed in others for $\gamma \geq 160$, and the question is open (Numerical observation 7.11 and Open problem S7.13); persistence with drifting frames and formation beyond this model are conjectures (Conjectures S7.12 and 7.13).

### S3.7 Blindness to shared error

By Lemma 3.6, a precision learned from agreement certifies mutual predictability, not accuracy. A receiver of class C1i, which fits its assigned message noise to the second moment of its innovations relative to its own estimate, is blind to every error it shares with its senders (C1g learns on grounded rounds and is not covered by the lemma), and a group with a large common error looks calibrated from inside. Receivers that read law headers do not learn from innovations at all, and by part (b) of Proposition S3.6 the innovation covariance they claim counts the shared world innovation twice, although their actual innovations do not contain it. A record that the members write and read is blind in the same way: its top-down innovations do not change when the members share an error with it, and in a collective closed under attention, reading it slows the grounding-driven correction of that error (Lemma 12.6).

### S3.8 Echo chambers in the literature

In Definition 3.7, small disagreement means mutually predictable beliefs. Nguyen (2020) distinguishes epistemic bubbles, which leave other voices out, from echo chambers, which actively exclude and discredit them; in our terms omission is closure of attention and discrediting is miscalibrated precision. Closure with literal fusion already yields small disagreement with large error, as Proposition 3.8, Lemma 3.9, and Theorem 3.10 show, and Lemma 3.6 shows why miscalibration then seals it.

By Proposition 3.11, echo chambers need misspecification. Madsen et al. (2018) report persistent echo chambers among Bayesian agents who trust senders fully, treat each received belief as fresh Gaussian evidence and drop partners outside a confidence window; in our terms these are literal receivers with withdrawal by disagreement. Belief sharing in federated inference (Friston et al., 2024), as we read it, also treats received beliefs as observations.

# S4 Additional results for Section 4: collectives

This section collects statements, examples and remarks of Section 4 that the main text cites without stating in full. Proofs are in Section S28.

## S4.1 Path blankets, shells and the Markov length

*Remark* S4.1 (Agent-level descriptions). Assigning a role to each whole agent, as constructions of nested blankets do (Kirchhoff et al., 2018; Palacios et al., 2020), coarsens the variables of the canonical partition (Definition 4.1). An interior agent that senses the shared world then reads an exterior variable: each sensing event is a leak activated by its reader, at a rate bounded by a function of the sensing frequency that vanishes with it (Dennis, 2026a). Channel-level roles move the sensing channel into $\mathsf{S}$ and remove the leak. The grounding tension of Assumption A11 survives as the leak of the world-conditional precision (Theorem 4.9), which caps contrast and extent (Theorem 4.14). We use agent-level descriptions only for layered partitions with one variable per agent (Proposition S4.3).

**Example S4.2** (The collider)**.** Let three scalar variables with $Q^{\Xi} = I$ follow $X_{i,t+1} = \Xi^i_t$, $X_{b,t+1} = \alpha_1 X_{i,t} + \alpha_2 X_{e,t} + \Xi^b_t$ and $X_{e,t+1} = \Xi^e_t$, with $\alpha_1\alpha_2 \neq 0$, $\mathsf{I} = \{i\}$, $\mathsf{B} = \{b\}$ and $\mathsf{E} = \{e\}$. Then $A_{\mathsf{IE}} = A_{\mathsf{EI}} = 0$ and the noise is diagonal, but $A^\top_{\mathsf{BI}}(Q^{\Xi}_{\mathsf{BB}})^{-1}A_{\mathsf{BE}} = \alpha_1\alpha_2$, and the path blanket fails on every window of two or more rounds, from every initial law. In each round the pair $(X_{i,t}, X_{e,t})$ given $X_{b,t+1}$ has squared partial correlation $\alpha_1^2\alpha_2^2/((1+\alpha_1^2)(1+\alpha_2^2))$, so

$$\iota_{\mathrm{path}} = \frac{1}{2}\ln\frac{(1+\alpha_1^2)(1+\alpha_2^2)}{1+\alpha_1^2+\alpha_2^2}, \tag{S13}$$

which is 0.1438 nats per round at $\alpha_1 = \alpha_2 = 1$ (Check S47.16).

**Proposition S4.3** (Two-hop shells)**.** *Let one variable per agent follow (23) with one-hop local coupling on the communication graph, $A_{ij} \neq 0$ only if $i = j$ or $i$ and $j$ are neighbors, and with per-agent noise.*

*(a) Every partition with $d(\mathsf{I},\mathsf{E}) \geq 3$ is an exact path blanket from every start with $X_{\mathsf{I},0} \perp X_{\mathsf{E},0} \mid X_{\mathsf{B},0}$, and its leak rate from the stationary start is zero when $\rho(A) < 1$.*

*(b) A partition with $d(\mathsf{I},\mathsf{E}) = 2$ is exact if and only if $\sum_{k\in\mathsf{B}} A^\top_{k\mathsf{I}}(Q^{\Xi}_{kk})^{-1}A_{k\mathsf{E}} = 0$; if some shell agent reads both sides, this fails outside a set of positive codimension.*

*(c) Both statements hold for a fixed time-varying schedule of one-hop couplings. For every set $\mathsf{I}$ the shell $\{k : 1 \leq d(k,\mathsf{I}) \leq 2\}$ satisfies (a), so the number of exact path blankets grows exponentially with the number of agents.*

Proof in Section S28; checked in Check S47.16. This is Fig. 3(b): a shell agent that reads both sides is a collider, and a second layer separates the readers of the interior from those of the exterior.

**Proposition S4.4** (Memory and instantaneous blankets)**.** *Suppose the path blanket holds on $[0,t]$ for every $t$.*

*(a) $X_{\mathsf{I},t} \perp X_{\mathsf{E},t} \mid X_{\mathsf{B},0:t}$, and the instantaneous leak is bounded by the memory of either side, $I(X_{\mathsf{I},t}; X_{\mathsf{E},t} \mid X_{\mathsf{B},t}) \leq \min\{I(X_{\mathsf{I},t}; X_{\mathsf{B},0:t-1} \mid X_{\mathsf{B},t}), I(X_{\mathsf{E},t}; X_{\mathsf{B},0:t-1} \mid X_{\mathsf{B},t})\}$.*

*(b) In the Gaussian case the blanket augmented by the filtered estimate $\hat{X}_{\mathsf{E},t} = \mathbb{E}[X_{\mathsf{E},t} \mid X_{\mathsf{B},0:t}]$ is an exact instantaneous blanket: $X_{\mathsf{I},t} \perp X_{\mathsf{E},t} \mid (X_{\mathsf{B},t}, \hat{X}_{\mathsf{E},t})$.*

(c) *If* $\rho(A) < 1$ *and* $AQ^{\Xi} = Q^{\Xi}A^{\top}$*, which makes the stationary chain time reversible, and if the noise is independent across the partition and* $A_{\mathsf{IE}} = A_{\mathsf{EI}} = 0$*, then the stationary precision is* $(Q^{\Xi})^{-1} - A^{\top}(Q^{\Xi})^{-1}A$ *and its interior-exterior block is* $-A_{\mathsf{BI}}^{\top}(Q^{\Xi}_{\mathsf{BB}})^{-1}A_{\mathsf{BE}}$*. The stationary instantaneous blanket then holds exactly when the collider condition of Theorem 4.3 does.*

Proof in Section S28. The present state of the shell does not screen interior from exterior because both remember the shell's past; the boundary's own filter of the exterior, the memory variable of Assumption A7, restores the blanket. A chain in which an interior variable reads a sensory variable that reads the exterior has an exact path blanket and a positive instantaneous leak (Theorem 4.23(d), Check S47.16). The condition $AQ^{\Xi} = Q^{\Xi}A^{\top}$ of part (c) is, for $\rho(A) < 1$, exactly time reversibility of the stationary chain: with $\Sigma$ its stationary covariance, reversibility is $A\Sigma = \Sigma A^{\top}$, and $Q^{\Xi} = \Sigma - A\Sigma A^{\top}$ gives $Q^{\Xi} = (I - A^2)\Sigma$ in one direction and $\Sigma = (I - A^2)^{-1}Q^{\Xi}$ in the other, so part (c) covers reversible processes only, and for noise independent across the four classes of the canonical partition these are collectives without active states whose sensory states drive the exterior in return (Remark S44.3). Agents that sense an exogenous world never form a reversible process, so we condition on the world, which removes its memory. The remaining stationary law has blankets whose thickness is measured by a length of its own.

**Theorem S4.5** (Thick shells)**.** *Let* $e \sim \mathcal{N}(0, \Sigma)$ *on the vertices of a finite or countable graph with hop distance* $d$*, with blocks of dimension at most* $K$*, sphere sizes* $\mathcal{S}(r) = \sup_i \#\{j : d(i,j) = r\}$*,* $0 < \lambda_{\min}(\Sigma) \le \lambda_{\max}(\Sigma) < \infty$ *and* $\|\Sigma_{ij}\| \le B_{\Sigma}e^{-m_{\Sigma}d(i,j)}$*, and put* $\Pi = \Sigma^{-1}$*. For* $0 \le m_{\Pi} < m_{\Sigma}$ *let* $\mathcal{C}(m_{\Pi}) = B_{\Sigma}\sum_{r\ge 1}\mathcal{S}(r)e^{-m_{\Sigma}r}(e^{m_{\Pi}r} - 1)$*.*

(a) *If* $\mathcal{C}(m_{\Pi}) < \lambda_{\min}(\Sigma)$*, then* $\|\Pi_{ij}\| \le B_{\Pi}e^{-m_{\Pi}d(i,j)}$ *with* $B_{\Pi} = (\lambda_{\min}(\Sigma) - \mathcal{C}(m_{\Pi}))^{-1}$*.*

(b) *For a partition* $(\mathsf{I}, \mathsf{B}, \mathsf{E})$ *with* $\mathsf{I}$ *finite put* $Z_{\mathsf{IE}} = KB_{\Pi}^2\sum_{i\in\mathsf{I}}\sum_{k\in\mathsf{E}}e^{-2m_{\Pi}d(i,k)}$*. If* $\lambda_{\max}(\Sigma)^2 Z_{\mathsf{IE}} < 1$*, then*

$$\iota_{\text{inst}} \le \frac{\lambda_{\max}(\Sigma)^2 Z_{\mathsf{IE}}}{2(1 - \lambda_{\max}(\Sigma)^2 Z_{\mathsf{IE}})}. \tag{S14}$$

(c) *If* $\mathcal{S}(r) \le B_{\mathcal{G}}r^{D-1}$ *for* $r \ge 1$*,* $d(\mathsf{I}, \mathsf{E}) \ge h$ *and* $\mathsf{I}$ *has at most* $n_{\partial}$ *vertices at each distance from* $\mathsf{E}$*, then* $Z_{\mathsf{IE}} \le B_{*}KB_{\Pi}^2 n_{\partial}(1+h)^{D-1}e^{-2m_{\Pi}h}$ *with* $B_{*}$ *depending only on* $B_{\mathcal{G}}$*,* $D$ *and* $m_{\Pi}$*, so the leak is* $O(n_{\partial}h^{D-1}e^{-2m_{\Pi}h})$*.*

Proof in Section S28, by the conjugation argument of Combes and Thomas (1973); checked in Check S47.18. The leak across a shell decays at twice the decay rate of the precision. The rate that (a) guarantees is conservative: 0.010 per hop on a ring where the precision decays at 0.94 per hop.

**Theorem S4.6** (An explicit Markov length)**.** *Let* $\Sigma$ *satisfy the hypotheses of Theorem S4.5, with* $\mathcal{S}(r) \le B_{\mathcal{G}}r^{D-1}$ *for* $r \ge 1$*.*

(a) *If* $\|\Pi_{ij}\| \le B_{\Pi}e^{-m_{\Pi}d(i,j)}$ *with* $m_{\Pi} > 0$*, then* $\xi_M \le 1/(2m_{\Pi})$*.*

(b) *With*

$$m_{*} = \min\Big\{\frac{m_{\Sigma}}{D+1}, \frac{\lambda_{\min}(\Sigma)m_{\Sigma}^{D+1}}{8e^{m_{\Sigma}}B_{\mathcal{G}}D!B_{\Sigma}}\Big\}, \tag{S15}$$

$\|\Pi_{ij}\| \le 2\lambda_{\min}(\Sigma)^{-1}e^{-m_{*}d(i,j)}$*, and therefore*

$$\xi_M \le \frac{1}{2m_{*}} = \max\Big\{\frac{D+1}{2m_{\Sigma}}, \frac{4e^{m_{\Sigma}}B_{\mathcal{G}}D!B_{\Sigma}}{\lambda_{\min}(\Sigma)m_{\Sigma}^{D+1}}\Big\}. \tag{S16}$$

(c) *Let the level-0 network have a time-invariant $W$ and a time-invariant law of the noise $\zeta(t)$, and satisfy the hypotheses of Theorem 8.9(b) and of its last clause, that is* (Lin), (Loc), ($C_r$) *and* (B) *of Assumption 8.1, with constants $\lambda < 1$, $q$ and $q_{\min}$, on a potential communication graph with $\mathcal{S}(r) \le B_{\mathcal{G}} r^{D-1}$ for $r \ge 1$. Then its stationary world-conditional covariance $\Sigma$ satisfies the hypotheses of Theorem S4.5 with the constants of Theorem 8.9(b),*

$$\|\Sigma_{ij}\| \le \frac{q\lambda^{2\lceil d(i,j)/2\rceil}}{1-\lambda^2} \le B_\Sigma e^{-m_\Sigma d(i,j)}, \qquad B_\Sigma = \frac{q}{1-\lambda^2}, \quad m_\Sigma = \ln\frac{1}{\lambda},$$

*and $\lambda_{\min}(\Sigma) \ge q_{\min}$, so that*

$$\xi_M \le \max\Big\{\frac{D+1}{2\ln(1/\lambda)}, \frac{4B_{\mathcal{G}} D! q}{q_{\min}\lambda(1-\lambda^2)(\ln(1/\lambda))^{D+1}}\Big\}.$$

Proof in Section S28. The bound depends on the decay of the covariance and on its conditioning $B_\Sigma/\lambda_{\min}(\Sigma)$. Inverse-closedness of exponentially localized matrices gives decay of the precision at some smaller rate (Jaffard, 1990), banded matrices have explicit rates (Demko et al., 1984), and (S15) makes the rate explicit for the matrices that arise here. As $\lambda \to 1$ the bound of (c) grows like $(1-\lambda)^{-(D+2)}$. No bound by the correlation length alone is possible.

*Remark* S4.7 (A corrected error). An earlier formulation of H1, which we had adopted, said that with a finite correlation length every region is an approximate blanket once its shell is thicker than $\xi$. Proposition 4.6 shows that this is false in both directions. The formulation conflated two rates: the leak across a shell decays at twice the decay rate of the precision (Theorem S4.5), and the precision and the covariance can decay at unrelated rates. In the hub family the mechanism is a hidden common cause: two listeners share the past broadcast of a source whose present state no longer carries it, so their errors form a moving-average field, and nothing in the present state of a shell screens the shared past. Contrast is bounded through $\xi_2$ (Theorem 4.14); the boundary is governed by $\xi_M$.

The hub family of Proposition 4.6(a) also shows that a bound on $\xi_M$ must depend on conditioning: its Markov length grows without bound with the anisotropy $q_{\text{src}}/q_{\text{lis}}$ of the local noise, which is $\text{cond}(\Sigma)$ once $q_{\text{lis}} \le (1-4a^2)q_{\text{src}}$. We expect conditioning to enter only through this anisotropy.

**Conjecture S4.8** (Markov length without conditioning)**.** *For time-invariant networks on $\mathbb{Z}^D$ with local noise and row-sum contraction, errors of dimension at most $K$ and noise anisotropy $q/q_{\min}$, there is a function $B$ with $\xi_M \le B(q/q_{\min}, K, D)(1+\xi)$. A family in this class with bounded $q/q_{\min}$, $K$ and $D$ along which $\xi_M/(1+\xi)$ is unbounded would refute it.*

A route to a proof through the conditional covariance of the previous state given the present one is described in Remark S28.2. The evidence near the massless limit, where the correlation length diverges, points the same way.

**Numerical observation S4.9** (The massless limit does not lengthen the Markov length)**.** (i) For agents on $\mathbb{Z}$ with two-dimensional errors, $W_{ii} = \begin{pmatrix} c & \tilde{a} \\ 0 & c \end{pmatrix}$ with $\tilde{a} \ge 0$, $W_{i,i\pm1} = aI$, $Q^\zeta = qI$ and leak $\varepsilon = 1-c-2a > 0$, the precision symbol was computed in closed form by symbolic algebra. As $\varepsilon, \tilde{a} \to 0$ with $\tilde{a}/\varepsilon$ fixed, Theorem S4.6(a) gives $\xi_M/\xi \le [2\,\text{Re}(1+i\tilde{a}/(2\varepsilon))^{1/2}]^{-1} \le \frac{1}{2}$ asymptotically, between 0.455 and 0.5 under row-sum contraction, which forces $\tilde{a} < 2\varepsilon$; at fixed $\tilde{a} > 0$, outside that class, $\xi_M$ stays bounded while $\xi \sim (a/\varepsilon)^{1/2}$ diverges. (ii) On heterogeneous and directed chains and rings of 200 scalar agents, $\xi_M/\xi$ stayed in $[0.18, 0.63]$ as $\xi$ grew to between 8 and 16 hops. (iii) In random translation-invariant two-component systems the ratio of the precision decay length to $\xi$, which bounds $2\xi_M/\xi$, was at most 1.93 with isotropic noise and grew with the noise anisotropy (Check S47.20).

## S4.2 Blanket existence selects nothing

The results so far give exact or approximate blankets to almost every set of agents. Collected, they show that blankets cannot be what selects a collective.

**Proposition S4.10** (Blanket existence selects nothing)**.** *(a) Under per-channel noise with separate channels, every set of agents is an exact path blanket under its canonical partition.*

*(b) Under one-hop local coupling with per-agent noise, every set with a two-hop shell is an exact path blanket, and for reversible dynamics, or normal dynamics with isotropic noise, also an exact stationary blanket.*

*(c) Under the hypotheses of Theorem S4.5, every region whose interior is finite and $h$ hops from its exterior is an approximate instantaneous blanket, with a leak bound that depends on the region only through $n_\partial$ and $h$.*

*(d) In a translation-invariant medium the path and instantaneous leaks of a region are invariant under translation of the region.*

*Consequently a rule that selects collectives by the existence of blankets admits every set under (a), every set with a two-hop shell under (b) and every region whose shell is thick enough for its boundary under (c), and no rule based on blankets or leaks separates a region from its translates in a homogeneous medium. The rule that minimizes the leak per unit of boundary at a fixed shell width of at least two hops is degenerate: path leaks vanish for every candidate under (b), and so do the instantaneous leaks of reversible dynamics and of normal dynamics with isotropic noise.*

Proof in Section S28. The degeneracy includes block media of large contrast (Proposition S4.19), in which every region with a two-hop shell has zero path leak, inside a block or straddling blocks. A blanket is a property of a declared model and does not by itself make a set of agents an agent (Bruineberg et al., 2022; Biehl et al., 2021); selection needs another criterion.

## S4.3 World-conditional law and diffusivity

*Remark* S4.11 (Exact fusion and records). The definition of the world-conditional law (Definition 4.7) applies verbatim to receivers that fuse exactly: given the world path, each member's error is a linear functional of local noises. Then $\Sigma^c$ no longer solves a Lyapunov equation in the members' states, as $\Sigma^{\mathrm{loc}}$ does in Lemma 2.13(b), because exact fusion keeps a filter state larger than $\mu_i$, and in the limit of complete sharing every member holds the same estimate and $\Sigma^c$ becomes singular (Open problem S4.29). Records enter the world-conditional law as nodes, so contrast is computed from the precision of the whole population of agents and records.

**Example S4.12** (Non-diffusive world-conditional precisions)**.** (a) Common listener. Let three agents have $W \geq 0$ and diagonal $Q^\zeta \succ 0$, let agents 1 and 2 attend no one, $W_{12} = W_{13} = W_{21} = W_{23} = 0$ with $W_{11}, W_{22} > 0$, and let agent 3 attend both, $W_{31}, W_{32} > 0$. Then $\Pi^c_{12} > 0$: conditioning on a common listener makes two independent sources anticorrelated, which is explaining away. (b) Signed weights. For $W = 0.2I + W_0$ with $(W_0)_{12} = (W_0)_{21} = 0.4$, $(W_0)_{23} = (W_0)_{32} = -0.3$ and all other entries zero, $W$ is symmetric with $\rho(W) = 0.7$, and with $Q^\zeta = I$, $\Pi^c = I - W^2$ has $\Pi^c_{13} = +0.12$. (c) Frequency. Among 2000 random directed $W \geq 0$ on four agents with row sums below one and $Q^\zeta = I$, 674 have a positive off-diagonal entry of $\Pi^c$ (365 under a second sampling scheme), while a directed relay line and a directed three-agent chain stay diffusive. The sign rule of (a) does not extend beyond three agents: when the two silent agents also feed separate listeners that share a third silent source, $\Pi^c_{12} < 0$ in 1328 of 2000 six-agent draws.

The proof of (a) is in Section S28; (b) is the computation $\Pi^c_{13} = -(W^2)_{13} = -(0.4)(-0.3)$, and the counts are described in Check S47.23. Non-reciprocal attention, which the agents' own choices produce, can therefore leave the diffusive class.

### S4.4 Contrast and baselines

*Remark* S4.13 (Which contrast). The inequality $r' \geq r$ of Lemma S4.30 can be strict by a large factor. For two equal blocks forming a closed population with uniform leak $\varepsilon^\Pi$ and an equitable cut of $\bar{a}^\partial$ per member, Lemma S4.30 gives every member of $C_1$ the marginal leak $\varepsilon^\Pi(\varepsilon^\Pi + 2\bar{a}^\partial)/(\varepsilon^\Pi + \bar{a}^\partial)$, close to $2\varepsilon^\Pi$ when $\varepsilon^\Pi \ll \bar{a}^\partial$. Then $r'(C_1) \geq (\varepsilon^\Pi + \lambda_2(L_{C_1}))(\varepsilon^\Pi + \bar{a}^\partial)/(\varepsilon^\Pi(\varepsilon^\Pi + 2\bar{a}^\partial))$, about $(\varepsilon^\Pi + \lambda_2(L_{C_1}))/(2\varepsilon^\Pi)$, while $r(C_1) = (\varepsilon^\Pi + \lambda_2(L_{C_1}))/(\varepsilon^\Pi + \bar{a}^\partial)$: an ungrounded exterior returns the cut coupling as internal correlation. We compute contrast from the whole population's precision, the conservative choice; the marginal contrast is an upper bound for an analyst who observes only the members. For records, the agents' marginal precision is diffusive whenever the precision with record nodes is (Lemma S4.30), and it was diffusive in the instances checked with noiseless records. With noisy records it can fail to be diffusive (Check S47.24); the precision with record nodes is then not diffusive either, contrast and the size bound do not apply, and such populations fall under Open problem S4.28.

*Remark* S4.14 (Why the moment bound is part of the definition). Including (M) in Definition 4.12 is a choice of definition, not a derived result. Without it the world-conditional precision does not exist, and its finite-time analog acquires a diverging direction: attention that amplifies a shared error makes it grow geometrically, and correlations spread behind a front (Example S8.1 and Propositions S8.6 and S8.7). Such profiles are rumors, not collectives (Remark S8.8). Without (M) only the finite-time statements of Sections 8.2 and 8.3 remain.

*Remark* S4.15 (A weaker convention for the baseline). A weaker convention compares a region only with regions of its own size in the comparison medium, $r_0 > r_{\mathrm{hom}}(|C|)$. It admits lower levels; we do not adopt it, and a result stated under it says so. The difference matters in finite closed populations, where the whole population is itself a region of large contrast when grounding is weak. In the two-group model of Section 7 at its base point, the comparison population of eight agents has contrast 7.954 as a whole while each group of the split has contrast 8.733 when in-group translations are exact, $\sigma_\alpha^2 = 0$, so the admissible levels form the window $(7.954, 8.733]$ and the significance of the split is 1.098 (Theorem 7.8); the matched-size baseline there is 1 and would admit levels down to 2.318 (Remark S7.5). With noisy angle observations, $\sigma_\alpha^2 = 0.01$, simulated group contrasts are about 2 percent higher and the admissible window is not computed (Numerical observation 7.10). Over the grid of 162 parameter points the significance lies between 1.009 and 1.415, and the split is a collective at every grid point (Numerical observation S7.6).

*Remark* S4.16 (Structure in the declared geometry is assumed, not found). This remark interprets the declared geometry. The comparison medium is built from it, so a partition that the declared geometry already contains is present in the comparison medium as well and cannot be found by contrast: a population whose declared geometry contains a partition is its own comparison medium for that partition. Two disjoint cliques of $n \geq 2$ agents, with precision coupling $a^\Pi_{ij} = a > 0$ inside each and uniform leak $\varepsilon^\Pi$, show both outcomes. Each clique has cut zero and $\lambda_2 = na$, so $r = 1 + na/\varepsilon^\Pi$. Declared as an interaction graph, the cliques are their own comparison medium, the baseline is at least the contrast of a clique, and no region is a collective. Declared without geometry, they are compared with the exchangeable population of coupling $(n-1)a/(2n-1)$, whose baseline $1 + 2n(n-1)a/((2n-1)\varepsilon^\Pi)$ is attained by the whole population, and each clique is a collective at every leak, with significance 1.162 at $n = 4$ and $a/\varepsilon^\Pi = 10$. A declared graph can still leave

something to be found: two cliques of six agents with coupling 0.2, joined by a perfect matching of coupling 0.05, at leak 0.1, have block contrast 8.667; with the graph declared, the baseline is 4.500, attained by the whole comparison population, and each clique is a collective with significance 1.926, while without geometry the baseline is 12.455 and neither clique is one (Check S47.25). Under the graph clause the topology is assumed and the heterogeneity of the weights on it is found. Under the exchangeable clause with the mean coupling and uniform leak, only regions denser than the population on average qualify, every member's internal coupling in a region of $n$ members among $N$ agents exceeding $(n-1)N/(n(N-1))$ times the mean total coupling of an agent, about half of it for pairs and about all of it for large regions (Proposition S13.7). Code domains of linear conformity on a ring can be found when the ring is declared and cannot when the ring without the walls between domains is declared (Example S12.24). Which declaration is right is a statement about what the model is meant to explain: a model whose declared couplings contain a partition cannot claim to have found it (Proposition 12.8).

The whole population is a region of a finite closed population, and whether it is a collective of its own comparison medium has an exact answer.

**Proposition S4.17** (When a closed population is a collective)**.** *Let a finite closed population of $N$ agents satisfy the moment bound (M), with diffusive positive definite world-conditional precision $\Pi = \operatorname{diag}(\varepsilon^{\Pi}) + L^{\Pi}$ and mean leak $\bar{\varepsilon}$, and let its comparison medium $\Pi_{\mathrm{hom}} = \bar{\varepsilon} I + L_{\mathrm{hom}}$ be obtained by replacing couplings and leaks by their means under clause (a), (b) or (c) of Definition 4.11. Write $\varkappa_{\mathrm{in}}(V) = \min_{f \perp \mathbf{1}} f^{\top}\Pi f / f^{\top} f$.*

*(a) If the whole population $V$ is a collective at some admissible level, then $\varkappa_{\mathrm{in}}(V) > \bar{\varepsilon} + \lambda_2(L_{\mathrm{hom}})$; with uniform leak this reads $\lambda_2(L^{\Pi}) > \lambda_2(L_{\mathrm{hom}})$.*

*(b) Let a group $\mathsf{G}$ of permutations of $V$ map the declared geometry to itself and act transitively on each set of pairs over which the comparison medium takes a mean, and, unless the leak is uniform, on $V$. Then $\Pi_{\mathrm{hom}} = |\mathsf{G}|^{-1} \sum_{g \in \mathsf{G}} P_g \Pi P_g^{\top}$, with $P_g$ the permutation matrices, $\varkappa_{\mathrm{in}}(V) \le \bar{\varepsilon} + \lambda_2(L_{\mathrm{hom}})$, and $V$ is a collective at no level. This applies to the exchangeable clause with the symmetric group, to the lattice clause on a torus with its translations, and to the graph clause for a declared graph whose automorphisms act transitively on its vertices and on its edges, such as a ring, a complete graph or a hypercube.*

*(c) Let the leak be uniform, $\varepsilon^{\Pi}$, and $\lambda_2(L^{\Pi}) > \lambda_2(L_{\mathrm{hom}})$. Then the comparison medium is connected, the largest grounding-free ratio $R^{\star}_{\mathrm{hom}}$ of its regions $C \subsetneq V$ with $|C| \ge 2$ that its couplings connect is finite, and $V$ is a collective at every level in $(\bar{r}_{\mathrm{hom}}, r(V)]$, a nonempty interval, whenever $\varepsilon^{\Pi} \le \lambda_2(L^{\Pi})/(R^{\star}_{\mathrm{hom}} - 1)$, and at every leak if $R^{\star}_{\mathrm{hom}} \le 1$. With the couplings fixed, $V$ is therefore a collective at some admissible level for some leak if and only if $\lambda_2(L^{\Pi}) > \lambda_2(L_{\mathrm{hom}})$, and at a given leak if and only if, in addition, $1 + \lambda_2(L^{\Pi})/\varepsilon^{\Pi}$ exceeds the contrast of every proper region of the comparison medium.*

*(d) The inequality of (c) can hold on a finite box. The path of four sites with couplings 0.9, 1.2 and 0.9, declared on its box of $\mathbb{Z}$, has $\lambda_2(L^{\Pi}) = 0.6$ against $2 - 2\cos(\pi/4) = 0.586$ for the path with the mean coupling 1, and with uniform leak its whole population is a collective exactly for $\varepsilon^{\Pi} < \frac{1}{3}$, below which its contrast $1 + 0.6/\varepsilon^{\Pi}$ exceeds that of an end pair of the comparison path.*

Proof in Section S28; checked in Check S47.26. Under the exchangeable clause with the mean coupling, on a torus and on a declared graph whose automorphisms are transitive on vertices and edges, a finite closed population is never a collective of its own comparison medium; on boxes and on other declared graphs, with uniform leak, it can be one only when it is more connected than its

comparison medium, and it then is one at weak grounding. In the two-grid example of Table 5 the union of the grids is the whole closed population, and by Theorem 4.18(i) its algebraic connectivity is at most $2\bar{a}^{\partial} = 0.0033$, against 0.0620 for its comparison box, so the union is a collective at no leak (Remark S4.23).

**Open problem S4.18** (Comparison media with record nodes)**.** Define the comparison medium of a population whose world-conditional precision includes record nodes (Remark S4.11). The clauses of Definition 4.11 take means over a declared geometry of the agents, so every record node needs a declared position, and the sets of pairs over which couplings that involve records are averaged must be declared with it. Until then the baseline is defined only for populations whose declared geometry covers every node of the precision.

## S4.5 Lattices, block media and lengths

**Proposition S4.19** (Block media)**.** *(a) Let $\mathbb{Z}^2$ be partitioned into square blocks of side $\ell_b$, with coupling $a_{\text{in}}$ between neighbors inside a block and $a_{\text{out}}$ between neighbors in different blocks, and uniform leak $\varepsilon^{\Pi}$. A block $C$ has $\lambda_2(L_C) = a_{\text{in}}(2 - 2\cos(\pi/\ell_b))$ and $\text{cut}(C)/n = 4a_{\text{out}}/\ell_b$, so*

$$r_\star(C) = \frac{\ell_b(2 - 2\cos(\pi/\ell_b))a_{\text{in}}}{4a_{\text{out}}} \approx \frac{\pi^2 a_{\text{in}}}{4\ell_b a_{\text{out}}}, \qquad r(C) = \frac{\varepsilon^{\Pi} + a_{\text{in}}(2 - 2\cos(\pi/\ell_b))}{\varepsilon^{\Pi} + 4a_{\text{out}}/\ell_b}.$$

*(b) For a partition of a finite population into connected regions $C_1, \dots, C_M$ and $\Pi = \varepsilon^{\Pi} I + L^{\Pi}$, the eigenvalues of $\Pi$ satisfy*

$$\lambda_M(\Pi) \le \varepsilon^{\Pi} + 2\max_j \frac{\text{cut}(C_j)}{|C_j|}, \qquad \lambda_{M+1}(\Pi) \ge \varepsilon^{\Pi} + \min_j \lambda_2(L_{C_j}), \tag{S17}$$

*so uniform contrast of a partition forces a gap after the $M$ slowest modes: $M$ slow collective modes separated from all internal modes.*

Proof in Section S28; checked in Check S47.27, where $6 \times 6$ blocks with $a_{\text{in}} = 1$ reach $r_\star = 402$ at $a_{\text{out}} = 10^{-3}$ but contrast only 3.66 at $\varepsilon^{\Pi} = 0.1$: grounding caps contrast. The converse of (b), that a spectral gap forces a partition into sets of high internal and low external conductance, is a theorem of graph partitioning for the normalized combinatorial Laplacian under a quantitative gap condition $\lambda_{k+1} \ge \text{poly}(k)\lambda_k^{1/4}$ (Oveis Gharan and Trevisan, 2014); its transfer to diffusive precisions with grounding has not been established. Contrast is the spectral form, written for precision-weighted message passing, of the bicriteria notion of a clustering (Kannan et al., 2004) and of the fast internal and slow aggregate modes of nearly decomposable systems (Simon and Ando, 1961).

**Numerical observation S4.20** (Observed baselines)**.** Exhaustive enumeration of the regions connected by the kernel on $\mathbb{Z}$, up to nine sites, and of all regions of a $4 \times 4$ window of $\mathbb{Z}^2$ gives the maxima over the searched regions reported in Table 4; those of $r$ are lower bounds on $\bar{r}_{\text{hom}}$. Both columns are far below the bound (35), whose largest value over $n$ is 17.5 for the two-dimensional kernel of the table, so baselines used in practice should be computed. In two dimensions every region examined has contrast at most one, the nearest-neighbor kernel reaching 1.000 at the $2 \times 2$ square; in one dimension the nearest-neighbor baseline tends to 2. At $c = 0$ the update matrix is bipartite, the world-conditional law splits into two independent sublattice media, and the most contrasted region is a pair of next-nearest neighbors; this parity artifact must be excluded or handled on each sublattice.

The searches are described in Check S47.28.

*Remark* S4.21 (The second-moment length and the decay length). For nearest-neighbor passing on $\mathbb{Z}^D$, Theorem 4.9(e) gives $\varsigma_d^2 = (4ca+8Da^2)/q = 4a\lambda/q$ and $\varepsilon^{\Pi} = (1-\lambda^2)/q$, so $\xi_2^2 = 2a\lambda/(1-\lambda^2) = (a/\varepsilon)\cdot 2\lambda/(1+\lambda)$, with $\varepsilon = 1-\lambda$ the leak of $W$ and $\lambda = c+2Da$. The covariance summed over hyperplanes decays along each axis at the rate $m$ with $\cosh m = 1+\varepsilon/(2a)$, in every dimension (Theorem 8.10), and $\xi_2 = (2\lambda/(1+\lambda))^{1/2}/(2\sinh(m/2)) \le 1/m$. So $\xi_2$ lies below both the Ornstein–Zernike length and $(a/\varepsilon)^{1/2}$ and agrees with both to leading order as $\varepsilon \to 0$; for nearest-neighbor passing every size statement made with $\xi_2$ remains true with either length of H4. For translation-invariant kernels of finite range $\xi_2 \le \xi_{\mathrm{OZ}}$ is proved in Proposition S8.27; beyond that class it is not established. The correlational coherence count (Definition 8.17) and the reach of information measure other quantities and are compared with $L^*$ in Table 9. The one-round diameter of Proposition S10.20 measures yet another and composes with the size bound instead of competing with it: it bounds the physical length of each coupling, while contrast bounds the extent in couplings (Corollary S10.21).

## S4.6 The two-grid example

**Numerical observation S4.22** (The baseline of the two-grid population)**.** Declare the population of Table 5 on its box of $6\times 12$ sites of $\mathbb{Z}^2$, with the grids side by side and the six cross edges joining them. Under the lattice clause of Definition 4.11 its comparison medium is the box with coupling $\bar{a}_{\mathrm{h}} = 0.91$ between horizontal neighbors, the mean over the 66 horizontal neighbor pairs, coupling $\bar{a}_{\mathrm{v}} = 1$ between vertical neighbors and uniform leak $\varepsilon^{\Pi}$; declared instead as its nearest-neighbor graph, under the graph clause, it has coupling $120.06/126 = 0.9529$ on every edge. In both cases, for every $\varepsilon^{\Pi} > 0$,

$$\bar{r}_{\mathrm{hom}} = \max\Big\{1+\frac{\lambda_2(L_{\mathrm{hom}})}{\varepsilon^{\Pi}}, \frac{\varepsilon^{\Pi}+\lambda_{\mathrm{sq}}}{\varepsilon^{\Pi}+c_{\mathrm{sq}}}\Big\}, \tag{S18}$$

the larger of the contrasts of the whole comparison box and of a $2\times 2$ square at its corner, with $(\lambda_2(L_{\mathrm{hom}}), \lambda_{\mathrm{sq}}, c_{\mathrm{sq}}) = (0.06201, 1.82, 0.955)$ under the lattice clause and $(0.06494, 1.906, 0.9529)$ under the graph clause; the two terms are equal at $\varepsilon^{\Pi} = 0.0738$ and $0.0697$. This was established by evaluating the 104355 connected regions of 2 to 8 sites and the complements of connected sets of 1 to 3 sites, by the bounds of Lemma S28.6 for regions of 9 to 71 sites, and by Lemma S28.3 for disconnected regions; for every competitor, the leaks at which it is dominated by the first term and those at which it is dominated by the second are each given by one inequality linear in the leak, and together they cover $(0,\infty)$. Consequently each grid is a collective at some admissible level exactly for $5.060\times 10^{-4} < \varepsilon^{\Pi} < 0.4223$ under the lattice clause and $5.375\times 10^{-4} < \varepsilon^{\Pi} < 0.3672$ under the graph clause, at the levels $(\bar{r}_{\mathrm{hom}}, r(C_1)]$, with largest significance 3.32 at $\varepsilon^{\Pi} = 0.0122$ (3.20 at 0.0126 under the graph clause); the level $r_0 = 2$ is admissible exactly for $\varepsilon^{\Pi} > \lambda_2(L_{\mathrm{hom}})$; and each $2\times 2$ square at a corner of the box, of contrast $(\varepsilon^{\Pi}+2)/(\varepsilon^{\Pi}+1)$, exceeds the baseline for $\varepsilon^{\Pi} > 0.0661$ (0.0694), with significance at most 1.049 (1.008), as does each $2\times 2$ square at an end of the seam, of contrast $(\varepsilon^{\Pi}+2)/(\varepsilon^{\Pi}+1.005)$, for $\varepsilon^{\Pi} > 0.0668$ (0.2261). Among the regions of the population itself, the connected regions of 2 to 8 sites, at leaks between $10^{-6}$ and $10^{3}$, the 1565 rectangles of 2 to 71 sites, at every leak, and the union were examined, and of these only the two grids and the eight squares just named exceed the baseline at some leak under either clause; the remaining regions of more than eight sites were not examined.

The computation is described in Check S47.31. The formula is not general: in a box of $12\times 3$ sites, blocks of full height at its ends compete with the corner square, and the same method leaves a gap between its bounds.

*Remark* S4.23 (Reading the bridge). Spectral contrast measures the separation of timescales between internal disagreement and the collective mode, and the criterion $r \geq r_0$ is hierarchical: at weak grounding it returns each block and also their union. A returned set is a collective only at an admissible level, above the baseline of Definition 4.11. Below the window the two kinds of criterion part in two different ways, which the closed forms of Theorem 4.18 locate for the two grids of Table 5 at $r_0 = 2$ and $\theta_0 = \frac{1}{2}$. Since $\varkappa_{\mathrm{out}}(C_1 \cup C_2) = \varepsilon^{\Pi}$ and $\varkappa_{\mathrm{in}}(C_1 \cup C_2) = \varepsilon^{\Pi} + \lambda_2(L^{\Pi})$, the criterion returns the union exactly when $\varepsilon^{\Pi} \leq \lambda_2(L^{\Pi})/(r_0 - 1)$, and here $\lambda_2(L^{\Pi}) = 0.003234$, inside the bracket $[0.003123, 0.003333]$ of (41). The correlation of the block averages, computed from the identity of (ii), decreases in the leak and equals $\frac{1}{2}$ at $\varepsilon^{\Pi} = 0.001617$, inside the bracket $[0.001564, 0.001667]$ that the two sides of (42) give. Below $\varepsilon^{\Pi} = 0.001617$ the block averages are correlated above $\frac{1}{2}$, so a correlation criterion does not separate the grids, and the criterion $r \geq 2$ also returns the union. Between 0.001617 and 0.003234 the criterion $r \geq 2$ still returns the union, while the block-average correlation is already below $\frac{1}{2}$: the correlation criterion separates the grids, and the spectral criterion returns them together with their union. Above 0.003234, up to 0.2646, both criteria return the two grids and not their union. The window (43), $0.0033 < \varepsilon^{\Pi} \leq 0.2646$, is the part of this range that the brackets certify, since its lower end $2\bar{a}^{\partial}$ is the upper end of the bracket on $\lambda_2(L^{\Pi})$. Above the window $r(C_1) < r_0$ and the criterion returns nothing.

Which returned sets are collectives is decided by the baseline, here that of the population declared on its $6 \times 12$ box (Numerical observation S4.22). The union is the whole closed population, and its algebraic connectivity, 0.0032, is far below the 0.0620 of its comparison box, so it is a collective at no leak (Proposition S4.17(a)); wherever the criterion returns it, the level 2 lies below the baseline, which exceeds 20 there. Each grid is a collective, at the levels between the baseline and its contrast, for $5.06 \times 10^{-4} < \varepsilon^{\Pi} < 0.4223$, with significance up to 3.32 near $\varepsilon^{\Pi} = 0.012$; below this range the whole comparison box has the larger contrast, above it a corner square of the comparison box does. The level $r_0 = 2$ is admissible only for $\varepsilon^{\Pi} > 0.0620$, the right part of the window, where both grids are collectives at level 2, so that there the sets which both criteria return are collectives. On the left part of the window, $0.0033 < \varepsilon^{\Pi} \leq 0.0620$, the criterion $r \geq 2$ returns the grids and not the union, but at a level below the baseline, which falls from 19.6 to 2 across this interval; the grids are collectives there only at levels between the baseline and their contrast. At strong grounding small compact pieces of the grids exceed the baseline by a few percent. These statements use the lattice clause; the graph clause moves the thresholds to $5.38 \times 10^{-4}$, 0.3672 and 0.0649 and changes no conclusion.

The ultrametric contrast, a difference of log correlations, vanishes at weak grounding, where every world-conditional correlation of a closed population tends to one; at strong grounding it reduces to a comparison of couplings, $\ln(a^{\mathrm{bn}}/a^{\partial}_{\max})$, a statement about the attention graph rather than about collective dynamics, while the criterion $r \geq r_0$ returns nothing. The two kinds of criterion agree only when grounding lies between the cut coupling per member and the internal stiffness. Likewise the ultrametric statement that in a homogeneous medium no set of intermediate size has positive ultrametric contrast (Proposition S10.4) reaches, by a criterion that is not equivalent, the conclusion that follows from Definitions 4.11 and 4.12 once Theorem 4.13 makes the baseline of a homogeneous lattice finite: such a medium contains no collective.

## S4.7 Contrast and coherence with the collective average

Contrast is a statement about relaxation rates, while H1 speaks of coherence. Members of a contrasted region should therefore also be coherent with the region's average, and up to a dependence on dimension they are.

**Proposition S4.24** (Contrast and coherence with the average)**.** *Let the leak be uniform and let $C$ be a region of $n$ members with average $\bar{e}_C = \frac{1}{n}\sum_{i\in C} e_i$ and standard deviations $\sigma_i = \Sigma_{ii}^{1/2}$.*

*(a)* $\mathrm{Var}(\phi_C^\top e) \geq 1/\varkappa_{\mathrm{out}}(C)$*, and* $\mathrm{Var}(f^\top e_C) \leq 1/\varkappa_{\mathrm{in}}(C)$ *for every unit vector $f$ on $C$ orthogonal to $\mathbf{1}_C$.*

*(b) The members' correlations with the average, weighted by their standard deviations, satisfy*

$$\frac{\sum_{i\in C}\sigma_i \,\mathrm{Corr}(e_i, \bar{e}_C)}{\sum_{i\in C}\sigma_i} \geq \Big(\frac{r(C)}{r(C)+n-1}\Big)^{1/2}. \tag{S19}$$

*(c) If $C$ is an isolated region, the whole of a closed population, then*

$$\frac{1}{n}\sum_{i\in C}\frac{1}{\mathrm{Corr}(e_i,\bar{e}_C)^2} = 1 + \varepsilon^{\Pi} T_C(\varepsilon^{\Pi}), \qquad T_C(\varepsilon^{\Pi}) = \sum_{k\geq 2}\frac{1}{\varepsilon^{\Pi}+\lambda_k(L_C)},$$

*and an interval of $n$ members with nearest-neighbor coupling at contrast $r$ has $\varepsilon^{\Pi}T_C(\varepsilon^{\Pi}) < \pi^2/(6(r-1))$ at every $n$.*

Proof in Section S28; checked in Check S47.32. Part (b) gives coherence for regions of order $r$ members or fewer, and (c) at every size in one dimension. In higher dimension coherence fails for large regions.

**Numerical observation S4.25** (Coherence at fixed contrast depends on dimension)**.** For isolated $\ell_b \times \ell_b$ boxes at contrast $r = 5$, $\varepsilon^{\Pi}T_C$ is 1.40, 1.72, 2.01 and 2.30 at $\ell_b = 8$, 16, 32 and 64, with increments 0.32, 0.29 and 0.28 per doubling against the limit $(\pi/8)\ln 2 = 0.27$, the growth rate of $\pi \ln \ell_b/(2(r-1))$. For $\ell_b \times \ell_b \times \ell_b$ boxes it is 2.86, 5.91, 11.39 and 21.81 at $\ell_b = 4$, 8, 16 and 32, a linear growth whose ratio to the Riemann-sum estimate $0.2527\pi^2\ell_b/(r-1)$ is 1.15, 1.19, 1.14 and 1.09.

A collective selected at fixed contrast is therefore coherent with its average in one dimension, marginally so in two, and not in three once it is large: its collective mode carries a vanishing share of each member's variance. The correlational coherence count of Section 8 (Definition 8.17) shows the same dependence on dimension from the other side (Corollary 8.19).

## S4.8 Nested and multiplex collectives

**Proposition S4.26** (Nesting and multiplexity)**.** *(a) Blankets nest: if $C_k \subseteq C$ and $\mathcal{V}(C_k)$ denotes the variables of the members of $C_k$, then $\mathsf{S}(C) \cap \mathcal{V}(C_k) \subseteq \mathsf{S}(C_k)$ and $\mathsf{A}(C) \cap \mathcal{V}(C_k) \subseteq \mathsf{S}(C_k) \cup \mathsf{A}(C_k)$. Exactness of the path blanket at one level neither implies nor is implied by exactness at the other.*

*(b) For one modality, the ultrametric classes of Definition 4.17, at any heights, are nested or disjoint, and the sets of positive ultrametric contrast, which are such classes, form a rooted tree under inclusion. For several modalities with scales $\vartheta_c > 0$, $\max_c u_{(c)}/\vartheta_c$ is an ultrametric whose balls, which we call composite classes, are intersections of balls of the modalities and are nested or disjoint; ultrametric classes of different modalities can cross, and $\min_c u_{(c)}$ is not an ultrametric in general.*

(c) *In the joint process the canonical partition of a modality-c collective puts in its blanket every channel of another modality through which a member reads or drives a non-member, and the modality-restricted partition is then not a path blanket. The split partition is an exact path blanket, given the remaining conditions of Theorem 4.3, exactly when the within-member couplings between the modality-c internal coordinates and the other coordinates and channels of members vanish in both directions.*

Proof in Section S28. Nesting of ultrametric classes follows from the definition by hierarchical clustering and is not an empirical finding. Spectral contrast nests only as far as baselines allow. The criterion $r \geq r_0$ returns nested sets at weak grounding, through the hierarchy of timescales of Theorem 4.18(iv), but a nest in one medium needs every set of the nest above the baseline, and a nest whose outermost set is the whole closed population is excluded except in media more connected than their comparison medium (Proposition S4.17). Numerical observation S4.27 gives a nest in one medium and a hierarchy across levels in the same population. A member who integrates modalities, a citizen whose economic and linguistic beliefs interact, is a channel through which a linguistic community and a market leak into each other; the size of such cross-modal leaks is computed in Section 10.

**Numerical observation S4.27** (Nesting in one medium and across levels)**.** Place the nested medium of Proposition 10.7 with $b = 2$, $n_{\mathrm{H}} = 4$, $A_{\mathrm{h}} = 1$ and $\mathfrak{q} = 0.2$ on the sites $0, \dots, 15$ of $\mathbb{Z}$, its blocks being the dyadic intervals, read its weights as the precision couplings $a^{\Pi}_{ij}$, take the precision leak $\varepsilon^{\Pi} = 0.1$, and declare the boxes of 16, 8 and 4 sites as the geometries of the coarse media of levels 0, 1 and 2, with the lattice clause of Definition 4.11. Enumerating every region of each comparison box gives the level-0 baseline 2.424, while the blocks of 2, 4 and 8 agents have contrasts 6.035, 3.378 and 1.667. Each block of four agents and the two pairs inside it therefore form a nested collective of depth two in one medium at every level in $(2.424, 3.378]$, and the blocks of eight agents are not collectives of that medium. Judged level by level, the blocks of 2, 4 and 8 agents have the same contrasts in the coarse media of levels 0, 1 and 2, whose baselines are 2.424, 1.789 and 1.301, so the three block sizes are collectives of their levels and form a hierarchy with $l_{\max} = 3$, with admissible levels in $(2.424, 6.035]$, $(1.789, 3.378]$ and $(1.301, 1.667]$; the first and the last of these are disjoint, so the hierarchy needs a level declared for each coarse medium.

## S4.9 Scope and reading of contrast

**Open problem S4.28** (Contrast for non-diffusive precisions)**.** Define contrast, and prove a size bound, for world-conditional precisions with positive off-diagonal entries, as produced by non-reciprocal attention. The problem is shared with Sections 5 and 8; until it is solved contrast is not defined for such populations. The available substitute is the mutual-information selector used with the tests of Proposition 5.18, and closure of the collective average under non-reversible drift is controlled by the drift-based contrast of Theorem 5.4, which is not the precision-based contrast: Proposition 5.5 bounds the difference of the corresponding rates by the stationary probability current, and the two contrasts differ in addition through the cut couplings, which the inward rate of Definition 4.10 removes and the drift-based inward rate keeps, so that they can differ even for reversible dynamics.

**Open problem S4.29** (Exact fusion)**.** Decide whether the world-conditional precision of receivers that fuse exactly is diffusive, and define contrast on members' residual disagreements when complete sharing makes $\Sigma^c$ singular.

The rates of Definition 4.10 have a dynamical and a statistical reading. Dynamically, for the reversible diffusion with drift $-\Pi^c$, whose stationary covariance is $\Sigma^c$, $\varkappa_{\text{out}}$ is the rate at which the members' average relaxes and $\varkappa_{\text{in}}$ the slowest relaxation rate of internal disagreement, so $r \gg 1$ separates timescales. Statistically, $1/\varkappa_{\text{out}}$ is the variance of the collective coordinate $\phi_C^\top e$ given all orthogonal coordinates, outsiders' included, and $\varkappa_{\text{in}}$ is the smallest precision of a disagreement mode once couplings to outsiders are removed. A region with $r \gg 1$ thus has a freely fluctuating collective coordinate and pinned internal disagreements, which is H1's requirement that coherence inside be much stronger than across the boundary. For non-reversible dynamics drift and precision differ, closure of the average is controlled by the contrast of the drift (Theorem 5.4 and Proposition 5.5), and each statement names the contrast it uses. The two contrasts differ even without circulation: the drift-based inward rate keeps the couplings to outsiders on the diagonal, so for reversible drift it lies between $\varkappa_{\text{in}}(C)$ and $\varkappa_{\text{in}}(C) + \max_{i\in C} a_i^\partial$, the outward rates agree, and the drift-based contrast is at least $r(C)$. An analyst who sees only the members computes yet another quantity.

**Lemma S4.30** (Members' marginal precision)**.** *Let $\Pi = \operatorname{diag}(\varepsilon^\Pi) + L^\Pi$ be diffusive and positive definite on a finite set $V$, let $C \subseteq V$ and $\bar{C} = V \setminus C$, and let $\mathcal{A}_{C\bar{C}}$ be the matrix of couplings $a^\Pi_{ik}$, $i \in C$, $k \in \bar{C}$. The precision of the members' marginal law, $(\Sigma_{CC})^{-1} = \Pi_{CC} - \mathcal{A}_{C\bar{C}}\Pi^{-1}_{\bar{C}\bar{C}}\mathcal{A}_{\bar{C}C}$, is diffusive. Its leaks $\varepsilon^\Pi_C + \mathcal{A}_{C\bar{C}}\Pi^{-1}_{\bar{C}\bar{C}}\varepsilon^\Pi_{\bar{C}}$ lie between $\varepsilon^\Pi_C$ and $\varepsilon^\Pi_C + a^\partial_C$ entrywise, with $a^\partial_C$ the vector of cut couplings of the members, and its couplings $a^\Pi_{ij} + (\mathcal{A}_{C\bar{C}}\Pi^{-1}_{\bar{C}\bar{C}}\mathcal{A}_{\bar{C}C})_{ij}$ are at least $a^\Pi_{ij}$. The contrast $r'(C)$ computed from it satisfies $r'(C) \geq r(C)$.*

Proof in Section S28; checked in Check S47.24. Marginalizing outsiders is a Kron reduction of the coupling graph.

The clauses of Definition 4.11 contain the earlier conventions as special cases: a translation-invariant medium on $\mathbb{Z}^D$ or on a torus is its own comparison medium, a heterogeneous medium on $\mathbb{Z}^D$ is compared with the translation-invariant medium of its spatially averaged couplings and leak, and a finite population without geometry is compared with an exchangeable population of its size. A declaration states what the model takes as given, and the choice matters (Remark S4.16). In the two-group model, clause (c) gives the population without frame mismatch in which every receiver attends $\kappa$ of the others uniformly at random, with exact translations and its own calibration (Lemma 7.7 and Theorem 7.8). The exchangeable population with the split's own mean coupling and leak would have the lower baseline 7.628 at the base point, against 7.954 (Remark S4.15), so the declared mechanism gives the more demanding comparison there (Check S47.25). The same holds at each of the 162 points of the grid of Numerical observation S7.6, where the declared mechanism gives the higher baseline by 0.5 to 11.2 percent and the split is a collective under both comparisons (Check S47.25). With uniform leak a region has $r(C) = (\varepsilon^\Pi + \lambda_2(L_C))/(\varepsilon^\Pi + \operatorname{cut}(C)/n)$, a region that its couplings do not connect has $r(C) \leq 1$, and a closed population has $r(V) = 1 + \lambda_2(L^\Pi)/\varepsilon^\Pi \geq 1$, so the baseline of a finite closed comparison medium with uniform leak is attained by its whole population or by a region that its couplings connect (Lemma S28.3 in Section S28).

## S4.10 Notes on extent and the bridge

**Open problem S4.31** (Sharp constant and diameter)**.** Decide whether $\pi/\sqrt{12}$ is the supremum of $\operatorname{sd}_d(C)(r(C)-1)^{1/2}/\xi_{2,d}$ over regions and media, compute exact homogeneous baselines beyond finite searches, and bound the diameter of contrast-selected regions, which the root-mean-square extent does not control.

In the notation of Definition 4.17, if $\mathcal{U}(C) > 0$, every world-conditional canonical correlation across the boundary of $C$ is at most $e^{-\varpi_{\text{out}}(C)}$, while any two members are joined by a chain of

links with canonical correlation at least $e^{-u_{\mathrm{in}}(C)}$. Single linkage is used because its dendrogram is an ultrametric (Carlsson and Mémoli, 2010), which makes nesting automatic (Proposition S4.26). Conditioning on the world is necessary when local noise is not large against the world's innovations: a common mode from those innovations bounds every unconditional correlation from below, so under the hypotheses of Proposition S10.27 the unconditional version of $\mathcal{U}$ is at most $\ln(1+\max_i \Sigma^c_{ii}/\Sigma_{\mathrm{cm}})$ for every nonempty proper subset, however strong its barriers, where $\Sigma_{\mathrm{cm}}$ is the variance of the common mode. In the scalar level-0 network $\Sigma_{\mathrm{cm}} = Q/(1-\lambda^2)$, with $Q$ the variance of the world innovation and $\lambda$ the common row sum of $W$, and the cap is $\ln(1+(1-\lambda^2)\max_i \Sigma^c_{ii}/Q)$. It tends to zero as grounding weakens only when $(1-\lambda^2)\max_i \Sigma^c_{ii} \to 0$, as in extended media; in a finite closed population it does not vanish. For $N$ agents with $W = \lambda P$, $P$ symmetric and doubly stochastic, and local noise covariance $Q^\zeta = qI$, the uniform mode of $\Sigma^c$ alone contributes $q/(N(1-\lambda^2))$ to every diagonal entry, so the cap is at least $\ln(1+q/(NQ))$ at every $\lambda$, and it tends to this value as $\lambda \to 1$ when $P$ is irreducible and aperiodic; it is of order $\ln(1+q/(nQ))$ for nearly isolated blocks of $n$ members while $1-\lambda^2$ is large against the coupling between the blocks and small against the spectral gap within them. The cap grows with $q/Q$: it forces every unconditional contrast to be small only when local noise is not large against the world innovation, $q/(NQ) \ll 1$ at weak grounding, and when $q/Q$ is large the barriers remain visible in unconditional correlations. On the ring of Check S47.116 at $q/Q = 100$ the unconditional contrast of one arc at leak 0.002 is 1.445 against the world-conditional 2.120. The world-conditional contrast has no such cap and grows without bound as the barriers strengthen at fixed grounding.

**Open problem S4.32** (The bridge beyond two equal blocks)**.** Prove explicit two-sided bounds of the form (41) and (42) for unequal blocks, for more than two blocks and for blocks coupled to an exterior, where the common mode is no longer an exact eigenvector of $\Pi$.

# S5 Additional results for Section 5: closure, agency and interaction

This section collects statements, examples and remarks of Section 5 that the main text cites without stating in full. Proofs are in Section S29.

## S5.1 Memory and closure of the average

Part (b) of Theorem 5.3 sharpens a reading of the boundary-interior split: the condition that the eliminated agents be unobservable from the retained ones, $W_{R\bar{R}}W^l_{\bar{R}\bar{R}} = 0$ for all $l \geq 0$, contains $l = 0$ and so says that the retained agents do not hear the eliminated ones at all. The summary of a collective whose boundary hears its interior is therefore never sufficient, $\mathfrak{d}_{\mathrm{cl}} > 0$, although its memory in the sense of $\mathfrak{d}_{\mathrm{M}}$ may still vanish: for $A = \left(\begin{smallmatrix} a & b \\ 0 & 0 \end{smallmatrix}\right)$ with $|a| < 1$ and $b \neq 0$, $Q^{\mathfrak{n}} = I$ and $\mathsf{P} = \left(\begin{smallmatrix} 1 & 0 \end{smallmatrix}\right)$ the hidden coordinate is white noise, the retained one is an exact first-order autoregression with $\mathfrak{d}_{\mathrm{M}} = 0$, and $\mathfrak{d}_{\mathrm{cl}} = \frac{1}{2}\ln(1+b^2)$ (Check S47.36; see also Example S12.3). Part (d) describes, for an exogenous input path, the Markov augmentation that is needed only when $\mathfrak{d}_{\mathrm{M}} > 0$: the filter of the hidden coordinates given the retained path, which is the role that an active-inference agent's internal states play. Part (d) is an open-loop statement: an input path generated by feedback from the retained or hidden coordinates, as the active states of a blanket are, violates its hypothesis that the path is known, deterministic or independent of $(x_0, \mathfrak{n})$, and neither the filter nor the order bound is established here under feedback. Without input, part (b) is equivalent to exact specification of a two-level hierarchical model whose autonomous top level is the coarse variable (Proposition S12.2), and a Gaussian Markov model of the coarse variable alone is misspecified at a relative entropy rate of at least its Markov defect, which is at most $\mathfrak{d}_{\mathrm{cl}}$ (Lemma S12.4). Asked of a meta-agent's exterior,

with the blanket as input, the same question decides whether the meta-agent has an abstract physics (Theorem 11.8).

*Remark* S5.1 (Minimal memory). The memory in (46) is the linear system with state $x_\perp$, input $(y, x_{\mathsf{B}}, \mathfrak{n}^\perp)$ and output $A_{y\perp}x_\perp$. Its input enters $x_\perp$ through the identity, so every hidden mode is reachable, and the least dimension of a linear memory that reproduces (46) exactly is the rank of the observability matrix of $(A_{y\perp}, A_{\perp\perp})$, equivalently of the block Hankel matrix $(A_{y\perp}A_{\perp\perp}^{k+l})_{k,l\geq 0}$, by minimal-realization theory (Ho and Kalman, 1966). For finite reversible Markov chains the same block elimination gives the memory equation of a partition statistic, which closes for every initial law exactly when the partition is lumpable in the sense of Kemeny and Snell (1960); this is the form of the identity in the author's earlier analysis (Dennis, 2026d).

**Open problem S5.2** (Minimal stochastic memory)**.** Determine the least dimension of a Markov augmentation of $y$ in law, the order of a minimal stochastic realization of the retained process, which can be smaller than $\dim x - \dim y$ and smaller than the dimension of Remark S5.1.

Part (a) of Theorem 5.4 says that, for the reversible diffusion with drift $\Pi^c_{CC}$ and the exterior held fixed, the contrast that selects a collective also makes its average close. The average is Markov to leading order, with slow rate $\varpi_1 \approx \varkappa_{\mathrm{out}}$ and memory of relative size $\Xi$, when internal disagreement relaxes much faster than the average, $r(C) \gg 1$, or when leakage is spread evenly over the members, $\sigma_C$ small, and the two effects multiply. For a boundary layer of $n_\partial$ members, each with cut coupling $\bar{a}^\partial$, and a uniform leak $\varepsilon^\Pi$, $\varkappa_{\mathrm{out}} = \varepsilon^\Pi + \bar{a}^\partial n_\partial/n$ and $\sigma_C = \bar{a}^\partial[(n_\partial/n)(1-n_\partial/n)]^{1/2}$. Part (b) covers non-reversible attention, for which the drift is no longer the precision; its contrast is the drift-based $r^M$ and not the selecting contrast $r(C)$, Proposition 5.5 compares the rates of the two, and a closure theorem in terms of $r(C)$ for non-reversible drift is open (Open problem S5.4). The dispersion that controls memory is that of the row and column sums of the whole drift, and non-normality enters through the circulation $\sigma_{\mathrm{circ}}$, not through a multiplicative constant. For the reversible drift $M = \Pi^c_{CC}$ of part (a), with the exterior held fixed, $\varkappa^M_{\mathrm{out}} = \varkappa_{\mathrm{out}}$ and $\varkappa^M_{\mathrm{in}} \geq \varkappa_{\mathrm{in}}$, because $\varkappa^M_{\mathrm{in}}$ keeps the cut couplings on the diagonal, and part (b) gives a bound of the form of part (a) with cruder constants; in random high-contrast regions the bound (50) was used to at most 1.2 percent (Check S47.37). Lemma 11.10 extends both parts to any unit coordinate in place of $\phi_C$. Theorem 13.8 gives the joint form for a population partitioned into several collectives, each block's exterior being the other blocks, and Proposition S12.5 bounds through the same contrast the misspecification of a member that models the collective average as an Ornstein–Uhlenbeck latent cause.

**Example S5.3** (The symmetric-dispersion conjecture is false)**.** An earlier formulation of this closure statement conjectured that, for a drift whose symmetric part is diffusive, $\Xi \leq c_*(\sigma_{\mathrm{sym}}/(\varkappa^M_{\mathrm{in}} - \varkappa^M_{\mathrm{out}}))^2\varkappa^M_{\mathrm{out}}/\varkappa^M_{\mathrm{in}}$, with $c_*$ depending only on $\|M - M^\top\|/\varkappa^M_{\mathrm{in}}$. The conjecture is false. Take a directed attention pattern $a_{ij} \geq 0$ on $C$ whose imbalance between attention received and attention given is not identically zero, and leaks

$$\varepsilon_i = \varepsilon_0 + \frac{1}{2}\Big(\sum_j a_{ji} - \sum_j a_{ij}\Big) + \frac{1}{2}\max_k\Big(\sum_j a_{kj} - \sum_j a_{jk}\Big) \geq \varepsilon_0 > 0. \tag{S20}$$

Then $M_{\mathrm{sym}}\mathbf{1}$ is constant, so $\sigma_{\mathrm{sym}} = 0$ and the conjectured bound vanishes, while $\sigma_{\mathrm{row}} = \sigma_{\mathrm{col}} = \sigma_{\mathrm{circ}} > 0$ and $\Gamma(0) = -\sigma^2_{\mathrm{circ}} \neq 0$. In random instances with drift-based contrast between 1.6 and 2.6 the Markov error is of order $10^{-2}$, and in a family on eight members whose symmetric coupling grows against a fixed directed pattern it stays positive as the contrast grows to about 130, below the bound of Theorem 5.4(b)(iii), whose hypothesis holds there (Check S47.38). The correct statement is Theorem 5.4(b), whose dispersions add the circulation to $\sigma_{\mathrm{sym}}$.

In Proposition 5.5, $\Pi$ is the stationary precision of the dynamics with drift $M$. For a reversible population with the exterior held fixed it is the block $\Pi^c_{CC}$ of the world-conditional precision, which part (a) of Theorem 5.4 uses. The contrast of a collective is defined on the precision (Definition 4.10); closure of its average under non-reversible drift is controlled by the drift-based contrast with row and column dispersions. Proposition 5.5 compares rates and not contrasts: $\varkappa^M_{\text{out}}$ and $\phi_C^\top \Pi \phi_C$ differ by at most $\|J\|\|(I - \phi_C\phi_C^\top)\Pi\phi_C|$, and $\varkappa^M_{\text{in}}$ differs from $\lambda_{\min}(Z\Pi Z^\top)$ by at most $\|J\|\|\Pi\|$. The second quantity is not the inward rate of Definition 4.10, which removes the cut couplings from the diagonal: for $J = 0$ and $M = \Pi^c_{CC}$, the drift of the members of $C$ with the exterior held fixed, $\varkappa^M_{\text{in}} = \lambda_{\min}(Z\Pi^c_{CC}Z^\top)$ satisfies $\varkappa_{\text{in}} \le \varkappa^M_{\text{in}} \le \varkappa_{\text{in}} + \max_{i\in C} a^\partial_i$ and $r^M \ge r(C)$, the cut couplings account for the gap, and the drift-based contrast can exceed the selecting contrast with no circulation at all. On a path of three members with unit couplings, leaks 0.05 and one end member carrying a cut coupling 1, $\varkappa_{\text{in}} = 1.05$ and $r(C) = 2.739$, whereas $\varkappa^M_{\text{in}} = 1.501$ and $r^M = 3.917$ (Check S47.39). For strongly non-reversible attention the difference is not small: in random non-reversible regions the two values of the outward rate differed by up to 62 percent (Check S47.39). Contrast itself is defined only for diffusive precisions, which non-reciprocal attention need not produce (Open problem S4.28).

**Open problem S5.4** (Closure in discrete time)**.** Prove a version of Theorem 5.4 for the discrete-time level-0 recursion, which needs a bound on $\|ZWZ^\top\|$ that the symmetric part of $I - W$ does not provide, and a closure theorem stated for the precision-based contrast of Definition 4.10 when the dynamics is strongly non-reversible, where the precision of a region given a fixed exterior and the block of the population's precision differ. The reversible case for one coordinate is Lemma 11.11; open are the precision-based contrast of Definition 4.10 and non-reversible recursions.

Lemma 11.11 answers the first part for reversible recursions, among them the level-0 recursion of a collective with symmetric $W_{CC}$ and isotropic local noise, with bounds through drift rates and through the rates of the members' stationary precision with the exterior held fixed, and Proposition S11.10 gives the exact memory terms of a level-0 collective average. Non-reversible recursions, and the block of the population's world-conditional precision on which contrast is defined, remain open.

## S5.2 Inferential closure

**Proposition S5.5** (Literal consensus counts the prior repeatedly)**.** *Let the world be scalar with prior $\mathcal{N}(m_\circ, \Lambda_\circ^{-1})$, let member $i$ sense $s_i = \eta + v_i$ with precision $\Lambda^s_i > 0$, let the $n \ge 2$ members form a connected undirected graph with adjacency matrix* Adj*, and let each round update every member's natural parameters by $(h, \Lambda) \leftarrow (I + \text{Adj})(h, \Lambda)$ from $h_i(0) = \Lambda_\circ m_\circ + \Lambda^s_i s_i$ and $\Lambda_i(0) = \Lambda_\circ + \Lambda^s_i$, which is the level-1 reading of every neighbor's posterior over a noiseless channel with its posted precision as the law header. Let $\mathsf{v}$ be the unit Perron vector of $I +$ Adj. Then:*

*(a) every member's mean converges to $\bar m^{\text{lit}} = \mathsf{v}^\top h(0)/\mathsf{v}^\top\Lambda(0)$, a pool weighted by eigenvector centrality;*

*(b) the believed precision after $n_c$ rounds grows like $\rho(I + \text{Adj})^{n_c}\mathsf{v}_i\mathsf{v}^\top\Lambda(0)$ and diverges;*

*(c) $\bar m^{\text{lit}}$ equals the union-model posterior mean for all data if and only if $\Lambda_\circ = 0$ and the graph is regular; with $\Lambda_\circ > 0$ on a regular graph the prior receives the weight $n\Lambda_\circ/(n\Lambda_\circ + \sum_j \Lambda^s_j)$ instead of $\Lambda_\circ/(\Lambda_\circ + \sum_j \Lambda^s_j)$, so that the common prior is counted $n$ times;*

*(d) for $\Lambda_\circ > 0$ the mean-squared error of the limit is $\mathsf{w}_\circ^2/\Lambda_\circ + \sum_j \mathsf{w}_j^2/\Lambda_j^s$, with $\mathsf{w}_j = \mathsf{v}_j\Lambda_j^s/\mathsf{v}^\top\Lambda(0)$ and $\mathsf{w}_\circ = \Lambda_\circ \sum_j \mathsf{v}_j/\mathsf{v}^\top\Lambda(0)$, at least the union posterior's $1/(\Lambda_\circ + \sum_j \Lambda_j^s)$, while the believed variance tends to zero.*

*Proof in Section S29; checked in Check S47.40.*

**Open problem S5.6** (Exact fusion at a finite number of rounds)**.** Determine inferential closure under exact, cross-covariance-aware fusion of posterior messages after a finite number of rounds: in a static Gaussian world, the number of rounds after which every member holds the union posterior, relative to the diameter of the graph, and in a dynamic world the rate. For agents with Gaussian private signals of a static state who exchange estimates with their neighbors and update by Bayes's rule, Mossel et al. (2016) state in their abstract that every agent reaches the full-information belief within $2n\,\mathrm{diam}(\mathcal{G})$ rounds for $n$ agents, and within $\mathrm{diam}(\mathcal{G})$ rounds on trees and distance-transitive graphs; whether their protocol coincides with the exact fusion of posterior messages considered here is not settled by that statement.

*Remark* S5.7 (The Gaussian location family)*.* A mean-pooled belief closes exactly under common-message dynamics only in a Gaussian location family: in the author's formation document, if the members' laws lie in a regular exponential family and relax their natural parameters toward a common message, and the velocity of the pooled mean parameter of a two-member block is a function of the pooled mean for every message, then the laws are Gaussian with one covariance (Dennis, 2026b). The scope there is relaxation toward a common message, not filtering of private data. The same family returns in Theorem 5.10(d), where beliefs that differ only by translation make deliberation an exact group posterior.

## S5.3 Bounds on the agency defect

*Remark* S5.8 (Heterogeneous precisions)*.* With a common belief and member-specific precisions $\gamma_i$, the cycle ratio of Theorem 5.10(a) through the null policies is $\exp(-(\gamma_j - \gamma_i)I_{q_C}(o_{A'}; o_{B'} \mid o_{A_{-ij}}))$. Member $i$'s conditional is a logit at the mean precision $\bar\gamma$ of the cost $(\gamma_i/\bar\gamma)G_C$, and by (S114) the part of $(\gamma_i/\bar\gamma - 1)G_C$ that depends on $\pi_i$ is $(\gamma_i/\bar\gamma - 1)[-I_{q_C}(\eta_C; o_A) + \psi_A^{(i)}(q_C) + I_{q_C}(o_A; o_{A_{-i}})]$, with $A = \mathfrak{B}_i(\pi_i)$: a tilt of member $i$'s habits plus a cross term. Heterogeneous precisions therefore act as a redundancy disagreement of size at most $|\gamma_i/\bar\gamma - 1| \max I_{q_C}(o_A; o_{A_{-i}})$, Theorem 5.11 applies with $\bar\gamma$ in place of $\gamma$ and the tilts included in $\Phi$, and the single precision of (P3) matters only when the members' inputs are redundant (Check S47.42).

The three bounds of Theorem 5.11 cover different regimes. The tree bound holds at every precision and grows with the number of profiles; with the full defect it is the per-profile bound of Candogan et al. (2013, Corollary 5.2(ii) of the arXiv version) with the argument of the hyperbolic tangent halved, because one member's logit probabilities in the two games differ by at most the factor $e^{\gamma\sigma}$ rather than the factor $e^{2\gamma\sigma}$ of their Lemma 5.1, and here it holds in total variation. The contraction bound does not depend on the number of profiles and is linear in $\gamma\sigma_{\max}$ but needs weak redundancy relative to $1/\gamma$; the spectral bound needs $e^{\gamma\sigma_{\max}} - 1$ below a gap that can be exponentially small in $\gamma$. In the checked instances no bound was violated and neither the contraction nor the spectral bound dominated the other, while at $\gamma = 12$ all three were vacuous or inapplicable at a distance of 0.040 (Check S47.42). Because $q_C$ is arbitrary in Lemma 5.9, every bound may be minimized over the declared belief, and by Theorem 5.10(b) the part of the defect that no declared belief removes is the members' disagreement with one another about redundancy. No bound uniform in $\gamma$ can be linear in $\sigma_{\max}$ alone: when the members agree with one another but not with $q_C$, $\mathbb{P}^\infty$ is the Gibbs law of a potential that differs from $\Phi$ by a perturbation of size of order $\sigma_{\max}$, and at

large $\gamma$ such a perturbation moves a Gibbs law between nearly degenerate minima by an amount of order one.

**Open problem S5.9** (A uniform agency bound)**.** Find a bound on the distance between the deliberation law and a group posterior that is uniform in $\gamma$ and polynomial in the number of members for weak redundancy disagreement, or a family showing that the factor $|\mathcal{A}| - 1$ of the tree bound is needed for perturbations without a potential part.

**Proposition S5.10** (Belief dispersion)**.** *Let the slots be linear Gaussian, $o_A = \mathsf{H}_A \eta_C + \mathsf{n}_A$ with $\mathsf{n}_A \sim \mathcal{N}(0, \mathsf{R}_A)$, $\mathsf{R}_A \succ 0$, independent across slots, and let $q_i = \mathcal{N}(m_i, P_i)$ and $q_C = \mathcal{N}(m_C, P_C)$, with covariance dispersion $\mathrm{Th}_i = \|\ln(P_C^{-1/2} P_i P_C^{-1/2})\|$ in the operator norm, the Thompson distance on the cone of positive definite matrices. Then for every profile, with $A = \mathfrak{B}_i(\pi_i)$,*

$$|\mathcal{R}_i(\pi)| \le \min\Big\{\dim(o_A)\mathrm{Th}_i, \max_{q \in \{q_i, q_C\}} I_q(o_A; o_{A_{-i}})\Big\}, \qquad \sigma_i \le 2 \max_{\pi_i} \dim(o_{\mathfrak{B}_i(\pi_i)})\mathrm{Th}_i. \tag{S21}$$

*The means do not enter. Proof in Section S29; checked in Check S47.42.*

## S5.4 Agency index and obstructions to agency

**Proposition S5.11** (Properties of the agency index)**.** *(a) $0 \le \mathfrak{A}_{\mathcal{X}_{\mathrm{dec}}}(C) \le 1$; $\mathfrak{A}_{\mathcal{X}_{\mathrm{dec}}}(C)$ is nondecreasing in the class $\mathfrak{G}$, and invariant under relabeling of the members when $\mathfrak{G}$ is closed under relabeling; and if $\mathfrak{G}$ allows free joint habits, $\mathfrak{A} = 1$ at every single decision state, so the class must be restricted, for instance to slot-product preferences, the belief and slot model of the collective's closure model, and one precision over the declared states.*

*(b) Under the exact cases of Theorem 5.10(c) and (d), $\mathfrak{A} = 1$ for every class that contains the group model with tilted habits named there.*

*(c) At one decision state, if $\mathfrak{G}$ contains the group model of $\mathbb{P}_\Phi$ (belief $q_C$, the members' slot preferences, precision $\gamma$ and product habits with free weights), then for every $B_{\mathrm{KL}} \ge D_{\mathrm{KL}}(\mathbb{P}^\infty \| \mathbb{P}_\Phi)$ and $B_{\mathrm{TV}} \ge \mathrm{TV}(\mathbb{P}^\infty, \mathbb{P}_\Phi)$ with $\mathsf{c}_\Phi = D_{\mathrm{KL}}(\mathbb{P}_\Phi \| E) - \gamma \operatorname{osc}(\Phi) B_{\mathrm{TV}} > 0$,*

$$1 - \mathfrak{A} \le \frac{D_{\mathrm{KL}}(\mathbb{P}^\infty \| \mathbb{P}_\Phi)}{D_{\mathrm{KL}}(\mathbb{P}^\infty \| E)} \le \frac{B_{\mathrm{KL}}}{B_{\mathrm{KL}} + \mathsf{c}_\Phi}, \tag{S22}$$

*where $B_{\mathrm{KL}}$ and $B_{\mathrm{TV}}$ may be taken from Theorem 5.11.*
*Proof in Section S29; checked in Check S47.43.*

**Proposition S5.12** (Synchronous deliberation)**.** *If two members with full-support habits and finite expected free energies (P3) revise simultaneously, each from its own conditional given the other's previous policy, the unique stationary law is a product law, whatever the values of their expected free energies. Hence synchronous deliberation of two members reproduces no group posterior with product habits and $\gamma_C > 0$ whose slots carry redundant information, $I_{q_C}(o_{A'}; o_{B'}) > 0$ for some policies. Proof in Section S29; checked in Check S47.41.*

**Conjecture S5.13** (Synchronous deliberation of three or more members)**.** *For three or more data-sharing members satisfying (P0) to (P3) who revise simultaneously, and some pair whose slots carry redundant information, the stationary law differs from the group posterior $\prod_i E_i e^{-\gamma G_C}$ for every $\gamma > 0$. An instance with equality would refute it. In a checked three-member instance the synchronous law is not a product law and differs from the group posterior by total variation 0.033 (Check S47.41).*

**Example S5.14** (Without data sharing, redundancy defeats agency)**.** Members who do not expect to receive their co-members' data evaluate $G_i(\pi_i) = -I_{q_C}(\eta_C; o_{\mathfrak{B}_i(\pi_i)}) + \psi^{(i)}_{\mathfrak{B}_i(\pi_i)}(q_C)$, a function of their own policy only. Their profile is an exact potential game with the separable potential $\sum_i G_i$, so the stationary law of deliberation is a product law, while every group model with product habits, a slot-product preference and $\gamma_C > 0$ gives two members whose slots, opened from the null policies, share information a negative log odds ratio (Proposition S5.19). The deliberation law $\prod_i E_i e^{-\gamma G_i}/\mathsf{Z}_i$ is then a group posterior of that kind only for a model with $\gamma_C = 0$ and re-tilted habits $E_i e^{-\gamma G_i}/\mathsf{Z}_i$, which is the habit-only model of Definition 5.12 only when the members act by habit; for two members reading identical scalar inputs the two laws are at total variation 0.057 (Check S47.41). A collective at any contrast level whose members do not share data, hold redundant slots and depart from their habits therefore fails agency relative to every class of such group models whose habits are held fixed at $\prod_i E_i$ or whose precision, apart from the habit-only model that Definition 5.12 requires, is bounded below by a positive constant. Relative to a class with free product habits that admits $\gamma_C = 0$ the same law has agency index 1, so the failure is a statement about the declared class.

**Example S5.15** (An unrepresented driver)**.** Let a common driver $\mathsf{d}$ of both members' slot costs, such as an exterior price or a shared reliability, vary given the collective's coarse state, and let the members satisfy (P0) to (P4) at each value of $\mathsf{d}$. At each value the deliberation law is the group posterior, with log odds ratio $-\gamma I < 0$. Because both members' costs move with the driver, the law given the coarse state, a mixture over $\mathsf{d}$, associates their choices positively: in a checked instance each component has log odds ratio $-0.13$, while mixtures over two cost shifts have log odds ratios between $+0.20$ and $+3.94$ (Check S47.41). No group model with product habits, a slot-product preference and $\gamma_C \geq 0$ has a positive log odds ratio between epistemic slots opened from the null policies (Proposition S5.19), so agency requires that the decision state be a function of the collective's coarse state: agency requires closure.

### S5.5 Joint performative acts

These obstructions (Proposition S5.12 and Examples S5.14 and S5.15) concern collectives whose joint policies only gather information. A collective whose active states change the world faces a different calculation.

**Definition S5.16** (Group model with joint performative acts)**.** Member $i$'s policy is a pair $\pi_i = (u_i, \pi_i^{\mathrm{epi}})$ of an intervention $u_i$ in a finite set $\mathcal{U}_i$ of vectors containing 0 and an epistemic slot choice $\pi_i^{\mathrm{epi}}$, whose slots observe the current external state $\eta$ before the acts take effect (Proposition S2.6). The external state moves as

$$\eta' = F\eta + \sum_i \mathcal{B}_i u_i + \mathcal{B}_{\mathrm{int}}(u) + w, \qquad w \sim \mathcal{N}(0, Q), \tag{S23}$$

with an interaction term $\mathcal{B}_{\mathrm{int}}$, which vanishes for additive effects, and the preference-relevant outcomes are $o^\star = \mathsf{H}_\star \eta' + \mathsf{n}_\star$ with Gaussian noise $\mathsf{n}_\star$. Member $i$ believes $\eta \sim \mathcal{N}(m_i, P_i)$, models the effect of $u_j$ on $o^\star$ as $\mathsf{M}_j^{(i)}$, the truth being $\mathsf{M}_j = \mathsf{H}_\star \mathcal{B}_j$, and prefers $o^\star \sim \mathcal{N}(o_i^\star, (\Lambda_i^\star)^{-1})$. The collective active state is $\sum_i \mathcal{B}_i u_i + \mathcal{B}_{\mathrm{int}}(u)$. The group model has belief $\mathcal{N}(m_C, P_C)$, preference $\mathcal{N}(o_C^\star, (\Lambda_C^\star)^{-1})$, the true effects, one precision and product habits.

Interventions shift means and leave covariances unchanged, so a member's expected free energy splits into an epistemic part that depends on the slot choices only and an act part that depends

on the interventions only. Writing $\|v\|^2_{\Lambda^\star} = v^\top \Lambda^\star v$ and $\langle v, v' \rangle_{\Lambda^\star} = v^\top \Lambda^\star v'$, the act part is, up to constants,

$$G_i^{\mathrm{act}}(u) = \frac{1}{2}\|\mathsf{z}_i + f^{(i)}(u)\|^2_{\Lambda_i^\star}, \qquad \mathsf{z}_i = \mathsf{H}_\star F m_i - o_i^\star, \qquad f^{(i)}(u) = \sum_j \mathsf{M}_j^{(i)} u_j + \mathsf{H}_\star \mathcal{B}_{\mathrm{int}}(u), \quad \text{(S24)}$$

and the group's act part $G_C^{\mathrm{act}}$ is the same expression with $\mathsf{z}_C$, $\Lambda_C^\star$ and the true effects.

**Theorem S5.17** (Potential structure of joint acts)**.** *Under (P2) to (P4):*

*(a) (Additive effects.) If $\mathcal{B}_{\mathrm{int}} = 0$, the act component is an exact potential game if and only if, for all members $i \neq j$ and all realized differences of their interventions,*

$$(u_i' - u_i)^\top \mathsf{M}_i^{(i)\top} \Lambda_i^\star \mathsf{M}_j^{(i)} (u_j' - u_j) = (u_i' - u_i)^\top \mathsf{M}_i^{(j)\top} \Lambda_j^\star \mathsf{M}_j^{(j)} (u_j' - u_j), \quad \text{(S25)}$$

*that is, when the members agree on how their acts interact in preference space; the four-cycle ratio of the deliberation chain is $\exp(-\gamma(\partial^2_{ij} G_j^{\mathrm{act}} - \partial^2_{ij} G_i^{\mathrm{act}}))$, with $\partial^2_{ij} G_k^{\mathrm{act}}$ the left or right side of (S25) for $k = i$ or $k = j$.*

*(b) (Shared precision and effect models.) If $\mathcal{B}_{\mathrm{int}} = 0$ and the members share $\Lambda_i^\star = \Lambda^\star$ and the true effect models, while their targets and beliefs differ, then with $\Lambda_C^\star = \Lambda^\star$ and any $\mathsf{z}_C$,*

$$G_i^{\mathrm{act}}(u) = \Phi^{\mathrm{act}}(u) + (\textit{a function of } u_{-i}), \qquad \Phi^{\mathrm{act}}(u) = G_C^{\mathrm{act}}(u) + \sum_j \langle \mathsf{z}_j - \mathsf{z}_C, \mathsf{M}_j u_j \rangle_{\Lambda^\star}, \quad \text{(S26)}$$

*so the act component is an exact potential game whose potential is the group expected free energy plus linear own-act terms. If the epistemic component has a potential equal to its group expected free energy plus own-policy terms, as under Theorem 5.10(c) and (d), the stationary law of deliberation is the group posterior of a group model with tilted habits.*

*(c) (Interacting effects.) If the members share $\Lambda^\star$ and the true effect models, the four-cycle sum for members $i$ and $j$ is*

$$\mathcal{L}_{ij} = -\langle \mathsf{z}_i - \mathsf{z}_j, \mathsf{H}_\star \partial^2_{ij} \mathcal{B}_{\mathrm{int}} \rangle_{\Lambda^\star}, \quad \text{(S27)}$$

*so the game is an exact potential game if and only if the members agree on their displacements $\mathsf{z}_i$, predicted outcome minus target, along every joint-effect direction $\mathsf{H}_\star \partial^2_{ij} \mathcal{B}_{\mathrm{int}}$. With a common displacement $\mathsf{z}_i = \mathsf{z}_C$ and $\Lambda_C^\star = \Lambda^\star$ the potential is $G_C^{\mathrm{act}}$, whatever $\mathcal{B}_{\mathrm{int}}$.*

*Proof in Section S29; checked in Check S47.44.*

With a shared preference precision and agreement on how acts combine, disagreement about where to steer the world, whether it comes from different targets or from different estimates of the present state, is absorbed into each member's bias toward its own act, and the collective acts as one agent over joint performative acts. With interacting effects this survives only if the members agree on their displacements along every joint effect, which common targets do not ensure when the members' estimates of the present state differ. For a meta-agent whose members run the consensus filter on its coarse exterior, part (b) applies with the meta-agent's predictive law as the declared belief, and the tilts of the habits are bounded through the members' consensus deviations, which vanish as the number of rounds grows under a symmetric, doubly stochastic and nonnegative consensus matrix with $\rho_\perp < 1$ (Theorem 11.13(v)).

**Example S5.18** (Interacting interventions break the potential)**.** Two members with $u_i \in \{0,1\}$ hold a joint act $\mathcal{B}_{\mathrm{int}}(u) = u_1 u_2 \mathsf{v}_{12}$ that works only when both act, such as a commitment that needs two keys, share $\Lambda^\star$ and the true effect models, and have displacements with $\langle \mathsf{z}_1 - \mathsf{z}_2, \mathsf{H}_\star \mathsf{v}_{12} \rangle_{\Lambda^\star} \neq 0$. The cycle log ratio is $\gamma \langle \mathsf{z}_1 - \mathsf{z}_2, \mathsf{H}_\star \mathsf{v}_{12} \rangle_{\Lambda^\star} \neq 0$, so deliberation is not reversible and the members' expected free energies have no potential. The stationary law then depends on the deliberation protocol: in instances with three acts per member, changing the members' selection probabilities from $(0.5, 0.5)$ to $(0.85, 0.15)$ moves the stationary law by up to 0.30 in total variation, against at most $1.4 \times 10^{-14}$ for the game with common displacements (Check S47.44). With a potential, every selection rule with full support gives the same Gibbs law; without one, who deliberates more often decides what the collective does.

**Proposition S5.19** (The sign of coordination)**.** *Under a group posterior $\mathbb{P}_C \propto \prod_i E_i e^{-\gamma_C G_C}$ with product habits, the log odds ratio of two members' choices with the others fixed is*

$$\ln \frac{\mathbb{P}_C(\pi_i', \pi_j')\mathbb{P}_C(\pi_i, \pi_j)}{\mathbb{P}_C(\pi_i', \pi_j)\mathbb{P}_C(\pi_i, \pi_j')} = -\gamma_C \partial_{ij}^2 G_C.$$

*For epistemic slots opened from the null policies it is $-\gamma_C I_{q_C}(o_{A'}; o_{B'} \mid o_{A_{-ij}}) \leq 0$ (Dennis, 2026a); for additive acts it is $-\gamma_C \langle \mathsf{M}_i(u_i' - u_i), \mathsf{M}_j(u_j' - u_j) \rangle_{\Lambda_C^\star}$, of either sign. Proof in Section S29; checked in Check S47.44.*

The sign obstruction behind the failures of group agency in the earlier report, and behind Example S5.15, is therefore specific to collectives whose joint policies only gather information: relative to the null policies, a group posterior can represent positively coordinated choices only when the choices change the world (Check S47.44). The reference to the null policies matters, because between two non-null epistemic alternatives the second difference $\partial_{ij}^2 G_C$ is a signed combination of four conditional mutual informations, its sign depends on the labeling of the alternatives, and the log odds ratio can be positive: for $\eta \in \mathbb{R}^2$ with identity prior covariance and slots of unit noise variance, a first member choosing between a slot on the first coordinate $(\pi_i)$ and a slot on the second $(\pi_i')$, and a second member choosing between the null policy $(\pi_j)$ and a slot on the first coordinate $(\pi_j')$, with zero costs, the log odds ratio of Proposition S5.19 is $\frac{1}{2}\gamma_C \ln \frac{4}{3} > 0$. This gives H1's agency property content beyond the redundancy of boundary inputs.

*Remark* S5.20 (Consensus and agency). Under the dynamic consensus of Proposition 5.7(b) every member's posterior covariance equals the centralized one at every step, so the members' beliefs about the collective's external states differ only in their means. By Theorem 5.10(d) the stationary law of deliberation is then exactly a group posterior, with habits tilted by the members' mean differences, at every number of consensus rounds: finite consensus changes the effective habits, not the existence of agency. Under literal consensus, in the scalar world of Proposition S5.5, the members' precisions after $n_c$ rounds are the entries $[(I + \mathrm{Adj})^{n_c} \Lambda(0)]_i$. They coincide at every number of rounds $n_c \geq 0$ exactly when the graph is regular and the initial precisions are equal, each then being $(1+d)^{n_c} \Lambda_1(0)$ on a graph of degree $d$; for $n_c \geq 1$ alone it suffices that the graph is regular and the entries of $(I + \mathrm{Adj})\Lambda(0)$ are equal, as they are on the complete graph for every $\Lambda(0)$. In either case the members then share one covariance and Theorem 5.10(d) again gives an exact group posterior. Whenever the precisions differ, the dispersion of Proposition S5.10 is positive for at least one member, whatever Gaussian $q_C$ is declared, and Theorem 5.11 applies.

**Open problem S5.21** (Mixed epistemic and performative policies)**.** State agency for collectives whose policies mix epistemic slots and performative acts when interventions change what slots reveal, for instance by changing the precision of observations or the world's noise, where the additive split behind (S24) fails.

## S5.6 Identity and records

In Definition 5.13, conditioning on the exterior's exogenous input excludes persistence borrowed from a persistent environment, which Krakauer et al. (2020) call environmentally driven individuality, and the noises and initial states of external records extend the exclusion to archives that outsiders write from their own inputs. The exterior path itself is not the conditioning set, because an exterior that hears the members carries a copy of their states: conditioning on it would remove the members' own persistence, the masking that Example S5.25 exhibits for the interaction order parameter. For a stationary scalar statistic with autoregression coefficient 0.9 and unit innovation variance, heard by an exterior variable $x_{t+1} = 0.5x_t + Y_t + \nu_t$ with unit noise variance, $I(Y_0; Y_5) = 0.214$ nats, which is also its value given the exterior's inputs, whereas given the exterior path it is $8 \times 10^{-5}$ nats. An archive that an outsider writes from what it hears of the members is likewise a function of the members' states, and the persistence it carries is not removed. Internal and boundary records are collective variables, so the persistence they carry counts.

**Conjecture S5.22** (Conformity gives long identity without records)**.** *If a collective statistic is held by a bistable conformity dynamic, in which members relax toward the local majority in a double-well effective potential, and newcomers are naive ($\mathsf{f} = 0$) but conform, the identity time grows exponentially in the number of members, unlike the identity time $1/(\varepsilon + g)$ that Proposition 5.14 gives without a record and with naive newcomers. A model of this kind with identity time bounded by a polynomial in $n$ would refute it. If it holds, Proposition 5.14 describes collectives without conformity.*

Records also bear on the boundary. With channel-level roles, an archive written by members and read by outsiders is a boundary record and leaks nothing (Proposition 4.4). A description that keeps whole agents and marginalizes records turns the same archive into a lagged coupling from interior to exterior, the writer's counterpart of the leak that interior sensing of the shared world creates in agent-level descriptions (Section 4).

**Proposition S5.23** (Writer-activated leak through a record)**.** *Let an interior agent whose state $\mu_{\mathrm{in}}(t)$ is white noise of variance $q_{\mathrm{wr}}$ write a record, $z(t+1) = (1-\delta_z)z(t) + \omega\mu_{\mathrm{in}}(t) + \zeta_z(t)$, with a loss $0 < \delta_z \le 1$ per round that does not depend on the write weight $\omega$ and $\operatorname{Var}\zeta_z = q_{\mathrm{rec}}$, and let an exterior agent read it, $\mu_{\mathrm{ex}}(t+1) = c_{\mathrm{ex}}\mu_{\mathrm{ex}}(t) + \beta z(t) + \zeta_{\mathrm{ex}}(t)$, with $|c_{\mathrm{ex}}| < 1$ and $\operatorname{Var}\zeta_{\mathrm{ex}} = q_{\mathrm{rd}}$, all noises independent. In the description that keeps the two agents and marginalizes the record, the leak rate is*

$$\iota_{\mathrm{path}} = \frac{1}{2}\ln\frac{\mathsf{a}_1 + (\mathsf{a}_1^2 - 4\mathsf{a}_\times^2)^{1/2}}{\mathsf{a}_0 + (\mathsf{a}_0^2 - 4\mathsf{a}_\times^2)^{1/2}}, \tag{S28}$$

*with $\mathsf{a}_0 = q_{\mathrm{rd}}(1 + (1-\delta_z)^2) + \beta^2 q_{\mathrm{rec}}$, $\mathsf{a}_1 = \mathsf{a}_0 + \beta^2\omega^2 q_{\mathrm{wr}}$ and $\mathsf{a}_\times = q_{\mathrm{rd}}(1-\delta_z)$; it is positive whenever $\beta\omega \neq 0$. With $q_{\mathrm{rec}} = 0$ and $\mathrm{snr} = \beta^2\omega^2 q_{\mathrm{wr}}/q_{\mathrm{rd}}$,*

$$\iota_{\mathrm{path}} = \ln\frac{(\delta_z^2 + \mathrm{snr})^{1/2} + ((2-\delta_z)^2 + \mathrm{snr})^{1/2}}{2} = \frac{1}{2}\Big((\delta_z^2 + \mathrm{snr})^{1/2} - \delta_z\Big)\Big(1 + O\big((\delta_z^2 + \mathrm{snr})^{1/2}\big)\Big), \tag{S29}$$

*which is close to $\mathrm{snr}/(2\delta_z(2-\delta_z))$, second order in the coupling $\beta\omega$, when $\mathrm{snr} \ll \delta_z^2$, and to $\mathrm{snr}^{1/2}/2$, first order, when $\delta_z^2 \ll \mathrm{snr} \ll 1$. Proof in Section S29; checked in Check S47.46.*

The records of Definition 2.9 have loss $\delta_z = \omega + \delta \ge \omega$, so $\mathrm{snr} \le \beta^2\delta_z^2 q_{\mathrm{wr}}/q_{\mathrm{rd}}$ and they stay in the quadratic regime whenever $\beta^2 q_{\mathrm{wr}} \ll q_{\mathrm{rd}}$; an accumulating archive, whose loss tends to zero at a fixed write weight, leaks at first order. Keeping the record and assigning it to the blanket removes the leak, as channel-level roles require: on a layered chain whose interior agent writes a record that an exterior agent reads, the leak is $8.4 \times 10^{-4}$ with the record marginalized, 0 with the record kept in the blanket and 0.028 with the record placed in the exterior (Check S47.46).

## S5.7 Interaction order parameter and tests

**Example S5.24** (Conditioning on an agent-driven world creates dependence)**.** Three scalar agents neither sense the world nor exchange messages; each has private internal noise, and two of them act on it, $\eta(t+1) = F\eta(t) + \mathcal{B}_1\mu_1(t) + \mathcal{B}_2\mu_2(t) + w(t)$. No causal path joins the agents, and (63) vanishes. Conditioning instead on the world path and the initial state gives $1.39 \times 10^{-3}$ nats over six rounds in the instance with $F = 0.9$, agent states $\mu_i(t+1) = 0.4\mu_i(t) + 0.5p_i(t)$ driven by independent standard normal private noises $p_i$, act gains $\mathcal{B}_1 = 0.30$ and $\mathcal{B}_2 = 0.25$, and unit variances of the world innovations and of the initial states: the world path is a common effect of both actors' private noises, and conditioning on it couples them, which is explaining away as in the collider of Example S4.2. The magnitude belongs to the instance: it grows approximately as the fourth power of a common factor on the two act gains while that factor is at most one, and a second instance with larger gains gives 0.20 nats. With the acts replaced by exogenous forcing, both versions vanish (Check S47.47).

**Example S5.25** (Conditioning on an agent-driven world hides interaction)**.** Let agent 1 act on the world and agent 3 sense the world and nothing else. Agent 3 depends on agent 1 only through the world, and the interaction is real: over six rounds the conditional mutual information of their paths given the shared drive is 0.614 nats in the instance with world coefficient 0.8, $\mu_1(t+1) = 0.6\mu_1(t) + p_1(t)$ acting on the world with gain 0.7, $\mu_3(t+1) = 0.4\mu_3(t) + 0.5\eta(t) + p_3(t)$, and unit variances of the private noises, the world innovations and the initial states, a value that depends on these gains. Given the world path and the initial state it is 0 to round-off, as Proposition 5.16(b) requires (Check S47.47).

The declarations of Table 6 have reasons. The deletion and freezing nulls, made separating by $\mathfrak{N}_{\text{eff}}$ when acts are present, give $\Omega_{\text{int}} = 0$ by construction, the deletion nulls also give contrast at most one in populations without records (Proposition 5.16(a),(d)), and almost every coupling configuration gives $\Omega_{\text{int}} > 0$ (Proposition 5.16(c)). Differing from them is necessary for any claim of interaction and is achieved by any communication at all, so they are weak. They delete every inter-agent channel of the population, so differing from them, like $\Omega_{\text{int}}(C) > 0$, certifies interaction in the population and not interaction among the members of a candidate $C$, which can depend on one another through a non-member alone; the member-restricted order parameter of Definition 5.15 isolates the latter. Partner randomization at matched rates keeps the amount of communication and destroys its arrangement; it tests whether the structure is produced by whom the agents choose to hear, which makes it the decisive differ null. The exact-fusion null removes only a misspecification and keeps the mechanism, so a structure that vanishes under it is attributed to misreading of joint structure (Section 3), an echo-chamber artifact when overvaluation drives it, and not to communication as action. Read as a null to differ from, it would reward misspecification; read as a persistence clause, it asks that the structure not be an artifact of misreading, which is the reading that hypothesis H2 supports and the earlier report adopted (Dennis, 2026a). The explanation nulls do not remove interaction: agents that share a niche may hear each other because their messages are informative, so agreement with $\mathfrak{N}_{\text{w}}$ or $\mathfrak{N}_{\text{code}}$ restricts the claim to communication within access groups or code groups. Forgetting is part of the agents' declared dynamics, not a misspecification, so $\mathfrak{N}_{\text{fg}}$ is a diagnostic. Under Dirichlet habits the comparison with $\mathfrak{N}_{\text{hab}}$ is stated almost surely about limit frequencies or with a declared pseudo-count and threshold.

Part (a) of Proposition 5.18 is Wilks's theorem (Wilks, 1938) applied to the conditional cross-covariance block. The restriction to $U_i$ and $U_j$ is needed for the statistic to exist: with $\mu_i(0)$ kept in the path, the residual covariance given the shared drive is singular and its log-determinant is undefined. At moderate sample sizes the test is liberal, and replacing $n_{\text{rep}}$ by the finite-sample multiplier $n_{\text{rep}} - 1 - \dim \mathcal{X}_{\text{sh}} - (\dim U_i + \dim U_j + 1)/2$ brought the rejection rate near its nominal

level in a check. The multiplier has the form of Bartlett's factor for Wilks's criterion in a regression of $U_i$ on $U_j$ and the shared drive, with $\dim U_j$ hypothesis and $n_{\mathrm{rep}} - 1 - \dim \mathcal{X}_{\mathrm{sh}} - \dim U_j$ error degrees of freedom; no proof of the correction is given here, and its support in this paper is numerical (Check S47.48). Survival of $\mathfrak{N}_{\mathrm{ex}}$ is decided by recomputing the selector under exact fusion on the same parameter set, which in model analysis is deterministic. For directed attention the world-conditional precision need not be diffusive and contrast is then undefined (Open problem S4.28); the selector of part (b) is then the largest ratio of mean within-block to mean between-block world-conditional mutual information over balanced bipartitions. With eight agents in two groups and attention capacity $\kappa = 2$, its observed value 4.65 exceeded the 95 percent quantile 3.02 of 999 quenched partner randomizations ($p = 0.007$), while the largest null value, 11.3, shows that random graphs sometimes split, which is why the comparison must be distributional (Check S47.48).

**Open problem S5.26** (Positivity under selected attention)**.** Prove generic positivity of $\Omega_{\mathrm{int}}$, as in Proposition 5.16(c), when attention is selected by expected free energy and therefore depends on the agents' states.

**Open problem S5.27** (An unrecorded shared drive)**.** Find an estimator of (63) when the shared drive is not recorded. A partial route exists for one Gaussian vector sampled independently. With an observed block and a latent block of coordinates, indexed by o and h, and joint precision $K$, the precision of the observed marginal is $K_{\mathrm{oo}} - K_{\mathrm{oh}} K_{\mathrm{hh}}^{-1} K_{\mathrm{ho}}$, the conditional precision given the latent coordinates minus a term of rank at most the number of latent coordinates, and Chandrasekaran et al. (2012) show that the two terms are locally identifiable from the observed covariance, and are recovered by a convex program, when the conditional precision is sparse and the latent effect is spread over most of the observed coordinates, in the quantitative sense of their transversality condition. For a static or stationary world without performative acts, in which the members' equal-time covariance is the world-conditional covariance plus a common-mode term of low rank, this locally identifies the world-conditional precision from members' statistics alone under those conditions. For (63), which is a function of the conditional covariance $K_{\mathrm{oo}}^{-1}$ when the observed block stacks the coordinates of the members' window paths that are not functions of the shared drive and the latent block is the whole shared drive, these conditions are not verified here: sender-side broadcast noises and record noises add a latent coordinate for every sender or record in every round, heard only by that channel's audience, so the latent term need be neither of low rank nor spread, and successive windows of one path are not independent samples. The estimator is open in that case, when the conditional precision is not sparse, when the latent term, which for a uniform common mode lies along the vector $\Pi^{c}\mathbf{1}$ of the members' leaks, is concentrated on few agents, and when acts drive the world. With performative acts the world innovations can be recovered from the world and act paths only within a structural model, and a test that conditions on observable variables without the collider bias of Example S5.24 is not available. Section S25 treats the qualitative part, whether (63) is positive, in a linear-Gaussian network with an unrecorded driver: records under one regime do not settle it without restrictions on the driver, since the stationary covariance identifies nothing about interaction (Theorem S25.3) and the recorded path law cannot separate interaction from a driver of full rank on the interacting agents (Theorem S25.5(b)), whereas under the invariance conditions (I1) to (I3) a change of the recorded law under a channel intervention certifies that it is positive (Theorem S25.9(a)). The model with interaction and a colored driver together is open (Open problem S25.14), and the estimator asked for here remains open.

## S5.8 Status of H1

*Remark* S5.28 (What is proved and what is declared). Parts (a), (b) and (d) of Theorem 5.20 are theorems in the linear-Gaussian framework with fixed or exogenously scheduled couplings, (c) is a theorem for continuous-time linear dynamics with the exterior held fixed, and (e) for finite policy sets under the deliberation protocol. Closure for the discrete-time level-0 recursion is settled only for one coordinate of a reversible recursion with the exterior held fixed (Lemma 11.11); non-reversible recursions and the precision-based contrast of Definition 4.10, in discrete time and under strong non-reversibility, are open (Open problem S5.4), and contrast is undefined for non-diffusive precisions (Open problem S4.28). What makes $C$ a meta-agent, the grade of Definition 5.19 with the declared types of Table 6, is a definition; in particular, differing from partner randomization and surviving exact fusion are requirements, not consequences of (a) to (e). The split of the two-group formation model is reached from the declared start by the frozen-calibration mean-field map, a theorem about that map with the stochastic step from the full model open (Theorem 7.6(a),(b) and Open problem 7.14). That the split is a collective at admissible levels is a separate claim: it is a theorem at $\gamma = \infty$ with exact in-group translations for every level in the window of Theorem 7.8(c), a window that is nonempty by computation at the grid points of Numerical observation S7.6, and at finite policy precision it is a computation at the base points of Numerical observation S7.7; online calibration from a symmetric start reaches its attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell, counted at a cutoff that makes the counts upper bounds on those at the cutoff of the mean-field map and with reruns of a cell differing by up to three replicates, and is open (Numerical observation 7.11 and Open problem S7.13). By proof the split differs from the deletion nulls, and from the freezing null only when the initial states that frozen messages carry are deterministic or conditioned on (Proposition 7.9(a)); by computation it differs from partner randomization (Section 7). From a start law exchangeable given the world path it fails the translation-aware form of the exact-fusion persistence clause, in which receivers know their translation errors, as Proposition 7.12 proves; for full cross-covariance-aware exact fusion in that model this conclusion is heuristic (Section 7.8), and persistence with drifting frames and formation beyond this model are conjectures (Conjectures S7.12 and 7.13). Above the threshold $\gamma_T$ it coincides with the partition predicted from codes alone. Under Definition 5.19 it is therefore a collective attributed to misreading of joint structure, not a meta-agent, a verdict proved for the translation-aware null and heuristic for full exact fusion. No population analyzed here, synthetic or empirical, is shown to meet every clause of Definition 5.19: parts (a) to (e) establish or bound the clauses one at a time, under hypotheses that are not shown to hold together, and their joint satisfiability is Open problem S5.29. Agency is independent of the other properties: it depends on the protocol and on the members' agreement about redundancy, which contrast does not constrain.

## S5.9 Commentary on closure

The coarse variable of Definition 5.1 must be constrained, because an unconstrained $\mathsf{P}$ whose rows span a subspace invariant under $A^\top$, for instance slow left eigenvectors, satisfies $\mathsf{P}A = \bar{A}\mathsf{P}$ for some $\bar{A}$ and is exactly closed (Dennis, 2026a): closure of some summary selects nothing. The definition asks for small memory and nothing more. Small memory, statistical sufficiency of the summary and a slow retained spectrum are pairwise distinct conditions: for reversible finite chains, four-state examples have an exactly closed summary without a slow spectrum, a slow retained mode without closure, and a vanishing memory residual without sufficiency (Dennis, 2026d). Sufficiency fails for the natural summary whenever the boundary reads the interior (Theorem 5.3(b)), and timescale separation is what contrast already measures; what closure adds to contrast is the alignment of the

declared variable with the slow mode, which is what bounds its memory (Theorem 5.4).

Theorem 5.2 is the discrete-time form of the projection of Zwanzig (1961) and Mori (1965): a coarse variable obeys its own equation exactly, at the price of a memory kernel and colored noise, both of which depend on the complement. This is Assumption A7 in exact form, and the same identity carries the other closure statements. For the block coordinates of a reversible population the kernel of part (b) integrates to the static correction of Theorem 4.18(ii) (Theorem 13.8(b)), and in the linear-Gaussian case part (b) identifies the memory kernel, the initial-state term and the color of the force that the elimination of internal states in the adiabatic reduction of Friston (2019) drops (Proposition S13.10).

In Proposition 5.7, as $n_c \to \infty$ the deviation vanishes, which is the exact statement of the distributed Kalman filter. The members' average is exact before that limit: with any consensus matrix that preserves sums it equals the steady-state centralized filter pathwise at every number of rounds, and only the members' deviations from it depend on $n_c$ (Proposition 11.12(b)). The world enters (53) through $\alpha_{cl}$, the memory of the centralized filter, and through $\|\bar{P}\|$: a slowly varying, weakly observed world needs more consensus rounds per world step, which is Assumption A6 in quantitative form. Before consensus converges a member claims more precision than the data it holds supports, because it multiplies its current estimate of the average contribution by $n$. A receiver that reads a neighbor's posterior mean literally, as an independent observation of $\eta$ whose variance is the neighbor's posted posterior variance, reads at level 1 in the sense of Definition 2.4, the posted variance acting as a law header, and consensus of that kind never closes.

The dependence of inferential closure on the interpretation level (Propositions S5.5 and 5.7) is therefore categorical. Consensus on information contributions, which is the information form of exact fusion, closes at the geometric rate of the consensus matrix; literal consensus at level 1 closes at no rate, with a bias toward central members in the mean, the persuasion bias of DeMarzo et al. (2003), and divergent overconfidence. On regular graphs the prior is counted $n$ times, as in the product of posteriors of Proposition 5.6(b). Exact fusion of posterior messages lies between the two cases. The contrast persists beyond the static world: forming a collective belief at every step as the product of the members' posteriors is overconfident and, when stable, less accurate than the exact filter (Proposition S11.9), a record rebuilt each round from its readers' posteriors has a claimed precision that grows geometrically under the product rule (Proposition S12.7), and across levels the persuasion bias multiplies while the information form stays exact (Proposition S13.17).

### S5.10 Commentary on agency

The index of Definition 5.12 measures as-if agency, representability by a collective expected free energy, not efficacy. With two members and two policies each, product habits and a nonnegative precision fit every full-support joint law whose log odds ratio has the sign of $-\partial^2_{ij} G_C$, three parameters for three degrees of freedom. With free product habits and $\gamma_C \geq 0$, every product law $\prod_i p_i$ is the group posterior with $\gamma_C = 0$ and habits $p_i$, and so has index 1 for any number of members and policies; the same holds when the admissible precisions of the models other than the habit-only model are positive but not bounded away from zero, because the group posteriors with habits $p_i$ converge to $\prod_i p_i$ as $\gamma_C \to 0$ and the infimum in (61) is zero. The failures of agency in Example S5.14 and Proposition S5.12, whose laws are product laws, are therefore failures relative to classes whose habits are held fixed at $\prod_i E_i$ or whose precision, apart from the habit-only model that Definition 5.12 requires, is bounded below by a positive constant. With habits held fixed the index discriminates already for two members (Check S47.43). Each protocol hypothesis removes an obstruction, as the results of Section S5.4 show.

Read as a statement about institutions, which is an interpretation and not a result, Theorems 5.10

and S5.17 give a criterion an institution can be checked against: shared reports and announced intentions (P2), one decision maker at a time (P4), agreement on how redundant the members' inputs are, and, for joint acts, agreement on how acts combine. Simultaneous voting fails the protocol (Proposition S5.12). Whether particular cultures, markets or polities meet the criterion is a question for Section 10.

### S5.11 Commentary on identity and interaction

The model of Proposition 5.14 is the record model of Definition 2.9 in continuous time with turnover: under the dictionary $\varepsilon \mapsto \varepsilon + g(1-\mathsf{f})$, $\beta \mapsto \beta + g\mathsf{f}$, with $\omega$ and $\delta$ unchanged, $\mathsf{A}$ is minus the matrix $\mathsf{R}$ of the reduced record model with $\upsilon = 0$, so that, when $\beta + g\mathsf{f} > 0$ as Proposition 8.11 requires of the read weight, $\tau_{\mathrm{id}}^{-1}$ is its effective leak (Proposition 8.11 and Remark S8.16). Without a record, identity through complete turnover needs nearly perfect transmission at entry: at $g = 1$ and $\varepsilon = 0.01$ the ratio of identity time to residence time is 0.99, 1.96, 9.09 and 50.0 at $\mathsf{f} = 0$, 0.5, 0.9 and 0.99. A durable medium that newcomers read faster than it is rewritten gives identity times of hundreds of residence times, up to 989 at $\beta = 100$, $\omega = 0.01$, $\mathsf{f} = 0.9$ and $\delta = 0.001$ (Check S47.45). This is the correlation time of H4 extended by stigmergic media (Proposition 8.12), stated as identity. Corollary 12.5 brackets the gain in identity time that a durable record gives over newcomers who copy the members' mean, and bounds it below for every decay; for a grounded coordinate in place of a convention, the memory that a member-written record buys is paid for in shared error (Theorem 12.7).

The shared drive of Definition 5.15 does not contain the private inputs of non-members, so for a proper subset $C$ the statistic $\Omega_{\mathrm{int}}(C)$ registers every dependence that interaction in the population produces between members, including a common drive that an outsider supplies. Two members that never hear each other and both read one outsider, with world coefficient 0.9, sensing gain 0.3, read gain 0.4 and unit noise variances, have $\Omega_{\mathrm{int}} = 0.029$ nats over six rounds, and the member-restricted order parameter is zero (Check S47.47). Positivity of $\Omega_{\mathrm{int}}(C)$ therefore certifies interaction in the population that reaches the members of $C$, and a claim that the members of $C$ interact with one another rests on $\Omega_{\mathrm{int}}(C \mid \zeta_{V\setminus C}) > 0$. The member-restricted order parameter vanishes whenever every member's path is a measurable function of $(\mathcal{X}_{\mathrm{sh}}, \zeta_{V\setminus C}, \zeta_i)$, by the independence of the inputs as in Proposition 5.16(a), which is the case when no member is heard by another member, directly or through a record, an act or a relay of non-members; a dependence between two members that a relaying outsider carries from one to the other is not removed, because it originates in a member's private input. For the whole population the two statistics coincide.

### S5.12 Open problem and reading of H1

**Open problem S5.29** (Existence of a meta-agent)**.** Exhibit one linear-Gaussian population and one set $C$ of its agents that meets every clause of Definition 5.19 at a stated grade with $r_0 > 1$ and $\mathfrak{A}_0 > 0$, that is, a collective at level $r_0$ that is closed at stated tolerances for a declared coarse variable, differs from every applicable null of type D, including partner randomization at a stated level, survives exact fusion and has agency index at least $\mathfrak{A}_0$; or prove that the clauses are jointly unsatisfiable in a stated class of populations.

Hypothesis H1 in final form (Definition 5.19 and Theorem 5.20) therefore reads as follows. A meta-agent is a set of agents whose world-conditional precision shows contrast above a declared baseline, which in a population without records already separates it from the deletion nulls (Theorem 5.20(b) proves the separation only there, so with records it is a separate requirement) and, for reversible dynamics with evenly spread leakage, makes its average close to leading order, and which in addition

is closed for a declared coarse variable, differs from partner randomization, survives exact fusion and has an agency index above a declared level. The index is maximal when members who share data and intentions deliberate asynchronously with beliefs that differ only in location, and close to maximal when they nearly agree about the redundancy of what they hear.

# S6 Additional results for Section 6: coherence

This section collects statements, examples and remarks of Section 6 that the main text cites without stating in full. Proofs are in Section S30.

## S6.1 Fidelity

Fidelity decides how far content can travel before nothing of it remains. Its loss along a path is governed by strong data processing, whose coefficient has a closed form for Gaussian channels. We write $\sigma_1(M) \ge \sigma_2(M) \ge \cdots$ for the singular values of a matrix $M$.

**Proposition S6.1** (Gaussian strong data processing)**.** *Let $X$ be a centered Gaussian vector with nonsingular covariance, and let the channel $\mathsf{K}$ be $Y = BX + n_{\mathsf{K}}$ with $n_{\mathsf{K}} \sim \mathcal{N}(0, R_{\mathsf{K}})$ independent of $X$ and $R_{\mathsf{K}} \succ 0$. Put $M_{XY} = \operatorname{Cov}(Y)^{-1/2} B \operatorname{Cov}(X)^{1/2}$, whose singular values are the canonical correlations of $(X, Y)$. Then*

$$s_{\mathrm{KL}}(P_X, \mathsf{K}) := \sup_{0 < D_{\mathrm{KL}}(Q_X \| P_X) < \infty} \frac{D_{\mathrm{KL}}(Q_X \mathsf{K} \| P_X \mathsf{K})}{D_{\mathrm{KL}}(Q_X \| P_X)} = \sigma_1(M_{XY})^2, \tag{S30}$$

*attained by mean shifts along the top canonical direction. Consequently $I(U;Y) \le \sigma_1(M_{XY})^2 I(U;X)$ whenever $U$, $X$, $Y$ form a Markov chain, and the constant is best possible.*

For a Gaussian source and an additive Gaussian channel the coefficient is known in substance (Section S16.5), while for non-Gaussian pairs the mutual-information constant can exceed the squared maximal correlation (Anantharam et al., 2013), which we read in abstract. Our proof uses only the decay of relative entropy along the Ornstein–Uhlenbeck semigroup, and it composes along relays.

**Proposition S6.2** (Relay paths)**.** *Let $X_0, X_1, \dots, X_L$ be a Markov chain of centered, jointly Gaussian vectors with nonsingular covariances and nonsingular conditional covariances $\operatorname{Cov}(X_k \mid X_{k-1})$, and let $M_k = \operatorname{Cov}(X_k)^{-1/2} \operatorname{Cov}(X_k, X_{k-1}) \operatorname{Cov}(X_{k-1})^{-1/2}$.*

*(a) $\operatorname{Cov}(X_L)^{-1/2} \operatorname{Cov}(X_L, X_0) \operatorname{Cov}(X_0)^{-1/2} = M_L M_{L-1} \cdots M_1$, so the top canonical correlation of $(X_0, X_L)$ is at most $\prod_k \sigma_1(M_k)$.*

*(b) The channel from $X_0$ to $X_L$ has $s_{\mathrm{KL}} = \sigma_1(M_L \cdots M_1)^2 \le \prod_k \sigma_1(M_k)^2$, and $I(U; X_L) \le \big(\prod_k \sigma_1(M_k)^2\big) I(U; X_0)$ whenever $U$, $X_0, \dots, X_L$ form a Markov chain.*

*(c) For scalar chains the correlation of $X_0$ and $X_L$ is the product of the correlations of consecutive variables, and*

$$I(X_0; X_L) = -\tfrac{1}{2} \ln\big(1 - \operatorname{Corr}(X_0, X_L)^2\big).$$

Equality in (a) holds when the top left singular vector of each $M_k$ is a top right singular vector of $M_{k+1}$. Proofs in Section S30.1; checks in Checks S47.49 and S47.50. The coefficient of a hop depends on the law of what enters it, so the rate along a path is set by what relays send: about a static world, relays that do not sense lose information doubly exponentially, exponentially or only algebraically

as they forward posterior means, content scaled to unit variance or unbiased estimates, and in a persistent world the per-hop factor of pure relays with gain $g$ is $\psi = F^2 g/(1 - F^2(1-g)) \le F^2$, which tends to one as $F \to 1$, so pure relays become massless only in that limit (Section S30.1).

In a network the same contraction acts hop by hop, and with the light cone and the clustering bound (Theorems 8.5 and 8.9) it bounds what an agent holds about anything injected elsewhere.

**Proposition S6.3** (Information decays at twice the mass)**.** *Let scalar agents in a linear-Gaussian world run a level-0 network with exact code models.*

*(a) Let the connected part of their errors obey*

$$e(t+1) = W(t)e(t) + \zeta(t), \qquad \lambda = \sup_t \max_i \sum_j |W_{ij}(t)| < 1,$$

*under* (Lin), (G) *and* (Loc) *(Assumption 8.1), each $\zeta_i(t)$ containing a component of variance at least $q_{\min} > 0$ independent of everything else. Let a scalar $U$ of finite variance, independent of the world, the initial errors and the other noise components jointly, enter only through agent $j$ at rounds $t' \le s$, as $\zeta_j(t') = g_j(t')U + \zeta_j'(t')$ with $|g_j(t')| \le \bar{g}$. Then for $t > s$ and $Z$ the connected error $e_i(t)$ or the mean $m_i(t)$,*

$$I(U; Z) \le \frac{\bar{g}^2 \operatorname{Var}(U)}{2q_{\min}(1-\lambda)^2} \lambda^{2\max\{d(i,j), t-1-s\}}.$$

*(b) On the infinite ring with $W_{ii} = c \ge 0$, $W_{ij} = a > 0$ for neighbors and $\lambda = c + 2a = 1 - \varepsilon < 1$, local noise of variance $q$ and a stationary scalar world, the information $I$ that $m_i(t)$ carries about the whole sequence of local noises injected at $j$ satisfies*

$$\lim_{d(i,j)\to\infty} \frac{-\ln I}{d(i,j)} = 2m, \qquad \cosh m = 1 + \frac{\varepsilon}{2a}.$$

Information is quadratic in influence: a source entering at $j$ moves $m_i$ through an entry of the resolvent of $W$, which decays at the mass, so the information length is $\xi/2$ in the Ornstein–Zernike form of Theorem 8.10. On the ring the proof bounds the prefactor between a multiple of $1/d(i,j)$ and a constant; a saddle-point estimate suggests $d(i,j)^{-1/2}$, which improves the fit of the check but is not proved. Proof in Section S30.1; checked in Check S47.51.

## S6.2 Withdrawal across complementary access

The example below complements Theorem 6.4, in the notation of Definition 6.2.

**Example S6.4** (Unique information keeps an inconsistent edge)**.** Let group $A$ sense only $\eta_1$ and group $B$ both coordinates, with $\sigma_\eta^2 = 1$, $R_c = 0.1$, sensing noise $R^s = 1$ in $A$ and $R^s = 0.5$ in $B$, $\kappa = 1$ and `C1g` calibration per coordinate. A receiver in $A$ keeps attending a cross-group sender with mistranslation Rot(0.3), Rot(1.0) or Rot($\pi/2$): the believed values of the cross edge are 0.640, 0.370 and 0.241, against 0.153 for the in-group sender (Check S47.52). The inconsistency sits on an attended edge, so withdrawal can fail when a cut separates complementary access to the world.

## S6.3 Flatness, coded consistency and frustration

The consensus space Cons($\Theta$) and the holonomies are those of Definition 6.1 and Theorem 6.5, and the frustration $\mathfrak{F}$, its operator $\mathcal{L}$, the precisions $\varpi_{ij}$ and the metrics $M_i$ are those of Proposition 6.6.

**Proposition S6.5** (True translations are flat)**.** *Let every agent encode one world through an invertible encoding, linear ($\mu_i = U_i m_i$) or general ($\mathcal{E}_i$). The true translations $U_i U_j^{-1}$, in general $\mathcal{E}_i \circ \mathcal{E}_j^{-1}$, have identity holonomy around every closed walk. On a weakly connected attention graph their consensus space is $\{(U_i x)_{i \in V} : x \in \mathbb{R}^K\}$, of dimension $K$, and the least eigenvalue of their frustration form (Proposition 6.6) is zero. Hence positive least frustration of the translations agents use arises only from learned translations that differ from the true ones, from generative models that are not invertible encodings of the true world, or from agents whose internal spaces do not encode one common space.*

Consensus about $\eta$ is thus always possible in principle, and the consistency component of H3 measures the error of learned translations, which grounded learning removes on attended edges (Proposition S7.1). Much of that error is also invisible: in the tangent learning model of Dennis (2026c) with an isotropic anchor, no loop prior and grounded messages on every edge, private encodings near a common frame leave first-order learned holonomy only on cycles whose edges carry codes with different kernels, such as cycles read through different modalities, and to first order in the frame error none of it is detectable from messages (Section S30.3). Between two language communities, sets of agents that share one encoding, the true translations are one fixed map per direction on all bridges, so the wall between them is flat in the sense of Dennis (2026c). Proof in Section S30.3.

**Open problem S6.6** (Frustration for general translations)**.** For translations in $\mathrm{GL}(K)$ no continuous conjugation-invariant function of the holonomy vanishes only at the identity, since the identity is a limit of conjugates of every unipotent matrix (Dennis, 2026c). Find frustration bounds under a declared metric, for example one built from the receivers' covariances, and the correlation functions of transported averaging under frustration beyond the bottom of the spectrum.

With rank-limited codes only what is talked about must agree, and the holonomy criterion gives way to a sheaf-theoretic one.

**Proposition S6.7** (Coded consistency)**.** *Let a public code $T_c$ of rank $r_c$ be shared, and let receiver $i$ compare the message $T_c \mu_j$ with its prediction $T_c \Theta_{ij}^{-1} \mu_i$. Coded consensus is the space $H^0(\mathfrak{S}) = \ker \delta_{\mathfrak{S}}$ of global sections of the cellular sheaf $\mathfrak{S}$ on $\mathcal{G}_\mathrm{a}$ with vertex stalks $\mathbb{R}^K$, edge stalks $\mathbb{R}^{r_c}$, restriction maps $T_c \Theta_{ij}^{-1}$ on the receiver's side and $T_c$ on the sender's side, and coboundary $(\delta_{\mathfrak{S}} \mu)_{j \to i} = T_c \Theta_{ij}^{-1} \mu_i - T_c \mu_j$.*

*(a)* $\dim H^0(\mathfrak{S}) \geq NK - |\mathcal{E}_\mathrm{a}| r_c$.

*(b) The holonomy criterion does not characterize coded consensus: on the directed three-cycle with $K = 2$, $T_c = (1\ 0)$ and translations that rotate by $0.3$, $0.5$ and $\pi/2 - 0.8$, whose holonomy is a quarter turn,* $\mathrm{Cons}(\Theta) = 0$ *while* $\dim H^0(\mathfrak{S}) = 3$.

The coded frustration $\|\delta_{\mathfrak{S}} \mu\|^2$ replaces $\mathfrak{F}$, and narrow codes impose fewer consistency constraints, so low fidelity buys tolerance of diverse frames. This is the discourse sheaf of Hansen and Ghrist (2021), with spectral theory in Hansen and Ghrist (2019); we read their abstracts. Which over-constrained subgraphs raise $\dim H^0(\mathfrak{S})$ above $\max\{NK - |\mathcal{E}_\mathrm{a}| r_c, 0\}$ is open; in 200 random coded instances it exceeded that value in 4 (Check S47.54). Proof in Section S30.3.

Frustration also supplies a mass to agents that average transported content without sensing, a role that grounding plays otherwise.

**Proposition S6.8** (Frustration acts like grounding)**.** *Let the translations be orthogonal, the precisions equal to one and $M_i = I$, let $\Delta_\mathrm{a}$ be the maximal degree of $\mathcal{G}_\mathrm{a}$, in- and out-edges counted, and*

*consider transported averaging* $\mu(t+1) = (1-\varepsilon)(I-\alpha\mathcal{L})\mu(t) + \zeta(t)$ *with leak* $\varepsilon \in [0,1)$ *and step* $0 < \alpha \le 1/(2\Delta_{\mathrm{a}})$. *Then* $\|(1-\varepsilon)(I-\alpha\mathcal{L})\|_2 = (1-\varepsilon)(1-\alpha\lambda_{\min}(\mathcal{L}))$, *so every configuration contracts at least at the effective leak* $\varepsilon_{\mathrm{cal}} = \varepsilon + (1-\varepsilon)\alpha\lambda_{\min}(\mathcal{L})$. *On a cycle whose holonomy has smallest eigen-angle* $\phi$ *the gain* $\alpha\lambda_{\min}(\mathcal{L})$ *is* $4\alpha\sin^2(\phi/(2N))$, *a finite-size effect.*

Pure listeners ($\varepsilon = 0$) in a frustrated network therefore still have a mass; reading it as a correlation length $\xi \approx \sqrt{a/\varepsilon_{\mathrm{cal}}}$ is a heuristic for the slowest mode only. Uniform frustration acts like a magnetic field: on a $24 \times 24$ torus with SO(2) translations of uniform flux per plaquette, with each lattice edge attended in one direction, the least eigenvalue of $\mathcal{L}$ equals the flux per plaquette to within 1.1 percent, and the lowest level has complex multiplicity equal to the number of flux quanta, twice that as a real eigenspace of $\mathcal{L}$ (Check S47.54). Proof in Section S30.3.

**Proposition S6.9** (Attended frustration)**.** *Let frames and translations be orthogonal, and let only edges with* $\|\Upsilon_e - I\| \le \delta_{\mathrm{att}}$ *in the spectral norm be attended. Every attended closed walk* $\mathfrak{c}$ *has* $\|\operatorname{Hol}(\mathfrak{c}) - I\| \le |\mathfrak{c}|\delta_{\mathrm{att}}$, *and every attended cycle graph, with unit precisions and* $M_i = I$, *has* $\lambda_{\min}(\mathcal{L}) \le (\pi\delta_{\mathrm{att}}/2)^2$, *whatever its length.*

The converse fails silently: a ring whose edges carry $\mathrm{Rot}(\phi_{\mathrm{e}})$ with $2\sin(\phi_{\mathrm{e}}/2) \le \delta_{\mathrm{att}}$ keeps every edge attended while its holonomy, a rotation by $N\phi_{\mathrm{e}}$, can be large. With grounding we expect such holonomy to be invisible to every member once the cycle is much longer than $\xi$, because information decays at twice the mass (Proposition S6.3); this is a heuristic. Proof in Section S30.3.

## S6.4 Excluding and exhibiting amplification

The notation is that of Definition 6.7 and Theorem 6.8.

**Proposition S6.10** (What excludes amplification)**.** *Each of the following forces* $\rho(A_{\mathfrak{c}}) \le 1$, *with strict inequality where stated.*

*(a) A common prior covariance,* $P_j = P$ *for all* $j$: *then* $\mathfrak{r}_e = \mathrm{snr}_e/(1+\mathrm{snr}_e) \le 1$, *with strict inequality for the cycle if every* $\hat{R}_i \succ 0$.

*(b) A small spread of priors: with* $\mathrm{sp}(P\|P') = \ln\lambda_{\max}(P'^{-1}P)$, *it suffices that* $\sum_{(i\to j)\in\mathfrak{c}} \mathrm{sp}(P_i\|P_j) \le 2\sum_{e\in\mathfrak{c}} \ln(1+\mathrm{snr}_e^{-1})$. *With the Thompson distance* $\mathrm{dist}_{\mathrm{T}}(P,P') = \max\{\mathrm{sp}(P\|P'), \mathrm{sp}(P'\|P)\}$, *priors within* $\delta_{\mathrm{T}}$ *of a common* $P$ *satisfy this when* $\delta_{\mathrm{T}} \le N^{-1}\sum_e \ln(1+\mathrm{snr}_e^{-1})$.

*(c) Assigned noise covering the sender's coded uncertainty,* $\hat{R}_i \succeq X_{ii}$ *for every sender: then* $\mathfrak{r}_e < 1$ *on every edge and* $\rho(A_{\mathfrak{c}}) < 1$ *for any priors and codes. Level-1 receivers satisfy this when the posted covariance is the sender's prior.*

*(d) A common code read with common noise,* $H_i = H$ *and* $\hat{R}_i = \hat{R} \succ 0$: *then* $\rho(A_{\mathfrak{c}}) \le \prod_j \|Y_j(I+Y_j)^{-1}\| < 1$ *with* $Y_j = \hat{R}^{-1/2}HP_jH^{\top}\hat{R}^{-1/2}$, *for any priors; likewise for a common code of rank one read with sender-specific positive scalar noises.*

Only the spread from sender to receiver enters (b), and the noise terms are what make a positive spread admissible; for noiseless codes no spread condition suffices (Example S6.11(ii)). A common code of rank at least two read with sender-specific noise covariances can amplify without bound: with common code $I_2$, the priors of Example S6.11(i) at $\alpha_a = \alpha_b = \alpha$, and assigned noise covariances $\mathrm{diag}(r, M)$ for sender $b$ and $\mathrm{diag}(M, r)$ for sender $a$, the spectral radius tends as $M \to \infty$ to the value $\alpha^2 x/(1+r)^2$ of the coordinate codes of the example, which is unbounded in $x$; at $\alpha = 0.9$, $x = 4$ and $r = 0.1$ it is 1.36, 2.36 and 2.64 at $M = 1$, 10 and 100 (Check S47.55). A large assigned

noise on one coordinate turns the common code into an effective coordinate code. For codes of rank one the cycle eigenvalue is a product of cosines in the receivers' prior metrics, noise factors at most one, and the prior-mismatch factor $\prod_i \|\boldsymbol{h}_i\|_{P_i}/\|\boldsymbol{h}_i\|_{P_{i^+}}$, where $\boldsymbol{h}_i^\top$ is sender $i$'s code and $i^+$ its receiver (Lemma S30.2): amplification needs some sender to believe its coded content more variable than its receiver does.

**Example S6.11** (Amplifying cycles)**.** (i) Two receivers with priors $P_a = \begin{pmatrix} 1 & \alpha_a\sqrt{x} \\ \alpha_a\sqrt{x} & x \end{pmatrix}$ and $P_b = \begin{pmatrix} x & \alpha_b\sqrt{x} \\ \alpha_b\sqrt{x} & 1 \end{pmatrix}$ exchange messages, $b$ sending the first coordinate to $a$ and $a$ the second to $b$, with channel noise $R_c$. Each believes the coordinate it does not hear to be highly variable and correlated with the one it hears, and the cycle eigenvalue is $\alpha_a\alpha_b x/(1+R_c)^2$, which is 2.68 at $\alpha_a = \alpha_b = 0.9$, $x = 4$, $R_c = 0.1$. Among 3000 random three-cycles in two dimensions with noiseless codes of rank one the largest spectral radius was 2.60 with heterogeneous priors and at most 0.9987 with a common prior. (ii) Noiseless cycles amplify at arbitrarily small spread. For $0 < u < \frac{1}{2}$ let $P = \begin{pmatrix} 1 & 1-u \\ 1-u & 1 \end{pmatrix}$, $P_a = D_a P D_a$, $P_b = D_b P D_b$ with $D_a = \mathrm{diag}(1, 1+2u)$, $D_b = \mathrm{diag}(1+2u, 1)$, and let $a$ send the second coordinate to $b$ and $b$ the first to $a$, noiselessly. Then $\rho(A_{\mathfrak{c}}) = (1-u)^2(1+2u)^2 > 1$, while

$$\mathrm{dist}_{\mathrm{T}}(P_a, P) = \mathrm{dist}_{\mathrm{T}}(P_b, P) = \ln(1+2u) + \operatorname{arccosh}\Big(1 + \frac{2u}{(2-u)(1+2u)}\Big) = \sqrt{2u}(1+O(\sqrt{u}));$$

at $u = 0.002$ the spectral radius is 1.003988 at Thompson distance 0.0671. Computations in Section S30.4; checked in Check S47.55.

Heterogeneous prior structure is not itself an error: it arises from heterogeneous information, and amplification survives when every prior is exactly right.

**Numerical observation S6.12** (Amplification without misspecified priors)**.** When every prior is the exact posterior of one common prior given private linear-Gaussian data, level-0 cycles with codes of rank one still amplify: over 20000 random draws the largest spectral radius was 6.26 for noiseless codes and 5.48 with channel noise 0.05 (Check S47.55).

*Remark* S6.13 (Where amplification sits). Amplification acts on a product of gain-code maps, which are neither translations nor invertible, and it occurs at identity encodings with correct codes, where translations are flat (Proposition S6.5); it is therefore not inconsistency. It occurs in a single circulation with independent errors and concerns the size of the transferred content, not the precision assigned to it, so it is not double counting (Proposition 3.4). We classify it under calibration: it needs an edge on which the sender's coded prior variance exceeds the receiver's predictive variance of the message (Theorem 6.8), and assigned noise covering the sender's coded uncertainty excludes it (Proposition S6.10(c)). Since it survives exact priors (Numerical observation S6.12), the remedy is calibrated reading of senders' uncertainty, as at level 1, not common priors.

**Open problem S6.14** (Amplification beyond noiseless circulation)**.** Characterize amplification for noisy circulation, where $A_{1N}$ in the round-trip map is not idempotent, and for dynamic cycles in which senders post posterior rather than prior covariances. For a common code of rank at least two read with sender-specific noise covariances, amplification occurs and is unbounded (the instance following Proposition S6.10); what remains open there is a condition on the assigned noise covariances, weaker than the common noise of Proposition S6.10(d), that excludes it.

## S6.5 Adoption rule and conjectures on code domains

The code game is that of Definition 6.9, its pairwise form and the potential $\Phi_{\text{code}}$ are those of Proposition 6.10(a), and the stochastically stable configuration is that of Theorem 6.11(a).

**Proposition S6.15** (Adoption rule)**.** *In the code game with uniform habits, agent $i$'s policy posterior is $q_i(\sigma) \propto \exp(-\gamma G_i(\sigma))$, so that, with $u_i^+$ partners using $+1$,*

$$q_i(+1) = \operatorname{sig}(\gamma[\mathcal{F}_i(u_i^+) - \mathcal{F}_i(\mathsf{d}_i - u_i^+)] + 2\gamma h_i), \qquad \operatorname{sig}(z) = \frac{1}{1+e^{-z}}.$$

*As a function of the local frequency $x = u_i^+/\mathsf{d}_i$:*

*(a) the adoption curve is nondecreasing and symmetric, $q_i(+1 \mid x, h_i) = 1 - q_i(+1 \mid 1-x, -h_i)$;*

*(b) for a value affine in $u$ with slope $J$, $q_i(+1) = \operatorname{sig}(\gamma J \mathsf{d}_i(2x-1) + 2\gamma h_i)$, with slope $\gamma J \mathsf{d}_i/2$ at $x = \frac{1}{2}$; with $h_i = 0$ the curve lies above the diagonal immediately to the majority side of $\frac{1}{2}$ if and only if $\gamma J \mathsf{d}_i > 2$, and then the map $x \mapsto q_i(+1 \mid x)$ has stable fixed points $x_\pm = 1 - x_\mp$, an unstable one at $\frac{1}{2}$, and $x_- = e^{-\gamma J \mathsf{d}_i}(1 + o(1))$ as $\gamma J \mathsf{d}_i \to \infty$;*

*(c) at finite $\gamma$ the curve has an innovation floor, $q_i(+1 \mid x = 0) = \operatorname{sig}(-\gamma[\mathcal{F}_i(\mathsf{d}_i) - \mathcal{F}_i(0)] + 2\gamma h_i) > 0$, which acts as a leak, and as $\gamma \to \infty$ it tends to the deterministic best response wherever $G_i(+1) \neq G_i(-1)$ and to $\frac{1}{2}$ where $G_i(+1) = G_i(-1)$. When $h_i = 0$ and $\mathcal{F}_i$ is strictly increasing, as (67) is without a binding capacity, the best response is the majority rule, with $\frac{1}{2}$ at a tie; under a binding capacity $\mathcal{F}_i(u_i^+) = \mathcal{F}_i(\mathsf{d}_i - u_i^+)$ can hold away from a tie, and the limit is $\frac{1}{2}$ there as well.*

No preference for conformity is assumed: the nonlinearity comes from the logistic policy posterior and from the growth of epistemic value with the number of matched partners. The fixed points in (b) are self-consistent frequencies of a large population in which every agent sees the population frequency, and Proposition S10.12 compares the rule with the power rule of models of dialect formation. Proofs in Section S30.5. In the mean field in which partners and record report the same frequency, a record of the code that the members write and read with cavity priors adds its read weight to the slope in (b), so the adoption map becomes bistable at lower policy precision (Proposition S12.21).

**Conjecture S6.16** (Code domains under random pulls)**.** *Consider the pairwise code game on a communication graph of dimension $D$ with nearest-neighbor attention, coupling $J$ and independent $\mathcal{N}(0, h^2)$ pulls, at its stochastically stable configuration, let $h/J \to 0$, and write $L_{\text{dom}}$ for the typical linear size of a domain.*

*(a) For $D = 2$ no code spans a large population, and, under the identification of $L_{\text{dom}}$ with the correlation length of Ding and Wirth (2023), $\ln L_{\text{dom}} \asymp (J/h)^{4/3}$.*

*(b) For $D \geq 3$ one code spans the population.*

*(c) For codes in a continuous group, such as rotations of an encoding with a mismatch cost proportional to one minus the cosine of the angle between neighboring codes, as the calibrated noise of Theorem 6.4(a) is for rotations, $L_{\text{dom}}$ is of order $(J/h)^{2/(4-D)}$ for $D < 4$, and no code spans the population for $D \leq 4$.*

*(d) If translations are learned with persistent independent noise in a continuous group, consistent-translation domains are finite on planar communication graphs, while on three-dimensional or richer graphs global consistency is possible at small noise.*

The corresponding statements for random-field models are theorems or established heuristics, collected in Section S16.10 at the level at which we read them (Imry and Ma, 1975; Aizenman and Wehr, 1989, 1990; Bricmont and Kupiainen, 1988; Abbe et al., 2018). Ding and Wirth (2023) define the correlation length of the two-dimensional random-field Ising model as the box size at which the influences of the random field and of the boundary condition on the magnetization are comparable, and prove that at zero temperature it grows like $\exp(\Theta((J/h)^{4/3}))$; Ding et al. (2026) extend this below the critical temperature. We read both abstracts. Identifying that length with the size of code domains is our heuristic, so the exponent 4/3 in (a) is conjectured under that declared identification, and the Imry–Ma estimate $\ln L_{\mathrm{dom}} \asymp (J/h)^2$ is neither confirmed nor refuted by what we read. All parts remain conjectures, because the theorems concern equilibrium measures of models onto which the code game maps only in the weak-message limit, and part (d) concerns translations whose errors come from coupled learning, not from independent noise on edges. Under Assumption A8 the dimension is that of the communication graph, and graphs with long-range edges are expected to behave as high-dimensional.

**Conjecture S6.17** (Surface tension and coarsening)**.** *For the pairwise code game on $\mathbb{Z}^2$ without pulls, asynchronous logit code dynamics at large policy precision coarsens from disordered initial conditions: domain walls move by curvature, and the typical domain size grows as $t^{1/2}$.*

Without pulls the stationary law of the pairwise game is the Ising Gibbs measure with neighboring codes coupled at strength $\gamma J/2$, which on $\mathbb{Z}^2$ has two ordered phases when $\gamma J/2$ exceeds $\frac{1}{2}\ln(1+\sqrt{2})$ (Onsager, 1944); the conjecture concerns how boundaries present initially disappear. Curvature-driven motion of walls is the classical picture of coarsening (Allen and Cahn, 1979; Bray, 1994), of which we read only records and an abstract, and Burridge (2017) obtains growth as the square root of time for dialect boundaries. In one dimension the innovation floor keeps walls at density $1/(1+e^{\gamma J})$ (Theorem 6.11(c)).

With several features the concavity of Lemma S30.3 couples their walls; an isogloss is the boundary of one linguistic feature.

**Conjecture S6.18** (Withdrawal bundles isoglosses)**.** *Let agents choose codes for several binary features, and let the mismatch cost of an edge be a concave function of the number of features mismatched on it, as for* C1 *calibration with or without withdrawal (Lemma S30.3). Then in the stochastically stable configuration walls of different features attract, so isoglosses bundle; with additive costs they are independent.*

On a chain with two features, $J = 3$ and unit Gaussian pulls, a wall of the second feature sits on the same edge as a wall of the first with probability 0.057 under additive cost, as independence predicts, and 0.366 and 0.662 when two mismatches cost $1.5J$ and $1.1J$, against 0.066 and 0.090 under independence (Check S47.58). Moving a wall onto an edge that carries a wall of the other feature saves the difference between two single costs and one double cost; a proof needs the statistics of the joint minimizer, which we do not have. The Séguy curves that walls produce are in Proposition S10.13.

## S6.6 Placement on the grounding trade-off

The setting is the ring of Section 6.6, and the curve is that of Fig. 7.

Where agents settle on the curve is set by how they assign precision. Precision on a channel is a belief, optimized by free-energy minimization on a slower timescale, and that optimization reads the wrong statistic.

**Proposition S6.19** (Where free-energy minimization places a level-0 agent)**.** *Let a level-0 receiver on the ring hold its claimed prior variance $\hat{P}^-$ fixed and choose its message precision $\varpi_y \le 1/R_c$ to minimize the time-averaged variational free energy of a neighbor's message, scored by its predictive density $\mathcal{N}(m_i, \hat{P}^- + 1/\varpi_y)$.*

*(a) The minimizer is moment matching: $1/\varpi_y = \max\{R_c, \mathbb{E}(y_{ij} - m_i)^2 - \hat{P}^-\}$.*

*(b) For neighbors $i$ and $j$, $\mathbb{E}(y_{ij} - m_i)^2 = 2(\Sigma^c_{ii} - \Sigma^c_{ij}) + R_c$, which contains no shared error, and the assigned message noise sits at its floor $R_c$ whenever $2(\Sigma^c_{ii} - \Sigma^c_{ij}) \le \hat{P}^-$.*

*(c) If the receiver also sets its sensing noise by moment matching of its sensory innovations $s_i - m_i$, it assigns the noise $R^s + \mathbb{E}e_i^2 - \hat{P}^-$, which exceeds $R^s$ whenever its claimed prior variance lies below its actual prior mean-square error.*

A receiver that calibrates its message precision from social innovations is pushed toward trusting neighbors as much as the channel allows, and one that calibrates its senses the same way reads the shared error as sensor noise, senses less and raises the shared error further. Proof in Section S30.6.

**Numerical observation S6.20** (Placement on the trade-off)**.** At $F = 0.99$, $Q = 0.02$, $R^s = 1$ and $R_c = 0.1$, where $P_0 = 1.005$ and an isolated agent's Kalman error is 0.1415 (Check S47.59), the social optimum over common gains has $\xi = 0.96$ and error 0.115. The correlation-aware best response, in which each agent fuses its own estimate, observation and neighbors' messages by the best linear unbiased rule given the stationary joint covariance including the common mode, iterated to a symmetric fixed point whose existence and uniqueness are not proved, has $\xi = 1.05$ and error 0.123. Level-0 agents that set message precision by Proposition S6.19(a) settle at the noise floor with $\xi = 2.64$ and error 0.350; setting sensing precision the same way, they assign it 0.72 against the true 1.0 and settle at $\xi = 2.93$ and error 0.425. Along the family of Fig. 7 a message's receiver fidelity, conditioned on the receiver's signals through the current round and on its neighbors' earlier messages, is 0.012 nats at $\varpi_y = \varpi_s$ and falls to 0.002 nats at $\varpi_y = 10\varpi_s$ and below $10^{-4}$ nats at $\varpi_y = 100\varpi_s$, while its believed fidelity rises from 0.04 nats at $\varpi_y = \varpi_s$ to 1.53 nats at $\varpi_y = 1000\varpi_s$.

Individual free-energy minimization thus lands on the long-$\xi$, low-accuracy side of the social optimum, slightly for correlation-aware agents and far for level-0 agents. The same floor appears in the report of November 7 (Dennis, 2026a) as a plateau of the coupling that learned noise models produce, on which every sufficiently correlated partner ties.

**Conjecture S6.21** (An externality of sensing)**.** *For correlation-aware agents on translation-invariant graphs, the symmetric myopic Nash equilibrium of the gain game has a smaller sensing gain and a longer correlation length than the social optimum over common gains.*

The reason we expect is that an agent's sensing makes its messages informative to its neighbors, a benefit its own expected free energy does not count; only the parameter point of Numerical observation S6.20 has been computed. The direction matches informational free-riding and herding in economics (Banerjee, 1992; Bikhchandani et al., 1992; Galeotti and Goyal, 2010), which we know from records and abstracts.

*Remark* S6.22 (Consequence for hypothesis H4)*.* In a medium with uniform leak a collective at level $r_0$ has root-mean-square extent at most $\xi_2/\sqrt{r_0 - 1}$ per coordinate (Corollary 4.16), with $\xi_2 \le \xi$ on translation-invariant media with finite-range kernels (Remark S4.21), while a long $\xi$ costs shared error (Theorem 6.12). The two bounds are proved in different settings, so we read them together only as an interpretation: large accurate collectives need more than homogeneous

local message passing, such as specialized sensors, aggregation hierarchies or records that raise the effective persistence (Section 8), and long-range coherence produced by over-weighting messages is a symptom of misspecification.

Of these remedies, records that the members both write and read do not escape the tension in the shared part of the error of a grounded coordinate: in the level-0 network of a collective closed under attention such a record raises the shared-error floor (Theorem 12.7), and under the conditions of Proposition S12.20 positive precision assigned to it raises the common-mode part of every member's stationary mean-square error. The total error rises when the local-noise part is a small share of it, as in every instance computed in Numerical observation S12.19(a), and can fall otherwise: at weak sensing and strong message precision the instances of Numerical observation S12.19(c) show a fall of 1 to 4 percent before the error rises again.

### S6.7 What coherence contributes to formation

Hypothesis H3 says that collective interiors are regions of high fidelity, calibrated precision and consistent translation, and that inconsistency and low fidelity concentrate at boundaries. Section 6 supports a qualified version. In the static two-group model attention withdraws from inconsistent edges when receivers are calibrated, capacity is scarce relative to consistent alternatives, and the cost of mistranslation exceeds the information advantage of the cross-group sender (Theorem 6.4); C0 receivers cannot withdraw (Proposition 6.3) and C2 receivers have no reason to. Withdrawn edges are those of low calibrated precision, so boundaries are low-fidelity in the calibrated sense. Withdrawal can fail across complementary access to the world (Example S6.4) and does not see inconsistency spread thinly along attended cycles (Proposition S6.9). Because true translations are flat (Proposition S6.5), the inconsistency that drives withdrawal is learning error, which grounded learning confines to unattended edges (Proposition S7.1). Calibration learned from social innovations is blind to shared error (Lemma 3.6), so an interior can be consistent and mutually calibrated and still be an echo chamber; the accuracy side of H3 must be measured with grounding statistics, and the grounding tension prices coherence that reaches far (Theorem 6.12 and Numerical observation S6.20).

Whether withdrawal, learning and attention together produce collectives is the subject of Section 7. In its two-group model the chain from calibrated precision through attention and couplings to contrast is closed by theorem only at $\gamma = \infty$ with noiseless angle observations, $\sigma_\alpha^2 = 0$: there the declared start is itself the split, and Theorem 7.8, a theorem about the world-conditional error law at the split and not about the map, shows that the two groups are exactly the collectives at every level of its admissible window (see also Theorem 7.6(e)). For $\gamma_S < \gamma < \infty$ reachability of the split branch is a theorem about the frozen-calibration mean-field map (Theorem 7.6(a),(b)), and the contrast of the reached state is computed at the base point (Numerical observation S7.7); the stochastic-approximation step from the full model to that map is open (Open problem 7.14), and online calibration from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell, with reruns of a cell differing by up to three replicates (Numerical observation 7.11 and Open problem S7.13). These collectives are a calibration-level effect of misreading joint structure: receivers that know their translation errors pay no mistranslation cost, so the split fails the translation-aware form of the exact-fusion persistence clause, as Proposition 7.12 proves, while for full cross-covariance-aware exact fusion, in which receivers also model the cross-covariances of their errors, this conclusion is heuristic in this model. Above the threshold $\gamma_T$ of Theorem 7.6(b) the split coincides with the code partition. Persistence under exact fusion with drifting frames (Conjecture S7.12) and formation beyond the two-group model (Conjecture 7.13) are conjectures.

# S7 Additional results for Section 7: formation

This section collects statements, examples and remarks of Section 7 that the main text cites without stating in full. Proofs are in Section S31.

## S7.1 Status summary from the section opening

The result is partial. In the two-group model at $\gamma = \infty$ with noiseless angle observations, $\sigma_\alpha^2 = 0$, clauses (T1) to (T3) and (T5) of Definition 7.1 are theorems, the contrast clause (T3) at the levels of the window of Theorem 7.8(c), which is nonempty at every computed parameter point (Numerical observation S7.6), and the declared start is then itself the split (Theorem 7.6(e), Theorem 7.8). For $\gamma_S < \gamma < \infty$ reachability of the split branch is a theorem about the frozen-calibration mean-field map (Theorem 7.6(a),(b)), with the stochastic-approximation step from the full model to the map open (Open problem 7.14), and the contrast of the reached state is computed at the base point (Numerical observation S7.7). Clause (T4) fails for the translation-aware null of type S, so the model yields no formation result in the sense of Definition 7.1. Formation there is a calibration-level effect of misreading joint structure: receivers that know their translation errors pay no mistranslation cost, so from a start law exchangeable given the world path the model's collectives fail the translation-aware form of the exact-fusion persistence clause of the null family, as Proposition 7.12 proves, while for full cross-covariance-aware exact fusion in this model the conclusion is heuristic; above a threshold $\gamma_T$ they coincide with the code partition. The split differs from the deletion nulls by proof, and from the frozen-environment null only when the initial states that frozen messages carry are deterministic or conditioned on (Proposition 7.9(a)). Persistence under exact fusion with drifting frames is a conjecture (Conjecture S7.12). Online calibration from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell, with reruns of a cell differing by up to three replicates, and is open (Numerical observation 7.11 and Open problem S7.13). Formation beyond this model is a conjecture (Conjecture 7.13).

## S7.2 Two mechanisms: flattening and bistability

Two facts make formation plausible before a population is modeled: grounded learning corrects translations only on attended edges, and attention follows calibrated precision, which learning raises.

**Proposition S7.1** (Grounded learning flattens the attended subgraph)**.** *Let codes have full rank, and suppose that every learned translation $\Theta_{ij}(t)$ (Section 2.7) converges, with limit $U_i U_j^{-1}$ on the event that receiver $i$ attends sender $j$ infinitely often with grounding. Then on the event that the limiting holonomy of a cycle (Theorem 6.5) differs from the identity, some edge of the cycle is attended only finitely often: the subgraph of edges attended infinitely often is asymptotically flat.*

*Proof.* If every edge of a cycle $i_0, i_1, \dots, i_L = i_0$ is attended infinitely often, each $\Theta_{i_{l+1} i_l}$ converges to $U_{i_{l+1}} U_{i_l}^{-1}$, and by continuity of matrix products the holonomy $\Theta_{i_0 i_{L-1}} \cdots \Theta_{i_1 i_0}$ converges to $U_{i_0} U_{i_{L-1}}^{-1} \cdots U_{i_1} U_{i_0}^{-1} = I$. □

The hypothesis holds for grounded least squares at full rank (Proposition S2.3) and fails for a learner that regresses on an attenuated code (Proposition 2.7(c)). Persistent inconsistency thus sits on edges that end up unattended, the consistency part of hypothesis H3. Which edges these are depends on attention and learning together; in the simplest instance one receiver of capacity one chooses each round between an in-group sender, read correctly, and a cross sender whose frame drifts, in the static model of Definition 6.2 with C1g calibration. Let $\mathcal{A}(\mathsf{p})$ be the believed

value (Definition 3.1) of the cross sender minus that of the in-group sender when the cross sender is attended with frequency $\mathsf{p}$, in the expected-variance mean field of grounded angle learning (Section S31); attention feeds learning, so $\mathcal{A}$ is continuous and strictly increasing on $(0, 1]$ with a finite limit $\mathcal{A}(0)$. With $[\cdot]$ the indicator of a statement, attention at policy precision $\gamma$ follows $\mathsf{p} = 1/(1 + e^{-\gamma\mathcal{A}(\mathsf{p})})$, and at $\gamma = \infty$ it follows $\mathsf{p} = [\mathcal{A}(\mathsf{p}) > 0]$.

**Proposition S7.2** (Bistability of attention and learning)**.** *In the single-edge mean field at $\gamma = \infty$ let the attention frequency relax toward the attention rule, $\dot{\mathsf{p}} = [\mathcal{A}(\mathsf{p}) > 0] - \mathsf{p}$. The split state $\mathsf{p} = 0$ is an attracting rest point if and only if $\mathcal{A}(0) < 0$, and the merged state $\mathsf{p} = 1$ is one if and only if $\mathcal{A}(1) > 0$. When $\mathcal{A}(0) < 0 < \mathcal{A}(1)$ both attract, and the unique root $\mathsf{p}^\dagger$ of $\mathcal{A}$ separates their basins.*

Proof in Section S31. The mean field replaces the random variance recursion of intermittent observation by its expected value. An unattended edge is never learned and stays unattractive, so a cut reinforces itself and the outcome depends on initial attention. At finite $\gamma$ exploration of order $e^{-\gamma|\mathcal{A}(0)|}$ competes with the cut: in a computed instance there is one fixed point, merged, for $\gamma \le 50$ and three for $\gamma \ge 80$ (Check S47.60).

### S7.3 Status summary after the mean-field theorem

Theorem 7.6 concerns the frozen-calibration mean-field map; the step from the full model to the map is open (Open problem 7.14). The split branch it reaches is a calibration-level effect of misreading joint structure, and above $\gamma_T$ it coincides with the code partition. That the blocks of the split are collectives is a theorem only at $\gamma = \infty$ with $\sigma_\alpha^2 = 0$ (Theorem 7.8) and a computation at the base point for finite $\gamma$ (Numerical observation S7.7). The split fails the translation-aware form of the exact-fusion persistence clause, as Proposition 7.12 proves; for full cross-covariance-aware exact fusion this conclusion is heuristic (Section 7.8). Persistence with drifting frames and formation beyond the model are conjectures (Conjectures S7.12 and 7.13), and online calibration from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in only 0 to 7 of 8 replicates per cell, with reruns of a cell differing by up to three replicates (Numerical observation 7.11 and Open problem S7.13).

### S7.4 Fixed points, thresholds and bridged states

**Numerical observation S7.3** (Fixed points and thresholds at the base point)**.** At the base point $\mathsf{p}^* = 0.0503$, $\Delta\mathcal{V}(\frac{1}{2}) = 0.0592$ and $\Delta\mathcal{V}_{\max} = 0.313$, so $\gamma_S = 68.2$ and $\gamma_M = 6.84$; the symmetric fixed points are those of Table S4. Continued in $\gamma$, they form a merged and a split branch joined by an unstable middle branch through folds at $\gamma = 60.05$ and 105.9. The merged branch loses stability to symmetry-breaking perturbations at $\gamma = 87.9$, before its fold: for $87.9 < \gamma < 105.9$ its symmetry-breaking multiplier lies between one and 1.026, its value at the fold. The split branch is stable from $\gamma = 60.11$. Conditions (i) and (ii) of Theorem 7.6(b) fail at $\gamma = 240$ and hold at 244 and every larger value computed, while a single learned bridge survives at 240, so (b) is sharp to within 2 percent here. Computed by bisection, $\gamma_T = 242.1$ at the base point, and $\gamma_T = 244.2$, 861.1, 3255.6 and 12684 at $\mathsf{p}^* = 0.05$, 0.025, 0.0125 and 0.00625 (varied through $h$), so that $\mathsf{p}^{*2}\gamma_T = 0.611$, 0.538, 0.509 and 0.495; the product decreases toward the lower bound $\mathsf{m}_\kappa^2/\mathsf{c}_\mathcal{V}^+ = 0.483$ of Theorem 7.6(b) and is 0.484 at $\mathsf{p}^* = 2 \times 10^{-4}$. The upper bound $2\kappa^2/(e\mathsf{c}_\mathcal{V}\mathsf{p}^{*2})$ is about 580 times too large. Self-consistent calibration moves the split branch at $\gamma = 80$ from $x = 0.9659$ to 0.9656 (Check S47.63).

Figure S1 draws the symmetric fixed points of Table S4 as continuous branches in $\gamma$, together with the thresholds of Theorem 7.6 and the range of $\gamma$ in which bridged states exist.

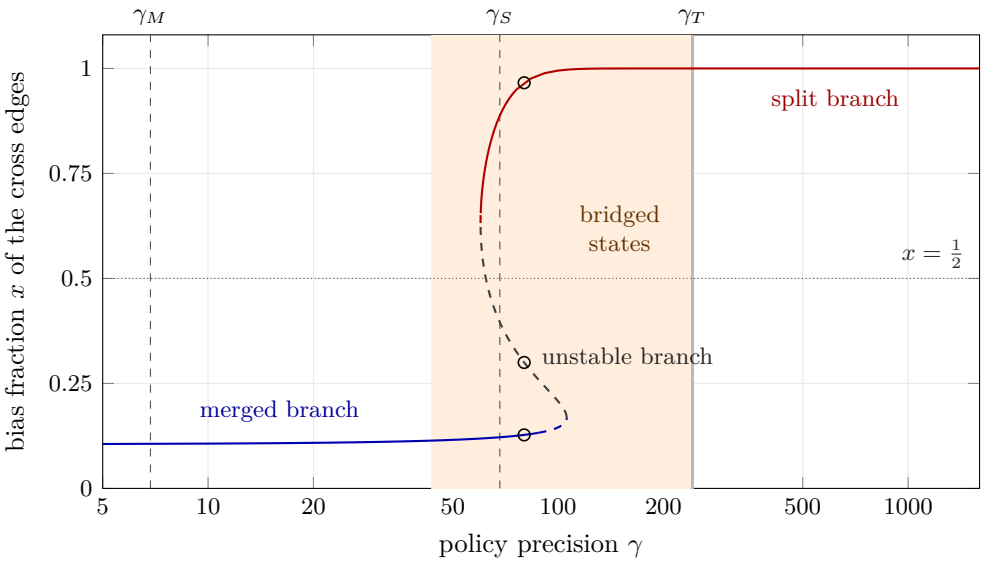



Figure S1: **Mean-field phase diagram at the base point.** Symmetric fixed points of the map (72) under Assumption 7.4, shown as the bias fraction $x$ of the cross edges against the policy precision $\gamma$ and computed with the exact set softmax over the 35 attended sets. The merged branch (blue) and the split branch (red) are joined by an unstable middle branch through folds at $\gamma = 105.9$ and $\gamma = 60.05$. Curves are solid where both multipliers of Theorem 7.6(d) lie below one and dashed where one of them exceeds one, so the merged branch loses stability to symmetry-breaking perturbations at $\gamma = 87.9$, before its fold. Circles mark the three fixed points at $\gamma = 80$ of Table S4, $x = 0.1274$, 0.2999 and 0.9659. Dashed vertical lines mark $\gamma_M = 6.84$ and $\gamma_S = 68.2$ of (73), and the gray bar marks the global-split threshold $\gamma_T$, which lies between 240 and 244. Bridged states (Numerical observation S7.4), stable fixed points of the per-receiver map with one to three of the four cross edges learned, exist in the shaded range $43.4 < \gamma < 241.0$, whose upper end is the fold of the single learned bridge.

**Numerical observation S7.4** (Bridged states)**.** Off the symmetric subspace the map has stable fixed points in which a receiver keeps $n_\mathrm{b}$ of its $n$ cross edges learned and attended. At the base point, continuation in $\gamma$ of the states found at $\gamma = 80$ gives them for $43.4 < \gamma < 241.0$: $n_\mathrm{b} = 3$ from its fold at $\gamma = 43.41$ to 111.3, $n_\mathrm{b} = 2$ from 46.69 to 151.2 and $n_\mathrm{b} = 1$ from 51.54 to its fold at 240.98, where the last learned bridge collapses, while the merged state, $n_\mathrm{b} = 4$, is stable up to $\gamma = 87.9$. Bridged states thus exist already below $\gamma = 60.05$, where the split branch begins on the symmetric subspace, and none is found above $\gamma = 241.0$ or, by iteration from bridged starts, at any $\gamma$ from 400 to 3200. Bridges must be nucleated: at $\gamma = 160$ an edge started at $x \le 0.1$ stays learned and one started at $x \ge 0.2$ collapses. One reciprocal bridge per receiver at $\gamma = 160$ gives group contrast 4.51 and union contrast 1.76, from the isotropic part of a covariance that departs by 0.44 in relative norm from blocks proportional to $I_2$, so these values are indicative only (Check S47.64).

A learned bridge persists because it is the only near-equal alternative to the in-group senders. Bridged states are what flattening leaves out: their attended subgraph is flat, but no unattended cut separates the groups.

Table S4: Symmetric fixed points of the mean-field map at the base point: bias fraction $x$ of the cross edges with the cross attention per edge in parentheses, and the symmetric and symmetry-breaking multipliers. Classes refer to the symmetric subspace. Off it, the merged point is unstable to symmetry-breaking perturbations for $87.9 < \gamma < 105.9$, and bridged states exist for $43.4 < \gamma < 241.0$ (Numerical observation S7.4 and Fig. S1).

| $\gamma$ | fixed points $x$ (cross attention) | multipliers | class |
|---|---|---|---|
| 5 to 60 | 0.106 to 0.119 (0.43 to 0.37) | 0.953 to 0.981 | merged only |
| 80 | 0.127 (0.344) | 0.976, 0.994 | bistable |
| | 0.300 (0.117) | 1.014, 1.027 (unstable) | |
| | 0.966 ($1.8 \times 10^{-3}$) | 0.995, 0.995 | |
| 120 | 0.9992 ($3.9 \times 10^{-5}$) | 0.995, 0.995 | split |
| 160 to 1600 | 1.0000 ($9.0 \times 10^{-7}$ to $2 \times 10^{-65}$) | 0.995, 0.995 | split |

## S7.5 Baseline convention, parameter grid and finite policy precision

*Remark* S7.5 (A weaker convention: the matched-size baseline). The weaker convention of Remark S4.15, which compares a region only with regions of its own size, gives $r_{\mathrm{hom}}(n) = 1$ here (Theorem 7.8(b)) and would admit every level above $r_{\mathrm{sub}} = 2.318$. We do not use it: at such levels the comparison medium itself has qualifying regions (at level 3, its whole population and its eight seven-member regions), so a selection there would not separate the split from a population without frames.

**Numerical observation S7.6** (The parameter grid)**.** On the grid of 162 points with $F$ equal to 0.9, 0.95, 0.98, 0.99, 0.995 or 0.999, $Q = 1 - F^2$, $R^s \in \{0.25, 1, 4\}$, $R_c \in \{0.01, 0.05, 0.5\}$, $n \in \{2, 4, 8\}$ and $\kappa = n - 1$, the calibration conditions of the split and of the comparison medium each have one root at every point, and both are nondegenerate. Everywhere $r_{\mathrm{sub}} < \bar{r}_{\mathrm{hom}} < r(A)$, the split contrast lies in $[1.24, 60.5]$ and the significance between 1.009 and 1.415, so every window is nonempty, some narrower than one percent of the level, and by Theorem 7.8(e) every grid point lies in an open set on which the split at $\gamma = \infty$ with $\sigma_\alpha^2 = 0$ is a collective at admissible levels (Check S47.66).

**Numerical observation S7.7** (Exploration at finite policy precision)**.** In the frozen-slow model with self-consistent calibration, the split branch at $\gamma = 80$, 120 and 160 has group contrast 8.6763, 8.7317 and 8.7329, union contrast at most 1.008, and a diffusive precision of the form $\Sigma^{\mathrm{sc}} \otimes I_2$ to within $3.1 \times 10^{-14}$ after rotating group $B$ by the remaining cross error; at $\gamma = 80$ only $A$ and $B$ among the 247 regions exceed 7.954. On the merged branch no region exceeds 7.954 at $\gamma = 10$, 40 or 80. Treating the mean cross weight $Fk_\times \mathsf{p}_\times$ ($k_\times$ the gain of an attended cross message, $\mathsf{p}_\times$ the cross attention per edge) as a symmetric coupling estimates $r(A) \approx (1 - \lambda_\perp^2)/(1 - \lambda^2 + 2n\lambda F k_\times \mathsf{p}_\times)$, 8.698 against 8.676 at $\gamma = 80$ (Check S47.67).

The split keeps its contrast while this cut coupling is small against the leak $1 - \lambda^2$, and by Lemma 7.5(a) exploration on the split branch is at most $\kappa e^{-\gamma \Delta \mathcal{V}(1/2)}$.

*Remark* S7.8 (Openness at finite policy precision). Theorem 7.8(e) proves openness at $\gamma = \infty$. At finite $\gamma$ openness is a sketch (Section S31), whose unproved step is that near the split the errors reduce to a diffusive scalar precision of the form $\Sigma^{\mathrm{sc}} \otimes I_2$, verified at the base point (Numerical observation S7.7) and not proved; its other hypothesis, nondegeneracy of the calibration roots, is verified on the grid (Numerical observation S7.6).

Table S5: Contrasts at the base point and the regions selected at admissible levels $r_0 \in (7.954, 8.733]$; the split at $\gamma = 80$ is selected for $r_0 \leq 8.676$, and for $\gamma \geq 160$ the contrasts equal those at $\gamma = \infty$ to four digits. In the row of the deletion and freezing nulls, the freezing null is taken with initial states that are deterministic or conditioned on (Proposition 7.9(a)). Dashes mark order parameters not computed.

| State | $r(A)$ | $r(A \cup B)$ | $\Omega_{\text{int}}$ across groups | selected |
|---|---|---|---|---|
| Split, $\gamma = \infty$ | 8.733 | 1.000 | 0 | $A$ and $B$ |
| Split, $\gamma = 80$ | 8.676 | 1.008 | – | $A$ and $B$ |
| Merged, $\gamma = 40$ | 1.019 | 7.801 | – | none |
| Deletion and freezing nulls | 1 | 1 | 0 | none |
| Partner randomization, unlearned | 1.351 | 3.953 | positive | none |
| Partner randomization, learned | 1.004 | 7.868 | positive | none |
| Comparison medium | 1.000 | 7.954 | – | none |

## S7.6 Partner randomization and the placement of the nulls

**Numerical observation S7.9** (Partner randomization and the other states)**.** At the base point, partner randomization at the matched rate of $\kappa$ partners per round, drawn uniformly from the $2n-1$ others, with idealized calibration solved at the null's own state, gives the contrasts of Table S5 with unlearned cross translations and with translations learned at the null's attention rate. In neither does any region exceed 7.954, and the cross-group world-conditional covariance is nonzero, so $\Omega_{\text{int}} > 0$ across groups. At the split, $\Omega_{\text{int}}$ of a within-group pair is 0.024, 0.140 and 0.510 nats over windows of two, four and eight rounds, and that of a cross-group pair is below $1.5 \times 10^{-14}$ (Check S47.68).

Two remarks place these comparisons within Definition 5.17; the second, on the explanation nulls, is in Section S7.11. Of the remaining nulls of type D, $\mathfrak{N}_{\text{eff}}$ and $\mathfrak{N}_{\text{inf}}$ are not applicable, since the model has no acts, and the habit null $\mathfrak{N}_{\text{hab}}$ is applicable and coincides with partner randomization redrawn every round with learned translations, because at $\gamma = 0$ with the uniform habits of Definition 7.2(iv) each receiver draws its $\kappa$ senders uniformly. For partner randomization redrawn every round the stationary second moments of the null, and with them the contrast of every region, are deterministic; the test of Proposition 5.18(b) with the largest contrast of a region as its statistic then has $p = 1/(n_{\text{null}} + 1)$, because that contrast is larger at the split (8.733 at $\gamma = \infty$ and 8.676 at $\gamma = 80$ against at most 7.868), and it rejects at level $\alpha_{\text{pr}}$ when $n_{\text{null}} \geq 1/\alpha_{\text{pr}} - 1$. Partner randomization drawn once, under which the selected partition is random, and the test on contrasts estimated from finite samples are not computed.

## S7.7 Reachability in the full model

In the runs of Numerical observation 7.11 and Table 8, early uniform attention lets cross translations be learned before calibration registers the mismatch, calibration noise acts as exploration, and at large $\gamma$ an in-group edge with an unlucky high estimate is dropped and never sampled again, its slot going to a cross edge (Check S47.70). The counts are not monotone in $\alpha_{\text{cal}}$ at $\gamma = 160$, where the rise from 2 to 7 and the fall to 1 exceed the sampling error of eight replicates (two-sided Fisher exact tests, $p = 0.04$ and 0.01, uncorrected for the number of comparisons); the remaining differences between nonzero cells, and the differences between reruns of a cell, do not.

## S7.8 Abundant capacity and the grounding tension

**Proposition S7.10** (Abundant capacity)**.** *If $\kappa \geq n$, every receiver attends at least $\kappa - n + 1$ cross senders in every round at every $\gamma$, so no attention cut separates the groups. Let calibrated precisions have the frozen form of Assumption 7.4(a), with any constants $\hat{\sigma}^2_{\text{in}} > 0$ and $D_{\text{cal}} > 0$. Then at $\gamma = \infty$ with noiseless angle observations, $\sigma^2_\alpha = 0$, from the declared start, each receiver attends in every round the $\kappa - n + 1$ cross senders it chose by tie-breaking in the first round, and their bias fractions converge to $\mathsf{p}^*/(\mathsf{p}^* + 1)$, which is below $\frac{1}{2}$ exactly when $\mathsf{p}^* < 1$: for $\mathsf{p}^* < 1$ they become learned bridges, and for $\mathsf{p}^* \geq 1$ they stay attended without being learned.*

Proof in Section S31. With reciprocal learned bridges at $\kappa = n = 4$ the group contrast is 2.46 and the union's 5.14, against 8.73 and 1.00 at $\kappa = n - 1$ (Check S47.71). Scarce capacity is needed as in the static withdrawal threshold of Theorem 6.4.

*Remark* S7.11 (The grounding tension in formation). For $\kappa = n-1$ the split contrast is $(1-\lambda^2_\perp)/(1-\lambda^2)$, whose denominator, the leak $1 - F^2(1 - k^s)^2$, shrinks as the sensing gain $k^s$ falls, while the withdrawal signal $D_{\text{cal}}$ of Lemma 7.3 vanishes linearly with $k^s$. Weak grounding therefore tends to raise contrast and weakens the signal that forms the cut: at $F = 0.99$, raising $R^s$ from 1 to 4 raises the split contrast from 8.73 to 14.33, lowers $D_{\text{cal}}/\hat{\sigma}^2_{\text{in}}$ from 4.16 to 1.41 and raises $\gamma_S$ from 68 to 154 (Check S47.72). High-contrast collectives of weakly grounded members need higher policy precision to form, which is Assumption A11, the grounding tension of Theorem 6.12, seen from the formation side.

## S7.9 Further conjecture and open problems

**Conjecture S7.12** (Drifting frames survive exact fusion)**.** *Let the group frames of the two-group model drift, the frame angles of $A$ and $B$ performing independent random walks, so that true cross translations change while in-group translations stay the identity. Then on an open set of parameters the split is selected at an admissible level under the exact-fusion null: it survives that null.*

The uncomputed basis is that an exact-fusion receiver that rarely attends a cross edge has a broad posterior over its translation, and a Gaussian message read through a rotation of unknown angle with spread $\mathsf{v}$ loses information about $\eta$ by an amount positive and increasing in $\mathsf{v}$. An instance with drifting frames in which exact-fusion receivers attend cross edges at the merged rate at every $\gamma$ would refute it.

**Open problem S7.13** (Online calibration)**.** With C1g calibration learned online, give conditions on $(\alpha, \alpha_{\text{cal}}, \gamma, \mathsf{p}^*)$ and on the rule for unattended edges under which the split is reached from the symmetric start with probability tending to one as the gains vanish, or show that bridged states keep positive probability.

**Open problem S7.14** (Self-consistent calibration along trajectories)**.** Replace the frozen constants of Assumption 7.4 by bounds uniform over configurations, so that Theorem 7.6(b) holds for the self-consistent mean field along whole trajectories.

**Open problem S7.15** (Access, heterogeneity and geometry)**.** Extend the analysis to anisotropic access, where a cut can separate complementary information, to heterogeneous senders, and to groups on a graph with geometry, where contrast meets the second-moment length $\xi_2$ and the largest extent $L^*$ of hypothesis H4 (the question of Open problem S8.28).

### S7.10 Moment bound and contrast conventions

*Remark* S7.16 (The moment bound holds on every path). After the first round every realization of (70) has block row sums $F(1-k_i^s) \le F(1-k_{\min}^s) < 1$, with $k_{\min}^s = 1/(R^s(1/Q + 1/R^s + \kappa/R_c))$, because rotations have norm one (Section S31). Row-sum contraction ($C_r$) and the uniform form of the moment bound (M) of Assumption 8.1 therefore hold on every path, and no cycle amplifies in the sense of Theorem 6.8.

*Remark* S7.17 (Second moments and two-dimensional errors). As Definition 4.10 declares, contrast is computed from the scalar precision $(\Sigma^{\mathrm{sc}})^{-1}$ when the world-conditional covariance has the per-coordinate form $\Sigma^c = \Sigma^{\mathrm{sc}} \otimes I_2$ (agent indices first), and from second moments when the conditional law is a Gaussian mixture, as it is here for $\kappa < n-1$. At finite $\gamma$ the mistranslation forcing also makes the conditional covariance depend on the world path, so we use $\Sigma^c = \mathbb{E}[\mathrm{Cov}(e(t) \mid \eta)]$. For other two-dimensional states contrast is not defined, and reported numbers name the reduction used.

### S7.11 Explanation nulls and merged regimes

The explanation nulls limit attribution. The world-and-access null predicts no partition, since all agents sense alike. The code null predicts the frame classes $\{A, B\}$: above $\gamma_T$ the partition is predictable from the codes alone and communication adds the contrast, while between $\gamma_S$ and $\gamma_T$ split and bridged states coexist for the same codes, and stable merged states as well up to $\gamma = 87.9$ (Fig. S1). Under Definition 5.17 the split above $\gamma_T$ is therefore code aligned.

Each failure of formation in the two-group model (Section 7.8) marks a hypothesis the positive results need. For $\gamma < \gamma_M$ every cross translation is learned. At the base point the symmetric map is merged only up to its fold at $\gamma = 60.05$, while bridged states of the per-receiver map, found by continuation, appear already at $\gamma = 43.4$ (Numerical observation S7.4); at the smaller mismatch $\vartheta = \pi/4$, where $\Delta\mathcal{V}(\frac{1}{2})$ is smaller, the symmetric map is merged only up to $\gamma \approx 80$. A C0 receiver that assigns every sender the same noise attends uniformly (Proposition 6.3), $\mathsf{p}_j = \kappa/(2n-1)$, so by (72) every cross translation ends learned whenever $\kappa/(2n-1) > \mathsf{p}^*$. Two failures are structural (Propositions S7.10 and 7.12).

### S7.12 Remarks on the formation conjecture

Regarding Conjecture 7.13, in the two-group model the attention and learning statements of the first two regimes hold for the frozen-calibration mean-field map (Theorem 7.6) and the third is observed (Numerical observation S7.4). A population of this class in which, for arbitrarily large $\gamma$, a stable configuration keeps a learned and attended cross-class edge would refute the conjecture; where it holds, its collectives are attributed to misreading unless they survive exact fusion. Linear conformity to a self-constituted prior cannot form collectives (Proposition 12.8); a route that does not rest on misreading, through conformity of codes with member-written code records and a pragmatic coordination value, is stated as Conjecture S12.25.

## S8 Additional results for Section 8: scale

This section collects statements, examples and remarks of Section 8 that the main text cites without stating in full. Proofs are in Section S32.

## S8.1 Amplifying profiles and the common-code conjecture

**Example S8.1** (An amplifying attention profile)**.** In the report (Dennis, 2026a), two level-1 receivers in a world $\eta \in \mathbb{R}^2$ with $F = 0.9I$ and $Q = I$, sender-specific effective codes $(3, 0)$ and $(0, 3)$, sensing rows $(3, 6)$ and $(6, 3)$ and attention capacity $\kappa = 2$ have sensing while listening to the partner as the strict best response of expected-free-energy attention, and the frozen mean map has the simple real dominant eigenvalue $\rho = 1.0613266751$ and block row sums 1.0999. By Lemma 8.2(c) no stationary law exists, and $\Sigma(t)$ grows like $\rho^{2t}$ (Proposition S8.6): (M) fails in both forms. The report describes a second instance, its CEX-1, which alternates two frozen profiles at capacity $\kappa = 1$, each with spectral radius below one, and for which it reports growth by the factor 1.0703173057 per round, so $(\mathrm{C}_\rho)$ for each frozen profile does not give (M) for a switching schedule. CEX-1 is a separate instance from the capacity-two system specified above: $F = 0.99I$, $Q = 0.1I$, the same two codes, sensing rows $(-5, 10)$ and $(10, -5)$, and a schedule in which both agents listen in one round and sense in the next. The report's summary does not state these parameters, which are taken from its working record, and the factor was recomputed with them (Check S47.101); in the capacity-two system itself, restricted to capacity one, every period-two alternation of two frozen profiles contracts at level 1, the slowest at the rate 0.9752822296 per round (Check S47.101). The capacity-two construction uses sender-specific codes by its specification, which the checks rebuild (Checks S47.79 and S47.101); for CEX-1 the report's phrase is "mismatched metrics" (Dennis, 2026a), and the instance recomputed in Check S47.101 has the sender-specific codes $(3, 0)$ and $(0, 3)$.

**Conjecture S8.2** (No amplification with one code)**.** *At identity encodings with one modality and one channel noise, in an isotropic world $F = fI_K$ with $0 < f < 1$, every frozen or periodic attention profile of level-1 receivers at its covariance fixed point has a mean map over one period of length $p$ with spectral radius at most $f^p$. Such profiles then satisfy* (M) *in uniform form, and frozen ones satisfy* $(\mathrm{C}_\rho)$ *at every rate above $f$.*

Proposition 9.1 proves two cases, both with a rank-one code: frozen profiles with symmetric listening, and profiles without sensing on any listening graph, frozen or periodic. Otherwise the evidence is numerical (Check S47.77), and one frozen or periodic common-code profile growing faster than $f$ would refute the conjecture.

## S8.2 Asynchronous clocks and routing

**Proposition S8.3** (Poisson clocks)**.** *Under the independent Poisson clocks of rate $\nu$ of Section 8.2, let $\bar{J}$ be an influence bound (Definition 8.4), $\bar{J}_{\mathrm{diag}} = \max_i \bar{J}_{ii}$, $g = \max_i \sum_{k \neq i} \bar{J}_{ik}$ and $\bar{J}_{\mathrm{off}} = \bar{J} - \mathrm{diag}(\bar{J})$. With $J_{ij}(t) = \partial \mu_i(t)/\partial \mu_j(0)$ and $d = d(i, j) \geq 1$,*

$$\mathbb{E}\|J_{ij}(t)\| \leq e^{-(1-\bar{J}_{\mathrm{diag}})\nu t} \sum_{n \geq d} \frac{(\nu t)^n}{n!} (\bar{J}_{\mathrm{off}}^n)_{ij} \leq e^{-(1-\bar{J}_{\mathrm{diag}})\nu t} \sum_{n \geq d} \frac{(g\nu t)^n}{n!}. \tag{S31}$$

*Consequently $\mathbb{E}\|J_{ij}(t)\| \leq e^{-(1-\bar{J}_{\mathrm{diag}})\nu t}(eg\nu t/d)^d/(\sqrt{2\pi d}(1 - g\nu t/(d+1)))$ when $g\nu t < d + 1$, and $\mathbb{E}\|J_{ij}(t)\| \leq \exp(-\varphi d + \nu t(ge^\varphi + \bar{J}_{\mathrm{diag}} - 1))$ for every $\varphi > 0$, so influence spreads at most at the velocity $v_{\mathrm{LR}} = \max\{0, \inf_{\varphi > 0} \nu(ge^\varphi + \bar{J}_{\mathrm{diag}} - 1)/\varphi\}$ hops per unit time, in the sense that the bound decays exponentially in $t$ along $d \geq vt$ for every $v > v_{\mathrm{LR}}$; the velocity is $eg\nu$ when $\bar{J}_{\mathrm{diag}} = 1$, and zero when $g + \bar{J}_{\mathrm{diag}} < 1$, where the infimum is $-\infty$. Without any smoothness, the probability that $\mu_i(t)$ depends on $\mu_j(0)$ is at most $(\Delta\nu t)^d/(d!(1 - \Delta\nu t/(d+1)))$ when $\Delta\nu t < d + 1$. For linear scalar updates with nonnegative coefficients and constant diagonal $\bar{J}_{ii} = \bar{J}_{\mathrm{diag}}$, the first inequality in* (S31) *is an equality.*

Proof in Section S32; checked on a ring of 40 agents (Check S47.75). The directed copy chain, in which agent $m$ copies agent $m-1$ at its events ($\bar{J}_{\mathrm{diag}} = 0$, $g = 1$), attains the first inequality of (S31), $\mathbb{E}J_{ij}(t) = e^{-\nu t}(\nu t)^d/d!$, while the probability that a signal held at the source has crossed $d$ hops by time $t$ equals the last member $e^{-\nu t}\sum_{n\geq d}(\nu t)^n/n!$ of (S31); its velocity $\nu$ equals $v_{\mathrm{LR}}$.

**Example S8.4** (A routing transition)**.** In the routing model of Proposition 8.6, agents $0, 1, \ldots, n$ sit on a line, and each relay $m \geq 1$ attends one of two persistent sources with contents $\pm 1$, with $G_m(\pm) = \mp\alpha_0\mu_{m-1}/2$, preferring the source that agrees with what it heard. Averaged routing gives $\mu_m = \tanh(\gamma\alpha_0\mu_{m-1}/2)$, and $\bar{A} = 0$, $\mathrm{Lip} = \alpha_0/2$, $\mathrm{osc} = 2$ give $\bar{J}_{m,m-1}(\gamma) = \gamma\alpha_0/2$, the derivative at the symmetric state. For $\gamma\alpha_0 < 2$ influence decays as $(\gamma\alpha_0/2)^n$; for $\gamma\alpha_0 > 2$ the sign of $\mu_0$ propagates without decay to the fixed points $\pm\mu^*$, $\mu^* = \tanh(\gamma\alpha_0\mu^*/2)$. With sampled routing and message noise of standard deviation $\sigma$, the relay choices form a binary symmetric Markov chain with flip probability $p(\gamma) = \mathbb{E}[(1 + e^{\gamma\alpha_0(1+\sigma Z)})^{-1}]$, $Z$ standard normal, so the routing bit has correlation length $-1/\ln(1-2p)$, 21.5 hops at $\sigma = 0.5$ and $\gamma = \infty$, where $p = \Pr(Z < -2)$ (Check S47.76), far beyond the linear correlation length of the continuous content.

*Remark* S8.5 (Infinite precision). At $\gamma = \infty$ causality survives, since Theorem 8.5(a) and the support bound of Proposition S8.3 use only locality, but the amplitude bound fails: the update is discontinuous where two policies' expected free energies tie, and in Example S8.4 a perturbation of $10^{-6}$ at the head moves the state 20 hops away by order one. Gumbel noise of scale $1/\gamma$ in evaluating expected free energy makes the argmax reproduce the softmax, restoring Proposition 8.6. Above $\gamma_D$ the routed dynamics can have several attractors, consistent with the several stable attention motifs found by Dennis (2026a).

### S8.3 Unstable profiles and rumors

**Proposition S8.6** (Rank-one dominance)**.** *Let $W$ be time invariant with a simple real eigenvalue $\rho = \rho(W) > 1$ that strictly dominates every other eigenvalue in modulus, with right and left eigenvectors $v$ and $u$, $u^\top v = 1$, let $Q^\zeta \succ 0$ be constant and let $\Sigma(0) \succeq 0$. Then*

$$\rho^{-2t}\Sigma(t) \to c_\infty vv^\top, \qquad c_\infty = \frac{u^\top Q^\zeta u}{\rho^2 - 1} + u^\top\Sigma(0)u > 0. \tag{S32}$$

*Consequently the top canonical correlation of $e_i(t)$ and $e_j(t)$ tends to one for every pair with nonzero blocks $v_i$ and $v_j$, and $\Sigma(t)^{-1}$ has one eigenvalue of order $\rho^{-2t}$, with eigenvector tending to $v/|v|$. If every other eigenvalue of $W$ lies in the open unit disk, the remaining eigenvalues of $\Sigma(t)^{-1}$ stay bounded away from zero.*

Proof in Section S32; checked on Example S8.1 (Check S47.79). The unstable profile turns the agents' errors into one shared error. How far it reaches when the amplification sits at one site of a stable medium can be computed exactly: on $\mathbb{Z}$ let $W_{xx} = c$ for $x \neq 0$, $W_{00} = c + b_0$ and $W_{xy} = a$ for $|x - y| = 1$, with $c \geq 0$, $a > 0$, $b_0 > 0$ and $\lambda = c + 2a < 1$.

**Proposition S8.7** (An amplifying site and its front)**.** *In this chain, with $Q^\zeta = qI$ and $\Sigma(0) = 0$:*

*(a) $W$ has exactly one eigenvalue outside $[c - 2a, c + 2a]$, namely $\rho = c + \sqrt{4a^2 + b_0^2}$, with eigenvector $v_x = e^{-m_\rho|x|}$ and $m_\rho = \operatorname{arcsinh}(b_0/(2a)) = \operatorname{arccosh}((\rho - c)/(2a))$. The profile is unstable exactly when $\rho > 1$.*

*(b) For $\rho \neq 1$ and $t \geq 1$, exactly, $\Sigma(t) = A(t)P_v + B(t)$, where $P_v = vv^\top/|v|^2$, $|v|^2 = \coth m_\rho$, $A(t) = q(\rho^{2t} - 1)/(\rho^2 - 1)$, $B(t)P_v = 0$ and $q(I - P_v) \preceq B(t) \preceq \frac{q}{1-\lambda^2}(I - P_v)$.*

*(c) For $\rho > 1$ and $t \geq 1$, with $\mathfrak{s}_x(t) = (1-\lambda^2)A(t)v_x^2/(q|v|^2)$,*

$$\frac{\mathfrak{s}_x - 1}{\mathfrak{s}_x + 1} \leq \mathrm{Corr}(e_x(t), e_0(t)) \leq \Big(\frac{\mathfrak{s}_x}{1-\lambda^2}\Big)^{1/2} + \frac{|v|q^{1/2}}{(1-\lambda^2)A(t)^{1/2}}. \tag{S33}$$

*(d) For $\rho > 1$, $\theta \in (0,1)$ and $t \geq 1$ the front $x_\theta(t) = \max\{x \geq 0 : \mathrm{Corr}(e_x(t), e_0(t)) \geq \theta\}$ satisfies $x_\theta(t) = v_{\mathrm{f}} t + O_\theta(1)$ with $v_{\mathrm{f}} = \ln \rho / m_\rho < 1$ hop per round.*

*(e) For any finite amplifying core embedded in the same medium with a simple dominant real eigenvalue $\rho > 1$, the eigenvector decays outside the core at the rate $m_\rho = \mathrm{arccosh}((\rho - c)/(2a))$, the decay rate of the medium's Green function at the value $\rho$. As $\rho \downarrow 1$, $m_\rho$ decreases to the medium's Ornstein–Zernike rate $m$, $\cosh m = 1 + \varepsilon/(2a)$ with $\varepsilon = 1 - \lambda$ (Theorem 8.10), so the unstable mode is never wider than the medium's correlation length.*

Proof in Section S32; checked on chains of 801 and 401 agents (Check S47.80). The amplifying site recruits its neighbors into one shared error at less than one hop per round, and the recruited mode is never wider than the stable medium allows.

*Remark* S8.8 (Rumors are not collectives). Contrast needs the stationary world-conditional precision (Definition 4.10), which does not exist without (M), and its finite-time analog is degenerate: $\Sigma(t)^{-1}$ has one eigenvalue of order $\rho^{-2t}$ along the unstable mode $v$ (Proposition S8.6), so a criterion that admitted the weighted coordinate $v$ would assign diverging contrast to the amplifying structure. The definition of a collective therefore includes (M) (Definition 4.12 and Remark S4.14), and we call an amplifying profile a rumor, an echo chamber whose shared error grows like $\rho^t$; the name is a definition, not a derived result. For rumors the scales change meaning: after $T$ rounds correlation extends at most $\min\{2T, 2v_\varphi T + O(1)\}$ hops, a localized front advances at $v_{\mathrm{f}} < 1$ hop per round, and $\tau$ is replaced by the amplification time $1/\ln \rho$, so the correlated extent is limited by the lifetime of the profile, which ends when attention is re-selected, the linear regime ends or grounding resets the errors. Level-1 receivers cannot detect the instability, because their claimed covariances follow recursions that do not involve their means.

## S8.4 Refinements of the clustering bound and the Ornstein–Zernike form

**Proposition S8.9** (World conditioning and correlated noise). *(a) Under* (Lin), (Loc) *and* ($\mathrm{C}_\tau$), *if the local noise is correlated across agents with $\|Q^\zeta_{kl}\| \leq q e^{-d(k,l)/\xi_w}$, where $\xi_w$ is the correlation length of the noise, then $\|\Sigma_{ij}\| \leq q(\lceil d/2 \rceil + (1-\lambda^2)^{-1}) \max(\lambda, e^{-1/\xi_w})^d$ with $d = d(i,j)$: the decay rate is at least $\min(\ln(1/\lambda), 1/\xi_w)$.*

*(b) Let identical scalar agents with exact code models sense one scalar world $\eta_{t+1} = F\eta_t + w_t$, $\mathrm{Var}(w_t) = Q$, with a common sensing gain, and let nobody communicate. Then the world-conditional covariance is diagonal, while the unconditional correlation of the errors of any two distinct agents is $Q/(Q+q)$, with $q$ the local noise variance, whatever their distance.*

Proof in Section S32. Unconditional correlation measures the shared world as well as communication, so every scale bound of Section 8 concerns the world-conditional law.

**Proposition S8.10** (Spectral form). *Assume* (Lin), (Loc), *block-diagonal local noise with blocks $\preceq qI$, and* ($\mathrm{C}_\rho$). *Then $\|\Sigma_{ij}\| \leq K_A^2 q f^{2\lceil d(i,j)/2 \rceil}/(1-f^2)$. If $W$ is normal, the bound holds with $K_A = 1$ and $f = \rho(W)$. Under* (B) *it follows that $\xi \leq \tau$: a correlation length in hops cannot exceed the correlation time in rounds, and the physical length is at most $\ell\tau$.*

Proof in Section S32; checked on signed rings, where the row-sum bound is unusable (Check S47.81). This is the classical counterpart of exponential clustering from a Lieb–Robinson bound and a spectral gap (Hastings, 2004; Nachtergaele and Sims, 2006; Hastings and Koma, 2006), and it applies with signed weights and in the common-code regime, where ($C_r$) fails. Attention that is reciprocal in a weak sense gives a much better rate, without translation invariance.

**Theorem S8.11** (Diffusive clustering)**.** *Let $P$ be a stochastic matrix, reversible with respect to a positive measure $\varpi$ ($\varpi_i P_{ij} = \varpi_j P_{ji}$) with $\varpi_{\max} = \sup_i \varpi_i < \infty$, and with $P_{ij} > 0$ only if $i = j$ or $(i,j) \in \mathcal{E}$. Assume* (Lin)*, block-diagonal local noise with blocks $\preceq qI$, and $\|W_{ij}(t)\| \le \lambda P_{ij}$ for all $i$, $j$, $t$, with $\lambda < 1$. Then the covariance $\Sigma = \Sigma(t)$ of the process started in the infinite past, or at any time from a zero start, satisfies for every $\vartheta \in (0,1)$ and every $t$*

$$\|\Sigma_{ij}\| \le \frac{2q\varpi_{\max}}{\sqrt{\varpi_i \varpi_j}} \cdot \frac{\exp\bigl(-\vartheta d(i,j)\sqrt{2\ln(1/\lambda)}\bigr)}{1-\lambda^{2(1-\vartheta)}}. \tag{S34}$$

Proof in Section S32; checked on a torus with random conductances (Check S47.82). For $\lambda > e^{-2}$, in particular for small leaks, the rate $\sqrt{2\ln(1/\lambda)} \approx \sqrt{2\varepsilon}$ improves on the row-sum rate $\ln(1/\lambda)$ by the square root characteristic of diffusion, on any graph, and for $\lambda \le e^{-2}$ the row-sum rate is at least as large; the diffusive bound needs domination by one reversible chain, not reversibility of $W$, and covers time-varying schedules dominated by that chain. On the ring with $c = 0$ the exact rate is $\operatorname{arccosh}(1/\lambda) = \sqrt{2\ln(1/\lambda)}(1 + O(\ln(1/\lambda)))$, so the rate is sharp to leading order. The proof uses the Chebyshev-polynomial argument with which Carne (1985) proved the estimate of Varopoulos (1985), known to us from a review; we prove the form we use in full. Heterogeneous gains give rates between those of two homogeneous media.

**Proposition S8.12** (Heterogeneous media)**.** *Let scalar agents occupy $\mathbb{Z}^D$ or a finite box of $\mathbb{Z}^D$ at positions $x_i$, attend only nearest neighbors, and satisfy* (Lin) *and* (B) *with local noise variances in $[q_{\min}, q_{\max}]$. For $\boldsymbol{\varphi} \in \mathbb{R}^D$ put $\bar\psi(\boldsymbol{\varphi}) = \sup_{t,i} \sum_j |W_{ij}(t)| e^{\boldsymbol{\varphi}\cdot(x_i - x_j)}$ and $\Psi(\boldsymbol{\varphi}) = \bar\psi(\boldsymbol{\varphi})\bar\psi(-\boldsymbol{\varphi})$.*

*(a) For every $\boldsymbol{\varphi}$ with $\Psi(\boldsymbol{\varphi}) < 1$, $|\Sigma_{ij}| \le q_{\max} e^{-\boldsymbol{\varphi}\cdot(x_i - x_j)}/(1 - \Psi(\boldsymbol{\varphi}))$, for the stationary covariance and at any time from a zero start. On a box with open ends the supremum in $\bar\psi$ includes the rows of the boundary agents, which have a single neighbor along some axis. On a periodic box of side $L$, $\bar\psi$ is computed for the $L$-periodic medium on $\mathbb{Z}^D$ that covers the box, and $\Sigma_{ij}$ is the sum over $\mathbf{w} \in \mathbb{Z}^D$ of that medium's covariances between $x_i$ and the images $x_j + L\mathbf{w}$, each of which obeys the bound with its own admissible $\boldsymbol{\varphi}$. On $\mathbb{Z}^D$, if $W(t) \ge 0$, every agent gives equal weights $a_{id}(t)$ to its two neighbors along each axis $d$, the local leaks $\varepsilon_i(t) = 1 - \sum_j W_{ij}(t)$ are at least $\varepsilon_{\min} > 0$ and the weights $a_{id}(t)$ at most $a_{\max}$, then the decay rate is at least $m_{\mathrm{up}} = \operatorname{arccosh}(1 + \varepsilon_{\min}/(2a_{\max}))$ per hop along a coordinate axis and at least $\sqrt{D}\operatorname{arccosh}(1 + \varepsilon_{\min}/(2Da_{\max}))$ per unit Euclidean length along the diagonal. For a tilt along its one-sided axis $d$, a boundary agent of an open box has the tilted row sum $1 - \varepsilon_i + a_{id}(e^{\pm\varphi_d} - 1)$, which for one sign exceeds the two-neighbor value $1 - \varepsilon_i + 2a_{id}(\cosh\varphi_d - 1)$, so on an open box the admissible tilts, and with them the guaranteed rate, can lie below $m_{\mathrm{up}}$.*

*(b) If $W(t) \ge W_{\mathrm{low}}$ entrywise for all $t$, where $(W_{\mathrm{low}})_{ii} = c_{\min} \ge 0$ and $(W_{\mathrm{low}})_{ij} = a_{\min} > 0$ for neighbors, then $\Sigma \ge q_{\min}(I - W_{\mathrm{low}}^2)^{-1}$ entrywise, the covariance of the homogeneous medium $(c_{\min}, a_{\min})$ on the same vertex set. Where every agent has $2D$ neighbors, on $\mathbb{Z}^D$ or a periodic box, the leak $\varepsilon_{\mathrm{low}} = 1 - c_{\min} - 2Da_{\min}$ of that medium is at least every local leak; a boundary agent of an open box, with fewer neighbors, can have a larger one.*

(c) *On $\mathbb{Z}$, let $W(t) \geq 0$ give equal weights at most $a_{\max}$ to the two neighbors of every agent, with local leaks at least $\varepsilon_{\min} > 0$, as in (a), and let $W(t) \geq W_{\text{low}}$ as in (b). With $m_{\text{low}} = \operatorname{arccosh}(1 + \varepsilon_{\text{low}}/(2a_{\min}))$, correlations decay at a rate between $m_{\text{up}}$ and $m_{\text{low}}$, so $1/m_{\text{low}} \leq \xi \leq 1/m_{\text{up}}$.*

(d) *For a frozen $W \geq 0$ on a finite population, $\min_i(1-\varepsilon_i) \leq \rho(W) \leq \max_i(1-\varepsilon_i)$, so $-1/\ln(1-\varepsilon_{\max}) \leq \tau \leq -1/\ln(1-\varepsilon_{\min})$.*

(e) *On any graph with $W \geq 0$, if $\lambda' = \max_i[W_{ii} + \sum_{j\neq i} \max(W_{ij}, W_{ji})] < 1$, then Theorem S8.11 applies with the symmetric chain $P_{ij} = \max(W_{ij}, W_{ji})/\lambda'$ for $i \neq j$, uniform $\varpi$, and the rate $\sqrt{2\ln(1/\lambda')}$.*

Proof in Section S32; checked on rings of 300 agents (Check S47.83). The bounds hold for time-varying schedules, attention chosen by expected free energy included, when the schedule is independent of the noises and the extremes are taken over the whole schedule; the lower bound needs every edge attended in every round. For small leaks the ratio of the two rates, about $[(\varepsilon_{\text{low}}/a_{\min})/(\varepsilon_{\min}/a_{\max})]^{1/2}$, measures the heterogeneity.

*Remark* S8.13 (The Ornstein–Zernike form). Theorem 8.10 proves that every direction decays at least at the rate $m(\mathbf{u})$ and that hyperplane sums decay exactly at $m_1$. That $m(\mathbf{u})$ is the exact pointwise rate, with the prefactor of the Ornstein–Zernike form $r^{-(D-1)/2}e^{-m(\mathbf{u})r}$ (Ornstein and Zernike, 1914), follows from the Green-function asymptotics of finite-range walks with drift (Ney and Spitzer, 1966), applied to the walk tilted by the maximizer $\boldsymbol{\varphi}^*$ of the support function, which is aperiodic when $c > 0$; we have read that theorem only in the restatement of Ballu (2024), which gives the power and the rate but not the constant. A local central limit computation suggests

$$\Sigma(r\mathbf{u}) \simeq \frac{q}{2} e^{-m(\mathbf{u})r} \frac{|v|^{(D-3)/2}}{(2\pi r)^{(D-1)/2}\big(\det H \cdot \mathbf{u}^\top H^{-1}\mathbf{u}\big)^{1/2}}, \tag{S35}$$

with $v = \nabla\psi(\boldsymbol{\varphi}^*)$ and $H = \nabla^2\psi(\boldsymbol{\varphi}^*)$, but its passage from the local limit theorem to the sum over the number of steps is heuristic, so (S35) is not a result of this manuscript; numerically the covariance approaches it (Check S47.85). The rigorous Ornstein–Zernike theory of Campanino et al. (2003), for Ising models above the critical temperature, is not used here.

## S8.5 Persistence, diffusive scaling and records

**Proposition S8.14** (Persistence of collective variables)**.** *Assume* (M) *and* (Lin) *with $W$ and $Q^\zeta \succ 0$ time invariant, and let $\tilde{W} = \Sigma^{-1/2} W \Sigma^{1/2}$. For all readout vectors $\mathbf{v}$ and $\mathbf{v}'$ and lags $h \geq 0$,*

$$\big|\operatorname{Corr}(\mathbf{v}'^\top e(t+h), \mathbf{v}^\top e(t))\big| \leq \|\tilde{W}^h\| \leq \|\tilde{W}\|^h, \qquad \|\tilde{W}\|^2 = 1 - \lambda_{\min}(\Sigma^{-1/2} Q^\zeta \Sigma^{-1/2}) < 1, \tag{S36}$$

*and $\|\tilde{W}^h\|^{1/h} \to \rho(W)$. The lifetime $T^*(\theta)$, the largest lag at which some pair of linear readouts keeps correlation at least $\theta$, equals $\max\{h : \|\tilde{W}^h\| \geq \theta\}$. Hence $T^*(\theta) \leq \ln(1/\theta)/\ln(1/\|\tilde{W}\|)$, and $T^*(\theta)/\ln(1/\theta) \to \tau$ as $\theta \to 0$.*

Proof in Section S32; for strongly non-normal $W$ the one-step bound can be loose (Check S47.87). The lifetime of a collective variable is set by the slowest mode of the whole system of agents and media, which is why records can lengthen it. On lattices, time and space are tied.

**Proposition S8.15** (Diffusive scaling)**.** *In the setting of Theorem 8.10:*

(a) *$\rho(W) = \lambda$, so $\tau = -1/\ln\lambda$, and $\xi_2^2/\tau = 2a\lambda\ln(1/\lambda)/(1-\lambda^2) \to a$ as $\varepsilon \to 0$. In physical units $\ell^2\xi_2^2 \simeq D_{\text{diff}}\tau$ with the diffusion constant $D_{\text{diff}} = a\ell^2$ per round: the dynamic exponent is two.*

*(b) If $c > 0$, the local autocovariance $\Sigma(0;h) = \mathrm{Cov}(e_x(t+h), e_x(t))$, which is*

$$\Sigma(0;h) = \int_{[-\pi,\pi]^D} \frac{q\hat{W}(p)^h}{1-\hat{W}(p)^2} \frac{\mathrm{d}p}{(2\pi)^D}, \tag{S37}$$

*satisfies, as $h \to \infty$ at fixed $\varepsilon$,*

$$\Sigma(0;h) \simeq \frac{q\lambda^h}{1-\lambda^2} \Big(\frac{\lambda}{4\pi a h}\Big)^{D/2}. \tag{S38}$$

Proof in Section S32; checked in one and two dimensions (Check S47.88). The relative error of (S38) is about $D/(2\varepsilon h)$ for small $\varepsilon$, so the asymptote applies for lags well beyond the correlation time. Spatial reach grows only as the square root of memory, and building correlations over a distance $L$ by local averaging takes about $L^2/D_{\mathrm{diff}}$ rounds, more than the light-cone time $L/(2\ell)$ once $L$ exceeds a fraction of one hop. Up to the factor $\varepsilon$, the resolvent $(I-W)^{-1}$ that sets the leading decay in Theorem 8.10(a) is the total-influence matrix of the model of Friedkin and Johnsen (1990), the leak playing stubbornness; (84) is the standard massive lattice Green's function (Section 17).

*Remark* S8.16 (Special cases). In the notation of Proposition 8.11, one record read by the mean mode with $\upsilon = \delta = 0$ gives $\bar{e}(t+1) = (1-\varepsilon-\beta)\bar{e}(t) + \beta z(t)$, $z(t+1) = (1-\omega)z(t) + \omega\bar{e}(t)$, whose first-order extension factor $1+\mathcal{R}$ is the lower end of the interval of Proposition 8.12(b): at $\varepsilon = 0.01$, $\beta = 0.2$, $\omega = 0.1$ the secular root gives $\tau = 306.24$ rounds, against 300 to first order and 99.5 without the record. The identity time of a collective whose members turn over (Proposition 5.14) is the continuous-time case: its rate matrix (62) equals $-\mathsf{R}$ for one record with $\upsilon = 0$ when forgetting plus the turnover rate times the share of a newcomer's content not copied from the record plays $\varepsilon$, the read rate plus the turnover rate times the copied share plays $\beta$, the members' write rate plays $\omega$ and the record's decay plays $\delta$, so, when the read rate plus the turnover rate times the copied share is positive, as Proposition 8.11 requires of the read weight, the identity rate $\tau_{\mathrm{id}}^{-1}$ is the secular root $\varepsilon_{\mathrm{rec}}$ (Check S47.89). When that sum is zero the record is unread, the function $g_z$ is affine with the single root $\varepsilon'$, and Proposition 5.14(a) gives the rates directly: the members' mean decorrelates at the rate that plays $\varepsilon$, and the pair at the smaller of that rate and $\omega+\delta$.

*Remark* S8.17 (Global records). In the record model of Proposition 8.13, a record read by all $N$ members with a uniform kernel couples only to the mean mode, since its kernel's symbol vanishes at every nonzero wavevector; for $p \neq 0$ the agents' symbol is $1-\varepsilon'-\beta-a\sum_d 2(1-\cos p_d)$. A global archive therefore acts on every deviation from the collective mean as an extra leak $\beta$, so fluctuations about the mean decorrelate over $\xi^2 \approx a/(\varepsilon'+\beta)$ while the mean has the memory of Proposition 8.12. The archive is a hub: it adds to every covariance a component $[\hat{\Sigma}_{00}(0) - q/(1-(1-\varepsilon'-\beta)^2)]/N$, the excess of the term at wavevector zero over the continuation of the branch $p \neq 0$ and about $\hat{\Sigma}_{00}(0)/N$ when $\varepsilon' + \mathcal{R}\delta \ll \beta\omega/(\beta+\omega)$ with $\mathcal{R} = \beta/\omega$ (the leak $\varepsilon'$ small beside both $\beta$ and $\omega$, and $\delta$ small beside $\omega^2/(\beta+\omega)$; Check S47.91), that does not decay with distance (compare the exogenous hub of Corollary 8.19(e), whose common correlation does not fall with $N$; here the component falls as $1/N$, see Proposition S12.17), which world conditioning does not remove, because agents write the archive.

For a closed population with a durable global record at the record noise that makes the dynamics reversible, Proposition S12.17 gives this component in closed form, together with the extra leak $\beta$ of the deviations. Its world-conditional cost falls like $1/N$, and the cost of a member-written archive that does not shrink with $N$ is the world-driven common mode bounded in Theorem 12.7.

## S8.6 Fronts with long-range links

**Proposition S8.18** (Fronts). *In the setting of Section 8.7, let $Y_t$ be the walk with step law $W/\lambda$, so that the normalized influence beyond distance $r$, $\sum_{|x|\ge r}(W^t)(x)/\lambda^t$, equals $\Pr(|Y_t|\ge r)$.*

*(a)* $\Pr(|Y_t|\ge r)\le C_1 t r^{-\sigma}$ *if* $\sigma<2$, *and* $\le C_1(tr^{-\sigma}+tr^{-2})$ *if* $\sigma>2$.

*(b) If* $\sigma<2$ *and* $r\ge c_2 t^{1/\sigma}$, *then* $\Pr(|Y_t|\ge r)\ge c_1 t r^{-\sigma}$.

*(c) For* $\sigma\neq 2$ *and every small enough threshold* $\theta$, *the front* $r_\theta(t)=\min\{r:\Pr(|Y_t|\ge r)\le\theta\}$ *satisfies* $r_\theta(t)\asymp t^{1/\min(\sigma,2)}$: *superballistic for* $\sigma<1$, *linear at* $\sigma=1$, *sublinear for* $\sigma>1$. *At* $\sigma=2$, *(a) holds with* $tr^{-2}\ln(er)$ *on the right and the front is* $O((t\ln t)^{1/2})$. *Influence therefore moves at finite speed if and only if* $\sigma\ge 1$.

*(d) Let* $\mathsf{J}(y)\le K_{\mathsf{J}}|y|^{-D-\sigma}$ *for* $y\neq 0$. *For any influence bound with* $\bar J_{ij}\le c\mathbf{1}\{i=j\}+a\mathsf{J}(x_i-x_j)$, $(\bar J^t)_{ij}\le K_{\mathsf{J}}t^{1+D+\sigma}g_{\mathrm{tot}}^t|x_i-x_j|^{-D-\sigma}$ *with* $g_{\mathrm{tot}}=c+a$: *synchronous rounds have no strict light cone in the lattice metric, and when* $g_{\mathrm{tot}}>1$ *the region of influence above a fixed level lies inside the logarithmic cone* $t\gtrsim(D+\sigma)\ln r/\ln g_{\mathrm{tot}}$.

Proof in Section S32; fitted front exponents lie within 7 percent of $1/\min(\sigma,2)$, with shortfalls of 5 and 6 percent at $\sigma=0.5$ and $\sigma=3$ that depend on the fit window (Check S47.93); the exponents themselves are those proved in (c). The threshold $\sigma=1$ coincides with the free-particle light cone of quantum systems with interactions decaying as $r^{-\alpha}$, $\alpha>D+1$ (Tran et al., 2020), because linear message passing propagates a single tangent vector; the many-body thresholds of that hierarchy have no analog here (Kuwahara and Saito, 2020), and the logarithmic cone of (d) matches the causal region reported for classical long-range lattices (Métivier et al., 2014). On a fixed graph with random long links, Theorem 8.9 applies in the graph metric, whose chemical distance in long-range percolation with connection probabilities decaying as $|x-y|^{-D-\sigma}$ and $0<\sigma<D$ grows like $(\log|x-y|)^{\Delta+o(1)}$ on the infinite cluster, with $1/\Delta=\log_2(2D/(D+\sigma))$ (Biskup, 2004); outside that range of $\sigma$ this result does not apply.

## S8.7 Broadcasting on trees and condensation

In the linear-Gaussian model the decisive structure is branching. Consider a rooted tree with $X_{\mathrm{root}}\sim\mathcal{N}(0,1)$ and, on each edge $e$ from a parent $u$ to a child $c$, $X_c=\alpha_e X_u+(1-\alpha_e^2)^{1/2}Z_c$ with independent standard normal $Z_c$ and edge correlations $\alpha_e\in(0,1)$, the vertices at depth $n$ being observed with noise variance $\sigma_{\mathrm{obs}}^2\ge 0$. Let $\mathcal{V}(u)$ be the noise variance of the equivalent Gaussian observation of $X_u$ carried by the observed descendants of $u$.

**Theorem S8.19** (Gaussian broadcasting on trees). *(a)* $\mathcal{V}(v)=\sigma_{\mathrm{obs}}^2$ *at the vertices* $v$ *of level* $n$, $\mathcal{V}(v)=\infty$ *at childless vertices above level* $n$ *(depth less than* $n$*), which carry no observation,* $\mathcal{V}(u)^{-1}=\sum_c \alpha_{e_c}^2/(1-\alpha_{e_c}^2+\mathcal{V}(c))$ *over the children* $c$ *of every other vertex* $u$ *above level* $n$, *a term with* $\mathcal{V}(c)=\infty$ *being zero, and* $I(X_{\mathrm{root}};\text{level } n)=\frac12\ln(1+1/\mathcal{V}(\mathrm{root}))$.

*(b) For constant* $\alpha$, $\mathcal{V}(u)=\alpha^{2|u|}\hat{\mathcal{V}}(u)$, *where* $|u|$ *is the depth and* $\hat{\mathcal{V}}(\mathrm{root})$ *is the effective resistance between the root and a ground node in the network with edge resistances* $(1-\alpha^2)\alpha^{-2|e|}$, $|e|$ *the depth of the child endpoint, in which each level-n vertex is joined to the ground by a resistance* $\sigma_{\mathrm{obs}}^2\alpha^{-2n}$ *(shorted to it when* $\sigma_{\mathrm{obs}}=0$*).*

*(c) On the b-ary tree with exact leaves,* $\mathcal{V}_n=(1-\alpha^2)\sum_{k=1}^{n}(b\alpha^2)^{-k}$, *so* $\lim_n I_n=\frac12\ln(1+\frac{b\alpha^2-1}{1-\alpha^2})$ *if* $b\alpha^2>1$, $I_n\simeq 1/(2n(1-\alpha^2))$ *if* $b\alpha^2=1$, *and* $I_n\simeq(1-b\alpha^2)(b\alpha^2)^n/(2(1-\alpha^2))$ *if* $b\alpha^2<1$.

*The Kesten–Stigum condition $b\alpha^2 > 1$ is exactly the reconstruction threshold for Gaussian channels, and linear estimation is optimal.*

*(d) On a general locally finite tree with exact leaves, reconstruction ($\lim_n I_n > 0$) holds if the branching number satisfies $\mathrm{br}(T)\alpha^2 > 1$ and fails if $\mathrm{br}(T)\alpha^2 < 1$.*

Proof in Section S32; checked against direct Gaussian conditioning (Check S47.94). For a general channel on the $b$-ary tree, reconstruction holds when $b$ times the squared second eigenvalue of the channel exceeds one, a condition named for the limit theorems of Kesten and Stigum (1966); it is tight for the binary symmetric channel (Bleher et al., 1995; Evans et al., 2000) and not for some other channels (Mossel, 2001). Part (c) adds the Gaussian case with an exact electrical identity, and part (d) uses, for the two strict cases only, the transience criterion of Lyons (1990) for the biased walk, read in a review and in the restatement of Lyons and Pemantle (1992). The information that reaches a level of the tree percolates.

**Theorem S8.20** (Gaussian information percolation on trees)**.** *In the tree model, let a source $X_{\mathrm{src}}$ be jointly Gaussian with $X_{\mathrm{root}}$, with $X_{\mathrm{src}} \to X_{\mathrm{root}} \to$ (observed level). Then*

$$I(X_{\mathrm{src}}; \textit{observed level}) \le \mathrm{perc}_{\alpha^2}(\mathrm{root} \to \textit{observed level}) \cdot I(X_{\mathrm{src}}; X_{\mathrm{root}}), \tag{S39}$$

*where $\mathrm{perc}_{\alpha^2}$ is bond percolation with edge $e$ open with probability $\alpha_e^2$, computed by $\mathfrak{p}(u) = 1 - \prod_c (1 - \alpha_{e_c}^2 \mathfrak{p}(c))$ above the observed level, with $\mathfrak{p}(v) = 1$ at the vertices of the observed level and the empty product, hence $\mathfrak{p}(v) = 0$, at childless vertices above it.*

Proof in Section S32; checked on 400 random trees (Check S47.95). The theorem extends the percolation bound of Polyanskiy and Wu (2017) to Gaussian channels, whose input-independent coefficient is one, through the Gaussian contraction $\alpha_e^2$; on trees it is tight at the level of thresholds, since percolation with $p = \alpha^2$ on the $b$-ary tree survives if and only if $b\alpha^2 > 1$. On graphs of subexponential growth, by contrast, aggregate information has short reach.

**Proposition S8.21** (Aggregate information on slowly growing graphs)**.** *If $\lambda_{\min}(\Sigma) \ge \sigma_{\mathrm{e}} > 0$ and $\|\Sigma_{0j}\| \le B_\Sigma e^{-m_\Sigma d(0,j)}$, then the squared top canonical correlation between $e_0$ and all $e_j$ with $d(0,j) \ge r$ is at most $(B_\Sigma^2/(\sigma_{\mathrm{e}} \lambda_{\min}(\Sigma_{00}))) \sum_{j: d(0,j) \ge r} e^{-2m_\Sigma d(0,j)}$, which tends to zero exponentially in $r$ on graphs whose spheres grow subexponentially.*

Proof in Section S32. On trees and other expanding graphs the sum need not converge, and an audience retains information about a distant source although each member's information decays. That reach does not make a collective.

**Proposition S8.22** (Reach lies outside the diffusive class)**.** *(a) Let the tree-indexed Gaussian law have constant edge correlation $\alpha$ and branching $b$. Its precision has entries $-\alpha/(1-\alpha^2)$ on tree edges, and its row sums are $(1 - b\alpha)/(1+\alpha)$ at internal vertices, $(1 + \alpha - b\alpha)/(1+\alpha)$ at the root and $1/(1+\alpha)$ at the leaves. For depth at least two it is diffusive if and only if $b\alpha \le 1$; at depth one there is no internal vertex and it is diffusive if and only if $(b-1)\alpha \le 1$. The Kesten–Stigum regime $b\alpha^2 > 1$ implies $b\alpha > 1$, so at depth at least two it is never diffusive in these coordinates.*

*(b) For the homogeneous diffusive medium $\Pi = \varepsilon^\Pi I + a^\Pi L$ on the $(b+1)$-regular tree, $b \ge 2$, $\varepsilon^\Pi \ge 0$, with $L$ the graph Laplacian, $\Sigma(o,v) \le b^{-d(o,v)} \Sigma(o,o)$, and the squared canonical correlation between $x_o$ and the sphere of radius $n$ about $o$ is at most $\frac{b+1}{b} b^{-n} \Sigma(o,o)(\varepsilon^\Pi + 2(b+1)a^\Pi)$: there is no reconstruction, and aggregate information decays by the factor $1/b$ per level.*

Proof in Section S32. Reach on trees is produced by directed broadcasting with $b\alpha^2 > 1$, hence $b\alpha > 1$, which a diffusive precision cannot represent in plain coordinates, and a diffusive tree medium has no reach. A broadcast hierarchy can therefore spread shared content without forming a collective in the sense of Definition 4.12. With this caution we state the condensation picture as a conjecture.

**Conjecture S8.23** (Condensation of coherent information)**.** *In a population whose effective communication structure has branching number $b_{\text{eff}}$, the growth rate per hop of the number of receivers that carry independent channel noise, and whose links contract squared correlation by $\alpha^2$, information about a common source reaches arbitrary hop distance if $b_{\text{eff}}\alpha^2 > 1$ and is confined within the information length $1/\ln(1/(b_{\text{eff}}\alpha^2))$ if $b_{\text{eff}}\alpha^2 < 1$.*

On trees the conjecture is Theorems S8.19 and S8.20; a family with $b_{\text{eff}}\alpha^2$ bounded away from one on one side whose information behaves as on the other would refute it. On low-dimensional lattices with linear relays a single receiver's information decays at twice the mass (Proposition S6.3) and branching from a point is subexponential, so condensation there would need long-range links, nonlinear processing such as the routing bit of Example S8.4 or majority processing (Makur et al., 2020), or persistent memory. As interpretations, span of control times link fidelity must exceed one in a hierarchical institution, and students per teacher times copying fidelity must exceed one for cultural transmission across generations, a tree in time.

## S8.8 Abstraction

Content that crosses many hops passes through many codes. A receiver with prior covariance $P$ that reads a code $H$ with noise covariance $\hat{R}$ moves its mean by the gain-code map $\mathcal{K}H$, $\mathcal{K} = PH^\top(HPH^\top + \hat{R})^{-1}$, which for noiseless codes is the $P^{-1}$-orthogonal projection onto the range of $PH^\top$ (Definition 6.7); frames rotate which world directions a code carries. With a common prior metric each noiseless hop is therefore an orthogonal projection. Let $V_1, \dots, V_J$ be subspaces of a finite-dimensional real inner-product space, the receivers' common prior metric, with orthogonal projections $P_{V_j}$, intersection $V_\cap = \bigcap_j V_j$ and projection $P_\cap$.

**Theorem S8.24** (Relays converge to the common ground)**.** *(a) Cyclic order. The product over one cycle of hops, $A_{\text{cyc}} = P_{V_J} \cdots P_{V_1}$, satisfies $c_{\text{cyc}} = \|A_{\text{cyc}}(I - P_\cap)\| < 1$ and $A_{\text{cyc}}^n - P_\cap = (A_{\text{cyc}}(I - P_\cap))^n$, so $\|A_{\text{cyc}}^n - P_\cap\| \le c_{\text{cyc}}^n$.*

*(b) Quasi-periodic order. If every block of $p_0$ consecutive hops uses every subspace, the content after $n$ hops satisfies $\|x_n - P_\cap x_0\| \le c_*^{\lfloor n/p_0 \rfloor}\|x_0 - P_\cap x_0\|$, with $c_* < 1$ the largest of $\|A_w(I - P_\cap)\|$ over the finitely many words $w$ of length $p_0$ that contain every index.*

*(c) Noisy messages. If instead each hop applies a self-adjoint map with spectrum in $[0, \bar{a}]$ for some $\bar{a} < 1$, every product of $n$ hops has norm at most $\bar{a}^n$, and all content decays.*

Proof in Section S32; checked with three codes in six dimensions (Check S47.97). The content that survives an arbitrarily long relay with fixed codes is the common ground $V_\cap$, the directions that every code along the path transmits, measured in the receivers' common metric. Codimensions add for subspaces in general position, so generically $\dim V_\cap = \max(0, K - \sum_j (K - \dim V_j))$, small as soon as several distinct codes or frames occur. Part (a) proves only $c_{\text{cyc}} < 1$; we use no explicit formula for it.

*Remark* S8.25 (Rates from the literature). For two subspaces the rate is exact, $\|(P_{V_2}P_{V_1})^n - P_\cap\| = c^{2n-1}$ with $c$ the cosine of the Friedrichs angle (Friedrichs, 1937) between $V_1 \cap V_\cap^\perp$ and $V_2 \cap V_\cap^\perp$: the upper bound is due to Aronszajn (1950) and the equality to Kayalar and Weinert (1988), both as

stated by Badea et al. (2012), and our computation reproduces it (Check S47.97). Cyclic products converge to $P_\cap$ in any Hilbert space by Halperin's theorem (Halperin, 1962), as stated by Badea et al. (2012), and in finite dimension products in any order in which each subspace recurs infinitely often converge (Amemiya and Ando, 1965; Práger, 1960), with no uniform rate when the gaps between recurrences grow. In infinite dimension norm convergence can fail for three subspaces (Kopecká and Paszkiewicz, 2017), and cyclic convergence is geometric exactly when the relevant cosine is below one (Badea et al., 2012).

With random codes, frames and noise levels the relevant object is the product of random hop maps. For independent, identically distributed hop maps $A_m$ with $\mathbb{E}\ln^+\|A_m\| < \infty$ the Lyapunov exponents $\Gamma_1 \geq \Gamma_2 \geq \cdots$ exist, with $\frac{1}{n}\ln\sigma_k(A_n \cdots A_1) \to \Gamma_k$ almost surely (Furstenberg and Kesten, 1960; Oseledets, 1968), and $\Gamma_1 > \Gamma_2$ for invertible maps whose generated semigroup is strongly irreducible and contracting (Guivarc'h and Raugi, 1985; Gol'dsheid and Margulis, 1989); we cite these results at the level of their records and restatements. Then $\sigma_2/\sigma_1 \approx e^{-n(\Gamma_1-\Gamma_2)}$, so the content that survives many hops is one-dimensional, with abstraction length $1/(\Gamma_1 - \Gamma_2)$ hops. In a computed example with rank-two codes and random frames, pure relays have an abstraction length of 1.70 hops, so that the second singular value of a typical path product falls below one percent of the first after about eight hops ($\ln 100$ divided by the gap), with wide scatter across realizations, while receivers that keep half of their own belief at each hop have an abstraction length of about 34 hops (Check S47.98). The pure-relay maps of this example have rank two and are not invertible, and the cited theorems reach them through a reduction. With identity prior the hop map is $A_m = U_m G U_m^\top$, with $U_m$ the $6 \times 2$ frame and $G$ a fixed positive diagonal $2 \times 2$ matrix, so $A_n \cdots A_1 = U_n (G M_n) \cdots (G M_2) G U_1^\top$ with $M_m = U_m^\top U_{m-1}$, and the nonzero singular values of the path product are those of a product of $2 \times 2$ factors $GM_m$. For independent Haar frames these factors are independent, identically distributed and almost surely invertible, and the support of their law contains $GM$ for every $2 \times 2$ contraction $M$, so the semigroup they generate is strongly irreducible and contracting. The receivers that keep part of their own belief have invertible hop maps.

**Conjecture S8.26** (Scale up, rank down)**.** *In relay networks with diversity of codes or frames, the effective rank of the content shared across hop distance $n$, the number of singular values of the path product above a fixed fraction of the largest, is nonincreasing in $n$ and tends to the multiplicity of the top Lyapunov exponent of the path cocycle, generically one. A collective of linear size $L$ therefore shares content of rank about $\#\{k : \Gamma_1 - \Gamma_k \leq \ell/L\}$.*

The limit statement holds for fixed codes in a common metric, with limiting rank $\dim V_\cap$ (Theorem S8.24), and for independent invertible hop maps under the cited conditions, with rank one. It is open for networks with many paths, where path products add and ranks can rise, for correlated or learned codes, for heterogeneous priors, where the contraction itself can fail (Theorem 6.8), and for non-invertible pure relays that do not reduce, as those of the example above do, to invertible factors of the code rank, for instance relays whose code ranks differ along the path. The routing bit of Example S8.4 is the nonlinear extreme, in which one bit survives where continuous content does not. Proposition S13.15 proves the conjecture for stationary relays of meta-level hops that average several members' projections in a common metric, with limiting rank $\dim V_\cap$ when $V_\cap \neq \{0\}$, and shows that its reading as meta-agents communicating at lower rank than their members fails at one hop, since averaging parallel paths raises the rank. Aggregated codes lose rank across levels only along directions on which the members' frames disagree, and only under aggregation in public coordinates (Proposition S13.14).

### S8.9 Relations between the lengths

With Theorem 8.10, two lengths now describe a lattice medium: the Ornstein–Zernike length of the decay, and the second-moment length $\xi_2$ that bounds the extent of collectives (Theorem 4.14), defined in Section 4 from the world-conditional precision. The next proposition relates them.

**Proposition S8.27** (Relations between the lengths)**.** *Let the world-conditional precision on $\mathbb{Z}^D$ be translation invariant and diffusive, with symbol $\hat{\Pi}(p) = \varepsilon^{\Pi} + \sum_x a^{\Pi}(x)(1 - \cos p \cdot x)$, leak $\varepsilon^{\Pi} > 0$ and $\sum_x |x|^2 a^{\Pi}(x) < \infty$, and let $\xi_{2,d}^2 = \varsigma_d^2/(2\varepsilon^{\Pi})$ with $\varsigma_d^2 = \sum_x a^{\Pi}(x) x_d^2$.*

*(a) The covariance-side length coincides with it: $\sum_x x_d^2 \Sigma(x)/(2\sum_x \Sigma(x)) = \xi_{2,d}^2$.*

*(b) If the couplings have finite range $\ell_d^{\Pi} = \max\{|x_d| : a^{\Pi}(x) > 0\}$ in direction $d$ and $\varsigma_d > 0$, then $m_d$ is the positive root of $\sum_x a^{\Pi}(x)(\cosh(m x_d) - 1) = \varepsilon^{\Pi}$, and*

$$\xi_{2,d} \le \xi_{\mathrm{OZ},d} \le \xi_{2,d}\sqrt{\cosh(\ell_d^{\Pi}/\xi_{2,d})}. \tag{S40}$$

*(c) For nearest-neighbor couplings $a^{\Pi}$, $\xi_{2,d} = 1/(2\sinh(m/2))$ with $\cosh m = 1 + \varepsilon^{\Pi}/(2a^{\Pi})$, so $\xi_{\mathrm{OZ}} - \xi_2 = 1/(24\xi_{\mathrm{OZ}}) + O(\xi_{\mathrm{OZ}}^{-3})$. For the medium of Theorem 8.10, with $\cosh m = 1 + \varepsilon/(2a)$ and $\lambda = c + 2Da$,*

$$\xi_2 = \sqrt{\frac{2\lambda}{1+\lambda}} \cdot \frac{1}{2\sinh(m/2)} \le \frac{1}{m} = \xi_{\mathrm{OZ}}, \qquad \xi_2^2 = \frac{a}{\varepsilon} \cdot \frac{2\lambda}{1+\lambda} \le \frac{a}{\varepsilon}. \tag{S41}$$

Proof in Section S32; checked on random kernels of range one to four (Check S47.86). For the medium of Theorem 8.10 the world-conditional precision is diffusive with couplings of range two (Theorem 4.9), so every size statement made with $\xi_2$ remains true with $\xi_{\mathrm{OZ}}$ or $\sqrt{a/\varepsilon}$, and the three agree to leading order as $\varepsilon \to 0$. The inequality $\xi_2 \le \xi_{\mathrm{OZ}}$ is proved only for translation-invariant couplings of finite range. Both lengths grow with the range of communication: doubling every coupling distance at fixed leak doubles $\xi_2$.

### S8.10 Endogenous coupling and leak

**Open problem S8.28** (Endogenous coupling and leak)**.** Determine whether attention selected by expected free energy, together with grounding, drives the message weight $a$ and the leak $\varepsilon$ toward the regime in which the largest extent $L^*$ of collectives is comparable to $\xi_2$, or keeps collectives far below that bound. The formation model of Section 7 makes $a$ and $\varepsilon$ endogenous but has no geometry, so neither $\xi_2$ nor $L^*$ is defined in it.

## S9 Additional results for Section 9: the minimal model

This section collects statements, examples and remarks of Section 9 that the main text cites without stating in full. Proofs are in Section S33.

### S9.1 The earlier model as a special case

The report's results can be used here only after translation, and the translation decides which of them survive when every agent encodes the world in the same coordinates. The earlier model is the linear-Gaussian case of the agent model of Definition 2.1 with an exogenous world, one round

Table S6: Dictionary between the earlier model of Dennis (2026a) and the framework. Indices are receiver first in both.

| Earlier model | Framework |
|---|---|
| world $x_t$ with transition $F$, innovation covariance $Q_x$ and stationary covariance $P_x$; access sets of sensing channels | external states $\eta_t$ with $F$, $Q$ and $P_0$; sensing $s_i = H_i^s \eta + v_i$ (Definition 2.1) |
| fixed private frame in $\mathrm{GL}^+(K)$; displayed belief about the framed world | linear encoding $U_i$; internal state $\mu_i = U_i m_i$ |
| public displayed code of rank $r_c$; reference code of sender $j$ | public code $T_c$ of rank $r_c$; effective code $H_j = T_c U_j$ (Proposition 2.11) |
| dictionary learned by receiver $i$ for sender $j$, with true value $U_i U_j^{-1}$ | learned translation $\Theta_{ij}$; model of the sender's code $\hat{H}_{ij} = T_c \Theta_{ij}^{-1} U_i$ |
| receiver that reads the law header (called level 0 there) | interpretation level 1, assigned noise $R_c + H_j P_j H_j^\top$ (Definition 2.4) |
| receiver that learns its noise model from innovations | level 0 with calibration class C1i (Definition 2.6) |
| fusion with the stacked joint error covariance | exact fusion |
| prior-lift echo; reply under the literal preference to be predicted | feedback of being understood; of being predicted (Proposition S2.8) |
| listening offset; alignment precision | prior preference for receiving a message; precision of the preference to predict incoming messages |
| capacity, habits, policy precision | $\kappa$, $E_i(\pi)$, $\gamma$ |
| agent-only path blanket of full agent states; forbidden mass | agent-level description of a path blanket (Definition 4.2 and Remark S4.1); leak frequency |
| moment bound on decision-time belief means | hypothesis (M) (Assumption 8.1) |
| null models and attribution rule | null family with declared types (Definition 5.17) |

per world step, linear encodings $U_i \in \mathrm{GL}^+(K)$ and a single modality in every computed instance. Table S6 translates its objects. Two rows need comment. The report's default receiver adds the sender's posted coded covariance, its law header, to the channel noise and calls the result level 0; in the framework this receiver reads at interpretation level 1 (Definition 2.4), by the correspondence of Definition 2.6, and we use the framework's names throughout. The report's blankets are partitions of agents, which in the framework are coarse descriptions of the channel-level partition (Definition 4.1 and Remark S4.1); a member's sensing of the shared world is a sensory channel of the collective and a leak only in the coarse description.

By Proposition 2.11, encodings enter the framework only through the effective codes $H_j = T_c U_j$ and the receivers' models $\hat{H}_{ij}$ of them, which sorts the earlier results by how much of the encoding structure they use.

**Definition S9.1** (Classes of dependence on encodings)**.** A result about the earlier model is of class I if it holds when all senders of a modality share one effective code, $H_j = T_c$, and hence at identity encodings; of class C if it needs sender-specific effective codes, which identity encodings supply only through several modalities or role-specific codes; of class F if it needs errors of learned translations caused by differences between frames, so that it has no counterpart at identity encodings; and of class G if it is a statement about Markov chains or linear-Gaussian systems that does not use the communication model.

Every result the report proved with known translations is of class I or C, read off the codes it uses. Table S7 assigns the classes. Results of classes I and G hold in the framework with the report's scope, class C results need sender-specific codes, and class F results lose their mechanism

Table S7: Main results of Dennis (2026a), their class (Definition S9.1) and their place in the framework.

| Result of the report | Class | In the framework |
|---|---|---|
| Exact-fusion listening paths at four agents | I at the exchangeable point, C elsewhere | fail partner randomization (Section S9.2) |
| Level-0 bilayers at six agents; nested instance at eight | I | misreading artifacts (Definition 5.17) |
| Relay chains under lock-out, pruning or targeting | C | explained by access; unreachable |
| No fed bilayer among enumerated four-agent schedules | C; mechanisms I | conjectured in general by the report |
| Single-hop evidence relay cannot feed an interior | I | holds |
| Innovation bias, its sizes, covariance intersection, exact fusion | I | overvaluation (Propositions 3.4 and S3.6) |
| Sufficient conditions for the moment bound | I | Proposition 8.3 |
| Failure of the moment bound by amplification | C | Lemma S9.4, Proposition 9.1, and Example S8.1 |
| Complete split graph, specialization, echo chambers, gradient chains | I; stars at declared points C | hold; echo chambers as in Definition 3.7 |
| Reciprocal pairs, directed paths and cycles, closed triangles | F | Remark S9.3 |
| Product form, path blankets, leak bound, instantaneous blankets, nesting | G | Theorems 4.3 and 4.23 |
| Closure of community summaries | G | Definition 5.1 |
| Separability of the group expected free energy | G | Theorem 5.10 and Proposition S5.19 |
| Being predicted is not being understood | I | Proposition S2.8 |
| Replicator flow of habits; strict vertices; random limits | I | holds where attention depends on covariances only |
| Relays as mediated couplings; direct-source dominance | I | holds |

at identity encodings and belong to the gauge-theoretic setting whose constructions the report borrowed (Dennis, 2026c).

## S9.2 Message-fed bilayers and why none qualifies

The report's target was a blanket of agents whose interior learns about the exterior through messages alone. Its definition makes precise what such a structure would have to satisfy, and the reasons none qualifies say what the framework must ask instead.

**Definition S9.2** (Message-fed bilayer, after Dennis (2026a))**.** A message-fed bilayer is a periodic orbit of the best-response map at $\gamma = \infty$ in which every scheduled action is a strict minimizer of expected free energy, together with a partition of the agents into interior, active, sensory and exterior sets $(\mathsf{I}, \mathsf{A}, \mathsf{S}, \mathsf{E})$ with $\mathsf{I}$ and $\mathsf{E}$ nonempty and an assignment of the blocks of $\eta$ to an interior side and an exterior side, such that (i) no delivery, echo, reply or sensing event realizes a coupling $\mathsf{E} \to \mathsf{I}$, $\mathsf{I} \to \mathsf{E}$, $\mathsf{I} \to \mathsf{S}$ or $\mathsf{E} \to \mathsf{A}$; (ii) no agent of $\mathsf{I} \cup \mathsf{A}$ senses an exterior-side block; and (iii) some interior agent hears a message and has true posterior error variance, computed from the actual joint

Table S8: The four variants in which Dennis (2026a) found message-fed bilayers. The error column gives the interior's true posterior error variance on an exterior-side block as a fraction of its prior variance.

| Variant | Error | Held by | Why it does not qualify |
|---|---|---|---|
| Exact fusion, four agents | 0.48 to 0.67 | a negative listening offset; the ambiguity term; certain defect beliefs | partner randomization yields a fed path in 12 of 27 graphs; at one access-structured point the split is the access partition; reached from none of 61 starts |
| Level 0 with learned noise or agreement tracking, six agents; a nested instance at eight | 0.70 to 0.89 | cheap listening; believed information 3 to 150 times the true | absent under exact fusion; reached from none of 120 (nested: 80) seeded starts |
| Relay chains $\mathsf{S} \to \mathsf{A} \to \mathsf{I}$ (two hops), four to six agents | 0.76 to 0.85; pruned 0.936 to 0.999 | forbidden deliveries locked out or pruned; access-structured world | sides are the access classes; the frozen-environment null reproduces the chains; the lock-out does not arise from a broadcast start |
| Literal preference to be predicted, four agents | 0.95 at best | targeted addressing; a pessimistic prior over untried partners | sides are the access partition; not reached from the untried partner state |

law, below its prior variance on an exterior-side block. In clause (i) a delivery from a sender $j$ to a receiver $i$ realizes the coupling $j \to i$, and its echo, in which $i$ returns to $j$ in its own code what it understood, counts as a coupling $i \to j$.

Clause (ii) is sensory privacy of the interior, and clause (iii) asks that messages, not sensing, inform it. Table S8 lists the four variants in which the report found such orbits on open parameter sets; each adds a commitment that the report's canonical model does not make, and each fails attribution or reachability.

Without such additions the report found none: with level-1 receivers at four agents, exhaustive enumerations of schedules of period at most two at the declared points, and of period at most three in an aligned sub-instance, contain no message-fed bilayer, and nonexistence in general is its conjecture. It identifies three mechanisms. An agent that neither senses nor hears about a block keeps it at its prior and, at equal prior signal-to-noise ratios, finds it its best sensing option, which draws a poorly informed interior to the shared block, also under exact fusion. In the epistemic limit (no preference to predict messages, no listening offset), listening to a sender of code gain $\vartheta_j$ and posted variance $\sigma_j^2$ beats sensing the shared block with noise variance $\sigma_s^2$ exactly when $1/\vartheta_j^2 + \sigma_j^2 < \sigma_s^2$, for every receiver at once, so two agents with clean channels both stop sensing it; the rule fails under exact fusion. A receiver that holds only its prior, of variance $\sigma_p^2$, and hears with gain $k$, through code gain $\vartheta_j$ and unit channel noise, a sender whose mean is its conditional mean of the world and whose true error variance is $\sigma_e^2$, ends at true error variance $\sigma_p^2 - 2k(\sigma_p^2 - \sigma_e^2) + k^2(\sigma_p^2 - \sigma_e^2 + 1/\vartheta_j^2)$, above $\sigma_p^2$ for an uninformed sender; exact fusion removes this relay poisoning. For single-hop relays of tagged observations the report proves an exact obstruction for every population size, period and precision: if the two sides of the world are independent, then in any partition satisfying clauses (i) and (ii) of Definition S9.2, where the interior hears only from $\mathsf{I} \cup \mathsf{A}$ because the echo of a delivery from $\mathsf{S}$ would be a coupling $\mathsf{I} \to \mathsf{S}$, every interior agent's posterior on the exterior-side blocks equals its prior at every round. The relay chains of Table S8 pass through an active agent in two hops and are not covered by this obstruction.

The framework settles one of the report's open choices and changes its target. Under hypothesis H2, misreading joint structure is the failure that exact fusion removes, so the exact-fusion null is a clause a structure must survive (Definition 5.17), which was the report's working reading. The level-0 bilayers and the nested instance are then artifacts of misreading (Section 3), and the exact-fusion paths, which survive by construction, fail partner randomization and are held by a prior preference for being addressed that the framework would have to declare. More basically, the framework does not ask for a bilayer. Blanket existence selects nothing (Proposition S4.10): under per-channel noise every set of agents is an exact path blanket under its canonical partition, and in layered descriptions with one variable per agent every region with a two-hop shell is one (Proposition S4.3). Sensory privacy forces interiors that sense no exterior-side block, and in the exact-fusion paths, the nested instance and both kinds of relay chain the interior senses nothing at all. Under channel-level roles a member's sensing of the shared world is a sensory channel of the collective (Definition 4.1), and an interior that never senses has no grounded data from which to identify its sources' codes (Proposition 2.7); the codes it learns are regressions on its own beliefs (Proposition S2.4). A collective is a region of high contrast on the world-conditional precision above a declared baseline, with the moment bound in its definition (Definition 4.12). The framework's formation result concerns that object and is narrower than the report's question. In the two-group model of Section 7, at policy precision $\gamma = \infty$ with noiseless angle observations, $\sigma_\alpha^2 = 0$, the chain from calibrated precision through attention to contrast and convergence, clauses (T1) to (T3) and (T5) of Definition 7.1, consists of theorems, the contrast clause at the levels of the window of Theorem 7.8(c), which is nonempty at every computed parameter point (Numerical observation S7.6), and the declared start is then itself the split (Theorem 7.6(e), Theorem 7.8). For $\gamma_S < \gamma < \infty$ reachability of the split branch is a theorem about the frozen-calibration mean-field map (Theorem 7.6(a),(b)); the step from the full stochastic model to that map is open (Open problem 7.14), and the contrast of the reached state is computed at the base point (Numerical observation S7.7). Online calibration from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell, with reruns of a cell differing by up to three replicates, and is open (Numerical observation 7.11 and Open problem S7.13). The collectives produced are a calibration-level effect of misreading joint structure, which, above a threshold in policy precision, coincide with the partition by codes. They provably fail the translation-aware form of the exact-fusion persistence clause: when receivers know their translation errors, no state of the null attends along the split (Proposition 7.12), so clause (T4) fails for that null of type S and the model yields no formation result in the sense of Definition 7.1. For full cross-covariance-aware exact fusion, whose receivers also model the correlations of message errors, this conclusion is heuristic in that model. Persistence under exact fusion with drifting frames (Conjecture S7.12) and formation beyond that model (Conjecture 7.13) are conjectures.

### S9.3 Attention structures

Attention chosen by expected free energy produced most of the structures the report found, and their class decides whether they survive at identity encodings. When every receiver ranks senders in one strict common order, as law headers make them do at $\gamma \to \infty$, capacities equal $\kappa$, all agents are candidates and senders broadcast, the best-response delivery graph is the complete split graph $K_\kappa \vee \overline{K}_{N-\kappa}$: the top $\kappa+1$ agents attend each other and the others attend the top $\kappa$. Its average local clustering tends to one while its global transitivity is about $3(\kappa-1)/N$, and with echoes counted it admits no blanket with nonempty interior and exterior. One receiver facing two complementary senders locks in on one of them for every $\gamma$ exactly when, besides no-invasion conditions, the variance $\sigma_{\mathrm{al}}^2$ of its preferred innovation lies below the logarithmic mean $(S_B - S_A)/\log(S_B/S_A)$ of the two

innovation variances; with many receivers the lock-ins are independent, partner randomization reproduces the law of the delivery graph, and the supports are disjoint stars. Level-0 receivers with learned noise or agreement tracking at capacity two form strict triangles, pairs and chains in which nobody senses the shared block and whose listeners end at 1.10 to 1.26 times their prior error while believing they gain 0.22 to 0.47 nats per round; none survives exact fusion. These are echo chambers in the sense of Definition 3.7, which exact inference does not form (Proposition 3.11). Under exact fusion the listening paths of Table S8 are gradients of knowledge, each agent listening to its most similar better-informed neighbor because the ambiguity term prefers the partner whose error is most correlated with its own.

With Dirichlet habits no invariant law exists, and where attention depends only on covariances condensation can mean only almost-sure convergence to random limits, conditionally on a limit-set theorem (Benaïm, 1999) that the report cites without checking it against the source; with true codes the averaged fitness has no pair term. The reciprocal pairs, directed paths and cycles of the report's replicator analysis, and the closed triangles selected by relays of translation data, are of class F. The next remark shows why for the pairs.

*Remark* S9.3 (Reciprocity needs frames). In the report's mutual replicator system, the mean-field equation of moving-average habits in a scalar, content-free reduction with capacity one, a reciprocal and a nonreciprocal delivered edge differ in fitness only through the dyad's mean-square echo defect, and a larger defect lowers fitness. To first order in the frames, the translation that $k$ has learned for $j$ has error $\phi_{kj} = (1 - g_{kj})\phi^0_{kj} + \phi^{\mathrm{n}}_{kj}$, with learning gain $g_{kj} \in [0, 1]$, prior errors $\phi^0_{kj} = -u$ and $\phi^0_{jk} = u$ set by the difference $u$ of the two frames' generators, and learning noises $\phi^{\mathrm{n}}_{kj}$, independent of each other and of the frames, of variance $hg_{kj}(1 - g_{kj})\sigma_0^2$, where $\sigma_0^2$ is the receivers' prior variance about the partner's code, $h = 1$ for calibrated learners and $h \to \frac{1}{2}$ under matched forgetting. The defect is proportional to $\phi_{kj} + \phi_{jk} = (g_{kj} - g_{jk})u + \phi^{\mathrm{n}}_{kj} + \phi^{\mathrm{n}}_{jk}$, whose mean square is

$$\Xi_{jk} = \mathbb{E}[u^2](g_{kj} - g_{jk})^2 + h\sigma_0^2\big[g_{kj}(1 - g_{kj}) + g_{jk}(1 - g_{jk})\big]. \tag{S42}$$

Under the frame-consistent prior $\mathbb{E}[u^2] = \sigma_0^2$, a reciprocal edge has gains $(g, g)$ and a nonreciprocal one $(g, 0)$, so $\Xi(g, g) - \Xi(g, 0) = -\sigma_0^2 g[g - h(1 - g)]$, and reciprocity is favored exactly when $g/(1 - g) > h$, that is, when learning is fast. At identity encodings $u \equiv 0$ and the first term of (S42) vanishes. If receivers still keep a prior variance $\sigma_0^2 > 0$ about partners' codes, the difference is $h\sigma_0^2 g(1 - g) > 0$ for $0 < g < 1$, so reciprocity is disfavored at every gain and the report's stable reciprocal dyads have no instance; under the frame-consistent prior, whose variance is twice that of the frame generators and hence zero, $\Xi \equiv 0$ and the echo selects nothing. The derivative $\partial\Xi/\partial g_{jk} = h\sigma_0^2(1 - 2g)$ at equal gains is the same with and without frames, so the report's sign rule for the response of a sender's fitness to its own listening does not depend on frames. The algebra was checked symbolically (Check S47.103).

## S9.4 Double counting and amplification

The report's level-1 receivers read a sender's posted predictive mean as an observation of the world whose error, of the posted covariance, is independent of their own. In a dynamic world the common innovation cancels in the difference of two predictions but is counted twice in the claimed innovation covariance, and the overconfidence of double counting adds to the excess; Proposition S3.6(b) states the report's identity. In the report's reciprocal sense-and-listen pairs the claimed own covariance is 0.35 to 0.65 of the actual one and the claimed innovation covariance 3.0 to 13.4 times the actual one. Covariance intersection (Julier and Uhlmann, 1997), which fuses estimates with unknown cross-covariance, leaves the innovation inflated by factors 2.6 to 9.5, because it carries no information

about the cross-covariance of two agents' predictions, while exact fusion never raises a true error but needs the joint error covariance of all agents (Remark S3.16). This is the overvaluation case of misreading joint structure (Section 3).

The report derives the moment bound (M) of Section 9.1 under a small-gain condition, for acyclic cumulative listening, in a decoupled regime and under exact recognition, and shows that it fails at level 1. The failure starts in a single round.

**Lemma S9.4** (One round of mutual listening, after Dennis (2026a))**.** *Let agents* 1 *and* 2*, with decision-time covariances* $P_1, P_2 \succ 0$*, listen only to each other at level 1, with known rank-one codes* $H_1, H_2 \in \mathbb{R}^{1\times K}$*,* $K \geq 2$*, and channel noise variance* $R_c \geq 0$*, and let neither sense. The reception step of the round multiplies the homogeneous part of their disagreement* $e_1 - e_2$ *by*

$$\mathbb{D} = I - \mathcal{K}_{12}H_2 - \mathcal{K}_{21}H_1, \qquad \mathcal{K}_{12} = \frac{P_1H_2^\top}{S_1}, \quad \mathcal{K}_{21} = \frac{P_2H_1^\top}{S_2},$$

*with the claimed innovation variances* $S_1 = H_2(P_1+P_2)H_2^\top + R_c$ *and* $S_2 = H_1(P_1+P_2)H_1^\top + R_c$*, and* $\rho(\mathbb{D}) > 1$ *exactly when* $(H_1P_1H_2^\top)(H_2P_2H_1^\top) > (H_2P_1H_2^\top)(H_1P_2H_1^\top)$*. The two sides are equal when* $H_1$ *and* $H_2$ *are proportional, so under a common code* $\rho(\mathbb{D}) \leq 1$ *and the round has no expanding mode.*

Proof in Section S33; the criterion and its common-code case were also checked numerically, and the criterion is specific to level 1 (Checks S47.99 and S47.100). Expected free energy selects such loops. In the two-agent profile of Example S8.1, with sender-specific codes $(3, 0)$ and $(0, 3)$, $F = 0.9I_2$, $Q = I_2$, oblique sensing rows $(3, 6)$ and $(6, 3)$ and capacity two, sensing while listening to the partner is the strict best response at every admissible state and the frozen mean map has spectral radius 1.0613266751 (Check S47.101); under Dirichlet habits the report derives divergence of second moments for every $\gamma \geq 0$, an argument we have not rechecked. The report describes a second instance, its CEX-1, in which alternating two individually stable profiles at capacity one grows by 1.0703173057 per round; CEX-1 is a separate instance from the capacity-two system above, with $F = 0.99I_2$, $Q = 0.1I_2$, the same two sender-specific codes, sensing rows $(-5, 10)$ and $(10, -5)$, and both agents listening in one round and sensing in the next. The report's summary does not state these parameters, which are taken from its working record, and the rate was recomputed with them (Check S47.101). In the two-agent instance above every period-two alternation of capacity-one profiles contracts (Check S47.101). Such amplifying profiles are rumors, not collectives (Remark S8.8). Both counterexamples use sender-specific codes, and one common code removes the amplification in two cases.

Parts (b) and (c) of Proposition 9.1 prove two cases of the conjecture that at identity encodings, with one modality and one channel noise, no frozen or periodic profile of level-1 receivers in an isotropic world grows faster than the world (Conjecture S8.2), both with a rank-one code: frozen profiles with symmetric listening, and profiles without sensing on any listening graph, frozen or periodic. In the computations that support the conjecture no common-code instance grew faster than $f$, against roughly a third of the instances with sender-specific codes (Check S47.77). With several modalities identity encodings give sender-specific codes again, so amplification can return in multiplex populations. Noisy sensing does not restore (M), since both agents of the two-agent counterexample sense every round. Noise-free observation of the world does, above a rate (Proposition 9.2).

### S9.5 Group agency and the two readings of "be predicted"

Hypothesis H1 asks that a collective act as one agent, and the preference to predict and be predicted, which we take from the duet for one, shapes whom agents address. The report examined both.

Its group-agency results are of class G. Let at least two outer-layer members of a collective decide independently, and let a group linear-Gaussian model give each boundary slot an outcome with noise conditionally independent given a group hidden variable. The interaction of the group expected free energy over two members' actions is the conditional mutual information of their slots plus a pragmatic term, which a slot-product preference removes; with product or uniform habits the group log odds ratio of the two actions is then $-\gamma_C$ times that mutual information, never positive, $\gamma_C$ being the group's policy precision. Independent members produce a product law, so a group model reproduces them only when their boundary inputs are conditionally independent under the group belief and their precisions match. It fails whenever interface members share inputs (mutual information 0.5649 nats and group log odds ratio $-0.9603$ for two members attending one exterior broadcaster in the report's instance), and without constraints as-if agency is vacuous. The framework's positive results concern another protocol. When members share data and announced intentions, have one precision and deliberate one at a time, deliberation is reversible with a Gibbs stationary law exactly when they agree on the redundancy of their inputs, and is then the posterior of a group model with tilted habits when their beliefs are translates of one law (Theorem 5.10); otherwise it stays within explicit distances of such a law (Theorem 5.11). With performative acts the sign obstruction is confined to epistemic slots opened from the null policies, and the group log odds ratio of two additive acts can have either sign (Proposition S5.19), and joint acts form an exact potential game when members share a preference precision and agree on how their acts combine (Theorem S5.17).

The preference to predict and be predicted has two formalizations (Proposition S2.8). The report's canonical echo, in which a receiver returns in its own code what it understood, scores being understood and is belief free; a literal preference to be predicted returns the receiver's innovation, which at identified codes is the coded disagreement $H_j(m_j - m_k)$ plus noise. At identity encodings with exact code models the echo error is pure noise, so being understood reduces to being heard, while being predicted keeps coded disagreement. The literal preference is a genuine prior preference when content is honest and fixed and replies update only partner models. In a scalar world its cost of addressing partner $k$ is the expected squared coded disagreement $\vartheta_j^2[(\sigma_j - \sigma_k)^2 + 2\sigma_j\sigma_k(1 - \mathrm{Corr}(e_j, e_k))]$, with $\sigma_j, \sigma_k$ the error standard deviations, so an informed sender finds partners that share its sources cheapest and uninformed partners most expensive. It adds a homophilic coupling seeded by the world, and under content choice the sender panders (Proposition S2.10). In the report's four-agent instance targeting makes fed relay chains strict for their receivers, but they are stable for their senders only under a pessimistic prior over untried partners and are not reached from the untried state; under a prior that knows the partners the preference stabilizes an informationless pair at 1.177 and 1.179 times the prior error. At identity encodings the choice is therefore between an attention-seeking sender and an agreement-seeking, homophilic one, since the misunderstanding that the echo measures exists only while code models are wrong. The framework leaves this choice open, together with whether a prior preference for being addressed, which holds the exact-fusion paths in place, belongs to the agents' preferences.

### S9.6 What grounding, records, performative acts and multiplexity change

The earlier model had grounding only as rounds that identify translations through a public referent unrelated to $\eta$, no persistent records (Heylighen, 2016), honest fixed content with no effect on the world, hence no performative acts in the sense of Austin (1962), and one modality in every computed instance, hence no multiplex structure (Kivelä et al., 2014). Tables S9 and S10 list the conclusions these omissions may change and where they now stand: three are answered within a stated scope, eleven in part, and three remain open.

Table S9: Results of Dennis (2026a) reopened by the framework's additions, and where they now stand: grounding and records. Performative acts and multiplexity follow in Table S10.

| Earlier result | Why it may change | Status in this manuscript |
|---|---|---|
| *Grounding* | | |
| Fed structures with non-sensing interiors | an interior that never senses cannot ground its sources' codes; the structures used known codes | in part: interior sensing is a sensory channel (Definition 4.1); grounding identifies codes (Proposition 2.7); recomputation open (Open problem S9.5) |
| Reciprocal pairs, cycles and triangles of the replicator analysis | their selecting defect comes from frame-induced translation errors | in part: never favored at identity encodings (Remark S9.3); the window under grounding with frames is open |
| Learned codes as regressions; the alignment preference read as synchrony | grounding changes the learning target | in part: grounded rounds remove the regression bias (Proposition S2.4); the readings at identity encodings (Proposition S2.8) |
| Teaching value and bridges | grounding gives a route to codes that bypasses teaching | open |
| Moment bound; mass from sensing gain | noisy sensing does not restore the bound | in part: noise-free grounding does when $\varpi_g < 1$ (Proposition 9.2); Proposition 8.3 and Remark S8.8; clustering under exact fusion and noisy joint-attention grounding open |
| *Records* | | |
| One-round lag, single-hop relay, direct-source dominance | a record can hold slow or accumulated quantities | in part: latency, delay and optimal persistence (Proposition S2.5); memory criterion (Proposition 8.12); listen, read or sense comparisons open |
| Blanket definitions, privacy, product form | a record written inside and read outside is a lagged coupling through the world | answered for path blankets: records are internal, boundary or external by their writers and readers (Definition 4.1 and Proposition 4.4); writer-activated leaks of agent-level descriptions (Proposition S5.23) |
| Double counting and tagging | records are common sources and can return one's own report | in part: the exact-fusion null carries provenance for records (Definition 5.17); circular reporting through records treated within stated scopes (Propositions S12.7 and S12.8); the provenance a record must carry when members keep their own past posteriors is open |
| Long-run objects; correlation time | records add slow persistent variables | in part: correlation time and length with records (Propositions 8.11 and 8.13); habit dynamics with records open |

Table S10: Results of Dennis (2026a) reopened by the framework's additions, continued from Table S9: performative acts and multiplexity.

| Earlier result | Why it may change | Status in this manuscript |
|---|---|---|
| *Performative acts* | | |
| Pandering; honest fixed content | acts move resources, so stakes diverge and content becomes strategic (Crawford and Sobel, 1982) | in part: performative channels (Proposition S2.6); pandering (Proposition S2.10); strategic content open |
| Group agency and nesting | a collective's active states can change $\eta$ | answered for agency (Definition S5.16, Theorem S5.17, Example S5.18, and Proposition S5.19); nesting with acts open |
| Nulls | deleting an act changes the world's dynamics | answered: informational and effect nulls and an order parameter conditioned on the shared exogenous drive (Definitions 5.15 and 5.17, Proposition 5.16, and Example S5.24) |
| *Multiplexity* | | |
| Moment bound with one code | several modalities give sender-specific codes again | in part: Proposition 9.1 and Conjecture S8.2; multiplex populations open |
| Blankets and nesting | overlapping collectives break the strict hierarchy that nesting needs | in part: modality-restricted and split partitions (Proposition S4.26); decoupled components and crossover distance (Propositions 10.2 and 10.3); multiplex nesting open |
| Listen versus sense; binding regime | capacities become modality specific | open |
| Learned dissonance and identification | edges are read through different modalities | in part: true translations are flat (Proposition S6.5); identification modulo stabilizers (Proposition S2.13); counts per modality pattern open |
| Complete split graph | rankings become modality specific | open |

Circular reporting through records (Table S9) is treated in Section 12 within stated scopes. In a static world a record rebuilt every round from its readers' posteriors has an overconfidence factor that grows without bound under every aggregation rule of Proposition S12.7 other than the exact one, and in a changing world a record that is initialized at the common prior and that memoryless members update by the prior-corrected union rule is the centralized Kalman filter started from that prior (Proposition S12.8); the provenance a record must carry when members also keep their own past posteriors remains open.

The calculation the report expected to change its verdict becomes, in the framework, a question about collectives rather than bilayers.

**Open problem S9.5** (Exact fusion in the minimal model)**.** With exact fusion, codes learned by grounding, identity encodings and interiors free to sense, does attention selected by expected free energy produce a collective in the sense of Definition 4.12 that differs from partner randomization and is reached from the common prior at finite $\gamma$ under Dirichlet habits? The report's proposed search for strict message-fed bilayers under exact fusion with six to eight agents in a two-niche world at capacity two is its agent-level analog, and its counterpart in the two-group model is Conjecture S7.12.

# S10 Additional results for Section 10: abstract collectives

This section collects statements, examples and remarks of Section 10 that the main text cites without stating in full. Proofs are in Section S34.

## S10.1 Multiplex populations: example, remark and conjecture

**Example S10.1** (Coupled contents share one light cone)**.** Let the world have two scalar contents with transition $F = \left(\begin{smallmatrix} F_{11} & f_\times \\ f_\times & F_{22} \end{smallmatrix}\right)$, and on a fixed schedule let agent 2 read agent 1 in modality 1 only and agent 3 read agent 2 in modality 2 only. Agents 1 and 3 are not joined in modality 1's graph, and the sensitivity $\partial e_3^{(1)}(s)/\partial e_1^{(1)}(t)$ vanishes at $s = t+1$, and at every $s$ when $f_\times = 0$; but for $s \geq t+2$ a path can change content inside agent 2 and again inside agent 3, each change carrying a factor of order $f_\times$, the order of the cross-content entries of the round map. In the instance of Check S47.105, with level-1 receivers, the sensitivity at $s = t+2$ is $0.4123 f_\times^2$ to leading order, a coefficient known only numerically. The light cone of Theorem 8.5 therefore holds only on the union graph $\bigcup_c \mathcal{G}_c$.

*Remark* S10.2 (Interpretation: multiplexity is a statement about intermediate scales). The separate correlation lengths of Assumption A9 are exact for dynamically decoupled contents (Proposition 10.2) and scale dependent when agents couple them (Proposition 10.3): since $d_\times$ grows only like $2\ln(1/|a_\times|)/(m_1^0 - m_2^0)$, separate lengths describe intermediate scales, and beyond $d_\times$ the shorter-range modality inherits the longer length. With $(a_1, \varepsilon_1) = (0.3, 0.1)$ and $(a_2, \varepsilon_2) = (0.3, 0.01)$ the crossover lies at 20.2 hops for $a_\times = 10^{-3}$ and at 11.7 hops for $a_\times = 5 \times 10^{-3}$.

**Conjecture S10.3** (Multiplex asymptotics)**.** *For finitely many modalities with finite-range translation-invariant blocks on $\mathbb{Z}^D$ and agent-local coupling, the long-distance decay of each block of the stationary covariance is governed by the singularity of the symbol of $(I - W^2)^{-1}$ nearest the real domain whose residue has a nonzero component in that block. Generically, every modality coupled to the longest-range one, directly or through others, inherits its decay rate beyond a crossover distance of order $\ln(1/|a_\times|)$ divided by the gap in rates.*

## S10.2 Homogeneous media, interfaces and bypass

**Proposition S10.4** (Homogeneous media have no borders)**.** *Fix a modality whose attention graph $\mathcal{G}_c$ is connected. If the weights of Definition 4.17, computed from world-conditional correlations, take a common value $\varpi^\circ$ on every edge of $\mathcal{G}_c$ and satisfy $\varpi_{ij} \geq \varpi^\circ$ for every pair, then every set $C$ with $2 \leq |C| < N$ has $\mathcal{U}(C) \leq 0$. The only sets of positive ultrametric contrast are singletons and the whole population.*

Proof in Section S34. The hypotheses hold on a homogeneous ring with positive self weight when nearest neighbors are the most correlated pairs (a ring of 120 agents has one merge height, Check S47.106), and fail at zero self weight, where correlations vanish at odd distances. This is the correlation counterpart of Theorem 4.13, and the two criteria are not equivalent (Remark S4.23). By Proposition S10.4, sets of positive ultrametric contrast and intermediate size need its hypotheses to fail, as heterogeneity can make them: barriers, nonlinear conventions (Section S10.4) or nested coupling (Proposition 10.7). Homogeneity alone does not secure the hypotheses, as the ring at zero self weight shows, and there the proposition is silent.

*Remark* S10.5 (Straight interfaces in two dimensions). For a straight barrier on the square lattice, in the notation of Proposition 10.4, Fourier transformation along the interface reduces each transverse mode $p$ exactly to the chain with leak $\varepsilon + 2a(1 - \cos p)$, so mode $p$ is transmitted with $\mathcal{T}$ evaluated at $\cosh m(p) = 1 + [\varepsilon + 2a(1 - \cos p)]/(2a)$. That the modes near $p = 0$, with the chain's $\xi$, govern long distances is the usual small-$p$ argument, which we have not checked numerically.

**Proposition S10.6** (Bypass through a coupled modality)**.** *On a finite agent set $V$ let $A_c = \varepsilon_c I + L_c$ for $c = 1, 2$, with $\varepsilon_c > 0$ and $L_c$ weighted graph Laplacians, let*

$$I - W = \begin{pmatrix} A_1 & -a_\times I \\ -a_\times I & A_2 \end{pmatrix}, \qquad a_\times^2 < \varepsilon_1 \varepsilon_2,$$

*and $\mathsf{G} = (I - W)^{-1}$ with blocks $\mathsf{G}_{cc'}$. Suppose that $L_1$ has no edge between a set $C \subset V$ and $V \setminus C$, a boundary in modality 1. Then for $x \in C$ and $y \notin C$,*

$$\mathsf{G}_{11}(x, y) = \sum_{n \geq 1} a_\times^{2n} [(A_1^{-1} A_2^{-1})^n A_1^{-1}](x, y) \geq a_\times^2 [A_1^{-1} A_2^{-1} A_1^{-1}](x, y) \geq 0.$$

*If $L_2$ also has no edge between $C$ and $V \setminus C$, then $\mathsf{G}_{11}(x, y) = 0$ for all such pairs. If instead $a_\times \neq 0$ and $L_2$ is connected, the lower bound is positive.*

Proof in Section S34. For symmetric $W$ with noise $qI$ the stationary covariance is $\frac{q}{2}[(I - W)^{-1} + (I + W)^{-1}]$, and $\mathsf{G}$ carries its long-range part when the second term decays faster, as in Proposition 10.4(c). A boundary present in only one of two coupled modalities is therefore sharp at short range, where the cross-cut value of $\mathsf{G}_{11}$ is of order $a_\times^2$, and invisible beyond the crossover distance of Proposition 10.3. On an open chain of 300 agents with the parameters of Remark S10.2 and a cut in modality 1 only, the ratio of same-side to cross-cut values is $1.9 \times 10^3$, 208 and 18.5 at 2 hops for $a_\times = 10^{-3}$, $3 \times 10^{-3}$ and $10^{-2}$, and 1.0 at 40 hops (Check S47.107).

*Remark* S10.7 (Interpretation: border effects). McCallum found, as secondary reports summarize (McCallum, 1995), that in 1988 Canadian provinces traded about twenty times more with one another than with US states of similar size and distance; Anderson and van Wincoop (2003) find with a theoretically founded gravity equation that national borders reduce trade between industrialized countries by 20 to 50 percent; and Engel and Rogers (1996) find that crossing the US–Canadian border adds as much relative-price volatility as 2500 miles of distance. Reading price dispersion

as the decorrelation of price messages, $1/\mathcal{T}$ and $d_b\ell$ of Proposition 10.4 correspond to the border effect and to Engel and Rogers's width, and the reading claims that a border is strong only when its resistance exceeds about $2\xi$ bonds of medium, that a border in one modality alone leaks at order $a_\times^2$, and that multiplying $\ell$ at fixed $a$, $\varepsilon$ and $n_b$ widens a border in kilometers without changing its factor. Testing these claims needs the structural corrections of Anderson and van Wincoop.

**Conjecture S10.8** (Alignment sharpens borders)**.** *Border effects in trade and in communication flows are larger where language, currency, legal and media boundaries coincide than where a border separates only one of them, as inside a currency union or along a language boundary inside a multilingual polity.*

## S10.3 Records and media

Writing, archives, ledgers and statutes let a collective remember beyond its members. The record model (Definition 2.9) and its results in Section 8 give their effect on the correlation time and length.

*Remark* S10.9 (Interpretation: Innis's biases of media). As secondary sources summarize it, Innis's thesis (Innis, 1950) is that durable media emphasize time and favor decentralized and religious organization, light media emphasize space and favor administration over territory, and empires endure when the two are balanced. A medium is a record (Definition 2.9) with decay $\delta_k$, reading range $\ell_{r,k}$ and write weight $\omega_k$. A single record whose reading displaces self weight lengthens the correlation time exactly when $\delta_k < \varepsilon$ (Proposition 8.12), and, comparing the asymptotic form of Proposition 8.13(b) with the length $(a/\varepsilon)^{1/2}$ without records, it lengthens the second-moment length to leading order as the leaks vanish exactly when $\omega_k \ell_{r,k}^2 \varepsilon > 2a\delta_k$. A durable local medium therefore slows a collective without widening it, a perishable medium read far away shortens its memory and widens it only when rewritten often, and a durable portable medium does both (Check S47.108). By the time and extent bounds of hypothesis H4 (Theorem 8.20), a collective large in space and time needs media that supply both, a reading of Innis's balance thesis rather than a derivation. The length criterion rests on the asymptotic statement Proposition 8.13(b), stated for $a > 0$ and reading that displaces self weight, and it inherits that statement's status; the extent bound is proved only when the world-conditional precision of agents and records jointly is diffusive, the hypothesis of Proposition 8.13(c), which noisy records break in a computed case (Check S47.90). Records also raise level-0 overconfidence: the mean of $n_y$ messages that share a record error of variance $\sigma_{\mathrm{rec}}^2$ and carry independent errors of variance $\sigma^2$ has variance $\sigma_{\mathrm{rec}}^2 + \sigma^2/n_y$, which a receiver that treats them as independent puts at $(\sigma_{\mathrm{rec}}^2 + \sigma^2)/n_y$.

*Remark* S10.10 (Interpretation: money as memory). A transfer is a performative message that also writes a record. A money balance is a record of rank one per agent, the net of its past transfers, read by every counterparty, with $\delta = 0$ unless inflation acts as a decay; it compresses a ledger to one number per agent and, by the logic of Proposition S10.16, loses nothing for decisions that depend on a counterparty's history only through its net balance. This is consistent with, and does not derive, the result of Kocherlakota (1998) that without commitment any allocation feasible with money is feasible with memory.

## S10.4 Cultures

Conventions are the clearest case of an abstract collective: a dialect, a set of norms or a notation is shared by those who use it and by no one else.

**Definition S10.11** (Culture)**.** A culture is a collective (Definition 4.12) of the language and convention modality (Table 10). Members' beliefs about code parameters and about others' policies

are its internal states, and the channels through which members such as bilinguals, translators and traders read or write messages in other codes are its sensory and active states. Following Assumption A10, the culture is the coarse-grained pattern of these beliefs and of the records that store them, not a membership list. Whether the code domains of Section 6 are collectives in this sense is open (Open problem S10.28). No result of this section shows that a culture in a stated model is a collective in the sense of Definition 4.12; the results below concern its borders and the ultrametric diagnostic.

We read the attention-patterning practices through which, according to Veissière et al. (2020), humans learn their culture's shared expectations as the learning of partner and population models from co-occurrence, and the concentration counts with which Constant et al. (2019) model regimes of expectations as records written by actions and erased by decay. A culture's boundary can be sharp because of barriers or because conventions are nonlinear, and expected free energy supplies the nonlinearity: for an agent with $\mathsf{d}_i$ partners, policy precision $\gamma$, no pull and a value affine in the number of matched partners with slope $J$, the adoption rule of the code game is logistic in the local frequency of a code, conformist exactly when $\gamma J \mathsf{d}_i > 2$, and at finite $\gamma$ has an innovation floor, a positive probability of adopting a code that no partner uses (Proposition S6.15). Models of dialect formation assume a conformity rule instead.

**Proposition S10.12** (Conformity from expected free energy against a power rule)**.** *Let* $q(x) = \mathrm{sig}(\gamma J \mathsf{d}(2x-1))$*, with* sig *the logistic function, be the adoption rule of Proposition S6.15(b) for an agent with* $\mathsf{d}$ *partners, a value affine in the number of matched partners with slope* $J > 0$*, and no pull, and let* $q_{\mathrm{p}}(x) = x^{b_{\mathrm{p}}}/(x^{b_{\mathrm{p}}} + (1-x)^{b_{\mathrm{p}}})$ *with* $b_{\mathrm{p}} > 1$ *be the power conformity rule.*

*(a)* $q_{\mathrm{p}}$ *has log-odds* $b_{\mathrm{p}} \ln(x/(1-x))$*; it is conformist for every* $b_{\mathrm{p}} > 1$ *and has no innovation floor,* $q_{\mathrm{p}}(0) = 0$*, whereas* $q(0) = \mathrm{sig}(-\gamma J \mathsf{d}) > 0$*.*

*(b)* $q$ *and* $q_{\mathrm{p}}$ *have the same value and slope at* $x = \frac{1}{2}$ *if and only if* $b_{\mathrm{p}} = \gamma J \mathsf{d}/2$*, and then* $q_{\mathrm{p}} < q$ *on* $(0, \frac{1}{2})$ *and* $q_{\mathrm{p}} > q$ *on* $(\frac{1}{2}, 1)$*: away from* $\frac{1}{2}$ *the power rule reinforces majorities more strongly.*

*(c) Let a finite open chain of agents attend their neighbors, and let one agent, drawn uniformly, revise at each step. The power rule is absorbed in one of the two uniform configurations with probability one, so it has no stationary state with walls. At finite* $\gamma$ *the rule* $q$*, with* $\mathsf{d} = \mathsf{d}_i$ *and the same slope* $J$ *for each agent, is logit revision in the pairwise code game, an exact potential game on the open chain (Proposition 6.10); its chain is irreducible, and its stationary law has independent walls with probability* $1/(1+e^{\gamma J})$ *(Proposition S10.13(c)).*

Proof in Section S34. The derived rule is nonlinear because the policy posterior is logistic and epistemic value grows with the number of matched partners; no preference for conformity is assumed. Its innovation floor acts as a leak, so at finite $\gamma$ it keeps a stationary pattern of domains, which on rings, and for the pairwise game on open chains, obeys the domain law of Theorem 6.11; in higher dimension the behavior is conjectural (Conjectures S6.16 and S6.17). The resulting geography is summarized by the Séguy curve.

**Proposition S10.13** (Séguy curves in leaky and walled regimes)**.** *Let the Séguy curve* $\mathcal{S}(d)$ *be the expected fraction of binary code features in which two members at distance* $d$ *differ.*

*(a) Leaky regime. If each of* $n_{\mathrm{f}}$ *features is read out as the sign of an independent copy of a stationary centered Gaussian field with correlation* $\Gamma(d)$ *at distance* $d$*, then* $\mathcal{S}(d) = \arccos \Gamma(d)/\pi$*, and the realized fraction converges to it almost surely as* $n_{\mathrm{f}} \to \infty$*. For* $\Gamma(d) = e^{-d/\xi}$*,* $\mathcal{S}(d) = \pi^{-1}(2d/\xi)^{1/2}(1 + O(d/\xi))$ *as* $d/\xi \to 0$*; for* $\Gamma(d) = e^{-d^2/(2\sigma_{\mathrm{g}}^2)}$*,* $\mathcal{S}(d) = (d/\pi\sigma_{\mathrm{g}})(1 + O(d^2/\sigma_{\mathrm{g}}^2))$*; and in every case* $\frac{1}{2} - \mathcal{S}(d) = \pi^{-1} \arcsin \Gamma(d) = \pi^{-1}\Gamma(d)(1 + O(\Gamma(d)^2))$*.*

Table S11: Séguy curves in one dimension. The walled regimes rise linearly. The approach to $\frac{1}{2}$ is exponential for the code game without pulls and for the leaky regime with exponential correlations, which rises as a square root. In sampled stochastically stable states with pulls the approach is faster than for independently placed walls, with a local rate that still grows at the largest sampled distances, so no exponential tail is established there. Coarsening rises linearly and approaches $\frac{1}{2}$ with a Gaussian tail.

| Dynamics | Stationary | Small $d$ | Approach to $\frac{1}{2}$ | Scale |
|---|---|---|---|---|
| Leaky linear field read by thresholds (Proposition S10.13(a)) | yes | $\pi^{-1}(2d/\xi)^{1/2}$ | $\pi^{-1}e^{-d/\xi}$ | $\xi$ |
| Pairwise code game, finite $\gamma$, no pulls (Proposition S10.13(c)) | yes | $d/(1+e^{\gamma J})$ | $\frac{1}{2}\tanh^{d}(\gamma J/2)$ | about $\frac{1}{2}e^{\gamma J}$ |
| Code game with pulls, stochastically stable state (Theorem 6.11) | yes, quenched | $dp_{\mathrm{w}}$ | faster than independent walls, local rate rising (sampled) | between $\frac{1}{2}(J/h)^2$ and $26(J/h)^2+7$ |
| Conformity without leak or pulls (Burridge, 2017) | no | linear | Gaussian tail | grows as $t^{1/2}$ |

*(b) Walled regime. For a stationary sequence of codes $(\sigma_x)_{x\in\mathbb{Z}}$ in $\{\pm 1\}$ with wall probability $p_{\mathrm{w}} = \Pr(\sigma_0 \neq \sigma_1)$ and $n_{\mathrm{w}}(d)$ walls among the $d$ bonds between 0 and $d$,*

$$dp_{\mathrm{w}} - \mathbb{E}\big[n_{\mathrm{w}}(d)(n_{\mathrm{w}}(d)-1)\big] \leq \Pr(\sigma_0 \neq \sigma_d) \leq dp_{\mathrm{w}}.$$

*Over a range of $d$ on which $\mathbb{E}[n_{\mathrm{w}}(d)(n_{\mathrm{w}}(d)-1)] = o(dp_{\mathrm{w}})$, as when walls are sparse and do not cluster, the curve rises linearly with slope $p_{\mathrm{w}}$.*

*(c) Code game without pulls. For the pairwise code game on an open chain with coupling $J > 0$ (Proposition 6.10(a)), $h_i = 0$ and finite $\gamma$, the stationary law of asynchronous logit revision has independent walls with probability $p_{\mathrm{w}} = 1/(1+e^{\gamma J})$, and $\mathcal{S}(d) = \frac{1}{2}(1-\tanh^{d}(\gamma J/2))$.*

*(d) Leaky dynamics make no borders. If the correlations $\Gamma_{ij} \geq 0$ of the field satisfy the hypotheses of Proposition S10.4, so do the correlations $\frac{2}{\pi}\arcsin\Gamma_{ij}$ of the sign readouts, which therefore have no set of positive ultrametric contrast other than singletons and the whole population.*

Proof in Section S34; checked by sampling and by exact minimizers of the pairwise potential (Check S47.109). Part (c) is the open-chain counterpart of Theorem 6.11(c), which treats the ring. Table S11 collects the regimes that can be told apart in one dimension, and Fig. S2 draws them at $\xi = \sigma_{\mathrm{g}} = 20$: on logarithmic axes only the leaky field rises with slope one half, and the approach to $\frac{1}{2}$ separates the code game without pulls, whose tail is exponential, from coarsening, whose tail is Gaussian.

*Remark* S10.14 (Interpretation: dialect geography). Quantitative studies since Séguy (1971) find aggregate linguistic variation growing sublinearly with distance, as Nerbonne (2010) reports in the abstract we read. In Burridge (2017), speakers with a memory of exposure and a conformity nonlinearity form boundaries that behave like interfaces with surface tension, and in periods of demographic stability and uniform density the expected linguistic distance has the arcsine form with a Gaussian correlator whose scale grows as the square root of time. Sublinear growth alone does not discriminate between the regimes of Table S11; the short-distance exponent and the approach

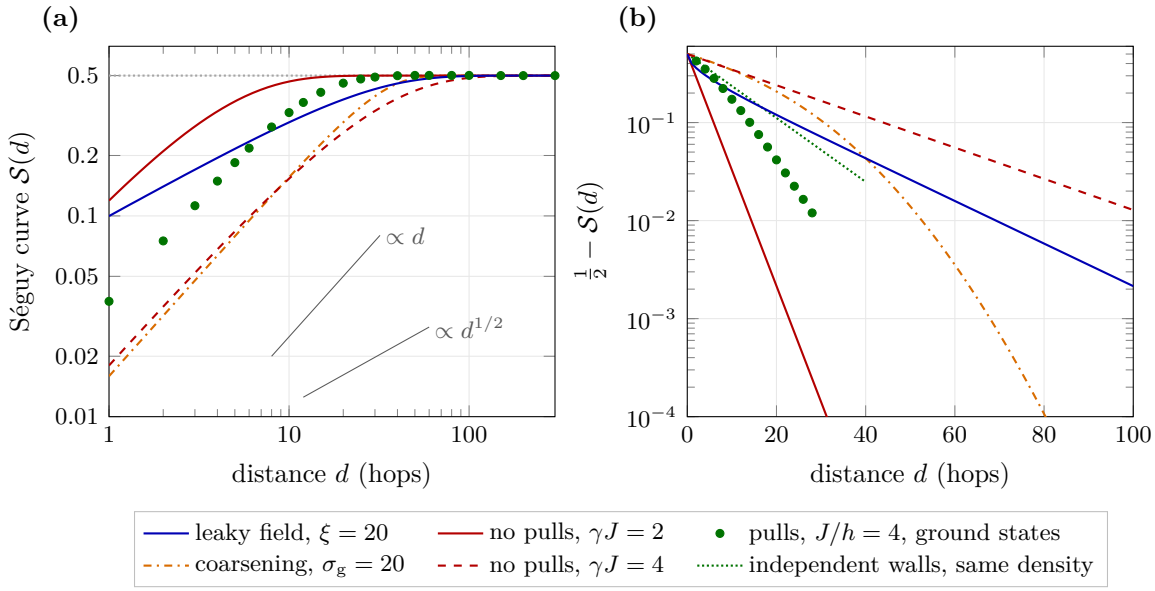



Figure S2: **Séguy curves in one dimension.** (a) The probability $\mathcal{S}(d)$ that two members at distance $d$ differ in a binary code feature, on logarithmic axes, for the regimes of Table S11 with $\xi = 20$ and Gaussian width $\sigma_g = \xi$. The leaky field read by thresholds, with $\Gamma(d) = e^{-d/\xi}$, rises as $d^{1/2}$; the Gaussian correlator $\Gamma(d) = e^{-d^2/(2\sigma_g^2)}$ of coarsening, the code game without pulls at $\gamma J = 2$ and 4 (Proposition S10.13(c)) and the stochastically stable state with Gaussian pulls at $J/h = 4$ rise linearly. (b) The approach to $\frac{1}{2}$. It is exponential, a straight line on these axes, for the code game without pulls and, at large $d$, for the leaky field; coarsening has a Gaussian tail; and the stochastically stable state approaches $\frac{1}{2}$ faster than independently placed walls with the same density (dotted), with a local rate that still grows at the largest sampled distances, so its tail is not established as exponential. A linear rise alone does not identify walls, since coarsening also rises linearly; the shape of the approach to $\frac{1}{2}$ separates them. The stochastically stable curve comes from exact ground states of 64 features on 40000 sites (Check S47.109); in (b) its points stop at $d = 28$, where their standard error reaches eight percent.

to $\frac{1}{2}$ do. Stationary walls need pulls, finite policy precision or barriers; homogeneous leaky linear dynamics make none (Proposition S10.13(d)), and coarsening is the limit $\gamma \to \infty$ of the derived rule without pulls.

Codes also have to compose. A language border is a place where translations between neighboring codes stop composing consistently, and the abelian case is exact.

**Proposition S10.15** (Consistency of translations and exchange rates)**.** *Let a connected graph carry an antisymmetric edge function $\alpha_{ij} = -\alpha_{ji}$: the logarithms of scalar translation factors $\Theta_{ij} > 0$ between members' codes, or log exchange rates between currencies.*

*(a) The following are equivalent: the sum of $\alpha$ around every cycle vanishes; $\alpha_{ij} = \bar{\alpha}_j - \bar{\alpha}_i$ for a potential $\bar{\alpha}$, unique up to an additive constant; the least-squares fit of a potential has zero residual.*

*(b) If $\alpha = B\bar{\alpha} + \phi$, with $B$ the edge-node incidence matrix and independent edge errors $\phi_{ij}$ of mean zero and variances $\sigma_{ij}^2$, then the holonomy* $\mathrm{Hol}(Z)$ *of an oriented cycle $Z$, the signed*

*sum of $\alpha$ around it, equals the signed sum of the $\phi_{ij}$ and the holonomy of the least-squares residual, and* $\mathbb{E}[\mathrm{Hol}(Z)^2] = \sum_{(i,j)\in Z} \sigma_{ij}^2$.

(c) *If receiver $i$ learns $\Theta_{ij}$ by least squares from $n_{ij}$ grounded co-occurrences of messages $\Theta_{ij}^* \eta_t + \nu_t$ with world values $\eta_t$, the $\nu_t$ independent $\mathcal{N}(0, \sigma_y^2)$ and independent of the $\eta_t$, then conditionally on the $\eta_t$ the estimate is Gaussian with mean $\Theta_{ij}^*$ and variance $\sigma_y^2 / \sum_t \eta_t^2$. If $n_{ij}^{-1} \sum_t \eta_t^2 \to \sigma_\eta^2 > 0$ almost surely, as it does when the $\eta_t$ are independent and identically distributed with $\mathbb{E}\eta_t^2 = \sigma_\eta^2$ or form a stationary ergodic sequence with that second moment, then $n_{ij}^{1/2}(\ln \Theta_{ij} - \ln \Theta_{ij}^*)$ converges in law to $\mathcal{N}(0, \sigma_y^2/((\Theta_{ij}^*)^2 \sigma_\eta^2))$ as $n_{ij} \to \infty$.*

Proof in Section S34; checked on a two-zone instance (Check S47.110). Part (c) is the scalar case of grounded learning (Proposition S2.3): where contact is rare, translations are poorly estimated, and cycles through such edges carry large holonomy. A shared code exists only where holonomy vanishes (Theorem 6.5), and true translations are flat (Proposition S6.5), so inconsistency is an error of learning. We read language borders as contact boundaries made visible as consistency boundaries, and bilinguals and translators as boundary agents whose precise translations lower the variance of boundary edges; bilinguals sit at the interfaces of monolingual domains in a model of language competition (Castelló et al., 2006), and translation inside firms is modeled in a related way by Crémer et al. (2007).

### S10.5 Economies: the Hayek dimension, informed trading and exchange rates

**Proposition S10.16** (The Hayek dimension)**.** *Let $\eta \sim \mathcal{N}(0, P_0)$ with $P_0 \succ 0$, let agent $i$ sense $s_i = H_i^s \eta + v_i$ with $v_i \sim \mathcal{N}(0, R_i^s)$, $R_i^s \succ 0$, independent of $\eta$, and let a public message $M\eta$ be available, with $M$ of rank $r_M$. If agent $i$'s decision depends on $\eta$ through $\psi_i^\top \eta$ with quadratic loss, then*

$$\mathrm{Var}(\psi_i^\top \eta \mid s_i, M\eta) = 0 \iff \psi_i \in \mathrm{row}(M).$$

*Hence the public message suffices for every agent's decision if and only if* $\mathrm{span}\{\psi_i\} \subseteq \mathrm{row}(M)$*, and the least number of public scalars that suffices is the Hayek dimension* $r_{\mathrm{Hay}} = \dim \mathrm{span}\{\psi_i\}$.

Proof in Section S34; checked in four dimensions (Check S47.112). Hayek's scarce material (Section 10.3) is the case $r_{\mathrm{Hay}} = 1$. In our reading, prices economize on communication to the extent that decision-relevant functionals are shared, and goods valued by buyer-specific attributes, whose functionals span more dimensions than there are prices, need bilateral communication of high rank; this agrees in sign with, and does not derive, the finding of Rauch (1999) that proximity, common language and colonial ties matter more for differentiated products than for goods traded on organized exchanges. The mechanism-design counterparts are Jordan (1982) and Mount and Reiter (1974), cited from their bibliographic records. For decisions about the state at a horizon in a dynamic world, the exact count is the dimension of the decision functionals carried back to the present, which does not shrink with the horizon when the dynamics is invertible, while in reversible worlds the effective count at a fixed tolerance is at most a number that falls to the number of slowest modes as the horizon grows (Theorem 14.7). When the units of each level of a hierarchy decide through aggregates of the decisions of the level below, the count is nonincreasing up the hierarchy, but a top-down message that must serve the members' own decisions needs the rank of those decisions (Proposition S13.16).

**Conjecture S10.17** (Informed trading as grounding)**.** *Informed trading is the grounding of the exchange modality, the part of its leak that makes the price informative about values. A market of pure price readers has the largest reach and carries no information about values, which is the*

*exchange form of the grounding tension (Theorem 6.12, Assumption A11), and by Proposition 10.5(c) a market in which many traders read the price at level 0 overreacts to it.*

The conjecture is our reading of the argument of Grossman and Stiglitz (1980), known to us from secondary summaries, that with costly information prices cannot be fully revealing in equilibrium, since no trader would then pay for information.

Exchange rates are translations between the codes in which values are expressed. With this identification, the only interpretive step here, the absence of cyclic arbitrage is the cocycle condition of Proposition S10.15(a), which holds exactly when all rates derive from currency values relative to a numeraire, and arbitrageurs, whose trades reduce holonomy, are boundary agents that enforce consistency. The split of noisy pairwise data into gradient and cycle parts is the combinatorial Hodge decomposition used for ranking (Jiang et al., 2011).

## S10.6 Nations: readings

Deutsch (1953), whose fourth chapter we read in the second edition, calls a people a larger group linked by complementary habits and facilities of communication, tests complementarity by communicative effectiveness and proposes measuring a channel's fidelity by the information lost in transmission. We read a people as a set with higher fidelity and code rank among members than across, a contrast statement in the language modality. The transaction-flow index of Savage and Deutsch (1960) and modularity (Newman, 2006) measure contrast in message rates rather than in coherence, and Propositions 10.4 and S10.6 translate the first into the second.

*Remark* S10.18 (Interpretation: readings of the polity criterion). Better communication lengthens $\xi$ and raises $\chi$; on the line it enlarges optimal polities when the fixed cost is small against $g\xi$ ($f/(g\xi)$ below about 1.07 under rule C and 0.37 under rule M), with $d^* \propto \xi^{1/2}$ in the small-cost limit, while nearer the thresholds of Proposition 10.6 a longer $\xi$ shortens the finite optimum. Above the threshold of the declared rule every finite polity costs more than the limit of unbounded ones, whose policy is unrelated to most members' preferences, a stylized regime of loosely governed polities about whose historical instances the model says nothing. The finding of Alesina and Spolaore that the equilibrium number of countries increases with economic integration has a multiplex reading: once the market extends beyond the polity, market size no longer needs a large polity. Conquest, conflict and contingency, which shape actual borders, are outside the model.

Statutes, registries and case law are durable media, and administration over territory needs messages of long range; in the reading of Remark S10.9 a polity extended in space and time needs both. Anderson's account of print capitalism, known to us from secondary summaries (Anderson, 1983), holds that print created mass reading publics in fewer languages than were spoken and made the nation imaginable. We read print as a durable medium of long reading range whose readership is bounded by a language and a book market, so that it lengthens $\xi$ and $\tau$ inside the readership and aligns boundaries across modalities, as a border sharp at long range needs (Proposition S10.6).

## S10.7 Cities

**Definition S10.19** (Co-presence and one-round sets)**.** The content of the co-presence modality is the location and activity of people; its messages are encounters, and its round is a day. A one-round set is a set of agents every pair of which can meet within one round, each member leaving from its home and returning to it within the round.

**Proposition S10.20** (One-round diameter)**.** *Let members' homes $x_i$ lie in $\mathbb{R}^D$, and let every member travel along straight lines at speed at most $v_{\mathrm{tr}}$ for at most $T_{\mathrm{tr}}$ per round. Then any two*

*members of a one-round set are within $v_{\mathrm{tr}}T_{\mathrm{tr}}$ of each other, so its diameter is at most $v_{\mathrm{tr}}T_{\mathrm{tr}}$, linear in the speed at a fixed budget. Equivalently, a member's daily reach, the set of points it can visit and return from within one round, is the ball of radius $v_{\mathrm{tr}}T_{\mathrm{tr}}/2$ about its home; two members can meet within a round if and only if their reaches intersect; and a one-round set is a set of pairwise intersecting reaches.*

Proof in Section S34. The daily reach is the one-round light cone of the co-presence modality, and a one-round set is defined without contrast, so it need not be a collective in the sense of Definition 4.12. Collectives that form over many rounds are bounded instead by the size bound of Corollary 4.16; the two quantities are not comparable (Table 9), but they compose.

**Corollary S10.21** (One-round reach and the extent of a collective)**.** *Let the world-conditional precision $\Pi^c$ of a modality be diffusive with uniform leak $\varepsilon^{\Pi}$, let $a^{\Pi}_{\max} = \max_i \sum_j a^{\Pi}_{ij}$, and let every coupling $a^{\Pi}_{ij} > 0$ join agents whose positions are within physical distance $\ell_{\Pi}$. Then the largest extent of Definition 4.15 satisfies, in every direction $d$ and at every admissible level $r_0$,*

$$L^*_d(r_0) \le \ell_{\Pi}\Big(\frac{a^{\Pi}_{\max}}{2\varepsilon^{\Pi}(r_0-1)}\Big)^{1/2}.$$

*In the reversible and normal classes of Theorem 4.9, couplings join agents with a common listener or a common source, so $\ell_{\Pi} \le 2\ell_{\mathrm{att}}$, with $\ell_{\mathrm{att}}$ the longest physical length of an attention link. In the co-presence modality an attention link is a meeting within one round, so $\ell_{\mathrm{att}} \le v_{\mathrm{tr}}T_{\mathrm{tr}}$ by Proposition S10.20, and $L^*_d(r_0) \le 2v_{\mathrm{tr}}T_{\mathrm{tr}}\big(a^{\Pi}_{\max}/(2\varepsilon^{\Pi}(r_0-1))\big)^{1/2}$.*

Proof in Section S34. The one-round reach bounds the physical length of a coupling, and contrast bounds the extent in units of that length.

*Remark* S10.22 (Interpretation: Marchetti's constant). Marchetti (1994) reports a mean daily travel time of about one hour across populations, pedestrian cities about 5 km across, and a sixfold to sevenfold linear expansion of cities with cars, which are six to seven times faster than walking; reviewing more than two dozen studies, Mokhtarian and Chen (2004) conclude that travel time is not constant except perhaps in aggregate. With $T_{\mathrm{tr}} \approx 1$ hour and 5 km per hour the bound of Proposition S10.20 is about 5 km, and the expansion with cars is its linear dependence on $v_{\mathrm{tr}}$; the bound then describes the population average, not each member.

Bettencourt et al. (2007) report urban quantities scaling as powers of population, with exponents near 1.2 for wealth creation and innovation and near 0.8 for infrastructure. Bettencourt's model, as Ribeiro and Rybski (2023) present it (Bettencourt, 2013), requires per-capita output to pay the cost of crossing the city, a condition that gives output proportional to $N^{7/6}$ and that we read as a one-round reach condition with a money budget in place of a time budget. Superlinear output also needs partners that carry different information: if $n_{\mathrm{p}}$ partners' estimates have errors of common variance $\sigma^2$ and common correlation $\bar{\Gamma} \ge 0$, their minimum-variance unbiased linear combination is the average, because $\mathbf{1}$ is an eigenvector of the error covariance, and its variance $\sigma^2[1+(n_{\mathrm{p}}-1)\bar{\Gamma}]/n_{\mathrm{p}}$ never falls below $\sigma^2\bar{\Gamma}$, the variance of the average of $1/\bar{\Gamma}$ independent estimates.

**Conjecture S10.23** (Redundancy limits urban superlinearity)**.** *Superlinear urban exponents are larger for outputs that depend on diverse, non-redundant information, and larger in urban systems whose diversity grows with size; outputs that depend on redundant information scale closer to linearly.*

Telecommunication adds edges whose cost does not depend on distance (Assumption A8), yet geography remains: the probability that two users of a Belgian mobile network are linked decays as

the inverse square of their distance (Lambiotte et al., 2008), while about one third of friendships in a large online network are independent of geography (Liben-Nowell et al., 2005). In our reading, collectives of modalities carried by such networks need not be territorial, while enforcement, transport and co-presence stay bounded by physical ranges.

### S10.8 Distance: elasticity, cutoffs and the distance effect

**Proposition S10.24** (Distance elasticity of coherence)**.** *Let coherence between locations at distance $d > 0$ be $\Psi(d/\xi)$ for a positive differentiable profile $\Psi$, and let its distance elasticity be* $\mathrm{el}(d) = -\mathrm{d}\ln\Psi(d/\xi)/\mathrm{d}\ln d = \mathrm{el}_\Psi(d/\xi)$*, with* $\mathrm{el}_\Psi(x) = -x\Psi'(x)/\Psi(x)$*.*

*(a) The elasticity at each fixed distance is the same for every $\xi > 0$ if and only if $\Psi$ is a power law, $\Psi(x) = \Psi(1)x^{-\alpha_{\mathrm{el}}}$, and then $\mathrm{el} \equiv \alpha_{\mathrm{el}}$.*

*(b) For the profile $\Psi = K_0$ of diffusion with leak in two dimensions, $\mathrm{el}_{K_0}(x) = xK_1(x)/K_0(x)$ is strictly increasing on $(0,\infty)$, with $\mathrm{el}_{K_0}(x) \to 0$ as $x \to 0$ and $\mathrm{el}_{K_0}(x) - x \to \frac{1}{2}$ as $x \to \infty$. Increasing $\xi$ therefore lowers the elasticity at every fixed distance.*

*(c) For an integral of exponential profiles against the weight $\xi^{-1-\alpha_{\mathrm{el}}}\mathrm{d}\xi$ on $(0,\infty)$, $\alpha_{\mathrm{el}} > 0$ (a $\sigma$-finite weight, not a probability density), the coherence at $d > 0$ is $\int_0^\infty e^{-d/\xi}\xi^{-1-\alpha_{\mathrm{el}}}\mathrm{d}\xi = d^{-\alpha_{\mathrm{el}}}\int_0^\infty \varrho^{\alpha_{\mathrm{el}}-1}e^{-\varrho}\mathrm{d}\varrho$, an exact power law with constant elasticity $\alpha_{\mathrm{el}}$.*

Proof in Section S34; checked numerically (Check S47.114). At a distance of 1000 km the elasticity of the profile $K_0$ is 4.47, 2.46, 1.43 and 0.90 for $\xi = 250$, 500, 1000 and 2000 km. Scale-free coherence arises from nested coupling, such as administrative and economic hierarchies may provide.

*Remark* S10.25 (Cutoffs and log-periodic modulation). In the notation of Proposition 10.7, with $\mathsf{c}_\infty = 1/(1-\mathfrak{q}) + 1/(b-1)$, the eigenvalues of an infinite hierarchy are $\mathsf{c}_\infty A_{\mathrm{h}}\mathfrak{q}^{l-1}$, and they fall below the precision leak $\varepsilon^\Pi$ at the level $l^* = 1 + \ln(\mathsf{c}_\infty A_{\mathrm{h}}/\varepsilon^\Pi)/\ln(1/\mathfrak{q})$. At finite leak the geometric law holds up to $l^*$ and, in a finite population, up to the levels where the constant mode $b^{-n_{\mathrm{H}}}/\varepsilon^\Pi$ takes over; multiplying all couplings by a factor leaves $\alpha_{\mathrm{H}}$ unchanged and moves $l^*$ outward by the logarithm of the factor divided by $\ln(1/\mathfrak{q})$. With $b = 2$, $n_{\mathrm{H}} = 40$, $\mathfrak{q} = 0.7$ and $\varepsilon^\Pi = 10^{-7}$ the per-level ratio is 0.714 at levels 1 to 5, as $1/(b\mathfrak{q})$ predicts. A small hierarchy ($b = 4$, five levels, $\mathfrak{q} = 0.5$, $\varepsilon^\Pi = 0.01$, on a line) does not reach the ratio 0.5: its per-level ratios are 0.58 to 0.65, and the local elasticity oscillates between 0.15 and 0.57 with the period of the hierarchy, a log-periodic modulation of the power law (Check S47.115).

**Conjecture S10.26** (What the framework says about the distance effect)**.** *(i) If the binding friction of trade were informational coherence with one scale that communication lengthened, distance elasticities at fixed distances would have fallen during the twentieth century (Proposition S10.24(b)) and would rise with distance in a cross-section; the persistence and mid-century rise found by Disdier and Head, and the constant long-distance elasticity Chaney takes as given, are inconsistent with this as the dominant mechanism. (ii) The persistence fits two readings: the modality that binds trade at long distances (transport, enforcement, trust) is not the one whose range grew, or coherence in the binding modality is scale free, nested (Proposition 10.7) or a superposition of scales (Proposition S10.24(c)), so that uniform improvements in range or coupling only move cutoffs. (iii) The nested reading predicts log-periodic modulation of elasticities at the scales of administrative and economic hierarchies, and border widths in kilometers that grow with communication range at fixed border factors.*

The framework does not explain the rise itself, which Berthelon and Freund find within industries and strongest for homogeneous goods, evidence they hold against a composition explanation; Chaney's mechanism is a scale-free alternative outside the information channel; and in our reading Steinwender's telegraph result agrees with the framework, a range increase in the price channel having raised coherence between markets.

### S10.9 World-conditional correlations in societies

Social data record the joint response of a population to shared shocks as well as the coherence that communication produces, which is why the ultrametric contrast of Definition 4.17 uses world-conditional correlations.

**Proposition S10.27** (Unconditional correlations cap the ultrametric contrast)**.** *Let the unconditional stationary covariance of a modality's scalar errors be* $\Sigma = \Sigma^c + \Sigma_{\mathrm{cm}}\mathbf{1}\mathbf{1}^\top$*, with the world-conditional covariance* $\Sigma^c$ *entrywise nonnegative and* $\Sigma_{\mathrm{cm}} > 0$ *the variance of the common mode. This is the case for the scalar level-0 network with* $W \geq 0$*, exact code models and equal sensing gains, whose row sums all equal* $\lambda$ *and whose common mode has variance* $\Sigma_{\mathrm{cm}} = Q/(1-\lambda^2)$ *(Lemma 2.13). Then every unconditional correlation is at least* $\Sigma_{\mathrm{cm}}/(\Sigma_{\mathrm{cm}} + \max_i \Sigma^c_{ii})$*, and every nonempty set* $C \neq V$ *has*

$$\mathcal{U}^{\mathrm{unc}}(C) \leq \ln\Big(1 + \frac{\max_i \Sigma^c_{ii}}{\Sigma_{\mathrm{cm}}}\Big),$$

*where* $\mathcal{U}^{\mathrm{unc}}$ *is the ultrametric contrast computed from unconditional instead of world-conditional correlations. In the level-0 network with* $W = \lambda P$*,* $P$ *a fixed irreducible aperiodic stochastic matrix with stationary vector* $\pi$*, and local noise covariance* $Q^\zeta = \mathrm{diag}(q_i) \neq 0$*,* $(1-\lambda^2)\Sigma^c_{ii} \to \pi^\top Q^\zeta \pi$ *for every* $i$ *as* $\lambda \to 1$ *(*$= q/N$ *for symmetric* $P$ *and* $Q^\zeta = qI$*), so the cap tends to* $\ln(1 + \pi^\top Q^\zeta \pi/Q) > 0$*; in the symmetric case this limit is* $\ln(1 + q/(NQ))$*, which is small when* $q/(NQ)$ *is small and vanishes only as* $N \to \infty$ *or* $q/Q \to 0$*. For* $W = \lambda P$ *with* $P$ *symmetric and doubly stochastic, irreducible or not, and* $Q^\zeta = qI$ *with* $q > 0$*, one has* $(1-\lambda^2)\Sigma^c_{ii} \geq q/N$ *for every* $i$ *and every* $\lambda \in (0,1)$*, so the cap is at least* $\ln(1 + q/(NQ))$ *at every level of grounding: in a finite closed population of this kind* $(1-\lambda^2)\max_i \Sigma^c_{ii}$ *does not tend to zero.*

Proof in Section S34. Unconditional correlations of regional price, opinion or usage series count the common response to shared shocks as coherence, so when $(1-\lambda^2)\max_i \Sigma^c_{ii} \ll Q$, that is, when local noise is not large against the world innovation ($q/(NQ) \ll 1$ at weak grounding), every unconditional contrast is small, whatever the barriers: on a ring of 60 agents with $q = Q = 1$ and two barriers with $n_b = 100$, the contrast of one arc is 3.37, 2.80 and 2.12 from world-conditional correlations at leaks $\varepsilon = 0.05$, 0.01 and 0.002, and 0.114, 0.072 and 0.041 from unconditional ones (Check S47.116). The limit $\ln(1 + q/(NQ))$ is taken at fixed coupling. When the barriers strengthen as grounding weakens, so that $1-\lambda^2$ stays large against the coupling between nearly isolated blocks of $n$ members and small against the spectral gap within them, $(1-\lambda^2)\Sigma^c_{ii}$ is of order $q/n$, the cap is of order $\ln(1 + q/(nQ))$, and the unconditional contrast of a block approaches it instead of zero: for two blocks of three with $q = Q = 1$ it is 0.286 under the cap 0.287 at $\lambda = 0.9999$, against $\ln(4/3) = 0.288$ (Check S47.116). The world-conditional contrast has no such cap and grows without bound as the barriers strengthen at fixed grounding. The cap scales with $q/Q$, and with $q/Q = 100$ on the same ring the unconditional contrast at $\varepsilon = 0.002$ is 1.445 against the world-conditional 2.120, so the barriers remain visible when local noise is large against the world innovation. Conditioning on the shared drive, or a recorded proxy for it, is the empirical counterpart of Definition 4.7. Without a recorded drive, the setting of Proposition S10.27 is a Gaussian graphical model with one latent

variable: by the Sherman–Morrison formula the unconditional precision is

$$\Sigma^{-1} = (\Sigma^c)^{-1} - \frac{(\Sigma^c)^{-1}\mathbf{1}\mathbf{1}^\top(\Sigma^c)^{-1}}{\Sigma_{\mathrm{cm}}^{-1} + \mathbf{1}^\top(\Sigma^c)^{-1}\mathbf{1}},$$

the world-conditional precision minus a rank-one term along the vector $(\Sigma^c)^{-1}\mathbf{1}$, which for a diffusive precision is the vector of its leaks, and for a stationary world the covariance of the beliefs themselves has the same form with a different common-mode variance, since beliefs and errors differ by a multiple of $\mathbf{1}$ that is a function of the world path. For a static or stationary world without performative acts, the world-conditional precision is locally identifiable from members' statistics alone under the sparse-plus-low-rank conditions of Chandrasekaran et al. (2012), a sparse conditional precision and a latent effect spread over most of the observed variables, in the quantitative sense of their transversality condition, and their convex estimator applies under those conditions; a diffusive precision on a sparse graph with comparable leaks has a sparse conditional precision and a rank-one term spread over the agents, but the quantitative condition is not verified for it here. The estimator is open when acts drive the world, when the conditional precision is not sparse, or when the leak vector $(\Sigma^c)^{-1}\mathbf{1}$, along which the rank-one term lies, is concentrated on few agents.

### S10.10 Graded meta-agents: discussion and open problems

Among the properties graded in Table 11, interaction necessity transfers most readily: with scalar errors, every causally separating null makes the precision of the members' states given the shared exogenous drive diagonal, so that no region has contrast above one for it (Proposition 5.16(d)). For the world-conditional precision on which collectives are selected the same is proved under the deletion nulls in populations without records, and in the two-group model under the frozen-environment null when the initial states that frozen messages carry are deterministic or conditioned on (Proposition 7.9(a)); in these cases a collective selected by contrast above one differs from the nulls. When members act on the world, as in polities and markets, the order parameter must condition on that drive, because conditioning on the world path hides interaction through acts and couples agents that never interact (Proposition 5.16(b), Example S5.24). The decisive tests of Definition 5.17, partner randomization at matched rates and the exact-fusion persistence clause, have been applied to no abstract collective. No abstract collective of this section is therefore shown to meet every clause of Definition 5.19: the entries of Table 11 are established one clause at a time, in different models, and neither their joint satisfaction in one population nor their joint unsatisfiability in a stated class is shown (Open problem S5.29).

Agency is furthest from the evidence. Nothing here makes a culture or a market behave as if it minimized a group expected free energy; a market has no collective preference, and agency, where it exists, sits in institutions such as courts, central banks and governments, which can be checked against the deliberation protocol of Definition 5.8, with shared reports and intentions and one decision maker at a time. Under it deliberation is reversible exactly when the members agree on how redundant their inputs are, its stationary law is a group posterior when a declared belief has the agreed redundancies, as for linear slots with additive noise and beliefs that are translates of one law (Theorem 5.10), and it lies within explicit distances of one otherwise (Theorem 5.11). Simultaneous decisions and unshared data fail the protocol (Proposition S5.12 and Example S5.14), and, relative to the null policies, a group posterior represents positively coordinated choices only when they change the world (Proposition S5.19 and Theorem S5.17); between two non-null epistemic alternatives the log odds ratio can be positive. Whether actual institutions meet these hypotheses is interpretation.

Nothing here derives the formation of a culture, a market or a polity. The manuscript's formation theorems (Theorems 7.6 and 7.8) concern the frozen-calibration mean-field map of the two-group model and the split it selects; the step from the full stochastic model to that map is open (Open problem 7.14), as is online calibration, which from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell of Table 8 after 6000 rounds, counted at a cutoff for a learned cross edge below the bias fraction $\frac{1}{2}$ used for the map, so that the counts are upper bounds on those at $\frac{1}{2}$, and differing by up to three replicates between reruns of one cell (Numerical observation 7.11 and Open problem S7.13). The collectives formed there are a calibration-level effect of misreading joint structure: receivers that know their translation errors pay no mistranslation cost, so the split fails the translation-aware form of the exact-fusion persistence clause, as Proposition 7.12 proves, while for full cross-covariance-aware exact fusion this conclusion is heuristic in the two-group model; above the threshold $\gamma_T$ of Theorem 7.6(b) the split coincides with the code partition. Persistence with drifting frames (Conjecture S7.12) and formation beyond the two-group model (Conjecture 7.13) are conjectures. Readings of cultural or political boundaries as formed collectives inherit these qualifications; the propositions here describe stationary structure under declared couplings, not its origin.

The framework assumes generative models about shared external states in which messages are grounded, and near-stationary dynamics on slowly changing graphs. It does not cover self-referential conventions without an external referent, such as fashions or asset prices driven by beliefs about beliefs; strategic communication, which is coarsened when preferences diverge (Crawford and Sobel, 1982); power, coercion and conflict; or breaks such as migration, conquest and epidemics. Membership by rule, such as citizenship, can diverge from the coarse-grained patterns of Assumption A10, and $\ell$, $a$ and $\varepsilon$ are inferred from proxies such as border effects, price correlations or Séguy curves only through a model.

**Open problem S10.28** (Contrast of code domains)**.** Do the code domains of the walled regimes (Proposition S10.13(b),(c); Theorem 6.11) have spectral contrast in the sense of Definition 4.10, so that consistency collectives are collectives in the manuscript's primary sense?

Example S12.24 answers the question for the declared precision of a belief network on a ring whose attention is cut at the walls: a domain has spectral contrast, every domain of at most half the ring is a collective at weak enough grounding when the ring is the declared geometry, and no domain is one when the ring without the walls is the declared graph. Whether the code game produces such domains from a disordered start is part of Conjecture S12.25.

**Open problem S10.29** (Attention across modalities)**.** When agents allocate one attention capacity $\kappa$ across modalities by expected free energy, does the allocation couple the modalities, generating $a_\times$, and does it align or misalign their boundaries?

For a single agent with decisions that each load on one content, Proposition S14.18(d) answers the single-agent part of the question for a Shannon budget $I_{\max}$ and pragmatic value alone: when the decision eigenvalues of different modalities differ, the optimal sensor does not mix modalities within channels, and its allocation across modalities is coupled only through a common water level. The single-agent case of a channel count $\kappa$ and of the full objective $G_h^{\mathrm{att}}$ of Definition 14.5 remains open, as do the questions whether that coupling generates $a_\times$ in a network and how it places boundaries.

**Open problem S10.30** (Agency of abstract collectives)**.** Find conditions under which an abstract collective with contrast and closure has agency in the sense of hypothesis H1, in particular whether institutions with collective active states make a polity or an economy behave as if minimizing a group expected free energy, and whether actual institutions meet the protocol hypotheses of Theorem 5.10.

### S10.11 Multiplex terminology

In the terminology of Kivelä et al. (2014) the population of Definition 10.1 is a multiplex network, whose layers share their nodes and differ in their edges. Table 10 lists the four modalities of this section. When the contents evolve independently and one agent senses several of them under a budget on the information it acquires, for decisions that each load on one content, the budget-optimal sensor reads each channel within one content when the decision eigenvalues of different contents differ, and its attention to different contents is coupled only through a common water level (Proposition S14.18(d)).

### S10.12 Composite classes, markets and polities

A composite class of type $M \subseteq \mathcal{M}$ is a closed ball of radius one of the ultrametric $\max_{c \in M} u_{(c)}/\vartheta_c$ of Proposition S4.26(b), with $u_{(c)}$ the ultrametric of modality $c$ and declared scales $\vartheta_c > 0$; its contrast profile is the vector of its ultrametric contrasts in the modalities of $M$. Composite classes of one type are nested or disjoint, ultrametric classes of different modalities can cross, and $\min_c u_{(c)}$ is not an ultrametric in general (Proposition S4.26(b)); these follow from single linkage and are not findings. Proposition S10.6 gives the dynamical reason why a composite class is sharp only where every member modality is.

**Definition S10.31** (Market)**.** A market is a collective (Definition 4.12) of the exchange modality (Table 10), read on members' valuations of the same goods. Public prices are codes of rank one for each good and market, transfers are performative messages that change holdings, and money balances are records. No result of this section shows that a market in a stated model is a collective in the sense of Definition 4.12; the results of Section 10 concern its borders and the ultrametric diagnostic.

**Definition S10.32** (Polity)**.** A polity is a composite class (Proposition S4.26(b)) whose type contains the law and enforcement modality (Table 10). Its promulgations are broadcast codes, usually records of small decay, and its rulings, registrations and sanctions are performative acts that change rights, holdings and the outcomes of other agents' policies, and so the risk term of their expected free energy. A nation-state, in the usage of this section, is a polity whose law boundary is aligned with its boundaries in the language, exchange and broadcast modalities.

### S10.13 The distance effect in trade: sources

Trade falls with distance, and the fall has not weakened as communication improved: in a meta-analysis of 1467 estimates the distance effect rose around the middle of the twentieth century and stayed high (Disdier and Head, 2008), the distance elasticity rose by about 10 percent after 1985, mostly within industries (Berthelon and Freund, 2008), and a constant long-distance elasticity follows if firm sizes are Pareto distributed and the squared distance of a firm's exports grows as a power of its size (Chaney, 2018). Yet the transatlantic telegraph of 1866 reduced the mean and the volatility of the difference between cotton prices in New York and Liverpool (Steinwender, 2018). We read these findings through the dependence of the elasticity of coherence on its scale (Section 10.5).

## S11 Additional results for Section 11: meta-agents

This section collects statements, examples and remarks of Section 11 that the main text cites without stating in full. Proofs are in Section S35.

## S11.1 Interpretation of the coarse variables

*Remark* S11.1 (Physical and abstract external states). This remark is an interpretation. Every coarse variable is a function of the joint process and in that sense physical; a member senses a price through the same kind of channel through which a cell senses a chemical potential. What changes with scale is which variable has a law of its own. For a meta-agent it is a function of many channels and agents, such as another collective's coarse state, a price or a record, abstract in that it need not be a world variable, and it becomes an object of the theory only when its closure is proved (Theorem 11.8) or bounded (Proposition 11.9). Base-level agents keep generative models of external physical states.

## S11.2 Coarse blankets: three failures of the sufficient conditions

**Example S11.2** (An equal-weight summary leaks)**.** Let the exterior $\eta_t \sim \mathcal{N}(0,1)$ be independent across rounds and read by two sensory channels $X_{k,t+1} = \eta_t + v_{k,t}$, $\operatorname{Var} v_{k,t} = R_k^s$, $k = 1, 2$; let the interior be $y_{t+1} = g_1 X_{1,t} + g_2 X_{2,t} + \mathfrak{n}_t$ with $\operatorname{Var} \mathfrak{n}_t = q > 0$; and declare $s = X_1 + X_2$. No channel reads both sides, so the fine path blanket holds, but the weak and strong coarse leak rates are both

$$\begin{aligned} \iota_{\text{path}} &= -\frac{1}{2} \ln\Big(1 - \frac{\operatorname{Cov}(y, \eta \mid s)^2}{\operatorname{Var}(y \mid s) \operatorname{Var}(\eta \mid s)}\Big), \\ \operatorname{Cov}(y, \eta \mid s) &= \frac{(g_1 - g_2)(R_2^s - R_1^s)}{4 + R_1^s + R_2^s}, \qquad \operatorname{Var}(\eta \mid s) = \frac{R_1^s + R_2^s}{4 + R_1^s + R_2^s}, \end{aligned} \tag{S43}$$

with $\operatorname{Var}(y \mid s) = g_1^2(1 + R_1^s) + g_2^2(1 + R_2^s) + 2g_1 g_2 + q - (g_1(2 + R_1^s) + g_2(2 + R_2^s))^2/(4 + R_1^s + R_2^s)$, the moments being those of $(y_{t+1}, \eta_{t-1}, s_t)$. The leak is positive whenever $g_1 \neq g_2$ and $R_1^s \neq R_2^s$; for members who weight by precision, $g_k = 1/R_k^s$, the conditional covariance is $(R_2^s - R_1^s)^2/(R_1^s R_2^s(4 + R_1^s + R_2^s))$. With these gains, the weights of the information-form statistic, at $R^s = (0.5, 2)$ and $q = 0.1$ the rate is 0.135600 nats per round, and at $R^s = (0.2, 5)$ it is 0.619053.

The computation is in Section S35; both rates agree to six digits with window computations from the path covariance (Check S47.117).

**Example S11.3** (Heterogeneous channel memories)**.** Let sensory channels with memories $\mathsf{d}_k$, $X_{k,t+1} = \mathsf{d}_k X_{k,t} + \eta_t + v_{k,t}$, read a scalar exterior state, with the information-form $s_C$ of weights $1/R_k^s$. Then $\mathsf{P}_{\mathsf{S}} A_{\mathsf{SS}} = (\mathsf{d}_k/R_k^s)_k$ is a multiple of $\mathsf{P}_{\mathsf{S}} = (1/R_k^s)_k$ only when all $\mathsf{d}_k$ are equal, so for unequal memories (a1) of Proposition 11.4 fails and $s_C$ reads the sensory residual. Members read their own channels, so the residual cannot be placed on the exterior side; on the interior side it makes $s_C$ read both sides, a coarse collider. In the three-member system of Check S47.117 the coarse leak on ten rounds from an independent start is $2 \times 10^{-14}$ with memories $(0.5, 0.5, 0.5)$, and 0.056 and 0.055 with memories $(0.2, 0.5, 0.8)$ and $(0, 0, 0.6)$. A meta-agent whose members' sensors integrate over different times has no exact coarse blanket at the information-form statistic.

**Example S11.4** (Reafference from one member's act)**.** Let the interior read only $s_C$, the exterior only $a_C = \alpha_1 + \alpha_2$, the sum of two active channels driven by the interior, and the single sensory channel the exterior and member 1's act, $X_{\mathsf{S},t+1} = \eta_t + 0.4\alpha_{1,t} + v_t$. The fine path blanket holds and the interior and exterior read the blanket only through $s_C$ and $a_C$, but (b3) of Proposition 11.4 fails: $s_C = X_{\mathsf{S}}$ reads the exterior and the residual $\alpha_1 - \alpha_2$, which lies on the interior side, and the coarse leak on ten rounds from an independent start is 0.113 nats. If the sensor reads the total act instead, (b3) holds and the leak vanishes (Check S47.117). Reading one member's act through a collective sensor is a coarse collider.

### S11.3 Abstract physics: aggregates and other collectives

**Example S11.5** (A price and many traders)**.** Let the exterior of a meta-agent consist of a price and $N$ outside traders,

$$p_{t+1} = \mathsf{f}_p p_t + \frac{\mathsf{b}_p}{N}\sum_j x_{j,t} + \mathsf{b}_C a_{C,t} + w_t, \qquad x_{j,t+1} = \alpha_j x_{j,t} + \mathsf{b}_x p_t + w_{j,t}, \tag{S44}$$

with persistences $0 \le \alpha_j \le \alpha_{\max} < 1$, independent Gaussian innovations of variances $q_p$ and $q_x$, and $\eta_C = (p, \bar{x})$ with $\bar{x} = N^{-1}\sum_j x_j$. Write $\bar{\alpha}$ and $\mathrm{Var}_N(\alpha)$ for the mean and the variance of the persistences and $\bar{\mathfrak{m}}_l = N^{-1}\sum_j \alpha_j^l$.

(a) For $N \ge 2$ the abstract physics is exact if and only if all $\alpha_j$ are equal: the partition into the price and the traders is lumpable exactly then.

(b) In general the stationary aggregate obeys exactly $\bar{x}_{t+1} = \bar{\alpha}\bar{x}_t + \mathsf{b}_x p_t + \sum_{l\ge 2}\Gamma_l p_{t-l} + \mathfrak{f}_t^{(N)} - \bar{\alpha}\mathfrak{f}_{t-1}^{(N)}$, with kernel $\Gamma_l = \mathsf{b}_x(\bar{\mathfrak{m}}_l - \bar{\alpha}\bar{\mathfrak{m}}_{l-1})$ and colored noise $\mathfrak{f}_t^{(N)} = N^{-1}\sum_j\sum_{l\ge 0}\alpha_j^l w_{j,t-l}$ of variance $(q_x/N^2)\sum_j(1-\alpha_j^2)^{-1} = O(1/N)$. The kernel has $\Gamma_1 = 0$, $\Gamma_l/\mathsf{b}_x \ge 0$ and $\sum_{l\ge 2}|\Gamma_l| \le |\mathsf{b}_x|\,\mathrm{Var}_N(\alpha)/(1-\alpha_{\max})^2$.

(c) If the empirical law of the $\alpha_j$ converges weakly to a law $\mathbb{Q}$ on $[0, \alpha_{\max}]$, with moments $\mathfrak{m}_l = \int \alpha^l \mathbb{Q}(\mathrm{d}\alpha)$, then $\Gamma_l \to \mathsf{b}_x(\mathfrak{m}_l - \mathfrak{m}_1\mathfrak{m}_{l-1})$; in particular $\Gamma_2 \to \mathsf{b}_x\,\mathrm{Var}_{\mathbb{Q}}(\alpha)$, which for $\mathsf{b}_x \ne 0$ is nonzero unless $\mathbb{Q}$ is a point mass.

With persistences evenly spaced on $[0.3, 0.7]$ the memory weight $\sum_{l\ge 2}|\Gamma_l|$ falls only from 0.0370 to 0.0297 between $N = 10$ and $N = 640$ (Check S47.123): the law of large numbers removes the noise of the aggregate, not its memory.

The proof is in Section S35.

*Remark* S11.6 (The information-form defect for aggregates). In Example S11.5, $\bar{Q} = \mathrm{diag}(q_p, q_x/N)$, and since $\det(I + X) \ge 1 + \mathrm{tr}\,X$ for positive semidefinite $X$, $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}} \ge \frac{1}{2}\ln(1 + N\mathfrak{e}_{\bar{x}}^{\mathrm{E}}\,\mathrm{Var}(\bar{x})/q_x)$, where $\mathrm{Var}(\bar{x}) \ge \mathsf{b}_x^2 q_p$ because $\bar{x}_{t+1}$ is $\mathsf{b}_x w_{t-1}$ plus terms independent of $w_{t-1}$. If $\mathsf{b}_x \ne 0$ and $\mathfrak{e}_{\bar{x}}^{\mathrm{E}}$ has a positive limit as $N \to \infty$, which the computed values suggest ($2.1 \times 10^{-4}$ at $N = 640$) and we have not proved, the information-form defect diverges at least like $\frac{1}{2}\ln N$, because the aggregate's own noise vanishes while its memory does not. For aggregates the memory clause of Definition 5.1 is therefore better stated with $\mathfrak{e}^{\mathrm{E}}$ or with kernel weights; long memory and closure by types are treated in Propositions S14.12 and S14.13.

**Example S11.7** (An environment of other collectives)**.** Let the exterior of $C$ contain a set $C'$ of eight other agents on a ring, with symmetric update $W_{C'C'} = \mathrm{diag}(c_i) + a_{\mathrm{in}}\mathrm{Adj}$, where Adj is the adjacency matrix of the ring, row sums 0.99 except at two opposite members that read $C$, whose rows sum to 0.97 because they place weight 0.02 on $C$'s act, isotropic noise and nonnegative spectrum, and let $\eta_C$ be the average of $C'$. By Lemma 11.11, the free dynamics of $\eta_C$ is Markov up to a bound set by the drift-based discrete contrast $r^{\mathrm{d}}$ of the average of $C'$ and by the dispersion $\sigma^{\mathrm{d}} = 0.0087$ of its members' total leaks. At $a_{\mathrm{in}} = 0.01$, 0.03, 0.1, 0.2 and 0.24, $r^{\mathrm{d}}$ is 1.06, 1.84, 4.57, 8.48 and 10.04, and the Markov error $\Xi^{\mathrm{d}}$ is $7.8 \times 10^{-3}$, $1.2 \times 10^{-3}$, $7.2 \times 10^{-5}$, $1.3 \times 10^{-5}$ and $8.1 \times 10^{-6}$. The bound of Lemma 11.11(a) is vacuous at the first value and exceeds $\Xi^{\mathrm{d}}$ by factors from 219 down to 54 at the others, and its precision-rate form (b) is smaller at every value. With $C$'s act as a white input the closure defect $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}}$ falls from $3.9 \times 10^{-4}$ to $4.6 \times 10^{-5}$ (Check S47.124).

A larger drift-based contrast of the other collective makes its average a more nearly closed variable of $C$'s abstract environment. That contrast is computed from the drift of $C'$ with $C$'s act held fixed; it is not the precision-based contrast of Definition 4.10, which selects collectives. The example is a quantitative instance, for one block, of the aggregation of nearly decomposable linear systems of Simon and Ando (1961).

## S11.4 Closure notions

The closure notions in use are related exactly.

**Proposition S11.8** (Closure notions)**.** *(a) (Defect criterion.) Let $x$ be a Gaussian vector with covariance $\Sigma_x \succ 0$, let $y' = \mathsf{G}x + \mathfrak{u}$ with $\mathfrak{u} \sim \mathcal{N}(0, Q^{\mathfrak{u}})$, $Q^{\mathfrak{u}} \succ 0$, independent of $x$, and let $\mathsf{c} = \mathcal{P}x$ with $\mathcal{P}$ of full row rank. Then $I(y'; x \mid \mathsf{c}) = \frac{1}{2} \ln\det(I + (Q^{\mathfrak{u}})^{-1}\mathsf{G}\Sigma_{x|\mathsf{c}}\mathsf{G}^\top)$ with $\Sigma_{x|\mathsf{c}} = \Sigma_x - \Sigma_x\mathcal{P}^\top(\mathcal{P}\Sigma_x\mathcal{P}^\top)^{-1}\mathcal{P}\Sigma_x$, and $I(y'; x \mid \mathsf{c}) = 0$ if and only if $\mathsf{G}$ vanishes on $\ker\mathcal{P}$. In particular $\mathfrak{d}_{\mathrm{cl}}^{C} = 0$ exactly under the conditions of Definition 11.6(i), and $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}} = 0$ exactly under those of (ii).*

*(b) (Dynamical dependence.) In the setting of Definition 5.1, with a Gaussian initial law and an input path that is known or independent of $(x_0, \mathfrak{n})$, let $\mathcal{Z}_t = (y_{0:t}, x_{\mathsf{B},0:t})$ and $P_t^{\mathrm{f}} = \mathrm{Var}(x_t \mid \mathcal{Z}_t)$. The dynamical dependence of $y$ on $x$ given the blanket path, $\mathfrak{d}_{\mathrm{dyn},t} = I(y_{t+1}; x_{0:t} \mid \mathcal{Z}_t)$, is $\frac{1}{2}\ln\det(I + (\mathsf{P}Q^{\mathfrak{n}}\mathsf{P}^\top)^{-1}\mathsf{P}AP_t^{\mathrm{f}}A^\top\mathsf{P}^\top)$. In the stationary case without input $\mathfrak{d}_{\mathrm{dyn},t} \le \mathfrak{d}_{\mathrm{cl}}$, and for $t \ge 1$, $\mathfrak{d}_{\mathrm{dyn},t} = 0$ if and only if $A_{y\perp} = 0$.*

*(c) (Consequence.) In the stationary case without input the following coincide: kernel invariance $A\ker\mathsf{P} \subseteq \ker\mathsf{P}$; $\mathfrak{d}_{\mathrm{cl}} = 0$; $\mathfrak{d}_{\mathrm{dyn},t} = 0$ for every $t \ge 1$; and one-step informational closure, $I(y_{t+1}; x_{0:t}) = I(y_{t+1}; y_{0:t})$ for every $t$. Their approximate forms differ: $\mathfrak{d}_{\mathrm{M}} \le \mathfrak{d}_{\mathrm{cl}}$ and $\mathfrak{d}_{\mathrm{dyn},t} \le \mathfrak{d}_{\mathrm{cl}}$, so the closure defect is the largest.*

Proof in Section S35; checked in Check S47.122. The quantity $\mathfrak{d}_{\mathrm{dyn},t}$ is the conditional transfer entropy whose vanishing Barnett and Seth (2023) call dynamical independence in the context of an environment, here the blanket path, and one-step informational closure, the macro past predicting the next macro state as well as the micro past does, is the notion of Rosas et al. (2024) in one-step form. In linear-Gaussian collectives in the stationary case without input, the exact notions coincide and only their approximate forms are ordered.

## S11.5 Literal and level-0 aggregation

**Proposition S11.9** (Literal aggregation is overconfident)**.** *(a) In a static world, literal consensus at level 1 counts a common prior $n$ times on regular graphs and makes the believed precision diverge (Proposition S5.5).*

*(b) In the setting of Proposition 11.12(b), let the collective belief be formed at every step as the product of the members' posteriors, each computed from a common predicted prior, so that the prediction is counted $n$ times. Its covariance converges to the fixed point $P_{\mathrm{b}}$ of $\Psi_n(P) = (n(\bar{F}P\bar{F}^\top + \bar{Q})^{-1} + \Lambda^s)^{-1}$, and $P_{\mathrm{b}} \preceq \bar{P}$: the literal collective is overconfident. If the resulting estimator is stable, $\rho(\bar{F}(I - P_{\mathrm{b}}\Lambda^s)) < 1$, its stationary error covariance satisfies $\mathcal{C}_{\mathrm{b}} \succeq \bar{P}$, so it is also less accurate than the exact one, and the expected relative entropy from the exact posterior to the literal belief is $\frac{1}{2}[\mathrm{tr}(P_{\mathrm{b}}^{-1}\mathcal{C}_{\mathrm{b}}) - \dim\eta_C + \ln\det P_{\mathrm{b}} - \ln\det\bar{P}]$.*

Proof in Section S35. In a two-dimensional instance the literal belief has variances $(0.023, 0.032)$ against exact $(0.181, 0.093)$ and true errors $(0.690, 0.210)$, at an expected relative entropy of 15.3 nats (Check S47.125).

Level-0 receivers fuse literally with fixed gains, so their average is a fixed linear filter of their inputs, and in general not the coarse filter. Its closure has an exact form.

**Proposition S11.10** (Level-0 collectives carry memory)**.** *Let the members of $C$ be scalar level-0 receivers (Definition 2.12) with exact code models, sensing gains $k_i^s$ and message gains $k_{ij}$ on senders inside and outside $C$. Write $k_i^\partial = \sum_{j\notin C} k_{ij}$, $k_j^{\mathrm{rec}} = \sum_{i\in C} k_{ij}$ and $k_j^{\mathrm{giv}} = \sum_{i\in C} k_{ji}$ for the attention that member $j$ receives from and gives to members, and bars for averages over $C$. The average $\bar m = n^{-1}\sum_{i\in C} m_i$ of the predicted means obeys*

$$\begin{aligned}\bar m(t+1) = F\Big[(1-\bar k^s-\bar k^\partial)\bar m + \frac{s_C}{n} - \frac{1}{n}\sum_{j\in C}(k_j^s + k_j^\partial - \bar k^s - \bar k^\partial)(m_j-\bar m)\\ + \frac{1}{n}\sum_{j\in C}(k_j^{\mathrm{rec}} - k_j^{\mathrm{giv}})(m_j-\bar m)\Big] + \bar\zeta_C(t),\end{aligned} \tag{S45}$$

*with the gain-weighted coarse sensory state $s_C = \sum_{i\in C}(\mathcal{K}_i^s s_i + \sum_{j\notin C}\mathcal{K}_{ij}y_{ij})$ and $\bar\zeta_C$ the average of the members' internal message noises. The first memory term is the dispersion of the members' total leaks $k_j^s + k_j^\partial$, the drift-level counterpart of $\sigma_C$ in Theorem 5.4(a), and the second is the imbalance between the attention a member receives and gives inside $C$, the circulation of Theorem 5.4(b)(iv). For $F \neq 0$ the average is exactly closed given $s_C$ if and only if $k_j^s + k_j^\partial + k_j^{\mathrm{giv}} - k_j^{\mathrm{rec}}$ is the same for every $j \in C$, that is, if and only if the update of the members' means has equal column sums (Definition 11.7 with one block); in particular when total leaks are equal and internal attention is balanced.*

Proof in Section S35; checked in Check S47.126, which includes a collective with unequal leaks and imbalanced attention whose average is exactly closed. Otherwise the average carries memory, which the contrast of $C$ bounds in the continuous-time reversible approximation (Theorem 5.20(c)).

### S11.6 Two firms and a price

**Example S11.11** (Two firms and a price)**.** A price and two firms $C_1$ and $C_2$ of $n = 4$ members each. The price follows $p_{t+1} = \mathsf{f}_p p_t + \mathsf{b}_p(a_{C_1,t} + a_{C_2,t}) + w_t$ with $\mathsf{f}_p = 0.9$, $\mathsf{b}_p = 0.2$ and $\operatorname{Var} w_t = 0.05$, where $a_{C_1}$ and $a_{C_2}$ are the firms' order flows, the sums of their members' acts. Each member has one sensory channel that reads the price, $X_{i,t+1} = p_t + v_{i,t}$, with noise variances $(0.5, 1, 2, 4)$ in $C_1$ and $(1, 1, 2, 2)$ in $C_2$, so that $\Lambda^s = 3.75$ and 3; an internal state made of a filter state and a stored copy of its firm's order flow; and an act channel $u_{i,t+1} = -\mathsf{g}_u\hat p_{i,t}$ + noise of variance 0.01, where $\hat p_{i,t}$ is the price coordinate of its filter state and $\mathsf{g}_u = 0.15$. Members run the consensus filter on the lazy ring $\mathcal{W} = \frac12 I + \frac14(\text{left} + \text{right shift})$, $\rho_\perp = 0.5$, with $n_c = 1$ unless stated, and their computation noise has variance $10^{-3}$ per coordinate except where $C_1$ is taken noiseless. Firm $C_2$ runs a bounded model: the price as an autoregression with coefficient $\mathsf{f}_p$, its own order flow as input with weight $\mathsf{b}_p$, and declared innovation variance 0.1; its filtered variance is $\bar P_2 = 0.12571$, the positive root of $2.43x^2 + 0.49x - 0.1 = 0$. Firm $C_1$ runs either the same bounded model or its exact abstract physics. By Theorem 11.13(ii), which needs only the protocol and therefore covers a bounded model, the average of $C_2$ is closed, so the coarse exterior $\eta_{C_1} = (p, s_{C_2}, y^\eta_{C_2}, y^\alpha_{C_2}, a_{C_2})$ obeys exactly

$$\begin{aligned} p' &= \mathsf{f}_p p + \mathsf{b}_p a_{C_2} + \mathsf{b}_p a_{C_1} + w, \qquad s'_{C_2} = 3p + \bar v,\\ y^{\eta\prime}_{C_2} &= \mathsf{f}_p\bar P_2 s_{C_2} + \mathsf{f}_p(1-3\bar P_2)y^\eta_{C_2} + \mathsf{b}_p y^\alpha_{C_2} + \text{noise},\\ y^{\alpha\prime}_{C_2} &= a_{C_2} + \text{noise}, \qquad a'_{C_2} = -4\mathsf{g}_u y^\eta_{C_2} + \text{noise},\end{aligned} \tag{S46}$$

Table S12: The properties of Theorem 11.13 in the two-firm model, for firm $C_1$ running its exact abstract physics or the bounded model. Leaks and relative entropies in nats; the first two rows give the leak with the sensory and active residuals placed with the interior, which bounds the strong coarse leak from above (Eq. (97)); variances refer to the price coordinate, the predictive variance is that of the current price given the firm's blanket path, and the rows on (iv) take the members of $C_1$ noiseless.

| Property | Exact model | Bounded model |
|---|---|---|
| (i) leak with residuals, independent start, 6 rounds | $1.1 \times 10^{-13}$ | $6.8 \times 10^{-13}$ |
| (i) leak with residuals, stationary start, 6 and 12 rounds | 0.1742, 0.1740 | 0.0915, 0.0915 |
| (i) equal-weight sensory summary: strong and weak coarse leak rates per round | 0.0439, 0.0439 | 0.0382, 0.0381 |
| (ii) closure of $y_{C_1}$: algebraic residual; $\mathfrak{d}_{\mathrm{cl}}^{C}$ | 0; $1 \times 10^{-11}$ | 0; $9 \times 10^{-16}$ |
| (iii) $\eta_{C_1}$ five-dimensional: residuals; $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}}$ | 0; $4 \times 10^{-14}$ | not used |
| (iii) price alone as coarse exterior: $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}}$; $\mathfrak{e}^{\mathrm{E}}$ | 0.0303; 0.0156 | 0.0298; 0.0152 |
| (iv) error variance of the member average | 0.11772 | 0.12734 |
| (iv) exact predictive variance, fine and coarse blanket | 0.11772, 0.11772 | 0.11772, 0.11772 |
| (iv) claimed variance; relative entropy to the exact law | 0.11772; below $2 \times 10^{-12}$ | 0.18982; 0.0743 |
| (iv) local filters with a product pool: error, claimed, relative entropy | | 0.12636, 0.07290, 0.1271 |

a five-dimensional abstract physics with input $a_{C_1}$: a price together with a competitor's sensory statistic, estimate, stored act and order flow. Only the price is sensed, so in condition (S) of Proposition 11.12 $\mathcal{H}_0 = \mathbf{1}$, $\mathsf{G}_\eta = e_1^\top$ and the information matrix of $\eta_{C_1}$ is the singular matrix $3.75 e_1 e_1^\top$; since $\rho(\bar{F}) = 0.618$, the pair $(\bar{F}, \Lambda^s)$ is detectable. The fine joint process has dimension 49 when $C_1$ runs the exact model and 33 when it runs the bounded one, with $\rho(A) = 0.797$ and 0.824. Table S12 reports the properties of Theorem 11.13 computed in this model (Check S47.127).

For the exact model the whole five-dimensional predictive covariance agrees with the error covariance of the member average and with the law obtained by conditioning the stationary path law on thirty rounds of the firm's blanket, to $4.5 \times 10^{-13}$. The variance 0.11772 is that of the current price; the filtered variance, of the price one round earlier, is 0.08167, the one-round latency of (iv). For agency, with acts $u_i \in \{-1, 0, 1\}$, $\Lambda^\star = 4$ about the target 0, $\gamma = 2$ and uniform habits, the logit chain over the 81 profiles has a stationary law within $1.1 \times 10^{-15}$ in total variation of the group posterior with tilted habits at $n_{\mathrm{c}} = 0$, 1 and 4, with largest tilt 0.249, 0.136 and 0.018; with preference precisions $(2, 3, 5, 6)$ it lies at total variation 0.06 to 0.13 from the shared-precision group posterior, as Theorem S5.17(a) predicts. Every property of H5a thus holds exactly for $C_1$ when it models its competitor's coarse states, and each fails by a computed amount when a hypothesis is removed (Table 12).

*Remark* S11.12 (A corporation and its market). This remark is an interpretation. In Example S11.11 the variables that close for a firm are the price together with its competitor's coarse states, so its abstract physics is a law for a market variable and for another organization's estimate and order flow. This is the author's corporation that senses an informational environment, made exact in one linear-Gaussian model with a fixed bounded model for the competitor; it is a worked example, not a result about markets. The reading that a person senses "the economy" in the same way would require the economy's aggregates to be closed variables at the person's timescale (Theorem 11.8 and Example S11.5) and is not established here; which variables a firm or a person should sense

is the subject of Section 14 (Examples S14.14 and S14.16 and Remark S14.15). Fields and Levin (2022) describe competent behavior at every scale as action in spaces other than physical space, and in the simulation of blankets of blankets of Palacios et al. (2020) a collective senses an average of exterior signals and acts through an average of its members' active states (their Eq. 14), the form that $(s_C, a_C)$ takes here.

### S11.7 Open problems

**Open problem S11.13** (An approximate H5a)**.** Bound the excess relative entropy of a meta-agent that filters an approximately closed coarse exterior with a Markov model in terms of the memory measures of Proposition 11.9, which would give property (iv) of Theorem 11.13 an error term, and bound the memory of the coarse internal state under level-0 protocols beyond the exact terms of Proposition S11.10. The two-firm model computes the first quantity (0.074 nats for a firm that ignores its competitor); without a bound, no approximate form of Theorem 11.13 is stated. Section S18 gives such an error term in the stable linear-Gaussian case with a stationary fine filter, positive definite record noise and no computation noise of the members, conditional on $c_- > 0$, and, without input and for the moment-matched declared model, in terms of the closure defect (Theorem S18.7 and Corollary S18.8); the first bound remains open beyond that case, apart from a finite-horizon bound that accumulates over the horizon (Proposition S39.6), and the second, on the memory of the coarse internal state, is not addressed there (Open problem S18.13).

**Open problem S11.14** (Mutually modeling meta-agents)**.** The minimal closed extension of a coarse exterior (Theorem 11.8(f)) can contain other meta-agents' internal states, as in Example S11.11. If two meta-agents modeled each other exactly and each observed the other's whole internal state, each filter would have to be larger than the other's. Decide when finite-dimensional pairs of mutually exact filters exist, given that only observable parts matter.

**Open problem S11.15** (A sharp leak constant)**.** In the checked system the constant $\mathfrak{c}_C$ of Proposition 11.5 exceeds the exact conditional-variance constant by a factor of about 20, and the bound exceeds the leak by four orders of magnitude. Find a bound, for instance through the coarse blanket's own filter, that is sharp in its constant as well as in its order.

**Open problem S11.16** (Nonlinear and discrete meta-agents)**.** State and prove versions of Theorems 11.3, 11.8 and 11.13 for discrete state spaces and nonlinear dynamics, the setting of renormalizing generative models (Remark S13.12), starting from lumpability of Markov chains and the closure notions of Rosas et al. (2024).

## S12 Additional results for Section 12: hierarchy

This section collects statements, examples and remarks of Section 12 that the main text cites without stating in full. Proofs are in Section S36.

### S12.1 The collective as a latent cause in a member's model

Let the members' variables $x$, their internal states as they appear in messages, records and acts, evolve as in Definition 5.1 without blanket input, $x_{t+1} = Ax_t + \mathfrak{n}_t$ with $\mathfrak{n}_t \sim \mathcal{N}(0, Q^{\mathfrak{n}})$ independent across rounds and $Q^{\mathfrak{n}} \succ 0$, and let $y_C = \mathsf{P}x$, with $\mathsf{P}$ of full row rank, be the collective's coarse internal state.

**Definition S12.1** (Two-level member model)**.** A member's two-level model of its collective is a linear-Gaussian model in coordinates $(y_C, x_\perp) = (\mathsf{P}x, Zx)$, for an invertible completion $\mathcal{Z} = \binom{\mathsf{P}}{Z}$,

$$y_{C,t+1} = A_C y_{C,t} + \mathfrak{n}_t^C, \qquad x_{\perp,t+1} = A_{\perp C} y_{C,t} + A_{\perp\perp} x_{\perp,t} + \mathfrak{n}_t^\perp, \tag{S47}$$

with $(\mathfrak{n}_t^C)$ and $(\mathfrak{n}_t^\perp)$ independent Gaussian white noises, in which the member's observations (messages, records, its own sensing) are linear functions of $x = \mathcal{Z}^{-1}(y_C, x_\perp)$ plus independent noise. The top level is autonomous and its prediction $A_{\perp C} y_C$ is an empirical prior on the level below, the structure of a hierarchical dynamic model (Friston, 2008). The latent-only model keeps the first equation: $y_C$ is a Gauss–Markov chain.

**Proposition S12.2** (Closure is exact specification)**.** *The following are equivalent: (i) $A \ker \mathsf{P} \subseteq \ker \mathsf{P}$, the closure condition of Theorem 5.3(b); (ii) for some completion $Z$ the members' process is a two-level model (S47) with $A_C = A_{yy}$ and $A_{\perp C} = A_{\perp y}$. When they hold, this two-level model is exact for every Gaussian initial law, and the path of $y_C$ is a Gauss–Markov chain.*

Proof in Section S36; with blanket input the equivalence is Theorem 5.3(b) itself. The latent-only model can be exact without closure.

**Example S12.3** (A Markov latent cause without closure)**.** Let $x \in \mathbb{R}^2$, $y_C = x_1$, $Q^{\mathfrak{n}} = I$ and $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ 0 & 0 \end{smallmatrix}\right)$ with $|A_{11}| < 1$ and $A_{12} \neq 0$. Then $A \ker \mathsf{P} \not\subseteq \ker \mathsf{P}$ and $\mathfrak{d}_{\mathrm{cl}} = \frac{1}{2}\ln(1 + A_{12}^2) > 0$, because $x_{2,t} = \mathfrak{n}_{2,t-1}$ is independent of $y_{C,t}$. Yet $y_{C,t+1} = A_{11} y_{C,t} + (A_{12}\mathfrak{n}_{2,t-1} + \mathfrak{n}_{1,t})$ with a bracket that is white and independent of $y_{C,0:t}$, so $y_C$ is a first-order autoregression with Markov defect zero: the latent-only model is exact and the two-level model is not.

The closure defect governs a member's model of the collective together with its members; the Markov defect governs its model of the collective coordinate alone, and it is the rate of misspecification. The quantity $\mathfrak{d}_{\mathrm{M}}^{\infty}$ of the next lemma is the Markov defect $\mathfrak{d}_{\mathrm{M}}$ of Definition 5.1 in the stationary state without input, which is defined there as the same limit.

**Lemma S12.4** (Misspecification of a Markov latent cause)**.** *Let $(y_t)_{t\in\mathbb{Z}}$ be a stationary Gaussian process in $\mathbb{R}^{d_y}$ with $S_\infty = \mathrm{Var}(y_{t+1} \mid y_{-\infty:t}) \succ 0$ and $S_1 = \mathrm{Var}(y_{t+1} \mid y_t)$. For a Gaussian Markov model $\mathbb{Q}$ with kernel $q(y_{t+1} \mid y_t)$ and the stationary marginal as initial law, the relative entropy rate $\bar{D}(\mathbb{P}\|\mathbb{Q}) = \lim_T T^{-1} D_{\mathrm{KL}}(\mathbb{P}_{0:T}\|\mathbb{Q}_{0:T})$ exists and*

$$\bar{D}(\mathbb{P}\|\mathbb{Q}) = \mathfrak{d}_{\mathrm{M}}^{\infty} + \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}\big(p(\cdot \mid y_0)\|q(\cdot \mid y_0)\big), \qquad \mathfrak{d}_{\mathrm{M}}^{\infty} = \lim_{t\to\infty} I(y_{t+1}; y_{0:t-1} \mid y_t) = \frac{1}{2}\ln\det(S_1 S_\infty^{-1}). \tag{S48}$$

*Its minimum over such models is $\mathfrak{d}_{\mathrm{M}}^{\infty}$, attained exactly by the moment-matched kernel $q = p(\cdot \mid y_t)$; if $y = \mathsf{P}x$ as above, $\mathfrak{d}_{\mathrm{M}}^{\infty} \leq \mathfrak{d}_{\mathrm{cl}}$; and observing $y$ through an independent linear-Gaussian channel can only lower the relative entropy of every window.*

Proof in Section S36; checked in Check S47.128. Under reversible drift contrast bounds the misspecification further.

**Proposition S12.5** (Contrast bounds the misspecification of the latent cause)**.** *Let the members of $C$ evolve as in Theorem 5.4(a), $\mathrm{d}x = -Mx\mathrm{d}t + \sqrt{2}\mathrm{d}\mathsf{B}$ with $M = \Pi_{CC}$ the block of a diffusive precision, and let $y = \phi_C^\top x$. Let $\varpi_1$ be the smallest eigenvalue of $M$, $\cos^2\alpha_C$ the squared norm of the projection of $\phi_C$ on its eigenspace, and $\mathbb{Q}$ the stationary Ornstein–Uhlenbeck law with rate $\varpi_1$ and variance $1/\varpi_1$, the member's Markov latent cause with the slow mode's parameters.*

*(a) The spectral densities*

$$\mathcal{S}_y(\mathrm{f}) = 2\phi_C^\top(\mathrm{f}^2 I + M^2)^{-1}\phi_C, \qquad \mathcal{S}_{\mathrm{OU}}(\mathrm{f}) = \frac{2}{\mathrm{f}^2 + \varpi_1^2},$$

*satisfy* $\cos^2\alpha_C \mathcal{S}_{\mathrm{OU}} \le \mathcal{S}_y \le \mathcal{S}_{\mathrm{OU}}$ *at every frequency* f.

*(b) For every set of* $m$ *distinct times the covariance matrices of the samples satisfy* $\cos^2\alpha_C C_{\mathbb{Q}} \preceq C_{\mathbb{P}} \preceq C_{\mathbb{Q}}$, *and*

$$D_{\mathrm{KL}}(\mathcal{N}(0, C_{\mathbb{P}}) \| \mathcal{N}(0, C_{\mathbb{Q}})) \le \frac{m}{2}(-\sin^2\alpha_C - \ln\cos^2\alpha_C) \le \frac{m\sin^4\alpha_C}{4\cos^2\alpha_C}. \tag{S49}$$

*(c) If* $r(C) > 1$, *then* $\tan\alpha_C \le \sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}}) = s/(r(C)-1)$, *with* $\sigma_C$ *the dispersion of Theorem 5.4(a) and* $s = \sigma_C/\varkappa_{\mathrm{out}}(C)$ *as in Theorem 5.20(c), and the relative entropy per sampled time is at most* $s^4/(4(r(C)-1)^2((r(C)-1)^2+s^2))$.

Proof in Section S36; checked in Check S47.128. Where contrast is high, a member that treats the collective average as an Ornstein–Uhlenbeck latent cause is misspecified only to fourth order in the dispersion of leakage, with no condition on that dispersion; for twelve members, three of them with cut coupling 0.3, at leak 0.02 and contrast 10.74, (c) gives $9.5 \times 10^{-5}$ nats per sampled time. Non-reversible drift is open (Open problem S12.29).

## S12.2 Predictive coding and mean-field message passing

**Proposition S12.6** (Predictive coding and mean-field message passing are exact in the means)**.** *In Definition 12.1 let* $\mathcal{F}(x)$ *be the negative logarithm of the joint density of the node variables and the data, as a function of* $x = (x_u)_u$.

*(a)* $\mathcal{F}(x) = \frac{1}{2}x^\top \mathsf{H} x - \mathsf{h}^\top x + const$ *is strictly convex, with Hessian* $\mathsf{H}$ *the posterior precision of* $x$, *and*

$$\nabla_u \mathcal{F} = \mathfrak{p}_u - \sum_{c\in\mathrm{ch}(u)} \mathsf{G}_c^\top \mathfrak{p}_c - (h_u^{\mathrm{o}} - \Lambda_u^{\mathrm{o}} x_u), \qquad \mathfrak{p}_u = \Psi_u^{-1}(x_u - \mathsf{G}_u x_{\mathrm{pa}(u)}), \quad \mathfrak{p}_{u_0} = \Lambda_{u_0}^{\downarrow} x_{u_0} - h_{u_0}^{\downarrow}, \tag{S50}$$

*a sum of precision-weighted prediction errors in which the top-down prediction* $\mathsf{G}_u x_{\mathrm{pa}(u)}$ *is weighted by the generative precision* $\Psi_u^{-1}$, *never by a posterior precision of the level above. Gradient descent with step* $0 < \alpha < 2/\lambda_{\max}(\mathsf{H})$ *converges to the exact posterior mean from every start.*

*(b) Coordinatewise minimization of the variational free energy over the mean-field family* $\prod_u q(x_u)$, *nodes updated in a fixed order, gives Gaussian factors with precisions* $\mathsf{H}_{uu}$ *whose means follow block Gauss–Seidel for* $\mathsf{H}x = \mathsf{h}$; *it converges to the exact posterior mean, and* $\mathsf{H}_{uu}^{-1} \preceq (\mathsf{H}^{-1})_{uu}$.

Proof in Section S36; checked in Check S47.129. In the two-pass form of Theorem 12.2 the upward message is a likelihood, equivalently a prediction error relative to the cavity prior; in predictive coding errors are computed against current estimates, with top-down predictions weighted by the generative precision, and that scheme is also exact. Double counting arises only when a posterior that already contains the receiver's data is used as an independent prior (Proposition 12.3). Hierarchical active inference avoids it because its updates are steps on one free energy in which every factor appears once; the ascending Bayesian model average of deep temporal models (Friston et al., 2017a) carries the mean-field approximation of (b), not a double count.

## S12.3 Persistent records, depth and shortcuts

**Proposition S12.7** (Persistent records in a static world)**.** *Let a static scalar latent $x \sim \mathcal{N}(m_\circ, 1/\Lambda_\circ)$ be observed privately by $n$ members, member $i$ holding a contribution with $h_i^\circ \mid x \sim \mathcal{N}(\Lambda_i^\circ x, \Lambda_i^\circ)$, independently, and put $\Lambda^\circ = \sum_i \Lambda_i^\circ$. A record holds natural parameters $(h_z, \Lambda_z)$, initially $(\Lambda_\circ m_\circ, \Lambda_\circ)$. Each round every member reads the record as its prior and writes its posterior $(h_z + h_i^\circ, \Lambda_z + \Lambda_i^\circ)$, and the record aggregates the writes by one of four rules: (E) exact, each member's contribution kept once; (U) the prior-corrected union rule $(h_z, \Lambda_z) \leftarrow \sum_i (h_z + h_i^\circ, \Lambda_z + \Lambda_i^\circ) - (n-1)(h_z, \Lambda_z)$; (A) the logarithmic pool with weights $1/n$; (P) the product of posteriors. After $t \geq 1$ rounds the record's precision, the mean-square error of its mean and its overconfidence factor are those of Table 13. Under (U) the record is exact after one round and then counts the data once more every round; under (A) it is underconfident for $t < n$ and overconfident for $t > n$; under (P) the claimed precision grows like $n^t$, and for $n \geq 3$ the limit mean is the posterior mean with the prior counted $n-1$ times, with error $((n-1)^2\Lambda_\circ + \Lambda^\circ)/((n-1)\Lambda_\circ + \Lambda^\circ)^2$ above the union value $1/(\Lambda_\circ + \Lambda^\circ)$.*

Proof in Section S36; checked in Check S47.130, where three members after eight rounds have overconfidence factors 1, 7.72, 2.48 and $4.59 \times 10^3$ under (E), (U), (A) and (P). Figure 13 follows the four rules over eight rounds: the mean-square error of the record's mean stays between $1/4$ and $1/3$, so the overconfidence is the growth of the claimed precision, linear under (U) and (A) and geometric under (P). In a changing world the cavity rule can hold without bookkeeping.

**Proposition S12.8** (The record as an exact parent in a changing world)**.** *Let $x_{t+1} = Ax_t + w_t^x$ with $w_t^x \sim \mathcal{N}(0, Q_x)$, and let member $i$ observe fresh data $o_{i,t} = H_i x_t + v_{i,t}$ with independent noises.*

*(a) If the record is initialized at the common prior of $x_0$, every member is memoryless, taking the record's prediction $\mathcal{N}(Am_z, AP_zA^\top + Q_x)$ as its prior and writing its posterior after its fresh data, and the record aggregates by rule (U) relative to its prediction, then the record equals the centralized Kalman filter started from that prior at every round; the cavity rule holds automatically, because a fresh datum is not yet in the record.*

*(b) Under the product of posteriors (P) the record counts its own prediction $n$ times per round. In a scalar world with $F = 0.99$, $Q = 0.02$, $n = 5$ and unit sensing noise its stationary claimed variance is 0.00483, the mean-square error of its mean 0.288, and the variance of the centralized Kalman filter 0.0528.*

Proof in Section S36; checked in Check S47.130. If members keep their own past posteriors and also read a record that already contains their past data, the space-time graph has cycles, and exactness needs a message from the record to each member with that member's past contributions removed (Open problem S12.30). Across levels the same accounting decides consistency.

**Proposition S12.9** (Depth and shortcuts)**.** *In a tree with a society root $S$, groups, and members with data:*

*(a) a member's exact top-down prior is the group's cavity with the member removed, pushed through the member's link, where the group's prior is the society's cavity with the group removed, so the member's information is removed at both levels and exact inference is consistent across levels;*

*(b) in a scalar tree of any depth with $\mathsf{G} = 1$ and $\Psi = 0$ on every link (the limit $\Psi \to 0$ of Definition 12.1, in which every node variable equals the root's) and literal reading at every level, each node taking its parent's literal posterior as its prior, a member's literal posterior*

*counts a datum* $\mathsf{L}+1-\mathsf{h}$ *times, where* $\mathsf{L}$ *is the number of levels above the member and* $\mathsf{h}$ *the height above the member of the lowest common ancestor of the member and the datum's node: its own datum* $\mathsf{L}+1$ *times, a group-mate's* $\mathsf{L}$ *times, and the prior once;*

*(c) a member* $i$ *that fuses, as independent priors, the group's cavity message and a message from the society that excludes only the member's own data, a shortcut to the top, counts the prior and every other datum twice: with* $\mathsf{G}=1$ *and* $\Psi=0$ *its believed precision is* $2(\Lambda_S^{\mathrm{post}}-\Lambda_i^{\mathrm{o}})+\Lambda_i^{\mathrm{o}}$ *against the exact* $\Lambda_S^{\mathrm{post}}$*, and the society's message is redundant given the group's cavity, the sufficient statistic of all data outside the member's subtree.*

Proof in Section S36; checked in Check S47.130. In four groups of five members with $\Lambda_\circ=0.2$ and unit information, literal reading at both levels has actual variance 0.0586 against the claimed 0.0382 and the exact 0.0495; at $\Lambda_\circ=0.5$ the shortcut's believed precision exceeds the exact one by 1.85, 1.98 and 2.00 for 6, 50 and 400 members, while its mean barely moves. Part (b) is the one-sweep form of the compounding of literal readings across levels (Proposition S13.17).

*Remark* S12.10 (Literal consensus and interpretation levels). Rule (P) is the hub form of literal consensus (Proposition S5.5): multiplying all $n$ members' posteriors is literal consensus on the complete graph with self-loops, whose believed precision grows like $\rho(I+\mathrm{Adj})^t=n^t$, and both count the common prior repeatedly, the persuasion bias of DeMarzo et al. (2003) with a hub in place of the network. All of these are level-1 readings (Definition 2.4): literal top-down reading reads a message whose provenance overlaps the receiver's data, literal consensus across levels reads every neighbor's posterior with its posted precision, and every double count in this section is the overvaluation case of Section 3.

*Remark* S12.11 (An interpretation, and the author's gauge-theoretic construction). This remark is an interpretation. A person who fuses, as independent evidence, a community newsletter that reports the community's view, including the letter that person wrote, is the literal reader of Proposition 12.3, and a newsletter rebuilt each issue from such readers' views grows overconfident as in Proposition S12.7. In the author's gauge-theoretic program, as secondary descriptions of those documents state it, a child's prior is set from the parent's belief transported into the child's frame with a blend factor, and the parent is a barycenter of the children's beliefs (Dennis, 2026c,b). Its linear-Gaussian analogue is literal top-down reading, whose exact alternative removes the child's own contribution from the parent's message; whether a covariance-averaging barycenter changes the size of the effect is not settled here.

## S12.4 Nested records and contrast with record nodes

The statements of this subsection are in the setting and notation of Section 12.3, the global record of (110).

**Proposition S12.12** (Nested records multiply memory). *Let* $\mathsf{R}=\mathsf{R}_0+\mathrm{diag}(\mathsf{e})$ *on nodes* $0$ *(the agents) and* $1,\dots,n_z$ *(records), with leaks* $\mathsf{e}=(\varepsilon,\delta_1,\dots,\delta_{n_z})\geq 0$*, where* $\mathsf{R}_0$ *has zero row sums, nonpositive off-diagonal entries, and nonzero entries in both directions along the edges of a tree and nowhere else. Let* $\mathsf{u}>0$ *be the left null vector of* $\mathsf{R}_0$ *with* $\mathsf{u}_0=1$*, so that* $\mathsf{u}_k$ *is the product of the read-to-write ratios* $\mathcal{R}$ *on the path from the agents to record* $k$*, let* $\bar{\mathsf{e}}_\mathsf{u}$ *and* $\mathrm{Var}_\mathsf{u}(\mathsf{e})$ *be the* $\mathsf{u}$*-weighted mean and variance of the leaks, and let* $g_0$ *be the smallest positive eigenvalue of* $\mathsf{R}_0$*. The smallest eigenvalue* $\varepsilon_{\mathrm{rec}}$ *of* $\mathsf{R}$ *is real, and*

$$\bar{\mathsf{e}}_\mathsf{u}-\frac{\mathrm{Var}_\mathsf{u}(\mathsf{e})}{g_0-\bar{\mathsf{e}}_\mathsf{u}}\leq\varepsilon_{\mathrm{rec}}\leq\bar{\mathsf{e}}_\mathsf{u}, \tag{S51}$$

*the lower bound when* $g_0 > \bar{\mathsf{e}}_{\mathsf{u}}$. *For the reduced record model of Proposition 8.11 with* $\upsilon_k = 0$, *a star, this gives the exact upper bound* $\varepsilon_{\mathrm{rec}} \le (\varepsilon + \sum_k \mathcal{R}_k\delta_k)/(1 + \sum_k \mathcal{R}_k)$, *the leading term of Proposition 8.12(e); for nested records, a chain in which record* $k$ *is read and written by record* $k-1$, *with record* 0 *the agents, it gives* $\varepsilon_{\mathrm{rec}} \le (\varepsilon + \mathcal{R}_1\delta_1 + \mathcal{R}_1\mathcal{R}_2\delta_2 + \cdots)/(1 + \mathcal{R}_1 + \mathcal{R}_1\mathcal{R}_2 + \cdots)$.

Proof in Section S36; checked in Check S47.131. With leaks only at the agents, each level of nested records multiplies the slow time by a product of read-to-write ratios. Drift is not precision, and the precision-based contrast sees records differently.

**Proposition S12.13** (Contrast with the record as a node)**.** *Embed* $C$ *in a population with symmetric couplings* $a_{ik} \ge 0$ *and leak* $\varepsilon$ *for every agent, the record belonging to* $C$ *only, with noise intensities* $\mathcal{Q} = \mathrm{diag}(q, \dots, q, q_z)$, *the last for the record. Put* $q_{\mathrm{rev}} = \omega q/(n\beta)$ *and* $\mathrm{cut}(C) = \sum_{i\in C, k\notin C} a_{ik}$.

*(a) If* $q_z = q_{\mathrm{rev}}$, *the drift* $M$ *of the population with the record satisfies* $M\mathcal{Q} = \mathcal{Q}M^\top$, *so* $\Sigma = M^{-1}\mathcal{Q}$ *and* $\Pi = \mathcal{Q}^{-1}M$, *which is diffusive, with member leaks* $\varepsilon/q$, *record leak* $n\beta\delta/(\omega q)$, *couplings* $a_{ik}/q$ *and member-record couplings* $\beta/q$.

*(b) The contrast of* $C' = C \cup \{z\}$ *computed from this precision, and that of* $C$ *without the record, are*

$$r(C') = \frac{(n+1)\min\Big\{\varepsilon + \beta + \lambda_2(L_C),\ \dfrac{\varepsilon + n^2\beta\delta/\omega}{n+1} + (n+1)\beta\Big\}}{n\varepsilon + n\beta\delta/\omega + \mathrm{cut}(C)}, \qquad r(C) = \frac{n(\varepsilon + \lambda_2(L_C))}{n\varepsilon + \mathrm{cut}(C)}. \tag{S52}$$

*For* $\delta = 0$ *and* $n\beta \ge \lambda_2(L_C) + \varepsilon n/(n+1)$, $r(C')/r(C) = (1 + 1/n)(1 + \beta/(\varepsilon + \lambda_2(L_C)))$, *which does not depend on* $\omega$.

*(c) The agents' marginal precision, with the record marginalized, is diffusive, with member leaks* $(\varepsilon + \beta\delta/(\omega + \delta))/q$ *and an added coupling* $\beta\omega/(nq(\omega + \delta))$ *between every pair of members, and the members' marginal contrast is* $r'(C) = (\varepsilon + \beta + \lambda_2(L_C))/(\varepsilon + \beta\delta/(\omega + \delta) + \mathrm{cut}(C)/n)$.

*(d) In discrete time (Definition 2.9 with* $\upsilon = 0$, *a uniform kernel on* $C$ *and symmetric agent weights) the same* $q_{\mathrm{rev}}$ *gives* $W\mathcal{Q} = \mathcal{Q}W^\top$, *and if* $W \ge 0$ *entrywise and* $\beta \le 1$ *the world-conditional precision of agents and record is diffusive, with record leak* $(\omega\varepsilon + \delta(2 - \omega - \delta))/q_z$ *and member leaks* $(\varepsilon(2 - \varepsilon) - \beta(\varepsilon - \delta))/q$.

Proof in Section S36; checked in Check S47.132, where a ring of eight with contrast 8.654 reaches 12.950 with the reversible record at $\beta = 0.2$ for both $\omega = 0.1$ and $\omega = 0.01$. Whether $C'$ is a collective needs a comparison medium for populations with record nodes, which Definition 4.11 does not declare (Open problems S4.18 and S12.28).

*Remark* S12.14 (Why the memory factor is absent from contrast). For reversible drift $M = \mathcal{Q}\Pi$, and the slowest rate is $\min_f f^\top\Pi f/f^\top\mathcal{Q}^{-1}f$. For a closed collective at $q_z = q_{\mathrm{rev}}$ the uniform vector over members and record gives $\mathbf{1}^\top\Pi\mathbf{1}/\mathbf{1}^\top\mathcal{Q}^{-1}\mathbf{1} = (\varepsilon + \mathcal{R}\delta)/(1 + \mathcal{R})$, the first-order effective leak: in the $\mathcal{Q}^{-1}$-weighted norm the record counts as $n\mathcal{R}$ members, while in the Euclidean norm of contrast it is one node and its memory enters only through the factor $1 + 1/n$. The memory factor $1 + \mathcal{R}$ is a drift quantity (Proposition 12.4 and Corollary 12.5); precision-based and drift-based separations coincide when all nodes have equal noise intensities and differ in general.

Away from the reversible noise the precision with the record node can leave the diffusive class, and for a noiseless record it always does.

**Example S12.15** (A noiseless global record)**.** Let $C$ be closed, $\delta = 0$ and $q_z = 0$. The stationary covariance is positive definite and, exactly,

$$\Pi_{CC} = \frac{(\varepsilon+\beta)I + L_C + \frac{\omega}{n}\mathbf{1}\mathbf{1}^\top}{q}, \qquad \Pi_{Cz} = -\frac{\varepsilon+\omega+\beta}{q}\mathbf{1}, \qquad \Pi_{zz} = \frac{n(\varepsilon+\omega)(\varepsilon+\omega+\beta)}{\omega q}. \tag{S53}$$

Every pair of members with $a_{ij} < \omega/n$, in particular every uncoupled pair, has the positive precision entry $(\omega/n - a_{ij})/q$, so the precision with the record node is not diffusive and contrast with records as nodes is undefined (Open problem S4.28). The members' marginal precision, $((\varepsilon+\beta)I + L_C - \frac{\omega\beta}{n(\varepsilon+\omega)}\mathbf{1}\mathbf{1}^\top)/q$, is diffusive. With outsiders both fail: for the ring of eight joined to eight outsiders in Numerical observation S12.16, with a noiseless record, the largest positive entry is $3.2 \times 10^{-3}$ of the largest entry in magnitude for the joint precision, and $2.5 \times 10^{-4}$ for the agents' marginal precision.

The closed form is derived in Section S36 and checked in Check S47.132; the agents' marginal precision (Lemma S4.30) is a safe substitute only in closed collectives. Near the reversible noise the sign pattern was computed, not proved.

**Numerical observation S12.16** (The diffusivity window of a record node)**.** For a ring of eight members (neighbor coupling 1, $\varepsilon = 0.02$, $\beta = 0.2$, $\omega = 0.1$, $q = 1$), closed or joined to a ring of eight outsiders by cross couplings 0.05 between matched members, the precision with the record node was computed at ratios $q_z/q_{\text{rev}}$ from 0 to 4 or 10, including 0.99, 1, 1.01, 1.1 and 2. For $\delta = 0$ it is diffusive only at $q_{\text{rev}}$: below it member-member entries turn positive, above it the record's leak turns negative for the closed collective and record-outsider entries turn positive with outsiders. For $\delta = 0.01$ the closed collective is diffusive at $q_z/q_{\text{rev}} = 1$, 1.01, 1.1 and 2 and not at 0.99 or 4, while with outsiders it is diffusive only at $q_{\text{rev}}$.

For a member-written global record in a population with outsiders, the realistic case, contrast with record nodes was defined only at the reversible noise in every instance computed, where a global record has an exact signature.

**Proposition S12.17** (The distance-independent component of a global record)**.** *For a closed population of $N$ members ($C = V$), $\delta = 0$ and $q_z = q_{\text{rev}}$,*

$$\Sigma_{CC} = \frac{q\beta}{N\varepsilon(\varepsilon+\beta)}\mathbf{1}\mathbf{1}^\top + q\big((\varepsilon+\beta)I + L\big)^{-1}, \tag{S54}$$

*a component common to all pairs plus the Green's function of mass $\varepsilon+\beta$; without the record, $\Sigma = q(\varepsilon I + L)^{-1}$ has no such component. For a noiseless record the variance of the members' mean is $q(\varepsilon+\omega)/(N\varepsilon(\varepsilon+\omega+\beta))$, below its value $q/(N\varepsilon)$ without the record.*

Proof in Section S36; checked in Check S47.132. This makes exact the hub component of a global archive described in Section 8.6: the record turns part of the collective mode's fluctuation into a distance-independent component. By (S54) the variance $q/(N\varepsilon)$ of the members' mean is the sum of the common term $q\beta/(N\varepsilon(\varepsilon+\beta))$ and the term $q/(N(\varepsilon+\beta))$ of the Green's function, so the common fraction is $\beta/(\varepsilon+\beta)$, more than half exactly when $\beta > \varepsilon$; for a noiseless record the same subtraction from the variance stated in the proposition gives the fraction $\omega\beta/((\varepsilon+\omega)(\varepsilon+\beta))$, close to one under the condition $\varepsilon \ll \beta\omega/(\beta+\omega)$ of Remark S8.17 and small when the leak dominates either rate. The component is invisible from inside by Lemma 3.6, and its world-conditional cost is of order $1/N$; the cost that does not shrink with $N$ is the world-driven common mode of Theorem 12.7.

## S12.5 The precision balance

How much weight a member should give a top-down prior is a Kalman question, with the cavity condition as its premise.

**Proposition S12.18** (The static echo trade-off)**.** *Let members' states be* $x_i = y_C + \mathfrak{e}_i$ *with* $\operatorname{Var}\mathfrak{e}_i = \psi$*, let member* $i$ *hold a private estimate* $\hat{x}_i = x_i + \mathsf{n}_i$ *with* $\operatorname{Var}\mathsf{n}_i = 1/\mathsf{J}$*, and let a top-down estimate* $m_C$ *of* $y_C$ *have an error* $e_C$ *with* $\operatorname{Var}e_C = 1/\mathsf{a}$*, common to all members and independent of their private data (the cavity condition), all errors independent. A member uses* $\tilde{x}_i = \mathsf{v}m_C + (1-\mathsf{v})\hat{x}_i$*.*

*(a)* $\operatorname{MSE}(\mathsf{v}) = \mathsf{v}^2(1/\mathsf{a}+\psi) + (1-\mathsf{v})^2/\mathsf{J}$*, uniquely minimized at* $\mathsf{v}^* = \Lambda_{\mathrm{td}}/(\Lambda_{\mathrm{td}}+\mathsf{J})$ *with* $\Lambda_{\mathrm{td}} = (1/\mathsf{a}+\psi)^{-1}$*, the cavity precision pushed through the link.*

*(b) For* $i \neq j$ *the shared error is* $\operatorname{Cov}(\tilde{x}_i - x_i, \tilde{x}_j - x_j) = \mathsf{v}^2/\mathsf{a}$*, and the disagreement is* $\mathbb{E}(\tilde{x}_i - \tilde{x}_j)^2 = 2(1-\mathsf{v})^2(\psi + 1/\mathsf{J})$*.*

*(c) As* $\mathsf{v} \to 1$ *the disagreement tends to* $0$ *and the mean-square error to* $1/\mathsf{a}+\psi$*, so the ratio of mean-square error to disagreement, the analogue for member-specific targets of the echo ratio of Definition 3.7, diverges: top-down precision assigned above the Kalman value trades disagreement for shared error.*

Proof in Section S36. With records of a grounded coordinate, in the setting and notation of Lemma 12.6 and Theorem 12.7, the trade-off was computed, not proved.

**Numerical observation S12.19** (Top-down precision on a member-written record)**.**

(a) Let eight level-0 members on the complete graph have $F = 0.99$, $Q = 0.02$, sensing noise variance 1, channel and record read noise variances 0.1, message precision $\varpi_{\mathrm{y}} = 2$ and $\omega = 0.05$, with claimed variances at the fixed point of Section 6.6, and let them assign the precision $\mathsf{r}_z$ to the record. As $\mathsf{r}_z$ rises through 0, 1, 3, 10 and 30, the mean-square error is 0.2815, 0.3426, 0.4495, 0.6631 and 0.8579, each within 0.021 of the shared floor $Q\mathsf{F}_{\mathrm{com}}$, and the echo ratio is 36.0, 46.7, 60.3, 49.0 and 22.8. The error is smallest at $\mathsf{r}_z = 0$ here and in three other parameter sets, and increases at $\mathsf{r}_z = 0$ in five cases on rings of 40 and 100 members.

(b) A single correlation-aware member that fuses its prior mean, sensing, neighbors' messages and the record by the best linear rule under the population's stationary law puts weight $-0.19$ to $-0.27$ on the record in four instances and gains 3.4 to 4.4 percent in posterior variance from it.

(c) With channel and record read noise variances 0.1, sensing noise variance 4 and message precision $\varpi_{\mathrm{y}} = 10$, the error falls below its value at $\mathsf{r}_z = 0$ before it rises: on the complete graph of eight with $F = 0.9$, $Q = 0.02$ and $\omega = 0.05$ from 0.111056 to 0.110038 at $\mathsf{r}_z = 10$, a fall of 0.92 percent; on the complete graph of eight with $F = 0.99$, $Q = 0.001$ and $\omega = 0.05$ from 0.052056 to 0.051185 at $\mathsf{r}_z = 10$, a fall of 1.67 percent; and on the ring of 40 with $F = 0.99$, $Q = 0.001$ and $\omega = 0.2$ from 0.050778 to 0.048964 at $\mathsf{r}_z = 3$, a fall of 3.57 percent. In each instance the common-mode part $Q\mathsf{F}_{\mathrm{com}}$ rises and the local-noise part, which is 7.1, 10.5 and 14.1 percent of the error at $\mathsf{r}_z = 0$ against 2.9 percent in (a), falls by more; at $\mathsf{r}_z = 30$ the error exceeds its value at $\mathsf{r}_z = 0$ in all three.

The computations are described in Check S47.133. In the instances of (b) the exact reader uses the record through its provenance, as a lagged copy of the shared error that can be subtracted, and not as an independent top-down prior.

**Proposition S12.20** (Top-down precision on a self-written record raises the common-mode error)**.** *In the setting of Theorem 12.7, let every member attend $d$ members of $C$ and assign the precisions $\varpi_{\mathrm{s}} > 0$ to its sensing, $\varpi_{\mathrm{y}}$ to each message and $\mathsf{r}_z \geq 0$ to the record, with gains $k^s = \varpi_{\mathrm{s}}/\hat{\Lambda}^+$, $k_{ij} = \varpi_{\mathrm{y}}/\hat{\Lambda}^+$ and $\beta = \mathsf{r}_z/\hat{\Lambda}^+$ computed from the claimed quantities that solve $\hat{\Lambda}^+ = 1/\hat{P}^- + \varpi_{\mathrm{s}} + d\varpi_{\mathrm{y}} + \mathsf{r}_z$ and $\hat{P}^- = F^2/\hat{\Lambda}^+ + Q$, as in Section 6.6. Then the sensing gain $k^s$ is strictly decreasing in $\mathsf{r}_z$, and for every $\mathsf{r}_z > 0$ with $\beta + \omega \leq 1$ the common-mode part $Q\mathsf{F}_{\mathrm{com}}$ of every member's stationary mean-square error is strictly larger than its value at $\mathsf{r}_z = 0$.*

Proof in Section S36. The statement concerns the common-mode part alone, which is the part that Theorem 12.7 controls. The total error adds a local-noise part, which a record of the members' mean can lower because it averages their independent noises. In Numerical observation S12.19(a) the local-noise part is 2.9 percent of the error at $\mathsf{r}_z = 0$ and the total rises with the floor; in the instances of Numerical observation S12.19(c), at weak sensing and strong message precision, it is 7 to 14 percent and the total falls by 1 to 4 percent before it rises. Positive precision on a self-written record of a grounded coordinate therefore costs shared error in every instance that the proposition covers, and whether it raises the total error depends on the share of local noise.

## S12.6 Conformity, records and code domains

**Proposition S12.21** (A record lowers the conformity threshold)**.** *In the code game with affine values, slope $J$ per matched attended partner and $\mathsf{d}$ partners (Proposition S6.15(b)), add a cavity-read record whose value enters $-G_i$ with slope $\beta_{\mathrm{r}}$ on the fraction of the other readers who use the agent's code. In the mean field in which partners and record report the same frequency $x$, $q_i(+1) = \mathrm{sig}(\gamma(J\mathsf{d} + \beta_{\mathrm{r}})(2x-1) + 2\gamma h_i)$; with $h_i = 0$ the adoption map is bistable if and only if $\gamma(J\mathsf{d} + \beta_{\mathrm{r}}) > 2$, and then $x_- = e^{-\gamma(J\mathsf{d}+\beta_{\mathrm{r}})}(1 + o(1))$ as $\gamma(J\mathsf{d} + \beta_{\mathrm{r}}) \to \infty$.*

*Proof.* The record adds $\beta_{\mathrm{r}} x$ to the value of $+1$ and $\beta_{\mathrm{r}}(1-x)$ to that of $-1$, a difference $\beta_{\mathrm{r}}(2x-1)$ that adds to the difference $J\mathsf{d}(2x-1)$ of the partners' values; Proposition S6.15(b) with $J\mathsf{d}$ replaced by $J\mathsf{d} + \beta_{\mathrm{r}}$ gives the rest. □

A record shared by a whole closed population favors one population-wide convention, which under the exchangeable clause or on a transitive declared graph is never a collective (Proposition S4.17); records per group favor group conventions. In the two-group model a preference for coordination supplies an attention cut by fiat.

*Remark* S12.22 (What the translation-aware null removes). In the two-group model (Definition 7.2) at $\gamma = \infty$ and $\kappa = n - 1$, add to the expected free energy of attention a pragmatic term $J_{\mathrm{p}} > 0$ for each attended sender of the receiver's own frame class, a slot-level prior preference for coordinated exchange that every null keeps. Under the translation-aware null of Proposition 7.12 every sender's calibrated noise at the split is $R_c + \mathrm{MSE}$, so all sets of $\kappa$ senders have the same believed value and the in-group set uniquely maximizes believed plus pragmatic value. In-group translations are exact, so the error law of Lemma 7.7, the contrast, the baseline and the admissible window of Theorem 7.8 are unchanged, and the split survives this null for every $J_{\mathrm{p}} > 0$. With $J_{\mathrm{p}} = 0$ it does not: by Proposition 7.12(b), in every state of the null and at every $\gamma$ the $n$ receivers outside the group of a sender of largest calibrated precision attend that cross sender with probability at least $\kappa/(2n-1)$, so the in-group attention of the split is not kept. The null removes the cost side of withdrawal, and a preference for coordination, which it keeps, supplies it at any strength. Because the pragmatic term rewards attention by frame class, the surviving split is the declared code partition, in agreement with the code-explanation null: this is not formation.

For full exact fusion the question is quantitative, and one receiver's answer was computed.

**Numerical observation S12.23** (One exact receiver at the split)**.** At the base point of the two-group model ($F = 0.99$, $Q = 1 - F^2$, $R^s = 1$, $R_c = 0.05$, $n = 4$, $\kappa = 3$; calibrated in-group noise 0.29872 by Lemma 7.3) and the stationary law of the literal split, a single receiver that retains its prior mean and current sensing and fuses translation-corrected messages by exact conditioning on the true joint law values three senders, over both coordinates, at 0.00443 nats when all are in-group, 0.01379 with one cross sender, 0.01932 with two and 0.02288 with three, while calibrated level-0 reading believes any three worth 0.40582 nats. With the pragmatic term of Remark S12.22 the split is a strict best response of this receiver if and only if $J_{\mathrm{p}} > \max_c[\mathcal{V}^*(c) - \mathcal{V}^*(0)]/c = 0.00936$ nats per matched partner, $\mathcal{V}^*(c)$ being the value with $c$ cross senders.

Three cross senders are worth about five times three in-group senders, which confirms, for one unilateral deviation at this point, the heuristic after Proposition 7.12 that exact receivers would prefer cross messages; the population law under full exact fusion is open (Open problem S12.27). Where attention is cut at the walls between code domains, the domains have contrast in a declared geometry.

**Example S12.24** (Code domains on a declared ring)**.** Let a belief network on a ring of $N$ agents have world-conditional precision $\Pi = \varepsilon^{\Pi} I + aL$, with precision leak $\varepsilon^{\Pi} > 0$, $L$ the ring Laplacian and $a > 0$ the common value of the couplings $a_{ij}^{\Pi}$ on its edges, and let attention be cut at $w \geq 1$ walls between code domains, the edge at each wall removed. A domain of $n$ sites is an isolated path with contrast $1 + \lambda_n/\varepsilon^{\Pi}$, $\lambda_n = 2a(1 - \cos(\pi/n))$. The declared geometry is the ring, so the comparison medium of Definition 4.11 is the homogeneous ring with mean coupling $\bar{a} = a(1 - w/N)$ and leak $\varepsilon^{\Pi}$, whose baseline is the larger of the whole ring's contrast $1 + 2\bar{a}(1 - \cos(2\pi/N))/\varepsilon^{\Pi}$ and an adjacent pair's $(\varepsilon^{\Pi} + 2\bar{a})/(\varepsilon^{\Pi} + \bar{a})$. A domain is therefore a collective at some admissible level if and only if

$$\lambda_n > 2\bar{a}\big(1 - \cos(2\pi/N)\big) \qquad \text{and} \qquad \Big(\lambda_n \geq \bar{a} \;\; \text{or} \;\; \varepsilon^{\Pi} < \frac{\lambda_n \bar{a}}{\bar{a} - \lambda_n}\Big), \tag{S55}$$

and the first condition holds for every domain of at most $N/2$ sites. At $N = 60$, $a = 1$ and $\varepsilon^{\Pi} = 0.002$, equal domains of 6, 10, 20 and 30 sites and a domain of 29 sites beside one of 31 give:

| Domain (walls) | Contrast | Baseline | Significance | Collective for |
|---|---|---|---|---|
| 6 (10) | 134.97 | 5.565 | 24.25 | $\varepsilon^{\Pi} < 0.39494$ |
| 10 (6) | 49.94 | 5.930 | 8.42 | $\varepsilon^{\Pi} < 0.10983$ |
| 20 (3) | 13.31 | 6.204 | 2.15 | $\varepsilon^{\Pi} < 0.025279$ |
| 30 (2) | 6.478 | 6.296 | 1.029 | $\varepsilon^{\Pi} < 0.011082$ |
| 29 (2) | 6.862 | 6.296 | 1.090 | $\varepsilon^{\Pi} < 0.011868$ |

The complementary domain of 31 sites fails the first condition and is never a collective. Under the exchangeable clause, which applies only to populations without declared geometry, the baselines are 848 to 984 and no domain of three or more sites qualifies, here and whenever the ring has at most $N/2$ walls (in particular when every domain has at least two sites). If the declared graph is the ring with the walls removed, the comparison medium is the population and no domain is a collective (Remark S4.16).

The derivation is in Section S36 and the numbers in Check S47.134. For this declared precision the example settles the question, left open in Section 10, whether code domains have contrast; whether the code game produces such a precision from a disordered start is the formation question.

**Conjecture S12.25** (Formation without misreading by the conformity loop)**.** *Let agents on a ring hold binary codes and scalar beliefs about a shared world. Codes are revised by asynchronous logit*

*dynamics of the code game (coordination value $J$, independent Gaussian pulls of scale $h$) with a local member-written code record read with cavity priors, and neighbors exchange belief messages only when their codes match, because a pragmatic coordination value $J_{\mathrm{p}}$ makes mismatched exchange worse than none. For large $\gamma$, large $J/h$ and $J_{\mathrm{p}}$ above the exact-fusion advantage of cross-wall senders:*

*(i) code domains form from disordered starts with mean length at least of order $(J/h)^2$, longer with records;*

*(ii) against the comparison medium of the declared ring, every domain that satisfies (S55), with $\lambda_n$ its internal rate in the belief network's world-conditional precision, in particular every domain of at most $N/2$ sites at precision leaks $\varepsilon^{\Pi}$ below $\lambda_n\bar{a}/(\bar{a}-\lambda_n)$ (at every leak if $\lambda_n \geq \bar{a}$), is a collective at admissible levels, while under the exchangeable comparison, which applies only without declared geometry, no domain of three or more sites qualifies when the ring has at most $N/2$ walls (in particular when every domain has at least two sites);*

*(iii) the domain partition differs from the deletion and partner-randomization nulls and survives the exact-fusion null;*

*(iv) it agrees with the code-explanation null, whose codes are themselves produced by communication.*

The basis is Theorem 6.11, Propositions 12.8 and S12.21, Remark S12.22, and Numerical observation S12.23 and Example S12.24, where clause (ii) holds exactly for the reversible nearest-neighbor precision. An instance in which exact-fusion receivers attend across walls with positive frequency for arbitrarily large $\gamma$, or in which no admissible level selects the domains, would refute it.

## S12.7 Dynamic hierarchies

*Remark* S12.26 (Dynamic hierarchies). When lower-level states have dynamics of their own the space-time graph is not a tree, and exact filtering needs a joint filter of all levels or provenance on every channel; records are exact parents only under the conditions of Proposition S12.8(a). The ordering of Theorem 12.9 lets a slow level act on faster ones as a slowly varying empirical prior, and Theorem 12.7 prices it: each factor of memory bought by reading a slower record is paid in shared error for grounded coordinates.

## S12.8 Open problems

**Open problem S12.27** (The two-group split under full exact fusion)**.** Compute the population law of the two-group split when every receiver fuses exactly (the null $\mathfrak{N}_{\mathrm{ex}}$ of Definition 5.17), and its contrast, with and without the pragmatic term of Remark S12.22; Numerical observation S12.23 settles only a unilateral deviation.

**Open problem S12.28** (Contrast with record nodes)**.** Characterize the record noises for which the precision with a global record node is diffusive, for $\delta > 0$ and with outsiders, where Numerical observation S12.16 found only the reversible value; decide whether a contrast for records can be defined in the $\mathcal{Q}^{-1}$-weighted norm of Remark S12.14 without breaking the size bound; and declare a comparison medium for populations of agents and record nodes.

**Open problem S12.29** (Non-reversible drift)**.** Extend Proposition S12.5 to the non-reversible drift of Theorem 5.4(b), where the spectral sandwich needs a replacement for normality.

**Open problem S12.30** (Provenance in dynamic hierarchies)**.** Determine the minimal provenance that a record must carry for exact hierarchical filtering when members keep their own dynamics and past posteriors (Proposition S12.8, Remark S12.26). The problem is narrowed and not closed by Section S21, which gives storage for links and records under which fusion is exact for one shared linear-Gaussian state with known sensing models (trees without process noise, stars, intermittent reports, graphs with cycles, turnover), with a matching lower bound only for intermittent reports from blind encoders under process noise (Theorem S21.3); the least storage on a graph with cycles (Open problem S21.10) and the dynamic tree of Definition 12.1 with a different variable at every node (Open problem S21.12) remain open.

Conjecture S12.25 is the other open item, the form that formation without misreading would take; a characterization of the instances in which the total error of Numerical observation S12.19 rises with the record precision is also open, since Proposition S12.20 settles the common-mode part only.

# S13 Additional results for Section 13: renormalization

This section collects statements, examples and remarks of Section 13 that the main text cites without stating in full. Proofs are in Section S37.

## S13.1 Decimation beyond the nearest-neighbor chain

**Proposition S13.1** (Nearest-neighbor passing on $\mathbb{Z}$)**.** *Let $W_{ii} = c > 0$ and $W_{ij} = a$ for neighbors on $\mathbb{Z}$, with $Q^{\zeta} = qI$ and $\varepsilon = 1 - c - 2a > 0$, so that $\Pi = (I - W^2)/q$ and, by Theorem 8.10(e), $\Sigma(x) = \mathsf{c}_+ e^{-m|x|} + \mathsf{c}_-(-1)^x e^{-m'|x|}$ with $\mathsf{c}_+ = q/(4a \sinh m)$, $\mathsf{c}_- = q/(4a \sinh m')$, $\cosh m = 1 + \varepsilon/(2a)$ and $\cosh m' = (1 + c)/(2a)$.*

*(a) The two-pole class $\{\mathsf{c}_+ e^{-\mu|x|} + \mathsf{c}_- \mathsf{p}^x e^{-\mu'|x|}\}$, $\mathsf{p} = \pm 1$, is closed under decimation by $b$: $(\mu, \mu', \mathsf{p}) \mapsto (b\mu, b\mu', \mathsf{p}^b)$, with $\mathsf{c}_\pm$ fixed.*

*(b) After $l \geq 1$ steps, with $\mu = b^l m$, $\mu' = b^l m'$ and $\mathsf{p} = (-1)^{b^l}$, the decimated precision is diffusive and has infinite range: its couplings decay geometrically with ratio $e^{-\operatorname{arccosh} u_*}$, where*

$$u_* = \frac{\mathsf{c}_+ \sinh\mu \cosh\mu' + \mathsf{c}_- \sinh\mu' \cosh\mu}{\mathsf{p}\mathsf{c}_+ \sinh\mu + \mathsf{c}_- \sinh\mu'} > 1.$$

*(c) For $|x| \geq 1$ the staggered term after $l$ steps is at most $(\sinh m / \sinh m') e^{-b^l(m'-m)}$ times the smooth one, so the coarse covariance converges in relative terms to that of a nearest-neighbor precision with mass $b^l m$, plus a nugget $\mathsf{c}_- \mathbf{1}_{\{x=0\}}$ of relative size $\sinh m / \sinh m' < 1$, which vanishes as $\varepsilon \to 0$.*

Proof in Section S37, including the step for odd $b$, where the staggered pole changes sign; checked in Check S47.137. The range-two structure and the staggered pole are irrelevant: the medium flows into the class of Theorem 13.4. In two dimensions finite-range classes do not close.

**Example S13.2** (Nearest-neighbor precisions do not close on $\mathbb{Z}^2$)**.** Let $\Pi$ be the nearest-neighbor precision on $\mathbb{Z}^2$, $\Pi_{ii} = \varepsilon^\Pi + 4a^\Pi$. Decimating the sites with $x_1 + x_2$ odd leaves the even sublattice, a copy of $\mathbb{Z}^2$ in the coordinates $((x_1+x_2)/2, (x_1-x_2)/2)$, with nearest-neighbor couplings $2(a^\Pi)^2/(\varepsilon^\Pi + 4a^\Pi)$, diagonal couplings $(a^\Pi)^2/(\varepsilon^\Pi + 4a^\Pi)$ and leak $\varepsilon^\Pi(\varepsilon^\Pi + 8a^\Pi)/(\varepsilon^\Pi + 4a^\Pi)$, so the nearest-neighbor class is left after one step. On a $16 \times 16$ torus with $\varepsilon^\Pi = 0.05$ and $a^\Pi = 1$, a second decimation, onto $2\mathbb{Z} \times 2\mathbb{Z}$, gives nonzero positive couplings at all 63 displacements of the coarse torus.

The first part counts common neighbors (Section S37), the second is a computation (Check S47.138).

**Numerical observation S13.3** (Decay of decimated kernels)**.** For the nearest-neighbor precision on $\mathbb{Z}^2$ with $a^{\Pi} = 1$, decimated onto $2\mathbb{Z} \times 2\mathbb{Z}$ on a $256 \times 256$ torus at $\varepsilon^{\Pi} = 0.5$, $0.05$, $0.005$ and $0.0005$, the coarse precision has no positive off-diagonal entry, and at coarse displacements up to five its axis couplings fall by factors between $0.025$ and $0.095$ per coarse step, between $0.090$ and $0.095$ from the second step on at the three smaller leaks; onto $4\mathbb{Z} \times 4\mathbb{Z}$ the factors at displacements up to five lie between $0.019$ and $0.129$. The factor rises slowly with distance: at $\varepsilon^{\Pi} = 0.0005$ the ratio of the couplings at coarse displacements nine and eight is $0.096$ onto $2\mathbb{Z} \times 2\mathbb{Z}$ and $0.135$ onto $4\mathbb{Z} \times 4\mathbb{Z}$. The decay per coarse step changes little between $\varepsilon^{\Pi} = 0.05$ and $0.0005$, by at most a quarter at each step; at $\varepsilon^{\Pi} = 0.5$, a leak comparable to the coupling, it is faster, and onto $4\mathbb{Z} \times 4\mathbb{Z}$ the factors there, at displacements up to five, lie between $0.019$ and $0.030$. We have not proved this; the eliminated sites form a random walk killed on the retained lattice, which suggests a rate that does not depend on the leak when the leak is small against the coupling.

## S13.2 Block averages, exponents and time

**Example S13.4** (Block averages leave the diffusive class)**.** Let $\Sigma(x) = \mathsf{c}_{\Sigma} e^{-m|x|}$, $\mathsf{c}_{\Sigma} = 1/(2a^{\Pi} \sinh m)$, be the nearest-neighbor field on $\mathbb{Z}$, and let $b \geq 2$. Then

$$\Sigma^{\mathsf{Avg}}(X) = \frac{\mathsf{c}_{\Sigma} \mathsf{g}_b}{b^2} (e^{-bm|X|} - \mathsf{k}_b \delta_{X0}), \qquad \mathsf{g}_b = \frac{\sinh^2(bm/2)}{\sinh^2(m/2)},$$

with $\mathsf{k}_b = 1 - \mathsf{g}_b^{-1}(b + 2\sum_{j=1}^{b-1}(b-j)e^{-jm}) \in (0,1)$, the covariance of an autoregressive moving-average field of order $(1,1)$. The precision of the block averages has the entries $\Pi^{\mathsf{Avg}}(n) = -\mathsf{c}_1(-\mathsf{h}_b)^{|n|-1}$ for $|n| \geq 1$, with $\mathsf{c}_1 > 0$, $\mathsf{h}_b = (\mathsf{e}_0 - (\mathsf{e}_0^2 - \mathsf{k}_b^2)^{1/2})/\mathsf{k}_b \in (0,1)$ and $\mathsf{e}_0 = \sinh bm - \mathsf{k}_b \cosh bm$, so it has positive entries, that is negative couplings, at every even distance, for every $\varepsilon^{\Pi} > 0$ and $b \geq 2$. As $m \to 0$, $\mathsf{h}_b \to (1 - (1 - \mathsf{t}_b^2)^{1/2})/\mathsf{t}_b$ with $\mathsf{t}_b = (b^2 - 1)/(2b^2 + 1)$, which is $0.1716$ for $b = 2$ and tends to $2 - \sqrt{3} = 0.2679$ as $b \to \infty$: the defect does not vanish near the massless point.

This form is the spatial counterpart of the temporal aggregation of an autoregression (Amemiya and Wu, 1972; Working, 1960), works cited from their records for attribution only. Proof in Section S37; checked in one and two dimensions in Check S47.140. The exact law of the natural meta-agent variables thus leaves the class on which contrast is defined (Open problem S4.28), even for reversible diffusive populations; the error $\mathcal{D}$ of the coarse medium is small for blocks with $\|\Pi_{\perp\Phi}\|^2 \ll \varkappa_{\mathrm{in}}^{\min} \varkappa_{\mathrm{out}}$, which high contrast alone does not ensure (Theorem 13.8(d), Section 13.7), and is not small without contrast (Example S13.9). The size bound of Theorem 4.14 concerns diffusive precisions and does not apply to the covariance-side length of block averages in Theorem 13.5(c).

**Proposition S13.5** (Fixed points, exponents and time)**.** *Let $\Pi$ be translation invariant and diffusive on $\mathbb{Z}^D$ with uniform leak $\varepsilon^{\Pi} > 0$, and let $\hat{\Pi} - \varepsilon^{\Pi}$ be positive away from $2\pi\mathbb{Z}^D$.*

*(a) If the couplings have finite second moments, decimation and block averaging send $\mathfrak{r}_{2,d} = 1/\xi_{2,d}^2$ to $b^2\mathfrak{r}_{2,d}(1 + O(b^2\mathfrak{r}_{2,d}))$: near the massless point the leak is relevant with $y_{\mathrm{RG}} = 2$. On a finite closed population (M) forces a positive mean leak, so the massless point is never reached.*

*(b) (Long-range couplings.) Let instead $\hat{\Pi}(p) = \varepsilon^{\Pi} + \mathsf{a}_{\sigma}|p|^{\sigma} + O(|p|^{\sigma'})$ near $p = 0$, with $0 < \sigma < \sigma' \leq 2$ and $\mathsf{a}_{\sigma} > 0$, and let $\hat{\Pi}$ be twice continuously differentiable away from $2\pi\mathbb{Z}^D$. Then the coarse symbols have the form $\varepsilon^{\Pi,(1)} + \mathsf{a}_{\sigma}^{(1)}|p'|^{\sigma} + O(|p'|^{\min(\sigma',2\sigma,2)})$, with $\varepsilon^{\Pi,(1)} = b^D\varepsilon^{\Pi}/(1 + \varepsilon^{\Pi}\mathsf{S}_1)$*

*and* $\mathsf{a}_\sigma^{(1)} = b^{D-\sigma}\mathsf{a}_\sigma/(1+\varepsilon^\Pi \mathsf{S}_1)^2$ *for decimation and* $\varepsilon^{\Pi,(1)} = b^D\varepsilon^\Pi$ *and* $\mathsf{a}_\sigma^{(1)} = b^{D-\sigma}\mathsf{a}_\sigma$ *for block averaging. The ratio* $\varepsilon^\Pi/\mathsf{a}_\sigma$ *is relevant with* $y_{\mathrm{RG}} = \sigma$*, the couplings have infinite second moment,* $\xi_2 = \infty$*, and the size bound of Corollary 4.16 is void.*

*(c) (Time.) Under the reversible diffusion with drift* $\Pi$*, each term* $k$ *of the decimated autocovariance relaxes at* $\hat{\Pi}((p'+2\pi k)/b)$*. The term* $k=0$ *relaxes at* $\varepsilon^\Pi + \sum_x a^\Pi(x)(p'\cdot x)^2/(2b^2) + o(|p'|^2 b^{-2})$*, which is* $\varepsilon^\Pi + \sum_d \varsigma_d^2 p_d'^2/(2b^2) + o(|p'|^2b^{-2})$ *when the mixed moments* $\sum_x a^\Pi(x)x_d x_{d'}$*,* $d \neq d'$*, vanish, so in the time unit* $b^2$ *the coupling is invariant and the leak is multiplied by* $b^2$ *(dynamic exponent* 2*, and* $\sigma$ *under (b)); for the nearest-neighbor kernel the other terms relax at rates at least* $\varepsilon^\Pi + 4a^\Pi \sin^2(\pi/(2b))$*. For the level-0 recursion one coarse round is* $b^2$ *base rounds.*

Proof in Section S37; part (b) is checked for the kernel $1/x^2$ on $\mathbb{Z}$ in Check S47.141. The exponent $y_{\mathrm{RG}} = \sigma$ is the thermal eigenvalue of the Gaussian fixed point for long-range couplings in the renormalization-group treatment of Fisher et al. (1972), a work cited from its record and abstract for attribution, where that fixed point governs the interacting model for $\sigma < D/2$; part (b) restates it for decimation and block averaging of a lattice precision with explicit remainder terms, and for the Gaussian fields considered here it holds for every $0 < \sigma < 2$. With $\tau \simeq \xi_2^2/a$ (Proposition S8.15(a)), the coarse length in block units and the coarse correlation time in units of $b^{2l}$ base rounds reach order one at the same level: space and time terminate together.

*Remark* S13.6 (The quartic moment). At the massless point the term $k=0$ of (118) multiplies the ratio of the quartic to the quadratic coefficient of $\hat{\Pi}$ by $b^{-2}$, which alone would make the shape of the kernel beyond its second moment irrelevant with $\Lambda_{\mathrm{RG}} = b^{-2}$. The aliased terms contribute to the quartic coefficient at the same order: in one dimension, with $\hat{\Pi}(p) = \alpha_2 p^2 + \alpha_4 p^4 + O(p^6)$ at $\varepsilon^\Pi = 0$, decimation gives the ratio $b^{-2}(\alpha_4/\alpha_2 - \alpha_2 \mathsf{S}_1^{(0)})$, so the term $k=0$ does not decide the flow of the shape. For the nearest-neighbor kernel the two contributions keep the ratio at its lattice value $-1/12$, as the closure of Theorem 13.4 requires.

### S13.3 Collectives across levels

**Proposition S13.7** (The exchangeable comparison admits only dense collectives)**.** *Let a finite closed population of* $N$ *agents with no declared geometry or coupling mechanism have precision* $\varepsilon^\Pi I + L^\Pi$ *with uniform precision leak* $\varepsilon^\Pi > 0$ *and mean total coupling* $\bar{d}$ *per agent, so that its exchangeable comparison medium has coupling* $\bar{a} = \bar{d}/(N-1)$ *between every pair. Then* $\bar{r}_{\mathrm{hom}} = 1 + N\bar{a}/\varepsilon^\Pi$*, and a region* $C$ *of* $n \geq 2$ *agents is a collective at some admissible level if and only if*

$$\lambda_2(L_C) - \frac{\mathrm{cut}(C)}{n} > N\bar{a} \qquad \textit{and} \qquad \varepsilon^\Pi > \frac{N\bar{a}\,\mathrm{cut}(C)/n}{\lambda_2(L_C) - \mathrm{cut}(C)/n - N\bar{a}}. \tag{S56}$$

*Since* $\lambda_2(L_C) \leq \frac{n}{n-1}\min_{i\in C} d_C(i)$*, with* $d_C(i)$ *the coupling of member* $i$ *to the other members, every member of a collective has* $d_C(i) > \frac{(n-1)N}{n(N-1)}\bar{d}$*: collectives are denser than the population on average.*

Proof in Section S37. Had the bridge population of Section 4.8 no declared geometry, $N\bar{a} = 3.382$ would exceed the gap 0.268 of each grid, and neither grid nor their union would ever be a collective; with its declared $6 \times 12$ box each grid is a collective on an interval of leaks, and the union, the whole closed population, never is (Proposition S4.17). Structure contained in a declared geometry is assumed rather than found (Remark S4.16).

**Proposition S13.8** (Nested collectives judged level by level)**.** *In the nested medium of Theorem 13.6, with the exchangeable comparison at every level and with* $\Delta_{\mathrm{H}}$ *and* $T_{\mathrm{H}}$ *as defined in Section 13.5:*

(a) *judged in the level-0 medium, blocks of level two or more are never collectives, and level-1 blocks are collectives at some admissible level if and only if* $\Delta_{\mathrm{H}}(b,\mathfrak{q},n_{\mathrm{H}})>0$ *and* $\varepsilon^{\Pi}>A_{\mathrm{h}}T_{\mathrm{H}}(b,\mathfrak{q},n_{\mathrm{H}})$*;*

(b) *for* $0\le l\le n_{\mathrm{H}}-2$*, the level-*$(l+1)$ *blocks are collectives of the level-*$l$ *coarse medium if and only if* $\Delta_{\mathrm{H}}(b,\mathfrak{q},n_{\mathrm{H}}-l)>0$ *and* $\varepsilon^{\Pi}/(A_{\mathrm{h}}\mathfrak{q}^{l})>T_{\mathrm{H}}(b,\mathfrak{q},n_{\mathrm{H}}-l)$*, and the whole population is never a collective of the level-*$(n_{\mathrm{H}}-1)$ *medium;*

(c) *for* $n_{\mathrm{H}}\ge 2$*,* $\Delta_{\mathrm{H}}(b,\mathfrak{q},m)>0$ *for every* $m=2,\dots,n_{\mathrm{H}}$ *if and only if* $\mathfrak{q}<b/(2b^{2}-1)$*, so nested collectives exist at every level* $1,\dots,n_{\mathrm{H}}-1$*, judged level by level, if and only if* $\mathfrak{q}<b/(2b^{2}-1)$ *and* $\varepsilon^{\Pi}>A_{\mathrm{h}}\max_{0\le l\le n_{\mathrm{H}}-2}\mathfrak{q}^{l}T_{\mathrm{H}}(b,\mathfrak{q},n_{\mathrm{H}}-l)$*;*

(d) *if* $\mathfrak{q}\ge 1/b$*, no block of any level is a collective at any leak, under either judgment; this contains the regime* $1/b<\mathfrak{q}<1$ *in which Proposition 10.7(b) gives power-law coherence;*

(e) *if* $\mathfrak{q}<1/b$ *and* $k<l$*, the level-*$(l+1)$ *blocks have contrast below the baseline of the level-*$k$ *medium, so a single level declared for all levels admits collectives at one level at most;*

(f) *if* $\mathfrak{q}\ge 1/(2b-1)$*, then* $\Delta_{\mathrm{H}}(b,\mathfrak{q},m)<0$ *for every* $m\ge 2$*, and the conclusion of (d) holds; if* $\mathfrak{q}<1/(2b-1)$*, then* $\Delta_{\mathrm{H}}(b,\mathfrak{q},m)$ *increases strictly with* $m\ge 2$ *to* $(2b-1)/(b-1)-2/(1-\mathfrak{q})>0$*, so with* $m^{*}$ *the least* $m\ge 2$ *at which it is positive, which is* 2 *exactly when* $\mathfrak{q}<b/(2b^{2}-1)$*, the level-*$(l+1)$ *blocks are collectives of the level-*$l$ *coarse medium at some leak if and only if* $l\le n_{\mathrm{H}}-m^{*}$*, and then at every leak above the threshold of (b).*

Proof in Section S37; checked in Check S47.144, where the medium $(b,n_{\mathrm{H}},A_{\mathrm{h}},\mathfrak{q})=(2,8,1,0.2)$ has collectives at every level for $\varepsilon^{\Pi}>0.634$. Block contrasts are the same in every coarse medium (Theorem 13.6(c)), while the excess of the exchangeable baseline over one falls like $\mathfrak{q}^{l}$; hence the dependence on $\varepsilon^{\Pi}/(A_{\mathrm{h}}\mathfrak{q}^{l})$ and the need for a declared level at each level. Without declared geometry, scale-free nested coherence and nested collectives exclude each other, and part (f) extends the exclusion below $1/b$: for $b=2$ the blocks of every level qualify at large enough leaks when $\mathfrak{q}<2/7$; when $2/7\le\mathfrak{q}<1/3$ the blocks of the top $m^{*}-2$ levels below the whole population never qualify while all lower levels do at large enough leaks, with $m^{*}=3$ at $\mathfrak{q}=0.3$ and $m^{*}=7$ at $\mathfrak{q}=0.33$; and when $\mathfrak{q}\ge 1/3$ no level qualifies. A declared geometry admits more: with the sixteen agents of the medium with $b=2$, $n_{\mathrm{H}}=4$, $A_{\mathrm{h}}=1$ at the sites $0,\dots,15$ of $\mathbb{Z}$ and the lattice clause at every level, exhaustive enumeration finds collectives at all three levels for $\mathfrak{q}\le 0.55$ at strong enough grounding; the general embedding is open. Figure 16 places the conditions of Proposition S13.8(c), (d) and (f) in the plane of $b$ and $\mathfrak{q}$ and shows, for the medium $(b,n_{\mathrm{H}},A_{\mathrm{h}},\mathfrak{q})=(2,8,1,0.2)$, how the leak thresholds fall like $\mathfrak{q}^{l}$ from 0.6336 at $l=0$.

## S13.4 Memory without contrast and the elimination of internal states

**Example S13.9** (Without contrast, memory is the leading effect)**.** In a homogeneous medium the Galerkin description, equivalently the Markov coarse network obtained by averaging the update weights, overstates the coarse coupling by the block factor. For the nearest-neighbor precision on $\mathbb{Z}$ and blocks of $b$ sites, the coarse medium has coupling $a^{\Pi}$ between adjacent blocks and leak $b\varepsilon^{\Pi}$, so its squared second-moment length in block units is $a^{\Pi}/(b\varepsilon^{\Pi})$, while the block averages have $a^{\Pi}/(b^{2}\varepsilon^{\Pi})+O(1)$ (Theorem 13.5(c)): the ratio tends to $b$. For the level-0 recursion with $W_{ii}=c$ and $W_{ij}=a$ between neighbors on $\mathbb{Z}$, averaging the weights over blocks gives the coarse message weight $a/b$ per round, while a nearest-neighbor coarse network with the same leak per base round reproduces the exact axis rate $bm$ of Theorem 13.5(b) only with the message weight $\varepsilon/(2(\cosh bm-1))=(a/b^{2})(1+O(\varepsilon))$.

Proof in Section S37; checked in one and two dimensions in Check S47.147. Within a block without contrast, internal disagreement relaxes no faster than content crosses the block, so treating each block as internally instantaneous transports content $b$ times too fast.

The grouping step $G$ of the adiabatic reduction described in Section 13.6 eliminates internal states. Let $x = (x_{\mathrm{ret}}, x_{\mathrm{int}})$ collect the states of a population of particles, $x_{\mathrm{int}}$ their internal states and $x_{\mathrm{ret}}$ all others, evolving as $\mathrm{d}x = -Mx\mathrm{d}t + \sqrt{2}\mathsf{N}\mathrm{d}\mathsf{B}_t$ with $\mathsf{N}$ block diagonal over the four blocks of the particular partition, as in Friston et al. (2023a), and every eigenvalue of $M$ with positive real part. Write $M_{\mathrm{rr}}, M_{\mathrm{ri}}, M_{\mathrm{ir}}, M_{\mathrm{ii}}$ and $\mathsf{N}_{\mathrm{r}}, \mathsf{N}_{\mathrm{i}}$ for the blocks, $\Sigma$ for the stationary covariance, $\mathsf{C}_{\mathrm{ir}} = \Sigma_{\mathrm{ir}}\Sigma_{\mathrm{rr}}^{-1}$ for the regression of internal on retained states, and $x_\perp = x_{\mathrm{int}} - \mathsf{C}_{\mathrm{ir}}x_{\mathrm{ret}}$. In the linear-Gaussian case $\mathsf{C}_{\mathrm{ir}}x_{\mathrm{ret}}$ is the conditional expectation that the elimination substitutes (Friston (2019) conditions on the particle's own blanket and asserts that the two coincide), and $x_\perp$ is what that work calls the intrinsic fluctuations.

**Proposition S13.10** (What the elimination of internal states drops)**.** *In this setting:*

*(a)* $\mathrm{d}x_{\mathrm{ret}} = -(M_{\mathrm{rr}}+M_{\mathrm{ri}}\mathsf{C}_{\mathrm{ir}})x_{\mathrm{ret}}\mathrm{d}t - M_{\mathrm{ri}}x_\perp\mathrm{d}t + \sqrt{2}\mathsf{N}_{\mathrm{r}}\mathrm{d}\mathsf{B}_{\mathrm{r}}$*, and at stationarity* $\mathrm{Cov}(x_\perp(t), x_{\mathrm{ret}}(t)) = 0$*;*

*(b) the residual is a state, not noise:* $\mathrm{d}x_\perp = -M_{\perp\mathrm{r}}x_{\mathrm{ret}}\mathrm{d}t - M_{\perp\perp}x_\perp\mathrm{d}t + \sqrt{2}(\mathsf{N}_{\mathrm{i}}\mathrm{d}\mathsf{B}_{\mathrm{i}} - \mathsf{C}_{\mathrm{ir}}\mathsf{N}_{\mathrm{r}}\mathrm{d}\mathsf{B}_{\mathrm{r}})$*, with* $M_{\perp\mathrm{r}} = M_{\mathrm{ir}} + M_{\mathrm{ii}}\mathsf{C}_{\mathrm{ir}} - \mathsf{C}_{\mathrm{ir}}M_{\mathrm{rr}} - \mathsf{C}_{\mathrm{ir}}M_{\mathrm{ri}}\mathsf{C}_{\mathrm{ir}}$ *and* $M_{\perp\perp} = M_{\mathrm{ii}} - \mathsf{C}_{\mathrm{ir}}M_{\mathrm{ri}}$*, so the retained states obey exactly*

$$\dot{x}_{\mathrm{ret}}(t) = -(M_{\mathrm{rr}} + M_{\mathrm{ri}}\mathsf{C}_{\mathrm{ir}})x_{\mathrm{ret}}(t) + \int_0^t \Gamma_{\mathrm{F}}(t-t')x_{\mathrm{ret}}(t')\mathrm{d}t' + \mathfrak{f}(t), \qquad \Gamma_{\mathrm{F}}(u) = M_{\mathrm{ri}}e^{-M_{\perp\perp}u}M_{\perp\mathrm{r}},$$

*the equation of Theorem 5.2(b) for the Mori complement* $x_\perp$*, where* $\mathfrak{f}$ *collects the white noise of* $x_{\mathrm{ret}}$*, the initial-state term* $-M_{\mathrm{ri}}e^{-M_{\perp\perp}t}x_\perp(0)$ *and a colored force; the elimination keeps the first term and replaces* $-M_{\mathrm{ri}}x_\perp$ *by white noise independent of the past of* $x_{\mathrm{ret}}$*, so it drops the kernel* $\Gamma_{\mathrm{F}}$*, the initial-state term and the color of the force;*

*(c) the retained process is Markov with a transition law that does not depend on the initial law if and only if* $M_{\mathrm{ri}} = 0$*, which fails for every particle whose active states read its internal states, the particles the construction is built for.*

Proof in Section S37; checked in Check S47.148. The size of what is dropped is observed, not proved.

**Numerical observation S13.11** (The error of the memoryless reduction)**.** In a random instance with three external, two sensory, two active and three internal states, with the internal rows of $M$ and their noise intensity multiplied by a speed factor $\mathsf{s}$, the largest deviation over lags $u \in [0, 3]$ between the exact stationary autocovariance of the retained states and that of the memoryless reduction, with white noise chosen to reproduce $\Sigma_{\mathrm{rr}}$, relative to $\|\Sigma_{\mathrm{rr}}\|$, is $2.6 \times 10^{-2}$, $2.2 \times 10^{-2}$, $1.0 \times 10^{-2}$, $3.7 \times 10^{-3}$ and $1.2 \times 10^{-3}$ at $\mathsf{s} = 1$, 3, 10, 30 and 100, and a second instance drawn independently behaves alike (Check S47.148). From $\mathsf{s} = 10$ on the error falls like $1/\mathsf{s}$, first order in the ratio of timescales, as the expansions in that ratio (van Kampen, 1985; Geigenmüller et al., 1983), known to us from their abstracts, lead one to expect; we have not proved a rate.

Three further steps add errors that Proposition S13.10 does not cover: the asserted independence of one particle's internal states from the blankets of others is a stationary blanket condition, which fails at second order in the coupling for non-normal drift (Section 17.2); the slow eigenmodes of each blanket are a second projection, with its own memory (Proposition 13.9); and correlations among the induced fluctuations are set aside. In a path formulation the exact replacement for the expected internal state given the blanket is the filter of Theorem 5.3(d), of which the elimination is the zero-memory truncation.

*Remark* S13.12 (Renormalizing generative models). The renormalizing generative models of Friston et al. (2025), read in their primary text, are discrete hierarchical models in which a state at one level generates the initial states and paths of a group of states below; groups do not overlap, so the latent factors of a level are conditionally independent given the level below by construction. Blocking is by tessellation, with a singular value decomposition of each local group (the main text keeps 32 principal vectors of $4 \times 4$ pixel groups, the caption of its Figure 4 at most 16) followed by quantization; higher levels encode sequences of sequences, which, as the authors observe, removes the Markov property of the lowest-level content. Applied to one neuroimaging dataset, $G$ and $R$ gave time constants growing by about 2.37 and spatial supports by about 2.13 per scale (Friston et al., 2021), figures read through a targeted extraction of the full text. In the parts of the two texts that we read we found no error bound for the reduction. We have not read the later treatment of these models by Zaghw et al. (2026). In our interpretation, renormalizability in these models is built into the generative architecture rather than derived from a given microscopic dynamics, whereas in a given linear-Gaussian network the retained modes carry the static error $\mathcal{D}$ and the memory $\Gamma$ (Theorem 13.8), both zero exactly when the precision leaves their span invariant; their temporal hierarchy of paths plays the role of the filter state of Theorem 5.3(d), a Markov augmentation that absorbs memory.

### S13.5 Codes, ranks and interpretation across levels

A meta-agent $J$ that speaks through its members can aggregate their public messages, $Y_J = \sum_{j \in J} \mathsf{w}_j y_j$, with weights $\mathsf{w}_j \geq 0$ that sum to one and the posted messages $y_j = H_j m_j + \nu_j$, where $H_j = T_c U_j$, $m_j = \eta + e_j$ and the channel noises $\nu_j \sim \mathcal{N}(0, R_c)$ are independent.

**Proposition S13.13** (Aggregated codes). *Write $\bar{U}_J = \sum_j \mathsf{w}_j U_j$, $H_J^{\mathrm{agg}} = T_c \bar{U}_J$, $\bar{e}_J = \sum_j \mathsf{w}_j e_j$ and $\tilde{e}_j = e_j - \bar{e}_J$. Then*

$$Y_J = H_J^{\mathrm{agg}}(\eta + \bar{e}_J) + \sum_{j \in J} \mathsf{w}_j (H_j - H_J^{\mathrm{agg}}) \tilde{e}_j + \sum_{j \in J} \mathsf{w}_j \nu_j .$$

*(a)* $\operatorname{rank} H_J^{\mathrm{agg}} \leq r_c$*: aggregation in public coordinates never raises the rank of a code, while concatenating the messages gives a code with row space $\sum_j \operatorname{row}(H_j)$, of rank up to $\min(K, |J| r_c)$.*

*(b) With a common effective code the middle term vanishes and the aggregate reads the collective belief $\eta + \bar{e}_J$ through that code. With diverse codes, uniform weights, uniform leak and a world-conditional covariance of the form $\Sigma^{\mathrm{sc}} \otimes I_K$, the world-conditional variance of the middle term, $\mathbb{E}|\cdot - \mathbb{E}[\cdot \mid \eta]|^2$, is at most $\max_j \|H_j - H_J^{\mathrm{agg}}\|^2 K(n-1)/(n \varkappa_{\mathrm{in}}(J))$, with $n = |J|$: contrast suppresses the disagreement term.*

*(c) A level-0 meta-receiver of $Y_J$, which assigns the noise $\sum_j \mathsf{w}_j^2 R_c$, ignores $\bar{e}_J$: for a scalar world, a common code $H$ and uniform weights, the second moment of $Y_J - H\eta$ given the world path is at least $H^2/(n \varkappa_{\mathrm{out}}(J)) + R_c/n$, so the believed precision exceeds the true one by at least the factor $1 + H^2/(\varkappa_{\mathrm{out}}(J) R_c)$, the more so the slower the collective's outward rate (Section 3).*

Proof in Section S37. Across levels, aggregation averages frames.

**Proposition S13.14** (Aggregated codes across levels). *Let the members' frames $U_j$ be independent draws of a law on $\mathrm{GL}(K)$ with finite second moment, and let each level average $b$ frames of the level below, so that the level-$l$ aggregate $\bar{U}^{(l)}$ averages $b^l$ independent frames. Then $\bar{U}^{(l)} \to \mathbb{E}U$ almost surely, $\mathbb{E}\|\bar{U}^{(l)} - \mathbb{E}U\|_F^2 = b^{-l} \mathbb{E}\|U - \mathbb{E}U\|_F^2$, and the singular values of $T_c \bar{U}^{(l)}$ converge to those of $T_c \mathbb{E}U$.*

(a) *Frames drawn from the Haar law on* $\mathrm{SO}(K)$ *or* $\mathrm{O}(K)$, $K \geq 2$, *have* $\mathbb{E}U = 0$*: the aggregate code vanishes, with* $\mathbb{E}\|T_c \bar{U}^{(l)}\|_F^2 = b^{-l}\|T_c\|_F^2$.

(b) *Rotations of* $\mathbb{R}^3$ *about* $\mathbf{e}_1$ *by angles* $\vartheta$ *with a symmetric law give* $\mathbb{E}U = \mathrm{diag}(1, \mathbb{E}\cos\vartheta, \mathbb{E}\cos\vartheta)$*; a code reading* $(\mathbf{e}_1, \mathbf{e}_2)$ *keeps the singular values* $(1, |\mathbb{E}\cos\vartheta|)$*, so its effective rank at a relative threshold* $\mathsf{f}_{\mathrm{th}}$ *drops from two to one exactly when* $|\mathbb{E}\cos\vartheta| < \mathsf{f}_{\mathrm{th}}$.

(c) *Rotations within the plane that the code reads attenuate both directions equally and keep the rank.*

(d) *Receivers that carry messages into world coordinates with true translations (Section 2) before aggregating aggregate identity frames and lose nothing.*

Proof in Section S37; checked in Check S47.149. Aggregate rank falls with scale only in directions on which members' frames disagree, and only under aggregation in public coordinates. Relays of meta-level messages have a rank of their own. In a common prior metric noiseless hops are orthogonal projections (Theorem S8.24); let a meta-level hop apply a convex combination $\bar{P} = \sum_j \mathsf{w}_j P_{V_j}$, $\mathsf{w}_j > 0$, as when each member of the receiving meta-agent relays one member of the sending one and the content is averaged.

**Proposition S13.15** (Meta-level relays)**.** *(a)* $\bar{P}$ *is symmetric with* $0 \preceq \bar{P} \preceq I$, $\bar{P}x = x$ *if and only if* $x \in V_\cap = \bigcap_j V_j$, *and* $\operatorname{rank}\bar{P} = \dim\sum_j V_j \geq \max_j \dim V_j$.

(b) *Along a stationary relay of meta-level hops,* $\sigma_k(\bar{P}^n)/\sigma_1(\bar{P}^n) = (\lambda_k(\bar{P})/\lambda_1(\bar{P}))^n$*, so the effective rank at any relative threshold is nonincreasing in* $n$*, and* $\|\bar{P}^n - P_\cap\| = \lambda_*^n$*, with* $\lambda_* < 1$ *the largest eigenvalue of* $\bar{P}$ *on* $V_\cap^\perp$.

*Hence Conjecture S8.26 holds for stationary averaged relays in a common metric, with limiting rank* $\dim V_\cap$ *when* $V_\cap \neq \{0\}$*, while in the reading that meta-agents communicate at lower rank than their members it fails at one hop, because averaging parallel paths raises the rank before relay depth lowers it.*

Proof in Section S37; in Check S47.150 twelve rank-two codes in $\mathbb{R}^4$ that share one direction give $\bar{P}$ of rank four and effective ranks $4, 3, 1, 1, \ldots$ along the relay. Decisions aggregate as codes do.

**Proposition S13.16** (The Hayek dimension under aggregation)**.** *In the setting of Proposition S10.16, let units* $C$ *of level* $l$ *have decision functionals* $\psi_C^{(l)}$*, and let each unit of level* $l+1$ *decide through an aggregate* $\psi_C^{(l+1)} = \sum_{C' \subset C} \mathsf{w}_{C'} \psi_{C'}^{(l)}$*, such as a total demand. Then* $r_{\mathrm{Hay}}^{(l+1)} = \dim\operatorname{span}\{\psi_C^{(l+1)}\} \leq \min\{r_{\mathrm{Hay}}^{(l)}, n^{(l+1)}\}$*, with* $n^{(l+1)}$ *the number of units of level* $l+1$*: the public rank that suffices between meta-agents is nonincreasing up the hierarchy. A public message of that rank suffices for member* $i$ *of a unit only if* $\psi_i \in \operatorname{span}\{\psi_C^{(l+1)}\}$*, so a meta-agent* $C$ *whose top-down message must serve its members' own decisions needs rank* $\dim\operatorname{span}\{\psi_i : i \in C\}$.

Proof in Section S37. In our interpretation, hierarchy economizes communication between meta-agents exactly to the extent that decisions aggregate linearly, and economizes nothing top-down unless members' decisions are themselves functions of aggregates. Interpretation levels compound.

**Proposition S13.17** (Literal reading compounds across levels)**.** *Let every level run the literal update of Proposition S5.5, the level-1 reading of posted precisions over noiseless channels.*

(a) *(Regular hierarchies.) Let* $n_{\mathrm{g}}$ *groups of* $n$ *members run literal consensus on connected* $\Delta_1$*-regular graphs, and one representative per group then on a connected* $\Delta_2$*-regular graph of*

*groups. As the numbers of rounds at both levels grow, every representative's mean converges to* $(N\Lambda_\circ m_\circ + \sum_i \Lambda_i^s s_i)/(N\Lambda_\circ + \sum_i \Lambda_i^s)$, $N = n_{\mathrm{g}} n$*: the common prior is counted* $N$ *times, as in flat literal consensus, and a regular hierarchy of group sizes* $n_1, \dots, n_L$ *counts it* $n_1 \cdots n_L = N$ *times. Believed precisions diverge at every level.*

(b) *(Unequal groups.) After* $n_{\mathrm{c},1}$ *rounds at level one, group* $J$ *enters level two with a weight proportional to* $\rho(I + \mathrm{Adj}_J)^{n_{\mathrm{c},1}} \mathsf{v}_{i_J}^{(J)} \mathsf{v}^{(J)\top} \Lambda_J(0)$ *times its level-two centrality, with* $\mathsf{v}^{(J)}$ *the unit Perron vector of* $I + \mathrm{Adj}_J$ *and* $i_J$ *the representative; the group with the largest spectral radius dominates the top-level belief exponentially in* $n_{\mathrm{c},1}$*, whatever its information, so the persuasion bias of Proposition S5.5 multiplies across levels.*

(c) *(Information form.) If every level runs consensus on information contributions (Proposition 5.7) and passes the exact sums* $\sum_{i \in J}(h_i^s, \Lambda_i^s)$ *upward, adding the common prior once at the top, the top-level belief is the union posterior for every hierarchy: the information form is closed under hierarchy, the posterior form is not.*

Proof in Section S37; checked on four groups of five and on six groups of three (Check S47.151). The overcounting is the overvaluation of Section 3; literal reading of top-down messages is treated in Section 12.

### S13.6 Network renormalization

For networks without coordinates, the Laplacian renormalization group of Villegas et al. (2023), whose procedure we read in its primary text, builds the density matrix $\varrho_{\tau_{\mathrm{L}}} = e^{-\tau_{\mathrm{L}} L} / \operatorname{tr} e^{-\tau_{\mathrm{L}} L}$ of a graph Laplacian at a diffusion time $\tau_{\mathrm{L}}$ (not the correlation time $\tau$), locates characteristic scales at peaks of the specific heat of its normalized entropy, keeps the modes with eigenvalues strictly below $1/\tau_{\mathrm{L}}$ (a mode whose eigenvalue equals $1/\tau_{\mathrm{L}}$ is discarded, the convention at equality that we fix for every statement and computation below), forms as many macronodes as kept modes by joining node pairs in descending order of $|(\varrho_{\tau_{\mathrm{L}}})_{ij}|$, and rescales time; other network renormalizations (García-Pérez et al., 2018; Garuccio et al., 2023; Gabrielli et al., 2025) are known to us from abstracts and a review. We apply the procedure to the coupling Laplacian $L^\Pi$ of the world-conditional precision.

**Proposition S13.18** (Laplacian renormalization does not see grounding)**.** *With uniform precision leak* $\varepsilon^\Pi$*,*

$$\frac{e^{-\tau_{\mathrm{L}}(\varepsilon^\Pi I + L^\Pi)}}{\operatorname{tr} e^{-\tau_{\mathrm{L}}(\varepsilon^\Pi I + L^\Pi)}} = \varrho_{\tau_{\mathrm{L}}},$$

*so the density matrix, the specific heat and the macronodes of the procedure do not depend on* $\varepsilon^\Pi$*; a cutoff applied to the spectrum of* $\varepsilon^\Pi I + L^\Pi$ *without subtracting its bottom removes every mode once* $\tau_{\mathrm{L}} > 1/\varepsilon^\Pi$*, an artifact. Contrast depends on* $\varepsilon^\Pi$*: a region* $C$ *has* $r(C) \geq r_0$ *if and only if* $\varepsilon^\Pi \leq (\lambda_2(L_C) - r_0 \operatorname{cut}(C)/|C|)/(r_0 - 1)$*, which never holds if* $\lambda_2(L_C) \leq r_0 \operatorname{cut}(C)/|C|$*. A macronode is therefore selected at level* $r_0$ *exactly for leaks below this threshold, and it is a collective only if it also exceeds the baseline of its comparison medium.*

**Proposition S13.19** (Agreement for two equitable blocks)**.** *Let two blocks of equal size form a population whose cut is equitable, every member having cut coupling* $\bar{a}^\partial$*, with internal gaps at least* $g_{\min} > 2\bar{a}^\partial$*. Then the spectrum of* $L^\Pi$ *is* $0$*,* $2\bar{a}^\partial$ *and the rest at least* $g_{\min}$*, and a renormalization step at any* $\tau_{\mathrm{L}}$ *with* $2\bar{a}^\partial < 1/\tau_{\mathrm{L}} \leq \lambda_3(L^\Pi)$ *and* $\tau_{\mathrm{L}}(\lambda_3(L^\Pi) - 2\bar{a}^\partial) > \ln N$ *returns exactly the two blocks; such* $\tau_{\mathrm{L}}$ *exist when* $g_{\min}/\bar{a}^\partial > 2(1 + \ln N)$*, a condition on the attention graph alone. When both gaps equal* $g$*, the criterion* $r \geq r_0$ *selects each block if and only if* $\varepsilon^\Pi \leq (g - r_0 \bar{a}^\partial)/(r_0 - 1)$ *and the union if and only*

*if* $\varepsilon^{\Pi} \le 2\bar{a}^{\partial}/(r_0-1)$*, so the two procedures agree exactly for* $2\bar{a}^{\partial}/(r_0-1) < \varepsilon^{\Pi} \le (g - r_0\bar{a}^{\partial})/(r_0-1)$*, an interval of leaks that contains the window of Theorem 4.18(iv), and disagree outside it.*

Proofs in Section S37. On the two grids of the bridge population of Section 4.8, which are not equitable, the procedure returns the grids at $\tau_{\mathrm{L}} = 11.2$ and 103 and one macronode at 928, as in the equitable case (Check S47.152). The union of two blocks that form a closed population is the whole population, never a collective under the exchangeable clause, on a torus or on an edge- and vertex-transitive declared graph (Proposition S4.17); there the criterion selects it only at levels below the baseline. With several scales a single step fails.

**Example S13.20** (Interleaved scales)**.** Let two stiff cliques $A$ and $A'$ of six agents (coupling 10) be joined by a perfect matching of coupling 1, each member of $A$ coupled to one member of $A'$, and let a soft $6 \times 6$ grid $B$ (coupling 1) be attached to them by two edges of coupling 0.001, one from each clique to one of two opposite corners of the grid. At $\varepsilon^{\Pi} = 0.01$ the contrasts are 59.4 for $A$ and $A'$, 197.7 for $A \cup A'$ and 27.6 for $B$, so the criterion $r \ge 2$ selects all four sets. The Laplacian procedure returns $A$ and $A'$ as macronodes only for some $\tau_{\mathrm{L}}$ in $(0.0167, 1)$, and not for $\tau_{\mathrm{L}} \in [0.283, 0.306)$ or $[0.387, 0.789)$; it returns $A \cup A'$ for $1 \le \tau_{\mathrm{L}} < 1/\lambda_2(L^{\Pi}) = 4503$, and $B$ for $1/\lambda_3(L^{\Pi}) = 3.73 \le \tau_{\mathrm{L}} < 4503$: no single diffusion time returns $A$, $A'$ and $B$ together, although every set that contrast selects appears somewhere along the flow.

One step of the procedure uses one global timescale and fails whenever one collective's internal mixing time exceeds another's escape time, while contrast uses each region's own rates. On a homogeneous torus the procedure still forms macronodes, chained by single linkage into one cluster, where contrast selects nothing; in the nested medium it returns the blocks of each level, while contrast selects them according to the leak (Check S47.152; the return of the level blocks is proved for $\mathfrak{q} < b$ and every $\tau_{\mathrm{L}}$ in Theorem S20.9(c)). Whether the sets of Example S13.20 are collectives depends on a comparison medium that we did not compute.

**Open problem S13.21** (Which regions of high ratio Laplacian renormalization returns)**.** With uniform leak, let $r_\star(C) = \lambda_2(L_C)|C|/\operatorname{cut}(C)$ be the grounding-free ratio of a region $C$. Characterize the regions $C$ that are macronodes of the Laplacian procedure at some $\tau_{\mathrm{L}} \in (1/\lambda_2(L_C), |C|/\operatorname{cut}(C))$. The conjecture that every region whose ratio $r_\star(C)$ exceeds a constant times $\ln N$ is one is false for every value of the constant (Proposition S20.10).

That conjecture is withdrawn as stated; its conclusion holds for two equitable blocks under the gap condition of Proposition S13.19, and a level-$l$ block of the nested medium with $\mathfrak{q} < b$ and $1 \le l < n_{\mathrm{H}}$ is returned at some $\tau_{\mathrm{L}}$ in the interval exactly when $r_\star(C) > b/(b - \mathfrak{q})$, a constant threshold without the factor $\ln N$ (Theorem S20.9(c)). On the complete graph it fails for a region that omits one vertex, which is not a local maximizer of $r_\star$ under the addition or removal of one site. The restriction to such maximizers (Open problem S20.13(c)) is not supported by computation either: a clique of 16 agents joined by one edge to a path of 256 sites is a strict local maximizer with $r_\star = 256$ and $\ln N = 5.61$, and no scanned $\tau_{\mathrm{L}}$ in the interval returned it (Check S47.152), which is numerical evidence on a grid and not a proof. Laplacian renormalization asks at which diffusion scales the attention graph has structure; contrast asks which regions are collectives at the given grounding.

## S13.7 Open problems

**Open problem S13.22** (Renormalization beyond the exact cases)**.** (a) Define contrast for the non-diffusive laws of block averages, which coarse-graining can produce (Example S13.4), extending Open

problem S4.28. (b) Prove discrete-time closure for non-reversible updates, beyond Lemma 11.11. (c) Determine the decay rate of decimated kernels in two and more dimensions (Numerical observation S13.3). (d) Compute baselines of nested media embedded in lattices beyond one instance. (e) Resolve Open problem S13.21 beyond two equitable blocks and the nested medium, and decide Conjecture S8.26 for networks with many paths and heterogeneous metrics. (f) Propagate finite-round consensus errors across levels in the information form. (g) Bound the error of nonlinear and discrete renormalization, such as that of Remark S13.12. (h) Prove the first-order rate of Numerical observation S13.11.

A partial result on the subject of this problem is in Section S18: for stable linear-Gaussian collectives in discrete time under small-gain coarse policies, Corollary S18.3 bounds the relative entropy between the retained controlled path law and a declared Markov coarse model uniformly over the policy class, and Theorem S18.11 composes the defects through levels. Neither result settles any of the items (a) to (h).

# S14 Additional results for Section 14: environments and selective sensing

This section collects statements, examples and remarks of Section 14 that the main text cites without stating in full. Proofs are in Section S38.

## S14.1 Summaries of the past

**Lemma S14.1** (Summaries of the past do no better than sensors of the present)**.** *Let $\eta$ be stationary Gauss–Markov and $\mathsf{Y} = \mathsf{S}_{\mathrm{m}}\eta_{t+h}$, with a selector $\mathsf{S}_{\mathrm{m}}$ of full row rank, and let $\mathcal{S}$ be a summary of the past that is jointly Gaussian with the process and conditionally independent of $(\eta_{t+u})_{u>0}$ given $(\eta_{t'})_{t'\le t}$, for example a linear function of the past plus independent noise, with $\dim \mathcal{S} \le \kappa$ and $I(\mathcal{S}; (\eta_{t'})_{t'\le t}) \le I_{\max}$. Then $I(\mathcal{S}; \mathsf{Y})$ is at most the value of Theorem 14.3(b) for $\mathsf{X} = \eta_t$.*

Proof in Section S38; checked in Check S47.155. For a fully observed Markov exterior the past-future canonical variates are therefore those of the pair $(\eta_t, \eta_{t+h})$, with $\Sigma_{\mathsf{X}} = P_0$ and $\Sigma_{\mathsf{XY}} = P_0(F^h)^\top \mathsf{S}_{\mathrm{m}}^\top$. An agent whose raw input is an observation process applies Theorem 14.3 with $\mathsf{X}$ a window of past observations and $\mathsf{Y}$ the observation at the horizon, the linear form of the past-future bottleneck of Creutzig et al. (2009). Predictive information, the mutual information between past and future of a time series, is defined by Bialek et al. (2001), and canonical variables of past and future give Markovian representations (Akaike, 1975), a work cited from its record.

## S14.2 Diffusive exteriors

**Proposition S14.2** (Diffusive exteriors: long horizons read coarse variables)**.** *Let the exterior be a field on a connected graph with at least two vertices and symmetric weighted Laplacian $L_{\mathrm{w}}$, $F = (1-\varepsilon_{\mathrm{w}})I - \mathsf{d}_{\mathrm{w}}L_{\mathrm{w}}$ with $\varepsilon_{\mathrm{w}} \in (0,1)$, $\mathsf{d}_{\mathrm{w}} > 0$ and $1 - \varepsilon_{\mathrm{w}} - \mathsf{d}_{\mathrm{w}}\lambda_{\max}(L_{\mathrm{w}}) > 0$, and $Q = Q_0 I$. Then the exterior is reversible, and at every horizon its canonical variates are, up to normalization, the projections on an orthonormal basis of Laplacian eigenvectors, ordered by increasing Laplacian eigenvalue: the slowest is the global average, with $\tau_1 = -1/\ln(1-\varepsilon_{\mathrm{w}})$, and the next is the projection on an eigenvector of the algebraic connectivity $\lambda_2(L_{\mathrm{w}})$, the second-smallest Laplacian eigenvalue, with $\tau_2 = -1/\ln(1-\varepsilon_{\mathrm{w}} - \mathsf{d}_{\mathrm{w}}\lambda_2(L_{\mathrm{w}}))$. Under a budget $I_{\max} > 0$, the budget-optimal sensor of Theorem 14.3(b) with $\kappa$ channels, for $\mathsf{X} = \eta_t$ and $\mathsf{Y} = \eta_{t+h}$, reads the lowest-frequency modes, at*

*most $\kappa$ of them and no more as $h$ grows (modes of equal Laplacian eigenvalue share one budget threshold), and only the global average once $h > I_{\max}/(\tau_2^{-1} - \tau_1^{-1})$.*

Proof in Section S38. In a diffusive exterior slow and coarse coincide: a long-horizon agent senses spatial averages and long-wavelength modes, which is the precise sense in which a slower agent has a more abstract environment. Whether the modes that a Gaussian coarse-graining retains are those a horizon-$h$ agent senses is a question for Section 13; in reversible diffusive exteriors both are the slow Laplacian modes.

### S14.3 Expected free energy and preferences

*Remark* S14.3 (The attention expected free energy of the framework is horizon-free). With world channels only and no preference over them, Proposition S2.7(a) scores attention by $-I(\eta; y_{\mathcal{J}})$, information about the present state. For a linear sensor of $\eta_t$ this is $\frac{1}{2}\sum_{\mathsf{j}} \ln(1-\mathsf{g}_{\mathsf{j}})$ by Lemma 14.2(i), a function of the captured fractions alone: under a binding Shannon budget every sensor that spends the budget has the same epistemic value, whatever it reads. Under a gain budget $|f| \le 1$ on one channel $s = f^\top \eta_t + v$, the value $\frac{1}{2}\ln(1 + f^\top P_0 f)$ is largest along the top eigenvector of $P_0$, the direction of largest variance. For two independent modes with $\lambda(F) = (0.95, 0.5)$ and innovation variances $(0.01, 1)$ the stationary variances are $(0.103, 1.333)$, so this sensor reads the fast mode, while the canonical sensor reads the slow mode at every horizon (Check S47.157). Selection by predictive information needs a horizon in the epistemic term, as in Definition 14.5, or a pragmatic term.

*Remark* S14.4 (The literal two-step expected free energy rewards complementary sensors). Score the two-step policy (sense at $t$, act, observe $o$ at $t+h$) by the expected free energy of Proposition S2.7, minus epistemic minus pragmatic value, with the path $\eta_{t:t+h}$ as hidden state, $(s, o)$ as outcomes and $R^{\mathrm{o}} \succ 0$. Since $s$ depends on the path through $\eta_t$ and, given $s$, the act is known and $o$ depends on the path through $\eta_{t+h}$, the epistemic term is $I(\eta_t; s) + I(\eta_{t+h}; o \mid s)$, and with $\tilde{o} = H^{\mathrm{o}}\eta_{t+h} + v^{\mathrm{o}}$ Gaussian conditioning gives

$$I(\eta_{t+h}; o \mid s) = \tfrac{1}{2}\ln\det\!\big(H^{\mathrm{o}}\,\mathrm{Cov}(\eta_{t+h} \mid s)(H^{\mathrm{o}})^\top + R^{\mathrm{o}}\big) - \tfrac{1}{2}\ln\det R^{\mathrm{o}},$$

a constant minus $I(s; \tilde{o})$. Under a binding budget the first term is fixed, and the epistemic term prefers the sensor least informative about the future outcome: a reading is worth less when the outcome will reveal the same thing later. This opposes the pragmatic term of Theorem 14.6(a), which dominates when preferences are precise. The predictive-information criterion of Theorem 14.3 follows from the expected free energy only when the hidden state of the agent's model for this decision is the state at the horizon and the later outcome $o$ is not scored in the epistemic term: with hidden state $\eta_{t+h}$ and outcomes $(s, o)$ the epistemic term is $I(s; \eta_{t+h}) + I(\eta_{t+h}; o \mid s)$, which by the display equals $I(s; \eta_{t+h} \mid \tilde{o})$ plus a constant and, as a function of the sensor, differs from the predictive information unless $H^{\mathrm{o}}F^h = 0$. We therefore declare the predictive information as the objective of Definition 14.5 and claim no more.

**Example S14.5** (A preference on a fast variable reverses the selection)**.** Take two independent modes, slow ($\lambda(F) = 0.95$, innovation variance 0.1) and fast ($\lambda(F) = 0.5$, innovation variance 1, stationary variance $\mathsf{p}_{\mathrm{f}} = 4/3$), $\mathsf{S}_{\mathrm{m}} = I$, one channel with $I_{\max} = 1$ nat, and $\Upsilon_{\mathrm{ctl}} = \mathrm{diag}(0, w_{\mathrm{p}})$ with $w_{\mathrm{p}} \in \{4, 40\}$, so that the preferred and controllable outcome is the fast mode. With $x$ the squared sine of the sensor's angle from the slow mode in whitened coordinates, $\mathsf{g} = 1 - e^{-2}$, $\sigma_{\mathrm{s}}^2 = 0.95^{2h}$ and $\sigma_{\mathrm{f}}^2 = 0.25^h$, up to a constant

$$G_h^{\mathrm{att}} = \tfrac{1}{2}\ln(1 - \mathsf{g}\sigma_{\mathrm{s}}^2 + \mathsf{g}x(\sigma_{\mathrm{s}}^2 - \sigma_{\mathrm{f}}^2)) - \tfrac{1}{2}\mathsf{g}w_{\mathrm{p}}\mathsf{p}_{\mathrm{f}}\sigma_{\mathrm{f}}^2 x,$$

which is concave in $x$, so the optimum reads a pure mode. With $w_{\mathrm{p}} = 4$ the slow mode is read at every $h \in \{1, \dots, 8\}$, by its epistemic value alone; with $w_{\mathrm{p}} = 40$ the fast mode is read at $h = 1, 2$ and the slow mode from $h = 3$ on (Check S47.157). The pragmatic optimum reads the fast mode and the epistemic optimum the slow mode at every $h$. At long horizons this agent reads a variable its preferences ignore, because $\mathsf{S}_{\mathrm{m}} = I$ makes it curious about everything; with the model target restricted to the fast mode the slow mode has no value (Proposition S14.6(ii)).

## S14.4 Selection principle and the cell model

**Proposition S14.6** (Selection principle)**.** *Let the exterior split into contents $\eta = (\eta^{(c)})_{c \in \mathcal{M}}$ as in Definition 10.1, let the agent hold channels $s^{\mathrm{old}}$ jointly Gaussian with $\eta$, and add a channel $s^{(c)} = H^{(c)}\eta_t^{(c)} + v^{(c)}$ with independent nondegenerate Gaussian noise. Under $G_h^{\mathrm{att}}$ its value is the sum of the epistemic increment $\mathsf{V}_{\mathrm{ep}} = I(s^{(c)}; \mathsf{S}_{\mathrm{m}}\eta_{t+h} \mid s^{\mathrm{old}})$ and the pragmatic increment $\mathsf{V}_{\mathrm{pr}} = \frac{1}{2}\operatorname{tr}(\Upsilon_{\mathrm{ctl}}[\operatorname{Cov}(\eta_{t+h} \mid s^{\mathrm{old}}) - \operatorname{Cov}(\eta_{t+h} \mid s^{\mathrm{old}}, s^{(c)})])$.*

*(i) (Coupling and predictability.) If $\operatorname{Cov}(s^{(c)}, \eta_{t+h} \mid s^{\mathrm{old}}) = 0$, both increments vanish. This covers no coupling, $H^{(c)} = 0$, and a content uncorrelated at lag $h$ with the exterior given what the agent already reads. Both increments are continuous in $H^{(c)}$.*

*(ii) (Relevance.) $\mathsf{V}_{\mathrm{pr}} = 0$ if and only if $\operatorname{Cov}(s^{(c)}, \mathbb{P}_{\mathrm{o}}H^{\mathrm{o}}\eta_{t+h} \mid s^{\mathrm{old}}) = 0$, which holds when the content neither enters the preferred outcomes nor correlates with what does, or when it enters them only along directions that acts cannot compensate and, given $s^{\mathrm{old}}$, is uncorrelated with what enters the compensable directions; $\mathsf{V}_{\mathrm{ep}} = 0$ if and only if $\operatorname{Cov}(s^{(c)}, \mathsf{S}_{\mathrm{m}}\eta_{t+h} \mid s^{\mathrm{old}}) = 0$.*

*(iii) (Too fast.) If content $c$ is a stationary Gauss–Markov block with transition $F^{(c)}$ and stationary covariance $P_0^{(c)}$, independent of the other contents and of the inputs of $s^{\mathrm{old}}$, then for every gain*

$$\mathsf{V}_{\mathrm{ep}} \le -\tfrac{1}{2}\sum_{\mathsf{j}} \ln(1 - \sigma_{\mathsf{j}}^{(c)}(h)^2), \qquad \mathsf{V}_{\mathrm{pr}} \le \tfrac{1}{2}\|\Upsilon_{\mathrm{ctl}}\| \operatorname{tr}(F^{(c)h} P_0^{(c)} (F^{(c)h})^{\top}),$$

*where $\sigma_{\mathsf{j}}^{(c)}(h)$ are the canonical correlations of the block at lag $h$. They are at most $\mathsf{c}^{(c)}\|F^{(c)h}\|$, with $\mathsf{c}^{(c)} = \|(P_0^{(c)})^{1/2}\|\|(P_0^{(c)})^{-1/2}\|$, so both bounds vanish geometrically in $h$.*

*(iv) (Constant at the agent's scale.) A content whose innovation is small against the sensor noise yields vanishing information per reading in steady state, whatever its persistence (Lemma S14.7).*

Proof in Section S38. Prediction fails for variables that relax much faster than the horizon and, per reading, for variables that barely change over the agent's lifetime. Curiosity values every predictable variable in the model, so selection by relevance needs a model target restricted to relevant variables or a budget under which pragmatic value dominates.

**Lemma S14.7** (Variables constant at the agent's scale)**.** *Let a scalar exterior variable follow $\eta_{t+1}^{\mathrm{g}} = F_{\mathrm{g}}\eta_t^{\mathrm{g}} + w_t^{\mathrm{g}}$ with $|F_{\mathrm{g}}| \le 1$ and innovation variance $Q_{\mathrm{g}} > 0$, and be read as $s_t = \eta_t^{\mathrm{g}} + v_t$ with $v_t \sim \mathcal{N}(0, R^s)$, $R^s > 0$. From any prior, the predicted variance $P_t^-$ converges to the positive root $P_\infty^-$ of $x^2 + x(R^s(1 - F_{\mathrm{g}}^2) - Q_{\mathrm{g}}) - Q_{\mathrm{g}}R^s = 0$, and the information that a new reading carries given the past readings, $\frac{1}{2}\ln(1 + P_t^-/R^s)$, converges to*

$$\tfrac{1}{2}\ln(1 + P_\infty^-/R^s) \le \tfrac{1}{2}\ln\Big(1 + \frac{Q_{\mathrm{g}}}{R^s} + \Big(\frac{Q_{\mathrm{g}}}{R^s}\Big)^{1/2}\Big);$$

*the information about $\eta^{\mathrm{g}}_{t+h}$ is no larger. For a constant ($Q_{\mathrm{g}} = 0$, $F_{\mathrm{g}} = 1$) with prior variance $P_{\mathrm{pr}}$, $n_{\mathrm{r}}$ readings carry $\frac{1}{2}\ln(1 + n_{\mathrm{r}}P_{\mathrm{pr}}/R^s)$ nats in total, with a rate per reading that tends to zero.*

Proof in Section S38; checked in Check S47.158. A cell senses its ligand through a diffusive medium, whose molecular noise sets the coupling of the chemical channel.

**Proposition S14.8** (Volume-time average in an ideal ligand gas)**.** *Let ligand molecules be independent Brownian particles in $\mathbb{R}^3$ with diffusion constant $D_{\mathrm{m}}$, at equilibrium with mean density $\bar{c}_{\mathrm{L}}$, so that their positions at each time form a Poisson process, and let a transparent ball $\mathbb{B}$ of radius $r_{\mathrm{cell}}$ and volume $|\mathbb{B}|$ average the number $\mathsf{N}_{\mathbb{B}}(t)$ of molecules inside it over $[0, T_{\mathrm{int}}]$: $\hat{c} = (|\mathbb{B}|T_{\mathrm{int}})^{-1}\int_0^{T_{\mathrm{int}}} \mathsf{N}_{\mathbb{B}}(t)\mathrm{d}t$. Then $\mathbb{E}\hat{c} = \bar{c}_{\mathrm{L}}$ and*

$$\frac{\operatorname{Var}\hat{c}}{\bar{c}_{\mathrm{L}}^2} = \frac{3}{5\pi D_{\mathrm{m}} r_{\mathrm{cell}} \bar{c}_{\mathrm{L}} T_{\mathrm{int}}}\Big[1 - \frac{5}{3}\Big(\frac{r_{\mathrm{cell}}^2}{\pi D_{\mathrm{m}} T_{\mathrm{int}}}\Big)^{1/2} + \mathfrak{q}_T\Big],$$

*with*

$$0 \le \mathfrak{q}_T \le \frac{5}{3\sqrt{\pi}}\frac{r_{\mathrm{cell}}^2}{D_{\mathrm{m}}T_{\mathrm{int}}} + \frac{5}{18\sqrt{\pi}}\Big(\frac{r_{\mathrm{cell}}^2}{D_{\mathrm{m}}T_{\mathrm{int}}}\Big)^{3/2}.$$

*In an ideal solution a small change of chemical potential, in units of $k_{\mathrm{B}}T_{\mathrm{K}}$ with $T_{\mathrm{K}}$ the absolute temperature, equals the relative change of concentration, so the same number is, to first order in the relative error, the variance of the cell's estimate of its ligand's chemical potential in units of $(k_{\mathrm{B}}T_{\mathrm{K}})^2$.*

Proof in Section S38; checked in Check S47.159. The leading term is the known formula for a sphere that counts the molecules inside its volume without hindering their diffusion: Mora and Wingreen (2010) give $3/(5\pi D_{\mathrm{m}} r_{\mathrm{cell}} \bar{c}_{\mathrm{L}} T_{\mathrm{int}})$, in our symbols, as their Eq. 10 and attribute it to Berg and Purcell (1977), whose abstract states a least fractional error of about $(T_{\mathrm{int}}\bar{c}_{\mathrm{L}} r_{\mathrm{cell}} D_{\mathrm{m}})^{-1/2}$. We take the attribution of the constant from the former and know the latter from its abstract. The appendix rederives the leading term; the finite-time correction $-\frac{5}{3}(r_{\mathrm{cell}}^2/(\pi D_{\mathrm{m}}T_{\mathrm{int}}))^{1/2}$ and the bound on $\mathfrak{q}_T$ are computed here. Bialek and Setayeshgar (2005) bound the relative root-mean-square error of a single receptor of linear size $a_{\mathrm{r}}$ below by $(\pi D_{\mathrm{m}} a_{\mathrm{r}} \bar{c}_{\mathrm{L}} T_{\mathrm{int}})^{-1/2}$ (their Eq. 35 in the preprint, in our symbols); sensor models change the constant, not the scaling.

**Example S14.9** (A cell that senses chemical potential and not gravity)**.** A cell of radius $r_{\mathrm{cell}} = 5$ $\mu$m has an exterior with three components: the ligand concentration $\bar{c}_{\mathrm{L}} + \eta^{\mathrm{L}}_t + c'_{\mathrm{L}}(x,t)$, where $\eta^{\mathrm{L}}$ is a slow signal (an Ornstein–Uhlenbeck process with correlation time $\tau_{\mathrm{L}} = 10$ s and a standard deviation of a few percent of $\bar{c}_{\mathrm{L}}$) and $c'_{\mathrm{L}}$ is the thermal fluctuation of Proposition S14.8; the component $\eta^{\mathrm{g}}$ of the gravitational acceleration along the cell's axis, which changes with the organism's posture; and constant offsets. The cell's acts affect outcomes that its preferences weight through $\eta^{\mathrm{L}}_{t+h}$ only, at a horizon of seconds. Chemoreception is the volume-time average of Proposition S14.8 with $T_{\mathrm{int}} = 1$ s, $D_{\mathrm{m}} = 500$ $\mu$m$^2$/s and $\bar{c}_{\mathrm{L}} = 1$ nM; the candidate gravity sensor is the equilibrium height of buoyant particles in a compartment of height $Z_{\mathrm{box}} = 10$ $\mu$m. The computed values (Check S47.160) are these.

(1) Coupling of the chemical channel. The relative error is $1.13 \times 10^{-2}$ in the leading term and $1.01 \times 10^{-2}$ exactly: since $r_{\mathrm{cell}}^2/D_{\mathrm{m}} = 0.05$ s, the finite-time factor on the variance is 0.81 (0.79 without the remainder $\mathfrak{q}_T = 0.020$). The values of this item use the leading term, which overstates the noise. A signal with standard deviation 5 (1) percent of $\bar{c}_{\mathrm{L}}$ has signal-to-noise ratio 19.7 (0.79) per reading, captured fraction 0.952 (0.441), and predictive information 0.755 (0.224) nats at $h = 1$ s and 0.069 (0.031) nats at $h = 10$ s; with the exact variance the ratios are 24.3 (0.97) and the captured fractions 0.960 (0.493). These ratios, fractions and

information values treat the one-second average as a sample of $\eta^{\mathrm{L}}$ at the end of the window plus the thermal noise, with $h$ measured from the end of the window. The reading averages the signal as well: with $a = T_{\mathrm{int}}/\tau_{\mathrm{L}} = 0.1$, the average of the Ornstein–Uhlenbeck signal has variance $2a^{-2}(a-1+e^{-a}) = 0.967$ times that of the signal and covariance $a^{-1}(1-e^{-a}) = 0.952$ times that variance with $\eta^{\mathrm{L}}$ at the end of the window, and the predictive information of the averaged reading is 0.651 (0.201) nats at $h = 1$ s and 0.064 (0.028) nats at $h = 10$ s in the leading term, and 0.664 (0.232) and 0.065 (0.032) nats with the exact variance.

(2) The thermal fluctuation is too fast. The autocorrelation of its volume average decays like $|\mathbb{B}|(4\pi D_{\mathrm{m}}u)^{-3/2}$ for $D_{\mathrm{m}}u \gg r_{\mathrm{cell}}^2$, and in the Gaussian approximation, with the exact variance of the reading, a one-second reading carries $2.0 \times 10^{-3}$ nats about the next one and $5.3 \times 10^{-7}$ nats about the one ten seconds ahead. These are computed values; the large-time forms are leading-order asymptotics without a remainder bound.

(3) Constant offsets ($\bar{c}_{\mathrm{L}}$, the magnitude of the acceleration) are learned once (Lemma S14.7).

(4) Coupling of gravity. A particle of buoyant mass $\mathsf{M}_{\mathrm{b}}$ has the truncated Boltzmann height law $\propto e^{-\mathsf{M}_{\mathrm{b}}\eta^{\mathrm{g}}\mathsf{z}/(k_{\mathrm{B}}T_{\mathrm{K}})}$ on $[0, Z_{\mathrm{box}}]$, with Fisher information $(\mathsf{M}_{\mathrm{b}}/(k_{\mathrm{B}}T_{\mathrm{K}}))^2\,\mathrm{Var}(\mathsf{z})$ about $\eta^{\mathrm{g}}$, so that, with $g$ the gravitational acceleration and $x_{\mathrm{g}} = \mathsf{M}_{\mathrm{b}}gZ_{\mathrm{box}}/(k_{\mathrm{B}}T_{\mathrm{K}})$, one independent reading at $\eta^{\mathrm{g}} = g$ has signal-to-noise ratio $g^2$ times this Fisher information, which equals $1 - (x_{\mathrm{g}}/2)^2/\sinh^2(x_{\mathrm{g}}/2)$, for changes of $\eta^{\mathrm{g}}$ of order $g$. This ratio is $x_{\mathrm{g}}^2/12$ to leading order for $x_{\mathrm{g}} \ll 1$, increases with $x_{\mathrm{g}}$ and stays below one. For a protein of 50 kDa with buoyancy factor 0.27 (partial specific volume 0.73 $\mathrm{cm}^3/\mathrm{g}$) at 310 K, $x_{\mathrm{g}} = 5.1 \times 10^{-7}$ and the ratio is $2.2 \times 10^{-14}$ per reading: about $4.5 \times 10^{13}$ independent readings give a ratio of one. A particle of radius 1 $\mu$m and density excess 500 $\mathrm{kg}/\mathrm{m}^3$ has $x_{\mathrm{g}} = 48$, and sedimentation dominates thermal motion: its ratio is one to within $4 \times 10^{-18}$, the value for a height that is exponentially distributed with the scale height $Z_{\mathrm{box}}/x_{\mathrm{g}}$, so that in this model no single particle reaches the ratio 19.7 of the 5 percent ligand signal per reading. The ratio measures small changes about $\eta^{\mathrm{g}} = g$. A reversal of $\eta^{\mathrm{g}}$ reflects the height law about the middle of the compartment, and the rule that reads the sign of $\eta^{\mathrm{g}}$ from the half in which the particle lies errs with probability $1/(1 + e^{x_{\mathrm{g}}/2})$, which is $3.8 \times 10^{-11}$ at $x_{\mathrm{g}} = 48$ and within $10^{-7}$ of one half for the protein.

(5) Relevance of gravity. In the model $\eta^{\mathrm{g}}$ is independent of $\eta^{\mathrm{L}}$ and absent from the preferred outcomes, so its pragmatic increment is zero (Proposition S14.6(ii)), and so is its epistemic increment if the model target is $\eta^{\mathrm{L}}$. It varies at the organism's timescale, so predictability is not what excludes it.

The ligand's chemical potential couples to the blanket above thermal noise, is predictable at the cell's horizon and enters its preferences; gravity fails the coupling clause, with a signal-to-noise ratio more than thirteen orders of magnitude below one per particle and reading, against a chemical estimate that integrates about $7.9 \times 10^3$ molecule counts, and it fails the relevance clause by construction. The two readings do not pool equally: the count $7.9 \times 10^3$ is $5\pi D_{\mathrm{m}}r_{\mathrm{cell}}\bar{c}_{\mathrm{L}}T_{\mathrm{int}}/3$, the inverse relative variance in the leading term, and with the exact variance the chemical reading averages about 315 ligand molecules in the ball over about 31 effective snapshots, some $10^4$ molecule-samples; a gravity reading that pooled as many independent protein heights would have ratio $2 \times 10^{-10}$, still nearly ten orders of magnitude below one. Item (4) is a known physical bound recovered: its number is the barometric exponent $x_{\mathrm{g}}$ of equilibrium statistical mechanics entered into a Fisher-information identity, no element of the active-inference framework changes it, and the framework contributes only its classification under the coupling clause of Proposition S14.6. Figure 18 draws items (1) and (4) against the radius of the buoyant particle: below $x_{\mathrm{g}} = 1$ the ratio of gravity sensing stays under

1/12, below both ligand signals, and the exact ratio $1-(x_{\mathrm{g}}/2)^2/\sinh^2(x_{\mathrm{g}}/2)$ passes the ratio 0.79 of the 1 percent signal at $x_{\mathrm{g}}=4.66$ and approaches one from below where sedimentation dominates, so that the small-$x_{\mathrm{g}}$ form $x_{\mathrm{g}}^2/12$ overstates it there.

*Remark* S14.10 (Interpretation: neurons, statocytes and vestibular cells). This remark is an interpretation, and its physiological facts are textbook statements that we did not check against primary sources. A neuron in a body does not sense gravity because a molecular sensor has the negligible coupling computed for the protein in item (4) of Example S14.9, and because its preferred outcomes, transmitter release and firing, do not depend on the direction of gravity. Cells that do sense gravity use dense bodies of the size and density excess of the sedimenting particle of that item, the statoliths of plants and the otoconia of the vestibular organs, for which one height reading has a ratio of one and gives the sign of the axial acceleration with negligible error, and serve organisms whose preferred outcomes, a growth direction or balance, depend on that direction. Whether a variable belongs to an agent's environment therefore depends on the agent's scale and preferences, and an organism, a meta-agent of cells, senses gravity through specialized members. The statement that nerve cells sense chemical potentials but not gravity holds for typical neurons in this sense and fails for vestibular hair cells, whose transduction is mechanical.

## S14.5 Aggregates as external states

**Definition S14.11** (Aggregates as external states)**.** Let a population have joint state $x_t$, the agents' states and the world, with linear dynamics $x_{t+1}=\mathbb{A}x_t+\mathfrak{n}_t$ and $\mathfrak{n}_t$ independent across rounds. A declared aggregate is $\mathsf{L}x_t$, with $\mathsf{L}$ of full row rank fixed in advance by an institution or a measurement procedure. It is exactly closed when $\ker\mathsf{L}$ is invariant under $\mathbb{A}$, so that $\mathsf{L}x_{t+1}=\bar{\mathbb{A}}\mathsf{L}x_t+\mathsf{L}\mathfrak{n}_t$ for some $\bar{\mathbb{A}}$ and every initial law (Theorem 11.8 without input); closed of order $d$ when $d$ further linear functionals $\mathsf{L}'x_t$ can be adjoined so that $(\mathsf{L}x_t,\mathsf{L}'x_t)$ is exactly closed, the least such $d$ being its minimal order $d^*$ in the sense of Definition 11.6(iii), which counts adjoined functionals of the joint state and bounds from above the tolerance-dependent closure order $\mathrm{ord}(\alpha_{\mathrm{M}})$ of Definition 5.1(iii), a count of functionals of the filter state (see the discussion after Theorem 11.8); and approximately closed when its memory is below a declared bound. For a scalar stationary aggregate $\bar{x}_t$ the lag-$u$ Markov error is

$$\Delta_u^{\mathrm{agg}}=\left|\mathrm{Corr}(\bar{x}_t,\bar{x}_{t+u})-\mathrm{Corr}(\bar{x}_t,\bar{x}_{t+1})^u\right|,$$

which compares with the lag-one correlation, unlike the Markov error $\Xi$ of Theorem 5.4, which compares with the slowest eigenvalue.

**Proposition S14.12** (Exact closure of aggregates)**.** *(a) (Polls and opinion averages.) For a linear recursion $x_{t+1}=Wx_t+\zeta_t$ on the agents, receiver first, the average $\bar{x}=N^{-1}\mathbf{1}^{\top}x$ is exactly closed if and only if all column sums of $W$ are equal, $\mathbf{1}^{\top}W=\bar{w}\mathbf{1}^{\top}$; then $\bar{x}_{t+1}=\bar{w}\bar{x}_t+N^{-1}\mathbf{1}^{\top}\zeta_t$. This is Definition 11.7 for the partition with one block: every agent carries the same total weight in all agents' updates, its own included.*

*(b) (Interchangeable types.) Let the agents form types $\mathfrak{t}=1,\dots,\mathfrak{T}$ of sizes $N_{\mathfrak{t}}$, with*

$$x_{i,t+1}=A_{\mathfrak{t}}x_{i,t}+\sum_{\mathfrak{t}'}\mathsf{B}_{\mathfrak{t}\mathfrak{t}'}\bar{x}_{\mathfrak{t}',t}+\mathsf{H}_{\mathfrak{t}}\eta_t+w_{\mathfrak{t},t}^{\mathrm{com}}+w_{i,t}^{\mathrm{id}}\qquad(i\in\mathfrak{t}),$$

*where $\bar{x}_{\mathfrak{t}}$ are the type averages, $w^{\mathrm{com}}$ common type shocks and $w_{i,t}^{\mathrm{id}}\sim\mathcal{N}(0,Q_{\mathrm{id}})$ idiosyncratic shocks, all independent, and $\eta$ is a Gauss–Markov world. Then $(\eta_t,\bar{x}_{1,t},\dots,\bar{x}_{\mathfrak{T},t})$ is exactly closed, with $\bar{x}_{\mathfrak{t},t+1}=A_{\mathfrak{t}}\bar{x}_{\mathfrak{t},t}+\sum_{\mathfrak{t}'}\mathsf{B}_{\mathfrak{t}\mathfrak{t}'}\bar{x}_{\mathfrak{t}',t}+\mathsf{H}_{\mathfrak{t}}\eta_t+w_{\mathfrak{t},t}^{\mathrm{com}}+\bar{w}_{\mathfrak{t},t}^{\mathrm{id}}$ and $\bar{w}_{\mathfrak{t},t}^{\mathrm{id}}\sim\mathcal{N}(0,Q_{\mathrm{id}}/N_{\mathfrak{t}})$. If the type-level transition $\mathbb{A}_{\mathrm{ty}}=\mathrm{diag}(A_{\mathfrak{t}})+[\mathsf{B}_{\mathfrak{t}\mathfrak{t}'}]$ has $\rho(\mathbb{A}_{\mathrm{ty}})<1$, the stationary deviation of the type*

*averages from their infinite-population limit, the same dynamics without $\bar{w}^{\mathrm{id}}$, is independent of that limit, with covariance*

$$\Sigma^{\mathrm{fl}} = \sum_{u\geq 0} \mathbb{A}^u_{\mathrm{ty}} \operatorname{diag}(Q_{\mathrm{id}}/N_{\mathfrak{t}})(\mathbb{A}^u_{\mathrm{ty}})^\top, \qquad \|\Sigma^{\mathrm{fl}}\| \leq \frac{\|Q_{\mathrm{id}}\|}{\min_{\mathfrak{t}} N_{\mathfrak{t}}} \sum_{u\geq 0} \|\mathbb{A}^u_{\mathrm{ty}}\|^2.$$

*(c) If in (b) two agents $i, i'$ of one type have own dynamics $A_i \neq A_{i'}$ in place of the common $A_{\mathfrak{t}}$, the type averages are not exactly closed.*

Proof in Section S38; checked in Check S47.161. For scalar states, (b) is Definition 11.7 for the partition into types with the world as one more block, and the law of large numbers adds an error whose variance is of order $1/\min_{\mathfrak{t}} N_{\mathfrak{t}}$, so that the deviation itself is of order $(\min_{\mathfrak{t}} N_{\mathfrak{t}})^{-1/2}$. Exact closure of an aggregate is the linear analog of lumpability of a partition of a Markov chain (Kemeny and Snell, 1960), which we know in restatement, and type homogeneity is its population form.

**Proposition S14.13** (Heterogeneous timescales give aggregates memory)**.** *(a) A stationary Gaussian scalar process with positive variance is Markov if and only if its autocovariance is $\mathsf{R}(u) = \mathsf{R}(0)\alpha_*^{|u|}$ for some $\alpha_*$; a scalar aggregate is therefore Markov exactly when $\Delta^{\mathrm{agg}}_u = 0$ for every $u$.*

*Let $x_{i,t+1} = \alpha_i x_{i,t} + w^{\mathrm{com}}_t + w^{\mathrm{id}}_{i,t}$ for $i \leq N$, with persistences $\alpha_i \in (-1,1)$, a common shock of variance $q_{\mathrm{com}}$ and idiosyncratic shocks of variance $q_{\mathrm{id}}$, all independent, $q_{\mathrm{com}} + q_{\mathrm{id}} > 0$; let $\mathbb{Q}$ be the empirical law of the $\alpha_i$ and $\bar{x}_t$ the stationary average.*

*(b) $\mathsf{R}(u) = \operatorname{Cov}(\bar{x}_t, \bar{x}_{t+u}) = \int \alpha^u \mathbb{M}(\mathrm{d}\alpha)$ for $u \geq 0$, with the positive measure*

$$\mathbb{M}(\mathrm{d}\alpha') = q_{\mathrm{com}}\Big(\int \frac{\mathbb{Q}(\mathrm{d}\alpha)}{1-\alpha\alpha'}\Big)\mathbb{Q}(\mathrm{d}\alpha') + \frac{q_{\mathrm{id}}}{N}\frac{\mathbb{Q}(\mathrm{d}\alpha')}{1-\alpha'^2}.$$

*(c) $\bar{x}$ is Markov if and only if all $\alpha_i$ are equal.*

*(d) If the $\alpha_i$ lie in $[\alpha_-, \alpha_+] \subset [0,1)$, then, with $\bar{\mathbb{M}} = \mathbb{M}/\mathbb{M}(\mathbb{R})$, for every $u$*

$$\Delta^{\mathrm{agg}}_u \leq \frac{\operatorname{Var}_{\bar{\mathbb{M}}}(\alpha)}{(1-\alpha_+)^3} \leq \frac{(\alpha_+ - \alpha_-)^2}{4(1-\alpha_+)^3}.$$

*(e) If the $\alpha_i$ take $\mathfrak{T}$ distinct values, the type averages are exactly closed and the minimal order of $\bar{x}$ is $d^* = \mathfrak{T} - 1$.*

*(f) (Long memory.) As $N \to \infty$ with the common shock only, let $\mathbb{Q}$ have density $(1-\alpha)^{\mathfrak{p}-1}\mathsf{f}_0(\alpha)$ on $[0,1)$, with $\mathsf{f}_0$ continuous on $[0,1]$, $\mathsf{f}_0(1) > 0$ and $\frac{1}{2} < \mathfrak{p} < 1$. Then $\mathfrak{m}_l = \int \alpha^l \mathbb{Q}(\mathrm{d}\alpha) \sim \mathsf{f}_0(1)\Gamma(\mathfrak{p})l^{-\mathfrak{p}}$, and the limiting aggregate $\bar{x}_t = \sum_{l\geq 0} \mathfrak{m}_l w^{\mathrm{com}}_{t-1-l}$ has*

$$\mathsf{R}(u) \sim q_{\mathrm{com}}\big(\mathsf{f}_0(1)\Gamma(\mathfrak{p})\big)^2 \mathrm{B}(1-\mathfrak{p}, 2\mathfrak{p}-1)u^{1-2\mathfrak{p}},$$

*with $\Gamma$ the Gamma function and $\mathrm{B}$ the Beta function, so $\sum_u \mathsf{R}(u) = \infty$: the aggregate has long memory although every agent is an autoregression of order one.*

Proof in Section S38; checked in Check S47.161. Parts (c) to (f) are Assumption A7 in aggregate form: an aggregate of heterogeneous members is not Markov, the memory it needs grows with the number of distinct timescales it mixes, and with a continuum of timescales reaching the unit root no finite memory suffices. Part (f) is a common-innovation case of the aggregation route to long memory

and is not new. The review of Leipus et al. (2013), whose first two sections we read, traces that route to Granger (1980), a paper we know from its record and from the review, for autoregressions with a random persistence that has a Beta law. For a mixing density proportional to $(1-\alpha)^{\beta}$ near one and common innovations of infinite variance, the review reports that the average normalized by $1/N$ converges to a moving average in the common innovations with coefficients $\mathbb{E}\alpha^{l} \sim \text{const} \cdot l^{-\beta-1}$, which are not summable for $\beta < 0$; with $\beta = \mathfrak{p} - 1$ these are the weights $\mathfrak{m}_l$ of (f). The review credits this limit to Puplinskaitė and Surgailis (2009), which we did not open, and notes that the general aggregation scheme of Zaffaroni (2004), which we did not open either, includes the case of common innovations. Part (f) is the Gaussian counterpart with finite variance, for which we give a self-contained proof with the explicit constant. Example S11.5 shows the input-driven form of the same effect: the memory kernel of an average of traders with heterogeneous persistences survives the law of large numbers. The third closure mechanism, timescale separation, is Theorem 5.4: an outside agent that reads the average of a collective inherits its bound on the Markov error $\Xi$, set by a dispersion of the members' leaks or of the row and column sums of the drift against the gap between the inward and outward rates of the precision-based contrast (reversible drift) or of the drift-based contrast (general drift). Part (d) is the discrete-time analog for uncoupled members, with the dispersion of the persistences against the gap $1 - \alpha_{+}$.

### S14.6 Firms and people

**Example S14.14** (A firm with a production lag reads one market aggregate)**.** The market state $\eta_t = (\eta_{1,t}, \eta_{2,t})$ has independent scalar autoregressive components with persistences $F_1 > F_2 > 0$, a slow trend in demand and a fast shock, and stationary variances $P_{0,1}$ and $P_{0,2}$; the value of the good is $\eta_t^{\text{val}} = \eta_{1,t} + \eta_{2,t}$. At the fully revealing price of Proposition 10.5(b) we model the spot price as $p_t = \eta_t^{\text{val}} + \mathsf{n}_t^{\text{p}}$, with $\mathsf{n}_t^{\text{p}}$ independent of the market state and of variance $(\sum_i 1/R_i^s)^{-1}$, the noise of the statistic that the price reveals, of order $1/N$ for $N$ traders. A firm sells at $t+h$ what it produces at $t$, with profit $p_{t+h}a - \frac{\mathsf{k}_a}{2}a^2$ for output $a$; its optimal output is $\mathbb{E}[p_{t+h} \mid \text{readings}]/\mathsf{k}_a$, and its excess loss over full information on $\eta_t$ is $\text{Var}(\mathbb{E}[p_{t+h} \mid \eta_t] \mid \text{readings})/(2\mathsf{k}_a)$. The decision functional of the present, $(F^h)^{\top}\mathbf{1} = (F_1^h, F_2^h)$, turns from $(1,1)$ toward $(1,0)$ as $h$ grows. The firm reads one aggregate, which we take to be noiseless: the spot price, functional $(1,1)$, neglecting the price noise $\mathsf{n}_t^{\text{p}}$, whose variance is of order $1/N$ (with it, $\mathcal{L}_{\text{spot}}$ is the formula in Section S38, and the comparison with the slow index below is that of the noiseless limit); a slow index such as a trend estimate or a long-dated contract, functional $(1,0)$; or a futures price of maturity $h$, the market forecast of the value at $t+h$, functional $(F_1^h, F_2^h)$. Up to the factor $1/(2\mathsf{k}_a)$ the excess losses are

$$\mathcal{L}_{\text{spot}}(h) = (F_1^h - F_2^h)^2 \frac{P_{0,1}P_{0,2}}{P_{0,1}+P_{0,2}}, \qquad \mathcal{L}_{\text{slow}}(h) = F_2^{2h}P_{0,2}, \qquad \mathcal{L}_{\text{fut}}(h) = 0,$$

so the futures price at the firm's own lag is sufficient. This holds by definition of the futures functional, for any dynamics, since the futures price is the conditional expectation that the output decision needs and a conditional expectation is sufficient for itself; Theorem 14.7(a) with $M = (F_1^h, F_2^h)$ states the same fact. That the futures price of the firm's own lag suffices is the separation theorem of production under price uncertainty, here in its risk-neutral form and by definition of the futures functional: for a competitive firm that can trade futures, production decisions depend only on the futures price and input costs, whatever the firm's subjective distribution of the future spot price (Feder et al., 1980), which we know from its record and abstract. The sufficiency of the futures price is therefore a known result recovered, and only the comparison of the other two readings is specific

to the declared two-mode exterior. The spot price beats the slow index exactly for $h < h^{\times}$, with

$$h^{\times} = \frac{\ln(1 + (1 + P_{0,2}/P_{0,1})^{1/2})}{\ln(F_1/F_2)}.$$

For $F = \mathrm{diag}(0.95, 0.6)$ and $P_0 = \mathrm{diag}(1, 2)$, $h^{\times} = 2.187$, and at $h = 4$ the information-optimal rank-one reading of the market state is the slow component $(1, 0)$, while the decision-optimal reading is the futures functional $(1, 0.159)$ (Check S47.163): the canonical variate is what to read for prediction in general, and the decision functional what to read for this decision. The price is treated as exogenous, without price impact.

*Remark* S14.15 (Interpretation: the firm as a meta-agent). This remark is an interpretation. Read as a meta-agent (Definition 11.1), the firm has as sensory states its members' readings of aggregates (prices, indices, futures, its own sales), as active states its output and prices, which enter the market as performative acts (Proposition S2.6), as internal states a belief about the market pooled by consensus on information contributions (Proposition 5.7), and as external states the market state and the other firms' aggregate output. Every exterior variable it reads is an aggregate, closed or carrying memory as Propositions S14.12 and S14.13 say, and its horizon and decision functionals fix which aggregates it should read. Whether its pooled belief and output meet the clauses of Theorem 11.13 is the subject of Section 11, and Example S11.11 computes them for two firms whose abstract physics contains the competitor's coarse states. A firm whose output is a non-negligible share of supply and that reads the price as a message about demand without removing its own act misreads its own act as evidence (Proposition S2.6(a)); removing it is the cavity rule of Theorem 12.2, and the size of the error depends on the price impact, which we have not modeled.

**Example S14.16** (A person senses the economy through one or two numbers)**.** The economy's state has four independent modes with correlation times $\tau = (40, 8, 2, 0.5)$ periods and innovation variances $(0.05, 0.2, 0.5, 1)$, so stationary variances $(1.03, 0.90, 0.79, 1.02)$, read in mode coordinates as trend, cycle, local prices and noise. A person has three equally weighted decisions $\mathfrak{d}$ with quadratic losses: consumption at horizon 1 with functional $(0.3, 0.3, 1, 0.6)$, a job decision at horizon 8 with $(1, 1, 0.3, 0)$, and saving at horizon 40 with $(1, 0.4, 0, 0)$. Its total excess loss is $\mathrm{tr}\,\mathsf{J} - \mathrm{tr}(\mathsf{G}_s\mathsf{J})$ with $\mathsf{J} = \sum_{\mathfrak{d}} P_0^{1/2}(F^{h_{\mathfrak{d}}})^{\top}\psi_{\mathfrak{d}}\psi_{\mathfrak{d}}^{\top}F^{h_{\mathfrak{d}}}P_0^{1/2}$, so under the Shannon budget $I(s;\eta_t) \le I_{\max}$ of rational inattention the optimal signal reads the leading eigenvectors of $\mathsf{J}$ by reverse water-filling (Theorem 14.3(c)), at most one channel per decision. The eigenvalues of $\mathsf{J}$ are $(1.118, 0.263, 0.016, 0)$. The first channel reads the functional $(1, 0.46, 0.31, 0.04)$ on (trend, cycle, local prices, noise), a slow aggregate, and the second $(0.31, -0.15, -1, -0.13)$, mainly local prices. With 0.05 or 0.3 nats per period the person reads the slow aggregate alone; the second channel enters above 0.72 nats, and the third, mainly the cycle, only above 3.50 nats (Check S47.164). The economy, for this person, is one or two numbers, chosen by the horizons and weights of its own decisions.

## S14.7 Sensing across levels and memberships

A person reads aggregates that meta-agents produce from their members' sensing, and belongs to several meta-agents at once. Two results bound what passes through such chains and how one budget is shared among memberships.

**Lemma S14.17** (Composition across levels)**.** *If* $\mathsf{X}_1 \to \mathsf{X}_2 \to \mathsf{X}_3$ *is a jointly Gaussian Markov chain with nonsingular covariances and* $\mathsf{C}_{12}$, $\mathsf{C}_{23}$ *and* $\mathsf{C}_{13}$ *are the whitened cross-covariances of* $(\mathsf{X}_1, \mathsf{X}_2)$, $(\mathsf{X}_2, \mathsf{X}_3)$ *and* $(\mathsf{X}_1, \mathsf{X}_3)$*, then* $\mathsf{C}_{13} = \mathsf{C}_{12}\mathsf{C}_{23}$*. Hence* $\sigma_{\mathsf{j}}(\mathsf{C}_{13}) \le \sigma_{\mathsf{j}}(\mathsf{C}_{12})\sigma_1(\mathsf{C}_{23})$ *and* $\mathrm{rank}\,\mathsf{C}_{13} \le \min(\mathrm{rank}\,\mathsf{C}_{12}, \mathrm{rank}\,\mathsf{C}_{23})$.

Proof in Section S38; checked in Check S47.165. Along a chain from the world through members' sensing and a meta-agent's aggregate to a person's reading of it, every canonical correlation between the world and the reading is at most the product of the stages' leading canonical correlations, and a broadcast code $T_c$ of rank $r_c$ passes at most $r_c$ canonical variates; this is consistent with Conjecture S8.26 and does not prove it, since that conjecture concerns hop distance. If a meta-agent's sensory statistic is linear in its members' readings (Definition 11.1), its exterior channels and their noises fix its budget, and its closure timescale sets its horizon, of order $1/\varpi_1$ for the average of Theorem 5.4, then Theorem 14.3 and Propositions S14.2 and 14.4 say which coarse exterior variables it should sense: the leading canonical variates at its horizon, slower and coarser for slower meta-agents. Whether a collective has such a statistic and horizon is the subject of Section 11.

**Proposition S14.18** (A person with physical and abstract channels)**.** *Let a person's decisions be* $\mathsf{Y} = (\psi_{\mathfrak{d}}^\top \eta_{t+h_{\mathfrak{d}}})_{\mathfrak{d}}$*, let it read the physical world through* $s^{\mathrm{ph}} = H^{\mathrm{ph}}\eta_t + v^{\mathrm{ph}}$ *and the aggregates of its memberships, one modality* $c \in \mathcal{M}_i \subseteq \mathcal{M}$ *each, through* $y^{\mathrm{agg}} = (T_c^{\mathrm{agg}}\eta_t + \nu_c)_c$*, all noises independent, and put* $\mathsf{Z}_h = \big[(F^{h_{\mathfrak{d}}})^\top \psi_{\mathfrak{d}}\big]_{\mathfrak{d}}$.

*(a)* $I(\mathsf{Y}; s^{\mathrm{ph}}, y^{\mathrm{agg}}) = I(\mathsf{Y}; y^{\mathrm{agg}}) + I(\mathsf{Y}; s^{\mathrm{ph}} \mid y^{\mathrm{agg}})$*, each term in closed form through conditional covariances.*

*(b) (Sufficiency.)* $I(\mathsf{Y}; s^{\mathrm{ph}} \mid y^{\mathrm{agg}}) = 0$ *if and only if* $\mathsf{Z}_h^\top \operatorname{Cov}(\eta_t \mid y^{\mathrm{agg}})(H^{\mathrm{ph}})^\top = 0$*. With noiseless aggregates,* $\operatorname{col}\mathsf{Z}_h \subseteq \operatorname{row} T^{\mathrm{agg}}$ *suffices, which is the condition of Theorem 14.7(a).*

*(c) (Long horizons.) For a scalar decision* $\mathsf{Y} = \psi^\top \eta_{t+h}$ *and every physical channel,* $I(\mathsf{Y}; s^{\mathrm{ph}} \mid y^{\mathrm{agg}}) \le -\frac{1}{2}\ln(1 - \mathfrak{r}_h^2)$ *with* $\mathfrak{r}_h^2 = \operatorname{Var}(\psi^\top F^h \eta_t \mid y^{\mathrm{agg}}) / \operatorname{Var}(\mathsf{Y} \mid y^{\mathrm{agg}})$*. In a reversible exterior with noiseless aggregates that read the* $r_{\mathrm{agg}}$ *slowest mode amplitudes, for* $h \ge 1$ *and* $\psi \ne 0$*,*

$$\mathfrak{r}_h^2 \le \frac{|\lambda_{r_{\mathrm{agg}}+1}(F)|^{2h} \sum_{\mathsf{j} > r_{\mathrm{agg}}} (\mathsf{e}_{\mathsf{j}}^\top P_0^{1/2} \psi)^2}{\psi^\top (P_0 - F^h P_0 (F^h)^\top)\psi},$$

*which decays geometrically in* $h$*.*

*(d) (Attention across memberships.) If* $F$ *and* $Q$ *are block diagonal across contents, as in Proposition 10.2(a), and each decision loads on one content, the decision matrix* $\mathsf{J}$ *of Example S14.16 is block diagonal across memberships. The budget-optimal sensor then reads channels within one membership's content each, when eigenvalues of different blocks differ, and the allocation across memberships is coupled only through the common water level: raising the decision eigenvalues of one membership leaves the capacity spent on every other membership's channels unchanged or lowers it.*

Proof in Section S38; checked in Check S47.166. For long-horizon decisions aggregates of the slow modes are therefore sufficient up to an explicit error, and a physical channel can add much only for short-horizon decisions. Part (d) answers the single-agent part of the open problem of attention across modalities (Open problem S10.29) for a Shannon budget and pragmatic value alone; the case of a channel count $\kappa$ and of the full objective $G_h^{\mathrm{att}}$ of Definition 14.5 remains open. In the case that (d) covers, with block-diagonal decision structure one budget shared across modalities does not mix them within channels, but it couples the amounts of attention through the water level. With decision eigenvalues $(2.0, 0.4)$ for a firm and $(1.2, 0.9)$ for a polity, a person attends only the firm's leading channel below 0.26 nats, adds the polity's two channels at 0.26 and 0.54 nats, and the firm's second channel only above 1.76 nats.

*Remark* S14.19 (Reading an aggregate one contributes to). When a person's own report enters an aggregate it reads, a poll it answered or a price at which it trades, a reader at interpretation level 0 counts its own information twice: in market form, by Proposition 10.5(c), its believed precision exceeds the true one by the precision of its prior and of its own signal. Lemma S14.17 assumes that the reading is a downstream stage of the chain; with own contributions the chain is a loop, and the correct rule removes the own contribution before reading, the cavity rule of Theorem 12.2. Literal reading of such an aggregate is the overvaluation case of Section 3, treated as literal top-down reading in Proposition 12.3.

*Remark* S14.20 (Interpretation: corporations and people sensing abstract environments). This remark is an interpretation of the results of Section 14 and of this section, which make the author's examples precise in linear-Gaussian models for sensors optimal under the declared objective of Definition 14.5; the one-step expected free energy of the framework does not imply that objective (Remarks S14.3 and S14.4), so each sentence below says what such a sensor reads and not what every agent of the framework senses. A corporation senses an informational environment of prices, indices and contracts because at its decision horizon the variables that are predictive and relevant to its acts are slow aggregates, which under the homogeneity conditions of Proposition S14.12 are closed processes that it can model as its exterior (Theorem 14.7 and Example S14.14). A person senses the economy through one or two numbers when its long-horizon decisions load on slow modes and its attention is limited (Example S14.16), and senses the physical world for its short-horizon decisions (Proposition S14.18(c)). A cell senses its ligand's chemical potential and not gravity because of coupling, horizon and relevance together (Example S14.9). In each case an abstract environment is the exterior seen through the leading canonical variates and decision functionals at the agent's horizon. Fields and Levin (2022) propose a related scale-free view, competent behavior in metabolic, transcriptional, morphological and spatial spaces framed by hierarchical active inference; Theorems 14.3, 14.6 and 14.7 give one quantitative criterion for which space an agent behaves in.

## S14.8 Open problems

**Open problem S14.21** (Sensor design for filtering agents)**.** For an agent that integrates its readings with a Kalman filter, which sensor of $\kappa$ channels maximizes $I(s_{\le t}; \eta_{t+h})$ in steady state? Lemma S14.1 settles summaries of a fully observed past, and Theorem 14.3 applied to a window of past observations settles a fixed window. With memory a sensor of one variable also informs others through the dynamics, so the answer is not the canonical variates in general: a single channel that reads a mixture of a slow and a fast mode yields a process whose past carries information about both. The problem couples the choice of sensor with the Riccati equation.

**Conjecture S14.22** (Selection beyond linear-Gaussian exteriors)**.** *For a stationary exterior with finite predictive information and an agent with a budget on $I(s;\eta_t)$, the optimal representations for prediction at horizon $h$, the information bottleneck between present and future, concentrate as $h$ grows on functions of the exterior that are invariant under the slowest part of the dynamics: eigenfunctions of the transfer operator of largest modulus, or their nonlinear analogs.*

For linear-Gaussian exteriors with diagonalizable $F$ and a gap in the moduli of its eigenvalues, the statement of Conjecture S14.22 follows from Theorem 14.3(b) and Proposition 14.4(b): the budgeted optimum reads the leading canonical variates of $(\eta_t, \eta_{t+h})$, and along the horizons at which it reads $\mathsf{m}$ of them, if $|\lambda_{\mathsf{m}}(F)| > |\lambda_{\mathsf{m}+1}(F)|$, the bound of Proposition 14.4(b) with $\kappa = \mathsf{m}$ places their span within an angle of order $(|\lambda_{\mathsf{m}+1}(F)|/|\lambda_{\mathsf{m}}(F)|)^h$ of the span of the $\mathsf{m}$ slowest mode amplitudes. Defective $F$ and counts $\mathsf{m}$ without such a gap are not covered; discrete and nonlinear versions, such as the renormalizing generative models of Friston et al. (2025), are covered by no

proof here. For field theories, the degrees of freedom that information-bottleneck compression keeps are reported to be the operators of lowest scaling dimension (Gordon et al., 2021), a result we know from its abstract that would tie the conjecture to the relevant variables of Section 13. Three narrower questions remain open: whether the number of variables read at a fixed budget is non-increasing in $h$ for non-reversible exteriors, where transient growth of a non-normal $F^h$ can raise a canonical correlation at short horizons, so that we expect counterexamples, although we did not search for them; whether, in a network whose agents' decision eigenvalues depend on the state of each modality, the water-level coupling of Proposition S14.18(d) generates a cross-modal coupling $a_\times$ in the sense of Proposition 10.3, the network part of the open problem of attention across modalities in Section 10, which stays open together with its single-agent case for a channel count $\kappa$ and the full objective $G_h^{\mathrm{att}}$; and which sensor model fits a given cell, since absorbing spheres and receptors with binding kinetics have constants other than those of Proposition S14.8 (Bialek and Setayeshgar, 2005).

## S15 Additional material for Section 16: predictions and tests

This section holds the extended discussion of Section 16 that the main text summarizes.

### S15.1 Proxies for collectives, lengths and memory

Every quantity that selects or measures a collective belongs to the world-conditional law of the agents' errors, their law given the world path and the noise of every broadcast heard by several receivers (Definition 4.7), so a test needs a record of the world path or of a proxy for it, outside the static case noted at the end of the next paragraph. From replicate windows or a long stationary series one estimates the world-conditional covariance $\Sigma^c$, with records written by agents as nodes of the law (Remark S4.11), and its inverse, the precision $\Pi^c$. If $\Pi^c$ is diffusive, with nonpositive off-diagonal entries and nonnegative row sums, the contrast $r(C) = \varkappa_{\mathrm{in}}(C)/\varkappa_{\mathrm{out}}(C)$ of a region, the slowest relaxation rate of its internal disagreement divided by that of its average, is defined (Definition 4.10). The baseline $\bar{r}_{\mathrm{hom}}$ is the supremum of contrast over regions of every size in a comparison medium declared with the model, which keeps the population's declared geometry and replaces its couplings and leaks by their means, or is the exchangeable state of a declared mechanism that generates the couplings of a population without geometry (Definition 4.11), and, under the moment bound, a collective is a region whose contrast reaches a level $r_0$ fixed in advance above the baseline and the leak heterogeneity, reported with its significance $r(C)/\bar{r}_{\mathrm{hom}}$ (Definition 4.12). Conditioning cannot be skipped: unconditional correlations count the common response to shared shocks as coherence and, when $\Sigma^c$ is entrywise nonnegative and the common mode is uniform, cap the ultrametric contrast of every proper subset (Proposition S10.27). For one arc of a ring of 60 agents with two barriers and variances $q = Q = 1$ of local noise and world innovation this contrast falls from 3.37 with conditioning to 0.114 without it at leak 0.05; the cap grows with $q/Q$, and at $q/Q = 100$ the unconditional value at the same leak is 2.470, so the suppression holds when local noise is not large against the world innovation (Check S47.116).

The interaction order parameter $\Omega_{\mathrm{int}}$ averages over pairs of members the conditional mutual information of their paths given the shared exogenous drive, which comprises the initial state, the world innovations and the record and sender-side broadcast noises (Definition 5.15). With the drive recorded in $n_{\mathrm{rep}}$ independent replicate windows, $2n_{\mathrm{rep}}$ times the plug-in Gaussian estimate for a pair, computed on the path coordinates that are not measurable with respect to the drive and with their conditional covariance nonsingular, is a likelihood-ratio statistic with a chi-squared limit, as $n_{\mathrm{rep}} \to \infty$ at fixed dimensions, under a causally separating null, and Monte Carlo $p$-values

against partner randomization are valid under exchangeability (Proposition 5.18). Conditioning on the world path instead fails when agents act on the world (Examples S5.24 and S5.25), and an estimator of $\Omega_{\text{int}}$ without a recorded drive is open (Open problem S5.27). The static case has a partial route: for a static or stationary world without performative acts, the equal-time precision $\Pi^c$ is locally identifiable from members' statistics alone under the sparse-plus-low-rank conditions of Chandrasekaran et al. (2012), which are not verified here for diffusive precisions; the estimator is open when acts drive the world, when the conditional precision is not sparse, or when the leaks are concentrated on few agents (Section S10.9).

The Markov length $\xi_M$ is the decay length, in hops, of the conditional mutual information between an agent and all agents beyond $h$ hops given those in between (Definition 4.5). It must be measured on its own: the correlation length does not bound it (Proposition 4.6), while, under the clustering and growth hypotheses of Theorem S4.6(a), half the inverse decay rate of $\Pi^c$ does. The correlation length $\xi$ is the decay length of standardized covariances (Definition 8.8). The second-moment length $\xi_{2,d} = (\varsigma_d^2/(2\varepsilon^{\Pi}))^{1/2}$ in coordinate $d$, with $\varsigma_d^2$ the largest second moment of an agent's couplings and $\varepsilon^{\Pi}$ a uniform leak, comes from $\Pi^c$ (Theorem 4.14) or, in translation-invariant diffusive media, from the second moment of the covariance (Proposition S8.27(a)). The correlation time $\tau = -1/\ln \rho(W)$, with $W$ the time-invariant error-transfer matrix of the level-0 network, is the limit of $T^*(\theta)/\ln(1/\theta)$ as $\theta \to 0$, where $T^*(\theta)$ is the longest lag at which some pair of linear readouts keeps correlation at least $\theta$ (Proposition S8.14).

A medium's effect on memory follows from four rates: the read weight $\beta_k$, write weight $\omega_k$ and decay $\delta_k$ of each record, and the agents' leak $\varepsilon$. When reading displaces self-weight and $\varepsilon$ lies below every $\omega_k + \delta_k$, records lengthen the memory of the collective mean exactly when $\sum_k \beta_k(\delta_k - \varepsilon)/(\omega_k + \delta_k - \varepsilon) < 0$ (Proposition 8.12).

## S15.2 Proxies for calibration, conventions and the quantities of H5

Calibration shows in behavior. The weight a receiver gives a sender's message reveals the precision it assigns, and its calibration class is read from whether that weight stays fixed (C0) or follows the sender's reliability, learned on rounds on which the receiver observes the referent (C1g) or from its own innovations (C1i) (Definition 2.6). In the static two-group model a calibrated receiver with at least $\kappa$ in-group senders available withdraws from every cross-group sender whose mistranslation cost exceeds its information advantage (Theorem 6.4); in the dynamic network a translation error $\phi$ costs $D_{\text{cal}}(1 - \cos\phi)$ in calibrated noise, with $D_{\text{cal}}$ twice the covariance of the sender's belief with the world (Lemma 7.3).

Conventions are measured through their domains. The Séguy curve, the expected fraction of binary features in which two members at distance $d$ differ, rises at short distance like $\pi^{-1}(2d/\xi)^{1/2}$ for thresholds of a leaky field with exponential correlation and linearly, with slope the wall probability, for walled codes whose walls are sparse and do not cluster (Proposition S10.13); on a ring the mean domain length of the code game is of order $(J/h)^2$, with $J$ the believed value lost per mismatched neighbor and $h$ the spread of idiosyncratic pulls (Theorem 6.11). Polity size needs the correlation volume $\chi$ of members' preferred policies, the fixed cost $f$ of a polity, the cost $g$ of a member matched to an uncorrelated reference and the declared policy rule: some finite polity has a lower per-capita cost than the limit that unbounded polities approach exactly when $f < g\chi$ under a central rule and $f < g\chi/2$ under the mean rule (Proposition 10.6). The weights $a$ and leaks $\varepsilon$ behind these quantities are inferred from border effects, price correlations or Séguy curves only through a declared model (Section 10.6).

The quantities of the extension behind hypothesis H5 have proxies of the same kind. A meta-agent's coarse states are declared linear functions of measured variables, such as the members'

summed orders, a poll or a price, and whether a declared coarse exterior has a law of its own is measured by its relative predictive defect, the share of its variance that the fine state predicts beyond the coarse state and act (Definition 11.6), or, for a scalar aggregate, by the lag-$u$ Markov error $\Delta_u^{\mathrm{agg}} = |\mathrm{Corr}(\bar{x}_t, \bar{x}_{t+u}) - \mathrm{Corr}(\bar{x}_t, \bar{x}_{t+1})^u|$ (Definition S14.11). Overconfidence is measured by the overconfidence factor, the mean-square error of an estimate times the precision it claims, which is one for a calibrated estimate (Proposition 12.3). A record's read weight $\beta$ and write weight $\omega$ are the weight that members give it and the rate at which it follows their mean, and the memory of the collective mode is the slowest relaxation time of the members' mean (Proposition 12.4). In a nested population the coupling ratio $\mathfrak{q}$ is the factor by which an agent's total coupling to the agents it first meets at a level is multiplied from one level to the next (Proposition 10.7, which assumes $1/b < \mathfrak{q} < 1$ at branching $b$, a range in which this coupling falls, only for the limit of its parts (b) and (c)), and, in a hierarchy whose coarse media have uniform leak and their units at distinct sites of $\mathbb{Z}^D$, as for cubic blocks of a translation-invariant precision on $\mathbb{Z}^D$, the second-moment length of a level is computed from its coarse medium in units of its blocks (Proposition 13.7). An agent's horizon $h$ is the lag between a reading and the outcome that the decision based on it affects, its capacity is a number of channels or a budget in nats, and what it senses is estimated from the loadings of its readings or acts on exterior variables (Definitions 14.1 and 14.5).

### S15.3 Qualifications of the formation predictions

Formation needs more qualification than a table cell of Table 17 holds. In the two-group model at $\gamma = \infty$ with noiseless angle observations the chain from calibrated precision through attention to contrast is proved, the contrast at the levels of the window of Theorem 7.8(c), which is nonempty at every computed parameter point (Numerical observation S7.6), and the declared start is then itself the split (Theorem 7.6(e)). For $\gamma_S < \gamma < \infty$ only reachability of the split branch is a theorem, about the frozen-calibration mean-field map (Theorem 7.6(a),(b)); the contrast of the reached state is computed at the base point (Numerical observation S7.7), and the stochastic-approximation step from the full model to the map is open (Open problem 7.14). Formation there is a calibration-level effect of misreading joint structure: receivers that know their translation errors pay no mistranslation cost, so from a start law exchangeable given the world path the split fails the translation-aware form of the exact-fusion persistence clause, which is proved (Proposition 7.12), while for full cross-covariance-aware exact fusion this conclusion is heuristic; above the threshold $\gamma_T$ the split coincides with the partition by codes. With reading noise specific to each receiver and at the levels of the window of Theorem 7.8(c), the split differs from the deletion nulls by proof, and the difference from the frozen-environment null is proved when the initial states that frozen messages carry are deterministic or conditioned on (Proposition 7.9(c)). Online calibration from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per cell, with reruns of a cell differing by up to three replicates, and is open (Numerical observation 7.11 and Open problem S7.13). Persistence under exact fusion with drifting frames is a conjecture, which would be refuted by an instance with drifting frames in which exact-fusion receivers attend cross edges at the merged rate at every policy precision (Conjecture S7.12); formation beyond the model is a conjecture, which would be refuted by a population of its class in which, at arbitrarily large policy precision, a stable configuration keeps a learned and attended cross-class edge (Conjecture 7.13).

## S16 Additional material for Section 17: related work

This section holds the extended discussion of Section 17 that the main text summarizes.

## S16.1 Formation status

Formation is where the simulations surveyed in Section 17.1 and our results meet. For two groups whose private frames differ by a rotation (Definition 7.2), we prove for the frozen-calibration mean-field map that, above a threshold in policy precision, grounded learning, calibrated precision and attention at finite capacity drive the population from a declared start to the split branch (Theorem 7.6). At the exact split, reached with top-$\kappa$ attention ($\gamma = \infty$) and exact in-group translations ($\sigma_\alpha^2 = 0$), the two groups are exactly the collectives at every level of the window of Theorem 7.8(c), a window of admissible levels that is nonempty at every point of the parameter grid we computed (Numerical observation S7.6); at finite policy precision their collective status is a numerical observation at one parameter point (Numerical observation S7.7). Roles are not supplied as priors, but the split follows a supplied difference of frames, and the result is qualified: the step from the stochastic model to the map is open (Open problem 7.14); calibration learned online from a symmetric start reaches the split's attention pattern (not necessarily a split selected by contrast) in 0 to 7 of 8 replicates per parameter cell, counts that differ by up to three between reruns of a cell, and is open (Numerical observation 7.11); and the collectives arise as a calibration-level effect of misreading joint structure and, at high policy precision, coincide with the partition by codes. From a start law exchangeable given the world path they fail the translation-aware form of the exact-fusion persistence clause (Proposition 7.12), while for full cross-covariance-aware exact fusion this conclusion is heuristic. Persistence with drifting frames (Conjecture S7.12) and formation beyond this model (Conjecture 7.13) are conjectures.

## S16.2 Communication models

In the duet of Friston and Frith (2015a), agents with a shared model synchronize, and in its sequel (Friston and Frith, 2015b) they learn one scalar map from narrative to articulation. Synchrony there comes from shared models rather than from a preference, so the preference to predict and be predicted that we take from the duet needs a formalization of its own; we give two, and at correct codes only being predicted contains coded agreement (Proposition S2.8). Agents that choose partners by expected free energy reach consensus, polarization or splits (Albarracin et al., 2022; Kastel et al., 2023). In federated inference (Friston et al., 2024) agents speak through a learned likelihood from beliefs to words in an assumed common frame, and language is presented as emerging in a shared econiche, the closest precedent we found for anchoring meaning in a shared world. Reading a partner's belief as an observation is literal reading in our terms (Definition 2.4), and Çatal et al. (2025) show that it produces echo chambers, which sharing likelihoods removes; this is the double counting of H2 in a discrete model. We know these works through their abstracts and secondary accounts. The deontic cues of Constant et al. (2019) are records in the sense of our record model (Definition 2.9), one mode of a transfer kernel that also covers direct and performative channels (Proposition 2.8); the word stigmergy is Grassé's (Grassé, 1959).

## S16.3 Lieb–Robinson bounds and clustering

Hypothesis H4 rests on bounds for the speed of influence and the decay of correlations. Lieb and Robinson (1972) proved that finite-range quantum spin dynamics have a finite group velocity, with classical analogs on lattices (Marchioro et al., 1978; Raz and Sims, 2009). A Lieb–Robinson bound with a spectral gap gives exponential clustering (Nachtergaele and Sims, 2006; Hastings and Koma, 2006), Hastings (2004) treats Markov dynamics on networks, correlations are created only at finite speed (Bravyi et al., 2006), and power-law interactions have a hierarchy of linear light cones (Tran et al., 2020). Our light cone (Theorem 8.5) is the discrete finite-range case, in which the cone is exact:

from a product law, agents more than $2t$ hops apart are independent given the world after $t$ rounds under any local rule, the exact counterpart of the statement of Bravyi et al. (2006). Our clustering theorem (Theorem 8.9) is the classical linear analog of exponential clustering, with the contraction from grounding and sensing gain in the role of the gap. These are known forms applied to a new object, and we claim no novelty for them as mathematics. We change the hypotheses: every length, time and volume statement assumes a stationary world-conditional law with finite second moments and a named contraction (Assumption 8.1), because attention selected by expected free energy with sender-specific codes can amplify; without the moment bound only the finite-time light-cone statements survive (Theorem 8.7), and we call amplifying profiles rumors (Remark S8.8). For link weights decaying as a power of distance, influence fronts have finite speed exactly when the power is at least $D + 1$, the threshold of the free-particle cone of Tran et al. (2020) (Proposition S8.18).

On $\mathbb{Z}^D$ the covariance decays at least at the Ornstein–Zernike rate (Ornstein and Zernike, 1914) (Theorem 8.10): along an axis the rate $m$ solves $\cosh m = 1 + \varepsilon/(2a)$, so for a small leak $\xi \approx \sqrt{a/\varepsilon}$ hops, a physical length $\ell\sqrt{a/\varepsilon}$. That the rate is exact pointwise, with the power-law prefactor of the full Ornstein–Zernike form, rests on Green-function asymptotics (Ney and Spitzer, 1966) that we know through a restatement (Ballu, 2024), and the constant of the prefactor on a heuristic step, so we state both only as a remark (Remark S8.13); the rigorous theory for Ising models above the critical temperature is that of Campanino et al. (2003). For weights dominated by a reversible chain we prove the form of the Carne–Varopoulos estimate that we use (Carne, 1985; Varopoulos, 1985) (Theorem S8.11). For precisions, inverse-closedness of localized matrices gives some positive decay rate (Jaffard, 1990), and our Combes–Thomas bound makes the rate explicit (Theorem S4.6). In quantum many-body physics the decay length of conditional mutual information diagnoses mixed-state phases (Sang and Hsieh, 2025) and is finite for Gibbs states above a threshold temperature (Kuwahara et al., 2020); we import it into blanket theory as the Markov length.

### S16.4 Walk sums, belief propagation and track fusion

Hypothesis H2 has precise precedents in estimation. Loopy Gaussian belief propagation converges on walk-summable models with correct means and variances that capture only part of the walks (Malioutov et al., 2006), correct means hold on any topology when it converges (Weiss and Freeman, 2001), and belief propagation is exact on trees (Pearl, 1988). In tracking, double counting of common information is classical: Bar-Shalom and Campo (1986) give the fused covariance of two tracks that share process noise and show that ignoring the dependence overstates the gain from fusion, covariance intersection handles unknown correlations (Julier and Uhlmann, 1997), and channel filters fuse exactly on trees (Grime and Durrant-Whyte, 1994). The ladder from literal reading through the extrinsic message of belief propagation to exact fusion is thus known. We place it inside each receiver's generative model, as a learned model of the sender with a calibration class and an attention policy (Definitions 2.4 and 2.6), and H2 becomes misreading of joint structure: literal receivers overvalue sources in a stated scope, which is double counting (Proposition 3.4), and can undervalue them through common-information cancellation and sender-side completion (Examples S3.8 and S3.9).

Consensus filters reproduce the central Kalman filter under exact average consensus at every step, uncorrelated sensor noise and a known number of nodes (Olfati-Saber, 2005, Theorem 1). Consensus on information contributions reaches the union-model posterior geometrically and recovers that theorem as a limit (Proposition 5.7), while literal consensus, in which each member reads its neighbors' posteriors at level 1 with their posted precisions as law headers, counts a common prior once per member on regular graphs (Proposition S5.5). For exact fusion at a finite number of rounds, which we leave open (Open problem S5.6), Mossel et al. (2016) state in their abstract

that Bayesian agents with Gaussian signals who exchange estimates reach the full-information belief within $2N \operatorname{diam}(\mathcal{G})$ rounds.

### S16.5 Strong data processing and broadcasting

Fidelity is governed by strong data-processing inequalities, which originate with Ahlswede and Gács (1976). Polyanskiy and Wu (2017) bound information flow in Bayesian networks by site percolation, following Evans and Schulman (1999), and Polyanskiy and Wu (2019) bound the mutual information of distant variables of a Markov random field by bond percolation. For a Gaussian source and an additive Gaussian channel the Kullback–Leibler and $\chi^2$ contraction coefficients coincide (Makur and Zheng, 2020), so our coefficient, the top squared canonical correlation (Proposition S6.1), is known in substance; we prove it directly and derive its consequences in the framework, namely products along relay paths (Proposition S6.2), decay of information at twice the mass (Proposition S6.3) and percolation bounds on the information that a protocol carries about a distant source (Theorem 8.16).

On trees, the binary symmetric channel is reconstructible exactly when its squared correlation exceeds the percolation threshold (Bleher et al., 1995; Evans et al., 2000), and the Kesten–Stigum condition (Kesten and Stigum, 1966) is sufficient in general but not always sharp (Mossel, 2001). For Gaussian broadcasting on $b$-ary trees with edge correlation $\alpha$ we prove that $b\alpha^2 > 1$ is necessary and sufficient, with the branching number deciding on general trees away from the critical case by the theorem of Lyons (1990), read in a review (Theorems S8.19 and S8.20). At depth two or more the regime $b\alpha^2 > 1$ lies, in plain coordinates, outside the diffusive class on which contrast is defined (Proposition S8.22), and condensation of coherent information on general networks is a conjecture (Conjecture S8.23). With power-constrained Gaussian relays, information decays only like $\log\log h / \log h$ over $h$ hops (Polyanskiy and Wu, 2016); whether relays that re-encode content, such as writing, extend a modality's reach beyond the linear correlation length is open.

### S16.6 Group synchronization

Consistency is a synchronization problem. Singer (2011) recovers angles from noisy offsets spectrally, Bandeira et al. (2013) relate the smallest eigenvalues of the connection Laplacian to the least frustration of an assignment by inequalities of Cheeger type, and Abbe et al. (2018) show that on $\mathbb{Z}^D$ distant group differences can be recovered at small noise for $D \geq 3$ but not for some continuous groups at $D = 2$. Frustration is the product of signs around a plaquette (Toulouse, 1977), balance of signed and gain graphs is the same condition on cycles (Harary, 1953; Zaslavsky, 1989), and with rank-limited maps agreement is a global section of a discourse sheaf (Hansen and Ghrist, 2021). Our holonomy criterion (Theorem 6.5), that consensus with a given content exists exactly when the cycle holonomies fix it, imports its form for directed closed walks on strongly connected graphs from the author's gauge-theoretic manuscript (Dennis, 2026c) and states it on the fundamental cycles of weakly connected graphs, where requiring every content to be fixed gives the balance test of gain graphs. We expect this reduction to fundamental cycles to be classical in the theory of local systems on graphs and claim no novelty for it as mathematics. What the framework adds is the origin of the maps: translations are learned from grounded co-occurrence and true translations are flat (Proposition S6.5), so holonomy diagnoses errors in learned codes, and frustration acts like grounding (Proposition S6.8). Whether learned translations inherit the dimension dependence that Abbe et al. (2018) prove for independent noise is open. Cycle amplification under heterogeneous priors is a calibration failure, distinct from holonomy and from double counting (Theorem 6.8), and the absence of cyclic arbitrage says that log exchange rates are a gradient, as in the combinatorial Hodge decomposition used for ranking (Jiang et al., 2011) (Proposition S10.15). Noiseless relays

of fixed codes in a common metric are alternating projections, which converge to the projection onto the intersection (von Neumann, 1950; Halperin, 1962) at rates studied by Kayalar and Weinert (1988) and Badea et al. (2012); we prove the rate we use directly (Theorem S8.24), and random hop maps suggest that the rank of shared content falls with scale (Conjecture S8.26).

### S16.7 Mori–Zwanzig and closure

Closure asks when a collective's coarse variables have dynamics of their own. Zwanzig (1961) and Mori (1965) derived exact equations for selected variables with memory kernels and fluctuating forces, minimal realization goes back to Ho and Kalman (1966), and aggregation to the nearly decomposable systems of Simon and Ando (1961) and the lumpable chains of Kemeny and Snell (1960). Information theory offers related order parameters: informational closure, which implies sufficiency under a Markovian system–environment structure (Krakauer et al., 2020), causal blankets (Rosas et al., 2020), which we know through secondary accounts, and dynamical independence (Barnett and Seth, 2023), whose definition we read in the primary text. Our elimination of a collective's interior, with its memory kernel, its Hankel-rank count of memory variables and its condition for vanishing memory (Theorems 5.2 and 5.3 and Remark S5.1), is linear Mori–Zwanzig in discrete time. What we add is a declared closure property and its link to contrast. A small memory residual, statistical sufficiency and a slow retained spectrum are distinct conditions (Dennis, 2026d); we require the first together with inferential closure (Definition 5.1). Contrast controls the memory of the collective average for reversible drift, and a drift-based contrast with row and column dispersions does so for non-reversible drift (Theorem 5.4); the symmetric dispersion alone does not suffice (Example S5.3). The drift-based rates differ from the corresponding Rayleigh quotients of the stationary precision by at most a term set by the stationary probability current (Proposition 5.5), and the drift-based inward rate also keeps the couplings to outsiders, so the drift-based contrast can exceed the selecting contrast with no current at all (Section S5.1).

Barnett and Seth (2023) call a macroscopic variable dynamically independent of the microscopic system, in the context of an environment, when a conditional transfer entropy from the microscopic past to the macroscopic future vanishes; they compute it for linear state-space systems and identify the condition with the informational closure of Bertschinger et al. (2006), which we know through their restatement. Rosas et al. (2024), whose main text we read through targeted extraction, define informational, causal and computational closure for stationary processes in discrete time, prove the first two equivalent for macroscopic variables that are functions of the microscopic process, and characterize closed levels by strong lumpability of causal states, which are maximally predictive and minimal (Shalizi and Crutchfield, 2001) and, according to an abstract, are macrostates (Shalizi and Moore, 2025); the framework has no environment, no blanket and no bound for approximate closure. For linear-Gaussian collectives with full-rank noise two of these notions, dynamical independence and informational closure in its one-step form, meet ours (Section 11 and Proposition S11.8): in the stationary case without input the dynamical dependence of a coarse variable is at most the closure defect of Definition 5.1, and in that case the exact forms of dynamical independence, one-step informational closure and the closure of Theorem 5.3(b) coincide.

Near decomposability, known to us from an abstract (Simon and Ando, 1961), a secondary account (Simon, 1962) and a record (Courtois, 1977), is the picture behind the nesting of spectral contrast at weak grounding, through a hierarchy of timescales (Proposition S4.26; Theorem 4.18(iv)). Fast variables are eliminated with corrections in the ratio of timescales (van Kampen, 1985; Geigenmüller et al., 1983; Haken and Wunderlin, 1982) and rigorously in the limit of infinite separation (Pavliotis and Stuart, 2008; Berglund and Gentz, 2006; Givon et al., 2004), works we know from records and abstracts. Whether any of them bounds the memory of coarse variables of our kind at a finite ratio

we have not checked; Theorem 5.4 and Sections 11 and 13 give such bounds through contrast.

### S16.8 Potential games and logit dynamics

Agency asks when members' separate choices act as the choices of one agent. A game is an exact potential game if and only if every four-cycle of profiles has zero total payoff change (Monderer and Shapley, 1996, Theorem 2.8), a statement we read in restatements (Arefizadeh et al., 2024). Asynchronous logit revision then has a Gibbs stationary law (Blume, 1993), which we know secondarily, its convergence depends on asynchrony (Alós-Ferrer and Netzer, 2010), and near-potential games have logit laws near a Gibbs law (Candogan et al., 2013). Wolpert and Tumer (2001) design members' payoffs so that individual optimization serves a world utility. Our agency theorem (Theorem 5.10) finds such a potential instead of designing it: under its protocol, in which members share belief and data, announce their policies, share one precision and deliberate asynchronously, each member's expected free energy has the group expected free energy as an exact potential, while synchronous deliberation of two members with redundant slots does not reproduce the group posterior (Proposition S5.12). With heterogeneous beliefs the deliberation law stays within explicit distances of a Gibbs law of a potential (Theorem 5.11), with a bound on the ratio of transition probabilities sharper than that of Candogan et al. (2013), and joint acts keep a potential when members agree on how their acts combine and lose it when interventions interact and the members' displacements, predicted outcome minus target, differ along a joint effect (Theorem S5.17 and Example S5.18). The code game with a common value is an exact potential game on rings and on complete graphs, and on open chains of at least three agents only in its pairwise form (Proposition 6.10). The stochastic-approximation step behind formation, of the kind treated by Benaïm (1999), is not carried out (Section 7).

### S16.9 Opinion dynamics and social learning

Repeated weighted averaging (DeGroot, 1974), known to us secondarily, and the influence process of Friedkin and Johnsen (1990), whose equilibrium is a resolvent of the influence matrix with DeGroot's model as a limit, are the basic linear models of opinion dynamics; bounded confidence fragments opinions (Hegselmann and Krause, 2002; Deffuant et al., 2000), a rule that senders who want to be predicted follow on the sender's side (Proposition S2.10 and Conjecture S2.11); and in noisy consensus the variance of deviations from the average, which Bamieh et al. (2012) call coherence, scales badly with network size in one and two dimensions. Our ring, with $W_{ii} = c$, $W_{ij} = a$ between neighbors and local noise $qI$, carries a Friedkin–Johnsen influence structure with grounding and forgetting as stubbornness. Its world-conditional covariance $q(I-W^2)^{-1} = \frac{q}{2}[(I-W)^{-1}+(I+W)^{-1}]$ decays at the rate $m$ of the total-influence matrix $(I-W)^{-1}$, $\cosh m = 1+\varepsilon/(2a)$, because the second term decays at least as fast, at $\cosh m' = (1+c)/(2a)$. So the physical correlation length $\ell\xi \approx \ell\sqrt{a/\varepsilon}$ is the Friedkin–Johnsen influence length (Theorem 8.10), and its square grows like the diffusion constant $a\ell^2$ times the correlation time (Proposition S8.15). Approximately exponential decay of influence has been reported for forced linear diffusion models (Murray-Watters et al., 2026), Friedkin–Johnsen dynamics have been written with a Green operator (Boudourides, 2026), and $\ell\sqrt{a/\varepsilon}$ is the decay length of the standard massive lattice Green's function, so we do not present it as new. The result of Bamieh et al. (2012) is the control-theoretic counterpart of our massless limit (Theorem 6.12).

Social learning names the overvaluation case of H2 several times: persuasion bias (DeMarzo et al., 2003), naive herding (Eyster and Rabin, 2010) and correlation neglect (Enke and Zimmermann, 2019), with conditions for the wisdom of DeGroot learners (Golub and Jackson, 2010). Madsen

et al. (2018) report echo chambers among Bayesian agents who treat received beliefs as fresh evidence and drop partners outside a confidence window; in our terms these are literal receivers with withdrawal by disagreement, consistent with our results that weak and self-sealed echo chambers require misspecification (Theorem 3.10 and Proposition 3.11). Nguyen (2020) separates epistemic bubbles, which omit outside voices, from echo chambers, which discredit them; in our terms these are closure of attention (Definition 3.7) and miscalibrated precision.

### S16.10 Disorder and convention domains

Imry and Ma (1975) argue that a weak random field destroys the order of systems with continuous symmetry below four dimensions, by the domain argument that bears their name; Aizenman and Wehr (1989, 1990) prove that quenched randomness suppresses continuous symmetry breaking up to four dimensions and leaves the two-dimensional random-field Ising model with a unique Gibbs state, whereas the three-dimensional model has a phase transition (Bricmont and Kupiainen, 1988). For the two-dimensional model, Ding and Wirth (2023) prove that the box size at which the influences of the field and of the boundary condition on the magnetization are comparable grows, at zero temperature, exponentially in the power $-4/3$ of the field strength relative to the coupling. Dialect variation grows sublinearly with distance in the studies since Séguy (1971) that Nerbonne (2010) reports, and Burridge (2017) models dialect boundaries as interfaces with surface tension produced by a conformity nonlinearity.

We derive that conformity: under calibrated reception and uniform habits, expected free energy gives a logistic adoption rule, conformist above a threshold in policy precision for an affine value without pull, with an innovation floor that acts as a leak; at equal frequencies it has the value and slope of the power conformity rule of dialect models at matched exponent, which has no such floor (Propositions S6.15 and S10.12). Because the code game with a common value is an exact potential game on rings (Proposition 6.10), the one-dimensional domain law, with the Imry–Ma scaling, is a theorem there (Theorem 6.11); on open chains the same bound holds for the pairwise game. In higher dimension the code game is an exact potential game only for special value functions, and coarsening, domain sizes under random pulls and the bundling of isoglosses remain conjectures (Conjectures S6.16 to S6.18); the two-dimensional conjecture may carry the exponent $4/3$ if domain size is identified with the length of Ding and Wirth (2023), a heuristic of ours and not their theorem. Sparse, unclustered walls make Séguy curves rise linearly with distance, against the square-root rise of a leaky field with exponential correlation (Proposition S10.13).

### S16.11 Abstract collectives in the social sciences

Deutsch (1953) defined a people by the complementarity of its members' communication, tested by how fast and accurately messages get through; we read the defining chapter in the 1966 edition. Savage and Deutsch (1960) compare transaction flows with a null model built from marginal totals, and modularity compares within-group links with a null model (Newman, 2006). These are the closest precedents for contrast and for declared nulls (Definitions 4.10 and 5.17); the transaction-flow index measures contrast in message rates rather than in coherence, and our barrier and bypass results connect the two (Propositions 10.4 and S10.6). The ultrametric contrast of societies uses world-conditional correlations, since under a common mode unconditional ones cap it (Definition 4.17 and Proposition S10.27).

Hayek (1945) held that dispersed knowledge is communicated by prices, and rational-expectations aggregation (Grossman, 1976; Hellwig, 1980; Grossman and Stiglitz, 1980) is cited for attribution only. A Walrasian price formed from private signals counts the common prior once per trader, while

a revealing price is sufficient (Proposition 10.5), and public prices suffice exactly when the decision functionals lie in their row space (Proposition S10.16). Innis's time- and space-biased media (Innis, 1950), known to us secondarily, map onto the decay and reading range of records (Propositions 8.11 and 8.13); the mapping is ours.

Borders reduce trade between industrialized countries by 20 to 50 percent (Anderson and van Wincoop, 2003), and the distance effect on trade has persisted (Disdier and Head, 2008). Single-scale diffusive coherence in two dimensions predicts that distance elasticities fall as communication lengthens the correlation length, contrary to that persistence, while nested coupling gives scale-free elasticities (Propositions 10.7 and S10.24). Alesina and Spolaore (1997) trade the gains of size against the costs of heterogeneity; our criterion compares a polity's fixed cost with the heterogeneity weight times the correlation volume of preferences, under two declared policy rules (Proposition 10.6). Marchetti's daily travel budget of about one hour (Marchetti, 1994), constant only in aggregate (Mokhtarian and Chen, 2004), bounds the diameter of one-round sets (Proposition S10.20), and the territorial-unity condition behind urban scaling (Bettencourt, 2013), known to us through Ribeiro and Rybski (2023), is a one-round reach condition with a money budget. We read the scale-free correlations of starling flocks (Cavagna et al., 2010) as the massless regime, where the size bound of Corollary 4.16 is vacuous. These mappings are interpretations, and the framework does not cover ungrounded conventions, strategic deception, power or non-stationary history (Section 10).

### S16.12 Hierarchical active inference

In hierarchical dynamic models (Friston, 2008) the causes of each level are generated by the level above, whose prediction is an empirical prior on the level below, and inversion is a gradient flow on free energy in which precision-weighted prediction errors ascend and predictions descend, the predictive coding of Rao and Ballard (1999), cited for attribution. Kiebel et al. (2008), read through targeted extraction, impose a separation of timescales in which a slow level is a control parameter of a fast one, and state that it is not a property of coupled neuronal systems as such. In the deep temporal models of Friston et al. (2017a) a higher state generates the initial state of a lower sequence, the ascending message is a Bayesian model average and the descending one an empirical prior, and for continuous states belief propagation and variational message passing reduce to predictive coding; Friston et al. (2017b) and Pezzulo et al. (2018), known to us from records, extend the architecture to planning and control.

As we read them, these hierarchies do not double count. Their updates are steps on one free energy of the joint model, in which each factor appears once, and on linear-Gaussian trees predictive coding and mean-field message passing reach the exact posterior means (Proposition S12.6), while belief propagation excludes the recipient's own message by the cavity rule (Pearl, 1988). In the two-pass form of exact inference on trees the upward message is a likelihood, equivalently a prediction error relative to the cavity prior; in predictive coding errors are computed against current estimates, with top-down predictions weighted by the generative precision, and that scheme is also exact. Double counting arises only when a posterior that already contains the receiver's data is used as an independent prior, which is level-1 reading of a message whose provenance overlaps the receiver's data, the overvaluation case of H2 (Definition 2.4 and Proposition 3.4).

In these hierarchies the level above is another part of one model. We found no treatment in which the top-down prior is built from the receivers' own messages, as when members read a summary of the collective that they wrote, and none that analyzes such a circular hierarchy with interpretation levels, calibration classes, records or a null family that includes exact fusion. Section 12 does so in linear-Gaussian models: a member's two-level model of its collective as a latent cause is correct exactly when the collective coordinate is closed, and literal top-down reading, done

once, overcounts by an overconfidence factor below two in a scalar tree, and by factors growing linearly or geometrically in the number of rounds when a persistent record is rebuilt from literal posteriors. That prediction errors between agents cannot detect shared error (Lemma 3.6) has no counterpart in the hierarchical literature we read. In the author's gauge-theoretic program, as secondary descriptions summarize it, the parent's belief is written into each child's prior with a blend factor (Dennis, 2026b), a circular case whose exact top-down message is the cavity message.

### S16.13 Cultural priors and top-down social signals

Cultural priors reach individuals through inference of other agents' expectations in shared practices (Veissière et al., 2020), known to us from its abstract, through conventional affordances that rest on shared expectations rather than on correlations in the environment (Ramstead et al., 2016), and through deontic cues that consolidate as repeated actions raise the counts of the environment's likelihood parameters (Constant et al., 2019), with niche and organism treated symmetrically by Constant et al. (2018). In our terms these priors travel through records (Definition 2.9) and behavior, and none of these works asks whether members' own contributions to the cues are counted twice when the cues are read back. Synergetics, cited here for the slaving principle (Haken, 1983), has order parameters that the parts produce and that enslave them, a circularity carried to closed equations for opinion shares by Weidlich (1991), and Galam (1986) renormalizes majority opinion up a hierarchy of committees; we know the last two secondarily. According to its abstract, Morris and Shin (2002) find that more precise public information raises welfare when agents have no socially valuable private information and has ambiguous effects otherwise, which bears on the precision balance of Theorem 6.12.

None of these is a linear-Gaussian theory with calibration tested against a null. In Section 12 a member-written record that does not decay lengthens the memory of the collective mode and extends identity under turnover, at the price, for a grounded coordinate, of a shared-error floor that is nondecreasing in the weight members give the record; the precision-based contrast computed with the record as a node, at the record noise that makes the joint dynamics reversible, gains a factor independent of the write weight when the record is read strongly enough, and it is undefined for a noiseless global record as soon as two members are uncoupled. Linear conformity to a self-constituted prior selects no partition that the declared couplings and readership do not already contain, and formation by conformity is a conjecture.

### S16.14 Renormalization in active inference

Friston (2019) builds scales with two operators, in the parts we read. A grouping operator forms a particular partition from the Jacobian, a partition the text calls one of many, and eliminates internal states by conditioning on the blankets, adding the remainder to the fluctuations; a reduction operator keeps the slow eigenmodes of each blanket's Jacobian and adds the fast ones to the fluctuations. The composite is called a renormalization in the working sense of Lin et al. (2017), known to us from its record. In our reading it assumes a sparse Jacobian read as a blanket, a heuristic partition with free thresholds, elimination given the particle's own blanket at one time, white intrinsic fluctuations independent across blankets, a spectral gap in each blanket's Jacobian with the fast modes of other blankets treated as noise, Gaussian higher-level noise by a central-limit argument, and slow and fast modes of different particles that decouple, with noise correlations neglected. Accuracy rests on the separation of timescales, with the slaving principle (Haken, 1983) and center manifold theory cited, and we found no error bound in the pages we read.

Friston et al. (2021), read through targeted extraction, apply the operators to one subject's

imaging data and report time constants growing by a factor of about 2.37 and spatial support by about 2.13 per scale, figures we have not verified, for a partition that they state is not unique. Friston et al. (2023a) call their fixed flow operator an adiabatic approximation and list particles of particles as an open question, without a renormalization theorem. Friston et al. (2023b) give a population at a slow scale its own partition into external, sensory, active and internal kinds, the closest precedent we found, in conceptual form, for coarse external states defined as slow variables of other collectives. The renormalizing generative models of Friston et al. (2025) build conditional independence into a discrete hierarchy with one parent per state and keep the leading singular variates of image blocks; they decline to benchmark the examples, although they report MNIST classification accuracy (95.1% on the full test set, 99.8% on the half with higher marginal likelihood), and, in the parts we read, give no bound on what the reduction loses.

What these constructions lack is an account of what the reduction drops, which Section 13 supplies for linear-Gaussian models. The elimination of internal states is the Mori projection of Theorem 5.2(b) with its memory kernel, initial slip and colored noise discarded; it is exact for every initial law only when the retained states do not read the internal ones, so never for a particle whose active states read its internal states, and in computed instances its error falls like the inverse ratio of timescales once that ratio is ten or more, which we observe without proof. For the linear reversible analogue of the adiabatic reduction the static error and the memory are bounded by spectral quantities of the partition, and for block averages through the dispersion of the cut over the internal stiffness, which high contrast alone does not make small (Theorem 13.8(d), Section 13.7). That no error bound accompanies the active-inference renormalization is a search result covering the papers above and the replies on blankets, not the whole literature; it does not cover two later works that we have not read, the treatment of renormalizing generative models by Zaghw et al. (2026) and the engineering perspective on active inference of de Vries (2026).

### S16.15 Block spins and network renormalization

Kadanoff (1966) takes the magnetizations of cells much smaller than the coherence length as collective variables, the block counterpart of our hierarchies ending where a coarse cell reaches the correlation length (Corollary 4.16 and Section 13), a connection of ours; the renormalization group (Wilson, 1971; Wilson and Kogut, 1974; Cardy, 1996), known to us from records and secondary accounts, is cited for the standard theory, and the exponent of the leak under long-range couplings (Proposition S13.5(b)) is the thermal eigenvalue of the long-range Gaussian fixed point of Fisher et al. (1972), cited from its record and abstract. Jona-Lasinio (2001) presents it as a limit theorem with a Gaussian fixed point and shows that the Gaussian block recursion of Dyson's hierarchical model (Dyson, 1969) is exact; our nested medium (Proposition 10.7) is a massive model of this type and renormalizes exactly (Section 13). Among information-theoretic formulations (Mehta and Schwab, 2014; Bény and Osborne, 2015; Koch-Janusz and Ringel, 2018), Gordon et al. (2021) state in their abstract that bottleneck compression keeps the operators of lowest scaling dimension, which ties the selection of sensed variables (Section S16.16) to renormalization, without saying whether Gaussian theories on graphs are covered.

For networks without coordinates, Laplacian renormalization defines supernodes through diffusion (Villegas et al., 2023), beside geometric and multiscale renormalization (García-Pérez et al., 2018; Garuccio et al., 2023; Gabrielli et al., 2025), and, according to the abstract of Dörfler and Bullo (2013), Kron reduction maps a Laplacian with self-loops to another such Laplacian, which is the decimation step of Lemma S4.30. A diffusive world-conditional precision with positive leaks, which Theorem 4.9 provides under stated conditions, is a massive Gaussian free field on the communication graph, where these operations are exact. Under Assumption 13.1, Section 13 gives exact flows of

the axis rate and the second-moment length, shows that block averages in general leave the diffusive class, so that contrast between meta-agents uses a coarse medium (Definition 4.19), and finds that Laplacian renormalization does not see a uniform precision leak, agrees with the criterion $r \geq r_0$ for a population formed by two blocks of equal size with an equitable cut and equal internal gap, under the gap condition of Proposition S13.19, only on the interval of leaks of that proposition, which contains the window of Theorem 4.18(iv), and in an example with interleaved scales returns the selected regions at no single diffusion time (Section S13.6).

### S16.16 Predictive information, information bottlenecks and rational inattention

Predictive information, the mutual information between past and future, measures the complexity of a process (Bialek et al., 2001), and the information bottleneck keeps what a compressed signal says about a relevance variable (Tishby et al., 2000). For jointly Gaussian variables the optimal bottleneck projects onto canonical-correlation directions that enter at critical trade-offs (Chechik et al., 2005, Theorem 3.1); the past-future bottleneck of Creutzig et al. (2009), read in its abstract, adds state dimensions at structural transitions, and Still (2014) treats predictive inference by the same method; and the optimal predictive representation of a driven oscillator rotates as the prediction lag grows (Sachdeva et al., 2021), read through targeted extraction. Rational inattention limits the capacity of the channel from observations to actions (Sims, 2003), and price-setting firms attend to firm-specific conditions when these are volatile or consequential (Maćkowiak and Wiederholt, 2009), a result we know from its abstract, as we know the review of Maćkowiak et al. (2023). Fields and Levin (2022, Section 7.2) state in words that organisms model only the principal components of their experience.

The selective-sensing results of Section 14 are thus anticipated in substance: for linear-Gaussian exteriors, information-optimal sensors read leading canonical variates between present and future with allocations of the water-filling form, and error-optimal ones read leading eigenvectors of a decision matrix with reverse water-filling. For the joint treatment of a cap on the number of channels and a budget we claim no priority: the budget form is the information curve of Chechik et al. (2005), whose Section 8 also names a dimension fixed in advance, and Section 14 records that Scharf and Thomas (1998) treat rank reduction and a rate budget together in canonical coordinates. What may be new is at most the dependence on horizon and control, for which the selection of slow features at one step has a precedent in Creutzig and Sprekeler (2008), read in its abstract, together with the remark that the one-step attention expected free energy over world channels is horizon-free (Proposition S2.7 and Remark S14.3); a selection principle of coupling, predictability and relevance; and conditions under which aggregates of many agents form a closed environment. Rank cascades concern compression, not products of relay maps, and neither prove nor refute Conjecture S8.26. The concentration constant of Section 14 is derived there, with Berg and Purcell (1977), known to us from its abstract, as context.

### S16.17 Organizations as information processors

Economics treats the firm as an information processor: team theory with its linear-quadratic-Gaussian case (Radner, 1962; Marschak and Radner, 1972), decentralized processing under delay (Radner, 1993), hierarchies as allocations of knowledge (Garicano, 2000), organizations as costly channels with a code (Arrow, 1974; Crémer et al., 2007) and design as information processing (Galbraith, 1974); organizations also scan their environment as interpretation systems (Daft and Weick, 1984). We know these works from their records, the formal code through a secondary account and the interpretation systems from an abstract, and cite them for attribution, as precedents for a firm whose environment is a market and for low-rank market signals (Propositions 10.5 and S10.16).

This supplement has worked examples in linear-Gaussian models (Examples S11.11 and S14.14), two firms trading on one price and a firm that reads a futures price at its production lag, whose readings for corporations are interpretations, and whether the linear-quadratic-Gaussian teams of Radner (1962) contain such statements we have not checked; the sufficiency of the futures price at the firm's own lag is a known result recovered, the separation theorem of production under price uncertainty in its risk-neutral form (Feder et al., 1980), which we know from its record and abstract.

## S17 Additional material for Section 18: the discussion

This section holds the extended discussion of Section 18 that the main text summarizes.

### S17.1 Other clauses of H1

Of the clauses of Hypothesis H1 other than the boundary clause, whose status is given in Section 18.1, the contrast clause, part (b), is a definition (Definition 4.12), and the clauses of part (c) on closure, on the deletion nulls and on agency are theorems in narrower settings (Theorem 5.20 and Remark S5.28). A linear coarse variable obeys an exact reduced equation whose memory, initial-state and colored-noise terms all vanish for every initial law exactly when the dynamics map the kernel of the coarse map into itself, which for a selection of agents means that the retained agents do not hear the eliminated ones, Assumption A7 in exact form (Theorems 5.2 and 5.3). Contrast above one bounds the memory of the collective average for the reversible continuous-time diffusion whose drift is the block on the collective of the diffusive world-conditional precision, with the exterior held fixed (Theorem 5.4). For drift that is not reversible and whose symmetric part $M_{\mathrm{sym}}$ is positive definite, a drift-based contrast above one, which is not the selecting contrast, bounds the memory kernel and, when the row and column dispersions of the drift are at most a quarter of the gap between the drift-based rates, the Markov error, and a closure theorem in terms of the selecting contrast for such drift is open (Theorem 5.4 and Open problem S5.4). The order parameter conditioned on the shared drive vanishes under causally separating nulls, and for scalar errors without records deletion, combined with $\mathfrak{N}_{\mathrm{eff}}$ of Table 6 when members act, leaves no region with contrast above one (Proposition 5.16). Each member minimizes only its own expected free energy (Assumption A5). Under an asynchronous protocol with shared data and intentions, deliberation is reversible exactly when the members agree on the redundancy of their inputs, and its stationary law is then a group posterior when some belief carries the agreed redundancies (Theorem 5.10); otherwise it lies within explicit distances of one (Theorem 5.11), and synchronous deliberation of two members with full-support habits and finite expected free energies, or deliberation without shared data, has a product stationary law, which with redundant inputs is the posterior of no group model with product habits and $\gamma_C > 0$ in the classes considered there; the failure of agency is thus relative to the declared class, and for members without shared data a class with free product habits that admits $\gamma_C = 0$ gives the same law agency index 1 (Proposition S5.12 and Example S5.14). Agency does not depend on contrast. A meta-agent is a definition (Definition 5.19), whose requirements to differ from partner randomization and to survive exact fusion are not consequences of these theorems, and no population analyzed in the manuscript, synthetic or empirical, is shown to meet every clause, so that whether the clauses can hold jointly is an open problem (Open problem S5.29): the split of the two-group model is attributed to misreading, for the reasons given under H3 in Section 18.1, and the decisive nulls have been applied to no abstract collective (Section 10.6).

## S17.2 Parts (b) to (d) of H5

Part (b) of Hypothesis H5, whose part (a) is treated in Section 18.1, holds in linear-Gaussian hierarchies. A member's two-level model of its collective is exact exactly when the collective coordinate is closed, a Markov model of the collective coordinate alone can be exact without closure, and its least rate of misspecification is the Markov defect, while under reversible drift contrast above one bounds the misspecification per sampled time to fourth order in the dispersion of leakage (Propositions S12.2 and S12.5, Example S12.3, and Lemma S12.4). On Gaussian trees exact inference sends likelihoods up and leave-one-out priors down, and predictive coding and mean-field message passing reach the exact means (Theorem 12.2 and Proposition S12.6), so, as we read it, hierarchical active inference does not double count. Double counting arises when a posterior that already contains the receiver's data is read as an independent prior, the overvaluation case of H2: in a scalar tree one reading leaves the reader overconfident by a factor below two, and a record rebuilt every round from its readers' posteriors has an overconfidence factor that is unbounded in the number of rounds under the union, pooling and product rules, reaching $4.59 \times 10^3$ for three members of unit information after eight rounds under the product rule (Propositions 12.3 and S12.7). A nondecaying record that members write and read lengthens the memory of the collective mode, at fixed exterior, by a factor between $1 + \beta/\omega$ and $1 + \beta/\omega + \varepsilon/\omega$ and extends identity under turnover, while for grounded coordinates it raises a shared-error floor that is nondecreasing in its read weight and invisible in social and top-down innovations (Proposition 12.4, Corollary 12.5, Theorem 12.7, and Lemma 12.6); the factor by which that floor grows is below the memory factor and approaches it in a persistent world with weak grounding (Theorem 12.7(e)). Linear conformity to a self-constituted prior selects no partition that its declared couplings and readership do not contain (Proposition 12.8), so it amplifies declared structure and forms none; formation by conformity, which would not rest on misreading, is a conjecture (Conjecture S12.25). Top-down precision on a self-written record of a grounded coordinate raises the common-mode part of every member's stationary error above its value at zero precision whenever $\beta + \omega \leq 1$ (Proposition S12.20), and in the computed instances the total error rises, or falls before it rises, according to the share of local noise (Numerical observation S12.19). Contrast with the record as a node was defined only at the reversible record noise in every instance computed with outsiders (Numerical observation S12.16).

Part (c) is proved for linear sensors of linear-Gaussian exteriors. Sensors with a rank limit or a Shannon budget that carry the most information about the exterior at a horizon read the leading canonical variates between present and future, with allocations of the water-filling form (Theorem 14.3), a form for which we claim no priority. In reversible exteriors the same slow variates are read at every horizon, under a budget and with distinct mode moduli no more of them as the horizon grows; in non-reversible diagonalizable ones with a gap in the mode moduli the optimum turns toward the slow mode amplitudes asymptotically, with an explicit bound that is informative only at long horizons; and in diffusive exteriors long horizons under a budget read coarse spatial averages (Propositions S14.2 and 14.4). Preferences select within this order and can reverse it, and the one-step attention expected free energy of the framework is horizon-free, so that selection by predictive information needs a horizon in the epistemic term or a pragmatic term (Theorem 14.6, Example S14.5, and Remark S14.3). An added channel has value only if it is correlated with the exterior at the agent's horizon given what the agent already reads, and pragmatic value only if it bears on outcomes that the agent's acts can compensate (Proposition S14.6). Aggregates are exactly closed for interchangeable types and keep memory when persistences differ, with long memory for the limiting aggregate of a continuum of persistences whose density diverges at the unit root like $(1-\alpha)^{\mathfrak{p}-1}$ with $\frac{1}{2} < \mathfrak{p} < 1$ (Propositions S14.12 and S14.13), and the effective Hayek dimension of a reversible exterior is bounded by a count that falls with the horizon while the exact one is

constant for invertible dynamics (Theorem 14.7). Beyond linear-Gaussian exteriors the selection is a conjecture (Conjecture S14.22).

Part (d) is proved for Gaussian coarse-graining under the moment bound and a diffusive precision with positive leaks (Assumption 13.1). Decimation and block averaging are exact maps of Gaussian laws; decimation closes on nearest-neighbor precisions on $\mathbb{Z}$ with a Chebyshev flow in which the leak is relevant with exponent 2, the axis rate and the second-moment length flow exactly in every dimension, block averages of the nearest-neighbor field on $\mathbb{Z}$ leave the diffusive class at every leak, and the nested medium renormalizes exactly, with a block's contrast, for $\mathfrak{q} < b$, the same in the coarse medium of every lower level (Theorems 13.4 to 13.6 and Example S13.4). The static error of the coarse medium is the time integral of the exact memory kernel of the block coordinates, and for uniform leak its norm is at most $\|\Pi_{\perp\Phi}\|^2/\varkappa_{\mathrm{in}}^{\min}$, with $\Pi_{\perp\Phi}$ the coupling of the block averages to the coordinates within blocks and $\varkappa_{\mathrm{in}}^{\min}$ the smallest internal rate of a block with at least two members (Lemma 13.3 and Theorem 13.8), so coarse-graining along collectives for which this bound is small against $\varkappa_{\mathrm{out}}$ makes memory a small correction and coarse-graining across them does not (Example S13.9). The size bound caps the depth of a hierarchy on a lattice, for decimated media in two and more dimensions only under an inequality proved in one dimension and checked for one decimation step in two (Proposition 13.7); without declared geometry, the nested medium has collectives at every level only for $\mathfrak{q} < b/(2b^2-1)$ and at no level for $\mathfrak{q} \geq 1/(2b-1)$, a range that contains the regime $1/b < \mathfrak{q} < 1$ of power-law coherence, so that scale-free nested coherence and nested collectives exclude each other there (Proposition S13.8). The framework is closed under coarse-graining exactly when the precision leaves the span of the block indicators invariant, approximately when $\|\Pi_{\perp\Phi}\|^2 \ll \varkappa_{\mathrm{in}}^{\min}\varkappa_{\mathrm{out}}$, which high contrast alone does not ensure because the bound on a block's defect also involves the largest coupling of one agent across its boundary, and not even approximately without contrast (Theorem 13.10 and Lemma 13.3). The decay of decimated kernels in two dimensions and the first-order rate of the memoryless reduction are numerical observations (Numerical observations S13.3 and S13.11), and which regions of high ratio Laplacian renormalization returns is an open problem, the conjecture that a ratio above a constant times $\ln N$ suffices being false for every constant (Open problem S13.21 and Proposition S20.10).

# Part II
# Extensions addressing open problems

## S18 Approximate meta-agents and controlled renormalization

This section addresses two open problems of the literature on active inference in multiscale systems. The first is collective agency, the question of when a collection warrants a model as an agent, taken here in its graded form, which asks for the prediction error and the control error of the collective model; the second is controlled renormalization, the question of what a coarse-grained model preserves of predictions and of responses to policies, and with what error. For linear-Gaussian collectives in discrete time the section establishes an exact identity and a uniform bound for the relative entropy between the retained controlled path law and a declared Markov coarse model over a class of small-gain coarse policies, a finite-memory model whose error decays geometrically in its order, bounds on the policy score and the policy posterior computed with the declared model, a recursion that composes defects through levels, and a bound on the excess relative entropy of a meta-agent that filters with the declared model. The last bound is conditional on a smallness condition, $c_- > 0$ in Theorem S18.7, which failed with the a priori constants on the generic instances

tried. Left open are the sharp constant and a bound without the smallness condition (Open problem S18.12), the memory of the coarse internal state (Open problem S18.13), and nonlinear or non-Gaussian dynamics, policies without the small-gain condition and the continuous-time limit (Open problem S18.14); persistence of the collective is not treated, and the results narrow Open problem S11.13 and Open problem S13.22 without closing either.

The abstract physics of Theorem 11.8, the meta-agent of Theorem 11.13 and the nested medium of Theorem 13.6 are exact statements: they hold when the coarse map closes. The error terms given here apply when closure fails by a declared amount, for the controlled law of the retained variables, for the belief of a meta-agent that filters with a Markov model, for its policy posterior, and through successive levels. Proofs are in Section S39 and checks in Section S47.14.

**Definition S18.1** (Controlled coarse model, defects, policy classes)**.** The fine process is $x_{t+1} = Ax_t + Bu_t + \mathfrak{n}_t$ on $\mathbb{R}^n$ with $\mathfrak{n}_t \sim \mathcal{N}(0, Q)$ independent, $Q \succ 0$, and $x_0 \sim \mathcal{N}(m_0, \Sigma_0)$ independent of the noise; $u_t \in \mathbb{R}^p$ is a coarse act and $B$ is the fine input matrix composed with a fixed lifting $\Lambda$ of coarse acts to fine acts. The retained variable is $y_t = Cx_t \in \mathbb{R}^k$ with $C$ of full row rank and $C\Sigma_0 C^\top \succ 0$. With a complement $x_\perp = Zx$, $x = C^\sharp y + Z^\sharp x_\perp$, the blocks $A_{yy}, A_{y\perp}, A_{\perp y}, A_{\perp\perp}$, $B_y = CB$, $B_\perp = ZB$ are those of Theorem 5.2 with $C$ in the place of $\mathsf{P}$ and the coarse act in the place of the blanket input, and $Q^y = CQC^\top$, $Q^\perp = ZQZ^\top$, $Q^{\perp y} = ZQC^\top$, $Q^{\perp|y} = Q^\perp - Q^{\perp y}(Q^y)^{-1}Q^{y\perp}$. A *declared coarse model* is a triple $(\bar A, \bar B, \bar Q)$ with $\bar Q \succ 0$ and an initial law $\hat P_0 = \mathcal{N}(\hat m_0, \hat\Sigma_0)$, $\hat\Sigma_0 \succ 0$; it asserts $y_{t+1} = \bar A y_t + \bar B u_t + \hat w_t$, $\hat w_t \sim \mathcal{N}(0, \bar Q)$. Its *closure defect* and *input defect* are

$$D = CA - \bar A C, \qquad E = CB - \bar B, \qquad \delta_t = Dx_t + Eu_t. \tag{S57}$$

A *causal coarse policy* is a sequence $u_t = \pi_t(y_{0:t}, \omega)$ of measurable maps with $\mathbb{E}|u_t|^2 < \infty$ under the fine process, where the randomization $\omega$ takes values in a standard Borel space, has the same law under the fine process and under the declared model, and is independent of $(x_0, \mathfrak{n})$ in the first and of the initial state and noise of the second; $P^\pi$ and $\hat P^\pi$ denote the laws of $(\omega, y_{0:T})$ under the fine process and under the declared model driven by the same maps. The class $\mathcal{P}(\kappa, \bar c)$ consists of the causal coarse policies with $|\pi_t| \le \kappa|y_t| + \bar c$, which contains the linear feedback laws $u = Ky + c$ with $\|K\| \le \kappa$, $|c| \le \bar c$. Throughout, $h(M) = \frac12(\operatorname{tr} M - k - \ln\det M)$ and $g(c) = c - 1 - \ln c$.

The model of Eq. (98) is the case $y = \eta_C$, $u = a_C$, $\bar A = \bar F$, and its $\bar Q$ is $Q^y$; here $\bar Q$ is declared and may differ from $Q^y$. For every causal coarse policy the conditional law of $x_t$ given $(y_{0:t}, \omega)$ is $\mathcal{N}(\hat x_t, P_t)$ with a covariance $P_t$ that no policy changes (Lemma S39.1); $P_t^\perp = ZP_tZ^\top$ is its unresolved block.

## S18.1 The retained controlled law

**Theorem S18.2** (Path relative entropy under interventions)**.** *For every causal coarse policy $\pi$ and every horizon $T$,*

$$D_{\mathrm{KL}}(P^\pi \| \hat P^\pi) = D_{\mathrm{KL}}(P_{y_0} \| \hat P_0) + \sum_{t=0}^{T-1} \Big[ h(\bar Q^{-1} Q^y) + \tfrac12 \mathbb{E}^\pi \|\delta_t\|^2_{\bar Q^{-1}} - \mathfrak{d}_{\mathrm{dyn},t} \Big], \tag{S58}$$

*where $\|v\|^2_{\bar Q^{-1}} = v^\top \bar Q^{-1} v$ and $\mathfrak{d}_{\mathrm{dyn},t} = \frac12 \ln\det(I + (Q^y)^{-1} CAP_tA^\top C^\top)$ has the form of the dynamical dependence of Proposition S11.8(b) along the coarse path; that proposition is stated for an input path that is known or independent of the initial state and the noise, and here the same expression is $I(y_{t+1}; x_t \mid y_{0:t}, \omega)$ under every causal coarse policy (Lemma S39.1). Each summand also equals $h(\bar Q^{-1} S_t) + \frac12 \mathbb{E}^\pi \|D\hat x_t + Eu_t\|^2_{\bar Q^{-1}}$ with $S_t = Q^y + CAP_tA^\top C^\top$.*

The relative entropy is finite because both laws have nondegenerate Gaussian one-step kernels on the same space, and it splits into a noise-calibration term, the energy of the defect along the controlled path, and a credit $\mathfrak{d}_{\mathrm{dyn},t}$ that the declared model earns because the unresolved fluctuation inflates the true innovation. Only the middle term depends on the policy.

**Corollary S18.3** (Uniform bound over a policy class)**.** *Let $\|A^l\| \le M\alpha^l$ and $\|A^l_{\perp\perp}\| \le M_\perp \alpha_\perp^l$ with $\alpha, \alpha_\perp < 1$, let $(\mathbb{E}|x_0|^2)^{1/2} \le \mu_0$, and let $g_\kappa = M\kappa\|B\|\|C\|/(1-\alpha) < 1$. Then, for every $\pi \in \mathcal{P}(\kappa, \bar{c})$ and every $T$,*

$$D_{\mathrm{KL}}(P^\pi \| \hat{P}^\pi) \le D_{\mathrm{KL}}(P_{y_0} \| \hat{P}_0) + T\Big[h(\bar{Q}^{-1}Q^y) + \frac{\bar{\mathfrak{d}}^2}{2\lambda_{\min}(\bar{Q})}\Big], \tag{S59}$$

*with $\bar{\mathfrak{d}} = \|A_{y\perp}\|\bar{X}_\perp + \|A_{yy} - \bar{A}\|\|C\|\bar{X} + \|E\|(\kappa\|C\|\bar{X} + \bar{c})$ and the explicit moment bounds $\bar{X}$, $\bar{X}_\perp$ of Eq. (S124). The hypothesis on $A_{\perp\perp}$ can be dropped, with $\bar{X}_\perp = \|Z\|\bar{X}$.*

This is a bound of the form $\sup_\pi d_T(P^{\Lambda\pi}C^{-1}, \hat{P}^\pi) \le \varepsilon_T$ for renormalization beyond exact closure, the form of statement that the problem of controlled renormalization asks for, with $d_T$ the relative entropy of coarse path laws, and with the class $\mathcal{P}(\kappa, \bar{c})$, the lifting and the set of initial laws fixed in advance. It covers stable linear-Gaussian collectives under small-gain policies. It bears on the subject of Open problem S13.22, renormalization beyond the exact cases, and settles none of the items (a) to (h) listed there. By Eq. (S58) the relative entropy grows linearly in $T$ whenever the bracket of the identity stays bounded away from zero. The small-gain condition $g_\kappa < 1$ is sufficient and not necessary in general, and it cannot be dropped altogether: in the family of Remark S39.2 a feedback gain with $g_\kappa > 1$ destabilizes the fine process and makes the defect energy grow geometrically in $t$.

**Proposition S18.4** (Dependence on the time step)**.** *Let $A = I + \mathsf{h}F$, $B = \mathsf{h}G$, $Q = \mathsf{h}\mathsf{q}$, $\bar{A} = I + \mathsf{h}\bar{F}$, $\bar{B} = \mathsf{h}\bar{G}$, $\bar{Q} = \mathsf{h}\bar{\mathsf{q}}$, and write $\mathsf{q}^y = C\mathsf{q}C^\top$, $D_{\mathrm{c}} = CF - \bar{F}C$, $E_{\mathrm{c}} = CG - \bar{G}$, on a physical horizon $\tau = T\mathsf{h}$, for policies in $\mathcal{P}(\kappa, \bar{c})$. If $\bar{\mathsf{q}} \ne \mathsf{q}^y$, then $\liminf_{\mathsf{h}\to 0} \mathsf{h} D_{\mathrm{KL}}(P^\pi \| \hat{P}^\pi) \ge \tau h(\bar{\mathsf{q}}^{-1}\mathsf{q}^y) > 0$. If $\bar{\mathsf{q}} = \mathsf{q}^y$, then $D_{\mathrm{KL}}(P^\pi \| \hat{P}^\pi) - D_{\mathrm{KL}}(P_{y_0} \| \hat{P}_0) \le \frac{\mathsf{h}}{2}\sum_{t<T} \mathbb{E}^\pi \|D_{\mathrm{c}}x_t + E_{\mathrm{c}}u_t\|^2_{(\mathsf{q}^y)^{-1}}$, which stays bounded as $\mathsf{h} \to 0$.*

At a fixed step the relative entropy is finite whatever the declared noise; as the step shrinks it diverges like $\tau/\mathsf{h}$ unless the declared diffusion matrix is the true one, the discrete form of the mutual singularity of diffusions with different quadratic variation. A coarse model meant to survive refinement of the time step must therefore match $\mathsf{q}^y$ exactly, and then only the drift defect is charged.

## S18.2 Memory of order $d$

**Proposition S18.5** (Kernel truncation need not converge)**.** *Let $n = 2$, $y = x_1$, $y_{t+1} = \mathsf{b}y_t + \mathsf{g}x_{2,t} + \mathfrak{n}_{1,t}$, $x_{2,t+1} = \mathsf{a}x_{2,t} + \mathfrak{n}_{2,t}$, $Q = I$, $|\mathsf{a}|, |\mathsf{b}| < 1$, $\mathsf{g} \ne 0$, without input, with $x_0 \sim \mathcal{N}(0, \Sigma_0)$ and $\mathrm{Var}(x_{2,0}) = V = (1-\mathsf{a}^2)^{-1}$. For the complement $x_\perp = x_2$ the memory kernels $\Gamma_l$ of Theorem 5.2 vanish for every $l \ge 1$, so truncating the kernel at any order $d$ and replacing the orthogonal force by white Gaussian noise of variance $\sigma^2 > 0$ gives the same first-order autoregression $\hat{y}_{t+1} = \mathsf{b}\hat{y}_t + \hat{w}_t$ for every $d$. Its relative entropy rate is*

$$\lim_{T\to\infty} \frac{1}{T} D_{\mathrm{KL}}(P\|\hat{P}) = \frac{1}{2}\Big[\frac{1+\mathsf{g}^2V}{\sigma^2} - 1 - \ln\frac{1+\mathsf{g}^2P_\infty}{\sigma^2}\Big] \ge \frac{1}{2}\ln\frac{1+\mathsf{g}^2V}{1+\mathsf{g}^2P_\infty}, \tag{S60}$$

*with equality at $\sigma^2 = 1 + \mathsf{g}^2V$, where $P_\infty \le V$ is the limit of the variance of $x_{2,t}$ given $y_{0:t}$, the positive solution of $P = 1 + \mathsf{a}^2P/(1+\mathsf{g}^2P)$. If $\mathsf{a} \ne 0$ then $P_\infty < V$ and the lower bound is positive. If $\mathsf{a} = 0$ then $P_\infty = V = 1$, the orthogonal force is white, and the rate vanishes at $\sigma^2 = 1 + \mathsf{g}^2$.*

The kernels depend on the complement, and the proposition concerns the complement $x_\perp = x_2$ only; truncation of the kernels of the Mori complement is not analyzed (Remark S39.3). The memory that matters for the law of the coarse path is that of its predictor, which the kernels do not see when the orthogonal force is colored. The finite-memory model that does converge truncates the filter for the unresolved coordinate: the *order-d model* predicts $y_{t+1}$ with the innovation covariance $S_t$ and the mean $A_{yy}y_t + B_y u_t + A_{y\perp}\hat{\xi}_t^{(d)}$, where $\hat{\xi}_t^{(d)}$ runs the exact recursion for $\mathbb{E}[x_{\perp,t} \mid y_{0:t}, \omega]$ over the window $t-d, \dots, t$ from the value 0 (Eq. (S125)). Order 0 is a memoryless model with $\bar{A} = A_{yy}$, and the untruncated recursion reproduces the coarse law under every causal coarse policy.

**Theorem S18.6** (Geometric decay in the order)**.** *Let $\mathfrak{q}_\perp = \lambda_{\min}(Q^{\perp|y})$, let $\bar{p} < \infty$ satisfy $\bar{p} \geq \sup_t \|P_t^\perp\|$, and let $\theta = 1 - \mathfrak{q}_\perp/\bar{p} \in [0,1)$ and $\sup_t \mathbb{E}^\pi |x_{\perp,t}|^2 \leq \bar{X}_\perp^2 < \infty$. For every causal coarse policy and every $t \geq d+1$, the one-step conditional relative entropy from the coarse law to the order-d model is $\frac{1}{2}\mathbb{E}^\pi \|A_{y\perp}\Psi_{t,d}\hat{\xi}_{t-d}\|^2_{S_t^{-1}}$ and*

$$\tfrac{1}{2}\mathbb{E}^\pi \big\|A_{y\perp}\Psi_{t,d}\hat{\xi}_{t-d}\big\|^2_{S_t^{-1}} \leq \theta^d \frac{\bar{X}_\perp^2}{\mathfrak{q}_\perp}\mathfrak{d}_{\mathrm{dyn},t}. \tag{S61}$$

The rate $\theta$ comes from the unresolved noise that the coarse path cannot reveal, through the transition matrix $\Phi = A_{\perp\perp} - GA_{y\perp}$ of the filter, and not from the spectral radius of $A_{\perp\perp}$: with $A_{\perp\perp} = 0$ and correlated noise the predictor can still depend on the entire past, as it does in the scalar instance of Check S47.168. The bound is the product of the forgetting factor, a signal-to-noise ratio of the unresolved coordinate and the dynamical dependence $\mathfrak{d}_{\mathrm{dyn},t}$ of Theorem S18.2. The restriction $t \geq d+1$ cannot be removed: at $t = d$ the window starts from the initial law, and for a sharp initial law the bound fails (Check S47.168).

### S18.3 The belief of a bounded meta-agent

Let the meta-agent read $s_{t+1} = Hy_t + v_t$, $v_t \sim \mathcal{N}(0,R)$, $R \succ 0$, as in Section 11.5 where $H = R = \Lambda^s$ (the case $\Lambda^s \succ 0$ of that section, which also allows a singular $\Lambda^s$), and let its policy be measurable in the record $\mathcal{F}_t = \sigma(s_{1:t})$ and independent randomization. Let $\rho(A) < 1$ and let the fine filter be stationary, that is, $\Sigma_0 = \Sigma^-$ for the fixed point $\Sigma^-$ of the predictive covariance recursion of the fine filter (Eq. (S123), Lemma S39.4), with predictive and filtered covariances $\Sigma^-$ and $\Sigma^{\mathrm{f}}$ of $x_t$ given $\mathcal{F}_t$ and $\mathcal{F}_{t+1}$ (conditioning on the randomization as well), so that the exact predictive law of $y_t$ given $\mathcal{F}_t$ is $\mathcal{N}(C\hat{x}_t^-, P_{\mathrm{e}})$, $P_{\mathrm{e}} = C\Sigma^- C^\top$. The meta-agent runs the stationary filter of the declared model, with predictive covariance $P_{\mathrm{b}}^-$ solving $P = \bar{A}\varphi_R(P)\bar{A}^\top + \bar{Q}$, $\varphi_R(P) = P - PH^\top(HPH^\top + R)^{-1}HP$, filtered covariance $P_{\mathrm{b}} = \varphi_R(P_{\mathrm{b}}^-)$, gain $L$ and belief $\mathcal{N}(\hat{\eta}_t^-, P_{\mathrm{b}}^-)$ about $y_t$ given $\mathcal{F}_t$. Its *excess relative entropy* is $\Delta\mathcal{F}_t = \mathbb{E}D_{\mathrm{KL}}(\mathcal{N}(C\hat{x}_t^-, P_{\mathrm{e}}) \| \mathcal{N}(\hat{\eta}_t^-, P_{\mathrm{b}}^-))$, the expected surprisal it pays beyond the exact predictive law of the current exterior. Under exact closure, with the centralized start and the stationary covariance, that law is the one parametrized in Theorem 11.13(iv); the response to the members' computation noise that appears in that part is not modeled here, where the meta-agent is its coarse filter.

**Theorem S18.7** (Excess relative entropy of a Markov filter)**.** *Let $(\bar{A}, H)$ be detectable, $\Delta = \bar{A}C\Sigma^{\mathrm{f}}D^\top + D\Sigma^{\mathrm{f}}C^\top\bar{A}^\top + D\Sigma^{\mathrm{f}}D^\top$, and let $c_- \leq 1 \leq c_+$ satisfy $c_-\bar{Q} \preceq Q^y + \Delta \preceq c_+\bar{Q}$ with $c_- > 0$. Then $c_- P_{\mathrm{b}}^- \preceq P_{\mathrm{e}} \preceq c_+ P_{\mathrm{b}}^-$, and for every record-measurable policy with $\limsup_{t\to\infty}(\mathbb{E}|\delta_t|^2)^{1/2} \leq \bar{\mathfrak{d}}$ and every initial value of the declared filter with $\mathbb{E}|C\hat{x}_0^- - \hat{\eta}_0^-|^2 < \infty$,*

$$\limsup_{t\to\infty} \Delta\mathcal{F}_t \leq \frac{k}{2}\max\{g(c_+), g(c_-)\} + \frac{\big(\mathcal{S}'\bar{\mathfrak{d}} + \mathfrak{c}\|P_{\mathrm{b}}\|\mathfrak{g}\sqrt{k}\mathcal{S}_2'\big)^2}{2\lambda_{\min}(P_{\mathrm{b}}^-)}, \tag{S62}$$

*where* $\Psi' = \bar{A}(I - LH)$, $\mathcal{S}' = \sum_{j\geq 0} \|\Psi'^j\|$, $\mathcal{S}'_2 = (\sum_{j\geq 0} \|\Psi'^j \bar{A}\|^2)^{1/2}$, $\mathfrak{c} = \max(c_+ - 1, 1 - c_-)$ *and* $\mathfrak{g}^2 = \|H^\top R^{-1}(c_+ H P_{\mathrm{b}}^- H^\top + R)R^{-1}H\|$. *The constants* $c_\pm = \lambda_{\max\,/\,\min}(\bar{Q}^{-1/2} Q^y \bar{Q}^{-1/2}) \pm \vartheta/\lambda_{\min}(\bar{Q})$, *truncated at* 1, *are admissible whenever the resulting* $c_-$ *is positive, with* $\vartheta = 2\|\bar{A}\|\varsigma_y\sigma_D + \sigma_D^2$, $\sigma_D^2 = \|D\Sigma_x D^\top\|$, $\varsigma_y^2 = \|C\Sigma_x C^\top\|$ *and* $\Sigma_x = A\Sigma_x A^\top + Q$. *A bound of the same form holds for the filtered law, with* $\Psi = (I - LH)\bar{A}$ *and its own constants (Eq. (S130)).*

**Corollary S18.8** (In terms of the closure defect)**.** *In the setting of Theorem S18.7 without input* ($B = 0$, $\bar{B} = 0$), *with* $\bar{Q} = Q^y$ *and the moment-matched* $\bar{A} = CA\Sigma_x C^\top (C\Sigma_x C^\top)^{-1}$, *the pair* $(\bar{A}, H)$ *is detectable for every* $H$, *and one has* $\sigma_D^2 \leq \|Q^y\|(e^{2\mathfrak{d}_{\mathrm{cl}}} - 1)$ *and* $\limsup_t \mathbb{E}|\delta_t|^2 \leq k\sigma_D^2$, *so that* $\bar{\mathfrak{d}} = \sqrt{k}\sigma_D$ *is admissible. Here* $\mathfrak{d}_{\mathrm{cl}} = I(y_{t+1}; x_t \mid y_t) = \frac{1}{2}\ln\det(I + (Q^y)^{-1} D\Sigma_x D^\top)$ *is the closure defect of Definition 5.1(i) in the stationary law of the fine process, in the Gaussian form of Proposition S11.8(a) and Proposition 11.9(b). Whenever* $\mathsf{x} = \vartheta/\lambda_{\min}(Q^y) < 1$, *the right side of Eq. (S62) with the a priori constants is at most* $\frac{k}{4}\mathsf{x}^2/(1-\mathsf{x}) + (\mathcal{S}'\sqrt{k}\sigma_D + \mathsf{x}\|P_{\mathrm{b}}\|\mathfrak{g}\sqrt{k}\mathcal{S}'_2)^2/(2\lambda_{\min}(Q^y))$. *This is* $O(\mathfrak{d}_{\mathrm{cl}})$ *as* $\mathfrak{d}_{\mathrm{cl}} \to 0$ *along every family of instances on which* $\|\bar{A}\|$, $\varsigma_y$, $\|Q^y\|$, $\lambda_{\min}(Q^y)^{-1}$, $\mathcal{S}'$, $\mathcal{S}'_2$, $\|P_{\mathrm{b}}\|$ *and* $\mathfrak{g}$ *stay bounded.*

The theorem is conditional: it asserts nothing unless $c_- > 0$, and with the a priori constants this hypothesis failed on all 30 generic instances of Check S47.169, while with the exact $\Delta$ it held on every instance tried. Where it applies, it is an error term of the kind that Open problem S11.13 asks for in property (iv) of Theorem 11.13, for a stable linear-Gaussian exterior, a positive definite record noise and a meta-agent identified with its coarse filter, without the computation noise of the members; through Corollary S18.8 it is expressed in the closure defect, one of the memory measures of Proposition 11.9, at the price of constants that depend on the declared filter. It is of second order in the defect, as the exact excess is on the scaling families of Check S47.169, and it separates a covariance part, the miscalibration of the claimed variance, from a mean part driven by the defect. In the two-firm model the coarse closed loop has eight variables and reproduces the 0.0743 nats of Table S12 as $0.04897 + 0.02535$, with noise variances reconstructed from the member-level description of Example S11.11; the theorem with the exact $\Delta$ gives 0.279, and with the a priori constant $\vartheta$ its smallness condition fails, since that firm declares twice the true price noise (Check S47.170). At finite horizon and without stationarity, the excess relative entropy of the time-varying filter of the declared model, started from the declared initial law, is at most the path relative entropy of Theorem S18.2 at horizon $t$ for the coarse policy that reads the record (Proposition S39.6).

## S18.4 Policy scores and policy posteriors

Let a policy be scored by $G(\pi) = \sum_{t=1}^{T} \frac{1}{2}\mathbb{E}^\pi \|y_t - o_t^\star\|^2_{\Lambda^\star}$, with $\Lambda^\star \succeq 0$ and deterministic targets $o_t^\star$, a quadratic score of the same form as the act part of the expected free energy for a Gaussian preference of precision $\Lambda^\star$ (Eq. (S24)), summed over the horizon; write $G_t$ for its time-$t$ term, and let $\hat{G}$, $\hat{G}_t$ be the same quantities under the declared model. In the linear-Gaussian setting the conditional covariances, and with them the epistemic part, do not depend on the policy in either model (Lemma S39.1).

**Proposition S18.9** (Score error)**.** *For affine feedback* $u_t = Ky_t + c_t$, *with* $\bar{A}_K = \bar{A} + \bar{B}K$, $\beta = \|(Q^y)^{1/2} - \bar{Q}^{1/2}\|_{\mathrm{F}}$ *and* $w_0 = W_2(P_{y_0}, \hat{P}_0)$, *the Wasserstein distance of the time-t coarse marginals obeys*

$$W_2(P^\pi_{y_t}, \hat{P}^\pi_{y_t}) \leq w_t := \|\bar{A}_K^t\| w_0 + \sum_{s<t} \|\bar{A}_K^{t-1-s}\| (\mathbb{E}^\pi|\delta_s|^2)^{1/2} + \beta\Big(\sum_{j<t} \|\bar{A}_K^j\|^2\Big)^{1/2}, \tag{S63}$$

*and* $|G_t - \hat{G}_t| \le a_t(2\hat{G}_t^{1/2} + a_t)$ *with* $a_t = (\|\Lambda^\star\|/2)^{1/2} w_t$. *For* $\kappa$*-Lipschitz state feedback the same holds with* $\|\bar{A}_K^j\|$ *replaced by* $(\|\bar{A}\| + \kappa\|\bar{B}\|)^j$ *and the last term by* $\beta \sum_{j<t}(\|\bar{A}\| + \kappa\|\bar{B}\|)^j$.

**Proposition S18.10** (Policy posterior)**.** *Let* $\Pi$ *be finite,* $\gamma > 0$, $G, \hat{G} : \Pi \to \mathbb{R}$, *and* $p \propto Ee^{-\gamma G}$, $\hat{p} \propto Ee^{-\gamma \hat{G}}$ *with common habits* $E : \Pi \to (0, \infty)$, *and* $\operatorname{osc} = \max_\Pi(\hat{G} - G) - \min_\Pi(\hat{G} - G) \le 2 \max_\Pi |\hat{G} - G|$. *Then* $|\ln(\hat{p}/p)| \le \gamma \operatorname{osc}$, $D_{\mathrm{KL}}(p\|\hat{p}) \le \gamma^2 \operatorname{osc}^2/8$, *and the total variation distance is at most* $\tanh(\gamma \operatorname{osc}/4)$. *The last bound is attained: for every value* $r \ge 0$ *of* $\gamma \operatorname{osc}$ *there is an instance with two policies and total variation distance* $\tanh(r/4)$.

The policy posterior of a bounded meta-agent therefore lies within total variation $\tanh(\gamma \max_\Pi |G - \hat{G}|/2)$ of the one it would form with the exact coarse law, and the score error is controlled by the same defect energy as the path law.

## S18.5 Composition through levels

**Theorem S18.11** (Level recursion)**.** *Let* $y^{(l)} = C_l y^{(l-1)}$, $y^{(0)} = x$, $C^{(l)} = C_l \cdots C_1$, *with declared models* $(\bar{A}_l, \bar{B}_l, \bar{Q}_l)$, $(\bar{A}_0, \bar{B}_0, \bar{Q}_0) = (A, B, Q)$, *local defects* $D_l = C_l \bar{A}_{l-1} - \bar{A}_l C_l$, $E_l = C_l \bar{B}_{l-1} - \bar{B}_l$, *and total defects* $D_l^{\mathrm{tot}} = C^{(l)} A - \bar{A}_l C^{(l)}$, $E_l^{\mathrm{tot}} = C^{(l)} B - \bar{B}_l$.

*(a)* $D_{l+1}^{\mathrm{tot}} = C_{l+1} D_l^{\mathrm{tot}} + D_{l+1} C^{(l)}$ *and* $E_{l+1}^{\mathrm{tot}} = C_{l+1} E_l^{\mathrm{tot}} + E_{l+1}$.

*(b) With* $\mathfrak{e}_l = \sup_t \|D_l^{\mathrm{tot}} x_t + E_l^{\mathrm{tot}} u_t\|_{L^2}$ *and* $\mathfrak{d}_l = \sup_t \|D_l y_t^{(l-1)} + E_l u_t\|_{L^2}$ *along the fine process under a fixed policy, the suprema being over* $0 \le t < T$ *and possibly infinite,* $\mathfrak{e}_{l+1} \le \|C_{l+1}\| \mathfrak{e}_l + \mathfrak{d}_{l+1}$. *If in addition* $C^{(l)}$ *has full row rank,* $C^{(l)} \Sigma_0 C^{(l)\top} \succ 0$, *the level-*$l$ *model has a nondegenerate Gaussian initial law, and the policy is causal in the top-level variable, then the path relative entropy at level* $l$ *obeys Eq. (S59) with* $Q^y = C^{(l)} Q C^{(l)\top}$, $\bar{Q} = \bar{Q}_l$ *and* $\bar{\mathfrak{d}} = \mathfrak{e}_l$.

*(c) Fix* $t$ *and a policy under which the laws below have finite second moments. With* $\varepsilon_l$ *the Wasserstein distance* $W_2$ *between the law of* $y_t^{(l)}$ *under the fine process and under the level-*$l$ *model, and* $\delta_{l+1}$ *the distance between the image under* $C_{l+1}$ *of the level-*$l$ *model law and the level-*$(l+1)$ *model law,* $\varepsilon_{l+1} \le \|C_{l+1}\| \varepsilon_l + \delta_{l+1}$.

The Lipschitz constant of the recursion is the norm of the coarse map, so with orthonormal coarse coordinates the errors add and stay bounded through any number of levels when the one-level errors are summable. In the nested medium of Theorem 13.6, in the discrete-time form of its part (a) with symmetric nested update weights, orthonormal block maps and the declared models $\bar{A}_l = C_l \bar{A}_{l-1} C_l^\top$, $\bar{B}_l = C_l \bar{B}_{l-1}$, every local defect vanishes and the recursion returns zero; for a symmetric perturbation of the update matrix of relative size 0.04 in an instance with $b = 2$ and three levels the three-level total stays within the bound and does not grow monotonically, which is possible because a coarse map can annihilate part of the defect below it (Check S47.171). Relative entropy has no triangle inequality, which is why (b) composes defects and not errors.

## S18.6 What remains open

**Open problem S18.12** (Sharp posterior bound)**.** On the scaling families of Check S47.169, the right side of Eq. (S62) exceeds the exact excess by a factor of about 2.3 to 3.7 with the exact $\Delta$ and of about 12 to 80 with $\vartheta$; for the two firms it is 0.279 against 0.0743 (Check S47.170). These numbers are specific to the instances checked. The theorem requires $c_- > 0$, and with the a priori constants this failed on all 30 generic instances of Check S47.169 and fails for a model whose declared noise is much larger than $Q^y$. Find the sharp constant, and a bound that needs no smallness condition, in terms of $\mathfrak{d}_{\mathrm{cl}}$ and the forgetting time of the declared filter.

**Open problem S18.13** (Memory of the coarse internal state)**.** The second half of Open problem S11.13, a bound on the memory of the coarse internal state under level-0 protocols beyond the exact terms of Proposition S11.10, is not addressed here: Theorem S18.7 treats the exterior as approximately closed and the meta-agent's own filter as exactly Markov.

**Open problem S18.14** (Beyond stable linear-Gaussian collectives)**.** Extend Theorems S18.2, S18.7 and S18.11 to nonlinear or non-Gaussian fine dynamics, to policy classes without the small-gain condition, where the bounds here grow geometrically in the horizon, and to the continuous-time limit of Proposition S18.4, which is stated here only at positive step.

# S19 Recursive formation with inherited dynamics

This section addresses the open problem of recursive formation with inherited dynamics in the literature on collective active inference: whether the collectives left by one formation event can form a collective one level up, and what law a formed group carries into that event. The starting point is the collectives of Theorem 7.8 as the first event leaves them, that is, at $\gamma = \infty$ with exact in-group translations and in the window of its part (c). What is established is the law a formed group hands to the next level, exact at that idealized split, together with the thresholds of a second event and the conditions that block it, both for the frozen-calibration mean-field map. The result is negative in part: the formation thresholds rise with level, at the computed parameter points the second event replaces the first-level collectives and does not nest them at a common level, a nested meta-agent is not established, and the verdict of Proposition S19.15 is conditional on admitting the translation-aware null of Proposition S19.11 among the nulls of type S of Definition 5.17. Left open are the full exact-fusion null at level 2, the stochastic model with calibration that tracks the forming law, competition among parents, a level-2 model with acts, and closure of the coarse state under turnover (Open problems S19.18 to S19.22).

The level-2 object is tested against the grade of Definition 5.19 and the hypotheses of Theorem 11.13, both fixed beforehand. The coarse map is the member average of a formed group, an exact marginalization at the idealized split.

## S19.1 A second formation event with inherited parameters

**Definition S19.1** (Nested population and aggregate channel)**.** Keep the constants of Definition 7.2. The population consists of $2M$ parents of $n$ agents; the members of parent $c$ share the frame $\mathrm{Rot}(\theta_c)$. The parents form two superunits of $M$ parents each, with mismatches $|\theta_c - \theta_{c'}| \in [0, \vartheta_{\mathrm{n}}]$ inside a superunit and in $[\vartheta_{\mathrm{f}}, \vartheta_{\max}] \subset (0, \pi]$ across, where $\vartheta_{\mathrm{n}} < \vartheta_{\mathrm{f}}/2$. The level-1 state is the split at the parents under the hypotheses of Lemma 7.7 with $\kappa = n - 1$. The aggregate channel is an added hypothesis: each round every agent reads $\kappa_2 \le M - 1$ parent aggregates $\bar{\mu}_c = n^{-1} \sum_{i \in c} \mu_i$ with read noise $R_c^{(2)}$ per coordinate, the set $J$ drawn among the $2M - 1$ other parents with probability proportional to $\exp(\gamma \frac{K}{2} \log(\hat{\Lambda}^+(J)/\Lambda^{(1)}))$, $\hat{\Lambda}^+(J) = \Lambda^{(1)} + \sum_{c \in J} \varpi_c$, $\Lambda^{(1)} = \Lambda_0 + (n-1)\varpi_{\mathrm{in}}$. Each parent edge has a translation learned as in Definition 7.2(vi) from the declared start, with bias fractions $y_c$ on near edges and $x_c$ on far edges. If an aggregate read averages $n$ independent member reads, $R_c^{(2)} = R_c/n$.

**Proposition S19.2** (What a formed group inherits)**.** *At the level-1 split of Definition S19.1, let $\bar{m}_A, \bar{s}_A$ be the member averages of a parent $A$ and $e_A = \bar{m}_A - \eta$.*

*(a) Exterior. $\bar{s}_A = \eta + \bar{v}_A$ with $\mathrm{Cov}(\bar{v}_A) = (R^s/n) I_2$, and $\eta$ keeps the law $(F, Q)$.*

(b) *Memory.* $\bar{m}_A(t+1) = F[(1-k^s)\bar{m}_A(t) + k^s \bar{s}_A(t)] + \bar{\zeta}_A(t)$ *with* $\mathrm{Cov}(\bar{\zeta}_A) = F^2\bar{k}^2(n-1)R_c/n \cdot I_2$, *white and independent across parents: a closed filter with the members' gain* $k^s$.

(c) *Belief covariance. In the stationary law* $e_A = b + \xi_A$ *with* $b(t+1) = \lambda b(t) - w_t$ *common to all parents,* $\mathrm{Var}_1(b) = Q/(1-\lambda^2)$, *and* $\xi_A$ *independent of* $b$ *and across parents,* $\mathrm{Var}_1(\xi_A) = q/(n(1-\lambda^2)) = \mathsf{s}_0/n + \mathsf{s}_1$, *with lag-*$\tau$ *correlation* $\lambda^\tau$. *Member deviations* $m_i - \bar{m}_A$ *are independent of* $(b, \xi_A)$, *with variance* $\mathsf{s}_0(1-1/n)$ *and multiplier* $\lambda_\perp = F(k^0 - \bar{k})$.

(d) *Frame.* $\bar{\mu}_A = \mathrm{Rot}(\theta_A)\bar{m}_A$. *For units* $c = 1, \dots, M$ *with distinct frames,* $M^{-1}\sum_c \mathrm{Rot}(\theta_c)\bar{m}_c = \rho\mathrm{Rot}(\bar{\theta})\bar{m} + M^{-1}\sum_c \mathrm{Rot}(\theta_c)(\bar{m}_c - \bar{m})$, *where* $\rho e^{\mathrm{i}\bar{\theta}} = M^{-1}\sum_c e^{\mathrm{i}\theta_c}$ *and* $\rho < 1$.

(e) *Outgoing channel. A reader of* $\bar{\mu}_A$ *with translation error* $\phi$ *has calibrated noise* $R_c^{(2)} + \mathrm{MSE}_A + D_{\mathrm{cal}}(1-\cos\phi)$, *with* $\mathrm{MSE}_A = Q/(1-\lambda^2) + q/(n(1-\lambda^2))$ *and the member value of* $D_{\mathrm{cal}}$.

Aggregation divides only $\mathsf{s}_0$ by $n$; the common lag and the correlated part pass to the parent unchanged. The capacity $\kappa_2$, the read noise $R_c^{(2)}$ and the choice of a separate channel are not inherited.

**Lemma S19.3** (Nested split law)**.** *Let units of sizes* $n_1, \dots, n_L \geq 2$ *be nested,* $N_j = n_1 \cdots n_j$, *and let every agent read all* $n_j - 1$ *sibling aggregates at every level* $j$ *with exact translations and read noise* $R_c/N_{j-1}$. *For* $\hat{P}^- > 0$ *and* $\varpi_j > 0$ *put* $\hat{\Lambda}^+ = 1/\hat{P}^- + 1/R^s + \sum_j (n_j - 1)\varpi_j$, $k^0 = 1/(\hat{P}^-\hat{\Lambda}^+)$, $k^s = 1/(R^s\hat{\Lambda}^+)$, $k_j = \varpi_j/\hat{\Lambda}^+$, $\lambda = F(1-k^s)$, $K_m = k^0 + \sum_{j\leq m}(n_j - 1)k_j$, $\mu_{m+1} = F(K_m - k_{m+1})$ *and* $q = F^2[(k^s)^2 R^s + \sum_j k_j^2(n_j - 1)R_c/N_{j-1}]$. *Then* $|\mu_{m+1}| \leq \lambda$, *and the error of the average of a level-*$m$ *unit has*

$$\mathrm{MSE}_m = \frac{Q}{1-\lambda^2} + \sum_{j=m}^{L-1} \frac{q(N_j^{-1} - N_{j+1}^{-1})}{1-\mu_{j+1}^2} + \frac{q}{N_L(1-\lambda^2)}, \qquad D_L = \frac{2QF^2k^s}{(1-F^2)(1-\lambda F)}, \tag{S64}$$

*where* $D_L = 2\,\mathrm{Cov}_1(\eta, \cdot)$ *for every unit average. The self-consistency equations* $\hat{P}^- = F^2/\hat{\Lambda}^+ + Q$, $\varpi_j = 1/(R_c/N_{j-1} + \mathrm{MSE}_{j-1})$ *have a solution, and every solution has* $\varpi_{\min} \leq \varpi_j \leq 1/Q$ *with* $\varpi_{\min} = 1/(R_c + P_0 + F^2(R^s + R_c)/(1-F^2))$.

**Theorem S19.4** (Level-2 thresholds)**.** *Freeze the calibration* $\varpi(\phi) = 1/(\hat{\sigma}^2 + D(1-\cos\phi))$ *and the base precision* $\Lambda > 0$ *of the aggregate channel, let* $0 < \mathsf{p}^* < 1$, *and consider the map* (72) *on the* $2M-1$ *parent edges of one receiver. Put*

$$\Delta\mathcal{V}_2(x) = \frac{K}{2}\log\Big(1 + \frac{\varpi(\vartheta_{\mathrm{n}}) - \varpi(x\vartheta_{\mathrm{f}})}{\Lambda + \kappa_2\varpi(0)}\Big), \quad \gamma_S^{(2)} = \frac{\log\frac{\kappa_2(1-\mathsf{p}^*)}{\mathsf{p}^*}}{\Delta\mathcal{V}_2(\frac{1}{2})}, \quad \gamma_M^{(2)} = \frac{\log\frac{\kappa_2}{(2M-1)\mathsf{p}^*}}{\frac{K}{2}\log\frac{\Lambda+\kappa_2\varpi(0)}{\Lambda+\kappa_2\varpi(\vartheta_{\max})}}, \tag{S65}$$

*and* $\bar{\mathsf{p}}_2 = \kappa_2 e^{-\gamma\Delta\mathcal{V}_2(1/2)}/(1+\kappa_2 e^{-\gamma\Delta\mathcal{V}_2(1/2)})$.

(a) *If* $\gamma > \gamma_S^{(2)}$, *the set where all far edges are unlearned is forward invariant and* $\liminf_t x_c(t) \geq \mathsf{p}^*/(\mathsf{p}^* + \bar{\mathsf{p}}_2) > \frac{1}{2}$ *there, whatever the near edges do.*

(b) *If in addition* $\kappa_2 = M-1$ *and* $M\bar{\mathsf{p}}_2 < 1-\mathsf{p}^*$, *then on that set every near edge has* $\limsup_t y_c(t) \leq \mathsf{p}^*/(\mathsf{p}^* + 1 - M\bar{\mathsf{p}}_2) < \frac{1}{2}$: *the superunits form. The extra condition follows from* $\gamma > \gamma_S^{(2)}$ *when* $\mathsf{p}^* \leq 1/(M+1)$.

(c) *If* $\gamma < \gamma_M^{(2)}$, *every edge is eventually learned and no level-2 partition forms.*

*The inherited constants at onset are* $\Lambda = \Lambda^{(1)}$, $\hat\sigma^2 = R_c^{(2)} + \mathrm{MSE}_A$, $D = D_{\mathrm{cal}}$*; at a level-*$\ell$ *event they are those of Lemma S19.3 at depth* $\ell - 1$.

The renormalized map sends the depth-$(\ell-1)$ solution of Lemma S19.3 to the triple $(\Lambda, \hat\sigma^2, D)$ and, through (S65), to the thresholds of level $\ell$. The triple is an exact summary for the thresholds, but its flow is not closed: the depth-$\ell$ solution recalibrates every lower $\varpi_j$.

**Corollary S19.5** (Thresholds grow with depth)**.** *At any self-consistent solution of depth* $L$*, with* $S_L = \sum_{j\le L}(n_j - 1)$ *and* $C = 2KF^2/(QR^s(1-F^2)^2)$*, the base precision of the next event satisfies* $\Lambda \ge 1/R^s + \varpi_{\min} S_L$*, and when the logarithms are positive*

$$\gamma_S \ge \frac{\Lambda^2}{C} \log \frac{\kappa(1-\mathsf{p}^*)}{\mathsf{p}^*}, \qquad \gamma_M \ge \frac{\Lambda^2}{C\kappa} \log \frac{\kappa}{(2M-1)\mathsf{p}^*}. \tag{S66}$$

*Moreover* $P_0(1 - 2F^2k^s/(1-F^2)) \le \mathrm{MSE}_m \le P_0 + q/(1-F^2)$ *with* $k^s \le 1/(1 + R^s\varpi_{\min}S_L)$ *and* $q \le F^2k^sR^s(1 + R_c/Q)$*. Hence the depth-*$L$ *solutions have no accumulation point with* $k^s > 0$*; as* $S_L \to \infty$ *the claimed variance* $1/\hat\Lambda^+$*, the grounding* $D_L$ *and every value gap tend to zero while every* $\mathrm{MSE}_m$ *tends to the prior variance* $P_0$*, and at fixed* $\gamma$*, with bounded unit sizes,* $\kappa_\ell = n_\ell - 1$ *and* $\mathsf{p}^* < \frac{1}{3}$*, only finitely many levels escape (c).*

**Numerical observation S19.6** (Three levels at the base point)**.** With $n_\ell = 4$, $\kappa_\ell = 3$ and the same supplied geometry $\vartheta_{\mathrm{n}} = 0$, $\vartheta_{\mathrm{f}} = \vartheta_{\max} = \pi/2$ substituted in (S65) at every level, the thresholds $(\gamma_M, \gamma_S)$ frozen at the split are $(6.84, 68.2)$, $(11.2, 109.6)$, $(16.1, 161.0)$ for levels 1 to 3, and $(8.18, 76.4)$, $(13.5, 128.4)$ for levels 2 and 3 frozen at onset; member mean-square error rises $0.249, 0.305, 0.346$ with depth, against 0.141 for an isolated Kalman agent (Check S47.173). A third level is reached in the mean field for $\gamma > 161.0$.

**Numerical observation S19.7** (Coarse partitions resolve first)**.** With inherited frames $\{0, \pi/16, \pi/8\}$ and $5\pi/8 + \{0, \pi/16, \pi/8\}$, $n = 4$, $M = 3$, $\kappa_2 = 2$, one $\gamma$ for both channels and level-2 constants inherited from the exact level-1 split, the superunits separate on the aggregate channel from $\gamma \approx 60$ ($\gamma_S^{(2)} = 113.9$), while on the individual channel the parents separate only between $\gamma = 900$ and $1300$ ($\gamma_S = 1853$ at mismatch $\pi/16$); below that the first event already produces the bridged superunit and no intermediate level exists (Check S47.174).

## S19.2 Obstructions and agency tests

The second event is driven by one inherited quantity, the dependence of the calibrated noise of an aggregate on the translation error. The statements below say when that dependence is too weak, what the event does to the first level, and how the superunit fares under a translation-aware form of the exact-fusion null, a change of representation and membership turnover.

**Lemma S19.8** (Aggregate of a frame-incoherent parent)**.** *Let the members of a parent have frames* $\mathrm{Rot}(\theta_i)$ *with exact in-group translations and the split law of Lemma 7.7, and write* $n^{-1}\sum_i e^{\mathrm{i}\theta_i} = \rho e^{\mathrm{i}\bar\theta}$*. A reader whose translation of* $\bar\mu_c = n^{-1}\sum_i \mathrm{Rot}(\theta_i) m_i$ *errs by* $\phi$ *from* $\bar\theta$ *calibrates, by the C1g rule of Definition 7.2(v),*

$$\hat\sigma_\rho^2(\phi) = \hat\sigma_\rho^2 + \rho D_{\mathrm{cal}}(1 - \cos\phi), \qquad \hat\sigma_\rho^2 = R_c^{(2)} + \rho^2 \mathrm{MSE}_A + (1-\rho^2)\Big(P_0 + \frac{\mathsf{s}_0}{n}\Big) - \rho(1-\rho)D_{\mathrm{cal}}, \tag{S67}$$

*with* $\mathsf{s}_0$ *as in Lemma 7.7, and* $\hat\sigma_\rho^2(\phi) \ge R_c^{(2)} + Q$*. At* $\rho = 1$ *this is Proposition S19.2(e); at* $\rho = 0$ *the calibration does not depend on* $\phi$*.*

**Theorem S19.9** (Obstructions to the second event)**.** *In Theorem S19.4 take* $\varpi(\phi) = 1/\hat{\sigma}_\rho^2(\phi)$ *with* $D = D_{\mathrm{cal}}$*, mismatches in* $[0, \vartheta_{\max}]$*,* $1 \le \kappa_2 \le 2M-1$*, and put*

$$B_2 = \frac{K\kappa_2 \rho D(1-\cos\vartheta_{\max})}{2\hat{\sigma}_\rho^2 \hat{\sigma}_\rho^2(\vartheta_{\max})\Lambda}, \qquad \ell_2 = \log\frac{\kappa_2}{(2M-1)\mathsf{p}^*}. \tag{S68}$$

*(a) At every state and every* $\gamma$ *each parent edge is attended with probability at least* $\kappa_2 e^{-\gamma B_2}/(2M-1)$*. If* $\gamma B_2 < \ell_2$*, every edge has* $\limsup_t x_c(t) < \frac{1}{2}$ *from every start, so no edge stays unlearned and* $\gamma_M^{(2)} \ge \ell_2/B_2$*.*

*(b) If* $\rho D(1-\cos\vartheta_{\max}) = 0$*, attention over parents is uniform at every* $\gamma \in [0,\infty]$ *and every bias fraction converges to* $\mathsf{p}^*/(\mathsf{p}^* + \kappa_2/(2M-1))$ *from every start.*

*(c) With the inherited constants,* $B_2 \le \rho C \kappa_2 (1-\cos\vartheta_{\max})/(2\Lambda^2)$ *with* $C$ *of Corollary S19.5, and* $\Lambda^{(1)} \ge 1/R^s + (n-1)\varpi_{\min}$*. When* $\ell_2 > 0$*, the conclusion of (a) therefore holds for every group size with* $(1/R^s + (n-1)\varpi_{\min})^2 > \gamma\rho C\kappa_2(1-\cos\vartheta_{\max})/(2\ell_2)$*.*

*(d) If* $\kappa_2 \ge M$*, every reader attends at least* $\kappa_2 - M + 1$ *far parents in every round at every* $\gamma$*, and for* $\mathsf{p}^* < (\kappa_2 - M + 1)/M$ *no fixed point has all far edges unlearned.*

**Proposition S19.10** (Contrast after the second event)**.** *At the depth-2 split of Lemma S19.3* ($n_1 = n$*,* $n_2 = M$) *the world-conditional precision of a superunit* $S$ *is* $(I - W^2)/q$*, with* $W$ *the entrywise nonnegative member matrix; it is diffusive with uniform leak* $(1-\lambda^2)/q$*, and*

$$r(S) = \frac{1 - \max\{\mu_1^2, \mu_2^2\}}{1-\lambda^2}, \qquad r(c \mid S) = \frac{1 - \lambda^2 + \mu_2^2 - \mu_1^2 + (\lambda^2 - \mu_2^2)/M}{1 - \lambda^2 + (M-1)(\lambda^2 - \mu_2^2)/M} \le \frac{M(1-\mu_1^2)}{(M-1)(1-\mu_2^2)}, \tag{S69}$$

*where* $r(c \mid S)$ *is the contrast of a parent inside its superunit. For the population of parent aggregates,* $r(S) = (1-\mu_2^2)/(1-\lambda^2)$*.*

**Proposition S19.11** (Translation-aware null at level 2)**.** *In Definition S19.1 let every reader know and remove the mistranslation of each aggregate read, with calibration on the current law.*

*(a) A read from which the mistranslation is removed has the law of an exactly translated one, with calibrated noise* $R_c^{(2)} + \mathrm{MSE}_c(t)$ *for every reader.*

*(b) From a start law invariant under permutations of the* $2M$ *parents and of the members inside each parent, with errors* $b + \xi_i$ *and* $\xi$ *independent of the world, all parent precisions are equal at all times, attention over parents is uniform at every* $\gamma \in [0,\infty]$*, and the law stays invariant. The conditional covariance of the parent aggregates is* $g_0 I + g_1 \mathbf{1}\mathbf{1}^\top$*, and at stationarity* $g_1 \ge 0$*; a union of* $m \ge 2$ *parents has contrast* $(\varepsilon + ma)/(\varepsilon + (2M-m)a)$ *with* $\varepsilon = 1/(g_0 + 2Mg_1)$*,* $a = g_1\varepsilon/g_0$*, which equals* $1$ *at* $m = M$*.*

*(c) Let the start law be invariant only under a group* $H$ *of permutations of the agents that maps parents to parents and is transitive on the agents, as at the depth-2 split, with errors as in (b). The first three claims of (b) hold with* $H$ *as the invariance group. The conditional covariance* $\Sigma_t$ *of the members and its average* $\bar{\Sigma}_t$ *over the permutations of (b) obey the same recursion, and* $\max_{ij} |(\Sigma_t - \bar{\Sigma}_t)_{ij}| \le F^{2t} \max_{ij} |(\Sigma_0 - \bar{\Sigma}_0)_{ij}|$*. Every stationary law invariant under* $H$ *therefore has the conditional covariance of (b).*

*Hence under this null no partition into superunits is selected, and a superunit has contrast* $1 < r_0$*.*

The null of Proposition S19.11 is not a row of Table 6. It is a partial form of $\mathfrak{N}_{\mathrm{ex}}$, introduced here: it removes the mistranslation of aggregate reads and keeps the literal fusion and the attention rule of Definition 7.2, so the cross-covariances of the aggregates are still misread. Whether it counts among the nulls of type S of Definition 5.17 is a choice that the definition does not make, and every conclusion drawn from it below is stated with that choice. Part (c) bounds the distance of the covariance from its symmetrized form and identifies the stationary laws with the symmetry of the split; convergence of the symmetrized covariance under the calibration loop is not proved, and is seen only in the computed orbit of Check S47.180.

**Proposition S19.12** (Change of representation)**.** *Let the retention of Definition 7.2(vi) pull each estimate toward a declared default $\hat{\alpha}^0_{cd}$, the model itself having $\hat{\alpha}^0 = 0$.*

*(a) The map (72) and its thresholds depend on the frames and the default only through the moduli $\vartheta^0_{cd} = |\theta_c - \theta_d - \hat{\alpha}^0_{cd}|$. The second moments at a given bias field depend on them only through the signed offsets $\theta_c - \theta_d - \hat{\alpha}^0_{cd}$, up to a common change of sign of all offsets. All of these are invariant under $\theta_c \mapsto \theta_c + \delta_c$, $\hat{\alpha}^0_{cd} \mapsto \hat{\alpha}^0_{cd} + \delta_c - \delta_d$.*

*(b) With the default held at 0 they are not invariant: $\delta_c = -\theta_c$ gives $\vartheta_{\max} = 0$ and, by Theorem S19.9(b), no event at any $\gamma$.*

*(c) An invertible recoding $T\bar{\mu}_c$ of the aggregate leaves closure and contrast unchanged. A recoding $T = \rho' \mathrm{Rot}(\delta)$ unknown to the readers shifts the mismatch by $\delta$ and replaces $D$ by $\rho' D$ and $\hat{\sigma}^2$ by $R^{(2)}_c + \rho'^2(\mathrm{MSE}_A - P_0 + D) + P_0 - \rho' D$.*

**Definition S19.13** (Turnover)**.** Each round, before fusion, every agent is replaced independently with probability $g$. The newcomer takes the position, frame and sensor of its predecessor; its memory is $f$ times the current parent aggregate plus independent zero-mean noise of finite variance; each translation is handed over with probability $f_\alpha$ and otherwise reset to the declared start. The fidelity $f$ is the analog of $\mathsf{f}$ in Proposition 5.14, with the current parent aggregate in place of the record.

**Proposition S19.14** (Membership turnover)**.** *Under Definition S19.13, with $\hat{g} = g(1 - f_\alpha)$:*

*(a) the mean-field map of the bias fractions recorded after the replacement step is (72) with $h$ replaced by $h_g = (1 - \hat{g})h$, so Theorems S19.4 and S19.9 hold with $\mathsf{p}^*(g) = (1 - h_g)/(\alpha h_g)$, increasing in $\hat{g}$. For $\hat{g} \geq g_c = 1 - 1/((1 + \alpha)h)$ every edge has $\liminf_t x_c(t) \geq \frac{1}{2}$ at every $\gamma$: no translation stays learned;*

*(b) at the depth-2 split with exact translations and fixed gains, the superunit mean is closed in conditional mean with multiplier $\lambda_g = \lambda(1 - g(1 - f))$. The part of $\bar{m}_S$ not explained by $\bar{s}_S$ has autocorrelation $\lambda^\tau_g$, the expected membership overlap is $(1 - g)^\tau$, and the coarse memory outlasts the membership exactly when $g > (1 - \lambda)/(1 - \lambda(1 - f))$.*

**Proposition S19.15** (Agency test of the superunit)**.** *For a superunit at the depth-2 split, with coarse variable $\bar{m}_S$ and blanket input $\bar{s}_S$:*

*(i) the closure defect of Definition 5.1 vanishes, and with $y = \bar{m}_S$ the conditional mutual information $I(y_{t+1}; y_{0:t-1} \mid y_t, \bar{s}_{S,0:t})$ of its input-conditional Markov defect vanishes for every $t$ and every initial law. This is the Markov defect of that definition when the initial law of the members is Gaussian and independent of the world and of the sensor noise; at the stationary law, where the members are correlated with the world, the independence proviso of the definition is not met and only the vanishing of this conditional quantity is claimed;*

*(ii) (H-suff) and (H-phys) of Theorem 11.13 hold, the detectability in (H-phys) because $\bar{F} = F$ and $\rho(F) < 1$, and (H-prot) fails, because the members run the level-0 fusion of Definition 7.2(v) and not the consensus filter of the pooled model. That theorem therefore does not apply to the superunit, and closure of the coarse state holds directly, by (i);*

*(iii) the model has no acts, so (H-act), the agency index, $\mathfrak{N}_{\mathrm{eff}}$ and $\mathfrak{N}_{\mathrm{inf}}$ are inapplicable;*

*(iv) under uniform partner draws redrawn every round (that variant of $\mathfrak{N}_{\mathrm{pr}}$, with which $\mathfrak{N}_{\mathrm{hab}}$, that is $\gamma = 0$, coincides here) the mean-field map has only the uniform fixed point of Theorem S19.9(b), which selects no superunit. The variant of $\mathfrak{N}_{\mathrm{pr}}$ with partners drawn once is not covered;*

*(v) the type-S clause fails in its translation-aware form, by Proposition S19.11.*

*With the translation-aware null admitted among the nulls of type S and evaluated at its stationary laws with the symmetry of the split, the superunit is not a meta-agent at any grade; that the null reaches such a law from the split is computed and not proved (Check S47.180). Under the full exact-fusion null $\mathfrak{N}_{\mathrm{ex}}$ the clause is open.*

**Numerical observation S19.16** (Base point)**.** For $n = 4$, $M = 3$, $\kappa_2 = 2$ (Checks S47.178 and S47.179) the superunit has $r(S) = 7.134$ against 6.066 for the whole population in the exchangeable state of Proposition S19.11, taken as the comparison medium. That state keeps the first-level parents and is invariant under permutations of the parents and of the members of a parent, not of all agents, so it is declared in the manner of clause (c) of Definition 4.11 and is not an instance of that clause; a medium exchangeable over all agents was not computed. An enumeration of all regions of that state finds no larger contrast, so 6.066 is the supremum of Definition 4.11 for this medium and the admissible levels are $r_0 \in (6.07, 7.13]$. A parent inside the superunit has $r(c \mid S) = 1.329$, down from 8.733 before the event: the second event replaces the first level and does not nest it at a common grade. With faithful replacement ($f = 1$) the contrast of the state recorded after fusion rises with $g$, from 7.13 to 9.92 at $g = 0.2$, while the state recorded after the replacement step keeps 7.13 at every $g$ (Check S47.180).

*Remark* S19.17 (Scope). The inherited law of Proposition S19.2 is exact at $\gamma = \infty$ and $\sigma_\alpha^2 = 0$. Theorems S19.4 and S19.9 and Proposition S19.14(a) concern the frozen-calibration mean-field map, as Theorem 7.6 does, and the size condition of Theorem S19.9(c) exceeds the exact threshold size by four orders of magnitude at the base point (Check S47.177). Lemma S19.3 idealizes in-unit translations above level 1 as exact; the residual bias changes member mean-square error by less than $2 \times 10^{-4}$ in relative terms at the base point (Check S47.175). The retained constants $(\Lambda, \hat{\sigma}^2, D)$ do not close under iteration, since the depth-$\ell$ solution recalibrates every lower precision.

**Open problem S19.18** (Exact fusion at level 2)**.** Under the full null $\mathfrak{N}_{\mathrm{ex}}$, in which readers also account for the cross-covariances of aggregates, decide whether any partition of the parents is selected with blocks at $r \geq r_0$. Proposition S19.11 settles the translation-aware form only, from exchangeable starts and at every stationary law with the symmetry of the split.

**Open problem S19.19** (Competing parents and sharp sizes)**.** Find thresholds for $\kappa_2 < M - 1$, where near parents compete for capacity, and a size condition in Theorem S19.9(c) of the order of the exact threshold sizes of Check S47.177.

**Open problem S19.20** (Calibration that tracks the forming law)**.** Show that the stochastic model with online calibration reaches the depth-2 split when $\gamma > \gamma_S^{(2)}$, or exhibit a start from which it does not, and find a closed extension of the map on $(\Lambda, \hat{\sigma}^2, D)$ (Open problems S7.14 and 7.14).

**Open problem S19.21** (A level-2 meta-agent)**.** Extend Definition S19.1 by acts and decision states so that the agency index, $\mathfrak{N}_{\text{eff}}$ and $\mathfrak{N}_{\text{inf}}$ apply, and either exhibit a nested collective formed by this mechanism that meets the grade of Definition 5.19, or prove that none survives the nulls of type S.

**Open problem S19.22** (Closure under turnover)**.** For $g > 0$ compute the closure defect of Definition 5.1 for $\bar{m}_S$, and treat newcomers that bring their own frames, which lower the coherence $\rho$ of Lemma S19.8.

# S20 Conflicting objectives, deliberation time and scale selection

This section addresses two open problems of the literature on collective active inference, conflicting objectives and collective policy, and scale and time laws: Theorems 5.11 and S5.17 decide when members with different targets form an exact potential game and bound the stationary deliberation law when they do not, without saying what conflict costs, how long deliberation takes or what can be identified from outside, and Section 13 coarse-grains along declared frames without saying which frame to declare. For quadratic preferences over additive acts (Section S20.1) the section proves that the antisymmetric part of the interaction matrix is the only obstruction to reversible deliberation on finite act sets and that on unbounded acts a reversible stationary law requires in addition a positive definite interaction matrix, that for two scalar members the existence of a stationary law depends on the protocol, a closed form for the cost of disagreement under shared precision and effect models, the spectral gap of heat-bath deliberation with tracking bounds for symmetric positive definite interaction, mixing and tracking bounds for finite policy sets under weak mutual influence, and that conflict is identified by reciprocity of the mean response or, under Langevin deliberation, by time-reversal asymmetry of lagged covariances, and under that protocol not by a stationary cross-section. For the stationary reversible diffusion of Theorem 13.8 (Section S20.2) it proves that a predictive-information criterion selects frames in the spectral gaps of $\Pi$, that in the nested medium with $\mathfrak{q} < b$ the selected frames are the levels, with relaxation times that for $\mathfrak{q} < 1$ in a deep hierarchy grow as a power of block size until the leak saturates them, that the Laplacian procedure of Section S13.6 returns the same levels, that no threshold of a constant times $\ln N$ on the grounding-free ratio suffices for a region to be returned by that procedure, and that on a fixed tree the block coordinates stay exactly Markov under time-dependent couplings, with level clocks in place of elapsed time. Left open (Section S20.3) are the heat-bath stationary law for nonsymmetric interaction beyond two scalar members, preferences that are not quadratic and interacting effects, simultaneous moves and members who learn about one another, institutions whose protocol is not declared, the endogenous choice of the medium, declared frames that are not invariant, precisions without common invariant frames, nonreversible drifts, and the characterization of the regions the Laplacian procedure returns; proofs are in Section S41 and numerical checks in Section S47.16.

## S20.1 Rotation, cost and time of deliberation

**Definition S20.1** (Quadratic deliberation)**.** Let members $i = 1, \dots, n$ choose acts $u_i \in \mathbb{R}^{d_i}$, $d_u = \sum_i d_i$, with additive effects ($\mathcal{B}_{\text{int}} = 0$) and act parts $G_i^{\text{act}}$ as in (S24), each $\mathsf{M}_i^{(i)}$ of full column rank. The gradient profile is

$$\nabla_{u_i} G_i^{\text{act}}(u) = \sum_j K_{ij} u_j + \mathsf{h}_i, \qquad K_{ij} = \mathsf{M}_i^{(i)\top} \Lambda_i^\star \mathsf{M}_j^{(i)}, \qquad \mathsf{h}_i = \mathsf{M}_i^{(i)\top} \Lambda_i^\star \mathsf{z}_i, \tag{S70}$$

with interaction matrix $K = (K_{ij})$, potential part $K_{\text{s}} = (K + K^\top)/2$, rotational part $K_{\text{a}} = (K - K^\top)/2$, $K_{\text{d}} = \text{blockdiag}(K_{ii}) \succ 0$ and $u^{\text{N}} = -K^{-1}\mathsf{h}$ when $K$ is invertible. Heat-bath deliberation

is (P4) of Definition 5.8 with flat habits: at each event a uniformly chosen member redraws $u_i \sim \mathcal{N}(-K_{ii}^{-1}(\sum_{j\neq i} K_{ij}u_j + \mathsf{h}_i), (\gamma K_{ii})^{-1})$. Langevin deliberation is $\mathrm{d}u = -(Ku+\mathsf{h})\mathrm{d}t + \sqrt{2/\gamma}\mathrm{d}\mathsf{W}_t$, every member descending its own gradient with independent noise.

**Theorem S20.2** (Reversibility and rotation)**.** *In the setting of Definition S20.1:*

*(a) For displacements $\Delta_i, \Delta_j$ of two members, the four-cycle sum of Theorem S5.17(a) is*

$$\mathcal{L}_{ij} = \Delta_i^\top (K_{ji}^\top - K_{ij})\Delta_j = -2\Delta_i^\top (K_{\mathrm{a}})_{ij}\Delta_j, \tag{S71}$$

*so every restriction to finite act sets is reversible if and only if $K_{\mathrm{a}} = 0$.*

*(b) Heat-bath deliberation has a reversible stationary law if and only if $K_{\mathrm{a}} = 0$ and $K \succ 0$; the law is then $\mathcal{N}(u^{\mathrm{N}}, (\gamma K)^{-1})$ and is the unique stationary law.*

*(c) Langevin deliberation has a stationary law if and only if every eigenvalue of $K$ has positive real part; the law is $\mathcal{N}(u^{\mathrm{N}}, \Sigma)$ with $K\Sigma + \Sigma K^\top = (2/\gamma)I$. Then $\mathsf{Q} = K\Sigma - \gamma^{-1}I$ is antisymmetric, $K = (\gamma^{-1}I + \mathsf{Q})\Sigma^{-1}$, the stationary current velocity is $-\mathsf{Q}\Sigma^{-1}(u - u^{\mathrm{N}})$, and its mean square in units of the noise,*

$$\gamma \operatorname{tr}(\mathsf{Q}\Sigma^{-1}\mathsf{Q}^\top) = \gamma \operatorname{tr}(K\Sigma K^\top) - \operatorname{tr} K \geq 0, \tag{S72}$$

*vanishes if and only if the process is reversible, if and only if $K_{\mathrm{a}} = 0$.*

The antisymmetric part of the interaction matrix is therefore the cycle sum of Theorem 5.10(a) in continuous acts, and it is the only obstruction to reversibility. This is the quadratic counterpart of the result of Heins et al. (2023) that binary active-inference agents updating asynchronously sample the Gibbs law of a spin glass, an equivalence that they show to rest on symmetric couplings, obtained from equal precisions within each pair, and to fail under synchronous updates. Unlike finite policy sets, unbounded acts add a second requirement, $K \succ 0$, and the two protocols respond differently to rotation.

**Proposition S20.3** (Two members with scalar acts)**.** *Let $n = 2$, $d_1 = d_2 = 1$ and $\chi = K_{12}K_{21}/(K_{11}K_{22})$. Langevin deliberation has a stationary law if and only if $\chi < 1$; heat-bath deliberation if and only if $|\chi| < 1$, and that law is the equal mixture of two Gaussian laws with mean $u^{\mathrm{N}}$, which is Gaussian if and only if $K_{12} = K_{21}$.*

*(a) (Rotation.) For $K = \left(\begin{smallmatrix} 1 & c \\ -c & 1 \end{smallmatrix}\right)$, $\mathsf{h} = 0$, the Langevin law is $\mathcal{N}(0, \gamma^{-1}I)$ for every $c$, with $\gamma \operatorname{tr}(\mathsf{Q}\Sigma^{-1}\mathsf{Q}^\top) = 2c^2$. The heat-bath law exists only for $|c| < 1$ and is $\frac{1}{2}\mathcal{N}(0, s\left(\begin{smallmatrix} 1 & -c \\ -c & 1 \end{smallmatrix}\right)) + \frac{1}{2}\mathcal{N}(0, s\left(\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right))$ with $s = 1/(\gamma(1-c^2))$; from every initial law with finite second moments the mean converges to 0 if $c^2 < 3$ and the variance diverges if $c^2 \geq 1$.*

*(b) (Escalation.) With shared effect models $\mathsf{M} = \left(\begin{smallmatrix} 1 & \alpha \\ \alpha & 1 \end{smallmatrix}\right)$ and precisions $\Lambda_1^\star = \operatorname{diag}(\epsilon, 1)$, $\Lambda_2^\star = \operatorname{diag}(1, \epsilon)$, $K$ is symmetric with $\chi = \alpha^2(1+\epsilon)^2/(\epsilon+\alpha^2)^2$; for $\alpha = 0.5$, $\epsilon = 0.1$, $\chi > 1$ and neither protocol has a stationary law. With shared precision and effect models $K = \mathsf{M}^\top\Lambda^\star\mathsf{M}$ is a Gram matrix, $0 \leq \chi \leq 1$, and both protocols have a stationary law unless the two effect directions are collinear ($\chi = 1$).*

Part (a) extends the protocol dependence of Example S5.18 to a closed form: rotation leaves the Langevin law untouched, inflates and then destroys the heat-bath law. Part (b) shows that an exact potential game need not have a group posterior: members who each care most about the outcome coordinate the other controls escalate without bound.

**Proposition S20.4** (Cost of disagreement)**.** *Let the members share $\Lambda^\star$ and the true effect models, with $\mathsf{M} = (\mathsf{M}_1, \dots, \mathsf{M}_n)$ of full column rank, and differ in their displacements $\mathsf{z}_i$, with mean $\bar{\mathsf{z}}$. Under either protocol the stationary law is $\mu = \mathcal{N}(u^{\mathrm{N}}, (\gamma K)^{-1})$, $K = \mathsf{M}^\top \Lambda^\star \mathsf{M}$, and*

$$\mathbb{E}_\mu \sum_i G_i^{\mathrm{act}} - \min_u \sum_i G_i^{\mathrm{act}}(u) = \frac{n}{2} \mathsf{w}^\top K^{-1} \mathsf{w} + \frac{n d_u}{2\gamma}, \qquad \mathsf{w}_i = \mathsf{M}_i^\top \Lambda^\star (\mathsf{z}_i - \bar{\mathsf{z}}). \tag{S73}$$

*The law $\mu$ is the group posterior with flat habits and displacement $\mathsf{z}_{\mathrm{N}} = -\mathsf{M} u^{\mathrm{N}}$, characterized on the range of the effects by $\mathsf{M}_j^\top \Lambda^\star \mathsf{z}_{\mathrm{N}} = \mathsf{M}_j^\top \Lambda^\star \mathsf{z}_j$ for all $j$. For two scalar acts with unit effect directions $e_i$ in the $\Lambda^\star$ inner product, $\cos\theta = \langle e_1, e_2 \rangle_{\Lambda^\star}$ and $\eta_i = \langle e_i, (\mathsf{z}_1 - \mathsf{z}_2)/2 \rangle_{\Lambda^\star}$,*

$$\frac{n}{2} \mathsf{w}^\top K^{-1} \mathsf{w} = \frac{\eta_1^2 + 2\cos\theta \eta_1 \eta_2 + \eta_2^2}{\sin^2 \theta}. \tag{S74}$$

The collective is an exact agent, but for a negotiated displacement in which each member's target prevails along its own effect directions, not for the mean target. The bias term is a quadratic counterpart of the simulation finding of Ruiz-Serra et al. (2025) that the ensemble-level expected free energy of an iterated game, the sum of the players' expected free energies, is not necessarily minimized at the equilibria the players reach. The correspondence is interpretive: their players choose among discrete actions simultaneously and learn beliefs about one another, and they report no closed form. The bias diverges as the two effects become collinear, and a single agent holding the summed objective at precision $\gamma$ would pay only $d_u/(2\gamma)$ in fluctuations.

**Theorem S20.5** (Deliberation time and tracking)**.** *Let $K = K^\top \succ 0$, $\mu = \mathcal{N}(u^{\mathrm{N}}, (\gamma K)^{-1})$, $\kappa = \lambda_{\min}(K_{\mathrm{d}}^{-1/2} K K_{\mathrm{d}}^{-1/2}) \in (0, 1]$, and let $\nu_t$ be the law after $t$ heat-bath events.*

*(a) In the transport distance $W_{2,K}$ of the norm $\|v\|_K^2 = v^\top K v$, $W_{2,K}(\nu_t, \mu) \le (1 - \kappa/n)^{t/2} W_{2,K}(\nu_0, \mu)$, and the mean obeys $\|\mathbb{E}_{\nu_t} u - u^{\mathrm{N}}\|_K \le (1 - \kappa/n)^t \|\mathbb{E}_{\nu_0} u - u^{\mathrm{N}}\|_K$ with equality along one direction.*

*(b) The spectral gap of the heat-bath kernel on $L^2(\mu)$ equals $\kappa/n$.*

*(c) For two members, $\kappa = 1 - \cos\theta_{\min}$ with $\cos\theta_{\min}$ the largest singular value of $K_{11}^{-1/2} K_{12} K_{22}^{-1/2}$; under shared $\Lambda^\star$ and effect models $\theta_{\min}$ is the smallest principal angle between the ranges of $(\Lambda^\star)^{1/2} \mathsf{M}_1$ and $(\Lambda^\star)^{1/2} \mathsf{M}_2$.*

*(d) If the loads move, $\mathsf{h} = \mathsf{h}_t$ at event $t$, with $\|u_{t+1}^{\mathrm{N}} - u_t^{\mathrm{N}}\|_K \le \delta$, then*

$$\begin{aligned} \|\mathbb{E}_{\nu_t} u - u_t^{\mathrm{N}}\|_K &\le (1 - \kappa/n)^t \|\mathbb{E}_{\nu_0} u - u_0^{\mathrm{N}}\|_K + \frac{n\delta}{\kappa}, \\ W_{2,K}(\nu_t, \mu_t) &\le (1 - \kappa/n)^{t/2} W_{2,K}(\nu_0, \mu_0) + \frac{2n\delta}{\kappa}, \end{aligned} \tag{S75}$$

*and the constant $n/\kappa$ in the first bound cannot be improved: for a constant drift of $K$-norm $\delta$ per event along the slowest direction and zero initial lag, the lag at event $t$ has $K$-norm $\delta(1 - (1 - \kappa/n)^t) n/\kappa$, which increases to $n\delta/\kappa$ without reaching it.*

Deliberation takes $1/\kappa$ rounds of $n$ events, and once the transient has decayed a collective whose Nash point moves at $K$-speed $v$ per round, that is $\delta = v/n$ per event, lags the moving group posterior by at most $v/\kappa$ in the mean and $2v/\kappa$ in $W_{2,K}$. Nearly collinear effects are therefore costly twice, in bias by (S74) and in time by (c). Finite policy sets admit analogous bounds without any potential.

**Proposition S20.6** (Finite policy sets)**.** *Under (P3) and (P4) with arbitrary $G_i$, let $\mathfrak{J}_{ij}$ be the largest total variation distance between member $i$'s conditionals at two profiles of the others that differ only in member $j$'s policy, $\mathfrak{D} = \max_j \sum_{i\neq j} \mathfrak{J}_{ij}$, and $W_1$ the transport distance of the Hamming metric on profiles. Then $\mathfrak{J}_{ij} \le \tanh(\gamma\omega_{ij}/4)$, with $\omega_{ij}$ the largest oscillation in $\pi_i$ of the change of $G_i$ under a change of $\pi_j$, and one event contracts $W_1$ by $\rho = 1-(1-\mathfrak{D})/n$. If $\mathfrak{D} < 1$,*

$$\mathrm{TV}(\nu_t, \mathbb{P}^\infty) \le W_1(\nu_t, \mathbb{P}^\infty) \le n\Big(1 - \frac{1-\mathfrak{D}}{n}\Big)^t, \tag{S76}$$

*and for a second profile of expected free energies $G'_i$ on the same policy sets, with $\omega_i = \max_{\pi_{-i}} \mathrm{osc}_{\pi_i}(G'_i - G_i)$,*

$$W_1(\mathbb{P}^\infty, \mathbb{P}'^\infty) \le \frac{\sum_i \tanh(\gamma\omega_i/4)}{1-\mathfrak{D}}. \tag{S77}$$

*If the expected free energies change between events, let $G_{i,t}$ be the profile in force at event $t$, $T_t$ its kernel, $\nu_{t+1} = \nu_t T_t$, $\mathfrak{D}_t$ and $\mathbb{P}^\infty_t$ the influence constant and the stationary law of $T_t$, and $\omega_{i,t} = \max_{\pi_{-i}} \mathrm{osc}_{\pi_i}(G_{i,t+1} - G_{i,t})$. If $\mathfrak{D}_t \le \bar{\mathfrak{D}} < 1$ and $\frac{1}{n}\sum_i \tanh(\gamma\omega_{i,t}/4) \le \bar\tau$ for all $t$, then for all $t$*

$$W_1(\nu_t, \mathbb{P}^\infty_t) \le \bar\rho^t W_1(\nu_0, \mathbb{P}^\infty_0) + \frac{n^2\bar\tau}{(1-\bar{\mathfrak{D}})^2}, \qquad \bar\rho = 1 - \frac{1-\bar{\mathfrak{D}}}{n}. \tag{S78}$$

The collective is within $\epsilon$ of its stationary law after $\ln(n/\epsilon)/(1-\mathfrak{D})$ rounds, and if each member's expected free energy moves by at most $\dot\omega/n$ in oscillation per event, that is $\dot\omega$ per round, the expected fraction of members whose policy differs from an optimally coupled draw of the current stationary law is at most $\gamma\dot\omega/(4(1-\bar{\mathfrak{D}})^2)$ once the transient $\bar\rho^t W_1(\nu_0, \mathbb{P}^\infty_0)/n$ has decayed. Equation (S77) extends (59) to any two profiles.

**Proposition S20.7** (What identifies conflict)**.** *(a) For Langevin deliberation, the interaction matrices with stationary law $\mathcal{N}(u^{\mathrm{N}}, \Sigma)$ are exactly $(\gamma^{-1}I + \mathsf{Q})\Sigma^{-1}$ with $\mathsf{Q}$ antisymmetric; the family contains one potential game, $K_0 = (\gamma\Sigma)^{-1}$, and for scalar acts every sufficiently small $\mathsf{Q}$ is realized by a quadratic deliberation. (b) Let $K$ be invertible. Under either protocol the existence of a stationary law does not depend on $\mathsf{h}$, and every stationary law with a finite first moment has mean $-K^{-1}\mathsf{h}$, so that $-\partial\mathbb{E}u/\partial\mathsf{h} = K^{-1}$; the game is a potential game if and only if this response is symmetric. (c) For Langevin deliberation the lagged covariance is $C(\tau) = e^{-K\tau}\Sigma$, so $K = -\dot C(0)\Sigma^{-1}$, $C(\tau)$ is symmetric for all $\tau$ if and only if $K_{\mathrm{a}} = 0$, and $K^{-1} = \gamma\Sigma(I + \gamma\mathsf{Q})^{-1}$.*

A stationary cross-section of a collective's acts can thus never certify that its members pursue one objective: under Langevin deliberation a group model always fits. Conflict is identified by reciprocity of responses to shifted targets or by time-reversal asymmetry of lagged statistics. This bears on Open problem S10.30 and on group models fitted from outside. Waade et al. (2025) fit an active-inference model to the choices of a simulated collective that aggregates its members' votes on a bandit task, and find a much lower fitted action precision when the members hold conflicting preferences. In the present setting disagreement about targets alone moves the fitted target and leaves the fitted precision at $\gamma$, while rotation under heat-bath deliberation inflates the variance of each act by the factor $1/(1-c^2)$ of Proposition S20.3(a), which a fitted group model would read as lower action precision. The comparison is interpretive: their finding comes from simulation, and their collective aggregates discrete votes probabilistically, which is neither protocol of Definition S20.1.

## S20.2 Selecting a scale

A declared frame is good at a horizon if what it discards does not help to predict. For the stationary reversible diffusion of Theorem 13.8, a frame $\Phi$ with orthonormal columns ($0 \le \mathrm{rank}\,\Phi \le N$), a

horizon $\tau > 0$ and a price $c > 0$ in nats per retained coordinate, put

$$J_{\tau,c}(\Phi) = I(x_0; x_\tau) - I(\Phi^\top x_0; x_\tau) + c \operatorname{rank} \Phi, \tag{S79}$$

the predictive information lost by the coarse description plus a resolution cost. The closure defect cannot replace the first two terms: $\mathcal{D} = 0$ on every invariant frame, including the empty one and every level of the nested medium, so the defect plus a cost always selects nothing (Remark S41.4). Let $\iota(s) = -\frac{1}{2}\ln(1 - e^{-2s})$, a decreasing involution of $(0, \infty)$, and $\varsigma(c) = \iota(c)$.

**Theorem S20.8** (Selected scales sit in spectral gaps)**.** *In the setting of Theorem 13.8, with eigenvalues $\varpi_1 \le \cdots \le \varpi_N$ of $\Pi$, $\varpi_0 = 0$ and $\varpi_{N+1} = \infty$:*

*(a) $I(x_0; x_\tau) = \sum_k \iota(\varpi_k \tau)$, and every frame of rank $M$ has $I(\Phi^\top x_0; x_\tau) \le \sum_{k \le M} \iota(\varpi_k \tau)$, with equality when $\operatorname{range} \Phi = \operatorname{range} \mathsf{V}_\mathrm{s}$ and, if $\varpi_M < \varpi_{M+1}$, only then.*

*(b) The minimum of $J_{\tau,c}$ is attained at rank*

$$M^*(\tau, c) = \#\{k : \varpi_k < \varsigma(c)/\tau\}, \tag{S80}$$

*uniquely unless some $\varpi_k \tau = \varsigma(c)$. The frames of rank $M$ on the slow eigenspace are the only minimizers exactly for*

$$\varsigma(c)/\varpi_{M+1} < \tau < \varsigma(c)/\varpi_M, \tag{S81}$$

*an interval that is nonempty if and only if $\varpi_M < \varpi_{M+1}$. For $1 \le M < N$ its logarithmic width is $\ln(\varpi_{M+1}/\varpi_M)$ whatever the price; the windows of $M = 0$ and $M = N$ are the half-lines $\tau > \varsigma(c)/\varpi_1$ and $\tau < \varsigma(c)/\varpi_N$.*

*(c) A declared frame with invariant range, $1 \le \operatorname{rank} \Phi < N$ and $\varkappa_\perp > \varkappa_\Phi$ minimizes $J_{\tau,c}$ over all frames if and only if $\varsigma(c)/\varkappa_\perp \le \tau \le \varsigma(c)/\varkappa_\Phi$. For the block frame of an equitable partition into fewer than $N$ regions with uniform leak and $\varkappa_\mathrm{in}^\mathrm{min} > 2\varkappa_\mathrm{out}^\mathrm{max} - \varepsilon^\Pi$, where $\varkappa_\mathrm{in}^\mathrm{min}$ is the minimum of $\varkappa_\mathrm{in}$ over the blocks with at least two members as in Theorem 13.8(d), this holds for all $\tau \in [\varsigma(c)/\varkappa_\mathrm{in}^\mathrm{min}, \varsigma(c)/(2\varkappa_\mathrm{out}^\mathrm{max} - \varepsilon^\Pi)]$.*

Scale is thus selected by the horizon: a rank is chosen over a range of horizons only if the spectrum of $\Pi$ has a gap there, and the range is the logarithm of the gap ratio. Part (c) connects this to Theorem 13.8(d): a partition into collectives of high contrast is the optimal description at every horizon between the internal relaxation time and the exchange time, up to the factor $\varsigma(c)$. With $\Pi = \varepsilon^\Pi I + L$, (S80) retains the Laplacian modes with eigenvalue strictly below $1/\tau_\mathrm{L} = \varsigma(c)/\tau - \varepsilon^\Pi$. This is the mode rule of the Laplacian renormalization group of Villegas et al. (2023) as Section S13.6 states it, with a mode at equality discarded by the convention fixed there; at equality both ranks minimize, so either convention is optimal. The rule is therefore information-optimal for the diffusion, with their diffusion time read as a prediction horizon; at $c = \iota(1) \approx 0.073$ and vanishing leak the two times coincide. The leak adds a cutoff that the Laplacian procedure, which does not depend on $\varepsilon^\Pi$ (Proposition S13.18), cannot see: nothing is worth retaining beyond $\tau = \varsigma(c)/\varepsilon^\Pi$.

**Theorem S20.9** (Selected levels and the scale-time law in the nested medium)**.** *In the nested medium of Theorem 13.6 with $\mathfrak{q} < b$, let $\mathfrak{g}_l$ be as in Proposition 10.7, $\tau_l = 1/(\varepsilon^\Pi + \mathfrak{g}_l)$ for $1 \le l \le n_\mathrm{H}$, $\tau_0 = 0$ and $\tau_{n_\mathrm{H}+1} = 1/\varepsilon^\Pi$.*

*(a) The spectrum of $\Pi$ is $\varepsilon^\Pi$, simple, and $\varepsilon^\Pi + \mathfrak{g}_l$ with multiplicity $(b-1)b^{n_\mathrm{H}-l}$, with $\mathfrak{g}_1 > \cdots > \mathfrak{g}_{n_\mathrm{H}}$. For $\varsigma(c)\tau_l < \tau < \varsigma(c)\tau_{l+1}$, $0 \le l \le n_\mathrm{H}$, the minimizers of $J_{\tau,c}$ are exactly the frames spanning the level-$l$ block indicators (the agents for $l = 0$, the population mean for $l = n_\mathrm{H}$); for $\tau > \varsigma(c)/\varepsilon^\Pi$ the minimizer is the empty frame. No other rank is selected on an interval of horizons.*

(b) *For* $\mathfrak{q} < 1$ *and* $n_{\mathrm{H}} \to \infty$ *at fixed* $l$, $\mathsf{g}_l$ *converges to* $\mathsf{g}_1^{\infty}\mathfrak{q}^{l-1}$ *with* $\mathsf{g}_1^{\infty} = A_{\mathrm{h}}(1/(1-\mathfrak{q}) + 1/(b-1))$ *and, with* $|C_l| = b^l$, *the relaxation time of level* $l$ *converges:*

$$\lim_{n_{\mathrm{H}}\to\infty} \tau_l = \frac{\mathfrak{q}|C_l|^{z_{\mathrm{H}}}}{\mathsf{g}_1^{\infty} + \varepsilon^{\Pi}\mathfrak{q}|C_l|^{z_{\mathrm{H}}}}, \qquad z_{\mathrm{H}} = \frac{\ln(1/\mathfrak{q})}{\ln b} : \tag{S82}$$

*the relaxation time grows as the power* $z_{\mathrm{H}}$ *of block size, which is the power* $y_{\mathrm{RG}}$ *of linear size under the convention of Theorem 13.6(b), until it saturates at* $1/\varepsilon^{\Pi}$ *where* $\mathfrak{x}_l$ *is of order one.*

(c) *For every* $\tau_{\mathrm{L}} > 0$ *the entries of* $e^{-\tau_{\mathrm{L}}L}$ *are positive, depend only on the level of the smallest common block and decrease strictly with it, and the procedure of Section S13.6 returns exactly the level-*$l$ *blocks for* $1/\mathsf{g}_l \le \tau_{\mathrm{L}} < 1/\mathsf{g}_{l+1}$ *(*$\mathsf{g}_0 = \infty$, $\mathsf{g}_{n_{\mathrm{H}}+1} = 0$*), whatever the order in which ties are broken. For a level-*$l$ *block* $C$ *with* $1 \le l < n_{\mathrm{H}}$ *and grounding-free ratio* $r_\star(C) = \lambda_2(L_C)|C|/\operatorname{cut}(C)$, *this window meets the interval* $(1/\lambda_2(L_C), |C|/\operatorname{cut}(C))$ *of Open problem S13.21 if and only if* $r_\star(C) > b/(b-\mathfrak{q})$.

The dynamic exponent of the nested medium equals its renormalization eigenvalue, as 2 and $\sigma$ do on the lattice (Proposition S13.5(c)), and each level is the selected description for a window of horizons whose logarithmic width tends to $\ln(1/\mathfrak{q})$ in a deep hierarchy with negligible leak. Part (c) turns the numerical statement of Section S13.6 about the nested medium into a theorem with a constant threshold and no logarithm. The entropy of the Laplacian density of Villegas et al. (2023) is not scale invariant here in their sense of a constant specific heat: its mean over a period $\ln(1/\mathfrak{q})$ of $\ln \tau_{\mathrm{L}}$ tends to $\ln b/\ln(1/\mathfrak{q}) = 1/z_{\mathrm{H}}$ (Proposition S41.5), and it is observed to oscillate within the period (Check S47.189). Outside the nested medium the real-space rule can fail.

**Proposition S20.10** (A region of large ratio that is never a macronode)**.** *On the complete graph on* $b \ge 3$ *vertices with unit couplings, so that* $N = b$, *a region* $C$ *of* $b-1$ *vertices has* $r_\star(C) = b-1$, *while for every* $\tau_{\mathrm{L}} > 0$ *and every order of breaking ties the procedure of Section S13.6 returns only the* $b$ *singletons (*$\tau_{\mathrm{L}} < 1/b$*) or the whole graph (*$\tau_{\mathrm{L}} \ge 1/b$*). Hence* $r_\star(C)/\ln N$ *is unbounded over regions that are never macronodes. The conjecture that, with uniform leak, every region* $C$ *whose ratio* $r_\star(C)$ *exceeds a constant times* $\ln N$ *is a macronode of the procedure at some* $\tau_{\mathrm{L}} \in (1/\lambda_2(L_C), |C|/\operatorname{cut}(C))$ *is therefore false for every value of the constant, which is why Open problem S13.21 asks for a characterization of the regions returned in that interval and not for a threshold.*

The same happens inside the nested medium for a level-one block with one site removed (Check S47.188). In both cases $C$ is not a local maximizer of $r_\star$: adding the missing site raises the ratio. Restricting the threshold conjecture to local maximizers does not rescue it on the available evidence: a clique joined by one edge to a long path is a strict local maximizer with $r_\star(C)/\ln N$ up to 95.2, and the procedure did not return it at any scanned diffusion time in the interval (Check S47.152), a computation on grids that does not exclude narrow windows and is not a proof. What is proved in the direction of the threshold conjecture is Theorem S20.9(c), for the level blocks of the nested medium, and Proposition S13.19, for two equitable blocks under its gap condition; the characterization that Open problem S13.21 asks for is open (Open problem S20.13).

**Proposition S20.11** (Level clocks without stationarity)**.** *Let the level couplings* $a_l^{\mathrm{H}}(t) \ge 0$ *and the leak* $\varepsilon^{\Pi}(t) > 0$ *of the nested medium be locally integrable functions of time on a fixed tree,* $\Pi(t) = \varepsilon^{\Pi}(t)I + L(t)$, *and let* $\mathrm{d}x = -\Pi(t)x\mathrm{d}t + \sqrt{2}\mathrm{d}\mathsf{B}_t$ *from an arbitrary initial law.*

(a) *At every level the block coordinates are exactly Markov,* $\mathrm{d}y = -\Pi_\Phi(t)y\mathrm{d}t + \sqrt{2}\mathrm{d}\mathsf{B}'_t$.

*(b) The component of $x$ in the level-$l$ eigenspace obeys $\mathbb{E}[x_t^{(l)} \mid x_s] = e^{-\Theta_l(s,t)} x_s^{(l)}$ with the clock*

$$\Theta_l(s,t) = \int_s^t (\varepsilon^{\Pi}(r) + \mathsf{g}_l(r))\mathrm{d}r, \qquad \mathsf{g}_l(r) = b^l a_l^{\mathrm{H}}(r) + \sum_{l'>l} (b^{l'} - b^{l'-1}) a_{l'}^{\mathrm{H}}(r), \tag{S83}$$

*and $\Theta_{n_{\mathrm{H}}+1} = \int_s^t \varepsilon^{\Pi}$ for the population mean, counted as level $n_{\mathrm{H}}+1$ with $\mathsf{g}_{n_{\mathrm{H}}+1} = 0$.*

*(c) If $x_s$ is Gaussian with $\mathrm{Cov}(x_s) = \sum_l v_l(s)\mathsf{E}_l$ on the level eigenprojections $\mathsf{E}_l$, every $v_l(s) > 0$, and $t > s$, the canonical correlations of $x_s$ and $x_t$ are $e^{-\tilde{\Theta}_l}$ with $\tilde{\Theta}_l = \Theta_l(s,t) + \frac{1}{2}\ln(v_l(t)/v_l(s))$ and $v_l(t) = e^{-2\Theta_l(s,t)} v_l(s) + 2\int_s^t e^{-2\Theta_l(r,t)}\mathrm{d}r$. If moreover $a_l^{\mathrm{H}}(r) > a_{l+1}^{\mathrm{H}}(r)$ on $[s,t]$ (with $a_{n_{\mathrm{H}}+1}^{\mathrm{H}} = 0$) and $v_l(s) \le v_{l+1}(s)$ for $1 \le l \le n_{\mathrm{H}}$, then $\tilde{\Theta}_1 > \cdots > \tilde{\Theta}_{n_{\mathrm{H}}+1}$ and, with $\tilde{\Theta}_0 = \infty$, the frames spanning the level-$l$ block indicators are the only minimizers of $I(x_s; x_t) - I(\Phi^\top x_s; x_t) + c\,\mathrm{rank}\,\Phi$ exactly when $\tilde{\Theta}_{l+1} < \varsigma(c) < \tilde{\Theta}_l$.*

On a fixed tree the scale-time statement therefore survives time-dependent couplings and finite horizons with elapsed time replaced by the level clocks, each a strictly increasing operational time for its level. What blocks a general version is non-commutation: for $\Pi(t)$ that do not share invariant frames, a fixed frame acquires the two-time kernel $\Pi_{\perp\Phi}(t)^\top \mathsf{U}_\perp(t,s)\Pi_{\perp\Phi}(s)$ and a co-moving frame is coupled to its complement through $\dot{\Phi}^\top \Phi_\perp$ (Remark S41.6).

### S20.3 What remains open

**Open problem S20.12** (Strategic deliberation)**.** (a) Determine the heat-bath stationary law and its existence threshold for nonsymmetric $K$ beyond two scalar members; Lemma S41.1 gives only the second-moment criterion. (b) Extend Proposition S20.4 and Theorem S20.5 to preferences that are not quadratic and to interacting effects (Theorem S5.17(c)). (c) Treat simultaneous moves and members who hold and learn beliefs about one another's internal states, the setting of Ruiz-Serra et al. (2025); Conjecture S5.13 is the first case. (d) Decide whether the reciprocity and time-reversal tests of Proposition S20.7 can be carried out on institutions, whose protocol is not declared; Open problem S10.30 stays open. (e) Bound mixing when $\mathfrak{D} \ge 1$, and close the gap between (S78) and the observed lag (Check S47.186).

**Open problem S20.13** (Endogenous and general scale selection)**.** (a) Theorem S20.8 selects a frame given the medium; it does not determine the medium. Whether attention selected by expected free energy drives $a$ and $\varepsilon$ to open spectral gaps, and where, is a variant of Open problem S8.28 and stays open. (b) Bound the excess of $J_{\tau,c}$ on a declared frame that is not invariant in terms of $\mathsf{t}_\Theta$. (c) Characterize the regions that the Laplacian procedure returns at some diffusion time in the interval of Open problem S13.21. In particular, decide whether a ratio $r_\star(C)$ above a constant times $\ln N$ suffices for regions that are local maximizers of $r_\star$ under the addition or removal of one site; the clique-and-path computations of Check S47.152 speak against this restriction on the grids scanned and do not disprove it. (d) Extend Proposition S20.11 to precisions $\Pi(t)$ without common invariant frames and to nonreversible drifts, where the optimal frames depend on the horizon.

## S21 Evidence provenance and institutional memory

This section addresses the open problem of evidence provenance and institutional memory: how a collective that communicates repeatedly through loops, shared records and changing membership can keep its confidence in line with its evidence, posed for records and hierarchies in Open problem S12.30. For one shared linear-Gaussian state with known sensing models it establishes storage for links and

records under which fusion equals the Kalman filter on the observations that can have arrived: one Gaussian pair per link suffices on trees without process noise and on stars at full reporting rate; with process noise, an invertible transition matrix and intermittent blind reports, a list of items indexed by time and as long as the reporting gap is needed and suffices; and on graphs with cycles, in the two settings treated, storage that does not grow with elapsed time suffices once members carry identifiers, whereas no deterministic rule without identifiers is exact for every value of the data on rings of two sizes. Under turnover the stored statistic belongs to the occupant of a position and must be reset with it, and in a static world the expected share of observations that never reaches the record has a closed form. Left open are hierarchies with a different variable at every node, the least storage of an exact protocol on a graph with cycles, encoders that know the model but report off a common schedule, the loss under turnover with process noise and sparse writes, which is computed only numerically, and source overlap or sensing models that are uncertain to the members, which are not treated.

A record that reposts posteriors counts shared evidence more than once, and Proposition S12.8 gives both the cure for memoryless members, when the record starts from the common prior, and, in a scalar example, the size of the error without it. Open problem S12.30 asks what a record must carry once members keep their own filters and histories; the results below answer it for one shared state only, so that problem is narrowed and not closed. The obstruction on cycles is removed by identifiers, by a ring size built into the rule or, with probability one, by random keys. Proofs and secondary results are in Section S42, and the computations in Section S47.17.

The state follows $x_{t+1} = Ax_t + w_t^x$ with $w_t^x \sim \mathcal{N}(0, Q_x)$ and $x_0 \sim \mathcal{N}(m_0, P_0)$, $P_0 \succ 0$. Members $i = 1, \dots, n$ sit at the nodes of a connected graph $G$ and observe $o_{i,t} = H_i x_t + v_{i,t}$ with $v_{i,t} \sim \mathcal{N}(0, R_i)$, $R_i \succ 0$, all noises independent. A pair $(\Lambda, h)$ stands for the density proportional to $\exp(h^\top x - \frac{1}{2} x^\top \Lambda x)$. The prior pair is $\pi_0 = (\Lambda_0, \Lambda_0 m_0)$ with $\Lambda_0 = P_0^{-1}$, the item of an observation is $e_{i,t} = (\Lambda_i^{\mathrm{o}}, h_{i,t}^{\mathrm{o}})$ with $\Lambda_i^{\mathrm{o}} = H_i^\top R_i^{-1} H_i$ and $h_{i,t}^{\mathrm{o}} = H_i^\top R_i^{-1} o_{i,t}$, and $\Pi$ is the one-step prediction of a pair (S143), defined when $A$ is invertible or $Q_x \succ 0$. This is the setting of Proposition S12.8, with the record as the hub of a star.

**Definition S21.1** (Provenance protocol)**.** The information set $D_v$ of node $v$ is empty at the start; an observation by $v$ at time $t$ adds the key $(v, t)$, and a report $u \to v$ over a link of $G$ replaces $D_v$ by $D_u \cup D_v$. A protocol assigns states to nodes and links, a message to every report and an update to every event. It is exact if after every event the output of each node $v$ is the pair of $p(x_t \mid o_{D_v})$. Its provenance statistic is what nodes and links store beyond the node outputs, measured in real numbers. A protocol is anonymous if nodes carry no identifiers and run one deterministic rule that does not depend on $G$. An encoder is blind if the message of a member is a function of its own data that does not depend on the sensing models, schedules or data of other members, and model-aware otherwise.

Exactness is equality with the Kalman filter (Kalman, 1960) run on the observations that can have reached the node.

**Theorem S21.2** (One pair per link on trees)**.** *Let $G$ be a tree.*

*(a) Let $Q_x = 0$. Nodes keep pairs $s_v$ about $x_0$ and links keep pairs $c_{uv}$, all initially $\pi_0$. An observation of $v$ at time $t$ adds $((A^t)^\top \Lambda_v^{\mathrm{o}} A^t, (A^t)^\top h_{v,t}^{\mathrm{o}})$ to $s_v$; a report $u \to v$ sets $s_v \leftarrow s_v + s_u - c_{uv}$ and then $c_{uv} \leftarrow s_u$. After any finite sequence of observations and reports, in any order,*

$$s_v = \pi_0 + \sum_{e \in D_v} e, \qquad c_{uv} = \pi_0 + \sum_{e \in D_u \cap D_v} e,$$

*so $s_v$ is the pair of $p(x_0 \mid o_{D_v})$, the posterior of $x_t = A^t x_0$ is its image under $A^t$, which has a pair when $A$ is invertible, and the protocol keeps $d(d+3)/2$ numbers per link. It is exact for $x_0$ for every $A$, and exact in the sense of Definition S21.1, that is for $x_t$, when $A$ is invertible.*

*(b) Let $G$ be a star with hub $z$, let $A$ be invertible or $Q_x \succ 0$, and let every member run the private filter $p_{i,t} = \Pi(p_{i,t-1}) + e_{i,t}$, $p_{i,0} = \pi_0$, and report $p_{i,t}$ at every step. The hub's recursion*

$$s_t = \Pi(s_{t-1}) + \sum_i (p_{i,t} - \Pi(c_{i,t-1})), \qquad c_{i,t} = p_{i,t}, \qquad s_0 = c_{i,0} = \pi_0,$$

*is the centralized Kalman filter. The same holds when members adopt $s_t$ after each round and $c_{i,t} = s_t$, the mechanism of Proposition S12.8(a).*

The link pair is a channel filter of decentralized fusion in the sense of Section S3.5. By the fusion identity (Lemma S42.1), exact fusion over a link needs only the posterior given the common information, provided the two private parts are conditionally independent given the state and the common part. Without process noise the proviso holds on a tree for every schedule. With process noise it holds on a star at full rate, and the stored pair must be predicted before it is subtracted (Remark S42.2 and Check S47.192). At a lower rate the rule is overconfident already for two members and two steps: it claims precision 43/15 against the exact 19/7, and the mean-square error of its mean is 231/215 times the claimed variance (Proposition S42.3). For $F = 0.99$, $Q = 0.02$ and five members the ratio of error to claim rises from 1.08 at every second step to 1.62 at every tenth (Check S47.193).

**Theorem S21.3** (Intermittent reports need the window ledger)**.** *Let $Q_x \succ 0$ and let $A$ be invertible. Member $i$ observes at the $k$ steps since its last report and then sends $\varphi_i(o_{i,1}, \dots, o_{i,k})$ with a blind encoder. The receiver holds a Gaussian law of the state at the last report with positive definite covariance and, for each step $\tau$ of the window, further evidence with likelihood $\exp(g_\tau^\top x_\tau - \frac{1}{2} x_\tau^\top D_\tau x_\tau)$, $D_\tau \succeq 0$, from other members. If the receiver can compute the exact posterior mean of $x_k$ for every value of the data and every choice of $(D_\tau, g_\tau)_{\tau \le k}$, then the window ledger $(h^{\circ}_{i,\tau})_{\tau \le k}$ is a function of the message. In particular a linear encoder has rank at least $k \operatorname{rank} H_i$. Conversely the ledger, $k \operatorname{rank} H_i$ numbers, suffices for the exact posterior.*

The provenance statistic of a dynamic record is therefore a list of items indexed by time, as long as the reporting gap. The bound fails for $Q_x = 0$, where Theorem S21.2(a) applies, and for $A = 0$ (Remark S42.5). Blindness is what it prices: on a common schedule, with a hub that holds nothing else, model-aware members can each send $d$ numbers per contact with no link storage (Proposition S42.6).

**Theorem S21.4** (No anonymous exact protocol on cycles)**.** *Let $Q_x = 0$, $A = I$ and $n \ge 3$. Identical members with $\Lambda^{\circ} \neq 0$ sit on the cycle $C_n$, observe once, and then exchange messages with both neighbors in synchronous rounds. No anonymous protocol, with states and messages of any dimension, outputs the exact pair $\pi_0 + \sum_s e_s$ at every node from some round on, for every value of the data and every labeling of the ports, on both $C_n$ and $C_{2n}$.*

The proof lifts an execution on $C_n$ to $C_{2n}$, where every node behaves as its image while the required precision differs by $n\Lambda^{\circ}$; with constant data the same comparison excludes any two ring sizes (Corollary S42.8). Storage plays no part in the argument, and the natural guess that no summary of bounded size can be exact on a cycle is false once members carry identifiers (Proposition S21.5). The absence of identifiers is not decisive either (Proposition S42.9): hop counters built for one ring size are exact on that size, and random keys are exact with probability one for every value of the

data. The theorem excludes one deterministic rule that is exact for all data on two sizes. The familiar local rules fail at once on the static triangle, where after $r$ rounds the product rule holds $3^{r-1}$ copies of each item and the channel rule of Theorem S21.2 counts $2r+1$ items in place of three (Proposition S42.7).

**Proposition S21.5** (Ledgers keyed by member and time)**.** *Let $G$ be connected, let members carry identifiers and share a clock, let every member observe at every step, and let one flooding round follow each step, in which every node sends each neighbor its fresh item and the items it first acquired in the previous round and discards duplicates by key $(s,\tau)$. Let $\delta(v,s)=\max(\operatorname{dist}(v,s)-1,0)$ and let $\operatorname{ecc}(v)$ be the eccentricity of $v$.*

*(a) The item $(s,\tau)$ first reaches $v$ in round $\tau+\delta(v,s)$, so $D_v(t)=\{(s,\tau):\tau\le t-\delta(v,s)\}$.*

*(b) If $A$ is invertible or $Q_x\succ 0$, a node that keeps a base pair for time $t-\operatorname{ecc}(v)+1$ and the later items computes the pair of $p(x_t\mid o_{D_v(t)})$ exactly. The filter uses $\sum_s(\operatorname{ecc}(v)-1-\delta(v,s))$ stored items; with the items held one more round to be forwarded, the node stores at most $n\max(\operatorname{ecc}(v)-1,1)$ items and one time per source, and at most $n$ items cross a link per round.*

*(c) A node that knows its distances to all sources can sum the items of each time slice on arrival, and then stores $\operatorname{ecc}(v)$ pairs, a distance table and, for one round, the at most $n-1$ items it has yet to forward.*

With identifiers, a window as long as the eccentricity of the node suffices on a graph with cycles, whatever the elapsed time. In a static world a counter and a cumulative pair per source suffice under any schedule (Proposition S42.11).

**Proposition S21.6** (Covariance intersection)**.** *Let $G$ be connected and let $A$ be invertible or $Q_x\succ 0$. At every step each member predicts, applies the Kalman update for its own observation with its claimed covariance, and replaces its pair by a convex combination $\sum_{j\in N[i]}\omega_{ij}(\Lambda_j,h_j)$ of the pairs in its closed neighborhood, with deterministic weights. Suppose that the initial means are deterministic and that the second-moment matrix $\mathbb{E}(m_i-x_0)(m_i-x_0)^\top$ of every member's initial error, called its error covariance below, is at most its claimed covariance in the positive semidefinite order.*

*(a) At every time the error covariance of every member is at most its claimed covariance $\Lambda_i^{-1}$, with no stored provenance.*

*(b) If the members are identical and share the prior, every claimed covariance equals that of a member that never communicates, for every graph and all weights. If all members start from $\pi_0$ and two members of one closed neighborhood have $\Lambda_j^{\circ}\neq 0$, the protocol is not exact.*

*(c) The error covariance of member $i$ is at least $\operatorname{Cov}(x_t\mid o_{D_i(t)})$, with $D_i(t)$ as in Proposition S21.5(a).*

*(d) For identical scalar members with $H=1$ that start from $\pi_0$, on the complete graph with weights $1/n$ and $Q>0$, the mean-square error of every member converges to $((1-K)^2Q+K^2R/n)/(1-(1-K)^2F^2)$, where $K=P_{\mathrm{loc}}/R$ and $P_{\mathrm{loc}}$ is the limit of the variance of a solitary filter.*

Covariance intersection, the fusion rule for unknown cross-covariance recalled in Section S3.5, is therefore safe without provenance, and for identical members the safety has a price: the claim is that of a solitary member whatever is heard (Numerical observation S21.7).

**Numerical observation S21.7** (Rings without provenance)**.** For $F = 0.99$, $Q = 0.02$, $R = 1$ on rings of $n = 3, 4, 5, 6, 8$ members with one exchange per step, the product rule claims variance 0.0095 with error 26.7 to 26.9 times the claim. The channel rule claims 0.0510 with error 1.52 times the claim at $n = 3$ and 1.12 at $n = 8$. Covariance intersection with equal weights claims 0.1239, with error 0.61 to 0.67 times the claim and 1.18 to 1.37 times the exact variance, where a solitary member has 1.77 to 2.25 (Checks S47.195 and S47.196).

Proposition 5.14 treats turnover for a convention that no member senses; the next proposition treats sensed evidence.

**Proposition S21.8** (Turnover)**.** *Let $Q_x = 0$ and $A = I$, and let the record $z$ be the hub of a star with $n$ positions. The occupant of position $i$ observes at every step and keeps its raw observations. At steps $k, 2k, \dots$ it writes summaries $y = S_i o$ of its unwritten observations, with $S_i$ of full row rank, each carrying information $\Lambda_i^S = H_i^\top S_i^\top (S_i R_i S_i^\top)^{-1} S_i H_i \preceq \Lambda_i^o$. At the end of every step, after the write if there is one, each occupant is replaced independently with probability $g \in (0, 1)$; a leaver takes its raw and unwritten observations, and the newcomer reads the record.*

*(a) Under the protocol of Theorem S21.2(a), resetting the link pair to the record's pair at the read preserves exactness. If the predecessor's link pair is kept, the newcomer's first report makes the record count twice every item that it held at the read and had not shared with the predecessor.*

*(b) At write times $t$ the record's pair is exact for what was written, and its precision $\Lambda_z(t)$ satisfies*

$$\mathbb{E}\Lambda_z(t) = \Lambda_0 + t\rho(g,k)\sum_i \Lambda_i^S, \qquad \rho(g,k) = \frac{1-(1-g)^k}{gk}, \tag{S84}$$

*with $\rho(g, 1) = 1$ and $\rho(g, k) \to 1$ as $g \to 0$.*

*(c) At a write time $t$, before the replacements of that step, the record together with the raw observations of the present occupants has precision $\Lambda_{\mathrm{coll}}(t)$ with expectation $\mathbb{E}\Lambda_z(t) + g^{-1}(1 - (1-g)^t)\sum_i(\Lambda_i^o - \Lambda_i^S)$. Its deficit against all observations ever made is $\sum_i[(t - N_i^W)\Lambda_i^o + (N_i^W - a_i)(\Lambda_i^o - \Lambda_i^S)]$, where $N_i^W$ counts the written observations of position $i$ and $a_i$ is the tenure of its occupant. The deficit vanishes for all $g$ and $k$ if leavers write on exit and summaries are lossless.*

Institutional memory has an exact price in this model: an expected fraction $1 - \rho(g, k)$ of the observations never reaches the record, and what rank-limited summaries discard of the rest survives only for the tenure of the member who saw it, $1/g$ steps on average. An exit write and a summary that spans the row space of $H_i^\top R_i^{-1}$ remove both losses. With process noise, full-rate writes of full pairs make turnover free once the link pair is reset (Remark S42.13); for sparser writes the loss is known only numerically.

**Numerical observation S21.9** (Turnover with process noise)**.** For $F = 0.99$, $Q = 0.02$, $R = 1$, $n = 5$ and a record that fuses exactly what it receives, the mean record variance at write times exceeds the centralized variance by factors 1.010, 1.029 and 1.039 at $g = 0.05$ and $k = 2, 5, 10$, and by 1.044, 1.128 and 1.187 at $g = 0.20$ (Check S47.199). These are Monte Carlo means; a rerun moved four of the six by 0.001 (Check S47.200).

Table S13 collects the storage of the schemes. All results concern one shared state. The tree of Definition 12.1 with $\Psi_u \succ 0$ has a different variable at every node, and its dynamic version is not treated here.

Table S13: Storage of provenance for $n$ members, state dimension $d$ and $T$ rounds; a pair is $d(d+3)/2$ numbers (Remark S42.14).

| Scheme | Storage | Exact when |
|---|---|---|
| Channel pairs | one pair per link | tree and $Q_x = 0$; star at full rate |
| Window ledger, blind | $k \operatorname{rank} H_i$ numbers per member | star, reports every $k$ steps |
| Model-aware summary | $d$ numbers per member, no link state | star, common schedule |
| Cumulative pairs per source | $n$ pairs and counters per node | any graph, $Q_x = 0$ |
| Ledger keyed by member and time | at most $n \max(\operatorname{ecc}(v) - 1, 1)$ items per node | any graph, one flooding round per step |
| Ledger keyed by observation | $nT$ items | any graph and schedule |
| Covariance intersection | none | consistent; not exact when two members of a closed neighborhood are informative |

**Open problem S21.10** (Minimal provenance with identifiers)**.** For $Q_x = 0$ on a graph with cycles and an arbitrary schedule, Proposition S42.11 is exact with $n$ pairs per node. Determine the least storage of an exact protocol as a function of $n$ and of the cycle structure of $G$, and whether growth with $n$ is necessary.

**Open problem S21.11** (Model-aware encoders off schedule)**.** Between the $d$ numbers of Proposition S42.6 and the $k \operatorname{rank} H_i$ numbers of Theorem S21.3, determine the least message dimension for exact fusion when encoders know the model but reports are asynchronous or the schedule is random.

**Open problem S21.12** (Hierarchies and lossy turnover)**.** Determine the provenance statistic for dynamic trees with distinct node variables, the part of Open problem S12.30 that the results of this section do not reach, and a closed form for the loss under turnover when $Q_x \succ 0$ and $k > 1$. Randomized rules without identifiers can be exact on cycles with probability one (Proposition S42.9(c)); whether a continuous deterministic anonymous rule can be exact with probability one from a round that depends on the data is open (Remark S42.10).

## S22 Learned interfaces, online codes and transfer

This section addresses the open problem of learned interfaces and shared meanings: how agents with different internal representations learn the codes and translations that make them mutually intelligible, how learnable calibration is told apart from information lost in the channel and from disagreement about objectives, and whether what is learned transfers to new partners. In the linear-Gaussian setting, under added assumptions that are stated with each result, it is established that the understanding error under Gaussian code error has a finite second moment only for $K \geq r_c + 2$ (Theorem S22.1), that a drifting code row cannot be tracked below a floor of order $q^{1/2}$ in its drift variance $q$ (Theorem S22.3), and that a code learned from messages alone under known symmetric content dynamics is identified exactly up to the orthogonal transformations commuting with those dynamics, a freedom that grounding reduces (Theorem S22.4 and Corollary S22.5). It is also

established that the receiver's risk separates channel loss, calibration and sender deviation only up to an interaction term, with a linear deviation indistinguishable from another code (Theorem S22.6 and Proposition S43.13), and that a hub code transfers to pairs that never interacted with an error bounded independently of the number of agents, whereas independently learned pairwise translations almost surely admit no nonzero consensus (Theorem S22.7). Left open are the closed form of the second moment where it is finite, tracking under attention that depends on calibration, convergence of any joint online algorithm for code and content, strategic deviations and nonlinear encoders (Section S22.6); the choice of an interface for a task, as distinct from its calibration, is not treated.

Section 2.5 shows what grounded pairs identify about a sender's code, the recursion Eq. (S2) learns it, and Definition 2.10 measures a learned translation by its mistranslation. This section follows the learned objects, in the linear-Gaussian setting, into the understanding error, online tracking, joint learning of code and content, the receiver's risk, and composed translations. Proofs are in Section S43 and checks in Section S47.18. Throughout, $\|\cdot\|$ is the Euclidean or operator norm, $\|\cdot\|_F$ the Frobenius norm, and $\kappa(P) = \lambda_{\max}(P)/\lambda_{\min}(P)$.

## S22.1 The understanding error under a random code model

Fix an edge and write $P = P_i \succ 0$ and $H = H_j \in \mathbb{R}^{r_c \times K}$, of full row rank. The message is $y = u + \nu$, where $u$ is the content the sender meant and $\nu \sim \mathcal{N}(0, R_c)$. The receiver holds the code model $\hat{H} = H + \sigma_H W$, with $\sigma_H > 0$ and $W$ a matrix of independent standard normal entries, reads $\hat{m} = m_i + O(y - \hat{H} m_i)$ with $O = P\hat{H}^\top (\hat{H} P \hat{H}^\top)^{-1}$, and incurs the understanding error $\psi = H\hat{m} - u$ of Eq. (S4). Open problem S2.9 concerns the second moment of $\psi$ under this code model. Whether that moment exists depends on the excess dimension $K - r_c$, and for small excess it does not; the open problem is posed with the dichotomy of the next theorem.

**Theorem S22.1** (Moments of the understanding error)**.** *Let $(m_i, u)$ have finite moments of all orders, let $\nu$ be independent of $(m_i, u)$, let $W$ be independent of $(m_i, u, \nu)$, and suppose $\Pr(y \neq 0) > 0$. Put $d = K - r_c + 1$.*

*(a) $\mathbb{E}\|\psi\|^p < \infty$ for $0 < p < d$, and $\mathbb{E}\|\psi\|^p = \infty$ for $p \geq d$. In particular $\mathbb{E}\|\psi\|^2 < \infty$ if and only if $K \geq r_c + 2$, and a square code has $\mathbb{E}\|\psi\| = \infty$.*

*(b) For $K \geq r_c + 2$,*

$$\mathbb{E}\|\psi\|^2 \leq 3\|H\|^2 \kappa(P) \Big[\mathbb{E}\|m_i\|^2 + \frac{\mathbb{E}\|u\|^2 + \operatorname{tr} R_c}{\sigma_H^2 (K - r_c - 1)}\Big] + 3\mathbb{E}\|u\|^2. \tag{S85}$$

*(c) With $t_0 = \|H\|\kappa(P)^{1/2}\|m_i\| + \|u\|$, for every $t > t_0$,*

$$\Pr(\|\psi\| > t \mid m_i, u, \nu) \leq \frac{1}{2^{d/2}\Gamma(d/2+1)} \Big(\frac{\|H\|\kappa(P)^{1/2}\|y\|}{\sigma_H (t - t_0)}\Big)^d. \tag{S86}$$

The mechanism is the inversion of $\hat{H} P \hat{H}^\top$: the lift $Oy$ is the reciprocal of a Gaussian vector in $d$ dimensions (Lemma S43.2), whose density is positive at the origin, so part (a) holds for every $\sigma_H > 0$. A regularized gain has bounded norm and is not affected (Theorem S22.6(d)). The first instance of Check S47.6 has $K = 3$ and $r_c = 2$; its second moment is infinite, with a logarithmic divergence that $10^5$ draws at $\sigma_H^2 = 0.02$ usually do not reveal (Check S47.202). For this reason Check S47.6 reports a truncated moment for that instance and an untruncated moment only for a second instance with $K = 6$ and $r_c = 2$. Open problem S2.9 has a finite answer only for $K \geq r_c + 2$, for a truncated moment, or for a regularized gain, and it is posed accordingly. For small $\sigma_H$ the next statement is sharper than (b).

**Proposition S22.2** (Second-order expansion). *Fix $(m_i, u)$ and let $\nu \sim \mathcal{N}(0, R_c)$ be independent of $W$. Put $S_0 = HPH^\top$, $O_0 = PH^\top S_0^{-1}$, $P_0^+ = P - O_0 HP$, $\hat{m}_0 = m_i + O_0(y - Hm_i)$ and $\bar{m}_0 = m_i + O_0(u - Hm_i)$, and let $s_{\min}(H)$ be the smallest singular value of $H$. For every $0 < \rho < s_{\min}(H)$, as $\sigma_H \to 0$,*

$$\begin{aligned}\mathbb{E}[\psi\psi^\top; \sigma_H\|W\|_F \le \rho] = R_c + \sigma_H^2\Big[&\mathbb{E}\|\hat{m}_0\|^2 I - \operatorname{tr}(P_0^+)(R_c S_0^{-1} + S_0^{-1}R_c) \\ &+ R_c O_0^\top O_0 + O_0^\top O_0 R_c\Big] + O(\sigma_H^4),\end{aligned} \tag{S87}$$

*where $\mathbb{E}\|\hat{m}_0\|^2 = \|\bar{m}_0\|^2 + \operatorname{tr}(O_0 R_c O_0^\top)$. If $K \ge r_c + 2$, the same expansion holds for $\mathbb{E}[\psi\psi^\top]$.*

The first term in the bracket is the one named in Section S2.4. The other three have the same order and are proportional to the channel noise. Their sign depends on the instance (Remark S43.4), so the share of the excess carried by the first term is not universal: it ranged from 0.95 to 1.15 in Check S47.203, and a scan of the closed form over random instances found values well outside that range (Check S47.208). In the two instances of Check S47.6, where the excess is estimated by Monte Carlo and so includes all orders in $\sigma_H$, and is that of the truncated moment in the instance with $K = r_c + 1$, the same ratio was between 0.69 and 0.98.

### S22.2 Learning and tracking a code online

For a fixed code the recursion Eq. (S2) is classical least squares, and Proposition S43.5 records what is used below. With $P_0^H = \Pi_0 \otimes R_c$ the error covariance stays of the form $\Pi_t \otimes R_c$; the code is learned exactly on the directions of content that keep occurring, which is persistent excitation in the terms of Proposition 2.7; and for centered Gaussian content with covariance $M_2 \succ 0$ the least-squares estimate from $T \ge K + 2$ grounded rounds has $\mathbb{E}\|\hat{H}_T - H\|_F^2 = \operatorname{tr}(R_c)\operatorname{tr}(M_2^{-1})/(T - K - 1)$, while the risk is infinite for $T \in \{K, K+1\}$, where the estimate is still unique. A drifting code cannot be learned once and for all.

**Theorem S22.3** (Tracking a drifting code). *Let one row $\ell_t \in \mathbb{R}^K$ of a code follow $\ell_{t+1} = \ell_t + w_t$ with $w_t \sim \mathcal{N}(0, qI_K)$, $q > 0$, from $\ell_0 \sim \mathcal{N}(\hat{\ell}_0^-, P_0^-)$ with $P_0^- \succ 0$, and let the receiver observe $y_t = \phi_t^\top \ell_t + \nu_t$ with $\nu_t \sim \mathcal{N}(0, \rho)$, $\rho > 0$, together with the regressor $\phi_t$. Assume that $\phi_t$ is conditionally independent of $\ell_t$ given the history $\mathcal{H}_t$ generated by $(\phi_s, y_s)_{s<t}$, and that $\nu_t$ and $w_t$ are independent of everything realized before them. Let $\hat{\ell}_t^-$ and $P_t^-$ be the conditional mean and covariance of $\ell_t$ given $\mathcal{H}_t$.*

*(a) If $\mathbb{E}[\phi_t\phi_t^\top] \preceq \tilde{M}$ for all $t$, where $\tilde{M}$ has eigenvalues $\tilde{\mu}_1, \dots, \tilde{\mu}_K$, then $\mathbb{E}P_t^- \succeq \bar{P}_t$ for all $t$, where*

$$\bar{P}_{t+1} = \left(\bar{P}_t^{-1} + \tilde{M}/\rho\right)^{-1} + qI, \qquad \bar{P}_0 = P_0^-, \tag{S88}$$

*and*

$$\liminf_{t\to\infty} \mathbb{E}\|\ell_t - \hat{\ell}_t^-\|^2 \ge \sum_{k=1}^K \left[\frac{q}{2} + \left(\frac{q^2}{4} + \frac{q\rho}{\tilde{\mu}_k}\right)^{1/2}\right] \ge \sum_{k=1}^K \left(\frac{q\rho}{\tilde{\mu}_k}\right)^{1/2}, \tag{S89}$$

*which is infinite if some $\tilde{\mu}_k = 0$.*

*(b) Let $\phi_t = b_t g_t$ be independent over rounds and independent of the code and the noise, with $b_t$ Bernoulli of mean $p_a$ and $g_t \sim \mathcal{N}(0, M_2)$, where $M_2 \succ 0$ has eigenvalues $\mu_k$, so that $\tilde{M} = p_a M_2$. Put $\gamma = \max_k (q\mu_k/(\rho p_a))^{1/2}$ and $\xi = \frac{1}{2}\sum_k (q\mu_k/(\rho p_a))^{1/2}$. If $\gamma + \xi < 1$, then*

$$\limsup_{t\to\infty} \mathbb{E}\|\ell_t - \hat{\ell}_t^-\|^2 \le (1 + \epsilon_q) \sum_{k=1}^K \left(\frac{q\rho}{\tilde{\mu}_k}\right)^{1/2}, \qquad \epsilon_q = \frac{\gamma + \xi}{1 - \gamma - \xi}. \tag{S90}$$

*(c) If $v^\top \phi_t = 0$ almost surely for all $t$ and $P_0^- = p_0 I$, then $P_t^- v = (p_0 + qt)v$. A run of $\tau$ rounds with $\phi_t = 0$ adds $q\tau I$ to the error covariance.*

The floor (a) needs only a bound on the second moment of what the receiver attends to and allows attention to depend on the receiver's history. By (b), in the Bernoulli–Gaussian attention model stated there, in which attention is independent of the code and of the noise, a constant-gain rule attains it up to the factor $1 + O(q^{1/2})$, and halving the attended fraction $p_{\mathrm{a}}$ then raises the floor by $\sqrt{2}$ to leading order. By (c) an edge left unattended for $\tau$ rounds returns with a code error of order $q\tau$, which bears on Open problem S7.13.

### S22.3 Learning code and content together

Without grounded pairs the receiver must infer the code and the content from the messages alone, a bilinear problem. Let the content follow the stationary process $\phi_{t+1} = F\phi_t + w_t$, $w_t \sim \mathcal{N}(0, Q)$, where $F$ is symmetric with spectral radius below one and spectral decomposition $F = \sum_{a=1}^{m} f_a \Pi_a$, with distinct eigenvalues $f_a$ and eigenprojectors $\Pi_a = V_a V_a^\top$ of rank $k_a$, and where $Q = \sum_a q_a \Pi_a$ with $q_a > 0$. Messages are $y_t = H\phi_t + \nu_t$, with $\nu_t \sim \mathcal{N}(0, R_c)$ independent over rounds and of the content, and $H \in \mathbb{R}^{r_c \times K}$ is arbitrary.

**Theorem S22.4** (Gauge freedom of ungrounded learning)**.** *Let the receiver know $(F, Q, R_c)$ and observe only the messages $(y_t)_{t \geq 0}$, and let $\mathcal{G}_F = \{T \in \mathrm{O}(K) : TF = FT\}$, which is isomorphic to $\prod_a \mathrm{O}(k_a)$.*

*(a) A code $H'$ gives the message process the same law as $H$ if and only if $H' = HT$ for some $T \in \mathcal{G}_F$.*

*(b) With $\varrho_a = \operatorname{rank}(HV_a)$, the set $H\mathcal{G}_F$ of such codes is a compact manifold of dimension*

$$\dim(H\mathcal{G}_F) = \sum_{a=1}^{m} \Big[ \frac{\varrho_a(\varrho_a - 1)}{2} + \varrho_a(k_a - \varrho_a) \Big]. \tag{S91}$$

*(c) $\mathcal{G}_F$ is the group of all $T \in \mathrm{GL}(K)$ for which $(T\phi_t)$ has the law of $(\phi_t)$, and $(H, \phi) \mapsto (HT^{-1}, T\phi)$ leaves every message unchanged.*

The freedom is the symmetry of the receiver's model of the content: a code can be told apart from its rotations only as far as the dynamics tell directions of content apart. If $F = fI$ the code is determined up to $\mathrm{O}(K)$, and if $F$ has simple spectrum, up to $2^K$ sign choices. At codes of full block rank the autocovariances have no further null directions (Proposition S43.10), so the code is locally identified on the quotient. The block ranks $\varrho_a$ are unrelated to the code-rate ratio $\varrho_c$ of the main text. Grounding removes what the dynamics leave.

**Corollary S22.5** (Grounding fixes the gauge)**.** *In the setting of Theorem S22.4, let the receiver also hold an independent sample of grounded pairs $(\phi_t, y_t)$, obtained under exact joint attention with an exactly observing receiver as in Proposition 2.7(b), with second moment $M_2 = \mathbb{E}[\phi_t \phi_t^\top]$, and let $\mathcal{P}_M$ be the orthogonal projector onto $\operatorname{col}(M_2)$. The codes compatible with both data sets are $\{HT : T \in \mathcal{G}_F,\ HT\mathcal{P}_M = H\mathcal{P}_M\}$. If $H$ is injective, this set is in bijection with $\prod_a \mathrm{O}(k_a - m_a)$, where $m_a = \operatorname{rank}(\Pi_a \mathcal{P}_M)$; the code is identified if and only if $m_a = k_a$ for every $a$, and the set is finite if and only if $k_a - m_a \leq 1$ for every $a$. Still for injective $H$: for $F = fI$ the residual freedom is $\mathrm{O}(K - \operatorname{rank} M_2)$, and for simple spectrum the code is identified if and only if $\mathcal{P}_M v_a \neq 0$ for every eigenvector $v_a$ of $F$.*

With isotropic dynamics the message stream does not change the condition for point identification, which remains $M_2 \succ 0$ as in Proposition 2.7. It does shrink the unidentified set: for injective $H$ the grounded pairs alone leave the affine family $\{H' : H'\mathcal{P}_M = H\mathcal{P}_M\}$, of dimension $r_c(K - \operatorname{rank} M_2)$, and the stream cuts it down to a copy of $\mathrm{O}(K - \operatorname{rank} M_2)$, which consists of two codes when $\operatorname{rank} M_2 = K - 1$ (Check S47.208). With simple spectrum and injective $H$, one grounded direction in general position identifies the whole code: the grounding a collective needs is set by the degeneracy of the shared dependence model. The assumptions that $F$ is symmetric and that $Q$ is isotropic on each eigenspace of $F$, $Q = \sum_a q_a \Pi_a$, which is stronger than commuting with $F$, are introduced here (Open problem S22.10).

### S22.4 What a receiver's risk is made of

The setting extends Proposition S26.1. Let the referent be $\eta \sim \mathcal{N}(m_i, P)$, so that the receiver's prior is correct, and let the message be $y = H\eta + h + \nu$ with $\nu \sim \mathcal{N}(0, R)$, where the sender's deviation $h$ has mean $\bar{h}$ and covariance $\Sigma_h$ and $H$ has full row rank. The receiver reads $\hat{m} = m_i + \hat{\mathcal{K}}(y - \hat{H}m_i)$ with the code model $\hat{H} = H + \Delta$ and a gain $\hat{\mathcal{K}}$ that is a measurable function of $\hat{H}$. Write $S^* = HPH^\top + R$, $\mathcal{K}^* = PH^\top S^{*-1}$ and $P^+ = P - \mathcal{K}^* S^* \mathcal{K}^{*\top}$ for the innovation covariance, gain and posterior covariance at the true code, and $\langle x, z\rangle_M = x^\top M z$, $\|x\|_M^2 = x^\top M x$ for a weight $M \succeq 0$.

**Theorem S22.6** (Decomposition of the receiver's risk)**.** *Let $(\eta, \nu)$, $h$ and $\hat{H}$ be independent, with $\mathbb{E}\|\hat{\mathcal{K}}\|_F^2 < \infty$ and $\mathbb{E}\|\hat{\mathcal{K}}\Delta\|_F^2 < \infty$.*

*(a) The risk is*

$$\mathbb{E}\|\hat{m} - \eta\|_M^2 = \operatorname{tr}(MP^+) + C_{\mathrm{cal}} + C_{\mathrm{obj}} + C_{\mathrm{int}}, \tag{S92}$$

*with $C_{\mathrm{cal}} = \mathbb{E}\operatorname{tr}\big(M(\hat{\mathcal{K}} - \mathcal{K}^*)S^*(\hat{\mathcal{K}} - \mathcal{K}^*)^\top\big) + \mathbb{E}\|\hat{\mathcal{K}}\Delta m_i\|_M^2$, $C_{\mathrm{obj}} = \mathbb{E}\operatorname{tr}\big(M\hat{\mathcal{K}}(\Sigma_h + \bar{h}\bar{h}^\top)\hat{\mathcal{K}}^\top\big)$ and $C_{\mathrm{int}} = -2\mathbb{E}\langle\hat{\mathcal{K}}\bar{h}, \hat{\mathcal{K}}\Delta m_i\rangle_M$. The channel term splits as $\operatorname{tr}(MP^+) = \operatorname{tr}(MP_0^+) + \operatorname{tr}(M(P^+ - P_0^+))$ with $P_0^+ = P - PH^\top(HPH^\top)^{-1}HP$, and both parts are nonnegative.*

*(b) The error is a sum $\hat{m} - \eta = e_{\mathrm{ch}} + e_{\mathrm{gain}} + e_{\mathrm{dev}} + e_{\mathrm{off}}$ of four vectors (Eq. (S163)), with $e_{\mathrm{off}} = \hat{\mathcal{K}}(\bar{h} - \Delta m_i)$, that are pairwise orthogonal in $L^2$ with weight $M$.*

*(c) For a given gain the offset depends on $(\bar{h}, \Delta)$ only through $\bar{h} - \Delta m_i$; the gain is itself a function of $\hat{H} = H + \Delta$, so the offset as a whole is not a function of $\bar{h} - \Delta m_i$ alone. If $\bar{h} = 0$ or $m_i = 0$ then $C_{\mathrm{int}} = 0$; otherwise it can have either sign. The terms that involve the offset satisfy*

$$C_{\mathrm{obj}} + \mathbb{E}\|\hat{\mathcal{K}}\Delta m_i\|_M^2 + C_{\mathrm{int}} = \mathbb{E}\operatorname{tr}(M\hat{\mathcal{K}}\Sigma_h\hat{\mathcal{K}}^\top) + \mathbb{E}\|e_{\mathrm{off}}\|_M^2, \tag{S93}$$

*so $C_{\mathrm{int}}$ cancels the mean part $\mathbb{E}\|\hat{\mathcal{K}}\bar{h}\|_M^2$ of $C_{\mathrm{obj}}$ and the second term of $C_{\mathrm{cal}}$ exactly if and only if $M^{1/2}\hat{\mathcal{K}}(\bar{h} - \Delta m_i) = 0$ almost surely, and it cancels all of $C_{\mathrm{obj}}$ if in addition $\mathbb{E}\operatorname{tr}(M\hat{\mathcal{K}}\Sigma_h\hat{\mathcal{K}}^\top) = 0$. For fixed $\Delta$ the mean of the innovation $y - \hat{H}m_i$ is an affine function of $m_i$, determined by its values at $K + 1$ affinely independent prior means and by no fewer.*

*(d) The moment conditions hold for the regularized gain $\hat{\mathcal{K}} = P\hat{H}^\top(\hat{H}P\hat{H}^\top + \hat{R})^{-1}$ with fixed $\hat{R} \succ 0$ whenever $\mathbb{E}\|\Delta\|_F^2 < \infty$, since $\|\hat{\mathcal{K}}\| \le \frac{1}{2}\|P^{1/2}\|\|\hat{R}^{-1/2}\|$. They fail for the receiver map $O$ under the code model of Theorem S22.1 when $K \le r_c + 1$.*

Channel loss, calibration error and sender deviation are therefore not three orthogonal pieces: the mean deviation and the code error meet in the single offset $e_{\mathrm{off}}$, and $C_{\mathrm{int}}$ can cancel the mean parts of their separate costs or double them. For a given gain a biased sender and a miscalibrated code enter the risk only through this offset, so the risk does not separate them, and neither do the

receiver's innovations at fewer than $K+1$ prior means. For an honest sender the calibration term has the second-order form Eq. (S164), which decays like $1/T$ for a fixed code. For a drifting code it is not removed by learning: given the receiver's history it is linear in the error covariance of the code, with a coefficient that is positive when $M \succ 0$ and the gain does not vanish (Eq. (S166)), and the mean of that covariance stays above the floor of Theorem S22.3, of order $q^{1/2}$. That the cost itself levels off at order $q^{1/2}$ is shown for its leading term only (Remark S43.12).

Learning sharpens the confounding (Proposition S43.13). If the deviation on grounded rounds is $h_t = D\phi_t + d + \zeta_t$ with centered Gaussian $\zeta_t$ of covariance $\Sigma_\zeta$, least squares with an intercept converges to $(H+D, d)$, the law of the data depends on the parameters only through $(H+D, d, R+\Sigma_\zeta)$, and the reading at the learned code is the conditional mean of the referent. A linear deviation is, to a learner, another code: $D = -H$ silences the channel while every message stays consistent with the learned code, and only a reference outside the grounded data separates bias from code (Remark S43.14).

### S22.5 Transfer along paths and through a hub

Let agent $i$ hold the frame $U_i \in \mathrm{GL}(K)$, let the edge $j \to i$ carry the learned translation $\Theta_{ij}$, and let $\Upsilon_{ij} = U_i^{-1}\Theta_{ij}U_j$ be its mistranslation (Definition 2.10). For a walk $\mathfrak{p} = (i_0 \to i_1 \to \cdots \to i_L)$ put $\Theta_\mathfrak{p} = \Theta_{i_L i_{L-1}} \cdots \Theta_{i_1 i_0}$, and define $\Upsilon_\mathfrak{p}$ in the same way.

**Theorem S22.7** (Composition of learned translations)**.** *(a) $U_{i_L}^{-1}\Theta_\mathfrak{p} U_{i_0} = \Upsilon_\mathfrak{p}$, and for a directed closed walk $\mathfrak{c}$ at $i_0$, $\mathrm{Hol}(\mathfrak{c}) = U_{i_0}\Upsilon_\mathfrak{c} U_{i_0}^{-1}$.*

*(b) Let the edges of $\mathfrak{p}$ be distinct, with independent mistranslations $\Upsilon_e = I + \Delta_e$ such that $\mathbb{E}\Delta_e = 0$ and $\mathbb{E}[\Delta_e^\top \Delta_e] = \delta_e^2 I$. Then $\mathbb{E}\Upsilon_\mathfrak{p} = I$, and for every $x \in \mathbb{R}^K$,*

$$\mathbb{E}\|(\Upsilon_\mathfrak{p} - I)x\|^2 = \Big(\prod_{e\in\mathfrak{p}}(1+\delta_e^2) - 1\Big)\|x\|^2. \tag{S94}$$

*Without the isotropy assumption the left side is still nondecreasing when the walk is extended.*

*(c) Let every agent hold one map $g_i \in \mathrm{GL}(K)$, an estimate of $U_i U_h^{-1}$ for a hub frame $U_h$ with spoke mistranslation $\Upsilon_i = U_i^{-1} g_i U_h$, and let $\Theta_{ij} = g_i g_j^{-1}$ for all pairs. Then every holonomy is the identity for every value of the maps, $\dim \mathrm{Cons}(\Theta) = K$, $\Upsilon_{ij} = \Upsilon_i \Upsilon_j^{-1}$, and if $\|\Upsilon_i - I\| \le \epsilon_i$ and $\|\Upsilon_k - I\| \le \epsilon_k < 1$, then*

$$\|\Upsilon_{ik} - I\| \le \frac{\epsilon_i + \epsilon_k}{1 - \epsilon_k}, \tag{S95}$$

*for every pair, whether or not $i$ and $k$ have interacted, and independently of the number $n$ of agents.*

*(d) On a strongly connected attention graph with $n \ge 2$ agents, let the translations on distinct edges be independent with laws absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^{K\times K}$. Then $\mathrm{Cons}(\Theta) = \{0\}$ almost surely. On a simple directed cycle with the mistranslations of (b), $\mathbb{E}\|(\Upsilon_\mathfrak{c} - I)x\|^2$ is given by Eq. (S94).*

Relayed error accumulates geometrically, as $L\delta^2$ to first order. By (d) independently learned pairwise translations almost surely admit no nonzero configuration in consensus, so the consensus space of Theorem 6.5 collapses however small the errors are. A hub code is flat for every realization of the learned maps, and (c) is the zero-shot statement: a new partner trains only its own spoke, and its mistranslation with every other agent is, to first order, the difference of two spoke errors. The

bound is deterministic because the inverse of a spoke with Gaussian error has no finite first moment (Remark S43.15); separate speaking and listening maps restore the moments and lose flatness (Remark S43.16). The pairwise design trains $n(n-1)$ maps and the hub design $n$; under a fixed budget of grounded rounds per agent and an assumed $1/T$ learning rate, a hub route with two trained legs has the smaller mean-square mistranslation once $n$ exceeds a small threshold (Corollary S43.17). Flatness does not imply correctness: the hub translations are the exact translations, in the sense of Proposition S6.5, of the frames $U_i\Upsilon_i$, which differ from the agents' own frames.

### S22.6 Open problems

**Open problem S22.8** (Exact second moment)**.** For $K \geq r_c + 2$, find $\mathbb{E}[\psi\psi^\top]$ in closed form as a function of $\sigma_H$. Proposition S22.2 gives it to second order and Eq. (S85) bounds it; neither is sharp at moderate $\sigma_H$.

**Open problem S22.9** (Attention that depends on calibration)**.** The floor of Theorem S22.3(a) holds for attention rules that depend on the receiver's history, but it is matched in (b) only for attention independent of the code. Determine the stationary tracking error when the attended fraction of an edge decreases with the receiver's error covariance on that edge, and decide whether the linear loss of calibration on unattended edges settles Open problem S7.13.

**Open problem S22.10** (Joint online learning)**.** Extend Theorem S22.4 to $F$ that is not symmetric, to $Q$ that is not of the form $\sum_a q_a \Pi_a$, whether or not it commutes with $F$, and to a receiver that must also learn $(F, Q)$. For a specific online algorithm that updates the code, the dependence model and the state estimate together, prove or refute convergence to the orbit $H\mathcal{G}_F$ from generic initial conditions.

**Open problem S22.11** (Strategic deviations)**.** In Proposition S43.13 the deviation $(D, d, \Sigma_\zeta)$ is fixed. Let the sender choose it to minimize its own quadratic loss, knowing that the receiver learns the code from grounded pairs. Characterize the equilibria in the linear-Gaussian setting and the reference a receiver needs in order to detect a deviation.

**Open problem S22.12** (Nonlinear representations)**.** Every result of this section uses linear codes and Gaussian laws. For nonlinear encoders, the transformations of the content space that preserve the law of the content process still leave the law of the messages unchanged, but whether they exhaust the freedom of ungrounded learning, and how composed nonlinear translations accumulate error, lie outside the setting.

## S23 Energetic viability

This section addresses the open problem of energetic viability: which energetic constraints select and maintain informational hierarchies, and how the dissipation and the information flow that a coarse description removes from view are to be accounted for. For the stationary linear-Gaussian joint process of Eq. (23), with every coordinate even under time reversal, we establish without physical assumptions the entropy production per round in closed form together with its reversibility criterion (Theorem S23.3), that a coarse linear description can only hide dissipation and a scalar one hides all of it (Theorem S23.5), and discrete-time second-law inequalities for the information that crosses a blanket (Theorem S23.8). Every statement in joules rests on three bridge premises that are declared and not derived (Assumptions S1 to S3) or, for the signal energy of Proposition S23.10(a), on a thermal-noise clause stated there, and under them the section gives necessary conditions only: a size

beyond which a collective of memoryless agents in which everyone listens to everyone has a negative margin, while the margin of a shared record converges (Theorem S23.12). Left open are a derivation of the premises from a physical model of the exterior, every sufficient condition for viability, the rate of a coarse description, which is supported by computation only (Numerical observation S23.7 and Open problem S23.14), agents with memory, and coordinates that are odd under time reversal (Open problems S23.15 to S23.17).

The main text counts information and never physical energy. The energy account given here has two layers that are kept apart. The first layer consists of theorems about the path law of the joint process: how irreversible it is, how much of that irreversibility a coarse description hides, and how much information crosses a blanket per round. No physics enters there. The second layer converts nats into joules through the three premises, and it yields necessary conditions for a collective to pay for its own channels. Proofs are in Section S44. Sans-serif symbols are local to this section: $\mathsf{f}$ is here the autoregressive coefficient of a recorded source and not the newcomer fidelity of Proposition 5.14, and the window covariance $\mathsf{T}_n$, which always carries its index, is distinct from the temperature $\mathsf{T}$.

## S23.1 Irreversibility of the joint process

**Assumption S23.1** (Stationary even process)**.** The joint process is Eq. (23) with $\rho(A) < 1$ and $Q^{\Xi} \succ 0$, started from its stationary law $\mathcal{N}(0, \Sigma)$, where $\Sigma = A\Sigma A^{\top} + Q^{\Xi}$. Every coordinate of $X$ is even under time reversal, so that the reverse of a path $(x_0, \dots, x_n)$ is $(x_n, \dots, x_0)$.

The parity convention is a modeling choice. Beliefs, records and messages are treated as positions; a model with momentum-like coordinates would flip their sign under reversal and change every formula below.

**Definition S23.2** (Window divergence and entropy production)**.** For a matrix $C$ of full row rank $m$ and the description $Y_t = CX_t$, the window divergence is

$$\mathsf{D}_n(C) = D_{\mathrm{KL}}\big(\mathrm{law}(Y_0, \dots, Y_n) \| \mathrm{law}(Y_n, \dots, Y_0)\big), \qquad n \geq 1, \tag{S96}$$

and $\mathsf{D}_n = \mathsf{D}_n(I)$. The entropy production per round is $\mathsf{ep} = \lim_n \mathsf{D}_n/n$, in nats, and the hidden dissipation of the description $Y$ is $\mathsf{ep} - \limsup_n \mathsf{D}_n(C)/n$.

The definition follows the relative-entropy expressions for dissipation. Kawai et al. (2007) equate the mean dissipated work of a Hamiltonian system driven between two equilibrium states with $k_{\mathrm{B}}\mathsf{T}$ times the relative entropy between the forward and backward phase-space densities, Gomez-Marin et al. (2007) give the corresponding expression in terms of path laws for stochastic dynamics, and Roldán and Parrondo (2010) use the rate $\lim_n \mathsf{D}_n/n$ for a stationary series. Here $\mathsf{D}_n$ is a property of the path law alone, and no physical dissipation is asserted before Section S23.4.

**Theorem S23.3** (Entropy production of the joint process)**.** *Under Assumption S23.1 put* $\mathsf{J} = A\Sigma - \Sigma A^{\top}$ *and* $\widetilde{Q}^{\Xi} = \Sigma - \Sigma A^{\top}\Sigma^{-1}A\Sigma$*. Then* $\mathsf{D}_n = n\mathsf{ep}$ *for every* $n \geq 1$*, with*

$$\mathsf{ep} = \mathrm{tr}\big((Q^{\Xi})^{-1}\mathsf{J}A^{\top}\big) = \tfrac{1}{2}\,\mathrm{tr}\big((Q^{\Xi})^{-1}(\widetilde{Q}^{\Xi} - Q^{\Xi})\big) + \tfrac{1}{2}\,\mathrm{tr}\big((Q^{\Xi})^{-1}\mathsf{J}\Sigma^{-1}\mathsf{J}^{\top}\big), \tag{S97}$$

*and both terms of the second expression are nonnegative. The following are equivalent:* $\mathsf{ep} = 0$*;* $\mathsf{J} = 0$*;* $AQ^{\Xi} = Q^{\Xi}A^{\top}$*; the stationary chain is time reversible. If* $A$ *is normal and* $Q^{\Xi} = qI$*, then* $\mathsf{ep} = \sum_k 2(\mathrm{Im}\,\lambda_k)^2/(1 - |\lambda_k|^2)$*, the sum running over the eigenvalues* $\lambda_k$ *of* $A$*.*

The matrix $\mathsf{J}$ is twice the antisymmetric part of the lag-one covariance, and the reversibility criterion is the hypothesis $AQ^{\Xi} = Q^{\Xi}A^{\top}$ of Proposition S4.4(c), so $\mathsf{ep}$ vanishes exactly on that hypothesis and is positive off it. In the normal case only the imaginary parts of the spectrum contribute: entropy is produced by circulation, and symmetric exchange produces none (Check S47.209).

**Corollary S23.4** (Division of labor at the blanket is irreversible)**.** *Under Assumption S23.1 let $Q^{\Xi}$ be block diagonal over the canonical partition $(\mathsf{I},\mathsf{S},\mathsf{A},\mathsf{E})$ of Definition 4.1, where a variable reads another when the corresponding entry of $A$ is nonzero. If $\mathsf{ep}=0$, then $\mathsf{A}=\varnothing$ and $A_{\mathsf{ES}}=Q^{\Xi}_{\mathsf{EE}}A^{\top}_{\mathsf{SE}}(Q^{\Xi}_{\mathsf{SS}})^{-1}$. Hence $\mathsf{ep}>0$ whenever the collective has an active state, and whenever it has sensory states none of which drives the exterior.*

With per-channel noise, the reversible case of Proposition S4.4(c) therefore has a purely sensory blanket in which every reading is reciprocated (Remark S44.3). When the noise is independent across the four classes, a collective that acts on its surroundings through states that do not read them, or senses them without pushing back, is irreversible by its wiring alone, whatever its parameter values. The restriction on the noise is needed: Remark S44.3 gives a reversible process with an active state whose noise is correlated with that of a sensory state.

## S23.2 What a coarse description hides

**Theorem S23.5** (Coarse descriptions hide dissipation)**.** *Under Assumption S23.1 let $C$ have full row rank $m$, and put $\Gamma_k=CA^k\Sigma C^{\top}$ and $d=\dim X$.*

*(a) $0\le \mathsf{D}_n(C)\le \mathsf{D}_{n+1}(C)$ and $\mathsf{D}_n(C)\le n\mathsf{ep}$ for every $n\ge 1$, so the hidden dissipation lies in $[0,\mathsf{ep}]$.*

*(b) Let $\mathsf{T}_n$ be the covariance of $(Y_0,\dots,Y_n)$, whose block $(s,t)$ is $\Gamma_{s-t}$ for $s\ge t$, and let $\mathsf{P}_n$ be the permutation that reverses the order of its $n+1$ blocks. Then $\mathsf{D}_n(C)=\frac{1}{2}\operatorname{tr}\big((\mathsf{P}_n\mathsf{T}_n\mathsf{P}_n^{\top})^{-1}\mathsf{T}_n\big)-\frac{1}{2}(n+1)m$.*

*(c) $\mathsf{D}_n(C)=0$ for all $n$ if and only if $\Gamma_k=\Gamma_k^{\top}$ for $k=1,\dots,d-1$.*

*(d) Every scalar description, $m=1$, has $\mathsf{D}_n(C)=0$ for all $n$, and its hidden dissipation is all of $\mathsf{ep}$.*

Part (a) is the data-processing inequality. For a stationary series it is the linear-Gaussian instance of the bound of Roldán and Parrondo (2010), whose Eq. (4) bounds the entropy production rate from below by the relative-entropy rate between an observed series and its reverse; Gomez-Marin et al. (2007) prove the analogous bound, their Eq. (8), for partial information about a process driven between two equilibrium states. Roldán and Parrondo (2010) also exhibit an observed series that is reversible while the system dissipates, the position of a discrete flashing ratchet in the limit of fast switching. Part (d) makes that case the rule for scalar linear order parameters: the mean belief of a collective, a single record, or any other scalar linear summary $CX$ carries no trace of irreversibility in its own statistics, however strongly the collective is driven. Among linear descriptions, irreversibility is visible only through at least two coordinates whose cross-covariances are asymmetric in the lag. Linearity matters here. A scalar nonlinear summary can be irreversible: for $Y=X_1+X_2^2$ in the directed ring of Example S23.6 at $c=0.3$, $a=0.5$ and $q=1$, the difference $\mathbb{E}[Y_t^2Y_{t+1}]-\mathbb{E}[Y_tY_{t+1}^2]$ has absolute value 0.5081, whereas it vanishes for every reversible stationary series (Check S47.214).

**Example S23.6** (Directed ring)**.** Three agents on a directed ring, $A=cI+aP$ with $P$ the cyclic shift, $|c|+|a|<1$ and $Q^{\Xi}=qI$, have $\mathsf{ep}=3a^2/(1-c^2+ac-a^2)$, independent of $q$. At $c=0.3$ and $a=0.5$ this is $25/27$ nats per round. The mean and each single agent are reversible, and a pair of agents retains about 0.35 nats per round, 37% of the total (Check S47.210).

**Numerical observation S23.7** (Rate of a coarse description)**.** Put

$$\bar{\mathsf{e}}=\frac{1}{4\pi}\int_{-\pi}^{\pi}\Big[\operatorname{tr}\big(\overline{S(\vartheta)}^{-1}S(\vartheta)\big)-m\Big]d\vartheta,\qquad S(\vartheta)=C\big(e^{i\vartheta}I-A\big)^{-1}Q^{\Xi}\big(e^{-i\vartheta}I-A^{\top}\big)^{-1}C^{\top}. \tag{S98}$$

In the two instances of Check S47.210 with $m = 2$ for which the integral was evaluated, the difference $\mathsf{e}_0 = n\bar{\mathsf{e}} - \mathsf{D}_n(C)$ takes the same value to five digits at $n = 32$, 64 and 128, which is the behavior $\mathsf{D}_n(C) = n\bar{\mathsf{e}} - \mathsf{e}_0 + o(1)$. The constant is $\mathsf{e}_0 = 0.08844$ in the directed ring, where the computed values of $\mathsf{D}_n(C)/n$ increase with $n$, and $\mathsf{e}_0 = -0.10697$ in the second instance, where they decrease. Four further random instances behave in the same way (Check S47.214). This is a numerical observation on six instances and is not proved; Open problem S23.14 asks for a proof.

### S23.3 Information flow across a blanket in discrete time

Split the coordinates into two blocks, $X = (X_{\mathrm{u}}, X_{\mathrm{v}})$, with noise independent across them, $Q^{\Xi} = \mathrm{diag}(Q^{\Xi}_{\mathrm{uu}}, Q^{\Xi}_{\mathrm{vv}})$. The transition density then factorizes, $k(x' \mid x) = k_{\mathrm{u}}(x'_{\mathrm{u}} \mid x) k_{\mathrm{v}}(x'_{\mathrm{v}} \mid x)$. With $X = X_t$ and $X' = X_{t+1}$ define

$$\mathsf{s}_{\mathrm{u}} = \mathbb{E} \ln \frac{k_{\mathrm{u}}(X'_{\mathrm{u}} \mid X_{\mathrm{u}}, X_{\mathrm{v}})}{k_{\mathrm{u}}(X_{\mathrm{u}} \mid X'_{\mathrm{u}}, X_{\mathrm{v}})}, \qquad \mathsf{ep}_{\mathrm{u}} = \mathbb{E} \ln \frac{k_{\mathrm{u}}(X'_{\mathrm{u}} \mid X)}{k_{\mathrm{u}}(X_{\mathrm{u}} \mid X')}, \tag{S99}$$

$$\mathsf{i}_{\mathrm{u}} = I(X'_{\mathrm{u}}; X_{\mathrm{v}}) - I(X_{\mathrm{u}}; X_{\mathrm{v}}), \qquad \mathsf{te}_{\mathrm{v}\to\mathrm{u}} = I(X'_{\mathrm{u}}; X_{\mathrm{v}} \mid X_{\mathrm{u}}), \tag{S100}$$

the entropy flow of u with its partner frozen, the partial entropy production, the information flow into u, and the transfer entropy.

**Theorem S23.8** (Discrete-time information flow)**.** *Under Assumption S23.1 with* $Q^{\Xi} = \mathrm{diag}(Q^{\Xi}_{\mathrm{uu}}, Q^{\Xi}_{\mathrm{vv}})$*:*

*(a)* $\mathsf{ep} = \mathsf{ep}_{\mathrm{u}} + \mathsf{ep}_{\mathrm{v}}$*, and* $\mathsf{ep}_{\mathrm{u}} \geq \ln |\det A_{\mathrm{uu}}|$ *when* $A_{\mathrm{uu}}$ *is invertible.*

*(b)* $\mathsf{s}_{\mathrm{u}} - \mathsf{i}_{\mathrm{u}} = D_{\mathrm{KL}}(P \| P^{\mathrm{fr}}) \geq 0$*, where* $P$ *is the law of* $(X_{\mathrm{u}}, X_{\mathrm{v}}, X'_{\mathrm{u}})$ *and* $P^{\mathrm{fr}}$ *has density* $\pi^{+}(x'_{\mathrm{u}}, x_{\mathrm{v}}) k_{\mathrm{u}}(x_{\mathrm{u}} \mid x'_{\mathrm{u}}, x_{\mathrm{v}})$*, with* $\pi^{+}$ *the density of* $(X'_{\mathrm{u}}, X_{\mathrm{v}})$*.*

*(c)* $\mathsf{i}_{\mathrm{u}} \leq \mathsf{te}_{\mathrm{v}\to\mathrm{u}} = \frac{1}{2} \ln \det\bigl(I + (Q^{\Xi}_{\mathrm{uu}})^{-1/2} A_{\mathrm{uv}} \Sigma_{\mathrm{v}|\mathrm{u}} A_{\mathrm{uv}}^{\top} (Q^{\Xi}_{\mathrm{uu}})^{-1/2}\bigr)$*, with* $\Sigma_{\mathrm{v}|\mathrm{u}} = \Sigma_{\mathrm{vv}} - \Sigma_{\mathrm{vu}} \Sigma_{\mathrm{uu}}^{-1} \Sigma_{\mathrm{uv}}$*.*

*(d)* $\mathsf{i}_{\mathrm{u}} + \mathsf{i}_{\mathrm{v}} = -\bigl[I(X'_{\mathrm{u}}; X'_{\mathrm{v}}) - I(X'_{\mathrm{u}}; X_{\mathrm{v}}) - I(X_{\mathrm{u}}; X'_{\mathrm{v}}) + I(X_{\mathrm{u}}; X_{\mathrm{v}})\bigr]$*.*

Part (b) is a discrete-time counterpart of the second law for one part of a bipartite Markov jump process, Eq. (11) of Horowitz and Esposito (2014), with the bath entropy evaluated at a fixed value of the partner as in Eq. (4) of Ito and Sagawa (2013); part (c) is the discrete-time analog of the comparison between information flow and transfer-entropy rate that Hartich et al. (2014) make for a bipartite sensor model. Three facts that Horowitz and Esposito (2014) prove for bipartite jump processes do not transfer. In their steady state the two information flows sum to zero, each part has nonnegative entropy production, and the entropy production is the sum of the two entropy flows, their Eqs. (11) and (17). In discrete time both blocks move in the same round, the sum in (d) is a mixed second difference that takes both signs, $\mathsf{ep}_{\mathrm{u}}$ can be negative, and $\mathsf{s}_{\mathrm{u}} + \mathsf{s}_{\mathrm{v}}$ differs from $\mathsf{ep}$ (Check S47.211). In the diffusive scaling $A = I + hG$, $Q^{\Xi} = hD$ of that check the defects $\mathsf{i}_{\mathrm{u}} + \mathsf{i}_{\mathrm{v}}$ and $\mathsf{s}_{\mathrm{u}} + \mathsf{s}_{\mathrm{v}} - \mathsf{ep}$ are of order $h^2$ per round in one instance, which we observe and do not prove.

**Corollary S23.9** (Flow across the blanket)**.** *Under Assumption S23.1 let* $\mathrm{u} = \mathsf{I} \cup \mathsf{B}$ *be the collective and* $\mathrm{v} = \mathsf{E}$ *its exterior in the canonical partition of Definition 4.1, with* $Q^{\Xi}$ *block diagonal over* $(\mathsf{I}, \mathsf{S}, \mathsf{A}, \mathsf{E})$*, and let* $\Sigma_{\mathsf{E}|\mathrm{u}} = \Sigma_{\mathsf{EE}} - \Sigma_{\mathsf{E}\mathrm{u}} \Sigma_{\mathrm{uu}}^{-1} \Sigma_{\mathrm{u}\mathsf{E}}$ *and* $\Sigma_{\mathsf{B}|\mathsf{E}} = \Sigma_{\mathsf{BB}} - \Sigma_{\mathsf{BE}} \Sigma_{\mathsf{EE}}^{-1} \Sigma_{\mathsf{EB}}$*. Then*

$$\begin{aligned} \mathsf{te}_{\mathsf{E}\to\mathrm{u}} &= \tfrac{1}{2} \ln \det\bigl(I + (Q^{\Xi}_{\mathsf{SS}})^{-1/2} A_{\mathsf{SE}} \Sigma_{\mathsf{E}|\mathrm{u}} A_{\mathsf{SE}}^{\top} (Q^{\Xi}_{\mathsf{SS}})^{-1/2}\bigr), \\ \mathsf{te}_{\mathrm{u}\to\mathsf{E}} &= \tfrac{1}{2} \ln \det\bigl(I + (Q^{\Xi}_{\mathsf{EE}})^{-1/2} A_{\mathsf{EB}} \Sigma_{\mathsf{B}|\mathsf{E}} A_{\mathsf{EB}}^{\top} (Q^{\Xi}_{\mathsf{EE}})^{-1/2}\bigr), \end{aligned} \tag{S101}$$

*and* $\mathsf{i}_{\mathrm{u}} \leq \min\{\mathsf{s}_{\mathrm{u}}, \mathsf{te}_{\mathsf{E}\to\mathrm{u}}\}$*,* $\mathsf{i}_{\mathsf{E}} \leq \min\{\mathsf{s}_{\mathsf{E}}, \mathsf{te}_{\mathrm{u}\to\mathsf{E}}\}$*.*

Everything the collective learns about its exterior in a round passes the sensory rows $A_{\mathsf{SE}}$, and everything the exterior learns about the collective passes the blanket columns $A_{\mathsf{EB}}$; each bound is the information of one use of a Gaussian channel given what the receiving side already holds.

### S23.4 The thermodynamic bridge

**Assumption S1** (Thermal implementation)**.** The update of a block u within a round is carried out by a physical process in contact with a heat bath at temperature $\mathsf{T}$ while the other block is held fixed, and it obeys local detailed balance: the mean heat released to the bath per round is $k_{\mathrm{B}}\mathsf{T}\mathsf{s}_{\mathrm{u}}$.

**Assumption S2** (Registers)**.** A register that acquires $I$ nats of mutual information about its input and is reset before its next use costs work of at least $k_{\mathrm{B}}\mathsf{T}I$ per use.

**Assumption S3** (Resource)**.** The free energy an agent can harvest per round is at most $\mathsf{x}k_{\mathrm{B}}\mathsf{T}$ times the mutual information $I^{\mathrm{pred}}$ between the data it stores in round $t$ and the world state $\eta_{t+1}$, with an exchange ratio $\mathsf{x} \geq 0$ that is a property of the environment.

These are bridges to physics and carry all the thermodynamic content. Assumption S1 is the local detailed balance of stochastic thermodynamics in the discrete-time form of Eq. (4) of Ito and Sagawa (2013), in which the bath entropy change of a transition is the logarithm of the ratio of the forward and backward transition probabilities with the other variables held fixed; Horowitz and Esposito (2014) assume the same condition for the rates of bipartite jump processes. With Theorem S23.8(b) it gives a heat of at least $k_{\mathrm{B}}\mathsf{T}\mathsf{i}_{\mathrm{u}}$ per round. Assumption S2 takes over inequality (4) of Sagawa and Ueda (2009), which bounds the total work of measurement and erasure with an isothermal memory of finitely many outcomes from below by $k_{\mathrm{B}}\mathsf{T}$ times the mutual information acquired, hence by $k_{\mathrm{B}}\mathsf{T}\ln 2$ for an error-free measurement with two equally likely outcomes; we do not verify that an implementation of the model's Gaussian channels meets its hypotheses. Assumption S3 follows the idea of Kolchinsky and Wolpert (2018) that the information a system has about its environment has value through its causal contribution to the system maintaining its own existence in a state of low entropy; its linear form and the ratio $\mathsf{x}$ are ours. Fields et al. (2024) add an explicit account of thermodynamic free energy to active inference and discuss its tradeoff with variational free energy, in a quantum information-theoretic setting that differs from ours.

**Proposition S23.10** (Cost of a channel use and of a record)**.** *(a) Under Assumption S2, a register that stores $y = x + \nu$ with $x \sim \mathcal{N}(0, \Sigma_x)$, $\Sigma_x \succeq 0$, and independent $\nu \sim \mathcal{N}(0, R)$, $R \succ 0$, costs at least $k_{\mathrm{B}}\mathsf{T}I(y;x) = \frac{1}{2}k_{\mathrm{B}}\mathsf{T}\ln\det(I + R^{-1/2}\Sigma_x R^{-1/2})$ per use. If the noise is thermal, meaning that the signal $x$ has energy $\frac{1}{2}k_{\mathrm{B}}\mathsf{T}x^{\top}R^{-1}x$, its mean energy is $E_{\mathrm{sig}} = \frac{1}{2}k_{\mathrm{B}}\mathsf{T}\operatorname{tr}(R^{-1}\Sigma_x) \geq k_{\mathrm{B}}\mathsf{T}I(y;x)$, and $E_{\mathrm{sig}} = \frac{1}{2}k_{\mathrm{B}}\mathsf{T}(e^{2I(y;x)} - 1)$ for a scalar channel.*

*(b) Let a source $e_{t+1} = \mathsf{f}e_t + w_t$ with $0 < |\mathsf{f}| < 1$ be written to a record $z_{t+1} = \omega e_t + \mathsf{r}z_t + \zeta_t^z$ with $\omega \neq 0$ and $|\mathsf{r}| < 1$, where $w_t \sim \mathcal{N}(0, q_e)$ and $\zeta_t^z \sim \mathcal{N}(0, q_z)$ with $q_e > 0$ and $q_z > 0$ are independent of each other, across rounds and of the past, and the pair $(z, e)$ is stationary. This is one site of Definition 2.9 with a single record, $\mathsf{r} = 1 - \omega - \delta$, messaging and reading switched off, $a = 0$ and $\beta = 0$, and $\mathsf{f} = 1 - \varepsilon$; the definition allows $q_z = 0$, which is excluded here. Let $\mathsf{s}_z$ and $\mathsf{i}_z$ be the quantities of Eqs. (S99) and (S100) for the blocks $\mathrm{u} = z$ and $\mathrm{v} = e$, let $P_e = q_e/(1 - \mathsf{f}^2)$ be the stationary variance of the source, and let $\varrho_{ze}$ be the stationary correlation of $z_t$ and $e_t$. Then*

$$\mathsf{s}_z \geq \mathsf{i}_z = \tfrac{1}{2}\ln\frac{1 - \varrho_{ze}^2}{1 - \varrho_{ze}^2/\mathsf{f}^2} \geq 0, \tag{S102}$$

*and* $\mathsf{i}_z \to 0$ *as* $|\mathsf{f}| \to 1$ *with* $P_e$, $\omega$, $\mathsf{r}$ *and* $q_z$ *held fixed, so that* $q_e = (1-\mathsf{f}^2)P_e \to 0$. *The limit is not claimed at fixed* $q_e$ *(Check S47.214).*

Under Assumption S2 a nat costs the receiver at least $k_\mathrm{B}\mathsf{T}$. Under the thermal clause a scalar channel costs the sender $k_\mathrm{B}\mathsf{T}(e^{2I}-1)/(2I)$ per nat, which is $k_\mathrm{B}\mathsf{T}$ at vanishing rate and grows exponentially with the rate, so many weak channels are cheaper per nat than one sharp channel. A sensor that exhausts the budget $I_{\max}$ of Theorem 14.3(b) costs, under Assumption S2, work of at least $k_\mathrm{B}\mathsf{T}I_{\max}$ per use. Under Assumption S1 a record releases at least $k_\mathrm{B}\mathsf{T}\mathsf{i}_z$ per round (Check S47.212): the bound is the price of tracking change and it vanishes in the limit of a frozen source of fixed variance. It is the same for any number of readers as long as reading feeds back neither on the source nor on the record, because the pair $(z, e)$ then keeps its law and its transition density. With feedback from readers the sign of $\mathsf{i}_z$ is not fixed by our argument.

### S23.5 Scale: complete graphs against records

**Definition S23.11** (Two architectures and their margins)**.** The world is scalar and stationary, $\eta_{t+1} = F\eta_t + w_t$ with $0 < |F| < 1$, $w_t \sim \mathcal{N}(0, Q)$, $Q > 0$, and $\eta_t \sim \mathcal{N}(0, P_0)$ with $P_0 = Q/(1-F^2)$. Each of $N \geq 2$ agents has a sensor $s_i = \eta_t + v_i$, $v_i \sim \mathcal{N}(0, R^s)$ with $R^s > 0$, and resets its registers every round. In the complete graph $K_N$ agent $i$ also stores $y_{ij} = s_j + \nu_{ij}$ for every $j \neq i$. In the record architecture agent $j$ writes $s_j + \nu^w_j$, the record holds $z = N^{-1}\sum_j (s_j + \nu^w_j)$, and agent $i$ stores $r_i = z + \nu^r_i$. All $\nu$ are $\mathcal{N}(0, R_c)$ with $R_c > 0$, and within a round $\eta_t$, $w_t$, the $v_i$ and all $\nu$ are mutually independent. The data of agent $i$ in round $t$ are $(s_i, (y_{ij})_{j\neq i})$ in $K_N$ and $(s_i, r_i)$ in the record architecture, and $I^{\mathrm{pred}}_K(N)$ and $I^{\mathrm{pred}}_R(N)$ are the mutual informations of these data with $\eta_{t+1}$. With $I_s = I(s_i;\eta_t)$, $I_m = I(s_j + \nu^w_j; s_j)$ and $I_r(N) = I(r_i; z)$, the information costs per agent and round are $\mathsf{c}_K(N) = I_s + (N-1)I_m$ and $\mathsf{c}_R(N) = I_s + I_m + I_r(N)$ nats, and the margins are $\mathsf{m}_\bullet(N) = \mathsf{x}I^{\mathrm{pred}}_\bullet(N) - \mathsf{c}_\bullet(N)$.

Under Assumptions S2 and S3, $k_\mathrm{B}\mathsf{T}\mathsf{m}_\bullet(N)$ bounds the free-energy surplus of an agent per round from above. A negative margin rules an architecture out; a positive margin does not rule it in.

**Theorem S23.12** (Linear cost against saturating benefit)**.** *In the setting of Definition S23.11 let* $I_\eta = I(\eta_t; \eta_{t+1}) = -\tfrac{1}{2}\ln(1-F^2)$.

*(a)* $I^{\mathrm{pred}}_K(N) = \tfrac{1}{2}\ln\big(P_0/(F^2 p_K(N) + Q)\big)$ *with* $p_K(N) = \big(1/P_0 + 1/R^s + (N-1)/(R^s + R_c)\big)^{-1}$*; it increases with* $N$ *and stays below* $I_\eta$.

*(b)* $I^{\mathrm{pred}}_\bullet(N) < \mathsf{c}_\bullet(N)$ *for both architectures, so* $\mathsf{m}_\bullet(N) < 0$ *for every* $N$ *when* $\mathsf{x} \leq 1$.

*(c)* $\mathsf{m}_K(N) < 0$ *for every* $N \geq N_K = 1 + (\mathsf{x}I_\eta - I_s)/I_m$.

*(d)* $\mathsf{m}_R(N)$ *converges to* $\mathsf{m}^\infty_R = \mathsf{x}I^{\mathrm{pred}}_{R,\infty} - I_s - I_m - \tfrac{1}{2}\ln(1 + P_0/R_c)$*, where* $I^{\mathrm{pred}}_{R,\infty} = \tfrac{1}{2}\ln\big(P_0/(F^2 p_\infty + Q)\big)$ *and* $p_\infty = (1/P_0 + 1/R^s + 1/R_c)^{-1}$*, and* $\mathsf{m}_R(N) > \mathsf{m}_K(N)$ *for every* $N > 2 + (\mathsf{x}I_\eta + I_r(1))/I_m$*, where* $I_r(1) = \tfrac{1}{2}\ln\big(1 + (P_0 + R^s + R_c)/R_c\big)$.

Part (b) plays the role of a second law in this accounting: when the exchange ratio is at most one, no amount of sensing or communication pays. Reading $\mathsf{x} \leq 1$ as an environment no further from equilibrium than the bath is an interpretation of Assumption S3 and is not derived. Part (c) bounds the size of a collective in which everyone listens to everyone, because the cost grows linearly in $N$ while the benefit is capped by the predictability $I_\eta$ of the world; the bound grows linearly with the exchange ratio. Part (d) says that a shared record removes the bound whenever $\mathsf{m}^\infty_R > 0$

and overtakes pairwise messaging beyond a finite size in every case. At $F = 0.9$, $Q = 0.19$, $R^s = 1$, $R_c = 0.5$ and $\mathsf{x} = 20$, $N_K = 21.2$, the complete graph has a negative margin from $N = 18$ on, the record architecture has the larger margin from $N = 7$ on, and $\mathsf{m}_R^\infty = 7.65$ (Check S47.213). The theorem compares memoryless agents; Open problem S23.16 asks what survives for agents that filter.

*Remark* S23.13 (Reading for expected free energy). The slot costs $g_{\mathrm{al}}$ and $g_{\mathrm{tx}}$ of Proposition S2.7 are free constants, fixed by the preference and by a declared cost of use. Under Assumption S2 an attended slot has a physical price of at least $k_{\mathrm{B}}\mathsf{T}$ times the information its register acquires, and in the setting of Definition S23.11 the proof of Theorem S23.12(b) bounds the information that an agent's data carry about $\eta_t$ by the information cost $\mathsf{c}_\bullet(N)$ of its registers. In that setting the information term and the price of a policy are thus tied together. This is an interpretive correspondence and does not derive the slot costs.

**Open problem S23.14** (Rate of a coarse description)**.** Prove that $\lim_n \mathsf{D}_n(C)/n$ exists for every $C$ of full row rank and equals $\bar{\mathsf{e}}$ of Eq. (S98), and determine, for given $m$, the descriptions that retain the largest fraction of $\mathsf{ep}$. Under Assumption S23.1 the symbol $S$ is continuous and positive definite on the circle, the window covariance $\mathsf{T}_n$ is a Hermitian block Toeplitz matrix with symbol $S$, and the reversed window has the symbol $\overline{S}$, so a limit theorem for traces of products of such matrices and their inverses would give the first claim. We have not checked such a theorem against a primary source, and the constant $\mathsf{e}_0$ of Numerical observation S23.7 would need a second-order statement.

**Open problem S23.15** (Attainability and the bridge)**.** The margins of Theorem S23.12 are one-sided. Exhibit a physical implementation of the model's Gaussian channels that attains $k_{\mathrm{B}}\mathsf{T}$ per nat, or prove a gap, and derive Assumption S3 with an explicit $\mathsf{x}$ from a model of an exterior held between two baths.

**Open problem S23.16** (Agents with memory)**.** Agents that filter keep their registers across rounds and acquire only innovation information. Decide whether a size bound of the form $N_K$ survives with the stationary innovation information in place of $I_m$, and compare the resulting cost with $\mathsf{ep}$ of the joint process of world and filters.

**Open problem S23.17** (Beyond even linear-Gaussian processes)**.** Extend Theorems S23.3 and S23.5 to processes with coordinates that are odd under time reversal and to the nonlinear chains of the earlier sections, including the deliberation chain of Theorem 5.10, and prove the order $h^2$ of the discrete-time defects of Theorem S23.8.

## S24 Reproducing collectives: fission, records and heritability

This section addresses the open problem of evolutionary individuality, the question of how collective organization becomes a unit of selection with heritable collective variation. It does so conditionally. The dynamics of this manuscript produce neither reproduction nor fitness, so both are declared as inputs; with them, the identity machinery of Proposition 5.14 gives the heritability of a convention as a function of the founder bottleneck, the fidelity of a copied record and the relaxation parameters (Proposition S24.2 and Theorem S24.3), the expected response to selection among collectives (Proposition S24.6), and, under a Gaussian closure that is assumed and not derived, the balance between selection and imperfect transmission (Proposition S24.8). Left open are fitness and reproduction as events derived from the dynamics, heritable variation in the organizational parameters, a threshold for discrete conventions, and tradeoffs between collective traits (Open problems S24.10 to S24.14).

The identity time of Definition 5.13 measures how long one collective keeps a convention while its members turn over. Evolutionary individuality asks for more: collectives that reproduce, heritable variation among them in a collective trait, and differences in reproductive success that depend on the trait. Reproduction is modeled below as a fission with a founder bottleneck and a copied record, and fitness as a number of daughters that the parental generation determines.

Throughout, the hypotheses of Proposition 5.14 hold, namely a turnover rate $g > 0$, newcomer noise of positive finite variance, $\omega > 0$ and $\det \mathsf{A} > 0$, and $\mathsf{A}$ and the newcomer fidelity $\mathsf{f} \in [0,1]$ are those of that proposition. The effective forgetting and read rates are $\tilde{\varepsilon} = \varepsilon + g(1-\mathsf{f})$ and $\tilde{\beta} = \beta + g\mathsf{f}$, as in Corollary 12.5, which writes the fidelity with the same symbol. We assume $\beta + g\mathsf{f} > 0$ throughout: the record is read, by the members or by the newcomers, so that by Proposition 5.14(a) the identity time $\tau_{\mathrm{id}}$ of the pair $(\bar{\mu}, z)$ is also that of the members' mean. Then $\tau_{\mathrm{id}} = 1/\varpi_-$, where $\varpi_\pm = \frac{1}{2}[\mathsf{p}_1 + \mathsf{p}_2 \pm ((\mathsf{p}_1 - \mathsf{p}_2)^2 + 4\omega\tilde{\beta})^{1/2}]$, $\mathsf{p}_1 = \varepsilon + \beta + g$ and $\mathsf{p}_2 = \omega + \delta$. Sans-serif symbols introduced below are local to this section. In particular $\mathsf{F}$, $\mathsf{m}$, $\mathsf{c}$, $\mathsf{N}_{\mathrm{fis}}$, $\mathsf{N}_{\mathrm{rel}}$ and $\mathsf{y}_*$ are unrelated to the symbols $\mathsf{F}_{\mathrm{com}}$, $\mathsf{m}_1$, $\mathsf{c}_1$, $\mathsf{N}$ and $\mathsf{y}$ of Section 12, and $\mathsf{f}$ is the newcomer fidelity and not one of the coefficients $\mathsf{f}_t$ of Lemma 12.6.

**Definition S24.1** (Fission with a copied record)**.** A parent is a collective of Proposition 5.14 with $n \geq 2$ positions, member states $\mu_1, \dots, \mu_n$ with mean $\bar{\mu}$, and record $z$, observed at the end of a generation of duration $\mathsf{T} > 0$. The parental information $\mathcal{H}$ comprises the states of all parents at that time and everything determined earlier. Fix a founder number $\mathsf{b} \in \{0, \dots, n\}$ and a copy fidelity $\mathsf{c} \in [0,1]$. Each daughter is formed independently in four steps.

(i) A set $S$ of $\mathsf{b}$ positions is drawn uniformly; these founders keep their states.

(ii) The daughter's record starts at $z_0' = \mathsf{c}z + \zeta$.

(iii) The $n - \mathsf{b}$ vacant positions are filled at once by recruits with states $\mathsf{f}z_0' + \nu_i$.

(iv) The daughter evolves for a time $\mathsf{T}$ under the dynamics of Proposition 5.14, with noise increments of zero conditional mean given $\mathcal{H}$ and the daughter's own past.

The noises $\zeta$ and $\nu_i$ have zero mean and variances $\mathsf{s}_z^2$ and $\mathsf{s}_\nu^2$, and $S$, $\zeta$ and the $\nu_i$ are independent of each other and of $\mathcal{H}$. In the variant without a record, with $\mathsf{b} \geq 1$, recruits start at $\mathsf{f}m_S + \nu_i$, where $m_S$ is the founders' mean, and the daughter relaxes as in the last sentence of Proposition 5.14(a). The number $\mathsf{W}_j \in \{0, 1, 2, \dots\}$ of daughters of parent $j$, its absolute fitness, is an input and is required to be $\mathcal{H}$-measurable.

The definition excludes selection on a daughter's own state during its generation, and it replaces sequential recruitment by an instantaneous refill. Write $\bar{\mathsf{b}} = \mathsf{b}/n$ for the founder fraction, $\mathsf{x} = (\bar{\mu}, z)^\top$ for a parent's collective state, $\mathsf{x}'(\mathsf{T})$ for a daughter's state at the end of its generation, and

$$\mathsf{F} = \begin{pmatrix} \bar{\mathsf{b}} & (1-\bar{\mathsf{b}})\mathsf{f}\mathsf{c} \\ 0 & \mathsf{c} \end{pmatrix}, \qquad \mathsf{M} = e^{\mathsf{T}\mathsf{A}}\mathsf{F}. \tag{S103}$$

**Proposition S24.2** (Transmission matrix and collective heritability)**.** *In the fission model of Definition S24.1:*

*(a)* $\mathbb{E}[\mathsf{x}'(\mathsf{T}) \mid \mathcal{H}] = \mathsf{M}\mathsf{x}$. *If the parents' states have covariance* $\Sigma$*, then* $\mathrm{Cov}(\mathsf{x}'(\mathsf{T}), \mathsf{x}) = \mathsf{M}\Sigma$*, and the regression of a trait* $\mathsf{a}^\top\mathsf{x}'(\mathsf{T})$ *of the daughter on* $\mathsf{a}^\top\mathsf{x}$ *has slope* $\mathsf{a}^\top\mathsf{M}\Sigma\mathsf{a}/(\mathsf{a}^\top\Sigma\mathsf{a})$ *when* $\mathsf{a}^\top\Sigma\mathsf{a} > 0$.

*(b)* $\mathsf{M}$ *has nonnegative entries, its spectral radius is*

$$\mathsf{h} = \tfrac{1}{2}\left[\mathsf{M}_{11} + \mathsf{M}_{22} + \left((\mathsf{M}_{11} - \mathsf{M}_{22})^2 + 4\mathsf{M}_{12}\mathsf{M}_{21}\right)^{1/2}\right] < 1, \tag{S104}$$

*and there is a nonzero* $\mathsf{l} \geq 0$ *with* $\mathsf{l}^\top \mathsf{M} = \mathsf{h}\mathsf{l}^\top$*. The heritable trait* $y = \mathsf{l}^\top \mathsf{x}$ *satisfies* $\mathbb{E}[y' \mid \mathcal{H}] = \mathsf{h}y$*, so its parent-offspring regression slope is* $\mathsf{h}$ *for every* $\Sigma$ *with* $\mathsf{l}^\top \Sigma \mathsf{l} > 0$*. For* $\mathsf{b} = 0$ *the heritable trait is the record, and for* $\mathsf{c} = 0$ *and* $\mathsf{b} \geq 1$ *it is the members' mean.*

*(c) If* $\mathsf{b} \geq 1$ *and* $\mathsf{c} > 0$*, then* $\mathsf{l}$ *is positive and unique up to scale,* $\mathsf{M}$ *has a positive right eigenvector* $r_{\mathsf{M}}$ *for* $\mathsf{h}$*, and the following holds along a line of descent in which every collective founds at least one daughter and the daughter followed is selected by a rule independent of the states and noises of all collectives on the line: the descendant* $\mathsf{x}^{(k)}$ *after* $k$ *generations satisfies* $\mathsf{h}^{-k}\mathbb{E}[\mathsf{a}^\top \mathsf{x}^{(k)} \mid \mathcal{H}] \to (\mathsf{a}^\top r_{\mathsf{M}})y/(\mathsf{l}^\top r_{\mathsf{M}})$ *for every* $\mathsf{a}$*. The limit is nonzero exactly when* $\mathsf{a}^\top r_{\mathsf{M}} \neq 0 \neq y$*.*

*(d) Without a record,* $\mathbb{E}[\bar{\mu}'(\mathsf{T}) \mid \mathcal{H}] = \mathsf{h}_0 \bar{\mu}$ *with* $\mathsf{h}_0 = (\bar{\mathsf{b}} + (1 - \bar{\mathsf{b}})\mathsf{f})e^{-\tilde{\varepsilon}\mathsf{T}}$*; put* $\mathsf{h}_0 = 0$ *for* $\mathsf{b} = 0$*.*

Proof in Section S45; checked in Check S47.218. The number $\mathsf{h}$ is a heritability in the regression sense, the slope of the offspring trait on the parent trait, for a convention measured from its neutral value 0. A general linear trait has a slope that depends on the parental covariance; the heritable trait is the combination of members' mean and record for which it does not, and by (c) the conditional mean of every linear trait with $\mathsf{a}^\top r_{\mathsf{M}} \neq 0$, in particular of every nonzero $\mathsf{a} \geq 0$, decays along a line of descent at the rate $\mathsf{h}$ per generation; the remaining traits decay faster (proof of (c)). Without a record the members are the only channel, and $\mathsf{h}_0 \leq e^{-\tilde{\varepsilon}\mathsf{T}}$ survives a long generation only when forgetting is slow and recruits copy the founders almost faithfully. The next theorem compares the two channels, with $\mathsf{u} = (\mathsf{p}_1 - \varpi_-)/(\varpi_+ - \varpi_-) \in (0, 1)$ and $\mathsf{k} = 1 + (1 - \bar{\mathsf{b}})\mathsf{f}\omega/(\mathsf{p}_1 - \varpi_-)$.

**Theorem S24.3** (The record as an inheritance channel)**.** *In the fission model of Definition S24.1:*

*(a) For every* $\mathsf{b}$*,* $\mathsf{c}$ *and* $\mathsf{T}$*,*

$$e^{-\mathsf{T}/\tau_{\mathrm{id}}} \max\{\mathsf{cu}, \bar{\mathsf{b}}(1 - \mathsf{u}), \min\{\bar{\mathsf{b}}, \mathsf{ck}\}\} \leq \mathsf{h} \leq e^{-\mathsf{T}/\tau_{\mathrm{id}}} \max\{\bar{\mathsf{b}}, \mathsf{ck}\}, \tag{S105}$$

$\mathsf{h}e^{\mathsf{T}/\tau_{\mathrm{id}}} \to (1 - \mathsf{u})\bar{\mathsf{b}} + \mathsf{uck}$ *as* $\mathsf{T} \to \infty$*, and* $\mathsf{h} \to \max\{\bar{\mathsf{b}}, \mathsf{c}\}$ *as* $\mathsf{T} \to 0$*.*

*(b)* $\min\{\delta, \tilde{\varepsilon}\} \leq 1/\tau_{\mathrm{id}} \leq \max\{\delta, \tilde{\varepsilon}\}$*, with strict inequalities when* $\delta \neq \tilde{\varepsilon}$*. A member-written record lengthens the identity time beyond its value* $1/\tilde{\varepsilon}$ *without a record exactly when* $\delta < \tilde{\varepsilon}$*.*

*(c) If* $\mathsf{c} > 0$*, then* $\mathsf{h} \geq \mathsf{cu}e^{-\mathsf{T}/\tau_{\mathrm{id}}}$ *for every* $\mathsf{b}$*, including* $\mathsf{b} = 0$*. If also* $\delta < \tilde{\varepsilon}$ *and* $\mathsf{b} \geq 1$*, then*

$$\frac{\mathsf{h}}{\mathsf{h}_0} \geq \frac{\mathsf{cu}}{\bar{\mathsf{b}} + (1 - \bar{\mathsf{b}})\mathsf{f}} e^{(\tilde{\varepsilon} - 1/\tau_{\mathrm{id}})\mathsf{T}} \geq \mathsf{cu}e^{(\tilde{\varepsilon} - 1/\tau_{\mathrm{id}})\mathsf{T}},$$

*which is unbounded in* $\mathsf{T}$*.*

*(d) If* $\delta \geq \tilde{\varepsilon}$ *and* $\mathsf{b} \geq 1$*, then* $\mathsf{h}/\mathsf{h}_0 \leq [\max\{\bar{\mathsf{b}}, \mathsf{ck}\}/(\bar{\mathsf{b}} + (1 - \bar{\mathsf{b}})\mathsf{f})]e^{-(1/\tau_{\mathrm{id}} - \tilde{\varepsilon})\mathsf{T}}$*, which tends to* 0 *as* $\mathsf{T} \to \infty$ *when* $\delta > \tilde{\varepsilon}$*. For every* $\mathsf{b} \geq 1$*,* $\lim_{\mathsf{T}\to\infty} \mathsf{T}^{-1} \log(\mathsf{h}/\mathsf{h}_0) = \tilde{\varepsilon} - 1/\tau_{\mathrm{id}}$*, which has the sign of* $\tilde{\varepsilon} - \delta$*. If the record is blank at birth,* $\mathsf{c} = 0$*, then* $\mathsf{h} = \bar{\mathsf{b}}[(1 - \mathsf{u})e^{-\varpi_-\mathsf{T}} + \mathsf{u}e^{-\varpi_+\mathsf{T}}] \leq \mathsf{b}/n$*.*

*(e) Let the record be written only by non-members whose writes do not depend on the members' states, the case of Proposition 5.14(b), and let recruits enter at* $\mathsf{f}z_{\mathrm{ext}}(0) + \nu_i$*, with no copy step, where* $z_{\mathrm{ext}}$ *is the record path. Assume that* $z_{\mathrm{ext}}$ *is locally integrable, that the founder set is uniform and the entry noises* $\nu_i$ *have mean zero conditionally on* $\mathcal{H}$ *and the whole path* $z_{\mathrm{ext}}$*,*

*and that the relaxation noise of the daughter has zero conditional mean increments given $\mathcal{H}$, $z_{\mathrm{ext}}$ and the daughter's past. Then $\mathbb{E}[\bar{\mu}'(\mathsf{T}) \mid \mathcal{H}, z_{\mathrm{ext}}]$ equals $\bar{\mathsf{b}} e^{-(\varepsilon+\beta+g)\mathsf{T}} \bar{\mu}$ plus a function of the record path, and the coefficient is at most $(\mathsf{b}/n)e^{-g\mathsf{T}}$ and at most $\mathsf{h}_0$.*

Proof in Section S45; checked in Check S47.216. Part (c) is the positive result of this section, within the declared fission model: a copied record sustains heritability with no founders at all, and its advantage over inheritance through the members grows exponentially with the generation time whenever the record decays more slowly than the collective forgets. Parts (d) and (e) mark the limits. A record that decays faster than the members forget is a liability at long generations, although it can still help at short ones when $\mathsf{c} > \bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f}$; with a record that is rebuilt and not copied, heritability is at most the founder fraction $\mathsf{b}/n$, although it can exceed the value $\mathsf{h}_0$ without a record, as Table S15 shows; and a record kept by an environment whose writes do not depend on the members gives persistence without heritable collective variation, the distinction that Definition 5.13 draws between the identity time and the conditional identity time given the exterior's exogenous input. Example S45.8 gives numbers: with naive recruits, two founders out of twenty and a generation of twenty residence times, $\mathsf{h} = 0.675$ with a copied record and $\mathsf{h}_0 = 1.7 \times 10^{-10}$ without one.

**Corollary S24.4** (Heritability horizon)**.** *Let $\mathsf{c} > 0$ and $0 < \mathsf{h}_* < \mathsf{cu}$. Then $\mathsf{h} \geq \mathsf{h}_*$ for every founder number whenever $\mathsf{T} \leq \tau_{\mathrm{id}} \log(\mathsf{cu}/\mathsf{h}_*)$. Without a record, for $\mathsf{b} \geq 1$ and $\tilde{\varepsilon} > 0$, $\mathsf{h}_0 \geq \mathsf{h}_*$ exactly when $\mathsf{T} \leq \tilde{\varepsilon}^{-1} \log((\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f})/\mathsf{h}_*)$.*

Proof in Section S45. Heritability concerns conditional means; variation among collectives comes from the noise of fission and relaxation.

**Proposition S24.5** (Sources of variation)**.** *Let $\mathsf{v}_{\mathrm{w}} = \frac{1}{n}\sum_i (\mu_i - \bar{\mu})^2$ be the parent's within-collective variance.*

*(a) The daughter's state at birth has conditional covariance*

$$\mathsf{N}_{\mathrm{fis}} = \begin{pmatrix} \dfrac{\bar{\mathsf{b}}(1-\bar{\mathsf{b}})}{n-1}\mathsf{v}_{\mathrm{w}} + \dfrac{1-\bar{\mathsf{b}}}{n}\mathsf{s}_\nu^2 + (1-\bar{\mathsf{b}})^2\mathsf{f}^2\mathsf{s}_z^2 & (1-\bar{\mathsf{b}})\mathsf{f}\mathsf{s}_z^2 \\ (1-\bar{\mathsf{b}})\mathsf{f}\mathsf{s}_z^2 & \mathsf{s}_z^2 \end{pmatrix} \tag{S106}$$

*given $\mathcal{H}$.*

*(b) $\mathrm{Cov}(\mathsf{x}'(\mathsf{T}) \mid \mathcal{H}) = e^{\mathsf{TA}}\mathsf{N}_{\mathrm{fis}}e^{\mathsf{TA}^\top} + \mathsf{N}_{\mathrm{rel}}$, where $\mathsf{N}_{\mathrm{rel}}$ is the covariance added by the relaxation noise, an input that Proposition 5.14 leaves unspecified. Let $\mathsf{V} = \mathrm{Var}(y)$ for a parent drawn from a parental distribution with finite second moments, let $\mathsf{V}' = \mathrm{Var}(y')$ for its daughter when every parent founds one daughter, and let $\mathsf{s}^2 = \mathbb{E}[\mathrm{Var}(y' \mid \mathcal{H})]$. Then $\mathsf{V}' = \mathsf{h}^2\mathsf{V} + \mathsf{s}^2$. If $\mathsf{s}^2$ is the same in every generation, $\mathsf{V}$ converges to $\mathsf{s}^2/(1-\mathsf{h}^2)$, and at that value with $\mathsf{s}^2 > 0$ the parent-offspring correlation of $y$ is $\mathsf{h}$.*

*(c) Without a record, $\mathrm{Var}(\bar{\mu}_0' \mid \mathcal{H}) = (\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f})^2 \frac{\mathsf{v}_{\mathrm{w}}}{\mathsf{b}} \frac{n-\mathsf{b}}{n-1} + \frac{1-\bar{\mathsf{b}}}{n}\mathsf{s}_\nu^2$, and $\mathsf{h}_0$ is nondecreasing in $\mathsf{b}$.*

Proof in Section S45; checked in Check S47.218. Table S14 sets conditions on a unit of selection beside these quantities; the list is this section's own summary and is not attributed to a source. The bottleneck deserves a comment. Without a record, a tighter bottleneck never raises heritability; it converts within-collective variance into variance among collectives, completely for one founder and faithful recruits. With a copied record the recruits read the record, and by Proposition S45.3 founders can lower heritability: at long generation times $\mathsf{h}$ decreases with $\mathsf{b}$ when $1/\tau_{\mathrm{id}} > \delta + \omega(1-\mathsf{cf})$

| Criterion | Model quantity | Status |
|---|---|---|
| Variation among collectives | $\mathsf{V}$, fed by $\mathsf{N}_{\text{fis}}$ and $\mathsf{N}_{\text{rel}}$ | Proposition S24.5 |
| Heritability | $\mathsf{h}$, the spectral radius of $\mathsf{M}$ | Proposition S24.2 and Theorem S24.3 |
| Fitness differences | $\mathsf{W}_j$, $\mathsf{d}_{\text{sel}}$ | declared input; Proposition S24.6 |
| Bottleneck | $\mathsf{b}$ | Propositions S24.5 and S45.3 |
| Integration | communication and shared record | interpretive |

Table S14: Conditions for selection among collectives and the quantities of the fission model that carry them.

and increases with $\mathsf{b}$ when the inequality is reversed, and for $\mathsf{c} = \mathsf{f} = 1$ the first case is the condition $\delta < \tilde{\varepsilon}$. The last row of the table is interpretive: the manuscript's integration is the mean-conserving communication and shared record that make the convention a collective variable, and no theorem here quantifies it.

**Proposition S24.6** (Response to selection among collectives)**.** *Let parents $j = 1, \dots, N$ with states $\mathsf{x}_j$ found $\mathsf{W}_j$ daughters each by the fission of Definition S24.1, with $\bar{\mathsf{W}} = \frac{1}{N}\sum_j \mathsf{W}_j > 0$ and relative fitness $\mathsf{w}_j = \mathsf{W}_j/\bar{\mathsf{W}}$. For a trait $\mathsf{a}^\top\mathsf{x}$ with parental mean $\bar{y}$ and mean $\bar{y}'$ over all daughters at the end of their generation,*

$$\mathbb{E}[\bar{y}' - \bar{y} \mid \mathcal{H}] = \mathrm{Cov}_j(\mathsf{w}_j, \mathsf{a}^\top\mathsf{M}\mathsf{x}_j) + \mathsf{a}^\top(\mathsf{M} - I)\bar{\mathsf{x}}, \tag{S107}$$

*with $\mathrm{Cov}_j$ the covariance over parents and $\bar{\mathsf{x}}$ their mean state. For the heritable trait, with selection differential $\mathsf{d}_{\text{sel}} = \mathrm{Cov}_j(\mathsf{w}_j, y_j)$,*

$$\mathbb{E}[\bar{y}' - \bar{y} \mid \mathcal{H}] = \mathsf{d}_{\text{sel}} - (1 - \mathsf{h})(\bar{y} + \mathsf{d}_{\text{sel}}) = \mathsf{h}\mathsf{d}_{\text{sel}} - (1 - \mathsf{h})\bar{y}, \tag{S108}$$

*the sum of the selection covariance and a transmission term. The expected response is positive exactly when $\mathsf{h}\mathsf{d}_{\text{sel}} > (1 - \mathsf{h})\bar{y}$, and with equal fitnesses in every generation $\mathbb{E}[\bar{y}_k] = \mathsf{h}^k\mathbb{E}[\bar{y}_0]$.*

Proof in Section S45; checked in Check S47.219. This is the Price decomposition (Price, 1970) for collectives, with the covariance identity proved as Lemma S45.4; Price's Eq. 4 writes the change of a gene frequency as the covariance of offspring number with frequency, divided by mean offspring number, plus an offspring-weighted mean of the change from parent to offspring, and he drops the second term for random meiosis and fertilization; Harman (2020) gives the history. The transmission term is always a regression toward the neutral value, because within-collective dynamics forget, and selection must outpace it. Iterating the decomposition requires a closure for the trait distribution, which the fission model does not provide: the daughter's trait given its parent need not be normal, and by (S106) its conditional variance depends on the parent's within-collective variance.

**Assumption S24.7** (Gaussian closure)**.** The population of collectives is infinite; the heritable trait is normal among parents with mean $\bar{y}$ and variance $\mathsf{V} \geq 0$; a daughter's trait given its parent's trait $y$ is normal with mean $\mathsf{h}y$, $0 \leq \mathsf{h} \leq 1$, and variance $\mathsf{s}^2 > 0$ independent of $y$; and a parent with trait $y$ founds daughters in proportion to $\mathsf{W}(y) = \exp(-(y - \mathsf{y}_*)^2/(2\mathsf{v}_\mathsf{s}))$ with $\mathsf{v}_\mathsf{s} > 0$.

**Proposition S24.8** (Selection-transmission balance without a threshold)**.** *Under Assumption S24.7 the trait remains normal, with*

$$\mathsf{V}' = \frac{\mathsf{h}^2\mathsf{V}\mathsf{v}_\mathsf{s}}{\mathsf{V} + \mathsf{v}_\mathsf{s}} + \mathsf{s}^2, \qquad \bar{y}' = \mathsf{h}\Big[\bar{y} + \frac{\mathsf{V}}{\mathsf{V} + \mathsf{v}_\mathsf{s}}(\mathsf{y}_* - \bar{y})\Big], \tag{S109}$$

*and from every initial condition the recursion converges to*

$$\hat{\mathsf{V}} = \tfrac{1}{2}\Big[\mathsf{s}^2 - \mathsf{v}_\mathrm{s}(1-\mathsf{h}^2) + \big((\mathsf{s}^2 - \mathsf{v}_\mathrm{s}(1-\mathsf{h}^2))^2 + 4\mathsf{s}^2\mathsf{v}_\mathrm{s}\big)^{1/2}\Big], \qquad \hat{y} = \frac{\mathsf{hi}}{1-\mathsf{h}+\mathsf{hi}}\mathsf{y}_*, \qquad \mathsf{i} = \frac{\hat{\mathsf{V}}}{\hat{\mathsf{V}}+\mathsf{v}_\mathrm{s}}. \tag{S110}$$

*The retained fraction* $\hat{y}/\mathsf{y}_*$ *is continuous and strictly increasing in* $\mathsf{h}$ *on* $[0,1]$*, vanishes at* $\mathsf{h}=0$*, is positive for every* $\mathsf{h}>0$ *and equals* $1$ *only at* $\mathsf{h}=1$*; it is at least* $\frac{1}{2}$ *exactly when* $\mathsf{hi} \geq 1-\mathsf{h}$*. For fixed* $0<\mathsf{h}<1$ *it is strictly increasing in* $\mathsf{s}^2$ *and tends to* $0$ *as* $\mathsf{s}^2 \to 0$*.*

Proof in Section S45; checked in Check S47.220. Only the conditional mean $\mathsf{h}y$ of the assumption is derived from the fission model. Within the linear-Gaussian closure there is no analog of an error threshold: no critical heritability or copy noise exists below or above which selection retains nothing, and the retained displacement degrades continuously as the record's fidelity falls (Lemma S45.5). Transmission noise is necessary for a lasting response, since it restores the variance that selection consumes, and its cost is the fitness load of (S179). A threshold does appear for a discrete convention with a bounded advantage $\mathsf{m}$ and retention probability $\mathsf{q}$, at $\mathsf{mq}=1$ (Proposition S45.7); that model lies outside the linear-Gaussian setting.

*Remark* S24.9 (Relation to tradeoff breaking). This remark is interpretive. Bourrat et al. (2022) argue that particle and collective fitness cannot decouple when both are computed over the same events, in the same environment and on the same timescale, and propose tradeoffs between traits, and their breaking, as the description of a transition in individuality; they treat tradeoff breaking as a marker only when it is observed in the context of a transition, since it can also occur outside one. The fission model is consistent with the first point only trivially, since its members are recruited and have no fitness of their own. It follows the second in being trait-based, and it has a single trait, so it cannot express a tradeoff or its breaking. Remark S45.9 gives the comparison and an interpretive candidate for a tradeoff, between the heritability of conventions and the shared-error floor of Theorem 12.7.

The theory does not supply the two events on which this section rests. Fitness is declared, reproduction is imposed, and the organizational parameters $\mathsf{f}$, $\omega$, $\beta$ and $\mathsf{c}$ that set the heritability are themselves fixed and carry no heritable variation.

**Open problem S24.10** (Endogenous fitness)**.** Derive the number of daughters of a collective from quantities the manuscript already defines, for example a decreasing function of the stationary tracking error of a grounded coordinate, and determine whether the resulting fitness is measurable with respect to the parental generation as Definition S24.1 requires.

**Open problem S24.11** (Reproduction as a derived event)**.** Find dynamics within the framework under which a collective divides into collectives of the same kind, with a founder number and a record copy that are outcomes and not parameters, and with sequential recruitment in place of the instantaneous refill.

**Open problem S24.12** (Heritable organization)**.** Let $\mathsf{f}$, $\omega$, $\beta$ or $\mathsf{c}$ vary among collectives and be transmitted at fission. The transmission is then no longer linear in the state, and the heritability of these parameters and the response of $\mathsf{h}$ itself to selection are not determined by the results above.

**Open problem S24.13** (Discrete conventions and a threshold)**.** For the bistable convention of Conjecture S5.22, derive the probability that a daughter retains its parent's convention as a function of $\mathsf{b}$, $\mathsf{c}$ and $n$, and decide whether the threshold of Proposition S45.7 follows.

**Open problem S24.14** (Tradeoffs and their breaking)**.** With two collective traits, a convention's heritability and a grounded coordinate's shared-error floor, both controlled by the record's read weight, determine whether declared fitness functions of the two define a tradeoff in the sense of Bourrat et al. (2022) and whether any change of organization within the framework breaks it.

## S25 Unrecorded drivers: what the recorded law identifies

This section addresses the open problem of empirical identification, which observations distinguish a proposed mechanism from its alternatives, in the form it takes for collective agency: the order parameter of Definition 5.15 conditions on the shared drive, and Open problem S5.27 asks for an estimator of it when that drive is not recorded. Inside the linear-Gaussian setting four things are established: the stationary covariance identifies nothing about interaction, and the split of inputs into shared and private is not in the law; between pure models, the path law separates interaction from a regular driver of low rank and cannot separate it from one of full rank on the interacting agents; a channel intervention that leaves the driver law and the loading unchanged certifies interaction with no rank or color condition on the driver; and scores computed from records under one regime cannot grade agency. Left open are the model in which interaction and a colored driver of low rank are both present, for which only local identifiability at one computed parameter point is available, sharp bounds and finite-sample tests, and a test of the invariance hypotheses of the intervention result (Open problems S25.14 to S25.16). Proofs are in Section S46 and checks in Section S47.21.

**Definition S25.1** (Recorded network with an unrecorded driver)**.** The recorded states $x_t \in \mathbb{R}^n$ of the members of $C$ follow

$$x_{t+1} = Ax_t + \mathsf{L}\mathsf{d}_t + \zeta_t, \qquad \zeta_t \sim \mathcal{N}(0, D) \text{ independent across rounds}, \qquad D \succ 0 \text{ diagonal}, \tag{S111}$$

with $\rho(A) < 1$, private inputs $\zeta$, a loading $\mathsf{L} \in \mathbb{R}^{n\times k}$ of full column rank, and an unrecorded stationary Gaussian driver $\mathsf{d}_t \in \mathbb{R}^k$ that is independent of $\zeta$ and belongs to the shared drive $\mathcal{X}_{\mathrm{sh}}$. The driver is regular when its one-step prediction error covariance $Q_{\mathsf{d}}$ is positive definite, and white when it is independent across rounds. The model is interaction-free when $A$ is diagonal and driver-free when there is no driver, the case $k = 0$, written $\mathsf{L} = 0$. The incidence set $\mathcal{E}(A)$ collects the agents $i$ with $A_{ij} \neq 0$ or $A_{ji} \neq 0$ for some $j \neq i$. The level-0 network of Definition 2.12 with scalar exact codes, read in belief coordinates with the world state unrecorded, is the instance $x = m$, $A = W$, $\mathsf{d} = \eta$, $\mathsf{L}_i = Fk_i^s$, $D = Q^\zeta$.

**Proposition S25.2** (Zero interaction is a diagonal coupling)**.** *In Definition S25.1 let $\mathcal{X}_{\mathrm{sh}}$ consist of $x_0$ and the driver path and let $\bar{t} \geq 2$. Then $\Omega_{\mathrm{int}}(C) = 0$ if and only if $A$ is diagonal.*

Whether the order parameter is positive, the qualitative part of what Open problem S5.27 asks to estimate, is therefore the question whether the off-diagonal part of $A$ can be read from the law of $x$ alone.

**Theorem S25.3** (The stationary covariance identifies nothing)**.** *Let $\Sigma \succ 0$ be any $n \times n$ covariance matrix.*

*(a) For every $a \in (-1, 1)$ and $0 < \delta \leq (1 - a^2)\lambda_{\min}(\Sigma)$, the interaction-free model with $A = aI$, $D = \delta I$, a white driver of unit covariance and a loading with $\mathsf{L}\mathsf{L}^\top = (1 - a^2)\Sigma - \delta I$ has stationary covariance $\Sigma$; at the upper limit of $\delta$ the driver has rank at most $n - 1$.*

*(b) The driver-free models with stationary covariance $\Sigma$ are exactly those with $D$ diagonal, $0 \prec D \preceq \Sigma$, and*

$$A = \Sigma^{1/2}\big(I - \Sigma^{-1/2} D \Sigma^{-1/2}\big)^{1/2} U \Sigma^{-1/2}, \qquad U \text{ orthogonal}, \tag{S112}$$

*and $A$ is not diagonal when $\Sigma$ is not.*

*(c) The identity covariance is the stationary covariance of the driver-free model $A = \sqrt{1-\delta}U$, $D = \delta I$ for every orthogonal $U$ and $\delta \in (0,1)$.*

Every covariance is thus produced both without interaction and without a driver, and uncorrelated members may interact. A bound $k$ on the driver rank restricts the construction of (a), which has equal memories $a$, to the factor-analytic set of covariances (Remark S46.1), which constrains $\Sigma$ and does not locate $A$.

**Proposition S25.4** (The classification of inputs is not in the law)**.** *Let $A$ be diagonal, $k = 1$, $\mathsf{d}_{t+1} = \phi \mathsf{d}_t + \omega_t$ with $|\phi| < 1$ and $\omega$ white Gaussian, independent of $\zeta$, with $\operatorname{Var} \omega_t > 0$, and let $\mathsf{L}_1 \mathsf{L}_2 \neq 0$. The joint law of $(x, \mathsf{d})$ is also the law of a network in which $\mathsf{d}$ is the state of an unrecorded agent that hears nobody and is heard by the members. With $\omega$ classified as exogenous forcing, so that $\mathcal{X}_{\mathrm{sh}}$ consists of $x_0$ and the driver path, $\Omega_{\mathrm{int}}(C) = 0$; with $\omega$ classified as that agent's private input, so that $\mathcal{X}_{\mathrm{sh}}$ consists of the initial states $(x_0, \mathsf{d}_0)$, $I(\mu_1[0{:}\bar{t}]; \mu_2[0{:}\bar{t}] \mid \mathcal{X}_{\mathrm{sh}}) > 0$ for $\bar{t} \geq 2$, and hence $\Omega_{\mathrm{int}}(C) > 0$. The member-restricted order parameter $\Omega_{\mathrm{int}}(C \mid \zeta_{V \setminus C})$ of Definition 5.15, which adds the private inputs of the non-members to the conditioning set, vanishes under the second classification; under the first there are no non-members, and the order parameter is $\Omega_{\mathrm{int}}(C) = 0$.*

The order parameter $\Omega_{\mathrm{int}}(C)$ is a functional of the law together with the declared split of inputs into shared and private (Definition 5.15); no analysis of records, however complete, replaces that declaration. In the example the quantity on which a claim of interaction among the members rests is zero under either classification, because the input $\omega$ is in its conditioning set in both, as shared forcing or as the private input of a non-member. That quantity requires a declaration of the same kind, namely which unrecorded variables are agents and what their private inputs are. The results below hold the declaration fixed, with the driver in the shared drive.

**Theorem S25.5** (Path law: rank against support)**.** *Let all members be recorded and let a driver-free model $(A, D)$ generate the recorded path law.*

*(a) If an interaction-free model $(A_0, D_0, \mathsf{L})$ with a regular driver of rank $k$ has the same recorded path law, then the range of $\mathsf{L}$ is the coordinate subspace of a set $T$ of $k$ agents, $T \supseteq \mathcal{E}(A)$, and $A_0$, $D_0$ agree with the diagonals of $A$, $D$ outside $T$.*

*(b) Conversely, for every $T \supseteq \mathcal{E}(A)$ and every stable diagonal $A_0$ that agrees with $A$ outside $T$, there is $c > 0$ such that for every diagonal $D_0$ with $D_0 = D$ outside $T$ and $0 \prec D_{0,T} \prec cI$ some regular driver of rank $|T|$ loading on $T$ gives the interaction-free model the same recorded path law.*

*(c) A regular driver of rank below $|\mathcal{E}(A)|$ never gives an interaction-free model the recorded law of $(A, D)$, and a regular driver whose loading range is not a coordinate subspace gives it the law of no driver-free model. In particular a regular driver of rank one acting on two or more agents is never confused with driver-free first-order interaction.*

Interaction can be explained away only by a driver that supplies a separate component to every agent taking part in it. A diffuse driver, with fewer components than agents affected, leaves a signature that no first-order network with private noise shares, because it makes the one-step innovations of the recorded process correlated across agents. The theorem compares pure models; it does not identify $A$ when interaction and a colored driver are both present.

**Proposition S25.6** (White drivers and bounds)**.** *Let the driver in Definition S25.1 be white with unit covariance, let $\mathcal{X}_{\mathrm{sh}}$ and the window $\bar{t} \geq 2$ be as in Proposition S25.2, and let $\Gamma_h = \operatorname{Cov}(x_{t+h}, x_t)$ for the stationary recorded process.*

*(a)* $A = \Gamma_1\Gamma_0^{-1}$ *and* $Q = D + \mathsf{L}\mathsf{L}^\top = \Gamma_0 - A\Gamma_0 A^\top$*, so* $A$*, and with it the event* $\Omega_{\mathrm{int}}(C) > 0$*, is identified.*

*(b) If* $k = 1$ *and at least three loadings are nonzero, then* $\mathsf{L}_i^2 = Q_{ij}Q_{il}/Q_{jl}$ *for distinct loaded agents,* $D$ *is identified, and so is* $\Omega_{\mathrm{int}}(C)$*.*

*(c) If only* $D_{ii} \geq d_i^- > 0$ *is known, for instance* $d_i^- = Q_{ii} - v_i$ *under a variance bound* $(\mathsf{L}\mathsf{L}^\top)_{ii} \leq v_i$*, then for* $\bar{t} \geq 2$

$$\Omega_{\mathrm{int}}(C) \geq \binom{n}{2}^{-1} \sum_{i<j} -\tfrac{1}{2}\log(1-\max(\varrho_{ij}^2, \varrho_{ji}^2)), \qquad \varrho_{ij}^2 = \frac{A_{ij}^2 d_j^-}{A_{ij}^2 d_j^- + \sum_{l\neq j} A_{il}^2 Q_{ll} + Q_{ii}}. \tag{S113}$$

A driver without memory enters only the innovation covariance, and lag-one structure isolates the coupling. Memory in the driver is what confounds, and the next statement shows the confusion at its smallest scale.

**Proposition S25.7** (Lags identify the modes and not their owner)**.** *Let the driver be first-order,* $\mathsf{d}_{t+1} = \Phi\mathsf{d}_t + \omega_t$*, with* $\rho(\Phi) < 1$ *and* $\omega$ *white Gaussian with covariance* $Q_\omega$*, independent of* $\zeta$*. The recorded spectral density extends from the unit circle to a rational function of* $z$ *whose finite poles lie among the eigenvalues of* $A$ *and of* $\Phi$ *and the reciprocals of the nonzero ones; reflection in the unit circle exchanges the nonzero eigenvalues with their reciprocals. For* $n = k = 1$*, with* $x_{t+1} = ax_t + \mathsf{d}_t + \zeta_t$*,* $\operatorname{Var}\zeta_t = D$*,* $\operatorname{Var}\omega_t = \beta^2$ *and* $0 < \phi < a < 1$*, the parameters* $(a, \phi, D, \beta^2)$ *and* $(\phi, a, D\phi/a, \beta^2 + D(a-\phi)(1-a\phi)/a)$ *give the same recorded law.*

A single recorded agent with long memory under a fast driver is indistinguishable from one with short memory under a slow driver.

**Proposition S25.8** (Static sparse and low-rank reading)**.** *Let the driver be constant within an epoch and drawn from* $\mathcal{N}(0, I_k)$ *independently across epochs, with one stationary snapshot recorded per epoch. With* $\Sigma^{\mathrm{loc}} = A\Sigma^{\mathrm{loc}}A^\top + D$*,* $\Pi^{\mathrm{loc}} = (\Sigma^{\mathrm{loc}})^{-1}$ *and* $M = (I - A)^{-1}\mathsf{L}$*, the recorded precision is* $\Pi^{\mathrm{loc}} - \mathsf{K}$ *with* $\mathsf{K} = \Pi^{\mathrm{loc}}M(I + M^\top\Pi^{\mathrm{loc}}M)^{-1}M^\top\Pi^{\mathrm{loc}} \succeq 0$ *of rank* $k$*. If* $\Pi^{\mathrm{loc}}$ *is not diagonal then* $\Omega_{\mathrm{int}}(C) > 0$*, for a window with* $\bar{t} \geq 2$ *inside an epoch and the shared drive of Proposition S25.2; the converse fails.*

This is the latent-variable model of Chandrasekaran et al. (2012), the static route recorded in Open problem S5.27, with conditional concentration $\Pi^{\mathrm{loc}}$ and low-rank term $\mathsf{K}$. Their Theorem 4.1 recovers the sign pattern, and so the support, of the first and the rank of the second by a convex program, with high probability, when the tangent spaces to the sparse and low-rank varieties meet transversally in the quantitative sense $\mu(\Omega)\xi(T)$ small, the number of independent samples is at least of order $n/\xi(T)^4$, and the smallest nonzero entry and singular value exceed thresholds that depend on the sample size (Remark S46.5); the result concerns independent Gaussian samples, and its conditions are imposed at the true model. For the order parameter it yields a one-sided certificate: a recovered off-diagonal entry of $\Pi^{\mathrm{loc}}$ implies interaction, while its support is that of the conditional precision (for normal isotropic networks the two-step graph, Theorem 4.9), which is not the communication graph and can be empty under interaction.

**Theorem S25.9** (Channel interventions)**.** *Let a channel intervention* $\mathfrak{N}$*, either a channel null of Definition 5.17 or the slot removal of part (b), replace* $(A, D)$ *in Eq. (S111) by diagonal* $(A^{\mathfrak{N}}, D^{\mathfrak{N}})$*, let the shared drive and the window of* $\Omega_{\mathrm{int}}(C)$ *be those of Proposition S25.2, let* $\Gamma_h$ *and* $\Gamma_h^{\mathfrak{N}}$ *be the stationary lagged covariances in the two regimes and* $\Sigma^{\mathrm{loc}} = A\Sigma^{\mathrm{loc}}A^\top + D$*, and assume: (I1) the*

*driver path has the same law in both regimes and is independent of the private inputs and of injected symbols; (I2) the loading* $\mathsf{L}$ *is the same; (I3) the intervention acts through cross gains only, so that* $(A^{\mathfrak{N}}, D^{\mathfrak{N}}) = (A, D)$ *when* $A$ *is diagonal.*

*(a) If the laws of any recorded subset of members differ between the regimes, then* $A$ *is not diagonal and* $\Omega_{\mathrm{int}}(C) > 0$.

*(b) If in addition* $(I - Az)^{-1}\mathsf{L} = (I - A^{\mathfrak{N}}z)^{-1}\mathsf{L}$, *then* $\Gamma_h - \Gamma_h^{\mathfrak{N}}$ *has the off-diagonal entries of* $A^h\Sigma^{\mathrm{loc}}$ *for every* $h \geq 0$, *and* $\Omega_{\mathrm{int}}(C) = 0$ *if and only if* $\Gamma_0 - \Gamma_0^{\mathfrak{N}}$ *and* $\Gamma_1 - \Gamma_1^{\mathfrak{N}}$ *are diagonal. The level-0 network with scalar exact codes and equal sensing gains satisfies the added condition when the message slots are removed and their weight returns to the prior prediction. This slot removal deletes the slots instead of filling them with noise and is not among the nulls tabulated in Definition 5.17:* $\mathfrak{N}_{\mathrm{lp}}$ *keeps the slots, and* $\mathfrak{N}_{\mathrm{rn}}$ *reassigns the deleted capacity to the remaining sources.*

*(c) If the symbol* $\xi_{ij}(t)$ *delivered to* $i$ *on slot* $j$ *is recorded, has variance* $s_{ij} > 0$, *is independent across rounds and of all other inputs, and the receiver keeps its gain, then* $A_{ij} = \mathrm{Cov}^{\mathfrak{N}}(x_i(t+1), \xi_{ij}(t))/s_{ij}$.

Part (a) needs no rank, color or recording condition: with no cross gains the intervention changes nothing, so a change certifies interaction. Part (b) is the difference formula, in which the driver cancels because its effect on the members is the same in both regimes; that condition fails for jamming with gains kept and for unequal sensing gains (Check S47.225). Part (c) uses the injected symbol as an instrument and identifies each gain from the receiver alone; a symbol that is correlated across rounds biases it (Check S47.225). Hypothesis (I1) fails when acts move the driver or when the unrecorded variable is an agent that hears a member, and (I2) fails when receivers recalibrate their sensing once messages are jammed.

**Definition S25.10** (Recorded grade)**.** Fix, before inspection, a scalar coarse variable $y = \mathsf{P}x$ of the stationary recorded states, a channel null $\mathfrak{N}$ and a window. The control score $\mathfrak{c}_{\mathfrak{N}}$ is the relative entropy, over the window, of the recorded law from its law under $\mathfrak{N}$; the recorded memory defect is $\hat{\mathfrak{d}}_{\mathrm{M}} = I(y_{t+1}; y_{-\infty:t-1} \mid y_t)$; the predictability is $R^2 = 1 - \mathrm{Var}(y_{t+1} \mid y_{-\infty:t})/\mathrm{Var}(y_{t+1})$; the persistence time is $\tau_y = \sum_{h\in\mathbb{Z}} \mathrm{Corr}(y_{t+h}, y_t)$.

**Proposition S25.11** (What a recorded grade certifies)**.** *(a) Under (I1) to (I3),* $\mathfrak{c}_{\mathfrak{N}} > 0$ *implies* $\Omega_{\mathrm{int}}(C) > 0$ *and that the full laws in the two regimes differ. When the null is applicable in the sense of Definition 5.17 and the selector is a function of the law, a difference of the full laws is necessary for the type D clause of Definition 5.19 at that null and does not establish it; a null whose intervention leaves the model unchanged is not applicable and imposes no requirement. This argument does not make* $\mathfrak{c}_{\mathfrak{N}} > 0$ *necessary for that clause, since the full laws can differ while their recorded marginals agree.*

*(b) If* $y_{t+1} = ay_t + \beta u_t + \mathfrak{n}_t$ *exactly, with* $|a| < 1$, *white Gaussian* $\mathfrak{n}$ *of variance* $q > 0$ *and an unrecorded stationary Gaussian input* $u$ *independent of* $\mathfrak{n}$, *then* $0 \leq \hat{\mathfrak{d}}_{\mathrm{M}} \leq \frac{1}{2}\log(1+\beta^2\,\mathrm{Var}(u_t)/q)$, *with* $\hat{\mathfrak{d}}_{\mathrm{M}} = 0$ *for white* $u$.

*(c) For every* $\varepsilon > 0$ *and* $\tau_0 < \infty$ *there is an interaction-free model, invariant under every channel null that satisfies (I1) to (I3), whose collective average has* $\hat{\mathfrak{d}}_{\mathrm{M}} \leq \varepsilon$, $R^2 \geq 1 - \varepsilon$ *and* $\tau_y \geq \tau_0$.

Prediction error and persistence time, computed from records, cannot grade agency: memoryless independent agents that sense a slow common driver pass both at any tolerance. Only the control

score separates them, and it certifies in one direction. When the blanket carries input, the memory clause of Definition 5.1 takes the Markov defect for the law of the coarse variable given the input path, and an exactly closed variable meets it with defect zero; with the input unrecorded only $\hat{\mathfrak{d}}_{\mathrm{M}}$ can be computed, and it can be positive for such a variable (the instance computed in Check S47.227), by at most the driver-to-noise term in (b). The grade of Definition 5.19 is accordingly not a functional of the recorded law: its memory clause is met by members that do not interact, on records and given the driver path alike, its clauses on nulls refer to interventions on the full model and, by the applicability clause of Definition 5.17, impose no requirement at a channel null that leaves an interaction-free model unchanged, and its agency clause refers to the habit law, which the policy law under one regime does not determine (Proposition S25.12(c)).

**Proposition S25.12** (Agent-level equivalence class)**.** *Let the level-0 gains come from assigned precisions, $k_i^s = \pi_i^s/\Pi_i$ and $k_{ij} = \pi_{ij}/\Pi_i$ with $\Pi_i = p_i + \pi_i^s + \sum_j \pi_{ij}$, and let $r_i$ and $\sigma_{ij}^2$ be the actual sensing and channel noise variances. The law of the beliefs depends on these parameters only through $W$, $b$ and $q_i = F^2(k_i^{s2} r_i + \sum_j k_{ij}^2 \sigma_{ij}^2)$. It is therefore unchanged by (a) the scaling $(p_i, \pi_i^s, \pi_{i\cdot}) \mapsto c_i(p_i, \pi_i^s, \pi_{i\cdot})$ with $c_i > 0$, and (b) any reallocation of $(r_i, \sigma_{i\cdot}^2)$ at fixed $q_i$. (c) A policy law $\mathbb{P}(\pi \mid X) \propto E(\pi)\exp(-\gamma G(\pi, X))$ is unchanged by $(E, \gamma, G) \mapsto (Ee^{-\gamma g}, \gamma', \gamma(G-g)/\gamma')$ for every $\gamma' > 0$ and every $g(\pi)$ independent of $X$, and by adding to $G$ a function of $X$ alone. Given $(W, b, q)$ in both regimes, which the law determines when the world is recorded (Corollary S46.8), and the policy law in both regimes, the classes are separated as follows: (a) for each member, by setting one of its assigned message precisions to a known positive value while its other precisions, not all zero, are held; (b) by jamming in turn each channel with $W_{ij} \neq 0$ with noise of known variance $s$ and the gain kept, which gives $D_{ii}^{\mathfrak{N}} - D_{ii} = W_{ij}^2(s - \sigma_{ij}^2)$, after which $r_i$ follows from $q_i$ when $k_i^s \neq 0$, variances attached to zero gains entering the law in neither regime; (c) by a known shift of $G$ at fixed habits together with the habit null, which fixes $\gamma$ when the shift separates two policies, then $E$, and $G$ up to an additive function of $X$.*

An effective linear law fixes neither confidence, nor the source of private noise, nor the division between preference and habit. Each is fixed by an intervention on the corresponding quantity, up to the additive function of $X$ in $G$, which no policy law fixes.

**Numerical observation S25.13** (Model recovery with a held-out intervention)**.** For a three-agent level-0 network with an unrecorded world, the collective model with a rank-one first-order driver is locally identified from the recorded window law, in the sense that the Jacobian of the window covariance has full rank 16 at the truth, and when a local search recovers the truth it predicts the law under jamming. Recovery depends on the start: the truth was reached from the starting point of the check and from 2 of 7 random starts, and the other 5 reached a second strict local minimum of the relative entropy, at $6.5 \times 10^{-3}$ nats with driver pole 0.58, whose prediction of the jammed law is off by 1.33 nats. Independent agents with a common driver and a single effective linear law both misfit the recorded law and mispredict the intervention (Check S47.228).

**Open problem S25.14** (Interaction and a colored driver together)**.** Determine the set of $(A, D, \mathsf{L}, \Phi)$ sharing a recorded path law when $A$ is not diagonal and the driver is first-order of rank $k < n$. Numerical observation S25.13 supports local identifiability for $n = 3$, $k = 1$ at one parameter point; global identifiability, the exceptional set, and the case of unrecorded members are open. The second local minimum found there has positive relative entropy, so it bears on recovery by local search and not on identifiability; whether a second parameter point shares the recorded law, as in the pole swap of Proposition S25.7, is not decided by the example.

**Open problem S25.15** (Sharp bounds and finite samples)**.** Find the exact range of $\Omega_{\mathrm{int}}(C)$ over the drivers of rank at most $k$ and variance at most $v$ compatible with a recorded law, of which Eq. (S113) is a lower estimate in the white case, and a finite-sample test of $\Omega_{\mathrm{int}}(C) = 0$ without a recorded drive, with its power against the alternatives of Theorem S25.5(b).

**Open problem S25.16** (Testing invariance)**.** Hypotheses (I1) and (I2) are assumed. Give a design, for instance graded jamming of single channels, whose recorded laws test them, and extend Theorem S25.9 to members whose acts move the driver.

# Part III
# Proofs

## S26 Proofs for Section 2

Throughout, Gaussian priors have positive-definite covariances and assigned noises are positive definite, so every innovation covariance $S$ is invertible. Divergences between Gaussian laws are $D_{\mathrm{KL}}(\mathcal{N}(m_a, P_a)\|\mathcal{N}(m_b, P_b)) = \frac{1}{2}[\mathrm{tr}(P_b^{-1}P_a) - K + (m_a - m_b)^\top P_b^{-1}(m_a - m_b) + \ln(\det P_b/\det P_a)]$.

*Proof of Proposition 2.2.* The update formulas are Gaussian conditioning.

(a) The change of the mean, $m_i^+ - m_i = P_i\hat{H}^\top S^{-1}(y - \hat{H}m_i)$, lies in $\mathrm{col}(P_i\hat{H}^\top)$, whose dimension is $\mathrm{rank}(P_i\hat{H}^\top) = \mathrm{rank}\,\hat{H}$ because $P_i$ is invertible. (b) $P_i - P_i^+ = \mathcal{K}\hat{H}P_i = P_i\hat{H}^\top S^{-1}\hat{H}P_i = G^\top G$ with $G = S^{-1/2}\hat{H}P_i$. Hence it is positive semidefinite, $\mathrm{rank}(G^\top G) = \mathrm{rank}\,G = \mathrm{rank}\,\hat{H}$, and $\mathrm{col}(G^\top G) = \mathrm{col}(G^\top) = \mathrm{col}(P_i\hat{H}^\top S^{-1/2}) = \mathrm{col}(P_i\hat{H}^\top)$. (c) The posterior mean of $f^\top\eta$ moves by $f^\top P_i\hat{H}^\top S^{-1}(y - \hat{H}m_i)$, which vanishes for every $y$ if and only if $f^\top P_i\hat{H}^\top = 0$; its variance falls by $f^\top(P_i - P_i^+)f = \|S^{-1/2}\hat{H}P_i f\|^2$, which vanishes if and only if $\hat{H}P_i f = 0$, the same condition. The precision rises by $\hat{H}^\top\hat{R}^{-1}\hat{H}$, the information form of the same update, whose rank is $\mathrm{rank}\,\hat{H}$. □

*Proof of Proposition 2.3.* (a) On a standard Borel space the prior disintegrates as $q = \int q(\cdot \mid y^\circ)q_T(\mathrm{d}y^\circ)$, with $T(\eta) = y^\circ$ holding $q(\cdot \mid y^\circ)$-almost surely. Write the likelihood as $g(y, T(\eta))$. For a Borel set $B$,

$$q(B \mid y) \propto \int_B g(y, T(\eta))q(\mathrm{d}\eta) = \int g(y, y^\circ)q(B \mid y^\circ)q_T(\mathrm{d}y^\circ),$$

and $g(y, y^\circ)q_T(\mathrm{d}y^\circ)$, normalized, is the posterior $q_T(\mathrm{d}y^\circ \mid y)$.

(b) Gaussian conditioning gives $\mathbb{E}_{q_i}[\eta \mid \hat{H}\eta = y^\circ] = m_i + P_i\hat{H}^\top(\hat{H}P_i\hat{H}^\top)^{-1}(y^\circ - \hat{H}m_i) = m_i + O_{ij}(y^\circ - \hat{H}m_i)$, and $\hat{H}O_{ij} = I$. Put $x^\circ = m_i + O_{ij}(y^\circ - \hat{H}m_i)$. Every $x$ with $\hat{H}x = y^\circ$ is $x^\circ + d$ with $d \in \ker\hat{H}$, and

$$\langle x^\circ - m_i, d\rangle_{\Lambda_i} = (y^\circ - \hat{H}m_i)^\top(\hat{H}P_i\hat{H}^\top)^{-1}\hat{H}P_i\Lambda_i d = (y^\circ - \hat{H}m_i)^\top(\hat{H}P_i\hat{H}^\top)^{-1}\hat{H}d = 0,$$

so $\|x - m_i\|_{\Lambda_i}^2 = \|x^\circ - m_i\|_{\Lambda_i}^2 + \|d\|_{\Lambda_i}^2$, which is smallest exactly at $d = 0$. Next, $O_{ij}\hat{H}P_i\hat{H}^\top S^{-1} = P_i\hat{H}^\top S^{-1} = \mathcal{K}$, and $O_{ij}(\hat{y}^\circ - \hat{H}m_i) = O_{ij}\hat{H}P_i\hat{H}^\top S^{-1}(y - \hat{H}m_i) = \mathcal{K}(y - \hat{H}m_i) = m_i^+ - m_i$. The decoded content is $\hat{H}m_i^+$, which is the stated expression.

(c) The chain rule for relative entropy, applied to the laws of the pair $(\eta, T(\eta))$, gives

$$D_{\mathrm{KL}}(q_j\|q_i) = D_{\mathrm{KL}}(T_\# q_j\|T_\# q_i) + \int D_{\mathrm{KL}}(q_j(\cdot \mid y^\circ)\|q_i(\cdot \mid y^\circ))(T_\# q_j)(\mathrm{d}y^\circ).$$

The reconstruction $\hat{q}_{ij}$ has marginal $T_\# q_j$ and conditional laws $q_i(\cdot \mid y^\circ)$, so the same chain rule gives $D_{\mathrm{KL}}(q_j\|\hat{q}_{ij}) = \int D_{\mathrm{KL}}(q_j(\cdot \mid y^\circ)\|q_i(\cdot \mid y^\circ))(T_\# q_j)(\mathrm{d}y^\circ)$, and subtracting gives (7). For a noisy

kernel, the joint laws $q_j(\mathrm{d}\eta)\mathsf{K}(\mathrm{d}y \mid \eta)$ and $q_i(\mathrm{d}\eta)\mathsf{K}(\mathrm{d}y \mid \eta)$ share the conditional law of $y$ given $\eta$, so their divergence equals $D_{\mathrm{KL}}(q_j\|q_i)$; decomposing it in the other order,

$$D_{\mathrm{KL}}(q_j\|q_i) = D_{\mathrm{KL}}(q_j\mathsf{K}\|q_i\mathsf{K}) + \int D_{\mathrm{KL}}(q_j(\cdot \mid y)\|q_i(\cdot \mid y))(q_j\mathsf{K})(\mathrm{d}y),$$

where $q_j(\cdot \mid y)$ is the posterior that the sender's law would give. Since $q_j = \int q_j(\cdot \mid y)(q_j\mathsf{K})(\mathrm{d}y)$ and $\hat{q}_{ij} = \int q_i(\cdot \mid y)(q_j\mathsf{K})(\mathrm{d}y)$ are mixtures with one mixing law, joint convexity of relative entropy bounds $D_{\mathrm{KL}}(q_j\|\hat{q}_{ij})$ by the integral, which is the claim.

(d) With a noiseless channel the update is conditioning on $H_j\eta = y$, so by (b) an honest message gives $m_i^+ = m_i + O_{ij}H_j(m_j - m_i)$; this is also the limit of the noisy update as $\hat{R} \to 0$, because $S \to H_jP_iH_j^\top$, which is invertible since $H_j$ has full row rank, and $\mathcal{K} \to O_{ij}$. Since $H_jO_{ij} = I$, $\mathcal{P}_{ij}^2 = O_{ij}(H_jO_{ij})H_j = \mathcal{P}_{ij}$. Its range is $\mathrm{col}(O_{ij}) = \mathrm{col}(P_iH_j^\top)$, because $H_j$ is onto, and its kernel is $\ker H_j$, because $O_{ij}$ has full column rank. The matrix $\Lambda_i\mathcal{P}_{ij} = H_j^\top(H_jP_iH_j^\top)^{-1}H_j$ is symmetric, so $\mathcal{P}_{ij}$ is self-adjoint for $\langle a, b\rangle_{\Lambda_i} = a^\top\Lambda_i b$, and its range and kernel are orthogonal in that inner product. □

*Proof of Proposition S2.1.* (a) Subtract $m_k^+ = m_k + \mathcal{K}_k(y - \hat{H}m_k)$ from $m_i^+ = m_i + \mathcal{K}_i(y - \hat{H}m_i)$ and write $y - \hat{H}m_i = (y - \hat{H}m_k) - \hat{H}(m_i - m_k)$.

(b) If $\mathcal{K}_i = \mathcal{K}_k$, then $\hat{H}\mathcal{K}_i = \hat{H}\mathcal{K}_k$. Since $\hat{H}\mathcal{K}_a = \hat{H}P_a\hat{H}^\top S_a^{-1} = I - \hat{R}S_a^{-1}$ and $\hat{R}$ is invertible, $S_i = S_k$, and then $P_i\hat{H}^\top = \mathcal{K}_iS_i = \mathcal{K}_kS_k = P_k\hat{H}^\top$. Conversely $S_a$ and $\mathcal{K}_a$ depend on $P_a$ only through $P_a\hat{H}^\top$.

(c) With $P_i = P_k = P$ and $\Lambda = P^{-1}$, the gains and posterior covariances coincide and $I - \mathcal{K}\hat{H} = P^+\Lambda$, so $m_i^+ - m_k^+ = P^+\Lambda(m_i - m_k)$ and $D_{\mathrm{KL}}(q_i^+\|q_k^+) = \frac{1}{2}(m_i - m_k)^\top\Lambda P^+\Lambda(m_i - m_k)$. By Proposition 2.2(b), $\Lambda - \Lambda P^+\Lambda = \Lambda(P - P^+)\Lambda = \hat{H}^\top S^{-1}\hat{H}$, and $D_{\mathrm{KL}}(q_i\|q_k) = \frac{1}{2}(m_i - m_k)^\top\Lambda(m_i - m_k)$.

(d) Write $\hat{p}(y \mid \eta)$ for the common likelihood and apply the chain rule for relative entropy to the joint laws $q_a(\mathrm{d}\eta)\hat{p}(y \mid \eta)$, $a \in \{i, k\}$, in both orders. Conditioning on $\eta$ first, the divergence of the likelihoods vanishes because they are equal, so the joint divergence is $D_{\mathrm{KL}}(q_i\|q_k)$; conditioning on $y$ first, it is $D_{\mathrm{KL}}(q_i(y)\|q_k(y)) + \mathbb{E}_{q_i(y)}D_{\mathrm{KL}}(q_i^+\|q_k^+)$.

(e) $\hat{y}_a^\circ = \hat{H}m_a^+ = \hat{H}m_a + \hat{H}P_a\hat{H}^\top S_a^{-1}(y - \hat{H}m_a) = \hat{H}m_a + (I - \hat{R}S_a^{-1})(y - \hat{H}m_a) = y - \hat{R}S_a^{-1}(y - \hat{H}m_a)$. If $\hat{H}$ has full row rank, $S_a \to \hat{H}P_a\hat{H}^\top$, which is invertible, so $\hat{R}S_a^{-1} \to 0$ and both decoded contents tend to $y$; the limiting posterior means are the noiseless updates of Proposition 2.3(b), which satisfy $\hat{H}m_a^+ = y$, so their difference lies in $\ker\hat{H}$. □

*Computation for Example S2.2.* Here $P_i\hat{H}^\top = (1, \varrho)^\top$, $P_k\hat{H}^\top = (1, -\varrho)^\top$ and $S_i = S_k = 1 + \sigma^2$, so $m_a^+ = P_a\hat{H}^\top y/(1 + \sigma^2)$ and $m_i^+ - m_k^+ = (0, 2\varrho y/(1 + \sigma^2))$. The posterior covariances are

$$P_i^+ = \frac{1}{1 + \sigma^2}\begin{pmatrix} \sigma^2 & \varrho\sigma^2 \\ \varrho\sigma^2 & 1 + \sigma^2 - \varrho^2 \end{pmatrix}, \qquad \Lambda_k^+ = \frac{1}{\sigma^2(1 - \varrho^2)}\begin{pmatrix} 1 + \sigma^2 - \varrho^2 & \varrho\sigma^2 \\ \varrho\sigma^2 & \sigma^2 \end{pmatrix},$$

and $P_k^+$ is $P_i^+$ with $-\varrho$ in place of $\varrho$, so $\det P_i^+ = \det P_k^+ = \sigma^2(1 - \varrho^2)/(1 + \sigma^2)$. The diagonal entries of $\Lambda_k^+P_i^+$ are both $(1 + \sigma^2 - \varrho^2 + \varrho^2\sigma^2)/((1 - \varrho^2)(1 + \sigma^2))$, and $1 + \sigma^2 - \varrho^2 + \varrho^2\sigma^2 = (1 + \sigma^2)(1 - \varrho^2) + 2\varrho^2\sigma^2$, so $\mathrm{tr}(\Lambda_k^+P_i^+) = 2 + 4\varrho^2\sigma^2/((1 - \varrho^2)(1 + \sigma^2))$ and the covariance part of the divergence is $2\varrho^2\sigma^2/((1 - \varrho^2)(1 + \sigma^2))$. The mean part is $\frac{1}{2}(2\varrho y/(1 + \sigma^2))^2[\Lambda_k^+]_{22} = 2\varrho^2y^2/((1 - \varrho^2)(1 + \sigma^2)^2)$. Their sum is (S1). The prior divergence is $2\varrho^2/(1 - \varrho^2)$ by the same formula with $P_i$ and $P_k$, and (S1) exceeds it if and only if $\sigma^2(1 + \sigma^2) + y^2 > (1 + \sigma^2)^2$, that is $y^2 > 1 + \sigma^2$. Taking the expectation with $\mathbb{E}y^2 = v$ gives the last statement, and $v = 1 + \sigma^2$ returns the prior divergence, as Proposition S2.1(d) requires. □

*Proof of Proposition 2.5.* (a) Here $p$ is a density of the conditional law of $y_{ij}$ given $(\eta, \mathcal{Z}_i)$ with respect to a $\sigma$-finite measure on messages that does not depend on $(\eta, \mathcal{Z}_i)$, and by Bayes' theorem for regular conditional laws the exact posterior given $(\mathcal{Z}_i, y)$ is $p(y \mid \eta, \mathcal{Z}_i)q_i(\mathrm{d}\eta)$ normalized. Let the normalizers $Z_{\hat{p}} = \int \hat{p}(y \mid \eta, \mathcal{Z}_i)q_i(\mathrm{d}\eta)$ and $Z_p = \int p(y \mid \eta, \mathcal{Z}_i)q_i(\mathrm{d}\eta)$ be positive and finite. The receiver's update has density $\hat{p}/Z_{\hat{p}}$ with respect to $q_i$ and the exact posterior has density $p/Z_p$; two probability measures with densities against $q_i$ coincide exactly when the densities agree $q_i$-almost everywhere, that is, when $\hat{p} = (Z_{\hat{p}}/Z_p)p$ for $q_i$-almost every $\eta$.

(b) The channel noise $\nu_{ij}$ is independent of $(\eta, \mathcal{Z}_i, m_j)$, and the characteristic function of $\mathcal{N}(0, R_c)$ never vanishes. Hence the law of $y_{ij}$ given $(\eta, \mathcal{Z}_i)$ is $\mathcal{N}(\hat{H}_{ij}\eta, \hat{R}_{ij})$ if and only if the law of $H_j m_j$ given $(\eta, \mathcal{Z}_i)$ is $\mathcal{N}(\hat{H}_{ij}\eta, \hat{R}_{ij} - R_c)$; for $\hat{R}_{ij} = R_c$ this is the point mass at $\hat{H}_{ij}\eta$. A true likelihood gives the exact posterior by (a), with factor one.

(c) The same deconvolution gives the first statement. In the static world write $\Lambda_j = \Lambda_{ij}^{\mathrm{sh}} + \Lambda_{j\setminus i}$, with $\Lambda_{j\setminus i}$ the precision of the sources $j$ observes and $i$ does not, and let $h^{\mathrm{sh}}$ and $h^{j\setminus i}$ be the information vectors of the shared and of these private sources, so that $m_j = P_j(\Lambda_0\bar{\eta} + h^{\mathrm{sh}} + h^{j\setminus i})$. A source $o_u = H_u^o\eta + v_u$ with $v_u \sim \mathcal{N}(0, R_u^o)$ contributes $(H_u^o)^\top (R_u^o)^{-1} o_u$, so $h^{\mathrm{sh}} = (\Lambda_{ij}^{\mathrm{sh}} - \Lambda_0)\eta + v^{\mathrm{sh}}$ and $h^{j\setminus i} = \Lambda_{j\setminus i}\eta + v^{j\setminus i}$, where $v^{\mathrm{sh}} \sim \mathcal{N}(0, \Lambda_{ij}^{\mathrm{sh}} - \Lambda_0)$ and $v^{j\setminus i} \sim \mathcal{N}(0, \Lambda_{j\setminus i})$ are independent of each other and of $\eta$. Since $i$ retains its raw data, $h^{\mathrm{sh}}$, and with it $v^{\mathrm{sh}}$ given $\eta$, is a function of $(\eta, \mathcal{Z}_i)$, while $v^{j\setminus i}$ is independent of $(\eta, \mathcal{Z}_i)$. Given $(\eta, \mathcal{Z}_i)$, $H_j m_j$ is therefore Gaussian with covariance $H_j P_j \Lambda_{j\setminus i} P_j H_j^\top$ and mean $H_j P_j(\Lambda_0\bar{\eta} + h^{\mathrm{sh}} + \Lambda_{j\setminus i}\eta)$, whose deviation from $H_j\eta = H_j P_j \Lambda_j \eta$ is

$$\delta = H_j P_j(\Lambda_0\bar{\eta} + h^{\mathrm{sh}} - \Lambda_{ij}^{\mathrm{sh}}\eta) = H_j P_j(\Lambda_0(\bar{\eta} - \eta) + v^{\mathrm{sh}}).$$

With $\hat{H}_{ij} = H_j$ and $\hat{P}_{ij} = P_j$ the level-1 likelihood is true exactly when $\delta = 0$ almost surely and the conditional covariance equals $H_j P_j H_j^\top$. The random vector $\delta$ is centered with covariance $H_j P_j(\Lambda_0 + \Lambda_{ij}^{\mathrm{sh}} - \Lambda_0)P_j H_j^\top = H_j P_j \Lambda_{ij}^{\mathrm{sh}} P_j H_j^\top$, so $\delta = 0$ almost surely if and only if $H_j P_j (\Lambda_{ij}^{\mathrm{sh}})^{1/2} = 0$, that is $H_j P_j \Lambda_{ij}^{\mathrm{sh}} = 0$. The covariance condition then holds, since $H_j P_j \Lambda_{j\setminus i} P_j H_j^\top = H_j P_j H_j^\top - H_j P_j \Lambda_{ij}^{\mathrm{sh}} P_j H_j^\top$. If $\Lambda_0 \succ 0$, then $H_j P_j \Lambda_{ij}^{\mathrm{sh}} P_j H_j^\top \succeq H_j P_j \Lambda_0 P_j H_j^\top$, which vanishes only if $H_j P_j = 0$, that is $H_j = 0$. At $\Lambda_0 = 0$ with $\Lambda_j \succ 0$ the same computation, carried out conditionally on $\eta$, gives $m_j = P_j(h^{\mathrm{sh}} + h^{j\setminus i})$ and $\delta = H_j P_j v^{\mathrm{sh}}$ with $v^{\mathrm{sh}} \sim \mathcal{N}(0, \Lambda_{ij}^{\mathrm{sh}})$, so the criterion is again $H_j P_j \Lambda_{ij}^{\mathrm{sh}} = 0$. In two dimensions with unit noise variances, a shared source reading $\eta_1$, a private source of $j$ reading $\eta_2$ and the code $H_j = (0, 1)$ give $P_j = I$ and $H_j P_j \Lambda_{ij}^{\mathrm{sh}} = 0$; if the private source reads $\eta_1 + \eta_2$ instead, then $P_j = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ and $H_j P_j \Lambda_{ij}^{\mathrm{sh}} = (-1, 0)$ although $H_j \Lambda_{ij}^{\mathrm{sh}} = 0$, and the coded error $v_2^{j\setminus i} - v_1^{\mathrm{sh}}$ has correlation $-1/\sqrt{2}$ with the shared noise. When $\Lambda_{j\setminus i} \succ 0$, an evidence message carries $\Lambda_{j\setminus i}^{-1} h^{j\setminus i} = \eta + \Lambda_{j\setminus i}^{-1} v^{j\setminus i}$; its error is independent of $(\eta, \mathcal{Z}_i)$ with covariance $\Lambda_{j\setminus i}^{-1}$, so with this header the coded error $H_j \Lambda_{j\setminus i}^{-1} v^{j\setminus i}$ has exactly the law that the level-1 likelihood assumes.

(d) Exactness relative to $\mathcal{Z}_i$ holds by construction. If $(\eta, y_{ij})$ is conditionally independent of $i$'s data given $\mathcal{Z}_i$, the posterior of $\eta$ given the data and $y_{ij}$ equals that given $(\mathcal{Z}_i, y_{ij})$. The numbers quoted after the proposition are exact Gaussian conditioning (Check S47.1). □

*Proof of Proposition 2.7.* (a) By the ergodic theorem $\hat{C}_T = T^{-1}\sum_{t<T} y_t \hat{\phi}_t^\top \to \mathbb{E}[y_t \hat{\phi}_t^\top] = H\mathbb{E}[\phi_t \hat{\phi}_t^\top] + \mathbb{E}[\nu_t \hat{\phi}_t^\top]$. The second term vanishes because $\nu_t$ is independent of $\mathcal{Z}_t$, and $\mathbb{E}[\phi_t \hat{\phi}_t^\top] = \mathbb{E}[\mathbb{E}[\phi_t \mid \mathcal{Z}_t]\hat{\phi}_t^\top] = M_2$ because $\hat{\phi}_t$ is $\mathcal{Z}_t$-measurable and the pair is content calibrated. Likewise $\hat{M}_T = T^{-1}\sum_{t<T} \hat{\phi}_t \hat{\phi}_t^\top \to M_2$. A random vector lies almost surely in the column space of its second-moment matrix, so $\mathrm{col}(\hat{M}_T) \subseteq \mathrm{col}(M_2)$, with equality for all large $T$ because $\hat{M}_T \to M_2$; hence $\hat{M}_T^+ \to M_2^+$ and $\hat{M}_T \hat{M}_T^+ = \mathcal{P}_M$ for large $T$. The least-squares estimates solve $\hat{H}\hat{M}_T = \hat{C}_T$, and the rows of $\hat{C}_T$ lie in $\mathrm{col}(\hat{M}_T)$, so every solution has the form $\hat{H}_T = \hat{C}_T \hat{M}_T^+ + A_T(I - \hat{M}_T \hat{M}_T^+)$ with $A_T$ arbitrary. For large $T$, $(I - \hat{M}_T \hat{M}_T^+)M_2 = 0$, so

$\hat{H}_T M_2 = \hat{C}_T \hat{M}_T^+ M_2 \to H M_2 M_2^+ M_2 = H M_2$, and the minimum-norm solution $\hat{C}_T \hat{M}_T^+$ tends to $H M_2 M_2^+ = H\mathcal{P}_M$.

(b) Given $\phi_t$, $y_t \sim \mathcal{N}(H\phi_t, R_c)$, and $\phi_t \in \mathrm{col}(\mathbb{E}[\phi_t \phi_t^\top])$ almost surely, so $H\phi_t = H\mathcal{P}_M \phi_t$ and the joint law depends on $H$ only through $H\mathcal{P}_M$. Conversely, if $(H - H')\phi_t = 0$ almost surely, then $(H - H')\mathbb{E}[\phi_t \phi_t^\top] = 0$, so $(H - H')\mathcal{P}_M = 0$; codes that differ on $\mathrm{col}(\mathbb{E}[\phi_t \phi_t^\top])$ therefore give different conditional means of $y_t$ on a set of positive probability.

(c) $\mathbb{E}[y_t(\phi_t + v_t^g)^\top] = H M_\phi$, because $v_t^g$ and $\nu_t$ are centered and independent of $\phi_t$, and $\mathbb{E}[(\phi_t + v_t^g)(\phi_t + v_t^g)^\top] = M_\phi + R^g$, which is invertible, so by the ergodic theorem least squares converges to $H M_\phi (M_\phi + R^g)^{-1}$. Under joint Gaussianity the calibrated statistic is $\mathbb{E}[\phi_t \mid \phi_t + v_t^g]$, so the pair is content calibrated with $\mathcal{Z}_t$ generated by $\phi_t + v_t^g$, and (a) applies; the statistic's second moment $M_\phi (M_\phi + R^g)^{-1} M_\phi$ is invertible because $M_\phi \succ 0$, so $\mathcal{P}_M = I$ and the limit is $H$. □

*Proof of Proposition S2.3.* (a) Since $X_t \mathrm{vec} H = \mathrm{vec}(H\hat{\phi}_t) = H\hat{\phi}_t$, the recursion is the Kalman filter for the static state $\mathrm{vec} H$ observed through $y_t = X_t \mathrm{vec} H + \tilde{\nu}_t$ with noise covariance $R_w$, and returns the posterior of that Gaussian regression model. With an exactly observing receiver under exact joint attention, $\hat{\phi}_t = \phi_t$ and $y_t = H\phi_t + \nu_t$ with $\nu_t \sim \mathcal{N}(0, R_c)$ independent of the regressors, so with $R_w = R_c$ the model is the true one.

(b) Each $\hat{\phi}_t$ lies almost surely in $\mathrm{col}(M_2)$, so the likelihood involves $\mathrm{vec} H$ only through $X_t \mathrm{vec} H = H\mathcal{P}_M \hat{\phi}_t$, that is through $\mathrm{vec}(H\mathcal{P}_M) = (\mathcal{P}_M \otimes I)\mathrm{vec} H$. Because $\mathcal{P}_M \otimes I$ and $(I - \mathcal{P}_M) \otimes I$ are complementary orthogonal projectors, a prior covariance that is a multiple of the identity makes $(\mathcal{P}_M \otimes I)\mathrm{vec} H$ and $((I - \mathcal{P}_M) \otimes I)\mathrm{vec} H$ independent, and the posterior of the second equals its prior.

(c) By (b) the posterior factorizes into the two blocks, and the second keeps its prior mean $H_0(I - \mathcal{P}_M)$. With $p^H I$ the prior covariance, $\sum_t X_t^\top R_w^{-1} X_t = T\hat{M}_T \otimes R_w^{-1}$ and $\sum_t X_t^\top R_w^{-1} y_t = T\mathrm{vec}(R_w^{-1}\hat{C}_T)$, so on the identified block the posterior mean is

$$\Big(\tfrac{1}{p^H T} I + \hat{M}_T \otimes R_w^{-1}\Big)^{-1} \Big(\tfrac{1}{p^H T}\hat{h}_0 + \mathrm{vec}\big(R_w^{-1}\hat{C}_T\big)\Big),$$

restricted to $\mathrm{col}(M_2) \otimes \mathbb{R}^{r_c}$. By the proof of Proposition 2.7(a), $\hat{M}_T \to M_2$, which is invertible on the identified block, and $\mathrm{vec}(R_w^{-1}\hat{C}_T) \to \mathrm{vec}(R_w^{-1} H M_2) = (M_2 \otimes R_w^{-1})\mathrm{vec} H$, so the identified block converges to $\mathrm{vec}(H\mathcal{P}_M)$.

(d) Take $T$ large enough that $\mathrm{col}(\hat{M}_T) = \mathrm{col}(M_2)$, and write $y_t = H\hat{\phi}_t + \tilde{\nu}_t$ with $\tilde{\nu}_t = H(\phi_t - \hat{\phi}_t) + \nu_t$, so that $\hat{C}_T = H\hat{M}_T + E_T$ with $E_T = T^{-1}\sum_t \tilde{\nu}_t \hat{\phi}_t^\top$. On the identified block let $A_T = (p^H T)^{-1} I + \hat{M}_T \otimes R_w^{-1}$, and let $B_T = \hat{M}_T^+ \otimes R_w$, the inverse of $\hat{M}_T \otimes R_w^{-1}$ there. Since $\mathrm{vec}(R_w^{-1} H \hat{M}_T) = (\hat{M}_T \otimes R_w^{-1})\mathrm{vec}(H\mathcal{P}_M)$, the formula of (c) gives

$$(\mathcal{P}_M \otimes I)\hat{h}_T - \mathrm{vec}(H\mathcal{P}_M) = A_T^{-1}\Big(\frac{(\mathcal{P}_M \otimes I)(\hat{h}_0 - \mathrm{vec} H)}{p^H T} + \mathrm{vec}\big(R_w^{-1} E_T\big)\Big).$$

Because $A_T^{-1} - B_T = -(p^H T)^{-1} A_T^{-1} B_T$ and $B_T \mathrm{vec}(R_w^{-1} E_T) = \mathrm{vec}(E_T \hat{M}_T^+)$,

$$(\mathcal{P}_M \otimes I)\hat{h}_T - \mathrm{vec}(H\mathcal{P}_M) = \mathrm{vec}\big(E_T \hat{M}_T^+\big) + O(T^{-1}) \quad \text{almost surely,}$$

since $A_T^{-1}$ and $B_T$ stay bounded on the block as $\hat{M}_T \to M_2$ and $E_T \to 0$ almost surely by the ergodic theorem. The working noise has canceled from the leading term. Content calibration gives $\mathbb{E}[\tilde{\nu}_t \mid \mathcal{Z}_t] = 0$ and $\mathbb{E}[\tilde{\nu}_t \tilde{\nu}_t^\top \mid \mathcal{Z}_t] = H\,\mathrm{Cov}(\phi_t \mid \mathcal{Z}_t) H^\top + R_c = R_{\mathrm{res}}$, constant by hypothesis, because $\nu_t$ is independent of $(\phi_t, \mathcal{Z}_t)$. For independent and identically distributed rounds the vectors $\mathrm{vec}(\tilde{\nu}_t \hat{\phi}_t^\top) = \hat{\phi}_t \otimes \tilde{\nu}_t$ are therefore independent and centered, and since $\hat{\phi}_t$ is $\mathcal{Z}_t$-measurable their

covariance is $\mathbb{E}[\hat\phi_t\hat\phi_t^\top\otimes\tilde\nu_t\tilde\nu_t^\top]=M_2\otimes R_{\mathrm{res}}$, finite because $\mathbb{E}[\|\hat\phi_t\|^2\|\tilde\nu_t\|^2]=\mathbb{E}\|\hat\phi_t\|^2\operatorname{tr}R_{\mathrm{res}}$. Multiplied by $T^{1/2}$ the remainder tends to zero, while $T^{1/2}\mathrm{vec}(E_T\hat M_T^+)=(\hat M_T^+\otimes I)T^{-1/2}\sum_t\hat\phi_t\otimes\tilde\nu_t$; the central limit theorem, $\hat M_T^+\to M_2^+$ and Slutsky's lemma give a centered Gaussian limit with covariance $(M_2^+\otimes I)(M_2\otimes R_{\mathrm{res}})(M_2^+\otimes I)=M_2^+\otimes R_{\mathrm{res}}$. □

*Proof of Proposition S2.4.* (a) As in the proof of Proposition 2.7(a), with $\mathbb{E}[y_t\hat\phi_t^\top]=HC_{\phi\hat\phi}$, the minimum-norm estimate converges to $HC_{\phi\hat\phi}M_2^+$. Since $\hat\phi_t\in\mathrm{col}(M_2)$ almost surely, $C_{\phi\hat\phi}-M_2=\mathbb{E}[(\phi_t-\hat\phi_t)\hat\phi_t^\top]$ vanishes on $\mathrm{col}(M_2)^\perp$, and $HC_{\phi\hat\phi}M_2^+=HM_2M_2^+$ holds if and only if $H(C_{\phi\hat\phi}-M_2)$ vanishes on $\mathrm{col}(M_2^+)=\mathrm{col}(M_2)$, that is if and only if $H\mathbb{E}[(\phi_t-\hat\phi_t)\hat\phi_t^\top]=0$. Content calibration makes the expectation zero.

(b) $C_{\phi\hat\phi}=\mathcal{K}_j^s\mathbb{E}[(\eta+v_j)(\eta+v_i)^\top](\mathcal{K}_i^s)^\top=\mathcal{K}_j^s(P_0+R_{ji}^s)(\mathcal{K}_i^s)^\top$ and $M_2=\mathcal{K}_i^s(P_0+R_i^s)(\mathcal{K}_i^s)^\top=P_0(\mathcal{K}_i^s)^\top$, because $\mathcal{K}_i^s(P_0+R_i^s)=P_0$. Both are invertible, and $C_{\phi\hat\phi}M_2^{-1}=\mathcal{K}_j^s(P_0+R_{ji}^s)P_0^{-1}$, which does not involve $R_i^s$. For scalars, $k_j^s(P_0+R_{ji}^s)/P_0-1=(R_{ji}^s-R_j^s)/(P_0+R_j^s)$.

(c) The grounding indicators are independent of everything else, so the pooled process is stationary and ergodic and its sample second moments converge to $p_g\mathbb{E}_{\mathrm{g}}[\cdot]+(1-p_g)\mathbb{E}_{\mathrm{u}}[\cdot]$, with $\mathbb{E}_{\mathrm{g}}[y_t\phi_t^\top]=HM_g$ and $\mathbb{E}_{\mathrm{g}}[\phi_t\phi_t^\top]=M_g$ on grounded rounds and $HC_{\phi\hat\phi}$ and $M_2$ on ungrounded rounds. For $p_g>0$ the matrix $p_gM_g+(1-p_g)M_2$ is positive definite, so the pooled normal equations give the limit. The grounded rounds alone form an exactly observed, content-calibrated sample of positive frequency with $M_g\succ 0$, so Proposition 2.7(a) gives the limit $H$. □

*Proof of Proposition 2.8.* For the linear system $X_{t+1}=\bar FX_t+\sum_j B_ja_j(t)+\bar w_t$, $s_i(t)=H_i^XX_t+\bar v_i(t)$, with the other agents' actions held fixed, $\partial X_t/\partial a_j(s)=\bar F^{t-s-1}B_j$ for $t>s$, so $\mathcal{X}_{ij}(t,s)=H_i^X\bar F^{t-s-1}B_j$. The block-diagonal $\bar F$ splits this into the three terms; the channel block has dynamics 0, whose power $0^{t-s-1}$ is the identity for $t-s=1$ and zero otherwise. Reading enters only $s_i$, not $\bar F$, so it leaves the external states unchanged. For a conserved direction, $g^\top\eta_{t+1}=g^\top F\eta_t+\sum_j g^\top B_j^\eta a_j^{\mathrm{e}}(t)+g^\top w_t=g^\top\eta_t+g^\top w_t$ whatever the acts, so acts move holdings among components without changing their total. Each term of (9) is a product of an active-to-external map, external dynamics and an external-to-sensory map, which is a path $a_j\to X\to s_i$. □

*Proof of Proposition S2.5.* (a) A write at round $s$ enters the record at round $s+1$ and is multiplied by the retention $1-\omega-\delta$ in each later round, so it reaches a reader at round $t\ge s+1$ with weight proportional to $(1-\omega-\delta)^{t-s-1}$. The normalized mean delay is $\sum_{n\ge1}n(1-\omega-\delta)^{n-1}/\sum_{n\ge1}(1-\omega-\delta)^{n-1}=1/(\omega+\delta)$, and the decay time is the $e$-folding time $-1/\ln(1-\omega-\delta)$. (b) A reader that attends the record at any later round receives the write with the weight given in (a), without having been present when it was written. The reading rounds are independent Bernoulli trials with success probability $p_r$, independent of what the writer does, so the first reading time is geometric with mean $1/p_r$. (c) In the communication graph augmented by the record, each reader is adjacent to the record. (d) Unrolling, $z_t=\omega\sum_{n\ge0}(1-\omega)^nu_{t-1-n}$. With $x=1-\omega$, $\mathrm{Cov}(z_t,\eta_t)=\omega\sum_n x^nF^{n+1}P_0=\omega FP_0/(1-xF)$, and, using $\sum_{n,n'\ge0}x^{n+n'}F^{|n-n'|}=(1+xF)/((1-x^2)(1-xF))$,

$$\mathrm{Var}(z_t)=\omega^2\Big[P_0\frac{1+xF}{(1-x^2)(1-xF)}+\frac{R^w}{1-x^2}\Big];$$

dividing $\mathrm{Cov}(z_t,\eta_t)^2$ by $P_0\,\mathrm{Var}(z_t)$ gives (S3). Write it as $F^2N(x)/\mathcal{D}(x)$ with $N=1-x^2$ and $\mathcal{D}=1-x^2F^2+(1-xF)^2/\mathrm{snr}$. The condition $N'\mathcal{D}=N\mathcal{D}'$ reduces to $(1-xF)(F-x)=x(1-F^2)\mathrm{snr}$, that is $Fx^2-Ax+F=0$. Its roots are positive with product 1 and sum $A/F>2$, so exactly one lies in $(0,1)$; the derivative of $N/\mathcal{D}$ is positive at $x=0$ and the correlation vanishes at $x=1$, so this root is the maximizer, and the equation forces $x<F$. The steady-state Kalman predictor obeys

$\hat{\eta}_{t+1|t} = F(1-k^*)\hat{\eta}_{t|t-1} + Fk^*u_t$, an exponential filter of the writes with discount $F(1-k^*)$, and it is the best linear predictor of $\eta_t$ from past writes. The squared correlation is invariant to rescaling the filter, so the discount $F(1-k^*)$ maximizes it over exponential filters and, by uniqueness of the interior stationary point, equals $1-\omega^*$. At the optimum $\mathrm{Corr}^2(z_t, \eta_t) = \mathrm{Var}(\hat{\eta}_{t|t-1})/P_0 = 1 - P^-/P_0$. □

*Proof of Proposition S2.6.* (a) The act is an input to the dynamics, chosen independently of $\eta_t$ given $i$'s information, so conditioning on it leaves $i$'s belief about $\eta_t$ unchanged, and the Kalman prediction with a known input is $\mathcal{N}(Fm_i + B_j^\eta u, FP_iF^\top + Q)$. A level-0 reading updates to $m_i^+ = m_i + \mathcal{K}(u - \hat{H}m_i)$ and $P_i^+ = P_i - \mathcal{K}\hat{H}P_i$ and predicts $Fm_i^+$ without the input; the difference of predicted means is $F\mathcal{K}(u - \hat{H}m_i) - B_j^\eta u$. (b) The act $u = H_j^{\mathrm{a}} m_j$ is a function of the sender's belief, so it enters the receiver's likelihood as a message with code $H_j^{\mathrm{a}}$, and it enters the world's dynamics through $B_j^\eta$; the receiver's model must contain both terms. (c) The policy posterior depends on $G_i(\pi)$ through $q_i(o \mid \pi) = \int p_i(o \mid X, \pi) q_i(\mathrm{d}X \mid \pi)$. A message that does not enter the dynamics changes only $i$'s belief about the external states. An act changes the transition to future states and hence $q_i(X \mid \pi)$ for future $X$; when the affected components enter $i$'s outcomes, this changes $q_i(o \mid \pi)$ and with it the pragmatic term $\mathbb{E} \ln \tilde{p}_i(o)$, even when the belief about the current state is unchanged. □

*Proof of Proposition S2.7.* When the information term uses the exact posterior of the agent's own model, $I_q((\eta, \alpha); o) = \mathbb{H}_q(o) - \mathbb{E}_q \mathbb{H}[p(o \mid \eta, \alpha)]$, and $-I_q - \mathbb{E}_q \ln \tilde{p}(o) = \mathbb{E}_q \ln q(o) - \mathbb{E}_q \ln \tilde{p}(o) + \mathbb{E}_q \mathbb{H}[p(o \mid \eta, \alpha)]$, which is the second form. For one Gaussian slot with conditional noise $R_o$, predictive covariance $S_o$ and predictive mean $\bar{o}$, the information is $\frac{1}{2} \ln \det(S_o R_o^{-1})$ and $\mathbb{E}[-\ln \mathcal{N}(o; 0, \Psi)] = \frac{1}{2} \mathrm{tr}(\Psi^{-1}(S_o + \bar{o}\bar{o}^\top)) + \frac{1}{2} \ln \det(2\pi\Psi)$; an empty slot contributes $-\ln p^\varnothing$ and a filled one $-\ln(1 - p^\varnothing)$ to the risk. (a) With known code and noise models the attended messages inform only $\eta$, their joint information under the receiver's model of the messages is $I_i(\eta; y_{\mathcal{J}})$, and each attended innovation has predictive mean zero and covariance $S_{ij}$ under that model; collecting the slot constants gives $g_{\mathrm{al}}$ per attended message. (b) Each addressed slot contributes its risk and minus its information, the silent slot contributes $-\ln p^\varnothing$, and the cost of use is added per addressee. (c) $I(\alpha; o_1, o_2) = I(\alpha; o_1) + I(\alpha; o_2 \mid o_1)$ and $I(\alpha; o_2 \mid o_1) = \mathbb{H}(o_2 \mid o_1) - \mathbb{H}(o_2 \mid \alpha) \le \mathbb{H}(o_2) - \mathbb{H}(o_2 \mid \alpha) = I(\alpha; o_2)$, where conditional independence gives $\mathbb{H}(o_2 \mid o_1, \alpha) = \mathbb{H}(o_2 \mid \alpha)$. □

*Proof of Proposition S2.8.* Since $\hat{H}_{ij} O_{ij} = I$, $\hat{H}_{ij}\hat{m}_{ij} = \hat{H}_{ij} m_i + (y_{ij} - \hat{H}_{ij} m_i) = y_{ij}$. Then $\psi_{ij} = H_j \hat{m}_{ij} - u_j = \hat{H}_{ij}\hat{m}_{ij} - (\hat{H}_{ij} - H_j)\hat{m}_{ij} - u_j = \nu_{ij} - (\hat{H}_{ij} - H_j)\hat{m}_{ij}$. Substituting $y_{ij} = H_j m_j + h_j + \nu_{ij}$ into $\varphi_{ij} = y_{ij} - \hat{H}_{ij} m_i$ and adding and subtracting $H_j m_i$ gives the second identity. For the posterior-mean reading at correct codes, $H_j m_i^+ - u_j = H_j m_i + H_j \mathcal{K}(y_{ij} - H_j m_i) - u_j$, and $H_j \mathcal{K} = I - \hat{R}S^{-1}$ gives $\hat{R}S^{-1}(H_j m_i - u_j) + (I - \hat{R}S^{-1})\nu_{ij}$. □

*Proof of Proposition S2.10.* (a) In $\varphi_{ij}$ the terms $H_j(m_j - m_i) + h_j$, $-(\hat{H}_{ij} - H_j)m_i$ and $\nu_{ij}$ are uncorrelated because the code error has mean zero and is independent of $m_i$ and $\nu_{ij}$. The first has mean $H_j(m_j - \bar{m}_{ji}) + h_j$ and covariance $H_j \bar{P}_{ji} H_j^\top$; for isotropic $\mathrm{vec}(\hat{H}_{ij} - H_j)$, $\mathbb{E}[(\hat{H}_{ij} - H_j) m_i m_i^\top (\hat{H}_{ij} - H_j)^\top] = \sigma_H^2 \,\mathrm{tr}(\mathbb{E}[m_i m_i^\top]) I = \sigma_H^2 (\|\bar{m}_{ji}\|^2 + \mathrm{tr}\,\bar{P}_{ji}) I$. (b) Write $\bar{o}_{ji} = H_j(m_j - \bar{m}_{ji})$. Given $m_i$ and the code error the reply is Gaussian with covariance $R_c$, so the ambiguity of the addressed slot is $\frac{1}{2} \ln \det(2\pi e R_c)$ and its information is $I_{ji}^{\mathrm{tx}}$ by the first identity in the proof of Proposition S2.7. The risk against the Gaussian preference needs only the second moment of the reply, which (a) gives: $\mathbb{E}[-\ln \mathcal{N}(\varphi_{ij}; 0, \Psi_{\mathrm{p}})] = \frac{1}{2}\|\bar{o}_{ji}\|^2_{\Psi_{\mathrm{p}}^{-1}} + \frac{1}{2}\mathrm{tr}(\Psi_{\mathrm{p}}^{-1} S_{ji}) + \frac{1}{2} \ln \det(2\pi\Psi_{\mathrm{p}})$. Adding the slot atoms and the cost of use as in Proposition S2.7(b),

$$G_j^{\mathrm{tx}}(\{i\}) - G_j^{\mathrm{tx}}(\text{silence}) = \tfrac{1}{2}\|\bar{o}_{ji}\|^2_{\Psi_{\mathrm{p}}^{-1}} + \tfrac{1}{2}\mathrm{tr}(\Psi_{\mathrm{p}}^{-1} S_{ji}) - I_{ji}^{\mathrm{tx}} + g_{\mathrm{tx}} = \tfrac{1}{2}\|\bar{o}_{ji}\|^2_{\Psi_{\mathrm{p}}^{-1}} - \Xi_{ji}^{\mathrm{ex}},$$

and $j$ addresses $i$ at $\gamma \to \infty$ exactly when this difference is negative. Given $m_i$, the reply is $\mathcal{N}(H_j(m_j - m_i), \sigma_H^2 \|m_i\|^2 I + R_c)$, so its predictive law is a mixture of Gaussians over $m_i \sim \mathcal{N}(\bar{m}_{ji}, \bar{P}_{ji})$ in which $m_j$ enters as a translation; entropy is invariant under translation, so $I_{ji}^{\mathrm{tx}}$ does not depend on $m_j$. Among laws with covariance $S_{ji}$ the Gaussian has the largest entropy, $\frac{1}{2}\ln\det(2\pi e S_{ji})$, which gives $I_{ji}^{\mathrm{tx}} \le \frac{1}{2}\ln\det(S_{ji}R_c^{-1})$ and $\Xi_{ji}^{\mathrm{ex}} \le \Xi_{ji}$; at $\sigma_H^2 = 0$ the mixture is the Gaussian $\mathcal{N}(\bar{o}_{ji}, S_{ji})$ and equality holds, which is Proposition S2.7(b) with $S_{ji}^{\mathrm{tx}} = S_{ji}$ and $R_{ji}^{\mathrm{tx}} = R_c$. Addressing therefore implies the inequality with $\Xi_{ji}$. The converse fails in the scalar instance $H_j = 1$, $\bar{m}_{ji} = 0$, $\bar{P}_{ji} = 4$, $\sigma_H^2 = 1$, $R_c = 0.05$, $\Psi_{\mathrm{p}} = 10$, $p^{\varnothing} = \frac{1}{2}$, $g_{\mathrm{use}} = 0$: there $\varphi_{ij} - m_j = -(1+\delta)m_i + \nu_{ij}$ with $\delta \sim \mathcal{N}(0, \sigma_H^2)$ has fourth cumulant $3\bar{P}_{ji}^2(4\sigma_H^2 + 2\sigma_H^4) > 0$, so the reply is not Gaussian, the entropy bound is strict and $\Xi_{ji}^{\mathrm{ex}} < \Xi_{ji} = -\frac{1}{2}(0.805) + \frac{1}{2}\ln 161 - \frac{1}{2}\ln(20\pi) > 0$; every disagreement with $\max(\Xi_{ji}^{\mathrm{ex}}, 0) \le \frac{1}{2}\|\bar{o}_{ji}\|^2_{\Psi_{\mathrm{p}}^{-1}} < \Xi_{ji}$ satisfies the inequality with $\Xi_{ji}$ while addressing does not lower the expected free energy (Check S47.6 gives the values). At finite $\gamma$ the log-odds against silence are $-\gamma$ times the difference above plus the habit log-odds, and the disagreement enters the difference only through $\frac{1}{2}\|\bar{o}_{ji}\|^2_{\Psi_{\mathrm{p}}^{-1}}$. Under being understood with known codes the outcome is $\nu_{ij}$, whose law does not depend on beliefs. (c) Only the risk depends on $h_j$; the most probable content minimizes $\frac{\gamma}{2}\|H_j(m_j - \bar{m}_{ji}) + h_j\|^2_{\Psi_{\mathrm{p}}^{-1}} + \frac{1}{2}\|h_j\|^2_{\Psi_{\mathrm{h}}^{-1}}$, a strictly convex quadratic with minimizer $h_j^* = -\Gamma H_j(m_j - \bar{m}_{ji})$, and $u_j^* = H_j m_j + h_j^*$. Under being understood at known codes the outcome does not depend on $h_j$, so the habit, maximal at $h_j = 0$, decides. (d) Under $j$'s model the updated partner mean is a martingale, $\mathbb{E}\bar{m}_{ji}^+ = \bar{m}_{ji}$, and by the law of total covariance $\mathrm{Cov}(\bar{m}_{ji}^+) = \bar{P}_{ji} - \bar{P}_{ji}^+$, the posterior covariance being deterministic in the linear-Gaussian model. Hence $\mathbb{E}[(m_j - \bar{m}_{ji}^+)(m_j - \bar{m}_{ji}^+)^\top] = (m_j - \bar{m}_{ji})(m_j - \bar{m}_{ji})^\top + \bar{P}_{ji} - \bar{P}_{ji}^+$ and $\mathbb{E}\|\bar{m}_{ji}^+\|^2 = \|\bar{m}_{ji}\|^2 + \operatorname{tr}\bar{P}_{ji} - \operatorname{tr}\bar{P}_{ji}^+$. By (a) the being-predicted risk is affine in $\bar{m}_{ji}$, in $(m_j - \bar{m}_{ji})(m_j - \bar{m}_{ji})^\top + \bar{P}_{ji}$ and in $\|\bar{m}_{ji}\|^2 + \operatorname{tr}\bar{P}_{ji}$, so with $m_j$ fixed its expectation is unchanged. When $m_j$ is updated from the same message, write $\iota = y_{ji} - H_i\bar{m}_{ji}$, which under $j$'s partner model has mean zero and covariance $S_{ji}^{\mathrm{p}}$. Then $\bar{m}_{ji}^+ = \bar{m}_{ji} + L_{ji}^{\mathrm{p}}\iota$, $\bar{P}_{ji}^+ = \bar{P}_{ji} - L_{ji}^{\mathrm{p}} S_{ji}^{\mathrm{p}} L_{ji}^{\mathrm{p}\top}$ and $m_j^+ = m_j + L_j\iota - L_j H_i d_{ji}$, so $m_j^+ - \bar{m}_{ji}^+ = (I - L_j H_i)d_{ji} + (L_j - L_{ji}^{\mathrm{p}})\iota$ and

$$\mathbb{E}\big[(m_j^+ - \bar{m}_{ji}^+)(m_j^+ - \bar{m}_{ji}^+)^\top\big] + \bar{P}_{ji}^+ = (I - L_j H_i)d_{ji}d_{ji}^\top(I - L_j H_i)^\top + (L_j - L_{ji}^{\mathrm{p}})S_{ji}^{\mathrm{p}}(L_j - L_{ji}^{\mathrm{p}})^\top + \bar{P}_{ji} - L_{ji}^{\mathrm{p}} S_{ji}^{\mathrm{p}} L_{ji}^{\mathrm{p}\top}.$$

Expanding the middle term and using $L_{ji}^{\mathrm{p}} S_{ji}^{\mathrm{p}} = \bar{P}_{ji} H_i^\top$ gives the stated identity. The term $\|\bar{m}_{ji}\|^2 + \operatorname{tr}\bar{P}_{ji}$ does not contain $m_j$ and keeps its expectation. By (a) with $h_j = 0$ the risk is $\frac{1}{2}\operatorname{tr}(\Psi_{\mathrm{p}}^{-1} H_j(d_{ji}d_{ji}^\top + \bar{P}_{ji})H_j^\top)$ plus terms whose expectation is unchanged, so its expected change is $\frac{1}{2}\operatorname{tr}(\Psi_{\mathrm{p}}^{-1} H_j \Delta_{ji} H_j^\top) = q(L_j) - \ell(L_j)$, where $q(L) = \frac{1}{2}\operatorname{tr}(\Psi_{\mathrm{p}}^{-1} H_j L(H_i d_{ji}d_{ji}^\top H_i^\top + S_{ji}^{\mathrm{p}})L^\top H_j^\top)$ collects the terms of $\Delta_{ji}$ quadratic in the gain and $\ell(L) = \operatorname{tr}(\Psi_{\mathrm{p}}^{-1} H_j L H_i(d_{ji}d_{ji}^\top + \bar{P}_{ji})H_j^\top)$ the linear ones. Both vanish when $H_j L = 0$, in particular at $L = 0$. Since $S_{ji}^{\mathrm{p}} \succeq R_c \succ 0$ and $\Psi_{\mathrm{p}} \succ 0$, $q(L) > 0$ whenever $H_j L \neq 0$, so on the ray $tL$ the change $t^2 q(L) - t\ell(L)$ vanishes for at most one $t \neq 0$. For $H_j \neq 0$ the change is therefore a polynomial in the entries of $L_j$ that is not identically zero, and its zero set is a proper algebraic subset. In the scalar case with unit codes the right side is $(1 - L_j)^2 d_{ji}^2 + (L_j^2 - 2L_j L_{ji}^{\mathrm{p}})(\bar{P}_{ji} + R_c) + \bar{P}_{ji}$, and subtracting $d_{ji}^2 + \bar{P}_{ji}$ gives the stated change; since $L_{ji}^{\mathrm{p}} \le 1$, both of its terms are nonpositive and at least one is negative for $0 < L_j < 2L_{ji}^{\mathrm{p}}$. □

*Proof of Proposition S2.12.* (a) By induction over rounds. If the internal filter holds $(U_i m_i, U_i P_i U_i^\top)$, prediction with dynamics $U_i F U_i^{-1}$ and innovation covariance $U_i Q U_i^\top$ gives $(U_i F m_i, U_i(F P_i F^\top + Q)U_i^\top)$, the image of the world-coordinate prediction. With sensing matrix $H_i^s U_i^{-1}$ the predicted

observation $H_i^s m_i$, the innovation and its covariance $S$ coincide with those in world coordinates, and the internal gain is $U_i P_i U_i^\top U_i^{-\top} H_i^{s\top} S^{-1} = U_i \mathcal{K}$, so the update is the image under $U_i$ of the world-coordinate update. (b) The messages are $T_c^{(i)} G^{-1} G U_i m_i = T_c^{(i)} U_i m_i$, and by (a) every world-coordinate belief is unchanged; every other quantity listed is a function of messages and world-coordinate beliefs. (c) $H\eta = (H\mathcal{T}^{-1})(\mathcal{T}\eta)$, so messages are unchanged and Gaussian updates commute with the change of variables; mutual informations and divergences are invariant under bijective maps of the state; $(\mathcal{T}\hat{m} - \mathcal{T}\eta)^\top (\mathcal{T} P_i \mathcal{T}^\top)^{-1} (\mathcal{T}\hat{m} - \mathcal{T}\eta) = (\hat{m} - \eta)^\top P_i^{-1} (\hat{m} - \eta)$; and expected free energies over sensory outcomes depend only on the predictive laws of those outcomes, which are unchanged. (d) The transformed likelihood is $\mathcal{N}(By; B\hat{H}\eta, B\hat{R}B^\top) = |\det B|^{-1} \mathcal{N}(y; \hat{H}\eta, \hat{R})$, a factor that does not depend on $\eta$, so posteriors and informations are unchanged. A preference density transformed as a density is $|\det B|^{-1} \tilde{p}(B^{-1} y')$, so $-\mathbb{E} \ln \tilde{p}$ over each attended message increases by $\ln|\det B|$ while the information term is unchanged; the atom at an empty slot is a probability and does not change. □

*Proof of Proposition 2.11.* A message is $y_{ij} = T_c U_j m_j + \nu_{ij} = H_j m_j + \nu_{ij}$, and a posted header is $T_c U_j P_j U_j^\top T_c^\top = H_j P_j H_j^\top$. A receiver at level 0 or level 1 uses $\hat{H}_{ij}$ and these quantities; one that computes in its internal coordinates performs the same update conjugated by $U_i$ (Proposition S2.12(a)), so its world-coordinate belief is the same. By induction over rounds, if every world-coordinate belief up to round $t$ depends on the encodings only through the effective codes and the code models, so do the messages of round $t$, the updates of round $t+1$, the conditional laws used by exact fusion and, for learned models, the data from which the code models are learned. The error recursion of Definition 2.12 contains $H_j$ and $\hat{H}_{ij}$ only, and the expected-free-energy terms of Proposition S2.7 are functions of predictive laws of messages, replies and echoes, which are functions of the above. Identity encodings give $H_j = T_c$, and any configuration with $H_j = T_c$ for every sender is then indistinguishable from them. The condition $H_j = T_c$ reads $T_c(U_j - I) = 0$; since $T_c$ has rank at most $r_c$, for $r_c < K$ it holds for $U_j = I + N$ with any nonzero $N$ whose columns lie in $\ker T_c$ and for which $I + N$ is invertible, for instance every such $N$ of operator norm below one. □

*Proof of Proposition S2.13.* Given the world-coordinate beliefs, which do not depend on the frame (Proposition S2.12(a)), the messages of $j$ are $T_c U_j m_j + \nu_{ij}$, so encodings with $T_c U_j' = T_c U_j$ give the same conditional law given $m_j$. If $X = T_c(U_j' - U_j) \neq 0$ and $\mathbb{E}[m_j m_j^\top] \succ 0$, then $\mathbb{E}\|X m_j\|^2 = \operatorname{tr}(X \mathbb{E}[m_j m_j^\top] X^\top) > 0$, so $X m_j \neq 0$ with positive probability and the conditional means of the messages differ. The matrices $X$ with $T_c X = 0$ are those whose $K$ columns lie in $\ker T_c$, of dimension $K - r_c$, so the set has dimension $K(K - r_c)$; with several modalities the columns must lie in the kernel of the stacked code, of dimension $K - \operatorname{rank}[T_c]_{c \in \mathcal{M}}$. Reception depends on the encoding only through the effective code (Proposition 2.11), so these mismatches cost nothing. □

The cost of a misidentified code, summarized in Section 2.7, is the following.

**Proposition S26.1** (Cost of a misidentified code)**.** *Let $\eta \sim \mathcal{N}(m_i, P_i)$ be receiver $i$'s correct prior, let the message be $y = H\eta + \nu$, $\nu \sim \mathcal{N}(0, R)$, with $H$ the true code, and let the receiver use $\hat{H}$. Write $\mathcal{K}^*, \hat{\mathcal{K}}$ for the two gains, $S^* = H P_i H^\top + R$, $\hat{S} = \hat{H} P_i \hat{H}^\top + R$, and $m^*, \hat{m}$ for the two posterior means.*

*(a) For every weight $M \succeq 0$, $\mathbb{E}\|\hat{m} - \eta\|_M^2 - \mathbb{E}\|m^* - \eta\|_M^2 = \operatorname{tr}\big(M(\hat{\mathcal{K}} - \mathcal{K}^*) S^* (\hat{\mathcal{K}} - \mathcal{K}^*)^\top\big) + \|\hat{\mathcal{K}}(\hat{H} - H) m_i\|_M^2$.*

*(b) With $M = P_i^{-1}$ and $m_i = 0$, listening with $\hat{H}$ beats silence if and only if $2 \operatorname{tr}(\hat{S}^{-1} H P_i \hat{H}^\top) > \operatorname{tr}(\hat{S}^{-1} S^* \hat{S}^{-1} \hat{H} P_i \hat{H}^\top)$.*

*(c) Under the hypotheses of (b), let the codes have rank one, $H = f^\top$ and $\hat H = \hat f^\top$, with noise variance $\sigma^2$, and write $\langle a, b\rangle_P = a^\top P_i b$. If $\hat f$ is $f$ rotated at fixed $P$-norm by a $P$-angle $\vartheta$, listening is worse than silence if and only if $\vartheta > 60°$, for every channel noise. For $P_i = pI$ in two dimensions and a noiseless channel the mean-square error is $(3 - 2\cos\vartheta)p$, against $2p$ in silence.*

*Proof of Proposition S26.1.* (a) Under the correct prior, $m^* = \mathbb{E}[\eta \mid y]$, and $\hat m - m^*$ is a function of $y$, so the cross term in $\mathbb{E}\|\hat m - \eta\|_M^2 = \mathbb{E}\|\hat m - m^*\|_M^2 + \mathbb{E}\|m^* - \eta\|_M^2$ vanishes. Writing $y - \hat H m_i = (y - Hm_i) - (\hat H - H)m_i$ gives $\hat m - m^* = (\hat{\mathcal{K}} - \mathcal{K}^*)(y - Hm_i) - \hat{\mathcal{K}}(\hat H - H)m_i$, and $y - Hm_i \sim \mathcal{N}(0, S^*)$. (b) With $m_i = 0$ the posterior error is $-(I - \hat{\mathcal{K}}H)\eta + \hat{\mathcal{K}}\nu$, so $\mathbb{E}\|\hat m - \eta\|_{P_i^{-1}}^2 = \operatorname{tr}(P_i^{-1}[(I - \hat{\mathcal{K}}H)P_i(I - \hat{\mathcal{K}}H)^\top + \hat{\mathcal{K}}R\hat{\mathcal{K}}^\top]) = K - 2\operatorname{tr}(\hat{\mathcal{K}}H) + \operatorname{tr}(P_i^{-1}\hat{\mathcal{K}}S^*\hat{\mathcal{K}}^\top)$, with $\operatorname{tr}(\hat{\mathcal{K}}H) = \operatorname{tr}(\hat S^{-1}HP_i\hat H^\top)$ and $P_i^{-1}\hat{\mathcal{K}} = \hat H^\top \hat S^{-1}$; silence gives $K$. (c) For rank-one codes $\hat S = \langle \hat f, \hat f\rangle_P + \sigma^2$ and $S^* = \langle f, f\rangle_P + \sigma^2$, and (b) reads $2\langle f, \hat f\rangle_P(\langle \hat f, \hat f\rangle_P + \sigma^2) > (\langle f, f\rangle_P + \sigma^2)\langle \hat f, \hat f\rangle_P$. With $\|\hat f\|_P = \|f\|_P$ and $\langle f, \hat f\rangle_P = \|f\|_P^2 \cos\vartheta$ this is $2\|f\|_P^2\cos\vartheta(\|f\|_P^2 + \sigma^2) > (\|f\|_P^2 + \sigma^2)\|f\|_P^2$, that is $\cos\vartheta > \frac12$. For $P_i = pI$, $K = 2$, unit vectors and no noise, $\hat{\mathcal{K}} = \hat f$ and the error is $-(I - \hat f f^\top)\eta$, with mean square $p\operatorname{tr}((I - \hat f f^\top)(I - f\hat f^\top)) = p(3 - 2\cos\vartheta)$; silence gives $2p$. □

*Proof of Lemma 2.13.* (a) Unrolling (14) from round $t_0$ gives

$$e(t) = W^{t-t_0}e(t_0) + \sum_{n=0}^{t-t_0-1} W^n\big(B^{\mathrm{mis}}\eta_{t-1-n} - \mathbf{1}\otimes w_{t-1-n} + \zeta(t-1-n)\big).$$

Since $\rho(W) < 1$, Gelfand's formula gives, for each $r' \in (\rho(W), 1)$, a constant $C'$ with $\|W^n\| \le C'r'^n$. The inputs have bounded second moments (the world is stationary when $B^{\mathrm{mis}} \neq 0$), so the series converge in mean square as $t_0 \to -\infty$ and $W^{t-t_0}e(t_0) \to 0$ in mean square for every initial law with bounded second moments; the limit is therefore the unique stationary solution with finite second moments. (b) In the level-0 network the world is exogenous, and its path determines every innovation $w_s = \eta_{s+1} - F\eta_s$, so $e^{\mathrm{com}}(t)$, a mean-square limit of linear functionals of the path, is measurable with respect to it. The local noises are functions of sensing and receiver-specific channel noises, which are independent of the world, so $e^{\mathrm{loc}}$ is independent of the world path and of $e^{\mathrm{com}}$, and the covariances add. The $\zeta(t-1-n)$ are independent with covariance $Q^\zeta$, which gives the series for $\Sigma^{\mathrm{loc}}$; shifting the summation index gives the Lyapunov equation, whose solution is unique because the map $X \mapsto X - WXW^\top$ has eigenvalues $1 - \lambda_a\lambda_b$ over pairs of eigenvalues of $W$, all nonzero. (c) If one random element is measurable with respect to a $\sigma$-algebra and another is independent of it, the conditional law of their sum given that $\sigma$-algebra is the law of the second shifted by the first. Apply this to the error path with the world path's $\sigma$-algebra, the common part as the first element and the connected part, which is Gaussian when the local noises are, as the second. (d) With exact code models $B^{\mathrm{mis}} = 0$ and $W\mathbf{1} = F(1 - k^s)\mathbf{1} = \lambda\mathbf{1}$, so $e^{\mathrm{com}}(t) = -\sum_{n\ge0}\lambda^n w_{t-1-n}\mathbf{1}$, with covariance $Q\sum_n \lambda^{2n}\mathbf{1}\mathbf{1}^\top = Q\mathbf{1}\mathbf{1}^\top/(1 - \lambda^2)$. □

## S27 Proofs for Section 3

Throughout, for jointly Gaussian vectors $I(\eta; y) = \frac12 \ln\det\operatorname{Cov}(y) - \frac12\ln\det\operatorname{Cov}(y \mid \eta)$. For $P \succ 0$ the map $X \mapsto \ln\det(I + PX) = \ln\det(I + P^{1/2}XP^{1/2})$ is nondecreasing on positive-semidefinite matrices in the Loewner order, and strictly increasing: if $X \preceq X'$ and $X \neq X'$, every eigenvalue of $P^{1/2}XP^{1/2}$ is at most the corresponding eigenvalue of $P^{1/2}X'P^{1/2}$ and the traces differ, so the determinant increases strictly. We use $\det(I + AB) = \det(I + BA)$ without comment.

*Proof of Proposition S3.1.* (a) At levels 0 and 1 the messages are modeled as conditionally independent given $\eta$, with likelihoods $\mathcal{N}(\hat{H}_{ij}\eta, \hat{R}_{ij})$ and $\mathcal{N}(\hat{H}_{ij}\eta, \hat{R}^1_{ij})$. Bayes' rule adds their precisions, so with $\Gamma_{ij}$ standing for $\Gamma^0_{ij}$ or $\Gamma^1_{ij}$ the modeled posterior covariance is $(\Lambda_i + \sum_{j\in\mathcal{J}} \Gamma_{ij})^{-1}$ whatever the messages, and the modeled mutual information is $\frac{1}{2}\ln\det P_i + \frac{1}{2}\ln\det(\Lambda_i + \sum_j \Gamma_{ij}) = \frac{1}{2}\ln\det(I + P_i\sum_j \Gamma_{ij})$. If $\hat{R}_{ij} = R_c$, then $\hat{R}_{ij} \preceq \hat{R}^1_{ij}$, so $\Gamma^0_{ij} \succeq \Gamma^1_{ij}$ (inversion reverses and congruence preserves the order) and $\mathcal{V}^0_i \geq \mathcal{V}^1_i$. (b) is read off (a).

(c) Fix the level, write $f(\mathcal{J})$ for its believed value and $\Lambda_\mathcal{J} = \Lambda_i + \sum_{k\in\mathcal{J}} \Gamma_{ik}$. For $j \notin \mathcal{J}$,

$$f(\mathcal{J}\cup\{j\}) - f(\mathcal{J}) = \tfrac{1}{2}\ln\det(\Lambda_\mathcal{J} + \Gamma_{ij}) - \tfrac{1}{2}\ln\det\Lambda_\mathcal{J} = \tfrac{1}{2}\ln\det(I + \Gamma^{1/2}_{ij}\Lambda^{-1}_\mathcal{J}\Gamma^{1/2}_{ij}) \geq 0.$$

If $\mathcal{J}\subseteq\mathcal{J}'$ and $j\notin\mathcal{J}'$, then $\Lambda_\mathcal{J} \preceq \Lambda_{\mathcal{J}'}$, hence $\Lambda^{-1}_{\mathcal{J}'} \preceq \Lambda^{-1}_\mathcal{J}$ and the gain at $\mathcal{J}'$ is at most the gain at $\mathcal{J}$. Nonnegative gains give monotonicity and decreasing gains give submodularity. Greedy attention starts from $\mathcal{J}_0 = \varnothing$ and adds at each step a sender of largest gain. Let $\mathcal{J}^*$ maximize $f$ over sets of at most $\kappa$ senders and put $\delta_l = f(\mathcal{J}^*) - f(\mathcal{J}_l)$. Monotonicity, then submodularity applied to the senders of $\mathcal{J}^*\setminus\mathcal{J}_l$ added one at a time, and then the greedy choice give

$$f(\mathcal{J}^*) \leq f(\mathcal{J}_l\cup\mathcal{J}^*) \leq f(\mathcal{J}_l) + \sum_{j\in\mathcal{J}^*\setminus\mathcal{J}_l} \big(f(\mathcal{J}_l\cup\{j\}) - f(\mathcal{J}_l)\big) \leq f(\mathcal{J}_l) + \kappa\big(f(\mathcal{J}_{l+1}) - f(\mathcal{J}_l)\big),$$

so $\delta_{l+1} \leq (1-1/\kappa)\delta_l$. Since $f(\varnothing) = 0$, $\delta_\kappa \leq (1-1/\kappa)^\kappa f(\mathcal{J}^*) \leq e^{-1} f(\mathcal{J}^*)$, which is the bound of Nemhauser et al. (1978). If at most $\kappa$ senders are available, greedy attention takes all of them and is optimal by monotonicity. □

*Proof of Lemma S3.3.* The Bayes error $\eta - m_j$ is uncorrelated with every square-integrable function of the sender's data, in particular with $m_j$, so $\mathrm{Cov}(\eta, m_j) = \mathrm{Cov}(m_j)$, and the law of total covariance, with the posterior covariance $P_j$ not depending on the data, gives $\mathrm{Cov}(m_j) = P_0 - P_j$. The pair $(\eta, m_j)$ is jointly Gaussian with $\mathbb{E}m_j = \bar{\eta}$, so $m_j - \bar{\eta} = (P_0 - P_j)P_0^{-1}(\eta - \bar{\eta}) + \tilde{m}_j$ with $\tilde{m}_j$ Gaussian, independent of $\eta$, and of covariance $(P_0 - P_j) - (P_0 - P_j)P_0^{-1}(P_0 - P_j) = (P_0 - P_j)P_0^{-1}P_j = P_j - P_jP_0^{-1}P_j$. Since $(P_0 - P_j)P_0^{-1} = I - P_jP_0^{-1}$, applying $H_j$ and adding the channel noise, which is independent of $(\eta, \tilde{m}_j)$, gives (S5) with $\tilde{\nu}_j = H_j\tilde{m}_j + \nu$. □

*Proof of Lemma 3.3.* (a) Substituting $o_u = H^s_u\eta + v_u$ into $\Lambda_k m_k = \Lambda_0\bar{\eta} + \sum_{u\in\mathcal{U}_k}(H^s_u)^\top(R^s_u)^{-1}o_u$ and using $\sum_{u\in\mathcal{U}_k}\Gamma_u = \Lambda_k - \Lambda_0$ gives

$$e_k = P_k\Big(\Lambda_0(\bar{\eta} - \eta) + \sum_{u\in\mathcal{U}_k}(H^s_u)^\top(R^s_u)^{-1}v_u\Big).$$

The terms are independent with covariances $\Lambda_0P_0\Lambda_0 = \Lambda_0$ and $\Gamma_u$, so $\mathrm{Var}(e_k) = P_k\Lambda_kP_k = P_k$. The errors $e_i$ and $e_j$ share only $\bar{\eta} - \eta$ and the noises of shared sources, so $\mathrm{Cov}(e_i, e_j) = P_i\Lambda^{\mathrm{sh}}_{ij}P_j$, and $\mathrm{Cov}(e_k, \eta) = -P_k\Lambda_0P_0 = -P_k$. In the symmetric scalar configuration $P_i = P_j = P$ and the correlation is $P(\Lambda_0 + \Gamma_{\mathrm{com}}) = \phi$.

(b) Write $\Lambda_j m_j = \Lambda_0\bar{\eta} + h^{\mathrm{sh}}_{ij} + h_{j\setminus i}$ with $h_{j\setminus i} = \sum_{u\in\mathcal{U}_j\setminus\mathcal{U}_i}(H^s_u)^\top(R^s_u)^{-1}o_u = \Lambda_{j\setminus i}\eta + \tilde{v}_{j\setminus i}$, where $\tilde{v}_{j\setminus i} \sim \mathcal{N}(0, \Lambda_{j\setminus i})$ involves only noises of sources outside $\mathcal{U}_i$ and is independent of $(\eta, \mathcal{Z}_i)$. Given $\mathcal{Z}_i$, the terms $\Lambda_0\bar{\eta}$ and $h^{\mathrm{sh}}_{ij}$ are known, and the shifted message is

$$\tilde{y}_{ij} := y_{ij} - H_jP_j(\Lambda_0\bar{\eta} + h^{\mathrm{sh}}_{ij}) = A_{ij}\eta + H_jP_j\tilde{v}_{j\setminus i} + \nu.$$

It is a linear-Gaussian observation of $\eta$ with code $A_{ij}$ and noise covariance $R^{\mathrm{ex}}_{ij} \succ 0$, conditionally independent of $\mathcal{Z}_i$ given $\eta$. As $y_{ij} - \tilde{y}_{ij}$ is $\mathcal{Z}_i$-measurable, $I(\eta; y_{ij} \mid \mathcal{Z}_i) = I(\eta; \tilde{y}_{ij} \mid \mathcal{Z}_i)$, the information of this observation under the prior $\mathcal{N}(m_i, P_i)$, which is (16).

(c) With $H_j$ invertible and $R_c = 0$, observing $y_{ij}$ given $\mathcal{Z}_i$ is observing $h_{j\setminus i}$, whose likelihood is proportional in $\eta$ to $\exp(\eta^\top h_{j\setminus i} - \frac{1}{2}\eta^\top \Lambda_{j\setminus i}\eta)$ (also when $\Lambda_{j\setminus i}$ is singular, since $h_{j\setminus i}$ then lies in its range). The update adds $\Lambda_{j\setminus i} = \Lambda_j - \Lambda_{ij}^{\mathrm{sh}}$ to the precision and $h_{j\setminus i} = \Lambda_j m_j - \Lambda_0\bar{\eta} - h_{ij}^{\mathrm{sh}}$ to the information vector $\Lambda_i m_i$, and the information gained is $\frac{1}{2}\ln\det((\Lambda_i + \Lambda_{j\setminus i})P_i)$.

(d) The exact update conditions $\mathcal{N}(m_i, P_i)$ on $\tilde{y}_{ij}$, an observation with code $A_{ij}$ and noise $R_{ij}^{\mathrm{ex}} \succ 0$, so by Proposition 2.2 its mean moves within $\mathrm{col}(P_i A_{ij}^\top) = \mathrm{col}(P_i \Lambda_{j\setminus i} P_j H_j^\top)$, while a literal update uses the code $H_j$. □

*Proof of Proposition S3.4.* Given $(\eta, m_i)$ the receiver's error $e_i = m_i - \eta$ is known, and $e_j \mid e_i \sim \mathcal{N}(\beta e_i, \sigma_j^2 - \sigma_{ij}^2/\sigma_i^2)$ with $\beta = \sigma_{ij}/\sigma_i^2$, independently of $\eta$. Hence

$$y_{ij} = \tilde{g}\eta + g\beta m_i + \tilde{\nu}, \qquad \tilde{g} = g(1-\beta), \qquad \mathrm{Var}(\tilde{\nu}) = R^{\mathrm{ex}} > 0,$$

an observation of $\eta$ given $m_i$ with code $\tilde{g}$ and noise $R^{\mathrm{ex}}$. Under the belief $\mathcal{N}(m_i, \sigma_i^2)$ it carries $\frac{1}{2}\ln(1 + \tilde{g}^2\sigma_i^2/R^{\mathrm{ex}})$, and $\tilde{g}^2\sigma_i^2 + R^{\mathrm{ex}} = S$ on expanding, so $\mathcal{V}_i^* = \frac{1}{2}\ln(S/R^{\mathrm{ex}})$. The level-1 model is the case $\sigma_{ij} = 0$, and level 0 uses the noise $R_c \le g^2\sigma_j^2 + R_c$.

Let $S_0$ and $R_0$ be the values of $S$ and $R^{\mathrm{ex}}$ at $\sigma_{ij} = 0$, so that $S = S_0 - 2g^2\sigma_{ij}$ and $R^{\mathrm{ex}} = R_0 - g^2\sigma_{ij}^2/\sigma_i^2$. Then $\mathcal{V}_i^* < \mathcal{V}_i^1$ if and only if $SR_0 < S_0 R^{\mathrm{ex}}$, and $SR_0 - S_0R^{\mathrm{ex}} = (g^2\sigma_{ij}/\sigma_i^2)(S_0\sigma_{ij} - 2\sigma_i^2 R_0)$, so the condition is $\sigma_{ij}(2\sigma_i^2 R_0 - S_0\sigma_{ij}) > 0$. For $\sigma_{ij} > 0$ this holds exactly when $\sigma_{ij} < 2\sigma_i^2 R_0/S_0 = \sigma_{\mathrm{b}}$. For $\sigma_{ij} < 0$ the second factor is positive, so the product is negative and $\mathcal{V}_i^* > \mathcal{V}_i^1$; for $\sigma_{ij} > \sigma_{\mathrm{b}}$ the second factor is negative and the same conclusion holds. For equal variances $\sigma^2$, $\sigma_{\mathrm{b}} = 2\sigma^2(g^2\sigma^2 + R_c)/(2g^2\sigma^2 + R_c) \ge \sigma^2 = \sigma_i\sigma_j$, so the band contains every admissible positive covariance. At $R_c = 0$, $\sigma_{\mathrm{b}} = 2\sigma_i^2\sigma_j^2/(\sigma_i^2 + \sigma_j^2)$, a harmonic mean, which is below the geometric mean $\sigma_i\sigma_j$ when $\sigma_i \ne \sigma_j$. If $\sigma_i < \sigma_j$, the value $\sigma_{ij} = \sigma_i^2$ is admissible and gives $\beta = 1$, $\tilde{g} = 0$, so $\mathcal{V}_i^* = 0$. As $\sigma_{ij} \to -\sigma_i\sigma_j$, $R^{\mathrm{ex}} \to R_c$ and $S \to g^2(\sigma_i + \sigma_j)^2 + R_c$, and the limit exceeds $\mathcal{V}_i^0$ because $(\sigma_i + \sigma_j)^2 > \sigma_i^2$. As $\sigma_{ij} \to \sigma_i\sigma_j$, $\mathcal{V}_i^* \to \frac{1}{2}\ln(1 + g^2(\sigma_j - \sigma_i)^2/R_c)$, which exceeds $\mathcal{V}_i^0$ exactly when $(\sigma_j - \sigma_i)^2 > \sigma_i^2$, that is $\sigma_j > 2\sigma_i$. □

*Proof of Proposition 3.4.* (a) As in Proposition S3.1(a), since $R_c \preceq R_c + H_j P_j H_j^\top$.

(b) Let $K = 1$ and let $g \ne 0$ be the code gain. If $\Lambda_{j\setminus i} = 0$, the message is a function of the receiver's data and of channel noise independent of $\eta$, so $\mathcal{V}_i^* = 0 < \mathcal{V}_i^1$. Otherwise $R_{ij}^{\mathrm{ex}} = g^2 P_j^2 \Lambda_{j\setminus i} + R_c > 0$, (16) gives $A_{ij} = g P_j \Lambda_{j\setminus i}$, and

$$\mathcal{V}_i^* = \tfrac{1}{2}\ln\Big(1 + \frac{g^2 P_j^2 \Lambda_{j\setminus i}^2 P_i}{g^2 P_j^2 \Lambda_{j\setminus i} + R_c}\Big), \qquad \mathcal{V}_i^1 = \tfrac{1}{2}\ln\Big(1 + \frac{g^2 P_i}{g^2 P_j + R_c}\Big).$$

Put $x = P_j\Lambda_{j\setminus i} = 1 - P_j\Lambda_{ij}^{\mathrm{sh}}$. Dividing by $g^2 P_i$ and clearing the positive denominators, $\mathcal{V}_i^* < \mathcal{V}_i^1$ is equivalent to $x g^2 P_j^2 \Lambda_{j\setminus i} + x^2 R_c < g^2 P_j^2 \Lambda_{j\setminus i} + R_c$, which holds because $\Lambda_{ij}^{\mathrm{sh}} \ge \Lambda_0 > 0$ forces $0 < x < 1$. In the flat-prior limit $x = 1$ exactly when no source is shared, and then the two sides coincide.

(c) Lemma 3.3(c) gives $\mathcal{V}_i^*$. Level 1 assigns the precision $H_j^\top(H_j P_j H_j^\top)^{-1} H_j = \Lambda_j$, so $\mathcal{V}_i^1 = \frac{1}{2}\ln\det(I + P_i\Lambda_j)$, and $\Lambda_j - \Lambda_{j\setminus i} = \Lambda_{ij}^{\mathrm{sh}} \succeq \Lambda_0 \succ 0$ gives the strict inequality.

(d) Each source $o_u$ splits into its information vector $h_u = (H_u^s)^\top (R_u^s)^{-1} o_u = \Gamma_u\eta + n_u$, $n_u \sim \mathcal{N}(0, \Gamma_u)$, and the residual $o_u - H_u^s\Gamma_u^+ h_u$, which equals $(I - H_u^s\Gamma_u^+(H_u^s)^\top(R_u^s)^{-1})v_u$ because $H_u^s$ vanishes on $\ker\Gamma_u$; the residual is uncorrelated with $n_u$, hence independent of $h_u$, of $\eta$ and of every other variable. Every belief, message and value is therefore unchanged when the raw data are replaced by the information vectors. In an aligned world the coordinates of $\eta$ under the prior, of the noises $n_u$ and of the channel noise are independent across the common basis, every posterior

covariance is diagonal, so the $d$th coordinate of each agent's mean depends only on $d$th coordinates, and each message coordinate reads one coordinate. The joint law of the world, the data and the messages factorizes over coordinates, and so do the level-0 and level-1 models, whose noises $R_c$ and $R_c + H_j P_j H_j^\top$ are diagonal. Every value is therefore a sum over coded coordinates of the values of scalar common-source worlds with proper priors, uncoded coordinates contribute nothing, and (a) and (b) apply coordinatewise.

(e) Here $\Lambda_i = \Lambda_{ij}^{\mathrm{sh}}$. Let $Y_j = P_j \Lambda_{j\setminus i} = I - P_j \Lambda_{ij}^{\mathrm{sh}}$, so $A_{ij} = H_j Y_j$, and $\Gamma^0 = H_j^\top R_c^{-1} H_j$. Since $R_{ij}^{\mathrm{ex}} \succeq R_c$, $A_{ij}^\top (R_{ij}^{\mathrm{ex}})^{-1} A_{ij} \preceq Y_j^\top \Gamma^0 Y_j$, and by monotonicity

$$\mathcal{V}_i^* \le \tfrac{1}{2} \ln\det(I + P_i Y_j^\top \Gamma^0 Y_j) = \tfrac{1}{2} \ln\det(I + (\Gamma^0)^{1/2} Y_j P_i Y_j^\top (\Gamma^0)^{1/2}).$$

Using $P_i \Lambda_{ij}^{\mathrm{sh}} = I$, $Y_j P_i Y_j^\top = P_i - 2P_j + P_j \Lambda_{ij}^{\mathrm{sh}} P_j \preceq P_i - P_j \preceq P_i$, because $\Lambda_{ij}^{\mathrm{sh}} \preceq \Lambda_j$ gives $P_j \Lambda_{ij}^{\mathrm{sh}} P_j \preceq P_j$. Hence $\mathcal{V}_i^* \le \frac{1}{2} \ln\det(I + (\Gamma^0)^{1/2} P_i (\Gamma^0)^{1/2}) = \mathcal{V}_i^0$. □

*Proof of Proposition S3.5.* Decoding $y_{ij}$ gives $m_j$, which level 1 reads as $\mathcal{N}(\eta, P_j)$ independent of its own information; Bayes' rule gives the precision $\Lambda_i + \Lambda_j$ and the stated mean. Because $\hat{P}_i^{(1)}(\Lambda_i + \Lambda_j) = I$, the error is $\hat{P}_i^{(1)}(\Lambda_i e_i + \Lambda_j e_j)$, and by Lemma 3.3(a) with $\Lambda_k P_k = I$ its covariance is $\hat{P}_i^{(1)}(\Lambda_i + \Lambda_j + 2\Lambda_{ij}^{\mathrm{sh}})\hat{P}_i^{(1)}$. Subtracting $\hat{P}_i^{(1)} = \hat{P}_i^{(1)}(\Lambda_i + \Lambda_j)\hat{P}_i^{(1)}$ leaves $2\hat{P}_i^{(1)} \Lambda_{ij}^{\mathrm{sh}} \hat{P}_i^{(1)}$. The exact posterior mean minimizes the mean-square-error matrix among estimators built from $(\mathcal{Z}_i, y_{ij})$, and its error covariance is $(\Lambda_i + \Lambda_{j\setminus i})^{-1}$ by Lemma 3.3(c). In the symmetric scalar configuration the claimed variance is $P/2$ and the actual one $P(1+\phi)/2$. □

*Proof of Proposition 3.5.* (a) In (16) with scalars, $A_{ij} = gP\Gamma_{\mathrm{pr}}$, $R_{ij}^{\mathrm{ex}} = g^2 P^2 \Gamma_{\mathrm{pr}} + R_c$ and $P\Gamma_{\mathrm{pr}} = 1 - \phi$, so after division by $g^2 P$, $A_{ij}^2 P / R_{ij}^{\mathrm{ex}} = (1-\phi)^2/(1 - \phi + \varrho_c)$. Levels 1 and 0 see the signal variance $g^2 P$ with noises $g^2 P + R_c$ and $R_c$. The expansion follows from $\ln(1+x) = x + O(x^2)$ with $x = (1-\phi)^2/\varrho_c + O((1-\phi)^3)$; at $\varrho_c = 0$, $\mathcal{V}_i^* = \frac{1}{2}\ln(2-\phi)$ and $\mathcal{V}_i^1 = \frac{1}{2}\ln 2$.

(b) Under a flat prior the errors are independent of $\eta$, with variances $P$ and covariance $\phi P$ by Lemma 3.3(a) at $\Lambda_0 = 0$, and the receiver's posterior is $\mathcal{N}(m_i, P)$. Proposition S3.4 with $\sigma_i^2 = \sigma_j^2 = P$ and $\sigma_{ij} = \phi P$ gives $S/(g^2 P) = 2(1-\phi) + \varrho_c$ and $R^{\mathrm{ex}}/(g^2 P) = 1 - \phi^2 + \varrho_c$.

(c) The messages $m_r + \nu_l$, $l = 1, \dots, n_{\mathrm{r}}$, depend on $\eta$ and the receiver's data only through $m_r$, and their average $m_r + \bar{\nu}$, $\mathrm{Var}(\bar{\nu}) = R_c/n_{\mathrm{r}}$, is sufficient for $m_r$. Lemma 3.3(b) with private precision $\Gamma_{\mathrm{rel}}$, unit code and noise $R_c/n_{\mathrm{r}}$ gives $A = P_r \Gamma_{\mathrm{rel}}$ and $R^{\mathrm{ex}} = P_r^2 \Gamma_{\mathrm{rel}} + R_c/n_{\mathrm{r}}$, hence the formula, which increases in $n_{\mathrm{r}}$ to the stated limit. Level 1 reads the $n_{\mathrm{r}}$ messages as independent observations, each with noise $P_r + R_c$. □

*Proof of Proposition S3.6.* (a) Let $m_k(t)$ be agent $k$'s filtered mean of $\eta_t$ and $e_k^+(t) = m_k(t) - \eta_t$. The update $m_k(t+1) = F m_k(t) + k_k^s(\eta_{t+1} + v_k(t+1) - F m_k(t))$ gives $e_k^+(t+1) = (1 - k_k^s)(F e_k^+(t) - w_t) + k_k^s v_k(t+1)$. The steady-state predicted variance $P_k^-$ solves $P_k^- = F^2(1-k_k^s)P_k^- + Q$ with $k_k^s = P_k^-/(P_k^- + R_k^s) \in (0,1)$, so $F^2(1-k_k^s) = 1 - Q/P_k^- < 1$ and hence $(1-k_i^s)(1-k_j^s)F^2 \le (1-k_i^s)F^2 < 1$. The sensing noises are independent of each other and of the world, so the stationary cross-covariance satisfies $\Sigma_{ij} = (1-k_i^s)(1-k_j^s)(F^2\Sigma_{ij} + Q)$, which is (S8), positive since $Q > 0$. Each steady-state filter is calibrated about its own error: its claimed filtered variance is $\Sigma_{kk}$. A level-1 receiver reading the belief message $y_{ij} = H_j m_j(t) + \nu$, $H_j \neq 0$, with the header $\Sigma_{jj}$ is therefore in the setting of Proposition S3.15 with $X = H_j \Sigma_{ij}$, and its actual posterior variance exceeds its claim by $2\mathcal{K}_0(1 - \mathcal{K}_0 H_j) H_j \Sigma_{ij} > 0$, since $0 < \mathcal{K}_0 H_j < 1$. An exact posterior claims a variance equal to the mean-square error of its mean, so level 1 is not exact.

(b) The receiver predicts the message as $H_j F m_i$, so its innovation is $y_{ij} - H_j F m_i = H_j F(e_j - e_i) + \nu$, in which the common innovation cancels; its covariance is $S_{ij}^{\mathrm{act}}$. The receiver models its own

predictive error with covariance $FP_iF^\top + Q$ and the message's predictive error, independent of its own, with the header $FP_jF^\top + Q$, which gives $S^{\mathrm{cl}}_{ij}$. Subtracting gives (S9). □

*Derivation of Example S3.8.* Here $\Lambda_i = \mathrm{diag}(\alpha_1, \alpha_2)$ and $\Lambda_j = \mathrm{diag}(\alpha_1, \alpha_2 + \Gamma_{\mathrm{pr}})$. With $H_j = (1,1)/\sqrt{2}$, $A_{ij} = H_jP_j\Lambda_{j\setminus i} = (0, \Gamma_{\mathrm{pr}}/(\alpha_2+\Gamma_{\mathrm{pr}}))/\sqrt{2}$ and $R^{\mathrm{ex}}_{ij} = \Gamma_{\mathrm{pr}}/(2(\alpha_2+\Gamma_{\mathrm{pr}})^2)$, so $A_{ij}P_iA_{ij}^\top/R^{\mathrm{ex}}_{ij} = \Gamma_{\mathrm{pr}}/\alpha_2$. Level 1 sees the signal variance $H_jP_iH_j^\top = \frac{1}{2}(\alpha_1^{-1} + \alpha_2^{-1})$ and the noise $H_jP_jH_j^\top = \frac{1}{2}(\alpha_1^{-1} + (\alpha_2+\Gamma_{\mathrm{pr}})^{-1})$. As $\alpha_2 \to 0$, $\mathcal{V}^*_i = \frac{1}{2}\ln(\Gamma_{\mathrm{pr}}/\alpha_2) + o(1)$ and $\mathcal{V}^1_i = \frac{1}{2}\ln(\alpha_2^{-1}/(\alpha_1^{-1}+\Gamma_{\mathrm{pr}}^{-1})) + o(1)$, whose difference tends to $\frac{1}{2}\ln((\alpha_1+\Gamma_{\mathrm{pr}})/\alpha_1)$. The numbers evaluate these formulas and, for $R_c = 0.1$, (16) and Proposition S3.1(a). □

*Derivation of Example S3.9.* Given $o_1 = \eta_1 + v_1$ with $\mathrm{Var}(v_1) = \Gamma_1^{-1}$, the sender's posterior mean is $m_j = (1,\psi)^\top\bar\Gamma_1 o_1$ with $\bar\Gamma_1 = \Gamma_1/(1+\Gamma_1)$, so $y_{ij} = \psi\bar\Gamma_1(\eta_1 + v_1) + \nu$. Given $\eta_1$, $y_{ij}$ is independent of $(\eta_2, o_2)$, so $I(\eta; y_{ij} \mid o_2) = I(\eta_1; y_{ij} \mid o_2)$, and with $\mathrm{Var}(\eta_1 \mid o_2) = 1 - \psi^2\Gamma_2/(1+\Gamma_2)$,

$$\mathcal{V}^*_i = \tfrac{1}{2}\ln\Big(1 + \frac{\psi^2\bar\Gamma_1^2\,\mathrm{Var}(\eta_1 \mid o_2)}{\psi^2\bar\Gamma_1^2\Gamma_1^{-1} + R_c}\Big).$$

Level 0 models $y_{ij} \sim \mathcal{N}(\eta_2, R_c)$ given $\eta$, with $\mathrm{Var}(\eta_2 \mid o_2) = (1+\Gamma_2)^{-1}$, so $\mathcal{V}^0_i = \frac{1}{2}\ln(1 + (1+\Gamma_2)^{-1}/R_c)$; level 1 adds the header $[P_j]_{22} = 1 - \psi^2\bar\Gamma_1$ to the noise. Letting $\Gamma_1, \Gamma_2 \to \infty$ gives (S11). For $H_j = I_2$ and $R_c = \sigma_c^2 I_2$, the mutual informations are continuous functions of the joint covariance as long as the conditional covariance of the message given $\eta$ and the receiver's data stays nonsingular, which $\sigma_c^2 > 0$ ensures. In the limit the receiver knows $\eta_2$ and the message is $(\eta_1, \psi\eta_1) + \nu$, an observation of $\eta_1$ with precision $(1+\psi^2)/\sigma_c^2$ under the variance $1-\psi^2$; level 0 believes it observes $\eta$ with noise $\sigma_c^2 I_2$ under posterior variances $1-\psi^2$ and 0. □

*Proof of Proposition 3.8.* Unrolling (18) gives $e(t) = \sum_{s\ge0} W^s(\zeta(t-1-s) - w_{t-1-s}\mathbf{1})$, convergent in mean square because $\rho(W) < 1$, with independent local and common parts (Lemma 2.13). Hence $\mathbb{E}e_i^2 \ge \mathbb{E}(e_i^{\mathrm{com}})^2 = Q\sum_{s\ge0}((W^s\mathbf{1})_i)^2$. The row sums of $W$ are $\lambda_i = F(1-k_i^s)$, and their minimum is $\lambda_{\min} = F(1-k_{\max}^s)$. Since $W \ge 0$ and $W\mathbf{1} \ge \lambda_{\min}\mathbf{1} \ge 0$ entrywise, induction gives $W^s\mathbf{1} \ge \lambda^s_{\min}\mathbf{1}$. The smallest row sum of a nonnegative matrix is at most its spectral radius, so $\lambda_{\min} \le \rho(W) < 1$, and summing the geometric series gives $\mathbb{E}e_i^2 \ge Q/(1-\lambda^2_{\min})$, which is (20). If $C$ is closed under attention, the rows of $W$ indexed by $C$ vanish outside $C$, so $e_C(t+1) = W_{CC}e_C(t) + \zeta_C(t) - w_t\mathbf{1}$ is a closed recursion, stationary when $\rho(W_{CC}) < 1$; $W$ is block triangular for the split of $V$ into $C$ and its complement, so $\rho(W_{CC}) \le \rho(W)$. The argument applies to $W_{CC}$, whose row sums are $F(1-k_i^s)$, $i \in C$. For pure listeners the bound is $Q/(1-F^2) = P_0$. □

*Proof of Lemma 3.9.* Write $\lambda = F(1-k^s)$ for the common row sum of $W_{CC} = \lambda M$, let $J_\varpi = \mathbf{1}\varpi^\top$ and $\tilde e = (I - J_\varpi)e_C$. Reversibility gives $\varpi^\top M = \varpi^\top$, and $M\mathbf{1} = \mathbf{1}$, so $J_\varpi M = MJ_\varpi = J_\varpi$ and $(I-J_\varpi)M = (M-J_\varpi)(I-J_\varpi)$. With $(I-J_\varpi)\mathbf{1} = 0$,

$$\tilde e(t+1) = \lambda(M - J_\varpi)\tilde e(t) + (I - J_\varpi)\zeta_C(t),$$

in which $w_t$ no longer appears. In the inner product $\langle x, x'\rangle_\varpi = \sum_i \varpi_i x_i x_i'$ reversibility makes $M$ self-adjoint and $J_\varpi$ the orthogonal projection onto the constants, so $M - J_\varpi$ is self-adjoint, vanishes on the constants and agrees with $M$ on $\{x : \varpi^\top x = 0\}$, which $M$ maps into itself; its norm is $\sigma_*$, and $\|I - J_\varpi\|_\varpi \le 1$. Since $\lambda\sigma_* < 1$, the stationary disagreement is $\sum_{s\ge0}\lambda^s(M-J_\varpi)^s(I-J_\varpi)\zeta_C(t-1-s)$, and independence across rounds gives

$$\mathbb{E}\|\tilde e\|^2_\varpi \le \sum_{s\ge0}(\lambda\sigma_*)^{2s}\mathbb{E}\|\zeta_C\|^2_\varpi = \frac{\bar q_\varpi}{1-\lambda^2\sigma_*^2},$$

which is the bound with $\lambda^2 = F^2(1-k^s)^2$. Finally $m_i - m_j = e_i - e_j = \tilde{e}_i - \tilde{e}_j$, and $(\tilde{e}_i - \tilde{e}_j)^2 \le 2\tilde{e}_i^2 + 2\tilde{e}_j^2 \le (4/\varpi_{\min})\|\tilde{e}\|_\varpi^2$ because $\varpi_i \tilde{e}_i^2 \le \|\tilde{e}\|_\varpi^2$. □

*Proof of Proposition S3.12.* (a) From $t_0$ on, $Y = m_C(t)$ is a linear function of $m_C(t_0)$ and of channel noises independent of $(\eta, m_C(t_0))$, so in a static world the data-processing inequality gives the claim. In a stationary scalar world write $\eta_t = F^{t-t_0}\eta_{t_0} + \bar{w}_t$, where $\bar{w}_t = \sum_{s=t_0}^{t-1} F^{t-1-s} w_s$ collects the innovations after $t_0$ and is independent of $(\eta_{t_0}, m_C(t_0))$ and of the channel noises, hence of $Y$. All variables are jointly Gaussian, $\mathrm{Cov}(Y, \eta_t) = F^{t-t_0}\,\mathrm{Cov}(Y, \eta_{t_0})$ and $\mathrm{Var}(\eta_t) = \mathrm{Var}(\eta_{t_0}) = P_0$, so the squared multiple correlations satisfy $\mathsf{R}^2(Y, \eta_t) = F^{2(t-t_0)}\mathsf{R}^2(Y, \eta_{t_0})$. For scalar $\eta$, $I(Y;\eta) = -\frac{1}{2}\ln(1 - \mathsf{R}^2(Y,\eta))$, and $x \mapsto -\ln(1-x)$ is convex on $[0,1)$ and vanishes at 0, so $-\ln(1 - F^{2(t-t_0)}x) \le -F^{2(t-t_0)}\ln(1-x)$ for $x \in [0,1)$. This gives $I(Y;\eta_t) \le F^{2(t-t_0)} I(Y;\eta_{t_0})$, and data processing gives $I(Y;\eta_{t_0}) \le I(m_C(t_0);\eta_{t_0})$.

(b) With $k^s = 0$ and $F = 1$, $e_C(t+1) = M e_C(t) + \zeta_C(t) - w_t \mathbf{1}$. Multiplying by $\varpi^\top$ and using $\varpi^\top M = \varpi^\top$, $\varpi^\top \mathbf{1} = 1$ gives the random walk, whose independent increments have variance $\sum_i \varpi_i^2 q_i + Q$. With $J_\varpi = \mathbf{1}\varpi^\top$, the disagreement satisfies $\tilde{e}(t+1) = (M - J_\varpi)\tilde{e}(t) + (I - J_\varpi)\zeta_C(t)$, as in the proof of Lemma 3.9. Since $M$ and $J_\varpi$ commute and $J_\varpi$ projects onto $\mathrm{span}(\mathbf{1})$ along $\{x : \varpi^\top x = 0\}$, the spectrum of $M - J_\varpi$ is that of $M$ with the simple eigenvalue 1 replaced by 0, so $\rho(M - J_\varpi) < 1$, $\|(M - J_\varpi)^s\|$ decays geometrically, and the second moments of $\tilde{e}(t)$ are bounded uniformly in $t$ for any initial law with finite second moments.

(c) Each member's message gains are $k_{ij}(t) = R_c^{-1}/\hat{\Lambda}(t+1) = (1 - c_t)/n_{\mathrm{in}}$, so $\zeta_i(t) = ((1 - c_t)/n_{\mathrm{in}}) \sum_j \nu_{ij}(t)$ has variance $(1-c_t)^2 R_c/n_{\mathrm{in}}$, and $1 - c_t = (n_{\mathrm{in}}/R_c)/\hat{\Lambda}(t+1)$ is of order $1/t$. Since $M$ is doubly stochastic, $\mathbf{1}^\top W_{CC}(t) = \mathbf{1}^\top$, and $\bar{e} = n^{-1}\mathbf{1}^\top e_C$ satisfies $\bar{e}(t+1) = \bar{e}(t) + n^{-1}\mathbf{1}^\top \zeta_C(t)$. The increments are independent, of mean zero, independent of $\bar{e}(t_0)$ and of summable variance, so $\bar{e}(t)$ is Cauchy in mean square and converges to some $\bar{e}_\infty$ with $\mathrm{Var}(\bar{e}_\infty) \ge \mathrm{Var}(\bar{e}(t_0)) = n^{-2}\mathbf{1}^\top \mathrm{Cov}(e_C(t_0))\mathbf{1}$. The matrices $W_{CC}(t)$ commute with $\mathbf{1}\mathbf{1}^\top$, so $\tilde{e} = (I - n^{-1}\mathbf{1}\mathbf{1}^\top)e_C$ satisfies $\tilde{e}(t+1) = W_{CC}(t)\tilde{e}(t) + (I - n^{-1}\mathbf{1}\mathbf{1}^\top)\zeta_C(t)$. On the invariant subspace $\mathbf{1}^\perp$ the normal matrix $M$ has eigenvalues $x \ne 1$ with $|x| \le 1$, hence $\mathrm{Re}\, x < 1$, and

$$|c + (1-c)x|^2 = 1 - 2c(1-c)(1 - \mathrm{Re}\, x) - (1-c)^2(1 - |x|^2) \le 1 - 2c(1-c)\,\mathrm{gap}(M),$$

with $\mathrm{gap}(M) = \min_{x \ne 1}(1 - \mathrm{Re}\, x) > 0$. As $W_{CC}(t)$ is normal, $\mathcal{X}_t = \mathbb{E}\|\tilde{e}(t)\|^2$ satisfies $\mathcal{X}_{t+1} \le (1 - A_t)\mathcal{X}_t + B_t$ with $A_t = 2c_t(1-c_t)\,\mathrm{gap}(M) \in [0,1]$ and $B_t = n(1-c_t)^2 R_c/n_{\mathrm{in}}$; the cross term vanishes because $\zeta_C(t)$ is independent of $\tilde{e}(t)$. Since $c_t \to 1$ and $1 - c_t$ is of order $1/t$, $\sum_t A_t = \infty$ and $B_t/A_t \to 0$. Given $\bar{\mathcal{X}} > 0$, for $t$ beyond some $t_1$ we have $B_t \le \bar{\mathcal{X}} A_t$, hence $(\mathcal{X}_{t+1} - \bar{\mathcal{X}})^+ \le (1 - A_t)(\mathcal{X}_t - \bar{\mathcal{X}})^+$ and $(\mathcal{X}_t - \bar{\mathcal{X}})^+ \le \exp(-\sum_{s=t_1}^{t-1} A_s)(\mathcal{X}_{t_1} - \bar{\mathcal{X}})^+ \to 0$. So $\limsup_t \mathcal{X}_t \le \bar{\mathcal{X}}$ for every $\bar{\mathcal{X}} > 0$, $\mathcal{X}_t \to 0$, and $e_i(t) = \bar{e}(t) + \tilde{e}_i(t) \to \bar{e}_\infty$ in mean square for every member. Finally $1/\hat{\Lambda}(t) = 1/(\hat{\Lambda}(t_0) + (t - t_0)n_{\mathrm{in}}/R_c)$, so $\frac{1}{2}\ln(P_0\hat{\Lambda}(t)) = \frac{1}{2}\ln t + O(1)$. □

*Derivations for Example S3.13.* By symmetry every member has the same claimed predicted variance and reads $n-1$ messages, each with assigned noise $R_c + \hat{P}^-$, so the stationary claimed predicted variance $\hat{P}_n^-$ is a fixed point of $f_n(p) = F^2/g_n(p) + Q$, with $g_n(p) = 1/p + 1/R_i^s + (n-1)/(R_c + p)$ the claimed precision after an update. The map $f_n$ is increasing in $p$ and strictly decreasing in $n$, and its derivative $F^2(p^{-2} + (n-1)(R_c + p)^{-2})/g_n(p)^2$ is at most $F^2 < 1$, because $g_n(p)^2 \ge (p^{-1} + (n-1)(R_c + p)^{-1})^2 \ge p^{-2} + (n-1)(R_c + p)^{-2}$. Hence $p - f_n(p)$ is strictly increasing, negative at $p = Q$ and positive for large $p$, so $\hat{P}_n^-$ is unique. Since $\hat{P}_n^- - f_{n+1}(\hat{P}_n^-) > \hat{P}_n^- - f_n(\hat{P}_n^-) = 0$ and $p - f_{n+1}(p)$ is strictly increasing, $\hat{P}_{n+1}^- < \hat{P}_n^-$. The claimed updated variance $\hat{P}_n^+ = (\hat{P}_n^- - Q)/F^2$ and the sensing gain $k_n^s = \hat{P}_n^+/R_i^s$ decrease in $n$ as well. The gains form a nonnegative $W$ with row sums $F(1 - k_n^s) < 1$ (Remark S2.14), so Proposition 3.8 gives $\mathbb{E}e_i^2 \ge Q/(1 - F^2(1-k_n^s)^2)$, a bound that increases with $n$. The group is closed, its sensing gain is common and its weights are symmetric, so

$W = F(1-k_n^s)M$ with $M$ symmetric and row-stochastic, hence reversible with respect to the uniform law, and Lemma 3.9 and Theorem 3.10 apply with $\alpha_{\mathcal{A}}$ equal to this bound. At $n = 2$, $\hat{P}_2^- = 0.0488$, $k_2^s = 0.0392$ and the bound is 0.116, above the isolated predicted error 0.1000, which is $\mathcal{A}_C^{\text{iso}}$ for every member. Because the bound increases with $n$, $\alpha_{\mathcal{A}} > \mathcal{A}_C^{\text{iso}}$ for every $n \geq 2$, and the group is a weak echo chamber. For the limit, $g_n(\hat{P}_n^-) \geq (n-1)/(R_c + \hat{P}_n^-)$ gives $\hat{P}_n^+ \leq (R_c + \hat{P}_n^-)/(n-1)$, and the fixed-point equation $\hat{P}_n^- = F^2\hat{P}_n^+ + Q$ gives $Q \leq \hat{P}_n^- \leq (Q + F^2 R_c/(n-1))/(1 - F^2/(n-1))$ for every $n \geq 2$. Hence $\hat{P}_n^- \to Q$, $k_n^s \leq (R_c + \hat{P}_n^-)/((n-1)R_i^s) \to 0$, the bound tends to $Q/(1-F^2) = P_0$, and the ratio of actual to claimed predicted variance has limit inferior at least $P_0/Q = 1/(1-F^2)$. □

*Proof of Proposition S3.15.* The update is $m_i^+ = m_i + \mathcal{K}(y - \hat{H}m_i)$, so $e^+ = e + \mathcal{K}\tilde{y}$ with the innovation $\tilde{y} = \tilde{\nu} - \hat{H}e$, and expanding the covariance of $(I - \mathcal{K}\hat{H})e + \mathcal{K}\tilde{\nu}$ gives (S12). (a) $\operatorname{Cov}(e, \tilde{y}) = X^\top - P\hat{H}^\top$ and $\operatorname{Cov}(\tilde{y}) = S^*$; completing the square, $\operatorname{Cov}(e + \mathcal{K}\tilde{y}) = (\mathcal{K} - \mathcal{K}^*)S^*(\mathcal{K} - \mathcal{K}^*)^\top + P - \operatorname{Cov}(e, \tilde{y})(S^*)^{-1}\operatorname{Cov}(\tilde{y}, e)$, which is smallest in the Loewner order at $\mathcal{K}^*$. (b) For $\mathcal{K}_0$, $(I - \mathcal{K}_0\hat{H})P\hat{H}^\top = \mathcal{K}_0(\hat{H}P\hat{H}^\top + R^{\text{a}}) - \mathcal{K}_0\hat{H}P\hat{H}^\top = \mathcal{K}_0 R^{\text{a}}$, so the first two terms of (S12) sum to $(I - \mathcal{K}_0\hat{H})P - \mathcal{K}_0 R^{\text{a}}\mathcal{K}_0^\top + \mathcal{K}_0 R^{\text{a}}\mathcal{K}_0^\top = (I - \mathcal{K}_0\hat{H})P$, and the remaining two terms are the excess. For scalars it equals $2P\hat{H}R^{\text{a}}X/(\hat{H}^2P + R^{\text{a}})^2$, which has the sign of $\hat{H}X$, and $\operatorname{Cov}(\tilde{\nu}/\hat{H}, e) = X/\hat{H}$ has the same sign. (c) The echo is a function of $m_i$ and of noise independent of $(\eta, m_i)$, so $I(\eta; y \mid m_i) = 0$, while level 0 computes the information of an observation of $\eta$ with code $\hat{H} \neq 0$ and noise $\hat{R}$ under the prior covariance $P$. □

## S28 Proofs for Section 4

The proofs follow the order of Section 4. Besides the lemmas below we use standard facts about symmetric and nonnegative matrices: the Courant–Fischer characterization of eigenvalues, the monotonicity of eigenvalues under addition of a positive semidefinite matrix, Cauchy interlacing, Gershgorin's theorem, the Perron–Frobenius theorem (Horn and Johnson, 2012), and the Schur test, which bounds the operator norm of a block matrix with block norms $b_{ij}$ by $(\max_i \sum_j b_{ij} \cdot \max_j \sum_i b_{ij})^{1/2}$.

**Lemma S28.1** (Gaussian conditional dependence)**.** *Let $(U, Y, Z)$ be jointly Gaussian with covariance $\Sigma$ and precision $J = \Sigma^{-1}$, where $U$ is finite dimensional and $Y$ and $Z$ are finite or countable families, $\Sigma$ being a bounded operator on $\ell^2$ with bounded inverse.*

*(a) $U \perp Z \mid Y$ if and only if $J_{UZ} = 0$.*

*(b) With $M = J_{UU}^{-1/2} J_{UZ} J_{ZZ}^{-1/2}$, $\|M\| < 1$, $I(U; Z \mid Y) = -\frac{1}{2}\ln\det(I - MM^\top)$, $\frac{1}{2}\|M\|_F^2 \leq I(U; Z \mid Y) \leq \|M\|_F^2/(2(1 - \|M\|^2))$ and $\|M\|_F^2 \leq \lambda_{\max}(\Sigma)^2\|J_{UZ}\|_F^2$.*

*Proof.* Conditional covariances of a Gaussian family are covariances of orthogonal projections in $L^2$, and for a bounded covariance with bounded inverse they are given by the block formulas of finite dimensions, with operator blocks of $J$ (Lauritzen, 1996). Given $(Y, Z)$, $U$ has covariance $J_{UU}^{-1}$; given $Y$ alone, $(U, Z)$ has precision $\begin{pmatrix} J_{UU} & J_{UZ} \\ J_{ZU} & J_{ZZ} \end{pmatrix}$, so $U$ has covariance $(J_{UU} - J_{UZ}J_{ZZ}^{-1}J_{ZU})^{-1}$. These covariances do not depend on the conditioning values, and they agree exactly when the conditional mean of $U$ given $(Y, Z)$ does not depend on $Z$, that is when $U \perp Z \mid Y$; they agree exactly when $J_{UZ}J_{ZZ}^{-1}J_{ZU} = 0$, that is when $J_{UZ} = 0$, since $J_{ZZ} \succ 0$. This is (a). For (b), with all determinants those of $\dim U \times \dim U$ matrices,

$$I(U; Z \mid Y) = h(U \mid Y) - h(U \mid Y, Z) = \frac{1}{2}\ln\frac{\det J_{UU}}{\det(J_{UU} - J_{UZ}J_{ZZ}^{-1}J_{ZU})} = -\frac{1}{2}\ln\det(I - MM^\top),$$

because $J_{UU} - J_{UZ}J_{ZZ}^{-1}J_{ZU} = J_{UU}^{1/2}(I - MM^\top)J_{UU}^{1/2}$. This matrix is positive definite, so $\|M\| < 1$. The two bounds follow from $x \le -\ln(1-x) \le x/(1-x)$ on $[0,1)$, applied to the eigenvalues of $MM^\top$, each at most $\|M\|^2$. Principal blocks satisfy $\lambda_{\min}(J_{UU}), \lambda_{\min}(J_{ZZ}) \ge \lambda_{\min}(J) = \lambda_{\max}(\Sigma)^{-1}$, which gives the last inequality. □

*Proof of Theorem 4.3.* (a) The path density is $p_{X_0}(x_0)\prod_{t<T}\mathcal{N}(x_{t+1}; Ax_t, Q^\Xi)$, so the path is Gaussian with a precision $J$ that is block tridiagonal in time: $J_{00} = \Sigma_0^{-1} + A^\top(Q^\Xi)^{-1}A$, $J_{tt} = (Q^\Xi)^{-1} + A^\top(Q^\Xi)^{-1}A$ for $1 \le t \le T-1$, $J_{TT} = (Q^\Xi)^{-1}$ and $J_{t+1,t} = -(Q^\Xi)^{-1}A = J_{t,t+1}^\top$. By Lemma S28.1(a) the path blanket holds exactly when every block of $J$ between an interior coordinate at some time and an exterior coordinate at some time vanishes. The time-diagonal blocks give $(\Sigma_0^{-1} + A^\top(Q^\Xi)^{-1}A)_{\mathsf{IE}} = 0$ at $t=0$, $((Q^\Xi)^{-1} + A^\top(Q^\Xi)^{-1}A)_{\mathsf{IE}} = 0$ at $1 \le t \le T-1$, a nonempty range because $T \ge 2$, and $((Q^\Xi)^{-1})_{\mathsf{IE}} = 0$ at $t = T$. The lag-one blocks pair the interior at $t+1$ with the exterior at $t$ and the exterior at $t+1$ with the interior at $t$, giving $((Q^\Xi)^{-1}A)_{\mathsf{IE}} = 0$ and $((Q^\Xi)^{-1}A)_{\mathsf{EI}} = 0$; blocks at larger lags vanish. The three time-diagonal conditions are equivalent to the first, fourth and fifth conditions of (24), and conversely (24) makes every listed block vanish. If $Q^\Xi$ is block diagonal for $(\mathsf{I}, \mathsf{B}, \mathsf{E})$, then $((Q^\Xi)^{-1}A)_{\mathsf{IE}} = (Q^\Xi_{\mathsf{II}})^{-1}A_{\mathsf{IE}}$ and $((Q^\Xi)^{-1}A)_{\mathsf{EI}} = (Q^\Xi_{\mathsf{EE}})^{-1}A_{\mathsf{EI}}$ vanish exactly when $A_{\mathsf{IE}} = 0$ and $A_{\mathsf{EI}} = 0$, and then

$$(A^\top(Q^\Xi)^{-1}A)_{\mathsf{IE}} = A_{\mathsf{II}}^\top(Q^\Xi_{\mathsf{II}})^{-1}A_{\mathsf{IE}} + A_{\mathsf{BI}}^\top(Q^\Xi_{\mathsf{BB}})^{-1}A_{\mathsf{BE}} + A_{\mathsf{EI}}^\top(Q^\Xi_{\mathsf{EE}})^{-1}A_{\mathsf{EE}} = A_{\mathsf{BI}}^\top(Q^\Xi_{\mathsf{BB}})^{-1}A_{\mathsf{BE}},$$

while $(\Sigma_0^{-1})_{\mathsf{IE}} = 0$ is the initial conditional independence by Lemma S28.1(a). Sufficiency needs no Gaussian start. Let $\mu_0$ be any initial law with $X_{\mathsf{I},0} \perp X_{\mathsf{E},0} \mid X_{\mathsf{B},0}$, and let $R$ be the law of a path whose initial slice has law $\mu_0$ and whose later slices are independent of it, with independent standard normal coordinates. Under $R$ the interior and exterior paths are conditionally independent given the blanket path. The path law has density $\prod_{t<T}\mathcal{N}(x_{t+1}; Ax_t, Q^\Xi)/\prod_{1\le t\le T}\mathcal{N}(x_t; 0, I)$ with respect to $R$, and under the first four conditions of (24) the exponent of the numerator has no term coupling an interior coordinate with an exterior one, so this density is a function of $(x_{\mathsf{I},0:T}, x_{\mathsf{B},0:T})$ times a function of $(x_{\mathsf{E},0:T}, x_{\mathsf{B},0:T})$. A density of this product form preserves the conditional independence, so the path blanket holds.

(b) By Definition 4.1 no internal variable reads an exterior one and no exterior variable reads an internal one, so $A_{\mathsf{IE}} = 0$ and $A_{\mathsf{EI}} = 0$; active variables read no exterior variable, so $A_{k\mathsf{E}} = 0$ for $k \in \mathsf{A}$. With per-channel noise $Q^\Xi_{\mathsf{BB}}$ is block diagonal over channels, and the collider term reduces to $\sum_{k\in\mathsf{S}} A_{k\mathsf{I}}^\top(Q^\Xi_{kk})^{-1}A_{k\mathsf{E}}$. A term is nonzero only if channel $k$ reads both an interior and an exterior variable. If a channel $k$ reads an interior variable $u$ and an exterior variable $u'$, the $(u,u')$ block of the sum contains $A_{ku}^\top(Q^\Xi_{kk})^{-1}A_{ku'}$, a polynomial in entries of $A$ that occur in no other term and that is not identically zero, because $(Q^\Xi_{kk})^{-1} \succ 0$. The sum is therefore a polynomial in the coefficients that is not identically zero, and its zero set has positive codimension. With separate channels every sensory channel reads one source, which lies on one side, and the sum vanishes.

(c) Under the first three conditions of (25), the sum $\sum_{t<T}\ln\mathcal{N}(x_{t+1}; Ax_t, Q^\Xi)$ has no term that couples an interior coordinate with an exterior one, so it equals $f(x_{\mathsf{I},0:T}, x_{\mathsf{B},0:T}) + g(x_{\mathsf{E},0:T}, x_{\mathsf{B},0:T})$. Fix a blanket path $x_\mathsf{B}$ and write $Y_\mathsf{I} = X_{\mathsf{I},0:T}$, $Y_\mathsf{E} = X_{\mathsf{E},0:T}$. Their conditional density is proportional to $p_{X_0}(y_\mathsf{I}(0), x_\mathsf{B}(0), y_\mathsf{E}(0))e^f e^g$, a function of $(y_\mathsf{I}(0), y_\mathsf{E}(0))$ times a function of $y_\mathsf{I}$ times a function of $y_\mathsf{E}$. Hence, given $x_\mathsf{B}$, $Y_\mathsf{I}(1{:}T) \perp Y_\mathsf{E} \mid Y_\mathsf{I}(0)$ and $Y_\mathsf{E}(1{:}T) \perp Y_\mathsf{I} \mid Y_\mathsf{E}(0)$, and the chain rule gives

$$\begin{aligned} I(Y_\mathsf{I}; Y_\mathsf{E} \mid x_\mathsf{B}) &= I(Y_\mathsf{I}(0); Y_\mathsf{E} \mid x_\mathsf{B}) = I(Y_\mathsf{I}(0); Y_\mathsf{E}(0) \mid x_\mathsf{B}) + I(Y_\mathsf{I}(0); Y_\mathsf{E}(1{:}T) \mid Y_\mathsf{E}(0), x_\mathsf{B}) \\ &= I(Y_\mathsf{I}(0); Y_\mathsf{E}(0) \mid x_\mathsf{B}). \end{aligned}$$

Averaging over $x_\mathsf{B}$ gives the initial-slice formula. For the bound, $I(U; Z \mid Y) \le I(U; Y, Z)$ and the chain rule give

$$\iota_T \le I(X_{\mathsf{I},0}; X_{\mathsf{B},0:T}, X_{\mathsf{E},0}) = I(X_{\mathsf{I},0}; X_{\mathsf{B},0}, X_{\mathsf{E},0}) + I(X_{\mathsf{I},0}; X_{\mathsf{B},1:T} \mid X_{\mathsf{B},0}, X_{\mathsf{E},0}).$$

Write $X_{\mathsf{B},1:T} = \mathcal{L}X_0 + \tilde{\Xi}_{\mathsf{B}}$, where $\mathcal{L}$ stacks the blanket rows of $A^t$, $t = 1, \dots, T$, and $\tilde{\Xi}_{\mathsf{B}} = \mathcal{J}_{\mathsf{B}}\Xi_{0:T-1}$ is the response of the blanket to the noise, independent of $X_0$. Given $(X_{\mathsf{B},0}, X_{\mathsf{E},0})$, $X_{\mathsf{I},0}$ is Gaussian with covariance at most $\|\Sigma_0\|I$, and the part of $\mathcal{L}X_0$ not driven by $X_{\mathsf{I},0}$ is known, so the Gaussian channel formula and $\ln\det(I+X) \le \operatorname{tr} X$ give

$$I(X_{\mathsf{I},0}; X_{\mathsf{B},1:T} \mid X_{\mathsf{B},0}, X_{\mathsf{E},0}) \le \tfrac{1}{2}\|\Sigma_0\| \|\operatorname{Cov}(\tilde{\Xi}_{\mathsf{B}})^{-1}\| \|\mathcal{L}\|_F^2.$$

The map $\mathcal{J}$ from $\Xi_{0:T-1}$ to $X_{1:T}$ at zero initial state is block lower triangular with identity diagonal blocks, and its inverse is block bidiagonal with blocks $I$ and $-A$, so its smallest singular value is at least $(1+\|A\|)^{-1}$. Since $\mathcal{J}_{\mathsf{B}}\mathcal{J}_{\mathsf{B}}^\top$ is a principal submatrix of $\mathcal{J}\mathcal{J}^\top$, Cauchy interlacing gives $\lambda_{\min}(\mathcal{J}_{\mathsf{B}}\mathcal{J}_{\mathsf{B}}^\top) \ge (1+\|A\|)^{-2}$, and $\operatorname{Cov}(\tilde{\Xi}_{\mathsf{B}}) = \mathcal{J}_{\mathsf{B}}(I \otimes Q^{\Xi})\mathcal{J}_{\mathsf{B}}^\top \succeq \lambda_{\min}(Q^{\Xi})\mathcal{J}_{\mathsf{B}}\mathcal{J}_{\mathsf{B}}^\top$. Finally $\|\mathcal{L}\|_F^2 \le \sum_{t\ge 1}\|A^t\|_F^2$, finite when $\rho(A) < 1$, which gives (26). □

*Proof of Example S4.2.* Here $A_{\mathsf{BI}}^\top (Q^{\Xi}_{\mathsf{BB}})^{-1} A_{\mathsf{BE}} = \alpha_1\alpha_2 \neq 0$. For $t \ge 1$ the triple formed by $X_{i,t}$, $X_{e,t}$ and $X_{b,t+1}$ is a function of the noises $\Xi^i_{t-1}$, $\Xi^e_{t-1}$ and $\Xi^b_t$, so for every initial law these triples, for $1 \le t \le T-1$, are independent of one another and of the variables of rounds 0, 1 and $T$ that they do not contain. The window leak is therefore the sum of a term that involves only round 0, the initial state and $X_{b,1}$, and of $T-1$ equal one-round terms $I(X_{i,t}; X_{e,t} \mid X_{b,t+1})$; on every window with $T \ge 2$ it is positive, and the leak rate is the one-round term whenever the round-0 term is finite. In one round, $X_{i,t}$ and $X_{e,t}$ are independent standard normal and $X_{b,t+1} = \alpha_1 X_{i,t} + \alpha_2 X_{e,t} + \Xi^b_t$, so their conditional covariance given $X_{b,t+1}$ is $I - vv^\top/(1+\alpha_1^2+\alpha_2^2)$ with $v = (\alpha_1, \alpha_2)$. The squared partial correlation is $\alpha_1^2\alpha_2^2/((1+\alpha_1^2)(1+\alpha_2^2))$, and since $(1+\alpha_1^2)(1+\alpha_2^2) - \alpha_1^2\alpha_2^2 = 1+\alpha_1^2+\alpha_2^2$, the conditional mutual information $-\frac{1}{2}\ln(1 - \alpha_1^2\alpha_2^2/((1+\alpha_1^2)(1+\alpha_2^2)))$ equals (S13). □

*Proof of Proposition S4.3.* If $d(\mathsf{I},\mathsf{E}) \ge 2$, one-hop coupling gives $A_{\mathsf{IE}} = 0$ and $A_{\mathsf{EI}} = 0$. A nonzero term $A_{k\mathsf{I}}^\top (Q^{\Xi}_{kk})^{-1} A_{k\mathsf{E}}$ requires an agent $k$ within one hop of some $i \in \mathsf{I}$ and of some $e \in \mathsf{E}$, hence $d(i,e) \le 2$. When $d(\mathsf{I},\mathsf{E}) \ge 3$ every term vanishes, and Theorem 4.3(a) gives exactness; the stationary start is Gaussian with covariance $\sum_n A^n Q^{\Xi}(A^\top)^n \succeq Q^{\Xi} \succ 0$, so Theorem 4.3(c) gives a zero leak rate. When $d(\mathsf{I},\mathsf{E}) = 2$ the collider term is the stated sum, and a shell agent that reads $i \in \mathsf{I}$ and $e \in \mathsf{E}$ contributes to the $(i,e)$ block a term in entries of $A$ that no other agent's term contains, so the argument of Theorem 4.3(b) applies. For a schedule $A_t$ the path precision has the blocks $\Sigma_0^{-1} + A_0^\top (Q^{\Xi})^{-1} A_0$, $(Q^{\Xi})^{-1} + A_t^\top (Q^{\Xi})^{-1} A_t$ and $-(Q^{\Xi})^{-1} A_t$ with the same sparsity, and the proof applies round by round. The shell $\{k : 1 \le d(k,\mathsf{I}) \le 2\}$ makes $d(\mathsf{I},\mathsf{E}) = 3$, and every subset $\mathsf{I}$ of the agents gives one such partition. □

*Proof of Proposition 4.4.* (a) Per-channel noise is independent across the partition, so by Theorem 4.3(a) we check $A_{\mathsf{IE}} = 0$, $A_{\mathsf{EI}} = 0$ and the collider term. An internal record reads only its member writers and is read only by member channels, so it reads and drives no exterior variable. A boundary record reads only its writers, which are collective, and lies in $\mathsf{A}$, which may drive exterior variables. A member channel that writes only internal or boundary records and reads nothing outside drives and reads only collective variables, so it lies in $\mathsf{I}$ and has no exterior neighbor. An exterior record is written by member channels, which lie in $\mathsf{A}$ or $\mathsf{S}$, and read by member channels that read nothing else, which lie in $\mathsf{S}$ and read one variable. Hence $A_{\mathsf{IE}} = A_{\mathsf{EI}} = 0$, record-reading channels contribute no collider term, and the collider condition is that of the remaining sensory channels, which is the condition of Theorem 4.3(b). The persistence of a record enters only the record's own row of $A$, on which no condition bears, and records are state variables of the one-step Markov process (23), so lags through them need no further condition.

(b) A record read by an outsider and placed in $\mathsf{I}$ makes an entry of $A_{\mathsf{EI}}$ nonzero. A record with writers on both sides placed in $\mathsf{B}$ makes its member writers, which then read nothing outside and

drive only collective variables, internal, while its other writers are exterior; the record reads an interior and an exterior variable in one round, and its collider term $A_{r\mathsf{I}}^\top (Q^\Xi_{rr})^{-1} A_{r\mathsf{E}}$ is nonzero for generic coefficients. An exterior record whose writers are all non-members, placed in $\mathsf{S}$, reads only exterior variables and itself, contributes no collider term, and leaves the conditions of Theorem 4.3(a) unchanged. □

*Proof of Proposition S4.4.* (a) The path blanket on $[0,t]$ marginalizes to $X_{\mathsf{I},t} \perp X_{\mathsf{E},t} \mid X_{\mathsf{B},0:t}$. By the chain rule,

$$I(X_{\mathsf{I},t}; X_{\mathsf{E},t} \mid X_{\mathsf{B},t}) \le I(X_{\mathsf{I},t}; X_{\mathsf{E},t}, X_{\mathsf{B},0:t-1} \mid X_{\mathsf{B},t}) = I(X_{\mathsf{I},t}; X_{\mathsf{B},0:t-1} \mid X_{\mathsf{B},t}) + I(X_{\mathsf{I},t}; X_{\mathsf{E},t} \mid X_{\mathsf{B},0:t}),$$

and the last term vanishes; exchanging interior and exterior gives the other bound. (b) By (a) and Gaussianity, the conditional density of $(X_{\mathsf{I},t}, X_{\mathsf{E},t})$ given the history $H = X_{\mathsf{B},0:t}$ is $p(x_{\mathsf{I}} \mid H)\mathcal{N}(x_{\mathsf{E}}; \hat{X}_{\mathsf{E},t}(H), C_{\mathsf{E}})$ with a covariance $C_{\mathsf{E}}$ that does not depend on $H$. Conditioning on $(X_{\mathsf{B},t}, \hat{X}_{\mathsf{E},t})$, a function of $H$, instead of on $H$ averages the first factor over $H$ and leaves the second unchanged, because it depends on $H$ only through $\hat{X}_{\mathsf{E},t}$; the result is a function of $x_{\mathsf{I}}$ times a function of $x_{\mathsf{E}}$. (c) From $AQ^\Xi = Q^\Xi A^\top$, induction gives $A^k Q^\Xi (A^\top)^k = A^{2k} Q^\Xi$, so $\Sigma = (I - A^2)^{-1} Q^\Xi$ and $A\Sigma = (I - A^2)^{-1} A Q^\Xi = \Sigma A^\top$, which is time reversibility of the stationary chain. Multiplying $AQ^\Xi = Q^\Xi A^\top$ by $(Q^\Xi)^{-1}$ on both sides gives $(Q^\Xi)^{-1} A = A^\top (Q^\Xi)^{-1}$, hence $(Q^\Xi)^{-1} A^2 = A^\top (Q^\Xi)^{-1} A$ and $\Pi = (Q^\Xi)^{-1}(I - A^2) = (Q^\Xi)^{-1} - A^\top (Q^\Xi)^{-1} A$. Under the block conditions the expansion in the proof of Theorem 4.3(a) gives $\Pi_{\mathsf{IE}} = -A^\top_{\mathsf{BI}} (Q^\Xi_{\mathsf{BB}})^{-1} A_{\mathsf{BE}}$, and Lemma S28.1(a) finishes the proof. □

*Proof of Theorem S4.5.* (a) Fix a vertex $j_0$ and $R > 0$, let $f(i) = \min\{d(i, j_0), R\}$, which is 1-Lipschitz and bounded, and let $\mathcal{T} = \mathrm{diag}(e^{m_\Pi f(i)} I)$ act blockwise; $\mathcal{T}$ and $\mathcal{T}^{-1}$ are bounded. The matrix $\mathcal{T}\Sigma\mathcal{T}^{-1} - \Sigma$ has blocks $\Sigma_{ij}(e^{m_\Pi (f(i) - f(j))} - 1)$, of norm at most $B_\Sigma e^{-m_\Sigma d(i,j)}(e^{m_\Pi d(i,j)} - 1)$ because $|e^x - 1| \le e^{|x|} - 1$ and $|f(i) - f(j)| \le d(i,j)$. Its block row and column sums are at most $\mathcal{C}(m_\Pi)$, so the Schur test bounds its norm by $\mathcal{C}(m_\Pi)$. Then $\mathcal{T}\Sigma\mathcal{T}^{-1} = \Sigma(I + \Sigma^{-1}(\mathcal{T}\Sigma\mathcal{T}^{-1} - \Sigma))$ with $\|\Sigma^{-1}(\mathcal{T}\Sigma\mathcal{T}^{-1} - \Sigma)\| \le \mathcal{C}(m_\Pi)/\lambda_{\min}(\Sigma) < 1$, so $\|(\mathcal{T}\Sigma\mathcal{T}^{-1})^{-1}\| \le (\lambda_{\min}(\Sigma) - \mathcal{C}(m_\Pi))^{-1} = B_\Pi$. The $(i, j_0)$ block of $(\mathcal{T}\Sigma\mathcal{T}^{-1})^{-1} = \mathcal{T}\Pi\mathcal{T}^{-1}$ is $e^{m_\Pi f(i)}\Pi_{ij_0}$, of norm at most $B_\Pi$, and letting $R \to \infty$ gives the claim. (b) Since $\mathsf{I}$ is finite, $e_{\mathsf{I}}$ is finite dimensional, and Lemma S28.1(b) applies with $U = e_{\mathsf{I}}$, $Y = e_{\mathsf{B}}$, $Z = e_{\mathsf{E}}$: $\|M\|_F^2 \le \lambda_{\max}(\Sigma)^2 \|\Pi_{\mathsf{IE}}\|_F^2 \le \lambda_{\max}(\Sigma)^2 Z_{\mathsf{IE}}$, since a block of size at most $K$ has $\|\Pi_{ik}\|_F^2 \le K\|\Pi_{ik}\|^2$. With $\sigma_l$ the singular values of $M$, $\sigma_l^2 \le \|M\|_F^2 < 1$, so $-\ln(1-\sigma_l^2) \le \sigma_l^2/(1 - \|M\|_F^2)$, and summing gives (S14). (c) For $i \in \mathsf{I}$, $\sum_{k \in \mathsf{E}} e^{-2m_\Pi d(i,k)} \le \Psi(d(i, \mathsf{E}))$ with $\Psi(t) = \sum_{r \ge t} \mathcal{S}(r) e^{-2m_\Pi r}$, and

$$\begin{aligned}\sum_{i \in \mathsf{I}} \Psi(d(i, \mathsf{E})) &\le n_\partial \sum_{t \ge h} \Psi(t) = n_\partial \sum_{r \ge h} (r - h + 1)\mathcal{S}(r) e^{-2m_\Pi r} \\ &\le n_\partial B_{\mathcal{G}} (1 + h)^{D-1} e^{-2m_\Pi h} \sum_{j \ge 0} (1 + j)^D e^{-2m_\Pi j},\end{aligned}$$

writing $r = h + j$ and using $h + j \le (1 + h)(1 + j)$. Take $B_* = B_{\mathcal{G}} \sum_{j \ge 0} (1 + j)^D e^{-2m_\Pi j}$. □

*Proof of Theorem S4.6.* (a) Fix a vertex $i$ and a width $h$ and apply Lemma S28.1(b) with $U = e_i$, $Y = e_{\mathsf{B}_h(i)}$ and $Z = e_{\mathsf{E}_h(i)}$. Then

$$\|M\|_F^2 \le \lambda_{\max}(\Sigma)^2 K \sum_{j \in \mathsf{E}_h(i)} \|\Pi_{ij}\|^2 \le \lambda_{\max}(\Sigma)^2 K B_\Pi^2 B_{\mathcal{G}} \sum_{r > h} r^{D-1} e^{-2m_\Pi r},$$

uniformly in $i$. For every $\xi' > 1/(2m_\Pi)$ the right side is at most a constant times $e^{-h/\xi'}$, and once it is at most $\frac{1}{4}$, $\iota(i,h) \le \frac{2}{3}\|M\|_F^2$. For the finitely many smaller $h$ we use a bound uniform in $i$:

$J_{UU} - J_{UZ}J_{ZZ}^{-1}J_{ZU}$ is the inverse of a conditional covariance, hence at least $\lambda_{\max}(\Sigma)^{-1}I$, while $J_{UU} \preceq \lambda_{\min}(\Sigma)^{-1}I$, so $I - MM^\top = J_{UU}^{-1/2}(J_{UU} - J_{UZ}J_{ZZ}^{-1}J_{ZU})J_{UU}^{-1/2} \succeq (\lambda_{\min}(\Sigma)/\lambda_{\max}(\Sigma))I$ and $\iota(i,h) \le \frac{K}{2}\ln(\lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma))$. Hence $\sup_i \iota(i,h) \le B'e^{-h/\xi'}$ for a constant $B'$ and all $h$, and $\xi_M \le 1/(2m_\Pi)$.

(b) Let $0 < m_\Pi \le m_\Sigma/(D+1)$ and $m' = m_\Sigma - m_\Pi \ge Dm_\Sigma/(D+1)$. Since $e^x - 1 \le xe^x$ for $x \ge 0$, $\mathcal{C}(m_\Pi) \le B_\Sigma B_\mathcal{G} m_\Pi \sum_{r\ge1} r^D e^{-m'r}$. For $x \in [r, r+1]$, $r^D e^{-m'r} \le e^{m'}x^D e^{-m'x}$, so $\sum_{r\ge1} r^D e^{-m'r} \le e^{m'}\int_0^\infty x^D e^{-m'x}\mathrm{d}x = e^{m'}D!m'^{-(D+1)}$. With $e^{m'} \le e^{m_\Sigma}$ and $m'^{-(D+1)} \le (1+1/D)^{D+1}m_\Sigma^{-(D+1)} \le 4m_\Sigma^{-(D+1)}$, because $(1+1/D)^{D+1} \le 4$ for $D \ge 1$,

$$\mathcal{C}(m_\Pi) \le 4B_\Sigma B_\mathcal{G} D! e^{m_\Sigma} m_\Pi m_\Sigma^{-(D+1)}.$$

The choice $m_\Pi = m_*$ of (S15) satisfies both constraints and makes $\mathcal{C}(m_*) \le \lambda_{\min}(\Sigma)/2$, so Theorem S4.5(a) gives $B_\Pi \le 2/\lambda_{\min}(\Sigma)$, and (a) gives (S16).

(c) The hypotheses are those of Theorem 8.9(b) and of its last clause: a time-invariant $W$ satisfies (Lin), the law of $\zeta(t)$ is time invariant by hypothesis, so that its covariance $Q^\zeta$ does not depend on $t$, and (B) gives block-diagonal local noise with blocks between $q_{\min}I$ and $qI$. We repeat the argument for the stationary covariance so that the constants are explicit. The covariance is $\Sigma = \sum_{n\ge0} W^n Q^\zeta (W^\top)^n$ by (27). By (Loc), $(W^n)_{ik} = 0$ when $d(i,k) > n$, and by ($C_r$) and induction $\sum_k \|(W^n)_{ik}\| \le \lambda^n$. Block $(i,j)$ of the $n$th term is $\sum_k (W^n)_{ik} Q^\zeta_{kk} (W^n)^\top_{jk}$, which vanishes unless $2n \ge d(i,j)$ and otherwise has norm at most $q\max_k \|(W^n)_{jk}\| \sum_k \|(W^n)_{ik}\| \le q\lambda^{2n}$. Summing over $n \ge \lceil d(i,j)/2\rceil$ gives $\|\Sigma_{ij}\| \le q\lambda^{2\lceil d(i,j)/2\rceil}/(1-\lambda^2) \le q\lambda^{d(i,j)}/(1-\lambda^2)$, which is (80), so $B_\Sigma = q/(1-\lambda^2)$, $m_\Sigma = \ln(1/\lambda)$ and $e^{m_\Sigma} = 1/\lambda$. Every term of the series is positive semidefinite and the term $n = 0$ is $Q^\zeta$, so $\Sigma \succeq Q^\zeta \succeq q_{\min}I$, the last clause of Theorem 8.9, and the Schur test gives $\lambda_{\max}(\Sigma) \le B_\Sigma \sum_{r\ge0} \mathcal{S}(r)\lambda^r < \infty$, with $\mathcal{S}(0) = 1$. Replacing $\lambda_{\min}(\Sigma)$ by its lower bound $q_{\min}$ in (S15) lowers $m_*$ and keeps $\mathcal{C}(m_*) \le q_{\min}/2 \le \lambda_{\min}(\Sigma)/2$, which gives the stated bound. □

*Remark* S28.2 (A dynamical route to Conjecture S4.8). From $\Sigma = W\Sigma W^\top + Q^\zeta$ and the Woodbury identity, $\Pi = (Q^\zeta)^{-1} - (Q^\zeta)^{-1}W\Sigma_{\leftarrow}W^\top(Q^\zeta)^{-1}$ with $\Sigma_{\leftarrow} = (\Pi + W^\top(Q^\zeta)^{-1}W)^{-1} = \mathrm{Cov}(e(t-1) \mid e(t))$: the long-range part of the precision is the conditional covariance of the previous state given the present one, pushed through one hop on each side. Since $0 \preceq \Pi \preceq (Q^\zeta)^{-1}$, the inverse of $\Sigma_{\leftarrow}$ lies between $\sigma_{\min}(W)^2/q$ and $(1+\|W\|^2)/q_{\min}$, a condition number free of $\lambda_{\max}(\Sigma)$, which is what diverges in the massless limit; for normal $W$ with $Q^\zeta = qI$, $\Sigma_{\leftarrow} = qI$ and the range-two precision of Theorem 4.9(b) follows. A proof of Conjecture S4.8 along this route needs decay of $\Sigma_{\leftarrow}$ from decay of $\Pi$ without the loss of a Combes–Thomas step.

*Proof of Proposition 4.6.* (a) In the stationary state $e_k^{\mathrm{src}}(t) = \zeta_k^{\mathrm{src}}(t-1)$ and $e_k^{\mathrm{lis}}(t) = a[\zeta_{k-1}^{\mathrm{src}}(t-2) + \zeta_k^{\mathrm{src}}(t-2)] + \zeta_k^{\mathrm{lis}}(t-1)$. At equal times sources are independent of one another and of every listener; listeners $k$ and $k+1$, two hops apart, share $\zeta_k^{\mathrm{src}}(t-2)$, so $\mathrm{Cov}(e_k^{\mathrm{lis}}, e_{k+1}^{\mathrm{lis}}) = a^2q_{\mathrm{src}}$ and $\mathrm{Var}(e_k^{\mathrm{lis}}) = 2a^2q_{\mathrm{src}} + q_{\mathrm{lis}}$; all other covariances vanish. The precision is block diagonal over sources and listeners, with $q_{\mathrm{src}}^{-1}I$ on sources. The listener covariance is the Laurent operator with diagonal $\sigma_0^2 = 2a^2q_{\mathrm{src}} + q_{\mathrm{lis}}$ and off-diagonal $\sigma_1^2 = a^2q_{\mathrm{src}}$, whose symbol $p \mapsto \sigma_0^2 + 2\sigma_1^2\cos p$ has range $[q_{\mathrm{lis}}, 4a^2q_{\mathrm{src}} + q_{\mathrm{lis}}]$. Its inverse $\Pi^{\mathrm{lis}}$ has entries

$$\Pi^{\mathrm{lis}}_{k,k+n} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{e^{inp}}{\sigma_0^2 + 2\sigma_1^2\cos p}\mathrm{d}p = \frac{(-1)^n e^{-m_\psi|n|}}{\sqrt{\sigma_0^4 - 4\sigma_1^4}}, \qquad 2\cosh m_\psi = \frac{\sigma_0^2}{\sigma_1^2} = 2 + \psi,$$

by the residue at the root $-e^{-m_\psi} \in (-1,0)$ of $\sigma_1^2x^2 + \sigma_0^2x + \sigma_1^2$, so $m_\psi = \mathrm{arccosh}(1+\psi/2)$ and $\sqrt{\sigma_0^4 - 4\sigma_1^4} = a^2q_{\mathrm{src}}\sqrt{\psi(4+\psi)}$. Sources drop out of every conditional mutual information,

and a source vertex has $\iota = 0$. For listener 0 and width $h$ the exterior listeners are those with $|k| \geq n_h = \lfloor h/2 \rfloor + 1$, and Lemma S28.1(b) applies to the listener precision. Every compression of $\Pi^{\mathrm{lis}}$ to a set of listeners has spectrum in $[(4a^2 q_{\mathrm{src}} + q_{\mathrm{lis}})^{-1}, q_{\mathrm{lis}}^{-1}]$, so $\|M\|_F^2 = (\Pi^{\mathrm{lis}}_{00})^{-1}\Pi^{\mathrm{lis}}_{0\mathsf{E}}(\Pi^{\mathrm{lis}}_{\mathsf{EE}})^{-1}\Pi^{\mathrm{lis}}_{\mathsf{E}0}$ lies between constants times $\sum_{|n|\geq n_h} e^{-2m_\psi |n|}$, which is of order $e^{-2m_\psi n_h}$. Since $\|M\| \to 0$, $\iota(0,h)$ is bounded above and below by constants times $e^{-2m_\psi n_h}$, and $h+1 \leq 2n_h \leq h+2$ gives $\xi_M = 1/m_\psi = 1/\operatorname{arccosh}(1+\psi/2)$. The expansion $\operatorname{arccosh}(1+x) = \sqrt{2x}(1+O(x))$ gives the asymptotics. The spectrum of $\Sigma$ is $\{q_{\mathrm{src}}\} \cup [q_{\mathrm{lis}}, 4a^2 q_{\mathrm{src}} + q_{\mathrm{lis}}]$, so $\operatorname{cond}(\Sigma) = q_{\mathrm{src}}/q_{\mathrm{lis}} = 1/(a^2\psi)$ when $4a^2 q_{\mathrm{src}} + q_{\mathrm{lis}} \leq q_{\mathrm{src}}$, that is when $\psi \leq (1-4a^2)/a^2$. Otherwise the largest eigenvalue is $4a^2 q_{\mathrm{src}} + q_{\mathrm{lis}}$, and $\operatorname{cond}(\Sigma) = 1 + 4/\psi$ as long as $q_{\mathrm{lis}} \leq q_{\mathrm{src}}$.

(b) The covariance of $h+2$ consecutive sites is the tridiagonal Toeplitz matrix $T_{h+2}$ with diagonal $1+\alpha^2$ and off-diagonal $\alpha$, so covariances vanish beyond one site. For a tridiagonal matrix the cofactor formula gives $|(T_n^{-1})_{1n}| = \alpha^{n-1}/\det T_n$ and $(T_n^{-1})_{11} = (T_n^{-1})_{nn} = \det T_{n-1}/\det T_n$, and the recursion $\det T_k = (1+\alpha^2)\det T_{k-1} - \alpha^2 \det T_{k-2}$ gives $\det T_k = \sum_{j=0}^{k} \alpha^{2j}$. The partial correlation of the end sites is therefore $\alpha^{n-1}/\det T_{n-1}$, which with $n = h+2$ is the stated value. The conditional mutual information of the end sites is $-\frac{1}{2}\ln(1-\mathrm{pc}_h^2) \geq \frac{1}{2}\mathrm{pc}_h^2$, with $\mathrm{pc}_h$ this partial correlation, and the half-line quantity is at least as large by the chain rule. For $\alpha < 1$, $\mathrm{pc}_h^2 \geq (1-\alpha^2)^2\alpha^{2h+2}$, so no bound $B'e^{-h/\xi'}$ holds with $\xi' < 1/(2\ln(1/\alpha))$; at $\alpha = 1$ the quantity is at least $\frac{1}{2}(h+2)^{-2}$, which decays slower than every exponential.

(c) The update matrix is symmetric, hence normal, and $Q^\zeta = qI$, so Theorem 4.9(c) gives a precision of range two; $\Pi^c_{i,\mathsf{E}_h(i)} = 0$ for $h \geq 2$, and Lemma S28.1(a) gives $\iota(i,h) = 0$, so $\xi_M = 0$. The divergence of the Ornstein–Zernike length as $\varepsilon \to 0$ is Theorem 8.10(c). □

*Proof of Proposition S4.10.* Part (a) is Theorem 4.3(b). In (b), path blankets are Proposition S4.3(a). The stationary precision is $(Q^\Xi)^{-1} - A^\top (Q^\Xi)^{-1} A$ for reversible dynamics (Proposition S4.4(c)) and $(I - AA^\top)/q$ for normal dynamics with $Q^\Xi = qI$ (as in Theorem 4.9(b), with $A$ for $W$ and $Q^\Xi$ for $Q^\zeta$); by the argument of Theorem 4.9(c) a block $(i,k)$ of either vanishes unless some agent is within one hop of both, so the interior-exterior block vanishes when $d(\mathsf{I},\mathsf{E}) \geq 3$, and Lemma S28.1(a) gives the stationary blanket. Part (c) is Theorem S4.5(c), and (d) follows from the translation invariance of the dynamics and of the stationary law. Under (a) and (b) every candidate of the stated classes thus has an exact blanket and zero path leak; under (c) the leak bound, and with it the tolerance that a thick enough shell certifies, depends on the region only through $n_\partial$ and $h$; and by (d) neither blankets nor leaks separate translates. In the reversible and normal isotropic cases the instantaneous leak of every region with a two-hop shell vanishes as well, which makes leak minimization degenerate. □

*Proof of Proposition 4.8.* In the level-0 network of Definition 2.12 every channel noise is specific to its receiver, so Definition 4.7 conditions on the world path alone, and the world-conditional law is the conditional law given the world path. By Lemma 2.13(a) and (c) this law is Gaussian, with mean $e^{\mathrm{com}}$ and the law of $e^{\mathrm{loc}}$ around it, and by Lemma 2.13(b) its covariance at one time is $\Sigma^{\mathrm{loc}}$, which is (27) and does not depend on the path. A sender-side noise shared by several receivers adds to row $i$ of (14) the term $F\mathcal{K}_{ij}$ times that noise for every attended sender $j$, a term of the same kind as the world innovation: it is independent of the world and of the local noises. The argument of Lemma 2.13(b) and (c), run with the $\sigma$-algebra generated by the world path and the shared noises in place of that of the world path, puts the response of the network to these terms, a series in powers of $W$ like (15), into the common part, which stays measurable for the enlarged conditioning, and leaves the connected part and its law unchanged. □

*Proof of Theorem 4.9.* (a) Induction on $WQ^\zeta = Q^\zeta W^\top$ gives $W^n Q^\zeta = Q^\zeta (W^\top)^n$ for every $n$, hence

$$W^n Q^\zeta (W^\top)^n = W^{2n} Q^\zeta, \qquad \Sigma^c = \sum_{n\geq 0} W^{2n} Q^\zeta = (I - W^2)^{-1} Q^\zeta.$$

As in the proof of Proposition S4.4(c), $(Q^\zeta)^{-1} W = W^\top (Q^\zeta)^{-1}$ and $\Pi^c = (Q^\zeta)^{-1} - W^\top (Q^\zeta)^{-1} W$, for block matrices as for scalar ones. For scalar errors the off-diagonal entries are $-\sum_k W_{ki} W_{kj}/q_k$ and the sum of row $i$ is $1/q_i - \sum_k W_{ki}\lambda_k/q_k$; the identity $(Q^\zeta)^{-1} W = W^\top (Q^\zeta)^{-1}$ reads $W_{ki}/q_k = W_{ik}/q_i$, which turns the row sum into $(1 - \sum_k W_{ik}\lambda_k)/q_i$. If $W \geq 0$ with row sums at most $\lambda$, the off-diagonal entries are nonpositive and $\sum_k W_{ik}\lambda_k \leq \lambda_i \max_k \lambda_k \leq \lambda^2$. (b) Normality gives $W^n (W^\top)^n = (WW^\top)^n$, and $\|WW^\top\| = \|W\|^2 = \rho(W)^2 < 1$ for normal $W$, so $\Sigma^c = q\sum_n (WW^\top)^n = q(I - WW^\top)^{-1}$. For scalar errors the off-diagonal entries are $-q^{-1}\sum_k W_{ik} W_{jk}$ and the sum of row $i$ is $(1 - (WW^\top \mathbf{1})_i)/q = (1 - \sum_k W_{ik}\lambda_k^{\mathrm{col}})/q$; with $W \geq 0$ diffusivity is nonnegativity of these sums, and $\sum_k W_{ik}\lambda_k^{\mathrm{col}} \leq \lambda_i \max_k \lambda_k^{\mathrm{col}}$. (c) With $Q^\zeta$ block diagonal, block $(i,j)$ of $W^\top (Q^\zeta)^{-1} W$ is $\sum_k W_{ki}^\top (Q^\zeta_{kk})^{-1} W_{kj}$ and block $(i,j)$ of $WW^\top$ is $\sum_k W_{ik} W_{jk}^\top$. A term of the first is nonzero only if $k$ attends both $i$ and $j$, and a term of the second only if $i$ and $j$ both attend $k$, self-weights included; either way $d(i,j) \leq 2$. When $d(\mathsf{I}, \mathsf{E}) \geq 3$, $\Pi^c_{\mathsf{IE}} = 0$, and Lemma S28.1(a) gives the stationary blanket. (d) The sum of row $i$ is $F(1 - k_i^s - \sum_j k_{ij}) + F\sum_j k_{ij} = F(1 - k_i^s)$, and reversibility $W_{ij} q_j = q_i W_{ji}$ is $k_{ij} q_j = q_i k_{ji}$. With a common sensing gain every row sum equals $\lambda$, so $\sum_k W_{ik}\lambda_k = \lambda^2$ in case (a); in case (b), $W\mathbf{1} = \lambda\mathbf{1}$ gives $W^\top \mathbf{1} = \lambda \mathbf{1}$, because a real normal matrix and its transpose have the same eigenvectors with conjugate eigenvalues, so every column sum is $\lambda$ as well. (e) $W$ is symmetric with $Q^\zeta = qI$, so (b) gives $\Pi^c = (I - W^2)/q$. In index form $(W^2)_{ij} = \sum_k W_{ik} W_{kj}$: for neighbors $i, j$ only $k \in \{i, j\}$ contributes, giving $2ca$; for sites two steps apart on one axis only the midpoint contributes, giving $a^2$; for sites displaced by one step in each of two coordinates the two common neighbors contribute, giving $2a^2$; and every row of $W^2$ sums to $\lambda^2$, which gives the leak. □

*Proof of Example S4.12(a).* Agents 1 and 2 are driven by independent noises, so $\Sigma^c_{12} = 0$. Stationarity gives $e_3(t) = \sum_{n\geq 0} W_{33}^n (W_{31} e_1(t-1-n) + W_{32} e_2(t-1-n) + \zeta_3(t-1-n))$, and since $\mathrm{Cov}(e_1(t), e_1(t-1-n)) = W_{11}^{n+1} \mathrm{Var}(e_1)$, every term of $\mathrm{Cov}(e_3, e_1) = \sum_n W_{33}^n W_{31} W_{11}^{n+1} \mathrm{Var}(e_1)$ is nonnegative and the term $n = 0$ is positive; likewise $\Sigma^c_{23} > 0$. The cofactor formula for the inverse of the $3 \times 3$ matrix $\Sigma^c$ gives

$$\Pi^c_{12} = \frac{\Sigma^c_{13}\Sigma^c_{23} - \Sigma^c_{12}\Sigma^c_{33}}{\det \Sigma^c} = \frac{\Sigma^c_{13}\Sigma^c_{23}}{\det \Sigma^c} > 0.$$

□

*Proof of Lemma S4.30.* The matrix $\Pi_{\bar{C}\bar{C}}$ is positive definite with nonpositive off-diagonal entries. Write it as $\mathsf{H} - \mathsf{N}$ with $\mathsf{H}$ diagonal and positive and $\mathsf{N} \geq 0$ entrywise. Since $\mathsf{H}^{-1/2} \Pi_{\bar{C}\bar{C}} \mathsf{H}^{-1/2} = I - \mathsf{H}^{-1/2} \mathsf{N} \mathsf{H}^{-1/2} \succ 0$ and the spectral radius of the nonnegative symmetric matrix $\mathsf{H}^{-1/2}\mathsf{N}\mathsf{H}^{-1/2}$ is its largest eigenvalue, $\rho(\mathsf{H}^{-1}\mathsf{N}) < 1$, and $\Pi^{-1}_{\bar{C}\bar{C}} = \sum_{l\geq 0} (\mathsf{H}^{-1}\mathsf{N})^l \mathsf{H}^{-1} \geq 0$ entrywise. Since $\Pi_{C\bar{C}} = -\mathcal{A}_{C\bar{C}} \leq 0$, the correction $\mathcal{A}_{C\bar{C}} \Pi^{-1}_{\bar{C}\bar{C}} \mathcal{A}_{\bar{C}C}$ is entrywise nonnegative, which gives the couplings. For the row sums, $\Pi_{CC}\mathbf{1}_C = \varepsilon^\Pi_C + a^\partial_C$, $\mathcal{A}_{\bar{C}C}\mathbf{1}_C = a^\partial_{\bar{C}}$, the vector of couplings of outsiders to $C$, and $\Pi_{\bar{C}\bar{C}}\mathbf{1}_{\bar{C}} = \varepsilon^\Pi_{\bar{C}} + a^\partial_{\bar{C}}$. Hence $\Pi^{-1}_{\bar{C}\bar{C}} a^\partial_{\bar{C}} = \mathbf{1}_{\bar{C}} - \Pi^{-1}_{\bar{C}\bar{C}} \varepsilon^\Pi_{\bar{C}}$ and, with $\mathcal{A}_{C\bar{C}}\mathbf{1}_{\bar{C}} = a^\partial_C$,

$$(\Sigma_{CC})^{-1}\mathbf{1}_C = \varepsilon^\Pi_C + a^\partial_C - \mathcal{A}_{C\bar{C}}(\mathbf{1}_{\bar{C}} - \Pi^{-1}_{\bar{C}\bar{C}}\varepsilon^\Pi_{\bar{C}}) = \varepsilon^\Pi_C + \mathcal{A}_{C\bar{C}}\Pi^{-1}_{\bar{C}\bar{C}}\varepsilon^\Pi_{\bar{C}}.$$

Both $\Pi^{-1}_{\bar{C}\bar{C}}\varepsilon^\Pi_{\bar{C}}$ and $\Pi^{-1}_{\bar{C}\bar{C}} a^\partial_{\bar{C}}$ are nonnegative and add to $\mathbf{1}_{\bar{C}}$, so $0 \leq \Pi^{-1}_{\bar{C}\bar{C}}\varepsilon^\Pi_{\bar{C}} \leq \mathbf{1}_{\bar{C}}$, which gives the bracket for the leaks. For the contrast, $\phi_C^\top (\Sigma_{CC})^{-1} \phi_C \leq \phi_C^\top \Pi_{CC} \phi_C = \varkappa_{\mathrm{out}}(C)$, and $(\Sigma_{CC})^{-1} -$

$(\mathrm{diag}(\varepsilon^{\Pi}_C) + L_C)$ is a nonnegative diagonal matrix plus a Laplacian of nonnegative weights, hence positive semidefinite, so the internal rate does not decrease. □

The amended baseline and the results on whole populations use contrast at uniform leak in the following form.

**Lemma S28.3** (Contrast with uniform leak)**.** *Let $\Pi = \varepsilon^{\Pi} I + L^{\Pi}$ be diffusive with uniform leak $\varepsilon^{\Pi} > 0$ on a finite population $V$, and let $C \subseteq V$ be a region of $n \geq 2$ members.*

*(a)* $r(C) = (\varepsilon^{\Pi} + \lambda_2(L_C))/(\varepsilon^{\Pi} + \mathrm{cut}(C)/n)$.

*(b) If the couplings inside $C$ do not connect $C$, then $r(C) \leq 1$, with equality only if* $\mathrm{cut}(C) = 0$.

*(c) The whole population has* $r(V) = 1 + \lambda_2(L^{\Pi})/\varepsilon^{\Pi} \geq 1$.

*(d) If* $\mathrm{cut}(C) > 0$, *then* $r(C) \leq \max\{1, r_\star(C)\}$, *and* $r(C) < r_\star(C)$ *when* $r_\star(C) > 1$.

*Hence the supremum (32) for a finite closed comparison medium with uniform leak is attained by its whole population or by a region that its couplings connect.*

*Proof.* (a) With uniform leak, (31) gives $\varkappa_{\mathrm{out}}(C) = \varepsilon^{\Pi} + \mathrm{cut}(C)/n$, and $\varkappa_{\mathrm{in}}(C)$ is $\varepsilon^{\Pi}$ plus the smallest Rayleigh quotient of $L_C$ on $\mathbf{1}_C^{\perp}$, which is $\lambda_2(L_C)$. (b) If the couplings inside $C$ do not connect it, the indicators of a connected component of $C$ and of the rest of $C$ both lie in the kernel of $L_C$, so $\lambda_2(L_C) = 0$ and $r(C) = \varepsilon^{\Pi}/(\varepsilon^{\Pi} + \mathrm{cut}(C)/n)$. (c) $\mathrm{cut}(V) = 0$. (d) With $x = \lambda_2(L_C)$ and $y = \mathrm{cut}(C)/n > 0$, $(\varepsilon^{\Pi} + x)/(\varepsilon^{\Pi} + y)$ is the convex combination of 1 and $x/y$ with the positive weights $\varepsilon^{\Pi}/(\varepsilon^{\Pi} + y)$ and $y/(\varepsilon^{\Pi} + y)$. The last statement follows from (b) and (c), since a region that its couplings do not connect has contrast at most $1 \leq r(V)$. □

*Derivations for Remark S4.16.* A clique of $n$ agents with coupling $a$ has $L_C = a(nI - \mathbf{1}\mathbf{1}^{\top})$, so $\lambda_2(L_C) = na$, and no coupling leaves it, so Lemma S28.3(a) gives $r = 1 + na/\varepsilon^{\Pi}$. Under the graph clause every declared edge carries $a$, so the comparison medium is the population itself, and its baseline is at least the contrast of each of its regions. Under the exchangeable clause the mean coupling over the $n(2n-1)$ pairs is $n(n-1)a/(n(2n-1)) = (n-1)a/(2n-1)$. In an exchangeable population of $N$ agents with coupling $\bar{a}$ and leak $\varepsilon^{\Pi}$, a region of $n' < N$ agents has $\lambda_2 = n'\bar{a}$ and cut $(N-n')\bar{a}$ per member, so its contrast $(\varepsilon^{\Pi} + n'\bar{a})/(\varepsilon^{\Pi} + (N-n')\bar{a})$ increases with $n'$ and is at most $(\varepsilon^{\Pi} + (N-1)\bar{a})/(\varepsilon^{\Pi} + \bar{a})$, which is below $1 + N\bar{a}/\varepsilon^{\Pi}$, the contrast of the whole population, because $\varepsilon^{\Pi}(\varepsilon^{\Pi} + (N-1)\bar{a}) < (\varepsilon^{\Pi} + N\bar{a})(\varepsilon^{\Pi} + \bar{a})$. The baseline is therefore $1 + N\bar{a}/\varepsilon^{\Pi}$, which for $N = 2n$ is $1 + 2n(n-1)a/((2n-1)\varepsilon^{\Pi})$, below $1 + na/\varepsilon^{\Pi}$ because $2(n-1) < 2n-1$. With uniform leak the admissible levels for a clique are then $(\bar{r}_{\mathrm{hom}}, 1 + na/\varepsilon^{\Pi}]$. The numbers for the cliques joined by a matching are computed in Check S47.25. □

*Proof of Proposition S4.17.* (a) The comparison medium is closed, so its whole population is one of its regions; its leak is uniform and its cut is zero, so Lemma S28.3(c) gives $\bar{r}_{\mathrm{hom}} \geq 1 + \lambda_2(L_{\mathrm{hom}})/\bar{\varepsilon}$. The population has $\mathrm{cut}(V) = 0$, so $\varkappa_{\mathrm{out}}(V) = \mathbf{1}^{\top}\Pi\mathbf{1}/N = \bar{\varepsilon}$ and $r(V) = \varkappa_{\mathrm{in}}(V)/\bar{\varepsilon}$, with $\varkappa_{\mathrm{in}}(V)$ the minimum of the Rayleigh quotient of $\Pi$ on $\mathbf{1}^{\perp}$ by (31). If $V$ is a collective at level $r_0$, then $r(V) \geq r_0 > \bar{r}_{\mathrm{hom}}$, which gives the inequality; with uniform leak $\varkappa_{\mathrm{in}}(V) = \varepsilon^{\Pi} + \lambda_2(L^{\Pi})$ and $\bar{\varepsilon} = \varepsilon^{\Pi}$.

(b) For a permutation $g$ of $V$ let $(P_g h)_i = h_{g^{-1}(i)}$ for vectors $h$, and put $\bar{\Pi} = |\mathsf{G}|^{-1} \sum_{g \in \mathsf{G}} P_g \Pi P_g^{\top}$, whose $(i,j)$ entry is the average of $\Pi_{g^{-1}(i)g^{-1}(j)}$ over $\mathsf{G}$. Let $i \neq j$. If $\{i,j\}$ lies in a set $\mathcal{S}$ of pairs on which $\mathsf{G}$ acts transitively, the orbit-stabilizer theorem gives every pair of $\mathcal{S}$ as $\{g^{-1}(i), g^{-1}(j)\}$ for the same number of elements $g$, so $\bar{\Pi}_{ij}$ is minus the mean coupling over $\mathcal{S}$, which is the coupling of the comparison medium. Otherwise $\{i,j\}$ lies in no averaged set, which happens only under

the graph clause for a pair outside $E_{\mathrm{d}}$; $\mathsf{G}$ maps it to pairs outside $E_{\mathrm{d}}$, all of coupling zero in $\Pi$, so $\bar{\Pi}_{ij} = 0$, as in the comparison medium. The row sums are $(\bar{\Pi}\mathbf{1})_i = |\mathsf{G}|^{-1} \sum_g \varepsilon^{\Pi}_{g^{-1}(i)}$, because $P_g^{\top}\mathbf{1} = \mathbf{1}$ and $\Pi\mathbf{1} = \varepsilon^{\Pi}$; they equal $\bar{\varepsilon}$ when $\mathsf{G}$ is transitive on $V$, by the same count, and when the leak is uniform. Hence $\bar{\Pi} = \Pi_{\mathrm{hom}}$. The map $\Pi \mapsto \varkappa_{\mathrm{in}}(V) = \min\{f^{\top}\Pi f : f \perp \mathbf{1}, \|f\| = 1\}$ is a minimum of linear functions and therefore concave, and it is invariant under $\Pi \mapsto P_g \Pi P_g^{\top}$, because $P_g$ is orthogonal and fixes $\mathbf{1}$. Thus $\varkappa_{\mathrm{in}}(V; \Pi_{\mathrm{hom}}) \geq |\mathsf{G}|^{-1} \sum_g \varkappa_{\mathrm{in}}(V; P_g \Pi P_g^{\top}) = \varkappa_{\mathrm{in}}(V; \Pi)$, while $\varkappa_{\mathrm{in}}(V; \Pi_{\mathrm{hom}}) = \bar{\varepsilon} + \lambda_2(L_{\mathrm{hom}})$, and (a) excludes $V$. The symmetric group acts transitively on agents and on pairs. On a torus the translations map the torus to itself and act transitively on the sites and on each displacement class, every pair of the class of $x$ being $\{y, y + x\}$ for a site $y$. For the graph clause the hypothesis is the transitivity of the automorphisms of the declared graph on its vertices and on its edges.

(c) Since $\lambda_2(L^{\Pi}) > \lambda_2(L_{\mathrm{hom}}) \geq 0$, the couplings of $\Pi$ connect $V$. Every pair of positive coupling in $\Pi$ has positive coupling in the comparison medium, whose coupling on a displacement class, on $E_{\mathrm{d}}$ or on all pairs is a mean of nonnegative numbers one of which is positive; so the comparison medium is connected, every proper region of it has a positive cut, and $R^{\star}_{\mathrm{hom}}$, a maximum over finitely many regions, is finite. Let $C \subsetneq V$ with $|C| \geq 2$ be a region of the comparison medium. If its couplings do not connect it, its contrast is at most 1 by Lemma S28.3(b), below $r(V) = 1 + \lambda_2(L^{\Pi})/\varepsilon^{\Pi}$. If they connect it and $r_{\star}(C) \leq 1$, its contrast is at most 1 by Lemma S28.3(d); if $r_{\star}(C) > 1$, its contrast is below $r_{\star}(C) \leq R^{\star}_{\mathrm{hom}}$, and $R^{\star}_{\mathrm{hom}} \leq r(V)$ when $\varepsilon^{\Pi} \leq \lambda_2(L^{\Pi})/(R^{\star}_{\mathrm{hom}} - 1)$. The whole comparison medium has contrast $1 + \lambda_2(L_{\mathrm{hom}})/\varepsilon^{\Pi} < r(V)$. The baseline, a maximum over finitely many regions, is therefore below $r(V)$, and since $\varepsilon^{\max}_V/\bar{\varepsilon}_V = 1 \leq \bar{r}_{\mathrm{hom}}$, every level in $(\bar{r}_{\mathrm{hom}}, r(V)]$ is admissible and $V$ is a collective at it. With the couplings fixed, (a) gives necessity at every leak and the bound just proved gives sufficiency at small leaks. At a given leak, $V$ is a collective at some admissible level exactly when $r(V) > \bar{r}_{\mathrm{hom}}$, and under the hypothesis $r(V)$ exceeds the contrast of the whole comparison medium, so only its proper regions remain to be compared.

(d) On vectors $(h_1, h_2, -h_2, -h_1)$ the Laplacian of the path acts as $\left(\begin{smallmatrix} 0.9 & -0.9 \\ -0.9 & 3.3 \end{smallmatrix}\right)$, whose eigenvalues are $2.1 \pm 1.5$, and $(1, -1, -1, 1)$ is an eigenvector with eigenvalue 1.8, so $\lambda_2(L^{\Pi}) = 0.6$. The comparison path has coupling 1 between neighbors, the mean over the three pairs at displacement one, and no other coupling, so $\lambda_2(L_{\mathrm{hom}}) = 2 - 2\cos(\pi/4)$. The regions of the comparison path that its couplings connect are the two end pairs, with $\lambda_2 = 2$ and cut $/n = \frac{1}{2}$, the middle pair, with 2 and 1, the two triples, with 1 and $\frac{1}{3}$, and the whole path; the other regions have contrast at most one. Clearing denominators, $1 + 0.6/\varepsilon^{\Pi}$ exceeds the contrast of an end pair exactly when $\varepsilon^{\Pi} < \frac{1}{3}$, of the middle pair when $\varepsilon^{\Pi} < \frac{3}{2}$, of a triple when $\varepsilon^{\Pi} < 3$, and that of the whole path at every leak. □

*Proof of Theorem 4.13.* With uniform leak, $\varkappa_{\mathrm{in}}(C) = \varepsilon^{\Pi} + \lambda_2(L_C)$ and $\varkappa_{\mathrm{out}}(C) = \varepsilon^{\Pi} + \mathrm{cut}(C)/n$, and $(\varepsilon^{\Pi} + x)/(\varepsilon^{\Pi} + y) \leq \max\{1, x/y\}$ for $x, y \geq 0$, which gives $r \leq \max\{1, r_{\star}\}$. Every line parallel to an axis that meets $C$ meets it in a finite nonempty set and so contains at least two lattice edges with exactly one endpoint in $C$, each of coupling at least $a_{\mathrm{nn}}$, and edges on different lines are different. With $\mathrm{pr}_d(C)$ the projection of $C$ along axis $d$, the number of such lines is $|\mathrm{pr}_d(C)|$, so $\mathrm{cut}(C) \geq 2a_{\mathrm{nn}} \sum_d |\mathrm{pr}_d(C)| \geq 2Da_{\mathrm{nn}} n^{(D-1)/D}$ by the inequality of arithmetic and geometric means and the discrete Loomis–Whitney inequality $n^{D-1} \leq \prod_d |\mathrm{pr}_d(C)|$ (Loomis and Whitney, 1949). Gershgorin's theorem gives $\lambda_2(L_C) \leq 2 \max_i \sum_{j \in C} a^{\Pi}_{ij} \leq 2\|a^{\Pi}\|_1$, which yields the first term of (35). For the second, Theorem 4.14 gives $\lambda_2(L_C) \leq \varsigma^2/(2 \max_d \mathrm{Var}_d(C))$. A sup-norm ball of radius $h_n = ((n/2)^{1/D} - 1)/2$ around the centroid $\bar{x}$ of $C$ contains at most $(2h_n + 1)^D = n/2$ lattice sites, so at least $n/2$ members lie farther than $h_n$ from $\bar{x}$, $\sum_d \mathrm{Var}_d(C) \geq h_n^2/2$ and $\max_d \mathrm{Var}_d(C) \geq ((n/2)^{1/D} - 1)^2/(8D)$. Dividing by $\mathrm{cut}(C)/n \geq 2Da_{\mathrm{nn}} n^{-1/D}$ gives the second term. For $n \leq 2^{D+1}$ the first term is at most $2^{(D+1)/D}\|a^{\Pi}\|_1/(Da_{\mathrm{nn}})$; for $n > 2^{D+1}$, $(n/2)^{1/D} - 1 \geq \frac{1}{2}(n/2)^{1/D}$ and the second term is at most $2^{3+2/D}\varsigma^2 n^{-1/D}/a_{\mathrm{nn}}$. Both bounds are

uniform in $\varepsilon^{\Pi}$. □

*Proof of Proposition S4.19.* (a) The Laplacian of an $\ell_b \times \ell_b$ grid is the Kronecker sum of two path Laplacians with eigenvalues $2 - 2\cos(\pi j/\ell_b)$, $j = 0, \dots, \ell_b - 1$, so $\lambda_2(L_C) = a_{\text{in}}(2 - 2\cos(\pi/\ell_b))$. A block has $4\ell_b$ edges to other blocks, one for each site on each side, so $\operatorname{cut}(C)/n = 4\ell_b a_{\text{out}}/\ell_b^2$. (b) A vector $f = \sum_j c_j \mathbf{1}_{C_j}$ in the span of the block indicators has

$$f^{\top} L^{\Pi} f = \sum_{\{i,k\} \text{ across blocks}} a^{\Pi}_{ik}(c_{j(i)} - c_{j(k)})^2 \le 2 \sum_j c_j^2 \operatorname{cut}(C_j) \le 2 \max_j \frac{\operatorname{cut}(C_j)}{|C_j|} \|f\|^2,$$

and Courant–Fischer on this $M$-dimensional space gives the upper bound. For the lower bound, $L^{\Pi} = \bigoplus_j L_{C_j} + L_{\times}$ with $L_{\times} \succeq 0$ the Laplacian of the couplings between blocks; the direct sum has exactly $M$ zero eigenvalues, because the blocks are connected, and next eigenvalue $\min_j \lambda_2(L_{C_j})$, and adding $L_{\times}$ lowers no eigenvalue. □

*Proof of Theorem 4.14.* The function $f = x_d - \bar{x}_{C,d}$ on $C$, zero elsewhere, is supported on $C$ and orthogonal to $\mathbf{1}_C$. By Courant–Fischer $\varkappa_{\text{in}}(C) \le f^{\top}(\operatorname{diag}(\varepsilon^{\Pi}) + L_C)f/f^{\top}f \le \varepsilon_C^{\max} + f^{\top}L_C f/f^{\top}f$ and $\lambda_2(L_C) \le f^{\top}L_C f/f^{\top}f$. Now

$$f^{\top} L_C f = \frac{1}{2} \sum_{i,j \in C} a^{\Pi}_{ij}(x_{i,d} - x_{j,d})^2 \le \frac{1}{2} \sum_{i \in C} \varsigma_d^2 = \frac{n\varsigma_d^2}{2}, \qquad f^{\top} f = n \operatorname{Var}_d(C),$$

which gives the first inequality of (37); dividing by $\varkappa_{\text{out}}(C) \ge \bar{\varepsilon}_C$ gives the second. For the metric form take $f = \operatorname{dist}(\cdot, j_0)$ minus its mean over $C$, for which $|f_i - f_j| \le \operatorname{dist}(i,j)$. For sharpness, an isolated box has $\varkappa_{\text{out}} = \varepsilon^{\Pi}$ and $r - 1 = \lambda_2(L_C)/\varepsilon^{\Pi}$ exactly. For a box of side $\ell_b$ with coupling $a^{\Pi}$, $\lambda_2(L_C) = 2a^{\Pi}(1 - \cos(\pi/\ell_b))$, $\operatorname{Var}_d(C) = (\ell_b^2 - 1)/12$ and $\xi_{2,d}^2 = a^{\Pi}/\varepsilon^{\Pi}$, so $\operatorname{Var}_d(C)(r-1)/\xi_{2,d}^2 = (\ell_b^2 - 1)(1 - \cos(\pi/\ell_b))/6 \to \pi^2/12$. □

*Proof of Corollary 4.16.* Let $r(C) \ge r_0$. The second inequality of (37) gives $r_0\bar{\varepsilon}_C - \varepsilon_C^{\max} \le \varsigma_d^2/(2\operatorname{Var}_d(C))$, and the left side is positive by (33), so $\operatorname{Var}_d(C) \le \varsigma_d^2/(2(r_0\bar{\varepsilon}_C - \varepsilon_C^{\max}))$. With uniform leak the right side is $\xi_{2,d}^2/(r_0 - 1)$, and the metric form is the same computation. With heterogeneous leaks $r_0\bar{\varepsilon}_C - \varepsilon_C^{\max} \ge r_0\varepsilon^{\Pi}_{\min} - \varepsilon^{\Pi}_{\max}$. Taking suprema over collectives gives the corollary. □

The proof of the bridge theorem uses two more facts.

**Lemma S28.4** (Kato–Temple)**.** *Let* $\mathsf{M}$ *be a real symmetric operator on a finite-dimensional space with smallest eigenvalues* $\lambda_1(\mathsf{M}) \le \lambda_2(\mathsf{M})$*, let* $u$ *be a unit vector with Rayleigh quotient* $\langle \mathsf{M} \rangle_u = u^{\top}\mathsf{M}u < \lambda_2(\mathsf{M})$*, and put* $\mathsf{r}_u = \|(\mathsf{M} - \langle \mathsf{M} \rangle_u)u\|$*. Then* $0 \le \langle \mathsf{M} \rangle_u - \lambda_1(\mathsf{M}) \le \mathsf{r}_u^2/(\lambda_2(\mathsf{M}) - \langle \mathsf{M} \rangle_u)$.

*Proof.* Write $\lambda_1, \lambda_2$ for $\lambda_1(\mathsf{M}), \lambda_2(\mathsf{M})$. The Rayleigh principle gives $\lambda_1 \le \langle \mathsf{M} \rangle_u$. The operator $(\mathsf{M} - \lambda_1)(\mathsf{M} - \lambda_2)$ is positive semidefinite, its eigenvalues being $(\lambda_k(\mathsf{M}) - \lambda_1)(\lambda_k(\mathsf{M}) - \lambda_2) \ge 0$, and since $(\mathsf{M} - \langle \mathsf{M} \rangle_u)u \perp u$, $\|\mathsf{M}u\|^2 = \mathsf{r}_u^2 + \langle \mathsf{M} \rangle_u^2$. Hence $0 \le u^{\top}(\mathsf{M} - \lambda_1)(\mathsf{M} - \lambda_2)u = \mathsf{r}_u^2 - (\langle \mathsf{M} \rangle_u - \lambda_1)(\lambda_2 - \langle \mathsf{M} \rangle_u)$. □

This is the inequality of Temple (1928) and Kato (1949).

**Lemma S28.5** (Effective resistance)**.** *For a connected population with Laplacian* $L^{\Pi}$ *and agents* $i \ne j$*, let* $R_{\text{eff}}(i,j) = (\mathbf{1}_{\{i\}} - \mathbf{1}_{\{j\}})^{\top}(L^{\Pi})^{+}(\mathbf{1}_{\{i\}} - \mathbf{1}_{\{j\}})$*. Then* $1/R_{\text{eff}}(i,j) = \min\{f^{\top}L^{\Pi}f : f_i = 1, f_j = 0\}$*. Consequently* $R_{\text{eff}}(i,j) \ge 1/\operatorname{cut}(C)$ *when* $i \in C$ *and* $j \notin C$*, and* $R_{\text{eff}}(i,j) \le 2/\lambda_2(L_C)$ *when* $i, j \in C$ *and* $C$ *is connected by its internal couplings.*

*Proof.* Since $L^\Pi \mathbf{1} = 0$, the constraint can be replaced by $f_i - f_j = 1$. With $b = \mathbf{1}_{\{i\}} - \mathbf{1}_{\{j\}}$, which lies in the range of $L^\Pi$, a minimizer satisfies $L^\Pi f = \mu b$, so $f = \mu (L^\Pi)^+ b$ up to a constant, the constraint gives $\mu = 1/R_{\mathrm{eff}}(i,j)$, and the minimum is $\mu^2 b^\top (L^\Pi)^+ b = 1/R_{\mathrm{eff}}(i,j)$. The indicator $f = \mathbf{1}_C$ is admissible when $i \in C$, $j \notin C$, and has $f^\top L^\Pi f = \mathrm{cut}(C)$, which gives the first consequence. For $i, j \in C$, $f^\top L^\Pi f \geq f_C^\top L_C f_C$ for every $f$, so the minimum over the population is at least the minimum for the couplings inside $C$, whose value is $1/(b_C^\top L_C^+ b_C)$ with $b_C$ the restriction of $b$; since $b_C \perp \mathbf{1}_C$ and $\|b_C\|^2 = 2$, $b_C^\top L_C^+ b_C \leq 2/\lambda_2(L_C)$. □

*Proof of Theorem 4.18.* (i) With uniform leak $\varkappa_{\mathrm{in}}(C_j) = \varepsilon^\Pi + g_j$ and $\varkappa_{\mathrm{out}}(C_j) = \varepsilon^\Pi + \bar{a}_j^\partial$. When two blocks form $V$, $\mathrm{cut}(V) = 0$, so $\varkappa_{\mathrm{out}}(V) = \varepsilon^\Pi$ and $\varkappa_{\mathrm{in}}(V) = \varepsilon^\Pi + \lambda_2(L^\Pi)$. The unit vector $u_- = (\phi_{C_1} - \phi_{C_2})/\sqrt{2}$ is orthogonal to $\mathbf{1}$, takes the values $\pm(2n)^{-1/2}$, and has $u_-^\top L^\Pi u_- = \mathrm{cut}(C_1) \cdot 4/(2n) = 2\bar{a}^\partial$, so $\lambda_2(L^\Pi) \leq 2\bar{a}^\partial$ by Courant–Fischer on $\mathbf{1}^\perp$. The vector $(L^\Pi - 2\bar{a}^\partial)u_-$ has entries $\pm 2(a_i^\partial - \bar{a}^\partial)/\sqrt{2n}$, with sign + on $C_1$ and − on $C_2$; its block sums vanish, so it is orthogonal to the span of $\Phi$, and its squared norm is $\frac{2}{n}\sum_i (a_i^\partial - \bar{a}^\partial)^2$. Since $L^\Pi \Phi v_+ = L^\Pi \mathbf{1}/\sqrt{N} = 0$ for $v_+ = (1,1)/\sqrt{2}$ and $L^\Pi u_-$ equals $2\bar{a}^\partial u_-$ plus this vector, $\sigma_{\mathrm{cut}}^2 = \|\Phi_\perp^\top L^\Pi \Phi\|_F^2$ equals that squared norm. On $\mathbf{1}^\perp$ the second eigenvalue of $L^\Pi$ is $\lambda_3(L^\Pi) \geq g_{\min}$ by Proposition S4.19(b) with $\varepsilon^\Pi = 0$. If $2\bar{a}^\partial < g_{\min}$, Lemma S28.4 on $\mathbf{1}^\perp$ with $u = u_-$ gives $2\bar{a}^\partial - \lambda_2(L^\Pi) \leq \sigma_{\mathrm{cut}}^2/(g_{\min} - 2\bar{a}^\partial)$, which is (41).

(ii) In the orthonormal basis $[\Phi, \Phi_\perp]$ the block formula for the inverse gives $\Phi^\top \Pi^{-1} \Phi = (\Pi_\Phi - \mathcal{D})^{-1}$, and $0 \preceq \mathcal{D} \preceq \|\Phi_\perp^\top \Pi \Phi\|^2 \lambda_{\min}(\Phi_\perp^\top \Pi \Phi_\perp)^{-1} I$ with $\|\cdot\| \leq \|\cdot\|_F$; the squared Frobenius norm is the sum of the squared norms of the columns $\Phi_\perp^\top \Pi \phi_{C_j}$, one per block, and for $M$ identical blocks in identical surroundings these are equal, so that $\sigma_{\mathrm{cut}}^2$ is $M$ times one of them. $\mathcal{D} = 0$ exactly when $\Phi_\perp^\top \Pi \Phi = 0$, that is when the span of $\Phi$ is invariant under $\Pi$; with uniform leak this means that $L^\Pi \mathbf{1}_{C_j}$ is constant on every block for every $j$, and since $(L^\Pi \mathbf{1}_{C_j})_i = -\sum_{k \in C_j} a_{ik}^\Pi$ for $i \notin C_j$ and $\sum_{l \neq j} \sum_{k \in C_l} a_{ik}^\Pi$ for $i \in C_j$, this is the equitable condition. For two equal blocks, $\Pi_\Phi = \begin{pmatrix} \varepsilon^\Pi + \bar{a}^\partial & -\bar{a}^\partial \\ -\bar{a}^\partial & \varepsilon^\Pi + \bar{a}^\partial \end{pmatrix}$ has eigenvectors $v_\pm = (1, \pm 1)/\sqrt{2}$ with eigenvalues $\varepsilon^\Pi$ and $\varepsilon^\Pi + 2\bar{a}^\partial$. Because $\Phi v_+ = \mathbf{1}/\sqrt{N}$ and $\Pi \mathbf{1} = \varepsilon^\Pi \mathbf{1}$, $\Phi_\perp^\top \Pi \Phi v_+ = 0$, so $\Phi_\perp^\top \Pi \Phi$ has rank at most one, its spectral and Frobenius norms coincide, and $\mathcal{D} = \mathcal{D}_{--} v_- v_-^\top$ with $0 \leq \mathcal{D}_{--} \leq \sigma_{\mathrm{cut}}^2/\lambda_{\min}(\Phi_\perp^\top \Pi \Phi_\perp)$. Here $\Phi_\perp^\top \Pi \Phi_\perp \succeq \varepsilon^\Pi I + \Phi_\perp^\top (L_{C_1} \oplus L_{C_2}) \Phi_\perp \succeq (\varepsilon^\Pi + g_{\min}) I$, since the columns of $\Phi_\perp$ are orthogonal to the kernel of $L_{C_1} \oplus L_{C_2}$, so $\mathcal{D}_{--} \leq \delta_{\mathrm{corr}}$. The matrix $\Phi^\top \Sigma \Phi$ then has eigenvalues $1/\varepsilon^\Pi$ on $v_+$ and $1/(\varepsilon^\Pi + 2\bar{a}^\partial - \mathcal{D}_{--})$ on $v_-$, and the correlation of $\phi_{C_1}^\top e$ and $\phi_{C_2}^\top e$, which is that of the block averages, is

$$\mathrm{Corr}(\bar{e}_1, \bar{e}_2) = \frac{2\bar{a}^\partial - \mathcal{D}_{--}}{2\varepsilon^\Pi + 2\bar{a}^\partial - \mathcal{D}_{--}},$$

decreasing in $\mathcal{D}_{--}$, which gives (42).

(iii)(a) For the closed connected population, $\Sigma = (\varepsilon^\Pi I + L^\Pi)^{-1} = \mathbf{1}\mathbf{1}^\top/(N\varepsilon^\Pi) + (L^\Pi)^+ + O(\varepsilon^\Pi)$ by the spectral decomposition. Write $\Sigma_{ij} = \alpha_0 + \alpha_{ij}$ with $\alpha_0 = 1/(N\varepsilon^\Pi)$, $\alpha_{ij} = O(1)$ and $\alpha_i = \alpha_{ii}$. Expanding in $1/\alpha_0$, $\mathrm{Corr}(e_i, e_j) = 1 - (\alpha_i + \alpha_j - 2\alpha_{ij})/(2\alpha_0) + O(\alpha_0^{-2})$, and $\alpha_i + \alpha_j - 2\alpha_{ij} = (L^\Pi)_{ii}^+ + (L^\Pi)_{jj}^+ - 2(L^\Pi)_{ij}^+ + O(\varepsilon^\Pi) = R_{\mathrm{eff}}(i,j) + O(\varepsilon^\Pi)$. Hence $\varpi_{ij} = \frac{N\varepsilon^\Pi}{2} R_{\mathrm{eff}}(i,j) + O((\varepsilon^\Pi)^2)$ uniformly over the finitely many pairs. The single-linkage ultrametric, $u_{\mathrm{in}}$ and $\varpi_{\mathrm{out}}$ change by no more than the largest change of the weights, and they scale with the weights, so $\mathcal{U}(C) = \frac{N\varepsilon^\Pi}{2} \mathcal{U}^R(C) + O((\varepsilon^\Pi)^2)$. By Lemma S28.5, the resistance version of $\varpi_{\mathrm{out}}(C)$ is at least $1/\mathrm{cut}(C)$, and the resistance version of $u_{\mathrm{in}}(C)$ is at most $\max_{i,j \in C} R_{\mathrm{eff}}(i,j) \leq 2/\lambda_2(L_C) = 2|C|/(r_\star(C)\,\mathrm{cut}(C))$, because a single-linkage height never exceeds the direct weight.

(iii)(b) Write $L^\Pi = \mathrm{Deg} - \mathcal{A}$ with Deg the diagonal of coupling sums and $\mathcal{A}$ the coupling matrix. Since $(\varepsilon^\Pi I + \mathrm{Deg})^{-1}\mathcal{A}$ is nonnegative with row sums below one, $\Sigma = \sum_{l \geq 0} ((\varepsilon^\Pi I + \mathrm{Deg})^{-1}\mathcal{A})^l (\varepsilon^\Pi I + \mathrm{Deg})^{-1}$ with nonnegative terms. As $\varepsilon^\Pi \to \infty$, $\Sigma_{ii} = (\varepsilon^\Pi)^{-1} + O((\varepsilon^\Pi)^{-2})$, a coupled pair has

$\mathrm{Corr}(e_i,e_j)=a_{ij}^{\Pi}/\varepsilon^{\Pi}+O((\varepsilon^{\Pi})^{-2})$ and $\varpi_{ij}=\ln(\varepsilon^{\Pi}/a_{ij}^{\Pi})+O(1/\varepsilon^{\Pi})$, and a pair at distance $l\geq 2$ in the coupling graph has $\mathrm{Corr}=O((\varepsilon^{\Pi})^{-l})$ and $\varpi_{ij}\geq l\ln\varepsilon^{\Pi}-O(1)$. For large $\varepsilon^{\Pi}$ every chain through a non-coupled pair is therefore worse than the best chain of coupled pairs, and the single-linkage height between $i$ and $j$ is $\ln\varepsilon^{\Pi}-\ln a^{\mathrm{bn}}(i,j)+O(1/\varepsilon^{\Pi})$, with $a^{\mathrm{bn}}(i,j)$ the largest, over paths from $i$ to $j$, of the smallest coupling on the path. Inside $C$ every pair is joined in a spanning tree of maximal weight through couplings at least $a^{\mathrm{bn}}(C)$; removing the tree edge of coupling $a^{\mathrm{bn}}(C)$ splits $C$ into two parts that no edge inside $C$ of larger coupling joins, by the cut property of such trees, and any path that leaves $C$ crosses a boundary coupling at most $a^{\partial}_{\max}(C)<a^{\mathrm{bn}}(C)$. Hence $u_{\mathrm{in}}(C)=\ln\varepsilon^{\Pi}-\ln a^{\mathrm{bn}}(C)+O(1/\varepsilon^{\Pi})$. The smallest direct weight across the boundary is at the boundary pair of largest coupling, $\varpi_{\mathrm{out}}(C)=\ln\varepsilon^{\Pi}-\ln a^{\partial}_{\max}(C)+O(1/\varepsilon^{\Pi})$. Meanwhile $r(C)=(\varepsilon^{\Pi}+\lambda_2(L_C))/(\varepsilon^{\Pi}+\mathrm{cut}(C)/|C|)\to 1$.

(iv) The statement on the blocks is $\varepsilon^{\Pi}+g\geq r_0(\varepsilon^{\Pi}+\bar a^{\partial})$ rearranged. The statements on the union are the bound $r(C_1\cup C_2)\leq 1+2\bar a^{\partial}/\varepsilon^{\Pi}$ of (i) and, when $2\bar a^{\partial}<g$, the lower bound of (41), each solved for $\varepsilon^{\Pi}$. The statements on the correlation are the two sides of (42) solved for $\varepsilon^{\Pi}$ at fixed $\delta_{\mathrm{corr}}$. For the closed form put $c=2\bar a^{\partial}-\sigma^2_{\mathrm{cut}}/g$. If $c\leq 0$ the condition holds trivially. If $c>0$, then $\delta_{\mathrm{corr}}\leq\sigma^2_{\mathrm{cut}}/g$ gives $2\bar a^{\partial}-\delta_{\mathrm{corr}}\geq c$, and since $x\mapsto x/(2\varepsilon^{\Pi}+x)$ increases on $x>0$, the left side of (42) is at least $c/(2\varepsilon^{\Pi}+c)$; the requirement $c/(2\varepsilon^{\Pi}+c)\leq\theta_0$ is $\varepsilon^{\Pi}\geq c(1-\theta_0)/(2\theta_0)$. On the window (43) the blocks are selected, the union is not, and the block averages are separated. □

The baseline of Numerical observation S4.22 rests on enumeration for small regions and on the following bounds for the others.

**Lemma S28.6** (Bounds for regions of a homogeneous box)**.** *Let a comparison medium be a box of $n_{\mathrm{row}}$ rows and $n_{\mathrm{col}}$ columns of $\mathbb{Z}^2$, $N=n_{\mathrm{row}}n_{\mathrm{col}}$, with coupling $a_{\mathrm{h}}$ between horizontal neighbors, $a_{\mathrm{v}}$ between vertical neighbors and no other coupling, and with Laplacian $L_{\mathrm{hom}}$. Let $S$ be a region of $n$ sites, $2\leq n\leq N-1$, put $k=N-n$ and $a_{\max}=\max\{a_{\mathrm{h}},a_{\mathrm{v}}\}$, and let $L_S$ be the Laplacian of the couplings inside $S$.*

*(U1)* $\lambda_2(L_S)\leq a_{\max}\mathsf{p}(n)/\mathsf{J}(n)$, *with* $\mathsf{p}(n)=2n-\lceil 2\sqrt{n}\rceil$ *and* $\mathsf{J}(n)=n^2/(2\pi)-n/6$.

*(U2)* $\lambda_2(L_S)\leq\lambda_{2+k}(L_{\mathrm{hom}})$, *the $(2+k)$th smallest eigenvalue of $L_{\mathrm{hom}}$.*

*(U3) If* $\mathsf{v}$ *is a unit eigenvector of $L_{\mathrm{hom}}$ for $\lambda_2(L_{\mathrm{hom}})$, $\mathsf{s}_k$ the sum of the $k$ largest $\mathsf{v}_i^2$ and $\mathsf{t}_k$ the sum of the $k$ largest $|\mathsf{v}_i|$, then $\lambda_2(L_S)\leq\lambda_2(L_{\mathrm{hom}})/(1-\mathsf{s}_k-\mathsf{t}_k^2/n)$ when the denominator is positive.*

*(U4)* $\lambda_2(L_S)\leq\frac{n}{n-1}(a_{\mathrm{h}}+a_{\mathrm{v}})$.

*(C)* $\mathrm{cut}(S)\geq\mathrm{cut}^{\mathrm{lb}}(n)$, *the smallest of the following numbers: $\mathsf{f}(n)$; when $n\geq n_{\mathrm{col}}$, $a_{\mathrm{v}}n_{\mathrm{col}}$, plus $a_{\mathrm{h}}$ if $n_{\mathrm{col}}$ does not divide $n$; when $n\geq n_{\mathrm{row}}$, $a_{\mathrm{h}}n_{\mathrm{row}}$, plus $a_{\mathrm{v}}$ if $n_{\mathrm{row}}$ does not divide $n$; and when $n\geq n_{\mathrm{row}}+n_{\mathrm{col}}-1$, $\mathsf{f}(N-n)$. Here $\mathsf{f}(m)$ is the minimum of $a_{\mathrm{h}}\mathsf{y}+a_{\mathrm{v}}\mathsf{x}$ over integers $1\leq\mathsf{y}\leq n_{\mathrm{row}}$ and $1\leq\mathsf{x}\leq n_{\mathrm{col}}$ with $\mathsf{xy}\geq m$, $\mathsf{y}(n_{\mathrm{col}}-1)\geq m$ and $\mathsf{x}(n_{\mathrm{row}}-1)\geq m$, and $+\infty$ if there are none.*

*Hence at uniform leak $\varepsilon^{\Pi}$ every region of $n$ sites has $r(S)\leq(\varepsilon^{\Pi}+\lambda^{\mathrm{ub}}(n))/(\varepsilon^{\Pi}+\mathrm{cut}^{\mathrm{lb}}(n)/n)$, with $\lambda^{\mathrm{ub}}(n)$ the smallest of the bounds (U1) to (U4).*

*Proof.* Give the sites coordinates $z_i=(x_i,y_i)$, column and row. (U1) The vector $f$ on $S$ with $f_i=x_i-\bar x_S$, $\bar x_S$ the mean column of $S$, is orthogonal to $\mathbf{1}_S$, $f^{\top}f=\sum_{i\in S}(x_i-\bar x_S)^2$, and $f^{\top}L_Sf=a_{\mathrm{h}}E_{\mathrm{h}}$, with $E_{\mathrm{h}}$ the number of horizontal neighbor pairs inside $S$, since such pairs differ by one in column and vertical pairs by zero. The row coordinate gives the quotient of $a_{\mathrm{v}}E_{\mathrm{v}}$ by $\sum_{i\in S}(y_i-\bar y_S)^2$ in the same way. By the mediant inequality $\min\{p_1/q_1,p_2/q_2\}\leq(p_1+p_2)/(q_1+q_2)$,

where a vanishing denominator comes with a vanishing numerator and the other quotient is used, $\lambda_2(L_S) \le a_{\max}(E_{\mathrm{h}} + E_{\mathrm{v}})/\mathsf{J}_S$ with $\mathsf{J}_S = \sum_{i\in S} |z_i - \bar{z}_S|^2$. If $S$ meets $\mathsf{y}$ rows and $\mathsf{x}$ columns, a row holding $m'$ sites of $S$ holds at most $m'-1$ horizontal pairs inside $S$, so $E_{\mathrm{h}} \le n - \mathsf{y}$, and likewise $E_{\mathrm{v}} \le n - \mathsf{x}$; since $\mathsf{xy} \ge n$, $\mathsf{x}+\mathsf{y} \ge 2\sqrt{n}$, and both are integers, so $E_{\mathrm{h}} + E_{\mathrm{v}} \le \mathsf{p}(n)$. For the inertia, put the unit square $Q_z$ centered at each site $z \in S$. These squares have disjoint interiors, $\int_{Q_z} |w-c|^2 \mathrm{d}w = |z-c|^2 + \frac{1}{6}$, and their union $A_S$ has area $n$. If $B$ is the disk of area $n$ centered at $c$, with radius $\varrho$, then $|w-c| \ge \varrho$ on $A_S \setminus B$, $|w-c| \le \varrho$ on $B \setminus A_S$, and the two sets have equal areas, so $\int_{A_S} |w-c|^2 \mathrm{d}w \ge \int_B |w-c|^2 \mathrm{d}w = n^2/(2\pi)$. With $c = \bar{z}_S$ this is $\mathsf{J}_S + n/6 \ge n^2/(2\pi)$, which gives (U1).

(U2) The principal submatrix of $L_{\mathrm{hom}}$ on $S$ is $L_S$ plus the diagonal matrix of the members' cut couplings, so $\lambda_2(L_S)$ is at most its second eigenvalue, and Cauchy interlacing bounds that by $\lambda_{2+N-n}(L_{\mathrm{hom}})$.

(U3) Put $f = \mathsf{v}|_S - \bar{\mathsf{v}}_S \mathbf{1}_S$, with $\bar{\mathsf{v}}_S$ the mean of $\mathsf{v}$ on $S$. Then $f \perp \mathbf{1}_S$, and $f^\top L_S f$ is the sum of $a_{ij}(\mathsf{v}_i - \mathsf{v}_j)^2$ over the neighbor pairs inside $S$, at most the same sum over all pairs, which is $\lambda_2(L_{\mathrm{hom}})$. Since $\mathsf{v} \perp \mathbf{1}$, $|\sum_{i\in S} \mathsf{v}_i| = |\sum_{i\notin S} \mathsf{v}_i| \le \mathsf{t}_k$, and $\sum_{i\in S} \mathsf{v}_i^2 \ge 1 - \mathsf{s}_k$, so $f^\top f = \sum_{i\in S} \mathsf{v}_i^2 - (\sum_{i\in S} \mathsf{v}_i)^2/n \ge 1 - \mathsf{s}_k - \mathsf{t}_k^2/n$.

(U4) For $i \in S$ the vector $\mathbf{1}_{\{i\}} - \mathbf{1}_S/n$ is orthogonal to $\mathbf{1}_S$, has squared norm $(n-1)/n$, and has quadratic form $(L_S)_{ii}$, the coupling of $i$ inside $S$. Take for $i$ the site of $S$ with the smallest column in the lowest row that meets $S$: it has no neighbor in $S$ to its left or below, so its coupling inside $S$ is at most $a_{\mathrm{h}} + a_{\mathrm{v}}$.

(C) A row that meets $S$ without being contained in it contains a horizontal pair with exactly one site in $S$, and such pairs in different rows are different; the same holds for columns and vertical pairs. If $S$ contains no full row and no full column, each of the $\mathsf{y}$ rows and $\mathsf{x}$ columns that it meets contributes to the cut, a row holds at most $n_{\mathrm{col}} - 1$ of its sites and a column at most $n_{\mathrm{row}} - 1$, and $\mathsf{xy} \ge n$, so $\mathrm{cut}(S) \ge a_{\mathrm{h}}\mathsf{y} + a_{\mathrm{v}}\mathsf{x} \ge \mathsf{f}(n)$. If $S$ contains a full row and no full column, then $n \ge n_{\mathrm{col}}$ and all $n_{\mathrm{col}}$ columns meet $S$ without being contained in it, so $\mathrm{cut}(S) \ge a_{\mathrm{v}} n_{\mathrm{col}}$, plus $a_{\mathrm{h}}$ when $n_{\mathrm{col}}$ does not divide $n$, since then some row that meets $S$ is not contained in it; the case of a full column and no full row is symmetric. If $S$ contains a full row and a full column, then $n \ge n_{\mathrm{row}} + n_{\mathrm{col}} - 1$ and $S$ meets every row and every column, so every row or column that meets the complement $\bar{S}$ of $S$ also meets $S$; then $\bar{S}$, with $N - n$ sites, is in the first case, and $\mathrm{cut}(S) = \mathrm{cut}(\bar{S}) \ge \mathsf{f}(N-n)$. The last statement holds because at uniform leak $r(S)$ increases with $\lambda_2(L_S)$ and decreases with $\mathrm{cut}(S)$, by Lemma S28.3(a). □

*Proof of Proposition S4.24.* (a) By the Cauchy–Schwarz inequality, $1 = (\phi_C^\top \phi_C)^2 \le (\phi_C^\top \Sigma \phi_C)(\phi_C^\top \Pi \phi_C)$. By Lemma S4.30, $(\Sigma_{CC})^{-1} \succeq \varepsilon^\Pi I + L_C$, so $\Sigma_{CC} \preceq (\varepsilon^\Pi I + L_C)^{-1}$, whose restriction to the orthogonal complement of $\mathbf{1}_C$ has norm $1/(\varepsilon^\Pi + \lambda_2(L_C)) = 1/\varkappa_{\mathrm{in}}(C)$. (b) $\sum_{i\in C} \sigma_i \mathrm{Corr}(e_i, \bar{e}_C) = \sum_i \mathrm{Cov}(e_i, \bar{e}_C)/\mathrm{sd}(\bar{e}_C) = n\,\mathrm{sd}(\bar{e}_C)$ and $\sum_i \sigma_i \le (n \operatorname{tr} \Sigma_{CC})^{1/2}$, so the left side of (S19) is at least $(\mathsf{V}/\operatorname{tr}\Sigma_{CC})^{1/2}$ with $\mathsf{V} = \mathrm{Var}(\phi_C^\top e) = n\,\mathrm{Var}(\bar{e}_C)$. Computing the trace in an orthonormal basis that contains $\phi_C$, part (a) gives $\operatorname{tr}\Sigma_{CC} \le \mathsf{V} + (n-1)/\varkappa_{\mathrm{in}}(C)$, and $\mathsf{V}/(\mathsf{V} + (n-1)/\varkappa_{\mathrm{in}})$ increases in $\mathsf{V} \ge 1/\varkappa_{\mathrm{out}}$, which gives $r/(r+n-1)$. (c) For an isolated region $\Sigma = (\varepsilon^\Pi I + L_C)^{-1}$, so $\mathrm{Cov}(e_i, \bar{e}_C) = \mathrm{Var}(\bar{e}_C) = 1/(n\varepsilon^\Pi)$ and $\Sigma_{ii} = 1/(n\varepsilon^\Pi) + \sum_{k\ge 2} v_k(i)^2/(\varepsilon^\Pi + \lambda_k(L_C))$ for orthonormal eigenvectors $v_k$. Hence $\mathrm{Corr}(e_i, \bar{e}_C)^{-2} = n\varepsilon^\Pi \Sigma_{ii}$, and averaging with $\sum_i v_k(i)^2 = 1$ gives the identity. For an interval with coupling $a^\Pi$, the Dirichlet principle of Lemma S28.5 gives $R_{\mathrm{eff}}(i,j) = |i-j|/a^\Pi$, the minimizer being linear between $i$ and $j$. Since $R_{\mathrm{eff}}(i,j) = L^+_{ii} + L^+_{jj} - 2L^+_{ij}$ and $L^+\mathbf{1} = 0$, $\sum_{i<j} R_{\mathrm{eff}}(i,j) = n \operatorname{tr} L_C^+$, so $\sum_{k\ge 2} 1/\lambda_k = \operatorname{tr} L_C^+ = n^{-1} \sum_{i<j} (j-i)/a^\Pi = (n^2-1)/(6a^\Pi)$. At contrast $r$, $\varepsilon^\Pi = \lambda_2(L_C)/(r-1)$ with $\lambda_2(L_C) = 2a^\Pi(1 - \cos(\pi/n)) \le a^\Pi \pi^2/n^2$, so $\varepsilon^\Pi T_C \le \varepsilon^\Pi \operatorname{tr} L_C^+ \le \pi^2(n^2-1)/(6n^2(r-1))$. □

*Proof of Lemma 4.20.* (a) For $\alpha \neq \beta$ the blocks are disjoint, so every term of $\Pi^{\mathcal{P}}_{\alpha\beta}$ is an off-diagonal entry of $\Pi$ and $\Pi^{\mathcal{P}}_{\alpha\beta} = -\operatorname{cut}(C_\alpha, C_\beta) \leq 0$; the row sums are $\sum_\beta \Pi^{\mathcal{P}}_{\alpha\beta} = \sum_{i\in C_\alpha} (\Pi \mathbf{1})_i = \sum_{i\in C_\alpha} \varepsilon^{\Pi}_i \geq 0$. The matrix $\Pi^{\mathcal{P}}$ is symmetric, and because the blocks are disjoint and nonempty, $\mathsf{P}^\top$ has full column rank, so $h \neq 0$ gives $h^\top \Pi^{\mathcal{P}} h = (\mathsf{P}^\top h)^\top \Pi (\mathsf{P}^\top h) > 0$. (b) In the orthonormal basis $[\Phi, \Phi_\perp]$ write $e = \Phi e_\Phi + \Phi_\perp e_\perp$, with $e_\Phi = \Phi^\top e$ and $e_\perp = \Phi_\perp^\top e$. The pair $(e_\Phi, e_\perp)$ is Gaussian with precision $\begin{pmatrix} \Phi^\top \Pi \Phi & \Phi^\top \Pi \Phi_\perp \\ \Phi_\perp^\top \Pi \Phi & \Phi_\perp^\top \Pi \Phi_\perp \end{pmatrix}$. Given $e_\perp = 0$, $e_\Phi$ is centered with precision $\Phi^\top \Pi \Phi$; without conditioning its precision is the Schur complement $\Phi^\top \Pi \Phi - \mathcal{D}$, with $\mathcal{D} \succeq 0$ because $\Phi_\perp^\top \Pi \Phi_\perp \succ 0$. Since $\mathsf{P} = \operatorname{diag}(n_\alpha)^{1/2} \Phi^\top$, the block averages are $\bar{e} = \operatorname{diag}(n_\alpha)^{-1/2} e_\Phi$, so their precision is $\operatorname{diag}(n_\alpha)^{1/2} \Phi^\top \Pi \Phi \operatorname{diag}(n_\alpha)^{1/2} = \Pi^{\mathcal{P}}$ given $e_\perp = 0$ and $\Pi^{\mathcal{P}} - \operatorname{diag}(n_\alpha)^{1/2} \mathcal{D} \operatorname{diag}(n_\alpha)^{1/2}$ without conditioning. The correction vanishes, $\mathcal{D} = 0$, exactly when $\Phi_\perp^\top \Pi \Phi = 0$, that is when $\Pi$ maps the span of $\Phi$, which is the span of the block indicators, into itself; with uniform leak this is the equitable condition, as in the proof of Theorem 4.18(ii). (c) Every block indicator of $\mathcal{P}_{l+1}$ is the sum of the block indicators of $\mathcal{P}_l$ that it contains, which is $\mathsf{P}_{l+1} = \mathsf{Q}_l \mathsf{P}_l$, and then $\mathsf{P}_{l+1} \Pi \mathsf{P}_{l+1}^\top = \mathsf{Q}_l \mathsf{P}_l \Pi \mathsf{P}_l^\top \mathsf{Q}_l^\top$. (d) With $n_\alpha = s$, $\Phi = s^{-1/2} \mathsf{P}^\top$, so $\Pi^{\mathcal{P}} = s \Phi^\top \Pi \Phi$, and contrast and the ratio of leaks are unchanged when a precision is multiplied by a positive constant. For unequal blocks $\Pi^{\mathcal{P}} = \operatorname{diag}(n_\alpha)^{1/2} \Phi^\top \Pi \Phi \operatorname{diag}(n_\alpha)^{1/2}$ is a congruence by a diagonal matrix that is not a multiple of the identity, which changes contrasts in general (Check S47.33). The orthonormal block precision $\Phi^\top \Pi \Phi$ can also leave the diffusive class: for a block of three agents on a path with unit couplings and a block of one agent coupled to an end of the path with coupling $\frac{1}{2}$, at leak 0.01, the first row sum of $\Phi^\top \Pi \Phi$ is $0.53/3 - 0.5/\sqrt{3} < 0$. □

*Proof of Proposition S4.26.* (a) Collective variables of $C_k$ are collective variables of $C$, so a variable that is not collective for $C$ is not collective for $C_k$. A member variable of $C_k$ in $\mathsf{S}(C)$ reads such a variable and lies in $\mathsf{S}(C_k)$; one in $\mathsf{A}(C)$ drives such a variable and lies in $\mathsf{A}(C_k)$ unless it reads outside $C_k$, in which case it lies in $\mathsf{S}(C_k)$. For exactness, use Theorem 4.3(b) with all other channels separate. A channel of a member of $C_1$ that superposes a message from a member of $C_2$, read only by members of $C$ and so internal to $C$, and a message from a non-member of $C$ reads two variables exterior to $C_1$, so it is no collider for $C_1$, but it is a collider for $C$. A channel of a member of $C_1$ that superposes messages from another member of $C_1$, internal to $C_1$, and from a member of $C_2$ is a collider for $C_1$, while for $C$ the channel and both sources are internal.

(b) Let $B(x, r)$ and $B(y, s)$ be closed $u$-balls with $r \leq s$ and a common point $z$. For $w \in B(x, r)$, $u(w, y) \leq \max\{u(w, x), u(x, z), u(z, y)\} \leq s$, so $B(x, r) \subseteq B(y, s)$; concatenating chains shows that $u$ is an ultrametric. A set $C$ with $\mathcal{U}(C) > 0$ is a ball: at any height $h \in [u_{\mathrm{in}}(C), \varpi_{\mathrm{out}}(C))$ its members are equivalent, and a chain from a member to a non-member contains a link across the boundary of weight at least $\varpi_{\mathrm{out}}(C) > h$. For several modalities, with $u_\vartheta = \max_c u_{(c)}/\vartheta_c$,

$$\frac{u_{(c)}(i,k)}{\vartheta_c} \leq \frac{\max\{u_{(c)}(i,j), u_{(c)}(j,k)\}}{\vartheta_c} \leq \max\{u_\vartheta(i,j), u_\vartheta(j,k)\},$$

and taking the maximum over $c$ shows that $u_\vartheta$ is an ultrametric, whose balls of radius one are the intersections $\bigcap_c B_{u_{(c)}}(i, \vartheta_c)$. For crossing, take three agents with $\varpi^{(1)}_{ij}$ and $\varpi^{(2)}_{jk}$ small and all other weights large: at a height between the small and the large weights, modality 1 has the ultrametric class $\{i, j\}$ and modality 2 the class $\{j, k\}$, and $\min_c u_{(c)}$ is small on $(i, j)$ and $(j, k)$ and large on $(i, k)$, violating the ultrametric inequality.

(c) The canonical partition is defined for channels of every modality, so a channel that reads or drives a non-member lies in the blanket whatever its modality. The modality-restricted partition places such a channel of another modality in the interior although it reads or drives an exterior

variable, which violates $A_{\mathsf{IE}} = 0$ or $A_{\mathsf{EI}} = 0$. In the split partition the only arrows between interior and exterior are the within-member couplings named in the statement, which are direct interior-exterior entries of $A$, and Theorem 4.3(a) applies. □

*Proof of Theorem 4.23.* Part (a) combines Theorem 4.3, Proposition 4.4 and Proposition S4.3; a one-hop shell in which no shell agent reads both sides has a vanishing collider sum. Part (b) applies Theorems S4.5 and S4.6 to the world-conditional law, which is Gaussian by Proposition 4.8, with the constants of Theorem S4.6(c) under time-invariant dynamics with local noise and row-sum contraction, and Theorem 4.9(c) to the reversible and normal isotropic cases. Part (c) is Proposition 4.6, and (e) is Proposition S4.10 with Definition 4.12.

For (d), the bound by memory, the restoration by the filtered estimate and the reversible case are Proposition S4.4. That the instantaneous blanket can fail is shown by one agent whose sensory channel $s$ reads a world variable $e$ and whose internal state $i$ reads $s$: $X_{e,t+1} = \varphi X_{e,t} + \Xi^e_t$ with $0 < \varphi < 1$, $X_{s,t+1} = X_{e,t} + \Xi^s_t$ and $X_{i,t+1} = X_{s,t} + \Xi^i_t$, with unit noise variances, $\mathsf{I} = \{i\}$, $\mathsf{B} = \{s\}$, $\mathsf{E} = \{e\}$. No variable reads both sides, so the path blanket holds by Theorem 4.3. In the stationary law, with $\sigma^2 = 1/(1-\varphi^2)$, $\mathrm{Cov}(X_{e,t}, X_{s,t}) = \varphi\sigma^2$, $\mathrm{Var}(X_{s,t}) = \sigma^2 + 1$, $\mathrm{Cov}(X_{i,t}, X_{s,t}) = \varphi\sigma^2$ and $\mathrm{Cov}(X_{i,t}, X_{e,t}) = \varphi^2\sigma^2$, since $X_{i,t}$ depends on the world through $X_{e,t-2}$. The partial covariance of $X_{i,t}$ and $X_{e,t}$ given $X_{s,t}$ is $\varphi^2\sigma^2 - \varphi^2\sigma^4/(\sigma^2+1) = \varphi^2\sigma^2/(\sigma^2+1) > 0$, so the instantaneous leak is positive; it is 0.039 nats at $\varphi = 0.9$ (Check S47.16). □

## S29 Proofs for Section 5

The proofs follow the order of Section 5. Two auxiliary lemmas, a four-cycle criterion for exact potential games and a bound on exponential tilts, are proved where they are first needed. For a matrix $X$, $\mathcal{F}(X) = \{z^*Xz : |z| = 1\}$ denotes its numerical range; it contains the eigenvalues of $X$, $\mathcal{F}(X^\top) = \mathcal{F}(X)$ for real $X$, and $\|(X-c)^{-1}\| \le 1/\operatorname{dist}(c, \mathcal{F}(X))$ for $c \notin \mathcal{F}(X)$, because $|(X-c)z| \ge |z^*(X-c)z|$ for unit $z$.

*Proof of Theorem 5.2.* In the coordinates $(y, x_\perp) = \mathcal{Z}x$ the recursion (45) reads $y_{t+1} = A_{yy}y_t + A_{y\perp}x_{\perp,t} + B_y x_{\mathsf{B},t} + \mathfrak{n}^y_t$ and $x_{\perp,t+1} = A_{\perp y}y_t + A_{\perp\perp}x_{\perp,t} + B_\perp x_{\mathsf{B},t} + \mathfrak{n}^\perp_t$. Variation of constants in the second equation gives

$$x_{\perp,t} = A^t_{\perp\perp}x_{\perp,0} + \sum_{s=0}^{t-1} A^{t-1-s}_{\perp\perp}(A_{\perp y}y_s + B_\perp x_{\mathsf{B},s} + \mathfrak{n}^\perp_s),$$

and substituting this in the first equation, with $l = t - s$, gives (46).

(a) In stationarity $\mathrm{Cov}(y_t, x_{\perp,t}) = \mathsf{P}\Sigma Z^\top = 0$, and $\mathfrak{n}^\perp_t = Z\mathfrak{n}_t$ is independent of $(y_t, x_{\perp,t})$, so $\Sigma_\perp = Z\Sigma Z^\top$ satisfies

$$\Sigma_\perp - A_{\perp\perp}\Sigma_\perp A^\top_{\perp\perp} = A_{\perp y}\mathsf{P}\Sigma\mathsf{P}^\top A^\top_{\perp y} + ZQ^{\mathfrak{n}}Z^\top \succeq ZQ^{\mathfrak{n}}Z^\top \succ 0,$$

the last because $Z$ has full row rank. For a left eigenvector $f^*$ of $A_{\perp\perp}$ with eigenvalue $\varpi$ this gives $(1-|\varpi|^2)f^*\Sigma_\perp f \ge f^*ZQ^{\mathfrak{n}}Z^\top f > 0$, so $|\varpi| < 1$. The noises $\mathfrak{n}_s$, $s \ge 0$, are independent of $x_0$, hence $\mathrm{Cov}(\mathfrak{f}_{t+1}, y_0) = A_{y\perp}A^t_{\perp\perp}Z\Sigma\mathsf{P}^\top = 0$.

(b) The same computation applies to $\dot{x}_\perp = -M_{\perp y}y - M_{\perp\perp}x_\perp + \text{noise}$, whose solution $x_\perp(t) = e^{-M_{\perp\perp}t}x_\perp(0) - \int_0^t e^{-M_{\perp\perp}(t-t')}M_{\perp y}y(t')\mathrm{d}t' + \text{noise}$ is substituted in $\dot{y} = -M_{yy}y - M_{y\perp}x_\perp + \text{noise}$. □

*Proof of Theorem 5.3.* (a) The proof of Theorem 5.2 uses no property of $A$, $B$ or $Q^{\mathfrak{n}}$ beyond linearity. For a coordinate selection take $\mathsf{P} = (I\ 0)$ and $Z = (0\ I)$ in the ordering $(R, \bar{R})$; then $A_{y\perp} = W_{R\bar{R}}$, $A_{\perp\perp} = W_{\bar{R}\bar{R}}$ and $A_{\perp y} = W_{\bar{R}R}$.

(b) As $x_\perp$ ranges over $\mathbb{R}^{\dim x-\dim y}$, $\mathcal{Z}^{-1}(0,x_\perp)$ ranges over $\ker \mathsf{P}$, and $A_{y\perp}x_\perp = \mathsf{P}A\mathcal{Z}^{-1}(0,x_\perp)$. Hence $A_{y\perp}=0$ exactly when $\mathsf{P}A$ vanishes on $\ker\mathsf{P}$, which is (i), whatever the complement. Every memory, initial-state and colored-noise term of (46) has $A_{y\perp}$ as its left factor, so (ii) implies (iii); conversely, the initial-state term at $t=0$ is $A_{y\perp}x_{\perp,0}$, which vanishes for every initial law only if $A_{y\perp}=0$. The equivalence of (i) and (iv) is proved in (c). For a coordinate selection, (ii) is $W_{R\bar{R}}=0$ by (a).

(c) Given $x_t$, $y_{t+1}\sim\mathcal{N}(\mathsf{P}Ax_t,\mathsf{P}Q^{\mathfrak{n}}\mathsf{P}^\top)$; given $y_t$, $x_t$ has covariance $\Sigma_{x|y}$, so $\mathrm{Cov}(y_{t+1}\mid y_t)=\mathsf{P}A\Sigma_{x|y}A^\top\mathsf{P}^\top+\mathsf{P}Q^{\mathfrak{n}}\mathsf{P}^\top$. Since $y_t$ is a function of $x_t$, $\mathfrak{d}_{\mathrm{cl}}=I(y_{t+1};x_t\mid y_t)$ is half the difference of the logarithms of the two conditional determinants, which is (47). The stationary covariance satisfies $\Sigma\succeq Q^{\mathfrak{n}}\succ 0$. Write $\Sigma_{x|y}=\Sigma^{1/2}(I-\mathsf{P}_U)\Sigma^{1/2}$, with $\mathsf{P}_U$ the orthogonal projector onto $U=\mathrm{range}(\Sigma^{1/2}\mathsf{P}^\top)$. A vector $f$ lies in $U^\perp$ exactly when $\mathsf{P}\Sigma^{1/2}f=0$, so the range of $\Sigma_{x|y}$ is $\Sigma^{1/2}U^\perp=\ker\mathsf{P}$. The matrix in the determinant is the identity plus a matrix similar to the positive semidefinite $(\mathsf{P}Q^{\mathfrak{n}}\mathsf{P}^\top)^{-1/2}\mathsf{P}A\Sigma_{x|y}A^\top\mathsf{P}^\top(\mathsf{P}Q^{\mathfrak{n}}\mathsf{P}^\top)^{-1/2}$, so $\mathfrak{d}_{\mathrm{cl}}=0$ exactly when $\mathsf{P}A\Sigma_{x|y}A^\top\mathsf{P}^\top=0$, that is, when $\mathsf{P}A$ vanishes on the range of $\Sigma_{x|y}$, which is $\ker\mathsf{P}$; this proves (i)$\Leftrightarrow$(iv). By the chain rule and the Markov property of $x$,

$$I(y_{t+1};y_{0:t-1}\mid y_t)\le I(y_{t+1};y_{0:t-1},x_t\mid y_t)=\mathfrak{d}_{\mathrm{cl}}+I(y_{t+1};y_{0:t-1}\mid x_t,y_t)=\mathfrak{d}_{\mathrm{cl}}.$$

This holds for every history length $t$. In the stationary state the left side equals $I(y_1;y_{-t:-1}\mid y_0)$, and lengthening the history by one round adds the nonnegative term $I(y_1;y_{-t-1}\mid y_{-t:0})$, so the left side is nondecreasing in $t$ and its limit $\mathfrak{d}_{\mathrm{M}}$ obeys the same bound.

(d) Because the input path is known and independent of $(x_0,\mathfrak{n})$, conditioning on it amounts to treating it as a deterministic affine term. For a Gaussian initial law, $(x_{\perp,t},y_{0:t})$ is then jointly Gaussian, and the law of $x_{\perp,t}$ given $(y_{0:t},x_{\mathsf{B},0:t})$ is $\mathcal{N}(\hat{x}_{\perp,t},\mathcal{V}_t)$. The pair $(y_{t+1},x_{\perp,t+1})$ is an affine function of $(y_t,x_{\perp,t},x_{\mathsf{B},t})$ plus noise independent of the past, so its conditional law given the past is Gaussian, with mean $(A_{yy}y_t+A_{y\perp}\hat{x}_{\perp,t}+B_yx_{\mathsf{B},t},A_{\perp y}y_t+A_{\perp\perp}\hat{x}_{\perp,t}+B_\perp x_{\mathsf{B},t})$ and a covariance built from $\mathcal{V}_t$ and $\mathcal{Z}Q^{\mathfrak{n}}\mathcal{Z}^\top$. Conditioning on the observed $y_{t+1}$ gives $(\hat{x}_{\perp,t+1},\mathcal{V}_{t+1})$; this is the Kalman recursion, and $\mathcal{V}_t$ does not depend on the data. The predictive law of $y_{t+1}$ is therefore a function of $(y_t,\hat{x}_{\perp,t})$ and of the input, and $\hat{x}_{\perp,t+1}$ is a function of these and of $y_{t+1}$, so the pair is a Markov state for the law of $y$ given the input path. Since $\hat{x}_{\perp,0}$ is a function of $y_0$, the process $(y_t,\hat{x}_{\perp,t})$ generates the same past as $y$, and adjoining the $\dim x-\dim y$ coordinates of $\hat{x}_{\perp,t}$ to $y$ gives a Markov process, whose memory residual vanishes. This bounds the closure order. □

*Proof of Theorem 5.4.* Use the orthonormal coordinates $y=\phi_C^\top x$ and $x_\perp=Zx$, so that $x=y\phi_C+Z^\top x_\perp$ and the noise stays standard. In these coordinates $M$ becomes

$$\tilde{M}=\begin{pmatrix}\varkappa_{\mathrm{out}}^M & f_{\mathrm{col}}^\top\\ f_{\mathrm{row}} & M_{\perp\perp}\end{pmatrix},\qquad M_{\perp\perp}=ZMZ^\top,$$

since $\phi_C^\top MZ^\top=(ZM^\top\phi_C)^\top$. In case (a) $M$ is symmetric, $\varkappa_{\mathrm{out}}^M=\varkappa_{\mathrm{out}}$, and $f_{\mathrm{row}}=f_{\mathrm{col}}=ZM\phi_C=Z(M-\varkappa_{\mathrm{out}})\phi_C$; its norm is $\sigma_C$, because $(M-\varkappa_{\mathrm{out}})\phi_C=(I-\phi_C\phi_C^\top)M\phi_C$ is orthogonal to $\phi_C$ and $Z^\top Z=I-\phi_C\phi_C^\top$, and its entries are $(\varepsilon_i^\Pi+a_i^\partial-\varkappa_{\mathrm{out}})/\sqrt{n}$, because the row sums of $\Pi_{CC}$ are $\varepsilon_i^\Pi+a_i^\partial$.

Kernel. By Theorem 5.2(b) the kernel is $\Gamma(u)=f_{\mathrm{col}}^\top e^{-M_{\perp\perp}u}f_{\mathrm{row}}$. Let $\varkappa_*$ be $\varkappa_{\mathrm{in}}^M$ in case (b) and $\varkappa_{\mathrm{in}}$ in case (a). Then $ZM_{\mathrm{sym}}Z^\top\succeq\varkappa_*I$: in case (b) by the definition of $\varkappa_{\mathrm{in}}^M$, and in case (a) because $ZMZ^\top=Z(\mathrm{diag}(\varepsilon^\Pi)+L_C)Z^\top+Z\,\mathrm{diag}(a^\partial)Z^\top$, where the first term is at least $\varkappa_{\mathrm{in}}I$ by the definition of $\varkappa_{\mathrm{in}}$ in Definition 4.10, the vectors $Z^\top z$ being orthogonal to $\mathbf{1}$ and supported on $C$, and the second is positive semidefinite. For every $z$, $\frac{\mathrm{d}}{\mathrm{d}t}|e^{-M_{\perp\perp}t}z|^2=-2(e^{-M_{\perp\perp}t}z)^\top ZM_{\mathrm{sym}}Z^\top(e^{-M_{\perp\perp}t}z)\le$

$-2\varkappa_*|e^{-M_{\perp\perp}t}z|^2$, so $\|e^{-M_{\perp\perp}u}\| \le e^{-\varkappa_* u}$ and $|\Gamma(u)| \le \sigma_{\mathrm{col}}\sigma_{\mathrm{row}}e^{-\varkappa_* u}$. In case (a), $M_{\perp\perp}$ is symmetric positive definite, so $\Gamma(u) = f_{\mathrm{row}}^\top e^{-M_{\perp\perp}u} f_{\mathrm{row}} \ge 0$. This proves (b)(i) and the kernel bound of (a).

Case (a), slow mode. Let $M$ have eigenvalues $\varpi_1 \le \varpi_2 \le \cdots \le \varpi_n$ with orthonormal eigenvectors $\mathsf{u}_1, \dots, \mathsf{u}_n$. By Courant–Fischer, $\varpi_2 \ge \min\{f^\top M f : |f| = 1, f \perp \phi_C\} = \lambda_{\min}(ZMZ^\top) \ge \varkappa_{\mathrm{in}}$, and $\varpi_1 \le \phi_C^\top M \phi_C = \varkappa_{\mathrm{out}} < \varkappa_{\mathrm{in}} \le \varpi_2$, so $\varpi_1$ is simple. Write $\phi_C = (\cos\angle)\mathsf{u}_1 + (\sin\angle)w$, with $\angle$ the angle between $\phi_C$ and $\mathsf{u}_1$ (the sign of $\mathsf{u}_1$ chosen so that $\cos\angle \ge 0$) and $w$ a unit vector orthogonal to $\mathsf{u}_1$. Then $(M-\varkappa_{\mathrm{out}})\phi_C = (\cos\angle)(\varpi_1 - \varkappa_{\mathrm{out}})\mathsf{u}_1 + (\sin\angle)(M-\varkappa_{\mathrm{out}})w$, the two terms are orthogonal, and $w$ lies in the span of eigenvectors with eigenvalues at least $\varpi_2 > \varkappa_{\mathrm{out}}$, so $\sigma_C^2 \ge \sin^2\angle(\varpi_2 - \varkappa_{\mathrm{out}})^2$ and $\sin\angle \le \sigma_C/(\varpi_2 - \varkappa_{\mathrm{out}}) \le \sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}})$. For the tangent, write $\mathsf{u}_1 = (\cos\angle)\phi_C + (\sin\angle)w'$ with $w'$ a unit vector orthogonal to $\phi_C$, and apply $Z$ to $M\mathsf{u}_1 = \varpi_1\mathsf{u}_1$: since $Z\phi_C = 0$ and $w' = Z^\top Z w'$, this reads $(\sin\angle)(M_{\perp\perp} - \varpi_1)Zw' = -(\cos\angle)f_{\mathrm{row}}$, with $M_{\perp\perp} = ZMZ^\top \succeq \varkappa_{\mathrm{in}} I$ and $|Zw'| = 1$. The left side has norm at least $(\sin\angle)(\varkappa_{\mathrm{in}} - \varpi_1) \ge (\sin\angle)(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}})$ and the right side has norm $(\cos\angle)\sigma_C$, so $\cos\angle > 0$ and $\tan\angle \le \sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}})$. The matrix $(M-\varpi_1)(M-\varpi_2)$ is positive semidefinite, its eigenvalues being $(\varpi_l - \varpi_1)(\varpi_l - \varpi_2) \ge 0$, and since $|M\phi_C|^2 = \varkappa_{\mathrm{out}}^2 + \sigma_C^2$,

$$0 \le \phi_C^\top (M-\varpi_1)(M-\varpi_2)\phi_C = \sigma_C^2 - (\varkappa_{\mathrm{out}} - \varpi_1)(\varpi_2 - \varkappa_{\mathrm{out}}),$$

so $0 \le \varkappa_{\mathrm{out}} - \varpi_1 \le \sigma_C^2/(\varpi_2 - \varkappa_{\mathrm{out}}) \le \sigma_C^2/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}})$. The angle bound is a scalar case of the sine theorem of Davis and Kahan (1970); both bounds are proved here directly.

Case (a), Markov error. The stationary covariance of $\mathrm{d}x = -Mx\mathrm{d}t + \sqrt{2}\mathsf{dB}$ solves $M\Sigma + \Sigma M = 2I$, so $\Sigma = M^{-1}$, and with $\mathsf{c}_l = \phi_C^\top \mathsf{u}_l$,

$$\mathrm{Cov}(y_{t+u}, y_t) = \phi_C^\top e^{-Mu} M^{-1} \phi_C = \sum_l \frac{\mathsf{c}_l^2}{\varpi_l} e^{-\varpi_l u}.$$

Let $\mathfrak{m} = \sum_{l\ge 2}(\mathsf{c}_l^2/\varpi_l)/\sum_l(\mathsf{c}_l^2/\varpi_l)$, the memory weight. The autocorrelation is $(1-\mathfrak{m})e^{-\varpi_1 u}$ plus $\mathfrak{m}$ times a convex combination of $e^{-\varpi_l u}$, $l \ge 2$, each in $[0, e^{-\varpi_1 u}]$, so $\Xi \le \mathfrak{m}$. Since $\sum_{l\ge 2}\mathsf{c}_l^2 = \sin^2\angle$ and $\mathsf{c}_1^2 = \cos^2\angle > 0$, $\mathfrak{m} \le (\sin^2\angle/\varpi_2)/(\cos^2\angle/\varpi_1) = \tan^2\angle \cdot \varpi_1/\varpi_2$. With $\tan\angle \le \sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}})$ and $\varpi_1/\varpi_2 \le \varkappa_{\mathrm{out}}/\varkappa_{\mathrm{in}}$, this gives (49), and no condition on $\sigma_C$ was used.

Case (b)(ii). Write $\mathsf{g} = \varkappa_{\mathrm{in}}^M - \varkappa_{\mathrm{out}}^M$ and $T_\perp = M_{\perp\perp} - \varkappa_{\mathrm{out}}^M I$. For a unit complex vector $z$, $\mathrm{Re}(z^* T_\perp z) = z^* Z M_{\mathrm{sym}} Z^\top z - \varkappa_{\mathrm{out}}^M \ge \mathsf{g}$, so $\|T_\perp^{-1}\| \le 1/\mathsf{g}$. A vector $(1, \mathsf{p})$ spans a right-invariant line of $\tilde{M}$ exactly when $T_\perp \mathsf{p} = \mathsf{p}(f_{\mathrm{col}}^\top \mathsf{p}) - f_{\mathrm{row}}$, that is, when $\mathsf{p}$ is a fixed point of $\mathcal{H}(\mathsf{p}) = T_\perp^{-1}(\mathsf{p}(f_{\mathrm{col}}^\top \mathsf{p}) - f_{\mathrm{row}})$. On the real ball of radius $\mathsf{r}_- = 2\sigma_{\mathrm{row}}/(\mathsf{g} + (\mathsf{g}^2 - 4\sigma_{\mathrm{row}}\sigma_{\mathrm{col}})^{1/2}) \le 2\sigma_{\mathrm{row}}/\mathsf{g}$, the smaller root of $\sigma_{\mathrm{col}}\mathsf{r}^2 - \mathsf{g}\mathsf{r} + \sigma_{\mathrm{row}} = 0$, one has $|\mathcal{H}(\mathsf{p})| \le (\sigma_{\mathrm{col}}\mathsf{r}_-^2 + \sigma_{\mathrm{row}})/\mathsf{g} = \mathsf{r}_-$, and since $\mathsf{p}(f_{\mathrm{col}}^\top \mathsf{p}) - \mathsf{p}'(f_{\mathrm{col}}^\top \mathsf{p}') = (\mathsf{p} - \mathsf{p}')(f_{\mathrm{col}}^\top \mathsf{p}) + \mathsf{p}'(f_{\mathrm{col}}^\top(\mathsf{p} - \mathsf{p}'))$, $\mathcal{H}$ is a contraction there with constant $2\sigma_{\mathrm{col}}\mathsf{r}_-/\mathsf{g} = 1 - (1 - 4\sigma_{\mathrm{row}}\sigma_{\mathrm{col}}/\mathsf{g}^2)^{1/2} < 1$. Its fixed point $\mathsf{p}$ is real, $\varpi_1 = \varkappa_{\mathrm{out}}^M + f_{\mathrm{col}}^\top \mathsf{p}$ is a real eigenvalue with $|\varpi_1 - \varkappa_{\mathrm{out}}^M| \le \sigma_{\mathrm{col}}\mathsf{r}_- \le 2\sigma_{\mathrm{row}}\sigma_{\mathrm{col}}/\mathsf{g}$, and the right eigenvector $\phi_C + Z^\top \mathsf{p}$ makes with $\phi_C$ an angle of tangent $|\mathsf{p}| \le 2\sigma_{\mathrm{row}}/\mathsf{g}$. With $\mathcal{L}_\mathsf{p} = \begin{pmatrix} 1 & 0 \\ \mathsf{p} & I \end{pmatrix}$,

$$\mathcal{L}_\mathsf{p}^{-1} \tilde{M} \mathcal{L}_\mathsf{p} = \begin{pmatrix} \varpi_1 & f_{\mathrm{col}}^\top \\ 0 & M_2 \end{pmatrix}, \qquad M_2 = M_{\perp\perp} - \mathsf{p} f_{\mathrm{col}}^\top,$$

and $\mathrm{Re}\,\mathcal{F}(M_2) \ge \varkappa_{\mathrm{in}}^M - |\mathsf{p}|\sigma_{\mathrm{col}} \ge \varkappa_{\mathrm{in}}^M - 2\sigma_{\mathrm{row}}\sigma_{\mathrm{col}}/\mathsf{g}$, which bounds the real parts of the other eigenvalues. Since $4\sigma_{\mathrm{row}}\sigma_{\mathrm{col}} < \mathsf{g}^2$, $\varpi_1 \le \varkappa_{\mathrm{out}}^M + 2\sigma_{\mathrm{row}}\sigma_{\mathrm{col}}/\mathsf{g} < \varkappa_{\mathrm{in}}^M - 2\sigma_{\mathrm{row}}\sigma_{\mathrm{col}}/\mathsf{g}$, so $\varpi_1$ is not an eigenvalue of $M_2$; it is simple and of smallest real part. The same argument for $\tilde{M}^\top$, with $f_{\mathrm{row}}$ and $f_{\mathrm{col}}$ exchanged and $M_{\perp\perp}^\top$, which has the same symmetric part, gives a left eigenvector at angle of tangent at most $2\sigma_{\mathrm{col}}/\mathsf{g}$ whose eigenvalue lies within $2\sigma_{\mathrm{row}}\sigma_{\mathrm{col}}/\mathsf{g}$ of $\varkappa_{\mathrm{out}}^M$ and is therefore $\varpi_1$. This is the invariant-subspace argument of Stewart (1973) in the form needed here.

Case (b)(iii). Let $\mathsf{k} = (M_2^\top - \varpi_1)^{-1} f_{\mathrm{col}}$ and $\mathcal{U}_\mathsf{k} = \begin{pmatrix} 1 & \mathsf{k}^\top \\ 0 & I \end{pmatrix}$. Then $(\mathcal{L}_\mathsf{p}\mathcal{U}_\mathsf{k})^{-1}\tilde{M}(\mathcal{L}_\mathsf{p}\mathcal{U}_\mathsf{k}) = \mathrm{diag}(\varpi_1, M_2)$, because $\varpi_1\mathsf{k}^\top + f_{\mathrm{col}}^\top - \mathsf{k}^\top M_2 = 0$, and

$$(\mathcal{L}_\mathsf{p}\mathcal{U}_\mathsf{k})^{-1} = \begin{pmatrix} 1+\mathsf{k}^\top\mathsf{p} & -\mathsf{k}^\top \\ -\mathsf{p} & I \end{pmatrix} =: \begin{pmatrix} \mathbf{a}^\top \\ \mathbf{G} \end{pmatrix}.$$

In the coordinates $(\tilde{y}, \tilde{x}) = (\mathcal{L}_\mathsf{p}\mathcal{U}_\mathsf{k})^{-1}(y, x_\perp)$ the drift is block diagonal, the noise covariance is $2(\mathcal{L}_\mathsf{p}\mathcal{U}_\mathsf{k})^{-1}(\mathcal{L}_\mathsf{p}\mathcal{U}_\mathsf{k})^{-\top}$, and $y = \tilde{y} + \mathsf{k}^\top\tilde{x}$. The real parts of the eigenvalues of $M$ are positive, because $M_{\mathrm{sym}} \succ 0$, so $\varpi_1 > 0$. The stationary moments are $\mathrm{Var}(\tilde{y}) = |\mathbf{a}|^2/\varpi_1$, $\mathbf{X} = \mathrm{Cov}(\tilde{x}, \tilde{y}) = 2(M_2 + \varpi_1)^{-1}\mathbf{G}\mathbf{a}$ and $\mathbf{V} = \mathrm{Var}(\tilde{x})$ with $M_2\mathbf{V} + \mathbf{V}M_2^\top = 2\mathbf{G}\mathbf{G}^\top$. Since $\mathbb{E}[\tilde{y}_{t+u} \mid \text{past}] = e^{-\varpi_1 u}\tilde{y}_t$ and $\mathbb{E}[\tilde{x}_{t+u} \mid \text{past}] = e^{-M_2 u}\tilde{x}_t$,

$$\mathrm{Corr}(y_t, y_{t+u}) = e^{-\varpi_1 u}\frac{\mathrm{Cov}(\tilde{y}, y)}{\mathrm{Var}(y)} + \mathfrak{r}(u), \qquad \mathfrak{r}(u) = \frac{\mathsf{k}^\top e^{-M_2 u}\,\mathrm{Cov}(\tilde{x}, y)}{\mathrm{Var}(y)},$$

and at $u = 0$ the two terms add to one, so $\mathrm{Corr}(y_t, y_{t+u}) - e^{-\varpi_1 u} = -\mathfrak{r}(0)e^{-\varpi_1 u} + \mathfrak{r}(u)$. The numerical range of $M_2$ lies in the right half plane, so $\|e^{-M_2 u}\| \le 1$, and $\Xi \le 2|\mathsf{k}||\mathbf{X} + \mathbf{V}\mathsf{k}|/\mathrm{Var}(y)$ because $\mathrm{Cov}(\tilde{x}, y) = \mathbf{X} + \mathbf{V}\mathsf{k}$. Under $\sigma_{\mathrm{row}}, \sigma_{\mathrm{col}} \le \mathsf{g}/4$ the following hold in turn: $|\mathsf{p}| \le 2\sigma_{\mathrm{row}}/\mathsf{g} \le \frac{1}{2}$; $|\varpi_1 - \varkappa_{\mathrm{out}}^M| \le \mathsf{g}/8$; $\mathrm{Re}\,\mathcal{F}(M_2) \ge \varkappa_{\mathrm{in}}^M - \mathsf{g}/8 \ge \frac{7}{8}\varkappa_{\mathrm{in}}^M$, using $\varkappa_{\mathrm{out}}^M = \phi_C^\top M_{\mathrm{sym}}\phi_C > 0$; $\mathrm{dist}(\varpi_1, \mathcal{F}(M_2^\top)) = \mathrm{dist}(\varpi_1, \mathcal{F}(M_2)) \ge \frac{3}{4}\mathsf{g}$, so $|\mathsf{k}| \le 4\sigma_{\mathrm{col}}/(3\mathsf{g}) \le \frac{1}{3}$ and $|\mathsf{k}||\mathsf{p}| \le \frac{1}{6}$; evaluating the Lyapunov equation for $\mathbf{V}$ on its top eigenvector, $\|\mathbf{V}\| \le \|\mathbf{G}\|^2/(\frac{7}{8}\varkappa_{\mathrm{in}}^M) = (1+|\mathsf{p}|^2)/(\frac{7}{8}\varkappa_{\mathrm{in}}^M) \le \frac{10}{7\varkappa_{\mathrm{in}}^M}$; $|\mathbf{G}\mathbf{a}| = |\mathsf{p}(1+\mathsf{k}^\top\mathsf{p}) + \mathsf{k}| \le (\frac{7}{3}\sigma_{\mathrm{row}} + \frac{4}{3}\sigma_{\mathrm{col}})/\mathsf{g}$; and $|\mathbf{X}| \le \frac{16}{7}|\mathbf{G}\mathbf{a}|/\varkappa_{\mathrm{in}}^M$, since $\mathrm{dist}(-\varpi_1, \mathcal{F}(M_2)) \ge \frac{7}{8}\varkappa_{\mathrm{in}}^M$. Hence

$$|\mathbf{X} + \mathbf{V}\mathsf{k}| \le \Big(\frac{16}{3}\sigma_{\mathrm{row}} + \frac{104}{21}\sigma_{\mathrm{col}}\Big)\frac{1}{\mathsf{g}\varkappa_{\mathrm{in}}^M} \le \frac{16}{3}\cdot\frac{\sigma_{\mathrm{row}} + \sigma_{\mathrm{col}}}{\mathsf{g}\varkappa_{\mathrm{in}}^M}.$$

For the variance: $\mathrm{Var}(\tilde{y}) \ge (1 - |\mathsf{k}||\mathsf{p}|)^2/\varpi_1 \ge (5/6)^2/\varpi_1$, and $\mathrm{Var}(\mathsf{k}^\top\tilde{x}) \le |\mathsf{k}|^2\|\mathbf{V}\| \le \frac{10}{63}(\varkappa_{\mathrm{in}}^M)^{-1} \le \frac{10}{63}\varpi_1^{-1}$, because $\varpi_1 \le \varkappa_{\mathrm{out}}^M + \mathsf{g}/8 < \varkappa_{\mathrm{in}}^M$; by the triangle inequality in $L^2$, $\mathrm{Var}(y) \ge (\frac{5}{6} - (\frac{10}{63})^{1/2})^2/\varpi_1 \ge 0.189/\varpi_1$. Altogether

$$\Xi \le 2\cdot\frac{4\sigma_{\mathrm{col}}}{3\mathsf{g}}\cdot\frac{16}{3}\cdot\frac{\sigma_{\mathrm{row}} + \sigma_{\mathrm{col}}}{\mathsf{g}\varkappa_{\mathrm{in}}^M}\cdot\frac{\varpi_1}{0.189} \le \frac{76\sigma_{\mathrm{col}}(\sigma_{\mathrm{col}} + \sigma_{\mathrm{row}})\varpi_1}{\mathsf{g}^2\varkappa_{\mathrm{in}}^M},$$

the numerical factor being $128/(9\cdot 0.189) < 75.3$. If in addition $\sigma_{\mathrm{row}}\sigma_{\mathrm{col}} \le \varkappa_{\mathrm{out}}^M\mathsf{g}/2$, then $\varpi_1 \le \varkappa_{\mathrm{out}}^M + 2\sigma_{\mathrm{row}}\sigma_{\mathrm{col}}/\mathsf{g} \le 2\varkappa_{\mathrm{out}}^M$, which gives the second bound.

Case (b)(iv). $M\mathbf{1} = M_{\mathrm{sym}}\mathbf{1} + M_{\mathrm{anti}}\mathbf{1}$ and $M^\top\mathbf{1} = M_{\mathrm{sym}}\mathbf{1} - M_{\mathrm{anti}}\mathbf{1}$, and $M_{\mathrm{anti}}\mathbf{1} \perp \mathbf{1}$ because $\mathbf{1}^\top M_{\mathrm{anti}}\mathbf{1} = 0$; hence $\sigma_{\mathrm{row}}, \sigma_{\mathrm{col}} \le \sigma_{\mathrm{sym}} + |M_{\mathrm{anti}}\phi_C|$, and $|M_{\mathrm{anti}}\phi_C| \le \|M_{\mathrm{anti}}\| = \frac{1}{2}\|M - M^\top\|$. For $M_{ii} = \varepsilon_i + \sum_j a_{ij}$ and $M_{ij} = -a_{ij}$, $(M\mathbf{1})_i = \varepsilon_i$ and $(M^\top\mathbf{1})_i = \varepsilon_i + \sum_j a_{ij} - \sum_j a_{ji}$, so $2M_{\mathrm{anti}}\mathbf{1}$ is the vector of differences between attention received and attention given, which sums to zero. □

*Proof of Proposition 5.5.* The Lyapunov equation of the stationary law is $M\Sigma + \Sigma M^\top = 2I$, so $M\Sigma = I + J$ and $M = (I + J)\Pi$, with $J$ antisymmetric by construction. Then $M_{\mathrm{sym}} = \frac{1}{2}[(I + J)\Pi + \Pi(I - J)] = \Pi + \frac{1}{2}(J\Pi - \Pi J)$. For a unit vector $f$, $f^\top\Pi J f = f^\top J^\top\Pi f = -f^\top J\Pi f$, so $f^\top M_{\mathrm{sym}} f - f^\top\Pi f = f^\top J\Pi f = (J^\top f)^\top\Pi f$. Since $f^\top J f = 0$, the vector $J^\top f$ is orthogonal to $f$, so $(J^\top f)^\top\Pi f = (J^\top f)^\top(I - ff^\top)\Pi f$, whose modulus is at most $|J^\top f||(I - ff^\top)\Pi f| \le \|J\||(I - ff^\top)\Pi f| \le \|J\||\Pi f|$. Two functions on the unit sphere of a subspace that differ everywhere by at most a given amount have minima that differ by at most that amount. □

*Proof of Proposition 5.6.* (a) The likelihoods are conditionally independent given $\eta_C$, so the union posterior is proportional to $p_\circ \prod_i p(s_i \mid \eta_C)$, and Bayes's rule for each member gives $p(s_i \mid \eta_C) \propto q_i/p_\circ$. In natural parameters $P_i^{-1} = P_\circ^{-1} + \Lambda_i^s$, so $\sum_i P_i^{-1} - (n-1)P_\circ^{-1} = P_\circ^{-1} + \sum_i \Lambda_i^s$, and the information vectors combine in the same way. (b) The pool with weights $1/n$ has natural parameters $\frac{1}{n}\sum_i(P_\circ^{-1} + \Lambda_i^s)$, and the product of posteriors $\sum_i(P_\circ^{-1} + \Lambda_i^s)$. □

*Proof of Proposition 5.7.* (a) Since $\mathcal{W}^{n_c}$ is entrywise nonnegative and the $\Lambda_j^s$ are positive semidefinite, $\Lambda_i \succeq \Lambda_\circ$. The union posterior has $h^* = \Lambda_\circ m_\circ + n\bar{h}^s$ and $\Lambda^* = \Lambda_\circ + n\bar{\Lambda}^s$, and $\mathcal{W}^{n_c}\mathbf{1} = \mathbf{1}$, so $h_i - h^* = n\sum_j(\mathcal{W}^{n_c})_{ij}(h_j^s - \bar{h}^s)$. The stacked vector $(h_j^s - \bar{h}^s)_j$ is orthogonal to the consensus subspace, on which the symmetric matrix $\mathcal{W}^{n_c}$ acts as $(\mathcal{W} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)^{n_c}$, with norm $\rho_\perp^{n_c}$; this gives (52), and the precisions are the same computation. For the means, $\Lambda_i(m_i - m^*) = (h_i - h^*) - (\Lambda_i - \Lambda^*)m^*$ and $\|\Lambda_i^{-1}\| \le \|\Lambda_\circ^{-1}\|$. For the relative entropy, with $x_k$ the eigenvalues of $X_i$, the covariance part of $D_{\mathrm{KL}}(q_i\|p^*)$ is $\frac{1}{2}\sum_k \mathsf{b}(x_k)$ with $\mathsf{b}(x) = (1+x)^{-1} - 1 + \ln(1+x)$; $\mathsf{b}(0) = 0$ and $|\mathsf{b}'(x)| = |x|/(1+x)^2 \le 4|x|$ for $x \ge -\frac{1}{2}$, so $0 \le \mathsf{b}(x) \le 2x^2$, and the mean part is $\frac{1}{2}(m_i - m^*)^\top\Lambda^*(m_i - m^*)$.

(b) In information form the steady-state centralized filter is $\hat{\eta}^{\mathrm{c}}_{t|t} = F\hat{\eta}^{\mathrm{c}}_{t-1|t-1} + \bar{P}(\sum_j h_j^s(t) - \Lambda^s F\hat{\eta}^{\mathrm{c}}_{t-1|t-1})$, because its gain is $\bar{P}(H^s)^\top(R^s)^{-1}$ for the stacked sensing, and member $i$'s update is the same with $n\tilde{h}_i(t)$ in place of $\sum_j h_j^s(t) = n\bar{h}^s(t)$. Subtracting,

$$\mathrm{dev}_i(t) = (I - \bar{P}\Lambda^s)F\,\mathrm{dev}_i(t-1) + n\bar{P}(\tilde{h}_i(t) - \bar{h}^s(t)), \qquad I - \bar{P}\Lambda^s = \bar{P}(\bar{P}^{-1} - \Lambda^s) = \bar{P}(\bar{P}^-)^{-1},$$

so $(I - \bar{P}\Lambda^s)F = A_{\mathrm{cl}}$. From $\mathrm{dev}(0) = 0$, $\mathrm{dev}_i(t) = \sum_{l=0}^{t-1} A_{\mathrm{cl}}^l n\bar{P}(\tilde{h}_i(t-l) - \bar{h}^s(t-l))$. The consensus rounds preserve the average and contract deviations from it, $(\sum_i|\tilde{h}_i - \bar{h}^s|^2)^{1/2} \le \rho_\perp^{n_c}(\sum_i|h_i^s - \bar{h}^s|^2)^{1/2}$, and summing the geometric series gives (53); Minkowski's inequality in $L^2$ gives the mean-square form. With $h_i^s = \Lambda_i^s\eta + (H_i^s)^\top(R_i^s)^{-1}v_i$, the noise terms are independent across members with covariances $\Lambda_i^s$ and independent of $\eta$, which gives the stated dispersion. The members' filters use the centralized covariances, so their beliefs share the covariance $\bar{P}$, and the relative entropy between two Gaussian laws with one covariance is half the squared Mahalanobis distance of their means. □

*Proof of Proposition S5.5.* A level-1 receiver that reads $m_j$ over a noiseless channel as an observation of $\eta$ whose error variance is the posted $1/\Lambda_j$ (Definition 2.4, with $\hat{H}_{ij} = 1$, $\hat{P}_{ij} = 1/\Lambda_j$ and $R_c = 0$) adds $\Lambda_j$ to its precision and $\Lambda_j m_j = h_j$ to its information vector, which is the update. The matrix $I + \mathrm{Adj}$ is symmetric, nonnegative and primitive (connected, with self-loops), so by Perron–Frobenius $(I + \mathrm{Adj})^{n_c}/\rho(I + \mathrm{Adj})^{n_c} \to \mathsf{v}\mathsf{v}^\top$ as the number of rounds $n_c \to \infty$, with $\mathsf{v} > 0$; this gives (a) and (b), since a connected graph on $n \ge 2$ vertices has minimum degree at least one, so every row sum of $I + \mathrm{Adj}$ is at least 2 and $\rho(I + \mathrm{Adj}) \ge 2$. For (c), compare the coefficients of $s_j$ in the two means: $\mathsf{v}_j\Lambda_j^s/\sum_l \mathsf{v}_l(\Lambda_\circ + \Lambda_l^s)$ against $\Lambda_j^s/(\Lambda_\circ + \sum_l \Lambda_l^s)$. Equality for every $j$ forces $\mathsf{v}$ to be constant, hence equal row sums of $I + \mathrm{Adj}$, a regular graph, and then the coefficients are $\Lambda_j^s/(n\Lambda_\circ + \sum_l \Lambda_l^s)$, equal to the union's only if $n\Lambda_\circ = \Lambda_\circ$, that is, $\Lambda_\circ = 0$ for $n \ge 2$; the converse is immediate, and the prior weights follow in the same way. For (d), the weights $\mathsf{w}_\circ, \mathsf{w}_j$ sum to one, so the limit minus $\eta$ is $\mathsf{w}_\circ(m_\circ - \eta) + \sum_j \mathsf{w}_j v_j$, with independent terms of variances $1/\Lambda_\circ$ and $1/\Lambda_j^s$; the union posterior mean has the least mean-squared error among all estimators, and its error is the posterior variance. □

*Proof of Lemma 5.9.* For slot sets $A$ and $A'$ opened by different members, conditional independence of the slots given $\eta_C$ gives $I(o_A; o_{A'} \mid \eta_C) = 0$, so the chain rule applied to $I(\eta_C, o_{A'}; o_A)$ in two orders gives, under every belief,

$$I(\eta_C; o_{A\cup A'}) = I(\eta_C; o_{A'}) + I(\eta_C; o_A) - I(o_A; o_{A'}). \tag{S114}$$

Apply (S114) to (55) under $q_i$ and to (54) under $q_C$, with $A = \mathfrak{B}_i(\pi_i)$ and $A' = A_{-i}$, and subtract. The terms that involve only $o_{A_{-i}}$, member $i$'s pragmatic terms for the others' slots and the others'

controller terms in $G_C$ do not depend on $\pi_i$; what remains is $\mathcal{T}_i(\pi_i)+\mathcal{R}_i(\pi)$. Hence $G_i-G_C-\mathcal{T}_i-\mathcal{R}_i$ is a function of $\pi_{-i}$, and so is $G_i-\Phi-\mathcal{R}_i$, which differs from it by $\sum_{j\neq i}\mathcal{T}_j$. □

**Lemma S29.1** (Four-cycle criterion, after Monderer and Shapley (1996)). *Let $G_i:\mathcal{A}\to\mathbb{R}$ for $i\in C$. If $\partial^2_{ij}G_i=\partial^2_{ij}G_j$ for all members $i\neq j$, all second differences that start from the null policies ($\pi_i=\pi_j=0$) and all policies of the others, then for any ordering $1,\dots,n$ of the members the function*

$$\Phi'(\pi)=\sum_{k=1}^{n}\big[G_k(\pi^{(k)})-G_k(\pi^{(k-1)})\big],\qquad \pi^{(k)}=(\pi_1,\dots,\pi_k,0,\dots,0),$$

*is an exact potential: $G_i-\Phi'$ does not depend on $\pi_i$. Conversely, the members of an exact potential game have $\partial^2_{ij}G_i=\partial^2_{ij}G_j$ for all second differences.*

*Proof.* Let $\mathsf{h}_k(\pi)=G_k(\pi)-G_k(0,\pi_{-k})$, so that $\Phi'(\pi)=\sum_k\mathsf{h}_k(\pi^{(k)})$. Fix $i$. The terms with $k<i$ do not involve $\pi_i$. For $k>i$ let $O_k$ be the policies $(\pi_1,\dots,\pi_{i-1},\pi_{i+1},\dots,\pi_{k-1},0,\dots)$ of the members other than $i$ and $k$. The hypothesis for the pair $(i,k)$, with $i$ moving from 0 to $\pi_i$ and $k$ from 0 to $\pi_k$ at the context $O_k$, reads

$$\mathsf{h}_k(\pi_i,\pi_k,O_k)-\mathsf{h}_k(0,\pi_k,O_k)=\mathsf{h}_i(\pi_i,\pi_k,O_k)-\mathsf{h}_i(\pi_i,0,O_k),$$

and the profiles on the right are $\pi^{(k)}$ and $\pi^{(k-1)}$. Summing over $k>i$, the right sides telescope to $\mathsf{h}_i(\pi)-\mathsf{h}_i(\pi^{(i)})$, while each $\mathsf{h}_k(0,\pi_k,O_k)$ does not involve $\pi_i$. Hence $\Phi'(\pi)-\mathsf{h}_i(\pi)$ does not depend on $\pi_i$, and neither does $G_i(\pi)-\mathsf{h}_i(\pi)=G_i(0,\pi_{-i})$. Conversely, if $G_i-\Phi'$ depends only on $\pi_{-i}$ for every $i$, its second difference in $(\pi_i,\pi_j)$ vanishes, so $\partial^2_{ij}G_i=\partial^2_{ij}\Phi'=\partial^2_{ij}G_j$. □

*Proof of Theorem 5.10.* (a) A move of member $i$ from $\pi_i$ to $\pi_i'$ at fixed $\pi_{-i}$ has probability

$$n^{-1}E_i(\pi_i')e^{-\gamma G_i(\pi_i',\pi_{-i})}/\mathsf{Z}_i(\pi_{-i}),$$

with a normalizer that depends on $\pi_{-i}$ only. Along the closed path, the four normalizers, the four habit factors and the factors $n^{-1}$ appear once in each direction and cancel, and the exponents give the forward-to-backward ratio $\exp(-\gamma(\partial^2_{ij}G_j-\partial^2_{ij}G_i))$. By Lemma 5.9, $\partial^2_{ij}G_i=\partial^2_{ij}\Phi+\partial^2_{ij}\mathcal{R}_i$, because a function of $\pi_{-i}$ has zero second difference in $(\pi_i,\pi_j)$, and likewise for $j$; this gives $\mathcal{L}_{ij}$. Through the null policies $\mathcal{R}_i(0,\cdot)=0$, so by the chain rule

$$\partial^2_{ij}\mathcal{R}_i=[I_{q_i}-I_{q_C}](o_{A'};o_{B'},o_{A_{-ij}})-[I_{q_i}-I_{q_C}](o_{A'};o_{A_{-ij}})=[I_{q_i}-I_{q_C}](o_{A'};o_{B'}\mid o_{A_{-ij}}),$$

and the same holds for $j$; subtracting gives (57), in which the terms in $q_C$ cancel.

(b) If the chain is reversible, multiplying the detailed-balance equations along any closed path shows that it has the same probability in both directions, which is the easy half of Kolmogorov's criterion (Kelly, 1979); so every $\mathcal{L}_{ij}$ through the null policies vanishes, which is agreement on redundancy. If the members agree, then $\partial^2_{ij}G_j-\partial^2_{ij}G_i=\mathcal{L}_{ij}=0$ for every second difference from the null policies, and Lemma S29.1 gives an exact potential $\Phi'$. If $\Phi'$ is an exact potential, a move of $i$ from $\pi$ to $\pi'=(\pi_i',\pi_{-i})$ satisfies detailed balance for $\mu\propto\prod_k E_k e^{-\gamma\Phi'}$: the ratio $\mu(\pi)\mathsf{T}(\pi,\pi')/(\mu(\pi')\mathsf{T}(\pi',\pi))$ equals $\exp(-\gamma[\Phi'(\pi)-\Phi'(\pi')+G_i(\pi')-G_i(\pi)])=1$, since $G_i-\Phi'$ does not depend on $\pi_i$. Finally an exact potential game has $\mathcal{L}_{ij}=0$ by the converse in Lemma S29.1.

(c) The chain is irreducible, since single moves with positive probability connect all profiles, and aperiodic, since it has self-loops, so its stationary law is unique and equals $\mu$. In the potential of Lemma S29.1, $G_k(\pi^{(k)})-G_k(\pi^{(k-1)})=-I_{q_k}(\eta_C;o_{A_k}\mid o_{A_1},\dots,o_{A_{k-1}})+\psi^{(k)}_{A_k}(q_k)$, because the pragmatic terms of the other slots cancel, and (S114) turns the conditional information into the

first and third terms of (58). Now let $q_C$ have the agreed conditional redundancies. By the chain rule, $I_{q_k}(o_{A_k}; o_{A_1}, \dots, o_{A_{k-1}}) = \sum_{l<k} I_{q_k}(o_{A_k}; o_{A_l} \mid o_{A_1}, \dots, o_{A_{l-1}})$, and each term is a conditional redundancy between the slots of members $k$ and $l$ in a context opened by other members, so it equals the corresponding term under $q_C$. The charges therefore add up to $\sum_k I_{q_C}(o_{A_k}; o_{A_1}, \dots, o_{A_{k-1}})$, which by (S114), applied $n-1$ times, is $-I_{q_C}(\eta_C; o_{\mathfrak{B}(\pi)}) + \sum_k I_{q_C}(\eta_C; o_{A_k})$. Hence $\Phi' = G_C + \sum_k \mathcal{T}_k$, with the $\mathcal{T}_k$ of Lemma 5.9, and $\mathbb{P}^\infty \propto \prod_k [E_k e^{-\gamma \mathcal{T}_k}] e^{-\gamma G_C}$. For two members the only redundancies are $I(o_{A_1}; o_{A_2})$ for all policies, on which the members agree, so $q_C = q_1$ has them.

(d) Write $o_A = \mathsf{H}_A \eta_C + \mathsf{n}_A$ with noise $\mathsf{n}_A$ independent of $\eta_C$, so that the law of the noise is the same under every belief. Under $q_i$ the pair $(\eta_C, o)$ is then the image of a pair distributed under $q$ by the translation $\eta_C \mapsto \eta_C + m_i$, $o_A \mapsto o_A + \mathsf{H}_A m_i$, which acts on each variable separately. Mutual informations and conditional mutual informations are invariant under bijective measurable maps of each argument, so every information term is the same under every $q_i$ and under every translate $q_C$. Then $\mathcal{R}_i \equiv 0$, $q_C$ has the agreed redundancies, the epistemic part of $\mathcal{T}_i$ vanishes, and (c) gives the group posterior. Under (P0), $q_i = q_C$ for every $i$, so for any slot likelihoods $\mathcal{R}_i \equiv 0$ and $\mathcal{T}_i \equiv 0$, and the chain rule for mutual information writes $-I_{q_C}(\eta_C; o_{\mathfrak{B}(\pi)}) = -I_{q_C}(\eta_C; o_{\mathfrak{B}_i(\pi_i)} \mid o_{A_{-i}}) - I_{q_C}(\eta_C; o_{A_{-i}})$, in which the second term depends on $\pi_{-i}$ only, so the part that depends on $\pi_i$ is the conditional information gain. □

**Lemma S29.2** (Tilts)**.** *Let $\mathbb{Q}$ be a law on a finite set and $g$ a function with oscillation* $\operatorname{osc} g$*, and let $\mathbb{Q}_g \propto \mathbb{Q} e^g$. Then* $\mathrm{TV}(\mathbb{Q}, \mathbb{Q}_g) \le \tanh(\operatorname{osc} g/4)$*, with equality for a law with masses* $(1 + e^{\operatorname{osc} g/2})^{-1}$ *and* $e^{\operatorname{osc} g/2}(1 + e^{\operatorname{osc} g/2})^{-1}$ *on a maximizer and a minimizer of $g$.*

*Proof.* Shift $g$ so that its maximum is $\frac{1}{2}\operatorname{osc} g$ and its minimum $-\frac{1}{2}\operatorname{osc} g$, and put $\mathsf{u} = e^{\operatorname{osc} g/2}$. For a set $S$ with $\mathbb{Q}(S) = \theta_S$, $\mathbb{Q}_g(S)$ is largest when $g$ is maximal on $S$ and minimal off $S$, so $\mathbb{Q}_g(S) - \theta_S \le \theta_S \mathsf{u}/(\theta_S \mathsf{u} + (1-\theta_S)\mathsf{u}^{-1}) - \theta_S = \mathsf{v}\theta_S(1-\theta_S)/(1+\mathsf{v}\theta_S)$ with $\mathsf{v} = \mathsf{u}^2 - 1$. The derivative in $\theta_S$ vanishes where $\mathsf{v}\theta_S^2 + 2\theta_S - 1 = 0$, at $\theta_S = 1/(1+\mathsf{u})$, where the value is $(\mathsf{u}-1)/(\mathsf{u}+1) = \tanh(\operatorname{osc} g/4)$. The two-point law attains it. □

*Proof of Theorem 5.11.* Ratio. By Lemma 5.9, member $i$'s conditional at $\pi_{-i}$ is its conditional in $\mathsf{T}_\Phi$ tilted by $e^{-\gamma \mathcal{R}_i(\cdot, \pi_{-i})}$, so the two probabilities of moving to $\pi_i'$ differ by the factor

$$e^{-\gamma \mathcal{R}_i(\pi_i', \pi_{-i})} / \mathbb{E} e^{-\gamma \mathcal{R}_i(\cdot, \pi_{-i})},$$

with the expectation under the reference conditional. Numerator and denominator both lie in $[e^{-\gamma \max \mathcal{R}_i}, e^{-\gamma \min \mathcal{R}_i}]$, so the factor lies in $[e^{-\gamma\sigma_i}, e^{\gamma\sigma_i}]$. An off-diagonal entry of $\mathsf{T}$ is $n^{-1}$ times one such probability, and a diagonal entry is an average of such probabilities, so both inherit the bound.

(a) By the Markov chain tree theorem (Anantharam and Tsoucas, 1989), the stationary probability of a profile is proportional to the sum, over spanning trees of the profiles directed toward it, of the products of the transition probabilities along their $|\mathcal{A}| - 1$ edges, all off the diagonal. $\mathsf{T}$ and $\mathsf{T}_\Phi$ have the same off-diagonal support, so each tree weight changes by a factor in $[e^{-(|\mathcal{A}|-1)\gamma\sigma_{\max}}, e^{(|\mathcal{A}|-1)\gamma\sigma_{\max}}]$, and so do the unnormalized stationary weights; after normalization $\ln(\mathbb{P}^\infty/\mathbb{P}_\Phi)$ takes values in an interval of length at most $2(|\mathcal{A}| - 1)\gamma\sigma_{\max}$ that contains 0. So $\mathbb{P}^\infty$ is a tilt of $\mathbb{P}_\Phi$ by a function of oscillation at most $2(|\mathcal{A}| - 1)\gamma\sigma_{\max}$, Lemma S29.2 gives the bound in total variation, and $D_{\mathrm{KL}}(\mathbb{P}^\infty \| \mathbb{P}_\Phi) \le \max \ln(\mathbb{P}^\infty/\mathbb{P}_\Phi)$.

(b) Let $d_{\mathrm{H}}$ be the Hamming distance between profiles and $\mathrm{Wass}_{\mathrm{H}}$ the corresponding Wasserstein distance between laws; $\mathrm{TV} \le \mathrm{Wass}_{\mathrm{H}}$ because $d_{\mathrm{H}}(\pi, \pi') \ge \mathbf{1}[\pi \ne \pi']$. Take two profiles that differ only in member $j$'s policy and run one step of $\mathsf{T}_\Phi$ from each with the same chosen member $i$. If $i = j$ the two conditionals coincide and the coupled draws agree, giving distance 0; if $i \ne j$, a maximal coupling of the two conditionals of $i$ disagrees with probability at most $\mathfrak{I}_{ij}$, giving distance

one plus that indicator. The expected distance is at most $1-(1-\mathfrak{D}_\Phi)/n$. Since $d_{\mathrm{H}}$ is the path metric of single-member changes, $\mathrm{Wass}_{\mathrm{H}}(\delta_\pi \mathsf{T}_\Phi, \delta_{\pi'}\mathsf{T}_\Phi) \le (1-(1-\mathfrak{D}_\Phi)/n) d_{\mathrm{H}}(\pi,\pi')$ for all profiles, and coupling initial laws optimally gives $\mathrm{Wass}_{\mathrm{H}}(\nu\mathsf{T}_\Phi, \nu'\mathsf{T}_\Phi) \le (1-(1-\mathfrak{D}_\Phi)/n)\mathrm{Wass}_{\mathrm{H}}(\nu,\nu')$; this is the path-coupling argument of Bubley and Dyer (1997). From one profile, a step of $\mathsf{T}$ and a step of $\mathsf{T}_\Phi$ with the same chosen member $i$ differ only through $i$'s conditional, which by the ratio argument is a tilt of the reference conditional by a function of oscillation at most $\gamma\sigma_i$, so by Lemma S29.2 $\mathrm{Wass}_{\mathrm{H}}(\nu\mathsf{T}, \nu\mathsf{T}_\Phi) \le n^{-1}\sum_i \tanh(\gamma\sigma_i/4)$. Then

$$\mathrm{Wass}_{\mathrm{H}}(\mathbb{P}^\infty, \mathbb{P}_\Phi) = \mathrm{Wass}_{\mathrm{H}}(\mathbb{P}^\infty\mathsf{T}, \mathbb{P}_\Phi\mathsf{T}_\Phi) \le \frac{1}{n}\sum_i \tanh\frac{\gamma\sigma_i}{4} + \Big(1-\frac{1-\mathfrak{D}_\Phi}{n}\Big)\mathrm{Wass}_{\mathrm{H}}(\mathbb{P}^\infty, \mathbb{P}_\Phi),$$

which is (59). Two conditionals of $i$ under $\mathbb{P}_\Phi$ at profiles that differ in $j$'s policy, from $\pi_j$ to $\pi'_j$, are tilts of each other by $-\gamma[\Phi(\cdot,\pi_j,\cdot) - \Phi(\cdot,\pi'_j,\cdot)]$, whose oscillation in $\pi_i$ is $\gamma\max|\partial^2_{ij}\Phi| = \gamma\max|\partial^2_{ij}G_C|$, because the tilts $\mathcal{T}_k$ are separable. A second difference of $G_C$ from the null policies is $I_{q_C}(o_{A'}; o_{B'} \mid o_{A_{-ij}}) \ge 0$ (proof of Proposition S5.19), and every second difference is the signed sum $\partial^2_{(\pi_i,\pi'_i),(\pi_j,\pi'_j)} = \partial^2_{(0,\pi'_i),(0,\pi'_j)} - \partial^2_{(0,\pi_i),(0,\pi'_j)} - \partial^2_{(0,\pi'_i),(0,\pi_j)} + \partial^2_{(0,\pi_i),(0,\pi_j)}$ of four of them, two with each sign; with two policies per member every second difference is one of them up to sign. Finally $\tanh x \le x$.

(c) $\mathsf{T}_\Phi$ is the average over members of the resampling operators, each the conditional expectation given $\pi_{-i}$ under $\mathbb{P}_\Phi$ and hence an orthogonal projection in $L^2(\mathbb{P}_\Phi)$; so $I-\mathsf{T}_\Phi$ is self-adjoint and positive, with smallest eigenvalue $\mathrm{gap}_\Phi$ on functions of mean zero. Write $\mathbb{P}^\infty = \mathbb{P}_\Phi(1+f)$ and $\mathsf{E} = \mathsf{T}-\mathsf{T}_\Phi$, and let $\mathsf{E}^*$ be the adjoint in $L^2(\mathbb{P}_\Phi)$, which represents the action of $\mathsf{E}$ on densities. Stationarity and the self-adjointness of $\mathsf{T}_\Phi$ give $f-\mathsf{T}_\Phi f = \mathsf{E}^*(1+f)$, with $\mathbb{E}_{\mathbb{P}_\Phi} f = 0$, and $f-\mathsf{T}_\Phi f$ also has mean zero. The entrywise bound $|\mathsf{E}| \le (e^{\gamma\sigma_{\max}}-1)\mathsf{T}_\Phi$ and the contraction of $\mathsf{T}_\Phi$ on $L^2(\mathbb{P}_\Phi)$ give $\|\mathsf{E}^*\| = \|\mathsf{E}\| \le e^{\gamma\sigma_{\max}}-1$, and $\mathsf{E}^*1 = \mathbb{P}_\Phi\mathsf{T}/\mathbb{P}_\Phi - 1$. Hence $\mathrm{gap}_\Phi\|f\| \le \|\mathsf{E}^*1\| + (e^{\gamma\sigma_{\max}}-1)\|f\|$, which is (60), with $\|\mathsf{E}^*1\| \le e^{\gamma\sigma_{\max}}-1$.

For the group posterior, the same arguments apply to the reference chain of $G_C$, whose resampling laws are tilted into the members' conditionals by $-\gamma(G_i - G_C)$. □

*Proof of Proposition S5.10.* The definition of $\mathrm{Th}_i$ gives $e^{-\mathrm{Th}_i}P_C \preceq P_i \preceq e^{\mathrm{Th}_i}P_C$. The posterior covariance after slots $O$, $P_{|O} = (P^{-1} + \mathsf{H}_O^\top\mathsf{R}_O^{-1}\mathsf{H}_O)^{-1}$, is monotone in $P$, and for $\mathsf{c} \ge 1$, $(\mathsf{c}P)^{-1} + \mathsf{H}_O^\top\mathsf{R}_O^{-1}\mathsf{H}_O \succeq \mathsf{c}^{-1}(P^{-1} + \mathsf{H}_O^\top\mathsf{R}_O^{-1}\mathsf{H}_O)$, so $(\mathsf{c}P)_{|O} \preceq \mathsf{c}P_{|O}$, and the reverse holds for $\mathsf{c} \le 1$. Hence $e^{-\mathrm{Th}_i}P_{C|O} \preceq P_{i|O} \preceq e^{\mathrm{Th}_i}P_{C|O}$. For Gaussian slots, $I_q(\eta_C; o_A \mid o_O) = \frac{1}{2}\ln\det(I + \mathsf{R}_A^{-1/2}\mathsf{H}_A P_{q|O}\mathsf{H}_A^\top\mathsf{R}_A^{-1/2})$ is monotone in $P_{q|O}$, and for a positive semidefinite $N$ of size $\dim o_A$, $\det(I+\mathsf{c}N)$ lies between $\det(I+N)$ and $\mathsf{c}^{\dim o_A}\det(I+N)$. So $I_{q_i}(\eta_C; o_A \mid o_O)$ and $I_{q_i}(\eta_C; o_A)$ differ from their values under $q_C$ by at most $\frac{1}{2}\dim(o_A)\mathrm{Th}_i$, and by (S114) their difference, the redundancy $I(o_A; o_O)$, differs by at most $\dim(o_A)\mathrm{Th}_i$. The second bound holds because both redundancies are nonnegative. Since $\mathrm{osc} \le 2\max|\mathcal{R}_i|$, the bound on $\sigma_i$ follows. The means do not enter because Gaussian mutual informations depend only on covariances. □

*Proof of Proposition S5.11.* (a) The habit-only model is admissible, so the infimum is at most the denominator and $0 \le \mathfrak{A} \le 1$. A larger class has a smaller infimum. Relabeling members permutes profiles, which leaves relative entropies unchanged, so the index is invariant when the class is closed under relabeling. With free joint habits, $E_C \propto \mathbb{P}e^{\gamma_C G_C}$ reproduces any full-support law $\mathbb{P}$ at one decision state. (b) The law $\mathbb{P}^\infty$ is then itself in the class. (c) The first inequality is the definition with the group model of $\mathbb{P}_\Phi$ admissible. For the second, $D_{\mathrm{KL}}(\mathbb{P}^\infty\|E) = D_{\mathrm{KL}}(\mathbb{P}^\infty\|\mathbb{P}_\Phi) + \mathbb{E}_{\mathbb{P}^\infty}\ln(\mathbb{P}_\Phi/E)$ and $\ln(\mathbb{P}_\Phi/E) = -\gamma\Phi - \ln\mathsf{Z}_\Phi$, so

$$\mathbb{E}_{\mathbb{P}^\infty}\ln\frac{\mathbb{P}_\Phi}{E} = D_{\mathrm{KL}}(\mathbb{P}_\Phi\|E) - \gamma(\mathbb{E}_{\mathbb{P}^\infty}\Phi - \mathbb{E}_{\mathbb{P}_\Phi}\Phi) \ge D_{\mathrm{KL}}(\mathbb{P}_\Phi\|E) - \gamma\,\mathrm{osc}(\Phi)B_{\mathrm{TV}} = \mathsf{c}_\Phi.$$

For $\mathsf{c}_\Phi > 0$ the map $x \mapsto x/(x+\mathsf{c}_\Phi)$ is increasing on $[0,\infty)$. Jensen's inequality gives $D_{\mathrm{KL}}(\mathbb{P}^\infty \| \mathbb{P}_\Phi) \le \ln(1 + \|\mathbb{P}^\infty/\mathbb{P}_\Phi - 1\|^2_{L^2(\mathbb{P}_\Phi)})$, so the bounds of Theorem 5.11 supply $B_{\mathrm{KL}}$ and $B_{\mathrm{TV}}$. □

*Proof of Proposition S5.12.* If the previous policies $(\pi_1, \pi_2)$ have a product law, the new ones depend on $\pi_2$ and on $\pi_1$ respectively, through independent randomness, so they are independent: the map induced on laws sends product laws to product laws. Restricted to them it is a continuous self-map of a product of simplices, which has a fixed point by Brouwer's theorem; a fixed product law is stationary. Each member's conditional is proportional to $E_i(\pi_i)e^{-\gamma G_i}$ with full-support habits and finite expected free energies, so every transition probability of the synchronous chain, a product of two such conditionals, is positive, and the chain has a unique stationary law. Positivity is needed: for the deterministic conditionals that copy the other member's previous policy, the chain exchanges the two policies and the uniform law on the profiles with equal policies is stationary without being a product law. A product law has log odds ratio zero, while by Proposition S5.19 a group posterior with product habits whose two slots from the null policies carry redundant information has log odds ratio $-\gamma_C I_{q_C}(o_{A'}; o_{B'}) < 0$ when $\gamma_C > 0$. □

*Proof of Theorem S5.17.* With intentions shared, member $i$'s outcome vector is $(o^\star, o_{\mathfrak{B}(\pi^{\mathrm{epi}})})$, and its expected free energy is minus the information gain about $(\eta, \eta')$ minus the expected log preference, the second form of the expected free energy of Section 2. Under its model the joint law of $(\eta, \eta', o^\star, o_{\mathfrak{B}})$ is Gaussian with a covariance that does not depend on $u$ and a mean affine in the effects of $u$; Gaussian information gains depend only on covariances, and the slots observe $\eta$ before the acts, so the information term and the slots' pragmatic terms depend on $\pi^{\mathrm{epi}}$ only. The predictive law of $o^\star$ is Gaussian with mean $\mathsf{H}_\star F m_i + f^{(i)}(u)$ and a covariance independent of $u$, so $-\mathbb{E}\ln\tilde{p}(o^\star)$ is $\frac{1}{2}\|\mathsf{H}_\star F m_i + f^{(i)}(u) - o_i^\star\|^2_{\Lambda_i^\star}$ plus a constant, which is (S24).

(a) For $\mathcal{B}_{\mathrm{int}} = 0$, $G_k^{\mathrm{act}}$ is a quadratic function of $u$ whose part bilinear in $(u_i, u_j)$ is

$$u_i^\top \mathsf{M}_i^{(k)\top} \Lambda_k^\star \mathsf{M}_j^{(k)} u_j, \qquad \text{so that} \qquad \partial^2_{ij} G_k^{\mathrm{act}} = (u_i' - u_i)^\top \mathsf{M}_i^{(k)\top} \Lambda_k^\star \mathsf{M}_j^{(k)} (u_j' - u_j).$$

The computation of the cycle ratio in the proof of Theorem 5.10(a) applies to any profile of expected free energies, and Lemma S29.1 gives the criterion. (b) Expanding, $\frac{1}{2}\|\mathsf{z}_i + y\|^2_{\Lambda^\star} = \frac{1}{2}\|\mathsf{z}_C + y\|^2_{\Lambda^\star} + \langle \mathsf{z}_i - \mathsf{z}_C, y\rangle_{\Lambda^\star} + \text{const}$ with $y = \sum_j \mathsf{M}_j u_j$, and $\langle \mathsf{z}_i - \mathsf{z}_C, y\rangle_{\Lambda^\star}$ splits into $\langle \mathsf{z}_i - \mathsf{z}_C, \mathsf{M}_i u_i\rangle_{\Lambda^\star}$ and terms in $u_{-i}$, which gives (S26). If the epistemic component has a potential $G_C^{\mathrm{epi}} + \sum_k \mathcal{T}_k$, then $G_i - (G_C^{\mathrm{epi}} + \sum_k \mathcal{T}_k) - \Phi^{\mathrm{act}}$ depends on the others' policies only, and the stationary law is the Gibbs law of Theorem 5.10(c) for the potential $G_C + \sum_k \mathcal{T}_k + \sum_j \langle \mathsf{z}_j - \mathsf{z}_C, \mathsf{M}_j u_j\rangle_{\Lambda^\star}$, the group expected free energy plus own-policy terms, which is the group posterior with habits tilted by these terms. (c) $G_i^{\mathrm{act}} - G_j^{\mathrm{act}} = \langle \mathsf{z}_i - \mathsf{z}_j, f(u)\rangle_{\Lambda^\star} + \text{const}$, whose second difference is $\langle \mathsf{z}_i - \mathsf{z}_j, \mathsf{H}_\star \partial^2_{ij}\mathcal{B}_{\mathrm{int}}\rangle_{\Lambda^\star}$, because the additive part of $f$ has zero second difference; the cycle sum is $\partial^2_{ij} G_j - \partial^2_{ij} G_i$, which is (S27). With a common displacement $\mathsf{z}_i = \mathsf{z}_C$ and $\Lambda_C^\star = \Lambda^\star$ every $G_i^{\mathrm{act}}$ equals $G_C^{\mathrm{act}}$ up to a constant. Common targets $o_i^\star$ alone do not give this, because $\mathsf{z}_i = \mathsf{H}_\star F m_i - o_i^\star$ also contains the member's predicted outcome. □

*Proof of Proposition S5.19.* The habits are a product and cancel in the odds ratio, which leaves $-\gamma_C[G_C(\pi_i', \pi_j') + G_C(\pi_i, \pi_j) - G_C(\pi_i', \pi_j) - G_C(\pi_i, \pi_j')] = -\gamma_C \partial^2_{ij} G_C$. For epistemic slots from the null policies, with $O = A_{-ij}$, the pragmatic terms are separable, and (S114) applied to $(A', B' \cup O)$ and to $(A', O)$ gives $-I(\eta_C; o_{A'\cup B'\cup O}) + I(\eta_C; o_{B'\cup O}) = -I(\eta_C; o_{A'}) + I(o_{A'}; o_{B'\cup O})$ and $-I(\eta_C; o_{A'\cup O}) + I(\eta_C; o_O) = -I(\eta_C; o_{A'}) + I(o_{A'}; o_O)$; subtracting, $\partial^2_{ij} G_C = I_{q_C}(o_{A'}; o_{B'\cup O}) - I_{q_C}(o_{A'}; o_O) = I_{q_C}(o_{A'}; o_{B'} \mid o_O)$. For additive acts the epistemic part does not depend on $u$ and the act part is the quadratic of (S24), whose second difference is the bilinear term. □

*Proof of Proposition 5.14.* (a) For each position, the expected change of $\mu_i$ in $\mathrm{d}t$ given the present is $[-\varepsilon\mu_i+\beta(z-\mu_i)+g(\mathsf{f}z-\mu_i)]\mathrm{d}t$ plus a communication term that sums to zero over the members, the term with $g$ being the expected jump at a replacement, which occurs at rate $g$ and takes $\mu_i$ to $\mathsf{f}z$ plus mean-zero noise. Averaging over positions gives the first row of (62), and the record's drift $\omega(\bar\mu-z)-\delta z$ gives the second. The noise terms are martingale increments, so $\mathbb{E}[(\bar\mu,z)_{t+u}\mid\text{process up to }t]=e^{u\mathsf{A}}(\bar\mu,z)_t$. With $\mathsf{p}_1=\varepsilon+\beta+g$, $\mathsf{p}_2=\omega+\delta$ and $\mathsf{E}=\omega(\beta+g\mathsf{f})\geq 0$, the eigenvalues of $\mathsf{A}$ are $-\varpi_\pm$ with $\varpi_\pm=\frac{1}{2}[(\mathsf{p}_1+\mathsf{p}_2)\pm((\mathsf{p}_1-\mathsf{p}_2)^2+4\mathsf{E})^{1/2}]$, real, and positive because $\operatorname{tr}\mathsf{A}<0$ and $\det\mathsf{A}>0$; the system is stable. The martingale part of $(\bar\mu,z)$ lies in the first coordinate, and in the stationary law its quadratic variation grows at a constant rate $\mathsf{q}$ that is at least $g$ times the newcomers' noise variance divided by $n$, because a replacement at one position moves $\bar\mu$ by $(\mathsf{f}z+\xi-\mu_i)/n$ with $\xi$ the newcomer's noise, independent of the present and of mean zero, and replacements occur at total rate $ng$. The stationary covariance $\mathsf{S}$ of $(\bar\mu,z)$ therefore solves $\mathsf{AS}+\mathsf{SA}^\top+\operatorname{diag}(\mathsf{q},0)=0$ with $\mathsf{q}>0$, and it is positive definite because $\mathsf{A}$ is stable and $\omega>0$ carries the noise of the first coordinate into the second. The projection of the state on the left eigenvector of $-\varpi_-$ is therefore a nondegenerate random variable, so $\varpi_-$ is the slowest rate at which the conditional expectation of the pair relaxes, and $\tau_{\mathrm{id}}^{-1}=\varpi_-$. When $\beta+g\mathsf{f}>0$, a right eigenvector of $-\varpi_-$ is $(\beta+g\mathsf{f},\mathsf{p}_1-\varpi_-)$, whose first component is nonzero, and $\varpi_-<\varpi_+$, so the conditional expectation of $\bar\mu$ alone also relaxes at the rate $\varpi_-$. When $\beta+g\mathsf{f}=0$ the first row of (62) is closed, $\mathbb{E}[\bar\mu_{t+u}\mid\text{process up to }t]=e^{-\mathsf{p}_1u}\bar\mu_t$ with $\mathsf{p}_1=\varepsilon+g$, and $\varpi_-=\min(\mathsf{p}_1,\mathsf{p}_2)$. For the comparison value, $\varpi_+-\mathsf{p}_1=\frac{1}{2}[(\mathsf{p}_2-\mathsf{p}_1)+((\mathsf{p}_1-\mathsf{p}_2)^2+4\mathsf{E})^{1/2}]\geq 0$ and $\varpi_-\varpi_+=\det\mathsf{A}=\mathsf{p}_1\mathsf{p}_2-\mathsf{E}$, so

$$\varpi_-=\frac{\mathsf{p}_1\mathsf{p}_2-\mathsf{E}}{\varpi_+}\leq\mathsf{p}_2-\frac{\mathsf{E}}{\mathsf{p}_1}=\delta+\frac{\omega(\varepsilon+g(1-\mathsf{f}))}{\varepsilon+\beta+g}.$$

When $\mathsf{p}_2\leq\mathsf{p}_1/2$, the inequality $(1+x)^{1/2}\leq 1+x/2$ for $x\geq 0$, with $x=4\mathsf{E}/(\mathsf{p}_1-\mathsf{p}_2)^2$, gives $\varpi_-\geq\mathsf{p}_2-\mathsf{E}/(\mathsf{p}_1-\mathsf{p}_2)$. Since $\mathsf{E}\leq\omega\mathsf{p}_1$ and $\mathsf{p}_1-\mathsf{p}_2\geq\mathsf{p}_1/2$, one has $\mathsf{E}/(\mathsf{p}_1-\mathsf{p}_2)-\mathsf{E}/\mathsf{p}_1=\mathsf{E}\mathsf{p}_2/(\mathsf{p}_1(\mathsf{p}_1-\mathsf{p}_2))\leq 2\omega\mathsf{p}_2/\mathsf{p}_1$. Hence $\varpi_-$ falls below $\mathsf{p}_2-\mathsf{E}/\mathsf{p}_1$ by at most $2\omega\mathsf{p}_2/\mathsf{p}_1$. Without a record, communication among the members conserves their mean and a replacement moves a position from $\mu_i$ to $\mathsf{f}\bar\mu$ plus mean-zero noise, so the expected change of $\bar\mu$ is $[-\varepsilon\bar\mu-g(1-\mathsf{f})\bar\mu]\mathrm{d}t$.

(b) By Proposition 4.4 a record written only by members is internal or boundary, a collective variable whose persistence Definition 5.13 counts. A record written only by non-members is external. When the writes do not depend on the members' states, the record's path is a function of the exterior's exogenous input $\mathcal{X}_{\mathrm{ext}}$ of Definition 5.13, and $\mathcal{X}_{\mathrm{ext}}$ is independent of the members' private noises, the replacement events and the newcomers' noises, which are the remaining exogenous inputs of the model. By the first row of (62), $\bar\mu_{t+u}=e^{-(\varepsilon+\beta+g)u}\bar\mu_t$ plus a functional of the record's path plus martingale increments after $t$, which have zero conditional mean given $\mathcal{X}_{\mathrm{ext}}$ and the past by that independence; so the conditional expectation of $\bar\mu_{t+u}$ given the past and $\mathcal{X}_{\mathrm{ext}}$ relaxes at the rate $\varepsilon+\beta+g$, and the conditional identity time of $\bar\mu$ is $1/(\varepsilon+\beta+g)\leq 1/g$. □

*Proof of Proposition S5.23.* With no blanket variables in the description, the leak rate is the mutual information rate of the stationary processes $\mu_{\mathrm{in}}$ and $\mu_{\mathrm{ex}}$. For jointly stationary Gaussian processes with spectral densities bounded and bounded away from zero, this rate is $-\frac{1}{2}\langle\ln(1-|\mathrm{coh}(\psi)|^2)\rangle_{\mathbb{T}}$, with coh the coherence and $\langle\cdot\rangle_{\mathbb{T}}$ the average over frequencies $\psi$ on the unit circle, a formula attributed to Gel'fand and Yaglom (1959). The exterior agent's own filter $(1-c_{\mathrm{ex}}\mathrm{L})^{-1}$, with L the lag, is invertible and leaves the coherence unchanged, so the rate equals that between $\mu_{\mathrm{in}}$ and $\beta z+\zeta_{\mathrm{ex}}$. With $\mathsf{D}(\psi)=|1-(1-\delta_z)e^{-i\psi}|^2=1+(1-\delta_z)^2-2(1-\delta_z)\cos\psi$, the spectral densities give

$$1-|\mathrm{coh}(\psi)|^2=\frac{\beta^2q_{\mathrm{rec}}+q_{\mathrm{rd}}\mathsf{D}(\psi)}{\beta^2(q_{\mathrm{rec}}+\omega^2q_{\mathrm{wr}})+q_{\mathrm{rd}}\mathsf{D}(\psi)}=\frac{\mathsf{a}_0-2\mathsf{a}_\times\cos\psi}{\mathsf{a}_1-2\mathsf{a}_\times\cos\psi}.$$

For $\mathsf{a} \geq 2|\mathsf{a}_\times|$, $\mathsf{a} - 2\mathsf{a}_\times \cos\psi = \mathsf{c}_\mathsf{a}|1 - \mathsf{r}_\mathsf{a} e^{\mathrm{i}\psi}|^2$ with $|\mathsf{r}_\mathsf{a}| \leq 1$ and $\mathsf{c}_\mathsf{a} = (\mathsf{a} + (\mathsf{a}^2 - 4\mathsf{a}_\times^2)^{1/2})/2$, and $\langle \ln|1 - \mathsf{r}_\mathsf{a} e^{\mathrm{i}\psi}|^2\rangle_\mathbb{T} = 0$, so $\langle \ln(\mathsf{a} - 2\mathsf{a}_\times \cos\psi)\rangle_\mathbb{T} = \ln \mathsf{c}_\mathsf{a}$. Both $\mathsf{a}_0$ and $\mathsf{a}_1$ exceed $2|\mathsf{a}_\times|$, since $1 + (1 - \delta_z)^2 \geq 2|1 - \delta_z|$; this gives (S28), which is positive when $\mathsf{a}_1 > \mathsf{a}_0$, that is, when $\beta\omega \neq 0$. With $q_{\mathrm{rec}} = 0$, $\mathsf{a}_0 + (\mathsf{a}_0^2 - 4\mathsf{a}_\times^2)^{1/2} = 2q_{\mathrm{rd}}$ and $\mathsf{a}_1^2 - 4\mathsf{a}_\times^2 = q_{\mathrm{rd}}^2(\delta_z^2 + \mathrm{snr})((2 - \delta_z)^2 + \mathrm{snr})$. Writing $\mathsf{s}_1 = (\delta_z^2 + \mathrm{snr})^{1/2}$ and $\mathsf{s}_2 = ((2 - \delta_z)^2 + \mathrm{snr})^{1/2}$, one has $\mathsf{a}_1/q_{\mathrm{rd}} = (\mathsf{s}_1^2 + \mathsf{s}_2^2)/2$, so the rate is $\ln((\mathsf{s}_1 + \mathsf{s}_2)/2)$, the first form of (S29). Since $\mathsf{s}_1 - \delta_z = \mathrm{snr}/(\mathsf{s}_1 + \delta_z)$ and $\mathsf{s}_2 - (2 - \delta_z) = \mathrm{snr}/(\mathsf{s}_2 + 2 - \delta_z)$, $(\mathsf{s}_1 + \mathsf{s}_2)/2 = 1 + \frac{1}{2}(\mathsf{s}_1 - \delta_z)(1 + (\mathsf{s}_1 + \delta_z)/(\mathsf{s}_2 + 2 - \delta_z))$; for $\delta_z \leq 1$ the denominator is at least 2, so the correction is at most $\mathsf{s}_1$, and $\ln(1 + x) = x(1 + O(x))$ with $x = O(\mathsf{s}_1)$ gives the second form. For $\mathrm{snr} \ll \delta_z^2$, the exact small-snr limit is $\frac{\mathrm{snr}}{4}(\frac{1}{\delta_z} + \frac{1}{2 - \delta_z}) = \mathrm{snr}/(2\delta_z(2 - \delta_z))$; for $\delta_z^2 \ll \mathrm{snr} \ll 1$, $\mathsf{s}_1 \approx \mathrm{snr}^{1/2}$. □

*Proof of Proposition 5.16.* (a) Given $\mathcal{X}_{\mathrm{sh}}$, the paths are functions of the private inputs $\zeta_i$, which are independent of each other and of $\mathcal{X}_{\mathrm{sh}}$, so the paths are conditionally independent; every pairwise term of (63) vanishes, and the conditional covariance of the members' states is block diagonal across agents, and so is its inverse.

(b) Because the member reads no message and no record, its path is a function of the world path, the initial state and its own private inputs. Because it neither acts, writes a record nor is read, its private inputs influence no other variable, so they are independent of the world path, the initial state and every other agent's variables. Given the world path and the initial state, its path is therefore conditionally independent of every other variable, and its terms in the statistic conditioned on them vanish.

(c) For a finite window with $X_0$ in the conditioning set, each path $\mu_i[0{:}\bar{t}]$ is $\sum_k L_{ik}\zeta_k[0{:}\bar{t}]$ plus a function of $\mathcal{X}_{\mathrm{sh}}$, with loadings $L_{ik}$ whose entries are polynomials in the coefficients, so the conditional cross-covariance $\mathrm{Cov}(\mu_i[0{:}\bar{t}], \mu_j[0{:}\bar{t}] \mid \mathcal{X}_{\mathrm{sh}}) = \sum_k L_{ik}\,\mathrm{Cov}(\zeta_k)L_{jk}^\top$ has polynomial entries. For Gaussian vectors the conditional mutual information is positive exactly when the conditional cross-covariance is nonzero. Take the directed path $j = k_0 \to k_1 \to \cdots \to k_\ell = i$ in the support, set every coefficient to zero except the path coefficients, which are set to one, and the loading of $\mu_j$ on its private input, which is kept. Then, given $\mathcal{X}_{\mathrm{sh}}$, $\mu_j(t)$ is a nonzero linear function $\mathsf{l}^\top\zeta_j(t-1)$ of its private input and $\mu_i(t + \ell) = \mu_j(t)$, so $\mathrm{Cov}(\mu_i(t + \ell), \mu_j(t) \mid \mathcal{X}_{\mathrm{sh}}) = \mathsf{l}^\top\,\mathrm{Cov}(\zeta_j(t-1))\mathsf{l} > 0$ by the positive variance of that input. Some entry of the conditional cross-covariance is thus a polynomial that is not identically zero, and its zero set is a proper real algebraic set, closed, of measure zero and with empty interior. For scalar states with independent scalar private inputs and coefficients that are nonnegative and positive on the path and on the loading, every term of $\sum_k L_{ik}\,\mathrm{Cov}(\zeta_k)L_{jk}^\top$ is nonnegative and the term through $j$'s input carried by the path is positive. For the stationary version, parametrize by the update matrix $W$ on its support with the private noise variances fixed and positive: the window covariances are rational in $W$ on the set $\{\rho(W) < 1\}$, solutions of a Lyapunov equation, and this set is open and star-shaped about 0, since $\rho(sW) = s\rho(W)$ for $s \in [0, 1]$, hence connected. A real-analytic function on a connected open set that vanishes on a set of positive measure vanishes identically, and the path matrix above is nilpotent, hence stable, with $\mathrm{Cov}(e_i(t + \ell), e_j(t)) = q_j \neq 0$, because no edge enters $j$.

(d) With $\Pi = \mathrm{diag}(\mathsf{d})$ on a region of $n$ members, $\varkappa_{\mathrm{out}} = \bar{\mathsf{d}}$, the mean of $\mathsf{d}$ over the region, and $\varkappa_{\mathrm{in}}$ is the smallest eigenvalue of the compression of $\mathrm{diag}(\mathsf{d})$ to $\mathbf{1}^\perp$, whose trace is $(n-1)\bar{\mathsf{d}}$; so $\varkappa_{\mathrm{in}} \leq \bar{\mathsf{d}} = \varkappa_{\mathrm{out}}$ and $r \leq 1$, with equality when $\mathsf{d}$ is constant. Under $\mathfrak{N}_{\mathrm{lp}}$ and $\mathfrak{N}_{\mathrm{rn}}$ every channel from another agent delivers noise that is either private to the receiver or a sender-side broadcast noise, and under $\mathfrak{N}_{\mathrm{eff}}$ the world is exogenous. Without records, in the stationary regime of the stable null dynamics each member's state is a measurable function of the world path, the broadcast noises and its own private inputs, the initial state being forgotten. The world-conditional law of Definition 4.7 conditions on the world path and the broadcast noises, so it is the law given $\mathcal{X}_{\mathrm{sh}}$, and the first part

applies. □

*Proof of Proposition 5.18.* (a) Let $U_i$ and $U_j$ be as in the statement and $Z_{\mathrm{sh}}$ the recorded drive, with maximum-likelihood covariance $S$ from the $n_{\mathrm{rep}}$ windows and $S_{\cdot|Z}$ the corresponding residual covariances. A coordinate of a path that is a function of the drive has zero residual, so it must be excluded for the residual covariances to be nonsingular; since $\mathrm{Cov}((U_i, U_j) \mid \mathcal{X}_{\mathrm{sh}}) \succ 0$ and $n_{\mathrm{rep}} > \dim \mathcal{X}_{\mathrm{sh}} + \dim U_i + \dim U_j + 1$, the residual covariances $S_{U_i|Z}$, $S_{U_j|Z}$ and $S_{U_iU_j|Z}$ are nonsingular with probability one, and the Gaussian regression model of $(U_i, U_j)$ on the drive is regular. Because the excluded coordinates are functions of the drive, $I(U_i; U_j \mid \mathcal{X}_{\mathrm{sh}})$ equals the conditional mutual information of the full paths. The maximized Gaussian log-likelihood of a regression model is $-\frac{n_{\mathrm{rep}}}{2} \ln \det$ of its residual covariance plus a constant that depends only on dimensions. Under the hypothesis, the joint density factorizes as $p(z)p(u_i \mid z)p(u_j \mid z)$ and its likelihood is maximized factor by factor, giving $-\frac{n_{\mathrm{rep}}}{2}[\ln\det S_Z + \ln\det S_{U_i|Z} + \ln\det S_{U_j|Z}]$; without the hypothesis the maximum is $-\frac{n_{\mathrm{rep}}}{2}[\ln\det S_Z + \ln\det S_{U_iU_j|Z}]$; the constants agree because the dimensions add up. Twice the difference is $n_{\mathrm{rep}} \ln(\det S_{U_i|Z} \det S_{U_j|Z} / \det S_{U_iU_j|Z}) = 2n_{\mathrm{rep}}\hat{I}$. The hypothesis sets the $\dim U_i \times \dim U_j$ block of the conditional cross-covariance to zero, a smooth constraint of that codimension in a regular model, so by Wilks (1938) the statistic converges in law to the chi-squared law with $\dim U_i \cdot \dim U_j$ degrees of freedom. Under a causally separating null the hypothesis holds by Proposition 5.16(a).

(b) Break ties uniformly at random. Under exchangeability the rank of $\mathcal{S}_0$ among $\mathcal{S}_0, \dots, \mathcal{S}_{n_{\mathrm{null}}}$, counted from the top, is uniform on $\{1, \dots, n_{\mathrm{null}} + 1\}$, and it is at most $1 + \#\{l \geq 1 : \mathcal{S}_l \geq \mathcal{S}_0\}$. The event $p \leq \alpha$ says that this count is at most $\alpha(n_{\mathrm{null}} + 1) - 1$, so it implies that the rank is at most $\lfloor \alpha(n_{\mathrm{null}} + 1) \rfloor$, an event of probability $\lfloor \alpha(n_{\mathrm{null}} + 1) \rfloor / (n_{\mathrm{null}} + 1) \leq \alpha$. □

*Proof of Theorem 5.20.* (a) The first sentence is Theorem 4.3(a) and (b): for the canonical partition with per-channel noise the first two conditions of (25) hold by construction, the start supplies the fourth, and the third, the collider condition, holds when no sensory channel reads both sides in one round and fails outside a set of coefficients of positive codimension when one does, so the converse holds only generically. The rest is Theorem 4.23(a) to (c) with Proposition S4.10. (b) The order parameter vanishes under the deletion nulls by Proposition 5.16(a), and in a population without records no region has contrast above one by Proposition 5.16(d), so under the nulls no region is selected at a level $r_0 > 1$. At every parameter of the given open set the full model selects $C$, a collective at the level $r_0$; the selected partitions therefore differ there, for every admissible level in $(1, r(C)]$, which is the clause of Definition 5.17. The positivity of $\Omega_{\mathrm{int}}$ is Proposition 5.16(c). (c) Apply Theorem 5.4(a) with $M = \Pi^{\mathrm{c}}_{CC}$, $\varkappa_{\mathrm{in}} = r(C)\varkappa_{\mathrm{out}}$ and $\sigma_C = s\varkappa_{\mathrm{out}}$: then $\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}} = (r(C) - 1)\varkappa_{\mathrm{out}} > 0$, $\sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}}) = s/(r(C) - 1)$ and $\varkappa_{\mathrm{out}}/\varkappa_{\mathrm{in}} = 1/r(C)$, so (49) becomes (64); the non-reversible case is Theorem 5.4(b) with Proposition 5.5. (d) is Propositions S5.5 and 5.7. (e) is Theorems 5.10, 5.11 and S5.17; for the last sentence, members without data sharing have expected free energies that depend on their own policies only, so the stationary law of deliberation is the product law $\prod_i E_i e^{-\gamma G_i}/\mathsf{Z}_i$, whose log odds ratios vanish, while by Proposition S5.19 a group posterior with product habits, a slot-product preference and $\gamma_C > 0$ has the negative log odds ratio $-\gamma_C I_{q_C}(o_{A'}; o_{B'} \mid o_{A_{-ij}})$ for two members whose slots, opened from the null policies, carry redundant information. □

## S30 Proofs for Section 6

This appendix proves the results of Section 6 in the order in which they appear. It also states the secondary results to which the section points, Lemmas S30.1 to S30.3 and Conjecture S30.7.

Notation is that of the section.

## S30.1 Components and fidelity

**Lemma S30.1** (One message)**.** *Let $j$'s data be jointly Gaussian with $\eta \sim \mathcal{N}(0, P_0)$, let $m = \mathbb{E}[\eta \mid$ data of $j]$ have posterior covariance $P$, so that $\operatorname{Cov}(m) = P_0 - P$, and let $y = H_j m + \nu$ with $\nu \sim \mathcal{N}(0, R_c)$ independent and $R_c \succ 0$. Then $y = H_j(P_0 - P)P_0^{-1}\eta + \nu'$ with $\nu'$ independent of $\eta$, $\operatorname{Cov}(\nu') = H_j(P - PP_0^{-1}P)H_j^\top + R_c$, and*

$$I(\eta; y) = \tfrac{1}{2} \ln\det(H_j(P_0 - P)H_j^\top + R_c) - \tfrac{1}{2} \ln\det(H_j(P - PP_0^{-1}P)H_j^\top + R_c). \tag{S115}$$

*This information is at most $\min\{I(\eta; m), I(m; y)\}$, where $I(m; y) = \frac{1}{2}\sum_k \ln(1 + x_k)$ over the at most $r_c$ nonzero eigenvalues $x_k$ of $R_c^{-1/2} H_j(P_0 - P)H_j^\top R_c^{-1/2}$.*

*Proof.* The Bayes error $\eta - m$ is orthogonal to every function of $j$'s data, so $\operatorname{Cov}(\eta, m) = \operatorname{Cov}(m) = P_0 - P$. The regression of $m$ on $\eta$ is therefore $(P_0 - P)P_0^{-1}\eta$, and its residual $\tilde{m}$ is independent of $\eta$, the pair being jointly Gaussian and uncorrelated, with covariance $(P_0 - P) - (P_0 - P)P_0^{-1}(P_0 - P) = P - PP_0^{-1}P$. Substituting $m = (P_0 - P)P_0^{-1}\eta + \tilde{m}$ into $y = H_j m + \nu$ gives the representation with $\nu' = H_j\tilde{m} + \nu$. For jointly Gaussian vectors $I(\eta; y) = \frac{1}{2}\ln\det\operatorname{Cov}(y) - \frac{1}{2}\ln\det\operatorname{Cov}(y \mid \eta)$, with $\operatorname{Cov}(y) = H_j(P_0 - P)H_j^\top + R_c$ and $\operatorname{Cov}(y \mid \eta) = \operatorname{Cov}(\nu')$, which is (S115). The variables $\eta$, $m$, $y$ form a Markov chain, so data processing gives both upper bounds, and $I(m; y) = \frac{1}{2}\ln\det(I + R_c^{-1/2} H_j \operatorname{Cov}(m) H_j^\top R_c^{-1/2})$ is the stated sum, with at most $\operatorname{rank} H_j \le r_c$ nonzero terms. □

*Proof of Proposition S6.1.* We use one cited fact. For the Ornstein–Uhlenbeck kernel $\mathsf{O}_\tau(x, \cdot) = \mathcal{N}(e^{-\tau}x, (1 - e^{-2\tau})I)$ in any dimension and every probability measure $Q$ with $D_{\mathrm{KL}}(Q\|\mathcal{N}(0, I)) < \infty$,

$$D_{\mathrm{KL}}(Q\mathsf{O}_\tau\|\mathcal{N}(0, I)) \le e^{-2\tau} D_{\mathrm{KL}}(Q\|\mathcal{N}(0, I)). \tag{S116}$$

This is the decay of relative entropy along the Ornstein–Uhlenbeck semigroup, equivalent to the Gaussian logarithmic Sobolev inequality with constant one (Gross, 1975); we use it in the form of Bakry et al. (2014, Theorem 5.2.1) as restated by Ledoux et al. (2015, Section 2.3).

Since $M_{XY}M_{XY}^\top = I - \operatorname{Cov}(Y)^{-1/2} R_{\mathsf{K}} \operatorname{Cov}(Y)^{-1/2} \prec I$, every canonical correlation is below one. Take a singular value decomposition $M_{XY} = V_0 D_0 W_0^\top$ and put $X' = W_0^\top \operatorname{Cov}(X)^{-1/2} X$ and $Y' = V_0^\top \operatorname{Cov}(Y)^{-1/2} Y$. Both are standard Gaussian vectors, $\operatorname{Cov}(Y', X') = D_0$ and $Y' \mid X' \sim \mathcal{N}(D_0 X', I - D_0 D_0^\top)$: with $\sigma_k = \sigma_k(M_{XY})$ and independent standard $Z_k$, $Y'_k = \sigma_k X'_k + (1 - \sigma_k^2)^{1/2} Z_k$ for $k \le \min\{\dim X, \dim Y\}$ and $Y'_k = Z_k$ beyond. Relative entropy is invariant under these bijections, so it suffices to treat $(X', Y')$. If $\sigma_1 = 0$ the output is independent of the input and every claim is trivial. Otherwise put $\tau_k = -\ln\sigma_k$, with $\tau_k = \infty$ when $\sigma_k = 0$ or $k > \min\{\dim X, \dim Y\}$. The channel $X' \mapsto Y'$ is the composition of three kernels: the isotropic kernel $\mathsf{O}_{\tau_1}$; the product kernel that applies to coordinate $k$ the one-dimensional Ornstein–Uhlenbeck kernel at time $\tau_k - \tau_1 \ge 0$, time $\infty$ meaning replacement by independent standard noise; and the map that keeps the first $\min\{\dim X, \dim Y\}$ coordinates and, if $\dim Y > \dim X$, appends independent standard coordinates. The composite is the channel because Ornstein–Uhlenbeck kernels compose additively in time: $e^{-\tau'}(e^{-\tau}x + (1 - e^{-2\tau})^{1/2} Z_1) + (1 - e^{-2\tau'})^{1/2} Z_2$ has the law $\mathsf{O}_{\tau+\tau'}(x, \cdot)$. The second and third kernels do not increase relative entropy, the second by data processing and the third because dropping coordinates is data processing and appending an independent common factor leaves relative entropy unchanged. The first contracts it by $e^{-2\tau_1} = \sigma_1^2$, by (S116). Hence $D_{\mathrm{KL}}(Q_X\mathsf{K}\|P_X\mathsf{K}) \le \sigma_1^2 D_{\mathrm{KL}}(Q_X\|P_X)$.

For attainment let $w_1$ be the top right singular vector of $M_{XY}$ and $\alpha \neq 0$ a scalar, and shift the mean along it:

$$Q_X = \mathcal{N}(\alpha \operatorname{Cov}(X)^{1/2} w_1, \operatorname{Cov}(X)), \qquad Q_X \mathsf{K} = \mathcal{N}(\alpha B \operatorname{Cov}(X)^{1/2} w_1, \operatorname{Cov}(Y)).$$

Then $D_{\mathrm{KL}}(Q_X \| P_X) = \alpha^2/2$ and $D_{\mathrm{KL}}(Q_X \mathsf{K} \| P_X \mathsf{K}) = \frac{\alpha^2}{2} \|M_{XY} w_1\|^2 = \frac{\alpha^2}{2} \sigma_1^2$.

For the mutual-information form, the Markov property gives $P_{Y|U} = P_{X|U} \mathsf{K}$ and $P_Y = P_X \mathsf{K}$, so $I(U;Y) = \mathbb{E}_U D_{\mathrm{KL}}(P_{Y|U} \| P_Y) \leq \sigma_1^2 \mathbb{E}_U D_{\mathrm{KL}}(P_{X|U} \| P_X) = \sigma_1^2 I(U;X)$. For $U = X_1' + Z'$ with $Z' \sim \mathcal{N}(0, 1/\alpha)$, $\alpha > 0$, independent of everything else, $I(U;X) = \frac{1}{2} \ln(1+\alpha)$ and $I(U;Y) = -\frac{1}{2} \ln(1 - \sigma_1^2 \alpha/(1+\alpha))$, and their ratio tends to $\sigma_1^2$ as $\alpha \to 0$, so the constant is best possible. □

*Proof of Proposition S6.2.* Write $\tilde{X}_k = \operatorname{Cov}(X_k)^{-1/2} X_k$. For a centered Gaussian Markov chain

$$\mathbb{E}[\tilde{X}_k \mid \tilde{X}_{k-1}, \ldots, \tilde{X}_0] = M_k \tilde{X}_{k-1},$$

so the tower property gives $\mathbb{E}[\tilde{X}_L \mid \tilde{X}_0] = M_L \cdots M_1 \tilde{X}_0$ and $\operatorname{Cov}(\tilde{X}_L, \tilde{X}_0) = M_L \cdots M_1$, which is (a), the operator norm being submultiplicative. The pair $(X_0, X_L)$ is jointly Gaussian, so $X_L$ is a linear function of $X_0$ plus independent Gaussian noise, whose covariance $\operatorname{Cov}(X_L \mid X_0) \succeq \operatorname{Cov}(X_L \mid X_{L-1}) \succ 0$ by the Markov property. Proposition S6.1 applies with $M_{X_0 X_L} = M_L \cdots M_1$, and $U$, $X_0$, $X_L$ form a Markov chain, which gives (b). For scalars (a) is the product of the correlations, and $I(X_0; X_L) = -\frac{1}{2} \ln(1 - \operatorname{Corr}(X_0, X_L)^2)$ for jointly Gaussian scalars, which is (c). □

Relay conventions. Let a source sense a static scalar world $\eta \sim \mathcal{N}(0, P_0)$ through noise of variance $R^s$, let every hop add independent channel noise of variance $R_c$, and let relays not sense. Under posterior-mean forwarding the message after $u$ hops is $y_u = m_{u-1} + n_u$, with $m_0 = \mathbb{E}[\eta \mid s]$ and $m_u = \mathbb{E}[\eta \mid y_u]$. Writing $v_u = \operatorname{Var}(m_u) = \operatorname{Cov}(\eta, m_u)$, the message has $\operatorname{Cov}(\eta, y_u) = v_{u-1}$ and $\operatorname{Var}(y_u) = v_{u-1} + R_c$, so $v_u = v_{u-1}^2/(v_{u-1} + R_c) \leq v_{u-1}^2/R_c$ and $I(\eta; y_u) = -\frac{1}{2} \ln(1 - v_u/P_0)$; once $v_u < R_c$ the information decays doubly exponentially. Under unit-variance forwarding every relay sends its content scaled to unit variance, so each hop multiplies the squared correlation with $\eta$ by $1/(1+R_c)$, starting from $P_0/(P_0 + R^s)$, and the information decays exponentially. Under unbiased forwarding every relay sends what it received, so after $u$ hops the message is $\eta$ plus noise of variance $R^s + uR_c$, and $I = \frac{1}{2} \ln(1 + P_0/(R^s + uR_c))$ decays like $P_0/(2uR_c)$. In a persistent world $\eta_{t+1} = F\eta_t + w_t$ with $|F| < 1$, let relay $k$ update $m_k(t+1) = F[(1-g)m_k(t) + g y_{k-1}(t)]$, where $y_{k-1}(t)$ is $m_{k-1}(t)$ plus independent channel noise. The innovation $w_t$ and the channel noise are independent of $m(t)$, so in stationarity $c_k = \operatorname{Cov}(\eta_t, m_k(t))$ satisfies $c_k = F^2[(1-g)c_k + g c_{k-1}]$, that is, $c_k = \psi c_{k-1}$ with $\psi = F^2 g/(1 - F^2(1-g))$. Since $g \leq 1$, $\psi \leq F^2$, and $\psi \to 1$ as $F \to 1$ at fixed $g > 0$.

*Proof of Proposition S6.3.* (a) Under (Lin) the schedule is independent of $U$ and of the noises, and $I(U;Z) \leq I(U; Z, (W(t))_t) = I(U; Z \mid (W(t))_t)$, so it suffices to prove the bound for a fixed schedule. Write $\Psi(t,t') = W(t-1) \cdots W(t')$, with $\Psi(t,t) = I$ (the transition product written $\Phi(t,t')$ in Section 8, the letter $\Phi$ being reserved here for the code potential). Unrolling the recursion, $e_i(t) = \mathfrak{a} U + Z_e$ with $\mathfrak{a} = \sum_{t' \leq s} \Psi_{ij}(t, t'+1) g_j(t')$, where $Z_e$ collects the initial errors and the other noise components; it is Gaussian by (G) and independent of $U$. The row-sum norm is submultiplicative, so $|\Psi_{ij}(t, t'+1)| \leq \lambda^{t-1-t'}$, and under (Loc) the entry vanishes when $t - 1 - t' < d(i,j)$, which is the light cone of Theorem 8.5. The exponents $t - 1 - t'$ that contribute are therefore at least $n_0 = \max\{d(i,j), t - 1 - s\}$, and $|\mathfrak{a}| \leq \bar{g} \sum_{n \geq n_0} \lambda^n = \bar{g} \lambda^{n_0}/(1-\lambda)$. The mean is $m_i(t) = \eta_t + e_i^{\mathrm{com}}(t) + e_i(t) = \mathfrak{a} U + Z_m$ with $Z_m = \eta_t + e_i^{\mathrm{com}}(t) + Z_e$, where the common mode $e_i^{\mathrm{com}}(t)$ of Lemma 2.13 is a linear function of the world path; $Z_m$ is Gaussian and independent

of $U$, because $U$ is independent of the world, the initial errors and the other noise components jointly. Both $Z_e$ and $Z_m$ contain, with coefficient one, the component of $\zeta_i(t-1)$ that is independent of everything else, so their variances are at least $q_{\min}$. For $Z$ Gaussian and independent of $U$, $I(U;\mathfrak{a}U+Z)=h(\mathfrak{a}U+Z)-h(Z)\le\frac{1}{2}\ln(1+\mathfrak{a}^2\operatorname{Var}(U)/\operatorname{Var}(Z))\le\mathfrak{a}^2\operatorname{Var}(U)/(2\operatorname{Var}(Z))$, where $h(\cdot)$ is differential entropy and a Gaussian maximizes it at fixed variance. Combining the bounds gives (a).

(b) On the ring $W$ is symmetric and nonnegative with $W\mathbf{1}=\lambda\mathbf{1}$, and $\lambda<1$ gives the stationary process. Place $i$ at 0 and $j$ at $d=d(i,j)$. The local noises injected at $j$ enter $m_i(t)$ only through $X_j=\sum_{n\ge0}(W^n)_{0d}\zeta_j(t-1-n)$, and $m_i(t)=X_j+Y_j$ with $Y_j$ Gaussian and independent of the noise sequence at $j$, so $m_i(t)$ depends on that sequence only through $X_j$. Hence

$$I=I(X_j;X_j+Y_j)=-\frac{1}{2}\ln\Big(1-\frac{qG(d)}{\operatorname{Var}(m_i(t))}\Big),\qquad G(d)=\sum_{n\ge0}\big((W^n)_{0d}\big)^2,$$

and since $\operatorname{Var}(m_i(t))$ does not depend on $d$, the rate of $I$ is that of $G(d)$. The entries $(W^n)_{0d}$ are nonnegative, and their sum over $n$ is the resolvent

$$\sum_{n\ge0}(W^n)_{0d}=\frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{e^{\mathrm{i}ud}}{1-c-2a\cos u}\mathrm{d}u=\frac{e^{-md}}{\sqrt{(1-c)^2-4a^2}},\qquad\cosh m=\frac{1-c}{2a}=1+\frac{\varepsilon}{2a},$$

by the standard evaluation of the integral by residues; $(1-c)^2>4a^2$ because $\lambda<1$. Upper bound: a sum of squares of nonnegative terms is at most the square of their sum, so $G(d)\le e^{-2md}/((1-c)^2-4a^2)$. Lower bound: fix $A_0>m/\ln(1/\lambda)$. Since $(W^n)_{0d}\le\lambda^n$, the terms with $n>A_0d$ sum to at most $\lambda^{A_0d}/(1-\lambda)$, which is $o(e^{-md})$, so for large $d$ the terms with $n\le A_0d$ sum to at least half the resolvent, and by the Cauchy–Schwarz inequality

$$G(d)\ge\frac{1}{A_0d+1}\Big(\sum_{n\le A_0d}(W^n)_{0d}\Big)^2\ge\frac{e^{-2md}}{4(A_0d+1)((1-c)^2-4a^2)}.$$

The two bounds give $-\ln G(d)/d\to2m$, with a prefactor between a multiple of $1/d$ and a constant. Since $I=qG(d)(1+o(1))/(2\operatorname{Var}(m_i(t)))$ as $d\to\infty$, the same holds for $I$. □

## S30.2 Calibration and withdrawal

*Proof of Proposition 6.3.* (a) The expected free energy of a policy is computed from the receiver's own predictive density $\int p_i(o\mid\eta,\pi)q_i(\eta)\mathrm{d}\eta$ of outcomes together with its prior preferences. Under C0 the likelihood of translated messages does not contain $\Upsilon_{ij}$, so the expected free energy depends on the mistranslations only through the belief $q_i$, which they influence through past data. (b) In the static two-group model the believed information of an attention set $\mathcal{J}$ is $\frac{K}{2}\ln(\Lambda_i(\mathcal{J})/\Lambda_i(\varnothing))$ with $\Lambda_i(\mathcal{J})=\sigma_\eta^{-2}+1/R_i^s+\sum_{j\in\mathcal{J}}g_j^2/\hat r_{ij}^0$, where $\hat r_{ij}^0$ is the fixed assigned noise, for example $R_c+g_j^2R_j^s$. This involves neither the data nor $\Upsilon_{ij}$, so the policy posterior, proportional to $E_i(\pi)\exp(-\gamma G_i(\pi))$, does not depend on the mistranslations at any $\gamma$, and two senders with equal assigned noise contribute equal terms. □

*Proof of Theorem 6.4.* (a) Write $\Upsilon=\Upsilon_{ij}$. The translated message is $\tilde y_{ij}=\Upsilon g_j(\eta+v_j)+\nu_{ij}$, so $\tilde y_{ij}-g_j\eta=(\Upsilon-I)g_j\eta+g_j\Upsilon v_j+\nu_{ij}$, a sum of independent terms. With $\Upsilon$ orthogonal, $\mathbb{E}\|\Upsilon v_j\|^2=KR_j^s$, and under C1g the population value $\hat r_{ij}=K^{-1}\mathbb{E}\|\tilde y_{ij}-g_j\eta\|^2$ equals $R_c+g_j^2R_j^s+g_j^2\sigma_\eta^2\|\Upsilon-I\|_F^2/K$. Under C1i the receiver's estimate is $\hat\eta_i=g_is_i$, whose error has variance $g_iR_i^s$ per coordinate and is uncorrelated with $\hat\eta_i$, so its model assigns the innovation $\tilde y_{ij}-g_j\hat\eta_i$ the variance $g_j^2g_iR_i^s+\hat r_{ij}$

per coordinate. The actual innovation is $g_j(\Upsilon - g_i I)\eta + g_j \Upsilon v_j + \nu_{ij} - g_j g_i v_i$, with second moment per coordinate $g_j^2\sigma_\eta^2\|\Upsilon - g_i I\|_F^2/K + g_j^2 R_j^s + R_c + g_j^2 g_i^2 R_i^s$. For orthogonal $\Upsilon$, $\|\Upsilon - g_i I\|_F^2 = K(1-g_i)^2 + g_i\|\Upsilon - I\|_F^2$, both sides being $K(1+g_i^2) - 2g_i \operatorname{tr}\Upsilon$, and $(1-g_i)^2\sigma_\eta^2 + g_i^2 R_i^s = g_i R_i^s$ by the definition of $g_i$. Matching the model value to the actual one gives $\hat{r}_{ij} = R_c + g_j^2 R_j^s + g_i g_j^2 \sigma_\eta^2 \|\Upsilon - I\|_F^2/K$.

(b) Under C1 the receiver's model treats attended messages as independent isotropic observations of $g_j\eta$ with noise variance $\hat{r}_{ij}$, so the believed information of an attention set $\mathcal{J}$ is $\frac{K}{2}\ln(\Lambda_i(\mathcal{J})/\Lambda_i(\varnothing))$ with $\Lambda_i(\mathcal{J}) = \sigma_\eta^{-2} + 1/R_i^s + \sum_{j\in\mathcal{J}} \varpi_{ij}$. This is strictly increasing in $\sum_{j\in\mathcal{J}} \varpi_{ij}$, and every $\varpi_{ij}$ is positive, so the maximizers over $|\mathcal{J}| \le \kappa$ consist of $\kappa$ largest calibrated precisions, ties broken arbitrarily, or of all senders if fewer are available.

(c) If $\varpi_{ib} < \varpi^{\text{in}}_{i(\kappa)}$, at least $\kappa$ in-group senders have calibrated precision above $\varpi_{ib}$, so by (b) the cross-group sender $b$ is not attended; with $\varpi_{ib} = g_b^2/(R_c + g_b^2 R_b^s + \hat{r}^{\text{mt}}_{ib})$ this condition is (65). If $\varpi_{ib} > \varpi^{\text{in}}_{i(\kappa)}$ for some cross-group sender $b$, every attention set of $\kappa$ in-group senders contains a sender of precision at most $\varpi^{\text{in}}_{i(\kappa)}$, and replacing it by $b$ raises the sum of precisions, so by (b) the receiver attends at least one cross-group sender. When no cross-group precision equals $\varpi^{\text{in}}_{i(\kappa)}$ the two cases are exhaustive, which gives the equivalence. With homogeneous sensing noise all in-group precisions equal $g^2/(R_c + g^2 R^s)$, the right side of (65) vanishes, and $\hat{r}^{\text{mt}}_{ib} > 0$ whenever $\Upsilon_{ib} \neq I$, because $g_i > 0$. □

### S30.3 Consistency

For a walk $p$ write $\Theta(p)$ for the ordered product of the translations along it, with inverses on steps against an edge.

*Proof of Theorem 6.5.* (a) We restate the proof of Dennis (2026c) for linear translations. If $\mu$ is in consensus, composing the edge relations around a directed closed walk $\mathfrak{c}$ at $i_0$ gives $\mu_{i_0} = \operatorname{Hol}(\mathfrak{c})\mu_{i_0}$. Conversely, let $v$ be fixed by all such holonomies and set $\mu_i = \Theta(p)v$ along a directed path $p$ from $i_0$ to $i$. A second path $p'$ gives the same value: strong connectivity supplies a directed path $p''$ from $i$ back to $i_0$, the products $\Theta(p'')\Theta(p)$ and $\Theta(p'')\Theta(p')$ are holonomies of directed closed walks at $i_0$ and both fix $v$, and $\Theta(p'')$ is invertible. Extending a directed path to $j$ by the edge $j \to i$ gives a directed path to $i$, so $\mu_i = \Theta_{ij}\mu_j$. The map $\mu \mapsto \mu_{i_0}$ is linear, and it is injective because every $\mu_i$ is determined by $\mu_{i_0}$ along a path.

(b) Let $\Theta_{\mathcal{T}}(i_0 \to i) = \Theta(p)$ for the path $p$ from $i_0$ to $i$ in $\mathcal{T}$. Each edge relation can be solved for either endpoint, so by induction along tree paths every consensus configuration satisfies $\mu_i = \Theta_{\mathcal{T}}(i_0 \to i)\mu_{i_0}$, and $\mu \mapsto \mu_{i_0}$ is injective. For a non-tree edge $e = (j \to i)$ the relation $\mu_i = \Theta_{ij}\mu_j$ then reads $\operatorname{Hol}_e v = v$ with $v = \mu_{i_0}$ and $\operatorname{Hol}_e = \Theta_{\mathcal{T}}(i_0 \to i)^{-1}\Theta_{ij}\Theta_{\mathcal{T}}(i_0 \to j)$, the holonomy of the closed walk described in the statement. Conversely, for $v$ fixed by every $\operatorname{Hol}_e$ the configuration $\mu_i = \Theta_{\mathcal{T}}(i_0 \to i)v$ satisfies the tree relations by construction and the non-tree relations by the fixed-point condition.

(c) The first statement restates (b); on strongly connected graphs both fixed spaces are images of $\operatorname{Cons}(\Theta)$ under the same injective map, so they coincide. If every $\operatorname{Hol}_e$ is the identity, put $g_i = \Theta_{\mathcal{T}}(i_0 \to i)$; then $\Theta_{ij} = g_i g_j^{-1}$ on tree edges by construction and on non-tree edges because $g_i^{-1}\Theta_{ij}g_j = \operatorname{Hol}_e = I$. Conversely, the holonomy of a coboundary telescopes to the identity around every closed walk. Since $\dim \operatorname{Cons}(\Theta) = \dim \bigcap_e \operatorname{Fix}(\operatorname{Hol}_e)$, full consensus holds exactly when every $\operatorname{Hol}_e$ is the identity. □

*Proof of Proposition S6.5.* Along a closed walk $i_0, i_1, \dots, i_L = i_0$ every step contributes $U_{i_u}U_{i_{u-1}}^{-1}$, whether it runs along an edge, contributing $\Theta_{i_u i_{u-1}}$, or against one, contributing $\Theta^{-1}_{i_{u-1} i_u} =$

$(U_{i_{u-1}}U_{i_u}^{-1})^{-1}$. The product telescopes to $U_{i_0}U_{i_0}^{-1} = I$, and for general encodings the cancellations $\mathcal{E}_{i_u}^{-1} \circ \mathcal{E}_{i_u}$ give the identity map. By Theorem 6.5(c) the consensus space has dimension $K$; the configurations $\mu_i = U_i x$ satisfy every edge relation and form a $K$-dimensional space, so they are all of it, and by Proposition 6.6(a) the least eigenvalue of the frustration form is zero. The last statement is the contrapositive: if the translations are the true ones and every agent encodes one true world invertibly, the least frustration vanishes. □

Learned holonomy. In the tangent learning model of Dennis (2026c), receiver $i$ learns a generator $\omega_e \in \mathfrak{gl}(K)$ of its translation $\Theta_{ij} = e^{\omega_e}$ on each edge $e = (j \to i)$ from grounded messages, whose law depends on $\omega_e$ only through $T_e\omega_e$, with $T_e$ the stacked public codes of the modalities read on $e$. The model is linear in the generators: a message on $e$ has mean $T_e(I - \omega_e)\bar{y}$ for a referent $\bar{y}$, the first-order form of the grounded message model at $\Theta_{ij} = e^{\omega_e}$, with Gaussian channel noise, and the prior on the generators is Gaussian. The convergence statement used here is part (ii) of the theorem on learning with a loop prior in the chapter on communication codes and learned alignment of Dennis (2026c). Its hypotheses are a positive-definite prior precision, referents drawn independently with a finite positive-definite second moment, observed source labels, and $\lceil \upsilon_e n \rceil$ independent exchanges on every edge $e$ with rates $\upsilon_e > 0$; its conclusion concerns the mean of the exact Gaussian posterior, so no step-size schedule enters, and holds almost surely as $n \to \infty$ by the strong law of large numbers. With an isotropic anchor, no loop prior and grounded messages on every edge, that posterior mean converges to $(I - \Pi_N)\omega^\star$, where $\omega^\star$ are the true generators and $\Pi_N$ is the orthogonal projection onto the generators that no message distinguishes from zero, those with $T_e\omega_e = 0$ on every edge; on edge $e$ it acts by left multiplication with the orthogonal projector $\Pi_e$ onto $\ker T_e$. For private encodings near a common frame, taken as the identity, the true generators are, to first order in $a_i = \log U_i$, the coboundary $\omega_e^\star = a_i - a_j$, which sums to zero around every directed closed walk. By the Baker–Campbell–Hausdorff formula the learned holonomy of a directed closed walk is therefore $I - \sum_e \Pi_e(a_i - a_j)$ to first order, the sum running over the edges $e = (j \to i)$ of the walk. Its first-order part vanishes when $\Pi_e$ is the same on every edge of the walk, as when all its edges are read through one modality. To first order in the frame error it is not detectable from messages: the learned generators differ from $\omega^\star$ by $-\Pi_e\omega_e^\star$ on each edge, and $T_e\Pi_e = 0$, so in the tangent model every message has the same law under the learned generators as under the true ones, whose translations are flat (Proposition S6.5). Beyond the tangent model the message means under $e^{(I-\Pi_e)\omega_e^\star}$ and $e^{\omega_e^\star}$ differ at second order, by $\frac{1}{2}T_e\omega_e^\star\Pi_e\omega_e^\star\bar{y}$ to leading order, so the statement does not extend to the exponentiated translations at that order. The first-order formula is checked in Check S47.53.

*Proof of Proposition 6.6.* (a) By the definition of $\mathcal{B}$, $\|\mathcal{B}\mu\|^2 = \mathfrak{F}(\mu)$. Since every $\varpi_{ij} > 0$ and $M_i \succ 0$, $\mathcal{B}\mu = 0$ exactly when $\mu_i = \Theta_{ij}\mu_j$ on every edge, so $\ker \mathcal{L} = \ker \mathcal{B} = \mathrm{Cons}(\Theta)$. By the Rayleigh principle $\lambda_{\min}(\mathcal{L}) = \min_{\|\mu\|=1} \mathfrak{F}(\mu)$, which is positive exactly when $\mathrm{Cons}(\Theta) = 0$, that is, by Theorem 6.5(b), when the $\mathrm{Hol}_e$ have no common nonzero fixed vector.

(b) An orthogonal change of internal coordinates $\mu_k \mapsto g_k\mu_k$ replaces $\Theta_{k+1,k}$ by $g_{k+1}\Theta_{k+1,k}g_k^{-1}$ and conjugates $\mathcal{L}$ by an orthogonal block-diagonal matrix, so it preserves the spectrum. Choosing $g_1 = I$ and $g_{k+1} = g_k\Theta_{k+1,k}^{-1}$ for $k = 1, \dots, N-1$ makes every translation the identity except the one on the edge from $N$ to 1, which becomes the holonomy Hol based at agent 1. Over $\mathbb{C}$ the orthogonal matrix Hol is diagonalized by a unitary matrix, applied at every agent, with eigenvalues $e^{\mathrm{i}\theta_r}$, $\theta_r \in (-\pi, \pi]$, and the frustration decouples into $K$ scalar rings in which one edge carries the phase $e^{\mathrm{i}\theta_r}$ and the others carry one. The diagonal unitary change of variables $\mu'_k = e^{\mathrm{i}(k-1)\theta_r/N}\mu_k$ gives every edge the phase $e^{\mathrm{i}\theta_r/N}$, and the resulting operator is translation invariant on $\mathbb{Z}/N$. Its eigenvectors are the waves $\mu'_k = e^{2\pi\mathrm{i}uk/N}$, $u = 0, \dots, N-1$, whose residual on every edge has squared

modulus $|e^{2\pi \mathrm{i}u/N} - e^{\mathrm{i}\theta_r/N}|^2 = 2 - 2\cos((2\pi u - \theta_r)/N)$ times $|\mu'_k|^2$, which gives the spectrum. For $u = 0$ the angle $(2\pi u - \theta_r)/N$ has modulus $|\theta_r|/N$, and for $1 \le u \le N-1$ it lies in $[\pi/N, 2\pi - \pi/N]$, at distance at least $\pi/N \ge |\theta_r|/N$ from $2\pi\mathbb{Z}$, so the least eigenvalue is $2 - 2\cos(\phi/N) = 4\sin^2(\phi/(2N))$.

(c) Lower bound. Take a simple cycle $\mathfrak{c} \in \mathcal{C}_i$ traversed as $i = i_0, i_1, \dots, i_L = i$, let $\Theta^{(u)}$ be the transport of step $u$ as in Definition 6.1(c) and $r_u = \mu_{i_u} - \Theta^{(u)}\mu_{i_{u-1}}$ its residual. Unrolling gives $(I - \mathrm{Hol}^{(i)}(\mathfrak{c}))\mu_i = \sum_{u=1}^{L} \Theta^{(L)} \cdots \Theta^{(u+1)} r_u$. The transports are orthogonal, so the residual of a step against an edge has the norm of that edge's residual and every product of transports has norm one; the Cauchy–Schwarz inequality gives $\|(I - \mathrm{Hol}^{(i)}(\mathfrak{c}))\mu_i\|^2 \le L \sum_u \|r_u\|^2 \le L\mathfrak{F}(\mu)/\varpi_{\min}$, since a simple cycle uses each edge once. Dividing by $L = |\mathfrak{c}|$ and averaging over $\mathcal{C}_i$ gives $\mu_i^\top \mathcal{D}_i \mu_i \le \mathfrak{F}(\mu)/\varpi_{\min}$ for every $i$. Choosing the agent with the largest $\|\mu_i\|$, for which $\|\mu_i\|^2 \ge \|\mu\|^2/N$, gives $\mathfrak{F}(\mu) \ge \varpi_{\min} N^{-1} \min_i \lambda_{\min}(\mathcal{D}_i) \|\mu\|^2$. Upper bound. For a unit vector $v$ the configuration $\mu_i = \Theta_{\mathcal{T}}(i_0 \to i)v$ has $\|\mu\|^2 = N$, zero residual on tree edges, and residual $\Theta_{\mathcal{T}}(i_0 \to i)(I - \mathrm{Hol}_e)v$, of norm $\|(I - \mathrm{Hol}_e)v\|$, on a non-tree edge $e = (j \to i)$; minimizing $\mathfrak{F}(\mu)/\|\mu\|^2$ over $v$ gives the bound. Single cycle. Take $\mathcal{C}_i$ to be the cycle itself for every $i$. The holonomies based at different agents are conjugate, and $(I - \mathrm{Hol})^\top (I - \mathrm{Hol}) = 2I - \mathrm{Hol} - \mathrm{Hol}^\top$ has the eigenvalues $2 - 2\cos\theta_r$, because Hol is normal, so the lower bound is $4\sin^2(\phi/2)/N^2$. With $\sin(\phi/(2N)) \le \phi/(2N)$ and $\sin(\phi/2) \ge \phi/\pi$ on $[0, \pi]$, the exact value $4\sin^2(\phi/(2N))$ of (b) is at most $\phi^2/N^2 \le (\pi^2/4) \cdot 4\sin^2(\phi/2)/N^2$. □

*Proof of Proposition S6.7.* (a) $\delta_{\mathfrak{S}}$ maps $\mathbb{R}^{NK}$ to $\mathbb{R}^{|\mathcal{E}_{\mathrm{a}}|r_c}$, so its kernel has dimension at least $NK - |\mathcal{E}_{\mathrm{a}}|r_c$. (b) The holonomy of the cycle is the rotation by $0.3 + 0.5 + \pi/2 - 0.8 = \pi/2$, which fixes no nonzero vector, so $\mathrm{Cons}(\Theta) = 0$ by Theorem 6.5. Number the agents so that the edges are $1 \to 2$, $2 \to 3$ and $3 \to 1$, and let $\alpha_1$, $\alpha_2$, $\alpha_3$ be the cosines of the rotation angles of $\Theta_{21}$, $\Theta_{32}$, $\Theta_{13}$, which lie in $(0, 1)$ because the angles lie in $(0, \pi/2)$. For a rotation by $\phi$, $(1\ 0)$ times its inverse is $(\cos\phi \quad \sin\phi)$, so the three rows of $\delta_{\mathfrak{S}}$, restricted to the first coordinates of $\mu_1$, $\mu_2$ and $\mu_3$, form the matrix with rows $(-1, \alpha_1, 0)$, $(0, -1, \alpha_2)$ and $(\alpha_3, 0, -1)$, whose determinant is $\alpha_1\alpha_2\alpha_3 - 1 \ne 0$. Hence $\delta_{\mathfrak{S}}$ has rank 3 and $\dim H^0(\mathfrak{S}) = 6 - 3 = 3$. □

*Proof of Proposition S6.8.* $\mathcal{L} = \mathcal{B}^\top \mathcal{B}$ is symmetric and positive semidefinite, and with orthogonal translations $\mu^\top \mathcal{L} \mu = \sum_e \|\mu_i - \Theta_{ij}\mu_j\|^2 \le 2\sum_e (\|\mu_i\|^2 + \|\mu_j\|^2) \le 2\Delta_{\mathrm{a}}\|\mu\|^2$, every agent lying on at most $\Delta_{\mathrm{a}}$ edges. So the spectrum of $\mathcal{L}$ lies in $[0, 2\Delta_{\mathrm{a}}]$ and that of $I - \alpha\mathcal{L}$ in $[0, 1]$, and the norm of $I - \alpha\mathcal{L}$ is its largest eigenvalue, $1 - \alpha\lambda_{\min}(\mathcal{L})$. Multiplying by $1 - \varepsilon$ gives the contraction factor and $\varepsilon_{\mathrm{cal}} = 1 - (1 - \varepsilon)(1 - \alpha\lambda_{\min}(\mathcal{L}))$. The value on the cycle is Proposition 6.6(b). □

*Proof of Proposition S6.9.* With orthogonal frames each translation is $\Theta_{ij} = U_i \Upsilon_{ij} U_j^{-1}$, so along a closed walk the intermediate frames cancel and $\mathrm{Hol}(\mathfrak{c}) = U_{i_0} \Upsilon^{(L)} \cdots \Upsilon^{(1)} U_{i_0}^{-1}$, where $\Upsilon^{(u)}$ is the mistranslation of step $u$, inverted on steps against an edge; inversion preserves $\|\Upsilon - I\|$ for orthogonal $\Upsilon$. Telescoping, $\Upsilon^{(L)} \cdots \Upsilon^{(1)} - I = \sum_u (\Upsilon^{(L)} \cdots \Upsilon^{(u+1)})(\Upsilon^{(u)} - I)$, and orthogonal factors and orthogonal conjugation preserve norms, so $\|\mathrm{Hol}(\mathfrak{c}) - I\| \le |\mathfrak{c}|\delta_{\mathrm{att}}$. An attended cycle graph of $N$ agents can be oriented as a directed cycle without changing its frustration, since an edge term $\|\mu_i - \Theta_{ij}\mu_j\|$ equals $\|\mu_j - \Theta_{ij}^{-1}\mu_i\|$ for orthogonal $\Theta_{ij}$. Let $\phi \in [0, \pi]$ be the smallest modulus of an eigen-angle of its holonomy. By Proposition 6.6(b), $\lambda_{\min}(\mathcal{L}) = 4\sin^2(\phi/(2N)) \le (\phi/N)^2$. Also $2\sin(\phi/2) \le \|\mathrm{Hol} - I\| \le N\delta_{\mathrm{att}}$, the norm being $2\sin$ of half the largest modulus of an eigen-angle, and $\phi \le \pi\sin(\phi/2)$ on $[0, \pi]$, so $\phi/N \le \pi\delta_{\mathrm{att}}/2$. □

## S30.4 Amplification

*Proof of Theorem 6.8.* (a) For the edge $e = (i \to j)$ put

$$\tilde{A}_e = P_j^{-1/2} A_{ji} P_i^{1/2} = [P_j^{1/2} H_i^\top S_{ji}^{-1/2}]\,[S_{ji}^{-1/2} H_i P_i^{1/2}].$$

Around the cycle the factors $P_k^{\pm 1/2}$ cancel, so $P_1^{-1/2} A_{\mathfrak{c}} P_1^{1/2} = \tilde{A}_{(N\to 1)} \cdots \tilde{A}_{(1\to 2)}$, and $\rho(A_{\mathfrak{c}}) \le \prod_e \|\tilde{A}_e\|$, because the spectral radius is invariant under similarity and bounded by any submultiplicative norm. The first bracket has squared norm $\lambda_{\max}(S_{ji}^{-1/2} X_{ji} S_{ji}^{-1/2}) = \mathrm{snr}_e/(1+\mathrm{snr}_e)$. When $\hat{R}_i \succ 0$, writing $X_{ji} = \hat{R}_i^{1/2} Y \hat{R}_i^{1/2}$ shows that $S_{ji}^{-1} X_{ji}$ is similar to $(I+Y)^{-1}Y$, whose eigenvalues are $x/(1+x)$ for the eigenvalues $x$ of $Y$, which identifies $\mathrm{snr}_e$. The second bracket has squared norm $\lambda_{\max}(S_{ji}^{-1/2} X_{ii} S_{ji}^{-1/2}) = \mathfrak{r}_e$. Since $\mathrm{snr}_e/(1+\mathrm{snr}_e) \le 1$, (66) follows.

(b) For noiseless codes $H_i \mathcal{K}_{ji} = H_i P_j H_i^\top S_{ji}^{-1} = I$, so every edge map is idempotent. A receiver $j$ whose mean is perturbed by $x_j$ and that hears a sender perturbed by $x_i$ moves to $(I - A_{ji})x_j + A_{ji} x_i$. Let agent 1 be perturbed by $y$ at the start of a circulation and let each agent update when it hears its predecessor. In the first circulation agents $2, \dots, N$ hold no perturbation; in later ones agent $k \ge 2$ holds $A_{k,k-1}$ applied to its predecessor's previous perturbation, and its own term drops out because $(I - A_{k,k-1}) A_{k,k-1} = 0$. In either case agent $k$ ends the circulation with $A_{k,k-1} \cdots A_{21} y$, and agent 1 ends it with $(I - A_{1N})y + A_{\mathfrak{c}} y = \bar{A}_{\mathfrak{c}} y$, so after $n$ circulations agent 1 holds $\bar{A}_{\mathfrak{c}}^n x$ when it started with $x$. Since $A_{1N}$ is idempotent and $A_{\mathfrak{c}} = A_{1N} A_{\mathfrak{c}}$, in the decomposition $\mathrm{ran}(A_{1N}) \oplus \ker(A_{1N})$ the map $\bar{A}_{\mathfrak{c}}$ is block upper triangular, with diagonal blocks $A_{\mathfrak{c}}$ restricted to $\mathrm{ran}(A_{1N})$ and the identity on $\ker(A_{1N})$, which is nonzero because the code read by agent 1 has rank below $K$. The eigenvectors of $A_{\mathfrak{c}}$ with nonzero eigenvalues lie in $\mathrm{ran}(A_{1N})$, so $\rho(\bar{A}_{\mathfrak{c}}) = \max\{\rho(A_{\mathfrak{c}}), 1\}$, and $\|\bar{A}_{\mathfrak{c}}^n\|$ grows geometrically exactly when this exceeds one. □

*Proof of Proposition S6.10.* (a) With $P_i = P_j$, $X_{ii} = X_{ji}$, so $\mathfrak{r}_e = \mathrm{snr}_e/(1+\mathrm{snr}_e)$, and (66) gives $\rho(A_{\mathfrak{c}}) \le \prod_e \mathrm{snr}_e/(1+\mathrm{snr}_e)$, which is at most one, and below one when every $\hat{R}_i \succ 0$.

(b) $P_i \preceq e^{\mathrm{sp}(P_i \| P_j)} P_j$, hence $X_{ii} \preceq e^{\mathrm{sp}(P_i\|P_j)} X_{ji}$ and $\mathfrak{r}_e \le e^{\mathrm{sp}(P_i\|P_j)} \mathrm{snr}_e/(1+\mathrm{snr}_e)$. By (66), $\rho(A_{\mathfrak{c}}) \le \prod_e e^{\mathrm{sp}(P_i\|P_j)/2} \mathrm{snr}_e/(1+\mathrm{snr}_e)$, which is at most one under the stated condition. If all priors lie within Thompson distance $\delta_{\mathrm{T}}$ of $P$, then $P_i \preceq e^{\delta_{\mathrm{T}}} P \preceq e^{2\delta_{\mathrm{T}}} P_j$, so $\mathrm{sp}(P_i \| P_j) \le 2\delta_{\mathrm{T}}$, and the condition holds when $N\delta_{\mathrm{T}} \le \sum_e \ln(1 + \mathrm{snr}_e^{-1})$.

(c) If $\hat{R}_i \succeq X_{ii}$, then $S_{ji} \succeq X_{ji} + X_{ii}$. Since $S_{ji} \succ 0$, the generalized Rayleigh quotient $\boldsymbol{w}^\top X_{ii} \boldsymbol{w} / \boldsymbol{w}^\top S_{ji} \boldsymbol{w}$ attains its maximum $\mathfrak{r}_e$ at some $\boldsymbol{w} \neq 0$. If $H_i^\top \boldsymbol{w} = 0$, then $\mathfrak{r}_e = 0$. Otherwise $\boldsymbol{w}^\top X_{ji} \boldsymbol{w} = \|P_j^{1/2} H_i^\top \boldsymbol{w}\|^2 > 0$, so $\boldsymbol{w}^\top X_{ii} \boldsymbol{w} < \boldsymbol{w}^\top (X_{ii} + X_{ji}) \boldsymbol{w} \le \boldsymbol{w}^\top S_{ji} \boldsymbol{w}$ and $\mathfrak{r}_e < 1$. By (66), $\rho(A_{\mathfrak{c}}) < 1$. A level-1 receiver assigns $\hat{R}_i = H_i P_i H_i^\top + R_c \succeq X_{ii}$ when the sender posts its prior covariance $P_i$.

(d) With a common code $H$ and noise $\hat{R}$, put $X_j = H P_j H^\top$ and $B_j = X_j (X_j + \hat{R})^{-1}$; then $H A_{j,j-1} = B_j H$, so $A_{\mathfrak{c}} = P_1 H^\top (X_1 + \hat{R})^{-1} B_N B_{N-1} \cdots B_2 H$. Since the nonzero spectra of $UV$ and $VU$ agree, with $U = P_1 H^\top (X_1 + \hat{R})^{-1}$ and $V = B_N \cdots B_2 H$, the nonzero spectrum of $A_{\mathfrak{c}}$ is that of $B_N B_{N-1} \cdots B_2 B_1$. Each $\hat{R}^{-1/2} B_j \hat{R}^{1/2} = Y_j (I + Y_j)^{-1}$ is symmetric with eigenvalues in $[0, 1)$, so $\rho(A_{\mathfrak{c}}) \le \prod_j \|Y_j (I + Y_j)^{-1}\| < 1$. For a common code $\boldsymbol{h}^\top$ of rank one with positive sender-specific scalar noises, $\boldsymbol{h}^\top A_{j,j-1} = x_j \boldsymbol{h}^\top$ with $x_j = \boldsymbol{h}^\top P_j \boldsymbol{h} / (\boldsymbol{h}^\top P_j \boldsymbol{h} + \hat{R}_{j-1}) \in [0, 1)$, and the only nonzero eigenvalue of $A_{\mathfrak{c}}$ is $\prod_j x_j < 1$. □

For codes of rank one the cycle eigenvalue factorizes.

**Lemma S30.2** (Codes of rank one)**.** *Let every sender $i$ on the cycle use a code $\boldsymbol{h}_i^\top$ of rank one with scalar assigned noise $\hat{R}_i \ge 0$, write $\langle \boldsymbol{x}, \boldsymbol{x}' \rangle_P = \boldsymbol{x}^\top P \boldsymbol{x}'$, $\|\boldsymbol{x}\|_P = \langle \boldsymbol{x}, \boldsymbol{x} \rangle_P^{1/2}$ and $\cos_P$ for the corresponding cosine, $j^-$ for the sender that $j$ reads and $i^+$ for the receiver that reads $i$. The cycle product has rank at most one, so its only eigenvalue that can be nonzero is its trace,*

$$\operatorname{tr} A_{\mathfrak{c}} = \prod_j \frac{\langle \boldsymbol{h}_j, \boldsymbol{h}_{j^-} \rangle_{P_j}}{\|\boldsymbol{h}_{j^-}\|_{P_j}^2 + \hat{R}_{j^-}} = \prod_j \cos_{P_j}(\boldsymbol{h}_j, \boldsymbol{h}_{j^-}) \cdot \mathfrak{M}_{\mathfrak{c}} \cdot \prod_i \frac{\|\boldsymbol{h}_i\|_{P_{i^+}}^2}{\|\boldsymbol{h}_i\|_{P_{i^+}}^2 + \hat{R}_i}, \qquad \mathfrak{M}_{\mathfrak{c}} = \prod_i \frac{\|\boldsymbol{h}_i\|_{P_i}}{\|\boldsymbol{h}_i\|_{P_{i^+}}}. \tag{S117}$$

*The factors of the first and last products have modulus at most one, and* $\mathfrak{M}_{\mathfrak{c}} = 1$ *for a common prior and for a common code. Hence* $|\operatorname{tr} A_{\mathfrak{c}}| > 1$ *requires* $\mathfrak{M}_{\mathfrak{c}} > 1$.

*Proof.* Each edge map has rank one, $A_{j,j^-} = P_j \boldsymbol{h}_{j^-} \boldsymbol{h}_{j^-}^\top / (\|\boldsymbol{h}_{j^-}\|_{P_j}^2 + \hat{R}_{j^-})$. A product of rank-one maps $\boldsymbol{a}_k \boldsymbol{b}_k^\top$ around a cycle has rank at most one, and its trace, which is its only possibly nonzero eigenvalue, is the product of the inner products $\boldsymbol{b}_{k+1}^\top \boldsymbol{a}_k$ of consecutive factors. Here these are $\boldsymbol{h}_j^\top P_j \boldsymbol{h}_{j^-} / (\|\boldsymbol{h}_{j^-}\|_{P_j}^2 + \hat{R}_{j^-})$, which gives the first form of (S117). Writing each factor as $\cos_{P_j}(\boldsymbol{h}_j, \boldsymbol{h}_{j^-}) \cdot \frac{\|\boldsymbol{h}_j\|_{P_j}}{\|\boldsymbol{h}_{j^-}\|_{P_j}} \cdot \frac{\|\boldsymbol{h}_{j^-}\|_{P_j}^2}{\|\boldsymbol{h}_{j^-}\|_{P_j}^2 + \hat{R}_{j^-}}$ and regrouping the last two products by the sender $i = j^-$, so that $j = i^+$, gives the second form. Cosines and noise factors have modulus at most one. For a common prior every ratio in $\mathfrak{M}_{\mathfrak{c}}$ is one, and for a common code $\mathfrak{M}_{\mathfrak{c}} = \prod_i \|\boldsymbol{h}\|_{P_i} / \prod_i \|\boldsymbol{h}\|_{P_{i^+}} = 1$, both products running over all agents of the cycle. □

Details of Example S6.11. In (i) the edge maps are $A_{ab} = P_a \boldsymbol{e}_1 \boldsymbol{e}_1^\top / (1 + R_c)$ and $A_{ba} = P_b \boldsymbol{e}_2 \boldsymbol{e}_2^\top / (1 + R_c)$, since $(P_a)_{11} = (P_b)_{22} = 1$, and Lemma S30.2 gives $\operatorname{tr} A_{\mathfrak{c}} = (P_a)_{21}(P_b)_{12}/(1+R_c)^2 = \alpha_a \alpha_b x/(1 + R_c)^2$. In (ii) the same lemma gives $\operatorname{tr} A_{\mathfrak{c}} = (P_a)_{21}(P_b)_{12} = (1-u)^2(1+2u)^2$, which exceeds one for $0 < u < \frac{1}{2}$ because $(1-u)(1+2u) = 1 + u(1-2u)$. The generalized eigenvalues $x$ of the pair $(P_a, P)$, the roots of $\det(P_a - xP) = 0$, satisfy $(1 + 2u - x)^2 = 4u^2 x/(1 - (1-u)^2)$; their product is $(1+2u)^2$, so they are $(1+2u)e^{\pm\vartheta}$ with $\cosh\vartheta = 1 + 2u/((2-u)(1+2u))$. Hence $\operatorname{dist}_{\mathrm{T}}(P_a, P) = \ln((1+2u)e^{\vartheta}) = \ln(1+2u) + \vartheta$, and the same holds for $P_b$ by exchanging the coordinates. As $u \to 0$, $\vartheta = \sqrt{2u}(1 + O(u))$ and $\ln(1+2u) = O(u)$, which gives the expansion.

## S30.5 Codes

**Lemma S30.3** (Value of matches)**.** *The value* (67) *is nondecreasing and concave in* $u$*, also under a capacity. For an edge on which* $n_{\mathrm{mis}}$ *of several binary features, each carried by one coordinate, are mismatched,* C1g *calibration gives the noise* $\hat{r}_0 + 4 n_{\mathrm{mis}} g^2 \sigma_\eta^2 / K$*; the believed value of the edge is convex and decreasing in* $n_{\mathrm{mis}}$*, and the value lost to mistranslation, also when capped at the value of the edge because the receiver withdraws, is concave, nondecreasing and zero at* $n_{\mathrm{mis}} = 0$*, hence subadditive.*

*Proof.* The argument of the logarithm in (67) is affine and increasing in $u$, because $\varpi^{=} > \varpi^{\neq}$, and the logarithm is concave and increasing; under a capacity, $u$ enters through the concave nondecreasing map $u \mapsto \min\{u, \kappa\}$, and a concave nondecreasing function of a concave function is concave. For the edge, the receiver models the message as $g\eta$ plus isotropic noise whose calibrated variance is, by Theorem 6.4(a) with $\|\Upsilon - I\|_F^2 = 4f$ for $f = n_{\mathrm{mis}}$, $\hat{r}(f) = \hat{r}_0 + 4 f g^2 \sigma_\eta^2 / K$ under C1g, and C1i multiplies the last term by $g_i$. The function $V(f) = \ln(1 + A_1/(A_2 + A_3 f))$ with positive constants $A_1 = g^2/\Lambda^{\mathrm{own}}$, $A_2 = \hat{r}_0$ and $A_3 = 4 g^2 \sigma_\eta^2 / K$ has

$$V'(f) = -\frac{A_1 A_3}{(A_2 + A_3 f)(A_2 + A_3 f + A_1)} < 0, \qquad V''(f) = \frac{A_1 A_3^2 (2(A_2 + A_3 f) + A_1)}{(A_2 + A_3 f)^2 (A_2 + A_3 f + A_1)^2} > 0,$$

so the edge value is convex and decreasing and the loss $V(0) - V(f)$ is concave and increasing with value zero at $f = 0$. The minimum of a concave function and a constant is concave, and a concave function $\ell$ on $[0, \infty)$ with $\ell(0) = 0$ satisfies $\ell(f + f') \le \ell(f) + \ell(f')$. □

*Proof of Proposition S6.15.* With uniform habits the policy posterior over the two codes is $q_i(\sigma) \propto \exp(\gamma \mathcal{F}_i(u_i(\sigma)) + \gamma h_i \sigma)$, and $u_i(-1) = \mathsf{d}_i - u_i^+$, which gives the logistic form. (a) $\mathcal{F}_i$ is nondecreasing, so the argument of sig is nondecreasing in $u_i^+$; the map $u_i^+ \mapsto \mathsf{d}_i - u_i^+$ changes the sign of

$\mathcal{F}_i(u_i^+) - \mathcal{F}_i(\mathsf{d}_i - u_i^+)$, and $\operatorname{sig}(-z) = 1 - \operatorname{sig}(z)$. (b) For $\mathcal{F}_i(u) = \text{const} + Ju$ the bracket is $J(2u_i^+ - \mathsf{d}_i) = J\mathsf{d}_i(2x-1)$. Put $\beta_0 = \gamma J \mathsf{d}_i$ and $\psi(x) = \operatorname{sig}(\beta_0(2x-1)) - x$. Then $\psi(\frac{1}{2}) = 0$, $\psi'(\frac{1}{2}) = \beta_0/2 - 1$, $\psi(1) < 0$, and $\psi$ is strictly concave on $[\frac{1}{2}, 1]$ because sig is strictly concave on $[0, \infty)$. If $\beta_0 > 2$, $\psi$ is positive just to the right of $\frac{1}{2}$ and has exactly one root $x_+$ in $(\frac{1}{2}, 1)$, with $\psi'(x_+) < 0$ by strict concavity; if $\beta_0 \le 2$, strict concavity gives $\psi(x) < \psi'(\frac{1}{2})(x - \frac{1}{2}) \le 0$ on $(\frac{1}{2}, 1]$, so there is no root. Since $\psi(1-x) = -\psi(x)$, the roots are symmetric about $\frac{1}{2}$. The map $x \mapsto \operatorname{sig}(\beta_0(2x-1))$ is increasing with derivative $1 + \psi'$, which is $\beta_0/2 > 1$ at $\frac{1}{2}$ and lies in $(0,1)$ at $x_\pm$, so $\frac{1}{2}$ is unstable and $x_\pm$ are stable. For $\beta_0 > 2\ln 4$, $\psi(0) > 0 > \psi(\frac{1}{4})$, so $x_- < \frac{1}{4}$ and $x_- = \operatorname{sig}(-\beta_0(1-2x_-)) \le e^{-\beta_0/2}$; then $\beta_0 x_- \to 0$ and $x_- e^{\beta_0} \to 1$. (c) At $x = 0$ the bracket is $\mathcal{F}_i(0) - \mathcal{F}_i(\mathsf{d}_i)$, and sig is positive. The argument of sig is $\gamma(G_i(-1) - G_i(+1))$, so as $\gamma \to \infty$ the probability tends to one, to $\frac{1}{2}$ or to zero as $G_i(-1) - G_i(+1) = \mathcal{F}_i(u_i^+) - \mathcal{F}_i(\mathsf{d}_i - u_i^+) + 2h_i$ is positive, zero or negative. For $h_i = 0$ and strictly increasing $\mathcal{F}_i$ this difference has the sign of $2u_i^+ - \mathsf{d}_i$, which is the majority rule; under a capacity $\kappa < \mathsf{d}_i$ the value is constant on $u \ge \kappa$, and the difference vanishes whenever $u_i^+ \ge \kappa$ and $\mathsf{d}_i - u_i^+ \ge \kappa$, for example at $\mathsf{d}_i = 3$, $\kappa = 1$ and $u_i^+ = 2$. □

*Proof of Proposition 6.10.* (a) For configurations $\sigma$ and $\sigma'$ that differ only at agent $i$, the terms of $\Phi_{\text{code}}$ that involve $\sigma_i$ are $\sum_j J_{ij}\mathbf{1}[\sigma_i \ne \sigma_j] - h_i\sigma_i$, by the symmetry of the couplings, so $G_i(\sigma_i') - G_i(\sigma_i) = \Phi_{\text{code}}(\sigma') - \Phi_{\text{code}}(\sigma)$ and $\Phi_{\text{code}}$ is an exact potential. The logit chain moves from $\sigma$ to $\sigma'$ with probability $N^{-1}e^{-\gamma G_i(\sigma_i')}/Z_i$ and back with probability $N^{-1}e^{-\gamma G_i(\sigma_i)}/Z_i$, with the same normalizer $Z_i$, which depends only on the other agents' codes. The ratio is $e^{-\gamma(\Phi_{\text{code}}(\sigma') - \Phi_{\text{code}}(\sigma))}$, which is detailed balance for the law proportional to $e^{-\gamma\Phi_{\text{code}}}$. Every single flip has positive probability, so the chain is irreducible on the finite configuration space and this law is its unique stationary law; as $\gamma \to \infty$ it concentrates on the minimizers of $\Phi_{\text{code}}$.

(b) For an agent with two partners $u_i(-\sigma_i) = 2 - u_i(\sigma_i)$, and $\mathcal{F}(u) - \mathcal{F}(2-u)$ takes the values $-(\mathcal{F}(2) - \mathcal{F}(0))$, $0$ and $\mathcal{F}(2) - \mathcal{F}(0)$ at $u = 0, 1, 2$, so $G_i(-\sigma_i) - G_i(\sigma_i) = (\mathcal{F}(2) - \mathcal{F}(0))(u_i(\sigma_i) - 1) + 2h_i\sigma_i$. In the pairwise game with coupling $J$ the same flip changes the number of mismatched edges at $i$ by $u_i(\sigma_i) - (2 - u_i(\sigma_i))$, so the cost changes by $2J(u_i(\sigma_i) - 1) + 2h_i\sigma_i$. The two agree in every configuration when $J = \frac{1}{2}(\mathcal{F}(2) - \mathcal{F}(0))$, so the two games differ by terms free of each agent's own code, and $\Phi_{\text{code}}$ is an exact potential for the code game.

(c) If an exact potential exists, the cost changes of the deviating agents along any closed path of unilateral deviations telescope in the potential and sum to zero. Take adjacent agents $i$ and $j$; let $p$ of $i$'s other partners use $+1$ and $\mathsf{d}_i - 1 - p$ use $-1$, and $p'$ and $\mathsf{d}_j - 1 - p'$ of $j$'s other partners likewise. Starting from $\sigma_i = \sigma_j = +1$, let $i$ flip, then $j$, then $i$ back, then $j$ back. With $\nabla\mathcal{F}(n) = \mathcal{F}(n+1) - \mathcal{F}(n)$ the four cost changes are $\mathcal{F}_i(p+1) - \mathcal{F}_i(\mathsf{d}_i - 1 - p) + 2h_i$, then $\mathcal{F}_j(p') - \mathcal{F}_j(\mathsf{d}_j - p') + 2h_j$, then $\mathcal{F}_i(\mathsf{d}_i - p) - \mathcal{F}_i(p) - 2h_i$, then $\mathcal{F}_j(\mathsf{d}_j - 1 - p') - \mathcal{F}_j(p'+1) - 2h_j$, and their sum is

$$\nabla\mathcal{F}_i(p) + \nabla\mathcal{F}_i(\mathsf{d}_i - 1 - p) - \nabla\mathcal{F}_j(p') - \nabla\mathcal{F}_j(\mathsf{d}_j - 1 - p'),$$

which is $\nabla^{\pm}\mathcal{F}_i(p) - \nabla^{\pm}\mathcal{F}_j(p')$ in the notation of (c). Let $i$ and $j$ have $r$ common partners. The pairs that occur are $p = a + b$ and $p' = a + b'$, where $a \in \{0, \dots, r\}$ counts the common partners that use $+1$ and $b \in \{0, \dots, \mathsf{d}_i - 1 - r\}$ and $b' \in \{0, \dots, \mathsf{d}_j - 1 - r\}$ count the remaining other partners of $i$ and of $j$ that use $+1$; $p$ and $p'$ range independently only when $r = 0$. If $i$ and $j$ have the same other partners, then $b = b' = 0$ and $p = p' = a$, and the sum vanishes in every configuration exactly when $\nabla^{\pm}\mathcal{F}_i(u) = \nabla^{\pm}\mathcal{F}_j(u)$ for every $u$. Otherwise one endpoint, say $i$, has another partner that is not a partner of $j$, so $\mathsf{d}_i - 1 - r \ge 1$. Fixing $a$ and $b'$ and varying $b$ shows that $\nabla^{\pm}\mathcal{F}_i$ is constant on $\{a, \dots, a + \mathsf{d}_i - 1 - r\}$; the intervals for consecutive $a$ overlap and cover $\{0, \dots, \mathsf{d}_i - 1\}$, so $\nabla^{\pm}\mathcal{F}_i$ is constant, and $\nabla^{\pm}\mathcal{F}_j(a + b')$ equals the same constant for all $a$ and $b'$, which exhaust $\{0, \dots, \mathsf{d}_j - 1\}$. Conversely, the conditions of (c) make the sum vanish for every attended edge

and every configuration, and for two agents that are not partners the four changes cancel in pairs because neither cost depends on the other's code; these are all the closed paths of length four, which is the sufficiency stated after the proposition. On a complete graph with a common value $\mathcal{F}$ any two agents have the same other partners, the condition holds trivially, and $u_i(\sigma_i) = n_{\sigma_i}(\sigma) - 1$. When $i$ moves from code $c$ to code $-c$ its cost changes by $\mathcal{F}(n_c - 1) - \mathcal{F}(n_{-c}) + 2h_i c$, and $\Phi$ of (c) changes by the same amount, since the sum over code $c$ loses its last term $\mathcal{F}(n_c - 1)$ and the sum over code $-c$ gains the term $\mathcal{F}(n_{-c})$; so $\Phi$ is an exact potential. For values $\mathrm{const}_i + Ju$ with a slope common to all agents, $G_i(\sigma)$ is $J$ times the number of $i$'s partners that use the other code, minus $h_i\sigma$, up to terms free of $\sigma$: a pairwise game. For a quadratic value $\alpha_0 + \alpha_1 u + \alpha_2 u^2$ on a graph in which every agent has $\mathsf{d}$ partners, $u_i(\sigma) = \frac{1}{2}(\mathsf{d} + \sigma\bar{\sigma}_i)$ with $\bar{\sigma}_i$ the sum of the codes of $i$'s partners, and since $\sigma^2 = 1$, $\mathcal{F}(u_i(\sigma)) = \mathrm{const} + \frac{1}{2}(\alpha_1 + \alpha_2\mathsf{d})\sigma\bar{\sigma}_i + \frac{1}{4}\alpha_2\bar{\sigma}_i^2$, whose last term does not involve $\sigma$; this is the pairwise game with $J = \alpha_1 + \alpha_2\mathsf{d}$. For the logarithmic value at $\mathsf{d}_i = 3$ without capacity, the quantity at $u = 0$ and at $u = 1$ differs by the third difference $\mathcal{F}(3) - 3\mathcal{F}(2) + 3\mathcal{F}(1) - \mathcal{F}(0)$, which equals the third derivative of $u \mapsto \frac{K}{2}\ln(1 + (u\varpi^{=} + (3-u)\varpi^{\neq})/\Lambda^{\mathrm{own}})$ at an intermediate point and is positive. With capacity one, $\mathcal{F}_i(u) = \mathcal{F}_i(1)$ for $u \geq 1$, so at $\mathsf{d}_i = 3$ the quantity is $\mathcal{F}_i(1) - \mathcal{F}_i(0) > 0$ at $u = 0$ and zero at $u = 1$. If an agent $i$ with three partners $j$, $k$ and $l$ has the same other partners as each of them, then the partners of $j$ are $i$, $k$ and $l$, and likewise for $k$ and $l$, so these four agents form a complete graph that no edge leaves, which is the whole graph when the graph is connected. In every other connected graph an agent with three partners has a partner with different other partners, on which edge the necessary condition demands a constant $\nabla^{\pm}\mathcal{F}_i$, and no exact potential exists. At an end of an open chain of at least three agents the end agent has one partner and its neighbor two, so the two do not have the same other partners; with $x_1 = \varpi^{=}/\Lambda^{\mathrm{own}} > x_0 = \varpi^{\neq}/\Lambda^{\mathrm{own}}$ the two constants are $K\ln\frac{1+x_1}{1+x_0}$ and $\frac{K}{2}\ln\frac{1+2x_1}{1+2x_0}$, and $(1+x_1)^2(1+2x_0) - (1+x_0)^2(1+2x_1) = (x_1 - x_0)(x_1 + x_0 + 2x_0x_1) > 0$, so they differ; with capacity one the neighbor's constant is $\frac{K}{2}\ln\frac{1+x_1}{1+x_0}$, half that of the end agent. □

The proof of Theorem 6.11 rests on three lemmas. On a ring or an open chain of $N$ agents, a wall at $i$ means $\sigma_i \neq \sigma_{i+1}$, a domain is a maximal run of consecutive agents with one code, and the aligned sum of a run of agents in a domain of sign $\sigma_{\mathsf{D}}$ is $\sigma_{\mathsf{D}}$ times the sum of their pulls. Let $\sigma^*$ minimize $\Phi_{\mathrm{code}}(\sigma) = J\sum_i \mathbf{1}[\sigma_i \neq \sigma_{i+1}] - \sum_i h_i\sigma_i$, the first sum running over the edges of the ring or chain.

**Lemma S30.4** (Structure of the minimizer)**.** *(a) A domain with walls on both sides has aligned sum at least $J$. (b) A proper initial or final run of such a domain has aligned sum at least* 0*. (c) Every run inside any domain has aligned sum at least $-J$.*

*Proof.* Flipping a run of agents inside a domain of sign $\sigma_{\mathsf{D}}$ changes the pull term by $2\sigma_{\mathsf{D}}$ times the run's sum of pulls and changes the number of walls; optimality of $\sigma^*$ makes the total change nonnegative. Flipping a whole domain with walls on both sides removes two walls, which gives (a). Flipping a proper initial or final run of it moves a wall, which gives (b). Flipping a run strictly inside a domain adds two walls, which gives aligned sum at least $-J$; a proper run that touches an end of an open chain adds one wall, which gives at least $-J/2$; flipping a whole end domain of a chain removes one wall, which gives at least $J/2$; and in a configuration without walls every proper run adds one or two walls while the whole population adds none, which gives at least $-J$ and 0. With (a) and (b) this gives (c) in every case. □

Let $X_0 = 0$ and $X_i = h_1 + \cdots + h_i$ for $1 \leq i \leq N$, the field walk. Its greedy $J$-hysteresis switches first at the first index $g_1$ at which the range $\max_{i' \leq i} X_{i'} - \min_{i' \leq i} X_{i'}$ reaches $J$. After a switch at which the walk has risen by $J$ above a running minimum (fallen by $J$ below a running maximum), the next phase tracks the running maximum (minimum) from the switch and switches when the walk falls $J$ below it (rises $J$ above it). Let $\Psi_N$ be the number of switches in $\{1, \ldots, N\}$.

**Lemma S30.5** (Counting walls)**.** *If no run of consecutive pulls sums to $\pm J$, then on the ring and on the open chain $\lfloor \Psi_N/2 \rfloor - 1 \le \#\mathrm{walls}(\sigma^*) \le 2\Psi_N + 4$.*

*Proof.* Let $x_1 < \cdots < x_L$ be the walls at agents $1, \dots, N-1$; on the ring the edge between $N$ and 1 may carry one more wall. For $1 < \alpha \le L$ the agents $x_{\alpha-1}+1, \dots, x_\alpha$ form a domain with walls on both sides, so Lemma S30.4(a) gives $\sigma_\alpha(X_{x_\alpha} - X_{x_{\alpha-1}}) \ge J$, with $\sigma_\alpha$ its sign. Signs alternate, so the wall levels $X_{x_\alpha}$ alternate between maxima and minima, and at a maximum wall $x_\alpha$ with $1 < \alpha < L$, $X_{x_\alpha} \ge X_{x_{\alpha\pm1}} + J$.

Upper bound. Let $x_\alpha$ be a maximum wall with $1 < \alpha < L$, and suppose no switch occurs in $(x_{\alpha-1}, x_{\alpha+1}]$. If none occurs at or before $x_{\alpha-1}$ either, the range of the walk on $[0, x_\alpha]$ is at least $J$, so $g_1 \le x_\alpha$, a contradiction. Otherwise let $g$ be the last switch at or before $x_{\alpha-1}$. If the phase after $g$ tracks a maximum, the running maximum is at least $X_{x_\alpha}$ by $x_\alpha$ and $X_{x_{\alpha+1}} \le X_{x_\alpha} - J$, so a switch occurs after $g$ and by $x_{\alpha+1}$, hence in $(x_{\alpha-1}, x_{\alpha+1}]$. If it tracks a minimum, the running minimum is at most $X_{x_{\alpha-1}}$ and $X_{x_\alpha} \ge X_{x_{\alpha-1}} + J$, so a switch occurs in $(x_{\alpha-1}, x_\alpha]$. Both contradict the assumption. The windows $(x_{\alpha-1}, x_{\alpha+1}]$ of the maximum walls are disjoint, and at least $\lfloor (L-2)/2 \rfloor$ maximum walls lie strictly between $x_1$ and $x_L$, so $\lfloor (L-2)/2 \rfloor \le \Psi_N$ and $L \le 2\Psi_N + 3$; the possible wall between $N$ and 1 adds one.

Lower bound. Between switches $g_k < g_{k+1} < g_{k+2}$ the walk completes reversals of at least $J$ in both directions. If the phase after $g_k$ tracks a maximum, attained at some $i' \in [g_k, g_{k+1})$, then $X_{g_{k+1}} - X_{i'} \le -J$, and the next phase gives $i'' \in [g_{k+1}, g_{k+2})$ with $X_{g_{k+2}} - X_{i''} \ge J$; the other case is symmetric. The runs of agents $i'+1, \dots, g_{k+1}$ and $i''+1, \dots, g_{k+2}$ lie in $\{g_k+1, \dots, g_{k+2}\}$, and by hypothesis their sums are strictly below $-J$ and strictly above $J$. By Lemma S30.4(c) a domain of sign $+1$ contains no run with sum below $-J$ and a domain of sign $-1$ none with sum above $J$, so the agents $g_k+1, \dots, g_{k+2}$ do not lie in one domain, and a wall sits between two consecutive agents of this window. The windows for $k = 1, 3, 5, \dots$ are disjoint, which gives at least $\lfloor (\Psi_N - 1)/2 \rfloor \ge \lfloor \Psi_N/2 \rfloor - 1$ walls. □

**Lemma S30.6** (Drawdown time)**.** *Let $(X_u)_{u \ge 0}$ be a random walk from 0 with independent $\mathcal{N}(0, h^2)$ steps, $B_u = \max_{u' \le u} X_{u'} - X_u$ its drawdown and $T_J = \min\{u : B_u \ge J\}$. Then $(J/h)^2 \le \mathbb{E}T_J \le 13(J/h)^2 + 3.25$.*

*Proof.* Lower bound. With $h_{u+1} = X_{u+1} - X_u$, $B_{u+1} = \max\{B_u - h_{u+1}, 0\}$, so $B_{u+1}^2 \le (B_u - h_{u+1})^2$ and $\mathbb{E}[B_{u+1}^2 \mid X_0, \dots, X_u] \le B_u^2 + h^2$; hence $B_u^2 - uh^2$ is a supermartingale. Optional stopping at $T_J \wedge n$ gives $\mathbb{E}B_{T_J \wedge n}^2 \le h^2 \mathbb{E}[T_J \wedge n]$, and letting $n \to \infty$, with Fatou's lemma on the left, monotone convergence on the right and $T_J < \infty$ almost surely by the upper bound, gives $J^2 \le \mathbb{E}B_{T_J}^2 \le h^2 \mathbb{E}T_J$. Upper bound. Split time into blocks of $n_0 = \lceil 4J^2/h^2 \rceil$ steps. A block whose increment is at most $-J$ completes a drawdown of $J$, and this happens independently across blocks with probability $\Pr(\mathcal{N}(0, n_0 h^2) \le -J) \ge \Pr(\mathcal{N}(0,1) \le -\frac{1}{2}) > 0.3085$. So $\mathbb{E}T_J \le n_0/0.3085 \le (4(J/h)^2 + 1)/0.3085 \le 13(J/h)^2 + 3.25$. □

The lemmas leave a factor of four between the bounds on the number of walls, and the data suggest that it is spurious.

**Conjecture S30.7** (Exact wall count)**.** *On an open chain the number of domains of the minimizer of the pairwise potential equals the number of switches of the greedy $J$-hysteresis or exceeds it by one, so that the mean domain length tends almost surely to $\mathbb{E}T_J$ of Lemma S30.6.*

In exact minimizers on chains of $4 \times 10^5$ agents the number of domains equaled the number of switches in four runs and exceeded it by one in the fifth (Check S47.57).

*Proof of Theorem 6.11.* (a) By Proposition 6.10(b) the game is the pairwise game with coupling $J$ on every edge of the ring, and Proposition 6.10(a) gives reversibility, the stationary law and its concentration. For distinct configurations $\sigma$ and $\sigma'$, $\Phi_{\mathrm{code}}(\sigma) - \Phi_{\mathrm{code}}(\sigma')$ is a multiple of $J$ plus $-2\sum_i h_i\sigma_i$ over the nonempty set where they differ, a variable with a continuous law, so with probability one no two of the finitely many configurations tie and the minimizer is unique.

(b) With probability one no run of pulls sums to $\pm J$, so Lemma S30.5 applies for every $N$. The switch indices of the greedy hysteresis are stopping times of the walk. After the first, each phase starts at the current value of the walk and ends when the walk has reversed by $J$ from the running extreme, so by the strong Markov property the periods between consecutive switches are independent, and by the symmetry of the steps each has the law of $T_J$ in Lemma S30.6. Since $\mathbb{E}T_J < \infty$, the renewal strong law of large numbers gives $\Psi_N/N \to 1/\mathbb{E}T_J$ almost surely. On the ring the number of domains is the number of walls, or one if there is none, so Lemma S30.5 gives $N/(2\Psi_N + 4) \le \bar{\ell}_N \le N/(\lfloor \Psi_N/2 \rfloor - 1)$ once $\Psi_N \ge 4$, hence $\frac{1}{2}\mathbb{E}T_J \le \liminf \bar{\ell}_N \le \limsup \bar{\ell}_N \le 2\mathbb{E}T_J$ almost surely, and Lemma S30.6 gives the stated bounds, since $2 \times 3.25 < 7$.

(c) At $h = 0$ the stationary law is proportional to $\exp(\frac{\gamma J}{2}\sum_i \sigma_i\sigma_{i+1})$, because $\mathbf{1}[\sigma_i \ne \sigma_{i+1}] = \frac{1}{2}(1 - \sigma_i\sigma_{i+1})$. The transfer matrix $\mathsf{M}$ with entries $e^{\gamma J\sigma\sigma'/2}$ has eigenvalues $2\cosh(\gamma J/2)$ and $2\sinh(\gamma J/2)$ with eigenvectors $(1,1)$ and $(1,-1)$, which $\mathrm{diag}(1,-1)$ exchanges. For agents at distance $d$, $\mathbb{E}[\sigma_i\sigma_j] = \mathrm{tr}(\mathrm{diag}(1,-1)\mathsf{M}^d\,\mathrm{diag}(1,-1)\mathsf{M}^{N-d})/\,\mathrm{tr}\,\mathsf{M}^N = (x_\gamma^d + x_\gamma^{N-d})/(1 + x_\gamma^N)$ with $x_\gamma = \tanh(\gamma J/2) \in [0,1)$, and $\Pr(\sigma_i \ne \sigma_j) = \frac{1}{2}(1 - \mathbb{E}[\sigma_i\sigma_j])$. The limits follow from $x_\gamma < 1$, and $\frac{1}{2}(1 - \tanh(\gamma J/2)) = 1/(1 + e^{\gamma J})$. □

## S30.6 The grounding tension

*Proof of Theorem 6.12.* (a) Since $\hat{\Lambda}^+ > \varpi_{\mathrm{s}} + 2\varpi_{\mathrm{y}}$, the gains satisfy $k^s + 2k^{\mathrm{y}} < 1$, and with $0 < F < 1$ this gives $c > 0$, $a > 0$ and $\lambda = F(1 - k^s) < 1$: (NN) and (C$_r$) hold. The gains are fixed, so $W$ is deterministic and time invariant, which gives (Lin), and it couples each agent only to itself and its two neighbors, which gives (Loc); the local noises are independent across agents with the constant variance $q$, the world innovation has the constant variance $Q$, and there is no mistranslation term, so the law of the local noise is time invariant and the moment hypotheses of Lemma 8.2(a) on the inputs hold, while the stationary law, given by the series of Theorem 8.9(b), involves no initial error. Hence (M) follows in stationary form by Theorem 8.9(b). By Lemma 2.13 the stationary error is the sum of a connected part, driven by the local noises, and an independent common mode, which on the ring, where $W\mathbf{1} = \lambda\mathbf{1}$, is

$$-\sum_{n\ge 0} w_{t-1-n} W^n \mathbf{1} = -\sum_{n\ge 0} \lambda^n w_{t-1-n}\mathbf{1},$$

with variance $Q/(1-\lambda^2) = Q/(\varepsilon(2-\varepsilon))$ at every agent. Since $W$ is symmetric, the connected part has covariance $\Sigma^c = q\sum_n W^{2n} = q(I - W^2)^{-1}$, and by Fourier analysis on $\mathbb{Z}$

$$\Sigma^c_{i,i+d} = \frac{q}{2\pi}\int_{-\pi}^{\pi} \frac{e^{\mathrm{i}ud}}{1 - w(u)^2}\mathrm{d}u, \qquad w(u) = c + 2a\cos u.$$

Write $(1-w^2)^{-1} = \frac{1}{2}[(1-w)^{-1} + (1+w)^{-1}]$ and use, for $A_1 > |A_2| > 0$, $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\mathrm{i}ud}(A_1 - A_2\cos u)^{-1}\mathrm{d}u = (\mathrm{sgn}A_2)^d e^{-m_{12}|d|}/\sqrt{A_1^2 - A_2^2}$ with $\cosh m_{12} = A_1/|A_2|$. This gives

$$\Sigma^c_{i,i+d} = \frac{q}{2}\Big[\frac{e^{-m|d|}}{\sqrt{(1-c)^2 - 4a^2}} + \frac{(-1)^d e^{-m'|d|}}{\sqrt{(1+c)^2 - 4a^2}}\Big], \qquad \cosh m = \frac{1-c}{2a}, \quad \cosh m' = \frac{1+c}{2a},$$

where $1+c>2a$ because $c>0$ and $a\le F/2<\frac{1}{2}$. At $d=0$, with $(1-c)^2-4a^2=(1-c-2a)(1-c+2a)=\varepsilon(1-c+2a)$, this is the second term of (68), and adding the common mode gives (ii).

(i) Since $c>0$, $m'>m$, so the first term dominates at large $d$ and the standardized connected covariance decays exactly at the rate $m$; hence $\xi=1/m$ by Definition 8.8, with $\cosh m=(1-c)/(2a)=1+\varepsilon/(2a)$, in agreement with Theorem 8.10. The gains give $a=F\varpi_{\mathrm{y}}/\hat{\Lambda}^+$ and $\varepsilon=1-F+F\varpi_{\mathrm{s}}/\hat{\Lambda}^+$, hence the formula for $a/\varepsilon$. As $F\to 1$ at fixed precisions $\hat{\Lambda}^+$ stays bounded, because $\hat{P}^-\ge Q$, so $(1-F)\hat{\Lambda}^+\to 0$. As $\varpi_{\mathrm{y}}\to\infty$ at $F<1$, $\hat{P}^-\ge Q$ keeps $1/\hat{P}^-$ bounded, so $\hat{\Lambda}^+/(2\varpi_{\mathrm{y}})\to 1$ and $a/\varepsilon\to F/(2(1-F))$, which makes $1+\varepsilon/(2a)\to 1/F$. At every assigned precision $\hat{\Lambda}^+\ge 2\varpi_{\mathrm{y}}$, so $a/\varepsilon\le F\varpi_{\mathrm{y}}/(2(1-F)\varpi_{\mathrm{y}})=F/(2(1-F))$, hence $1+\varepsilon/(2a)\ge 1/F$ and $\xi\le 1/\operatorname{arccosh}(1/F)$.

(iii) Since $0<\varepsilon<1$, $\varepsilon(2-\varepsilon)\le 2\varepsilon=4a(\cosh(1/\xi)-1)=8a\sinh^2(1/(2\xi))$, and $\sinh y\le y\cosh y$ for $y\ge 0$ because $\tanh y\le y$. Hence $\varepsilon(2-\varepsilon)\le 2a\xi^{-2}\cosh^2(1/(2\xi))$, which gives (69). The asymptotic forms follow from $\cosh^2(1/(2\xi))=1+O(\xi^{-2})$ and $a=Fk^{\mathrm{y}}\le F/2$.

(iv) As $\varpi_{\mathrm{y}}\to\infty$, $k^s=\varpi_{\mathrm{s}}/\hat{\Lambda}^+\to 0$ and $k^{\mathrm{y}}\to\frac{1}{2}$, so $c\to 0$, $a\to F/2$, $\varepsilon\to 1-F$ and $q\to F^2R_c/2$. The first term of (68) tends to $Q/(1-F^2)=P_0$ and the bracket to $2/\sqrt{1-F^2}$, which gives the limit.

(b) With exact code models and homogeneous sensing gain every row of $W$ sums to $F(1-k)=\lambda$, and by (NN) this is also the row-sum norm, so ($\mathrm{C}_r$) holds. The common mode is $-\sum_n\lambda^n w_{t-1-n}$ at every agent, with variance $Q/(1-\lambda^2)$, independent of the connected part (Lemma 2.13); this is the bound of Proposition 3.8. Under (Lin), (Loc), ($\mathrm{C}_r$) and (B), Theorem 8.9 gives $\xi\le 1/\ln(1/\lambda)$, so $\lambda^2\ge e^{-2/\xi}$ and $\mathbb{E}e_i^2\ge Q/(1-\lambda^2)\ge Q/(1-e^{-2/\xi})$, and $1-e^{-y}\le y$ gives the last inequality. □

*Proof of Proposition S6.19.* (a) The receiver's predictive density of a neighbor's message is $\mathcal{N}(m_i,S)$ with $S=\hat{P}^-+1/\varpi_{\mathrm{y}}$. At the exact posterior of its model the variational free energy of one message equals its negative log evidence, $\frac{1}{2}\ln(2\pi S)+(y_{ij}-m_i)^2/(2S)$, so the time average is $\frac{1}{2}\ln S$ plus the time average of $(y_{ij}-m_i)^2$ divided by $2S$, up to a constant; in stationarity that time average is $\mathbb{E}(y_{ij}-m_i)^2$. As a function of $S$ this decreases up to $S=\mathbb{E}(y_{ij}-m_i)^2$ and increases beyond, and the constraint $\varpi_{\mathrm{y}}\le 1/R_c$ is $S\ge\hat{P}^-+R_c$, which gives (a). (b) $y_{ij}-m_i=e_j-e_i+\nu_{ij}$. On the ring the common mode is the same at every agent, so $e_j-e_i$ is the difference of the connected parts, with second moment $2(\Sigma^c_{ii}-\Sigma^c_{ij})$ by translation invariance, and the channel noise adds $R_c$. Under the stated condition the target $\mathbb{E}(y_{ij}-m_i)^2-\hat{P}^-$ of (a) is at most $R_c$. (c) $s_i-m_i=v_i-e_i$ with $v_i$ independent of the prior error $e_i$, so $\mathbb{E}(s_i-m_i)^2=R^s+\mathbb{E}e_i^2$. The receiver's model gives this innovation the variance $\hat{P}^-+\hat{R}^s$, and moment matching sets the assigned sensing noise $\hat{R}^s=R^s+\mathbb{E}e_i^2-\hat{P}^-$, which exceeds $R^s$ exactly when $\mathbb{E}e_i^2>\hat{P}^-$. □

## S31 Proofs for Section 7

Notation is that of Section 7. Indicators are written $[\cdot]$, and $\|\cdot\|$ is the Euclidean norm of vectors and the induced norm of matrices.

### The single-edge model and the proof of Proposition S7.2

The world is static, $\eta\sim\mathcal{N}(0,P_0I_2)$ independently across rounds, as in Definition 6.2. A sender $j$ with sensing noise variance $R^s_j$ broadcasts its posterior mean $g_js_j$, $g_j=P_0/(P_0+R^s_j)$, read with noise of variance $R_c$, and the receiver's claimed prior-plus-sensing precision is $\Lambda_0$. The in-group sender $a$ is read with the correct translation, so its C1g precision is $\varpi_a=g_a^2/(R_c+g_a^2R^s_a)$ (Theorem 6.4). The angle error of the cross sender $b$ drifts by an independent Gaussian increment of variance $\sigma^2_{\mathrm{dr}}$ per

round, and on attended rounds the receiver obtains a grounded observation of the angle with noise variance $\sigma_\alpha^2$. Let $\mathsf{v}_-$ be the variance of the angle error before a round and $\mathsf{v}$ after it; an attended round maps $\mathsf{v}_-$ to $\mathsf{v}_-\sigma_\alpha^2/(\mathsf{v}_-+\sigma_\alpha^2)$. The expected-variance mean field replaces the attention indicator by its frequency $\mathsf{p}$, so that at stationarity $\mathsf{v}_- = \mathsf{v} + \sigma_{\mathrm{dr}}^2$ and $\mathsf{v} = (1-\mathsf{p})\mathsf{v}_- + \mathsf{p}\mathsf{v}_-\sigma_\alpha^2/(\mathsf{v}_- + \sigma_\alpha^2)$. Subtracting gives $\sigma_{\mathrm{dr}}^2(\mathsf{v}_- + \sigma_\alpha^2) = \mathsf{p}\mathsf{v}_-^2$, whose positive root yields

$$\mathsf{v}(\mathsf{p}) = \frac{\sigma_{\mathrm{dr}}^2 + (\sigma_{\mathrm{dr}}^4 + 4\mathsf{p}\sigma_{\mathrm{dr}}^2\sigma_\alpha^2)^{1/2}}{2\mathsf{p}} - \sigma_{\mathrm{dr}}^2 \qquad (\mathsf{p} > 0).$$

Differentiating $\mathsf{p}\mathsf{v}_-^2 - \sigma_{\mathrm{dr}}^2\mathsf{v}_- - \sigma_{\mathrm{dr}}^2\sigma_\alpha^2 = 0$ in $\mathsf{p}$ gives $\mathsf{v}_-^2 + (2\mathsf{p}\mathsf{v}_- - \sigma_{\mathrm{dr}}^2)\mathrm{d}\mathsf{v}_-/\mathrm{d}\mathsf{p} = 0$, and $2\mathsf{p}\mathsf{v}_- - \sigma_{\mathrm{dr}}^2 = (\sigma_{\mathrm{dr}}^4 + 4\mathsf{p}\sigma_{\mathrm{dr}}^2\sigma_\alpha^2)^{1/2} > 0$, so $\mathsf{v}_-$ and $\mathsf{v}$ decrease strictly on $(0,1]$, and $\mathsf{v} \to \infty$ as $\mathsf{p} \to 0$. For a centered Gaussian angle error $\phi$ of variance $\mathsf{v}$, $\mathbb{E}(1-\cos\phi) = 1 - e^{-\mathsf{v}/2}$ and $\|\mathrm{Rot}(\phi) - I\|_F^2 = 4(1-\cos\phi)$, so C1g calibration assigns the cross sender the noise $R_c + g_b^2 R_b^s + 2g_b^2 P_0(1-e^{-\mathsf{v}/2})$ (Theorem 6.4(a) with $K=2$) and the precision $\varpi_b(\mathsf{p}) = g_b^2/(R_c + g_b^2 R_b^s + 2g_b^2 P_0(1-e^{-\mathsf{v}(\mathsf{p})/2}))$. With $K=2$ the believed value of attending one sender is $\log(1+\varpi/\Lambda_0)$, and $\mathcal{A}(\mathsf{p}) = \log(1+\varpi_b(\mathsf{p})/\Lambda_0) - \log(1+\varpi_a/\Lambda_0)$ is continuous and strictly increasing on $(0,1]$, with the finite limit obtained by putting $e^{-\mathsf{v}/2} = 0$; we extend $\mathcal{A}$ to $\mathsf{p} = 0$ by this limit.

*Proof of Proposition S7.2.* The vector field is $f(\mathsf{p}) = [\mathcal{A}(\mathsf{p}) > 0] - \mathsf{p}$ on $[0,1]$. If $\mathcal{A}(0) < 0$, continuity gives $\mathcal{A} < 0$ on some $[0,\delta)$, where $f(\mathsf{p}) = -\mathsf{p}$; then 0 is a rest point that attracts $[0,\delta)$ exponentially. If $\mathcal{A}(0) \geq 0$, strict monotonicity gives $\mathcal{A} > 0$ on $(0,1]$, so $f = 1-\mathsf{p} > 0$ on $(0,1)$ and every trajectory from $(0,1]$ moves to 1; the point 0 is not attracting. Symmetrically, if $\mathcal{A}(1) > 0$ then $\mathcal{A} > 0$ on some $(1-\delta,1]$, where $f = 1-\mathsf{p}$ and 1 attracts; if $\mathcal{A}(1) \leq 0$ then $\mathcal{A} < 0$ on $[0,1)$, $f(1) = -1$ and $f = -\mathsf{p} < 0$ on $(0,1]$, so 1 is not a rest point. When $\mathcal{A}(0) < 0 < \mathcal{A}(1)$, $\mathcal{A}$ has a unique root $\mathsf{p}^\dagger \in (0,1)$, the field is $-\mathsf{p}$ on $[0,\mathsf{p}^\dagger]$ and $1-\mathsf{p}$ on $(\mathsf{p}^\dagger,1]$, and the basins of 0 and 1 are $[0,\mathsf{p}^\dagger]$ and $(\mathsf{p}^\dagger,1]$. □

*Proof of Lemma 7.3.* Write $\mathsf{R} = \mathrm{Rot}(\phi)$. Since $\tilde{y}_{ij} = \mathsf{R}m_j + \tilde{\nu}_{ij}$ and $m_j = \eta_t + e_j(t)$,

$$\tilde{y}_{ij} - \eta_t = (\mathsf{R} - I)\eta_t + \mathsf{R}e_j(t) + \tilde{\nu}_{ij},$$

with $\tilde{\nu}_{ij} \sim \mathcal{N}(0, R_c I_2)$ independent of $(e_j, \eta)$. A $2\times 2$ matrix that commutes with every rotation is of the form $aI + b\mathrm{Rot}(\pi/2)$, and one that also commutes with a reflection has $b = 0$; so invariance gives $\mathbb{E}[\eta_t\eta_t^\top] = P_0 I$, $\mathbb{E}[e_j e_j^\top] = \mathrm{MSE}_j I$ and $\mathbb{E}[e_j\eta_t^\top] = -\mathsf{c}_\eta I$ for a scalar $\mathsf{c}_\eta$. With $\|\mathsf{R} - I\|_F^2 = \mathrm{tr}(2I - \mathsf{R} - \mathsf{R}^\top) = 4(1-\cos\phi)$ and $(\mathsf{R}-I)^\top\mathsf{R} = I - \mathsf{R}$,

$$\mathbb{E}\|(\mathsf{R}-I)\eta_t\|^2 = 4P_0(1-\cos\phi), \qquad \mathbb{E}\|\mathsf{R}e_j\|^2 = 2\mathrm{MSE}_j,$$
$$2\mathbb{E}\langle(\mathsf{R}-I)\eta_t, \mathsf{R}e_j\rangle = 2\,\mathrm{tr}\big((I-\mathsf{R})\mathbb{E}[e_j\eta_t^\top]\big) = -4\mathsf{c}_\eta(1-\cos\phi),$$

and $\mathbb{E}\|\tilde{\nu}_{ij}\|^2 = 2R_c$, while the cross terms with $\tilde{\nu}_{ij}$ vanish. Summing and dividing by $K=2$ gives $\hat{\sigma}_{ij}^2 = R_c + \mathrm{MSE}_j + 2(P_0 - \mathsf{c}_\eta)(1-\cos\phi)$, and $P_0 - \mathsf{c}_\eta = \mathbb{E}[\eta_{t,1}(\eta_{t,1} + e_{j,1}(t))] = \mathrm{Cov}_1(\eta_t, m_j(t))$. For a random error independent of $(e_j, \eta, \tilde{\nu}_{ij})$, average over $\phi$.

At the split every coefficient in (70) for $j$'s group is a multiple of $I_2$ and the noises are isotropic, so the joint law is invariant under rotations and reflections and the coordinates evolve separately. In one coordinate the group's errors obey $e(t+1) = W(t)e(t) + \zeta(t) - w_t\mathbf{1}$ with $W(t) = F(k^0 I + \bar{k}\mathsf{J}(t))$, where $\mathsf{J}_{ik}(t) = [k \in J_i(t)]$ and $\zeta(t)$ collects the sensing and reading noises. Every realization has nonnegative entries and $W(t)\mathbf{1} = F(k^0 + \kappa\bar{k})\mathbf{1} = \lambda\mathbf{1}$, so unrolling as in Lemma 2.13,

$$e(t) = -\sum_{l\geq 0}\lambda^l w_{t-1-l}\mathbf{1} + \sum_{l\geq 0} W(t-1)\cdots W(t-l)\zeta(t-1-l),$$

both series converging in mean square, and the second sum is independent of the world innovations because the local noises and attention draws are. Since $\eta_t = \sum_{l\ge 0} F^l w_{t-1-l}$, $\mathsf{c}_\eta = Q\sum_l (\lambda F)^l = Q/(1-\lambda F)$, and

$$P_0 - \mathsf{c}_\eta = \frac{Q}{1-F^2} - \frac{Q}{1-\lambda F} = \frac{QF(F-\lambda)}{(1-F^2)(1-\lambda F)} = \frac{QF^2 k^s}{(1-F^2)(1-\lambda F)},$$

using $F - \lambda = Fk^s$. □

### The row-sum bound of Remark S7.16

In (70) the coefficient blocks of agent $i$ are $Fk_i^0 I$ and $Fk_{ij}\mathrm{Rot}(\phi_{ij})$, whose norms are $Fk_i^0$ and $Fk_{ij}$ because rotations have norm one, so every realization has block row sum $F(k_i^0 + \sum_{j\in J} k_{ij}) = F(1-k_i^s)$. After the first round $\hat{P}_i^- \ge Q$, and $\hat{\sigma}_{ij}^2 \ge R_c$ because $\tilde{\nu}_{ij}$ is independent of $m_j$ and $\eta_t$, so the claimed posterior precision is at most $1/Q + 1/R^s + \kappa/R_c$ and $k_i^s \ge k_{\min}^s = 1/(R^s(1/Q + 1/R^s + \kappa/R_c)) > 0$; put $\lambda_{\max} = F(1-k_{\min}^s) < 1$. Write $\mathsf{M}(t) = \max_i \|e_i(t)\|$ and let $\mathsf{Z}(t)$ be the largest over $i$ of the norm of the forcing and noise terms of agent $i$, $\|F\sum_{j\in J} k_{ij}((\mathrm{Rot}(\phi_{ij}) - I)\eta_t + \tilde{\nu}_{ij}(t)) + Fk_i^s v_i(t) - w_t\|$. Then $\mathsf{M}(t+1) \le \lambda_{\max}\mathsf{M}(t) + \mathsf{Z}(t)$ on every path, and $\mathsf{Z}(t) \le 2\|\eta_t\| + \|w_t\| + \sum_{i,j}\|\tilde{\nu}_{ij}(t)\| + \sum_i \|v_i(t)\|$ has second moments bounded uniformly in $t$ because $\eta$ is stationary and the noises are independent across rounds. By Minkowski's inequality $(\mathbb{E}\mathsf{M}(t)^2)^{1/2} \le \lambda_{\max}^t (\mathbb{E}\mathsf{M}(0)^2)^{1/2} + \sup_s(\mathbb{E}\mathsf{Z}(s)^2)^{1/2}/(1-\lambda_{\max})$, which is the uniform form of (M) on every attention and translation path. The edge map from $j$ to $i$ in (70) is $Fk_{ij}\mathrm{Rot}(\phi_{ij})$, of norm $Fk_{ij} < 1$, so every product of edge maps around a cycle has norm below one and no cycle amplifies.

### The mean-field recursion

Let $\mathsf{a} = [j \in J_i(t)]$. The learning step followed by the relaxation gives $\hat{\alpha}_{ij} \leftarrow h(\hat{\alpha}_{ij} + \alpha\mathsf{a}(\alpha_{ij}^\star + \nu_{ij}^\alpha - \hat{\alpha}_{ij}))$. For $i \in A$ and $j \in B$, $\alpha_{ij}^\star = -\vartheta$; substituting $\hat{\alpha}_{ij} = \phi_{ij} + \alpha_{ij}^\star$ gives $\phi_{ij} \leftarrow h(1-\alpha\mathsf{a})\phi_{ij} + h\alpha\mathsf{a}\nu_{ij}^\alpha + (1-h)\vartheta$, and dividing by $\vartheta$,

$$x_j(t+1) = h(1-\alpha\mathsf{a})x_j(t) + \alpha h\mathsf{a}\nu_{ij}^\alpha(t)/\vartheta + 1 - h.$$

Under Assumption 7.4(b), $\mathsf{p}_j(x)$ is the conditional probability of $\mathsf{a} = 1$ given the past, so the conditional mean of $x_j(t+1)$ is $\Phi_j(x(t))$, and $\mathsf{u}_j = h[-(\mathsf{a} - \mathsf{p}_j)x_j + \mathsf{a}\nu_{ij}^\alpha/\vartheta]$ has conditional mean zero and conditional variance $h^2[x_j^2\mathsf{p}_j(1-\mathsf{p}_j) + \mathsf{p}_j\sigma_\alpha^2/\vartheta^2]$, at most $\frac{1}{4} + \sigma_\alpha^2/\vartheta^2$ when $x_j \in [0,1]$. The second form of (72) follows from $1 - h = \alpha h\mathsf{p}^*$. Under Assumption 7.4(a) the set values of receiver $i$ involve only $\varpi_{\mathrm{in}}$ and $\varpi_\times(x_j\vartheta)$ for its own edges, so receivers decouple.

*Proof of Lemma 7.5.* Write $\mathcal{V}(J)$ for $\mathcal{V}_i^0(J)$, so that $q(J) \propto e^{\gamma\mathcal{V}(J)}$. Every precision is at most $\varpi_{\mathrm{in}}$, and $\varpi_\times(x\vartheta)$ is nonincreasing in $x \in [0,1]$ because $1 - \cos$ increases on $[0,\pi]$.

(a) Let $J \ni j$. It holds at most $\kappa - 1 \le n - 2$ in-group senders, so some in-group sender is outside $J$; let $j_0(J)$ be the smallest such and $J' = J \setminus \{j\} \cup \{j_0(J)\}$. Since $\sum_{k\in J}\varpi_{ik} \le \kappa\varpi_{\mathrm{in}}$,

$$\mathcal{V}(J') - \mathcal{V}(J) = \frac{K}{2}\log\Big(1 + \frac{\varpi_{\mathrm{in}} - \varpi_\times(x_j\vartheta)}{\Lambda_0 + \sum_{k\in J}\varpi_{ik}}\Big) \ge \Delta\mathcal{V}(x_j),$$

so $q(J) \le e^{-\gamma\Delta\mathcal{V}(x_j)}q(J')$. A set $J' \not\ni j$ arises in this way from at most $|J' \cap A| \le \kappa$ sets, namely $J' \setminus \{k\} \cup \{j\}$ with $k \in J' \cap A$. Summing over $J \ni j$ gives $\mathsf{p}_j \le \kappa e^{-\gamma\Delta\mathcal{V}(x_j)}(1-\mathsf{p}_j)$, which rearranges to (a).

(b) Every set value lies between $\frac{K}{2}\log((\Lambda_0 + \kappa\varpi_\times(\vartheta))/\Lambda_0)$ and $\frac{K}{2}\log((\Lambda_0 + \kappa\varpi_{\mathrm{in}})/\Lambda_0)$, an interval of length $\Delta\mathcal{V}_{\max}$, so the Gibbs weights differ by a factor at most $e^{\gamma\Delta\mathcal{V}_{\max}}$. A fraction $\binom{2n-2}{\kappa-1}/\binom{2n-1}{\kappa} = \kappa/(2n-1)$ of the sets contain $j$, which gives both bounds.

(c) For a Gibbs law, $\partial \mathsf{p}_j/\partial x_{j'} = \gamma \operatorname{Cov}_q([j \in J], \partial_{x_{j'}} \mathcal{V}(J))$. The derivative $\partial_{x_{j'}} \mathcal{V}(J)$ vanishes unless $j' \in J$, and then its modulus is at most $\frac{K}{2}\vartheta|\varpi_\times'|/\Lambda_0 \le \mathcal{L}$, because $|\mathrm{d}\varpi_\times(\phi)/\mathrm{d}\phi| = D_{\mathrm{cal}} \sin\phi/(\hat\sigma_{\mathrm{in}}^2 + D_{\mathrm{cal}}(1-\cos\phi))^2 \le D_{\mathrm{cal}}/(\hat\sigma_{\mathrm{in}}^2)^2$. Hence $|\operatorname{Cov}_q| \le \mathcal{L}(q(j, j' \in J) + \mathsf{p}_j \mathsf{p}_{j'}) \le 2\mathcal{L}\mathsf{p}_j$; sum over the $n$ cross edges. □

*Proof of Theorem 7.6.* (a) The cross senders of one receiver enter the set softmax symmetrically, so equal states give equal frequencies and the symmetric subspace is invariant. On it a set is characterized by the number $\mathsf{b} \in \{0, \dots, \kappa\}$ of its cross members; the class with $\mathsf{b}$ cross members has weight $\mathsf{w}_\mathsf{b}(x) = \binom{n-1}{\kappa-\mathsf{b}}\binom{n}{\mathsf{b}}(\Lambda_\mathsf{b}(x)/\Lambda_0)^{\gamma K/2}$ with $\Lambda_\mathsf{b}(x) = \Lambda_0 + (\kappa-\mathsf{b})\varpi_{\mathrm{in}} + \mathsf{b}\varpi_\times(x\vartheta)$, and $\mathsf{P}(x) = \mathbb{E}_x[\mathsf{b}]/n$ under the law proportional to $\mathsf{w}_\mathsf{b}$. Now $\frac{\mathrm{d}}{\mathrm{d}x}\log \mathsf{w}_\mathsf{b} = \frac{\gamma K}{2}\vartheta\varpi_\times'(x\vartheta)\mathsf{b}/\Lambda_\mathsf{b}(x)$, and $\mathsf{b}/\Lambda_\mathsf{b}$ increases in $\mathsf{b}$ while $\varpi_\times' \le 0$, so this derivative is nonincreasing in $\mathsf{b}$. Hence $\frac{\mathrm{d}}{\mathrm{d}x}\mathbb{E}_x[\mathsf{b}] = \operatorname{Cov}_x(\mathsf{b}, \frac{\mathrm{d}}{\mathrm{d}x}\log \mathsf{w}_\mathsf{b}) \le 0$, because a nondecreasing and a nonincreasing function of one random variable have nonpositive covariance (average the product of their increments over two independent copies of the variable). So $\mathsf{P}$ is nonincreasing and $g'(x) = h(1-\alpha\mathsf{P}(x)) - \alpha h x \mathsf{P}'(x) \ge 0$. Since $g(1) \le 1$ and $g$ is continuous and nondecreasing, $x(t) = g^t(1)$ is nonincreasing and converges to a fixed point; for every fixed point $\hat x \le 1$ induction gives $x(t) \ge \hat x$, so the limit is the largest fixed point $\bar x$. The fixed-point equation is $g(x) = x$ rearranged with $1 - h = \alpha h \mathsf{p}^*$. By Lemma 7.5(a), $\mathsf{P}(\frac12) \le \bar{\mathsf{p}}(\frac12)$, and $\bar{\mathsf{p}}(\frac12) < \mathsf{p}^*$ is equivalent to $\gamma > \gamma_S$. Then $\Psi(x) = x\mathsf{P}(x) - \mathsf{p}^*(1-x)$ is negative at $\frac12$ and equals $\mathsf{P}(1) > 0$ at 1, the last because every set has positive weight at finite $\gamma$, so it has a zero in $(\frac12, 1)$, whence $\bar x > \frac12$ and $\mathsf{P}(\bar x) \le \mathsf{P}(\frac12) < \mathsf{p}^*$.

(b) $\bar{\mathsf{p}}$ is nonincreasing because $\Delta\mathcal{V}$ is nondecreasing, so $\bar g' \ge 0$, and $\bar g(0) = 1 - h > 0$. By Lemma 7.5(a), $\Phi_j(x) \ge \bar g(x_j)$, so $x_j(t) \ge \bar g^t(x_j(0)) \ge \bar g^t(0)$, which increases to $\underline{x}$. Since $\bar g(y) - y = \alpha h[\mathsf{p}^*(1-y) - y\bar{\mathsf{p}}(y)]$, condition (i) is equivalent to $\bar g(y) > y$ on $[0, \frac12]$, that is, to $\underline{x} > \frac12$. A fixed point $x^*$ of $\Phi$ satisfies $x_j^* \ge \bar g(x_j^*)$, so $x_j^* \ge \underline{x}$. Every trajectory enters the box $[x', 1]^n$ after finitely many steps, and the box is invariant because $\bar g(x') > x'$ and $\Phi_j \le h x_j + 1 - h \le 1$. On the box $\mathsf{p}_j \le \bar{\mathsf{p}}(x')$, and by Lemma 7.5(c) the Jacobian, with entries $h(1-\alpha\mathsf{p}_j)[j = j'] - \alpha h x_j \partial \mathsf{p}_j/\partial x_{j'}$, has absolute row sums at most $h + 2\alpha h\gamma\mathcal{L}n\bar{\mathsf{p}}(x')$, which is below one exactly when $2\gamma\mathcal{L}n\bar{\mathsf{p}}(x') < (1-h)/(\alpha h) = \mathsf{p}^*$, condition (ii). By the mean-value inequality on the convex box, $\Phi$ contracts the maximum norm there, so it has a unique fixed point in the box that attracts the box, and hence all of $[0,1]^n$; every fixed point lies in $[\underline{x}, 1]^n \subseteq [x', 1]^n$, so it is the only one, and it is symmetric because $\Phi$ commutes with permutations of the cross senders. Its cross attention is at most $\bar{\mathsf{p}}(\underline{x})$.

For the explicit conditions, let

$$\mathsf{c}_\mathcal{V} = \frac{2\kappa D_{\mathrm{cal}}\vartheta^2}{(\kappa+1)\pi^2\hat\sigma_{\mathrm{in}}^2(\hat\sigma_{\mathrm{in}}^2 + 2D_{\mathrm{cal}})(\Lambda_0 + \kappa\varpi_{\mathrm{in}})}.$$

The bounds $\log(1+\mathsf{z}) \ge \mathsf{z}/(1+\mathsf{z})$ with $\mathsf{z} = (\varpi_{\mathrm{in}} - \varpi_\times)/(\Lambda_0 + \kappa\varpi_{\mathrm{in}}) \le 1/\kappa$, $\varpi_{\mathrm{in}} - \varpi_\times(\phi) \ge D_{\mathrm{cal}}(1-\cos\phi)/(\hat\sigma_{\mathrm{in}}^2(\hat\sigma_{\mathrm{in}}^2 + 2D_{\mathrm{cal}}))$ and $1 - \cos\phi = 2\sin^2(\phi/2) \ge 2\phi^2/\pi^2$ on $[0,\pi]$ give $\Delta\mathcal{V}(y) \ge \mathsf{c}_\mathcal{V} y^2$ for $K = 2$. Then $y\bar{\mathsf{p}}(y) \le \kappa y e^{-\gamma \mathsf{c}_\mathcal{V} y^2} \le \kappa(2e\gamma\mathsf{c}_\mathcal{V})^{-1/2}$. If $\gamma > 2\kappa^2/(e\mathsf{c}_\mathcal{V}\mathsf{p}^{*2})$ this is below $\mathsf{p}^*/2 \le \mathsf{p}^*(1-y)$ on $[0, \frac12]$, which is (i). If $\gamma > 8\kappa^2/(e\mathsf{c}_\mathcal{V}\mathsf{p}^{*2})$ it is below $\mathsf{p}^*/4 \le \mathsf{p}^*(1-y)$ on $[0, \frac34]$, so $\underline{x} > \frac34$, and $x' = \frac58$ is admissible in (ii) with $\bar{\mathsf{p}}(\frac58) \le \kappa e^{-c\gamma}$, $c = 25\mathsf{c}_\mathcal{V}/64$. Since $\gamma e^{-c\gamma/2} \le 2/(ec)$, condition (ii) holds when $(4\mathcal{L}n\kappa/(ec))e^{-c\gamma/2} < \mathsf{p}^*$. Both sufficient conditions persist as $\gamma$ grows, so

$$\gamma_T \le \max\Big\{\frac{8\kappa^2}{e\mathsf{c}_\mathcal{V}\mathsf{p}^{*2}},\ \frac{128}{25\mathsf{c}_\mathcal{V}}\log\frac{256\mathcal{L}n\kappa}{25e\mathsf{c}_\mathcal{V}\mathsf{p}^*}\Big\},$$

which is finite and, the second term being $O(\log(1/\mathsf{p}^*))$, is $O(\mathsf{p}^{*-2})$ as $\mathsf{p}^* \to 0$.

For the lower bound, $\log(1+\mathsf{z}) \le \mathsf{z}$, $\varpi_{\mathrm{in}} - \varpi_\times(\phi) = D_{\mathrm{cal}}(1-\cos\phi)/(\hat\sigma_{\mathrm{in}}^2(\hat\sigma_{\mathrm{in}}^2 + D_{\mathrm{cal}}(1-\cos\phi))) \le D_{\mathrm{cal}}(1-\cos\phi)/(\hat\sigma_{\mathrm{in}}^2)^2$ and $1 - \cos\phi \le \phi^2/2$ give $\Delta\mathcal{V}(y) \le \mathsf{c}_\mathcal{V}^+ y^2$ on $[0,1]$. Since $\mathsf{r} \mapsto \kappa e^{-\mathsf{r}}/(1+\kappa e^{-\mathsf{r}})$

is decreasing, $\bar{\mathsf{p}}(y) \geq \kappa e^{-\gamma \mathsf{c}_{\mathcal{V}}^{+} y^2}/(1+\kappa e^{-\gamma \mathsf{c}_{\mathcal{V}}^{+} y^2})$. Put $\gamma = \mathsf{m}_\kappa^2/(\mathsf{c}_{\mathcal{V}}^{+}\mathsf{p}^{*2})$ and $y = \mathsf{t}_\kappa/\sqrt{\gamma \mathsf{c}_{\mathcal{V}}^{+}} = \mathsf{t}_\kappa \mathsf{p}^*/\mathsf{m}_\kappa$, which lies in $[0, \frac{1}{2}]$ when $\mathsf{p}^* \leq \mathsf{m}_\kappa/(2\mathsf{t}_\kappa)$. Then

$$y\bar{\mathsf{p}}(y) \geq \frac{1}{\sqrt{\gamma \mathsf{c}_{\mathcal{V}}^{+}}} \frac{\kappa \mathsf{t}_\kappa e^{-\mathsf{t}_\kappa^2}}{1+\kappa e^{-\mathsf{t}_\kappa^2}} = \frac{\mathsf{m}_\kappa}{\sqrt{\gamma \mathsf{c}_{\mathcal{V}}^{+}}} = \mathsf{p}^* \geq \mathsf{p}^*(1-y),$$

so (i) fails at this $\gamma$, and $\gamma_T$, an infimum over thresholds beyond which (i) holds throughout, is at least this $\gamma$.

(c) By Lemma 7.5(b), $\Phi_j(x) \leq \hat{g}(x_j)$ with $\hat{g}(y) = h(1-\alpha \underline{\mathsf{p}})y + 1 - h$, which is nondecreasing and affine with slope below one. By induction $x_j(t) \leq \hat{g}^t(x_j(0))$, which converges to the fixed point $(1-h)/(1-h+\alpha h \underline{\mathsf{p}}) = \mathsf{p}^*/(\mathsf{p}^* + \underline{\mathsf{p}})$. This is below $\frac{1}{2}$ exactly when $\underline{\mathsf{p}} > \mathsf{p}^*$, that is, when $\gamma < \gamma_M$.

(d) Exchangeability of the cross senders gives, at a symmetric point, $\partial \mathsf{p}/\partial x = \mathsf{p}'_{\text{self}} I + \mathsf{p}'_{\text{cross}}(\mathbf{1}\mathbf{1}^\top - I)$, so the Jacobian $h(1-\alpha \mathsf{P})I - \alpha h \bar{x}(\partial \mathsf{p}/\partial x)$ is symmetric with eigenvalue $h(1-\alpha \mathsf{P}) - \alpha h \bar{x}(\mathsf{p}'_{\text{self}} + (n-1)\mathsf{p}'_{\text{cross}})$ on $\mathbf{1}$, where $\mathsf{p}'_{\text{self}} + (n-1)\mathsf{p}'_{\text{cross}} = \mathsf{P}'(\bar{x})$ so that this eigenvalue is $g'(\bar{x}) \geq 0$, and eigenvalue $h(1-\alpha \mathsf{P}) - \alpha h \bar{x}(\mathsf{p}'_{\text{self}} - \mathsf{p}'_{\text{cross}})$ on $\mathbf{1}^\perp$. Linear stability is spectral radius below one. Raising $x_j$ changes only the values of sets containing $j$, and lowers them: $\partial_{x_j} \mathcal{V}(J)$ vanishes for $J \not\ni j$ and is nonpositive for $J \ni j$. Hence $\mathsf{p}'_{\text{self}} = \gamma \operatorname{Cov}_q([j \in J], \partial_{x_j} \mathcal{V}(J)) = \gamma(1-\mathsf{p}_j)\mathbb{E}_q[[j \in J]\partial_{x_j}\mathcal{V}(J)] \leq 0$. The symmetry-breaking multiplier minus the symmetric one is $\alpha h \bar{x} n \mathsf{p}'_{\text{cross}}$, whose sign is not determined here.

(e) At $\gamma = \infty$ the attended sets maximize $\sum_{k \in J} \varpi_{ik}$. If $x_j > 0$ then $\varpi_\times(x_j \vartheta) < \varpi_{\text{in}}$, because $D_{\text{cal}} > 0$ and $x_j \vartheta \in (0, \pi]$, and $\kappa \leq n-1$ in-group senders are available, so $j$ is never attended and $\Phi_j(x) = h x_j + 1 - h$. If $x_j = 0$ then $\Phi_j(x) = 1 - h$ whether or not $j$ is attended. After one step every $x_j \geq 1 - h > 0$, and then $x_j(t) - 1 = h^{t-1}(x_j(1) - 1) \to 0$. The only fixed point is $x = 1$. □

*Proof of Lemma 7.7.* At $\gamma = \infty$ with nonzero cross errors, $\varpi_\times(\phi_{ij}) < \varpi_{\text{in}}$ by Lemma 7.3, so every attended set consists of $\kappa \leq n-1$ in-group senders, all with precision $\varpi_{\text{in}}$. The claimed precision $\Lambda_0 + \kappa \varpi_{\text{in}}$ is then the same for every set and round, the claimed prior variance is at its fixed point, and the gains are constants. In one coordinate, group $A$ obeys $e_A(t+1) = W(t) e_A(t) + \zeta(t) - w_t \mathbf{1}$ with $W(t) = F(k^0 I + \bar{k} \mathsf{J}(t))$, where the in-group attention matrix $\mathsf{J}(t)$ has $\kappa$ ones in each row and none on the diagonal, its rows are independent and uniformly distributed, it is independent across rounds, and $\zeta(t)$ has independent entries of variance $q$; every realization has $W(t)\mathbf{1} = \lambda \mathbf{1}$.

The errors of $A$ are functions of the world innovations and of $A$'s local noises and attention draws, those of $B$ of the world innovations and of $B$'s, and these three families are independent; so the groups are conditionally independent given the world path. As in the proof of Lemma 7.3, $e_A(t) = e^{\text{com}}(t) + e^{\text{loc}}(t)$ with $e^{\text{com}}(t) = -\sum_l \lambda^l w_{t-1-l} \mathbf{1}$, which the world path determines because $w_s = \eta_{s+1} - F\eta_s$, and $e^{\text{loc}}(t) = \sum_l W(t-1)\cdots W(t-l)\zeta(t-1-l)$, independent of the world. Given the world path, $e_A$ is $e^{\text{loc}}$ shifted by a known vector, so its conditional covariance is $\Sigma^{\text{sc}} = \operatorname{Cov}(e^{\text{loc}})$, independent of the path; this is the second-moment form of Remark S7.17.

Let $\mathcal{W}(\mathsf{X}) = \mathbb{E}[W(t)\mathsf{X}W(t)^\top]$ on symmetric matrices. Since $W(t)$ is independent of $e^{\text{loc}}(t)$ and $\zeta(t)$ of both, stationarity gives $\Sigma^{\text{sc}} = \mathcal{W}(\Sigma^{\text{sc}}) + qI$. The map $\mathcal{W}$ is positive. Products of $l$ transfer matrices are nonnegative with row sums $\lambda^l$, so the entries of $\mathcal{W}^l(I)$ are at most $\lambda^{2l}$ and $\|\mathcal{W}^l(I)\| \leq n\lambda^{2l}$. For symmetric $\mathsf{X}$, $-\|\mathsf{X}\| I \preceq \mathsf{X} \preceq \|\mathsf{X}\| I$, positivity gives $-\|\mathsf{X}\|\mathcal{W}^l(I) \preceq \mathcal{W}^l(\mathsf{X}) \preceq \|\mathsf{X}\|\mathcal{W}^l(I)$, and hence $\|\mathcal{W}^l(\mathsf{X})\| \leq n\lambda^{2l}\|\mathsf{X}\|$. The spectral radius of $\mathcal{W}$ is therefore at most $\lambda^2 < 1$, and $\Sigma^{\text{sc}} = \sum_l \mathcal{W}^l(qI)$ is the unique solution, with $\|\Sigma^{\text{sc}}\| \leq nq/(1-\lambda^2)$.

Now $\mathbb{E}\mathsf{J} = \mathsf{p}_{\text{in}}(\mathbf{1}\mathbf{1}^\top - I)$, and $\mathbb{E}[\mathsf{J}\mathsf{J}^\top] = (\kappa - (n-2)\mathsf{p}_{\text{in}}^2)I + (n-2)\mathsf{p}_{\text{in}}^2 \mathbf{1}\mathbf{1}^\top$, because each row has $\kappa$ ones and two distinct rows, being independent, share a given third column with probability $\mathsf{p}_{\text{in}}^2$, with $n-2$ such columns. Hence $\mathbb{E}[WW^\top] = \psi_\perp I + n^{-1}(\psi_{\mathbf{1}} - \psi_\perp)\mathbf{1}\mathbf{1}^\top$ with $\psi_\perp$ and $\psi_{\mathbf{1}} - \psi_\perp$ as stated, and $W\mathbf{1}\mathbf{1}^\top W^\top = \lambda^2 \mathbf{1}\mathbf{1}^\top$ for every realization. The span of $I$ and $\mathbf{1}\mathbf{1}^\top$ is invariant:

$\mathcal{W}(I) = \mathbb{E}[WW^\top]$ and $\mathcal{W}(\mathbf{1}\mathbf{1}^\top) = \lambda^2\mathbf{1}\mathbf{1}^\top$. On it $\mathcal{W}$ is triangular with eigenvalues $\psi_\perp$ and $\lambda^2$, so $\psi_\perp \le \lambda^2 < 1$, and $\psi_\perp \ge 0$ as the eigenvalue of the positive semidefinite matrix $\mathbb{E}[WW^\top]$ on $\mathbf{1}^\perp$. The ansatz $\Sigma^{\mathrm{sc}} = \mathsf{s}_0 I + \mathsf{s}_1\mathbf{1}\mathbf{1}^\top$ gives $\mathcal{W}(\Sigma^{\mathrm{sc}}) + qI = (\mathsf{s}_0\psi_\perp + q)I + (\mathsf{s}_0(\psi_{\mathbf{1}} - \psi_\perp)/n + \mathsf{s}_1\lambda^2)\mathbf{1}\mathbf{1}^\top$, and matching coefficients yields the stated $\mathsf{s}_0$ and $\mathsf{s}_1$, which by uniqueness give $\mathrm{Cov}(e^{\mathrm{loc}})$. Since $\psi_{\mathbf{1}} > \psi_\perp$, $\mathsf{s}_1 > 0$.

By the Sherman–Morrison formula $(\mathsf{s}_0 I + \mathsf{s}_1\mathbf{1}\mathbf{1}^\top)^{-1} = \mathsf{s}_0^{-1}(I - \frac{\mathsf{s}_1}{\mathsf{s}_0 + n\mathsf{s}_1}\mathbf{1}\mathbf{1}^\top)$, whose off-diagonal entries are $-a^\Pi < 0$ and whose row sums are $\varepsilon^\Pi = 1/(\mathsf{s}_0 + n\mathsf{s}_1)$; it equals $\varepsilon^\Pi I + a^\Pi(nI - \mathbf{1}\mathbf{1}^\top)$. The two coordinates evolve as independent copies, so $\Sigma_A^c = \Sigma^{\mathrm{sc}} \otimes I_2$. The mean-square error is $\mathrm{Var}(e_i^{\mathrm{com}}) + \mathrm{Var}(e_i^{\mathrm{loc}}) = Q/(1-\lambda^2) + \mathsf{s}_0 + \mathsf{s}_1$. For $\kappa = n - 1$, $\mathsf{J} = \mathbf{1}\mathbf{1}^\top - I$ is deterministic, $W$ is symmetric with eigenvalue $F(k^0 + (n-1)\bar{k}) = F(1 - k^s) = \lambda$ on $\mathbf{1}$ and $\lambda_\perp = F(k^0 - \bar{k})$ on $\mathbf{1}^\perp$, $\mathbb{E}[WW^\top] = W^2$, and the recursion is linear with Gaussian inputs, so the law is Gaussian. □

*Proof of Theorem 7.8.* (a) By Lemma 7.7 the scalar precision of the population is block diagonal with blocks $\varepsilon^\Pi I + a^\Pi L_{K_n}$, $L_{K_n} = nI - \mathbf{1}\mathbf{1}^\top$, uniform leak $\varepsilon^\Pi$ and no couplings between groups. By Definition 4.10, $\varkappa_{\mathrm{out}}(C)$ is the mean leak plus the cut per member and $\varkappa_{\mathrm{in}}(C)$ the smallest Rayleigh quotient of $\varepsilon^\Pi I + L_C$ on vectors supported in $C$ and orthogonal to $\mathbf{1}_C$, with $L_C$ the Laplacian of the couplings inside $C$. For $C = A$ the cut is zero, $\varkappa_{\mathrm{out}} = \varepsilon^\Pi$, and the second eigenvalue of $a^\Pi L_{K_n}$ is $na^\Pi$, so $r(A) = 1 + na^\Pi/\varepsilon^\Pi = 1 + n\mathsf{s}_1/\mathsf{s}_0 = 1 + (\psi_{\mathbf{1}} - \psi_\perp)/(1-\lambda^2)$, which is (74); for $\kappa = n - 1$ it equals $1 + (\lambda^2 - \lambda_\perp^2)/(1-\lambda^2) = (1 - \lambda_\perp^2)/(1-\lambda^2)$. For $S \subset A$ with $n_S$ members the internal couplings form a complete graph with second eigenvalue $n_S a^\Pi$, and each member has $n - n_S$ couplings leaving $S$, so $\varkappa_{\mathrm{in}} = \varepsilon^\Pi + n_S a^\Pi$ and $\varkappa_{\mathrm{out}} = \varepsilon^\Pi + (n - n_S)a^\Pi$; the ratio increases in $n_S$, so the largest value over proper subsets is attained at $n_S = n - 1$. If $C$ meets both groups, put $C_A = C \cap A$, $C_B = C \cap B$ and $f = \mathbf{1}_{C_A}/|C_A| - \mathbf{1}_{C_B}/|C_B|$; then $f \perp \mathbf{1}_C$ and $L_C f = 0$ because $L_C$ has no couplings between $C_A$ and $C_B$ and $f$ is constant on each, so $\varkappa_{\mathrm{in}} \le \varepsilon^\Pi \le \varkappa_{\mathrm{out}}$ and $r(C) \le 1$. For $C = A \cup B$ the cut is zero and $\varkappa_{\mathrm{in}} = \varkappa_{\mathrm{out}} = \varepsilon^\Pi$, so $r = 1$. Regions have at least two members, so for $n = 2$ every region other than $A$ and $B$ meets both groups.

(b) The comparison medium is one group of $2n$ agents each attending $\kappa \le 2n - 1$ others uniformly at random with exact translations, so Lemma 7.7 applies with $n$ replaced by $2n$ and gives the scalar precision $\varepsilon^h I + a^h(2nI - \mathbf{1}\mathbf{1}^\top)$ with $a^h > 0$. The computation of (a) gives $(\varepsilon^h + n_C a^h)/(\varepsilon^h + (2n - n_C)a^h)$ for a region of $n_C < 2n$ members, increasing in $n_C$ and equal to one at $n_C = n$, and $1 + 2na^h/\varepsilon^h$ for the whole population. Since $(\varepsilon^h + (2n-1)a^h)/(\varepsilon^h + a^h) < 1 + 2na^h/\varepsilon^h$, equivalent to $(2n-1)a^h\varepsilon^h < (2n+1)a^h\varepsilon^h + 2n(a^h)^2$, the supremum over regions of every size is attained by the whole population.

(c) The collectives at level $r_0$ are the regions with $r \ge r_0$ (Definition 4.12). By (a), $A$ and $B$ qualify when $r_0 \le r(A)$, and every other region has $r \le \max\{r_{\mathrm{sub}}, 1\} < r_0$. Leaks are uniform, so the ratio of the largest to the mean leak of every region is one, and $r_0 > \bar{r}_{\mathrm{hom}} \ge r_{\mathrm{hom}}(n) = 1$; the level is admissible. The moment bound holds because the split's law is stationary with finite second moments, and the precision is diffusive by Lemma 7.7.

(d) For fixed $\hat{\sigma}^2_{\mathrm{in}}$ the claimed prior variance is the unique positive root of $\Lambda_1(\hat{P}^-)^2 + (1 - F^2 - Q\Lambda_1)\hat{P}^- - Q = 0$ with $\Lambda_1 = 1/R^s + \kappa/\hat{\sigma}^2_{\mathrm{in}}$, obtained from $\hat{P}^- = F^2\hat{P}^-/(1 + \Lambda_1\hat{P}^-) + Q$; the product of the roots is $-Q/\Lambda_1 < 0$. It depends continuously on $\hat{\sigma}^2_{\mathrm{in}}$, and so do the gains and, by Lemma 7.7, MSE. The function $\Psi_{\mathrm{cal}} = R_c + \mathrm{MSE} - \hat{\sigma}^2_{\mathrm{in}}$ is positive at $\hat{\sigma}^2_{\mathrm{in}} = R_c$, since $\mathrm{MSE} \ge Q/(1-\lambda^2) > 0$. As $\hat{\sigma}^2_{\mathrm{in}} \to \infty$ the message gains vanish, $\lambda \le F$, $q \le F^2(R^s + \kappa R_c)$ and $\mathsf{s}_0 + \mathsf{s}_1 \le \|\Sigma^{\mathrm{sc}}\| \le nq/(1-F^2)$, so MSE stays bounded and $\Psi_{\mathrm{cal}} \to -\infty$. A root exists in $(R_c, \infty)$ by the intermediate value theorem. The comparison medium is the same computation with $2n$ agents.

(e) The root $\hat{P}^-$ is a real-analytic function of $(F, Q, R^s, \hat{\sigma}^2_{\mathrm{in}})$, being a simple root of a quadratic, the gains are rational in it with positive denominators, and $\lambda$, $\psi_{\mathbf{1}}$, $\psi_\perp$, $q$, $\mathsf{s}_0$ and $\mathsf{s}_1$ are rational with denominators $1 - \psi_\perp \ge 1 - \lambda^2 > 0$ and $1 - \lambda^2 > 0$. Hence $\Psi_{\mathrm{cal}}$, $r(A)$ and $r_{\mathrm{sub}}$ are real analytic in

$(\hat{\sigma}^2_{\text{in}}, F, Q, R^s, R_c)$, and the same holds for the comparison medium and $\bar{r}_{\text{hom}}$. At a nondegenerate root, $\partial\Psi_{\text{cal}}/\partial\hat{\sigma}^2_{\text{in}} \neq 0$, and the implicit function theorem gives a real-analytic branch of roots on a neighborhood. Along the two branches the three contrasts are continuous, so the strict inequalities persist on a smaller neighborhood, and so does nondegeneracy. □

### Openness at finite policy precision: a sketch

This supports Remark S7.8; it is a sketch, and the step that is not proved is named at the end. For fixed $n$ and $\kappa$, consider the parameter points $(F, Q, R^s, R_c, \vartheta, \alpha, h, \gamma)$ with $\gamma < \infty$ at which the calibration roots of the split and of the comparison medium are nondegenerate, conditions (i) and (ii) of Theorem 7.6(b) hold, the world-conditional precision of the frozen-slow model at the mean-field fixed point is diffusive with $A$ and $B$ above a declared level $r_0$ and every other region below it, and $r_0 > \bar{r}_{\text{hom}}$.

The calibration constants depend continuously on the parameters along the branches of Theorem 7.8(e). Conditions (i) and (ii) then define an open set. Fix $x' \in (\frac{1}{2}, 1)$. The pair of statements $y\bar{\mathsf{p}}(y) < \mathsf{p}^*(1-y)$ *for all* $y \in [0, x']$ and $2\gamma\mathcal{L}n\bar{\mathsf{p}}(x') < \mathsf{p}^*$ is equivalent to (i) and (ii) with this $x'$, because the first statement means $\bar{g}(y) > y$ on $[0, x']$, which holds exactly when $\underline{x} > x'$, since $\bar{g}(0) > 0$ and $\underline{x}$ is the smallest fixed point. Each statement is a strict inequality between functions continuous in $y$ and the parameters on a compact set of $y$, so each holds on an open set, and the set on which (i) and (ii) hold is the union over $x'$ of these open sets. Under (i) and (ii) the proof of Theorem 7.6(b) shows that $\Phi$ contracts the maximum norm on $[x', 1]^n$ with a constant continuous in the parameters, so its unique fixed point, and with it the attention law over sets, depends continuously on the parameters. The stationary second moments of the frozen-slow model solve a linear system whose operator is block triangular, with the world block contracting by $F^2$ and the error block contracting because of the row-sum bound of Remark S7.16, so the system has a unique solution, and the second moments, with $\Sigma^c$ of Remark S7.17, depend continuously on the attention law. Where $\Sigma^c$ has the form $\Sigma^{\text{sc}} \otimes I_2$ and $(\Sigma^{\text{sc}})^{-1}$ is diffusive, contrast is a continuous function of $(\Sigma^{\text{sc}})^{-1}$, so strict inequalities between contrasts persist on a neighborhood.

The step that is not proved is the form of $\Sigma^c$. At finite $\gamma$ cross edges are attended with small probability and read through rotations, so the form $\Sigma^{\text{sc}} \otimes I_2$ (after a rotation of group $B$) and the diffusivity of $(\Sigma^{\text{sc}})^{-1}$ near the split, where the cross-group entries of the precision are close to zero and could change sign, were verified numerically at the base point for $\gamma = 80$, 120 and 160 (Numerical observation S7.7) but not proved. Nondegeneracy of the calibration roots is verified on the grid of Numerical observation S7.6. The same continuity argument would give the open-set form of clause (T4).

*Proof of Proposition 7.9.* (a) Under each of the three nulls the path of every agent is a measurable function of the shared drive (the initial state and the world innovations) and of the agent's private inputs: its sensing noise, its reading noise (under label-preserving deletion, the noise the deleted channels deliver), its angle observations and its attention draws. Under the frozen-environment null the messages carry other agents' initial states, which belong to the shared drive. By Proposition 5.16(a), $\Omega_{\text{int}} = 0$, and the covariance of the agents' states given the shared drive, and its inverse, are block diagonal per agent. The world-conditional law of Definition 4.7 conditions on the world path and not on the initial state. Under the two deletion nulls the model has no records and no broadcast noise, each agent's error depends on its own past only through the coefficient $Fk_i^0 < 1$, and the initial state is forgotten, so the stationary world-conditional law is the law given the shared drive (Proposition 5.16(d)). Under the frozen-environment null a frozen message delivers its sender's initial state in every round, so the initial states are never forgotten, and the two laws

coincide only when the initial states are deterministic or are added to the conditioning set; otherwise two receivers that read the same sender share its initial state, and the law given the world path is not block diagonal. Under the stated condition the world-conditional covariance and precision are therefore block diagonal per agent, each block a multiple of $I_2$ by isotropy. For a diagonal scalar precision, $\varkappa_{\text{out}}(C)$ is the mean of the diagonal over $C$ and $\varkappa_{\text{in}}(C)$ is the smallest eigenvalue of the compression of the diagonal to $\mathbf{1}_C^{\perp}$, whose trace is $(|C|-1)$ times that mean; so $r(C) \le 1$.

(b) No agent acts on the world and reading noise is receiver specific, so the shared drive of Definition 5.15 consists of the initial state and the world innovations. Given it, a member of $A$ and a member of $B$ are functions of disjoint independent families of private inputs (Lemma 7.7), so their paths are conditionally independent and $\Omega_{\text{int}} = 0$. For $\kappa = n-1$ the dynamics is linear and Gaussian with $W \ge 0$ and $W_{ij} = F\bar{k} > 0$ for distinct members of a group. Every private input enters every error with a nonnegative coefficient, $e_j(1)$ contains $Fk^s v_j(0)$, and $e_i(2)$ contains $F\bar{k}e_j(1)$, so the conditional covariance of $e_i(2)$ and $e_j(1)$ is a sum of nonnegative multiples of $I_2$, one of them positive. For jointly Gaussian paths the conditional mutual information is positive exactly when the conditional cross-covariance is nonzero, so $\Omega_{\text{int}} > 0$ for every window containing rounds one and two.

(c) By Theorem 7.8(c) the split selects exactly $A$ and $B$; by (a), under the deletion nulls, and under the frozen-environment null with initial states that are deterministic or conditioned on, every region has $r \le 1 < r_0$. □

*Proof of Proposition S7.10.* A set of $\kappa$ senders contains at most $n-1$ members of the receiver's group, hence at least $\kappa - n + 1$ cross senders. For the second statement the calibrated precisions are $\varpi_{\text{in}}$ on in-group edges and $\varpi_{\times}(x_j\vartheta)$ on cross edges, the latter strictly decreasing in $x_j \in [0,1]$ because $D_{\text{cal}} > 0$, and $\sigma_\alpha^2 = 0$, so that each bias fraction follows $x_j \leftarrow h(1-\alpha[j \in J_i(t)])x_j + 1 - h$ exactly. At $\gamma = \infty$ the receiver attends its $n-1$ in-group senders, whose precision exceeds that of every cross edge with $x_j > 0$, together with $\kappa - n + 1$ cross senders of largest precision, that is, of smallest bias fraction. From the declared start every $x_j = 1$ and the first choice is made by tie-breaking. A chosen edge moves to $h(1-\alpha)x_j + 1 - h < x_j$, while an unchosen edge at $x_j = 1$ stays at 1, so the chosen edges remain the strict minimizers and are chosen in every round. Their bias fractions follow $y \mapsto h(1-\alpha)y + 1 - h$ and converge to $(1-h)/(1-h+\alpha h) = \mathsf{p}^*/(\mathsf{p}^*+1)$, which is positive, and below $\frac{1}{2}$ exactly when $\mathsf{p}^* < 1$. In the mean-field map itself the first-round tie is never broken: the symmetric state, with every cross edge attended at rate $(\kappa - n + 1)/n$, is a fixed point there, unstable at $\gamma = \infty$ because any asymmetry is resolved in favor of the more learned edges. □

*Proof of Proposition 7.12.* (a) A receiver that knows $\text{Rot}(\phi_{ij})$ applies $\text{Rot}(-\phi_{ij})$ to $\tilde{y}_{ij}$ and obtains $m_j + \text{Rot}(-\phi_{ij})\tilde{\nu}_{ij}$. Rotations preserve the law $\mathcal{N}(0, R_c I_2)$ and $\tilde{\nu}_{ij}$ is independent of everything else, so the corrected message has the joint law with $(\eta, e)$ of a message read with the correct translation, the mistranslation forcing in (70) vanishes, and translation learning no longer affects the dynamics. Its calibrated noise is $\frac{1}{K}\mathbb{E}\|e_j(t) + \text{Rot}(-\phi_{ij})\tilde{\nu}_{ij}(t)\|^2 = \text{MSE}_j + R_c$, which does not depend on the receiver and needs no invariance hypothesis.

(b) By (a) every receiver assigns sender $j$ the same precision $\varpi_j = 1/(R_c + \text{MSE}_j)$, so receiver $i$'s believed value of a set, $\mathcal{V}_i^0(J) = \frac{K}{2}\log(1 + \sum_{j\in J}\varpi_j/\Lambda_0)$ with its own claimed precision $\Lambda_0$, is an increasing function of $\sum_{j\in J}\varpi_j$, and all receivers rank sets alike. Fix a receiver $i$ and senders $j, k \ne i$ with $\varpi_j \ge \varpi_k$. Exchanging $k$ for $j$ maps the sets of $\kappa$ senders that contain $k$ and not $j$ bijectively onto those that contain $j$ and not $k$, and does not lower $\sum_{l\in J}\varpi_l$. At finite $\gamma$ it therefore does not lower the weight $e^{\gamma\mathcal{V}_i^0(J)}$; at $\gamma = \infty$ it maps maximizing sets to maximizing sets, among which the tie-breaking is uniform. In both cases the sets containing $j$ and not $k$ have at least the

probability of those containing $k$ and not $j$, and adding the probability of the sets that contain both gives $q_i(\{J : j \in J\}) \geq q_i(\{J : k \in J\})$. Let $j_1$ be a sender of largest precision. For a receiver $i \neq j_1$ the inclusion probabilities of the $2n-1$ possible senders sum to $\kappa$ and that of $j_1$ is the largest, so it is at least $\kappa/(2n-1)$. This holds in particular for the $n$ receivers of the group not containing $j_1$, for which $j_1$ is a cross sender.

(c) By (a) the corrected messages carry no mistranslation, so under the null the belief states evolve by (70) with every $\mathrm{Rot}(\phi_{ij})$ replaced by the identity and by the update of $\hat{P}_i^-$, and neither the frames nor the learned angles enter. Suppose that at round $t$ the law of $(e_i(t), \hat{P}_i^-(t))_i$ given the world path is invariant under all permutations of the $2n$ agents. Then $\mathrm{MSE}_j(t)$ is the same for every $j$, and by (a) all senders have the same calibrated precision. Every set of $\kappa$ senders then has the same believed value for a given receiver, so its policy posterior is uniform over sets at every $\gamma$, as is the tie-breaking at $\gamma = \infty$, and its gains depend on the receiver only through $\hat{P}_i^-(t)$. The update is therefore the same function of the state for every agent, with attention draws and noises that are independent and identically distributed across agents, so the law at round $t+1$ is again invariant. The base of the induction is the hypothesis of (c), which is conditional on the world path: invariance of the unconditional start law would not suffice, since an error whose sign follows a coordinate of $\eta_0$ for one agent only can be exchangeable unconditionally and not given the world path. Identical initial beliefs satisfy the hypothesis, because the errors $m_i(0) - \eta_0$ and the $\hat{P}_i^-(0)$ then coincide across agents under every conditioning. So does a start that jointly with $\eta_0$ is independent of the innovations $(w_t)_{t\geq 0}$ and has an invariant law given $\eta_0$: the world path is a function of $\eta_0$ and the innovations and determines both, so the law of such a start given the world path is its law given $\eta_0$. By induction from the start law the three assertions hold at every time. A permutation-invariant world-conditional second moment has equal diagonal blocks and equal off-diagonal blocks, so the contrast of a region depends only on its size; $A$ and $B$ then have the contrast of each of the $\binom{2n}{n} > 2$ regions of $n$ members, and at no level are they the selected regions. A stationary state of the trajectory has uniform attention, messages with the law of exactly translated ones, and calibrated noise solving $\hat{\sigma}^2 = R_c + \mathrm{MSE}(\hat{\sigma}^2)$, a solution of which exists by Theorem 7.8(d): it is the comparison medium of Theorem 7.8(b). By Theorem 7.8(b) every region then has contrast at most $\bar{r}_{\mathrm{hom}} < r_0$, so at every admissible level no region is selected and $\{A, B\}$ is not selected again. □

## S32 Proofs for Section 8

We prove the results of Section 8 in order. Block norms are spectral norms, $\Phi(t,s) = W(t-1)\cdots W(s)$, and we use repeatedly that block row sums of a product of block matrices are at most the products of the block row sums, and likewise for column sums: $\sum_j \|(AB)_{ij}\| \leq \sum_k \|A_{ik}\| \max_k \sum_j \|B_{kj}\|$.

*Proof of Lemma 8.2.* (a) The row-sum inequality and induction give $\sum_j \|\Phi_{ij}(t,s)\| \leq \lambda^{t-s}$. Write $e_i(t) = \sum_k \Phi_{ik}(t,0)e_k(0) + \sum_{u<t}\sum_k \Phi_{ik}(t,u+1)\zeta_k(u)$. Under (Lin) the noise terms are centered, independent of one another and of the first sum. With the weights $\|\Phi_{ik}(t,0)\|$ the Cauchy–Schwarz inequality gives

$$\mathbb{E}\Big\|\sum_k \Phi_{ik}(t,0)e_k(0)\Big\|^2 \leq \Big(\sum_k \|\Phi_{ik}(t,0)\|\Big)\sum_k \|\Phi_{ik}(t,0)\|\mathbb{E}\|e_k(0)\|^2 \leq \lambda^{2t}\sup_k \mathbb{E}\|e_k(0)\|^2,$$

and for block-diagonal noise, with $k_{\max}$ the largest block dimension,

$$\mathbb{E}\Big\|\sum_k \Phi_{ik}(t,u+1)\zeta_k(u)\Big\|^2 = \sum_k \mathrm{tr}\big(\Phi_{ik}Q_k^{\zeta}(u)\Phi_{ik}^{\top}\big) \leq qk_{\max}\Big(\sum_k \|\Phi_{ik}(t,u+1)\|\Big)^2 \leq qk_{\max}\lambda^{2(t-1-u)}.$$

Summing, $\mathbb{E}\|e_i(t)\|^2 \le \lambda^{2t}\sup_k \mathbb{E}\|e_k(0)\|^2 + qk_{\max}/(1-\lambda^2)$, which is finite and uniform in $i$ and $t$ whenever $\sup_k \mathbb{E}\|e_k(0)\|^2 < \infty$. For the uniform form of (M), write $\tilde{e}_i(t) = m_i(t) - \eta_t$ for the errors themselves, which obey $\tilde{e}(t+1) = W(t)\tilde{e}(t) + \zeta(t) + B^{\mathrm{mis}}\eta_t - \mathbf{1}\otimes w_t$ (Lemma 2.13). Then $\tilde{e}_i(t)$ is the sum of the initial term $\sum_k \Phi_{ik}(t,0)\tilde{e}_k(0)$, the noise terms of the second display and the input terms $\sum_k \Phi_{ik}(t,u+1)(B^{\mathrm{mis}}_k\eta_u - w_u)$, $u<t$. Because the schedule is independent of the initial errors, the first display holds with $\tilde{e}_k(0)$ in place of $e_k(0)$, so the initial term is at most $\lambda^t \sup_k(\mathbb{E}\|m_k(0)-\eta_0\|^2)^{1/2}$ in root mean square. The noise terms are centered and uncorrelated across rounds, so together they contribute at most $(qk_{\max}/(1-\lambda^2))^{1/2}$. Pathwise $\|\sum_k \Phi_{ik}(t,u+1)(B^{\mathrm{mis}}_k\eta_u - w_u)\| \le \lambda^{t-1-u}(b\|\eta_u\| + \|w_u\|)$ with $b = \sup_k \|B^{\mathrm{mis}}_k\|$, so by Minkowski's inequality the input terms contribute at most $(b\sup_u(\mathbb{E}\|\eta_u\|^2)^{1/2} + \sup_u(\mathbb{E}\|w_u\|^2)^{1/2})/(1-\lambda)$, where the first summand is absent when $B^{\mathrm{mis}} = 0$. Minkowski's inequality applied to the three contributions bounds $(\mathbb{E}\|\tilde{e}_i(t)\|^2)^{1/2}$ uniformly in $i$ and $t$; this is the uniform form of (M). For time-invariant $W$ and identically distributed $\zeta(t)$ the series $\sum_{n\ge 0} W^n\zeta(t-1-n)$ converges in mean square, blockwise with $\mathbb{E}\|(\cdot)_i\|^2 \le qk_{\max}/(1-\lambda^2)$, and its law does not depend on $t$, which defines the stationary law; for Gaussian noise the law of $\zeta(t)$ is fixed by $Q^\zeta$. For a schedule that depends on the states, on $N$ agents, write the recursion of the errors as $e(t+1) = W(t)e(t) + u(t)$, where $u(t)$ collects all inputs, local noise, world innovation and any mistranslation term, with $\sup_{k,t}\mathbb{E}\|u_k(t)\|^2 = U < \infty$. Every realized $W(t)$ has block row sums at most $\lambda$, so pathwise $\max_i \|e_i(t+1)\| \le \lambda \max_k \|e_k(t)\| + \max_k \|u_k(t)\|$, and by induction $\|e_i(t)\| \le \lambda^t \max_k \|e_k(0)\| + \sum_{s<t}\lambda^{t-1-s}\max_k\|u_k(s)\|$. Minkowski's inequality with $\mathbb{E}\max_k\|u_k(s)\|^2 \le \sum_k \mathbb{E}\|u_k(s)\|^2 \le NU$ gives $(\mathbb{E}\|e_i(t)\|^2)^{1/2} \le (\mathbb{E}\max_k\|e_k(0)\|^2)^{1/2} + (NU)^{1/2}/(1-\lambda)$, finite and uniform in $i$ and $t$. No independence between the schedule and the inputs is used.

(b) Let $b_{ij} = \|W_{ij}\|$. For a vector with blocks $x_j$, $\|(Wx)_i\| \le \sum_j b_{ij}\|x_j\|$, so $\|W\|_2 \le \|(b_{ij})\|_2 \le (\max_i \sum_j b_{ij} \cdot \max_j \sum_i b_{ij})^{1/2}$ by the Schur test. Hence $\|W^k\|_2 \le (\lambda_r\lambda_c)^{k/2}$, which is ($\mathrm{C}_\rho$) with $K_A = 1$ and $f = (\lambda_r\lambda_c)^{1/2}$.

(c) If $\rho(W) < 1$, Gelfand's formula for the bounded operator $W$ on $\ell^2(V)$ gives $\|W^n\|_2 \le K_A f^n$ for some $f<1$, and since $Q^\zeta$ is bounded the series $\sum_n W^n Q^\zeta W^{n\top}$ converges in operator norm to the covariance of the stationary solution $\sum_{n\ge 0} W^n\zeta(t-1-n)$, whose blocks are therefore finite. Conversely, let the population be finite. A stationary covariance satisfies $\Sigma = W\Sigma W^\top + Q^\zeta$. For a left eigenvector $y^*W = \Lambda y^*$ this gives $(1-|\Lambda|^2)y^*\Sigma y = y^*Q^\zeta y > 0$, so $|\Lambda| < 1$ for every eigenvalue $\Lambda$. The argument needs an eigenvector, which a bounded operator on an infinite population need not have at its spectral radius.

For scalar agents on $\mathbb{Z}^D$ under (TI), $W^m$ is the convolution with symbol $\hat{W}^m$, so by Parseval's identity $\mathrm{Var}((W^m\zeta(u))_0) = q\sum_x |(W^m)_{0x}|^2 = q(2\pi)^{-D}\int |\hat{W}(p)|^{2m}\mathrm{d}p$. If (76) holds, then $|\hat{W}| < 1$ almost everywhere and monotone convergence gives $\sum_{m\ge 0}(2\pi)^{-D}\int|\hat{W}|^{2m}\mathrm{d}p = (2\pi)^{-D}\int(1-|\hat{W}|^2)^{-1}\mathrm{d}p < \infty$, so $\sum_{n\ge 0} W^n\zeta(t-1-n)$ converges in mean square in every coordinate and defines a stationary law with the stated variance. Conversely, under a stationary law with finite second moments, $e(t) = W^n e(t-n) + \sum_{m<n} W^m\zeta(t-1-m)$ with $e(t-n)$ independent of the later noises, so $\mathrm{Var}(e_0(t)) \ge q\sum_{m<n}(2\pi)^{-D}\int|\hat{W}|^{2m}\mathrm{d}p$ for every $n$; letting $n\to\infty$, monotone convergence gives (76). For the lazy walk on $\mathbb{Z}^3$, $\hat{W}(p) = \frac{1}{4} + \frac{1}{4}\sum_d \cos p_d \in [-\frac{1}{2}, 1]$ attains one at $p=0$, so $\rho(W) = 1$, while $1-\hat{W}(p)^2 \ge \frac{1}{2}(1-\hat{W}(p))$ and $1-\hat{W}(p) \ge |p|^2/(2\pi^2)$ on $[-\pi,\pi]^3$, which makes the integrand at most $4\pi^2|p|^{-2}$, integrable in three dimensions. With $G(E) = \int_0^\infty e^{-Es}I_0(s)^3\mathrm{d}s$, the lattice Green function of the simple cubic lattice, partial fractions give the mean of the integrand as $2(G(3)+G(5)) = 1.4395$. On $\mathbb{Z}$ and $\mathbb{Z}^2$ a nearest-neighbor walk kernel with unit row sums has $1-\hat{W}(p)^2 \le 2(1-\hat{W}(p)) \le C|p|^2$ near $p=0$, and $|p|^{-2}$ is not integrable there. □

*Proof of Proposition 8.3.* (a) Remark S2.14 gives $W(t) \ge 0$ with row sums $F(1-k_i^s(t))$, hence the

bound on $\lambda$, and Lemma 8.2(a) gives (M), through its pathwise bound when the schedule depends on the states. A sender attended by $L$ receivers has column sum at least $F\sum_i k_{ij}$ over them, which grows with $L$; Corollary 8.19(e) shows the consequence. (b) Proposition 9.1(b) gives $\rho(W)\le f$ and $\|W^k\|_2 \le K_{\tilde M} f^k$, which is ($\mathrm{C}_\rho$), and Lemma 8.2(c) gives (M); the instances of Check S47.99 with block row sum above one show that ($\mathrm{C}_r$) can fail. (c) is the final assertion of Proposition 9.2, which applies when $\varpi_g < 1$, and (d) is Proposition 8.6(d). □

*Proof of Theorem 8.5.* (a) By induction on $t-s$, the case $t=s$ being trivial. Suppose the claim holds at $t$. The state $\mu_i(t+1) = \mathfrak{U}_i^{(t)}(\mu_{\mathcal{G}[i]}(t), \zeta_i(t))$ depends on $\mu_k(t)$ for $k\in\mathcal{G}[i]$, each a function of states $\mu_l(s)$ with $d(k,l)\le t-s$ and of noises $\zeta_l(u)$ with $d(k,l)\le t-1-u$. Since $d(i,k)\le 1$, these have $d(i,l)\le t+1-s$ and $d(i,l)\le (t+1)-1-u$, and the new noise $\zeta_i(t)$ has $d(i,i)=0=(t+1)-1-t$. If $d(i,j)>t-s$, $\mu_i(t)$ does not depend on $\mu_j(s)$, so $J_{ij}(t,s)=0$.

(b) By (a) with $s=0$, $\mu_i(t)$ is a function of the family $\mathfrak{F}_i = \{\mu_k(0) : d(i,k)\le t\}\cup\{\zeta_k(u) : u<t,\ d(i,k)\le t-1-u\}$, and $\mu_j(t)$ of $\mathfrak{F}_j$. A common initial state would give $d(i,j)\le 2t$, and a common $\zeta_k(u)$ would give $d(i,j)\le 2(t-1-u)<2t$. So if $d(i,j)>2t$ the families are disjoint, and given the world path their members are independent; functions of disjoint independent families are independent. For $\mu_i(t+h)$ the family has radius $t+h$ for initial states and $t+h-1-u$ for $\zeta_k(u)$; a common initial state needs $d(i,j)\le 2t+h$, and a common $\zeta_k(u)$ needs $u<t$ and then $d(i,j)\le 2t+h-2-2u$. So $d(i,j)>2t+h$ suffices.

(c) The chain rule gives $J_{ij}(t+1,s) = \sum_{k\in\mathcal{G}[i]} \partial_k \mathfrak{U}_i^{(t)} J_{kj}(t,s)$, so $\|J_{ij}(t+1,s)\| \le \sum_k \bar J_{ik}\|J_{kj}(t,s)\|$, and induction from $J(s,s)=I$ gives $(\bar J^{t-s})_{ij}$. Summing over $j$ uses $\sum_j(\bar J^n)_{ij}\le g_{\mathrm{tot}}^n$. If $(\bar J^n)_{kj}\le e^{-\varphi d(k,j)} g_\varphi^n$, then, since $-d(k,j)\le d(i,k)-d(i,j)$, $(\bar J^{n+1})_{ij}\le \sum_k \bar J_{ik} e^{-\varphi d(k,j)} g_\varphi^n \le e^{-\varphi d(i,j)} g_\varphi^{n+1}$. □

*Proof of Proposition S8.3.* Fix the clocks and the noises; almost surely $[0,t]$ contains finitely many events. At an event of $l$ the Jacobian of the event map has row $l$ equal to $(\partial_k \mathfrak{U}_l)_k$ and identity rows elsewhere. Let $x(0)$ be the $j$-th coordinate vector and, at each event of $l$, set $x_l \leftarrow \bar J_{ll} x_l + \sum_{k\ne l} \bar J_{lk} x_k$, the other coordinates unchanged; by induction over events, $\|J_{kj}(s)\| \le x_k(s)$ for all $k$ and $s$. Expanding the recursion, $x_i(t)$ is a sum over walks $(i_0=j, i_1, \dots, i_n=i)$ with $i_m\ne i_{m-1}$ and over increasing times $t_1<\cdots<t_n$, $t_m$ an event of $i_m$, of $\prod_{m=1}^n \bar J_{i_m i_{m-1}} \prod_{m=0}^n \bar J_{i_m i_m}^{N_m}$, where $N_m$ counts the other events of $i_m$ in $(t_m, t_{m+1})$, $t_0=0$, $t_{n+1}=t$. By the multivariate Mecke formula for independent Poisson processes, the expected sum over increasing tuples is $\nu^n \int_{0<t_1<\cdots<t_n<t} \mathbb{E}[\prod_m \bar J_{i_m i_m}^{N_m}]\mathrm{d}t_1\cdots\mathrm{d}t_n$, with the $N_m$ now counts of the original processes over disjoint intervals, independent and Poisson with means $\nu(t_{m+1}-t_m)$. Hence $\mathbb{E}\prod_m \bar J_{i_m i_m}^{N_m} = \prod_m e^{\nu(t_{m+1}-t_m)(\bar J_{i_m i_m}-1)} \le e^{-(1-\bar J_{\mathrm{diag}})\nu t}$. The time integral is $t^n/n!$, walks from $j$ to $i$ have $n\ge d$, and the sum of $\prod_m \bar J_{i_m i_{m-1}}$ over walks of length $n$ is $(\bar J_{\mathrm{off}}^n)_{ij}\le g^n$. This proves (S31). The explicit forms follow from $\sum_{n\ge d} x^n/n! \le (x^d/d!)\sum_{m\ge 0}(x/(d+1))^m$ with $d!\ge\sqrt{2\pi d}(d/e)^d$, and from $\sum_{n\ge d} x^n/n! \le e^{-\varphi d} e^{x e^\varphi}$ with $x=g\nu t$. For $v>v_{\mathrm{LR}}$ there is $\varphi>0$ with $c_\varphi = \nu(g e^\varphi + \bar J_{\mathrm{diag}} - 1)/\varphi < v$, and along $d\ge vt$ the second form is at most $e^{-\varphi(v-c_\varphi)t}$; when $g+\bar J_{\mathrm{diag}}<1$ the ratio $c_\varphi$ tends to $-\infty$ as $\varphi\downarrow 0$, so every $v>0$ qualifies. Without smoothness, $\mu_i(t)$ can depend on $\mu_j(0)$ only through a walk from $j$ to $i$ whose steps occur at increasing event times; the expected number of such chains is $\sum_{n\ge d} N_n(i,j)(\nu t)^n/n!$, with $N_n(i,j)\le \Delta^n$ walks of length $n$. For linear scalar updates with nonnegative coefficients $A$ and $\mathrm{diag}(A)=\bar J_{\mathrm{diag}} I$, an event of agent $l$ in $[s, s+\mathrm{d}s)$ replaces row $l$ of $J$ by $A_{l\cdot}J$, independently of $J(s)$, so $\frac{\mathrm{d}}{\mathrm{d}s}\mathbb{E}J = \nu(A-I)\mathbb{E}J$ and $\mathbb{E}J(t) = e^{\nu t(A-I)} = e^{-(1-\bar J_{\mathrm{diag}})\nu t} e^{\nu t A_{\mathrm{off}}}$, whose entries, all nonnegative, are the series in (S31). □

*Proof of Proposition 8.6.* For $q(\mathbf{g})_\pi \propto E_i(\pi) e^{-\gamma g_\pi}$ and $\mathbf{g}(s) = \mathbf{g} + s\boldsymbol{\Delta}$, $\frac{\mathrm{d}}{\mathrm{d}s} q_\pi = -\gamma q_\pi(\Delta_\pi - \mathbb{E}_q\Delta)$, so $\|\frac{\mathrm{d}}{\mathrm{d}s} q\|_1 = \gamma \mathbb{E}_q|\Delta - \mathbb{E}_q\Delta| \le \gamma(\mathrm{Var}_q\,\Delta)^{1/2} \le \frac{\gamma}{2}\mathrm{osc}(\Delta)$ by Popoviciu's inequality $\mathrm{Var}\le \mathrm{osc}^2/4$;

integrating over $s \in [0,1]$, $\|q(\mathbf{g}+\boldsymbol{\Delta}) - q(\mathbf{g})\|_1 \le \frac{\gamma}{2}\mathrm{osc}(\Delta)$. With $\Delta_\pi = G_i(\pi;\mu') - G_i(\pi;\mu)$ and $\delta = \mu' - \mu$, $\mathrm{osc}(\Delta) \le 2\sum_k \mathrm{Lip}_{ik}\|\delta_k\|$.

(a) The averaged update changes by $\sum_\pi q'_\pi[\mathfrak{U}_{i,\pi}(\mu') - \mathfrak{U}_{i,\pi}(\mu)] + \sum_\pi (q'_\pi - q_\pi)\mathfrak{U}_{i,\pi}(\mu)$. The first sum is at most $\sum_k \bar{A}_{ik}\|\delta_k\|$ by the mean value inequality on the convex domain. The second has weights of total zero, so it is at most $\frac{1}{2}\|q'-q\|_1 \max_{\pi,\pi'} \|\mathfrak{U}_{i,\pi}(\mu) - \mathfrak{U}_{i,\pi'}(\mu)\| \le \frac{\gamma}{2}\mathrm{osc}_i \sum_k \mathrm{Lip}_{ik}\|\delta_k\|$. The proofs of Theorem 8.5(c) and Proposition S8.3 use only such one-step bounds, with differences in place of derivatives.

(b) Given coupled states, draw each agent's two policies from the maximal coupling of $q_i(\cdot \mid \mu)$ and $q_i(\cdot \mid \mu')$, independently across agents, and share all other noise. The policies differ with probability $\frac{1}{2}\|q - q'\|_1 \le \frac{\gamma}{2}\sum_k \mathrm{Lip}_{ik}\|\delta_k\|$; when they agree the update difference is at most $\sum_k \bar{A}_{ik}\|\delta_k\|$, and otherwise at most $\mathrm{osc}_i + \sum_k \bar{A}_{ik}\|\delta_k\|$. Hence $\mathbb{E}[\|\mu_i^+ - \mu_i'^+\| \mid \mu, \mu'] \le \sum_k \bar{J}_{ik}(\gamma)\|\delta_k\|$. Take expectations and iterate, using $\bar{J}(\gamma) \ge 0$.

(c) A maximum of sums is at most the sum of the maxima, and Proposition S8.3 with $\bar{J}_{\mathrm{diag}}(\gamma) \le 1$ gives $v_{\mathrm{LR}} \le \inf_{\varphi>0} \nu g(\gamma) e^{\varphi}/\varphi = e g(\gamma)\nu$.

(d) By (a) the averaged map is a contraction with constant $g_{\mathrm{tot}}(\gamma)$ in $\max_i \|\cdot\|$, and the Banach fixed-point theorem gives a unique fixed point approached exponentially. By (b) the vector of expected coordinate distances contracts by $g_{\mathrm{tot}}(\gamma)$ per round in the maximum norm; on $N$ agents this bounds the Wasserstein distance between the laws at time $t$, for the cost $\sum_i \|\mu_i - \mu'_i\|$, by $N g_{\mathrm{tot}}(\gamma)^t$ times the initial one, so some power of the transition operator is a strict contraction on the complete space of laws with finite first moment on the closed domain. The operator maps this space to itself: the one-step law from one state has a finite first moment by hypothesis, and by (b) with finite $\mathrm{osc}_i$ the one-step laws from two states lie at Wasserstein distance at most $\sum_{i,k} \bar{J}_{ik}(\gamma)\|\delta_k\| < \infty$, so the one-step law from every state, and from every initial law with finite first moment, has a finite first moment. This gives a unique stationary law reached at an exponential rate. The condition $g_{\mathrm{tot}}(\gamma) < 1$ reads $\sum_k \bar{A}_{ik} + \frac{\gamma}{2}\mathrm{osc}_i \sum_k \mathrm{Lip}_{ik} < 1$ for every $i$, that is $\gamma < \gamma_D$. □

*Proof of Theorem 8.7.* (a) Write $e(t) = \Phi(t,0)e(0) + \sum_{u<t} \Phi(t,u+1)\zeta(u)$; under (Lin) the terms are independent and centered. Block $(i,j)$ of the term injected at $u$ is $\sum_k \Phi_{ik} Q_k^\zeta(u) \Phi_{jk}^\top$ with $n = t-1-u$ factors. By Theorem 8.5(a) applied to the linear maps, it vanishes unless some $k$ has $d(i,k) \le n$ and $d(j,k) \le n$, which forces $n \ge \lceil d/2 \rceil$. By Theorem 8.5(c), $\|\Phi_{jk}\| \le (\bar{J}^n)_{jk} \le e^{-\varphi d(j,k)} g_\varphi^n$, and induction with $e^{\varphi d(i,k)} \le e^{\varphi d(i,l)} e^{\varphi d(l,k)}$ gives $\sum_k (\bar{J}^n)_{ik} e^{\varphi d(i,k)} \le g_\varphi^n$. With $e^{-\varphi d(j,k)} \le e^{-\varphi d(i,j)} e^{\varphi d(i,k)}$,

$$\sum_k \|\Phi_{ik}\|\|\Phi_{jk}\| \le e^{-\varphi d(i,j)} g_\varphi^n \sum_k (\bar{J}^n)_{ik} e^{\varphi d(i,k)} \le e^{-\varphi d(i,j)} g_\varphi^{2n},$$

so the term is at most $q e^{-\varphi d} g_\varphi^{2n}$. Summing over $n$ from $\lceil d/2 \rceil$ to $t-1$ and adding the initial term, an injection with $n = t$ and $q_0$ in place of $q$, gives (77). For the lag, $\mathrm{Cov}(e(t+h), e(t)) = \Phi(t+h,t)\,\mathrm{Cov}(e(t))$ because later noise is independent of $e(t)$, so $\mathrm{Cov}(e_i(t+h), e_j(t)) = \sum_k \Phi_{ik}(t+h,t)\Sigma_{kj}(t)$. Write $B(d')$ for the bracket of (77) at distance $d'$; it is nonincreasing in $d'$, and $\|\Sigma_{kj}(t)\| \le e^{-\varphi d(k,j)} B(d(k,j))$. By Theorem 8.5(a), $\Phi_{ik}(t+h,t)$ vanishes unless $d(i,k) \le h$, and then $d(k,j) \ge (d-h)_+$, so $B(d(k,j)) \le B((d-h)_+)$. The same inequality as above gives $\sum_k \|\Phi_{ik}(t+h,t)\| e^{-\varphi d(k,j)} \le e^{-\varphi d(i,j)} g_\varphi^h$, hence the lagged covariance is at most $e^{-\varphi d} g_\varphi^h B((d-h)_+)$, which is (78) because $\mathbf{1}\{2t \ge (d-h)_+\} = \mathbf{1}\{2t+h \ge d\}$.

(b) $\Sigma_{ii}(t) \succeq Q_i^\zeta(t-1) \succeq q_{\min} I$ because the last injection is independent of everything else, so the top canonical correlation $\|\Sigma_{ii}^{-1/2}\Sigma_{ij}\Sigma_{jj}^{-1/2}\|$ is at most $\|\Sigma_{ij}\|/q_{\min}$. For $g_\varphi > 1$, $\sum_{n<t} g_\varphi^{2n} \le g_\varphi^{2t}/(g_\varphi^2 - 1)$, so the correlation is at most $K_\varphi e^{-\varphi d + 2t \ln g_\varphi} = K_\varphi \exp(-\varphi[d - 2v_\varphi t])$, which is below $\theta$ once $d > 2v_\varphi t + \varphi^{-1}\ln(K_\varphi/\theta)$. The bound $2t$ is the support statement.

(c) For $g_\varphi < 1$, $\sum_{n\ge\lceil d/2\rceil} g_\varphi^{2n} \le g_\varphi^d/(1-g_\varphi^2)$, and $g_\varphi^{2t} \le g_\varphi^d$ whenever $2t \ge d$, so the bound is at most $e^{-(\varphi+\ln(1/g_\varphi))d}(q/(1-g_\varphi^2)+q_0)$ for every $t$. □

*Proof of Proposition S8.6.* Let $r = \max_{k\ge2}|\nu_k| < \rho$ over the other eigenvalues $\nu_k$ of $W$. Then $W^n = \rho^n vu^\top + R_n$ with $R_n = W^n(I-vu^\top)$ and $\|R_n\| \le C_\delta(r+\delta)^n$ for every $\delta > 0$. In $\Sigma(t) = W^t\Sigma(0)W^{t\top} + \sum_{n<t} W^n Q^\zeta W^{n\top}$, the leading terms sum to $\rho^{2t}[u^\top\Sigma(0)u + u^\top Q^\zeta u(1-\rho^{-2t})/(\rho^2-1)]vv^\top$. The cross terms are $O(t\max\{\rho(r+\delta),1\}^t)$ and the remainders $O(t\max\{(r+\delta)^2,1\}^t)$; with $\delta < \rho - r$ both are $o(\rho^{2t})$. This proves (S32), and $c_\infty > 0$ because $Q^\zeta \succ 0$. For the readouts $\hat v_i^\top e_i$ with $\hat v_i = v_i/|v_i|$, covariances and variances are $\rho^{2t}c_\infty|v_i||v_j|(1+o(1))$ and $\rho^{2t}c_\infty|v_i|^2(1+o(1))$, so their correlation tends to one, and so does the top canonical correlation. The top eigenvalue of $\Sigma(t)$ is $c_\infty\rho^{2t}|v|^2(1+o(1))$ with eigenvector tending to $v/|v|$, which gives the eigenvalue of order $\rho^{-2t}$ of the precision. For the others, the Courant–Fischer theorem bounds $\lambda_2(\Sigma(t))$ by the largest Rayleigh quotient on $v^\perp$. There the leading terms vanish, and so do the cross terms, which have the form $\rho^n(va_n^\top + a_n v^\top)$ with $a_n = R_nQ^\zeta u$ and its analog for $\Sigma(0)$, since $x^\top v = 0$. What remains is bounded by $\sum_n \|R_n\|^2\|Q^\zeta\| + \|R_t\|^2\|\Sigma(0)\|$, which is bounded in $t$ when $r < 1$. So $\lambda_2(\Sigma(t))$ is bounded, and every eigenvalue of $\Sigma(t)^{-1}$ but the smallest is bounded away from zero. □

*Proof of Proposition S8.7.* (a) Away from the origin the eigenvalue equation is $\rho v_x = cv_x + a(v_{x+1}+v_{x-1})$, whose square-summable solutions are multiples of $e^{-m|x|}$ with $c+2a\cosh m = \rho$; at the origin, $\rho = c + b_0 + 2ae^{-m}$. Hence $2a\sinh m = b_0$ and $\rho = c + 2a\cosh(\operatorname{arcsinh}(b_0/(2a))) = c + \sqrt{4a^2+b_0^2}$. The operator $W$ is self-adjoint, $W \succeq (c-2a)I$ since $b_0 > 0$, and every two-dimensional subspace contains a unit vector with vanishing component at the origin, on which the Rayleigh quotient is that of the medium, at most $c+2a$. By the min-max principle at most one eigenvalue exceeds $c+2a$, and the spectrum of $W$ on $v^\perp$ lies in $[c-2a, c+2a]$.

(b) Write $W = \rho P_v + W_c$ with $W_c = (I-P_v)W(I-P_v)$, which commutes with $P_v$ and has norm at most $\max(|c-2a|, c+2a) = \lambda$, since $c \ge 0$. Then $W^{2n} = \rho^{2n}P_v + W_c^{2n}$, with the convention $W_c^0 = I - P_v$, and from $\Sigma(0) = 0$, $\Sigma(t) = q\sum_{n<t} W^{2n} = A(t)P_v + B(t)$ with $B(t) = q\sum_{n<t} W_c^{2n}$, so $q(I-P_v) \preceq B(t) \preceq q(1-\lambda^2)^{-1}(I-P_v)$. Also $|v|^2 = \sum_x e^{-2m_\rho|x|} = \coth m_\rho$.

(c) Since $W \ge 0$ entrywise, $\Sigma(t) \ge 0$ entrywise, so the lower bound holds trivially when $\mathfrak{s}_x \le 1$. Put $\beta = q/(1-\lambda^2)$, $a_1 = Av_x^2/|v|^2$ and $a_0 = A/|v|^2$, where $v_0 = 1 \ge v_x$. Since $|B_{xy}| \le \beta$, $\Sigma_{x0} \ge \sqrt{a_1a_0} - \beta$, $\Sigma_{xx} \le a_1 + \beta$ and $\Sigma_{00} \le a_0 + \beta$. For $\mathfrak{s}_x > 1$ the numerator is positive, the function $a_0 \mapsto (\sqrt{a_1a_0} - \beta)/\sqrt{(a_1+\beta)(a_0+\beta)}$ is increasing (its derivative in $\sqrt{a_0}$ has numerator $\beta(\sqrt{a_1}+\sqrt{a_0}) > 0$), and at $a_0 = a_1$ it equals $(a_1-\beta)/(a_1+\beta) = (\mathfrak{s}_x-1)/(\mathfrak{s}_x+1)$, which gives the lower bound. For the upper bound use $\Sigma_{xx} \ge q$, from the fresh noise, and $\Sigma_{00} \ge a_0$, from $B(t) \succeq 0$: Corr $\le (\sqrt{a_1a_0}+\beta)/\sqrt{qa_0}$, which is the right side of (S33).

(d) For $x \ge 0$, $\mathfrak{s}_x(t) = (1-\lambda^2)(\rho^{2t}-1)e^{-2m_\rho x}/((\rho^2-1)\coth m_\rho)$. The lower bound gives correlation at least $\theta$ when $\mathfrak{s}_x \ge (1+\theta)/(1-\theta)$, which, since $\rho^{2t}-1 \ge (1-\rho^{-2})\rho^{2t}$ for $t \ge 1$, holds for $x \le (t\ln\rho - c_1(\theta))/m_\rho$ with a constant $c_1(\theta)$. The upper bound gives correlation below $\theta$ when both of its terms are below $\theta/2$; the second is for $t \ge t_0(\theta)$ because $A(t) \to \infty$, and the first is when $\mathfrak{s}_x < \theta^2(1-\lambda^2)/4$, which holds for $x > (t\ln\rho + c_2(\theta))/m_\rho$ with a constant $c_2(\theta)$. Hence $x_\theta(t) = t\ln\rho/m_\rho + O_\theta(1)$. Finally $\rho = c + 2a\cosh m_\rho \le (c+2a)\cosh m_\rho < \cosh m_\rho < e^{m_\rho}$, so $\ln\rho < m_\rho$.

(e) Outside the core the eigenvector solves the medium's recurrence $\rho v_x = cv_x + a(v_{x+1}+v_{x-1})$, whose solutions are combinations of $e^{\pm m_\rho x}$ with $c + 2a\cosh m_\rho = \rho$; square summability selects the decaying one on each side. As $\rho \downarrow 1$, $(\rho - c)/(2a)$ decreases to $(1-c)/(2a) = 1 + \varepsilon/(2a)$, so $m_\rho$ decreases to the rate of Theorem 8.10(e). □

*Proof of Theorem 8.9.* (a) The support statement is Theorem 8.5(a) for linear maps, and the row-

sum bound is Lemma 8.2(a). (b) (M) is Lemma 8.2(a). Block $(i,j)$ of the $n$-th term of the series is $\sum_k \Phi_{ik} Q^\zeta_k \Phi^\top_{jk}$ with $\Phi = \Phi(t, t-n)$. It vanishes unless $d(i,k) \le n$ and $d(j,k) \le n$ for some $k$, which forces $n \ge \lceil d(i,j)/2 \rceil$, and its norm is at most $q \sum_k \|\Phi_{ik}\| \|\Phi_{jk}\| \le q(\max_k \|\Phi_{jk}\|) \sum_k \|\Phi_{ik}\| \le q\lambda^{2n}$. Summing the geometric series from $\lceil d/2 \rceil$ gives (80), with $\lambda^{2\lceil d/2 \rceil} \le \lambda^d = e^{-m_\Sigma d}$, and absolute convergence. Every term of the series is positive semidefinite and the term $n = 0$ is $Q^\zeta(t-1)$, so $\Sigma(t) \succeq Q^\zeta(t-1)$; under (B) this gives $\lambda_{\min}(\Sigma(t)) \ge q_{\min}$, $\Sigma_{ii} \succeq q_{\min} I$, hence $\|R_{ij}\| \le \|\Sigma_{ij}\|/q_{\min}$ and $\xi \le 1/\ln(1/\lambda)$. (c) The initial covariance is one more injection, $n = t - s$ steps back, with $q_0$ in place of $q$. (d) Later noise is independent of $e(t)$, so $\mathrm{Cov}(e(t+h), e(t)) = \Phi(t+h, t)\Sigma(t)$; block $(i,j)$ is at most $\sum_k \|\Phi_{ik}\| \|\Sigma_{kj}\|$ over $k$ with $d(i,k) \le h$, hence $d(k,j) \ge d(i,j) - h$, and the sum is at most $\lambda^h \cdot q\lambda^{(d-h)_+}/(1-\lambda^2) = q\lambda^{\max(h,d)}/(1-\lambda^2)$. □

*Proof of Proposition S8.9.* (a) Block $(i,j)$ of the $n$-th term is $\sum_{k,l} \Phi_{ik} Q^\zeta_{kl} \Phi^\top_{jl}$, with $d(i,k) \le n$ and $d(j,l) \le n$ on the support, so $d(k,l) \ge d - 2n$ and $\|Q^\zeta_{kl}\| \le q e^{-(d-2n)_+/\xi_w}$. Since $\sum_{k,l} \|\Phi_{ik}\| \|\Phi_{jl}\| \le \lambda^{2n}$, the term is at most $q\lambda^{2n} e^{-(d-2n)_+/\xi_w}$. Each of the at most $\lceil d/2 \rceil$ terms with $2n < d$ is at most $q \max(\lambda, e^{-1/\xi_w})^d$, and the terms with $2n \ge d$ sum to at most $q\lambda^d/(1-\lambda^2)$. (b) Without messages $W = \lambda I$ with $\lambda = F(1-k^s)$, so the connected part has the diagonal covariance $qI/(1-\lambda^2)$, and by Lemma 2.13(d) the common mode adds $Q\mathbf{1}\mathbf{1}^\top/(1-\lambda^2)$, independent of the connected part. Every error then has variance $(Q+q)/(1-\lambda^2)$, and any two distinct errors have covariance $Q/(1-\lambda^2)$. □

*Proof of Proposition S8.10.* The stationary covariance is $\sum_n W^n Q^\zeta W^{n\top}$. Block $(i,j)$ of the $n$-th term vanishes for $n < \lceil d(i,j)/2 \rceil$, as in the proof of Theorem 8.9, and its norm is at most $\|W^n Q^\zeta W^{n\top}\| \le q\|W^n\|^2 \le q K_A^2 f^{2n}$; sum the series. For normal $W$, $\|W^n\| = \rho(W)^n$. For every $f > \rho(W)$ there is $K_A$ with $\|W^n\| \le K_A f^n$ (Gelfand's formula), so with $\|R_{ij}\| \le \|\Sigma_{ij}\|/q_{\min}$ the decay rate of correlations is at least $\ln(1/f)$ for every $f > \rho(W)$, which gives $\xi \le \tau$. □

*Proof of Theorem S8.11.* We first prove Carne's bound. For $w \in \mathbb{R}$ and $z = \cos w$,

$$z^n = \mathbb{E}\cos(S_n w) = \sum_{k=-n}^{n} \Pr(S_n = k) T_{|k|}(z),$$

where $S_n$ is the simple random walk on $\mathbb{Z}$ and $T_k$ is the Chebyshev polynomial with $T_k(\cos w) = \cos(kw)$. Reversibility makes $P$ self-adjoint on $L^2(\varpi)$ with spectrum in $[-1,1]$, so $P^n = \sum_k \Pr(S_n = k) T_{|k|}(P)$ with $\|T_k(P)\|_{L^2(\varpi)} \le 1$, and $\langle \mathbf{1}_x, T_k(P)\mathbf{1}_y \rangle_\varpi = 0$ for $k < d(x,y)$ because $T_k$ has degree $k$ and $(P^l)_{xy} = 0$ for $l < d(x,y)$. Hence $\varpi_x P^n_{xy} = \langle \mathbf{1}_x, P^n \mathbf{1}_y \rangle_\varpi \le \Pr(|S_n| \ge d(x,y)) \sqrt{\varpi_x \varpi_y}$. By Markov's inequality applied to $e^{sS_n}$, with $\mathbb{E}e^{sS_n} = \cosh^n s \le e^{ns^2/2}$ and $s = d/n$, $\Pr(|S_n| \ge d) \le 2e^{-d^2/(2n)}$, so $P^n_{xy} \le 2\sqrt{\varpi_y/\varpi_x} e^{-d(x,y)^2/(2n)}$.

Clustering. By induction, $\|\Phi_{ik}(t, t-n)\| \le \lambda^n (P^n)_{ik}$, so the series of Theorem 8.9(b) for the process started in the infinite past gives $\|\Sigma_{ij}(t)\| \le q \sum_n \lambda^{2n} \sum_k P^n_{ik} P^n_{jk}$; from a zero start the sum over $n$ is finite and the same bound holds. Reversibility gives $P^n_{jk} = \varpi_k P^n_{kj}/\varpi_j$, so $\sum_k P^n_{ik} P^n_{jk} \le (\varpi_{\max}/\varpi_j) P^{2n}_{ij} \le 2\varpi_{\max}(\varpi_i \varpi_j)^{-1/2} e^{-d^2/(4n)}$. With $\varepsilon_\lambda = \ln(1/\lambda)$, the exponent $h(n) = 2n\varepsilon_\lambda + d^2/(4n)$ satisfies $h(n) \ge d\sqrt{2\varepsilon_\lambda}$ by the inequality between arithmetic and geometric means, hence $h(n) \ge \vartheta d\sqrt{2\varepsilon_\lambda} + (1-\vartheta) 2n\varepsilon_\lambda$, and the geometric sum over $n$ gives (S34). □

*Proof of Proposition S8.12.* (a) Let $(W_{\boldsymbol{\varphi}})_{ij} = |W_{ij}| e^{\boldsymbol{\varphi} \cdot (x_i - x_j)}$. The moduli of the entries of a product are at most the entries of the product of the moduli, and the tilts telescope along products, so the tilted moduli $|\Phi_{ik}(t, t-n)| e^{\boldsymbol{\varphi} \cdot (x_i - x_k)}$ are at most the entries of a product of $n$ matrices $W_{\boldsymbol{\varphi}}(s)$, whose row sums are at most $\bar\psi(\boldsymbol{\varphi})^n$. For scalar agents,

$$|\Sigma_{ij}| e^{\boldsymbol{\varphi} \cdot (x_i - x_j)} \le q_{\max} \sum_n \sum_k |\Phi_{ik}| e^{\boldsymbol{\varphi} \cdot (x_i - x_k)} |\Phi_{jk}| e^{-\boldsymbol{\varphi} \cdot (x_j - x_k)} \le q_{\max} \sum_n \bar\psi(\boldsymbol{\varphi})^n \bar\psi(-\boldsymbol{\varphi})^n,$$

using that the largest entry of a nonnegative row is at most its sum; the same holds for any finite sum, hence from a zero start. The telescoping uses single-valued positions, which $\mathbb{Z}^D$ and a box with open ends have, and the row sums are taken over all agents, boundary agents included. On a periodic box of side $L$ let $\tilde{W}(t)$ be the $L$-periodic medium on $\mathbb{Z}^D$ with $\tilde{W}_{\tilde{\imath}\tilde{\jmath}} = W_{ij}$ for neighboring lifts $\tilde{\imath}, \tilde{\jmath}$ of $i, j$, driven by independent noises with the periodically extended variances, and let $\tilde{\Phi}$ and $\tilde{\Sigma}$ be its propagator and covariance. Every walk on the box from $i$ to $k$ lifts uniquely to a walk from a fixed lift $\tilde{\imath}$ to one lift of $k$, so $\Phi_{ik} = \sum_{\mathbf{m}\in\mathbb{Z}^D} \tilde{\Phi}_{\tilde{\imath},\tilde{k}+L\mathbf{m}}$, and by periodicity $\tilde{\Phi}_{\tilde{\jmath},\tilde{k}+L\mathbf{m}'} = \tilde{\Phi}_{\tilde{\jmath}+L(\mathbf{m}-\mathbf{m}'),\tilde{k}+L\mathbf{m}}$. Hence

$$\Sigma_{ij} = \sum_n \sum_k Q^\zeta_k \Phi_{ik} \Phi_{jk} = \sum_{\mathbf{w}\in\mathbb{Z}^D} \sum_n \sum_{\tilde{k}} Q^\zeta_{\tilde{k}} \tilde{\Phi}_{\tilde{\imath}\tilde{k}} \tilde{\Phi}_{\tilde{\jmath}+L\mathbf{w},\tilde{k}} = \sum_{\mathbf{w}\in\mathbb{Z}^D} \tilde{\Sigma}_{\tilde{\imath},\tilde{\jmath}+L\mathbf{w}},$$

with $\tilde{\jmath}+L\mathbf{w}$ the lift of $j$ displaced by $L\mathbf{w}$. The rearrangement is justified by absolute convergence: $\sum_{\mathbf{w}} |\tilde{\Phi}_{\tilde{\jmath}+L\mathbf{w},\tilde{k}}| = \sum_{\mathbf{w}} |\tilde{\Phi}_{\tilde{\jmath},\tilde{k}-L\mathbf{w}}|$ is part of a row sum of the product of the moduli, hence at most $\bar{\psi}(0)^n$, and $\bar{\psi}(0)^2 \le \Psi(\boldsymbol{\varphi}) < 1$ by the Cauchy–Schwarz inequality applied to each row. Each term obeys the bound for $\tilde{W}$ with any admissible $\boldsymbol{\varphi}$. On $\mathbb{Z}^D$, for $W(t) \ge 0$ with equal weights $a_{id}$ to the two neighbors along axis $d$, $\sum_j W_{ij} e^{\boldsymbol{\varphi}\cdot(x_i - x_j)} = W_{ii} + \sum_d 2a_{id} \cosh\varphi_d = 1 - \varepsilon_i + 2\sum_d a_{id}(\cosh\varphi_d - 1)$, which is at most $1 - \varepsilon_{\min} + 2a_{\max}\sum_d(\cosh\varphi_d - 1)$ and is even in $\boldsymbol{\varphi}$. Along an axis, $\boldsymbol{\varphi} = m\mathbf{u}_d$ gives $\Psi < 1$ when $\cosh m < 1 + \varepsilon_{\min}/(2a_{\max})$; along the diagonal, $\varphi_d = s$ for all $d$ gives $\boldsymbol{\varphi}\cdot\mathbf{u} = \sqrt{D}s$ and $\Psi < 1$ when $\cosh s < 1 + \varepsilon_{\min}/(2Da_{\max})$.

(b) Products of nonnegative matrices are monotone in each factor, so $\Phi(t, t-n) \ge W_{\text{low}}^n$ entrywise, and with the diagonal $Q^\zeta \ge q_{\min} I$, $\Sigma \ge q_{\min}\sum_n W_{\text{low}}^{2n} = q_{\min}(I - W_{\text{low}}^2)^{-1}$. Each row of $W(t)$ dominates the corresponding row of $W_{\text{low}}$, so $1 - \varepsilon_i \ge c_{\min} + 2Da_{\min} = 1 - \varepsilon_{\text{low}}$ for an agent with $2D$ neighbors. A boundary agent of an open box with $n_i < 2D$ neighbors only has $1 - \varepsilon_i \ge c_{\min} + n_i a_{\min}$; for $W = W_{\text{low}}$ on an open chain the end leak is $1 - c_{\min} - a_{\min} > \varepsilon_{\text{low}}$.

(c) On $\mathbb{Z}$ the hypotheses of the second part of (a) and of (b) hold. By (a) at $\boldsymbol{\varphi} = 0$, $\Sigma_{ii} \le q_{\max}/(1 - \bar{\psi}(0)^2)$, and $\Sigma_{ii} \ge q_{\min}$. So $R_{ij} \le |\Sigma_{ij}|/q_{\min}$ decays at least at the rate $m_{\text{up}}$, and $R_{ij} \ge q_{\min}(1 - \bar{\psi}(0)^2)(I - W_{\text{low}}^2)^{-1}_{ij}/q_{\max}$ decays at most at the rate $m_{\text{low}}$ of the homogeneous medium, which in one dimension is exact by Theorem 8.10(e).

(d) For a nonnegative matrix the spectral radius lies between the smallest and the largest row sum (Perron–Frobenius).

(e) The definitions give $W_{ij} \le \lambda' P_{ij}$ for all $i$, $j$, with $P$ symmetric and stochastic and $P_{ii} = 1 - \sum_{j\ne i} P_{ij} \ge W_{ii}/\lambda' \ge 0$. A symmetric chain is reversible with respect to the uniform measure, which is the hypothesis of Theorem S8.11. □

*Proof of Theorem 8.10.* (a) $W$ is symmetric with $\|W\| = \sup_p |\hat{W}(p)| = \lambda < 1$, so $\Sigma = q\sum_n W^{2n} = q(I - W^2)^{-1}$, and $(I - W^2)^{-1} = \frac{1}{2}[(I - W)^{-1} + (I + W)^{-1}]$.

(b) Since $c, a \ge 0$ the kernel is nonnegative and $\sum_x W^n(x) e^{\boldsymbol{\varphi}\cdot x} = \psi(\boldsymbol{\varphi})^n$. By Tonelli's theorem $\sum_x \Sigma(x) e^{\boldsymbol{\varphi}\cdot x} = q\sum_n \psi(\boldsymbol{\varphi})^{2n} = q/(1 - \psi(\boldsymbol{\varphi})^2)$, and each nonnegative term is at most the sum. For $\boldsymbol{\varphi}$ in the interior of $\{\psi \le 1\}$, $-\ln\Sigma(x) \ge \boldsymbol{\varphi}\cdot x - \ln(q/(1 - \psi^2))$; take $\boldsymbol{\varphi}$ close to the maximizer of $\boldsymbol{\varphi}\cdot\mathbf{u}$.

(c) Summing a convolution power over the transverse coordinates gives the convolution power of the summed kernel, in which transverse moves become stays: $\sum_y W^n(t, y) = W_1^n(t)$, where $W_1$ is the one-dimensional kernel with self-weight $c' = c + 2(D-1)a$ and neighbor weight $a$, with the same $\lambda$. So the hyperplane sum is the one-dimensional covariance of (e) with $c'$, which decays at the rate $m_1$ with $\cosh m_1 = (1 - c')/(2a) = 1 + \varepsilon/(2a)$; its staggered part decays faster when $c' > 0$ and vanishes on odd $t$ when $c' = 0$, so in all cases $-\limsup t^{-1}\ln\sum_y \Sigma(t, y) = m_1$. For the support function along $\mathbf{u}_1$, the constraint $c + 2a\sum_d \cosh\varphi_d \le 1$ is least restrictive for $\varphi_1$ when $\varphi_d = 0$ for $d \ge 2$, which gives $\cosh\varphi_1 \le 1 + \varepsilon/(2a)$. Since $\cosh m - 1 = 2\sinh^2(m/2)$, $m_1 = 2\,\mathrm{arcsinh}\,\sqrt{\varepsilon/(4a)}$, and $\operatorname{arccosh}(1 + y) = \sqrt{2y}(1 - y/12 + O(y^2))$ with $y = \varepsilon/(2a)$ gives the expansion.

(d) The set $\{\psi \le 1\}$ is convex and invariant under permutations of the coordinates, and so is the objective $\boldsymbol{\varphi}\cdot\mathbf{u}$ for the diagonal $\mathbf{u}$; averaging a maximizer over permutations gives a symmetric maximizer, $\varphi_d = s$ with $c + 2aD\cosh s = 1$, that is $\cosh s = 1 + \varepsilon/(2Da)$, and $\boldsymbol{\varphi}\cdot\mathbf{u} = \sqrt{D}s$. The expansion of (c) with $y = \varepsilon/(2Da)$ gives $m(\mathbf{u}) = \sqrt{\varepsilon/a}(1 - \varepsilon/(24Da) + O(\varepsilon^2))$, and the anisotropy follows by subtraction.

(e) In one dimension $1 - \hat W(p) = 2a(\cosh m - \cos p)$ and $1 + \hat W(p) = 2a(\cosh m' + \cos p)$, with $\cosh m = (1-c)/(2a) = 1 + \varepsilon/(2a) > 1$ and $\cosh m' = (1+c)/(2a) > 1$ because $2a < 1 - c \le 1 + c$. The residue computation $\int_{-\pi}^{\pi} e^{ipx}(\cosh m - \cos p)^{-1}\mathrm{d}p/2\pi = e^{-m|x|}/\sinh m$, and the shift $p \to p + \pi$ for the second resolvent, give (82) through (a). On a ring of even length $N$ the covariance is the periodization $\sum_{j\in\mathbb{Z}} \Sigma(x + jN)$, and $\sum_j e^{-m|x+jN|} = \cosh(m(N/2 - |x|))/\sinh(mN/2)$ for $|x| \le N/2$, with $(-1)^{x+jN} = (-1)^x$.

(f) $\Sigma(0) = \frac{q}{2}\int[(1-\hat W)^{-1} + (1+\hat W)^{-1}]\mathrm{d}p/(2\pi)^D$. With $c > 0$, $1 + \hat W \ge 1 + c - 2Da = 2c + \varepsilon$, so the second integral is bounded. Write $1 - \hat W(p) = \varepsilon + ag(p)$ with $g(p) = 2\sum_d(1 - \cos p_d) = |p|^2(1 + O(|p|^2))$, and $g$ bounded below away from the origin. For $D = 1$ the claim follows from (e), since $\sinh m \sim \sqrt{\varepsilon/a}$ while $m'$ stays bounded away from zero. For $D = 2$, the difference between $\int_{|p|\le 1}(\varepsilon + ag)^{-1}$ and $\int_{|p|\le 1}(\varepsilon + a|p|^2)^{-1} = (\pi/a)\ln((\varepsilon + a)/\varepsilon)$ is bounded uniformly in $\varepsilon$, as is the integral over $|p| > 1$; dividing by $(2\pi)^2$ and multiplying by $q/2$ gives $(q/8\pi a)\ln(1/\varepsilon) + O(1)$. For $D \ge 3$, $(ag)^{-1}$ is integrable and monotone convergence gives a finite limit.

(g) By (b), $\Sigma$ decays exponentially, so $\hat\Sigma$ is smooth and $\sum_x |x|^2\Sigma(x) = -\Delta_p\hat\Sigma(0)$. From $\hat W(p) = \lambda - a|p|^2 + O(|p|^4)$, $\hat\Sigma(p) = q/(1-\lambda^2) - 2aq\lambda|p|^2/(1-\lambda^2)^2 + O(|p|^4)$, so $-\Delta_p\hat\Sigma(0) = 4Dqa\lambda/(1-\lambda^2)^2$, and $\sum_x \Sigma(x) = \hat\Sigma(0) = q/(1-\lambda^2)$. □

*Proof of Proposition S8.27.* (a) For a translation-invariant scalar field, $\hat\Sigma = 1/\hat\Pi$. By symmetry $\partial_d\hat\Pi(0) = 0$, and $\partial_d^2\hat\Pi(0) = \varsigma_d^2$, $\hat\Pi(0) = \varepsilon^\Pi$, so $\partial_d^2\hat\Sigma(0) = -\varsigma_d^2/(\varepsilon^\Pi)^2$. Hence $\sum_x x_d^2\Sigma(x) = \varsigma_d^2/(\varepsilon^\Pi)^2$ and $\sum_x \Sigma(x) = 1/\varepsilon^\Pi$, and the ratio is $2\xi_{2,d}^2$. The walk expansion $\Sigma = \sum_m \mathcal{A}^m/(\varepsilon^\Pi + A)^{m+1}$, with $\mathcal{A}$ the convolution by $a^\Pi$ and $A = \sum_x a^\Pi(x)$, shows that $\Sigma \ge 0$ has finite second moments, which justifies the differentiation.

(b) By Tonelli's theorem and the walk expansion, $\sum_x e^{\vartheta x_d}\Sigma(x) = (\varepsilon^\Pi - \sum_x a^\Pi(x)(\cosh(\vartheta x_d) - 1))^{-1}$ when the bracket is positive and $+\infty$ otherwise. The left side is a two-sided series in $e^\vartheta$ with nonnegative coefficients $g(t) = \sum_{x:x_d=t}\Sigma(x) = g(-t)$, finite for $|\vartheta| < m_d$ and tending to $+\infty$ as $\vartheta \uparrow m_d$, where $m_d$ is the positive root (it exists because the left side of the root equation is continuous, increasing, zero at zero and unbounded for couplings of finite range with $\varsigma_d > 0$). So the power series $\sum_{t\ge 0} g(t)z^t$ converges for $0 < z < e^{m_d}$ and diverges at $z = e^{m_d}$ by monotone convergence; its radius of convergence is $e^{m_d}$, and the Cauchy–Hadamard formula gives $\limsup g(t)^{1/t} = e^{-m_d}$, the rate of Definition 8.8(b). Comparing Taylor coefficients, $y^2/2 \le \cosh y - 1 \le (y^2/2)\cosh y$, so at $m = m_d$, $m^2\varsigma_d^2/2 \le \varepsilon^\Pi \le (m^2\varsigma_d^2/2)\cosh(\ell_d^\Pi m)$. The first inequality is $m_d \le 1/\xi_{2,d}$, and the second, with $\cosh(\ell_d^\Pi m_d) \le \cosh(\ell_d^\Pi/\xi_{2,d})$, gives $m_d \ge \xi_{2,d}^{-1}\cosh(\ell_d^\Pi/\xi_{2,d})^{-1/2}$.

(c) For nearest-neighbor couplings $\varsigma_d^2 = 2a^\Pi$, and the equation of (b) reads $2a^\Pi(\cosh m - 1) = \varepsilon^\Pi = 4a^\Pi\sinh^2(m/2)$, so $\xi_{2,d}^2 = a^\Pi/\varepsilon^\Pi = (2\sinh(m/2))^{-2}$; the expansion $2\sinh(m/2) = m(1 + m^2/24 + O(m^4))$ gives the difference. For the medium of Theorem 8.10, the world-conditional precision is $(I - W^2)/q$ with couplings $2ca/q$ at axis distance one, $a^2/q$ at axis distance two and $2a^2/q$ at the diagonal neighbors (Theorem 4.9), so $\varsigma_d^2 = (4ca + 8a^2 + 8(D-1)a^2)/q = 4a\lambda/q$ and $\varepsilon^\Pi = (1-\lambda^2)/q$, which gives $\xi_2^2 = 2a\lambda/(1-\lambda^2)$, in agreement with Theorem 8.10(g). The hyperplane rate is $m$ with $\cosh m = 1 + \varepsilon/(2a)$ by Theorem 8.10(c), and $\varepsilon = 2a(\cosh m - 1) = 4a\sinh^2(m/2)$, so $a/\varepsilon = (2\sinh(m/2))^{-2}$ and $\xi_2^2 = (2\lambda/(1+\lambda))(a/\varepsilon)$. The inequalities follow from $2\lambda/(1+\lambda) \le 1$ and $2\sinh(m/2) \ge m$. □

*Proof of Proposition S8.14.* $\mathrm{Cov}(e(t+h), e(t)) = W^h\Sigma$, so with $u = \Sigma^{1/2}\mathbf{v}$ and $u' = \Sigma^{1/2}\mathbf{v}'$ the

correlation is $u'^{\top}\tilde{W}^h u/(|u'||u|)$, whose supremum over $u$ and $u'$ is $\|\tilde{W}^h\|$; this supremum is attained, so $T^*(\theta) = \max\{h : \|\tilde{W}^h\| \geq \theta\}$. Also $\tilde{W}\tilde{W}^{\top} = \Sigma^{-1/2} W \Sigma W^{\top} \Sigma^{-1/2} = I - \Sigma^{-1/2} Q^{\zeta} \Sigma^{-1/2}$, which gives $\|\tilde{W}\|^2$. $\tilde{W}$ is similar to $W$, so $\rho(\tilde{W}) = \rho(W)$, and Gelfand's formula gives $\|\tilde{W}^h\|^{1/h} \to \rho(W)$. Hence $\rho(W)^h \leq \|\tilde{W}^h\| \leq C_\delta(\rho(W)+\delta)^h$ for every $\delta > 0$, so $T^*(\theta) \geq \ln(1/\theta)/\ln(1/\rho(W)) - 1$ and $T^*(\theta) \leq (\ln(1/\theta) + \ln C_\delta)/\ln(1/(\rho(W)+\delta))$. Dividing by $\ln(1/\theta)$ and letting $\theta \to 0$ and then $\delta \to 0$ gives the limit $\tau$; if $\rho(W) = 0$, $\tilde{W}$ is nilpotent, $T^*$ stays bounded and the limit is $0 = \tau$. □

*Proof of Proposition S8.15.* (a) $|\hat{W}(p)| \leq c + 2Da = \lambda$ with equality at $p = 0$, so the spectral radius of the convolution is $\lambda$. With Theorem 8.10(g), $\xi_2^2/\tau = 2a\lambda \ln(1/\lambda)/(1-\lambda^2)$, and $\ln(1/\lambda) = \varepsilon + O(\varepsilon^2)$, $1 - \lambda^2 = 2\varepsilon + O(\varepsilon^2)$. (b) $\mathrm{Cov}(e(t+h), e(t)) = W^h \Sigma$ gives (S37). With $c > 0$, $|\hat{W}(p)| = \lambda$ only at $p = 0$, because $\hat{W}(p) = -\lambda$ would need $c + 2a\sum_d \cos p_d = -c - 2Da$, impossible for $c > 0$; by continuity and compactness $|\hat{W}| \leq \lambda - \eta_0$ outside any neighborhood of the origin, for some $\eta_0 > 0$, and that region contributes $O((\lambda - \eta_0)^h)$. Near the origin $\hat{W}(p)^h = \lambda^h e^{-ha|p|^2/\lambda}(1 + O(h|p|^4))$ and $(1 - \hat{W}^2)^{-1} = (1-\lambda^2)^{-1} + O(|p|^2)$, and Laplace's method gives $\int e^{-ha|p|^2/\lambda} \mathrm{d}p/(2\pi)^D = (\lambda/4\pi a h)^{D/2}$ with relative error $O(1/h)$. The $O(|p|^2)$ term of $(1-\hat{W}^2)^{-1}$ is $-2\lambda a|p|^2/(1-\lambda^2)^2$, and the Gaussian weight has $\langle |p|^2 \rangle = D\lambda/(2ah)$, so this term contributes the relative correction $-D\lambda^2/((1-\lambda^2)h)$, about $-D/(2\varepsilon h)$ for small $\varepsilon$; the remaining $O(1/h)$ corrections have coefficients bounded as $\varepsilon \to 0$ at fixed $c$, so (S38) is accurate once $\varepsilon h \gg 1$. □

*Proof of Proposition 8.11.* (a) With $\mathcal{T} = \mathrm{diag}(1, (\beta_1/\omega_1)^{1/2}, \dots)$, $\mathcal{T}\mathsf{R}\mathcal{T}^{-1}$ is a symmetric arrowhead matrix with off-diagonal entries $-(\beta_k\omega_k)^{1/2}$, so the spectrum is real. (b) For $x \notin \{\omega_k + \delta_k\}$, the Schur complement of the diagonal record block gives

$$\det(\mathsf{R} - xI) = \prod_k (\omega_k + \delta_k - x)\Big[\varepsilon' + \sum_k \beta_k - x - \sum_k \frac{\beta_k \omega_k}{\omega_k + \delta_k - x}\Big] = g_z(x) \prod_k (\omega_k + \delta_k - x),$$

so the eigenvalues below $d_{\min} = \min_k(\omega_k + \delta_k)$ are the roots of $g_z$ there. Since $g_z'(x) = -1 - \sum_k \beta_k\omega_k/(\omega_k + \delta_k - x)^2 < 0$, $g_z(x) \to +\infty$ as $x \to -\infty$ and $g_z(x) \to -\infty$ as $x \uparrow d_{\min}$, there is exactly one root $x_0$ below $d_{\min}$, and every other eigenvalue is at least $d_{\min}$; so $x_0 = \lambda_{\min}(\mathsf{R})$. Since $\omega_k + \delta_k > 0$ for every $k$, the point 0 lies below $d_{\min}$, where $g_z$ decreases; hence $x_0 \geq 0$ if and only if $g_z(0) = \varepsilon' + \sum_k \beta_k\delta_k/(\omega_k + \delta_k) \geq 0$. (c) By the Perron–Frobenius theorem $\rho(I - \mathsf{R})$ is an eigenvalue of $I - \mathsf{R}$, and it is at least every real eigenvalue; as all eigenvalues are real, it is the largest, $1 - x_0$. (d) For $A \geq 0$ and a positive vector $y$, $\min_i (Ay)_i/y_i \leq \rho(A) \leq \max_i (Ay)_i/y_i$ (Collatz–Wielandt), so a positive eigenvector belongs to the spectral radius. Since the joint matrix is nonnegative, every agent's self-weight is at least $\sum_k \beta_k$ and $1 - \omega_k - \delta_k \geq 0$, so the reduced matrix $I - \mathsf{R}$ with $\upsilon_k = 0$, whose first diagonal entry $1 - \varepsilon - \sum_k \beta_k$ is at least an agent's self-weight minus $\sum_k \beta_k$, is nonnegative; it is irreducible, with a positive Perron vector $(y_0, y_1, \dots, y_{n_z})$. The vector $(y_0\mathbf{1}, y_1, \dots, y_{n_z})$ is an eigenvector of the joint matrix with eigenvalue $1 - x_0$: an agent's row gives $(1-\varepsilon)y_0 - \sum_k \beta_k y_0 + \sum_k \beta_k y_k$, using $W_0\mathbf{1} = (1-\varepsilon)\mathbf{1}$, and a record's row gives $\sum_i \omega_{k,i} y_0 + (1 - \omega_k - \delta_k)y_k$, using $\sum_i \omega_{k,i} = \omega_k$; both equal the corresponding rows of $(I - \mathsf{R})y$. It is positive, so $1 - x_0$ is the spectral radius. □

*Proof of Proposition 8.12.* (a) Put $m_- = \min\{\varepsilon, \delta_k\}$ and $m_+ = \max\{\varepsilon, \delta_k\}$, with $\varepsilon' = \varepsilon$. Every term of $g_z(m_-) = \varepsilon - m_- + \sum_k \beta_k(\delta_k - m_-)/(\omega_k + \delta_k - m_-)$ is nonnegative, so $x_0 \geq m_-$. If $m_+ < d_{\min}$, every term of $g_z(m_+)$ is nonpositive and $x_0 \leq m_+$; otherwise $x_0 < d_{\min} \leq m_+$. For one record with $\delta < \varepsilon$, $g_z(\delta) = \varepsilon - \delta > 0$, and either $g_z(\varepsilon) = \beta(\delta - \varepsilon)/(\omega + \delta - \varepsilon) < 0$ or $\varepsilon \geq \omega + \delta > x_0$; so $\delta < x_0 < \varepsilon$, and the case $\delta > \varepsilon$ is symmetric. For $\varepsilon < d_{\min}$, $x_0 < \varepsilon$ if and only if $g_z(\varepsilon) < 0$, and $g_z(\varepsilon)$ is the sum in (86); for $\varepsilon \geq d_{\min}$, $x_0 < d_{\min} \leq \varepsilon$. The same signs show that the inequalities are strict when every $\delta_k < \varepsilon$ or every $\delta_k > \varepsilon$. When $I - \mathsf{R} \geq 0$, $\tau = -1/\ln(1 - x_0)$ by Proposition 8.11(c), and $x_0 \geq m_-$ gives the cap on $\tau$.

(b) At $\delta = 0$ the root satisfies $x_0 = \varepsilon(\omega - x_0)/(\omega - x_0 + \beta)$. Since $u \mapsto u/(u+\beta)$ is increasing and $\omega - x_0 \le \omega$, $x_0 \le \varepsilon\omega/(\omega+\beta) = \varepsilon/(1+\mathcal{R})$; and $x_0 \ge \varepsilon(\omega - x_0)/(\omega+\beta)$ rearranges to $x_0 \ge \varepsilon\omega/(\omega+\beta+\varepsilon)$. Since $\partial g_z/\partial\delta = \beta\omega/(\omega+\delta-x)^2 > 0$ and $g_z$ decreases in $x$, the root increases with $\delta$.

(c) With $\varepsilon' = \varepsilon - \upsilon\beta$ and $\varepsilon < \omega + \delta$, $g_z(\varepsilon) = \beta[(\delta-\varepsilon)/(\omega+\delta-\varepsilon) - \upsilon]$, which is negative if and only if $(1-\upsilon)(\delta-\varepsilon) < \upsilon\omega$. If $\varepsilon \ge \omega + \delta$, then $\delta < \varepsilon$ and $x_0 < \omega + \delta \le \varepsilon$, consistent with the criterion. Several records are treated in the same way.

(d) If $0 \le A \le B$ entrywise, then $A^n \le B^n$, so $\|A^n\|_\infty \le \|B^n\|_\infty$ and Gelfand's formula gives $\rho(A) \le \rho(B)$; apply it to the matrix whose only nonzero entry is the record's self-weight. When reading displaces only leak, the agents keep their self-weight and gain the read weight, so the joint matrix dominates the agents-only matrix and $\rho \ge 1 - \varepsilon$, the case $\upsilon = 1$ of (c).

(e) At $(\varepsilon, \delta) = 0$, $g_z(0) = 0$ and $\partial_x g_z(0) = -(1 + \sum_k \mathcal{R}_k) \ne 0$; $g_z$ is analytic near the origin, with $\partial_\varepsilon g_z = 1$ and $\partial_{\delta_k} g_z = \beta_k\omega_k/(\omega_k + \delta_k - x)^2 = \mathcal{R}_k$ there. The implicit function theorem gives the first formula of (87). Since $\varepsilon + \sum_k \mathcal{R}_k\delta_k \ge c_0(\varepsilon + \sum_k \delta_k)$ with $c_0 > 0$, the quadratic error is $o(\varepsilon_{\mathrm{rec}})$, and $\tau = -1/\ln(1-\varepsilon_{\mathrm{rec}}) = \varepsilon_{\mathrm{rec}}^{-1}(1 + O(\varepsilon_{\mathrm{rec}}))$. □

*Proof of Proposition 8.13.* (a) For kernels symmetric under reflections $\hat{\mathsf{W}}(p)$ is real, and each of its entries is a Fourier sum of nonnegative real-space weights, so $|\hat{\mathsf{W}}(p)| \le \hat{\mathsf{W}}(0)$ entrywise and $\rho(\hat{\mathsf{W}}(p)) \le \rho(\hat{\mathsf{W}}(0)) = \rho(I - \mathsf{R}) < 1$. The stationary covariance of mode $p$ is therefore the unique solution $\hat{\Sigma}(p) = \mathcal{L}_p^{-1}(Q_z)$ of $\hat{\Sigma}(p) = \hat{\mathsf{W}}(p)\hat{\Sigma}(p)\hat{\mathsf{W}}(p)^\top + Q_z$, where $\mathcal{L}_p(X) = X - \hat{\mathsf{W}}(p)X\hat{\mathsf{W}}(p)^\top$ is invertible. The agents' row has $\hat{\mathsf{W}}(p)_{00} = 1 - \varepsilon' - \sum_k \beta_k - a\sum_d 2(1-\cos p_d)$ and $\hat{\mathsf{W}}(p)_{0k} = \beta_k \hat{\mathsf{K}}_k(p)$, with $\hat{\mathsf{K}}_k(p) = 1 - \ell_{r,k}^2|p|^2/2 + o(|p|^2)$ by the finite second moments and the symmetry of the kernels; this gives the expansion with $\mathsf{R}_2$, and $\hat{\mathsf{W}}$ is twice continuously differentiable near $p = 0$. The solution of the Lyapunov equation is a rational function of the entries of $\hat{\mathsf{W}}(p)$ without poles where $\rho(\hat{\mathsf{W}}(p)) < 1$, so $\hat{\Sigma}$ is twice continuously differentiable near the origin, with $\hat{\Sigma}(p) = \Sigma_0 - |p|^2\Sigma_2 + o(|p|^2)$; collecting the orders $|p|^0$ and $|p|^2$ gives (88), each solution being unique because $\rho(\hat{\mathsf{W}}_0) < 1$. The real-space transition matrix and the noise covariance are entrywise nonnegative, so the agents' covariance $\Sigma_{00}(x)$ is nonnegative and summable, with $\sum_x \Sigma_{00}(x) = (\Sigma_0)_{00}$. For a nonnegative summable sequence whose Fourier series is twice differentiable at the origin, Fatou's lemma applied to $(1 - \cos(p\cdot x))/|p|^2$ shows that its second moments are finite, and dominated convergence then gives $\sum_x |x|^2\Sigma_{00}(x) = -\Delta_p\hat{\Sigma}_{00}(0) = 2D(\Sigma_2)_{00}$. The definition $\xi_2^2 = \sum_x |x|^2\Sigma_{00}(x)/(2D\sum_x \Sigma_{00}(x))$ gives the formula.

(b) Let $x_0 = \varepsilon_{\mathrm{rec}}$ be the slow eigenvalue of $\mathsf{R}$, with right and left eigenvectors $r_0$ and $l_0$, $(r_0)_0 = 1$ and $l_0^\top r_0 = 1$. From the record rows of $\mathsf{R}r_0 = x_0 r_0$ and the record columns of $l_0^\top\mathsf{R} = x_0 l_0^\top$, $(r_0)_k = \omega_k/(\omega_k + \delta_k - x_0)$ and $(l_0)_k = (l_0)_0\beta_k/(\omega_k + \delta_k - x_0)$, and the normalization gives $(l_0)_0 = (1 + \sum_k \beta_k\omega_k/(\omega_k + \delta_k - x_0)^2)^{-1}$. Since only the first row of $\mathsf{R}_2$ is nonzero, $l_0^\top\mathsf{R}_2 r_0 = (l_0)_0(a + \sum_k \beta_k\ell_{r,k}^2(r_0)_k/2)$, which is (89); by first-order perturbation of a simple eigenvalue it is the $|p|^2$-coefficient of the slow eigenvalue of $I - \hat{\mathsf{W}}(p) = \mathsf{R} + |p|^2\mathsf{R}_2 + o(|p|^2)$.

At zero leaks with $\upsilon_k = 0$, $\mathsf{R}$ has zero row sums, so 0 is an eigenvalue with right eigenvector $\mathbf{1}$, and $I - \mathsf{R}$ is nonnegative and irreducible (its graph is a star), with a positive diagonal entry $1 - \sum_k \beta_k \ge 2Da > 0$. It is therefore primitive, and its other eigenvalues have modulus below one; equivalently, the other eigenvalues of $\mathsf{R}$ lie in a compact subset of $(0, 2)$. By continuity this persists for small leaks, so $x_0$ is a simple isolated eigenvalue, and the spectral projector $\mathsf{P}_0 = r_0 l_0^\top$, the vectors $r_0$, $l_0$ and the fast part $M_f = (I - \mathsf{R})(I - \mathsf{P}_0)$ depend continuously on the leaks. At zero leaks $\rho(M_f) < 1$, so some norm makes $M_f$ a strict contraction there, and by continuity it remains one, with a common factor below one, near zero leaks; the powers of $M_f$ therefore decay geometrically, uniformly near zero leaks. At zero leaks $r_0 = \mathbf{1}$ and $l_0 = (1, \mathcal{R}_1, \dots, \mathcal{R}_{n_z})^\top/(1 + \sum_k \mathcal{R}_k)$.

The map $\mathcal{L}(X) = X - \hat{\mathsf{W}}_0 X \hat{\mathsf{W}}_0^\top$ commutes with the four block projections $X \mapsto \mathsf{P}_a X \mathsf{P}_b^\top$, $\mathsf{P}_a, \mathsf{P}_b \in \{\mathsf{P}_0, I - \mathsf{P}_0\}$, because $\hat{\mathsf{W}}_0 = I - \mathsf{R}$ commutes with $\mathsf{P}_0$. On the slow-slow block it is multiplication by $1 - (1-x_0)^2 = x_0(2-x_0)$. On the other three blocks its inverse is a convergent series in $M_f$ (for example $Y \mapsto \sum_n (1-x_0)^n Y M_f^{n\top}$ on the slow-fast block, where $0 \le 1 - x_0 \le 1$), bounded uniformly near zero leaks by the geometric decay of the powers of $M_f$. With $\mathsf{q}_0 = l_0^\top Q_z l_0$, the first equation of (88) therefore gives

$$\Sigma_0 = \frac{\mathsf{q}_0}{x_0(2-x_0)} r_0 r_0^\top + B_0, \qquad \Sigma_0 l_0 = \frac{\mathsf{q}_0}{x_0(2-x_0)} r_0 + B_0 l_0,$$

with $B_0$ bounded. In the second equation the source $Y = \mathsf{R}_2 \Sigma_0 \hat{\mathsf{W}}_0^\top + \hat{\mathsf{W}}_0 \Sigma_0 \mathsf{R}_2^\top$ is $O(1/x_0)$, and its slow-slow block is $\mathsf{P}_0 Y \mathsf{P}_0^\top = (1-x_0) r_0 r_0^\top (l_0^\top \mathsf{R}_2 \Sigma_0 l_0 + l_0^\top \Sigma_0 \mathsf{R}_2^\top l_0) = r_0 r_0^\top [2(1-x_0)\mathcal{D}_{\text{eff}} \mathsf{q}_0 / (x_0(2-x_0)) + O(1)]$, using $l_0^\top \mathsf{R}_2 r_0 = \mathcal{D}_{\text{eff}}$. Dividing the slow-slow block by $x_0(2-x_0)$ and bounding the other blocks,

$$\Sigma_2 = \frac{2(1-x_0)\mathcal{D}_{\text{eff}} \mathsf{q}_0}{x_0^2 (2-x_0)^2} r_0 r_0^\top + O(1/x_0).$$

Since $(r_0)_0 = 1$ and $\mathsf{q}_0 \to q/(1+\sum_k \mathcal{R}_k)^2 + q_z \sum_k \mathcal{R}_k^2/(1+\sum_k \mathcal{R}_k)^2 > 0$, the agents' entries give

$$\xi_2^2 = \frac{(\Sigma_2)_{00}}{(\Sigma_0)_{00}} = \frac{2(1-x_0)}{2-x_0} \cdot \frac{\mathcal{D}_{\text{eff}}}{x_0}(1 + O(x_0)) = \frac{\mathcal{D}_{\text{eff}}}{x_0}(1 + O(x_0)),$$

the first form of (90), with constants depending on the fixed parameters. As the leaks vanish, $\omega_k/(\omega_k + \delta_k - x_0) \to 1$ and $\beta_k \omega_k/(\omega_k + \delta_k - x_0)^2 \to \mathcal{R}_k$, so $\mathcal{D}_{\text{eff}} \to (a + \sum_k \beta_k \ell_{r,k}^2/2)/(1+\sum_k \mathcal{R}_k) > 0$, and Proposition 8.12(e) gives the second form.

(c) Let $\Pi$ be the joint precision of agents and records, diffusive by hypothesis, and $S = \Pi_{AA} - \Pi_{AZ}\Pi_{ZZ}^{-1}\Pi_{ZA}$ the agents' marginal precision. The proof of Lemma S4.30, with the agents as members and the records as outsiders, uses only the sign pattern of $\Pi$ and the Neumann series of $\Pi_{ZZ}^{-1}$, which converges in operator norm on $\mathbb{Z}^D$ because $\Pi \succeq \|\Sigma\|^{-1} I$ and the diagonal of $\Pi$ is bounded; so $S$ is diffusive, its leaks are at least those of $\Pi$ on the agents, and its couplings among agents are at least those of $\Pi$. $S$ is translation invariant, hence has a uniform leak, and $\hat{S} = 1/\hat{\Sigma}_{00}$ is twice differentiable at the origin by (a); since its couplings are nonnegative, Fatou's lemma again gives them finite second moments, and Proposition S8.27(a) shows that the length of (a) is the second-moment length of $S$. For a region $C$ of agents, $\Pi_{ZZ}^{-1} \succ 0$ gives $S \preceq \Pi_{AA}$, so $\varkappa_{\text{out}}$ computed from $S$ is at most $\varkappa_{\text{out}}$ computed from $\Pi$; and since leaks and internal couplings only increase, the quadratic form that defines $\varkappa_{\text{in}}$ does not decrease on vectors supported on $C$. Hence the contrast computed from $S$ is at least the contrast computed from $\Pi$, which is at least $r_0$. Theorem 4.14 with uniform leak, applied to $S$, gives $r_S(C) \le 1 + \xi_2^2/\operatorname{Var}_d(C)$, hence $\operatorname{Var}_d(C) \le \xi_2^2/(r_S(C) - 1) \le \xi_2^2/(r_0 - 1)$. □

The long-range results rest on two lemmas about convolution powers of a probability $\mathsf{J}$ on $\mathbb{Z}^D \setminus \{0\}$; $\mathsf{J}^{*m}$ is the law of the sum $X_1 + \cdots + X_m$ of independent steps with law $\mathsf{J}$.

**Lemma S32.1** (Union bound)**.** *If $\mathsf{J}(y) \le K_{\mathsf{J}} |y|^{-D-\sigma}$ for $y \neq 0$, then $\mathsf{J}^{*m}(x) \le K_{\mathsf{J}} m^{1+D+\sigma} |x|^{-D-\sigma}$ for all $m \ge 1$ and $x \neq 0$.*

*Proof.* If $X_1 + \cdots + X_m = x$ then $\max_l |X_l| \ge |x|/m$. By exchangeability, $\Pr(X_1 + \cdots + X_m = x) \le m \Pr(X_1 + \cdots + X_m = x,\ |X_m| \ge |x|/m) = m \sum_{|y| \ge |x|/m} \mathsf{J}(y) \mathsf{J}^{*(m-1)}(x - y) \le m \sup_{|y| \ge |x|/m} \mathsf{J}(y)$, which is at most $K_{\mathsf{J}} m^{1+D+\sigma} |x|^{-D-\sigma}$. □

**Lemma S32.2** (Single big jump)**.** *If $\mathsf{J}(x) \sim \mathsf{J}_\infty |x|^{-D-\sigma}$ with $\sigma > 0$, then $\mathsf{J}^{*m}(x)/\mathsf{J}(x) \to m$ as $|x| \to \infty$, for each fixed $m$.*

*Proof.* By induction it suffices to show that if $F$ and $G$ are probabilities with $F(x) \sim \ell_F \mathsf{J}(x)$ and $G(x) \sim \ell_G \mathsf{J}(x)$, both dominated by $K|x|^{-D-\sigma}$, then $(F*G)(x) \sim (\ell_F+\ell_G)\mathsf{J}(x)$. Fix $\delta_1 \in (0, 1/2)$ and split $\sum_y F(y)G(x-y)$ over $\mathcal{A} = \{|y| \le \delta_1|x|\}$, $\mathcal{B} = \{|x-y| \le \delta_1|x|\}$ and the rest. On $\mathcal{A}$, $G(x-y)/\mathsf{J}(x)$ lies in $\ell_G[(1+\delta_1)^{-D-\sigma}, (1-\delta_1)^{-D-\sigma}](1+o(1))$ uniformly, and $F(\mathcal{A}) \to 1$; $\mathcal{B}$ is symmetric with $\ell_F$. The rest is at most $\sup_{|z|>\delta_1|x|} G(z) \cdot F(\{|y| > \delta_1|x|\}) = O(|x|^{-D-\sigma}|x|^{-\sigma}) = o(\mathsf{J}(x))$. Let $\delta_1 \to 0$. □

*Proof of Proposition 8.14.* $\Sigma = q\sum_n W^{2n}$ and $W^{2n} = \sum_{m=0}^{2n} \binom{2n}{m} c^{2n-m} a^m \mathsf{J}^{*m}$, so for $x \neq 0$

$$\frac{\Sigma(x)}{\mathsf{J}(x)} = q \sum_{n\ge1} \sum_{m=1}^{2n} \binom{2n}{m} c^{2n-m} a^m \frac{\mathsf{J}^{*m}(x)}{\mathsf{J}(x)}.$$

Each ratio tends to $m$ by Lemma S32.2, and is dominated by $K' m^{1+D+\sigma}$ by Lemma S32.1 together with $\mathsf{J}(x) \ge c_{\mathsf{J}}|x|^{-D-\sigma}$ for large $|x|$. Since $\sum_n \sum_m \binom{2n}{m} c^{2n-m} a^m m^{1+D+\sigma} \le \sum_n (2n)^{1+D+\sigma} \lambda^{2n} < \infty$, dominated convergence gives the limit $q\sum_n \sum_m \binom{2n}{m} c^{2n-m} a^m m = q\sum_n 2na\lambda^{2n-1} = 2qa\lambda/(1-\lambda^2)^2$. Without translation invariance, $\|\Sigma_{ij}\| \le q\sum_n \sum_k \|(W^n)_{ik}\|\|(W^n)_{jk}\|$, and $\|(W^n)_{ik}\|$ is dominated by the $n$-th convolution power of the symmetric majorant $c\mathbf{1}\{x=0\} + a\mathsf{J}(x)$, so the bound is the translation-invariant expression evaluated at $x_i - x_j$. □

*Proof of Proposition S8.18.* The normalized influence is $\Pr(|Y_t| \ge r)$ for the walk $Y_t$ whose steps $X_s$ have law $(c/\lambda)\mathbf{1}\{x=0\} + (a/\lambda)\mathsf{J}(x)$. (a) $\Pr(|Y_t| \ge r) \le t\Pr(|X| \ge r/2) + \Pr(|\sum_s X_s \mathbf{1}\{|X_s| < r/2\}| \ge r)$. The tail of $\mathsf{J}$ gives $\Pr(|X| \ge r/2) = O(r^{-\sigma})$. The truncated steps are symmetric and independent, so Chebyshev's inequality bounds the second term by $t\mathbb{E}[|X|^2\mathbf{1}\{|X| < r/2\}]/r^2$, and the truncated second moment is $O(r^{2-\sigma})$ for $\sigma < 2$, $O(\ln r)$ for $\sigma = 2$ and $O(1)$ for $\sigma > 2$. (b) Let $p_r = \Pr(|X| \ge 2r) \ge c_3 r^{-\sigma}$, and let $E_s$ be the event that $|X_s| \ge 2r$, that $|X_{s'}| < 2r$ for $s' \neq s$, and that $|\sum_{s'\neq s} X_{s'}| < r$. On $E_s$, $|Y_t| \ge r$, and the $E_s$ are disjoint. By independence and (a) applied to $t-1$ steps, once $r \ge c_2 t^{1/\sigma}$ with $c_2$ large, $\Pr(E_s) \ge p_r(\frac{1}{2} - tp_r) \ge p_r/4$, so $\Pr(|Y_t| \ge r) \ge tp_r/4 \ge c_1 t r^{-\sigma}$. (c) For $\sigma < 2$, (a) gives $r_\theta(t) \le (C_1 t/\theta)^{1/\sigma}$, and (b) at $r = Kt^{1/\sigma}$ with $c_2 \le K < (c_1/\theta)^{1/\sigma}$ gives $r_\theta(t) \ge Kt^{1/\sigma}$, which is possible for $\theta < c_1 c_2^{-\sigma}$. For $\sigma > 2$ the steps have finite variance; (a) gives $r_\theta(t) = O(t^{1/2})$ and the central limit theorem gives the matching lower bound. For $\sigma = 2$, (a) with the logarithm gives $r_\theta(t) = O((t\ln t)^{1/2})$. The front is at most linear exactly when $\sigma \ge 1$. (d) $(\bar{J}^t)(x) \le \sum_m \binom{t}{m} c^{t-m} a^m \mathsf{J}^{*m}(x)$, and Lemma S32.1 with $m \le t$ gives the bound; $(\bar{J}^t)_{ij}$ exceeds a level $\theta_1$ only if $t\ln g_{\mathrm{tot}} + (1+D+\sigma)\ln t \ge (D+\sigma)\ln|x_i - x_j| + \ln(\theta_1/K_{\mathsf{J}})$. □

*Proof of Theorem 8.16.* All randomness other than $X_{\mathrm{src}}$, namely the initial states of the other agents and the fresh $\zeta_{i,t}$, is independent and can be absorbed into the channels, so the run is a Bayesian network with the single source $X_{0,0}$. Theorem 5 of Polyanskiy and Wu (2017) bounds the strong data-processing coefficient of the channel from $X_{0,0}$ to $X_{\mathcal{S}}$ by $\mathrm{perc}(X_{0,0} \to \mathcal{S})$ for site percolation with each vertex open with the coefficient of its channel, and $I(X_{\mathrm{src}}; X_{\mathcal{S}}) \le s_{\mathrm{KL}}(P_{X_{\mathcal{S}}|X_{0,0}}) I(X_{\mathrm{src}}; X_{0,0})$ for the chain $X_{\mathrm{src}} \to X_{0,0} \to X_{\mathcal{S}}$ by the definition of the coefficient. Their Corollary 7 bounds the percolation probability by the sum over directed paths of the products of the open probabilities. A path from $X_{0,0}$ to $X_{i,t}$ makes $t$ steps, each a message step, open with probability at most $s_e$, or a memory step, open with probability at most $s_m$; it makes at least $d$ message steps; and there are at most $\binom{t}{k}\Delta^k$ paths with $k$ message steps. This gives the first bound. Summing over $t \ge k$ with $\sum_{t\ge k} \binom{t}{k} y^{t-k} = (1-y)^{-k-1}$ and then over $k \ge d$ gives (92). □

*Proof of Theorem S8.19.* (a) Given $X_u$ the subtrees of the children are independent, and by induction each child's subtree is equivalent to an observation $X_c + N_c$ with $N_c \sim \mathcal{N}(0, \mathcal{V}(c))$ independent of everything above $c$, since for Gaussian data the likelihood depends on the subtree only through

one linear statistic. Then $(X_c + N_c)/\alpha_{e_c} = X_u + ((1-\alpha_{e_c}^2)^{1/2} Z_c + N_c)/\alpha_{e_c}$ has noise variance $(1-\alpha_{e_c}^2 + \mathcal{V}(c))/\alpha_{e_c}^2$, and independent observations add precisions. A childless vertex above level $n$ has no observed descendant, so its subtree is the uninformative observation $\mathcal{V} = \infty$ and contributes nothing to its parent's sum. The root's posterior variance is $\mathcal{V}/(1+\mathcal{V})$, so $I = \frac{1}{2}\ln(1+1/\mathcal{V})$. (b) Substituting $\mathcal{V}(u) = \alpha^{2|u|}\hat{\mathcal{V}}(u)$, the recursion becomes $\hat{\mathcal{V}}(u)^{-1} = \sum_c ((1-\alpha^2)\alpha^{-2|c|} + \hat{\mathcal{V}}(c))^{-1}$, the series and parallel laws, with terminal values $\hat{\mathcal{V}} = \sigma_{\text{obs}}^2 \alpha^{-2n}$ at level $n$. (c) By symmetry level $k$ contributes $b^k$ equal resistances $(1-\alpha^2)\alpha^{-2k}$ in parallel, and the levels are in series, so $\mathcal{V}_n = (1-\alpha^2)\sum_{k=1}^n (b\alpha^2)^{-k}$; expanding $\frac{1}{2}\ln(1+1/\mathcal{V}_n)$ in the three regimes gives the limits. Because the data are Gaussian, the linear statistic is sufficient and linear estimation is optimal. (d) The resistance to the shorted level $n$ increases with $n$ to the resistance to infinity, and a locally finite network is transient exactly when that resistance is finite; by (b), $\lim_n I_n > 0$ exactly then. The conductances $\alpha^{2|e|}/(1-\alpha^2)$ are proportional to $(\alpha^{-2})^{-|e|}$, which define the biased walk of Lyons (1990) with bias $\alpha^{-2}$; by his theorem, as restated by Lyons and Pemantle (1992), that walk is transient if $\alpha^{-2} < \mathrm{br}(T)$ and recurrent if $\alpha^{-2} > \mathrm{br}(T)$. □

*Proof of Theorem S8.20.* Let $\mathfrak{g}(u) = 1/(1+\mathcal{V}(u))$, the squared correlation of $X_u$ with its subtree statistic. We show $\mathfrak{g}(u) \le \mathfrak{p}(u)$ by induction from the observed level, where $\mathfrak{g} = 1/(1+\sigma_{\text{obs}}^2) \le 1 = \mathfrak{p}$, and from the childless vertices above it, where $\mathfrak{g} = 0 = \mathfrak{p}$. With $h_c = \alpha_{e_c}^2 \mathfrak{g}(c)$, $1 - \alpha_{e_c}^2 + \mathcal{V}(c) = (1-h_c)/\mathfrak{g}(c)$, so the recursion of Theorem S8.19(a) reads $\mathcal{V}(u)^{-1} = \mathcal{T} := \sum_c h_c/(1-h_c)$ and $\mathfrak{g}(u) = \mathcal{T}/(1+\mathcal{T})$, increasing in each $h_c$. By induction $h_c \le \bar{h}_c := \alpha_{e_c}^2 \mathfrak{p}(c)$, so $\mathfrak{g}(u) \le \bar{\mathcal{T}}/(1+\bar{\mathcal{T}})$ with $\bar{\mathcal{T}} = \sum_c \bar{h}_c/(1-\bar{h}_c)$. With $u_c = 1/(1-\bar{h}_c) \ge 1$, $1 + \bar{\mathcal{T}} = 1 + \sum_c (u_c - 1) \le \prod_c u_c$, hence $\bar{\mathcal{T}}/(1+\bar{\mathcal{T}}) \le 1 - \prod_c (1-\bar{h}_c) = \mathfrak{p}(u)$. The observed level enters through the statistic $X_{\text{root}} + N$ with $N \sim \mathcal{N}(0, \mathcal{V}(\text{root}))$ independent of $(X_{\text{src}}, X_{\text{root}})$, which is sufficient, so the squared multiple correlation of $X_{\text{src}}$ with the observations is $\mathfrak{g}(\text{root})$ times that with $X_{\text{root}}$. Finally $f(x) = -\frac{1}{2}\ln(1-x)$ is convex with $f(0) = 0$, so $f(\mathfrak{g}x) \le \mathfrak{g} f(x)$ for $\mathfrak{g} \in [0,1]$, and $\mathfrak{g}(\text{root}) \le \mathfrak{p}(\text{root})$, the bond-percolation probability. □

*Proof of Proposition S8.21.* Let $Y = (e_j)_{d(0,j) \ge r}$. The squared top canonical correlation of $e_0$ and $Y$ is the largest eigenvalue of $\Sigma_{00}^{-1/2} \Sigma_{0Y} \Sigma_{YY}^{-1} \Sigma_{Y0} \Sigma_{00}^{-1/2}$, at most $\|\Sigma_{0Y}\|^2 / (\lambda_{\min}(\Sigma_{YY}) \lambda_{\min}(\Sigma_{00}))$. Interlacing gives $\lambda_{\min}(\Sigma_{YY}) \ge \lambda_{\min}(\Sigma) \ge \sigma_{\text{e}}$, and $\|\Sigma_{0Y}\|^2 \le \sum_j \|\Sigma_{0j}\|^2 \le B_\Sigma^2 \sum_j e^{-2m_\Sigma d(0,j)}$. If the spheres grow subexponentially, the tail sum decays exponentially in $r$. □

*Proof of Proposition S8.22.* (a) The joint density is $p(x_o) \prod_c p(x_c \mid x_{\text{parent}(c)})$ with conditional variance $1-\alpha^2$, so the precision has entries $-\alpha/(1-\alpha^2)$ on edges and diagonal entries $(1+b\alpha^2)/(1-\alpha^2)$ at internal vertices, $1 + b\alpha^2/(1-\alpha^2)$ at the root and $1/(1-\alpha^2)$ at the leaves. The row sums follow, for example $(1 + b\alpha^2 - \alpha - b\alpha)/(1-\alpha^2) = (1-b\alpha)/(1+\alpha)$ at internal vertices. Off-diagonal entries are nonpositive, so the precision is diffusive exactly when every row sum is nonnegative. The leaf sums are positive, and the root sum is nonnegative exactly when $(b-1)\alpha \le 1$. A tree of depth at least two has internal vertices, whose sums are nonnegative exactly when $b\alpha \le 1$, a condition that implies the one at the root; a tree of depth one has none, and the root condition $(b-1)\alpha \le 1$ alone decides. Since $\alpha < 1$, $b\alpha^2 > 1$ forces $b\alpha > 1$. (b) With $P = a^\Pi \mathcal{A}/(\varepsilon^\Pi + (b+1)a^\Pi)$, $\mathcal{A}$ the adjacency matrix, $P$ is the transition matrix of the simple random walk killed at rate $\varepsilon^\Pi/(\varepsilon^\Pi + (b+1)a^\Pi)$, and $\Sigma = (\varepsilon^\Pi + (b+1)a^\Pi)^{-1} \sum_m P^m$. The strong Markov property gives $\Sigma(o,v) = \Pr_o(\text{hit } v)\Sigma(v,v)$. Killing lowers hitting probabilities, and for the simple random walk on the $(b+1)$-regular tree the distance to $v$ decreases with probability $1/(b+1)$ and increases with probability $b/(b+1)$, so each unit decrease is ever achieved with probability $1/b$ and $\Pr_o(\text{hit } v) \le b^{-d(o,v)}$. By transitivity $\Sigma(v,v) = \Sigma(o,o)$, finite also at $\varepsilon^\Pi = 0$ by transience. The squared canonical correlation with the sphere $S_n$ is at most $\|\Sigma_{oS_n}\|^2 / (\lambda_{\min}(\Sigma_{S_nS_n})\Sigma(o,o))$, with $\|\Sigma_{oS_n}\|^2 \le |S_n| b^{-2n} \Sigma(o,o)^2$,

$|S_n| = (b+1)b^{n-1}$, and $\lambda_{\min}(\Sigma_{S_nS_n}) \ge 1/\lambda_{\max}(\Pi) \ge 1/(\varepsilon^{\Pi} + 2(b+1)a^{\Pi})$ by Gershgorin's theorem. □

*Proof of Theorem 8.18.* (a) The variable $Z$ ranges over all nondegenerate square-integrable variables, on an extension of the probability space if necessary, so a variable independent of the readouts is admissible and $\theta^* \ge 0$. Write $R = \mathrm{Corr}_C$ and $\underline{v} = \min_{p \in \mathbb{P}_C} p^\top R p$. If $\underline{v} = 0$, some $p \in \mathbb{P}_C$ has $p^\top X = 0$ almost surely, so $\sum_i p_i \operatorname{Corr}(X_i, Z) = 0$ for every $Z$, hence $\min_i \operatorname{Corr}(X_i, Z) \le 0$ and $\theta^* = 0$. Let $\underline{v} > 0$. A variable $Z$ with $\min_i \operatorname{Corr}(X_i, Z) > 0$ has a nonzero projection onto the linear span of the readouts, and the projection keeps every $\operatorname{Cov}(X_i, Z)$ and does not increase $\operatorname{Var} Z$, so it does not decrease any positive correlation; variables with nonpositive minimum do not affect a positive supremum. We may therefore take $Z = c^\top X$ with $c$ in the range of $R$. Normalizing $c^\top R c = 1$, $\operatorname{Corr}(X_i, Z) = (Rc)_i$, and the supremum of $\min_i (Rc)_i$ over the sphere $c^\top R c = 1$ agrees with the maximum over the ball $c^\top R c \le 1$ whenever the latter is positive, since scaling $c$ up increases a positive minimum. The maximum over the ball is $\max_{c^\top R c \le 1} \min_{p \in \mathbb{P}_C} p^\top R c$. Both sets are compact and convex and the objective is bilinear, so von Neumann's minimax theorem gives $\min_p \max_c p^\top R c = \min_p (p^\top R p)^{1/2}$, by the Cauchy–Schwarz inequality in the inner product of $R$. At the saddle point $c^* = p^*/(p^{*\top} R p^*)^{1/2}$, and the optimality conditions of the quadratic program on the simplex, $(Rp^*)_i \ge p^{*\top} R p^*$ with equality on the support of $p^*$, show that $\min_i (Rc^*)_i = (p^{*\top} R p^*)^{1/2} = \underline{v}^{1/2} > 0$, so $\theta^* = \underline{v}^{1/2}$ is positive and attained by $c^{*\top} X$. (b) By (a), $\theta^{*2} = \underline{v}$ in both cases; take $p$ uniform, then use the Rayleigh quotient and Gershgorin's theorem. (c) Apply (b) to the set of members with $\operatorname{Corr}(X_i, Z) \ge \theta$, whose best correlational coherence is at least $\theta$ and whose susceptibility is at most $\chi_C$. (d) With $|\mathbf{w}| = 1$, $Z = \mathbf{w}^\top X$ has $\operatorname{Corr}(X_i, Z) = (R\mathbf{w})_i / (\mathbf{w}^\top R \mathbf{w})^{1/2} = \lambda_{\max}^{1/2} w_i$. □

*Proof of Corollary 8.19.* (a) From the series $\Sigma = \sum_n \Phi Q^\zeta \Phi^\top$ with block-diagonal noise, $\sum_j \|\Sigma_{ij}\| \le q \sum_n \sum_k \|\Phi_{ik}\| \sum_j \|\Phi_{jk}\| \le q \sum_n \lambda_r^n \lambda_c^n$, by the row and column inequalities. Readout correlations satisfy $|(\mathrm{Corr}_C)_{ij}| \le \|\Sigma_{ij}\|/(\sigma_i \sigma_j)$ with $\sigma_i^2 = \mathbf{v}_i^\top \Sigma_{ii} \mathbf{v}_i \ge q_{\min}$, so $\chi_C \le q/(q_{\min}(1 - \lambda_r \lambda_c))$, and Theorem 8.18(b) concludes. (b) $\lambda_{\max}(\Sigma) \le \sum_n \|W^n Q^\zeta W^{n\top}\| \le K_A^2 q/(1 - f^2)$. The readout map has orthonormal columns in disjoint blocks and the standardization divides by $\sigma_i \ge q_{\min}^{1/2}$, so $\lambda_{\max}(\mathrm{Corr}_C) \le \lambda_{\max}(\Sigma)/q_{\min}$; and $1/(1 - f^2) = \tau_f(1 + O(1 - f))/2$. Since $\|W^k\| \le K_A f^k$ forces $\rho(W) \le f$, $\tau_f \ge \tau$. (c) $W^n = (\lambda^n/N)\mathbf{1}\mathbf{1}^\top$ for $n \ge 1$, so $\Sigma = qI + q b_N \mathbf{1}\mathbf{1}^\top$ with $b_N = \lambda^2/((1 - \lambda^2)N)$, all correlations equal $b_N/(1 + b_N)$, and by symmetry and convexity the uniform vector minimizes $p^\top \mathrm{Corr}_V p$. Then $N\theta^{*2} = (1 + N b_N)/(1 + b_N)$, which is also $\lambda_{\max}(\mathrm{Corr}_V)$, and $1 + N b_N = 1/(1 - \lambda^2)$. (d) $W \ge 0$ gives $\Sigma \ge 0$, so $\chi = \sum_x \Sigma(x)/\Sigma(0) = \hat{\Sigma}(0)/\Sigma(0) = q/((1 - \lambda^2)\Sigma(0))$; insert Theorem 8.10(f) and (g), with $1 - \lambda^2 \sim 2\varepsilon$ and $\xi_{\mathrm{OZ}} \sim (a/\varepsilon)^{1/2}$. For $D \ge 3$, $q/(1 - \lambda^2) = q\xi_2^2/(2a\lambda)$. (e) The hub's error is an autoregression independent of the leaves, $e_h(t+1) = c_h e_h(t) + \zeta_h(t)$, and each leaf has $e_\ell(t) = \sum_{n \ge 0} c^n [a e_h(t-1-n) + \zeta_\ell(t-1-n)]$. For two leaves, $\operatorname{Cov}(e_\ell, e_{\ell'}) = a^2 \sum_{n,n'} c^{n+n'} \operatorname{Cov}(e_h(t - 1 - n), e_h(t - 1 - n')) > 0$, the same for every pair and independent of $N$, as is $\operatorname{Var}(e_\ell)$; so all leaf-leaf correlations equal some $\theta_\star > 0$. As in (c), a set of $N$ leaves has $\theta^{*2} = \theta_\star + (1 - \theta_\star)/N > \theta^2$ when $\theta^2 < \theta_\star$, for every $N$. The hub's column of $W$ has Euclidean norm $(c_h^2 + N a^2)^{1/2}$, so $\|W\| \ge (c_h^2 + N a^2)^{1/2}$, and the case $k = 1$ of $(\mathrm{C}_\rho)$ gives $K_A f \ge \|W\|$; since $f < 1$, $K_A > a\sqrt{N}$, so the family satisfies neither hypothesis uniformly in $N$. □

*Proof of Theorem S8.24.* (a) Each $P_{V_j}$ fixes $V_\cap$ pointwise and is self-adjoint, so $P_{V_j} P_\cap = P_\cap = P_\cap P_{V_j}$, hence $A_{\mathrm{cyc}} P_\cap = P_\cap A_{\mathrm{cyc}} = P_\cap$. Expanding with these relations, $(A_{\mathrm{cyc}} - P_\cap)^n = A_{\mathrm{cyc}}^n - P_\cap$, and $A_{\mathrm{cyc}} - P_\cap = A_{\mathrm{cyc}}(I - P_\cap)$. If $x \perp V_\cap$, $|x| = 1$ and $|A_{\mathrm{cyc}} x| = 1$, then each projection in turn preserves the norm and therefore fixes the vector, so $x \in \bigcap_j V_j = V_\cap$, a contradiction; since $A_{\mathrm{cyc}}$ maps $V_\cap^\perp$ into itself and the unit sphere of $V_\cap^\perp$ is compact, $c_{\mathrm{cyc}} < 1$. (b) $V_\cap^\perp$ is invariant under every $P_{V_j}$, every

product fixes $P_\cap x_0$, and $x_n - P_\cap x_0$ is the product applied to $x_0 - P_\cap x_0 \in V_\cap^\perp$. Every complete block of $p_0$ hops is a word containing every index, whose norm on $V_\cap^\perp$ is less than one by the argument of (a), and there are finitely many such words; the remaining hops have norm at most one. (c) Each factor has norm at most $\bar{a}$. □

*Proof of Theorem 8.20.* (a) The independence statement is Theorem 8.5(b); the radius bound is Theorem 8.7(b); Definition 8.8 defines its lengths from the stationary covariance under (M) or from the covariance of the process started in the infinite past under the uniform form, so it defines no length when both forms fail; the failure of the stationary form is by definition the absence of a stationary law with finite second moments, and Definition 4.12 requires (M). (b) Under (B), standardized blocks satisfy $\|R_{ij}\| \le \|\Sigma_{ij}\|/q_{\min}$. Theorem 8.9 gives the first bound, Proposition S8.10 the second, and Theorem S8.11, whose prefactor is at most $2q\varpi_{\max}/\varpi_{\min}$, gives decay rate at least $\vartheta\sqrt{2\ln(1/\lambda)}$ for every $\vartheta < 1$, hence the third. The lattice statements are Theorem 8.10(c),(g) and Proposition S8.27(c). (c) is Proposition S8.27(b), Proposition 8.14 and Proposition S8.18(c). (d) is Proposition S8.14, Proposition S8.15(a), Proposition 8.11(c), Proposition 8.12(a),(e) and Proposition 8.13(b). (e) is Theorem 4.14 with uniform leak, Corollary 4.16, Proposition 8.13(c), Theorem 8.18(c) and Corollary 8.19(a),(b),(e). □

## S33 Proofs for Section 9

*Proof of Lemma S9.4.* At level 1 with known codes, agent 1 models the message $y_{12} = H_2 m_2 + \nu_{12}$ as $H_2\eta$ plus noise of variance $R_c + H_2 P_2 H_2^\top$, so its claimed innovation variance is $S_1$ and its posterior mean is $m_1^+ = m_1 + \mathcal{K}_{12}(y_{12} - H_2 m_1)$; likewise for agent 2. Writing $y_{12} = H_2\eta + H_2 e_2 + \nu_{12}$ and dropping the noise, the homogeneous errors satisfy $e_1^+ = e_1 + \mathcal{K}_{12}H_2(e_2 - e_1)$ and $e_2^+ = e_2 + \mathcal{K}_{21}H_1(e_1 - e_2)$, so $e_1^+ - e_2^+ = \mathbb{D}(e_1 - e_2)$.

Write $\langle H, H'\rangle_k = H P_k H'^\top$ for row vectors $H, H'$ and $k \in \{1,2\}$, an inner product because $P_k \succ 0$. The matrix $\mathcal{K}_{12}H_2 + \mathcal{K}_{21}H_1$ is the product of the $K \times 2$ matrix with columns $\mathcal{K}_{12}, \mathcal{K}_{21}$ and the $2 \times K$ matrix with rows $H_2, H_1$, so it has rank at most two and characteristic polynomial $x^{K-2}\det(xI_2 - \mathsf{N})$, where

$$\mathsf{N} = \begin{pmatrix} H_2\mathcal{K}_{12} & H_2\mathcal{K}_{21} \\ H_1\mathcal{K}_{12} & H_1\mathcal{K}_{21} \end{pmatrix} = \begin{pmatrix} \langle H_2, H_2\rangle_1/S_1 & \langle H_2, H_1\rangle_2/S_2 \\ \langle H_1, H_2\rangle_1/S_1 & \langle H_1, H_1\rangle_2/S_2 \end{pmatrix},$$

with $S_1 = \langle H_2, H_2\rangle_1 + \langle H_2, H_2\rangle_2 + R_c$ and $S_2 = \langle H_1, H_1\rangle_1 + \langle H_1, H_1\rangle_2 + R_c$. Hence the eigenvalues of $\mathbb{D}$ are 1, with multiplicity at least $K-2$, and $1-\mathsf{n}$ for the two eigenvalues $\mathsf{n}$ of $\mathsf{N}$. Every $\langle H_k, H_k\rangle_l$ is positive, so $0 < \operatorname{tr}\mathsf{N} < 2$, and

$$\det\mathsf{N} = \frac{\langle H_2, H_2\rangle_1\langle H_1, H_1\rangle_2 - \langle H_1, H_2\rangle_1\langle H_2, H_1\rangle_2}{S_1 S_2}.$$

If $\det\mathsf{N} < 0$, $\mathsf{N}$ has real eigenvalues of opposite signs, and the negative one gives an eigenvalue $1 - \mathsf{n} > 1$ of $\mathbb{D}$. If $\det\mathsf{N} \ge 0$ and the eigenvalues are real, both lie in $[0, \operatorname{tr}\mathsf{N}] \subset [0,2)$, so $|1-\mathsf{n}| \le 1$. If they are complex, their real part is $\frac{1}{2}\operatorname{tr}\mathsf{N}$ and $|1-\mathsf{n}|^2 = 1 - \operatorname{tr}\mathsf{N} + \det\mathsf{N}$, which is below 1 exactly when the numerator of $\det\mathsf{N}$ is below $\langle H_2, H_2\rangle_1 S_2 + \langle H_1, H_1\rangle_2 S_1$. By the Cauchy–Schwarz inequality for the two inner products, $\langle H_1, H_2\rangle_1^2 \le \langle H_1, H_1\rangle_1\langle H_2, H_2\rangle_1$ and $\langle H_2, H_1\rangle_2^2 \le \langle H_1, H_1\rangle_2\langle H_2, H_2\rangle_2$, so, writing $\mathsf{a} = \langle H_2, H_2\rangle_1\langle H_1, H_1\rangle_2$, $\mathsf{b}_1 = \langle H_1, H_1\rangle_1\langle H_2, H_2\rangle_1$ and $\mathsf{b}_2 = \langle H_1, H_1\rangle_2\langle H_2, H_2\rangle_2$,

$$\begin{aligned}\mathsf{a} - \langle H_1, H_2\rangle_1\langle H_2, H_1\rangle_2 &\le \mathsf{a} + (\mathsf{b}_1\mathsf{b}_2)^{1/2} \le \mathsf{a} + \tfrac{1}{2}(\mathsf{b}_1 + \mathsf{b}_2) \\ &< 2\mathsf{a} + \mathsf{b}_1 + \mathsf{b}_2 + R_c(\langle H_2, H_2\rangle_1 + \langle H_1, H_1\rangle_2) = \langle H_2, H_2\rangle_1 S_2 + \langle H_1, H_1\rangle_2 S_1,\end{aligned}$$

and $|1-\mathsf{n}|<1$. Hence $\rho(\mathbb{D})>1$ exactly when $\det \mathsf{N}<0$, which is the stated inequality. If $H_2=\vartheta H_1$, both sides of the inequality equal $\vartheta^2\langle H_1,H_1\rangle_1\langle H_1,H_1\rangle_2$. □

*Proof of Proposition 9.1.* We first check that (93) is the homogeneous part of level-1 reception. A level-1 receiver models $y_{ij}=H_j m_j+\nu_{ij}$ as $H_j\eta$ plus noise of variance $1+H_jP_jH_j^\top$, so its posterior precision is $\Lambda_i^+$, because $H_j^\top(1+H_jP_jH_j^\top)^{-1}H_j=\alpha_jT_c^\top T_c$, and its posterior mean satisfies

$$\Lambda_i^+ m_i^+=\Lambda_i m_i+(H_i^s)^\top(R_i^s)^{-1}s_i+\sum_{j\in J_i}H_j^\top(1+H_jP_jH_j^\top)^{-1}y_{ij}.$$

Subtracting $\Lambda_i^+\eta$ and using $s_i=H_i^s\eta+v_i$ and $y_{ij}=H_j\eta+H_je_j+\nu_{ij}$ gives $\Lambda_i^+e_i^+=\Lambda_ie_i+\sum_{j\in J_i}\alpha_jT_c^\top T_ce_j$ plus terms linear in the sensing and channel noises; dropping the noise terms gives (93). In gain form the same update has $W_{ii}^{\mathrm{rec}}=(\Lambda_i^+)^{-1}\Lambda_i=I-\mathcal{K}_i^sH_i^s-\sum_{j\in J_i}\mathcal{K}_{ij}H_j$ and $W_{ij}^{\mathrm{rec}}=\mathcal{K}_{ij}H_j$ for $j\in J_i$, with $\mathcal{K}_i^s=(\Lambda_i^+)^{-1}(H_i^s)^\top(R_i^s)^{-1}$ and $\mathcal{K}_{ij}=(\Lambda_i^+)^{-1}H_j^\top(1+H_jP_jH_j^\top)^{-1}$, so that $FW^{\mathrm{rec}}$ has the form of Definition 2.12 with exact code models.

Write $\mathbf{\Lambda}=\mathrm{diag}_i(\Lambda_i)$, $\mathbf{\Gamma}=\mathrm{diag}_i(\Gamma_i)$, $\boldsymbol{\alpha}=\mathrm{diag}_i(\alpha_iI_K)$, let $\mathcal{T}$ be the block matrix with blocks $\mathcal{T}_{ij}=\alpha_jT_c^\top T_c$ for $j\in J_i$ and zero otherwise, and let $\mathcal{D}=\mathrm{diag}_i(\bar\alpha_iT_c^\top T_c)$ with $\bar\alpha_i=\sum_{j\in J_i}\alpha_j$. Then $\mathbf{\Lambda}^+=\mathrm{diag}_i(\Lambda_i^+)=\mathbf{\Lambda}+\mathbf{\Gamma}+\mathcal{D}$ and $W^{\mathrm{rec}}=(\mathbf{\Lambda}^+)^{-1}(\mathbf{\Lambda}+\mathcal{T})$.

(a) Put $\tilde M=\boldsymbol{\alpha}\mathbf{\Lambda}^+$ and $X=\boldsymbol{\alpha}(\mathbf{\Lambda}+\mathcal{T})$. Since $\boldsymbol{\alpha}$ is a positive multiple of the identity on each block, it commutes with $\mathbf{\Lambda}^+$, so $\tilde M=\mathrm{diag}_i(\alpha_i\Lambda_i^+)\succ 0$ and $W^{\mathrm{rec}}=\tilde M^{-1}X$. The diagonal blocks of $X$ are $\alpha_i\Lambda_i$, and its off-diagonal blocks are $X_{ij}=\alpha_i\alpha_jT_c^\top T_c$ when $j\in J_i$ and zero otherwise. Under symmetric listening $X_{ij}=X_{ji}$, so $X$ is symmetric. Hence $\tilde MW^{\mathrm{rec}}=X$ is symmetric, which is self-adjointness for $\langle\cdot,\cdot\rangle_{\tilde M}$, and $W^{\mathrm{rec}}$ is similar to the symmetric matrix $\tilde M^{-1/2}X\tilde M^{-1/2}$, so it is diagonalizable with real spectrum. Let $Z=\tilde M-X$. Its diagonal blocks are $\alpha_i\Gamma_i+\alpha_i\bar\alpha_iT_c^\top T_c$ and its off-diagonal blocks are $-\alpha_i\alpha_jT_c^\top T_c$ on listening edges, so

$$Z=\mathrm{diag}_i(\alpha_i\Gamma_i)+\mathcal{L}\otimes T_c^\top T_c,\qquad e^\top Ze=\sum_i\alpha_ie_i^\top\Gamma_ie_i+\sum_{\{i,j\}}\alpha_i\alpha_j(T_ce_i-T_ce_j)^2\geq 0,$$

where $\mathcal{L}$ is the Laplacian of the listening graph with weight $\alpha_i\alpha_j$ on the edge $\{i,j\}$ and the second sum runs over its edges. The eigenvalues of $W^{\mathrm{rec}}=I-\tilde M^{-1}Z$ are one minus the eigenvalues of $\tilde M^{-1/2}Z\tilde M^{-1/2}\succeq 0$, so they are at most 1. They exceed $-1$ exactly when $\tilde M^{-1/2}(2\tilde M-Z)\tilde M^{-1/2}\succ 0$, and

$$2\tilde M-Z=\tilde M+X=2\,\mathrm{diag}_i(\alpha_i\Lambda_i)+\mathrm{diag}_i(\alpha_i\Gamma_i)+\mathcal{L}^+\otimes T_c^\top T_c,$$

where $\mathcal{L}^+$ is the signless Laplacian of the same weighted graph, positive semidefinite because $e^\top(\mathcal{L}^+\otimes T_c^\top T_c)e=\sum_{\{i,j\}}\alpha_i\alpha_j(T_ce_i+T_ce_j)^2$. Since every $\Lambda_i\succ 0$ and $\alpha_i>0$, $\tilde M+X\succ 0$. Finally $W^{\mathrm{rec}}e=e$ exactly when $Ze=0$, which for $Z\succeq 0$ is $e^\top Ze=0$; since $\alpha_i>0$ and $e_i^\top\Gamma_ie_i=\|(R_i^s)^{-1/2}H_i^se_i\|^2$, this is $e\in\mathcal{P}$.

(b) At a covariance fixed point of a frozen profile the covariances, and hence $W^{\mathrm{rec}}$, are the same in every round, and with $F=fI_K$ the prediction step multiplies every posterior error by $f$, so $W=(I_N\otimes F)W^{\mathrm{rec}}=fW^{\mathrm{rec}}$. In the norm $\|x\|_{\tilde M}=(x^\top\tilde Mx)^{1/2}$ the operator norm of a self-adjoint map equals its spectral radius, so $\|W^{\mathrm{rec}}\|_{\tilde M}\leq 1$ and $\|W^t\|_{\tilde M}\leq f^t$ for every $t\geq 0$. Since $\lambda_{\min}(\tilde M)\|x\|_2^2\leq\|x\|_{\tilde M}^2\leq\lambda_{\max}(\tilde M)\|x\|_2^2$, $\|W^t\|_2\leq K_{\tilde M}f^t$, and $\rho(W)\leq f$. By (a) the spectrum of $W^{\mathrm{rec}}$ lies in $(-1,1]$, so $\rho(W^{\mathrm{rec}})=1$ exactly when 1 is an eigenvalue, that is when $\mathcal{P}\neq\{0\}$, and $\rho(W)=f\rho(W^{\mathrm{rec}})$.

We characterize $\mathcal{P}=\{0\}$. If some agent $i$ has a nonzero vector in $\ker H_i^s\cap\ker T_c$, that vector placed at $i$, with zero at every other agent, lies in $\mathcal{P}$, since all coded values are then zero. If every such intersection is trivial and some connected component $V'$ of the listening graph contains no

agent with injective sensing, then for each $j \in V'$ the kernel $\ker H_j^s$ is nonzero and meets $\ker T_c$ trivially, so it contains a vector $e_j$ with $T_c e_j = 1$; with $e_j = 0$ outside $V'$, all coded values on $V'$ equal 1, no edge joins $V'$ to its complement, and $e \in \mathcal{P}$. Conversely, if every intersection is trivial and every component contains an agent $i_0$ with injective sensing, then $e \in \mathcal{P}$ forces $e_{i_0} = 0$, hence $T_c e_j = T_c e_{i_0} = 0$ for every $j$ in the component of $i_0$, hence $e_j \in \ker H_j^s \cap \ker T_c = \{0\}$; so $\mathcal{P} = \{0\}$.

For the covariance bound, the world-conditional law of the errors (Definition 4.7) is driven by the local noise alone, the common world innovation being fixed by the conditioning. Its stationary covariance is therefore $\Sigma^c = \sum_{t \geq 0} W^t Q^\zeta (W^t)^\top$, with $Q^\zeta = \operatorname{diag}_l(Q_l^\zeta)$ the block-diagonal covariance of the local noise (Definition 2.12), $Q_l^\zeta \preceq qI$; the series converges by the bound on $\|W^t\|_2$, which is the moment bound for the frozen profile. The block $(W^t Q^\zeta (W^t)^\top)_{ij} = \sum_l W_{il}^t Q_l^\zeta (W_{jl}^t)^\top$ vanishes unless some $l$ lies within $t$ hops of both $i$ and $j$, because $W_{il} \neq 0$ only if $l = i$ or $l \in J_i$; this needs $t \geq \lceil d(i,j)/2 \rceil$. Each block has norm at most $\|W^t Q^\zeta (W^t)^\top\|_2 \leq q K_{\tilde{M}}^2 f^{2t}$, and summing over $t \geq \lceil d(i,j)/2 \rceil$, with $f^{2\lceil d/2 \rceil} \leq f^d$, gives

$$\|\Sigma_{ij}^c\| \leq \sum_{t \geq \lceil d(i,j)/2 \rceil} q K_{\tilde{M}}^2 f^{2t} = \frac{q K_{\tilde{M}}^2 f^{2\lceil d(i,j)/2 \rceil}}{1 - f^2} \leq \frac{K_{\tilde{M}}^2 q f^{d(i,j)}}{1 - f^2}.$$

(c) With $\Gamma_i = 0$, $\Lambda_i^+ = \Lambda_i + \bar{\alpha}_i T_c^\top T_c$, so $(\Lambda_i^+)^{-1} \Lambda_i = I - \bar{\alpha}_i P_i^+ T_c^\top T_c$ with $P_i^+ = (\Lambda_i^+)^{-1}$, and (93) becomes, in terms of the coded errors $x_i = T_c e_i$,

$$e_i \mapsto e_i - \sum_{j \in J_i} \alpha_j P_i^+ T_c^\top (x_i - x_j).$$

The subspace $\mathcal{K} = \{e : x_i = 0 \text{ for every } i\}$ is therefore fixed pointwise, and the coded errors evolve by $x \mapsto \Psi x$ with $\Psi_{ij} = \alpha_j T_c P_i^+ T_c^\top > 0$ for $j \in J_i$, $\Psi_{ij} = 0$ for other $j \neq i$, and $\Psi_{ii} = 1 - \sum_{j \in J_i} \Psi_{ij}$. With $\sigma_i^2 = T_c P_i T_c^\top > 0$, the rank-one update $P_i^+ = P_i - \bar{\alpha}_i P_i T_c^\top T_c P_i / (1 + \bar{\alpha}_i \sigma_i^2)$, verified by multiplication with $\Lambda_i^+$, gives $T_c P_i^+ T_c^\top = \sigma_i^2 / (1 + \bar{\alpha}_i \sigma_i^2)$ and hence

$$\sum_{j \in J_i} \Psi_{ij} = \frac{\bar{\alpha}_i \sigma_i^2}{1 + \bar{\alpha}_i \sigma_i^2} < 1,$$

so $\Psi$ has nonnegative entries and unit row sums. The map $e \mapsto x$ from $\mathbb{R}^{NK}$ onto $\mathbb{R}^N$ is surjective because $T_c \neq 0$, and its kernel is $\mathcal{K}$, so it identifies the quotient $\mathbb{R}^{NK}/\mathcal{K}$ with $\mathbb{R}^N$, on which $W^{\mathrm{rec}}$ induces $\Psi$. In a basis that begins with a basis of $\mathcal{K}$, $W^{\mathrm{rec}}$ is block upper triangular with diagonal blocks the identity on $\mathcal{K}$ and $\Psi$, so its eigenvalues are 1, with multiplicity $N(K-1)$, together with the eigenvalues of $\Psi$. A row-stochastic matrix has 1 as an eigenvalue and all eigenvalues in the closed unit disk, so $\rho(W^{\mathrm{rec}}) = 1$; at a covariance fixed point in the world $F = fI_K$, $\rho(W) = f\rho(W^{\mathrm{rec}}) = f = \rho(F)$. For a profile of period $p$ along a periodic covariance orbit, the map over one period is $f^p W^{\mathrm{rec}}(p) \cdots W^{\mathrm{rec}}(1)$, whose factors are the reception maps of the successive rounds. Each factor fixes the same subspace $\mathcal{K}$ pointwise and induces a row-stochastic matrix $\Psi(t)$ on the same quotient, so in the same basis the product is block upper triangular with diagonal blocks the identity and $\Psi(p) \cdots \Psi(1)$, which is row stochastic; its spectral radius is therefore 1, and that of the period map is $f^p$. None of these steps used symmetry of listening or equality of the scales $\vartheta_j$. □

*Proof of Proposition 9.2.* Fix a round $t$. Let $\mathbb{1}_i \in \{0,1\}$ indicate a grounding event of agent $i$, and let $\mathfrak{F}_t$ be the history up to the decision time of round $t$ together with the round's actions. The indicators are independent of $\mathfrak{F}_t$, and the round's sensing, channel and world noises are independent

of $\mathfrak{F}_t$ and of the indicators. Covariance recursions do not depend on the data, so the posterior covariances $P_i^+$ and the next decision-time precisions $\Lambda_i(t+1)$ are functions of $\mathfrak{F}_t$ and the indicators. Updates only add precision, and the prediction step maps $\{P : 0 \preceq P \preceq P_0\}$ into $\{P : Q \preceq P \preceq P_0\}$ because $FP_0F^\top + Q = P_0$; by induction from $Q \preceq P_i(0) \preceq P_0$, every decision-time covariance satisfies $Q \preceq P_i(t) \preceq P_0$ and every posterior covariance $P_i^+ \preceq P_0$.

Three inequalities do the work. (i) Projection. Let $A_l$ be matrices, $Y_l \succ 0$ and $x_l$ vectors, indexed by $l$ in a finite set, with $\Lambda' = \sum_l A_l^\top Y_l^{-1} A_l \succ 0$, and let $\bar{x} = (\Lambda')^{-1} \sum_l A_l^\top Y_l^{-1} x_l$. Stacking $A = (A_l)_l$, $Y = \mathrm{diag}_l(Y_l)$ and $\tilde{x} = Y^{-1/2}(x_l)_l$ gives $\bar{x}^\top \Lambda' \bar{x} = \tilde{x}^\top Y^{-1/2} A (A^\top Y^{-1} A)^{-1} A^\top Y^{-1/2} \tilde{x}$, a quadratic form in an orthogonal projector, so $\bar{x}^\top \Lambda' \bar{x} \le \|\tilde{x}\|^2 = \sum_l x_l^\top Y_l^{-1} x_l$. For an agent that is not grounded, the homogeneous posterior error $\bar{e}_i^+$, the posterior error with all new noises set to zero, has this form with the prior ($A = I$, $Y = P_i$, $x = e_i$), the sensing ($A = H_i^s$, $Y = R_i^s$, $x = 0$) and the messages ($A = H_j$, $Y = \hat{R}_{ij}$, $x = H_j e_j$), because the receiver knows the codes. Hence, with $\mathcal{V}_i = e_i^\top \Lambda_i e_i$ at the decision time,

$$(\bar{e}_i^+)^\top \Lambda_i^+ \bar{e}_i^+ \le \mathcal{V}_i + \sum_{j \in J_i} (H_j e_j)^\top \hat{R}_{ij}^{-1} H_j e_j.$$

(ii) Header bound. With $X_j = R_c^{-1/2} H_j P_j^{1/2}$, $P_j^{1/2} H_j^\top \hat{R}_{ij}^{-1} H_j P_j^{1/2} = X_j^\top (I + X_j X_j^\top)^{-1} X_j$, whose eigenvalues are $\mathsf{u}/(1+\mathsf{u})$ for the eigenvalues $\mathsf{u}$ of $X_j X_j^\top \preceq R_c^{-1/2} H_j P_0 H_j^\top R_c^{-1/2}$, all at most $\mathrm{snr}_j$. As $\mathsf{u} \mapsto \mathsf{u}/(1+\mathsf{u})$ is increasing, $H_j^\top \hat{R}_{ij}^{-1} H_j \preceq \varpi_H \Lambda_j$, and each message term is at most $\varpi_H \mathcal{V}_j$. (iii) Prediction contraction. With $\Phi_i = F(P_i^+)^{1/2}$, $(P_i^+)^{1/2} F^\top (F P_i^+ F^\top + Q)^{-1} F (P_i^+)^{1/2} = \Phi_i^\top (\Phi_i \Phi_i^\top + Q)^{-1} \Phi_i$, whose largest eigenvalue is

$$1 - \lambda_{\min}\big((\Phi_i\Phi_i^\top + Q)^{-1/2} Q (\Phi_i\Phi_i^\top + Q)^{-1/2}\big) = 1 - \frac{1}{\lambda_{\max}\big(Q^{-1/2}(\Phi_i\Phi_i^\top + Q)Q^{-1/2}\big)} \le \varpi_Q,$$

because $\Phi_i\Phi_i^\top + Q = FP_i^+F^\top + Q \preceq FP_0F^\top + Q = P_0$. Since $\Lambda_i(t+1) = (FP_i^+F^\top + Q)^{-1}$ for an agent that is not grounded, $(F\bar{e}_i^+)^\top \Lambda_i(t+1) F\bar{e}_i^+ \le \varpi_Q (\bar{e}_i^+)^\top \Lambda_i^+ \bar{e}_i^+$.

The error recursion is $e(t+1) = \mathbb{A}_t e(t) + \zeta(t) - \mathbf{1}_N \otimes w_t$, where $(\mathbb{A}_t e(t))_i = F\bar{e}_i^+$ for agents that are not grounded and 0 for grounded ones, and $\zeta_i(t) = FP_i^+[(H_i^s)^\top (R_i^s)^{-1} v_i(t) + \sum_{j \in J_i} H_j^\top \hat{R}_{ij}^{-1} \nu_{ij}(t)]$, zero for grounded agents. By (i) to (iii),

$$\sum_i \|(\mathbb{A}_t e(t))_i\|^2_{\Lambda_i(t+1)} \le \varpi_Q \sum_i (1 - \mathbb{1}\!\!\!

If $\varpi_g < 1$, iteration gives $\sup_t \mathbb{E}\mathcal{V}(t) \le \max\{\mathbb{E}\mathcal{V}(0), N\operatorname{tr}(P_0)/(\lambda_{\min}(Q)(1-\varpi_g))\}$, which is finite because the initial errors have finite second moments and $P_i(0) \succeq Q$. Then $\mathbb{E}\|e_i(t)\|^2 \le \lambda_{\max}(P_0)\mathbb{E}\mathcal{V}(t)$, and $m_i(t) = \eta_t + e_i(t)$ with $\eta_t \sim \mathcal{N}(0, P_0)$ for every $t$ gives $\sup_t \mathbb{E}\|m_i(t)\|^2 \le 2\operatorname{tr}(P_0) + 2\lambda_{\max}(P_0)\sup_t \mathbb{E}\mathcal{V}(t) < \infty$. The connected part of the errors, the error less its world-conditional mean, has second moments no larger than those of the errors, so (M) holds in its uniform form. □

## S34 Proofs for Section 10

The proofs follow the order of Section 10. Two facts about M-matrices are used throughout. Let $M = D_0 - N_0$ be symmetric, with $D_0$ diagonal, $N_0 \ge 0$ entrywise with zero diagonal, and $M \succeq \delta D_0$ for some $\delta > 0$; in finite dimensions the last condition is positive definiteness, and on $\ell^2(\mathbb{Z})$ we also take $D_0$ bounded above and below by positive constants. Then $S = D_0^{-1/2} N_0 D_0^{-1/2}$ is symmetric and nonnegative with $S \preceq (1-\delta)I$, and since $|\langle f, Sf\rangle| \le \langle |f|, S|f|\rangle$, its norm is at most $1-\delta$. Hence $M^{-1} = D_0^{-1/2}\sum_{n\ge 0} S^n D_0^{-1/2} = \sum_{n\ge 0}(D_0^{-1}N_0)^n D_0^{-1}$, the series converging in norm, so $M^{-1} \ge 0$ entrywise, and $M^{-1}$ is positive at every entry when $N_0$ is irreducible, since some term is. If two such operators satisfy $M_1 \le M_2$ entrywise, then $M_1^{-1} - M_2^{-1} = M_1^{-1}(M_2 - M_1)M_2^{-1} \ge 0$.

*Proof of Proposition 10.2.* Consider first a level-0 receiver $i$ with any assigned noises. It predicts with $F$ and $Q$ and updates by one observation at a time; observations with independent noises can be processed sequentially, because under the receiver's own model the sequential and the joint Gaussian updates coincide. Suppose $P_i$ is block diagonal across contents. The prediction $FP_iF^\top + Q$ is block diagonal by (a). An observation whose model matrix $\hat{H}$ has nonzero columns only in block $c$, a sensing channel by (c) or a modality-$c$ message by block purity, has gain $\mathcal{K} = P_i^-\hat{H}^\top(\hat{H}P_i^-\hat{H}^\top + \hat{R})^{-1}$ with nonzero rows only in block $c$, and $(I - \mathcal{K}\hat{H})P_i^-$ differs from $P_i^-$ only in the $(c,c)$ block. By induction over rounds every posterior covariance and every gain is block diagonal, and under (d) the gains are deterministic. The error obeys $e_i^- = Fe_i - w_t$ in the prediction and $e_i^+ = e_i^- + \mathcal{K}(o - \hat{H}m_i^-)$ in an update. For a modality-$c$ message $o = H_j m_j + \nu = H_j(\eta + e_j) + \nu$, where $H_j$ reads only the $c$-block of $m_j$, so $o - \hat{H}m_i^- = (H_j - \hat{H})\eta + H_j e_j - \hat{H}e_i^- + \nu$ involves only the $c$-blocks of $\eta$, $e_j$ and $e_i^-$, and $\mathcal{K}$ writes only into block $c$; sensing updates are alike. Hence $e^{(c)}(t+1)$ is a linear function, with deterministic coefficients, of $e^{(c)}(t)$, of $\eta_t^{(c)}$ and $w_t^{(c)}$, and of the noises of content-$c$ channels, and the coefficient of $e_j^{(c)}$ in the update of $e_i^{(c)}$ vanishes unless $i = j$ or $i$ reads $j$ in modality $c$, an edge of $\mathcal{G}_c$. This is the recursion with transfer $W^{(c)}(t)$. The world blocks evolve independently by (a), the noises of different contents are independent by (c), and so are the initial quantities by (b); the processes $(e^{(c)}, \eta^{(c)})$, $c \in \mathcal{M}$, are therefore independent.

For exact fusion we prove by induction over rounds that every agent's estimate of $\eta_t^{(c)}$ is a linear functional of the content-$c$ primitives: the path of $\eta^{(c)}$, the noises of content-$c$ channels and the content-$c$ initial quantities. If this holds up to round $t$, every content-$c$ observation at round $t$, sensing or message, is such a functional. The primitives of different contents are independent, so the conditional law of $\eta_t^{(c)}$ given all of an agent's data equals its conditional law given the agent's content-$c$ data, which proves the claim at the next round. Posterior covariances are then block diagonal, and the error processes of different contents are independent. The results of Section 8 cited in the statement assume a linear error recursion with local noise on a fixed schedule, which each $e^{(c)}$ satisfies at level 0. □

*Proof of Proposition 10.3.* The update is symmetric and translation invariant with symbol $\hat{W}(k) = \left(\begin{smallmatrix} c_1 + 2a_1\cos k & a_\times \\ a_\times & c_2 + 2a_2\cos k \end{smallmatrix}\right)$. Since $\hat{W}(0) - \hat{W}(k)$ and $\hat{W}(k) - \hat{W}(\pi)$ are diagonal and positive semidefinite,

every eigenvalue of $\hat{W}(k)$ lies between the smallest eigenvalue of $\hat{W}(\pi)$ and the largest of $\hat{W}(0)$. Now $I-\hat{W}(0)=\left(\begin{smallmatrix}\varepsilon_1 & -a_\times\\ -a_\times & \varepsilon_2\end{smallmatrix}\right)\succ 0$ because $a_\times^2<\varepsilon_1\varepsilon_2$, and $I+\hat{W}(\pi)=\left(\begin{smallmatrix}2c_1+\varepsilon_1 & a_\times\\ a_\times & 2c_2+\varepsilon_2\end{smallmatrix}\right)\succ 0$ because $(2c_1+\varepsilon_1)(2c_2+\varepsilon_2)\ge\varepsilon_1\varepsilon_2>a_\times^2$. Hence $\rho(W)<1$, and since $W$ is symmetric and the noise is $qI$, the stationary covariance is $\Sigma=q\sum_{n\ge0}W^{2n}=q(I-W^2)^{-1}=\frac{q}{2}[(I-W)^{-1}+(I+W)^{-1}]$.

For $u=\cosh m>1$, two geometric series give $\sum_{d\in\mathbb{Z}}e^{-m|d|}e^{ikd}=\sinh m/(u-\cos k)$, so the Fourier coefficients of $1/(u-\cos k)$ are $e^{-m|d|}/\sinh m$, and, replacing $k$ by $k+\pi$, those of $1/(u+\cos k)$ are $(-1)^d e^{-m|d|}/\sinh m$. With $z=\cos k$, $[(I-\hat{W})^{-1}]_{11}=A_2(z)/(A_1(z)A_2(z)-a_\times^2)$. Since $A_c(z)=2a_c(\cosh m_c^0-z)$, the denominator is $4a_1a_2(z-u_-)(z-u_+)$, and because $A_2$ is affine, partial fractions give $\sum_\pm A_2(u_\pm)/[4a_1a_2(u_\pm-u_\mp)(z-u_\pm)]$, where $u_+\ne u_-$ because $\cosh m_1^0\ne\cosh m_2^0$. As $-1/(z-u_\pm)=1/(u_\pm-\cos k)$, this gives the non-alternating terms with the stated $\mathcal{A}_\pm$. In the same way $[(I+\hat{W})^{-1}]_{11}=B_2(z)/(B_1(z)B_2(z)-a_\times^2)=\sum_\pm B_2(w_\pm)/[4a_1a_2(w_\pm-w_\mp)(z-w_\pm)]$, with $w_+\ne w_-$ when $a_\times\ne0$, and $1/(z-w_\pm)=1/(\cosh m'_\pm+\cos k)$ gives the alternating terms.

(a) The roots of $A_1A_2=a_\times^2$ solve $(u-\cosh m_1^0)(u-\cosh m_2^0)=a_\times^2/(4a_1a_2)\ge0$, and the left side is negative strictly between $\cosh m_2^0$ and $\cosh m_1^0$, so $u_-\le\cosh m_2^0$ and $u_+\ge\cosh m_1^0$, strictly when $a_\times\ne0$. At $u=1$, $A_1A_2-a_\times^2=\varepsilon_1\varepsilon_2-a_\times^2>0$, so 1 lies outside $[u_-,u_+]$, an interval containing $[\cosh m_2^0,\cosh m_1^0]\subset(1,\infty)$; hence $u_->1$. Similarly $B_c(w)=2a_c(w-w_c^0)$ with $w_c^0=-(1+c_c)/(2a_c)<-1$, because $1+c_c-2a_c=2c_c+\varepsilon_c>0$; the roots $w_\pm$ bracket the $w_c^0$, and $B_1(-1)B_2(-1)-a_\times^2=(2c_1+\varepsilon_1)(2c_2+\varepsilon_2)-a_\times^2>0$ places $-1$ outside $[w_-,w_+]$, so $w_+<-1$.

(b) $A_2(u_+)=2a_2(\cosh m_2^0-u_+)<0$ and $u_+-u_->0$ give $\mathcal{A}_+>0$, and for $a_\times\ne0$, $A_2(u_-)=2a_2(\cosh m_2^0-u_-)>0$ and $u_--u_+<0$ give $\mathcal{A}_->0$. For $u\in[1,u_-]$, $A_c(u)\ge A_c(u_-)>0$, and $A_1(u)A_2(u)\ge a_\times^2$ because the quadratic $A_1A_2-a_\times^2$ is nonnegative to the left of its smaller root. Since $B_c(-u)=A_c(u)+2c_c$ and $c_1,c_2\ge0$, if $c_1+c_2>0$ then every $u\in[1,u_-]$ satisfies

$$B_1(-u)B_2(-u)-a_\times^2\ge 2c_1A_2(u)+2c_2A_1(u)+4c_1c_2>0.$$

So the quadratic $B_1B_2-a_\times^2$, which is negative strictly between its roots, is positive on $[-u_-,-1]$; both roots lie below $-1$, hence below $-u_-$, and $\cosh m'_\pm=-w_\pm>u_-=\cosh m_-$. The alternating terms and the term in $m_+>m_-$ therefore decay faster than $e^{-m_-|d|}$.

(c) Evaluating the quadratic at $u_-$ gives

$$\cosh m_2^0-u_-=\frac{a_\times^2}{4a_1a_2(\cosh m_1^0-u_-)}=\frac{a_\times^2}{4a_1a_2(\cosh m_1^0-\cosh m_2^0)}+O(a_\times^4),$$

and $u_+-\cosh m_1^0=O(a_\times^2)$. Hence $A_2(u_-)=a_\times^2/[2a_1(\cosh m_1^0-\cosh m_2^0)]+O(a_\times^4)$, $A_2(u_+)=-2a_2(\cosh m_1^0-\cosh m_2^0)+O(a_\times^2)$, $u_+-u_-=\cosh m_1^0-\cosh m_2^0+O(a_\times^2)$ and $\sinh m_\pm=\sinh m_{1,2}^0+O(a_\times^2)$. So $\mathcal{A}_+=1/(2a_1\sinh m_1^0)+O(a_\times^2)$ and $\mathcal{A}_-=a_\times^2/[8a_1^2a_2(\cosh m_1^0-\cosh m_2^0)^2\sinh m_2^0]+O(a_\times^4)$, whose ratio is the stated one. The two exponentials are equal where $\mathcal{A}_+e^{-m_+d}=\mathcal{A}_-e^{-m_-d}$, which is the first form of $d_\times$. Inserting the ratio and $m_+-m_-=m_1^0-m_2^0+O(a_\times^2)$ gives the second, the error being the product of the logarithm with the correction to the denominator. □

*Proof of Proposition S10.4.* For distinct $i,j$ every chain from $i$ to $j$ contains a link of weight at least $\varpi^\circ$, so $u(i,j)\ge\varpi^\circ$, and $u_{\text{in}}(C)\ge\varpi^\circ$ whenever $|C|\ge2$. Since $\mathcal{G}_c$ is connected and $C$ is a proper nonempty subset, some edge of $\mathcal{G}_c$ joins $C$ to its complement, and its weight $\varpi^\circ$ bounds $\varpi_{\text{out}}(C)$ from above. Hence $\mathcal{U}(C)\le0$. □

*Proof of Proposition 10.4.* (a) Read $\varepsilon I+L_b$ as the admittance matrix of a resistor network with conductance $a$ on every bond except $a_b$ on $(0,1)$, and conductance $\varepsilon$ from every node to ground; $\mathsf{G}(x,y)$ is the potential at $x$ when unit current enters at $y$, decaying at infinity. A semi-infinite uniform chain with first node 1 carries the decaying potentials $\phi_n=\phi_1e^{-m(n-1)}$, since $\varepsilon\phi_n+$

$a(2\phi_n - \phi_{n-1} - \phi_{n+1}) = 0$ for $n \geq 2$ requires $e^m + e^{-m} = 2 + \varepsilon/a$. The current entering node 1 is then $\phi_1[\varepsilon + a(1 - e^{-m})] = \phi_1 a(e^m - 1)$, so the input admittance is $Y = a(e^m - 1)$. When unit current enters at node 1, node 1 heads the right half-chain and is joined through $a_b$ to node 0, which heads the left half-chain; the admittance seen at node 1 is $Y + a_b Y/(a_b + Y)$, so $\mathsf{G}(1,1) = (a_b + Y)/[Y(2a_b + Y)]$, and the voltage divider gives $\mathsf{G}(0,1) = \mathsf{G}(1,1)a_b/(a_b + Y) = a_b/[Y(2a_b + Y)]$. No current enters at $x \leq 0$, so $\mathsf{G}(x,1) = \mathsf{G}(0,1)e^{-m|x|}$, and by the symmetry of $\mathsf{G}$ and the same argument on the right, $\mathsf{G}(x,y) = \mathsf{G}(0,1)e^{-m(|x|+y-1)}$ for $x \leq 0 < y$. For $n_b = 1$ this gives $\mathsf{G}^0(x,y) = e^{-m(y-x)}/(2a\sinh m)$. The ratio is $2ae^m\mathsf{G}(0,1)\sinh m$, and with $e^m(e^m - e^{-m}) = (e^m - 1)(e^m + 1)$ it equals $a_b(e^m + 1)/(2a_b + a(e^m - 1)) = (e^m + 1)/(2 + n_b(e^m - 1))$. Dividing numerator and denominator by $e^m - 1$ gives the second form.

(b) With $\mathfrak{s} = e^m - 1 = m + O(m^2)$, $\mathcal{T} = (1 + \mathfrak{s}/2)/(1 + n_b\mathfrak{s}/2) = 1 - (n_b - 1)\mathfrak{s}/2 + O(n_b^2\mathfrak{s}^2) = 1 - \frac{m}{2}(n_b - 1) + O(m^2 n_b^2)$. From the second form, $\mathcal{T} n_b m/2 = \frac{m}{2}\coth(m/2) \cdot n_b/(n_b + 2/(e^m - 1))$; the first factor tends to one as $m \to 0$, and the second tends to one when $mn_b \to \infty$, because $2/(e^m - 1) \leq 2/m$. Taking logarithms gives the form of $d_b$.

(c) The update $W = (1 - \varepsilon)I - L_b$ is symmetric with spectrum in $[1 - \varepsilon - 4a, 1 - \varepsilon]$, and $1 - \varepsilon - 4a = 2c + \varepsilon - 1 > -1$, so $\rho(W) < 1$ and $\Sigma = \frac{q}{2}[\mathsf{G} + (I + W)^{-1}]$, and likewise for $\Sigma^0$. Let $D_\pm = \mathrm{diag}((-1)^x)$ and let $\mathrm{Deg}_b$ be the diagonal matrix of weighted degrees. The chain is bipartite, so $D_\pm L_b D_\pm = 2\mathrm{Deg}_b - L_b$ and $\tilde{M} := D_\pm(I + W)D_\pm = (2 - \varepsilon)I - 2\mathrm{Deg}_b + L_b$, whose diagonal is $2 - \varepsilon - 2a = 2c + \varepsilon + 2a$ in the bulk and $2 - \varepsilon - a - a_b$, larger by $a - a_b \geq 0$, at the sites 0 and 1, and whose off-diagonal entries are $-a$, or $-a_b \geq -a$ on the barrier bond. The uniform operator $\tilde{M}^0 = (2c + \varepsilon)I + L_0$ therefore satisfies $\tilde{M}^0 \leq \tilde{M}$ entrywise. Both are of the form of the preliminary facts, with $\tilde{M} \succeq (2c+\varepsilon)I$ because $\tilde{M}$ is unitarily equivalent to $I+W$, and likewise $\tilde{M}^0$, so $0 \leq \tilde{M}^{-1} \leq (\tilde{M}^0)^{-1}$ entrywise. The symbol of $\tilde{M}^0$ is $2c + \varepsilon + 2a(1 - \cos k) = 2a(\cosh m' - \cos k)$ with $\cosh m' = 1 + (2c + \varepsilon)/(2a) = (1 + c)/(2a)$, using $2a + c + \varepsilon = 1$, so $(\tilde{M}^0)^{-1}(x,y) = e^{-m'|x-y|}/(2a\sinh m')$ by the Fourier fact in the previous proof. Since $(I + W)^{-1}(x,y) = (-1)^{x-y}\tilde{M}^{-1}(x,y)$, we get $|(I+W)^{-1}(x,y)| \leq e^{-m'(y-x)}/(2a\sinh m')$, with equality, and the sign $(-1)^{x-y}$, for the uniform chain. Also $m' > m$, because $(1 + c)/(2a) > (1 - c)/(2a) = \cosh m$ when $c > 0$. Write $g = \mathsf{G}^0(x,y) > 0$, so that $e^{-m'(y-x)}/(2a\sinh m') = r_{xy}g$. By (a), $\Sigma(x,y) = \frac{q}{2}g(\mathcal{T} + \mathfrak{e}_1)$ and $\Sigma^0(x,y) = \frac{q}{2}g(1 + \mathfrak{e}_0)$ with $|\mathfrak{e}_0|, |\mathfrak{e}_1| \leq r_{xy}$, hence

$$\Big|\frac{\Sigma(x,y)}{\Sigma^0(x,y)} - \mathcal{T}\Big| = \frac{|\mathfrak{e}_1 - \mathcal{T}\mathfrak{e}_0|}{|1 + \mathfrak{e}_0|} \leq \frac{(1 + \mathcal{T})r_{xy}}{1 - r_{xy}} \leq \frac{2r_{xy}}{1 - r_{xy}},$$

since $\mathcal{T} \leq 1$ for $n_b \geq 1$. □

*Proof of Proposition S10.6.* The matrix $I - W$ is positive definite, because $A_1 \succeq \varepsilon_1 I$, $A_2 \succeq \varepsilon_2 I$ and its Schur complement satisfies $A_1 - a_\times^2 A_2^{-1} \succeq (\varepsilon_1 - a_\times^2/\varepsilon_2)I \succ 0$. Block inversion gives $\mathsf{G}_{11} = (A_1 - a_\times^2 A_2^{-1})^{-1} = \sum_{n\geq 0}(a_\times^2 A_1^{-1}A_2^{-1})^n A_1^{-1}$, where the series converges because $\|a_\times^2 A_1^{-1}A_2^{-1}\|_2 \leq a_\times^2/(\varepsilon_1\varepsilon_2) < 1$. Each $A_c$ is positive definite with nonpositive off-diagonal entries, so $A_c^{-1} \geq 0$ entrywise and every term is nonnegative. Since $L_1$ has no edge across $C$, $A_1$ and $A_1^{-1}$ are block diagonal with respect to $(C, V\setminus C)$, so the term $n = 0$ vanishes across the cut, which gives the identity and the bounds. If $A_2$ is also block diagonal, every term is, and $\mathsf{G}_{11}$ vanishes across the cut. If $L_2$ is connected, $A_2$ is an irreducible nonsingular M-matrix and $A_2^{-1}$ is entrywise positive, and the diagonal entries of the positive definite matrix $A_1^{-1}$ are positive; since all terms of the sum are nonnegative, $[A_1^{-1}A_2^{-1}A_1^{-1}](x,y) = \sum_{x'\in C, y'\notin C} A_1^{-1}(x,x')A_2^{-1}(x',y')A_1^{-1}(y',y) \geq A_1^{-1}(x,x)A_2^{-1}(x,y)A_1^{-1}(y,y) > 0$, and the lower bound $a_\times^2[A_1^{-1}A_2^{-1}A_1^{-1}](x,y)$ is positive when $a_\times \neq 0$. At $a_\times = 0$ it vanishes, as does $\mathsf{G}_{11}(x,y)$. □

*Proof of Proposition S10.12.* (a) $\ln[q_\mathrm{p}/(1-q_\mathrm{p})] = b_\mathrm{p}\ln[x/(1-x)]$, so $q_\mathrm{p}(\frac{1}{2}) = \frac{1}{2}$ and, since $\mathrm{sig}'(0) = \frac{1}{4}$,

the slope of $q_{\mathrm{p}}$ at $\frac{1}{2}$ is $b_{\mathrm{p}} \cdot \frac{1}{4} \cdot 4 = b_{\mathrm{p}} > 1$, which puts $q_{\mathrm{p}}$ above the diagonal immediately to the right of $\frac{1}{2}$. Further $q_{\mathrm{p}}(0) = 0$ for $b_{\mathrm{p}} > 0$, while $q(0) = \mathrm{sig}(-\gamma J\mathsf{d}) > 0$ because sig is positive.

(b) Both rules equal $\frac{1}{2}$ at $x = \frac{1}{2}$, and the slope of $q$ there is $2\gamma J\mathsf{d}\,\mathrm{sig}'(0) = \gamma J\mathsf{d}/2$, so the slopes agree exactly when $b_{\mathrm{p}} = \gamma J\mathsf{d}/2$. Then the log-odds of $q$, $2b_{\mathrm{p}}(2x-1)$, is the tangent at $\frac{1}{2}$ of the log-odds $\ell_{\mathrm{p}}(x) = b_{\mathrm{p}} \ln[x/(1-x)]$ of $q_{\mathrm{p}}$. Since $\ell_{\mathrm{p}}''(x) = b_{\mathrm{p}}[(1-x)^{-2} - x^{-2}]$ is negative on $(0, \frac{1}{2})$ and positive on $(\frac{1}{2}, 1)$, $\ell_{\mathrm{p}}$ lies strictly below its tangent on $(0, \frac{1}{2})$ and strictly above it on $(\frac{1}{2}, 1)$, and sig is increasing.

(c) An end agent of the chain has one partner, so its local frequency is 0 or 1 and the power rule copies its partner; an interior agent has $x \in \{0, \frac{1}{2}, 1\}$ and adopts $+1$ with probability 0, $\frac{1}{2}$ or 1. The two uniform configurations are therefore absorbing, and no revision increases the number of walls. In any other configuration take the leftmost wall, between the agents $i$ and $i+1$. If $i$ is the end agent, its revision removes the wall and creates none; otherwise agent $i$ sees partners of both codes and with probability $\frac{1}{2}$ switches, which moves the wall one bond to the left and creates no other wall. From every configuration with walls, a finite sequence of revisions, each of positive probability, therefore lowers the number of walls by one, and the finite chain is absorbed in a uniform configuration with probability one. Under $q$ at finite $\gamma$ every change of a single code has positive probability, since sig takes values in $(0,1)$, so the chain is irreducible. With values affine in the number of matched partners and a slope $J$ common to all agents, as here, the code game is an exact potential game on every graph (Proposition 6.10(c)), here the pairwise game with coupling $J$ on every edge of the chain, and $q$ is its logit revision with a uniformly drawn reviser, so the stationary law is the Gibbs law of Proposition 6.10(a), described in Proposition S10.13(c). □

*Proof of Proposition S10.13.* (a) Write $X = \langle Z, \mathsf{u}_1\rangle$ and $Y = \langle Z, \mathsf{u}_2\rangle$ for an isotropic standard Gaussian vector $Z \in \mathbb{R}^2$ and unit vectors with $\langle \mathsf{u}_1, \mathsf{u}_2\rangle = \Gamma$. The signs of $X$ and $Y$ differ exactly when the line orthogonal to $Z$ separates $\mathsf{u}_1$ from $\mathsf{u}_2$; the direction of $Z$ is uniform, so the probability is the angle $\arccos\Gamma$ divided by $\pi$, which is Sheppard's formula (Sheppard, 1899). Linearity of expectation gives $\mathcal{S}(d)$, and the strong law of large numbers over independent features gives almost sure convergence. The expansions follow from $\arccos(1-\mathsf{h}) = (2\mathsf{h})^{1/2}(1+O(\mathsf{h}))$ as $\mathsf{h} \to 0$, with $1 - e^{-d/\xi} = (d/\xi)(1+O(d/\xi))$ and $1 - e^{-d^2/(2\sigma_{\mathrm{g}}^2)} = (d^2/2\sigma_{\mathrm{g}}^2)(1+O(d^2/\sigma_{\mathrm{g}}^2))$, and from $\arccos\Gamma = \frac{\pi}{2} - \arcsin\Gamma$ with $\arcsin\Gamma = \Gamma(1+O(\Gamma^2))$.

(b) $\sigma_0 \neq \sigma_d$ exactly when $n_{\mathrm{w}}(d)$ is odd, and every integer $n \geq 0$ satisfies $n - n(n-1) \leq \mathbf{1}[n \text{ odd}] \leq n$: both sides equal $n$ for $n \in \{0,1\}$, and for $n \geq 2$ the left side is $n(2-n) \leq 0$. Taking expectations and using $\mathbb{E}n_{\mathrm{w}}(d) = dp_{\mathrm{w}}$, which follows from stationarity, gives the bounds; the linear rise follows when the second term is $o(dp_{\mathrm{w}})$.

(c) By Proposition 6.10(a), asynchronous logit revision of the pairwise game with coupling $J$ on the edges of the chain is reversible, and at $h_i = 0$ its stationary law is proportional to $\exp(-\gamma J \sum_x \mathbf{1}[\sigma_x \neq \sigma_{x+1}])$. On an open chain the map from configurations to the first code and the bond variables $\sigma_x\sigma_{x+1}$ is a bijection, so the bond variables are independent, each equal to $-1$ with probability $e^{-\gamma J}/(1+e^{-\gamma J}) = 1/(1+e^{\gamma J})$. Then $\mathbb{E}[\sigma_0\sigma_d] = \left(1 - 2/(1+e^{\gamma J})\right)^d = \tanh^d(\gamma J/2)$, and $\Pr(\sigma_0 \neq \sigma_d) = \frac{1}{2}(1 - \mathbb{E}[\sigma_0\sigma_d])$.

(d) For a standard Gaussian pair with correlation $\Gamma$, $\mathbb{E}[\operatorname{sign} X \operatorname{sign} Y] = 1 - 2\arccos(\Gamma)/\pi = \frac{2}{\pi}\arcsin\Gamma$, which is strictly increasing in $\Gamma$ and nonnegative for $\Gamma \geq 0$. The readout weights $-\ln(\frac{2}{\pi}\arcsin\Gamma_{ij})$ are therefore a strictly increasing function of the weights $\varpi_{ij} = -\ln\Gamma_{ij}$ of the field. The hypotheses of Proposition S10.4, a common value on the edges and nothing smaller elsewhere, are preserved by a strictly increasing transformation, and the conclusion follows from that proposition. □

*Proof of Proposition S10.15.* (a) A potential makes the sums around cycles telescope to zero. Con-

versely, if every cycle sum vanishes, fix a root and define $\bar{\alpha}$ at each node by summing $\alpha$ along any path from the root; the vanishing cycle sums make the value independent of the path, and two potentials differ by a constant on the connected graph. The least-squares residual is zero exactly when $\alpha$ lies in the range of $B$, that is, when a potential exists.

(b) $B\bar{\alpha}$ has zero holonomy by (a), so only $\phi$ contributes; the least-squares residual differs from $\alpha$ by an element of the range of $B$, which has zero holonomy. The holonomy is a signed sum of independent terms of mean zero, so its second moment is the sum of their variances.

(c) The least-squares estimate through the origin is $\Theta_{ij} = \sum_t \eta_t y_t / \sum_t \eta_t^2 = \Theta^*_{ij} + \sum_t \eta_t \nu_t / \sum_t \eta_t^2$, where $y_t$ are the messages; given the $\eta_t$ it is Gaussian with mean $\Theta^*_{ij}$ and variance $\sigma_y^2 / \sum_t \eta_t^2$. Given the $\eta_t$, the law of $n_{ij}^{1/2}(\Theta_{ij} - \Theta^*_{ij})$ is therefore exactly $\mathcal{N}(0, \sigma_y^2 n_{ij} / \sum_t \eta_t^2)$. The almost sure convergence $n_{ij}^{-1} \sum_t \eta_t^2 \to \sigma_\eta^2$, which for independent and identically distributed $\eta_t$ is the strong law of large numbers and for a stationary ergodic sequence is the ergodic theorem, makes this conditional law converge to $\mathcal{N}(0, \sigma_y^2/\sigma_\eta^2)$ for almost every sequence of world values, and by dominated convergence of characteristic functions so does the unconditional law. The event $\Theta_{ij} > 0$, on which the logarithm is defined, has probability tending to one, and the delta method with the logarithm, differentiable at $\Theta^*_{ij} > 0$, gives the stated limit. □

*Proof of Proposition 10.5.* (a) With only its own signal, agent $i$ has $\mathbb{E}_i \eta = (\bar{\eta}_0/P_0 + s_i/R_i^s)/(1/P_0 + 1/R_i^s)$ and $\operatorname{Var}_i \eta = 1/(1/P_0 + 1/R_i^s)$, so its demand is $A_{\mathrm{ra}}^{-1}[\bar{\eta}_0/P_0 + s_i/R_i^s - (1/P_0 + 1/R_i^s)p]$. Setting the sum of demands to zero gives $p^\circ$. With $R_i^s = R^s$, $p^\circ = (\bar{\eta}_0/P_0 + \bar{s}/R^s)/(1/P_0 + 1/R^s)$ with $\bar{s} = N^{-1} \sum_i s_i$, and $\bar{s} \to \eta$ almost surely by the strong law of large numbers, conditionally on $\eta$.

(b) By Gaussian conjugacy the pooled posterior is $\mathcal{N}(\hat{\eta}_{\mathrm{pool}}, 1/\Lambda_{\mathrm{pool}})$, and it depends on the signals only through $\sum_j s_j/R_j^s$. If $p$ is an invertible function of this statistic, the tower property gives $\mathbb{E}[\eta \mid s_i, p] = \mathbb{E}[\mathbb{E}[\eta \mid s_1, \ldots, s_N] \mid s_i, p] = \hat{\eta}_{\mathrm{pool}}$, and the same argument applies to the whole posterior law. If every agent conditions on $p = \hat{\eta}_{\mathrm{pool}}$, every agent holds the pooled posterior and demands $(\hat{\eta}_{\mathrm{pool}} - p)\Lambda_{\mathrm{pool}}/A_{\mathrm{ra}} = 0$, so the market clears.

(c) The level-0 posterior is the Gaussian update of the prior $\mathcal{N}(\bar{\eta}_0, P_0)$ by $s_i$ and $p$ treated as independent observations with precisions $1/R_i^s$ and $h_p$, which has the stated mean and precision. The price errs by $\hat{\eta}_{\mathrm{pool}} - \eta$, of variance $1/\Lambda_{\mathrm{pool}}$, and by (b) the true posterior given $(s_i, p)$ has precision $\Lambda_{\mathrm{pool}}$, so the excess of the believed precision at $h_p = \Lambda_{\mathrm{pool}}$ is $1/P_0 + 1/R_i^s$. □

*Proof of Proposition S10.16.* If $\psi_i = M^\top \mathsf{c}_1$ for some vector $\mathsf{c}_1$, then $\psi_i^\top \eta = \mathsf{c}_1^\top M \eta$ is a function of the conditioning variables and its conditional variance is zero. Conversely, for jointly Gaussian variables a conditional variance vanishes exactly when the variable is almost surely an affine function of the conditioning variables, so $\psi_i^\top \eta = \mathsf{c}_1^\top M\eta + \mathsf{c}_2^\top (H_i^s \eta + v_i) + \mathsf{c}_0$ almost surely for some vectors $\mathsf{c}_1$, $\mathsf{c}_2$ and a constant $\mathsf{c}_0$. Taking the covariance of both sides with $v_i$, which is independent of $\eta$, gives $0 = R_i^s \mathsf{c}_2$, so $\mathsf{c}_2 = 0$. Then $(\psi_i - M^\top \mathsf{c}_1)^\top \eta$ is almost surely constant, and $P_0 \succ 0$ forces $\psi_i = M^\top \mathsf{c}_1$. A public message suffices for every agent if and only if every $\psi_i$ lies in $\operatorname{row}(M)$, so $\operatorname{rank} M \geq \dim \operatorname{span}\{\psi_i\}$, with equality for the matrix whose rows are a basis of $\operatorname{span}\{\psi_i\}$. □

*Proof of Proposition 10.6.* (a) Under rule C, stationarity gives $\mathbb{E}(\varphi_x - \varphi_{x_0})^2 = 2\sigma_\varphi^2(1 - \Gamma(x - x_0))$, so a member at $x$ pays $g(1 - \Gamma(x - x_0))$; summing over $C$, adding $f$ and dividing by $|C|$ gives $\mathcal{C}_{\mathrm{C}}$. Since $\Gamma \geq 0$, every finite sum $\sum_{x \in C} \Gamma(x - x_0)$ is at most $\chi$, so $\mathcal{C}_{\mathrm{C}} \to g$ whenever $|C| \to \infty$. Further, $\mathcal{C}_{\mathrm{C}}(C) < g$ if and only if $\sum_{x \in C} \Gamma(x - x_0) > f/g$; the supremum of these sums over finite $C \ni x_0$ is $\chi$, so such a $C$ exists if $f < g\chi$, and none exists if $f \geq g\chi$. At fixed size, $\mathcal{C}_{\mathrm{C}}$ decreases in $\sum_{x \in C} \Gamma(x - x_0)$, which the sites of largest $\Gamma(\cdot - x_0)$ maximize.

(b) Under rule M the policy is $\bar\varphi_C = |C|^{-1}\sum_{x\in C}\varphi_x$, and with $\mu_\varphi = \mathbb{E}\varphi_x$,

$$\sum_{x\in C}\mathbb{E}(\varphi_x-\bar\varphi_C)^2 = \sum_{x\in C}\mathbb{E}(\varphi_x-\mu_\varphi)^2 - |C|\operatorname{Var}(\bar\varphi_C) = |C|\sigma_\varphi^2 - \frac{\sigma_\varphi^2}{|C|}\sum_{x,y\in C}\Gamma(x-y).$$

Multiplying by $g/(2\sigma_\varphi^2)$, adding $f$ and dividing by $|C|$ gives $\mathcal{C}_{\mathrm{M}}$. For each $x$, $\sum_{y\in C}\Gamma(x-y)\le\chi$, so $\chi_C\le\chi$ and $\mathcal{C}_{\mathrm{M}}\to g/2$ whenever $|C|\to\infty$. Further, $\mathcal{C}_{\mathrm{M}}(C)<g/2$ if and only if $\chi_C>2f/g$. For boxes $C_n$ of side $n$, $\chi_{C_n}=\sum_z\Gamma(z)|C_n\cap(C_n-z)|/|C_n|\to\chi$ by dominated convergence, since each ratio of cardinalities tends to one; so if $f<g\chi/2$ some box has $\chi_C>2f/g$, and if $f\ge g\chi/2$ no finite $C$ does.

(c) Rule C: for $C=[-d/2,d/2]$ with $x_0=0$, $\int_C\Gamma = 2\xi(1-e^{-d/2\xi})$, so with $\varrho=d/(2\xi)$, $(\mathcal{C}_{\mathrm{C}}-g)/g = [\bar f_{\mathrm{C}}/2-1+e^{-\varrho}]/\varrho$. Its derivative in $\varrho$ has the sign of $1-\bar f_{\mathrm{C}}/2-e^{-\varrho}(1+\varrho)$, and $e^{-\varrho}(1+\varrho)$ decreases from 1 to 0 on $(0,\infty)$ (its derivative is $-\varrho e^{-\varrho}$), so the derivative vanishes at exactly one point $\varrho_{\mathrm{C}}$ when $0<\bar f_{\mathrm{C}}<2$. There the bracket equals $-\varrho_{\mathrm{C}}e^{-\varrho_{\mathrm{C}}}<0$; the bracket divided by $\varrho$ tends to $+\infty$ as $\varrho\to0$ and to 0 from below as $\varrho\to\infty$, so $\varrho_{\mathrm{C}}$ is the global minimizer and the minimum lies below $g$. If $\bar f_{\mathrm{C}}\ge2$ the bracket is positive and the derivative negative for every $\varrho$. The expansion $e^{-\varrho}(1+\varrho)=1-\varrho^2/2+O(\varrho^3)$ gives $\varrho_{\mathrm{C}}=\bar f_{\mathrm{C}}^{1/2}(1+O(\bar f_{\mathrm{C}}^{1/2}))$, that is $d^*_{\mathrm{C}}=2(f\xi/g)^{1/2}(1+O(\bar f_{\mathrm{C}}^{1/2}))$, and the root equation can be written $\varrho_{\mathrm{C}}-\ln(1+\varrho_{\mathrm{C}})=\ln[1/(1-\bar f_{\mathrm{C}}/2)]$, so $\varrho_{\mathrm{C}}$ diverges like the logarithm of $1/(1-\bar f_{\mathrm{C}}/2)$.

Rule M: $\int_0^d\int_0^d e^{-|x-y|/\xi}\mathrm{d}x\mathrm{d}y = 2\xi d-2\xi^2(1-e^{-d/\xi})$, so with $\varrho=d/\xi$, $2\mathcal{C}_{\mathrm{M}}/g = 1+[\bar f_{\mathrm{M}}-2+2(1-e^{-\varrho})/\varrho]/\varrho$. The derivative of the bracket divided by $\varrho$ vanishes exactly when $2-\bar f_{\mathrm{M}}=\psi_{\mathrm{M}}(\varrho)$. Now $\varrho^2\psi'_{\mathrm{M}}(\varrho)=e^{-\varrho}(4+4\varrho+2\varrho^2)-4$, which vanishes at $\varrho=0$ and has derivative $-2\varrho^2e^{-\varrho}<0$, so $\psi_{\mathrm{M}}$ decreases strictly from 2 to 0 on $(0,\infty)$ and the root $\varrho_{\mathrm{M}}$ is unique for $0<\bar f_{\mathrm{M}}<2$. The bracket divided by $\varrho$ tends to $+\infty$ as $\varrho\to0$ and to 0 from below as $\varrho\to\infty$, so the stationary point is the global minimizer and the minimum lies below $g/2$; if $\bar f_{\mathrm{M}}\ge2$ the bracket is positive for all $\varrho$. The expansions $\psi_{\mathrm{M}}(\varrho)=2-\varrho^2/3+O(\varrho^3)$ and $\psi_{\mathrm{M}}(\varrho)=4/\varrho+O(e^{-\varrho})$ give $\varrho_{\mathrm{M}}=(3\bar f_{\mathrm{M}})^{1/2}(1+O(\bar f_{\mathrm{M}}^{1/2}))$ and $\varrho_{\mathrm{M}}=4(2-\bar f_{\mathrm{M}})^{-1}(1+o(1))$, which are the stated limits.

(d) Put $\bar f=f/(g\xi)\in(0,1)$, so that $\bar f_{\mathrm{C}}=\bar f$ and $\bar f_{\mathrm{M}}=2\bar f<2$, and write $\varrho=\varrho_{\mathrm{C}}$. The root equation of rule C gives $2-\bar f_{\mathrm{M}}=2-2\bar f=4e^{-\varrho}(1+\varrho)-2$. Since $\psi_{\mathrm{M}}$ is strictly decreasing, $d^*_{\mathrm{M}}>d^*_{\mathrm{C}}$, that is $\varrho_{\mathrm{M}}>2\varrho$, holds if and only if $\psi_{\mathrm{M}}(2\varrho)>4e^{-\varrho}(1+\varrho)-2$. Multiplying the difference of the two sides by $\varrho e^{2\varrho}/2>0$ gives

$$\varrho e^{2\varrho}+e^{2\varrho}-1-\varrho-2\varrho(1+\varrho)e^{\varrho} = \sum_{\mathsf{n}\ge3}c_{\mathsf{n}}\varrho^{\mathsf{n}},\qquad c_{\mathsf{n}}=\frac{1}{(\mathsf{n}-1)!}\Big(2^{\mathsf{n}-1}+\frac{2^{\mathsf{n}}}{\mathsf{n}}-2\mathsf{n}\Big),$$

the coefficients of $\varrho^0$, $\varrho^1$ and $\varrho^2$ being zero. Here $c_3=\frac13$, and $c_{\mathsf{n}}>0$ for $\mathsf{n}\ge4$ because $2^{\mathsf{n}-1}\ge2\mathsf{n}$, so the series is positive for $\varrho>0$ and $d^*_{\mathrm{M}}>d^*_{\mathrm{C}}$. As $f\to0$, the expansions of (c) give $d^*_{\mathrm{M}}/d^*_{\mathrm{C}}\to(6f\xi/g)^{1/2}/(2(f\xi/g)^{1/2})=\sqrt6/2$; as $f\to g\xi$, $\bar f_{\mathrm{M}}\to2$ and $d^*_{\mathrm{M}}\to\infty$ while $d^*_{\mathrm{C}}$ tends to its finite value at $\bar f_{\mathrm{C}}=1$. □

*Proof of Proposition S10.20.* A member with home $x_i$ can visit a point $x$ and return within a round exactly when $2|x-x_i|\le v_{\mathrm{tr}}T_{\mathrm{tr}}$, which identifies the daily reach. Two members meet at $x$ within a round exactly when $x$ lies in both reaches. If they meet, $|x_i-x_j|\le|x_i-x|+|x-x_j|\le v_{\mathrm{tr}}T_{\mathrm{tr}}$; conversely, if $|x_i-x_j|\le v_{\mathrm{tr}}T_{\mathrm{tr}}$, the midpoint of their homes lies in both reaches. □

*Proof of Corollary S10.21.* Positions differ by at most $\ell_\Pi$ in every coordinate along every coupling, so $\varsigma_d^2=\max_i\sum_j a^\Pi_{ij}(x_{j,d}-x_{i,d})^2\le\ell_\Pi^2a^\Pi_{\max}$ and $\xi_{2,d}^2=\varsigma_d^2/(2\varepsilon^\Pi)\le\ell_\Pi^2a^\Pi_{\max}/(2\varepsilon^\Pi)$; Corollary 4.16 gives $L^*_d(r_0)\le\xi_{2,d}/(r_0-1)^{1/2}$. By the coupling formulas of Theorem 4.9, a coupling between agents

$i$ and $j$ is nonzero in the reversible class only if some agent $k$ attends both, and in the normal class only if both attend some agent $k$, self weights included; by the triangle inequality through $k$ the physical distance between $i$ and $j$ is then at most $2\ell_{\mathrm{att}}$. The last bound uses Proposition S10.20. $\square$

*Proof of Proposition S10.24.* (a) If $\Psi(x) = \Psi(1)x^{-\alpha_{\mathrm{el}}}$, then $\mathrm{el}_\Psi \equiv \alpha_{\mathrm{el}}$. Conversely, if for every $d$ the value $\mathrm{el}_\Psi(d/\xi)$ does not depend on $\xi > 0$, then $\mathrm{el}_\Psi$ is constant on $(0,\infty)$, say equal to $\alpha_{\mathrm{el}}$, and integrating $\Psi'/\Psi = -\alpha_{\mathrm{el}}/x$ gives $\Psi(x) = \Psi(1)x^{-\alpha_{\mathrm{el}}}$.

(b) We use the representation $K_\nu(x) = \int_0^\infty e^{-x\cosh \mathsf{s}} \cosh(\nu \mathsf{s}) \mathrm{d}\mathsf{s}$ for $x > 0$ (Olver et al., 2010, eq. 10.32.9). Differentiating under the integral gives $K_0' = -K_1$ and $K_0'' = \int_0^\infty e^{-x\cosh \mathsf{s}} \cosh^2 \mathsf{s}\, \mathrm{d}\mathsf{s}$. Since $\frac{\mathrm{d}}{\mathrm{d}\mathsf{s}}[-e^{-x\cosh \mathsf{s}} \sinh \mathsf{s}] = (x\sinh^2 \mathsf{s} - \cosh \mathsf{s}) e^{-x\cosh \mathsf{s}}$ and the bracket vanishes at both ends, $xK_0'' + K_0' - xK_0 = \int_0^\infty (x\sinh^2 \mathsf{s} - \cosh \mathsf{s}) e^{-x\cosh \mathsf{s}} \mathrm{d}\mathsf{s} = 0$. With $\mathrm{el}_{K_0} = -xK_0'/K_0$,

$$\mathrm{el}_{K_0}' = -\frac{K_0' + xK_0''}{K_0} + x\Big(\frac{K_0'}{K_0}\Big)^2 = -x + \frac{\mathrm{el}_{K_0}^2}{x} = \frac{\mathrm{el}_{K_0}^2 - x^2}{x}.$$

Since $K_1(x) - K_0(x) = \int_0^\infty (\cosh \mathsf{s} - 1) e^{-x\cosh \mathsf{s}} \mathrm{d}\mathsf{s} > 0$, we have $\mathrm{el}_{K_0}(x) > x > 0$, hence $\mathrm{el}_{K_0}' > 0$. As $x \to 0$, the substitution $\mathsf{v} = x\sinh \mathsf{s}$ gives $xK_1(x) = \int_0^\infty e^{-(x^2+\mathsf{v}^2)^{1/2}} \mathrm{d}\mathsf{v} \to 1$ by monotone convergence, while $K_0(x) \geq e^{-1} \operatorname{arccosh}(1/x) \to \infty$, because the integrand is at least $e^{-1}$ where $\cosh \mathsf{s} \leq 1/x$; so $\mathrm{el}_{K_0}(x) \to 0$. As $x \to \infty$, the expansion $K_\nu(x) = (\pi/2x)^{1/2} e^{-x}[1 + (4\nu^2 - 1)/(8x) + O(x^{-2})]$ (Olver et al., 2010, eqs. 10.40.2 and 10.17.1) gives $K_1/K_0 = 1 + 1/(2x) + O(x^{-2})$, so $\mathrm{el}_{K_0}(x) - x \to \frac{1}{2}$. Finally $\mathrm{el}(d) = \mathrm{el}_{K_0}(d/\xi)$, and $d/\xi$ decreases as $\xi$ increases.

(c) The substitution $\varrho = d/\xi$ turns $\int_0^\infty e^{-d/\xi} \xi^{-1-\alpha_{\mathrm{el}}} \mathrm{d}\xi$ into $d^{-\alpha_{\mathrm{el}}} \int_0^\infty \varrho^{\alpha_{\mathrm{el}}-1} e^{-\varrho} \mathrm{d}\varrho$ for every $d > 0$, which is finite for $\alpha_{\mathrm{el}} > 0$; the weight is not integrable at $\xi = 0$, and the profile diverges as $d \to 0$. $\square$

*Proof of Proposition 10.7.* (a) Let $\mathsf{B}_l(i,i') = b^{-l}\mathbf{1}[i,i' \text{ lie in the same level-}l \text{ block}]$ for $0 \leq l \leq n_{\mathrm{H}}$, the orthogonal projector onto functions constant on level-$l$ blocks, with $\mathsf{B}_0 = I$. For $1 \leq l \leq n_{\mathrm{H}}$, the range of $\mathsf{B}_{l-1} - \mathsf{B}_l$ consists of functions constant on level-$(l-1)$ blocks with zero sum on every level-$l$ block, and it is spanned by such functions supported on single level-$l$ blocks. Let $f$ be one of them, supported on the block $\mathfrak{b}$. For $i \in \mathfrak{b}$, the cousins of $i$ at levels below $l$ lie in $i$'s level-$(l-1)$ block and share the value $f_i$; the cousins at level $l$ are the rest of $\mathfrak{b}$ and carry in total $0 - b^{l-1} f_i$; and the cousins at higher levels lie outside $\mathfrak{b}$. Hence

$$(Lf)_i = \sum_{l'=1}^{n_{\mathrm{H}}} A_{\mathrm{h}} \mathfrak{q}^{l'-1} f_i - \sum_{l'<l} A_{\mathrm{h}} \mathfrak{q}^{l'-1} f_i + a_l^{\mathrm{H}} b^{l-1} f_i = \mathsf{g}_l f_i,$$

using $a_l^{\mathrm{H}} b^{l-1} = A_{\mathrm{h}} \mathfrak{q}^{l-1}/(b-1)$. For $i \notin \mathfrak{b}$, all members of $\mathfrak{b}$ are cousins of $i$ at one level, so $(Lf)_i$ is a multiple of $\sum_{\mathfrak{b}} f = 0$. Constants have eigenvalue 0, and $\mathsf{B}_{n_{\mathrm{H}}} + \sum_{l=1}^{n_{\mathrm{H}}} (\mathsf{B}_{l-1} - \mathsf{B}_l) = I$, so

$$\mathsf{G} = \frac{\mathsf{B}_{n_{\mathrm{H}}}}{\varepsilon^{\Pi}} + \sum_{l=1}^{n_{\mathrm{H}}} \frac{\mathsf{B}_{l-1} - \mathsf{B}_l}{\varepsilon^{\Pi} + \mathsf{g}_l}.$$

For a pair at level $l \geq 1$, $\mathsf{B}_{l''}(i,i') = b^{-l''}$ for $l'' \geq l$ and 0 for $l'' < l$, so $(\mathsf{B}_{l'-1} - \mathsf{B}_{l'})(i,i')$ is $(b-1)b^{-l'}$ for $l' > l$, $-b^{-l}$ for $l' = l$ and 0 for $l' < l$; on the diagonal it is $(b-1)b^{-l'}$ for every $l' \geq 1$. This gives (a).

(b) As $n_{\mathrm{H}} \to \infty$ at fixed $\varepsilon^{\Pi} > 0$, the constant term $b^{-n_{\mathrm{H}}}/\varepsilon^{\Pi}$ tends to zero, $\mathsf{g}_l$ increases to $\mathsf{c}_\infty A_{\mathrm{h}} \mathfrak{q}^{l-1}$ with $\mathsf{c}_\infty = 1/(1-\mathfrak{q}) + 1/(b-1)$ (here $\mathfrak{q} < 1$ is used), and the terms $(b-1)b^{-l'}/(\varepsilon^{\Pi} + \mathsf{g}_{l'})$ are dominated by the summable $(b-1)b^{-l'}/\varepsilon^{\Pi}$, so the sums converge to their infinite versions. As

$\varepsilon^{\Pi} \to 0$ these increase to sums with terms $(b-1)b^{-l'}/(\mathsf{c}_\infty A_{\mathrm{h}}\mathfrak{q}^{l'-1}) = (b-1)\mathfrak{q}(b\mathfrak{q})^{-l'}/(\mathsf{c}_\infty A_{\mathrm{h}})$, which are summable because $b\mathfrak{q} > 1$; by monotone convergence the limits are

$$\mathsf{G}(l) \to \frac{\mathfrak{q}}{\mathsf{c}_\infty A_{\mathrm{h}}}\Big[(b-1)\sum_{l'>l}(b\mathfrak{q})^{-l'} - (b\mathfrak{q})^{-l}\Big] = \frac{\mathfrak{q}}{\mathsf{c}_\infty A_{\mathrm{h}}}\frac{b(1-\mathfrak{q})}{b\mathfrak{q}-1}(b\mathfrak{q})^{-l}, \qquad \mathsf{G}(0) \to \frac{\mathfrak{q}}{\mathsf{c}_\infty A_{\mathrm{h}}}\frac{b-1}{b\mathfrak{q}-1},$$

and their ratio is the stated limit.

(c) With $l - 1 = D_e \ln(d_l/d_1)/\ln b$, $(b\mathfrak{q})^{-l} = (b\mathfrak{q})^{-1}(d_l/d_1)^{-D_e \ln(b\mathfrak{q})/\ln b}$. The limit in (b) does not involve $A_{\mathrm{h}}$, and a change of the unit of distance multiplies $d_l$ and $d_1$ by the same factor. □

*Proof of Proposition S10.27.* For scalar errors the unconditional correlation is

$$\Gamma_{ij} = \frac{\Sigma_{\mathrm{cm}} + \Sigma^c_{ij}}{\big((\Sigma_{\mathrm{cm}} + \Sigma^c_{ii})(\Sigma_{\mathrm{cm}} + \Sigma^c_{jj})\big)^{1/2}} \geq \frac{\Sigma_{\mathrm{cm}}}{\Sigma_{\mathrm{cm}} + \max_k \Sigma^c_{kk}},$$

because $\Sigma^c_{ij} \geq 0$. Every weight $-\ln \Gamma_{ij}$ is therefore at most $\ln(1 + \max_k \Sigma^c_{kk}/\Sigma_{\mathrm{cm}})$. For a nonempty $C \neq V$ this bounds $\varpi_{\mathrm{out}}(C)$, a minimum over a nonempty set of pairs, while $u_{\mathrm{in}}(C) \geq 0$. For the level-0 network, Lemma 2.13 splits the error into the common response to the world innovations and a part driven by local noise, independent of each other; with homogeneous row sums $W\mathbf{1} = \lambda\mathbf{1}$, the common part is $-\sum_{n\geq 0}\lambda^n w_{t-1-n}\mathbf{1}$, with variance $Q/(1-\lambda^2)$, and $W \geq 0$ makes $\Sigma^c = \sum_n W^n \operatorname{diag}(q_i)(W^\top)^n$ entrywise nonnegative for local noise variances $q_i$. The cap equals $\ln(1 + (1-\lambda^2)\max_k \Sigma^c_{kk}/Q)$. For $W = \lambda P$ with $P$ irreducible, aperiodic and stochastic, $P^n = \mathbf{1}\pi^\top + R_n$ with $\|R_n\| \leq c\rho^n$ for some $c$ and $\rho < 1$ that depend on $P$ alone, so

$$\Sigma^c = \sum_{n\geq 0}\lambda^{2n}P^n Q^\zeta (P^\top)^n = \frac{\pi^\top Q^\zeta \pi}{1-\lambda^2}\mathbf{1}\mathbf{1}^\top + B_\lambda, \qquad B_\lambda = \sum_{n\geq 0}\lambda^{2n}\big(\mathbf{1}\pi^\top Q^\zeta R_n^\top + R_n Q^\zeta \pi \mathbf{1}^\top + R_n Q^\zeta R_n^\top\big),$$

and $\|B_\lambda\|$ is bounded uniformly in $\lambda \leq 1$ by a convergent geometric series. Hence $(1-\lambda^2)\Sigma^c_{kk} \to \pi^\top Q^\zeta \pi$ for every $k$, and the cap tends to $\ln(1 + \pi^\top Q^\zeta \pi/Q)$. For symmetric $P$ the stationary vector is $\mathbf{1}/N$, and $Q^\zeta = qI$ gives $\pi^\top Q^\zeta \pi = q/N$. Aperiodicity is needed: for the swap of two agents with $Q^\zeta = \operatorname{diag}(q_1, q_2)$ the limit is $(q_1+q_2)/2$ against $\pi^\top Q^\zeta \pi = (q_1+q_2)/4$. For the bound at every $\lambda$, let $P$ be symmetric and doubly stochastic and $Q^\zeta = qI$, and write $P = \sum_k \mu_k v_k v_k^\top$ with orthonormal eigenvectors, $v_1 = \mathbf{1}/\sqrt{N}$, $\mu_1 = 1$ and $|\mu_k| \leq 1$. Then $\Sigma^c = q\sum_{n\geq 0}\lambda^{2n}P^{2n} = q\sum_k (1-\lambda^2\mu_k^2)^{-1} v_k v_k^\top$, every term has a nonnegative diagonal, and the term $k = 1$ contributes $q/(N(1-\lambda^2))$ to every diagonal entry, so $(1-\lambda^2)\Sigma^c_{ii} \geq q/N$ for every $i$ and every $\lambda \in (0,1)$; neither irreducibility nor aperiodicity is used. □

## S35 Proofs for Section 11

The proofs follow the order of Section 11. Conditional independence and leaks of Gaussian paths are read from precisions by two standard facts, which the proof of Theorem 4.3 also uses. Let $(\mathsf{U}, Y, Z)$ be a nondegenerate Gaussian vector with precision $J$. Then $\mathsf{U} \perp Z \mid Y$ if and only if $J_{\mathsf{U}Z} = 0$, and with $M = J_{\mathsf{U}\mathsf{U}}^{-1/2} J_{\mathsf{U}Z} J_{ZZ}^{-1/2}$,

$$\|M\| < 1, \qquad I(\mathsf{U}; Z \mid Y) = -\frac{1}{2}\ln\det(I - MM^\top), \qquad \frac{1}{2}\|M\|_F^2 \leq I(\mathsf{U}; Z \mid Y) \leq \frac{\|M\|_F^2}{2(1-\|M\|^2)}. \tag{S118}$$

Both hold because the law of $(\mathsf{U}, Z)$ given $Y$ has precision equal to the $(\mathsf{U}, Z)$ block of $J$: a Gaussian vector has independent components exactly when its precision is block diagonal, and $I(\mathsf{U}; Z \mid Y) =$

$\frac{1}{2}\ln\det J_{\mathsf{UU}} - \frac{1}{2}\ln\det(J_{\mathsf{UU}} - J_{\mathsf{U}Z}J_{ZZ}^{-1}J_{Z\mathsf{U}})$ with $I - MM^\top = J_{\mathsf{UU}}^{-1/2}(J_{\mathsf{UU}} - J_{\mathsf{U}Z}J_{ZZ}^{-1}J_{Z\mathsf{U}})J_{\mathsf{UU}}^{-1/2} \succ 0$; the bounds follow from $x \le -\ln(1-x) \le x/(1-x)$ on $[0,1)$, applied to the eigenvalues of $MM^\top$. Path precisions have the block-tridiagonal form of the proof of Theorem 4.3(a). One auxiliary lemma, on the steady state of the coarse filter, is proved where it is first needed.

*Proof of Theorem 11.3.* (a) The map $\mathcal{S}$ is invertible and does not depend on the round, so $\tilde{X} = \mathcal{S}X$ follows (23) with $\tilde{A} = \mathcal{S}A\mathcal{S}^{-1}$, noise $\mathcal{S}\Xi_t$ of covariance $\tilde{Q} = \mathcal{S}Q^{\Xi}\mathcal{S}^\top \succ 0$, independent across rounds, and a Gaussian start with covariance $\mathcal{S}\Sigma_0\mathcal{S}^\top \succ 0$. The proof of Theorem 4.3(a) uses nothing else, so it applies to $\tilde{X}$ and the partition $(\mathsf{I}', \mathsf{B}', \mathsf{E}')$, and its independent-noise form gives (95). Conditional independence of $(X_{\mathsf{I}}, \mathfrak{r}_{\mathrm{in}})$ and $(\mathfrak{r}_{\mathrm{out}}, X_{\mathsf{E}})$ given $b_C$ implies that of their components $X_{\mathsf{I}}$ and $X_{\mathsf{E}}$, and functions of conditionally independent variables are conditionally independent; this gives the strong and the weak coarse blankets.

(b) The path of $(X_{\mathsf{I}}, b_C, X_{\mathsf{E}})$ is a marginal of the nondegenerate Gaussian path of $(X_{\mathsf{I}}, b_C, \mathfrak{r}, X_{\mathsf{E}})$, so its precision is the Schur complement $J/J_{\mathfrak{r}\mathfrak{r}}$, whose interior-exterior block is $J_{\mathsf{IE}} - J_{\mathsf{I}\mathfrak{r}}J_{\mathfrak{r}\mathfrak{r}}^{-1}J_{\mathfrak{r}\mathsf{E}}$. The first Gaussian fact above gives the criterion, and (S118), applied to the precision $J^{\mathrm{b}}$ of the paths of $(X_{\mathsf{I}}, X_{\mathsf{E}})$ given $b_{C,0:T}$, gives the leak. The change of coordinates from $X_{\mathsf{B}}$ to $(b_C, \mathfrak{r})$ is block diagonal over the partition and the identity on $X_{\mathsf{I}}$ and $X_{\mathsf{E}}$, so the interior-exterior block of the path precision is the same in the coordinates $(X_{\mathsf{I}}, X_{\mathsf{B}}, X_{\mathsf{E}})$; under the fine path blanket it vanishes by Theorem 4.3(a). □

*Proof of Proposition 11.4.* (a) Invertibility: if $c^\top\mathcal{H}^\top R_{\mathsf{S}}^{-1} + d^\top \mathsf{Z}_{\mathsf{S}} = 0$, right multiplication by $\mathcal{H}$ gives $c^\top\Lambda^s = 0$ with $\Lambda^s \succ 0$, so $c = 0$ and then $d = 0$, and the numbers of rows add to $\dim X_{\mathsf{S}}$; right multiplication by $Q^{\Xi}_{\mathsf{AA}}\mathsf{P}_{\mathrm{A}}^\top$ shows in the same way that $(\mathsf{P}_{\mathrm{A}}; \mathsf{Z}_{\mathrm{A}})$ is invertible. Write $(\mathsf{P}_{\mathrm{A}}^+ \ \mathsf{Z}_{\mathrm{A}}^+)$ for its inverse, so that $X_{\mathsf{A}} = \mathsf{P}_{\mathrm{A}}^+ a_C + \mathsf{Z}_{\mathrm{A}}^+\mathfrak{r}_{\mathrm{A}}$, $\mathsf{P}_{\mathrm{A}}\mathsf{P}_{\mathrm{A}}^+ = I$ and $\mathsf{P}_{\mathrm{A}}\mathsf{Z}_{\mathrm{A}}^+ = 0$.

Noise: with per-channel noise $Q^{\Xi}$ is block diagonal over $(\mathsf{I}, \mathsf{S}, \mathsf{A}, \mathsf{E})$, and $\mathrm{Cov}(\mathsf{Z}_{\mathsf{S}}\Xi^{\mathsf{S}}, \mathcal{H}^\top R_{\mathsf{S}}^{-1}\Xi^{\mathsf{S}}) = \mathsf{Z}_{\mathsf{S}}\mathcal{H} = 0$ and $\mathrm{Cov}(\mathsf{Z}_{\mathrm{A}}\Xi^{\mathsf{A}}, \mathsf{P}_{\mathrm{A}}\Xi^{\mathsf{A}}) = \mathsf{Z}_{\mathrm{A}}Q^{\Xi}_{\mathsf{AA}}\mathsf{P}_{\mathrm{A}}^\top = 0$. Hence $\tilde{Q}$ is block diagonal for $\mathsf{I}' = (X_{\mathsf{I}}, \mathfrak{r}_{\mathsf{S}}, \mathfrak{r}_{\mathrm{A}})$, $\mathsf{B}' = (s_C, a_C)$ and $\mathsf{E}' = X_{\mathsf{E}}$, with $\tilde{Q}_{\mathsf{B}'\mathsf{B}'} = \mathrm{diag}(\Lambda^s, \mathsf{P}_{\mathrm{A}}Q^{\Xi}_{\mathsf{AA}}\mathsf{P}_{\mathrm{A}}^\top)$.

No variable of $\mathsf{I}'$ reads the exterior: internal and active variables read no exterior variable (Definition 4.1), and by (a1) $\mathfrak{r}_{\mathsf{S},t+1} = \mathsf{Z}_{\mathsf{S}}A_{\mathsf{SS}}X_{\mathsf{S},t} + \mathsf{Z}_{\mathsf{S}}\mathcal{H}\mathsf{G}_{\mathrm{E}}X_{\mathsf{E},t} + \mathsf{Z}_{\mathsf{S}}\Xi_t^{\mathsf{S}}$ with $\mathsf{Z}_{\mathsf{S}}\mathcal{H} = 0$. The exterior reads no variable of $\mathsf{I}'$: it reads no internal variable, no sensory channel by (a2), and the active channels only through $A_{\mathsf{EA}}X_{\mathsf{A}} = \mathsf{L}_{\mathrm{E}}\mathsf{P}_{\mathrm{A}}(\mathsf{P}_{\mathrm{A}}^+ a_C + \mathsf{Z}_{\mathrm{A}}^+\mathfrak{r}_{\mathrm{A}}) = \mathsf{L}_{\mathrm{E}}a_C$. Hence $\tilde{A}_{\mathsf{I}'\mathsf{E}'} = 0$ and $\tilde{A}_{\mathsf{E}'\mathsf{I}'} = 0$. By (a1), $s_{C,t+1} = \mathsf{P}_{\mathsf{S}}A_{\mathsf{SS}}X_{\mathsf{S},t} + \Lambda^s\mathsf{G}_{\mathrm{E}}X_{\mathsf{E},t} + \mathcal{H}^\top R_{\mathsf{S}}^{-1}\Xi_t^{\mathsf{S}} = \mathcal{C}s_{C,t} + \Lambda^s\mathsf{G}_{\mathrm{E}}X_{\mathsf{E},t} + \mathcal{H}^\top R_{\mathsf{S}}^{-1}\Xi_t^{\mathsf{S}}$, so $s_C$ reads no variable of $\mathsf{I}'$, while $a_C$ reads no exterior variable. With $\tilde{Q}_{\mathsf{B}'\mathsf{B}'}$ block diagonal the collider term is $\tilde{A}_{s\mathsf{I}'}^\top(\Lambda^s)^{-1}\tilde{A}_{s\mathsf{E}'} + \tilde{A}_{a\mathsf{I}'}^\top(\mathsf{P}_{\mathrm{A}}Q^{\Xi}_{\mathsf{AA}}\mathsf{P}_{\mathrm{A}}^\top)^{-1}\tilde{A}_{a\mathsf{E}'} = 0$, since $\tilde{A}_{s\mathsf{I}'} = 0$ and $\tilde{A}_{a\mathsf{E}'} = 0$. These conclusions hold whatever the blocks $\tilde{A}_{\mathsf{I}'\mathsf{B}'}$, $\tilde{A}_{\mathsf{I}'\mathsf{I}'}$, $\tilde{A}_{a\mathsf{I}'}$ and $\tilde{A}_{a\mathsf{B}'}$ are. With (a3), Theorem 11.3(a) applies. For memoryless channels $\mathfrak{r}_{\mathsf{S},t+1} = \mathsf{Z}_{\mathsf{S}}\Xi_t^{\mathsf{S}}$, independent of the exterior and, its covariance with the noise of $s_C$ being $\mathsf{Z}_{\mathsf{S}}\mathcal{H} = 0$, of $s_C$.

(b) By (b1) to (b3) and Definition 4.1,

$$X_{\mathsf{I},t+1} = A_{\mathsf{II}}X_{\mathsf{I},t} + \mathcal{B}_{\mathsf{I}}b_{C,t} + \Xi_t^{\mathsf{I}}, \qquad X_{\mathsf{E},t+1} = A_{\mathsf{EE}}X_{\mathsf{E},t} + \mathcal{B}_{\mathsf{E}}b_{C,t} + \Xi_t^{\mathsf{E}},$$
$$b_{C,t+1} = \mathsf{P}_{\mathrm{B}}A_{\mathsf{BI}}X_{\mathsf{I},t} + \mathcal{A}_{bb}b_{C,t} + \mathsf{P}_{\mathrm{B}}A_{\mathsf{BE}}X_{\mathsf{E},t} + \mathsf{P}_{\mathrm{B}}\Xi_t^{\mathsf{B}}.$$

The reduced triple at $t+1$ is an affine function of the triple at $t$ plus noise independent of $X_{0:t}$, so by induction its path law is that of a linear-Gaussian chain started from the marginal of $X_0$, whose covariance is positive definite. Its noise $(\Xi^{\mathsf{I}}, \mathsf{P}_{\mathrm{B}}\Xi^{\mathsf{B}}, \Xi^{\mathsf{E}})$ is independent across the three blocks and across rounds, with positive definite covariance. Theorem 4.3(a) in its independent-noise form applies to the reduced chain: its first two conditions hold because interior and exterior do not read each other, the third is (b4) and the fourth is (b5). If $s_C$ reads no interior variable and $a_C$ no

exterior one, $\mathsf{P}_{\mathrm{B}} A_{\mathsf{BI}}$ has a zero $s_C$ block and $\mathsf{P}_{\mathrm{B}} A_{\mathsf{BE}}$ a zero $a_C$ block, and with a block-diagonal noise covariance of $b_C$ the product in (b4) vanishes.

(c) The rows of $\mathsf{Z}$ span the left null space of $R_{\mathsf{S}}\mathsf{P}_{\mathrm{S}}^{\top}$, so $\ker \mathsf{Z} \supseteq \operatorname{range}(R_{\mathsf{S}}\mathsf{P}_{\mathrm{S}}^{\top})$, and both spaces have dimension $\operatorname{rank} \mathsf{P}_{\mathrm{S}}$; they are equal. The residual reads the exterior through $\mathsf{Z}A_{\mathsf{SE}} = \mathsf{Z}\mathcal{H}\mathsf{G}_{\mathrm{E}}$, which vanishes if and only if $\mathsf{Z}\mathcal{H} = 0$, because $\mathsf{G}_{\mathrm{E}}$ has full row rank. Now $\mathsf{Z}\mathcal{H} = 0$ if and only if $\operatorname{range} \mathcal{H} \subseteq \operatorname{range}(R_{\mathsf{S}}\mathsf{P}_{\mathrm{S}}^{\top})$, that is, $\operatorname{range}(R_{\mathsf{S}}^{-1}\mathcal{H}) \subseteq \operatorname{range}(\mathsf{P}_{\mathrm{S}}^{\top})$, which is the stated inclusion of row spaces. □

*Computation for Example S11.2.* The joint process is $(y, X_1, X_2, \eta)$ with $A_{yX_k} = g_k$, $A_{X_k\eta} = 1$ and all other entries zero. For $t \geq 1$ the blocks $G_t = (\eta_{t-1}, X_{1,t}, X_{2,t}, y_{t+1})$ are functions of the disjoint sets of noises $(\eta_{t-1}, v_{1,t-1}, v_{2,t-1}, \mathfrak{n}_t)$ and are therefore independent, and $s_t$ lies in $G_t$. The conditional mutual information of the interior and exterior paths given the path of $s$ is therefore a sum over blocks, two boundary terms plus $T-1$ equal terms $I(y_{t+1}; \eta_{t-1} \mid s_t)$, as in Example S4.2, and the rate is the one-block term. In one block $\operatorname{Var}\eta = 1$, $\operatorname{Cov}(X_k, \eta) = 1$, $\operatorname{Var} X_k = 1 + R_k^s$ and $\operatorname{Cov}(X_1, X_2) = 1$, so $\operatorname{Var} s = 4 + R_1^s + R_2^s$, $\operatorname{Cov}(s, \eta) = 2$, $\operatorname{Cov}(y, \eta) = g_1 + g_2$, $\operatorname{Cov}(y, s) = g_1(2 + R_1^s) + g_2(2 + R_2^s)$ and $\operatorname{Var} y = g_1^2(1 + R_1^s) + g_2^2(1 + R_2^s) + 2g_1g_2 + q$. The partial covariance is $(g_1+g_2) - 2(g_1(2+R_1^s)+g_2(2+R_2^s))/(4+R_1^s+R_2^s) = (g_1-g_2)(R_2^s-R_1^s)/(4+R_1^s+R_2^s)$, the conditional variances follow in the same way, and the Gaussian formula for the conditional mutual information of two scalars gives (S43). The interior is $y$ itself and the exterior is $\eta$, so the weak and the strong leaks coincide. □

*Proof of Proposition 11.5.* The path precision of $\tilde{X}$ on $[0, T]$ is $J = \mathcal{J}^{-\top}\mathcal{Q}^{-1}\mathcal{J}^{-1}$, where $(\mathcal{J}^{-1}\tilde{X})_0 = \tilde{X}_0$, $(\mathcal{J}^{-1}\tilde{X})_t = \tilde{X}_t - \tilde{A}\tilde{X}_{t-1}$ for $t \geq 1$, and $\mathcal{Q} = \operatorname{diag}(\tilde{\Sigma}_0, \tilde{Q}, \ldots, \tilde{Q})$. Let $\mathsf{U} = X_{\mathsf{I}',0:T}$ and $Z = X_{\mathsf{E}',0:T}$. By block diagonality of $\tilde{Q}$, the same-round blocks of $J_{\mathsf{U}Z}$ are $(\tilde{\Sigma}_0^{-1})_{\mathsf{I}'\mathsf{E}'} + \mathsf{D}_{\mathrm{c}} = \mathsf{D}_{\mathrm{c}}$ at $t = 0$, $\mathsf{D}_{\mathrm{c}}$ for $1 \leq t \leq T-1$ and $(\tilde{Q}^{-1})_{\mathsf{I}'\mathsf{E}'} = 0$ at $t = T$, and its lag blocks are $-\mathsf{D}_1$ (interior at $t+1$, exterior at $t$) and $-\mathsf{D}_2^{\top}$ (interior at $t$, exterior at $t+1$). Hence $\|J_{\mathsf{U}Z}\| \leq \mathsf{k}$ and $\|J_{\mathsf{U}Z}\|_F^2 = T\mathsf{k}_F^2$. For a path vector $x$ supported on the $\mathsf{I}'$ coordinates, $x^{\top}J_{\mathsf{U}\mathsf{U}}x = |\mathcal{Q}^{-1/2}\mathcal{J}^{-1}x|^2$, and dropping the nonnegative contributions of the $\mathsf{B}'$ and $\mathsf{E}'$ rows gives $x^{\top}J_{\mathsf{U}\mathsf{U}}x \geq \mathfrak{v}_{\mathsf{I}'}^{-1}\sum_{t=0}^{T}|(\mathcal{D}x)_t|^2$, with $(\mathcal{D}x)_0 = x_0$ and $(\mathcal{D}x)_t = x_t - \tilde{A}_{\mathsf{I}'\mathsf{I}'}x_{t-1}$. The inverse of $\mathcal{D}$ is block lower triangular Toeplitz with blocks $\tilde{A}^l_{\mathsf{I}'\mathsf{I}'}$, whose operator norm is at most $\mathfrak{s}_{\mathsf{I}'}$, so $\lambda_{\min}(J_{\mathsf{U}\mathsf{U}}) \geq 1/(\mathfrak{v}_{\mathsf{I}'}\mathfrak{s}_{\mathsf{I}'}^2)$; likewise $\lambda_{\min}(J_{ZZ}) \geq 1/(\mathfrak{v}_{\mathsf{E}'}\mathfrak{s}_{\mathsf{E}'}^2)$. With $M = J_{\mathsf{U}\mathsf{U}}^{-1/2}J_{\mathsf{U}Z}J_{ZZ}^{-1/2}$, $\|M\| \leq \mathfrak{c}_C\mathsf{k} < 1$ and $\|M\|_F^2 \leq \mathfrak{c}_C^2 T\mathsf{k}_F^2$, and (S118) gives $I(\mathsf{U}; Z \mid b_{C,0:T}) \leq \|M\|_F^2/(2(1-\|M\|^2))$, the second inequality of (97). The first is monotonicity of conditional mutual information under taking components. Under (95) the three blocks vanish, and each is linear in $\tilde{A}_{\mathsf{I}'\mathsf{E}'}$, $\tilde{A}_{\mathsf{E}'\mathsf{I}'}$ and $\tilde{A}_{\mathsf{B}'\mathsf{I}'}^{\top}\tilde{Q}_{\mathsf{B}'}^{-1}\tilde{A}_{\mathsf{B}'\mathsf{E}'}$ when the other blocks are fixed. □

*Proof of Proposition S11.8.* (a) Given $x$, $y' \sim \mathcal{N}(\mathsf{G}x, Q^{\mathfrak{u}})$; given $\mathsf{c}$, $x$ has covariance $\Sigma_{x|\mathsf{c}}$ and $\mathfrak{u}$ is independent of $x$, so $\operatorname{Cov}(y' \mid \mathsf{c}) = Q^{\mathfrak{u}} + \mathsf{G}\Sigma_{x|\mathsf{c}}\mathsf{G}^{\top}$. Since $\mathsf{c}$ is a function of $x$, $I(y'; x \mid \mathsf{c}) = h(y' \mid \mathsf{c}) - h(y' \mid x)$, the stated half log-determinant. The matrix in the determinant is the identity plus a matrix similar to a positive semidefinite one, so the information vanishes exactly when $\mathsf{G}\Sigma_{x|\mathsf{c}}\mathsf{G}^{\top} = 0$, that is, when $\mathsf{G}$ vanishes on the range of $\Sigma_{x|\mathsf{c}}$. Writing $\Sigma_{x|\mathsf{c}} = \Sigma_x^{1/2}(I - \mathsf{P}_{\mathcal{L}})\Sigma_x^{1/2}$ with $\mathsf{P}_{\mathcal{L}}$ the orthogonal projector onto $\mathcal{L} = \operatorname{range}(\Sigma_x^{1/2}\mathcal{P}^{\top})$, a vector $f$ lies in $\mathcal{L}^{\perp}$ exactly when $\mathcal{P}\Sigma_x^{1/2}f = 0$, so the range of $\Sigma_{x|\mathsf{c}}$ is $\Sigma_x^{1/2}\mathcal{L}^{\perp} = \ker \mathcal{P}$, as in the proof of Theorem 5.3(c). A linear map vanishing on $\ker \mathcal{P}$ factors through $\mathcal{P}$. For $\mathfrak{d}_{\mathrm{cl}}^C$ take $x = (X_{\mathsf{I},t}, X_{\mathsf{B},t})$, $\mathsf{G} = \mathsf{P}_{\mathrm{I}}(A_{\mathsf{II}}\ A_{\mathsf{IB}})$, $\mathcal{P} = \operatorname{diag}(\mathsf{P}_{\mathrm{I}}, \mathsf{P}_{\mathrm{B}})$ and $\mathfrak{u} = \mathsf{P}_{\mathrm{I}}\Xi_t^{\mathsf{I}}$, using that the interior reads no exterior variable; for $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}}$ take $x = (X_{\mathsf{E},t}, X_{\mathsf{B},t})$, $\mathsf{G} = \mathsf{P}_{\mathrm{E}}(A_{\mathsf{EE}}\ A_{\mathsf{EB}})$, $\mathsf{c} = (\eta_{C,t}, a_{C,t})$ and $\mathfrak{u} = \mathsf{P}_{\mathrm{E}}\Xi_t^{\mathsf{E}}$. In both cases $\Sigma_x \succ 0$ because the stationary covariance satisfies $\Sigma \succeq Q^{\Xi} \succ 0$, and $\mathsf{G}$ vanishes on $\ker \mathcal{P}$ exactly under the conditions of Definition 11.6(i) or (ii).

(b) Given $x_t$ and the input, $y_{t+1} = \mathsf{P}Ax_t + \mathsf{P}Bx_{\mathsf{B},t} + \mathsf{P}\mathfrak{n}_t$ with $\mathfrak{n}_t$ independent of $(x_{0:t}, \mathcal{Z}_t)$, so $\mathfrak{d}_{\mathrm{dyn},t} = I(y_{t+1}; x_t \mid \mathcal{Z}_t)$. The conditional law of $x_t$ given $\mathcal{Z}_t$ is Gaussian with the data-independent covariance $P_t^{\mathrm{f}}$ (Theorem 5.3(d)), which gives the formula as in (a). Without input, $\mathfrak{d}_{\mathrm{cl}}$ is the same expression with $P_t^{\mathrm{f}}$ replaced by $\mathrm{Var}(x_t \mid y_t) \succeq P_t^{\mathrm{f}}$, and $X \mapsto \ln\det(I + S^{-1}X)$ is nondecreasing on positive semidefinite $X$ for $S \succ 0$; hence $\mathfrak{d}_{\mathrm{dyn},t} \le \mathfrak{d}_{\mathrm{cl}}$. In the coordinates $(y, x_\perp)$ of Theorem 5.2, $y_t$ is known given $\mathcal{Z}_t$, so $\mathsf{P}AP_t^{\mathrm{f}}A^\top\mathsf{P}^\top = A_{y\perp}\,\mathrm{Var}(x_{\perp,t} \mid \mathcal{Z}_t)A_{y\perp}^\top$. Let $\mathfrak{G}$ be generated by $x_0$, $\mathfrak{n}_{0:t-2}$, $\mathfrak{n}_{t-1}^y = \mathsf{P}\mathfrak{n}_{t-1}$ and the input. Then $\mathcal{Z}_t$ is $\mathfrak{G}$-measurable, because $y_t$ involves $\mathfrak{n}_{t-1}$ only through $\mathfrak{n}_{t-1}^y$, and given $\mathfrak{G}$, $x_{\perp,t}$ is a known vector plus $\mathfrak{n}_{t-1}^\perp$. By the law of total variance $\mathrm{Var}(x_{\perp,t} \mid \mathcal{Z}_t) \succeq \mathrm{Var}(\mathfrak{n}_{t-1}^\perp \mid \mathfrak{n}_{t-1}^y) \succ 0$, the last because $Q^{\mathfrak{n}} \succ 0$. Hence $\mathfrak{d}_{\mathrm{dyn},t} = 0$ if and only if $A_{y\perp} = 0$.

(c) Kernel invariance is $A_{y\perp} = 0$ and is equivalent to $\mathfrak{d}_{\mathrm{cl}} = 0$ by Theorem 5.3(b), and to $\mathfrak{d}_{\mathrm{dyn},t} = 0$ for $t \ge 1$ by (b). Without input $y_{0:t}$ is a function of $x_{0:t}$, so $I(y_{t+1}; x_{0:t}) - I(y_{t+1}; y_{0:t}) = \mathfrak{d}_{\mathrm{dyn},t}$, and one-step informational closure is $\mathfrak{d}_{\mathrm{dyn},t} = 0$ for every $t$. The inequality $\mathfrak{d}_{\mathrm{M}} \le \mathfrak{d}_{\mathrm{cl}}$ is Theorem 5.3(c), and $\mathfrak{d}_{\mathrm{dyn},t} \le \mathfrak{d}_{\mathrm{cl}}$ is (b). □

*Proof of Theorem 11.8.* The exterior obeys $X_{\mathsf{E},t+1} = A_{\mathsf{EE}}X_{\mathsf{E},t} + A_{\mathsf{EB}}X_{\mathsf{B},t} + \Xi_t^{\mathsf{E}}$, the recursion of Definition 5.1 with $x = X_{\mathsf{E}}$, $A = A_{\mathsf{EE}}$, $B = A_{\mathsf{EB}}$ and the blanket as input.

(a)⇔(b). Complete $\mathsf{P}_{\mathrm{E}}$ to an invertible matrix and write the exterior in the coordinates $(\eta_C, x_\perp)$. Theorem 5.2 and Theorem 5.3(a) hold for arbitrary matrices and inputs, and by Theorem 5.3(b) the memory, initial-state and colored-noise terms vanish for every initial law if and only if $A_{y\perp} = 0$, that is, if and only if $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}$ vanishes on $\ker\mathsf{P}_{\mathrm{E}}$, which is $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}} = \bar{F}\mathsf{P}_{\mathrm{E}}$ for some $\bar{F}$; then $A_{yy} = \bar{F}$. The remaining input term is $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EB}}X_{\mathsf{B},t}$. For $t \ge 1$ the blanket has a nondegenerate Gaussian law, because $\mathrm{Cov}(X_t) \succeq Q^{\Xi} \succ 0$, so this term equals $\bar{B}a_{C,t}$ almost surely if and only if $\mathsf{P}_{\mathrm{E}}A_{\mathsf{ES}} = 0$ and $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EA}} = \bar{B}\mathsf{P}_{\mathrm{A}}$.

(a)⇔(c). $\ker\mathsf{P}_{\mathrm{E}}$ is invariant under $A_{\mathsf{EE}}$ if and only if $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}$ vanishes on $\ker\mathsf{P}_{\mathrm{E}}$, if and only if $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}} = \bar{F}\mathsf{P}_{\mathrm{E}}$ for some $\bar{F}$, because $\mathsf{P}_{\mathrm{E}}$ is onto. Then $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}^l = \bar{F}^l\mathsf{P}_{\mathrm{E}}$ by induction, so $\mathrm{rank}\,\mathcal{O} = \mathrm{rank}\,\mathsf{P}_{\mathrm{E}}$; conversely $\mathrm{rank}\,\mathcal{O} = \mathrm{rank}\,\mathsf{P}_{\mathrm{E}}$ forces $\mathrm{row}(\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}) \subseteq \mathrm{row}\,\mathcal{O} = \mathrm{row}\,\mathsf{P}_{\mathrm{E}}$. Likewise $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EA}}$ vanishes on $\ker\mathsf{P}_{\mathrm{A}}$ if and only if it equals $\bar{B}\mathsf{P}_{\mathrm{A}}$ for some $\bar{B}$.

(a)⇔(d). This is Proposition S11.8(a) for $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}}$. The kernel of the map $(X_{\mathsf{E}}, X_{\mathsf{B}}) \mapsto (\eta_C, a_C)$ is the product of $\ker\mathsf{P}_{\mathrm{E}}$ and the set of blanket vectors whose active part lies in $\ker\mathsf{P}_{\mathrm{A}}$, and the two factors can be varied separately, so $\mathsf{P}_{\mathrm{E}}(A_{\mathsf{EE}}\ A_{\mathsf{EB}})$ vanishes on it if and only if $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}$ vanishes on $\ker\mathsf{P}_{\mathrm{E}}$, $\mathsf{P}_{\mathrm{E}}A_{\mathsf{ES}} = 0$ and $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EA}}$ vanishes on $\ker\mathsf{P}_{\mathrm{A}}$.

(e) The entry $(\alpha, v)$ of $\mathsf{P}A_{\mathsf{EE}}$ is $\sum_{u\in\mathfrak{B}_\alpha}(A_{\mathsf{EE}})_{uv}$, and that of $\bar{F}\mathsf{P}$ is $\bar{F}_{\alpha\beta}$ for $v \in \mathfrak{B}_\beta$; they agree for all $\alpha$ and $v$ exactly under Definition 11.7.

(f) Let $\mathcal{R}_l = \sum_{j\le l}\mathrm{row}(\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}^j)$. Then $\mathcal{R}_{l+1} = \mathrm{row}\,\mathsf{P}_{\mathrm{E}} + \mathcal{R}_lA_{\mathsf{EE}}$, the sequence is nondecreasing and stabilizes at $l = \dim X_{\mathsf{E}} - 1$ by the Cayley–Hamilton theorem, and its limit $\mathcal{R}^* = \mathrm{row}\,\mathcal{O}$ is invariant; every invariant space that contains $\mathrm{row}\,\mathsf{P}_{\mathrm{E}}$ contains every $\mathcal{R}_l$. A coarse exterior $\mathsf{P}^*X_{\mathsf{E}}$ that contains $\eta_C$ and satisfies the first condition of (a) has an invariant row space containing $\mathrm{row}\,\mathsf{P}_{\mathrm{E}}$, hence containing $\mathcal{R}^*$. By (a)⇔(c) with $\mathsf{P}^*$ in place of $\mathsf{P}_{\mathrm{E}}$, a coarse exterior with row space $\mathcal{R}^*$ satisfies the first condition, and it has an exact abstract physics exactly under the stated input conditions. These conditions concern every row of $\mathsf{P}^*$, so if they fail on $\mathcal{R}^*$ they fail on every larger row space. For the order, take $\mathsf{P}_{\mathrm{E}} = (I\ 0)$ in coordinates $(\eta_C, x_\perp)$, so that $\mathrm{row}(\mathsf{P}_{\mathrm{E}})A_{\mathsf{EE}} = \mathrm{row}(A_{yy}\ A_{y\perp})$ and $(0\ v)A_{\mathsf{EE}} = (vA_{\perp y}\ vA_{\perp\perp})$. By induction $\mathcal{R}_l = \mathrm{row}\,\mathsf{P}_{\mathrm{E}} \oplus \{(0\ v) : v \in \mathcal{K}_l\}$ with $\mathcal{K}_l = \sum_{j<l}\mathrm{row}(A_{y\perp}A_{\perp\perp}^j)$, whose limit is the observability row space of $(A_{y\perp}, A_{\perp\perp})$; hence the least number of adjoined coordinates is $\mathrm{rank}\,\mathcal{O} - \mathrm{rank}\,\mathsf{P}_{\mathrm{E}}$. Finally $\ker\mathcal{O} = \bigcap_l\ker(\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}^l)$ is invariant and lies in $\ker\mathsf{P}_{\mathrm{E}}$, and an invariant $\mathcal{V} \subseteq \ker\mathsf{P}_{\mathrm{E}}$ satisfies $\mathsf{P}_{\mathrm{E}}A_{\mathsf{EE}}^l\mathcal{V} \subseteq \mathsf{P}_{\mathrm{E}}\mathcal{V} = 0$ for every $l$. □

*Proof of Lemma 11.10.* (a) In the proof of Theorem 5.4 the vector $\phi_C$ enters only through $|\phi_C| = 1$

and $Z^\top Z = I - \phi_C \phi_C^\top$: the coordinates $(y, x_\perp)$, the block form of $\tilde{M}$, the kernel, the invariant-line argument of (b)(ii) and the stationary moments of (b)(iii) use nothing else. The diffusive structure is used only to show $ZMZ^\top \succeq \varkappa_{\text{in}} I$ in case (a), which becomes the definition of $\varkappa_{\text{in}}(M, \phi)$, and to read $\sigma_C$ through row sums, which the definition $\sigma(M, \phi) = |(M - \varkappa_{\text{out}}(M, \phi))\phi|$ makes unnecessary. The inequality of (b)(iv) uses only $\phi^\top M_{\text{anti}} \phi = 0$.

(b) Write $\varkappa_{\text{out}}$, $\varkappa_{\text{in}}$ and $\sigma$ for the rates (101) of $(M, \phi)$. By Courant–Fischer, $\varpi_1 \le \varkappa_{\text{out}} < \varkappa_{\text{in}} \le \varpi_2$, so $\varpi_1$ is simple. In the orthonormal basis $(\phi, Z^\top)$ write the eigenvector of $\varpi_1$ as $(x_\phi, x_\perp)$. The second block row of the eigenvalue equation is $f x_\phi + (ZMZ^\top - \varpi_1) x_\perp = 0$ with $f = ZM\phi$, $|f| = \sigma$, and $ZMZ^\top - \varpi_1 \succeq (\varkappa_{\text{in}} - \varkappa_{\text{out}}) I \succ 0$. Hence $x_\phi \neq 0$, since otherwise $x_\perp = 0$, and $\tan \angle(\phi, \mathsf{u}_1) = |x_\perp|/|x_\phi| \le \sigma/(\varkappa_{\text{in}} - \varkappa_{\text{out}})$. The stationary covariance is $M^{-1}$, and with $\mathsf{c}_l = \phi^\top \mathsf{u}_l$ for orthonormal eigenvectors $\mathsf{u}_l$, $\operatorname{Cov}(y_{t+u}, y_t) = \sum_l (\mathsf{c}_l^2/\varpi_l) e^{-\varpi_l u}$. As in the proof of Theorem 5.4(a), $\Xi$ is at most the memory weight $\mathfrak{m} = \sum_{l \ge 2} (\mathsf{c}_l^2/\varpi_l) / \sum_l (\mathsf{c}_l^2/\varpi_l)$, and $\mathfrak{m} \le \tan^2 \angle(\phi, \mathsf{u}_1) \varpi_1/\varpi_2 \le (\sigma/(\varkappa_{\text{in}} - \varkappa_{\text{out}}))^2 \varkappa_{\text{out}}/\varkappa_{\text{in}}$. For $M = \Pi_{CC}$ and $\phi = \phi_C$, $\varkappa_{\text{in}}(M, \phi_C) \ge \varkappa_{\text{in}}(C)$, and both factors of (102) decrease as $\varkappa_{\text{in}}$ increases on $\varkappa_{\text{in}} > \varkappa_{\text{out}}$, so when $\varkappa_{\text{in}}(C) > \varkappa_{\text{out}}(C)$ its right-hand side is at most that of (49). □

*Proof of Lemma 11.11.* Reversibility gives $(Q^{\mathfrak{n}})^{-1/2} A (Q^{\mathfrak{n}})^{1/2} = (Q^{\mathfrak{n}})^{1/2} A^\top (Q^{\mathfrak{n}})^{-1/2}$, so $\tilde{A}$ is symmetric, with eigenvalues $1 - \varpi_l$ and orthonormal eigenvectors $\mathsf{u}_l$. The whitened state $\tilde{x} = (Q^{\mathfrak{n}})^{-1/2} x$ obeys $\tilde{x}_{t+1} = \tilde{A} \tilde{x}_t + \tilde{\mathfrak{n}}_t$ with identity noise covariance, its stationary covariance is $(I - \tilde{A}^2)^{-1}$, and $y = |(Q^{\mathfrak{n}})^{1/2} f| \phi^\top \tilde{x}$. With $\mathsf{c}_l = \phi^\top \mathsf{u}_l$ and $w_l = \mathsf{c}_l^2/(1 - (1 - \varpi_l)^2)$,

$$\operatorname{Corr}(y_t, y_{t+u}) = \frac{\sum_l w_l (1 - \varpi_l)^u}{\sum_l w_l},$$

and the stationary precision is $\Pi = (Q^{\mathfrak{n}})^{-1} - A^\top (Q^{\mathfrak{n}})^{-1} A = (Q^{\mathfrak{n}})^{-1/2} (I - \tilde{A}^2) (Q^{\mathfrak{n}})^{-1/2}$, as in the proof of Theorem 4.9(a).

(a) The eigenvalues $\varpi_l$ of $I - \tilde{A}$ lie in $(0, 1]$. The argument of Lemma 11.10(b) applied to $I - \tilde{A}$ gives $\varpi_1 \le \varkappa_{\text{out}}^{\text{d}} < \varkappa_{\text{in}}^{\text{d}} \le \varpi_2$, simplicity and the tangent bound. The matrix $(I - \tilde{A} - \varpi_1)(I - \tilde{A} - \varpi_2)$ is positive semidefinite and $|(I - \tilde{A})\phi|^2 = (\varkappa_{\text{out}}^{\text{d}})^2 + (\sigma^{\text{d}})^2$, so $0 \le (\sigma^{\text{d}})^2 - (\varkappa_{\text{out}}^{\text{d}} - \varpi_1)(\varpi_2 - \varkappa_{\text{out}}^{\text{d}})$, which gives the bracket on $\varpi_1$. The terms $(1 - \varpi_l)^u$ with $l \ge 2$ lie in $[0, (1 - \varpi_1)^u]$, so $\operatorname{Corr}(y_t, y_{t+u}) - (1 - \varpi_1)^u \in [-\mathfrak{m}(1 - \varpi_1)^u, 0]$ with $\mathfrak{m} = \sum_{l \ge 2} w_l / \sum_l w_l$, and $\Xi^{\text{d}} \le \mathfrak{m} \le \sum_{l \ge 2} w_l / w_1 \le \tan^2 \angle(\phi, \mathsf{u}_1) \varpi_1 (2 - \varpi_1)/(\varpi_2 (2 - \varpi_2))$, because $1 - (1 - \varpi_l)^2 = \varpi_l (2 - \varpi_l) \ge \varpi_2 (2 - \varpi_2)$ for $l \ge 2$. The function $x \mapsto x(2 - x)$ increases on $[0, 1]$; with $\varpi_1 \le \varkappa_{\text{out}}^{\text{d}} \le 1$ and $\varkappa_{\text{in}}^{\text{d}} \le \varpi_2 \le 1$ this gives the second inequality of (103), and $(2 - \varkappa_{\text{out}}^{\text{d}})/(2 - \varkappa_{\text{in}}^{\text{d}}) \le 2$ the third.

(b) The matrix $I - \tilde{A}^2$ has the eigenvectors $\mathsf{u}_l$ and the eigenvalues $\varpi_l (2 - \varpi_l)$, which are ordered as the $\varpi_l$ because these lie in $(0, 1]$. The argument of Lemma 11.10(b) applied to $I - \tilde{A}^2$ gives the tangent bound with its rates, and $\mathfrak{m} \le \tan^2 \angle(\phi, \mathsf{u}_1) \varpi_1 (2 - \varpi_1)/(\varpi_2 (2 - \varpi_2))$, as in (a), is at most $(\sigma^\Pi/(\varkappa_{\text{in}}^\Pi - \varkappa_{\text{out}}^\Pi))^2 \varkappa_{\text{out}}^\Pi/\varkappa_{\text{in}}^\Pi$ by Courant–Fischer for $I - \tilde{A}^2$.

(c) With the largest eigenvalue of $A$ equal to $\rho(A)$, $\varpi_1 = 1 - \rho(A)$ is still the smallest eigenvalue of $I - \tilde{A}$, so Courant–Fischer and the tangent bound are unchanged. Each $(1 - \varpi_l)^u$ with $l \ge 2$ lies in $[-(1 - \varpi_1)^u, (1 - \varpi_1)^u]$, so $\operatorname{Corr}(y_t, y_{t+u}) - (1 - \varpi_1)^u \in [-2\mathfrak{m}(1 - \varpi_1)^u, 0]$ and $\Xi^{\text{d}} \le 2\mathfrak{m}$, and $w_l \le \mathsf{c}_l^2/(1 - \max_{l' \ge 2} (1 - \varpi_{l'})^2)$ for $l \ge 2$.

For the level-0 statement, $A = W_{CC}$ and $Q^{\mathfrak{n}} = qI$ give $\tilde{A} = W_{CC}$ and $\phi = \phi_C$; then $(I - W_{CC})\phi_C$ has entries $n^{-1/2}(1 - \sum_{j \in C} W_{ij})$, whose mean is $n^{-1/2} \varkappa_{\text{out}}^{\text{d}}$ and whose deviation from the mean has norm $\sigma^{\text{d}}$. □

*Proof of Proposition 11.9.* (a) By Theorem 5.2, $\Gamma_l = A_{y\perp} A_{\perp\perp}^{l-1} A_{\perp y}$ and $\Gamma_l^{\mathsf{B}} = A_{y\perp} A_{\perp\perp}^{l-1} B_\perp$; take norms and sum the geometric series.

(b) The formula is Proposition S11.8(a), and $\ln\det(I+X) \le \operatorname{tr} X$ for $X$ similar to a positive semidefinite matrix. The exterior reads no interior variable and $\bar{w}_t$ is independent of $X_t$, so $\mathbb{E}[\eta_{C,t+1} \mid X_t] = \mathsf{G}x$ and $\mathbb{E}[\eta_{C,t+1} \mid \mathsf{c}] = \mathsf{G}\mathbb{E}[x \mid \mathsf{c}]$; their difference $\mathsf{G}(x - \mathbb{E}[x \mid \mathsf{c}])$ has covariance $\mathsf{G}\Sigma_{x|\mathsf{c}}\mathsf{G}^\top$, and stationarity gives $\operatorname{Var}(\eta_{C,t+1,\alpha}) = \operatorname{Var}(\eta_{C,\alpha})$.

(c) For symmetric $M_{\mathsf{E}}$, $\sigma_{\text{row}} = \sigma_{\text{col}} = \sigma(M_{\mathsf{E}}, \phi)$, and (102) with $\varkappa_{\text{in}} - \varkappa_{\text{out}} = \varkappa_{\text{out}}(r_\phi - 1)$, $\varkappa_{\text{in}} = r_\phi \varkappa_{\text{out}}$ and $\sigma = \hat{\sigma}\varkappa_{\text{out}}$ gives the first bound. For general $M_{\mathsf{E}}$, Lemma 11.10(a) makes Theorem 5.4(b)(iii) available for $\phi$. Its hypotheses $\sigma_{\text{row}}, \sigma_{\text{col}} \le (\varkappa^M_{\text{in}} - \varkappa^M_{\text{out}})/4$ and $\sigma_{\text{row}}\sigma_{\text{col}} \le \varkappa^M_{\text{out}}(\varkappa^M_{\text{in}} - \varkappa^M_{\text{out}})/2$ follow from $\hat{\sigma} \le (r_\phi - 1)/4$ and $\hat{\sigma}^2 \le (r_\phi - 1)/2$, and its second bound, with $\sigma_{\text{col}}(\sigma_{\text{col}} + \sigma_{\text{row}}) \le 2\hat{\sigma}^2(\varkappa^M_{\text{out}})^2$, gives $\Xi \le 152 \cdot 2\hat{\sigma}^2(\varkappa^M_{\text{out}})^3/((\varkappa^M_{\text{out}})^3(r_\phi - 1)^2 r_\phi)$.

(d) The last inequality of (103), rewritten in the same way. □

*Proof of Example S11.5.* (a) The coarse map has rows $(1, 0, \dots, 0)$ and $(0, N^{-1}\mathbf{1}^\top)$, whose row space is that of the block-sum matrix of the blocks $\{p\}$ and $\{x_1, \dots, x_N\}$. The total weight that the traders place on trader $j$ is $\alpha_j$, the only entry of column $x_j$ in the trader rows, so Definition 11.7 for the pair (traders, traders) holds if and only if all $\alpha_j$ are equal. The weight that the price places on each trader is $\mathsf{b}_p/N$, the total weight that the traders place on the price is $N\mathsf{b}_x$, and the act enters only the price, so the other pairs and the input conditions of Theorem 11.8(a) hold for any persistences. (b) In the stationary regime $x_{j,t+1} = \sum_{l\ge 0} \alpha_j^l(\mathsf{b}_x p_{t-l} + w_{j,t-l})$, so $\bar{x}_{t+1} = \mathsf{b}_x \sum_{l\ge 0} \bar{\mathfrak{m}}_l p_{t-l} + \mathfrak{f}_t^{(N)}$. Subtracting $\bar{\alpha}\bar{x}_t$ gives the stated equation with $\bar{\mathfrak{m}}_0 = 1$, and $\bar{\mathfrak{m}}_1 = \bar{\alpha}$ gives $\Gamma_1 = 0$. The noise variance follows from the independence of the $w_{j,t}$. Since $\sum_j(\alpha_j - \bar{\alpha}) = 0$,

$$\bar{\mathfrak{m}}_l - \bar{\alpha}\bar{\mathfrak{m}}_{l-1} = \frac{1}{N}\sum_j(\alpha_j - \bar{\alpha})\alpha_j^{l-1} = \frac{1}{N}\sum_j(\alpha_j - \bar{\alpha})(\alpha_j^{l-1} - \bar{\alpha}^{l-1}),$$

whose terms are nonnegative because $\alpha \mapsto \alpha^{l-1}$ is nondecreasing on $[0, 1)$, and by the mean value theorem each is at most $(l-1)\alpha_{\max}^{l-2}(\alpha_j - \bar{\alpha})^2$; finally $\sum_{l\ge 2}(l-1)\alpha_{\max}^{l-2} = (1 - \alpha_{\max})^{-2}$. (c) $\bar{\mathfrak{m}}_l$ and $\bar{\alpha} = \bar{\mathfrak{m}}_1$ are moments of the empirical law, and moments of laws on a compact interval converge under weak convergence, here to $\mathfrak{m}_l$ and $\mathfrak{m}_1$; $\mathfrak{m}_2 - \mathfrak{m}_1^2 = \operatorname{Var}_{\mathbb{Q}}(\alpha)$. □

The coarse filter has a unique steady state with a stable closed loop. We record the facts used below.

**Lemma S35.1** (Steady state of the coarse filter)**.** *Let $\bar{Q} \succ 0$ and $\Lambda^s \succeq 0$, let $(\bar{F}, \Lambda^s)$ be detectable, that is, $\rho(\bar{F} - \mathsf{L}\Lambda^s) < 1$ for some matrix $\mathsf{L}$, as when $\rho(\bar{F}) < 1$ (take $\mathsf{L} = 0$) or $\Lambda^s \succ 0$ (take $\mathsf{L} = \bar{F}(\Lambda^s)^{-1}$), and let $\Psi(P) = ((\bar{F}P\bar{F}^\top + \bar{Q})^{-1} + \Lambda^s)^{-1}$ on positive semidefinite matrices. Then $\Psi$ is monotone, $\Psi(P) \preceq \bar{F}P\bar{F}^\top + \bar{Q}$, it has a unique fixed point $\bar{P}$, the iterates $\Psi^k(P)$ converge to $\bar{P}$ from every positive semidefinite $P$, and with $\bar{P}^- = \bar{F}\bar{P}\bar{F}^\top + \bar{Q}$ the closed-loop matrix $\bar{F}(I - \bar{P}\Lambda^s)$ has spectral radius below one.*

*Proof.* Monotonicity and the bound are immediate from the formula. Write $\mathfrak{g}(X) = (X^{-1} + \Lambda^s)^{-1} = (I + X\Lambda^s)^{-1}X$ and $\mathsf{A}_X = (I + X\Lambda^s)^{-1} = I - \mathfrak{g}(X)\Lambda^s$ for $X \succ 0$, so that $\Psi(P) = \mathfrak{g}(\bar{F}P\bar{F}^\top + \bar{Q})$. A direct computation, using $\mathfrak{g}(X) = X\mathsf{A}_X^\top$, $\mathsf{A}_X - I = -\mathfrak{g}(X)\Lambda^s$ and $\mathsf{A}_X^\top - I = -\Lambda^s\mathfrak{g}(X)$, gives

$$\mathfrak{g}(X) - \mathfrak{g}(Y) = \mathsf{A}_Y(X - Y)\mathsf{A}_X^\top.$$

Here $I + X\Lambda^s$ is invertible because the eigenvalues of $X\Lambda^s$ are those of $X^{1/2}\Lambda^s X^{1/2} \succeq 0$. A bound: for every matrix $\mathsf{L}$ and every $X \succ 0$, with $\mathsf{D} = \mathsf{L} - \bar{F}\mathfrak{g}(X)$,

$$(\bar{F} - \mathsf{L}\Lambda^s)X(\bar{F} - \mathsf{L}\Lambda^s)^\top + \mathsf{L}\Lambda^s\mathsf{L}^\top = \bar{F}\mathfrak{g}(X)\bar{F}^\top + \mathsf{D}(\Lambda^s X\Lambda^s + \Lambda^s)\mathsf{D}^\top \succeq \bar{F}\mathfrak{g}(X)\bar{F}^\top.$$

Indeed, at $\mathsf{D}=0$ the left side is $\bar{F}(\mathsf{A}_X X \mathsf{A}_X^\top + \mathfrak{g}(X)\Lambda^s\mathfrak{g}(X))\bar{F}^\top = \bar{F}\mathfrak{g}(X)(X^{-1}+\Lambda^s)\mathfrak{g}(X)\bar{F}^\top = \bar{F}\mathfrak{g}(X)\bar{F}^\top$, and the terms linear in $\mathsf{D}$ are $-\mathsf{D}\Lambda^s X\mathsf{A}_X^\top\bar{F}^\top + \mathsf{D}\Lambda^s\mathfrak{g}(X)\bar{F}^\top = 0$ and its transpose. Existence: write $\Psi^-(X) = \bar{F}\mathfrak{g}(X)\bar{F}^\top + \bar{Q}$ for the map of predicted covariances, which is monotone and satisfies $\Psi^{k+1}(P) = \mathfrak{g}((\Psi^-)^k(\bar{F}P\bar{F}^\top + \bar{Q}))$; if $P$ is a fixed point of $\Psi$ then $\bar{F}P\bar{F}^\top + \bar{Q}$ is a fixed point of $\Psi^-$, and if $X$ is a fixed point of $\Psi^-$ then $\mathfrak{g}(X)$ is a fixed point of $\Psi$ and $X = \bar{F}\mathfrak{g}(X)\bar{F}^\top + \bar{Q}$ is its predicted covariance, so the fixed points of the two maps correspond one to one. Fix $\mathsf{L}$ with $\rho(\bar{F} - \mathsf{L}\Lambda^s) < 1$ and, for $c \geq 0$, let $\mathsf{S}_c \succeq \bar{Q} + cI$ be the solution of $\mathsf{S} = (\bar{F} - \mathsf{L}\Lambda^s)\mathsf{S}(\bar{F} - \mathsf{L}\Lambda^s)^\top + \mathsf{L}\Lambda^s\mathsf{L}^\top + \bar{Q} + cI$. The bound gives $\Psi^-(\mathsf{S}_c) \preceq \mathsf{S}_c - cI \preceq \mathsf{S}_c$, so the iterates $(\Psi^-)^k(\mathsf{S}_c)$ decrease, while $(\Psi^-)^k(\bar{Q})$ increase and stay below $\mathsf{S}_0$; the limits are fixed points of $\Psi^-$ by continuity. Stability: for a fixed point $\bar{P}$ with $\bar{P}^- = \bar{F}\bar{P}\bar{F}^\top + \bar{Q} \succeq \bar{Q} \succ 0$, the error identity of the information-form update, $\bar{P} = \mathsf{A}_{\bar{P}^-}\bar{P}^-\mathsf{A}_{\bar{P}^-}^\top + \bar{P}\Lambda^s\bar{P}$, gives $\bar{P}^- = \mathsf{K}\bar{P}^-\mathsf{K}^\top + \bar{F}\bar{P}\Lambda^s\bar{P}\bar{F}^\top + \bar{Q}$ with $\mathsf{K} = \bar{F}\mathsf{A}_{\bar{P}^-} = \bar{F}(I - \bar{P}\Lambda^s)$. For a left eigenvector $f^*$ of $\mathsf{K}$ with eigenvalue $\varpi$ this gives $(1-|\varpi|^2)f^*\bar{P}^-f \geq f^*\bar{Q}f > 0$, so $\rho(\mathsf{K}) < 1$. Uniqueness: if $\bar{P}_1$ and $\bar{P}_2$ are fixed points with predicted covariances $X_1$ and $X_2$, the identity gives $X_1 - X_2 = \bar{F}(\mathfrak{g}(X_1) - \mathfrak{g}(X_2))\bar{F}^\top = \mathsf{K}_2(X_1 - X_2)\mathsf{K}_1^\top$, with $\mathsf{K}_j = \bar{F}\mathsf{A}_{X_j}$ the closed-loop matrices of the two fixed points, which are stable by the previous step, so $X_1 - X_2 = \mathsf{K}_2^k(X_1 - X_2)(\mathsf{K}_1^\top)^k \to 0$; hence $X_1 = X_2$ and $\bar{P}_1 = \bar{P}_2$. Convergence: for $P \succeq 0$ let $X_0 = \bar{F}P\bar{F}^\top + \bar{Q} \succeq \bar{Q}$ and $c \geq \|X_0\|$, so that $\mathsf{S}_c \succeq X_0$. Then $(\Psi^-)^k(\bar{Q}) \preceq (\Psi^-)^k(X_0) \preceq (\Psi^-)^k(\mathsf{S}_c)$, both bounds converge to the unique fixed point $\bar{P}^-$ of $\Psi^-$, and $\Psi^{k+1}(P) = \mathfrak{g}((\Psi^-)^k(X_0)) \to \mathfrak{g}(\bar{P}^-) = \bar{P}$ by continuity of $\mathfrak{g}$. □

*Proof of Proposition 11.12.* (a) The primitive randomness that reaches $\eta_C$ or a collective variable splits into $\mathfrak{W}_1 = (\eta_{C,0}, (\bar{w}_t), (\bar{v}_t))$, with $\bar{v}_t = \mathcal{H}^\top R_{\mathsf{S}}^{-1}\Xi_t^{\mathsf{S}}$, and $\mathfrak{W}_2$, made of the initial collective variables, the sensory residual noises $\mathsf{Z}_{\mathrm{S}}\Xi_t^{\mathsf{S}}$ and the interior and act noises. The remaining exterior randomness affects neither $\eta_C$, by (X), nor any collective variable, because collective variables read the exterior only through sensory channels, which by (S) read only $\eta_C$. By Theorem 11.8(b) and (S), $\eta_{C,t+1} = \bar{F}\eta_{C,t} + \bar{B}a_{C,t} + \bar{w}_t$ and $s_{C,t+1} = \Lambda^s\eta_{C,t} + \bar{v}_t$. By the proof of Proposition 11.4(a), whose dynamical hypotheses (a1), with $\mathcal{H}_0$ for $\mathcal{H}$ and $\mathsf{G}_{\mathrm{E}} = \mathsf{G}_\eta\mathsf{P}_{\mathrm{E}}$, and (a2) are contained in (S) and (X), and whose statistic $s_C^0 = \mathcal{H}_0^\top R_{\mathsf{S}}^{-1}X_{\mathsf{S}}$ determines and is determined by $s_C = \mathsf{G}_\eta^\top s_C^0$ because $\mathsf{G}_\eta^\top$ is injective, the variables of $\mathsf{I}'$ and the coarse act read only variables of $\mathsf{I}'$ and $\mathsf{B}'$, and the memoryless sensory residual is $\mathfrak{r}_{\mathrm{S},t+1} = \mathsf{Z}_{\mathrm{S}}\Xi_t^{\mathsf{S}}$. Hence $a_{C,t}$ is a function of $s_{C,1:t}$ and $\mathfrak{W}_2$. By per-channel noise, (Ind) and $\mathsf{Z}_{\mathrm{S}}\mathcal{H} = \mathsf{Z}_{\mathrm{S}}\mathcal{H}_0\mathsf{G}_\eta = 0$, $\mathfrak{W}_2$ is independent of $\mathfrak{W}_1$ given $\mathcal{I}_0$.

Order the variables as $\eta_{C,0}, a_{C,0}, s_{C,1}, \eta_{C,1}, a_{C,1}, s_{C,2}, \ldots$ and factor their joint density given $\mathcal{I}_0$ into conditional densities. The factor of $s_{C,t+1}$ is $\mathcal{N}(\Lambda^s\eta_{C,t}, \Lambda^s)$ and that of $\eta_{C,t+1}$ is $\mathcal{N}(\bar{F}\eta_{C,t} + \bar{B}a_{C,t}, \bar{Q})$, because $\bar{v}_t$ and $\bar{w}_t$ are independent of everything earlier in the order and of $\mathfrak{W}_2$. For the factor of $a_{C,t}$: the earlier variables depend on $\mathfrak{W}_2$ only through $a_{C,0:t-1}$, so the conditional law of $\mathfrak{W}_2$ given them is its law given $\mathcal{I}_0$ restricted by the values of $a_{C,0:t-1}$ as functions of $s_{C,1:t}$ and $\mathfrak{W}_2$, and it does not involve $\eta_{C,0:t}$. The factor of $a_{C,t}$ is therefore a function of $(s_{C,1:t}, a_{C,0:t-1})$ alone. Conditioning on $\mathcal{F}_t$, the act factors are constants, and the conditional law of $\eta_{C,t}$ is that of the linear-Gaussian state-space model with the realized acts as known inputs. Gaussian conditioning in information form gives (104): the observation $s_{C,t}$ of $\eta_{C,t-1}$ adds $\Lambda^s$ to the precision and $s_{C,t}$ to the information vector, and the prediction uses $(\bar{F}, \bar{B}, \bar{Q})$. When $\Lambda^s$ is singular the law $\mathcal{N}(\Lambda^s\eta_{C,t}, \Lambda^s)$ has no density, and the factorization is carried out with $s_C^0$ in the place of $s_C$: its factor is $\mathcal{N}(\Lambda_0\mathsf{G}_\eta\eta_{C,t}, \Lambda_0)$ with $\Lambda_0 = \mathcal{H}_0^\top R_{\mathsf{S}}^{-1}\mathcal{H}_0 \succ 0$, and conditioning on $s_{C,t}^0$ adds $\mathsf{G}_\eta^\top\Lambda_0\mathsf{G}_\eta = \Lambda^s$ to the precision and $\mathsf{G}_\eta^\top s_{C,t}^0 = s_{C,t}$ to the information vector, so (104) is unchanged. The fine sensory path is $(s_C, \mathfrak{r}_{\mathrm{S}})$ with $\mathfrak{r}_{\mathrm{S}}$ a function of $\mathfrak{W}_2$, and internal states are functions of $\mathcal{I}_0$, $s_C$ and $\mathfrak{W}_2$; adding them to the conditioning adds factors that, by the same argument, do not involve $\eta_C$, and leaves the law unchanged.

(b) The steady-state covariances exist by Lemma S35.1. Since $\mathbf{1}^\top\mathcal{W} = \mathbf{1}^\top$, $\frac{1}{n}\sum_i n\tilde{h}_i = (\mathbf{1}^\top\mathcal{W}^{n_{\mathrm{c}}} \otimes$

$I)h^s = (\mathbf{1}^\top \otimes I)h^s = s_C$. The update (105) is affine with coefficients and input common to all members, so the average obeys $y_{C,t} = \bar{F}[y_{C,t-1} + \bar{P}(s_{C,t} - \Lambda^s y_{C,t-1})] + \bar{B}a_{C,t-1}$, the steady-state form of (104) with $y_{C,t}$ in the place of $\hat{\eta}_t^-$; from a common start equal to $\hat{\eta}_0^-$ with $P_0^- = \bar{P}^-$, the time-varying filter is the steady-state one and the two coincide. Subtracting the averaged recursion from a member's gives the deviation recursion, in which the input cancels. For symmetric, doubly stochastic and nonnegative $\mathcal{W}$, consensus rounds contract deviations of the contributions from their average by $\rho_\perp^{n_c}$, and summing the deviation recursion from $\mathrm{dev}(0) = 0$ with a geometric bound on the powers of the stable matrix $\bar{F}(I - \bar{P}\Lambda^s)$ gives the bound, as in the proof of Proposition 5.7(b).

(c) All members use the covariance $\bar{P}^-$. □

*Proof of Proposition S11.9.* (a) is Proposition S5.5. (b) Let $X = \bar{F}P\bar{F}^\top + \bar{Q}$ be the covariance of the common predicted prior. Each member's posterior, formed from that prior and its own contribution, has precision $X^{-1} + \Lambda_i^s$, so the product of the $n$ posteriors has precision $nX^{-1} + \Lambda^s$. The map $\Psi_n$ is the map $\Psi$ of Lemma S35.1 for the model $(\bar{F}/\sqrt{n}, \bar{Q}/n, \Lambda^s)$, which is detectable because $\rho(\bar{F}/\sqrt{n} - (\mathsf{L}/\sqrt{n})\Lambda^s) < 1$ whenever $\rho(\bar{F} - \mathsf{L}\Lambda^s) < 1$, so it has a unique fixed point $P_\mathrm{b}$, which attracts every start. Since $\Psi_n(P) \preceq \Psi_1(P)$ for $n \geq 1$ and $\Psi_1$ is monotone, $P_\mathrm{b} = \Psi_n(P_\mathrm{b}) \preceq \Psi_1(P_\mathrm{b})$, the iterates $\Psi_1^k(P_\mathrm{b})$ increase, and they converge to $\bar{P}$ by Lemma S35.1; hence $P_\mathrm{b} \preceq \bar{P}$. The lemma applied to the modified model makes $\bar{F}(I - P_\mathrm{b}\Lambda^s)/\sqrt{n}$ stable but not $\bar{F}(I - P_\mathrm{b}\Lambda^s)$ itself, which is why stability is assumed for the remaining claims; it holds, for instance, in the scalar case with $|\bar{F}| < 1$, where the closed loop is $\bar{F}n/(n + \Lambda^s(\bar{F}^2 P_\mathrm{b} + \bar{Q}))$. The literal estimator is a linear function of the coarse data. At every round the filtered covariance of the exact filter is the least error covariance among all estimators of $\eta_{C,t-1}$ from the same data, and it converges to $\bar{P}$ by Lemma S35.1; when the literal estimator is stable its error covariance converges to $\mathcal{C}_\mathrm{b}$, so $\mathcal{C}_\mathrm{b} \succeq \bar{P}$. The literal error is the exact error plus the difference of the two means, which is a function of the data and therefore uncorrelated with the exact error; hence the expected squared Mahalanobis term is $\mathrm{tr}(P_\mathrm{b}^{-1}(\mathcal{C}_\mathrm{b} - \bar{P}))$, and the Gaussian formula for the relative entropy, $\frac{1}{2}[\mathrm{tr}(P_\mathrm{b}^{-1}\bar{P}) + (\text{Mahalanobis term}) - \dim \eta_C + \ln\det P_\mathrm{b} - \ln\det \bar{P}]$, gives the expression. □

*Proof of Proposition S11.10.* With exact codes, $\mathcal{K}_{ij}y_{ij} = k_{ij}m_j + \mathcal{K}_{ij}\nu_{ij}$ for senders $j \in C$, and the update of Definition 2.12 reads $m_i(t+1) = F[(1 - k_i^s - k_i^\partial - k_i^{\mathrm{giv}})m_i + \sum_{j\in C} k_{ij}m_j + \mathcal{K}_i^s s_i + \sum_{j\notin C}\mathcal{K}_{ij}y_{ij} + \sum_{j\in C}\mathcal{K}_{ij}\nu_{ij}]$. Averaging over $i \in C$, $\frac{1}{n}\sum_{i,j\in C} k_{ij}m_j - \frac{1}{n}\sum_i k_i^{\mathrm{giv}}m_i = \frac{1}{n}\sum_j (k_j^{\mathrm{rec}} - k_j^{\mathrm{giv}})m_j = \frac{1}{n}\sum_j (k_j^{\mathrm{rec}} - k_j^{\mathrm{giv}})(m_j - \bar{m})$, because $\sum_j (k_j^{\mathrm{rec}} - k_j^{\mathrm{giv}}) = 0$, and $\frac{1}{n}\sum_i (1 - k_i^s - k_i^\partial)m_i = (1 - \bar{k}^s - \bar{k}^\partial)\bar{m} - \frac{1}{n}\sum_i (k_i^s + k_i^\partial - \bar{k}^s - \bar{k}^\partial)(m_i - \bar{m})$; this is (S45). The memory terms together are $\frac{1}{n}\sum_j \mathsf{e}_j(m_j - \bar{m})$ with $\mathsf{e}_j = k_j^{\mathrm{rec}} - k_j^{\mathrm{giv}} - k_j^s - k_j^\partial$, up to a constant that multiplies $\sum_j (m_j - \bar{m}) = 0$. The members' inputs enter the average only through $s_C$, and the memory terms enter it multiplied by $F$, so for $F \neq 0$, by Definition 11.6(i), the average is exactly closed given $s_C$ if and only if this functional vanishes for every vector of deviations, that is, if and only if $\mathsf{e}_j$ is the same for every $j$. The update of the means has column sums $F(1 - k_j^s - k_j^\partial - k_j^{\mathrm{giv}} + k_j^{\mathrm{rec}}) = F(1 + \mathsf{e}_j)$, so this is equality of column sums. For $F = 0$ the means do not depend on their past and the average is closed whatever the gains. □

*Proof of Theorem 11.13.* (i) Condition (S) gives (a1) with memoryless channels, with $\mathcal{H}_0$ for $\mathcal{H}$, and (H-suff) contains (a2), so Proposition 11.4(a) gives the exact blanket for the statistic $\mathcal{H}_0^\top R_\mathsf{S}^{-1} X_\mathsf{S}$, and hence for $s_C$, its image under the injective map $\mathsf{G}_\eta^\top$, which has the same paths up to a fixed injective linear map. From the stationary start, Theorem 4.3(c) applies to the transformed process $\tilde{X}$, whose first three conditions in (95) hold, whose noise is independent across $(\mathsf{I}', \mathsf{B}', \mathsf{E}')$, whose start is Gaussian with positive definite covariance and for which $\rho(\tilde{A}) = \rho(A) < 1$: the window leak of $\tilde{X}$ is bounded in $T$, and $\iota_T^C$ is at most that leak. The perturbative statement is Proposition 11.5.

(ii) The members' filter states follow (105) in the indexing $\mu_{i,t+1} = \hat{\eta}_t^{(i)}$, and their stored acts follow $\alpha_{i,t+1} = a_{C,t}$, each plus computation noise; the filter state at round $t+1$ uses the stored act $\alpha_{i,t} = a_{C,t-1}$. By the computation in the proof of Proposition 11.12(b), the average of the right sides is a function of $y_C$ and $b_C$, which gives the stated recursion, so $\mathsf{P}_\mathrm{I} A_{\mathsf{II}} = \bar{A}_y \mathsf{P}_\mathrm{I}$ and $\mathsf{P}_\mathrm{I} A_{\mathsf{IB}} = \bar{B}_y \mathsf{P}_\mathrm{B}$ with $\mathsf{P}_\mathrm{B} = \mathrm{diag}(\mathcal{H}^\top R_\mathsf{S}^{-1}, \mathsf{P}_\mathrm{A})$. When $\mathsf{G}_\eta \neq I$ this map is not of full row rank, and the defect criterion is applied with $\mathsf{P}_\mathrm{B}^0 = \mathrm{diag}(\mathcal{H}_0^\top R_\mathsf{S}^{-1}, \mathsf{P}_\mathrm{A})$, which has full row rank: $\mathsf{P}_\mathrm{B} = \mathrm{diag}(\mathsf{G}_\eta^\top, I)\mathsf{P}_\mathrm{B}^0$ with $\mathsf{G}_\eta^\top$ injective, so $\ker \mathsf{P}_\mathrm{B} = \ker \mathsf{P}_\mathrm{B}^0$, conditioning on $(y_C, b_C)$ is conditioning on $\mathsf{c} = \mathrm{diag}(\mathsf{P}_\mathrm{I}, \mathsf{P}_\mathrm{B}^0)(X_\mathsf{I}, X_\mathsf{B})$, and the map $(\mathsf{P}_\mathrm{I} A_{\mathsf{II}}, \mathsf{P}_\mathrm{I} A_{\mathsf{IB}})$ vanishes on $\ker \mathrm{diag}(\mathsf{P}_\mathrm{I}, \mathsf{P}_\mathrm{B}^0)$. Proposition S11.8(a) with $\mathcal{P} = \mathrm{diag}(\mathsf{P}_\mathrm{I}, \mathsf{P}_\mathrm{B}^0)$ then gives $\mathfrak{d}_\mathrm{cl}^C = 0$. The argument uses only the form of the update (105), a matrix $\mathcal{W}$ that preserves sums and coefficients common to all members, so it holds for any fixed matrices in place of $(\bar{F}, \bar{P}, \bar{B}, \Lambda^s)$.

(iii) is Theorem 11.8 and Proposition 11.9.

(iv) Let $\hat{\eta}_t^-$ be computed by the steady-state filter started at the members' common start. Subtracting the steady-state form of (104) from the recursion of (ii) shows that $y_{C,t+1}^\eta - \hat{\eta}_t^-$ is $\bar{F}(I - \bar{P}\Lambda^s)(y_{C,t}^\eta - \hat{\eta}_{t-1}^-)$ plus the averaged computation noise of the filter states and $\bar{B}$ times that of the stored acts, and the matrix is stable by Lemma S35.1. Proposition 11.12(a), which holds at every level of the computation noise because that noise belongs to $\mathfrak{W}_2$, gives the parametrization by $(\hat{\eta}_t^-, \bar{P}^-)$ from the centralized start with $P_0^- = \bar{P}^-$ and the equality with the law given the fine blanket and interior paths. From that start the difference is zero at the first round, so it equals $\sum_{l \geq 0} \mathsf{K}^l \mathfrak{n}_{t-l}$ with $\mathsf{K} = \bar{F}(I - \bar{P}\Lambda^s)$ and $\mathfrak{n}$ the driving noise, the sum running over the rounds since the start; with $\mathsf{n}^2 = \sup_t \mathbb{E}|\mathfrak{n}_t|^2$, the triangle inequality in mean square bounds its root mean square by $\mathsf{n} \sum_{l \geq 0} \|\mathsf{K}^l\|$, uniformly in $t$, and $\mathsf{n}$ vanishes with the computation noise. For the deviations, subtract the average recursion of (ii) from the member recursions: with $\mathrm{dev}_{i,t} = \mu_{i,t} - y_{C,t}^\eta$ and computation noises $\nu_{i,t}^\mu$ and $\nu_{i,t}^\alpha$ of the filter state and the stored act, $\alpha_{i,t} - y_{C,t}^\alpha = \nu_{i,t-1}^\alpha - \bar{\nu}_{t-1}^\alpha$ because every member stores the same $a_{C,t-1}$, and

$$\mathrm{dev}_{i,t+1} = \mathsf{K}\mathrm{dev}_{i,t} + n\bar{F}\bar{P}(\tilde{h}_i(t) - \bar{h}^s(t)) + (\nu_{i,t}^\mu - \bar{\nu}_t^\mu) + \bar{B}(\nu_{i,t-1}^\alpha - \bar{\nu}_{t-1}^\alpha),$$

where bars denote averages over members. Without computation noise this is the recursion of Proposition 11.12(b); with it, the last two terms are the stated additional driving terms, and by linearity the deviation is the sum of the noiseless deviation and the response of $\mathsf{K}$ to those terms, which does not involve $\mathcal{W}$ or $n_\mathrm{c}$.

(v) Interventions shift means and leave covariances unchanged, so with a common $P$ each member's expected free energy over acts is the act part of the expected free energy in the joint-act model of Definition S5.16, with $\bar{F}$ for $F$ and the effects $\mathsf{M}_i$. With a shared $\Lambda^\star$, the true effects and additive effects, Theorem S5.17(b) gives the exact potential $\Phi^\mathrm{act} = G_C^\mathrm{act} + \sum_j \langle \mathsf{z}_j - \mathsf{z}_C, \mathsf{M}_j u_j \rangle_{\Lambda^\star}$ for any $\mathsf{z}_C$. Take $\mathsf{z}_C = \mathsf{H}_\star \bar{F} m_C - o^\star$; with common targets, $\mathsf{z}_j - \mathsf{z}_C = \mathsf{H}_\star \bar{F}(m_j - m_C)$, so the linear terms are the $\mathcal{T}_j$ of (106). Under (P3) and (P4) the deliberation chain is the asynchronous logit chain of an exact potential game, reversible with stationary law proportional to $\prod_i E_i e^{-\gamma \Phi^\mathrm{act}} = \prod_i (E_i e^{-\gamma \mathcal{T}_i}) e^{-\gamma G_C^\mathrm{act}}$, by the argument of Theorem 5.10(c), which uses only the potential; this is the stated group posterior. The bound on $\mathcal{T}_i$ is the Cauchy–Schwarz inequality, $|\langle a, b \rangle_{\Lambda^\star}| \leq \|\Lambda^\star\| |a| |b|$, with $|\mathsf{H}_\star \bar{F}(m_i - m_C)| \leq \|\mathsf{H}_\star \bar{F}\| |m_i - m_C|$. Under (iv) the members' means are $\hat{\eta}_t^{(i)}$ and their average is $y_C^\eta$, the predictive mean up to the response to the computation noise, so $|m_i - m_C| = |\mathrm{dev}_i|$. □

## S36 Proofs for Section 12

The proofs follow the order of Section 12. Throughout, $D_\mathrm{KL}$ between Gaussian laws is evaluated with the formula $D_\mathrm{KL}(\mathcal{N}(0, C_1) \| \mathcal{N}(0, C_2)) = \frac{1}{2}[\mathrm{tr}(C_2^{-1} C_1) - m - \ln \det(C_2^{-1} C_1)]$, and the function

$\mathrm{g}(x) = -x - \ln(1-x) = \sum_{k\ge 2} x^k/k$ on $[0,1)$ is increasing and satisfies $\mathrm{g}(x) \le x^2/(2(1-x))$.

### The collective as a latent cause

*Proof of Proposition S12.2.* (i)⇒(ii). Let the rows of $Z$ be a basis of the row vectors $v$ with $vQ^{\mathfrak{n}}\mathsf{P}^\top = 0$; there are $\dim x - \dim y$ of them because $Q^{\mathfrak{n}}\mathsf{P}^\top$ has full column rank. If $\alpha\mathsf{P} + \zeta Z = 0$, right multiplication by $Q^{\mathfrak{n}}\mathsf{P}^\top$ gives $\alpha\mathsf{P}Q^{\mathfrak{n}}\mathsf{P}^\top = 0$, so $\alpha = 0$ and then $\zeta = 0$: $\mathcal{Z}$ is invertible. The noises $\mathsf{P}\mathfrak{n}_t$ and $Z\mathfrak{n}_t$ are jointly Gaussian with cross-covariance $\mathsf{P}Q^{\mathfrak{n}}Z^\top = 0$, hence independent, and both are white. By Theorem 5.3(b), (i) is equivalent to $A_{y\perp} = 0$ for every completion, so in these coordinates $y_{t+1} = A_{yy}y_t + \mathsf{P}\mathfrak{n}_t$ and $x_{\perp,t+1} = A_{\perp y}y_t + A_{\perp\perp}x_{\perp,t} + Z\mathfrak{n}_t$, which is (S47); observations that are linear in $x$ are linear in $(y_C, x_\perp)$. This is an identity of the process, so the model is exact for every Gaussian initial law, and $y_C$ is driven by its own state and a white noise independent of its past, hence Gauss–Markov.

(ii)⇒(i). In the coordinates of (ii) the true conditional mean of $y_{t+1}$ given $(y_t, x_{\perp,t})$ is $A_{yy}y_t + A_{y\perp}x_{\perp,t}$ and the model's is $A_C y_t$. For $t \ge 1$ the law of $x_t$ has a positive density on $\mathbb{R}^{\dim x}$ because $Q^{\mathfrak{n}} \succ 0$, so the two linear functions agree everywhere, $A_{y\perp} = 0$, and (i) follows from Theorem 5.3(b). □

*Proof of Lemma S12.4.* The initial laws agree, so by the chain rule

$$D_{\mathrm{KL}}(\mathbb{P}_{0:T}\|\mathbb{Q}_{0:T}) = \sum_{t=0}^{T-1} \mathbb{E}_{\mathbb{P}} \ln \frac{p(y_{t+1} \mid y_{0:t})}{q(y_{t+1} \mid y_t)},$$

and each term splits as

$$\mathbb{E}\ln\frac{p(y_{t+1}\mid y_{0:t})}{p(y_{t+1}\mid y_t)} + \mathbb{E}\ln\frac{p(y_{t+1}\mid y_t)}{q(y_{t+1}\mid y_t)} = I(y_{t+1}; y_{0:t-1} \mid y_t) + \mathbb{E}D_{\mathrm{KL}}(p(\cdot\mid y_t)\|q(\cdot\mid y_t)).$$

The second term does not depend on $t$ by stationarity; it is nonnegative and vanishes if and only if $q = p(\cdot \mid y_t)$ almost surely. Let $S^{(k)} = \mathrm{Var}(y_{t+1} \mid y_{t-k+1:t})$ be the one-step prediction error covariance from $k$ past values, which by stationarity does not depend on $t$, so that $S^{(1)} = S_1$. For Gaussian laws the first term is $\frac{1}{2}\ln\det(S_1(S^{(t+1)})^{-1})$. As $k$ grows, $S^{(k)}$ decreases in the Loewner order and converges to $S_\infty \succ 0$, because orthogonal projections onto an increasing sequence of closed subspaces converge strongly to the projection onto the closure of their union; $\ln\det$ is continuous on positive definite matrices, so the terms converge to $\mathfrak{d}_{\mathrm{M}}^{\infty}$, and Cesàro averaging gives (S48) and the minimum. Given $x_t$, $y_{t+1} = \mathsf{P}(Ax_t + \mathfrak{n}_t)$ is independent of $y_{0:t-1}$, so by the chain rule $I(y_{t+1}; y_{0:t-1} \mid y_t) \le I(y_{t+1}; x_t, y_{0:t-1} \mid y_t) = I(y_{t+1}; x_t \mid y_t) + I(y_{t+1}; y_{0:t-1} \mid x_t, y_t) = \mathfrak{d}_{\mathrm{cl}}$, and the limit inherits the bound. Finally, the laws of an observation window under $\mathbb{P}$ and under $\mathbb{Q}$ are the images of the laws of the corresponding window of $y$ under one Markov kernel, so the relative entropy can only decrease (data processing). □

*Proof of Proposition S12.5.* (a) For symmetric $M \succ 0$ the stationary covariance solves $M\Sigma + \Sigma M = 2I$, so $\Sigma = M^{-1}$, and $\mathrm{Cov}(x_{t+h}, x_t) = e^{-Mh}M^{-1}$ for $h \ge 0$. Its Fourier transform is

$$\int_{\mathbb{R}} e^{-M|h|}M^{-1}e^{-\mathrm{i}\mathrm{f}h}\mathrm{d}h = M^{-1}\left[(M + \mathrm{i}\mathrm{f}I)^{-1} + (M - \mathrm{i}\mathrm{f}I)^{-1}\right] = 2(\mathrm{f}^2 I + M^2)^{-1},$$

so the autocovariance of $y$ is $(2\pi)^{-1}\int \mathcal{S}_y(\mathrm{f})e^{\mathrm{i}\mathrm{f}h}\mathrm{df}$; likewise for $\mathbb{Q}$, whose autocovariance is $e^{-\varpi_1|h|}/\varpi_1$. Expanding $\phi_C = \sum_k c_k \mathsf{v}_k$ in an orthonormal eigenbasis of $M$ with eigenvalues $\lambda_k \ge \varpi_1$,

$$\frac{\mathcal{S}_y(\mathrm{f})}{\mathcal{S}_{\mathrm{OU}}(\mathrm{f})} = \sum_k c_k^2 \frac{\mathrm{f}^2 + \varpi_1^2}{\mathrm{f}^2 + \lambda_k^2},$$

in which every factor lies in $(0,1]$ and equals 1 on the eigenspace of $\varpi_1$; since $\sum_k c_k^2 = 1$, the ratio lies in $[\cos^2\alpha_C, 1]$.

(b) For $\mathsf{a} \in \mathbb{R}^m$ and times $t_1, \dots, t_m$, $\mathsf{a}^\top C \mathsf{a} = (2\pi)^{-1} \int \mathcal{S}(\mathrm{f}) |\sum_l \mathsf{a}_l e^{\mathrm{i}\mathrm{f}t_l}|^2 \mathrm{df}$ for either law, so the pointwise sandwich of (a) passes to the covariance matrices. $C_{\mathbb{Q}} \succ 0$ because $\mathcal{S}_{\mathrm{OU}} > 0$ and exponentials at distinct times are linearly independent. The eigenvalues $\nu_j$ of $C_{\mathbb{Q}}^{-1} C_{\mathbb{P}}$ therefore lie in $[\cos^2\alpha_C, 1]$, the relative entropy is $\frac{1}{2}\sum_j (\nu_j - 1 - \ln \nu_j)$, and $\nu \mapsto \nu - 1 - \ln \nu$ decreases on $(0,1]$, which gives the first bound of (S49) as $\frac{m}{2}\mathsf{g}(\sin^2\alpha_C)$; the second is $\mathsf{g}(x) \le x^2/(2(1-x))$.

(c) $\Pi_{CC} = \mathrm{diag}(\varepsilon_C^{\Pi}) + L_C + \mathrm{diag}(a^{\partial})$ with cut couplings $a^{\partial} \ge 0$, so for $Z$ with orthonormal rows spanning $\mathbf{1}_C^{\perp}$, $\lambda_{\min}(Z \Pi_{CC} Z^\top) \ge \varkappa_{\mathrm{in}}(C)$ by Definition 4.10, while $\phi_C^\top \Pi_{CC} \phi_C = \varkappa_{\mathrm{out}}(C)$ and $|(\Pi_{CC} - \varkappa_{\mathrm{out}})\phi_C| = \sigma_C$. If $r(C) > 1$, Lemma 11.10(b) applies to $(M, \phi_C)$ and gives $\tan\alpha_C \le \sigma_C / (\lambda_{\min}(ZMZ^\top) - \varkappa_{\mathrm{out}}) \le \sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}}) = t$, with $t = s/(r(C)-1)$ because $\varkappa_{\mathrm{in}} = r(C)\varkappa_{\mathrm{out}}$. Then $\sin^2\alpha_C = \tan^2\alpha_C/(1+\tan^2\alpha_C) \le t^2/(1+t^2)$, and since $\mathsf{g}$ increases, the relative entropy per sampled time is at most $\frac{1}{2}\mathsf{g}(t^2/(1+t^2)) \le t^4/(4(1+t^2))$, which is the stated bound. □

## Exact messages and predictive coding

*Proof of Theorem 12.2.* Write the joint density of node variables and data as $p(x_{u_0}) \prod_{u \ne u_0} p(x_u \mid x_{\mathrm{pa}(u)}) \prod_u \ell_u(x_u)$, with $\ell_u$ the data likelihood at $u$, and let

$$L_u(x_u) = \int \prod_{v \in \mathcal{T}_u \setminus u} p(x_v \mid x_{\mathrm{pa}(v)}) \prod_{v \in \mathcal{T}_u} \ell_v(x_v) \mathrm{d}x_{\mathcal{T}_u \setminus u},$$

the likelihood of the data in $\mathcal{T}_u$ given $x_u$ up to a factor free of the variables. Since $\mathcal{T}_u \setminus u$ is the disjoint union of the children's subtrees, Fubini's theorem gives $L_u = \ell_u \prod_{c \in \mathrm{ch}(u)} \lambda_c$ with $\lambda_c(x_u) = \int p(x_c \mid x_u) L_c(x_c) \mathrm{d}x_c$. If $L_c \propto \exp(h^\top x_c - \frac{1}{2} x_c^\top \Lambda x_c)$, the exponent of the integrand is $-\frac{1}{2} x_c^\top (\Psi_c^{-1} + \Lambda) x_c + x_c^\top (\Psi_c^{-1} \mathsf{G}_c x_u + h) - \frac{1}{2} x_u^\top \mathsf{G}_c^\top \Psi_c^{-1} \mathsf{G}_c x_u$, and integrating out $x_c$ leaves

$$\lambda_c(x_u) \propto \exp\Big(-\frac{1}{2} x_u^\top \mathsf{G}_c^\top \big[\Psi_c^{-1} - \Psi_c^{-1}(\Psi_c^{-1} + \Lambda)^{-1} \Psi_c^{-1}\big] \mathsf{G}_c x_u + x_u^\top \mathsf{G}_c^\top \Psi_c^{-1} (\Psi_c^{-1} + \Lambda)^{-1} h\Big).$$

From $(\Psi_c^{-1} + \Lambda)\Psi_c = I + \Lambda\Psi_c$ we get $\Psi_c^{-1}(\Psi_c^{-1} + \Lambda)^{-1} = (I + \Lambda\Psi_c)^{-1}$ and $\Psi_c^{-1} - (I + \Lambda\Psi_c)^{-1}\Psi_c^{-1} = (I + \Lambda\Psi_c)^{-1}\Lambda$, which give the stated $(\Lambda_c^{\uparrow}, h_c^{\uparrow})$; $I + \Lambda\Psi_c$ is invertible because $\Lambda\Psi_c$ is similar to $\Psi_c^{1/2}\Lambda\Psi_c^{1/2} \succeq 0$. Induction from the leaves, where $L_u = \ell_u$, proves (a).

For (b), the factorization separates the variables and data of $\mathcal{T}_u \setminus u$ from all others given $x_u$, so $p(x_u \mid \text{all data}) \propto p(x_u \mid \text{data outside } \mathcal{T}_u) L_u(x_u)$. It remains to show that $(\Lambda_u^{\downarrow}, h_u^{\downarrow})$ is the information of $p(x_u \mid \text{data outside } \mathcal{T}_u)$, which holds at the root, where there are no such data and the prior is proper. Let it hold at $u$ and let $c \in \mathrm{ch}(u)$. The data outside $\mathcal{T}_c$ are those outside $\mathcal{T}_u$, those at $u$ and those in the other children's subtrees, so $p(x_u \mid \text{data outside } \mathcal{T}_c) \propto p(x_u \mid \text{data outside } \mathcal{T}_u) \ell_u(x_u) \prod_{c' \ne c} \lambda_{c'}(x_u) = p(x_u \mid \text{all data}) / \lambda_c(x_u)$ up to normalization, a proper Gaussian law whose information is the cavity. Since $x_c$ depends on the data outside $\mathcal{T}_c$ only through $x_u$, $p(x_c \mid \text{data outside } \mathcal{T}_c) = \int p(x_c \mid x_u) p(x_u \mid \text{data outside } \mathcal{T}_c) \mathrm{d}x_u$, which is the stated Gaussian law; induction from the root proves (b).

(c) From $\Lambda_u^{\mathrm{post}} = \Lambda_u^{\downarrow} + \Lambda_u^{\mathcal{T}}$, $m_u^{\mathrm{post}} = (\Lambda_u^{\mathrm{post}})^{-1}(\Lambda_u^{\downarrow} m_u^{\downarrow} + h_u^{\mathcal{T}}) = m_u^{\downarrow} + (\Lambda_u^{\mathrm{post}})^{-1}(h_u^{\mathcal{T}} - \Lambda_u^{\mathcal{T}} m_u^{\downarrow})$, and $h_u^{\mathcal{T}} - \Lambda_u^{\mathcal{T}} m_u^{\downarrow}$ splits over the direct data and the children by (107); the two expressions follow from the definitions of $(\Lambda_u^{\mathrm{o}}, h_u^{\mathrm{o}})$ and $(\Lambda_c^{\uparrow}, h_c^{\uparrow})$. Given $m_u^{\downarrow}$ and $\Lambda_c^{\uparrow}$, $h_c^{\uparrow}$ is the prediction error plus $\Lambda_c^{\uparrow} m_u^{\downarrow}$. □

*Proof of Proposition S12.6.* (a) Up to a constant, $\mathcal{F}(x) = \frac{1}{2} x_{u_0}^\top \Lambda_{u_0}^{\downarrow} x_{u_0} - h_{u_0}^{\downarrow\top} x_{u_0} + \sum_{u \ne u_0} \frac{1}{2}(x_u - \mathsf{G}_u x_{\mathrm{pa}(u)})^\top \Psi_u^{-1} (x_u - \mathsf{G}_u x_{\mathrm{pa}(u)}) - \sum_u (h_u^{\mathrm{o}\top} x_u - \frac{1}{2} x_u^\top \Lambda_u^{\mathrm{o}} x_u)$. Its Hessian $\mathsf{H}$ dominates the Hessian of the first two groups of terms, which is the precision of the joint prior of $x$, a nondegenerate Gaussian

law because the root prior is proper and every link noise is nondegenerate; so $\mathsf{H} \succ 0$. The posterior density is proportional to $e^{-\mathcal{F}}$, hence Gaussian with precision $\mathsf{H}$ and mean the unique minimizer $\mathsf{H}^{-1}\mathsf{h}$. Differentiating the link term of $u$ gives $\mathfrak{p}_u$, the link terms of its children give $-\mathsf{G}_c^\top \mathfrak{p}_c$, the data give $-(h_u^\circ - \Lambda_u^\circ x_u)$ and the root prior gives $\mathfrak{p}_{u_0}$, which is (S50). Gradient descent maps the error $x - \mathsf{H}^{-1}\mathsf{h}$ to $(I - \alpha\mathsf{H})(x - \mathsf{H}^{-1}\mathsf{h})$, a contraction for $0 < \alpha < 2/\lambda_{\max}(\mathsf{H})$.

(b) Given the other factors, the free-energy-minimizing factor is $q(x_u) \propto \exp(\mathbb{E}_{q_{-u}}[-\mathcal{F}])$; since $\mathcal{F}$ is quadratic, this is $\exp(-\frac{1}{2}x_u^\top \mathsf{H}_{uu} x_u + x_u^\top(\mathsf{h}_u - \sum_{v\neq u}\mathsf{H}_{uv}\mu_v))$ up to a constant, Gaussian with precision $\mathsf{H}_{uu}$ and mean $\mathsf{H}_{uu}^{-1}(\mathsf{h}_u - \sum_{v\neq u}\mathsf{H}_{uv}\mu_v)$, so updating the nodes in a fixed order is block Gauss–Seidel for $\mathsf{H}x = \mathsf{h}$. Write $\mathsf{H} = D - L - L^\top$ in that order, with $D$ block diagonal and $L$ strictly block lower triangular. A sweep maps the error $e$ to $e' = e - y$ with $y = (D - L)^{-1}\mathsf{H}e$, and since $y^\top \mathsf{H} e = y^\top(D - L)y = \frac{1}{2}y^\top(D + \mathsf{H})y$,

$$e'^\top \mathsf{H} e' = e^\top \mathsf{H} e - 2y^\top \mathsf{H} e + y^\top \mathsf{H} y = e^\top \mathsf{H} e - y^\top D y.$$

$D \succ 0$ and $y \neq 0$ when $e \neq 0$, so the sweep strictly decreases the $\mathsf{H}$-norm on the unit sphere and, by compactness, contracts it by a factor below one; its fixed point solves $\mathsf{H}x = \mathsf{h}$. Finally $(\mathsf{H}^{-1})_{uu}$ is the inverse of the Schur complement $\mathsf{H}_{uu} - \mathsf{H}_{u,-u}\mathsf{H}_{-u,-u}^{-1}\mathsf{H}_{-u,u} \preceq \mathsf{H}_{uu}$, so $(\mathsf{H}^{-1})_{uu} \succeq \mathsf{H}_{uu}^{-1}$. □

## Literal reading and persistent records

*Proof of Proposition 12.3.* (a) By Theorem 12.2(b), $\Lambda_c^{\text{post}} = \Lambda_c^{\mathcal{T}} + (\mathsf{G}_c(\Lambda_{u\setminus c}^{\text{cav}})^{-1}\mathsf{G}_c^\top + \Psi_c)^{-1}$. Since $\Lambda_u^{\text{post}} = \Lambda_{u\setminus c}^{\text{cav}} + \Lambda_c^{\uparrow} \succeq \Lambda_{u\setminus c}^{\text{cav}}$, inversion gives $(\Lambda_u^{\text{post}})^{-1} \preceq (\Lambda_{u\setminus c}^{\text{cav}})^{-1}$, so $\mathsf{G}_c(\Lambda_u^{\text{post}})^{-1}\mathsf{G}_c^\top + \Psi_c \preceq \mathsf{G}_c(\Lambda_{u\setminus c}^{\text{cav}})^{-1}\mathsf{G}_c^\top + \Psi_c$, and inverting once more reverses the order; the two inverses agree only if the two matrices do.

(b) Let $e_c$ be the error of the cavity mean of $x_C$ (variance $1/\mathsf{a}$), $\mathfrak{e}_u = x_u - x_C$ (variance $\psi$) and $\mathsf{n}$ the error of the child's data estimate $\hat{x}_u = h_u^{\mathcal{T}}/\mathsf{J}$ (variance $1/\mathsf{J}$), independent by the tree structure, and let $\mathsf{s} = \mathsf{a} + \Lambda_u^{\uparrow}$ with $\Lambda_u^{\uparrow} = \mathsf{J}/(1 + \mathsf{J}\psi)$. The upward message is a likelihood for $x_C$ centered at $\hat{x}_u$ with variance $\psi + 1/\mathsf{J}$, so the parent's posterior mean satisfies $m_C - x_u = (\mathsf{a}e_c - \mathsf{a}\mathfrak{e}_u + \Lambda_u^{\uparrow}\mathsf{n})/\mathsf{s}$. The literal child has prior precision $\tilde{\pi} = (1/\mathsf{s} + \psi)^{-1}$ and mean error $[\tilde{\pi}(m_C - x_u) + \mathsf{J}\mathsf{n}]/(\tilde{\pi} + \mathsf{J})$. Put $\mathsf{D}_0 = \mathsf{a}(1 + \mathsf{J}\psi) + \mathsf{J}$ and $\mathsf{E}_0 = (1 + \mathsf{J}\psi)(1 + \mathsf{a}\psi) + \mathsf{J}\psi$; then $\mathsf{s} = \mathsf{D}_0/(1 + \mathsf{J}\psi)$, $\tilde{\pi} = \mathsf{D}_0/\mathsf{E}_0$, $\tilde{\pi}/\mathsf{s} = (1 + \mathsf{J}\psi)/\mathsf{E}_0$, $\tilde{\pi}\Lambda_u^{\uparrow}/\mathsf{s} = \mathsf{J}/\mathsf{E}_0$, $\mathsf{E}_0 + 1 = (1 + \mathsf{J}\psi)(2 + \mathsf{a}\psi)$ and $\tilde{\pi} + \mathsf{J} = (1 + \mathsf{J}\psi)(\mathsf{a} + \mathsf{J}(2 + \mathsf{a}\psi))/\mathsf{E}_0$. Summing squared coefficients times variances,

$$(\tilde{\pi} + \mathsf{J})^2 \mathrm{MSE}_1 = \Big(\frac{\tilde{\pi}}{\mathsf{s}}\Big)^2 \mathsf{a}^2\Big(\frac{1}{\mathsf{a}} + \psi\Big) + \frac{1}{\mathsf{J}}\Big(\frac{\tilde{\pi}\Lambda_u^{\uparrow}}{\mathsf{s}} + \mathsf{J}\Big)^2 = \frac{(1 + \mathsf{J}\psi)^2\big[\mathsf{a}(1 + \mathsf{a}\psi) + \mathsf{J}(2 + \mathsf{a}\psi)^2\big]}{\mathsf{E}_0^2},$$

which is (108). The claimed precision is $\mathsf{J} + \tilde{\pi} = (1 + \mathsf{J}\psi)(\mathsf{a} + \mathsf{J}(2 + \mathsf{a}\psi))/\mathsf{E}_0$, so $\mathrm{oc}_1 = [\mathsf{a}(1 + \mathsf{a}\psi) + \mathsf{J}(2 + \mathsf{a}\psi)^2](1 + \mathsf{J}\psi)/[(\mathsf{a} + \mathsf{J}(2 + \mathsf{a}\psi))\mathsf{E}_0]$. Substitute $\mathsf{a} = (1 - \mathsf{w})\mathsf{s}$, $\psi = \mathsf{t}/\mathsf{s}$ and $\mathsf{J} = \mathsf{w}\mathsf{s}/(1 - \mathsf{w}\mathsf{t})$, which follow from the definitions of $\mathsf{w}$ and $\mathsf{t}$ and from $\Lambda_u^{\uparrow} = \mathsf{J}/(1 + \mathsf{J}\psi)$; then $\mathsf{a}\psi = (1 - \mathsf{w})\mathsf{t}$, $1 + \mathsf{J}\psi = 1/(1 - \mathsf{w}\mathsf{t})$ and $\mathsf{E}_0 = (1 + \mathsf{t})/(1 - \mathsf{w}\mathsf{t})$. Multiplying the remaining numerator and denominator by $(1 - \mathsf{w}\mathsf{t})/\mathsf{s}$ turns them into $(1 - \mathsf{w})(1 + (1 - \mathsf{w})\mathsf{t})(1 - \mathsf{w}\mathsf{t}) + \mathsf{w}(2 + (1 - \mathsf{w})\mathsf{t})^2 = 1 + 3\mathsf{w} + \mathsf{t}(1 - \mathsf{w})(1 + 2\mathsf{w})$ and $(1 - \mathsf{w})(1 - \mathsf{w}\mathsf{t}) + \mathsf{w}(2 + (1 - \mathsf{w})\mathsf{t}) = 1 + \mathsf{w}$, which gives (109). The bound $\mathsf{t} < 1/\mathsf{w}$ reads $\psi < (1 + \mathsf{J}\psi)/\mathsf{J}$, which always holds. With $A_0 = 1 + 3\mathsf{w}$ and $B_0 = (1 - \mathsf{w})(1 + 2\mathsf{w})$ the derivative of $(A_0 + \mathsf{t}B_0)/(1 + \mathsf{t})$ has the sign of $B_0 - A_0 = -2\mathsf{w}(1 + \mathsf{w}) < 0$; at $\mathsf{t} = 1/\mathsf{w}$ both numerator and denominator of (109) equal $(1 + \mathsf{w})^2/\mathsf{w}$; and $(1 + 3\mathsf{w})/(1 + \mathsf{w}) < 2$ because $\mathsf{w} < 1$, which holds since $\mathsf{a} > 0$.

(c) At $\psi = 0$, $\Lambda_u^{\uparrow} = \mathsf{J}$, $\mathrm{MSE}_1 = (\mathsf{a} + 4\mathsf{J})/(\mathsf{a} + 2\mathsf{J})^2$ and the exact posterior precision is $\mathsf{a} + \mathsf{J}$; with $\mathsf{a} = \mathsf{J}(1 - \mathsf{w})/\mathsf{w}$ the ratio $(\mathsf{a} + 4\mathsf{J})(\mathsf{a} + \mathsf{J})/(\mathsf{a} + 2\mathsf{J})^2$ is $(1 + 3\mathsf{w})/(1 + \mathsf{w})^2$, whose derivative has the sign of $1 - 3\mathsf{w}$, so its maximum is $9/8$ at $\mathsf{w} = 1/3$. With $n$ children of information $\mathsf{J}$ under a prior $\Lambda_\circ$, $\Lambda_C^{\text{post}} = \Lambda_\circ + n\mathsf{J}$, so $\mathsf{w} = \mathsf{J}/(\Lambda_\circ + n\mathsf{J})$, and at $\Lambda_\circ = 0$, $(1 + 3/n)/(1 + 1/n) = (n + 3)/(n + 1)$. □

*Proof of Proposition S12.7.* Let $S_h = \sum_i h_i^{\mathrm{o}}$. Rule (E) holds the union posterior $(\Lambda_\circ m_\circ + S_h, \Lambda_\circ + \Lambda^{\mathrm{o}})$ from the first round on, because the contributions are conditionally independent given $x$. The other rules map natural parameters affinely: (U) $(h_z, \Lambda_z) \mapsto (h_z + S_h, \Lambda_z + \Lambda^{\mathrm{o}})$, (A) $(h_z, \Lambda_z) \mapsto (h_z + S_h/n, \Lambda_z + \Lambda^{\mathrm{o}}/n)$ and (P) $(h_z, \Lambda_z) \mapsto (nh_z + S_h, n\Lambda_z + \Lambda^{\mathrm{o}})$. Solving, $\Lambda_z(t) = w_\circ \Lambda_\circ + w_{\mathrm{d}} \Lambda^{\mathrm{o}}$ and $h_z(t) = w_\circ \Lambda_\circ m_\circ + w_{\mathrm{d}} S_h$ with $(w_\circ, w_{\mathrm{d}}) = (1, t)$, $(1, t/n)$ and $(n^t, (n^t - 1)/(n - 1))$ under (U), (A) and (P). The error of the mean $h_z/\Lambda_z$ is $[w_\circ \Lambda_\circ (m_\circ - x) + w_{\mathrm{d}}(S_h - \Lambda^{\mathrm{o}} x)]/\Lambda_z$, because the coefficient of $x$ cancels, $w_\circ \Lambda_\circ + w_{\mathrm{d}} \Lambda^{\mathrm{o}} = \Lambda_z$; the two terms are independent with variances $\Lambda_\circ$ and $\Lambda^{\mathrm{o}}$, so the mean-square error is $(w_\circ^2 \Lambda_\circ + w_{\mathrm{d}}^2 \Lambda^{\mathrm{o}})/\Lambda_z^2$, which gives Table 13, and oc = MSE $\cdot\, \Lambda_z$. Under (A), oc $< 1$ if and only if $t^2/n^2 < t/n$, that is $t < n$. Under (P), dividing by $n^t$, the mean tends to $((n-1)\Lambda_\circ m_\circ + S_h)/((n-1)\Lambda_\circ + \Lambda^{\mathrm{o}})$ and the error to the stated limit; a linear estimator whose weights are not proportional to the precisions $(\Lambda_\circ, \Lambda^{\mathrm{o}})$ has error above the union value, which is the case for $n \geq 3$. □

*Proof of Proposition S12.8.* (a) By induction: let the record hold the centralized posterior $\mathcal{N}(m_z, P_z)$ of $x_t$, as it does initially with the common prior. Its prediction $\mathcal{N}(Am_z, P^-)$, $P^- = AP_zA^\top + Q_x$, is the centralized predictive law of $x_{t+1}$, and member $i$'s posterior has information $((P^-)^{-1} + \Lambda_i^{\mathrm{o}}, (P^-)^{-1} A m_z + h_i^{\mathrm{o}})$. Rule (U) relative to the prediction returns $((P^-)^{-1} + \sum_i \Lambda_i^{\mathrm{o}}, (P^-)^{-1} A m_z + \sum_i h_i^{\mathrm{o}})$, the centralized update, because the members' fresh data are conditionally independent given $x_{t+1}$.

(b) In the scalar world $x_{t+1} = Fx_t + w_t$ with $n$ members sensing $s_i = x_{t+1} + v_i$, $\mathrm{Var}\, v_i = R$, each member's posterior has precision $1/P^- + 1/R$ with $P^- = F^2/\Lambda_z + Q$ the record's own prediction variance, and the product has $\Lambda_z = n(1/P^- + 1/R)$, whose stationary solution gives the claimed variance $1/\Lambda_z$. The record's mean is $[nFm_z/P^- + \sum_i s_i/R]/\Lambda_z$; substituting $s_i = x_{t+1} + v_i$ and $x_t = (x_{t+1} - w_t)/F$, the coefficient of $x_{t+1}$ cancels because $n/P^- + n/R = \Lambda_z$, and the error obeys $e_z(t+1) = \mathsf{b}_1 e_z(t) - (\mathsf{b}_1/F) w_t + \mathsf{b}_2 \bar{v}_{t+1}$ with $\mathsf{b}_1 = nF/(P^- \Lambda_z)$, $\mathsf{b}_2 = n/(R\Lambda_z)$ and $\bar{v}$ the average sensing noise, of variance $R/n$. Its stationary variance is $((\mathsf{b}_1/F)^2 Q + \mathsf{b}_2^2 R/n)/(1 - \mathsf{b}_1^2)$. At $F = 0.99$, $Q = 0.02$, $n = 5$, $R = 1$ the fixed point is $\Lambda_z = 207.18$, so the claimed variance is 0.00483, $\mathsf{b}_1 = 0.9661$, $\mathsf{b}_2 = 0.02413$ and the actual variance is 0.2876; the centralized filter's stationary variance solves $P = 1/(1/(F^2 P + Q) + n/R)$, giving 0.0528. □

*Proof of Proposition S12.9.* (a) Theorem 12.2(b) at the group and at the member. (b) With $\mathsf{G} = 1$ and $\Psi = 0$ the push-forward through a link leaves natural parameters unchanged, so a node's literal posterior is its parent's literal posterior plus the information of the data in its own subtree. By induction from the root, whose literal posterior is the prior plus all data, the literal posterior of the member is the prior plus, for each ancestor $a$ of the member (the member included), the data of $\mathcal{T}_a$. A datum at node $v$ lies in $\mathcal{T}_a$ exactly when $a$ is a common ancestor of the member and $v$, and there are $\mathsf{L} + 1 - \mathsf{h}$ of these. (c) With $\mathsf{G} = 1$ and $\Psi = 0$ all node variables coincide, so the group's cavity with the member removed and the society's message that excludes the member's data are both the posterior given all data except the member's, with precision $\Lambda_S^{\mathrm{post}} - \Lambda_i^{\mathrm{o}}$ and the same mean, and fusing them with the member's datum gives the stated precision. By Theorem 12.2(b) the group's cavity is the law given all data outside the member's subtree, so any further message built from those data is redundant. □

## Records

*Proof of Proposition 12.4.* (a) Let $M$ act on $(f, \zeta)$, $f$ on the members and $\zeta$ on the record: $(Mf)_i = (\varepsilon + \beta) f_i + (L_C f)_i - \beta\zeta$ and $(M(f,\zeta))_z = (\omega + \delta)\zeta - \frac{\omega}{n} \mathbf{1}^\top f$. For $f = g \perp \mathbf{1}$ and $\zeta = 0$, $M(g, 0) = (((\varepsilon + \beta) I + L_C) g, 0)$, so this subspace is invariant with eigenvalues $\varepsilon + \beta + \lambda_k(L_C)$, $k \geq 2$. On the

span of $(\mathbf{1},0)$ and $(0,1)$, $M(\alpha\mathbf{1},\zeta)=(((\varepsilon+\beta)\alpha-\beta\zeta)\mathbf{1},-\omega\alpha+(\omega+\delta)\zeta)$, which is $\mathsf{R}$. The dimensions add to $n+1$. Without the record $M=\varepsilon I+L_C$.

(b) $\mathsf{R}$ is similar, by $\operatorname{diag}(1,(\omega/\beta)^{1/2})$, to a symmetric matrix with $(1,1)$ entry $\varepsilon+\beta$, so $\mathsf{r}_-\le\varepsilon+\beta\le\varepsilon+\beta+\lambda_k(L_C)$: the smallest rate is $\mathsf{r}_-$, the second smallest is $\min\{\mathsf{r}_+,\varepsilon+\beta+\lambda_2(L_C)\}$, and without the record the ratio is $(\varepsilon+\lambda_2(L_C))/\varepsilon$.

(c) The bracket on $\varepsilon/\mathsf{r}_-$ is Proposition 8.12(b), a statement about the matrix $\mathsf{R}$. With $\mathsf{r}_-\le\varepsilon/(1+\mathcal{R})$, $\mathsf{r}_+=\varepsilon+\beta+\omega-\mathsf{r}_-\ge\beta+\omega+\varepsilon\mathcal{R}/(1+\mathcal{R})$, which the hypothesis makes at least $\varepsilon+\beta+\lambda_2(L_C)$; then $\mathsf{g}_{\text{sep}}=(\varepsilon/\mathsf{r}_-)(1+\beta/(\varepsilon+\lambda_2(L_C)))$ and the bracket gives the bounds.

(d) Now $\mathsf{g}_{\text{sep}}=(\varepsilon/\mathsf{r}_-)\mathsf{r}_+/(\varepsilon+\lambda_2(L_C))$, and dividing by $\mathsf{g}_\times=(1+\mathcal{R})(\varepsilon+\beta+\lambda_2(L_C))/(\varepsilon+\lambda_2(L_C))$ gives (112); its second factor is below one, and $\varepsilon/\mathsf{r}_-\le1+\mathcal{R}+\varepsilon/\omega$ for every $\delta\ge0$ by Proposition 8.12(b). For $\delta=0$, $\varepsilon/\mathsf{r}_-\to1+\mathcal{R}$ by the same bracket and $\mathsf{r}_+\to\beta+\omega$ as $\varepsilon\to0$.

(e) Proposition S12.12 for one record gives $\mathsf{r}_-\le(\varepsilon+\mathcal{R}\delta)/(1+\mathcal{R})$.

For the condition stated after the proposition, let $\delta=0$ and $\beta>0$, so that $\mathsf{r}_+\mathsf{r}_-=\det\mathsf{R}=\varepsilon\omega$. If $\mathsf{r}_+\ge\varepsilon+\beta+\lambda_2(L_C)$, then $\mathsf{g}_{\text{sep}}=(\varepsilon/\mathsf{r}_-)(1+\beta/(\varepsilon+\lambda_2(L_C)))>1$, because $\varepsilon/\mathsf{r}_-\ge1+\mathcal{R}$ by Proposition 8.12(b). Otherwise $\mathsf{g}_{\text{sep}}=\varepsilon\mathsf{r}_+/(\mathsf{r}_-(\varepsilon+\lambda_2(L_C)))=\mathsf{r}_+^2/(\omega(\varepsilon+\lambda_2(L_C)))$, which is below one exactly when $\mathsf{r}_+<\mathsf{r}_\circ=(\omega(\varepsilon+\lambda_2(L_C)))^{1/2}$. The inequality $\mathsf{r}_+<\mathsf{r}_\circ$ already places $\mathsf{r}_+$ in this second case: the symmetric matrix similar to $\mathsf{R}$ in (b) has the diagonal entry $\omega$, so $\mathsf{r}_+\ge\omega$, and $\omega<\mathsf{r}_\circ$ gives $\omega<\varepsilon+\lambda_2(L_C)$ and then $\mathsf{r}_\circ<\varepsilon+\lambda_2(L_C)$. By the formula of (a), $\partial\mathsf{r}_+/\partial\beta=\frac12[1+(\varepsilon+\beta+\omega)/((\varepsilon+\beta-\omega)^2+4\beta\omega)^{1/2}]\ge1$, since $(\varepsilon+\beta+\omega)^2-(\varepsilon+\beta-\omega)^2-4\beta\omega=4\varepsilon\omega$, and $\mathsf{r}_+\to\max\{\varepsilon,\omega\}$ as $\beta\to0$. The characteristic polynomial $r^2-(\varepsilon+\beta+\omega)r+\varepsilon\omega$ of $\mathsf{R}$ vanishes at $r=\mathsf{r}_\circ$ exactly when $\beta=(\mathsf{r}_\circ-\varepsilon)(\mathsf{r}_\circ-\omega)/\mathsf{r}_\circ$, and $\mathsf{r}_\circ$ is then the larger root, the other being $\varepsilon\omega/\mathsf{r}_\circ<\mathsf{r}_\circ$. Hence $\mathsf{g}_{\text{sep}}<1$ exactly for $0<\beta<(\mathsf{r}_\circ-\varepsilon)(\mathsf{r}_\circ-\omega)/\mathsf{r}_\circ$, a nonempty range exactly when $\max\{\varepsilon,\omega\}<\mathsf{r}_\circ$, and when $\varepsilon<\omega<\varepsilon+\lambda_2(L_C)$, $\mathsf{g}_{\text{sep}}\to\omega^2/(\omega(\varepsilon+\lambda_2(L_C)))=\omega/(\varepsilon+\lambda_2(L_C))$ as $\beta\to0$. □

*Proof of Corollary 12.5.* By Proposition 5.14(a), under its hypotheses $g>0$, newcomer noise of positive finite variance and $\det\mathsf{A}>0$, the inverse identity time $\tau_{\text{id}}^{-1}$ of the pair $(\bar\mu,z)$, and of $\bar\mu$ alone when $\tilde\beta>0$, is the smaller eigenvalue of $-\mathsf{A}$, whose first row is $(\varepsilon+\beta+g,-(\beta+g\mathsf{f}))=(\tilde\varepsilon+\tilde\beta,-\tilde\beta)$ and whose second row is $(-\omega,\omega+\delta)$, so that $-\mathsf{A}$ is the matrix $\mathsf{R}$ with $(\varepsilon,\beta)$ replaced by $(\tilde\varepsilon,\tilde\beta)$; without the record $\tau_{\text{id}}^{-1}=\tilde\varepsilon$. The ratio of identity times is $\tilde\varepsilon/\varepsilon_{\text{rec}}(\tilde\varepsilon,\tilde\beta,\omega,\delta)$, bracketed for $\delta=0$ by Proposition 8.12(b) and bounded below for $\delta\ge0$ by Proposition S12.12; both need $\tilde\beta>0$, and $\det\mathsf{A}=\tilde\varepsilon(\omega+\delta)+\tilde\beta\delta>0$ gives $\tilde\varepsilon>0$ when $\delta=0$. If $\tilde\beta=0$, then $-\mathsf{A}$ is triangular and, by the definition $\varepsilon_{\text{rec}}=\lambda_{\min}(\mathsf{R})$ and not by the secular equation of Proposition 8.11(b), which assumes positive read weights, $\varepsilon_{\text{rec}}=\min\{\tilde\varepsilon,\omega+\delta\}$ with $\tilde\varepsilon>0$, the ratio is $\max\{1,\tilde\varepsilon/(\omega+\delta)\}$, and both claims hold with $\tilde{\mathcal{R}}=0$. In that case $g>0$ forces $\beta=0$ and $\mathsf{f}=0$, so by Proposition 5.14(a) the members' mean decorrelates at the rate $\varepsilon+g=\tilde\varepsilon$ with the record, which is its rate without it, and a ratio above 1 is the rate $\omega+\delta$ of the record alone. □

*Proof of Lemma 12.6.* (a) Substitute $m_k=\eta+e_k$ and $z=\eta+e_z$ into the innovations, as in Lemma 3.6.

(b) With exact codes and attention closed in $C$, a member's row of the error-transfer matrix over members and record sums to $F(1-k^s)$, with weight $F\beta$ on the record and $F(1-k^s-\beta)$ on the members, while the record's row puts $F\omega/n$ on each member and $F(1-\omega)$ on itself. Hence $(\mathbf{1}_C,0)\mapsto F((1-k^s-\beta)\mathbf{1}_C,\omega)$ and $(0,1)\mapsto F(\beta\mathbf{1}_C,1-\omega)$: the span is invariant with matrix $F\mathsf{N}$, and a common shift evolves as stated. Induction gives $\mathsf{f}_t^z\ge\mathsf{f}_t\ge0$, because $\mathsf{f}_{t+1}^z-\mathsf{f}_{t+1}=F[(1-\omega-\beta)(\mathsf{f}_t^z-\mathsf{f}_t)+k^s\mathsf{f}_t]$ with $1-\omega-\beta\ge0$; then $\mathsf{f}_{t+1}=F[(1-k^s)\mathsf{f}_t+\beta(\mathsf{f}_t^z-\mathsf{f}_t)]\ge F(1-k^s)\mathsf{f}_t$. For $\beta>0$, $\mathsf{f}_t=F^t(\mathsf{c}_1\mathsf{m}_1^t+\mathsf{c}_2\mathsf{m}_2^t)$ with $\mathsf{c}_1>0$ and $|\mathsf{m}_2|<\mathsf{m}_1$ (proof of Theorem 12.7), so $\mathsf{f}_t^{1/t}\to F\mathsf{m}_1$. Since $\mathsf{N}=I-\mathsf{R}'$, with $\mathsf{R}'$ the

matrix of the reduced record model at leak $k^s$ and $\delta = 0$, Proposition 8.12(b) gives the bracket on $1 - \mathsf{m}_1$. □

*Proof of Theorem 12.7.* With exact codes, Lemma 2.13 applies to the error recursion of members and record, in which the world innovation enters every node with weight $-1$; by the invariance in Lemma 12.6(b), every member's common-mode error is $e_i^{\mathrm{com}}(t) = -\sum_{s\ge 0} \mathsf{f}_s w_{t-1-s}$, with variance $Q\mathsf{F}_{\mathrm{com}}$, and it is independent of the connected part, so $\mathbb{E}e_i^2 \ge Q\mathsf{F}_{\mathrm{com}}$.

(a) The discriminant of $\mathsf{N}$ is $(k^s + \beta - \omega)^2 + 4\beta\omega > 0$, so its eigenvalues are real and distinct, with right eigenvectors $(\beta, \mathsf{m}_j - 1 + k^s + \beta)^\top$. Writing $(1,1)^\top$ in this basis gives $\mathsf{c}_1 + \mathsf{c}_2 = 1$ and $\mathsf{c}_1\mathsf{m}_1 + \mathsf{c}_2\mathsf{m}_2 = 1 - k^s$, whence the coefficients, and $\mathsf{f}_s = F^s(\mathsf{c}_1\mathsf{m}_1^s + \mathsf{c}_2\mathsf{m}_2^s)$; summing the geometric series gives (113). The Perron root $\mathsf{m}_1$ of the nonnegative matrix $\mathsf{N}$ is at least its smallest row sum $1 - k^s$ and at least its largest diagonal entry, so $\mathsf{c}_2 \ge 0$; $\mathsf{m}_2 = \operatorname{tr}\mathsf{N} - \mathsf{m}_1 \le (2 - k^s - \beta - \omega) - (1 - \omega) = 1 - k^s - \beta < 1 - k^s$, so $\mathsf{c}_1 > 0$; and $|\mathsf{m}_2| < \mathsf{m}_1 \le 1$, since the largest row sum is 1 and $\operatorname{tr}\mathsf{N} \ge 1 - \omega > 0$ unless $\omega = 1$, which forces $\beta = 0$.

(b) $\mathsf{f}_s \ge (F(1 - k^s))^s$ by Lemma 12.6(b), and $\mathsf{f}_2 > F^2(1 - k^s)^2$ when $\beta > 0$ and $k^s > 0$, because $\mathsf{f}_1^z - \mathsf{f}_1 = Fk^s > 0$.

(c) Differentiate the recursions $\mathsf{f}_{t+1} = F[(1 - k^s - \beta)\mathsf{f}_t + \beta\mathsf{f}_t^z]$ and $\mathsf{f}_{t+1}^z = F[\omega\mathsf{f}_t + (1 - \omega)\mathsf{f}_t^z]$: $\partial_\beta \mathsf{f}_{t+1} = F[(\mathsf{f}_t^z - \mathsf{f}_t) + (1 - k^s - \beta)\partial_\beta\mathsf{f}_t + \beta\partial_\beta\mathsf{f}_t^z]$ and $\partial_\beta\mathsf{f}_{t+1}^z = F[\omega\partial_\beta\mathsf{f}_t + (1-\omega)\partial_\beta\mathsf{f}_t^z]$ have nonnegative coefficients, a nonnegative source and zero initial values, so $\partial_\beta\mathsf{f}_t \ge 0$; likewise $\partial_\omega\mathsf{f}_{t+1}^z = F[(\mathsf{f}_t - \mathsf{f}_t^z) + \omega\partial_\omega\mathsf{f}_t + (1-\omega)\partial_\omega\mathsf{f}_t^z]$ and $\partial_\omega\mathsf{f}_{t+1} = F[(1 - k^s - \beta)\partial_\omega\mathsf{f}_t + \beta\partial_\omega\mathsf{f}_t^z]$ give $\partial_\omega\mathsf{f}_t \le 0$. Each term $\mathsf{f}_t^2$ is therefore monotone, and so is their sum.

(d) The three terms of (113) are nonnegative, and $1 - \mathsf{m}_1 = \varepsilon_{\mathrm{rec}} \le k^s/(1 + \mathcal{R})$ by Lemma 12.6(b), so $1 - F^2\mathsf{m}_1^2 \le 1 - F^2(1 - k^s/(1 + \mathcal{R}))^2$; as $k^s \to 0$, $\mathsf{m}_1 \to 1$ and $\mathsf{m}_2 \to 1 - \beta - \omega$, so $\mathsf{c}_1 \to 1$.

(e) If $k^s = 0$, then $\mathsf{N}\mathbf{1} = \mathbf{1}$, $\mathsf{f}_s = F^s$ and $\mathsf{F}_{\mathrm{com}} = 1/(1 - F^2) < (1 + \mathcal{R})/(1 - F^2)$. Let $k^s > 0$ and put $\mathsf{p}_{\mathrm{m}} = F(1-k^s)$, $\mathsf{p}_{\mathrm{z}} = F(1-\beta-\omega)$, $\mathsf{p}_\times = F^2\beta k^s$ and $\mathsf{p}_{\mathrm{A}} = (F - \mathsf{p}_{\mathrm{m}})(F - \mathsf{p}_{\mathrm{z}}) = F^2k^s(\beta+\omega)$. Then $0 \le \mathsf{p}_{\mathrm{m}}, \mathsf{p}_{\mathrm{z}} < F$, $\mathsf{p}_{\mathrm{A}} - \mathsf{p}_\times = F^2k^s\omega > 0$ and $1 + \mathcal{R} = \mathsf{p}_{\mathrm{A}}/(\mathsf{p}_{\mathrm{A}} - \mathsf{p}_\times)$. Since $\operatorname{tr}\mathsf{N} = 2 - k^s - \beta - \omega$ and $\det\mathsf{N} = (1-k^s)(1-\beta-\omega) - \beta k^s$, the numbers $F\mathsf{m}_1$ and $F\mathsf{m}_2$ are the roots of $x^2 - (\mathsf{p}_{\mathrm{m}} + \mathsf{p}_{\mathrm{z}})x + \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}} - \mathsf{p}_\times$, so by (a) $\mathsf{f}_{s+2} = (\mathsf{p}_{\mathrm{m}} + \mathsf{p}_{\mathrm{z}})\mathsf{f}_{s+1} - (\mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}} - \mathsf{p}_\times)\mathsf{f}_s$, with $\mathsf{f}_0 = 1$ and $\mathsf{f}_1 = \mathsf{p}_{\mathrm{m}}$. Multiplying this recursion by $\mathsf{f}_{s+2}$, $\mathsf{f}_{s+1}$ and $\mathsf{f}_s$ and summing over $s \ge 0$ gives three linear equations for the sums $\sum_s \mathsf{f}_s\mathsf{f}_{s+j}$, $j = 0, 1, 2$, whose solution is

$$\mathsf{F}_{\mathrm{com}} = \frac{(1 - \mathsf{p}_{\mathrm{z}}^2)(1 - \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}}) - \mathsf{p}_\times(1 + \mathsf{p}_{\mathrm{z}}^2)}{(1 - \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}} + \mathsf{p}_\times)\mathsf{T}_-(\mathsf{p}_\times)\mathsf{T}_+(\mathsf{p}_\times)}, \qquad \mathsf{T}_\mp(x) = (1 \mp \mathsf{p}_{\mathrm{m}})(1 \mp \mathsf{p}_{\mathrm{z}}) - x,$$

where $\mathsf{T}_\mp(\mathsf{p}_\times) = \det(I \mp F\mathsf{N}) = (1 \mp F\mathsf{m}_1)(1 \mp F\mathsf{m}_2) > 0$ and $1 - \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}} + \mathsf{p}_\times = 1 - F^2\mathsf{m}_1\mathsf{m}_2 > 0$; at $\beta = 0$ it reduces to $1/(1 - \mathsf{p}_{\mathrm{m}}^2)$. After multiplication by the positive denominators, the claim $(1 - \mathsf{p}_{\mathrm{m}}^2)\mathsf{F}_{\mathrm{com}} < \mathsf{p}_{\mathrm{A}}/(\mathsf{p}_{\mathrm{A}} - \mathsf{p}_\times)$ reads $\mathsf{K}(\mathsf{p}_\times) > 0$, where

$$\mathsf{K}(x) = \mathsf{p}_{\mathrm{A}}(1 - \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}} + x)\mathsf{T}_-(x)\mathsf{T}_+(x) - (\mathsf{p}_{\mathrm{A}} - x)(1 - \mathsf{p}_{\mathrm{m}}^2)\big[(1 - \mathsf{p}_{\mathrm{z}}^2)(1 - \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}}) - x(1 + \mathsf{p}_{\mathrm{z}}^2)\big].$$

Since $\mathsf{T}_-(0)\mathsf{T}_+(0) = (1 - \mathsf{p}_{\mathrm{m}}^2)(1 - \mathsf{p}_{\mathrm{z}}^2)$, $\mathsf{K}(0) = 0$, and expanding gives $\mathsf{K}(x) = x\mathsf{K}_1(x)$ with

$$\mathsf{K}_1(x) = \mathsf{p}_{\mathrm{A}}x^2 - \big[\mathsf{p}_{\mathrm{A}}(1 + 3\mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}}) + (1 - \mathsf{p}_{\mathrm{m}}^2)(1 + \mathsf{p}_{\mathrm{z}}^2)\big]x + (1 - \mathsf{p}_{\mathrm{z}}^2)\big[(1 - \mathsf{p}_{\mathrm{m}}^2)(1 - \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}}) - 2\mathsf{p}_{\mathrm{A}}\mathsf{p}_{\mathrm{m}}^2\big].$$

Now $(1 - \mathsf{p}_{\mathrm{m}})(1 - \mathsf{p}_{\mathrm{z}}) - \mathsf{p}_{\mathrm{A}} = (1 - F)(1 + F - \mathsf{p}_{\mathrm{m}} - \mathsf{p}_{\mathrm{z}}) > 0$, so $\mathsf{p}_{\mathrm{A}} < (1 - \mathsf{p}_{\mathrm{m}})(1 - \mathsf{p}_{\mathrm{z}}) \le 1$, and the vertex of the convex quadratic $\mathsf{K}_1$ lies at $\frac{1}{2}(1 + 3\mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}}) + (1 - \mathsf{p}_{\mathrm{m}}^2)(1 + \mathsf{p}_{\mathrm{z}}^2)/(2\mathsf{p}_{\mathrm{A}}) > \frac{1}{2} + (1 + \mathsf{p}_{\mathrm{m}})/(2(1 - \mathsf{p}_{\mathrm{z}})) \ge 1 > \mathsf{p}_{\mathrm{A}}$, so $\mathsf{K}_1$ decreases on $[0, \mathsf{p}_{\mathrm{A}}]$. At the right end, $\mathsf{K}_1(\mathsf{p}_{\mathrm{A}}) = \mathsf{K}(\mathsf{p}_{\mathrm{A}})/\mathsf{p}_{\mathrm{A}} = (1 - \mathsf{p}_{\mathrm{m}}\mathsf{p}_{\mathrm{z}} + \mathsf{p}_{\mathrm{A}})\mathsf{T}_-(\mathsf{p}_{\mathrm{A}})\mathsf{T}_+(\mathsf{p}_{\mathrm{A}}) > 0$, because $\mathsf{T}_-(\mathsf{p}_{\mathrm{A}}) = (1 - F)(1 + F - \mathsf{p}_{\mathrm{m}} - \mathsf{p}_{\mathrm{z}})$ and $\mathsf{T}_+(\mathsf{p}_{\mathrm{A}}) \ge 1 - \mathsf{p}_{\mathrm{A}}$. Hence $\mathsf{K}_1 > 0$ on $[0, \mathsf{p}_{\mathrm{A}}]$, and $\mathsf{K}(\mathsf{p}_\times) = \mathsf{p}_\times\mathsf{K}_1(\mathsf{p}_\times) > 0$ because $0 < \mathsf{p}_\times < \mathsf{p}_{\mathrm{A}}$. For the limit, write $1 - F = \mathsf{y}'k^s$ with $\mathsf{y}' \to \mathsf{y}$. The numerator of the closed form tends to $(\beta+\omega)^2(2 - \beta - \omega)$,

$1-\mathsf{p_m p_z}+\mathsf{p}_\times$ to $\beta+\omega$ and $\mathsf{T}_+(\mathsf{p}_\times)$ to $2(2-\beta-\omega)$, while $1-\mathsf{p}_\mathrm{m}^2=2(1+\mathsf{y}')k^s+O((k^s)^2)$ and $\mathsf{T}_-(\mathsf{p}_\times)=(\omega+\mathsf{y}'(\beta+\omega))k^s+O((k^s)^2)$. Hence $(1-\mathsf{p}_\mathrm{m}^2)\mathsf{F}_\mathrm{com}\to(1+\mathsf{y})(\beta+\omega)/(\omega+\mathsf{y}(\beta+\omega))$, which is the stated limit. □

*Proof of Proposition S12.12.* $G=-\mathsf{R}_0$ is the generator of a Markov chain on a tree with positive rates in both directions along every edge; such a chain satisfies detailed balance with respect to its stationary vector, which is $\mathsf{u}$ up to scale, so $D_\mathsf{u}\mathsf{R}_0$ is symmetric, and so is $D_\mathsf{u}\mathsf{R}$. Hence $S=D_\mathsf{u}^{1/2}\mathsf{R}D_\mathsf{u}^{-1/2}$ is symmetric with the eigenvalues of $\mathsf{R}$, which are real; $S=S_0+E$ with $S_0=D_\mathsf{u}^{1/2}\mathsf{R}_0D_\mathsf{u}^{-1/2}\succeq 0$, whose null vector $\hat{\mathbf{1}}=D_\mathsf{u}^{1/2}\mathbf{1}/|D_\mathsf{u}^{1/2}\mathbf{1}|$ is simple, with next eigenvalue $g_0$, and $E=\mathrm{diag}(\mathsf{e})\succeq 0$. Since $\mathsf{R}\mathbf{1}=\mathsf{e}$, $\hat{\mathbf{1}}^\top S\hat{\mathbf{1}}=\mathbf{1}^\top D_\mathsf{u}\mathsf{e}/\mathbf{1}^\top D_\mathsf{u}\mathbf{1}=\bar{\mathsf{e}}_\mathsf{u}$, the upper bound. For a unit vector $\alpha\hat{\mathbf{1}}+w$ with $w\perp\hat{\mathbf{1}}$, $S\hat{\mathbf{1}}=E\hat{\mathbf{1}}=\bar{\mathsf{e}}_\mathsf{u}\hat{\mathbf{1}}+b$ with $b\perp\hat{\mathbf{1}}$ and $|b|^2=\mathrm{Var}_\mathsf{u}(\mathsf{e})$, and $w^\top Sw\geq g_0|w|^2$, so the Rayleigh quotient is at least $\alpha^2\bar{\mathsf{e}}_\mathsf{u}-2|\alpha||b||w|+g_0|w|^2$. Its minimum over $\alpha^2+|w|^2=1$ is the smaller eigenvalue of $\left(\begin{smallmatrix}\bar{\mathsf{e}}_\mathsf{u} & -|b|\\ -|b| & g_0\end{smallmatrix}\right)$, which is at least $\bar{\mathsf{e}}_\mathsf{u}-|b|^2/(g_0-\bar{\mathsf{e}}_\mathsf{u})$ when $g_0>\bar{\mathsf{e}}_\mathsf{u}$, by $\sqrt{x^2+y}\leq x+y/(2x)$. Along an edge on which the lower node $m$ reads the upper node $l$ with weight $\beta$ and writes it with weight $\omega$, detailed balance reads $\mathsf{u}_m\beta=\mathsf{u}_l\omega$, so $\mathsf{u}_l/\mathsf{u}_m$ is that edge's $\mathcal{R}$. A star and a chain give the two stated forms. □

*Proof of Proposition S12.13.* (a) The only entries of $M\mathcal{Q}$ that are not symmetric by construction are $(i,z)$ and $(z,i)$ for $i\in C$, equal to $-\beta q_z$ and $-(\omega/n)q$, which agree if and only if $q_z=q_\mathrm{rev}$. Then $M(M^{-1}\mathcal{Q})+(M^{-1}\mathcal{Q})M^\top=\mathcal{Q}+M^{-1}(M\mathcal{Q})=2\mathcal{Q}$, the Lyapunov equation of (110), whose solution is unique because $M$ is stable, and $M^{-1}\mathcal{Q}$ is symmetric. The entries of $\Pi=\mathcal{Q}^{-1}M$ are read off: members' rows sum to $(\varepsilon+\beta-\beta)/q$, the record's row to $(\omega+\delta)/q_\mathrm{rev}-n\beta/q=n\beta\delta/(\omega q)$, and the member-record entry is $-\beta/q$.

(b) $\varkappa_\mathrm{out}(C')=(\sum_{v\in C'}\varepsilon_v^\Pi+\mathrm{cut}(C'))/(n+1)$ with $\mathrm{cut}(C')=\mathrm{cut}(C)/q$, which is the denominator. Every $f\perp\mathbf{1}_{C'}$ supported on $C'$ is $f=(\alpha\mathbf{1}_C+g,-n\alpha)$ with $g\perp\mathbf{1}_C$, and

$$qf^\top(\mathrm{diag}(\varepsilon^\Pi)+L_{C'})f=\varepsilon(n\alpha^2+|g|^2)+\frac{n^3\beta\delta}{\omega}\alpha^2+g^\top L_Cg+\beta(n(n+1)^2\alpha^2+|g|^2),$$

while $|f|^2=n(n+1)\alpha^2+|g|^2$, with no cross terms because $\mathbf{1}_C^\top g=0$ and $L_C\mathbf{1}_C=0$. The minimum of the ratio is the smaller of the minima over the two pieces, $(\varepsilon+\beta+\lambda_2(L_C))/q$ and $(\varepsilon n+n^3\beta\delta/\omega+n(n+1)^2\beta)/(qn(n+1))$, which gives (S52); $r(C)$ is the same computation without the record. For $\delta=0$ the first piece is the smaller exactly when $n\beta\geq\lambda_2(L_C)+\varepsilon n/(n+1)$, and the ratio is $(n+1)(\varepsilon+\beta+\lambda_2(L_C))/(n(\varepsilon+\lambda_2(L_C)))$.

(c) Marginalizing the record subtracts $\Pi_{Cz}\Pi_{zz}^{-1}\Pi_{zC}=(\beta/q)^2\omega q/(n\beta(\omega+\delta))\mathbf{1}_C\mathbf{1}_C^\top=\beta\omega/(nq(\omega+\delta))\mathbf{1}_C\mathbf{1}_C^\top$, an added coupling between every pair of members, and lowers each member's row sum by $\beta\omega/(q(\omega+\delta))$, from $(\varepsilon+\beta)/q$ to $(\varepsilon+\beta\delta/(\omega+\delta))/q$. On $\mathbf{1}_C^\perp$ the added complete-graph Laplacian acts as $\beta\omega/(q(\omega+\delta))$, so $\varkappa_\mathrm{in}=(\varepsilon+\beta+\lambda_2(L_C))/q$, and $\varkappa_\mathrm{out}=(\varepsilon+\beta\delta/(\omega+\delta)+\mathrm{cut}(C)/n)/q$.

(d) Reversibility follows as in (a), since $W_{iz}=\beta$ and $W_{zi}=\omega/n$. The joint matrix $W\geq 0$ is irreducible on $C\cup\{z\}$ and has row sums at most one, below one at the agents, so $\rho(W)<1$, and Theorem 4.9(a) gives couplings $\sum_k W_{ki}W_{kj}/q_k\geq 0$ and leaks $(1-\sum_k W_{ik}\lambda_k)/q_i$ with row sums $\lambda_k=1-\varepsilon$ for agents and $1-\delta$ for the record. For the record, $(1-\omega-\delta)(1-\delta)+\omega(1-\varepsilon)=1-\omega\varepsilon-\delta(2-\omega-\delta)$; for a member, $(1-\varepsilon-\beta)(1-\varepsilon)+\beta(1-\delta)=1-\varepsilon(2-\varepsilon)+\beta(\varepsilon-\delta)$, and $\varepsilon(2-\varepsilon)-\beta(\varepsilon-\delta)\geq\varepsilon(1-\varepsilon)\geq 0$ for $\beta\leq 1$. □

*Derivation of Example S12.15.* The deviations $g\perp\mathbf{1}$ of the members do not feel the record, so they form an Ornstein–Uhlenbeck process with drift $(\varepsilon+\beta)I+L_C$ and noise intensity $2q$, and $\Pi$ restricted to them is $((\varepsilon+\beta)I+L_C)/q$. In the coordinates $u=\mathbf{1}^\top e/\sqrt{n}$ and $z$ the drift is $\left(\begin{smallmatrix}\varepsilon+\beta & -\beta\sqrt{n}\\ -\omega/\sqrt{n} & \omega\end{smallmatrix}\right)$ with noise $\mathrm{diag}(2q,0)$. The Lyapunov equation gives $s_{zz}=s_{uz}/\sqrt{n}$, $s_{uz}=\omega s_{uu}/(\sqrt{n}(\varepsilon+\omega))$ and

$s_{uu} = q(\varepsilon+\omega)/(\varepsilon(\varepsilon+\omega+\beta))$, so $\operatorname{Var}\bar{e} = s_{uu}/n$ and $\operatorname{Cov}(\bar{e}, z) = \operatorname{Var} z = \omega \operatorname{Var}\bar{e}/(\varepsilon+\omega)$; the determinant $\omega\varepsilon s_{uu}^2/(n(\varepsilon+\omega)^2)$ is positive. Inverting, $\Pi_{uu} = (\varepsilon+\omega+\beta)/q$, $\Pi_{uz} = -\sqrt{n}(\varepsilon+\omega+\beta)/q$ and $\Pi_{zz} = n(\varepsilon+\omega)(\varepsilon+\omega+\beta)/(\omega q)$. Reassembling with $L_C\mathbf{1} = 0$, $\Pi_{CC} = ((\varepsilon+\beta)I + L_C)/q + [(\varepsilon+\omega+\beta)-(\varepsilon+\beta)]\mathbf{1}\mathbf{1}^\top/(nq)$, which is (S53), and the off-diagonal member entries are $(\omega/n - a_{ij})/q$. The marginal is $\Pi_{CC} - \Pi_{Cz}\Pi_{zz}^{-1}\Pi_{zC}$, in which the rank-one correction is $(\omega/n)(\varepsilon+\omega+\beta)/(\varepsilon+\omega)\mathbf{1}\mathbf{1}^\top/q$; it has off-diagonal entries $-a_{ij}/q - \omega\beta/(nq(\varepsilon+\omega)) \le 0$ and row sums $\varepsilon(\varepsilon+\omega+\beta)/(q(\varepsilon+\omega)) > 0$. □

*Proof of Proposition S12.17.* By Proposition S12.13(c) with $n = N$, no cut and $\delta = 0$, the members' precision is $q^{-1}((\varepsilon+\beta)I + L - (\beta/N)\mathbf{1}\mathbf{1}^\top)$, which acts as $\varepsilon/q$ on $\mathbf{1}$ and as $((\varepsilon+\beta)I+L)/q$ on $\mathbf{1}^\perp$. With $P_\parallel = \mathbf{1}\mathbf{1}^\top/N$ and $((\varepsilon+\beta)I+L)^{-1}P_\parallel = P_\parallel/(\varepsilon+\beta)$, $\Sigma_{CC} = q[P_\parallel/\varepsilon + ((\varepsilon+\beta)I+L)^{-1} - P_\parallel/(\varepsilon+\beta)]$, which is (S54). The noiseless statement is $\operatorname{Var}\bar{e}$ in the derivation of Example S12.15 with $n = N$; without the record the mean is an Ornstein–Uhlenbeck process with rate $\varepsilon$ and noise intensity $2q/N$, of variance $q/(N\varepsilon)$, and $(\varepsilon+\omega)/(\varepsilon+\omega+\beta) < 1$. □

## Precision balance and conformity

*Proof of Proposition S12.18.* $\tilde{x}_i - x_i = \mathsf{v}(e_C - \mathfrak{e}_i) + (1-\mathsf{v})\mathsf{n}_i$ with independent terms, which gives the mean-square error and, for $i \neq j$, the shared error $\mathsf{v}^2 \operatorname{Var} e_C$; $\tilde{x}_i - \tilde{x}_j = (1-\mathsf{v})(\mathfrak{e}_i - \mathfrak{e}_j + \mathsf{n}_i - \mathsf{n}_j)$ gives the disagreement. The mean-square error is a strictly convex quadratic in $\mathsf{v}$ whose stationary point is $\mathsf{v}^*$, and the limits in (c) are read off. □

*Proof of Proposition S12.20.* Eliminating $\hat{P}^-$ gives $\mathsf{h}(\hat{\Lambda}^+) = \varpi_\mathrm{s} + d\varpi_\mathrm{y} + \mathsf{r}_z$ with $\mathsf{h}(\Lambda) = \Lambda - \Lambda/(F^2 + Q\Lambda) = \Lambda(F^2 + Q\Lambda - 1)/(F^2 + Q\Lambda)$. The function $\mathsf{h}$ is nonpositive on $(0, (1-F^2)/Q]$, and beyond that point $F^2 + Q\Lambda > 1$, so $\mathsf{h}'(\Lambda) = 1 - F^2/(F^2+Q\Lambda)^2 > 1 - F^2 > 0$ and $\mathsf{h}$ increases without bound. Since the right-hand side is positive, the solution $\hat{\Lambda}^+$ is unique and strictly increasing in $\mathsf{r}_z$, and $k^s = \varpi_\mathrm{s}/\hat{\Lambda}^+$ is strictly decreasing. The weight that a member leaves on its own predicted mean is $1 - k^s - \sum_j k_{ij} - \beta = 1/(\hat{P}^-\hat{\Lambda}^+) > 0$, so with $\beta + \omega \le 1$ the hypotheses of Lemma 12.6(b) hold. Write $k_0^s$ for the sensing gain at $\mathsf{r}_z = 0$. There $\beta = 0$, the members' errors do not involve the record, $\mathsf{f}_t = (F(1-k_0^s))^t$ and $\mathsf{F}_\mathrm{com} = 1/(1 - F^2(1-k_0^s)^2)$. For $\mathsf{r}_z > 0$ the gains satisfy $\beta > 0$ and $0 < k^s < k_0^s$, and Theorem 12.7(b) gives

$$\mathsf{F}_\mathrm{com} > \frac{1}{1 - F^2(1-k^s)^2} > \frac{1}{1 - F^2(1-k_0^s)^2}.$$

By the first paragraph of the proof of Theorem 12.7, $Q\mathsf{F}_\mathrm{com}$ is the variance of the common-mode error, which is independent of the remaining part of every member's error, so it is the common-mode part of the mean-square error. □

*Proof of Proposition 12.8.* (a) For agent $i$ in group $\mathcal{K}_\alpha$, $\mathsf{x}_i - \bar{z}_{\alpha,-i} = (n_\alpha - 1)^{-1}\sum_{j\in\mathcal{K}_\alpha, j\neq i}(\mathsf{x}_i - \mathsf{x}_j)$, so $\partial_{\mathsf{x}_i} G_i$ at $\mathsf{x} = \mathsf{x}_i$ equals $2\sum_j J_{ij}(\mathsf{x}_i - \mathsf{x}_j) + 2\mathsf{a}_\mathrm{p}(\mathsf{x}_i - \mathsf{x}_i^0) + \frac{2\beta_\mathrm{r}}{n_\alpha - 1}\sum_{j\in\mathcal{K}_\alpha, j\neq i}(\mathsf{x}_i - \mathsf{x}_j) = \partial_{\mathsf{x}_i}\mathcal{V}$. Both are quadratic in $\mathsf{x}_i$ with the same leading coefficient $\sum_j J_{ij} + \mathsf{a}_\mathrm{p} + \beta_\mathrm{r}$, so $G_i - \mathcal{V}$ does not depend on $\mathsf{x}_i$. If the record's mean includes the agent's own convention, the record term is $\beta_\mathrm{r}((n_\alpha - 1)/n_\alpha)^2(\mathsf{x}_i - \bar{z}_{\alpha,-i})^2$, the same form with $\beta_\mathrm{r}$ scaled by $(n_\alpha - 1)^2/n_\alpha^2$, so the argument applies with the group coupling $\beta_\mathrm{r}(n_\alpha - 1)/n_\alpha^2$.

(b) By (a) the full conditional of $\mathsf{x}_i$ under $e^{-\gamma\mathcal{V}}$ is $q_i$, so the random-scan Gibbs sampler leaves $e^{-\gamma\mathcal{V}}$ invariant; $\mathcal{V}$ is a positive definite quadratic, because $\mathsf{a}_\mathrm{p} > 0$, plus a linear term, so $e^{-\gamma\mathcal{V}}$ is Gaussian with the stated precision, whose off-diagonal entries are nonpositive and whose row sums are $2\gamma\mathsf{a}_\mathrm{p}$. For uniqueness, run two copies with the same choices of agent and the same Gaussian innovations. The conditional mean is linear and the conditional variance does not depend on the state, so their difference $d$ moves deterministically, $d \mapsto d - e_i(\Pi d)_i/\Pi_{ii}$ when agent $i$ moves, and $\|d\|_\Pi^2$

falls by $(\Pi d)_i^2/\Pi_{ii}$. Averaging over the uniform choice among the $N$ agents, $\mathbb{E}\|d'\|_\Pi^2 \le (1-\mathsf{c}/N)\|d\|_\Pi^2$ with $\mathsf{c} = \lambda_{\min}(D_\Pi^{-1/2}\Pi D_\Pi^{-1/2}) > 0$, $D_\Pi = \mathrm{diag}(\Pi)$, because $(\Pi d)^\top D_\Pi^{-1}(\Pi d) \ge \mathsf{c}\|d\|_\Pi^2$. So $\|d_t\|_\Pi$ never increases and $\mathbb{E}[\|d_t\|_\Pi^2 \mid d_0] \le (1-\mathsf{c}/N)^t\|d_0\|_\Pi^2$, hence $d_t \to 0$ almost surely from every pair of initial states. If both copies start from stationary laws, under any coupling of the initial states, then for every bounded Lipschitz function $f$ the difference of the expectations of $f$ does not depend on $t$ and is at most $\mathbb{E}\min\{2\sup|f|, \mathrm{Lip}(f)|d_t|\} \to 0$, so the two laws coincide.

(c) The leak is uniform, and the couplings inside $\mathcal{K}_\alpha$ are those of $2\gamma(L_{J,\alpha} + \frac{\beta_\mathrm{r}}{n_\alpha - 1}L_\alpha^{\mathrm{cpl}})$, where $L_\alpha^{\mathrm{cpl}} = n_\alpha I - \mathbf{1}\mathbf{1}^\top$ acts as $n_\alpha$ on $\mathbf{1}^\perp$; so $\varkappa_{\mathrm{in}} = 2\gamma(\mathsf{a}_\mathrm{p} + \lambda_2(L_{J,\alpha}) + \beta_\mathrm{r} n_\alpha/(n_\alpha - 1))$. Readership couplings join only members of one group, so $\mathrm{cut}(\mathcal{K}_\alpha) = 2\gamma\,\mathrm{cut}_J(\mathcal{K}_\alpha)$ and $\varkappa_{\mathrm{out}} = 2\gamma(\mathsf{a}_\mathrm{p} + \mathrm{cut}_J(\mathcal{K}_\alpha)/n_\alpha)$.

(d) By (b) the precision depends only on $J$, $\mathsf{a}_\mathrm{p}$, $\beta_\mathrm{r}$, the readership groups and the common factor $2\gamma$, which cancels in contrast; the native conventions enter only the mean. If the comparison medium is the population itself, the baseline is the supremum of the population's own contrasts, so no region exceeds it and no level above it is reached; under the graph clause with equal couplings on the declared edges and uniform leak, replacing couplings and leak by their means returns $\Pi$. The other two statements are Proposition S4.17 and Proposition S13.7, with the example of Remark S4.16.

(e) Under the deletion nulls $\Pi = 2\gamma\mathsf{a}_\mathrm{p} I$, so every region has $\varkappa_{\mathrm{in}} = \varkappa_{\mathrm{out}}$. Under redraws renewed independently every round, exchangeable in law and independent of the conventions, with equal native conventions $\mathsf{x}_i^0 = \mathsf{x}^0$, round $s$ applies a random affine map $f_s$, the maps of different rounds are independent and identically distributed, and the transition law is equivariant under relabeling of the agents. Couple two copies with the same redraws, choices and innovations: when agent $i$ moves, $d_i \mapsto \sum_j w_{ij} d_j$ with $w_{ij} \ge 0$ and $\sum_j w_{ij} = (\sum_j J_{ij} + \beta_\mathrm{r})/(\sum_j J_{ij} + \mathsf{a}_\mathrm{p} + \beta_\mathrm{r}) < 1$ for the couplings drawn in that round, and the other coordinates are unchanged. So $\max_i|d_i|$ never increases, and over any stretch of rounds in which every agent moves at least once it falls at least by the factor $\mathsf{c}' < 1$, the largest coefficient sum used in the stretch, whatever the order of the moves. Cut the rounds into consecutive stretches, each ending at the first round by which every agent has moved since the previous stretch ended. Their lengths are independent with the coupon-collector law, of mean $N(1 + \frac{1}{2} + \cdots + \frac{1}{N})$, and their factors $\mathsf{c}'_1, \mathsf{c}'_2, \ldots$ are independent and identically distributed with values in $[0, 1)$; by the strong law of large numbers, applied to the lengths and to the logarithms of the factors, whose mean lies in $[-\infty, 0)$, the product of the factors of the stretches completed by round $t$ is at most $e^{-\mathsf{c}'' t}$ for some $\mathsf{c}'' > 0$ and all large $t$, almost surely. Hence $d_t \to 0$ almost surely from every pair of initial states and, as in (b), there is at most one stationary law. For existence, iterate backward, $Y_t = f_1 \circ \cdots \circ f_t(\mathsf{x}^0\mathbf{1})$. The sup-norm Lipschitz constant of $f_1 \circ \cdots \circ f_t$ is at most the same product of stretch factors, because the contraction over a stretch does not depend on the order of its moves. The deterministic part of every map fixes $\mathsf{x}^0\mathbf{1}$, since the coefficients of an update, the pull included, sum to one, so $|f_{t+1}(\mathsf{x}^0\mathbf{1}) - \mathsf{x}^0\mathbf{1}|_\infty$ is the size of one Gaussian innovation of variance at most $1/(2\gamma\mathsf{a}_\mathrm{p})$, which by the Borel–Cantelli lemma is at most $2(\ln t/(2\gamma\mathsf{a}_\mathrm{p}))^{1/2}$ for all large $t$, almost surely. Therefore $\sum_t |Y_{t+1} - Y_t|_\infty < \infty$, $Y_t$ converges almost surely to some $Y_\infty$, and $Y_\infty = f_1(Y'_\infty)$ with $Y'_\infty = \lim_t f_2 \circ \cdots \circ f_t(\mathsf{x}^0\mathbf{1})$, which has the law of $Y_\infty$ and is independent of $f_1$; so the law of $Y_\infty$ is stationary. Its image under a relabeling is stationary, hence equal to it. Under the exact-fusion null nothing changes: conventions are not inferred from evidence, preferences are kept and attention is exogenous, so the law, and every selected structure, is the same, and by (d) it is predicted by the declared attention and readership. □

*Derivation of Example S12.24.* With uniform precision leak $\varepsilon^\Pi$, a region $C$ of $m$ sites has $r(C) = (\varepsilon^\Pi + \lambda_2(L_C))/(\varepsilon^\Pi + \mathrm{cut}(C)/m)$ by Definition 4.10, and a region that is disconnected in the coupling graph has $\lambda_2(L_C) = 0$ and $r(C) \le 1$. In the comparison ring, with coupling $\bar{a}$ on each of its $N$ edges, the connected regions are arcs and the whole ring. An arc of $m \le N - 1$ sites has $\bar{\lambda}_m = 2\bar{a}(1 - \cos(\pi/m))$ and two boundary edges, cut $= 2\bar{a}$; the whole ring has $r = 1 + 2\bar{a}(1 - \cos(2\pi/N))/\varepsilon^\Pi \ge 1$.

For $m \geq 3$, $r(\text{pair}) - r(\text{arc}_m)$ has the sign of $(\bar{a}+2\bar{a}/m-\bar{\lambda}_m)\varepsilon^{\Pi}+\bar{a}(4\bar{a}/m-\bar{\lambda}_m)$, and both coefficients are positive because $\bar{\lambda}_m \leq \bar{\lambda}_3 = \bar{a}$ and $\bar{\lambda}_m \leq \bar{a}\pi^2/m^2 < 4\bar{a}/m$. The baseline is therefore the larger of the whole ring's contrast and the adjacent pair's $(\varepsilon^{\Pi}+2\bar{a})/(\varepsilon^{\Pi}+\bar{a})$. In the population a domain cut at both ends is an isolated path with cut $= 0$ and contrast $1+\lambda_n/\varepsilon^{\Pi}$, and with uniform leak an admissible level exists exactly when this exceeds the baseline. It beats the whole ring if and only if $\lambda_n > 2\bar{a}(1-\cos(2\pi/N))$, and the pair if and only if $(\varepsilon^{\Pi}+\lambda_n)(\varepsilon^{\Pi}+\bar{a}) > \varepsilon^{\Pi}(\varepsilon^{\Pi}+2\bar{a})$, that is $\varepsilon^{\Pi}(\bar{a}-\lambda_n) < \lambda_n\bar{a}$, which is (S55). For $n \leq N/2$, $\lambda_n \geq 2a(1-\cos(2\pi/N)) > 2\bar{a}(1-\cos(2\pi/N))$ because $w \geq 1$. Under the exchangeable clause the comparison population has coupling $\bar{a}_{\text{ex}} = 2a(N-w)/(N(N-1))$ between every pair, a region of $k$ agents has contrast $(\varepsilon^{\Pi}+k\bar{a}_{\text{ex}})/(\varepsilon^{\Pi}+(N-k)\bar{a}_{\text{ex}})$, increasing in $k$, and the baseline is $1+N\bar{a}_{\text{ex}}/\varepsilon^{\Pi}$; a domain of $n \geq 3$ sites has $\lambda_n \leq a < N\bar{a}_{\text{ex}} = 2a(N-w)/(N-1)$ whenever $w \leq N/2$, in particular when every domain has at least two sites. The bound on the number of walls is needed: $a > N\bar{a}_{\text{ex}}$ if and only if $w > (N+1)/2$, and then a domain of three sites, with $\lambda_3 = a$, exceeds the exchangeable baseline. If the declared graph omits the wall edges, the comparison medium is the population itself, and the first case of Proposition 12.8(d) applies. □

### Three levels

*Proof of Theorem 12.9.* (a) Three subspaces are invariant. On vectors supported on one group's members, orthogonal to that group's constant vector and zero on the records, $M$ acts as $(\varepsilon+\beta_1)I+L$, because records read and are read only through group means. On vectors $(\alpha_g \mathbf{1}_{C_g}, \zeta_g)_g$ with $\sum_g \alpha_g = \sum_g \zeta_g = 0$ and zero society coordinate, the society record reads only $\sum_g \zeta_g$, so the subspace is invariant and each pair $(\alpha_g, \zeta_g)$ evolves by $\mathsf{R}_{\text{grp}}$. On the vectors constant on each type (members, group records, society record) $M$ acts as $\mathsf{R}_{\text{soc}}$. The dimensions $N_{\text{g}}(n-1)+2(N_{\text{g}}-1)+3$ add to $N_{\text{g}}n+N_{\text{g}}+1$.

(b) The first family is the spectrum of a symmetric matrix. Because the read and write weights are positive, $\mathsf{R}_{\text{grp}}$ and $\mathsf{R}_{\text{soc}}$ are tridiagonal with positive products $\beta_1\omega_1$ and $\beta_2\omega_2$ of opposite off-diagonal entries, hence similar to symmetric matrices, and they are irreducible Z-matrices with nonnegative row sums ($\varepsilon$ and $\delta_1+\beta_2$; $\varepsilon$, $\delta_1$ and $\delta_2$), at least one positive, hence nonsingular M-matrices, whose eigenvalues have positive real parts. The positivity of the write weights cannot simply be dropped: with $\omega_2 = \delta_2 = 0$ the society record has zero drift, the last row of $\mathsf{R}_{\text{soc}}$ vanishes, and 0 is an eigenvalue for every $\varepsilon > 0$.

(c) $\mathsf{R}_{\text{soc}}$ is $\mathsf{R}_0 + \text{diag}(\varepsilon, \delta_1, \delta_2)$ on a chain, with $\mathsf{u} = (1, \mathcal{R}_1, \mathcal{R}_1\mathcal{R}_2)$, so Proposition S12.12 gives the upper bound. When $\delta_1 = \delta_2 = 0$ its lower bound is $\bar{\mathsf{e}}_{\mathsf{u}} - O(\varepsilon^2)$, because $\text{Var}_{\mathsf{u}}(\mathsf{e}) = O(\varepsilon^2)$ and $g_0$ does not depend on $\varepsilon$, so the slowest rate is $\varepsilon/(1+\mathcal{R}_1+\mathcal{R}_1\mathcal{R}_2)+O(\varepsilon^2)$. For a two-by-two matrix with positive real eigenvalues $\lambda_- \leq \lambda_+$, $\lambda_- = \det/\lambda_+$ with $\lambda_+ \in [\text{tr}/2, \text{tr}]$, which gives the bracket, and $\det \mathsf{R}_{\text{grp}} = (\varepsilon+\beta_1)(\omega_1+\delta_1+\beta_2)-\beta_1\omega_1$ is the stated expression. The slowest member-deviation rate is $\varepsilon+\beta_1+\lambda_2(L)$, the slower group-level rate lies in $[\det \mathsf{R}_{\text{grp}}/\operatorname{tr}\mathsf{R}_{\text{grp}}, 2\det \mathsf{R}_{\text{grp}}/\operatorname{tr}\mathsf{R}_{\text{grp}}]$ and the slowest society rate is at most the bound above, so the displayed condition orders the slowest rates of the three families. With $\delta_1 > 0$ the bound itself, not its value $\varepsilon/(1+\mathcal{R}_1+\mathcal{R}_1\mathcal{R}_2)$ at $\delta_1 = \delta_2 = 0$, has to be small: Check S47.135 gives an instance in which the latter is far below $\det \mathsf{R}_{\text{grp}}/\operatorname{tr}\mathsf{R}_{\text{grp}}$ while the slowest society rate equals the slower group-level rate. □

## S37 Proofs for Section 13

The proofs follow the order of Section 13. On $\mathbb{Z}^D$ the symbol of $f$ is $\hat{f}(p) = \sum_x f(x)e^{-ip\cdot x}$, $p \in [-\pi, \pi]^D$, and Poisson summation for a subsampled function reads $\sum_{x'} f(bx')e^{-ip'\cdot x'} = b^{-D}\sum_{k\in\mathcal{I}_b} \hat{f}((p'+2\pi k)/b)$.

*Proof of Lemma 13.3.* (a) In the orthonormal basis $[\Phi, \Phi_\perp]$ the precision has the blocks $\Pi_\Phi$, $\Pi_{\perp\Phi}^\top$, $\Pi_{\perp\Phi}$ and $\Pi_{\perp\perp}$, and the block-inverse formula gives $(\Phi^\top \Sigma \Phi)^{-1} = \Pi_\Phi - \Pi_{\perp\Phi}^\top \Pi_{\perp\perp}^{-1} \Pi_{\perp\Phi} = \Pi_\Phi - \mathcal{D}$. As a congruence of $\Pi_{\perp\perp}^{-1} \succ 0$, $\mathcal{D} \succeq 0$, with $\mathcal{D} = 0$ if and only if $\Pi_{\perp\Phi} = 0$, and $\Pi_{\perp\perp}^{-1} \preceq \varkappa_\perp^{-1} I$ gives $\mathcal{D} \preceq \varkappa_\perp^{-1} \Pi_{\perp\Phi}^\top \Pi_{\perp\Phi} \preceq (\|\Pi_{\perp\Phi}\|^2/\varkappa_\perp) I$. The condition $\Pi_{\perp\Phi} = 0$ says $\Pi\Phi = \Phi\Pi_\Phi$, that $\Pi$ maps range $\Phi$ into itself. With uniform leak, $\Pi \mathbf{1}_Y = \varepsilon^\Pi \mathbf{1}_Y + L^\Pi \mathbf{1}_Y$, and $(L^\Pi \mathbf{1}_Y)_x$ is $-\sum_{y \in Y} a_{xy}^\Pi$ for $x \notin Y$ and $\sum_{z \notin Y} a_{xz}^\Pi$ for $x \in Y$; it is constant on every block for every $Y$ exactly when every member of $X$ has the same total coupling to $Y$, for all $X \neq Y$.

(b) $\mathcal{D}_{XY} = (\Pi_{\perp\Phi} \mathbf{e}_X)^\top \Pi_{\perp\perp}^{-1} (\Pi_{\perp\Phi} \mathbf{e}_Y)$, and $\Pi_{\perp\Phi} \mathbf{e}_X = \Phi_\perp^\top \Pi \phi_X$ has the norm $|(I - \Phi\Phi^\top)\Pi\phi_X| = \sigma_X$. The Cauchy–Schwarz inequality in the inner product of $\Pi_{\perp\perp}^{-1} \preceq \varkappa_\perp^{-1} I$ gives the bound, and $\sum_X \sigma_X^2 = \|\Pi_{\perp\Phi}\|_F^2 = \sigma_{\text{cut}}^2$; the spectral norm is at least the norm of every column and at most the Frobenius norm. For uniform leak, $(I - \Phi\Phi^\top)$ annihilates $\varepsilon^\Pi \phi_X$ and is a contraction, so $\sigma_X \leq |L^\Pi \phi_X|$. By the formula for $L^\Pi \mathbf{1}_X$ in (a), with $d_i$ the total coupling of agent $i$ across the boundary of $X$, to the other blocks for $i \in X$ and to $X$ for $i \notin X$, $|L^\Pi \phi_X|^2 = |X|^{-1} \sum_i d_i^2 \leq d_{\max}(X) |X|^{-1} \sum_i d_i = 2 d_{\max}(X) \operatorname{cut}(X)/|X|$, since the members and the nonmembers each contribute $\operatorname{cut}(X)$ to $\sum_i d_i$, and $\operatorname{cut}(X)/|X| = \varkappa_{\text{out}}(X) - \varepsilon^\Pi$.

(c) With $\mathsf{n} = (|X|^{1/2})_X$, $\Phi \mathsf{n} = \mathbf{1}$ and $\Pi \mathbf{1} = \varepsilon^\Pi \mathbf{1}$, so $\Pi_{\perp\Phi} \mathsf{n} = \varepsilon^\Pi \Phi_\perp^\top \mathbf{1} = 0$ and $\mathcal{D}\mathsf{n} = 0$. The block averages are $\bar{e} = \operatorname{diag}(\mathsf{n})^{-1} y$, with exact precision $\operatorname{diag}(\mathsf{n})(\Pi_\Phi - \mathcal{D}) \operatorname{diag}(\mathsf{n}) = \mathsf{P}\Pi\mathsf{P}^\top - \operatorname{diag}(\mathsf{n}) \mathcal{D} \operatorname{diag}(\mathsf{n})$, and $\operatorname{diag}(\mathsf{n}) \mathcal{D} \operatorname{diag}(\mathsf{n}) \mathbf{1} = \operatorname{diag}(\mathsf{n}) \mathcal{D} \mathsf{n} = 0$.

(d) For $X \neq Y$, $(\mathsf{P}\Pi\mathsf{P}^\top)_{XY} = \sum_{x \in X, y \in Y} \Pi_{xy} = -\operatorname{cut}(X, Y) \leq 0$, and $\sum_Y (\mathsf{P}\Pi\mathsf{P}^\top)_{XY} = \sum_{x \in X} (\Pi \mathbf{1})_x = \sum_{x \in X} \varepsilon_x^\Pi \geq 0$. The blocks are disjoint and nonempty, so $\mathsf{P}^\top u \neq 0$ for $u \neq 0$ and $u^\top \mathsf{P}\Pi\mathsf{P}^\top u > 0$. For blocks of equal size $s$, $\Pi_\Phi = s^{-1} \mathsf{P}\Pi\mathsf{P}^\top$. □

*Proof of Theorem 13.4.* The eliminated sites form the segments $\{kb + 1, \dots, kb + b - 1\}$, each coupled only to its two flanking retained sites, so $\Pi_{\bar{\mathcal{R}}\bar{\mathcal{R}}}$ is block diagonal with blocks $\mathsf{S} = a^\Pi \operatorname{tridiag}(-1, 2\cosh m, -1)$ of size $b-1$, and the Schur complement couples only consecutive retained sites. The Dirichlet Green function is $(\mathsf{S}^{-1})_{ij} = \sinh(im) \sinh((b-j)m)/(a^\Pi \sinh m \sinh bm)$ for $1 \leq i \leq j \leq b-1$, extended symmetrically. As a function of $i$ it satisfies $2 g_i \cosh m = g_{i-1} + g_{i+1}$ away from $i = j$ and vanishes at $i = 0$ and $i = b$, and at $i = j$ the recursion leaves the remainder $[\sinh((j+1)m) \sinh((b-j)m) - \sinh(jm) \sinh((b-j-1)m)]/(\sinh m \sinh bm) = 1$, by $\sinh A \sinh B = \frac{1}{2}(\cosh(A+B) - \cosh(A-B))$. The coupling between the retained sites $0$ and $b$ is $(a^\Pi)^2 (\mathsf{S}^{-1})_{1,b-1} = a^\Pi \sinh m / \sinh bm$, and each of the two adjacent segments lowers the diagonal entry by $(a^\Pi)^2 (\mathsf{S}^{-1})_{11} = a^\Pi \sinh((b-1)m)/\sinh bm$, so $\Pi_{00}^{(1)} = 2a^\Pi \cosh m - 2a^\Pi \sinh((b-1)m)/\sinh bm$. Writing $\Pi_{00}^{(1)} = 2a^{\Pi,(1)} \cosh m^{(1)}$ and using $\sinh((b-1)m) = \sinh bm \cosh m - \cosh bm \sinh m$ gives $\cosh m^{(1)} = \cosh bm$, and then $\varepsilon^{\Pi,(1)} = 2a^{\Pi,(1)}(\cosh bm - 1) = 2a^\Pi \sinh m \tanh(bm/2)$. With $\cosh m = 1 + \mathfrak{x}/2$ this is (117); $\mathsf{Ch}_b'(1) = b^2$ gives $\Lambda_{\text{RG}} = b^2$ at $\mathfrak{x} = 0$, and $m^{(l)} = b^l m$ makes $\mathfrak{x}^{(l)} = 2(\cosh(b^l m) - 1)$ increase to infinity for every $m > 0$. The lengths: $\xi_{\text{OZ}} = 1/m$ and, for nearest-neighbor couplings, $\xi_2 = 1/(2\sinh(m/2))$ (Proposition S8.27(c)); with $1/(4\sinh^2(y/2)) = y^{-2} - 1/12 + O(y^2)$, $(\xi_2^{(1)})^2 = (bm)^{-2} - 1/12 + O(m^2) = \xi_2^2/b^2 - (1-b^{-2})/12 + O(m^2)$. Decimation is a marginalization, so the site variance is preserved, and indeed $1/(2a^{\Pi,(1)} \sinh m^{(1)}) = 1/(2a^\Pi \sinh m)$; finally $\sinh m / \sinh bm \to 1/b$ as $m \to 0$. □

*Proof of Proposition S13.1.* (a) Decimation evaluates $\Sigma$ at $bx'$, and $\mathsf{p}^{bx'} = (\mathsf{p}^b)^{x'}$.

(b) Write $\mathsf{h} = \sinh \mu$, $\mathsf{g} = \cosh \mu$, $\mathsf{h}' = \sinh \mu'$, $\mathsf{g}' = \cosh \mu'$ and $u = \cos p$. The symbols of $e^{-\mu|x|}$ and of $\mathsf{p}^x e^{-\mu'|x|}$ are $\mathsf{h}/(\mathsf{g} - u)$ and $\mathsf{h}'/(\mathsf{g}' - \mathsf{p}u)$, so

$$\hat{\Sigma} = \frac{(\mathsf{p}\mathsf{c}_+ \mathsf{h} + \mathsf{c}_- \mathsf{h}')(u_* - u)}{(\mathsf{g} - u)(\mathsf{g}' - \mathsf{p}u)}, \qquad \hat{\Pi} = \frac{(\mathsf{g} - u)(\mathsf{g}' - \mathsf{p}u)}{(\mathsf{p}\mathsf{c}_+ \mathsf{h} + \mathsf{c}_- \mathsf{h}')(u_* - u)}.$$

Dividing the quadratic numerator by $u_* - u$ leaves an affine function of $u = \cos p$, a precision of range one, plus the constant $(\mathsf{g} - u_*)(\mathsf{g}' - \mathsf{p}u_*)$ times $1/(u_* - u)$, whose Fourier coefficients are proportional to $e^{-\operatorname{arccosh}(u_*)|x|}$ when $u_* > 1$; the constant is nonzero because $u_*$ differs from $\mathsf{g}$ and from $\mathsf{pg}'$, as follows. For $\mathsf{p} = 1$, $u_*$ is a convex combination of $\mathsf{g}'$ and $\mathsf{g}$, which are distinct and above one. For $\mathsf{p} = -1$, which occurs for odd $b$, the function $x \mapsto \sinh(Bx)/\sinh x$ is increasing on $(0, \infty)$ for $B > 1$, since its logarithmic derivative is $(\varphi(Bx) - \varphi(x))/x$ with $\varphi(y) = y \coth y$, and $\varphi'(y) = (\sinh y \cosh y - y)/\sinh^2 y > 0$, and $m' > m$ because $\cosh m' - \cosh m = c/a > 0$. Hence, with $\mathsf{c}_-/\mathsf{c}_+ = \sinh m/\sinh m'$,

$$\frac{\mathsf{c}_-\mathsf{h}'}{\mathsf{c}_+\mathsf{h}} = \frac{\sinh(b^l m')/\sinh m'}{\sinh(b^l m)/\sinh m} > 1, \qquad u_* - \mathsf{g} = \frac{\mathsf{c}_+\mathsf{h}(\mathsf{g} + \mathsf{g}')}{\mathsf{c}_-\mathsf{h}' - \mathsf{c}_+\mathsf{h}} > 0,$$

so $u_* > \mathsf{g} > 1$ and the factor $\mathsf{pc}_+\mathsf{h} + \mathsf{c}_-\mathsf{h}'$ is positive. The precision of passing is diffusive (Theorem 4.9(e)), so its Kron reductions on rings whose length is a multiple of $b^l$ are diffusive (Lemma S4.30); on $\mathbb{Z}$ the entries are limits of those on rings, and nonpositive off-diagonal entries and nonnegative row sums survive the limit.

(c) By (a) the staggered term after $l$ steps is $\mathsf{c}_-\mathsf{p}^x e^{-b^l m'|x|}$; for $|x| \geq 1$ its ratio to $\mathsf{c}_+ e^{-b^l m|x|}$ is at most $(\mathsf{c}_-/\mathsf{c}_+)e^{-b^l(m'-m)}$, and at $x = 0$ it is $\mathsf{c}_-$. A pure exponential $\mathsf{c}_+ e^{-\mu|x|}$ is the covariance of a nearest-neighbor precision with mass $\mu$ (Theorem 13.4), and $\sinh m/\sinh m' \to 0$ as $\varepsilon \to 0$, since $m \to 0$ while $m'$ stays positive. □

*Proof of the first part of Example S13.2.* Odd sites have only even neighbors, so $\Pi_{\mathrm{oo}} = (\varepsilon^\Pi + 4a^\Pi)I$ and the Schur complement onto the even sites is $(\varepsilon^\Pi + 4a^\Pi)I - (a^\Pi)^2 N_{\mathrm{eo}} N_{\mathrm{oe}}/(\varepsilon^\Pi + 4a^\Pi)$, where $(N_{\mathrm{eo}} N_{\mathrm{oe}})_{xy}$ counts the odd sites adjacent to both $x$ and $y$: four for $x = y$, two for $y - x \in \{(\pm 1, \pm 1)\}$, one for $y - x \in \{(\pm 2, 0), (0, \pm 2)\}$, and none otherwise. In the rotated coordinates the first displacements are nearest neighbors and the second diagonal neighbors, and the leak is $(\varepsilon^\Pi + 4a^\Pi) - 16(a^\Pi)^2/(\varepsilon^\Pi + 4a^\Pi) = \varepsilon^\Pi(\varepsilon^\Pi + 8a^\Pi)/(\varepsilon^\Pi + 4a^\Pi)$. □

*Proof of Theorem 13.5.* (a) Decimation is Poisson summation. For averaging, $\bar{e}_{x'} = b^{-D} \sum_{v \in \mathcal{I}_b} e_{bx'+v}$, so $\mathrm{Cov}(\bar{e}_0, \bar{e}_{x'}) = g(bx')$ with $g(x) = b^{-2D} \sum_{u,v \in \mathcal{I}_b} \Sigma(x + v - u)$, the convolution of $\Sigma$ with the autocorrelation of the box, whose symbol is $b^{-2D}|\sum_{v \in \mathcal{I}_b} e^{ip \cdot v}|^2 = F_b(p)$; subsampling $g$ gives the formula.

(b) Decimation. By Poisson summation in the transverse coordinates, the sum of the decimated covariance over the coarse hyperplane $\{x'_1 = t'\}$ is $b^{-(D-1)} \sum_{k_\perp} \tilde{\Sigma}(bt', 2\pi k_\perp/b)$, with $k_\perp \in \{0, \dots, b-1\}^{D-1}$ and $\tilde{\Sigma}(t, p_\perp) = \sum_{x_\perp} \Sigma(t, x_\perp) e^{-ip_\perp \cdot x_\perp}$, the covariance in $t$ of the one-dimensional field with symbol $\hat{\Pi}(\cdot, p_\perp)$. For the nearest-neighbor precision, $\hat{\Pi}(p_1, p_\perp) = \varepsilon^\Pi + 2a^\Pi \sum_{d \geq 2}(1 - \cos p_d) + 2a^\Pi(1 - \cos p_1)$, so $\tilde{\Sigma}(t, p_\perp)$ is a pure exponential in $|t|$ with rate $m(p_\perp)$, $\cosh m(p_\perp) = 1 + (\varepsilon^\Pi + 2a^\Pi \sum_{d \geq 2}(1 - \cos p_d))/(2a^\Pi)$, which exceeds $m_1 = m(0)$ for every $p_\perp = 2\pi k_\perp/b$ with $k_\perp \neq 0$. For passing, $\hat{\Pi} = (1 - \hat{W}^2)/q$ with $\hat{W} = c + 2a \sum_d \cos p_d$, and $\tilde{\Sigma}(\cdot, p_\perp)$ is the covariance of Theorem 8.10(e) with $c$ replaced by $c(p_\perp) = c + 2a \sum_{d \geq 2} \cos p_d$: its smooth rate solves $\cosh m = (1 - c(p_\perp))/(2a)$, which is smallest at $p_\perp = 0$, where it is $m_1$, and its staggered rate solves $\cosh m'(p_\perp) = (1 + c(p_\perp))/(2a)$, above $\cosh m_1 = (1 - c - 2a(D-1))/(2a)$ for every $p_\perp$ because $c > 0$. In both cases the term $k_\perp = 0$, a positive multiple of $e^{-m_1|t|}$ plus a faster staggered part, decays at the rate $m_1$ per site, hence $bm_1$ per coarse step, and every other term decays faster. Averaging. Summed over the coarse transverse positions and the transverse offsets, every fine transverse displacement occurs exactly $b^{D-1}$ times, so the hyperplane sum of the averaged field is $b^{-D-1} \sum_{u_1, v_1 = 0}^{b-1} \tilde{\Sigma}(bt' + v_1 - u_1, 0)$. For $t' \geq 1$ every argument is positive, so the sum is a positive multiple of $e^{-bm_1 t'}$ plus a term of the faster rate $bm'_1$. The effective ratio follows from $\cosh m_1 = 1 + \varepsilon/(2a)$, with $\cosh(bm_1) - 1 = b^2(\cosh m_1 - 1)(1 + O(m_1^2))$.

(c) Expand at $p'=0$. For decimation,

$$\hat\Sigma^{\mathsf{Dec}}(0)=b^{-D}(\hat\Sigma(0)+\mathsf{S}_1),\qquad -\partial^2_{p'_d}\hat\Sigma^{\mathsf{Dec}}(0)=b^{-D-2}(-\partial^2_{p_d}\hat\Sigma(0)+\mathsf{S}_{2,d}),$$

with $-\partial^2_{p_d}\hat\Sigma(0)=2\xi^2_{2,d}\hat\Sigma(0)$ by Proposition S8.27(a); the covariance-side length is half the ratio of the two. The decimated precision is diffusive, being a Kron reduction on tori (Lemma S4.30) and a limit of those on $\mathbb{Z}^D$. Its symbol $1/\hat\Sigma^{\mathsf{Dec}}$ is twice differentiable, because $\hat\Pi$ is (finite second moments) and $\hat\Pi\ge\varepsilon^\Pi>0$, so by Fatou's lemma its nonnegative couplings have finite second moments, and Proposition S8.27(a) identifies its second-moment length with the covariance-side one. Averaging: $F_b(2\pi k/b)=0$ for $k\neq 0$, so $\hat\Sigma^{\mathsf{Avg}}(0)=b^{-D}\hat\Sigma(0)$. With $\sin^2(p/2)/(b^2\sin^2(p/(2b)))=1-p^2(1-b^{-2})/12+O(p^4)$, the term $k=0$ contributes $\hat\Sigma(0)[1-p'^2_d((1-b^{-2})/12+\xi^2_{2,d}/b^2)]$ in coordinate $d$ up to fourth order; a term $k=k_d\mathbf{e}_d$ contributes $(p'^2_d/4)\hat\Sigma(2\pi k_d\mathbf{e}_d/b)/(b^2\sin^2(\pi k_d/b))+O(p'^4)$; and terms with a nonzero coordinate other than $d$ carry the factor $\sin^2(p'_{d'}/2)$ and do not enter $\partial^2_{p'_d}$ at 0. Collecting terms with $\hat\Sigma(0)=1/\varepsilon^\Pi$ gives the formula. For the constants, decimation gives $(\xi^{\mathsf{Dec}}_{2,d})^2-\xi^2_{2,d}/b^2=(\varepsilon^\Pi\mathsf{S}_{2,d}-\varsigma_d^2\mathsf{S}_1)/(2b^2(1+\varepsilon^\Pi\mathsf{S}_1))$, using $2\xi^2_{2,d}\varepsilon^\Pi=\varsigma_d^2$, which tends to $-\varsigma_d^2\mathsf{S}_1^{(0)}/(2b^2)$ because $\mathsf{S}_1$ and $\mathsf{S}_{2,d}$ stay bounded as $\varepsilon^\Pi\to 0$, $\hat\Pi-\varepsilon^\Pi$ being positive at the points $2\pi k/b$, $k\neq 0$; for averaging the last term is $O(\varepsilon^\Pi)$ for the same reason. For the nearest-neighbor kernel in one dimension, $\mathsf{S}_1^{(0)}=\sum_{k=1}^{b-1}(4a^\Pi\sin^2(\pi k/b))^{-1}=(b^2-1)/(12a^\Pi)$ and $\varsigma^2=2a^\Pi$, which give $-(1-b^{-2})/12$, and the classical sum $\sum_{k=1}^{b-1}\sin^{-4}(\pi k/b)=(b^4+10b^2-11)/45$ shows that the last term of the averaging formula is then $(\varepsilon^\Pi/a^\Pi)(b^4+10b^2-11)/(720b^2)+O((\varepsilon^\Pi)^2)$. □

*Proof of Example S13.4.* For $X\ge 1$ every argument $bX+v-u$ with $u,v\in\{0,\dots,b-1\}$ is positive, so $\Sigma^{\mathsf{Avg}}(X)=b^{-2}\mathsf{c}_\Sigma e^{-bmX}\sum_{u,v}e^{-m(v-u)}$, and $\sum_{u,v}e^{-m(v-u)}=\sum_u e^{mu}\sum_v e^{-mv}=\mathsf{g}_b$. At $X=0$ the sum is $\sum_{u,v}e^{-m|v-u|}=b+2\sum_{j=1}^{b-1}(b-j)e^{-jm}$, which is positive and smaller than $\mathsf{g}_b$, because pairs with $u>v$ contribute $e^{-m(u-v)}<e^{m(u-v)}$; this gives $\mathsf{k}_b\in(0,1)$. With $\sum_X e^{-bm|X|}e^{-ipX}=\sinh bm/(\cosh bm-\cos p)$, the symbol of $\Sigma^{\mathsf{Avg}}$ is a positive multiple of $(\mathsf{e}_0+\mathsf{k}_b\cos p)/(\cosh bm-\cos p)$. It is positive for every $p$, so $\mathsf{e}_0>\mathsf{k}_b$, and the precision symbol is a positive multiple of $(\cosh bm-\cos p)/(\mathsf{e}_0+\mathsf{k}_b\cos p)$. The Fourier coefficients of $1/(\mathsf{e}_0+\mathsf{k}_b\cos p)$ are $(-\mathsf{h}_b)^{|n|}/(\mathsf{e}_0^2-\mathsf{k}_b^2)^{1/2}$, where $\mathsf{h}_b\in(0,1)$ solves $(1+\mathsf{h}_b^2)/(2\mathsf{h}_b)=\mathsf{e}_0/\mathsf{k}_b$, by the geometric series. Multiplying by $\cosh bm-\cos p$ gives, for $|n|\ge 1$, the coefficient $(-\mathsf{h}_b)^{|n|}[\cosh bm+(\mathsf{h}_b^{-1}+\mathsf{h}_b)/2]/(\mathsf{e}_0^2-\mathsf{k}_b^2)^{1/2}=-(-\mathsf{h}_b)^{|n|-1}[\mathsf{h}_b\cosh bm+(1+\mathsf{h}_b^2)/2]/(\mathsf{e}_0^2-\mathsf{k}_b^2)^{1/2}$, which is $-\mathsf{c}_1(-\mathsf{h}_b)^{|n|-1}$ with $\mathsf{c}_1>0$ once the positive multiple is included. The covariance symbol is a positive ratio of trigonometric polynomials of degree one, $(\mathsf{e}_0+\mathsf{k}_b\cos p)\propto|1+\mathsf{h}_be^{ip}|^2$ over $\cosh bm-\cos p\propto|1-e^{-bm}e^{ip}|^2$, so the block averages form an autoregressive moving-average field of order $(1,1)$. As $m\to 0$, $\mathsf{g}_b\to b^2$ and $\mathsf{g}_b-\sum_{u,v}e^{-m|v-u|}=\sum_{u>v}2\sinh(m(u-v))=mb(b^2-1)/3+O(m^3)$, so $\mathsf{k}_b=m(b^2-1)/(3b)+O(m^2)$, $\mathsf{e}_0=bm-\mathsf{k}_b+O(m^2)$ and $\mathsf{k}_b/\mathsf{e}_0\to\mathsf{t}_b$, which gives the limit of $\mathsf{h}_b$; $\mathsf{t}_2=1/3$ gives $3-2\sqrt{2}=0.1716$, and $\mathsf{t}_b\to 1/2$ gives $2-\sqrt{3}$. □

*Proof of Proposition S13.5.* (a) By Theorem 13.5(c), $(\xi^{(1)}_{2,d})^2=\xi^2_{2,d}/b^2+c_{b,D}+O(\varepsilon^\Pi)$ for both maps, with $c_{b,D}$ independent of the leak and $\varepsilon^\Pi=\varsigma_d^2\mathfrak{x}_{2,d}/2$; inverting gives $\mathfrak{x}^{(1)}_{2,d}=b^2\mathfrak{x}_{2,d}/(1+O(b^2\mathfrak{x}_{2,d}))$. If every leak of a finite closed population vanished, $\Pi\mathbf{1}=0$ and $\Pi$ would be singular, contradicting (M).

(b) Near $q=0$, $\hat\Sigma(q)=1/\hat\Pi(q)=1/\varepsilon^\Pi-\mathsf{a}_\sigma|q|^\sigma/(\varepsilon^\Pi)^2+O(|q|^{\min(\sigma',2\sigma)})$. Away from $2\pi\mathbb{Z}^D$, $\hat\Sigma$ is twice continuously differentiable, so $\mathsf{S}(p')=\sum_{k\in\mathcal{I}_b\setminus\{0\}}\hat\Sigma((p'+2\pi k)/b)$ is twice continuously differentiable near $p'=0$; it is even, because $\hat\Sigma$ is even and $2\pi$-periodic and $k\mapsto -k$ permutes the nonzero classes modulo $b$, so $\mathsf{S}(p')=\mathsf{S}_1+O(|p'|^2)$. Hence $b^D\hat\Sigma^{\mathsf{Dec}}(p')=1/\varepsilon^\Pi+\mathsf{S}_1-b^{-\sigma}\mathsf{a}_\sigma|p'|^\sigma/(\varepsilon^\Pi)^2+$

$O(|p'|^{\min(\sigma',2\sigma,2)})$, and inverting,

$$\hat{\Pi}^{\mathsf{Dec}}(p') = \frac{b^D\varepsilon^\Pi}{1+\varepsilon^\Pi \mathsf{S}_1} + \frac{b^{D-\sigma}\mathsf{a}_\sigma}{(1+\varepsilon^\Pi \mathsf{S}_1)^2}|p'|^\sigma + O(|p'|^{\min(\sigma',2\sigma,2)}).$$

For averaging, each term $k \neq 0$ carries the factor $F_b((p'+2\pi k)/b) = O(|p'|^2)$, because some coordinate has $k_d \neq 0$ and $\sin^2((p'_d+2\pi k_d)/2) = \sin^2(p'_d/2)$, and the term $k=0$ carries $F_b(p'/b) = 1+O(|p'|^2)$; so $b^D\hat{\Sigma}^{\mathsf{Avg}}(p') = 1/\varepsilon^\Pi - b^{-\sigma}\mathsf{a}_\sigma|p'|^\sigma/(\varepsilon^\Pi)^2 + O(|p'|^{\min(\sigma',2\sigma,2)})$, which inverts to the stated coefficients. The ratio $\varepsilon^\Pi/\mathsf{a}_\sigma$ is therefore multiplied by $b^\sigma(1+\varepsilon^\Pi\mathsf{S}_1)$ or by $b^\sigma$, and $\mathsf{S}_1$ stays bounded as $\varepsilon^\Pi \to 0$ because $\hat{\Pi} - \varepsilon^\Pi$ is positive at the points $2\pi k/b$, $k \neq 0$; so $\Lambda_{\mathrm{RG}} = b^\sigma$ at the massless point. If the couplings had a finite second moment, then $\hat{\Pi}(p) - \varepsilon^\Pi = \sum_x a^\Pi(x)(1-\cos p\cdot x) \leq \frac{1}{2}|p|^2\sum_x a^\Pi(x)|x|^2$, contradicting $\mathsf{a}_\sigma|p|^\sigma \gg |p|^2$ as $p \to 0$; so $\varsigma^2 = \infty$ and $\xi_2 = \infty$.

(c) Under the reversible diffusion the stationary autocovariance at lag $u$ has the symbol $e^{-u\hat{\Pi}(p)}\hat{\Sigma}(p)$, and decimation acts on it by the alias sum of (118), so the term $k$ relaxes at $\hat{\Pi}((p'+2\pi k)/b)$. For $k=0$, $\hat{\Pi}(p'/b) = \varepsilon^\Pi + \sum_x a^\Pi(x)(1-\cos(p'\cdot x/b)) = \varepsilon^\Pi + \sum_x a^\Pi(x)(p'\cdot x)^2/(2b^2) + o(|p'|^2 b^{-2})$, by dominated convergence, since $|1-\cos\theta-\theta^2/2| \leq \theta^2\min\{1,\theta^2/24\}$ and the couplings have finite second moments; the quadratic term is $\sum_d \varsigma_d^2 p_d'^2/(2b^2)$ when the mixed moments vanish. In the time unit $b^2$ the rate is $b^2\varepsilon^\Pi + \sum_x a^\Pi(x)(p'\cdot x)^2/2 + o(|p'|^2)$; under (b) the same holds with $b^\sigma$ and $\mathsf{a}_\sigma|p'|^\sigma$. For $k \neq 0$ and $p' \in [-\pi,\pi]^D$, a coordinate with $k_d \neq 0$ has $(p'_d+2\pi k_d)/b \in [\pi/b, 2\pi-\pi/b]$, so for the nearest-neighbor kernel $\hat{\Pi} - \varepsilon^\Pi \geq 2a^\Pi(1-\cos(\pi/b)) = 4a^\Pi\sin^2(\pi/(2b))$. □

*Derivation for Remark S13.6.* At $\varepsilon^\Pi = 0$ write $u = \hat{\Pi}(p'/b) = \alpha_2 p'^2/b^2 + \alpha_4 p'^4/b^4 + O(p'^6)$ and $\mathsf{S}(p') = \mathsf{S}_1^{(0)} + O(p'^2)$ as in the proof of Proposition S13.5(b). Then $\hat{\Pi}^{\mathsf{Dec}} = b^D u/(1+u\mathsf{S}) = b^D(u - u^2\mathsf{S}_1^{(0)} + O(p'^6))$, whose coefficients of $p'^2$ and $p'^4$ are $b^{-2}\alpha_2$ and $b^{-4}(\alpha_4 - \alpha_2^2\mathsf{S}_1^{(0)})$, up to the factor $b^D$. For the nearest-neighbor kernel $\alpha_2 = a^\Pi$, $\alpha_4 = -a^\Pi/12$ and $\mathsf{S}_1^{(0)} = (b^2-1)/(12a^\Pi)$, so the ratio is $b^{-2}(-1/12 - (b^2-1)/12) = -1/12$. □

*Proof of Theorem 13.6.* (a) For $x$ in a level-1 block $X$ and another level-1 block $Y$, $\sum_{y\in Y} a^{\mathrm{H}}(x,y) = ba^{\mathrm{H}}_{l(X,Y)}$ does not depend on $x$, with $l(X,Y)$ the level of the smallest common block. Hence $\Pi\mathbf{1}_Y$ is constant on every level-1 block, range $\Phi$ is $\Pi$-invariant and $\Pi_{\perp\Phi} = 0$, so $\mathcal{D} = 0$ (Lemma 13.3), and $\mathrm{d}y = -\Phi^\top\Pi(\Phi y + \Phi_\perp\Phi_\perp^\top x)\mathrm{d}t + \sqrt{2}\Phi^\top\mathrm{d}B_t$ with $\Phi^\top\Pi\Phi_\perp = \Pi_{\perp\Phi}^\top = 0$. Symmetric weights with the nested structure have block sums constant on level-1 blocks, so the partition is lumpable for them (Definition 11.7), and Theorem 13.10(b) gives a closed recursion for the block averages.

(b) $(\Pi_\Phi)_{XY} = b^{-1}\sum_{x\in X, y\in Y}\Pi_{xy} = -ba^{\mathrm{H}}_l$ for blocks whose smallest common block has level $l$, which is level $l-1$ in the coarse medium, and $ba^{\mathrm{H}}_l = (\mathfrak{q}A_{\mathrm{h}})\mathfrak{q}^{(l-1)-1}/((b-1)b^{(l-1)-1})$; the row sums are $b^{-1}\sum_{x\in X}(\Pi\mathbf{1})_x = \varepsilon^\Pi$. So the amplitude becomes $\mathfrak{q}A_{\mathrm{h}}$ and $\mathfrak{x}_l$ is multiplied by $1/\mathfrak{q}$. If lengths grow by $b^{1/D_e}$ per level, $1/\mathfrak{q} = (b^{1/D_e})^{y_{\mathrm{RG}}}$ defines $y_{\mathrm{RG}}$, and $D_e - y_{\mathrm{RG}} = D_e\ln(b\mathfrak{q})/\ln b = \alpha_{\mathrm{H}}$ by Proposition 10.7(c); the massless lattice field has the Green function $|x|^{2-D_e}$.

(c) Each member of a level-$(l+1)$ block $C$ has total coupling $A_{\mathrm{h}}\mathfrak{q}^{l'-1}$ to its level-$l'$ cousins, so $\mathrm{cut}(C)/|C| = A_{\mathrm{h}}\sum_{l'=l+2}^{n_{\mathrm{H}}}\mathfrak{q}^{l'-1}$. Inside $C$, a vector that is constant on the level-$(j-1)$ sub-blocks of one level-$j$ block, sums to zero over it and vanishes elsewhere in $C$ is an eigenvector of $L_C$ with eigenvalue $g_j = a^{\mathrm{H}}_j b^j + \sum_{l'=j+1}^{l+1} a^{\mathrm{H}}_{l'} b^{l'-1}(b-1) = A_{\mathrm{h}}\mathfrak{q}^{j-1}b/(b-1) + \sum_{l'=j+1}^{l+1} A_{\mathrm{h}}\mathfrak{q}^{l'-1}$: members of the same sub-block contribute nothing, those of other sub-blocks of the same level-$j$ block contribute $a^{\mathrm{H}}_j b^j$ because the vector sums to zero there, and those outside the level-$j$ block contribute their couplings. These vectors, over $j = 1,\ldots,l+1$, span $\mathbf{1}_C^\perp$, and $g_j - g_{j+1} = A_{\mathrm{h}}\mathfrak{q}^{j-1}(b-\mathfrak{q})/(b-1) > 0$, so $\lambda_2(L_C) = g_{l+1} = bA_{\mathrm{h}}\mathfrak{q}^l/(b-1)$, and with uniform leak $r(C) = (\varepsilon^\Pi + \lambda_2(L_C))/(\varepsilon^\Pi + \mathrm{cut}(C)/|C|)$. By (b) the level-$k$ medium is nested with amplitude $A_{\mathrm{h}}\mathfrak{q}^k$, in which $C$ is a block of level $l+1-k$,

and the same formula gives the same value; for equal blocks the orthonormal and the block-sum conventions give the same contrast (Lemma 13.3(d)).

(d) As $n_{\mathrm{H}} \to \infty$ the denominator of (120) is $\varepsilon^{\Pi} + A_{\mathrm{h}}\mathfrak{q}^{l+1}/(1-\mathfrak{q})$, and dividing by $A_{\mathrm{h}}\mathfrak{q}^{l}$ gives the formula. After clearing denominators $r_l \geq r_0$ reads $\mathfrak{x}_l(r_0-1) \leq b/(b-1) - r_0\mathfrak{q}/(1-\mathfrak{q})$, whose right side is positive exactly when $r_0 < r_\infty$; since $\mathfrak{x}_l = \mathfrak{x}_0\mathfrak{q}^{-l}$ increases with $l$, the solutions are $l \leq l^*$. □

*Proof of Proposition 13.7.* (a) With uniform leak Theorem 4.14 gives $\mathrm{Var}_d(S) \leq \xi_{2,d}^2/(r_0-1)$ in every coordinate, so $\sum_d \mathrm{Var}_d(S) \leq D \max_d \xi_{2,d}^2/(r_0-1)$, while $\sum_d \mathrm{Var}_d(S) = (2|S|^2)^{-1}\sum_{i,j\in S}|x_i - x_j|^2 \geq (|S|-1)/(2|S|) \geq \frac{1}{4}$, since distinct lattice points are at distance at least one.

(b) By Theorem 13.4 the decimated precision is nearest-neighbor with mass $b^l m$, so $\xi_2^{(l)} = 1/(2\sinh(b^l m/2)) \leq 1/(b^l m)$, because $\sinh y \geq y$. The condition of (a), $1/(4\sinh^2(b^l m/2)) \geq (r_0-1)/4$, is $b^l \leq (2/m)\operatorname{arcsinh}((r_0-1)^{-1/2})$, and $\operatorname{arcsinh} y \leq y$.

(c) In one dimension $\xi_2^{(l)} = 1/(2\sinh(b^l m/2)) \leq 1/(2b^l \sinh(m/2)) = \xi_2/b^l$ by (b), because sinh is convex on $[0,\infty)$ and vanishes at 0. Insert $\xi_{2,d}^{(l)} \leq \xi_{2,d}/b^l$ into (a). For passing, $\xi_2 \leq (a/\varepsilon)^{1/2}$ (Proposition S8.27(c)).

(d) Let $X_0 = \{0,\dots,B-1\}^D$, and for $x \in X_0$ let $k_d(x,z) = \lfloor (x_d+z_d)/B \rfloor$ be the block displacement in coordinate $d$ of the pair $(x, x+z)$. By Lemma 13.3(d) the coarse medium is translation invariant on the coarse lattice, with leak $B^D\varepsilon^{\Pi}$ and second moment $\sum_{x\in X_0}\sum_z a^{\Pi}(z)k_d(x,z)^2$ in coordinate $d$. For $z_d \geq 0$ write $z_d = \mathsf{j}B + s$ with $0 \leq s < B$; then $k_d = \mathsf{j} + \mathbf{1}\{x_d \geq B-s\}$, so $\sum_{x_d=0}^{B-1} k_d^2 = (B-s)\mathsf{j}^2 + s(\mathsf{j}+1)^2 = (z_d^2 + s(B-s))/B$, and the same holds for $z_d < 0$ by symmetry. The other $D-1$ coordinates of $x$ contribute the factor $B^{D-1}$, so the second moment is $B^{D-2}\sum_z a^{\Pi}(z)(z_d^2 + s_d(z)(B-s_d(z)))$, and dividing by twice the leak gives the formula. The inequality is $s(B-s) \leq (B-1)z_d^2$: for $1 \leq |z_d| < B$, $s = |z_d|$ and $B - |z_d| \leq (B-1)|z_d|$, with equality only at $|z_d| = 1$; for $|z_d| \geq B$, $s(B-s) \leq B^2/4 < (B-1)z_d^2$; for $z_d = 0$ both sides vanish. Block indicators of level $l+1$ are sums of those of level $l$, so coarse media compose and the level-$l$ medium is that of cubic blocks of side $b^l$; then $(\xi_{2,d}^{(l)})^2 \leq \xi_{2,d}^2/b^l$, with equality for nearest-neighbor couplings, and (a) gives $b^l \leq 4D\max_d \xi_{2,d}^2/(r_0-1)$. □

*Proof of Proposition S13.7.* In the exchangeable medium a region of $n'$ agents has $\lambda_2 = n'\bar{a}$ and cut $/n' = (N-n')\bar{a}$, so its contrast $(\varepsilon^{\Pi}+n'\bar{a})/(\varepsilon^{\Pi}+(N-n')\bar{a})$ increases with $n'$, and the supremum is the whole population, $1+N\bar{a}/\varepsilon^{\Pi}$. With uniform leak $r(C) = (\varepsilon^{\Pi}+\lambda_2(L_C))/(\varepsilon^{\Pi}+\mathsf{c})$, $\mathsf{c} = \mathrm{cut}(C)/n$, and since $\varepsilon_C^{\max}/\bar{\varepsilon}_C = 1 < \bar{r}_{\mathrm{hom}}$, $C$ is a collective at some admissible level exactly when $r(C) > \bar{r}_{\mathrm{hom}}$, that is $\varepsilon^{\Pi}(\varepsilon^{\Pi}+\lambda_2(L_C)) > (\varepsilon^{\Pi}+\mathsf{c})(\varepsilon^{\Pi}+N\bar{a})$, or $\varepsilon^{\Pi}(\lambda_2(L_C) - \mathsf{c} - N\bar{a}) > \mathsf{c}N\bar{a} \geq 0$, which is (S56). The vector $\mathbf{e}_i - \mathbf{1}_C/n$ is orthogonal to $\mathbf{1}_C$, has squared norm $(n-1)/n$ and quadratic form $d_C(i)$, so $\lambda_2(L_C) \leq \frac{n}{n-1}\min_i d_C(i)$, and $\lambda_2(L_C) > N\bar{a} = \frac{N}{N-1}\bar{d}$ gives the last claim. □

*Proof of Proposition S13.8.* By Theorem 13.6(b) and Lemma 13.3(d), the level-$l$ coarse medium is, up to a positive factor, the nested medium on $M = b^m$ units, $m = n_{\mathrm{H}} - l$, with amplitude $A_l = A_{\mathrm{h}}\mathfrak{q}^l$ and leak $\varepsilon^{\Pi}$. Its level-1 blocks have $\lambda_2 = bA_l/(b-1)$ and $\mathsf{c} = \mathrm{cut}/b = A_l(S_m - 1)$ (Theorem 13.6(c)), every unit has total coupling $A_l S_m$, and the exchangeable comparison has the pair coupling $A_l S_m/(M-1)$. By Proposition S13.7 the blocks are collectives if and only if $\varepsilon^{\Pi}(\lambda_2 - \mathsf{c} - MA_lS_m/(M-1)) > \mathsf{c}MA_lS_m/(M-1)$, and $\lambda_2 - \mathsf{c} - MA_lS_m/(M-1) = A_l\Delta_{\mathrm{H}}(b,\mathfrak{q},m)$ and $\mathsf{c}MA_lS_m/(M-1) = A_l^2 MS_m(S_m-1)/(M-1)$; this is (b). For $m = 1$ the medium is exchangeable, hence its own comparison medium, and no region exceeds its baseline.

(a) The level-1 blocks are (b) at $l = 0$. Let $C$ be a block of level $l+1 \geq 2$, judged at level 0. If $l+1 \leq n_{\mathrm{H}} - 1$, then $\lambda_2(L_C) \leq g_{l+1} = bA_{\mathrm{h}}\mathfrak{q}^l/(b-1)$, an eigenvalue of $L_C$ on $\mathbf{1}_C^{\perp}$ by the proof of Theorem 13.6(c), while $\mathrm{cut}(C)/|C| \geq A_{\mathrm{h}}\mathfrak{q}^{l+1}$ and $N\bar{a} > \bar{d} = A_{\mathrm{h}}S_{n_{\mathrm{H}}} \geq A_{\mathrm{h}}\sum_{j=0}^{l+1}\mathfrak{q}^j$, so

$\lambda_2(L_C) - \mathrm{cut}(C)/|C| - N\bar{a} < A_\mathrm{h}[\mathfrak{q}^l/(b-1) - 2\mathfrak{q}^{l+1} - 1] \le A_\mathrm{h}[\mathfrak{q}^l(1-2\mathfrak{q}) - 1] < 0$ for every $\mathfrak{q} > 0$. If $C$ is the whole population, $\lambda_2(L) \le \frac{N}{N-1}\bar{d} = N\bar{a}$ by the degree bound of Proposition S13.7. In both cases (S56) fails.

(c) $\Delta_\mathrm{H} = \frac{2b-1}{b-1} - S_m(2 + \frac{1}{M-1})$. For $m = 2$, $\Delta_\mathrm{H} > 0$ reads $\frac{b}{b-1} - \frac{b^2}{b^2-1} > \mathfrak{q}(1 + \frac{b^2}{b^2-1})$, that is $\mathfrak{q} < b/(2b^2-1)$. For $m \ge 3$, $S_m < 1/(1-\mathfrak{q})$ and $1/(M-1) \le 1/(b^3-1)$, so $\Delta_\mathrm{H} > \frac{2b-1}{b-1} - \frac{2b^3-1}{(1-\mathfrak{q})(b^3-1)}$. At $\mathfrak{q} = b/(2b^2-1)$, $1-\mathfrak{q} = (2b+1)(b-1)/(2b^2-1)$, and the right side is nonnegative if and only if $(2b^2-1)(2b^3-1) \le (4b^2-1)(b^3-1)$, which is $b^2(b-2) \ge 0$; it increases as $\mathfrak{q}$ decreases. So $\mathfrak{q} < b/(2b^2-1)$ gives $\Delta_\mathrm{H} > 0$ for every $m \ge 2$, and $m = 2$ shows the converse. The second statement combines this with (b).

(d) For $m \ge 2$, $S_m \ge 1+\mathfrak{q}$ and $2+1/(M-1) > 2$, so $\Delta_\mathrm{H} < \frac{2b-1}{b-1} - 2 - 2\mathfrak{q} \le \frac{2b-1}{b-1} - 2 - \frac{2}{b} = \frac{2-b}{b(b-1)} \le 0$ when $\mathfrak{q} \ge 1/b$; then (a) and (b) exclude every level.

(e) $r_l - 1 = (\lambda_2 - \mathsf{c})/(\varepsilon^\Pi + \mathsf{c}) \le \lambda_2/\varepsilon^\Pi = bA_\mathrm{h}\mathfrak{q}^l/((b-1)\varepsilon^\Pi)$, while the baseline of the level-$k$ medium satisfies $\bar{r}^{(k)}_\mathrm{hom} - 1 = \frac{M_k}{M_k-1}A_\mathrm{h}\mathfrak{q}^k S_{n_\mathrm{H}-k}/\varepsilon^\Pi > A_\mathrm{h}\mathfrak{q}^k/\varepsilon^\Pi$, and $b\mathfrak{q}^{l-k}/(b-1) \le b\mathfrak{q}/(b-1) < 1/(b-1) \le 1$ for $l > k$ when $\mathfrak{q} < 1/b$. A level admissible at level $k$ exceeds $\bar{r}^{(k)}_\mathrm{hom}$, hence every $r_l$ with $l > k$.

(f) Put $\mathfrak{q}_* = 1/(2b-1)$. For $m \ge 2$, $S_m$ increases strictly with $\mathfrak{q}$, so $\Delta_\mathrm{H} = \frac{2b-1}{b-1} - 2S_m - \frac{S_m}{M-1}$ decreases strictly with $\mathfrak{q}$, and for $\mathfrak{q} < 1$, $2S_m = 2(1-\mathfrak{q}^m)/(1-\mathfrak{q})$ gives $\Delta_\mathrm{H} = \frac{2b-1}{b-1} - \frac{2}{1-\mathfrak{q}} + \frac{2\mathfrak{q}^m}{1-\mathfrak{q}} - \frac{S_m}{M-1}$, which tends to $\frac{2b-1}{b-1} - \frac{2}{1-\mathfrak{q}}$ as $m \to \infty$, a limit that is positive exactly when $\mathfrak{q} < \mathfrak{q}_*$. At $\mathfrak{q}_*$, $2/(1-\mathfrak{q}_*) = (2b-1)/(b-1)$, so $\Delta_\mathrm{H}(b, \mathfrak{q}_*, m) = \frac{(2b-1)^{1-m}}{b-1} - \frac{S_m}{b^m-1}$; since $2b-1 \ge b+1$, $(2b-1)^{m-1} \ge (b+1)^{m-1} \ge \sum_{j=0}^{m-1} b^j = (b^m-1)/(b-1)$, and $S_m > 1$, this is negative for every $m \ge 2$, and so is $\Delta_\mathrm{H}(b, \mathfrak{q}, m)$ for every $\mathfrak{q} \ge \mathfrak{q}_*$; then (a) and (b) exclude every level. For the increase in $m$, let $0 < \mathfrak{q} \le \mathfrak{q}_*$, $m \ge 2$ and $M = b^m$. Then

$$\Delta_\mathrm{H}(b, \mathfrak{q}, m+1) - \Delta_\mathrm{H}(b, \mathfrak{q}, m) = \frac{S_m}{M-1} - \frac{S_{m+1}}{bM-1} - 2\mathfrak{q}^m = \frac{(b-1)MS_m - \mathfrak{q}^m(M-1)(2bM-1)}{(M-1)(bM-1)},$$

and $S_m/\mathfrak{q}^m = \sum_{j=1}^m \mathfrak{q}^{-j} \ge (2b-1)^m + (2b-1)^{m-1} = 2b(2b-1)^{m-1}$, so the numerator is at least $\mathfrak{q}^m[2b(b-1)(2b-1)^{m-1}M - (M-1)(2bM-1)]$. For $m = 2$ the bracket equals $(b-1)[2b^3(b-2)+b+1] > 0$. For $m \ge 3$, $(b-1)(2b-1)^{m-1}/b^m = \frac{b-1}{2b-1}(\frac{2b-1}{b})^m \ge \frac{(b-1)(2b-1)^2}{b^3} = 4 - \frac{8}{b} + \frac{5}{b^2} - \frac{1}{b^3} \ge \frac{9}{8}$, the last expression being increasing in $b \ge 2$, so the bracket is at least $\frac{9}{4}bM^2 - (M-1)(2bM-1) > 0$. Hence $\Delta_\mathrm{H}$ increases strictly with $m \ge 2$ when $\mathfrak{q} \le \mathfrak{q}_*$ (which gives the negativity at $\mathfrak{q}_*$ a second time, the limit there being zero). For $\mathfrak{q} < \mathfrak{q}_*$ the $m \ge 2$ with $\Delta_\mathrm{H}(b, \mathfrak{q}, m) > 0$ are therefore exactly those with $m \ge m^*$, with $m^* = 2$ exactly when $\mathfrak{q} < b/(2b^2-1)$ by (c), and (b) with $m = n_\mathrm{H} - l$ gives the statement. □

*Proof of Theorem 13.8.* (a) is Lemma 13.3(a). (b) is Theorem 5.2(b) in the coordinates $(y, \Phi_\perp^\top x)$, in which the drift has the blocks $\Pi_\Phi$, $\Pi_{\perp\Phi}^\top$, $\Pi_{\perp\Phi}$ and $\Pi_{\perp\perp}$; $\Gamma(u)$ is a congruence of $e^{-\Pi_{\perp\perp}u} \succeq 0$, $\|e^{-\Pi_{\perp\perp}u}\| = e^{-\varkappa_\perp u}$, and $\int_0^\infty e^{-\Pi_{\perp\perp}u}\mathrm{d}u = \Pi_{\perp\perp}^{-1}$.

(c) By the Courant–Fischer theorem, $\varpi_M \le$ max of the Rayleigh quotient over range $\Phi$, which is $\varkappa_\Phi$, and $\varpi_{M+1} \ge$ min over range $\Phi_\perp$, which is $\varkappa_\perp$. From $\Pi\mathsf{V}_\mathrm{f} = \mathsf{V}_\mathrm{f}\,\mathrm{diag}(\varpi_\mathrm{f})$ and $\Pi\Phi = \Phi\Pi_\Phi + \Phi_\perp\Pi_{\perp\Phi}$, the matrix $\mathsf{Q}_\Theta = \mathsf{V}_\mathrm{f}^\top\Phi$ solves $\mathrm{diag}(\varpi_\mathrm{f})\mathsf{Q}_\Theta - \mathsf{Q}_\Theta\Pi_\Phi = \mathsf{V}_\mathrm{f}^\top\Phi_\perp\Pi_{\perp\Phi}$. With the midpoint $\mathsf{c}_\mathrm{m} = (\varkappa_\perp + \varkappa_\Phi)/2$, $\mathsf{M}_1 = \mathrm{diag}(\varpi_\mathrm{f}) - \mathsf{c}_\mathrm{m}I$ and $\mathsf{M}_2 = \mathsf{c}_\mathrm{m}I - \Pi_\Phi$ are at least $\frac{1}{2}(\varkappa_\perp - \varkappa_\Phi)$, the equation reads $\mathsf{M}_1\mathsf{Q}_\Theta + \mathsf{Q}_\Theta\mathsf{M}_2 = \mathsf{V}_\mathrm{f}^\top\Phi_\perp\Pi_{\perp\Phi}$, and its solution $\mathsf{Q}_\Theta = \int_0^\infty e^{-t\mathsf{M}_1}\mathsf{V}_\mathrm{f}^\top\Phi_\perp\Pi_{\perp\Phi}e^{-t\mathsf{M}_2}\mathrm{d}t$ has norm at most $\|\Pi_{\perp\Phi}\|/(\varkappa_\perp - \varkappa_\Phi)$; $\|\mathsf{Q}_\Theta\| = \sin\Theta$. The coordinates $z = \mathsf{V}^\top x$ obey $\mathrm{d}z = -\mathrm{diag}(\varpi)z\mathrm{d}t + \sqrt{2}\mathrm{d}(\mathsf{V}^\top\mathsf{B})$ and are independent, and $y = \mathsf{T}z_\mathrm{s} + \Phi^\top\mathsf{V}_\mathrm{f}z_\mathrm{f}$; the smallest singular value of $\mathsf{T}$ is $\cos\Theta > 0$. $\mathrm{Cov}(y_\mathrm{f}) = \mathsf{Q}_\Theta^\top\mathrm{diag}(\varpi_\mathrm{f})^{-1}\mathsf{Q}_\Theta \preceq \varkappa_\perp^{-1}\mathsf{Q}_\Theta^\top\mathsf{Q}_\Theta$, and $\mathrm{Cov}(y)^{-1} = \Pi_\Phi - \mathcal{D} \preceq \varkappa_\Phi I$, which gives the relative weight. For the drift, $\Phi^\top\Pi\mathsf{V}_\mathrm{s} = \mathsf{T}\,\mathrm{diag}(\varpi_\mathrm{s})$ and $\Phi^\top\Pi = \Pi_\Phi\Phi^\top + \Pi_{\perp\Phi}^\top\Phi_\perp^\top$ give $\mathsf{T}\,\mathrm{diag}(\varpi_\mathrm{s})\mathsf{T}^{-1} - \Pi_\Phi = \Pi_{\perp\Phi}^\top(\Phi_\perp^\top\mathsf{V}_\mathrm{s})\mathsf{T}^{-1}$, with $\|\Phi_\perp^\top\mathsf{V}_\mathrm{s}\| = \sin\Theta$ and $\|\mathsf{T}^{-1}\| = 1/\cos\Theta$.

(d) For $f$ orthogonal to the block constants, with $f_{C_j}$ its restriction to $C_j$ extended by zero, $f^\top \Pi f \geq \sum_j f_{C_j}^\top (\operatorname{diag}(\varepsilon^\Pi) + L_{C_j}) f_{C_j} \geq \varkappa_{\text{in}}^{\min} |f|^2$, since the couplings across blocks add a positive semidefinite Laplacian and each $f_{C_j}$ is orthogonal to $\mathbf{1}_{C_j}$, hence admissible in the minimum that defines $\varkappa_{\text{in}}(C_j)$ (Definition 4.10) and zero on a region with one member, so that only the regions with at least two members enter the minimum. No uniformity of the leak is used. For $f = \Phi \mathsf{y}$, $f^\top \operatorname{diag}(\varepsilon^\Pi) f = \sum_j \bar{\varepsilon}_{C_j} \mathsf{y}_j^2$ and $f^\top L^\Pi f = \sum_{j<k} \operatorname{cut}(C_j, C_k)(\mathsf{y}_j/n_j^{1/2} - \mathsf{y}_k/n_k^{1/2})^2 \leq 2\sum_j (\operatorname{cut}(C_j)/n_j)\mathsf{y}_j^2$, and $\operatorname{cut}(C_j)/n_j = \varkappa_{\text{out}}(C_j) - \bar{\varepsilon}_{C_j}$, so that $f^\top \Pi f \leq \sum_j (2\varkappa_{\text{out}}(C_j) - \bar{\varepsilon}_{C_j})\mathsf{y}_j^2$. The spectral norm is at most the Frobenius norm. For common rates, $\varkappa_\perp - \varkappa_\Phi \geq \varkappa_{\text{in}} - 2\varkappa_{\text{out}}$, so $\mathsf{t}_\Theta \leq \|\Pi_{\perp\Phi}\|/(\varkappa_{\text{in}} - 2\varkappa_{\text{out}})$, and $\varkappa_\Phi/\varkappa_\perp \leq 2\varkappa_{\text{out}}/\varkappa_{\text{in}} = 2/r$. □

*Proof of Proposition 13.9.* $\Pi_{C_jC_j}$ has nonpositive off-diagonal entries and a connected graph, so $\mathsf{c}_\mathrm{p} I - \Pi_{C_jC_j}$ is nonnegative and irreducible for large $\mathsf{c}_\mathrm{p}$, and its Perron vector, the eigenvector of the smallest eigenvalue of $\Pi_{C_jC_j}$, is positive.

(a) $\Pi_{\perp\Phi^\mathsf{P}} = 0$ if and only if $\Pi\Phi^\mathsf{P}\mathbf{e}_j \in \operatorname{range}\Phi^\mathsf{P}$ for every $j$. The $C_j$ block of $\Pi\Phi^\mathsf{P}\mathbf{e}_j$ is $\Pi_{C_jC_j}\mathsf{u}_j$, proportional to $\mathsf{u}_j$, and its $C_k$ block is $\Pi_{C_kC_j}\mathsf{u}_j$. If it vanishes, $\mathcal{D} = 0$ and $\Gamma = 0$ by Theorem 13.8(a),(b), and $\mathfrak{f}$ reduces to $\sqrt{2}\Phi^{\mathsf{P}\top}\mathrm{d}\mathsf{B}_t$, the increment of a standard Brownian motion, so AR0 is exact. Conversely AR0 has stationary covariance $\Pi_{\Phi^\mathsf{P}}^{-1}$, which equals $\operatorname{Cov}(y) = (\Pi_{\Phi^\mathsf{P}} - \mathcal{D}^\mathsf{P})^{-1}$ only if $\mathcal{D}^\mathsf{P} = 0$, that is $\Pi_{\perp\Phi^\mathsf{P}} = 0$.

(b) The first statements are Theorem 13.8 in the frame $\Phi^\mathsf{P}$, whose complement inside each particle holds that particle's other modes. A stationary process $\mathrm{d}y = -\mathsf{A}_\mathrm{M} y\mathrm{d}t + \sqrt{2}\mathsf{G}_\mathrm{M}^{1/2}\mathrm{d}\mathsf{W}_t$ has autocovariance $e^{-\mathsf{A}_\mathrm{M}u}\operatorname{Cov}(y)$, while the exact autocovariance $\Phi^{\mathsf{P}\top}e^{-\Pi u}\Pi^{-1}\Phi^\mathsf{P}$ has derivative $-\Phi^{\mathsf{P}\top}\Phi^\mathsf{P} = -I$ at $u = 0$. Matching requires $\mathsf{A}_\mathrm{M}\operatorname{Cov}(y) = I$, and the Lyapunov equation $\mathsf{A}_\mathrm{M}\operatorname{Cov}(y) + \operatorname{Cov}(y)\mathsf{A}_\mathrm{M}^\top = 2\mathsf{G}_\mathrm{M}$ then gives $\mathsf{G}_\mathrm{M} = I$.

(c) $\Pi\mathbf{1} = \varepsilon^\Pi\mathbf{1}$ and $\Pi \succeq \varepsilon^\Pi I$, so $\mathbf{1}/N^{1/2}$ is a slowest mode; it is a combination of block indicators and lies in range $\Phi^\mathsf{P}$ only if every $\mathsf{u}_j$ is proportional to $\mathbf{1}_{C_j}$. The Rayleigh quotient gives $\lambda_{\min}(\Pi_{\Phi^\mathsf{P}}) \geq \varepsilon^\Pi$. Let $\lambda = \varepsilon^\Pi$, $\Pi v = \lambda v$ and $v = \Phi^\mathsf{P}\mathsf{a} + \Phi_\perp^\mathsf{P}\mathsf{b}$. Since $\lambda \leq \lambda_{\min}(\Pi_{\Phi^\mathsf{P}}) < \varkappa_\perp$, the second block row gives $\mathsf{b} = -(\Pi_{\perp\perp} - \lambda)^{-1}\Pi_{\perp\Phi^\mathsf{P}}\mathsf{a}$ with $\mathsf{a} \neq 0$, and the first gives $[\Pi_{\Phi^\mathsf{P}} - \Pi_{\perp\Phi^\mathsf{P}}^\top(\Pi_{\perp\perp} - \lambda)^{-1}\Pi_{\perp\Phi^\mathsf{P}}]\mathsf{a} = \lambda\mathsf{a}$; hence $\lambda \geq \lambda_{\min}(\Pi_{\Phi^\mathsf{P}}) - \|\Pi_{\perp\Phi^\mathsf{P}}\|^2/(\varkappa_\perp - \lambda)$, and $\varkappa_\perp - \lambda \geq \varkappa_\perp - \lambda_{\min}(\Pi_{\Phi^\mathsf{P}})$. For block averages, $\Pi_\Phi \mathsf{n} = \Phi^\top\Pi\mathbf{1} = \varepsilon^\Pi\mathsf{n}$ with $\mathsf{n} = (|X|^{1/2})_X$.

(d) $(\Pi_\Phi)_{XX} = \varkappa_{\text{out}}(X) = |X|^{-1}\sum_{x\in X}(\varepsilon_x^\Pi + a_x^\partial)$, so for $M$ blocks of $N/M$ agents $M^{-1}\operatorname{tr}\Pi_\Phi = N^{-1}\sum_i(\varepsilon_i^\Pi + a_i^\partial) \leq N^{-1}\sum_i \Pi_{ii}$, with equality only if no agent is coupled inside its block; $\mathcal{D} \succeq 0$ gives the second inequality. A block average has variance $\phi_X^\top\Sigma\phi_X/|X| \leq \lambda_{\max}(\Sigma)/|X|$. □

*Proof of Example S13.9.* The coarse medium of blocks of $b$ consecutive sites has the coupling $\operatorname{cut}(X, X\pm 1) = a^\Pi$, one edge, and the leak $b\varepsilon^\Pi$ (Lemma 13.3(d)), so its second moment in block units is $2a^\Pi$ and its squared length $a^\Pi/(b\varepsilon^\Pi)$. For the nearest-neighbor kernel $\xi_2^2 = a^\Pi/\varepsilon^\Pi$ (Section 4.7), and Theorem 13.5(c) gives the exact length $\xi_2^2/b^2 + O(1)$. For the recursion, $W^{(1)} = \operatorname{diag}(|X|)^{-1}\mathsf{P}W\mathsf{P}^\top$ has self weight $c + 2a(b-1)/b$ and neighbor weight $a/b$. A nearest-neighbor coarse network with message weight $\mathsf{a}_\mathrm{c}$ and leak $\varepsilon$ per round has the axis rate $\mathsf{m}_\mathrm{c}$ with $\cosh\mathsf{m}_\mathrm{c} = 1 + \varepsilon/(2\mathsf{a}_\mathrm{c})$ (Theorem 8.10(c)); equating it with $bm$ gives $\mathsf{a}_\mathrm{c} = \varepsilon/(2(\cosh bm - 1))$, and $\cosh bm - 1 = b^2(\cosh m - 1)(1 + O(m^2)) = b^2\varepsilon(1 + O(\varepsilon))/(2a)$. □

*Proof of Proposition S13.10.* (a) Substitute $x_{\text{int}} = \mathsf{C}_{\text{ir}}x_{\text{ret}} + x_\perp$ into $\mathrm{d}x_{\text{ret}} = -(M_{\text{rr}}x_{\text{ret}} + M_{\text{ri}}x_{\text{int}})\mathrm{d}t + \sqrt{2}\mathsf{N}_\mathrm{r}\mathrm{d}\mathsf{B}_\mathrm{r}$. At stationarity $\operatorname{Cov}(x_\perp, x_{\text{ret}}) = \Sigma_{\text{ir}} - \mathsf{C}_{\text{ir}}\Sigma_{\text{rr}} = 0$.

(b) Itô's rule for the linear map $x_\perp = x_{\text{int}} - \mathsf{C}_{\text{ir}}x_{\text{ret}}$ gives $\mathrm{d}x_\perp = \mathrm{d}x_{\text{int}} - \mathsf{C}_{\text{ir}}\mathrm{d}x_{\text{ret}}$; substituting both rows and $x_{\text{int}} = \mathsf{C}_{\text{ir}}x_{\text{ret}} + x_\perp$ and collecting terms gives the stated drift, the lower block row of $\mathcal{T}M\mathcal{T}^{-1}$ with $\mathcal{T} = \left(\begin{smallmatrix} I & 0 \\ -\mathsf{C}_{\text{ir}} & I \end{smallmatrix}\right)$. By variation of constants $x_\perp(t) = e^{-M_{\perp\perp}t}x_\perp(0) - \int_0^t e^{-M_{\perp\perp}(t-t')}M_{\perp\mathrm{r}}x_{\text{ret}}(t')\mathrm{d}t' +$

$\sqrt{2}\int_0^t e^{-M_{\perp\perp}(t-t')}(\mathsf{N}_{\mathrm{i}}\mathrm{d}\mathsf{B}_{\mathrm{i}}-\mathsf{C}_{\mathrm{ir}}\mathsf{N}_{\mathrm{r}}\mathrm{d}\mathsf{B}_{\mathrm{r}})(t')$, and inserting it into (a) gives the integro-differential equation with $\Gamma_{\mathrm{F}}$. By (a), $x_\perp$ is the Mori complement of Theorem 5.2(a).

(c) If $M_{\mathrm{ri}}=0$ the retained rows do not contain $x_{\mathrm{int}}$, so $x_{\mathrm{ret}}$ solves an autonomous linear equation and is Markov. If $M_{\mathrm{ri}}z\neq 0$ for some $z$, start from $x_{\mathrm{ret}}(0)=0$ with $x_{\mathrm{int}}(0)=0$ or $x_{\mathrm{int}}(0)=z$: the mean of $x_{\mathrm{ret}}(t)$ is $[e^{-Mt}]_{\mathrm{ri}}z=-tM_{\mathrm{ri}}z+O(t^2)$ in the second case and zero in the first, so no transition law of $x_{\mathrm{ret}}$ alone reproduces both. Active states are retained, so $M_{\mathrm{ai}}\neq 0$ gives $M_{\mathrm{ri}}\neq 0$. □

*Proof of Proposition S13.13.* Substitute $m_j=\eta+\bar{e}_J+\tilde{e}_j$ and use $\sum_j \mathsf{w}_j H_j=H_J^{\mathrm{agg}}$ and $\sum_j \mathsf{w}_j\tilde{e}_j=0$. (a) $\operatorname{rank}T_c\bar{U}_J\le\operatorname{rank}T_c=r_c$, and the stacked code has row space $\sum_j\operatorname{row}(H_j)$. (b) With a common code $H$, $\sum_j \mathsf{w}_j H\tilde{e}_j=0$. With uniform weights the Cauchy–Schwarz inequality gives $|\sum_j n^{-1}(H_j-H_J^{\mathrm{agg}})\tilde{e}_j|^2\le n^{-1}\sum_j\|H_j-H_J^{\mathrm{agg}}\|^2\cdot n^{-1}\sum_j|\tilde{e}_j|^2$. The fluctuations of $\tilde{e}_j$ about their world-conditional means are those of the connected parts, and $\sum_j\operatorname{Var}(\tilde{e}_j\mid\eta)=K\operatorname{tr}((I-\mathbf{1}\mathbf{1}^\top/n)\Sigma^{\mathrm{sc}}_{JJ})\le K(n-1)/\varkappa_{\mathrm{in}}(J)$, because every unit vector orthogonal to $\mathbf{1}_J$ has variance at most $1/\varkappa_{\mathrm{in}}(J)$ (Proposition S4.24(a)). (c) The term $H\bar{e}_J$ has world-conditional variance $H^2\operatorname{Var}(\phi_J^\top e)/n\ge H^2/(n\varkappa_{\mathrm{out}}(J))$ by Proposition S4.24(a), the channel noises are independent of it with variance $R_c/n$, and the second moment is at least the variance. The level-0 receiver assigns the variance $R_c/n$, so its believed precision $n/R_c$ exceeds the true precision $n/(H^2/\varkappa_{\mathrm{out}}(J)+R_c)$ at least by the stated factor. □

*Proof of Proposition S13.14.* The strong law of large numbers and the variance of an average of independent matrices give the first statements, and singular values are 1-Lipschitz in the operator norm (Weyl's inequality). (a) Left invariance of the Haar law gives $V\mathbb{E}U=\mathbb{E}U$ for every rotation $V$, and for $K\ge 2$ the only vector fixed by every rotation is 0; apply this to each column. With $\mathbb{E}U=0$, $\mathbb{E}\|T_c\bar{U}^{(l)}\|_F^2=b^{-l}\mathbb{E}\|T_cU\|_F^2=b^{-l}\|T_c\|_F^2$, since the Frobenius norm is invariant under right multiplication by an orthogonal matrix. (b), (c) $\mathbb{E}\sin\vartheta=0$ by symmetry, which gives $\mathbb{E}U$; the code with rows $\mathbf{e}_1^\top$ and $\mathbf{e}_2^\top$ then has $T_c\mathbb{E}U$ with singular values $(1,|\mathbb{E}\cos\vartheta|)$ for rotations about $\mathbf{e}_1$ and $(|\mathbb{E}\cos\vartheta|,|\mathbb{E}\cos\vartheta|)$ for rotations about $\mathbf{e}_3$. (d) A message translated with the true translation has the identity as its effective frame (Section 2). □

*Proof of Proposition S13.15.* (a) $x^\top\bar{P}x=\sum_j\mathsf{w}_j\|P_{V_j}x\|^2\le\|x\|^2$, with equality if and only if $x\in V_j$ for every $j$, and for $0\preceq\bar{P}\preceq I$ the equality $\bar{P}x=x$ is equivalent to $x^\top\bar{P}x=\|x\|^2$. Also $x^\top\bar{P}x=0$ if and only if every $P_{V_j}x=0$, so $\ker\bar{P}=(\sum_j V_j)^\perp$. (b) $\bar{P}^n$ is symmetric positive semidefinite with eigenvalues $\lambda_k(\bar{P})^n$, which are its singular values, and the ratios $(\lambda_k/\lambda_1)^n$ do not increase with $n$. $V_\cap$ is the eigenspace of the eigenvalue one, $V_\cap^\perp$ is invariant, and on it the largest eigenvalue $\lambda_*$ is below one by (a), so $\bar{P}^n-P_\cap=(\bar{P}(I-P_\cap))^n$ has norm $\lambda_*^n$. The effective rank tends to the multiplicity of $\lambda_1$, which is $\dim V_\cap$ when $V_\cap\neq\{0\}$; one meta-level hop has rank $\dim\sum_j V_j$. □

*Proof of Proposition S13.16.* Each aggregate $\psi_C^{(l+1)}$ lies in the span of its summands, so the span of the functionals of level $l+1$ lies in that of level $l$, and its dimension is at most the number of its generators. The sufficiency statements are Proposition S10.16 applied to each decision, with a public message whose rows span the functionals of level $l+1$, respectively those of the members. □

*Proof of Proposition S13.17.* (a) $I+\mathrm{Adj}$ of a connected $\Delta$-regular graph is primitive, with Perron root $1+\Delta$ and Perron vector $\mathbf{1}/n^{1/2}$, so $(I+\mathrm{Adj})^{n_{\mathrm{c}}}/(1+\Delta)^{n_{\mathrm{c}}}\to\mathbf{1}\mathbf{1}^\top/n$ as $n_{\mathrm{c}}\to\infty$. After $n_{\mathrm{c},1}$ rounds every member of group $J$, the representative included, holds natural parameters $(1+\Delta_1)^{n_{\mathrm{c},1}}n^{-1}(\mathbf{1}^\top h_J(0),\mathbf{1}^\top\Lambda_J(0))(1+o(1))$, with the same factor for every group. At level two Proposition S5.5(a) on a regular graph gives the limit mean $\sum_J\mathbf{1}^\top h_J(0)/\sum_J\mathbf{1}^\top\Lambda_J(0)$, in which every member contributes $\Lambda_\circ m_\circ$ and $\Lambda_\circ$ once. Induction on the number of levels gives the product, and the believed precisions grow geometrically at every level by Proposition S5.5(b). (b) For a connected group, $(I+\mathrm{Adj}_J)^{n_{\mathrm{c},1}}=\rho(I+\mathrm{Adj}_J)^{n_{\mathrm{c},1}}\mathsf{v}^{(J)}\mathsf{v}^{(J)\top}(1+o(1))$, and the level-two limit mean is the ratio

of the entering natural parameters weighted by the level-two Perron vector (Proposition S5.5(a)). (c) Contributions add exactly, so the top level holds $(\Lambda_\circ m_\circ + \sum_i h_i^s, \Lambda_\circ + \sum_i \Lambda_i^s)$, the natural parameters of the union posterior, as in Proposition 5.7. □

*Proof of Proposition S13.18.* $e^{-\tau_L(\varepsilon^\Pi I + L^\Pi)} = e^{-\tau_L \varepsilon^\Pi} e^{-\tau_L L^\Pi}$, and the scalar factor cancels in the normalization; the specific heat and the macronodes are functions of $\varrho_{\tau_L}$. The spectrum of $\varepsilon^\Pi I + L^\Pi$ is that of $L^\Pi$ shifted by $\varepsilon^\Pi$, so a cutoff $1/\tau_L < \varepsilon^\Pi$ applied to it keeps no mode. With uniform leak $\varkappa_{\text{in}}(C) = \varepsilon^\Pi + \lambda_2(L_C)$ and $\varkappa_{\text{out}}(C) = \varepsilon^\Pi + \text{cut}(C)/|C|$, so $r(C) \geq r_0$ reads $\varepsilon^\Pi(r_0 - 1) \leq \lambda_2(L_C) - r_0\,\text{cut}(C)/|C|$. □

*Proof of Proposition S13.19.* For $i \in C_1$, $(L^\Pi u_-)_i = \sum_{j \in C_2} a^\Pi_{ij} \cdot 2/N^{1/2} = 2\bar{a}^\partial (u_-)_i$ with $u_- = (\mathbf{1}_{C_1} - \mathbf{1}_{C_2})/N^{1/2}$, and likewise on $C_2$, so $u_-$ is an eigenvector with eigenvalue $2\bar{a}^\partial$. The span of $\mathbf{1}$ and $u_-$ is invariant, hence so is its complement, the vectors orthogonal to both block indicators, on which $L^\Pi \succeq L_{C_1} \oplus L_{C_2} \succeq g_{\min}$. With unit eigenvectors $\mathsf{v}_k$, $(\varrho_{\tau_L})_{ij}\,\text{tr}\, e^{-\tau_L L^\Pi} = N^{-1}(1 \pm e^{-\tau_L \lambda_2}) + \sum_{k \geq 3} e^{-\tau_L \lambda_k} \mathsf{v}_k(i)\mathsf{v}_k(j)$, with the sign + within blocks and − across, and by the Cauchy–Schwarz inequality the last sum is at most $e^{-\tau_L \lambda_3}$ in absolute value. Every pair within a block therefore exceeds every pair across in absolute value when $2e^{-\tau_L \lambda_2}/N > 2e^{-\tau_L \lambda_3}$, that is $\tau_L(\lambda_3 - \lambda_2) > \ln N$. Exactly two modes are kept, the mode $\lambda_3$ being discarded also when $1/\tau_L = \lambda_3$ because the rule of Section S13.6 is strict, and joining pairs in descending order completes both blocks before any pair across, stopping at the two blocks. Taking $\tau_L$ just below $1/(2\bar{a}^\partial)$, the conditions hold when $\lambda_3/(2\bar{a}^\partial) > 1 + \ln N$, for which $g_{\min}/\bar{a}^\partial > 2(1 + \ln N)$ suffices. The contrast statements are Theorem 4.18(i) and (iv), with $\lambda_2(L^\Pi) = 2\bar{a}^\partial$ exactly for an equitable cut. □

*Proof of Theorem 13.10.* (a) Lemma 13.3, with $\|\Pi_{\perp\Phi}\|^2/\varkappa_\perp \leq \|\Pi_{\perp\Phi}\|^2/\varkappa^{\min}_{\text{in}} \leq \sigma^2_{\text{cut}}/\varkappa^{\min}_{\text{in}}$ and $\varkappa_\perp \geq \varkappa^{\min}_{\text{in}}$ from Theorem 13.8(d), which does not require a uniform leak; the statement on the leak is Lemma 13.3(c).

(b) $\bar{e} = \text{diag}(|X|)^{-1}\mathsf{P}e$ and $e = \mathsf{P}^\top \bar{e} + e_\perp$ with $\mathsf{P}e_\perp = 0$, so $\bar{e}(t+1) = \text{diag}(|X|)^{-1}\mathsf{P}(We(t) + \zeta(t)) = W^{(1)}\bar{e}(t) + \text{diag}(|X|)^{-1}\mathsf{P}We_\perp(t) + \bar{\zeta}(t)$. The entries of $W^{(1)}$ are $|X|^{-1}\sum_{x \in X, y \in Y} W_{xy}$ and its row sums are $|X|^{-1}\sum_{x \in X}(W\mathbf{1})_x$. Under (B) the local noises are independent across agents with variances at most $q$, so the $\bar{\zeta}_X = |X|^{-1}\sum_{x \in X}\zeta_x$, averages over disjoint sets, are independent across blocks with variances $|X|^{-2}\sum_{x \in X}\text{Var}(\zeta_x) \leq q/|X|$, with equality when every variance equals $q$, and $\bar{e}$ is a linear function of $e$. The memory term vanishes for every $e_\perp \in \ker \mathsf{P}$ if and only if $\mathsf{P}W \ker \mathsf{P} = 0$, that is $\mathsf{P}W = \bar{W}\mathsf{P}$ for some $\bar{W}$, which is lumpability; then $\bar{e}(t+1) = \text{diag}(|X|)^{-1}\bar{W}\,\text{diag}(|X|)\bar{e}(t) + \bar{\zeta}(t)$.

(c) For blocks of size $s$, $\Phi\mathbf{1}_S/|S|^{1/2} = \mathbf{1}_{\bigcup S}/|\bigcup S|^{1/2}$, so the outward rate of $S$ in $\Pi_\Phi$ is $\varkappa_{\text{out}}(\bigcup S)$. With $\Phi_S$ the block frame of the blocks of $S$, $(\Phi_S^\top L_{\bigcup S}\Phi_S)_{XY} = -\,\text{cut}(X,Y)/s$ for $X \neq Y$ and $(\Phi_S^\top L_{\bigcup S}\Phi_S)_{XX} = \sum_{Y \in S\setminus\{X\}}\text{cut}(X,Y)/s$, and the coarse leaks equal $\varepsilon^\Pi$, so the internal operator of $S$ is $\varepsilon^\Pi I + \Phi_S^\top L_{\bigcup S}\Phi_S$. Hence $\varkappa^{(1)}_{\text{in}}(S)$ is the minimum of the Rayleigh quotient of $\varepsilon^\Pi I + L_{\bigcup S}$ over the vectors of $\mathbf{1}^\perp_{\bigcup S}$ that are constant on blocks, a subspace of the set over which $\varkappa_{\text{in}}(\bigcup S)$ is the minimum; the two minima agree exactly when the subspace contains a minimizer. Contrast is invariant under scaling, so the coarse medium $s\Pi_\Phi$ gives the same values, with uniform leak. The collective statement is Definition 4.12 applied to the coarse medium. Under the exchangeable clause the fine baseline is $1 + \frac{N}{N-1}\bar{d}/\varepsilon^\Pi$ (Proposition S13.7); the coarse medium in the orthonormal convention has $M = N/s$ units, leak $\varepsilon^\Pi$ and mean total coupling per unit $M^{-1}\sum_X \text{cut}(X)/s = \bar{c}$, so its baseline is $1 + \frac{M}{M-1}\bar{c}/\varepsilon^\Pi$, and $\frac{M(N-1)}{(M-1)N} = \frac{N-1}{N-s}$. The nested statement is Theorem 13.6(c).

(d) Propositions S13.13, S13.14 and S13.17.

(e) Averaging the record equation of Definition 2.9 over $X$ gives the coarse record equation. The read term of the agents' equation there averages to $\beta_k|X|^{-1}\sum_{x \in X}\sum_y \mathsf{K}_k(x-y)z_{k,y}$, which is $\beta_k \bar{z}_{k,X}$

for local records and $\beta_k \sum_Y \mathsf{K}_k^{(1)}(X,Y)\bar{z}_{k,Y}$ when the record field is constant on blocks; the self term averages exactly and the message term is that of (b); record noises are independent across sites. On $\mathbb{Z}^D$ with cubic blocks the mean of the block averages is the mean of all agents, and likewise for the records, so the mode of wavevector zero, which obeys the reduced model of Proposition 8.11, is the same at every level.

(f) Examples S13.4 and S13.9, Lemma 11.11, and Propositions S13.7 and S13.8. □

## S38 Proofs for Section 14

The proofs follow the order of Section 14. In the setting of Definition 14.1 put $\tilde{\mathsf{X}} = \Sigma_{\mathsf{X}}^{-1/2}\mathsf{X}$ and $\tilde{\mathsf{Y}} = \Sigma_{\mathsf{Y}}^{-1/2}\mathsf{Y}$, so that $\mathrm{Cov}(\tilde{\mathsf{X}}) = I$, $\mathrm{Cov}(\tilde{\mathsf{X}},\tilde{\mathsf{Y}}) = \mathsf{C}$ and $s = \tilde{H}\tilde{\mathsf{X}} + v$. For a symmetric matrix, $\lambda_{\mathsf{j}}^{\downarrow}$ and $\lambda_{\mathsf{j}}^{\uparrow}$ are its eigenvalues in decreasing and in increasing order. An orthogonal change of a sensor's channels, $\tilde{H} \mapsto \mathsf{O}\tilde{H}$, leaves $\mathsf{G}_s$ unchanged, and after a singular value decomposition the channels beyond the rank of $\tilde{H}$ are pure noise; we may therefore take $\kappa \le d_{\mathsf{X}}$ whenever convenient.

**Lemma S38.1** (Eigenvalue inequalities). *(a) (Interlacing.) For symmetric $\mathsf{S} \in \mathbb{R}^{d\times d}$ and $\mathsf{U} \in \mathbb{R}^{d\times \mathsf{m}}$ with $\mathsf{U}^\top\mathsf{U} = I_{\mathsf{m}}$, $\lambda_{\mathsf{j}}^{\downarrow}(\mathsf{U}^\top\mathsf{S}\mathsf{U}) \le \lambda_{\mathsf{j}}^{\downarrow}(\mathsf{S})$ and $\lambda_{\mathsf{j}}^{\uparrow}(\mathsf{U}^\top\mathsf{S}\mathsf{U}) \ge \lambda_{\mathsf{j}}^{\uparrow}(\mathsf{S})$ for $\mathsf{j} \le \mathsf{m}$.*

*(b) (Ky Fan.) $\max\{\mathrm{tr}(\mathsf{U}^\top\mathsf{S}\mathsf{U} : \mathsf{U}^\top\mathsf{U} = I_{\mathsf{m}}\} = \sum_{\mathsf{j}\le\mathsf{m}} \lambda_{\mathsf{j}}^{\downarrow}(\mathsf{S})$, and if $\lambda_{\mathsf{m}}^{\downarrow}(\mathsf{S}) > \lambda_{\mathsf{m}+1}^{\downarrow}(\mathsf{S})$ the maximizers are the $\mathsf{U}$ whose columns span the top-$\mathsf{m}$ eigenspace.*

*(c) (Concave majorization.) Let $x, y \in \mathbb{R}^d$ be non-increasing, with $\sum_{\mathsf{j}\le\mathsf{m}} x_{\mathsf{j}} \le \sum_{\mathsf{j}\le\mathsf{m}} y_{\mathsf{j}}$ for $\mathsf{m} < d$ and $\sum_{\mathsf{j}\le d} x_{\mathsf{j}} = \sum_{\mathsf{j}\le d} y_{\mathsf{j}}$, and let $\varphi$ be concave and differentiable on an open interval that contains all entries. Then $\sum_{\mathsf{j}} \varphi(x_{\mathsf{j}}) \ge \sum_{\mathsf{j}} \varphi(y_{\mathsf{j}})$.*

*(d) (Determinant of a sum.) For $\mathsf{S}_1, \mathsf{S}_2 \succeq 0$ in $\mathbb{R}^{d\times d}$, $\det(\mathsf{S}_1 + \mathsf{S}_2) \ge \prod_{\mathsf{j}} (\lambda_{\mathsf{j}}^{\downarrow}(\mathsf{S}_1) + \lambda_{\mathsf{j}}^{\downarrow}(\mathsf{S}_2))$.*

*(e) (Trace bound.) For $\mathsf{G} \succeq 0$ of rank at most $\kappa$ with eigenvalues $\mathsf{g}_1 \ge \cdots \ge \mathsf{g}_\kappa$ and symmetric $\mathsf{J}$, $\mathrm{tr}(\mathsf{G}\mathsf{J}) \le \sum_{\mathsf{j}\le\kappa} \mathsf{g}_{\mathsf{j}} \lambda_{\mathsf{j}}^{\downarrow}(\mathsf{J})$, with equality when the eigenvector of each positive $\mathsf{g}_{\mathsf{j}}$ is an eigenvector of $\lambda_{\mathsf{j}}^{\downarrow}(\mathsf{J})$.*

*(f) (Determinant bound for sensors.) For $\tilde{H} \in \mathbb{R}^{\kappa\times d}$, $\kappa \le d$, with singular values $\mathsf{d}_1 \ge \cdots \ge \mathsf{d}_\kappa \ge 0$, and $\mathsf{S} \succeq 0$ in $\mathbb{R}^{d\times d}$, $\det(I_\kappa + \tilde{H}\mathsf{S}\tilde{H}^\top) \ge \prod_{\mathsf{j}\le\kappa}(1 + \mathsf{d}_{\mathsf{j}}^2 \lambda_{\mathsf{j}}^{\uparrow}(\mathsf{S}))$, with equality when $\tilde{H} = \mathsf{O}\,\mathrm{diag}(\mathsf{d})[\mathsf{f}_1, \ldots, \mathsf{f}_\kappa]^\top$, $\mathsf{O}$ orthogonal and $\mathsf{f}_{\mathsf{j}}$ orthonormal eigenvectors of $\lambda_{\mathsf{j}}^{\uparrow}(\mathsf{S})$.*

*Proof.* (a) By the Courant–Fischer theorem (Horn and Johnson, 2012), $\lambda_{\mathsf{j}}^{\downarrow}(\mathsf{U}^\top\mathsf{S}\mathsf{U})$ is the maximum over j-dimensional subspaces $\mathcal{V} \subseteq \mathbb{R}^{\mathsf{m}}$ of $\min\{x^\top\mathsf{U}^\top\mathsf{S}\mathsf{U}x : x \in \mathcal{V}, |x| = 1\}$; since $\mathsf{U}\mathcal{V}$ is a j-dimensional subspace of $\mathbb{R}^d$ and $|\mathsf{U}x| = |x|$, this is at most $\lambda_{\mathsf{j}}^{\downarrow}(\mathsf{S})$. Applied to $-\mathsf{S}$, the same argument gives the second inequality.

(b) Write $\mathsf{S} = \sum_{\mathsf{j}} \mathsf{s}_{\mathsf{j}} \mathsf{b}_{\mathsf{j}} \mathsf{b}_{\mathsf{j}}^\top$ with $\mathsf{s}_{\mathsf{j}} = \lambda_{\mathsf{j}}^{\downarrow}(\mathsf{S})$ and orthonormal $\mathsf{b}_{\mathsf{j}}$, and put $\mathsf{y}_{\mathsf{j}} = |\mathsf{U}^\top\mathsf{b}_{\mathsf{j}}|^2 \in [0,1]$, so that $\sum_{\mathsf{j}} \mathsf{y}_{\mathsf{j}} = \mathrm{tr}(\mathsf{U}\mathsf{U}^\top) = \mathsf{m}$ and $\mathrm{tr}(\mathsf{U}^\top\mathsf{S}\mathsf{U}) = \sum_{\mathsf{j}} \mathsf{s}_{\mathsf{j}}\mathsf{y}_{\mathsf{j}}$. Since $\sum_{\mathsf{j}\le\mathsf{m}}(1 - \mathsf{y}_{\mathsf{j}}) = \sum_{\mathsf{j}>\mathsf{m}} \mathsf{y}_{\mathsf{j}}$,

$$\sum_{\mathsf{j}} \mathsf{s}_{\mathsf{j}}\mathsf{y}_{\mathsf{j}} - \sum_{\mathsf{j}\le\mathsf{m}} \mathsf{s}_{\mathsf{j}} = -\sum_{\mathsf{j}\le\mathsf{m}} \mathsf{s}_{\mathsf{j}}(1 - \mathsf{y}_{\mathsf{j}}) + \sum_{\mathsf{j}>\mathsf{m}} \mathsf{s}_{\mathsf{j}}\mathsf{y}_{\mathsf{j}} \le (\mathsf{s}_{\mathsf{m}+1} - \mathsf{s}_{\mathsf{m}}) \sum_{\mathsf{j}>\mathsf{m}} \mathsf{y}_{\mathsf{j}} \le 0.$$

Equality holds for $\mathsf{U} = [\mathsf{b}_1, \ldots, \mathsf{b}_{\mathsf{m}}]$, and under the gap it forces $\sum_{\mathsf{j}>\mathsf{m}} \mathsf{y}_{\mathsf{j}} = 0$, so the columns of $\mathsf{U}$ span the top-$\mathsf{m}$ eigenspace.

(c) Put $\mathsf{a}_\mathsf{j} = \int_0^1 \varphi'((1-\vartheta)y_\mathsf{j} + \vartheta x_\mathsf{j})\mathrm{d}\vartheta$, so that $\varphi(x_\mathsf{j}) - \varphi(y_\mathsf{j}) = \mathsf{a}_\mathsf{j}(x_\mathsf{j} - y_\mathsf{j})$. For each $\vartheta$ the point $(1-\vartheta)y_\mathsf{j} + \vartheta x_\mathsf{j}$ is non-increasing in $\mathsf{j}$ and $\varphi'$ is non-increasing, so $\mathsf{a}_1 \le \mathsf{a}_2 \le \cdots \le \mathsf{a}_d$. With $\mathsf{t}_\mathsf{m} = \sum_{\mathsf{j}\le\mathsf{m}}(x_\mathsf{j} - y_\mathsf{j}) \le 0$ and $\mathsf{t}_d = 0$, Abel summation gives $\sum_\mathsf{j} \mathsf{a}_\mathsf{j}(x_\mathsf{j} - y_\mathsf{j}) = \sum_{\mathsf{m}<d}(\mathsf{a}_\mathsf{m} - \mathsf{a}_{\mathsf{m}+1})\mathsf{t}_\mathsf{m} \ge 0$.

(d) Let $x = \lambda^\downarrow(\mathsf{S}_1 + \mathsf{S}_2)$ and $y_\mathsf{j} = \lambda_\mathsf{j}^\downarrow(\mathsf{S}_1) + \lambda_\mathsf{j}^\downarrow(\mathsf{S}_2)$, both non-increasing. By (b), $\sum_{\mathsf{j}\le\mathsf{m}} x_\mathsf{j} = \max_\mathsf{U} \operatorname{tr}(\mathsf{U}^\top(\mathsf{S}_1+\mathsf{S}_2)\mathsf{U}) \le \sum_{\mathsf{j}\le\mathsf{m}} y_\mathsf{j}$, with equality at $\mathsf{m} = d$, where both sides are traces. If $y_d = 0$ the right side of (d) vanishes. Otherwise $x_d = \operatorname{tr}(\mathsf{S}_1+\mathsf{S}_2) - \sum_{\mathsf{j}<d} x_\mathsf{j} \ge y_d > 0$, all entries are positive, and (c) with $\varphi = \ln$ gives $\sum_\mathsf{j} \ln x_\mathsf{j} \ge \sum_\mathsf{j} \ln y_\mathsf{j}$.

(e) Write $\mathsf{G} = \sum_{\mathsf{j}\le\kappa} \mathsf{g}_\mathsf{j}\mathsf{f}_\mathsf{j}\mathsf{f}_\mathsf{j}^\top$ with orthonormal $\mathsf{f}_\mathsf{j}$, and put $\mathsf{g}_{\kappa+1} = 0$. Abel summation and (b) give

$$\operatorname{tr}(\mathsf{G}\mathsf{J}) = \sum_{\mathsf{m}\le\kappa} (\mathsf{g}_\mathsf{m} - \mathsf{g}_{\mathsf{m}+1}) \sum_{\mathsf{j}\le\mathsf{m}} \mathsf{f}_\mathsf{j}^\top \mathsf{J}\mathsf{f}_\mathsf{j} \le \sum_{\mathsf{m}\le\kappa} (\mathsf{g}_\mathsf{m} - \mathsf{g}_{\mathsf{m}+1}) \sum_{\mathsf{j}\le\mathsf{m}} \lambda_\mathsf{j}^\downarrow(\mathsf{J}) = \sum_{\mathsf{j}\le\kappa} \mathsf{g}_\mathsf{j} \lambda_\mathsf{j}^\downarrow(\mathsf{J}),$$

with equality in the stated case.

(f) Factors with $\mathsf{d}_\mathsf{j} = 0$ equal one on both sides, so let $\kappa'$ be the number of positive singular values and write $\tilde{H} = \mathsf{O}'\mathsf{H}_\mathrm{d}\mathsf{U}^\top$ with $\mathsf{O}'$ and $\mathsf{U}$ having $\kappa'$ orthonormal columns and $\mathsf{H}_\mathrm{d} = \operatorname{diag}(\mathsf{d}_1, \dots, \mathsf{d}_{\kappa'})$. By Sylvester's identity, $\det(I_\kappa + \tilde{H}\mathsf{S}\tilde{H}^\top) = \det(I + \mathsf{H}_\mathrm{d}\mathsf{N}\mathsf{H}_\mathrm{d}) = \det(\mathsf{H}_\mathrm{d})^2 \det(\mathsf{H}_\mathrm{d}^{-2} + \mathsf{N})$ with $\mathsf{N} = \mathsf{U}^\top\mathsf{S}\mathsf{U}$. By (d), $\det(\mathsf{H}_\mathrm{d}^{-2} + \mathsf{N}) \ge \prod_\mathsf{j}(\lambda_\mathsf{j}^\downarrow(\mathsf{H}_\mathrm{d}^{-2}) + \lambda_\mathsf{j}^\downarrow(\mathsf{N}))$; since $\lambda_\mathsf{j}^\downarrow(\mathsf{H}_\mathrm{d}^{-2}) = \mathsf{d}_{\kappa'+1-\mathsf{j}}^{-2}$ and $\lambda_\mathsf{j}^\downarrow(\mathsf{N}) = \lambda_{\kappa'+1-\mathsf{j}}^\uparrow(\mathsf{N})$, the product is $\prod_\mathsf{j}(\mathsf{d}_\mathsf{j}^{-2} + \lambda_\mathsf{j}^\uparrow(\mathsf{N}))$, and multiplying by $\det(\mathsf{H}_\mathrm{d})^2$ gives $\prod_\mathsf{j}(1 + \mathsf{d}_\mathsf{j}^2\lambda_\mathsf{j}^\uparrow(\mathsf{N}))$. By (a), $\lambda_\mathsf{j}^\uparrow(\mathsf{N}) \ge \lambda_\mathsf{j}^\uparrow(\mathsf{S})$. In the equality case $\mathsf{N} = \operatorname{diag}(\lambda_\mathsf{j}^\uparrow(\mathsf{S}))$ and the determinant is the product. □

*Proof of Lemma 14.2.* Gaussian conditioning gives $\mathbb{E}[\tilde{\mathsf{X}} \mid s] = \tilde{H}^\top(I + \tilde{H}\tilde{H}^\top)^{-1}s$, whose covariance is $\mathsf{G}_s$, so $\operatorname{Cov}(\tilde{\mathsf{X}} \mid s) = I - \mathsf{G}_s$. With the singular value decomposition $\tilde{H} = \mathsf{O}\operatorname{diag}(\mathsf{d})\mathsf{U}^\top$, $\mathsf{G}_s = \mathsf{U}\operatorname{diag}(\mathsf{d}_\mathsf{j}^2/(1+\mathsf{d}_\mathsf{j}^2))\mathsf{U}^\top$, which gives the captured fractions, and (i) is $I(s;\tilde{\mathsf{X}}) = h(\tilde{\mathsf{X}}) - h(\tilde{\mathsf{X}} \mid s)$. (ii) Since $v$ is independent of $(\mathsf{X}, \mathsf{Y})$, $\mathbb{E}[\tilde{\mathsf{Y}} \mid \tilde{\mathsf{X}}, s] = \mathsf{C}^\top\tilde{\mathsf{X}}$, hence $\mathbb{E}[\tilde{\mathsf{Y}} \mid s] = \mathsf{C}^\top\mathbb{E}[\tilde{\mathsf{X}} \mid s]$, whose covariance is $\mathsf{C}^\top\mathsf{G}_s\mathsf{C}$; Gaussian conditional covariances do not depend on the data, so $\operatorname{Cov}(\tilde{\mathsf{Y}} \mid s) = I - \mathsf{C}^\top\mathsf{G}_s\mathsf{C}$, and $I(s;\mathsf{Y}) = h(\tilde{\mathsf{Y}}) - h(\tilde{\mathsf{Y}} \mid s)$. Since $s$ and $\mathsf{Y}$ are conditionally independent given $\mathsf{X}$, the chain rule also gives $I(s;\mathsf{Y}) = I(s;\mathsf{X}) - I(s;\mathsf{X} \mid \mathsf{Y})$, and with $\operatorname{Cov}(\tilde{\mathsf{X}} \mid \mathsf{Y}) = I - \mathsf{C}\mathsf{C}^\top$,

$$I(s;\mathsf{Y}) = \tfrac{1}{2}\ln\det(I + \tilde{H}\tilde{H}^\top) - \tfrac{1}{2}\ln\det(I + \tilde{H}(I - \mathsf{C}\mathsf{C}^\top)\tilde{H}^\top). \tag{S119}$$

(iii) $\operatorname{Cov}(\mathsf{Y} \mid s) = \Sigma_\mathsf{Y}^{1/2}(I - \mathsf{C}^\top\mathsf{G}_s\mathsf{C})\Sigma_\mathsf{Y}^{1/2}$ and $\Sigma_\mathsf{Y}^{1/2}\mathsf{C}^\top = \Sigma_{\mathsf{Y}\mathsf{X}}\Sigma_\mathsf{X}^{-1/2}$. (iv) Captured covariances have the stated form by the decomposition above. Conversely, for $\mathsf{G} = \sum_{\mathsf{j}\le\kappa} \mathsf{g}_\mathsf{j}\mathsf{f}_\mathsf{j}\mathsf{f}_\mathsf{j}^\top$ with $\mathsf{g}_\mathsf{j} \in [0,1)$ and orthonormal $\mathsf{f}_\mathsf{j}$, the whitened sensor with rows $(\mathsf{g}_\mathsf{j}/(1-\mathsf{g}_\mathsf{j}))^{1/2}\mathsf{f}_\mathsf{j}^\top$ has captured covariance $\mathsf{G}$. □

*Proof of Theorem 14.3.* (a) Apply Lemma S38.1(f) to the second determinant in (S119), with $\mathsf{S} = I - \mathsf{C}\mathsf{C}^\top$, whose eigenvalues in increasing order are $1-\sigma_1^2 \le 1-\sigma_2^2 \le \dots$. Since $\det(I + \tilde{H}\tilde{H}^\top) = \prod_\mathsf{j}(1 + \mathsf{d}_\mathsf{j}^2)$,

$$I(s;\mathsf{Y}) \le \sum_{\mathsf{j}\le\kappa} \tfrac{1}{2}\ln\frac{1+\mathsf{d}_\mathsf{j}^2}{1+\mathsf{d}_\mathsf{j}^2(1-\sigma_\mathsf{j}^2)} = -\tfrac{1}{2}\sum_{\mathsf{j}\le\kappa}\ln(1 - \mathsf{g}_\mathsf{j}\sigma_\mathsf{j}^2),$$

because $\mathsf{g}_\mathsf{j} = \mathsf{d}_\mathsf{j}^2/(1+\mathsf{d}_\mathsf{j}^2)$. The equality case of Lemma S38.1(f) is the sensor whose $\mathsf{j}$-th row of $\tilde{H}$ is $\mathsf{d}_\mathsf{j}\mathsf{e}_\mathsf{j}^\top$.

(b) Let $\mathsf{c}_\mathsf{j} = -\tfrac{1}{2}\ln(1-\mathsf{g}_\mathsf{j}) \ge 0$ be the capacity spent on channel $\mathsf{j}$. By (a) and Lemma 14.2(i), every sensor satisfies $I(s;\mathsf{Y}) \le \sum_{\mathsf{j}\le\kappa} \mathcal{Y}_\mathsf{j}(\mathsf{c}_\mathsf{j})$ with $\sum_\mathsf{j} \mathsf{c}_\mathsf{j} = I(s;\mathsf{X})$ and $\mathcal{Y}_\mathsf{j}(x) = -\tfrac{1}{2}\ln(1 - \sigma_\mathsf{j}^2(1-e^{-2x}))$, and the canonical sensor attains this bound. The optimum is therefore the maximum of $\sum_{\mathsf{j}\le\kappa} \mathcal{Y}_\mathsf{j}(\mathsf{c}_\mathsf{j})$ over $\mathsf{c}_\mathsf{j} \ge 0$ with $\sum_\mathsf{j} \mathsf{c}_\mathsf{j} \le I_{\max}$, provided the maximizer is non-increasing in $\mathsf{j}$, as the one found

below is, so that its pairing with the ordered correlations in (a) costs nothing. For $0 < \sigma_j < 1$ the derivative is

$$\mathcal{Y}_j'(x) = \frac{\sigma_j^2}{(1-\sigma_j^2)e^{2x} + \sigma_j^2},$$

which is positive and decreasing, so each $\mathcal{Y}_j$ is increasing and concave ($\mathcal{Y}_j \equiv 0$ when $\sigma_j = 0$). For a separable concave objective on this polytope the Karush–Kuhn–Tucker conditions are necessary and sufficient: there is $v > 0$ with $\mathcal{Y}_j'(c_j) = v$ when $c_j > 0$ and $\mathcal{Y}_j'(0) = \sigma_j^2 \le v$ when $c_j = 0$, and the budget binds because $\mathcal{Y}_j' > 0$. Solving gives $e^{2c_j} = \sigma_j^2(1-v)/(v(1-\sigma_j^2))$ for $\sigma_j^2 > v$, then $g_j = 1 - e^{-2c_j} = (\sigma_j^2 - v)/(\sigma_j^2(1-v))$ and $1 - g_j\sigma_j^2 = (1-\sigma_j^2)/(1-v)$, which is the stated value. The capacity used is a continuous, strictly decreasing function of $v \in (0, \sigma_1^2)$ that tends to infinity as $v \to 0$ and to zero as $v \to \sigma_1^2$, so $v$ is unique and decreases as $I_{\max}$ grows; variate $m$ is active exactly when $v < \sigma_m^2$, that is, when the capacity used at $v = \sigma_m^2$, which is $\mathcal{I}_m$, is below $I_{\max}$.

(c) The bound and its equality case are Lemma S38.1(e). Under the budget, maximize $\sum_j \lambda_j(J)(1 - e^{-2c_j})$, again separable and concave: for active channels $2\lambda_j(J)e^{-2c_j} = 2v'$, so $c_j = \frac{1}{2}\ln(\lambda_j(J)/v')$, $g_j = 1 - v'/\lambda_j(J)$, and the value is $\sum_j(\lambda_j(J) - v')^+$. The weighted error is Lemma 14.2(iii).

(d) $J_{\Sigma_Y^{-1}} = \Sigma_X^{-1/2}\Sigma_{XY}\Sigma_Y^{-1}\Sigma_{YX}\Sigma_X^{-1/2} = CC^\top$. Let $u_I(v)$ and $u_E(v)$ be the capacities that the information and the error allocations use at level $v$, namely the sums over $\sigma_j^2 > v$ of $\frac{1}{2}\ln\frac{\sigma_j^2(1-v)}{v(1-\sigma_j^2)}$ and of $\frac{1}{2}\ln\frac{\sigma_j^2}{v}$. Since $(1-v)/(1-\sigma_j^2) \ge 1$ for $\sigma_j^2 \ge v$, $u_I \ge u_E$ pointwise. Both decrease, so at a common budget $u_E(v_I) \le u_I(v_I) = u_E(v_E)$ forces $v_I \ge v_E$, and the information criterion's active set is contained in the error criterion's. For another weight, Proposition 14.4(c) gives an instance in which $J_I$ and $CC^\top$ are diagonal in one basis and their leading eigenvectors differ. Two symmetric matrices with a common orthonormal eigenbasis commute, so when $J_\Upsilon CC^\top \ne CC^\top J_\Upsilon$ no orthonormal eigenbasis of $J_\Upsilon$ consists of eigenvectors of $CC^\top$, which are the left singular vectors of $C$.

(e) For a noiseless sensor, $G_s = UU^\top$ with $U$ an orthonormal basis of the row space of $\tilde{H}$, and by Sylvester's identity $I(s;Y) = -\frac{1}{2}\ln\det(I_\kappa - U^\top CC^\top U)$. By Lemma S38.1(a), $\lambda_j^{\downarrow}(U^\top CC^\top U) \le \sigma_j^2$, which gives the bound. Equality forces each factor $1 - \lambda_j^{\downarrow}(U^\top CC^\top U) \ge 1 - \sigma_j^2 > 0$ to be an equality, so $\operatorname{tr}(U^\top CC^\top U) = \sum_{j\le\kappa}\sigma_j^2$, and Lemma S38.1(b) under the gap identifies the span. The error statement is Lemma 14.2(iii) with $\operatorname{tr}(U^\top J_\Upsilon U) \le \sum_{j\le\kappa}\lambda_j(J_\Upsilon)$. □

*Proof of Lemma S14.1.* Let $Z = \mathbb{E}[\eta_t \mid \mathcal{S}]$, a linear function of $\mathcal{S}$. By the tower property $\mathbb{E}[\eta_t \mid Z] = Z$, so $G_Z = \operatorname{Cov}(P_0^{-1/2}Z)$ satisfies $\operatorname{Cov}(P_0^{-1/2}\eta_t \mid Z) = I - G_Z$, $0 \preceq G_Z \preceq I$ and $\operatorname{rank} G_Z \le \kappa$. By the conditional independence of $\mathcal{S}$ and the Markov property, $\mathbb{E}[\tilde{Y} \mid \mathcal{S}] = \mathbb{E}[\mathbb{E}[\tilde{Y} \mid (\eta_{t'})_{t'\le t}] \mid \mathcal{S}] = \mathbb{E}[\mathbb{E}[\tilde{Y} \mid \eta_t] \mid \mathcal{S}] = C^\top P_0^{-1/2}Z$, so $\operatorname{Cov}(\tilde{Y} \mid \mathcal{S}) = I - C^\top G_Z C$ and $I(\mathcal{S};Y) = -\frac{1}{2}\ln\det(I - C^\top G_Z C)$. Data processing gives $I(Z;\eta_t) \le I(\mathcal{S};\eta_t) \le I(\mathcal{S};(\eta_{t'})_{t'\le t}) \le I_{\max}$, and $I(Z;\eta_t) = -\frac{1}{2}\ln\det(I - G_Z)$ is finite, so $G_Z \prec I$. By Lemma 14.2(iv) a sensor of $\eta_t$ with captured covariance $G_Z$ exists; it has the same information about $Y$ as $\mathcal{S}$, uses at most $I_{\max}$, and Theorem 14.3(b) bounds it. □

*Proof of Proposition 14.4.* (a) Detailed balance gives $F^\top = P_0^{1/2}F^\top P_0^{-1/2} = P_0^{-1/2}(P_0F^\top)P_0^{-1/2} = P_0^{-1/2}FP_0^{1/2} = F$, and $\operatorname{Cov}(\eta_t, \eta_{t+h}) = P_0(F^h)^\top$ gives $C(h) = (F^h)^\top = F^h$. With $F = \sum_j \lambda_j(F)e_j e_j^\top$, the singular values of $F^h$ are $|\lambda_j(F)|^h$ with singular vectors $e_j$, and $u_j = P_0^{-1/2}e_j$ satisfies $u_j^\top F = e_j^\top F P_0^{-1/2} = \lambda_j(F)u_j^\top$ and $u_j^\top P_0 u_j = 1$. The noiseless statement is Theorem 14.3(e), and the activation condition is Theorem 14.3(b). With $\sigma_j(h) = e^{-h/\tau_j}$, the $j$-th term of $\mathcal{I}_m(h)$ is $h(\tau_m^{-1} - \tau_j^{-1}) + \frac{1}{2}\ln\frac{1-e^{-2h/\tau_m}}{1-e^{-2h/\tau_j}}$, whose derivative in $h$ is $\mathcal{Z}(\tau_m^{-1}) - \mathcal{Z}(\tau_j^{-1})$ with $\mathcal{Z}(y) = y/(1-e^{-2yh}) = (2h)^{-1}x/(1-e^{-x})$, $x = 2yh$. The function $x/(1-e^{-x})$ increases on $x > 0$ (its derivative has

numerator $1-e^{-x}(1+x)>0$), and $\tau_{\mathsf{m}}^{-1}\ge\tau_{\mathsf{j}}^{-1}$ for $\mathsf{j}<\mathsf{m}$, so each term is non-decreasing in $h$. The logarithm is non-negative because $\tau_{\mathsf{m}}\le\tau_{\mathsf{j}}$, which gives the linear lower bound; the last claim is the case $\mathsf{m}=2$.

(b) Write $\mathsf{F}^h=\mathsf{M}_h+\mathsf{N}_h$ with $\mathsf{M}_h=P_0^{-1/2}F^h\mathbb{P}_{\mathrm{slow}}P_0^{1/2}$ and $\mathsf{N}_h=P_0^{-1/2}F^h(I-\mathbb{P}_{\mathrm{slow}})P_0^{1/2}$, so that $\|\mathsf{N}_h\|\le\mathfrak{a}_h$ and $\sigma_\kappa(\mathsf{M}_h)\ge\mathfrak{b}_h$. A top-$\kappa$ canonical subspace of $\mathsf{X}$ is spanned by top-$\kappa$ left singular vectors of $\mathsf{C}(h)=(\mathsf{F}^h)^\top$, that is, by top-$\kappa$ right singular vectors of $\mathsf{F}^h$, the columns of some $\mathsf{Z}_\kappa$ with $(\mathsf{F}^h)^\top\mathsf{W}_\kappa=\mathsf{Z}_\kappa\mathsf{T}_\kappa$, where $\mathsf{W}_\kappa$ holds the left singular vectors and $\mathsf{T}_\kappa$ the top $\kappa$ singular values. Since $F^h\mathbb{P}_{\mathrm{slow}}=\mathbb{P}_{\mathrm{slow}}F^h$ and the slow eigenvalues are nonzero, the row space of $F^h\mathbb{P}_{\mathrm{slow}}$ is $\mathcal{U}_\kappa$ and that of $\mathsf{M}_h$ is $P_0^{1/2}\mathcal{U}_\kappa$. Let $\mathbb{P}_\perp$ project orthogonally onto the complement of $P_0^{1/2}\mathcal{U}_\kappa$, so $\mathbb{P}_\perp\mathsf{M}_h^\top=0$. Then $\mathbb{P}_\perp\mathsf{Z}_\kappa=\mathbb{P}_\perp\mathsf{N}_h^\top\mathsf{W}_\kappa\mathsf{T}_\kappa^{-1}$, and Weyl's inequality $\sigma_\kappa(\mathsf{F}^h)\ge\sigma_\kappa(\mathsf{M}_h)-\|\mathsf{N}_h\|$ gives $\sin\vartheta_h=\|\mathbb{P}_\perp\mathsf{Z}_\kappa\|\le\mathfrak{a}_h/(\mathfrak{b}_h-\mathfrak{a}_h)$. (Since $\sigma_{\kappa+1}(\mathsf{F}^h)\le\|\mathsf{N}_h\|$, the subspace is unique once $\mathfrak{a}_h<\mathfrak{b}_h/2$.) With $F=\mathsf{Q}\,\mathrm{diag}(\lambda(F))\mathsf{Q}^{-1}$, the slow and fast columns $\mathsf{Q}_{\mathrm{s}},\mathsf{Q}_{\mathrm{f}}$ of the eigenvector matrix $\mathsf{Q}$ and rows $\mathsf{U}_{\mathrm{s}},\mathsf{U}_{\mathrm{f}}$ of $\mathsf{Q}^{-1}$ give $F^h(I-\mathbb{P}_{\mathrm{slow}})=\mathsf{Q}_{\mathrm{f}}\,\mathrm{diag}(\lambda_{\mathrm{fast}}(F))^h\mathsf{U}_{\mathrm{f}}$ and $F^h\mathbb{P}_{\mathrm{slow}}=\mathsf{Q}_{\mathrm{s}}\,\mathrm{diag}(\lambda_{\mathrm{slow}}(F))^h\mathsf{U}_{\mathrm{s}}$, hence

$$\mathfrak{a}_h\le\mathsf{c}_P\|\mathsf{Q}\|\|\mathsf{Q}^{-1}\||\lambda_{\kappa+1}(F)|^h,\qquad\mathfrak{b}_h\ge\sigma_\kappa(\mathsf{Q}_{\mathrm{s}})\sigma_\kappa(\mathsf{U}_{\mathrm{s}})|\lambda_\kappa(F)|^h/\mathsf{c}_P,\tag{S120}$$

which gives the rate. A canonical direction $z\in P_0^{1/2}\mathcal{U}_\kappa$ is the functional $z^\top P_0^{-1/2}\eta_t$ with $P_0^{-1/2}z\in\mathcal{U}_\kappa$, a combination of slow mode amplitudes. For the error criterion the optimal readings are the top right singular vectors of $\Upsilon^{1/2}F^hP_0^{1/2}$ (Theorem 14.3(e)), and the same splitting applies.

(c) Here $P_0=Q(I-F^2)^{-1}$ shares the eigenbasis of $F$, $\mathsf{J}_I=P_0^{1/2}(F^h)^\top F^hP_0^{1/2}$ is diagonal with entries $\lambda_{\mathsf{j}}(F)^{2h}\mathsf{p}_{\mathsf{j}}$, and Theorem 14.3(e) applies, so the rank-one error optimum reads a mode of largest entry. The ratio of the entries of the fast and the slow mode, $(\mathsf{p}_{\mathrm{f}}/\mathsf{p}_{\mathrm{s}})(|\lambda_{\mathrm{f}}(F)|/|\lambda_{\mathrm{s}}(F)|)^{2h}$, exceeds one at $h=0$, decreases strictly in $h$ and equals one at $h^*$, which gives the ranking of the two modes on either side of $h^*$. The canonical correlations are $|\lambda_{\mathsf{j}}(F)|^h$, larger for the slow mode than for the fast mode at every $h\ge1$. With only these two modes the larger entry and the larger correlation are the maxima, which gives the modes read. With a third mode neither maximum need lie in the pair: for $\lambda(F)=(0.97,0.9,0.6)$ and $\mathsf{p}=(0.3384,5,1.5625)$ the entries at $h=1$ are $(0.318,4.05,0.5625)$, and the error optimum reads the second mode, and a third mode with $|\lambda_{\mathsf{j}}(F)|>|\lambda_{\mathrm{s}}(F)|$ has the largest canonical correlation at every $h\ge1$. □

*Proof of Proposition S14.2.* $F$ is symmetric and commutes with $Q=Q_0I$, so $P_0=Q_0(I-F^2)^{-1}$ commutes with $F$ and $FP_0=P_0F^\top$. The eigenvalues $1-\varepsilon_{\mathrm{w}}-\mathsf{d}_{\mathrm{w}}\lambda(L_{\mathrm{w}})$ of $F$ lie in $(0,1)$ and decrease with the Laplacian eigenvalue, so the order by modulus is the order by frequency, with the constant vector, of Laplacian eigenvalue zero, first. Proposition 14.4(a) gives the rest; repeated Laplacian eigenvalues do no harm, since modes of equal modulus share one threshold and the terms of $\mathcal{I}_{\mathsf{m}}(h)$ that pair them vanish, and $\tau_2<\tau_1$ because the graph is connected. □

*Proof of Theorem 14.6.* (a) Write $\hat\eta=\mathbb{E}[\eta_{t+h}\mid s]$ and $\Sigma_s=\mathrm{Cov}(\eta_{t+h}\mid s)$. Given $s$, $o-o^*=(H^{\mathrm{o}}\hat\eta+B^{\mathrm{o}}a-o^*)+H^{\mathrm{o}}(\eta_{t+h}-\hat\eta)+v^{\mathrm{o}}$, and the last two terms have conditional mean zero, so

$$2\mathbb{E}[-\ln\tilde p(o)\mid s]-\ln\det(2\pi\Psi_{\mathrm{o}})=\|H^{\mathrm{o}}\hat\eta+B^{\mathrm{o}}a-o^*\|^2_{\Psi_{\mathrm{o}}^{-1}}+\mathrm{tr}\big(\Psi_{\mathrm{o}}^{-1}(H^{\mathrm{o}}\Sigma_s(H^{\mathrm{o}})^\top+R^{\mathrm{o}})\big).$$

Weighted least squares minimizes the first term by $B^{\mathrm{o}}a^*=-\mathbb{P}_{\mathrm{o}}(H^{\mathrm{o}}\hat\eta-o^*)$, which is the stated $a^*$, with residual $\|(I-\mathbb{P}_{\mathrm{o}})(H^{\mathrm{o}}\hat\eta-o^*)\|^2_{\Psi_{\mathrm{o}}^{-1}}$; since the act may depend on $s$ arbitrarily, the minimal expectation is the expectation of this conditional minimum. The matrix $\Psi_{\mathrm{o}}^{-1}\mathbb{P}_{\mathrm{o}}$ is symmetric and $\mathbb{P}_{\mathrm{o}}$ is idempotent, so $\mathbb{P}_{\mathrm{o}}$ is an orthogonal projector in the $\Psi_{\mathrm{o}}^{-1}$ inner product, $\|x\|^2=\|\mathbb{P}_{\mathrm{o}}x\|^2+\|(I-\mathbb{P}_{\mathrm{o}})x\|^2$ in that norm, and $\mathbb{P}_{\mathrm{o}}^\top\Psi_{\mathrm{o}}^{-1}\mathbb{P}_{\mathrm{o}}=\Psi_{\mathrm{o}}^{-1}\mathbb{P}_{\mathrm{o}}\succeq0$. Since $\mathbb{E}\hat\eta=0$ and $\mathrm{Cov}(\hat\eta)=P_0-\Sigma_s$, the expected residual is a constant plus $\mathrm{tr}\big(\Psi_{\mathrm{o}}^{-1}(I-\mathbb{P}_{\mathrm{o}})H^{\mathrm{o}}(P_0-\Sigma_s)(H^{\mathrm{o}})^\top(I-\mathbb{P}_{\mathrm{o}})^\top\big)$, and adding $\mathrm{tr}(\Psi_{\mathrm{o}}^{-1}H^{\mathrm{o}}\Sigma_s(H^{\mathrm{o}})^\top)$ leaves, by the Pythagorean identity, the sensor-dependent part $\mathrm{tr}(\Psi_{\mathrm{o}}^{-1}\mathbb{P}_{\mathrm{o}}H^{\mathrm{o}}\Sigma_s(H^{\mathrm{o}})^\top\mathbb{P}_{\mathrm{o}}^\top)=\mathrm{tr}(\Upsilon_{\mathrm{ctl}}\Sigma_s)$. The

rank bound is $\operatorname{rank}\mathbb{P}_{\mathrm{o}} = \operatorname{rank} B^{\mathrm{o}}$. With no sensor $\Sigma_s = P_0$, and $P_0 - \Sigma_s = F^h P_0^{1/2} \mathsf{G}_s P_0^{1/2} (F^h)^\top$ by the identity in the proof of Lemma 14.2(iii) with $\Sigma_{\mathsf{XY}} = P_0 (F^h)^\top$, which gives the pragmatic value $\frac{1}{2}\operatorname{tr}(\mathsf{G}_s \mathsf{J}_{\mathrm{ctl}})$. Finally $P_0 - \Sigma_s = \operatorname{Cov}(\eta_{t+h}, s)\operatorname{Cov}(s)^{-1}\operatorname{Cov}(s, \eta_{t+h})$, so $\operatorname{tr}(\Upsilon_{\mathrm{ctl}}(P_0 - \Sigma_s)) = \operatorname{tr}(\Psi_{\mathrm{o}}^{-1} K K^\top)$ with $K = \operatorname{Cov}(\mathbb{P}_{\mathrm{o}} H^{\mathrm{o}} \eta_{t+h}, s)\operatorname{Cov}(s)^{-1/2}$, which vanishes if and only if $K = 0$.

(b) Theorem 14.3(c) with $\mathsf{J} = \mathsf{J}_{\mathrm{ctl}}$, whose rank is at most $\operatorname{rank}\Upsilon_{\mathrm{ctl}} \le \operatorname{rank} B^{\mathrm{o}}$; reverse water-filling gives capacity only to positive eigenvalues.

(c) By (a), $G_h^{\mathrm{att}} = \mathfrak{c}_0 + \frac{1}{2}\operatorname{tr}(\Upsilon_{\mathrm{ctl}} P_0) - I(s; \mathsf{S}_{\mathrm{m}} \eta_{t+h}) - \frac{1}{2}\operatorname{tr}(\mathsf{G}_s \mathsf{J}_{\mathrm{ctl}})$. By Theorem 14.3(a) and Lemma S38.1(e), $-I(s; \mathsf{S}_{\mathrm{m}} \eta_{t+h}) \ge \sum_{\mathsf{j} \le \kappa} \frac{1}{2}\ln(1 - \mathsf{g}_{\mathsf{j}} \sigma_{\mathsf{j}}^2)$ and $-\frac{1}{2}\operatorname{tr}(\mathsf{G}_s \mathsf{J}_{\mathrm{ctl}}) \ge -\frac{1}{2}\sum_{\mathsf{j} \le \kappa} \mathsf{g}_{\mathsf{j}} \lambda_{\mathsf{j}}(\mathsf{J}_{\mathrm{ctl}})$, both with the sorted captured fractions; under the hypothesis the sensor whose j-th channel reads $\mathsf{e}_{\mathsf{j}}$ attains both. Minimizing the right side over the captured fractions is separable in the capacities $\mathsf{c}_{\mathsf{j}} = -\frac{1}{2}\ln(1 - \mathsf{g}_{\mathsf{j}})$, with per-channel marginal value

$$\frac{\sigma_{\mathsf{j}}^2 e^{-2\mathsf{c}_{\mathsf{j}}}}{1 - \sigma_{\mathsf{j}}^2 + \sigma_{\mathsf{j}}^2 e^{-2\mathsf{c}_{\mathsf{j}}}} + \lambda_{\mathsf{j}}(\mathsf{J}_{\mathrm{ctl}}) e^{-2\mathsf{c}_{\mathsf{j}}},$$

which decreases in $\mathsf{c}_{\mathsf{j}}$ and does not decrease in $(\sigma_{\mathsf{j}}^2, \lambda_{\mathsf{j}}(\mathsf{J}_{\mathrm{ctl}}))$. The objective is concave in the capacities, so the Karush–Kuhn–Tucker conditions characterize the optimum, and if channel $\mathsf{j}+1$ is active with marginal value equal to the multiplier, channel j, whose marginal value function lies above, receives at least as much capacity. The active set is therefore a leading segment with non-increasing capacities; the same argument applies to each of the three objectives. □

*Proof of Proposition S14.6.* The value splits as stated by the chain rule $I(s^{\mathrm{old}}, s^{(c)}; \mathsf{S}_{\mathrm{m}} \eta_{t+h}) = I(s^{\mathrm{old}}; \mathsf{S}_{\mathrm{m}} \eta_{t+h}) + \mathsf{V}_{\mathrm{ep}}$ and by the identity $\min_{a(\cdot)} \mathbb{E}[-\ln \tilde{p}(o)] = \mathfrak{c}_0 + \frac{1}{2}\operatorname{tr}(\Upsilon_{\mathrm{ctl}} \operatorname{Cov}(\eta_{t+h} \mid s))$ of the proof of Theorem 14.6(a), whose derivation uses only that the readings $s$ are jointly Gaussian with $\eta_{t+h}$ and independent of $v^{\mathrm{o}}$. (i) For jointly Gaussian variables a vanishing conditional covariance is conditional independence, so $s^{(c)}$ is independent of $\eta_{t+h}$ given $s^{\mathrm{old}}$; the conditional information vanishes and the conditional covariance of $\eta_{t+h}$ does not change. Both increments are continuous functions of the joint covariance, which is continuous in $H^{(c)}$, because the nondegenerate noise keeps $\operatorname{Cov}(s^{(c)} \mid s^{\mathrm{old}})$ invertible. (ii) As in the proof of Theorem 14.6(a), conditionally on $s^{\mathrm{old}}$, $2\mathsf{V}_{\mathrm{pr}} = \operatorname{tr}(\Psi_{\mathrm{o}}^{-1} K K^\top)$ with $K = \operatorname{Cov}(\mathbb{P}_{\mathrm{o}} H^{\mathrm{o}} \eta_{t+h}, s^{(c)} \mid s^{\mathrm{old}})\operatorname{Cov}(s^{(c)} \mid s^{\mathrm{old}})^{-1/2}$, zero if and only if $K = 0$; the epistemic part is Gaussian conditional independence. (iii) By independence $I(s^{(c)}; s^{\mathrm{old}}) = 0$, so $\mathsf{V}_{\mathrm{ep}} = I(s^{(c)}; \mathsf{S}_{\mathrm{m}} \eta_{t+h}, s^{\mathrm{old}}) \le I(\eta_t^{(c)}; \eta_{t+h}, s^{\mathrm{old}}) = I(\eta_t^{(c)}; \eta_{t+h}^{(c)})$, using data processing and the fact that $(\eta_{t+h}, s^{\mathrm{old}})$ consists of $\eta_{t+h}^{(c)}$ and variables independent of content $c$; the last quantity is the noiseless value $-\frac{1}{2}\sum_{\mathsf{j}} \ln(1 - \sigma_{\mathsf{j}}^{(c)}(h)^2)$ of the block. For the pragmatic part, independence gives $\operatorname{Cov}(\eta_{t+h}, s^{(c)} \mid s^{\mathrm{old}}) = \operatorname{Cov}(\eta_{t+h}, s^{(c)})$ and $\operatorname{Cov}(s^{(c)} \mid s^{\mathrm{old}}) = \operatorname{Cov}(s^{(c)})$, so the reduction of the covariance is $\operatorname{Cov}(\mathbb{E}[\eta_{t+h} \mid s^{(c)}]) \preceq \operatorname{Cov}(\mathbb{E}[\eta_{t+h} \mid \eta_t^{(c)}])$, the embedding of $F^{(c)h} P_0^{(c)} (F^{(c)h})^\top$, and $\operatorname{tr}(\Upsilon_{\mathrm{ctl}} X) \le \|\Upsilon_{\mathrm{ctl}}\| \operatorname{tr} X$ for $X \succeq 0$. The bound on $\sigma_{\mathsf{j}}^{(c)}(h)$ is the norm of $(P_0^{(c)})^{1/2} (F^{(c)h})^\top (P_0^{(c)})^{-1/2}$, which vanishes geometrically because $\rho(F^{(c)}) < 1$: by the independence of the contents and $P_0 \succ 0$, $F^{(c)h}$ is the diagonal block of $F^h$ for content $c$, so $\rho(F^{(c)}) \le \rho(F) < 1$. (iv) is Lemma S14.7. □

*Proof of Lemma S14.7.* The predicted variance obeys $P_{t+1}^- = f(P_t^-)$ with $f(x) = F_{\mathrm{g}}^2 x R^s/(x + R^s) + Q_{\mathrm{g}}$, increasing and concave on $[0, \infty)$ with $f(0) = Q_{\mathrm{g}} > 0$ and $f(x) - x \to -\infty$. Hence $f(x) - x$ has exactly one positive root, $f(x) > x$ below it and $f(x) < x$ above it, and the iteration converges monotonically from any prior. The fixed-point equation $x(x + R^s) = F_{\mathrm{g}}^2 x R^s + Q_{\mathrm{g}}(x + R^s)$ is the stated quadratic, whose positive root $\frac{1}{2}(-(\mathsf{b} - Q_{\mathrm{g}}) + ((\mathsf{b} - Q_{\mathrm{g}})^2 + 4 Q_{\mathrm{g}} R^s)^{1/2})$, $\mathsf{b} = R^s(1 - F_{\mathrm{g}}^2)$, does not increase in $\mathsf{b}$; at $\mathsf{b} = 0$ it is $\frac{1}{2}(Q_{\mathrm{g}} + (Q_{\mathrm{g}}^2 + 4 Q_{\mathrm{g}} R^s)^{1/2}) \le Q_{\mathrm{g}} + (Q_{\mathrm{g}} R^s)^{1/2}$. The innovation $s_t - \mathbb{E}[s_t \mid s_{<t}]$ has variance $P_t^- + R^s$, and $R^s$ given $\eta_t^{\mathrm{g}}$, which gives the information. Given the past

readings, $s_t \to \eta_t^{\mathrm{g}} \to \eta_{t+h}^{\mathrm{g}}$ is a Markov chain, so data processing bounds the information about the future. For a constant, the predicted variance after $n_{\mathrm{r}}$ readings is $(1/P_{\mathrm{pr}} + n_{\mathrm{r}}/R^s)^{-1}$, and the total information is one half the logarithm of the ratio of $P_{\mathrm{pr}}$ to it. $\square$

*Proof of Proposition S14.8.* Let $\mathsf{k}_u(z) = (4\pi D_{\mathrm{m}} u)^{-3/2} e^{-|z|^2/(4D_{\mathrm{m}}u)}$ be the heat kernel, and $\mathfrak{K}(u) = \int_{\mathbb{B}}\int_{\mathbb{B}} \mathsf{k}_u(x-y)\mathrm{d}x\mathrm{d}y$ for $u > 0$, $\mathfrak{K}(0) = |\mathbb{B}|$. Restrict first to the particles that start in a ball $B_{R_0}$ of radius $R_0$: their number $\mathsf{N}_0$ is Poisson with mean $\bar{c}_{\mathrm{L}}|B_{R_0}|$, each starts uniformly in $B_{R_0}$, and their paths are independent. For bounded functionals $\mathsf{a}_{\mathsf{m}}, \mathsf{b}_{\mathsf{m}}$ of the path of particle $\mathsf{m}$, $\mathbb{E}[\sum_{\mathsf{m}} \mathsf{a}_{\mathsf{m}} \sum_{\mathsf{m}} \mathsf{b}_{\mathsf{m}}] = \mathbb{E}[\mathsf{N}_0]\mathbb{E}[\mathsf{a}_1\mathsf{b}_1] + \mathbb{E}[\mathsf{N}_0(\mathsf{N}_0-1)]\mathbb{E}\mathsf{a}_1\mathbb{E}\mathsf{b}_1$ with $\mathbb{E}[\mathsf{N}_0(\mathsf{N}_0-1)] = \mathbb{E}[\mathsf{N}_0]^2$, so the covariance of the sums is $\mathbb{E}[\mathsf{N}_0]\mathbb{E}[\mathsf{a}_1\mathsf{b}_1]$. With $\mathsf{a}$ and $\mathsf{b}$ the indicators that the particle lies in a set $A$ at time $t$ and in $A'$ at time $t' \geq t$, and $R_0 \to \infty$,

$$\mathrm{Cov}\big(\mathsf{N}_A(t), \mathsf{N}_{A'}(t')\big) = \bar{c}_{\mathrm{L}} \int \Big[\int_A \mathsf{k}_t(y-x) \int_{A'} \mathsf{k}_{t'-t}(y'-y)\mathrm{d}y'\mathrm{d}y\Big]\mathrm{d}x = \bar{c}_{\mathrm{L}} \int_A \int_{A'} \mathsf{k}_{t'-t}(y'-y)\mathrm{d}y'\mathrm{d}y.$$

Integrating over $[0, T_{\mathrm{int}}]^2$ gives $\mathrm{Var}\,\hat{c} = 2\bar{c}_{\mathrm{L}}(|\mathbb{B}|T_{\mathrm{int}})^{-2}\int_0^{T_{\mathrm{int}}}(T_{\mathrm{int}}-u)\mathfrak{K}(u)\mathrm{d}u$, and $\mathbb{E}\hat{c} = \bar{c}_{\mathrm{L}}$ is stationarity. Leading term: $\int_0^\infty \mathsf{k}_u(z)\mathrm{d}u = 1/(4\pi D_{\mathrm{m}}|z|)$, so $\int_0^\infty \mathfrak{K} = (4\pi D_{\mathrm{m}})^{-1}\int_{\mathbb{B}}\int_{\mathbb{B}}|x-y|^{-1}$. For a function $\mathsf{f}$ of the distance, $\int_{\mathbb{B}}\int_{\mathbb{B}} \mathsf{f}(|x-y|) = \int_0^{2r_{\mathrm{cell}}} 4\pi z^2 V_\cap(z)\mathsf{f}(z)\mathrm{d}z$, with the lens volume $V_\cap(z) = \frac{\pi}{12}(4r_{\mathrm{cell}}+z)(2r_{\mathrm{cell}}-z)^2$; with $z = 2r_{\mathrm{cell}}\vartheta$,

$$\int_0^{2r_{\mathrm{cell}}} zV_\cap(z)\mathrm{d}z = \frac{32\pi r_{\mathrm{cell}}^5}{12}\int_0^1 \vartheta(2+\vartheta)(1-\vartheta)^2\mathrm{d}\vartheta = \frac{8\pi r_{\mathrm{cell}}^5}{15},$$

so $\int_{\mathbb{B}}\int_{\mathbb{B}}|x-y|^{-1} = 32\pi^2 r_{\mathrm{cell}}^5/15 = 6|\mathbb{B}|^2/(5r_{\mathrm{cell}})$ and $2\bar{c}_{\mathrm{L}}(|\mathbb{B}|^2T_{\mathrm{int}})^{-1}\int_0^\infty \mathfrak{K} = 3\bar{c}_{\mathrm{L}}/(5\pi D_{\mathrm{m}} r_{\mathrm{cell}} T_{\mathrm{int}})$. Corrections: since $|x-y| \leq 2r_{\mathrm{cell}}$ on $\mathbb{B}\times\mathbb{B}$, $\exp(-|x-y|^2/(4D_{\mathrm{m}}u)) \geq 1 - r_{\mathrm{cell}}^2/(D_{\mathrm{m}}u)$, which gives

$$\frac{|\mathbb{B}|^2}{(4\pi D_{\mathrm{m}}u)^{3/2}}\Big(1 - \frac{r_{\mathrm{cell}}^2}{D_{\mathrm{m}}u}\Big) \leq \mathfrak{K}(u) \leq \frac{|\mathbb{B}|^2}{(4\pi D_{\mathrm{m}}u)^{3/2}}.$$

Write

$$\mathrm{Var}\,\hat{c} = \frac{2\bar{c}_{\mathrm{L}}}{|\mathbb{B}|^2 T_{\mathrm{int}}}\Big[\int_0^\infty \mathfrak{K} - \int_{T_{\mathrm{int}}}^\infty \mathfrak{K}\Big] - \frac{2\bar{c}_{\mathrm{L}}}{|\mathbb{B}|^2 T_{\mathrm{int}}^2}\int_0^{T_{\mathrm{int}}} u\mathfrak{K}(u)\mathrm{d}u.$$

The bounds give, splitting the second integral at $u = r_{\mathrm{cell}}^2/D_{\mathrm{m}}$ and using the lower bound only above that point,

$$\int_{T_{\mathrm{int}}}^\infty \mathfrak{K} = \frac{2|\mathbb{B}|^2}{(4\pi D_{\mathrm{m}})^{3/2} T_{\mathrm{int}}^{1/2}} - \mathrm{rem}_2, \qquad \int_0^{T_{\mathrm{int}}} u\mathfrak{K}(u)\mathrm{d}u = \frac{2|\mathbb{B}|^2 T_{\mathrm{int}}^{1/2}}{(4\pi D_{\mathrm{m}})^{3/2}} - \mathrm{rem}_3,$$

with

$$0 \leq \mathrm{rem}_2 \leq \frac{2|\mathbb{B}|^2 r_{\mathrm{cell}}^2}{3(4\pi D_{\mathrm{m}})^{3/2} D_{\mathrm{m}} T_{\mathrm{int}}^{3/2}}, \qquad 0 \leq \mathrm{rem}_3 \leq \frac{4|\mathbb{B}|^2 r_{\mathrm{cell}}}{(4\pi D_{\mathrm{m}})^{3/2} D_{\mathrm{m}}^{1/2}}.$$

The two terms of order $T_{\mathrm{int}}^{-3/2}$ sum to $-8\bar{c}_{\mathrm{L}}(4\pi D_{\mathrm{m}} T_{\mathrm{int}})^{-3/2}$, which relative to the leading term is $-\frac{5}{3}(r_{\mathrm{cell}}^2/(\pi D_{\mathrm{m}} T_{\mathrm{int}}))^{1/2}$, and $\mathrm{rem}_2$ and $\mathrm{rem}_3$ enter with positive sign and give the two parts of the bound on $\mathfrak{q}_T$. In an ideal solution the chemical potential is a constant plus $k_{\mathrm{B}}T_{\mathrm{K}}$ times the logarithm of the concentration, so its small changes in units of $k_{\mathrm{B}}T_{\mathrm{K}}$ are relative changes of concentration. $\square$

*Proof of Proposition S14.12.* (a) By Definition S14.11, $\bar{x}$ is exactly closed if and only if $\ker \mathbf{1}^\top$ is invariant under $W$, that is, $\mathbf{1}^\top W x = 0$ for every $x \perp \mathbf{1}$, that is, $W^\top \mathbf{1} \in \mathrm{span}(\mathbf{1})$; then $N^{-1}\mathbf{1}^\top W x = \bar{w}\bar{x}$. (b) Averaging the agents' equation over type $\mathfrak{t}$ gives the type equation, because

the coupling depends on the other agents only through type averages and $A_{\mathfrak{t}}$ is common to the type; the averaged idiosyncratic noise is independent of the past, with covariance $Q_{\mathrm{id}}/N_{\mathfrak{t}}$, and the world is autonomous. The deviation from the infinite-population limit obeys $\bar{x}^{\mathrm{fl}}_{t+1} = \mathbb{A}_{\mathrm{ty}}\bar{x}^{\mathrm{fl}}_t + \bar{w}^{\mathrm{id}}_t$, driven by noise independent of the world and of the common shocks, which gives $\Sigma^{\mathrm{fl}}$ and its bound. (c) For agents $i, i'$ of type $\mathfrak{t}$ with own dynamics $A_i \neq A_{i'}$, the state with $x_i = e$, $x_{i'} = -e$ and all other coordinates zero lies in the kernel of the map to type averages, and its image has type-$\mathfrak{t}$ average $(A_i - A_{i'})e/N_{\mathfrak{t}}$, nonzero for a suitable $e$; the kernel is not invariant. □

*Proof of Proposition S14.13.* (a) If $\mathsf{R}(u) = \mathsf{R}(0)\alpha_*^{|u|}$, then $\mathrm{Cov}(x_{t+1} - \alpha_* x_t, x_{t-l}) = \mathsf{R}(l+1) - \alpha_*\mathsf{R}(l) = 0$ for $l \geq 0$, so the Gaussian residual is independent of the past and the process is an autoregression of order one. If the process is Markov, $\mathbb{E}[x_{t+u} \mid x_{\leq t}] = (\mathsf{R}(u)/\mathsf{R}(0))x_t$, and the tower property applied to $\mathbb{E}[x_{t+u+1} \mid x_{\leq t}]$ gives $\mathsf{R}(u+1)/\mathsf{R}(0) = (\mathsf{R}(1)/\mathsf{R}(0))(\mathsf{R}(u)/\mathsf{R}(0))$, so the autocovariance is geometric. A geometric autocorrelation is $\mathrm{Corr}(\bar{x}_t, \bar{x}_{t+1})^u$.

(b) The stationary solution is $x_{i,t} = \sum_{l\geq 0} \alpha_i^l (w^{\mathrm{com}}_{t-1-l} + w^{\mathrm{id}}_{i,t-1-l})$, so $\bar{x}_t = \sum_l \bar{\mathfrak{m}}_l w^{\mathrm{com}}_{t-1-l} + N^{-1}\sum_i \sum_l \alpha_i^l w^{\mathrm{id}}_{i,t-1-l}$ with $\bar{\mathfrak{m}}_l = N^{-1}\sum_i \alpha_i^l$, and

$$\mathsf{R}(u) = q_{\mathrm{com}} \sum_{l\geq 0} \bar{\mathfrak{m}}_l \bar{\mathfrak{m}}_{l+u} + \frac{q_{\mathrm{id}}}{N^2} \sum_i \frac{\alpha_i^u}{1-\alpha_i^2}, \qquad \sum_{l\geq 0} \bar{\mathfrak{m}}_l \bar{\mathfrak{m}}_{l+u} = \frac{1}{N^2} \sum_{i,i'} \frac{\alpha_{i'}^u}{1-\alpha_i\alpha_{i'}},$$

which is the stated measure; its density with respect to $\mathbb{Q}$ is positive because $|\alpha\alpha'| < 1$.

(c) By (a), $\bar{x}$ is Markov if and only if $\int \alpha^u \bar{\mathbb{M}}(\mathrm{d}\alpha) = \alpha_*^u$ for all $u \geq 0$ and some $\alpha_*$, with $\bar{\mathbb{M}} = \mathbb{M}/\mathbb{M}(\mathbb{R})$; the probability measure $\bar{\mathbb{M}}$ on $[-1,1]$ then has the moments of the point mass at $\alpha_*$, and moments determine such a measure, so $\bar{\mathbb{M}}$ is that point mass. Since $\mathbb{M}$ charges every $\alpha_i$ (as $q_{\mathrm{com}} + q_{\mathrm{id}} > 0$), all $\alpha_i$ equal $\alpha_*$. Equal persistences give an autoregression of order one.

(d) With $\bar{\alpha} = \int \alpha \bar{\mathbb{M}}(\mathrm{d}\alpha) = \mathrm{Corr}(\bar{x}_t, \bar{x}_{t+1})$ and $\varphi(\alpha) = \alpha^u$, convex on $[0,1)$, Taylor's theorem gives

$$0 \leq \int \varphi \mathrm{d}\bar{\mathbb{M}} - \varphi(\bar{\alpha}) = \int [\varphi(\alpha) - \varphi(\bar{\alpha}) - \varphi'(\bar{\alpha})(\alpha - \bar{\alpha})]\bar{\mathbb{M}}(\mathrm{d}\alpha) \leq \tfrac{1}{2}u(u-1)\alpha_+^{u-2}\,\mathrm{Var}_{\bar{\mathbb{M}}}(\alpha),$$

for $u \geq 2$ (the error vanishes for $u \leq 1$), and $\frac{1}{2}u(u-1)x^{u-2} \leq \sum_{\mathsf{m}\geq 2} \frac{1}{2}\mathsf{m}(\mathsf{m}-1)x^{\mathsf{m}-2} = (1-x)^{-3}$. A law on an interval of length $\alpha_+ - \alpha_-$ has variance at most a quarter of its square.

(e) The type averages obey (b) of Proposition S14.12 with $\mathsf{B} = 0$ and $A_{\mathfrak{t}}$ the common persistence, and $\bar{x}$ is a fixed linear combination of them; adjoining $\mathfrak{T} - 1$ of them to $\bar{x}$ spans all, so $\bar{x}$ is closed of order $\mathfrak{T} - 1$. No lower order suffices. By Theorem 11.8(f), applied without input as in Definition S14.11, every exactly closed extension of $\bar{x}$ has a row space that contains the row space of the observability matrix $\mathcal{O}$ of $(N^{-1}\mathbf{1}^\top, \mathrm{diag}(\alpha_i))$, whose rows are $N^{-1}(\alpha_1^l, \ldots, \alpha_N^l)$ for $0 \leq l < N$. The columns of $\mathcal{O}$ that belong to agents of one type coincide, and its $\mathfrak{T}$ distinct columns are those of a Vandermonde matrix with distinct nodes, multiplied by $N^{-1}$, so $\mathrm{rank}\,\mathcal{O} = \mathfrak{T}$ and $d^* = \mathrm{rank}\,\mathcal{O} - 1 = \mathfrak{T} - 1$.

(f) Put $\alpha = 1 - y/l$: $l^{\mathfrak{p}}\mathfrak{m}_l = \int_0^l (1-y/l)^l y^{\mathfrak{p}-1}\mathsf{f}_0(1-y/l)\mathrm{d}y \to \mathsf{f}_0(1)\Gamma(\mathfrak{p})$ by dominated convergence, with dominating function $\|\mathsf{f}_0\|_\infty e^{-y}y^{\mathfrak{p}-1}$; the same bound gives $\mathfrak{m}_l \leq \|\mathsf{f}_0\|_\infty \Gamma(\mathfrak{p}) l^{-\mathfrak{p}}$ for $l \geq 1$. Since $\sum_l \mathfrak{m}_l^2 < \infty$, the limiting aggregate $\sum_l \mathfrak{m}_l w^{\mathrm{com}}_{t-1-l}$ exists in mean square, with $\mathsf{R}(u)/q_{\mathrm{com}} = \sum_{l\geq 0} \mathfrak{m}_l \mathfrak{m}_{l+u}$. The term $l = 0$ is $\mathfrak{m}_u = O(u^{-\mathfrak{p}}) = o(u^{1-2\mathfrak{p}})$ because $\mathfrak{p} < 1$. For $l \geq 1$, $u^{2\mathfrak{p}-1}\mathfrak{m}_l\mathfrak{m}_{l+u} = \int_{(l-1)/u}^{l/u} \mathsf{h}_u(x)\mathrm{d}x$ with the step function $\mathsf{h}_u(x) = u^{\mathfrak{p}}\mathfrak{m}_{\lceil ux\rceil} u^{\mathfrak{p}}\mathfrak{m}_{\lceil ux\rceil+u}$, which converges pointwise to $(\mathsf{f}_0(1)\Gamma(\mathfrak{p}))^2 x^{-\mathfrak{p}}(1+x)^{-\mathfrak{p}}$ and, because $\lceil ux\rceil \geq ux$, is dominated by a constant multiple of $x^{-\mathfrak{p}}(1+x)^{-\mathfrak{p}}$, integrable on $(0,\infty)$ for $\frac{1}{2} < \mathfrak{p} < 1$. Dominated convergence and $\int_0^\infty x^{-\mathfrak{p}}(1+x)^{-\mathfrak{p}}\mathrm{d}x = \mathrm{B}(1-\mathfrak{p}, 2\mathfrak{p}-1)$ give the claim, and $1 - 2\mathfrak{p} > -1$ makes the autocovariance non-summable. □

*Proof of Theorem 14.7.* (a) $\psi_i^\top \eta_{t+h} = ((F^h)^\top \psi_i)^\top \eta_t + \mathsf{f}_h$, with $\mathsf{f}_h$ a combination of future innovations, independent of $(\eta_t, v_i)$. Hence $\operatorname{Var}(\psi_i^\top \eta_{t+h} \mid s_i, M\eta_t) = \operatorname{Var}(((F^h)^\top \psi_i)^\top \eta_t \mid s_i, M\eta_t) + \operatorname{Var}(\mathsf{f}_h)$, and the first term vanishes if and only if $(F^h)^\top \psi_i \in \operatorname{row}(M)$ by Proposition S10.16. The dimension of a linear image of a fixed subspace does not increase under further maps and is preserved by invertible ones, and $(F^{h+1})^\top = F^\top (F^h)^\top$.

(b) Ignoring private signals, the total excess loss of a noiseless message $M\eta_t$ is $\sum_i \operatorname{Var}(\psi_i^\top F^h \eta_t \mid M\eta_t)$, which equals $\operatorname{tr}(\mathsf{D}_h(I - \mathbb{P}_M))$ by Lemma 14.2(iii) with a noiseless sensor, $\mathbb{P}_M$ being the orthogonal projector onto the row space of $MP_0^{1/2}$; and Lemma S38.1(b) bounds $\operatorname{tr}(\mathsf{D}_h \mathbb{P}_M)$ by $\sum_{\mathsf{j} \le r_M} \lambda_{\mathsf{j}}(\mathsf{D}_h)$. Under detailed balance $P_0^{1/2}(F^h)^\top = \mathsf{F}^h P_0^{1/2}$, so $\mathsf{e}_{\mathsf{j}}^\top \mathsf{D}_h \mathsf{e}_{\mathsf{j}} = \lambda_{\mathsf{j}}(F)^{2h} \mathsf{w}_{\mathsf{j}}$, $\operatorname{tr} \mathsf{D}_h = \sum_{\mathsf{j}} \lambda_{\mathsf{j}}(F)^{2h} \mathsf{w}_{\mathsf{j}}$, and the message reading $\mathsf{e}_1^\top P_0^{-1/2} \eta_t, \dots, \mathsf{e}_{r_M}^\top P_0^{-1/2} \eta_t$ leaves $\sum_{\mathsf{j} > r_M} \lambda_{\mathsf{j}}(F)^{2h} \mathsf{w}_{\mathsf{j}}$. Bounding the numerator by $|\lambda_{r_M+1}(F)|^{2h} \sum_{\mathsf{j} > r_M} \mathsf{w}_{\mathsf{j}}$ and the denominator below by $|\lambda_1(F)|^{2h} \mathsf{w}_1$, which is positive because $\lambda_1(F) \neq 0$ and $\mathsf{w}_1 > 0$, gives the ratio bound. For the count, whose threshold is defined because $h \ge 1$, let $r_M$ be the number of modes at or above the threshold; every other mode has $|\lambda_{\mathsf{j}}(F)|^{2h} < |\lambda_1(F)|^{2h} \mathfrak{e} \mathsf{w}_1 / \sum_{\mathsf{j}'} \mathsf{w}_{\mathsf{j}'}$, so the excess is below $\mathfrak{e} |\lambda_1(F)|^{2h} \mathsf{w}_1 \le \mathfrak{e} \operatorname{tr} \mathsf{D}_h$. As $h \to \infty$ the threshold tends to $|\lambda_1(F)|$.

(c) $F^h = \sum_{\mathsf{j}} \lambda_{\mathsf{j}}(F)^h \mathsf{r}_{\mathsf{j}} \mathsf{u}_{\mathsf{j}}^\top$, so $\psi^\top F^h \eta_t = \sum_{\mathsf{j}} \lambda_{\mathsf{j}}(F)^h (\psi^\top \mathsf{r}_{\mathsf{j}}) \mathsf{u}_{\mathsf{j}}^\top \eta_t$. Given the slow amplitudes, which determine the terms with $\mathsf{j} \le r_M$ (conjugate pairs are not split because $|\lambda_{r_M}(F)| > |\lambda_{r_M+1}(F)|$), the conditional variance is at most the variance of the remainder $\sum_{\mathsf{j} > r_M} \lambda_{\mathsf{j}}(F)^h (\psi^\top \mathsf{r}_{\mathsf{j}}) \mathsf{u}_{\mathsf{j}}^\top \eta_t$, whose standard deviation is at most the sum of the terms' root-mean-square values, $|\lambda_{\mathsf{j}}(F)|^h |\psi^\top \mathsf{r}_{\mathsf{j}}| (\mathsf{u}_{\mathsf{j}}^* P_0 \mathsf{u}_{\mathsf{j}})^{1/2}$. When $\lambda_1(F)$ is real, simple and strictly dominant, $\operatorname{Var}(\psi^\top F^h \eta_t) = \lambda_1(F)^{2h} (\psi^\top \mathsf{r}_1)^2 \mathsf{u}_1^\top P_0 \mathsf{u}_1 (1 + o(1))$. □

*Derivations for Example S14.14.* With $\mathsf{f}_1 = F_1^h$ and $\mathsf{f}_2 = F_2^h$, the decision variable is $\mathsf{f}_1 \eta_1 + \mathsf{f}_2 \eta_2 = \frac{\mathsf{f}_1 + \mathsf{f}_2}{2}(\eta_1 + \eta_2) + \frac{\mathsf{f}_1 - \mathsf{f}_2}{2}(\eta_1 - \eta_2)$, and $\operatorname{Var}(\eta_1 - \eta_2 \mid \eta_1 + \eta_2) = 4P_{0,1}P_{0,2}/(P_{0,1} + P_{0,2})$ gives $\mathcal{L}_{\text{spot}}$; $\mathcal{L}_{\text{slow}}$ is the variance of the unread term $\mathsf{f}_2 \eta_2$, and the futures price reads the decision variable itself. The ratio $\mathcal{L}_{\text{spot}}/\mathcal{L}_{\text{slow}} = (\mathsf{f}_1/\mathsf{f}_2 - 1)^2 P_{0,1}/(P_{0,1} + P_{0,2})$ increases with $h$ and equals one where $\mathsf{f}_1/\mathsf{f}_2 = (F_1/F_2)^h = 1 + (1 + P_{0,2}/P_{0,1})^{1/2}$, which is $h^\times$. With price noise $\mathsf{n}_t^{\mathrm{p}}$ the spot loss is $\mathsf{f}_1^2 P_{0,1} + \mathsf{f}_2^2 P_{0,2} - (\mathsf{f}_1 P_{0,1} + \mathsf{f}_2 P_{0,2})^2/(P_{0,1} + P_{0,2} + \operatorname{Var}(\mathsf{n}_t^{\mathrm{p}}))$. The optimal output and the excess loss follow from $\mathbb{E}[p_{t+h} \mid \eta_t] = \mathsf{f}_1 \eta_{1,t} + \mathsf{f}_2 \eta_{2,t}$: the expected profit at output $\mathbb{E}[p_{t+h} \mid \text{readings}]/\mathsf{k}_a$ is $\mathbb{E}[\mathbb{E}[p_{t+h} \mid \text{readings}]^2]/(2\mathsf{k}_a)$, and the readings are functions of $\eta_t$ plus independent noise. □

*Proof of Lemma S14.17.* Since $\mathbb{E}[\mathsf{X}_1 \mid \mathsf{X}_2, \mathsf{X}_3] = \mathbb{E}[\mathsf{X}_1 \mid \mathsf{X}_2] = \Sigma_{12}\Sigma_{22}^{-1}\mathsf{X}_2$, $\Sigma_{13} = \mathbb{E}[\mathbb{E}[\mathsf{X}_1 \mid \mathsf{X}_2]\mathsf{X}_3^\top] = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{23}$, and whitening gives $\mathsf{C}_{13} = \mathsf{C}_{12}\mathsf{C}_{23}$. Then $\mathsf{C}_{13}\mathsf{C}_{13}^\top = \mathsf{C}_{12}\mathsf{C}_{23}\mathsf{C}_{23}^\top\mathsf{C}_{12}^\top \preceq \sigma_1(\mathsf{C}_{23})^2 \mathsf{C}_{12}\mathsf{C}_{12}^\top$, and eigenvalues are monotone in the Loewner order (Horn and Johnson, 2012). The rank bound is that of a product. □

*Proof of Proposition S14.18.* (a) is the chain rule, and for jointly Gaussian variables $I(\mathsf{Y}; s \mid y)$ is $\frac{1}{2} \ln\det \operatorname{Cov}(\mathsf{Y} \mid y) - \frac{1}{2} \ln\det \operatorname{Cov}(\mathsf{Y} \mid s, y)$. (b) Gaussian conditional independence holds exactly when $\operatorname{Cov}(\mathsf{Y}, s^{\mathrm{ph}} \mid y^{\mathrm{agg}}) = 0$, and this covariance is $\mathsf{Z}_h^\top \operatorname{Cov}(\eta_t \mid y^{\mathrm{agg}})(H^{\mathrm{ph}})^\top$ because the future innovations in $\mathsf{Y}$ are independent of $(\eta_t, s^{\mathrm{ph}}, y^{\mathrm{agg}})$. With noiseless aggregates $\operatorname{Cov}(\eta_t \mid T^{\mathrm{agg}}\eta_t)(T^{\mathrm{agg}})^\top = 0$, so columns of $\mathsf{Z}_h$ in the row space of $T^{\mathrm{agg}}$ are annihilated. (c) Write $\mathsf{Y} = \mathsf{z}^\top \eta_t + \mathsf{f}_h$ with $\mathsf{z} = (F^h)^\top \psi$ and $\mathsf{f}_h$ independent. The squared partial correlation of $\mathsf{Y}$ with $s^{\mathrm{ph}}$ given $y = y^{\mathrm{agg}}$ is $\operatorname{Cov}(\mathsf{Y}, s \mid y) \operatorname{Cov}(s \mid y)^{-1} \operatorname{Cov}(s, \mathsf{Y} \mid y) / \operatorname{Var}(\mathsf{Y} \mid y)$; its numerator equals that for $\mathsf{z}^\top \eta_t$, which is at most $\operatorname{Var}(\mathsf{z}^\top \eta_t \mid y)$, and $I(\mathsf{Y}; s \mid y)$ is minus one half the logarithm of one minus it. In the reversible case with $y$ reading $\mathsf{e}_1^\top P_0^{-1/2} \eta_t, \dots, \mathsf{e}_{r_{\mathrm{agg}}}^\top P_0^{-1/2} \eta_t$, the identity $F^h = P_0^{1/2} \mathsf{F}^h P_0^{-1/2}$ gives $\operatorname{Var}(\mathsf{z}^\top \eta_t \mid y) = \sum_{\mathsf{j} > r_{\mathrm{agg}}} \lambda_{\mathsf{j}}(F)^{2h} (\mathsf{e}_{\mathsf{j}}^\top P_0^{1/2} \psi)^2$, and $\operatorname{Var}(\mathsf{Y} \mid y) \ge \operatorname{Var}(\mathsf{Y} \mid \eta_t) = \psi^\top (P_0 - F^h P_0 (F^h)^\top) \psi = \psi^\top P_0^{1/2} (I - \mathsf{F}^{2h}) P_0^{1/2} \psi$, which is positive for $h \ge 1$ and $\psi \neq 0$ because the symmetric $\mathsf{F}$ has spectral radius below one; at $h = 0$ it vanishes and the displayed bound does not apply. (d) Under the

hypotheses $P_0$ and $F^h$ are block diagonal and each $(F^{h_\mathfrak{d}})^\top \psi_\mathfrak{d}$ is supported on one block, so $\mathsf{J}$ is block diagonal; a block-diagonal symmetric matrix whose blocks have no common eigenvalue has block-pure eigenvectors. Reverse water-filling (Theorem 14.3(c)) gives channel $\mathsf{j}$ the capacity $\frac{1}{2}\ln^+(\lambda_\mathsf{j}(\mathsf{J})/\mathsf{v}')$; the capacity used increases in every $\lambda_\mathsf{j}(\mathsf{J})$ and decreases in $\mathsf{v}'$, so raising some eigenvalues at a fixed budget raises $\mathsf{v}'$ and lowers or keeps the capacity of every other channel. □

## S39 Proofs for Section S18

Throughout, $|v|$ is the Euclidean norm, $\|M\|$ the spectral norm, $\|M\|_{\mathrm{F}}$ the Frobenius norm, $\|X\|_{L^2} = (\mathbb{E}|X|^2)^{1/2}$, and $\preceq$ the order of symmetric matrices. The matrices $C^\sharp$ and $Z^\sharp$ are the blocks of the inverse of $\binom{C}{Z}$, so that $CC^\sharp = I$, $ZZ^\sharp = I$, $CZ^\sharp = 0$, $ZC^\sharp = 0$ and $C^\sharp C + Z^\sharp Z = I$, and $A_{yy} = CAC^\sharp$, $A_{y\perp} = CAZ^\sharp$, $A_{\perp y} = ZAC^\sharp$, $A_{\perp\perp} = ZAZ^\sharp$. The randomization $\omega$ of a causal coarse policy takes values in a standard Borel space, has the same law under the fine process and under the declared model, and is independent of $(x_0, \mathfrak{n})$ in the first and of $(\hat{y}_0, \hat{w})$ in the second; the second moments $\mathbb{E}|u_t|^2$ are finite under the fine process. Regular conditional laws exist because all spaces are standard Borel. For a symmetric $S \succeq 0$ with $CSC^\top \succ 0$ put

$$\Pi_C(S) = S - SC^\top (CSC^\top)^{-1} CS, \tag{S121}$$

the covariance of a Gaussian vector with covariance $S$ given its image under $C$; it satisfies $C\Pi_C(S) = 0$ and $0 \preceq \Pi_C(S) \preceq S$, and it is nondecreasing in $S$, because $\Pi_C(S) = \min_K (I - KC)S(I - KC)^\top$.

**Lemma S39.1** (Separation). *(a) For every causal coarse policy and every $t \geq 0$, the conditional law of $x_t$ given $\mathcal{G}_t = \sigma(y_{0:t}, \omega)$ is $\mathcal{N}(\hat{x}_t, P_t)$, where $\hat{x}_t$ is a measurable function of $(y_{0:t}, \omega)$ with $C\hat{x}_t = y_t$ and*

$$P_0 = \Pi_C(\Sigma_0), \qquad P_{t+1} = \Pi_C(AP_tA^\top + Q) \tag{S122}$$

*is deterministic and the same for all causal coarse policies, with $CP_t = 0$. The conditional law of $y_{t+1}$ given $\mathcal{G}_t$ is $\mathcal{N}(CA\hat{x}_t + CBu_t, S_t)$ with $S_t = Q^y + CAP_tA^\top C^\top \succ 0$.*

*(b) Let $s_{t+1} = Hy_t + v_t$ with $v_t \sim \mathcal{N}(0, R)$ independent of each other and of $(x_0, \mathfrak{n}, \omega)$, $R \succ 0$, and let $u_t = \pi_t(s_{1:t}, \omega)$. Then the conditional laws of $x_t$ given $(s_{1:t}, \omega)$ and given $(s_{1:t+1}, \omega)$ are $\mathcal{N}(\hat{x}_t^-, \Sigma_t^-)$ and $\mathcal{N}(\hat{x}_t^{\mathrm{f}}, \Sigma_t^{\mathrm{f}})$ with*

$$\Sigma_0^- = \Sigma_0, \qquad \Sigma_t^{\mathrm{f}} = \Sigma_t^- - \Sigma_t^- C^\top H^\top (HC\Sigma_t^- C^\top H^\top + R)^{-1} HC\Sigma_t^-, \qquad \Sigma_{t+1}^- = A\Sigma_t^{\mathrm{f}} A^\top + Q, \tag{S123}$$

*which no such policy changes, and $\hat{x}_{t+1}^- = A\hat{x}_t^{\mathrm{f}} + Bu_t$.*

*Proof.* (a) The proof is by induction on $t$. Since $x_0 \sim \mathcal{N}(m_0, \Sigma_0)$ is independent of $\omega$ and $C\Sigma_0 C^\top \succ 0$, Gaussian conditioning on $y_0 = Cx_0$ gives the law $\mathcal{N}(m_0 + \Sigma_0 C^\top (C\Sigma_0 C^\top)^{-1}(y_0 - Cm_0), \Pi_C(\Sigma_0))$, and conditioning on $\omega$ as well changes nothing. Assume the claim at $t$. By construction $x_t$ and $y_{0:t}$ are measurable functions of $(x_0, \mathfrak{n}_{0:t-1}, \omega)$, so $\mathfrak{n}_t$ is independent of $(x_t, \mathcal{G}_t)$, and $u_t$ is $\mathcal{G}_t$-measurable. Given $\mathcal{G}_t$ the vector $x_{t+1} = Ax_t + Bu_t + \mathfrak{n}_t$ is therefore Gaussian with mean $A\hat{x}_t + Bu_t$ and covariance $AP_tA^\top + Q$, and its image $y_{t+1} = Cx_{t+1}$ is Gaussian with mean $CA\hat{x}_t + CBu_t$ and covariance $C(AP_tA^\top + Q)C^\top = S_t \succeq Q^y \succ 0$. Conditioning the pair $(x_{t+1}, y_{t+1})$, which is jointly Gaussian given $\mathcal{G}_t$, on $y_{t+1}$ gives a Gaussian law with covariance $\Pi_C(AP_tA^\top + Q)$ and a mean that is an affine function of $y_{t+1}$ with $\mathcal{G}_t$-measurable coefficients. Since $\mathcal{G}_{t+1} = \mathcal{G}_t \vee \sigma(y_{t+1})$, this is the claim at $t+1$. The identities $C\hat{x}_t = y_t$ and $CP_t = 0$ hold because $Cx_t = y_t$ is $\mathcal{G}_t$-measurable.

(b) The same induction applies with the two conditioning steps of Eq. (S123). Given $(s_{1:t}, \omega)$ and the induction hypothesis $x_t \sim \mathcal{N}(\hat{x}_t^-, \Sigma_t^-)$, the reading $s_{t+1} = HCx_t + v_t$ is jointly Gaussian

with $x_t$, because $v_t$ is independent of $(x_t, s_{1:t}, \omega)$, and conditioning on it gives $\mathcal{N}(\hat{x}_t^{\mathrm{f}}, \Sigma_t^{\mathrm{f}})$. The act $u_t$ is measurable in $(s_{1:t}, \omega)$ and $\mathfrak{n}_t$ is independent of $(x_t, s_{1:t+1}, \omega)$, so $x_{t+1}$ given $(s_{1:t+1}, \omega)$ is $\mathcal{N}(A\hat{x}_t^{\mathrm{f}} + Bu_t, A\Sigma_t^{\mathrm{f}}A^\top + Q)$. □

*Proof of Theorem S18.2.* Both $P^\pi$ and $\hat{P}^\pi$ are laws of $(\omega, y_0, \dots, y_T)$ on the same standard Borel space. Under $P^\pi$ the randomization has its law, $y_0 \sim P_{y_0} = \mathcal{N}(Cm_0, C\Sigma_0 C^\top)$ is independent of it, and by Lemma S39.1(a) the kernel of $y_{t+1}$ given $(\omega, y_{0:t})$ is $\mathcal{N}(CA\hat{x}_t + CBu_t, S_t)$. Under $\hat{P}^\pi$ the randomization has the same law, $y_0 \sim \hat{P}_0$ is independent of it, and the kernel of $y_{t+1}$ given $(\omega, y_{0:t})$ is $\mathcal{N}(\bar{A}y_t + \bar{B}u_t, \bar{Q})$ with the same $u_t = \pi_t(y_{0:t}, \omega)$. The chain rule for relative entropy, which holds with values in $[0, \infty]$ on standard Borel spaces, gives

$$D_{\mathrm{KL}}(P^\pi \| \hat{P}^\pi) = D_{\mathrm{KL}}(P_{y_0} \| \hat{P}_0) + \sum_{t=0}^{T-1} \mathbb{E}^\pi D_{\mathrm{KL}}\Big(\mathcal{N}(CA\hat{x}_t + CBu_t, S_t) \Big\| \mathcal{N}(\bar{A}y_t + \bar{B}u_t, \bar{Q})\Big),$$

the term of $\omega$ being zero. The relative entropy of $\mathcal{N}(m, S)$ from $\mathcal{N}(\hat{m}, \bar{Q})$ is $h(\bar{Q}^{-1}S) + \frac{1}{2}\|m - \hat{m}\|^2_{\bar{Q}^{-1}}$, and $CA\hat{x}_t - \bar{A}y_t = (CA - \bar{A}C)\hat{x}_t = D\hat{x}_t$ because $y_t = C\hat{x}_t$. Hence the summand is $h(\bar{Q}^{-1}S_t) + \frac{1}{2}\mathbb{E}^\pi\|D\hat{x}_t + Eu_t\|^2_{\bar{Q}^{-1}}$, the second form of the statement; it is finite because $\mathbb{E}|u_t|^2 < \infty$ and, by induction on the recursion, $\mathbb{E}|x_t|^2 < \infty$.

For the first form write $\delta_t = D\hat{x}_t + Eu_t + D(x_t - \hat{x}_t)$. Given $\mathcal{G}_t$ the last term has mean zero and covariance $DP_tD^\top$, and the first two are $\mathcal{G}_t$-measurable, so $\mathbb{E}^\pi\|\delta_t\|^2_{\bar{Q}^{-1}} = \mathbb{E}^\pi\|D\hat{x}_t + Eu_t\|^2_{\bar{Q}^{-1}} + \operatorname{tr}(\bar{Q}^{-1}DP_tD^\top)$. Since $CP_t = 0$, $DP_tD^\top = CAP_tA^\top C^\top = S_t - Q^y$. Finally

$$h(\bar{Q}^{-1}S_t) = h(\bar{Q}^{-1}Q^y) + \tfrac{1}{2}\operatorname{tr}(\bar{Q}^{-1}CAP_tA^\top C^\top) - \tfrac{1}{2}\ln\det((Q^y)^{-1}S_t),$$

and the term subtracted on the right is $\mathfrak{d}_{\mathrm{dyn},t}$. Adding the two displays gives Eq. (S58). The expression for $\mathfrak{d}_{\mathrm{dyn},t}$ is that of Proposition S11.8(b) with $\mathsf{P} = C$, $Q^{\mathfrak{n}} = Q$ and $P_t^{\mathrm{f}} = P_t$; by Lemma S39.1(a) it is also the conditional mutual information $I(y_{t+1}; x_t \mid y_{0:t}, \omega)$ under every causal coarse policy. □

*Proof of Corollary S18.3.* The moment bounds of the statement are

$$\bar{X} = \frac{1}{1-g_\kappa}\Big[M\mu_0 + \frac{M\|B\|\bar{c}}{1-\alpha} + M\Big(\frac{\operatorname{tr}Q}{1-\alpha^2}\Big)^{1/2}\Big],$$
$$\bar{X}_\perp = \min\Big\{\|Z\|\bar{X},\ M_\perp\|Z\|\mu_0 + \frac{M_\perp}{1-\alpha_\perp}\big[(\|A_{\perp y}\| + \kappa\|B_\perp\|)\|C\|\bar{X} + \|B_\perp\|\bar{c}\big] + M_\perp\Big(\frac{\operatorname{tr}Q^\perp}{1-\alpha_\perp^2}\Big)^{1/2}\Big\}, \tag{S124}$$

and the claim is $\sup_t\|x_t\|_{L^2} \le \bar{X}$ and $\sup_t\|x_{\perp,t}\|_{L^2} \le \bar{X}_\perp$ for every $\pi \in \mathcal{P}(\kappa, \bar{c})$. The first entry of the minimum does not use the hypothesis on $A_{\perp\perp}$, which can therefore be dropped with $\bar{X}_\perp = \|Z\|\bar{X}$.

Let $a_t = \|x_t\|_{L^2}$. From $x_t = A^t x_0 + \sum_{s<t} A^{t-1-s}(Bu_s + \mathfrak{n}_s)$, the bound $|u_s| \le \kappa\|C\||x_s| + \bar{c}$, the independence of the noise terms and $\operatorname{tr}(A^jQA^{j\top}) \le \|A^j\|^2\operatorname{tr}Q$,

$$a_t \le M\mu_0 + \frac{M\|B\|}{1-\alpha}\Big(\kappa\|C\|\max_{s<t}a_s + \bar{c}\Big) + M\Big(\frac{\operatorname{tr}Q}{1-\alpha^2}\Big)^{1/2} = (1-g_\kappa)\bar{X} + g_\kappa\max_{s<t}a_s.$$

Since $M \ge 1$ and $g_\kappa < 1$, $a_0 \le \mu_0 \le \bar{X}$, and induction gives $a_t \le \bar{X}$ for all $t$. Hence $\|x_{\perp,t}\|_{L^2} \le \|Z\|\bar{X}$. The second entry of the minimum follows in the same way from $x_{\perp,t+1} = A_{\perp\perp}x_{\perp,t} + A_{\perp y}y_t + B_\perp u_t + Z\mathfrak{n}_t$, with $\|y_s\|_{L^2} \le \|C\|\bar{X}$ and $\|u_s\|_{L^2} \le \kappa\|C\|\bar{X} + \bar{c}$.

Since $x = C^\sharp y + Z^\sharp x_\perp$, $Dx = (A_{yy} - \bar{A})y + A_{y\perp}x_\perp$, so $\|\delta_t\|_{L^2} \le \|A_{yy} - \bar{A}\|\|C\|\bar{X} + \|A_{y\perp}\|\bar{X}_\perp + \|E\|(\kappa\|C\|\bar{X} + \bar{c}) = \bar{\mathfrak{d}}$. Inserting $\mathbb{E}^\pi\|\delta_t\|^2_{\bar{Q}^{-1}} \le \bar{\mathfrak{d}}^2/\lambda_{\min}(\bar{Q})$ and $\mathfrak{d}_{\mathrm{dyn},t} \ge 0$ in Eq. (S58) gives Eq. (S59). □

*Remark* S39.2 (The small-gain condition). Let $n = 2$, $A = \mathsf{a}I$ with $0 < \mathsf{a} < 1$, $Q = I$, $C = (1 \;\; 0)$, $B = (1 \;\; 0)^\top$, $\bar{A} = \mathsf{a}$, $\bar{B} = \frac{1}{2}$ and $u_t = \kappa y_t$ with $\kappa > 0$. Then $D = 0$, $E = \frac{1}{2}$, $\delta_t = \frac{1}{2}\kappa y_t$, $M = 1$, $\alpha = \mathsf{a}$ and $g_\kappa = \kappa/(1-\mathsf{a})$. The closed loop is $y_{t+1} = (\mathsf{a}+\kappa)y_t + \mathfrak{n}_{1,t}$, so $\mathbb{E}\delta_t^2 = \frac{1}{4}\kappa^2[(\mathsf{a}+\kappa)^{2t}\mathbb{E}y_0^2 + \sum_{j<t}(\mathsf{a}+\kappa)^{2j}]$, which is bounded in $t$ exactly when $g_\kappa < 1$ and grows like $(\mathsf{a}+\kappa)^{2t}$ when $g_\kappa > 1$. In this family the condition $g_\kappa < 1$ of Corollary S18.3 is therefore necessary for a bound linear in $T$. In general it is sufficient and not necessary, since it ignores the sign of the feedback.

*Proof of Proposition S18.4.* With the stated scaling, $D = CA - \bar{A}C = \mathsf{h}D_{\mathrm{c}}$, $E = \mathsf{h}E_{\mathrm{c}}$, $Q^y = \mathsf{h}\mathsf{q}^y$ and $\bar{Q}^{-1}Q^y = \bar{\mathsf{q}}^{-1}\mathsf{q}^y$, and the horizons are the integers $T = \tau/\mathsf{h}$. Because $CP_t = 0$, $CAP_tA^\top C^\top = \mathsf{h}^2 CFP_tF^\top C^\top$, so that $\bar{Q}^{-1}S_t = \bar{\mathsf{q}}^{-1}(\mathsf{q}^y + \mathsf{h}CFP_tF^\top C^\top)$.

Lower bound. By the second form of the summand in Theorem S18.2, $D_{\mathrm{KL}}(P^\pi\|\hat{P}^\pi) \geq \sum_{t<T} h(\bar{Q}^{-1}S_t)$ for every causal coarse policy. By Eq. (S122) and the monotonicity of $\Pi_C$, $P_t$ is at most the covariance of the uncontrolled state, so for $t \leq T$

$$\|P_t\| \leq \|A\|^{2t}\|\Sigma_0\| + \mathsf{h}\|\mathsf{q}\| \sum_{j<t} \|A\|^{2j} \leq e^{2\|F\|\tau}(\|\Sigma_0\| + \tau\|\mathsf{q}\|),$$

because $\|A\| \leq e^{\mathsf{h}\|F\|}$. Hence $\bar{Q}^{-1}S_t \to \bar{\mathsf{q}}^{-1}\mathsf{q}^y$ uniformly in $t \leq T$ as $\mathsf{h} \to 0$, and since $h$ is continuous on matrices with positive determinant, $\mathsf{h}\sum_{t<T} h(\bar{Q}^{-1}S_t) \to \tau h(\bar{\mathsf{q}}^{-1}\mathsf{q}^y)$. The limit is positive when $\bar{\mathsf{q}} \neq \mathsf{q}^y$, because $h(M) = \frac{1}{2}\sum_i g(\lambda_i)$ over the eigenvalues of $M$, which are positive here, and $g$ vanishes only at 1. The bound holds uniformly over the causal coarse policies, which may depend on $\mathsf{h}$.

Upper bound. When $\bar{\mathsf{q}} = \mathsf{q}^y$ the first term of the summand in Eq. (S58) vanishes, and $\mathfrak{d}_{\mathrm{dyn},t} \geq 0$, so the summand is at most $\frac{1}{2}\mathbb{E}^\pi\|\mathsf{h}D_{\mathrm{c}}x_t + \mathsf{h}E_{\mathrm{c}}u_t\|^2_{(\mathsf{h}\mathsf{q}^y)^{-1}} = \frac{\mathsf{h}}{2}\mathbb{E}^\pi\|D_{\mathrm{c}}x_t + E_{\mathrm{c}}u_t\|^2_{(\mathsf{q}^y)^{-1}}$. For the boundedness let $a_t = \|x_t\|_{L^2}$, $\mathsf{c}_1 = \|F\| + \kappa\|G\|\|C\|$ and $\mathsf{h} \leq 1$. Since $\mathfrak{n}_t$ is independent of $(x_t, u_t)$ and $|u_t| \leq \kappa\|C\||x_t| + \bar{c}$,

$$a_{t+1}^2 \leq \big((1 + \mathsf{h}\mathsf{c}_1)a_t + \mathsf{h}\|G\|\bar{c}\big)^2 + \mathsf{h}\,\mathrm{tr}\,\mathsf{q} \leq (1+\mathsf{h})(1+\mathsf{h}\mathsf{c}_1)^2 a_t^2 + \mathsf{h}\big[(1+\mathsf{h})\|G\|^2\bar{c}^2 + \mathrm{tr}\,\mathsf{q}\big],$$

by $(p+q)^2 \leq (1+\mathsf{h})p^2 + (1+\mathsf{h}^{-1})q^2$. With $(1+\mathsf{h})(1+\mathsf{h}\mathsf{c}_1)^2 \leq e^{\mathsf{h}(1+2\mathsf{c}_1)}$ this gives, for all $t \leq T$,

$$a_t^2 \leq \bar{X}_\tau^2 := e^{(1+2\mathsf{c}_1)\tau}(\mu_0^2 + \tau[2\|G\|^2\bar{c}^2 + \mathrm{tr}\,\mathsf{q}]),$$

and therefore

$$D_{\mathrm{KL}}(P^\pi\|\hat{P}^\pi) - D_{\mathrm{KL}}(P_{y_0}\|\hat{P}_0) \leq \frac{\tau}{2\lambda_{\min}(\mathsf{q}^y)}\Big(\|D_{\mathrm{c}}\|\bar{X}_\tau + \|E_{\mathrm{c}}\|(\kappa\|C\|\bar{X}_\tau + \bar{c})\Big)^2$$

for every $\pi \in \mathcal{P}(\kappa, \bar{c})$ and every $\mathsf{h} \leq 1$, a bound that does not depend on $\mathsf{h}$. No stability or small-gain hypothesis is used on a fixed physical horizon. □

*Proof of Proposition S18.5.* Here $C = (1 \;\; 0)$, $Z = (0 \;\; 1)$, $A_{yy} = \mathsf{b}$, $A_{y\perp} = \mathsf{g}$, $A_{\perp y} = 0$ and $A_{\perp\perp} = \mathsf{a}$, so $\Gamma_l = A_{y\perp}A_{\perp\perp}^{l-1}A_{\perp y} = 0$ for every $l \geq 1$, and the orthogonal force of Theorem 5.2 is $\mathfrak{f}_{t+1} = \mathsf{g}x_{2,t} + \mathfrak{n}_{1,t}$, of stationary variance $1 + \mathsf{g}^2V$ and of covariance $\mathsf{g}^2\mathsf{a}^jV$ at lag $j \geq 1$. Truncating the kernel at any order and replacing $\mathfrak{f}$ by white Gaussian noise of variance $\sigma^2 > 0$ gives the declared model $\bar{A} = \mathsf{b}$, $\bar{Q} = \sigma^2$, without input, for which $D = CA - \bar{A}C = (0 \;\; \mathsf{g})$.

Let $x_0 \sim \mathcal{N}(0, \Sigma_0)$ with $\mathrm{Var}(x_{2,0}) = V$, so that $x_{2,t}$ is stationary with mean zero. By Lemma S39.1(a), $P_t$ has the single nonzero entry $P_t^\perp = \mathrm{Var}(x_{2,t} \mid y_{0:t})$, and Eq. (S122) reads

$$P_{t+1}^\perp = f(P_t^\perp), \qquad f(P) = 1 + \frac{\mathsf{a}^2 P}{1 + \mathsf{g}^2 P},$$

while $S_t = 1 + \mathsf{g}^2 P_t^{\perp}$. The map $f$ is nondecreasing, concave and bounded on $[0,\infty)$ with $f(0) = 1$, so $f(P) - P$ has a single zero $P_\infty > 0$, is positive below it and negative above it, and $P_t^{\perp}$ converges monotonically to $P_\infty$. Since $V = \mathsf{a}^2 V + 1$, $f(V) - V = -\mathsf{a}^2\mathsf{g}^2 V^2/(1+\mathsf{g}^2 V)$, which is negative when $\mathsf{a} \neq 0$ and zero when $\mathsf{a} = 0$; hence $P_\infty < V$ if $\mathsf{a} \neq 0$ and $P_\infty = V = 1$ if $\mathsf{a} = 0$.

By the second form of the summand in Theorem S18.2 and $\mathbb{E}\hat{x}_{2,t}^2 = V - P_t^{\perp}$, the summand at time $t$ is

$$h(S_t/\sigma^2) + \frac{\mathsf{g}^2(V - P_t^{\perp})}{2\sigma^2} = \frac{1}{2}\Big[\frac{1+\mathsf{g}^2 V}{\sigma^2} - 1 - \ln\frac{1+\mathsf{g}^2 P_t^{\perp}}{\sigma^2}\Big].$$

The initial term of Eq. (S58) is finite, so the rate $\lim_{T\to\infty} T^{-1} D_{\mathrm{KL}}(P\|\hat{P})$ exists and equals the right side with $P_\infty$ in place of $P_t^{\perp}$. As a function of $\sigma^2$ this is minimal at $\sigma^2 = 1 + \mathsf{g}^2 V$, where it equals $\frac{1}{2}\ln((1+\mathsf{g}^2 V)/(1+\mathsf{g}^2 P_\infty))$. This is positive when $\mathsf{a} \neq 0$. When $\mathsf{a} = 0$ the force $\mathsf{f}$ is white, the coarse process is a first-order autoregression and the rate at $\sigma^2 = 1 + \mathsf{g}^2$ is zero. □

*Remark* S39.3 (Scope of Proposition S18.5). The kernels of Theorem 5.2 depend on the complement $Z$. The proposition concerns the complement $x_\perp = x_2$, for which $A_{\perp y} = 0$. For the complement of part (a) of that theorem, chosen so that $x_\perp$ and $y$ are uncorrelated in the stationary law, the kernels of this example do not vanish in general, and the truncation of those kernels is not analyzed here.

The filter of the unresolved coordinate, used in Theorem S18.6, is the following. Let $\hat{\xi}_t = Z\hat{x}_t = \mathbb{E}[x_{\perp,t} \mid \mathcal{G}_t]$ and $P_t^{\perp} = ZP_tZ^{\top}$. Since $CP_t = 0$, $P_t = Z^{\sharp}P_t^{\perp}Z^{\sharp\top}$, so $S_t = Q^y + A_{y\perp}P_t^{\perp}A_{y\perp}^{\top}$ and $CA\hat{x}_t = A_{yy}y_t + A_{y\perp}\hat{\xi}_t$. Given $\mathcal{G}_t$, the pair $(x_{\perp,t+1}, y_{t+1})$ is Gaussian with cross-covariance $A_{\perp\perp}P_t^{\perp}A_{y\perp}^{\top} + Q^{\perp y}$, and conditioning on $y_{t+1}$ gives, with

$$G_t = (A_{\perp\perp}P_t^{\perp}A_{y\perp}^{\top} + Q^{\perp y})S_t^{-1}, \qquad \Phi_t = A_{\perp\perp} - G_t A_{y\perp},$$

the recursion

$$\hat{\xi}_{t+1} = \Phi_t\hat{\xi}_t + A_{\perp y}y_t + B_\perp u_t + G_t(y_{t+1} - A_{yy}y_t - B_y u_t). \tag{S125}$$

The error $\epsilon_t = x_{\perp,t} - \hat{\xi}_t$ obeys $\epsilon_{t+1} = \Phi_t\epsilon_t + Z\mathfrak{n}_t - G_tC\mathfrak{n}_t$, with $\mathfrak{n}_t$ independent of $\epsilon_t$, so

$$P_{t+1}^{\perp} = \Phi_t P_t^{\perp}\Phi_t^{\top} + \mathrm{Cov}(Z\mathfrak{n}_t - G_tC\mathfrak{n}_t) \succeq \Phi_t P_t^{\perp}\Phi_t^{\top} + Q^{\perp|y}, \tag{S126}$$

because $Q^{\perp} - GQ^{y\perp} - Q^{\perp y}G^{\top} + GQ^yG^{\top} \succeq Q^{\perp|y}$ for every matrix $G$. The order-$d$ model is defined for $t \geq d$: $\hat{\xi}_t^{(d)}$ is the value at time $t$ of the recursion Eq. (S125), with the exact $\Phi_s$ and $G_s$, started at time $t-d$ from the value 0. It is a function of $(y_{t-d:t}, u_{t-d:t-1})$, and

$$\hat{\xi}_t - \hat{\xi}_t^{(d)} = \Psi_{t,d}\hat{\xi}_{t-d}, \qquad \Psi_{t,d} = \Phi_{t-1}\Phi_{t-2}\cdots\Phi_{t-d}, \qquad \Psi_{t,0} = I.$$

*Proof of Theorem S18.6.* The kernel of the coarse law is $\mathcal{N}(A_{yy}y_t + B_y u_t + A_{y\perp}\hat{\xi}_t, S_t)$ and that of the order-$d$ model is $\mathcal{N}(A_{yy}y_t + B_y u_t + A_{y\perp}\hat{\xi}_t^{(d)}, S_t)$, so their expected relative entropy is $\frac{1}{2}\mathbb{E}^{\pi}\|A_{y\perp}\Psi_{t,d}\hat{\xi}_{t-d}\|^2_{S_t^{-1}}$.

Since $Q \succ 0$, $Q^{\perp|y} \succ 0$ and $\mathsf{q}_\perp > 0$. By Eq. (S126), $P_s^{\perp} \succeq \mathsf{q}_\perp I$ for every $s \geq 1$, and for every $s \geq 0$, using $P_{s+1}^{\perp} \preceq \bar{p}I$,

$$\Phi_s P_s^{\perp}\Phi_s^{\top} \preceq P_{s+1}^{\perp} - \mathsf{q}_\perp I \preceq \Big(1 - \frac{\mathsf{q}_\perp}{\bar{p}}\Big)P_{s+1}^{\perp} = \theta P_{s+1}^{\perp},$$

with $\theta \in [0,1)$ because $\bar{p} \geq \|P_1^{\perp}\| \geq \mathsf{q}_\perp$. Iterating over $s = t-d, \ldots, t-1$ gives $\Psi_{t,d}P_{t-d}^{\perp}\Psi_{t,d}^{\top} \preceq \theta^d P_t^{\perp}$. Next, $\mathbb{E}^{\pi}[\hat{\xi}_s\hat{\xi}_s^{\top}] \preceq \mathbb{E}^{\pi}[x_{\perp,s}x_{\perp,s}^{\top}] \preceq \bar{X}_\perp^2 I$, and for $s = t-d \geq 1$ the right side is at most $(\bar{X}_\perp^2/\mathsf{q}_\perp)P_{t-d}^{\perp}$. Therefore

$$\mathbb{E}^{\pi}\|A_{y\perp}\Psi_{t,d}\hat{\xi}_{t-d}\|^2_{S_t^{-1}} \leq \frac{\bar{X}_\perp^2}{\mathsf{q}_\perp}\,\mathrm{tr}(S_t^{-1}A_{y\perp}\Psi_{t,d}P_{t-d}^{\perp}\Psi_{t,d}^{\top}A_{y\perp}^{\top}) \leq \theta^d\frac{\bar{X}_\perp^2}{\mathsf{q}_\perp}\,\mathrm{tr}(S_t^{-1}A_{y\perp}P_t^{\perp}A_{y\perp}^{\top}).$$

If $\lambda_i \geq 0$ are the eigenvalues of $(Q^y)^{-1/2} A_{y\perp} P_t^{\perp} A_{y\perp}^{\top} (Q^y)^{-1/2}$, the last trace is $\sum_i \lambda_i/(1+\lambda_i) \leq \sum_i \ln(1+\lambda_i) = 2\mathfrak{d}_{\mathrm{dyn},t}$, which gives Eq. (S61).

The restriction $t \geq d+1$ is needed. At $t = d$ the window starts at time 0, where $P_0^{\perp} = Z\Pi_C(\Sigma_0)Z^{\top}$ can be arbitrarily small while $\hat{\xi}_0$ is not, and the bound then fails (Check S47.168). If $A_{\perp\perp} = 0$ then $\Phi_t = -Q^{\perp y} S_t^{-1} A_{y\perp}$, which is nonzero when $Q^{\perp y} S_t^{-1} A_{y\perp} \neq 0$. The predictor of the coarse path can then depend on the entire past although every kernel $\Gamma_l$ with $l \geq 2$ vanishes; it does so in the scalar instance of Check S47.168, where the stationary value of $\Phi_t$ is a nonzero number. □

**Lemma S39.4** (Comparison for the filter equation)**.** *Let $\mathsf{F}$ be a square matrix, $\mathsf{J}$ a matrix with as many columns, $R \succ 0$, and for $P \succeq 0$ let $\varphi(P) = P - P\mathsf{J}^{\top}(\mathsf{J}P\mathsf{J}^{\top} + R)^{-1}\mathsf{J}P$ and $\mathcal{T}_N(P) = \mathsf{F}\varphi(P)\mathsf{F}^{\top} + N$; write $\varphi = \varphi_R$ when the dependence on $R$ matters.*

*(a) $\varphi(P) = \min_K[(I-K\mathsf{J})P(I-K\mathsf{J})^{\top} + KRK^{\top}]$ and $\mathsf{F}\varphi(P)\mathsf{F}^{\top} = \min_K[(\mathsf{F}-K\mathsf{J})P(\mathsf{F}-K\mathsf{J})^{\top} + KRK^{\top}]$ in the order of symmetric matrices. Hence $0 \preceq \varphi(P) \preceq P$, and $\varphi_R(P)$ is nondecreasing in $P$ and in $R$.*

*(b) $\varphi_R(\mu P) = \mu\varphi_{R/\mu}(P)$ for $\mu > 0$; hence $\varphi_R(\mu P) \preceq \mu\varphi_R(P)$ for $\mu \geq 1$ and $\varphi_R(\mu P) \succeq \mu\varphi_R(P)$ for $0 < \mu \leq 1$.*

*(c) If $X_i \succeq 0$ solve $X_i = \mathcal{T}_{N_i}(X_i)$ for $i = 1, 2$, with $N_2 \succ 0$ and $N_1 \preceq cN_2$ for a constant $c \geq 1$, then $X_1 \preceq cX_2$. In particular, for $N \succ 0$ the equation $X = \mathcal{T}_N(X)$ has at most one solution $X \succeq 0$.*

*(d) If $(\mathsf{F}, \mathsf{J})$ is detectable and $N \succ 0$, the solution $X \succeq 0$ of $X = \mathcal{T}_N(X)$ exists, $X \succeq N$, and with $L = X\mathsf{J}^{\top}(\mathsf{J}X\mathsf{J}^{\top} + R)^{-1} = \varphi(X)\mathsf{J}^{\top}R^{-1}$ the matrices $\mathsf{F}(I-L\mathsf{J})$ and $(I-L\mathsf{J})\mathsf{F}$ have spectral radius less than 1. If $\rho(\mathsf{F}) < 1$, then $X \preceq Y$ for the solution $Y$ of $Y = \mathsf{F}Y\mathsf{F}^{\top} + N$.*

*Proof.* (a) With $\Sigma = \mathsf{J}P\mathsf{J}^{\top} + R \succ 0$ and $K_\star = P\mathsf{J}^{\top}\Sigma^{-1}$, expanding the square gives $(I-K\mathsf{J})P(I-K\mathsf{J})^{\top} + KRK^{\top} = \varphi(P) + (K-K_\star)\Sigma(K-K_\star)^{\top}$, and in the same way $(\mathsf{F}-K\mathsf{J})P(\mathsf{F}-K\mathsf{J})^{\top} + KRK^{\top} = \mathsf{F}\varphi(P)\mathsf{F}^{\top} + (K-\mathsf{F}K_\star)\Sigma(K-\mathsf{F}K_\star)^{\top}$. For fixed $K$ the left sides are nondecreasing in $P$ and in $R$, and a minimum of nondecreasing maps that is attained is nondecreasing. Taking $K = 0$ gives $\varphi(P) \preceq P$.

(b) $\varphi_R(\mu P) = \mu P - \mu^2 P\mathsf{J}^{\top}(\mu\mathsf{J}P\mathsf{J}^{\top} + R)^{-1}\mathsf{J}P = \mu[P - P\mathsf{J}^{\top}(\mathsf{J}P\mathsf{J}^{\top} + R/\mu)^{-1}\mathsf{J}P]$, and (a) compares $\varphi_{R/\mu}$ with $\varphi_R$.

(c) From $X_2 = \mathcal{T}_{N_2}(X_2) \succeq N_2 \succ 0$, the set of $\lambda \geq 1$ with $X_1 \preceq \lambda c X_2$ is nonempty and closed; let $\lambda_\star$ be its least element and suppose $\lambda_\star > 1$. By (a) and (b) with $\mu = \lambda_\star c \geq 1$,

$$X_1 = \mathcal{T}_{N_1}(X_1) \preceq \mathsf{F}\varphi(\lambda_\star c X_2)\mathsf{F}^{\top} + N_1 \preceq \lambda_\star c\mathsf{F}\varphi(X_2)\mathsf{F}^{\top} + cN_2 = \lambda_\star c X_2 - (\lambda_\star - 1)cN_2.$$

With $\nu = \lambda_{\min}(N_2)/\|X_2\| \in (0, 1]$ one has $N_2 \succeq \nu X_2$, so $X_1 \preceq (\lambda_\star - (\lambda_\star - 1)\nu)cX_2$ with $1 \leq \lambda_\star - (\lambda_\star - 1)\nu < \lambda_\star$, which contradicts the choice of $\lambda_\star$. Hence $\lambda_\star = 1$. Uniqueness follows by applying this twice with $N_1 = N_2 = N$ and $c = 1$.

(d) Let $K_0$ be such that $\rho(\mathsf{F} - K_0\mathsf{J}) < 1$ and let $Y_0 \succeq 0$ solve $Y_0 = (\mathsf{F}-K_0\mathsf{J})Y_0(\mathsf{F}-K_0\mathsf{J})^{\top} + K_0RK_0^{\top} + N$. The iterates $X^{(0)} = 0$, $X^{(j+1)} = \mathcal{T}_N(X^{(j)})$ are nondecreasing, because $X^{(1)} = N \succeq X^{(0)}$ and $\mathcal{T}_N$ is nondecreasing, and they satisfy $X^{(j)} \preceq Y_0$ by induction, because (a) gives $\mathcal{T}_N(P) \preceq (\mathsf{F}-K_0\mathsf{J})P(\mathsf{F}-K_0\mathsf{J})^{\top} + K_0RK_0^{\top} + N$. They converge, and the limit solves $X = \mathcal{T}_N(X)$ by continuity of $\varphi$. If $\rho(\mathsf{F}) < 1$ one may take $K_0 = 0$, which gives $X \preceq Y$. By (a) with $K = \mathsf{F}L$, $X = \mathsf{F}(I-L\mathsf{J})X(I-L\mathsf{J})^{\top}\mathsf{F}^{\top} + \mathsf{F}LRL^{\top}\mathsf{F}^{\top} + N$. If $v^*\mathsf{F}(I-L\mathsf{J}) = \lambda v^*$ with $v \neq 0$, then $(1-|\lambda|^2)v^*Xv \geq v^*Nv > 0$, so $|\lambda| < 1$; and $(I-L\mathsf{J})\mathsf{F}$ has the same nonzero eigenvalues. Finally $\varphi(X)\mathsf{J}^{\top} = X\mathsf{J}^{\top}\Sigma^{-1}(\Sigma - \mathsf{J}X\mathsf{J}^{\top}) = X\mathsf{J}^{\top}\Sigma^{-1}R$ with $\Sigma = \mathsf{J}X\mathsf{J}^{\top} + R$. □

The fine filter of Lemma S39.1(b) is the case $\mathsf{F} = A$, $\mathsf{J} = HC$, $N = Q$, and the declared filter is the case $\mathsf{F} = \bar{A}$, $\mathsf{J} = H$, $N = \bar{Q}$, for which $\varphi = \varphi_R$ of the main text. Since $\rho(A) < 1$, Lemma S39.4(d) gives a unique $\Sigma^- \succeq 0$ with $\Sigma^- = A\Sigma^{\mathrm{f}}A^\top + Q$, where $\Sigma^{\mathrm{f}}$ is obtained from $\Sigma^-$ by the middle formula of Eq. (S123), and $\Sigma^{\mathrm{f}} \preceq \Sigma^- \preceq \Sigma_x$. That the fine filter is stationary means $\Sigma_0 = \Sigma^-$, so that $\Sigma_t^- = \Sigma^-$ and $\Sigma_t^{\mathrm{f}} = \Sigma^{\mathrm{f}}$ for all $t$. Detectability of $(\bar{A}, H)$ gives $P_{\mathrm{b}}^-$, with $P_{\mathrm{b}}^- \succeq \bar{Q} \succ 0$, $P_{\mathrm{b}} = \varphi_R(P_{\mathrm{b}}^-) = ((P_{\mathrm{b}}^-)^{-1} + H^\top R^{-1} H)^{-1} \succ 0$, $L = P_{\mathrm{b}}^- H^\top (HP_{\mathrm{b}}^- H^\top + R)^{-1} = P_{\mathrm{b}} H^\top R^{-1}$, and $\rho(\Psi') = \rho(\Psi) < 1$ for $\Psi' = \bar{A}(I - LH)$ and $\Psi = (I - LH)\bar{A}$. The declared filter is $\hat{\eta}_{t+1}^- = \bar{A}(\hat{\eta}_t^- + L(s_{t+1} - H\hat{\eta}_t^-)) + \bar{B}u_t$, started from any $\hat{\eta}_0^-$ with $\mathbb{E}|C\hat{x}_0^- - \hat{\eta}_0^-|^2 < \infty$.

*Proof of Theorem S18.7.* Fixed-point identity. By Lemma S39.1(b) in the stationary regime, $P_{\mathrm{e}} = C\Sigma^- C^\top = CA\Sigma^{\mathrm{f}}A^\top C^\top + Q^y$. Inserting $CA = \bar{A}C + D$ gives $CA\Sigma^{\mathrm{f}}A^\top C^\top = \bar{A}C\Sigma^{\mathrm{f}}C^\top\bar{A}^\top + \Delta$, and $C\Sigma^{\mathrm{f}}C^\top = P_{\mathrm{e}} - P_{\mathrm{e}}H^\top(HP_{\mathrm{e}}H^\top + R)^{-1}HP_{\mathrm{e}} = \varphi_R(P_{\mathrm{e}})$. Hence

$$P_{\mathrm{e}} = \bar{A}\varphi_R(P_{\mathrm{e}})\bar{A}^\top + Q^y + \Delta, \tag{S127}$$

the filter equation of the declared model with $\bar{Q}$ replaced by $Q^y + \Delta$, which is symmetric and satisfies $Q^y + \Delta \succeq c_-\bar{Q} \succ 0$.

Sandwich. Lemma S39.4(c) with $X_1 = P_{\mathrm{e}}$, $N_1 = Q^y + \Delta$, $X_2 = P_{\mathrm{b}}^-$, $N_2 = \bar{Q}$ and $c = c_+$ gives $P_{\mathrm{e}} \preceq c_+ P_{\mathrm{b}}^-$. With $X_1 = P_{\mathrm{b}}^-$, $N_1 = \bar{Q}$, $X_2 = P_{\mathrm{e}}$, $N_2 = Q^y + \Delta$ and $c = 1/c_- \geq 1$ it gives $P_{\mathrm{b}}^- \preceq P_{\mathrm{e}}/c_-$. This step needs $c_- > 0$.

Decomposition. Let $e_t = C\hat{x}_t^- - \hat{\eta}_t^-$. Both laws in $\Delta\mathcal{F}_t$ are Gaussian, so

$$\Delta\mathcal{F}_t = h\big((P_{\mathrm{b}}^-)^{-1}P_{\mathrm{e}}\big) + \tfrac{1}{2}\mathbb{E}\|e_t\|^2_{(P_{\mathrm{b}}^-)^{-1}}. \tag{S128}$$

The eigenvalues of $(P_{\mathrm{b}}^-)^{-1}P_{\mathrm{e}}$ lie in $[c_-, c_+]$ and $g$ is decreasing on $(0,1]$ and increasing on $[1,\infty)$, so the first term is at most $\frac{k}{2}\max\{g(c_+), g(c_-)\}$.

Error recursion. The innovation $\iota_t = s_{t+1} - HC\hat{x}_t^- = HC(x_t - \hat{x}_t^-) + v_t$ is, given $(s_{1:t}, \omega)$, Gaussian with mean zero and covariance $\Sigma_\iota = HP_{\mathrm{e}}H^\top + R$; the $\iota_t$ are therefore independent and identically distributed. With $L_{\mathrm{e}} = P_{\mathrm{e}}H^\top\Sigma_\iota^{-1}$, $C\hat{x}_t^{\mathrm{f}} = C\hat{x}_t^- + L_{\mathrm{e}}\iota_t$, and $C\hat{x}_{t+1}^- = CA\hat{x}_t^{\mathrm{f}} + CBu_t = \bar{A}C\hat{x}_t^{\mathrm{f}} + \bar{B}u_t + d_t$ with $d_t = D\hat{x}_t^{\mathrm{f}} + Eu_t$. Since $s_{t+1} - H\hat{\eta}_t^- = \iota_t + He_t$, subtracting the declared filter gives

$$e_{t+1} = \Psi' e_t + \bar{A}(L_{\mathrm{e}} - L)\iota_t + d_t. \tag{S129}$$

The act $u_t$ is measurable in $(s_{1:t}, \omega)$, so $d_t = \mathbb{E}[\delta_t \mid s_{1:t+1}, \omega]$ and $\|d_t\|_{L^2} \leq \|\delta_t\|_{L^2}$.

Gain difference. By Lemma S39.4(d) applied to Eq. (S127), $L_{\mathrm{e}} = \varphi_R(P_{\mathrm{e}})H^\top R^{-1}$, so $L_{\mathrm{e}} - L = WH^\top R^{-1}$ with $W = \varphi_R(P_{\mathrm{e}}) - \varphi_R(P_{\mathrm{b}}^-)$. By the sandwich and Lemma S39.4(a),(b), $c_- P_{\mathrm{b}} \preceq \varphi_R(P_{\mathrm{e}}) \preceq c_+ P_{\mathrm{b}}$, hence $-\mathfrak{c}P_{\mathrm{b}} \preceq W \preceq \mathfrak{c}P_{\mathrm{b}}$ and $\|W\| \leq \mathfrak{c}\|P_{\mathrm{b}}\|$. Since $\Sigma_\iota \preceq c_+ HP_{\mathrm{b}}^- H^\top + R$,

$$\operatorname{tr}\big((L_{\mathrm{e}} - L)\Sigma_\iota(L_{\mathrm{e}} - L)^\top\big) = \operatorname{tr}\big(WH^\top R^{-1}\Sigma_\iota R^{-1}HW\big) \leq k\mathfrak{c}^2\|P_{\mathrm{b}}\|^2\mathfrak{g}^2.$$

Conclusion. Solving Eq. (S129) and using the independence of the innovations,

$$\|e_t\|_{L^2} \leq \|\Psi'^t\|\|e_0\|_{L^2} + \sum_{s<t}\|\Psi'^{t-1-s}\|\|\delta_s\|_{L^2} + \Big(\sum_{j<t}\|\Psi'^j\bar{A}\|^2\Big)^{1/2}\sqrt{k}\mathfrak{c}\|P_{\mathrm{b}}\|\mathfrak{g}.$$

Since $\rho(\Psi') < 1$, the sums $\mathcal{S}'$ and $\mathcal{S}'_2$ are finite, the first term tends to zero, and

$$\limsup_t \sum_{s<t}\|\Psi'^{t-1-s}\|\|\delta_s\|_{L^2} \leq \mathcal{S}' \limsup_s \|\delta_s\|_{L^2} \leq \mathcal{S}'\bar{\mathfrak{d}}.$$

With $\|e_t\|^2_{(P_{\mathrm{b}}^-)^{-1}} \leq |e_t|^2/\lambda_{\min}(P_{\mathrm{b}}^-)$ and Eq. (S128) this is Eq. (S62).

A priori constants. Since $\Sigma^{\mathrm{f}} \preceq \Sigma_x$, $\|D\Sigma^{\mathrm{f}}D^\top\| \le \sigma_D^2$ and $\|C\Sigma^{\mathrm{f}}D^\top\| \le \|C\Sigma^{\mathrm{f}}C^\top\|^{1/2}\|D\Sigma^{\mathrm{f}}D^\top\|^{1/2} \le \varsigma_y\sigma_D$, so $\|\Delta\| \le \vartheta$ and $Q^y - \vartheta I \preceq Q^y + \Delta \preceq Q^y + \vartheta I$. With $\vartheta I \preceq (\vartheta/\lambda_{\min}(\bar{Q}))\bar{Q}$ the constants of the statement satisfy $c_-\bar{Q} \preceq Q^y + \Delta \preceq c_+\bar{Q}$; they are admissible when the resulting $c_-$ is positive, and the theorem gives nothing otherwise. □

**Proposition S39.5** (Filtered law)**.** *Under the hypotheses of Theorem S18.7, the exact law of $y_t$ given $(s_{1:t+1}, \omega)$ is $\mathcal{N}(C\hat{x}_t^{\mathrm{f}}, \varphi_R(P_{\mathrm{e}}))$, the declared filter holds $\mathcal{N}(\hat{\eta}_t^{\mathrm{f}}, P_{\mathrm{b}})$ with $\hat{\eta}_t^{\mathrm{f}} = \hat{\eta}_t^- + L(s_{t+1} - H\hat{\eta}_t^-)$, one has $c_-P_{\mathrm{b}} \preceq \varphi_R(P_{\mathrm{e}}) \preceq c_+P_{\mathrm{b}}$, and*

$$\limsup_{t\to\infty} \mathbb{E} D_{\mathrm{KL}}\Big(\mathcal{N}(C\hat{x}_t^{\mathrm{f}}, \varphi_R(P_{\mathrm{e}}))\Big\|\mathcal{N}(\hat{\eta}_t^{\mathrm{f}}, P_{\mathrm{b}})\Big) \le \frac{k}{2}\max\{g(c_+), g(c_-)\} + \frac{\big(\mathcal{S}^{\mathrm{f}}\bar{\mathfrak{d}} + \mathfrak{c}\|P_{\mathrm{b}}\|\mathfrak{g}\sqrt{k}\mathcal{S}_2^{\mathrm{f}}\big)^2}{2\lambda_{\min}(P_{\mathrm{b}})}, \tag{S130}$$

*with $\mathcal{S}^{\mathrm{f}} = \sum_{j\ge0}\|\Psi^j(I - LH)\|$ and $\mathcal{S}_2^{\mathrm{f}} = (\sum_{j\ge0}\|\Psi^j\|^2)^{1/2}$.*

*Proof.* The covariance statement is the sandwich for $\varphi_R(P_{\mathrm{e}})$ from the previous proof. Let $e_t^{\mathrm{f}} = C\hat{x}_t^{\mathrm{f}} - \hat{\eta}_t^{\mathrm{f}} = (I - LH)e_t + (L_{\mathrm{e}} - L)\iota_t$. By Eq. (S129),

$$e_{t+1}^{\mathrm{f}} = \Psi e_t^{\mathrm{f}} + (I - LH)d_t + (L_{\mathrm{e}} - L)\iota_{t+1},$$

so $e_t^{\mathrm{f}} = \Psi^t(I - LH)e_0 + \sum_{s<t}\Psi^{t-1-s}(I - LH)d_s + \sum_{s\le t}\Psi^{t-s}(L_{\mathrm{e}} - L)\iota_s$, and the three terms are bounded as before, with $\lambda_{\min}(P_{\mathrm{b}})$ in place of $\lambda_{\min}(P_{\mathrm{b}}^-)$ because the reference covariance is now $P_{\mathrm{b}}$. □

The constants of Eq. (S130) differ from those of Eq. (S62): $\lambda_{\min}(P_{\mathrm{b}}) \le \lambda_{\min}(P_{\mathrm{b}}^-)$, and the factor $I - LH$ cannot be removed from $\mathcal{S}^{\mathrm{f}}$ at no cost, since its spectral norm can exceed 1. Whether Eq. (S62) holds verbatim for the filtered law with $\Psi$ in place of $\Psi'$ was neither proved nor refuted.

*Proof of Corollary S18.8.* Without input, $B = 0$, $\bar{B} = 0$ and $\delta_t = Dx_t$. Write $\Sigma_y = C\Sigma_xC^\top$ and $\Sigma_{x|y} = \Pi_C(\Sigma_x)$. The moment-matched $\bar{A} = CA\Sigma_xC^\top\Sigma_y^{-1}$ gives $D\Sigma_xC^\top = CA\Sigma_xC^\top - \bar{A}\Sigma_y = 0$, and therefore

$$D\Sigma_xD^\top = CA\Sigma_xA^\top C^\top - CA\Sigma_xC^\top\Sigma_y^{-1}C\Sigma_xA^\top C^\top = CA\Sigma_{x|y}A^\top C^\top.$$

In the stationary law, $x_t$ given $y_t$ is Gaussian with covariance $\Sigma_{x|y}$ and $y_{t+1} = CAx_t + C\mathfrak{n}_t$, so the closure defect $\mathfrak{d}_{\mathrm{cl}} = I(y_{t+1}; x_t \mid y_t)$ of Definition 5.1(i), here without a blanket variable, is

$$\mathfrak{d}_{\mathrm{cl}} = \tfrac{1}{2}\ln\det(I + (Q^y)^{-1}D\Sigma_xD^\top),$$

the Gaussian form of Proposition S11.8(a) and Proposition 11.9(b). The largest eigenvalue of $(Q^y)^{-1/2}D\Sigma_xD^\top(Q^y)^{-1/2}$ is at least $\sigma_D^2/\|Q^y\|$, and the determinant is at least 1 plus that eigenvalue, so $e^{2\mathfrak{d}_{\mathrm{cl}}} \ge 1 + \sigma_D^2/\|Q^y\|$, which is the first claim.

Defect energy. The fine filter is stationary, so $\mathrm{Cov}(x_0) = \Sigma^- \preceq \Sigma_x$, and by induction $\mathrm{Cov}(x_t) \preceq \Sigma_x$ for all $t$, because $A\Sigma_xA^\top + Q = \Sigma_x$. The mean of $x_t$ is $A^tm_0$, which tends to zero. Hence $\limsup_t \mathbb{E}|Dx_t|^2 \le \mathrm{tr}(D\Sigma_xD^\top) \le k\sigma_D^2$, and $\bar{\mathfrak{d}} = \sqrt{k}\sigma_D$ is admissible in Theorem S18.7. If $m_0 = 0$ the bound holds for every $t$.

Detectability. Since $D\Sigma_xC^\top = 0$, $\Sigma_y = CA\Sigma_xA^\top C^\top + Q^y = \bar{A}\Sigma_y\bar{A}^\top + D\Sigma_xD^\top + Q^y$ with $Q^y \succ 0$, so $\rho(\bar{A}) < 1$ and $(\bar{A}, H)$ is detectable for every $H$.

Constants. With $\bar{Q} = Q^y$ the a priori constants of Theorem S18.7 are $c_\pm = 1 \pm \mathsf{x}$, which are admissible when $\mathsf{x} < 1$, and $\mathfrak{c} = \mathsf{x}$. From the power series, $g(1 + \mathsf{x}) = \mathsf{x} - \ln(1 + \mathsf{x}) \le \mathsf{x}^2/2$ and $g(1 - \mathsf{x}) = \sum_{j\ge2}\mathsf{x}^j/j \le \mathsf{x}^2/(2(1 - \mathsf{x}))$, so $\frac{k}{2}\max\{g(c_+), g(c_-)\} \le \frac{k}{4}\mathsf{x}^2/(1 - \mathsf{x})$. By Lemma S39.4(d), $\lambda_{\min}(P_{\mathrm{b}}^-) \ge \lambda_{\min}(Q^y)$. Inserting these in Eq. (S62) gives the stated bound.

Order. By the first claim $\sigma_D^2 = O(\mathfrak{d}_{\mathrm{cl}})$, hence $\vartheta = O(\mathfrak{d}_{\mathrm{cl}}^{1/2})$ and $\varkappa^2 = O(\mathfrak{d}_{\mathrm{cl}})$, and the bound is $O(\mathfrak{d}_{\mathrm{cl}})$ along every family of instances on which $\|\bar{A}\|$, $\varsigma_y$, $\|Q^y\|$, $\lambda_{\min}(Q^y)^{-1}$, $\mathcal{S}'$, $\mathcal{S}_2'$, $\|P_{\mathrm{b}}\|$ and $\mathfrak{g}$ stay bounded. Without such a restriction the constants can grow as the defect shrinks, and no order is claimed. □

**Proposition S39.6** (Finite horizon without stationarity)**.** *Let the meta-agent run the time-varying filter of the declared model from the declared initial law: $\hat{P}_0^- = \hat{\Sigma}_0$, $\hat{\eta}_0^- = \hat{m}_0$, $L_t = \hat{P}_t^- H^\top (H\hat{P}_t^- H^\top + R)^{-1}$, $\hat{P}_{t+1}^- = \bar{A}\varphi_R(\hat{P}_t^-)\bar{A}^\top + \bar{Q}$ and $\hat{\eta}_{t+1}^- = \bar{A}(\hat{\eta}_t^- + L_t(s_{t+1} - H\hat{\eta}_t^-)) + \bar{B}u_t$. Let the policy be $u_t = \pi_t(s_{1:t}, \omega)$ with $\mathbb{E}|u_t|^2 < \infty$. Without any hypothesis of stability, stationarity or detectability, for every $t \geq 0$,*

$$\mathbb{E} D_{\mathrm{KL}}\Big(P(y_t \in \cdot \mid s_{1:t}, \omega)\Big\|\mathcal{N}(\hat{\eta}_t^-, \hat{P}_t^-)\Big) \leq D_{\mathrm{KL}}(P^{\pi'}\|\hat{P}^{\pi'}), \tag{S131}$$

*where the right side is the path relative entropy of Theorem S18.2 at horizon $T = t$ for the causal coarse policy $\pi'_j(y_{0:j}, \omega') = \pi_j(Hy_0 + v_0, \ldots, Hy_{j-1} + v_{j-1}, \omega)$ with randomization $\omega' = (\omega, v_{0:t-1})$. At $t = 0$ the two sides are equal.*

*Proof.* Let $\mathbb{P}$ and $\hat{\mathbb{P}}$ be the laws of $(\omega, s_{1:t}, y_t)$ under the fine process and under the declared model read through the same channel and driven by the same maps. By Lemma S39.1(b) applied to the declared model, with $(A, B, Q, C)$ replaced by $(\bar{A}, \bar{B}, \bar{Q}, I)$, the conditional law of $y_t$ given $(\omega, s_{1:t})$ under $\hat{\mathbb{P}}$ is $\mathcal{N}(\hat{\eta}_t^-, \hat{P}_t^-)$. The chain rule gives

$$D_{\mathrm{KL}}(\mathbb{P}\|\hat{\mathbb{P}}) = D_{\mathrm{KL}}(\mathbb{P}_{(\omega,s_{1:t})}\|\hat{\mathbb{P}}_{(\omega,s_{1:t})}) + \mathbb{E} D_{\mathrm{KL}}\Big(P(y_t \in \cdot \mid s_{1:t}, \omega)\Big\|\mathcal{N}(\hat{\eta}_t^-, \hat{P}_t^-)\Big),$$

and the first term on the right is nonnegative. The vector $(\omega, s_{1:t}, y_t)$ is the same measurable function of $(\omega', y_{0:t})$ under both laws, $\omega'$ has the same law under both and is independent of $(x_0, \mathfrak{n})$ and of $(\hat{y}_0, \hat{w})$, and $\pi'$ is a causal coarse policy. Relative entropy does not increase under a measurable map, so $D_{\mathrm{KL}}(\mathbb{P}\|\hat{\mathbb{P}}) \leq D_{\mathrm{KL}}(P^{\pi'}\|\hat{P}^{\pi'})$. At $t = 0$ both sides of Eq. (S131) are $D_{\mathrm{KL}}(P_{y_0}\|\hat{P}_0)$. □

*Proof of Proposition S18.9.* Here $\Lambda^\star \succeq 0$ is symmetric, the offsets $c_t$ and the targets $o_t^\star$ are deterministic, $G_t = \frac{1}{2}\mathbb{E}^\pi\|y_t - o_t^\star\|^2_{\Lambda^\star}$ and $\hat{G}_t$ is the same expression under the declared model. Under the fine process $y_{t+1} = \bar{A}_K y_t + \bar{B}c_t + \delta_t + C\mathfrak{n}_t$. Let $\zeta_t = (Q^y)^{-1/2}C\mathfrak{n}_t$, which are independent standard normal vectors, and define on the same probability space $\hat{y}_{t+1} = \bar{A}_K\hat{y}_t + \bar{B}c_t + \bar{Q}^{1/2}\zeta_t$, where $\hat{y}_0$ is drawn from the conditional law of an optimal coupling of $(P_{y_0}, \hat{P}_0)$ given $y_0$, independently of everything else. Then $\hat{y}_0$ is independent of $(\zeta_t)$, so $\hat{y}$ has the law of the declared model under the same feedback, and $\|y_0 - \hat{y}_0\|_{L^2} = w_0$. The difference $\Delta_t = y_t - \hat{y}_t$ obeys $\Delta_{t+1} = \bar{A}_K\Delta_t + \delta_t + \Gamma\zeta_t$ with $\Gamma = (Q^y)^{1/2} - \bar{Q}^{1/2}$, hence

$$\Delta_t = \bar{A}_K^t\Delta_0 + \sum_{s<t}\bar{A}_K^{t-1-s}\delta_s + \sum_{s<t}\bar{A}_K^{t-1-s}\Gamma\zeta_s.$$

The last sum has squared $L^2$ norm $\sum_{s<t}\|\bar{A}_K^{t-1-s}\Gamma\|_{\mathrm{F}}^2 \leq \beta^2\sum_{j<t}\|\bar{A}_K^j\|^2$, and the triangle inequality in $L^2$ gives $W_2(P_{y_t}^\pi, \hat{P}_{y_t}^\pi) \leq \|\Delta_t\|_{L^2} \leq w_t$.

For the score, $G_t^{1/2}$ and $\hat{G}_t^{1/2}$ are the $L^2$ norms of $(\Lambda^\star/2)^{1/2}(y_t - o_t^\star)$ and $(\Lambda^\star/2)^{1/2}(\hat{y}_t - o_t^\star)$, so $|G_t^{1/2} - \hat{G}_t^{1/2}| \leq (\|\Lambda^\star\|/2)^{1/2}\|\Delta_t\|_{L^2} \leq a_t$ and $|G_t - \hat{G}_t| = |G_t^{1/2} - \hat{G}_t^{1/2}|(G_t^{1/2} + \hat{G}_t^{1/2}) \leq a_t(2\hat{G}_t^{1/2} + a_t)$. Summing over $t$ bounds $|G(\pi) - \hat{G}(\pi)|$.

For state feedback $u_t = k_t(y_t)$ with $|k_t(y) - k_t(y')| \leq \kappa|y - y'|$, the same coupling gives $\Delta_{t+1} = \bar{A}\Delta_t + \bar{B}(k_t(y_t) - k_t(\hat{y}_t)) + \delta_t + \Gamma\zeta_t$, so with $\ell = \|\bar{A}\| + \kappa\|\bar{B}\|$ and $\|\Gamma\zeta_t\|_{L^2} = \beta$, $\|\Delta_{t+1}\|_{L^2} \leq \ell\|\Delta_t\|_{L^2} + \|\delta_t\|_{L^2} + \beta$, and induction gives $\|\Delta_t\|_{L^2} \leq \ell^t w_0 + \sum_{s<t}\ell^{t-1-s}(\|\delta_s\|_{L^2} + \beta)$. □

*Proof of Proposition S18.10.* Here $\gamma > 0$ and $E(\pi) > 0$ for every $\pi \in \Pi$. Let $f = \hat{G} - G$ and $R = \gamma \operatorname{osc}$. Then $\hat{p}/p = e^{-\gamma f}/\mathbb{E}_p e^{-\gamma f}$.

Log-ratio. Since $-\gamma \max f \le \ln \mathbb{E}_p e^{-\gamma f} \le -\gamma \min f$, $\ln(\hat{p}(\pi)/p(\pi)) = -\gamma f(\pi) - \ln \mathbb{E}_p e^{-\gamma f}$ lies between $-\gamma(f(\pi) - \min f)$ and $\gamma(\max f - f(\pi))$, hence in $[-R, R]$.

Relative entropy. $D_{\mathrm{KL}}(p\|\hat{p}) = \gamma \mathbb{E}_p f + \ln \mathbb{E}_p e^{-\gamma f} = \psi(-\gamma)$ with $\psi(s) = \ln \mathbb{E}_p e^{s(f - \mathbb{E}_p f)}$. One has $\psi(0) = 0$, $\psi'(0) = 0$, and $\psi''(s)$ is the variance of $f$ under the law proportional to $pe^{sf}$, which is at most $\operatorname{osc}^2/4$ because $f$ takes values in an interval of length osc. Taylor's formula gives $\psi(-\gamma) \le \gamma^2 \operatorname{osc}^2/8$.

Total variation. The distance is $\max_{S \subseteq \Pi}(\hat{p}(S) - p(S))$. With the weights $w = e^{-\gamma(f - \max f)} \in [1, e^R]$ and $a = p(S)$,

$$\hat{p}(S) = \frac{\sum_S pw}{\sum_S pw + \sum_{\Pi \setminus S} pw} \le \frac{ae^R}{ae^R + 1 - a},$$

because the ratio increases with the first sum and decreases with the second. The function $a \mapsto ae^R/(ae^R + 1 - a) - a$ on $[0, 1]$ is maximal at $a = 1/(1 + e^{R/2})$, where it equals $(e^{R/2} - 1)/(e^{R/2} + 1) = \tanh(R/4)$. The value is attained: for two policies with equal habits, $\gamma(G(\pi_1) - G(\pi_2)) = R/2$ and $\gamma(\hat{G}(\pi_1) - \hat{G}(\pi_2)) = -R/2$, one has $p = (a, 1 - a)$ and $\hat{p} = (1 - a, a)$ with $a = 1/(1 + e^{R/2})$, and $1 - 2a = \tanh(R/4)$. □

*Proof of Theorem S18.11.* (a) With $C^{(0)} = I$, $D_0^{\mathrm{tot}} = 0$ and $E_0^{\mathrm{tot}} = 0$,

$$D_{l+1}^{\mathrm{tot}} = C_{l+1}(D_l^{\mathrm{tot}} + \bar{A}_l C^{(l)}) - \bar{A}_{l+1} C_{l+1} C^{(l)} = C_{l+1} D_l^{\mathrm{tot}} + D_{l+1} C^{(l)},$$

and $E_{l+1}^{\mathrm{tot}} = C_{l+1}(E_l^{\mathrm{tot}} + \bar{B}_l) - \bar{B}_{l+1} = C_{l+1} E_l^{\mathrm{tot}} + E_{l+1}$.

(b) By (a) and $y_t^{(l)} = C^{(l)} x_t$, $D_{l+1}^{\mathrm{tot}} x_t + E_{l+1}^{\mathrm{tot}} u_t = C_{l+1}(D_l^{\mathrm{tot}} x_t + E_l^{\mathrm{tot}} u_t) + D_{l+1} y_t^{(l)} + E_{l+1} u_t$, and the triangle inequality in $L^2$ gives the recursion. A policy that is causal in the top-level variable is a causal coarse policy for $C = C^{(l)}$ at every level $l$, since the top-level variable is a linear image of $y^{(l)}$. If $C^{(l)}$ has full row rank and $C^{(l)} \Sigma_0 C^{(l)\top} \succ 0$, Theorem S18.2 applies to the retained variable $y^{(l)}$ and the declared model $(\bar{A}_l, \bar{B}_l, \bar{Q}_l)$, with $\delta_t = D_l^{\mathrm{tot}} x_t + E_l^{\mathrm{tot}} u_t$; dropping $\mathfrak{d}_{\mathrm{dyn},t} \ge 0$ and using $\mathbb{E}|\delta_t|^2 \le \mathfrak{e}_l^2$ gives Eq. (S59) with $Q^y = C^{(l)} Q C^{(l)\top}$, $\bar{Q} = \bar{Q}_l$ and $\bar{\mathfrak{d}} = \mathfrak{e}_l$.

(c) Let $\mu_l$ and $\nu_l$ be the laws of $y_t^{(l)}$ under the fine process and under the level-$l$ model, both of finite second moment, and let $C_{l+1}\mu_l$ denote the image law. Then $\mu_{l+1} = C_{l+1}\mu_l$, the image of a coupling of $(\mu_l, \nu_l)$ under $C_{l+1}$ applied to both entries is a coupling of the images, so $W_2(C_{l+1}\mu_l, C_{l+1}\nu_l) \le \|C_{l+1}\| \varepsilon_l$, and the triangle inequality for $W_2$ gives $\varepsilon_{l+1} \le \|C_{l+1}\| \varepsilon_l + W_2(C_{l+1}\nu_l, \nu_{l+1})$. The statement is metric and uses nothing about the models beyond finite second moments. □

# S40 Proofs for Section S19

## S40.1 Proofs for the second formation event

*Proof of Proposition S19.2.* Work in common world coordinates, in which an exactly translated message from member $j$ reads $m_j + \varepsilon_{ij}$ with $\mathrm{Cov}(\varepsilon_{ij}) = R_c I_2$. At the split with $\kappa = n - 1$ and $\gamma = \infty$ every member $i$ of $A$ attends all other members of $A$ and nobody else, so by Definition 7.2(v) all members have the same gains, $k^0 + k^s + (n - 1)\bar{k} = 1$, and

$$m_i(t + 1) = F\Big[k^0 m_i + k^s s_i + \bar{k} \sum_{j \in A \setminus \{i\}} (m_j + \varepsilon_{ij})\Big]. \tag{S132}$$

(a) Averaging $s_i = \eta + v_i$ over $A$ gives $\bar{s}_A = \eta + \bar{v}_A$ with $\mathrm{Cov}(\bar{v}_A) = (R^s/n) I_2$, and no agent acts on $\eta$. (b) Averaging (S132) over $i \in A$ and using $\sum_i \sum_{j \ne i} m_j = n(n - 1)\bar{m}_A$ gives the stated recursion

with $k^0 + (n-1)\bar{k} = 1 - k^s$ and $\bar{\zeta}_A = F\bar{k}n^{-1}\sum_i \sum_{j\neq i} \varepsilon_{ij}$, a sum of $n(n-1)$ independent reads, so $\mathrm{Cov}(\bar{\zeta}_A) = F^2\bar{k}^2(n-1)R_c/n \cdot I_2$; reads are independent across rounds and parents. (c) Subtracting $\eta_{t+1} = F\eta_t + w_t$ from (b),

$$e_A(t+1) = \lambda e_A(t) - w_t + \nu_A(t), \qquad \nu_A = Fk^s\bar{v}_A + \bar{\zeta}_A,$$
$$\mathrm{Var}_1(\nu_A) = \frac{F^2(k^s)^2R^s + F^2\bar{k}^2(n-1)R_c}{n} = \frac{q}{n}.$$

Let $b$ and $\xi_A$ be the stationary solutions of $b(t+1) = \lambda b(t) - w_t$ and $\xi_A(t+1) = \lambda\xi_A(t) + \nu_A(t)$. Their sum solves the recursion of $e_A$, which has a unique stationary law because $|\lambda| < 1$. The drivers $w$ and $\nu_A$, $A$ ranging over parents, are independent Gaussian sequences, hence so are $b$ and the $\xi_A$, with $\mathrm{Var}_1(b) = Q/(1-\lambda^2)$, $\mathrm{Var}_1(\xi_A) = q/(n(1-\lambda^2))$ and lag-$\tau$ correlation $\lambda^\tau$. With $\psi_{\mathbf{1}} = \lambda^2$ and $\psi_\perp = \lambda_\perp^2$ at $\kappa = n-1$ in Lemma 7.7,

$$\frac{\mathsf{s}_0}{n} + \mathsf{s}_1 = \frac{\mathsf{s}_0}{n}\Big(1 + \frac{\lambda^2 - \lambda_\perp^2}{1-\lambda^2}\Big) = \frac{\mathsf{s}_0(1-\lambda_\perp^2)}{n(1-\lambda^2)} = \frac{q}{n(1-\lambda^2)}.$$

For the deviation $d_i = m_i - \bar{m}_A$, subtract (b) from (S132) and use $\sum_{j\neq i} m_j - (n-1)\bar{m}_A = -d_i$: $d_i(t+1) = F(k^0 - \bar{k})d_i(t) + \nu_i - \bar{\nu}$, where $\nu_i$ is the noise of member $i$, of variance $q$ per coordinate and independent across members. The deviation $\nu_i - \bar{\nu}$ of independent identically distributed Gaussian variables from their mean is independent of the mean and has variance $q(1-1/n)$, which gives the variance $q(1-1/n)/(1-\lambda_\perp^2) = \mathsf{s}_0(1-1/n)$ and the independence from $(b, \xi_A)$. (d) $\mu_i = \mathrm{Rot}(\theta_A)m_i$ for every $i \in A$, and averaging is linear. For distinct frames write $\bar{m}_c = \bar{m} + (\bar{m}_c - \bar{m})$ and identify $\mathrm{Rot}(\theta)$ with multiplication by $e^{\mathrm{i}\theta}$ on $\mathbb{C}$; then $M^{-1}\sum_c \mathrm{Rot}(\theta_c) = \rho\,\mathrm{Rot}(\bar{\theta})$, and $\rho < 1$ by the strict triangle inequality unless all $e^{\mathrm{i}\theta_c}$ coincide. (e) The recursion of $(e_A, \eta)$ has scalar coefficients and isotropic Gaussian noise, so its stationary law is invariant under rotations and reflections and Lemma 7.3 applies to the sender state $\bar{m}_A$ with read noise $R_c^{(2)}$. Its constant is $2\,\mathrm{Cov}_1(\eta, \bar{m}_A) = 2n^{-1}\sum_{i\in A}\mathrm{Cov}_1(\eta, m_i) = D_{\mathrm{cal}}$, and $\mathrm{MSE}_A = \mathrm{Var}_1(b) + \mathrm{Var}_1(\xi_A)$ by (c). □

*Proof of Lemma S19.3.* For a member $i$ let $a^{(j)}$ be the average belief of the level-$j$ unit containing $i$, so $a^{(0)} = m_i$. The aggregates of the $n_j - 1$ sibling units at level $j$ sum to $n_j a^{(j)} - a^{(j-1)}$, hence in common coordinates

$$m_i(t+1) = F\Big[k^0 m_i + k^s\eta + \sum_{j=1}^{L} k_j\big(n_j a^{(j)} - a^{(j-1)}\big)\Big] + \nu_i, \tag{S133}$$

where $\nu_i$ collects $Fk^s v_i$ and $F$ times the weighted read noises, is independent across members and rounds, and has variance $q$ per coordinate. Average (S133) over the level-$m$ unit of $i$. For $j \le m$ the averages of $a^{(j)}$ and $a^{(j-1)}$ over that unit both equal $a^{(m)}$, and for $j > m$ they are unchanged, so

$$a^{(m)}(t+1) = F\Big[K_m a^{(m)} + k^s\eta + \sum_{j>m} k_j\big(n_j a^{(j)} - a^{(j-1)}\big)\Big] + \bar{\nu}^{(m)}.$$

Subtracting the same identity at $m+1$, the terms $j > m+1$ cancel and the term $j = m+1$ leaves $(K_m - k_{m+1})a^{(m)} - (K_{m+1} - n_{m+1}k_{m+1})a^{(m+1)}$. Since $K_{m+1} - n_{m+1}k_{m+1} = K_m - k_{m+1}$, the difference $\delta^{(m)} = a^{(m)} - a^{(m+1)}$ obeys $\delta^{(m)}(t+1) = \mu_{m+1}\delta^{(m)}(t) + \bar{\nu}^{(m)} - \bar{\nu}^{(m+1)}$. At $m = L$ there are no terms $j > L$ and $K_L = 1 - k^s$, so $e^{(L)} = a^{(L)} - \eta$ obeys $e^{(L)}(t+1) = \lambda e^{(L)}(t) - w_t + \bar{\nu}^{(L)}$. The gains are positive and sum to one, so $-(1-k^s) \le -k_{m+1} \le K_m - k_{m+1} \le K_m \le 1 - k^s$ and $|\mu_{m+1}| \le \lambda < F$.

Averages of independent variables over nested sets satisfy $\mathrm{Cov}_1(\bar\nu^{(m)},\bar\nu^{(m')})=q/N_{m'}$ for $m\le m'$. Therefore $\bar\nu^{(m)}-\bar\nu^{(m+1)}$ has variance $q(N_m^{-1}-N_{m+1}^{-1})$, the differences for distinct $m$ are uncorrelated with one another and with $\bar\nu^{(L)}$, and all are independent of $w$. The components $\delta^{(m)},\dots,\delta^{(L-1)},e^{(L)}$ thus follow decoupled stable scalar recursions with mutually uncorrelated white drivers, and their stationary joint covariance is diagonal with entries $q(N_j^{-1}-N_{j+1}^{-1})/(1-\mu_{j+1}^2)$ and $Q/(1-\lambda^2)+q/(N_L(1-\lambda^2))$. Summing over $e^{(m)}=\sum_{j=m}^{L-1}\delta^{(j)}+e^{(L)}$ gives $\mathrm{MSE}_m$. Only $e^{(L)}$ is correlated with $\eta$: $g=\mathrm{Cov}_1(\eta,e^{(L)})$ solves $g=F\lambda g-Q$, so $2\,\mathrm{Cov}_1(\eta,a^{(m)})=2(P_0-Q/(1-\lambda F))=2QF^2k^s/((1-F^2)(1-\lambda F))$, using $P_0=Q/(1-F^2)$ and $F-\lambda=Fk^s$.

For the self-consistency equations, note three bounds valid for every positive $(\hat P^-,\varpi)$. First $\hat\Lambda^+\ge 1/R^s$, so $Q\le F^2/\hat\Lambda^++Q\le Q+F^2R^s$. Second $\mathrm{MSE}_{j-1}\ge Q/(1-\lambda^2)\ge Q$, so $1/(R_c/N_{j-1}+\mathrm{MSE}_{j-1})\le 1/Q$. Third, the weights $N_j^{-1}-N_{j+1}^{-1}$ and $N_L^{-1}$ sum to $N_m^{-1}\le 1$, the denominators are at least $1-F^2$, and $q\le F^2(R^s+R_c)$ because $k^s\le 1$, $k_j\le 1$ and $\sum_j(n_j-1)k_j\le 1$; hence $\mathrm{MSE}_{j-1}\le P_0+F^2(R^s+R_c)/(1-F^2)$ and $1/(R_c/N_{j-1}+\mathrm{MSE}_{j-1})\ge\varpi_{\min}$. The map $(\hat P^-,\varpi)\mapsto(F^2/\hat\Lambda^++Q,(1/(R_c/N_{j-1}+\mathrm{MSE}_{j-1}))_j)$ is continuous on the box $[Q,Q+F^2R^s]\times[\varpi_{\min},1/Q]^L$, since $1-\mu^2\ge 1-F^2$ there, and maps it into itself. Brouwer's fixed-point theorem gives a solution, and the same bounds applied to any solution give $\varpi_{\min}\le\varpi_j\le 1/Q$. The calibration equation is Lemma 7.3 at $\phi=0$ for the sender state $a^{(j-1)}$, whose law is isotropic as in the proof of Proposition S19.2(e). □

*Proof of Theorem S19.4.* Fix a receiver. Its $2M-1$ senders are $M-1$ near parents, with mismatch in $[0,\vartheta_{\mathrm n}]$, and $M$ far parents, with mismatch $\theta_j\in[\vartheta_{\mathrm f},\vartheta_{\max}]\subset(0,\pi]$. Bias fractions stay in $[0,1]$ under (72), the calibrated precision of sender $l$ is $\varpi_l=\varpi(x_l\theta_l)$, the function $\varpi$ is decreasing on $[0,\pi]$, and a set $J$ has value $\mathcal V(J)=(K/2)\log(\Lambda^+(J)/\Lambda)$ with $\Lambda^+(J)=\Lambda+\sum_{l\in J}\varpi_l\le\Lambda+\kappa_2\varpi(0)$ and softmax weight proportional to $e^{\gamma\mathcal V(J)}$.

(a) Let $j$ be far with $x_j\ge\frac12$, and let $J\ni j$. Since $\kappa_2\le M-1$, the set $J$ misses at least one near parent; let $j'$ be the first of them and $J'=(J\setminus\{j\})\cup\{j'\}$. Then $\varpi_{j'}\ge\varpi(\vartheta_{\mathrm n})$ and $\varpi_j\le\varpi(x_j\vartheta_{\mathrm f})\le\varpi(\vartheta_{\mathrm f}/2)<\varpi(\vartheta_{\mathrm n})$, because $\vartheta_{\mathrm n}<\vartheta_{\mathrm f}/2$, so

$$\mathcal V(J')-\mathcal V(J)=\frac{K}{2}\log\Big(1+\frac{\varpi_{j'}-\varpi_j}{\Lambda^+(J)}\Big)\ge\Delta\mathcal V_2(\tfrac12).$$

The map $J\mapsto J'$ takes sets containing $j$ to sets not containing $j$ and is at most $\kappa_2$-to-one, since $J$ is recovered from $J'$ and the choice of $j'\in J'$. Summing softmax weights, $p_j\le\kappa_2e^{-\gamma\Delta\mathcal V_2(1/2)}(1-p_j)$, that is, $p_j\le\bar{\mathsf p}_2$. With $1-h=\alpha h\mathsf p^*$,

$$\Phi_j(x)-\tfrac12\ge h(1-\alpha\bar{\mathsf p}_2)\tfrac12+1-h-\tfrac12=\tfrac{\alpha h}{2}(\mathsf p^*-\bar{\mathsf p}_2),$$

which is positive exactly when $\kappa_2e^{-\gamma\Delta\mathcal V_2(1/2)}<\mathsf p^*/(1-\mathsf p^*)$, that is, when $\gamma>\gamma_S^{(2)}$. The argument used no property of the near edges, so the set where all far edges have $x_j\ge\frac12$ is forward invariant. On it $x_j(t+1)\ge h(1-\alpha\bar{\mathsf p}_2)x_j(t)+1-h$, an increasing affine contraction with fixed point $(1-h)/(1-h+\alpha h\bar{\mathsf p}_2)=\mathsf p^*/(\mathsf p^*+\bar{\mathsf p}_2)>\frac12$, and comparison gives the lower limit.

(b) Let $\kappa_2=M-1$ and let $c$ be near. A set of $M-1$ senders that misses $c$ is drawn from $M-2$ near and $M$ far parents and therefore contains a far parent, so on the invariant set $1-p_c\le\sum_{j\ \mathrm{far}}p_j\le M\bar{\mathsf p}_2$. Hence $y_c(t+1)\le h(1-\alpha(1-M\bar{\mathsf p}_2))y_c(t)+1-h$, with fixed point $\mathsf p^*/(\mathsf p^*+1-M\bar{\mathsf p}_2)$, which is below $\frac12$ exactly when $M\bar{\mathsf p}_2<1-\mathsf p^*$. Writing $u=\kappa_2e^{-\gamma\Delta\mathcal V_2(1/2)}$, the last condition reads $u<(1-\mathsf p^*)/(M-1+\mathsf p^*)$, whereas $\gamma>\gamma_S^{(2)}$ reads $u<\mathsf p^*/(1-\mathsf p^*)$. The second implies the first when $\mathsf p^*(M-1+\mathsf p^*)\le(1-\mathsf p^*)^2$, that is, when $\mathsf p^*(M+1)\le1$.

(c) All decoding errors lie in $[0,\vartheta_{\max}]$, so every set has value between $(K/2)\log((\Lambda+\kappa_2\varpi(\vartheta_{\max}))/\Lambda)$ and $(K/2)\log((\Lambda+\kappa_2\varpi(0))/\Lambda)$, and softmax weight at least $e^{-\gamma\Delta}/\binom{2M-1}{\kappa_2}$ with $\Delta$ the

denominator of $\gamma_M^{(2)}$ in (S65). Each sender lies in $\binom{2M-2}{\kappa_2-1}$ sets, so $p_j \ge \kappa_2 e^{-\gamma\Delta}/(2M-1) =: p_{\min}$ for every edge and every state. Then $x_j(t+1) \le h(1-\alpha p_{\min})x_j(t)+1-h$ and $\limsup_t x_j(t) \le \mathsf{p}^*/(\mathsf{p}^*+p_{\min})$, which is below $\frac{1}{2}$ exactly when $p_{\min} > \mathsf{p}^*$, that is, when $\gamma < \gamma_M^{(2)}$.

For the constants, the receiver's precision before parent messages at the level-1 split is $\hat\Lambda^+ = \Lambda_0+(n-1)\varpi_{\text{in}} = \Lambda^{(1)}$, and Proposition S19.2(e) gives the calibrated noise of a parent broadcast, hence $\hat\sigma^2$ and $D$. At depth $\ell-1$ the same reading of Lemma S19.3 gives $\Lambda = \hat\Lambda^+$, $\hat\sigma^2 = R_c/N_{\ell-1} + \text{MSE}_{\ell-1}$ and $D = D_{\ell-1}$. □

*Proof of Corollary S19.5.* At a depth-$L$ solution the next event has $\Lambda = \hat\Lambda^+ \ge 1/R^s + \sum_j (n_j - 1)\varpi_j \ge 1/R^s + \varpi_{\min} S_L$ by Lemma S19.3, $\hat\sigma^2 \ge \text{MSE}_L \ge Q/(1-\lambda^2) \ge Q$, and $D = D_L \le 2QF^2k^s/(1-F^2)^2$ with $k^s = 1/(R^s\Lambda)$, because $1-\lambda F \ge 1-F^2$. Since $\log(1+z) \le z$ and $\varpi$ is decreasing on $[0,\pi]$,

$$\Delta\mathcal{V}_2(\tfrac{1}{2}) \le \frac{K}{2}\frac{\varpi(0)-\varpi(\pi)}{\Lambda} = \frac{K}{2}\frac{2D}{\hat\sigma^2(\hat\sigma^2+2D)\Lambda} \le \frac{KD}{Q^2\Lambda} \le \frac{2KF^2}{QR^s(1-F^2)^2\Lambda^2} = \frac{C}{\Lambda^2},$$

and in the same way the denominator of $\gamma_M$ is at most $(K/2)\kappa(\varpi(0)-\varpi(\vartheta_{\max}))/\Lambda \le \kappa C/\Lambda^2$. Dividing positive logarithms by these bounds gives (S66).

For the errors, $\lambda^2 \ge F^2(1-2k^s)$ gives $1-\lambda^2 \le (1-F^2)(1+2F^2k^s/(1-F^2))$, hence $\text{MSE}_m \ge Q/(1-\lambda^2) \ge P_0(1-2F^2k^s/(1-F^2))$. In (S64) the first term is below $P_0$ and the remaining weights sum to $N_m^{-1} \le 1$ over denominators at least $1-F^2$, hence $\text{MSE}_m \le P_0 + q/(1-F^2)$. The bound on $k^s = 1/(R^s\hat\Lambda^+)$ is the bound on $\Lambda$. Finally $k_j = \varpi_j R^s k^s \le R^s k^s/Q$ and $\sum_j (n_j-1)k_j \le 1$ give $\sum_j k_j^2(n_j-1)R_c/N_{j-1} \le R^s k^s R_c/Q$, and $(k^s)^2R^s \le k^s R^s$, so $q \le F^2 k^s R^s(1+R_c/Q)$.

Since $S_L \ge L$, these bounds give $k^s \to 0$ along any sequence of solutions with $S_L \to \infty$, so no accumulation point has $k^s > 0$; then $1/\hat\Lambda^+ = R^s k^s \to 0$, $D_L \to 0$, $\Delta\mathcal{V}_2 \le C/\Lambda^2 \to 0$, $q \to 0$ and $\text{MSE}_m \to P_0$. For the last claim take $M = n_\ell \le n_{\max}$ and $\kappa_\ell = n_\ell - 1$. Then $\kappa_\ell/((2M-1)\mathsf{p}^*) \ge 1/(3\mathsf{p}^*) > 1$, and (S66) with $\Lambda \ge \varpi_{\min}(\ell-1)$ gives $\gamma_M^{(\ell)} \ge \varpi_{\min}^2(\ell-1)^2\log(1/(3\mathsf{p}^*))/(C(n_{\max}-1))$, which exceeds any fixed $\gamma$ for all but finitely many $\ell$; at those levels Theorem S19.4(c) applies. □

## S40.2 Proofs for the obstructions and the agency tests

Throughout, $\text{Var}_1$ and $\text{Cov}_1$ denote variance and covariance per coordinate, rotations of the plane are identified with unit complex numbers, and $\xi_i$ is the part of the error $e_i = m_i - \eta = b + \xi_i$ that is independent of the world.

*Proof of Lemma S19.8.* Write $m_i = \bar m + d_i$ with $\bar m = n^{-1}\sum_i m_i$. Since $n^{-1}\sum_i \text{Rot}(\theta_i) = \rho\text{Rot}(\bar\theta)$, the aggregate is $\bar\mu_c = \rho\text{Rot}(\bar\theta)\bar m + n^{-1}\sum_i \text{Rot}(\theta_i)d_i$, and a translation that errs by $\phi$ returns

$$X = \rho\text{Rot}(\phi)\bar m + \frac{1}{n}\sum_i \text{Rot}(\phi+\theta_i-\bar\theta)d_i, \tag{S134}$$

plus read noise of variance $R_c^{(2)}$ per coordinate, independent of everything else. C1g sets the calibrated noise to $R_c^{(2)} + \mathbb{E}\|X-\eta\|^2/K$.

At the split of Lemma 7.7 the errors are $e_i = b + \xi_i$ with $\xi_{t+1} = W\xi_t + \zeta_t$, where $W = F[k^0 I + k_1(\mathbf{1}\mathbf{1}^\top - I)]$ is symmetric and $\text{Cov}(\zeta_t) = qI$ per coordinate. Hence $\text{Cov}_1(\xi) = q(I-W^2)^{-1}$ commutes with $\mathbf{1}\mathbf{1}^\top/n$, and $d = (I - \mathbf{1}\mathbf{1}^\top/n)\xi$ has $\text{Cov}_1(d_i,d_j) = \mathsf{s}_0(\delta_{ij} - 1/n)$ with $\mathsf{s}_0 = q/(1-\mu_1^2)$ the constant of Lemma 7.7, is uncorrelated with $\bar\xi$, and is independent of $b$ and $\eta$. The two terms of

(S134) are therefore uncorrelated and the second is uncorrelated with $\eta$. Because the coordinates are isotropic, $\mathbb{E}[d_i^\top \mathrm{Rot}(\beta) d_j] = 2\cos\beta\,\mathrm{Cov}_1(d_i, d_j)$, so the second term has variance per coordinate

$$\frac{\mathsf{s}_0}{n^2}\sum_{i,j}\cos(\theta_i-\theta_j)\Big(\delta_{ij}-\frac{1}{n}\Big) = \frac{\mathsf{s}_0}{n^2}(n-n\rho^2) = \frac{\mathsf{s}_0(1-\rho^2)}{n},$$

by $\sum_{i,j}\cos(\theta_i-\theta_j) = |\sum_i e^{\mathrm{i}\theta_i}|^2 = n^2\rho^2$. For the first term, $\mathrm{Cov}_1(\eta, \bar{m}) = \mathrm{Cov}_1(\eta, m_i) = D_{\mathrm{cal}}/2$ and $\mathrm{MSE}_A = \mathrm{Var}_1(\bar{m}) + P_0 - D_{\mathrm{cal}}$, so

$$\frac{1}{K}\mathbb{E}\|\rho\mathrm{Rot}(\phi)\bar{m} - \eta\|^2 = \rho^2(\mathrm{MSE}_A - P_0 + D_{\mathrm{cal}}) + P_0 - \rho D_{\mathrm{cal}}\cos\phi.$$

Adding the three contributions and writing $-\rho D_{\mathrm{cal}}\cos\phi = -\rho D_{\mathrm{cal}} + \rho D_{\mathrm{cal}}(1-\cos\phi)$ gives (S67). At $\rho = 1$ the constant is $R_c^{(2)} + \mathrm{MSE}_A$; at $\rho = 0$ the term in $\phi$ is absent. For the lower bound, $X$ at round $t$ is a function of data up to round $t-1$, and $\eta_t = F\eta_{t-1} + w_{t-1}$ with $w_{t-1}$ independent of those data, so $\mathbb{E}\|X - \eta_t\|^2 \geq \mathbb{E}\|w_{t-1}\|^2 = KQ$. □

*Proof of Theorem S19.9.* Write $\varpi_0 = 1/\hat{\sigma}_\rho^2$, $\varpi_{\mathrm{m}} = 1/\hat{\sigma}_\rho^2(\vartheta_{\max})$ and $\Delta = \rho D(1-\cos\vartheta_{\max})$. Translation errors lie in $[0, \vartheta_{\max}] \subset [0, \pi]$, where the cosine decreases, so every parent edge has $\varpi_{\mathrm{m}} \leq \varpi_c \leq \varpi_0$ at every state, and every $\kappa_2$-set $J$ has $\Lambda + \kappa_2\varpi_{\mathrm{m}} \leq \hat{\Lambda}^+(J) \leq \Lambda + \kappa_2\varpi_0$. The values of two sets therefore differ by at most

$$\Delta\mathcal{V}_{\max}^{(2)} = \frac{K}{2}\log\frac{\Lambda+\kappa_2\varpi_0}{\Lambda+\kappa_2\varpi_{\mathrm{m}}} \leq \frac{K\kappa_2(\varpi_0-\varpi_{\mathrm{m}})}{2\Lambda} = \frac{K\kappa_2\Delta}{2\hat{\sigma}_\rho^2\hat{\sigma}_\rho^2(\vartheta_{\max})\Lambda} = B_2, \tag{S135}$$

by $\log(1+u) \leq u$ and $\varpi_0 - \varpi_{\mathrm{m}} = \Delta/(\hat{\sigma}_\rho^2\hat{\sigma}_\rho^2(\vartheta_{\max}))$.

(a) Of the $\binom{2M-1}{\kappa_2}$ sets, $\binom{2M-2}{\kappa_2-1}$ contain a given parent $c$. Bounding the weights of these from below by the least weight and all weights from above by the greatest,

$$\mathsf{p}_c = \frac{\sum_{J\ni c} e^{\gamma\mathcal{V}(J)}}{\sum_J e^{\gamma\mathcal{V}(J)}} \geq \frac{\kappa_2}{2M-1}e^{-\gamma\Delta\mathcal{V}_{\max}^{(2)}} \geq \frac{\kappa_2}{2M-1}e^{-\gamma B_2} =: \underline{\mathsf{p}}.$$

The map (72) is $x_c \mapsto h(1-\alpha\mathsf{p}_c)x_c + 1 - h$ and decreases in $\mathsf{p}_c$, so $x_c(t+1) \leq h(1-\alpha\underline{\mathsf{p}})x_c(t) + 1 - h$ along every orbit and $\limsup_t x_c(t) \leq (1-h)/(1-h+\alpha h\underline{\mathsf{p}}) = \mathsf{p}^*/(\mathsf{p}^* + \underline{\mathsf{p}})$. This is below $\frac{1}{2}$ exactly when $\underline{\mathsf{p}} > \mathsf{p}^*$, that is, when $\gamma B_2 < \ell_2$. The threshold of Theorem S19.4(c) is $\gamma_M^{(2)} = \ell_2/\Delta\mathcal{V}_{\max}^{(2)}$, and (S135) gives $\gamma_M^{(2)} \geq \ell_2/B_2$.

(b) If $\Delta = 0$ then $\varpi_c = \varpi_0$ on every edge at every state, $\mathcal{V}$ is constant on $\kappa_2$-sets, and the attention law is uniform for every finite $\gamma$ and, with ties broken uniformly, at $\gamma = \infty$. Then $\mathsf{p}_c = \kappa_2/(2M-1)$ for every edge, the coordinates of the map decouple, and each is an affine contraction with slope $h(1-\alpha\mathsf{p}_c) \in [0,1)$ and fixed point $\mathsf{p}^*/(\mathsf{p}^* + \kappa_2/(2M-1))$.

(c) By Lemma S19.8, $\hat{\sigma}_\rho^2(\vartheta_{\max}) \geq \hat{\sigma}_\rho^2 \geq Q$. The inherited coefficient is $D = D_{\mathrm{cal}} = 2QF^2k^s/((1-F^2)(1-\lambda F))$ with $k^s = 1/(R^s\Lambda)$ and $\lambda = F(1-k^s) \leq F$, so $D \leq 2QF^2/((1-F^2)^2R^s\Lambda)$ and

$$B_2 \leq \frac{K\kappa_2\rho(1-\cos\vartheta_{\max})}{2Q^2\Lambda}\cdot\frac{2QF^2}{(1-F^2)^2R^s\Lambda} = \frac{\rho C\kappa_2(1-\cos\vartheta_{\max})}{2\Lambda^2}.$$

The base precision is $\Lambda = \Lambda^{(1)} = 1/\hat{P}^- + 1/R^s + (n-1)\varpi_{\mathrm{in}} \geq 1/R^s + (n-1)\varpi_{\min}$ by Lemma S19.3. For $\ell_2 > 0$ the stated inequality gives $\gamma B_2 \leq \gamma\rho C\kappa_2(1-\cos\vartheta_{\max})/(2\Lambda^2) < \ell_2$, and (a) applies; for $\ell_2 \leq 0$ the hypothesis $\gamma B_2 < \ell_2$ of (a) cannot hold.

(d) A reader has $M-1$ near parents and $M$ far ones, so a set of $\kappa_2 \geq M$ parents contains at least $\kappa_2 - M + 1$ far parents and $\sum_{c\ \mathrm{far}}\mathsf{p}_c \geq \kappa_2 - M + 1$ at every state and every $\gamma$. Some far edge then has $\mathsf{p}_c \geq (\kappa_2 - M + 1)/M$. At a fixed point $x_c = \mathsf{p}^*/(\mathsf{p}^* + \mathsf{p}_c)$, which is below $\frac{1}{2}$ when $\mathsf{p}_c > \mathsf{p}^*$; hence that edge is learned whenever $\mathsf{p}^* < (\kappa_2 - M + 1)/M$. □

*Proof of Proposition S19.10.* Order the $nM$ members of $S$ by parent, let $P_B$ be the orthogonal projector onto vectors constant on parents and $P_J = \mathbf{1}\mathbf{1}^\top/(nM)$. At the depth-2 split a member gives weight $Fk^0$ to itself, $Fk_1$ to each of its $n-1$ parent mates and $Fk_2/n$ to each member of the $M-1$ other parents of $S$, and nothing to the other superunit. The member matrix is therefore symmetric with nonnegative entries and

$$W = \mu_1(I - P_B) + \mu_2(P_B - P_J) + \lambda P_J, \qquad \mu_1 = F(k^0 - k_1), \quad \mu_2 = F(k^0 + (n-1)k_1 - k_2),$$

with $\lambda = F(k^0 + (n-1)k_1 + (M-1)k_2) = F(1-k^s)$, in agreement with Lemma S19.3. The world-independent parts obey $\xi_{t+1} = W\xi_t + \zeta_t$ with $\mathrm{Cov}_1(\zeta_t) = qI$, since each member adds its own sensing and read noises. As $W$ is symmetric with spectrum in $(-1, 1)$, the stationary covariance is $\Sigma^c = q\sum_{k\geq 0} W^{2k} = q(I - W^2)^{-1}$, and $\Pi = (I - W^2)/q$. Its off-diagonal entries are $-(W^2)_{ij}/q \leq 0$ because $W$ is entrywise nonnegative, so $\Pi$ is diffusive, and $\Pi\mathbf{1} = (1-\lambda^2)\mathbf{1}/q$ gives the uniform leak $\varepsilon = (1-\lambda^2)/q$.

No coupling leaves $S$, so $\mathrm{cut}(S) = 0$ and $r(S) = (\varepsilon + \lambda_2(L_S))/\varepsilon$ with $L_S = \Pi - \varepsilon I$. On the two nonconstant eigenspaces $L_S$ equals $(\lambda^2 - \mu_2^2)/q$ and $(\lambda^2 - \mu_1^2)/q$, both nonnegative because $L_S$ is a Laplacian, so $\lambda_2(L_S) = (\lambda^2 - \max\{\mu_1^2, \mu_2^2\})/q$ and the first formula of (S69) follows.

For a parent $c$, the entries of $W^2 = \mu_1^2(I - P_B) + \mu_2^2(P_B - P_J) + \lambda^2 P_J$ are $w_{\mathrm{in}} = (\mu_2^2 - \mu_1^2)/n + (\lambda^2 - \mu_2^2)/(nM)$ between distinct members of $c$ and $w_{\mathrm{x}} = (\lambda^2 - \mu_2^2)/(nM)$ between members of different parents. The Laplacian of $c$ is $(w_{\mathrm{in}}/q)(nI - \mathbf{1}\mathbf{1}^\top)$ with $\lambda_2 = nw_{\mathrm{in}}/q$, and $\mathrm{cut}(c)/|c| = (M-1)nw_{\mathrm{x}}/q$. Inserting these and multiplying by $q$ gives the second formula. Its numerator equals $1 - \mu_1^2 - (M-1)(\lambda^2 - \mu_2^2)/M \leq 1 - \mu_1^2$, and its denominator equals $[(1-\lambda^2) + (M-1)(1-\mu_2^2)]/M \geq (M-1)(1-\mu_2^2)/M$, which is the stated bound.

For the parent aggregates, with $A$ the $M \times nM$ averaging matrix, $A\Sigma^c A^\top = (q/n)[(1 - \mu_2^2)^{-1}(I - \mathbf{1}\mathbf{1}^\top/M) + (1-\lambda^2)^{-1}\mathbf{1}\mathbf{1}^\top/M]$, whose inverse has leak $n(1-\lambda^2)/q$ and Laplacian eigenvalue $n(\lambda^2 - \mu_2^2)/q$; the ratio is $(1-\mu_2^2)/(1-\lambda^2)$. □

*Proof of Proposition S19.11.* (a) A reader that knows the mistranslation $\phi$ of a read applies $\mathrm{Rot}(-\phi)$ to the translated message. The result is $\bar{m}_c$ plus the rotated read noise, which is isotropic and so has the law of the unrotated noise. C1g on the current law then gives $R_c^{(2)} + \mathbb{E}\|\bar{m}_c - \eta\|^2/K = R_c^{(2)} + \mathrm{MSE}_c(t)$, with no dependence on the reader.

(b) Let $G$ be the group generated by permutations of the $2M$ parents and of the members inside each parent, and suppose the law at round $t$ is $G$-invariant with $e_i = b + \xi_i$, $\xi$ independent of the world. Invariance gives a common $\mathrm{MSE}_c(t)$ and common member quantities, hence by (a) one assigned precision $\varpi_2(t)$ for every aggregate read, one $\varpi_1(t)$ inside parents and one prior variance $\hat{P}^-(t)$. Every $\kappa_2$-set has the value $\frac{K}{2}\log((\Lambda + \kappa_2\varpi_2)/\Lambda_0)$, so attention is uniform on $\kappa_2$-sets for every $\gamma \in [0, \infty]$, with ties broken uniformly, and it is independent of the state. The gains $k^0, k^s, k_1, k_2$ are common and do not depend on the attended set, and they sum to one with multiplicities $1, 1, n-1, \kappa_2$. Therefore

$$e_i(t+1) = F\Big[k^0 e_i + k^s v_i + k_1 \sum_{j\sim i}(e_j + \nu_{ij}) + k_2 \sum_{c\in J_i}(\bar{e}_c + \nu_{ic})\Big] - w_t = \lambda b_t - w_t + \xi_i(t+1),$$

with $\lambda = F(1-k^s)$ and $\xi(t+1) = W_t\xi(t) + \zeta_t$, where $W_t$ is the random nonnegative matrix of the drawn sets and $\zeta_t$ collects the sensing and read noises. The law of $(W_t, \zeta_t)$ is $G$-invariant and independent of the world and of the past, so the law at round $t+1$ is $G$-invariant with the same decomposition. Induction from the start gives the first three claims.

The covariance $\Sigma_t = \mathrm{Cov}_1(\xi(t))$, averaged over attention draws, obeys $\Sigma_{t+1} = \mathbb{E}[W_t\Sigma_t W_t^\top] + q_t I$. The rows of $W_t$ are nonnegative and sum to $\lambda < 1$, so this linear map contracts the entrywise

maximum by $\lambda^2$; at fixed gains it has a unique fixed point, which is the limit from $\Sigma = 0$ and is therefore entrywise nonnegative. The covariance of the parent aggregates commutes with all permutations of the parents and so equals $g_0 I + g_1 \mathbf{1}\mathbf{1}^\top$, with $g_0 > 0$ and, at stationarity, $g_1 \geq 0$. Its inverse is $g_0^{-1}[I - g_1 \mathbf{1}\mathbf{1}^\top/(g_0 + 2Mg_1)]$: the row sums are $\varepsilon = 1/(g_0 + 2Mg_1)$ and the off-diagonal entries are $-a$ with $a = g_1\varepsilon/g_0 \geq 0$, so the precision is diffusive with uniform leak. A union $C$ of $m \geq 2$ parents has $L_C = a(mI - \mathbf{1}\mathbf{1}^\top)$, $\lambda_2(L_C) = ma$ and $\mathrm{cut}(C)/|C| = (2M - m)a$, which gives the stated contrast and the value 1 at $m = M$. Every partition into two unions of $M$ parents has both blocks at contrast 1, and Definition 4.12 requires $r_0 > 1$ under a uniform leak, so no such partition is selected.

(c) Transitivity of $H$ on the agents gives a common member mean-square error, a common $\hat{P}^-(t)$ and, since $H$ maps parents to parents and is transitive on them, a common $\mathrm{MSE}_c(t)$. The induction of (b) used invariance only through these facts, and the update with common gains and uniform attention commutes with every element of $G \supseteq H$, so the law stays $H$-invariant with the same decomposition and the first three claims follow. Let $P_g$ be the permutation matrix of $g \in G$ and $\bar{\Sigma}_t = |G|^{-1} \sum_{g \in G} P_g \Sigma_t P_g^\top$. The matrix $P_g W_t P_g^\top$ has the law of $W_t$, so averaging the recursion $\Sigma_{t+1} = \mathbb{E}[W_t \Sigma_t W_t^\top] + q_t I$ over $G$ gives $\bar{\Sigma}_{t+1} = \mathbb{E}[W_t \bar{\Sigma}_t W_t^\top] + q_t I$ with the same gains, and

$$\Sigma_{t+1} - \bar{\Sigma}_{t+1} = \mathbb{E}[W_t(\Sigma_t - \bar{\Sigma}_t)W_t^\top].$$

The rows of $W_t$ are nonnegative and sum to $\lambda_t = F(1 - k_t^s) \leq F$, so the entrywise maximum of the difference contracts by $F^2$ in each round. At a stationary law the difference is constant in $t$ and hence zero, so the covariance is $G$-invariant and the remaining claims of (b) apply to it. The depth-2 split is invariant under the permutations of the members inside each parent, of the parents inside each superunit and of the two superunits, which generate a group that maps parents to parents and is transitive on the agents, and its errors have the form $b + \xi_i$ with a common $b$ by Lemma S19.3. □

*Proof of Proposition S19.12.* (a) With a default, retention reads $\hat{\alpha}_{cd} \leftarrow \hat{\alpha}^0_{cd} + h(\hat{\alpha}_{cd} - \hat{\alpha}^0_{cd})$. Put $u_{cd} = \hat{\alpha}_{cd} - \hat{\alpha}^0_{cd}$ and let $\vartheta^0_{cd} = \alpha^\star_{cd} - \hat{\alpha}^0_{cd}$ be the signed default error, where $\alpha^\star_{cd}$ is the frame difference. The two steps of Definition 7.2(vi) become $u \leftarrow u + \alpha(\vartheta^0 + \nu^\alpha - u)$ on attended rounds and $u \leftarrow hu$ every round. A translated read is $\mathrm{Rot}(\phi)\bar{m}_d$ plus isotropic noise with $\phi = u - \vartheta^0$ (Eq. (70)), and the calibrated noise depends on the frames through $\cos\phi$ only. Beliefs, attention and the variables $u$ thus evolve by a law in which the frames and the default enter only through $\vartheta^0$. The substitution $\theta_c \mapsto \theta_c + \delta_c$ shifts $\alpha^\star_{cd}$ by the same amount as the stated shift of $\hat{\alpha}^0_{cd}$, so it fixes $\vartheta^0$; with the estimates shifted alike it fixes $u$, and the law is unchanged. The map (72) with frozen calibration involves the errors through the precisions $\varpi(x_c|\vartheta^0_c|)$ only, which are even in the error, so the map and its thresholds depend on the moduli $|\vartheta^0_{cd}|$. A reflection of the plane maps the model with errors $\vartheta^0$ to the model with errors $-\vartheta^0$ and preserves second moments, which are therefore unchanged when all errors change sign together. The relative signs are not removed: a reader that fuses two reads with errors $\phi_1, \phi_2$ at a common gain $k$ has in its error the term $k(\mathrm{Rot}(\phi_1) + \mathrm{Rot}(\phi_2) - 2I)\eta$, whose mean square per coordinate is $2k^2 P_0[3 - 2\cos\phi_1 - 2\cos\phi_2 + \cos(\phi_1 - \phi_2)]$ and differs between $\phi_2 = \phi_1$ and $\phi_2 = -\phi_1$ (Check S47.180).

(b) With $\hat{\alpha}^0 = 0$, $\vartheta^0_{cd}$ is the frame difference, and the substitution $\delta_c = -\theta_c$ sets every frame difference to zero. Then $\vartheta_{\max} = 0$ and Theorem S19.9(b) applies at every $\gamma$, while the original population may satisfy the hypotheses of Theorem S19.4(a) and (b).

(c) The closure defect is a conditional mutual information and is invariant under a bijection of the coarse variable; the contrast is computed from the world-conditional precision of the members and does not involve the coarse map. If the aggregate is sent as $\rho'\mathrm{Rot}(\delta)\bar{\mu}_c$ and the reader translates as before, the read is $\rho'\mathrm{Rot}(\phi + \delta)\bar{m}_c$ plus noise, and the computation in the proof of Lemma S19.8,

with $\rho'$ for $\rho$ and no deviation term, gives the calibrated noise $R_c^{(2)} + \rho'^2(\mathrm{MSE}_A - P_0 + D) + P_0 - \rho' D + \rho' D(1 - \cos(\phi + \delta))$. □

*Proof of Proposition S19.14.* (a) Record the bias fraction $x_c$ of an edge after the replacement step, which is the state that enters attention and fusion. Given the attention probability $\mathsf{p}_c$, learning and retention take the mean to $h(1 - \alpha\mathsf{p}_c)x_c + 1 - h$. The reader is then replaced with probability $g$ and the estimate reset with conditional probability $1 - f_\alpha$; the declared start has $x = 1$, so the mean after replacement is

$$(1 - \hat{g})\big[h(1 - \alpha\mathsf{p}_c)x_c + 1 - h\big] + \hat{g} = h_g(1 - \alpha\mathsf{p}_c)x_c + 1 - h_g,$$

which is (72) with $h_g$ for $h$. The proofs of Theorems S19.4 and S19.9 use the map only through its form, so they hold with $\mathsf{p}^*(g) = (1 - h_g)/(\alpha h_g)$, which increases as $h_g$ decreases. Since $\mathsf{p}_c \le 1$, every orbit has $x_c(t+1) \ge h_g(1 - \alpha)x_c(t) + 1 - h_g$ and $\liminf_t x_c(t) \ge \mathsf{p}^*(g)/(\mathsf{p}^*(g) + 1)$, which is at least $\frac{1}{2}$ exactly when $\mathsf{p}^*(g) \ge 1$, that is, $h_g \le 1/(1 + \alpha)$, that is, $\hat{g} \ge g_c$.

(b) Let $m^+$ denote the memories after replacement. Given the state before it, $\mathbb{E}[m_i^+] = (1 - g)m_i + gf\bar{m}_{c(i)}$, so $\mathbb{E}[\bar{m}_S^+] = (1 - g(1 - f))\bar{m}_S$. The member matrix of the split is symmetric (proof of Proposition S19.10), so its column sums equal its row sums and, pathwise,

$$\bar{m}_S(t+1) = F\big[(1 - k^s)\bar{m}_S^+(t) + k^s\bar{s}_S(t)\big] + \bar{\zeta}_t,$$

with $\bar{\zeta}_t$ the average read noise. Hence $y_t = \bar{m}_S(t)$ obeys $y_{t+1} = \lambda_g y_t + Fk^s\bar{s}_S(t) + \chi_t$, where $\chi_t = F(1 - k^s)(\bar{m}_S^+ - \mathbb{E}[\bar{m}_S^+ \mid \text{state}]) + \bar{\zeta}_t$ has zero mean given the state and the inputs up to round $t$ and finite variance. Split $y = y^s + y^\chi$ into the linear response to the input path and $y^\chi_{t+1} = \lambda_g y^\chi_t + \chi_t$. Since $y^\chi_t$ is a function of $\chi_u$, $u < t$, the martingale-difference property gives $\mathbb{E}[\chi_{t+k}y^\chi_t] = 0$ for $k \ge 0$, so $\mathrm{Cov}(y^\chi_{t+\tau}, y^\chi_t) = \lambda_g^\tau \mathrm{Var}(y^\chi_t)$, and at stationarity the autocorrelation is $\lambda_g^\tau$. A member present at round $t$ is still present at $t + \tau$ with probability $(1 - g)^\tau$. Finally $\lambda(1 - g(1 - f)) > 1 - g$ rearranges to $g(1 - \lambda(1 - f)) > 1 - \lambda$. □

*Proof of Proposition S19.15.* (i) By the proof of Proposition S19.14(b) at $g = 0$, $y_{t+1} = \lambda y_t + Fk^s\bar{s}_S(t) + \bar{\zeta}_t$, with $\bar{\zeta}_t$ Gaussian and independent of the state, the inputs and the past. Given $y_t$ and the blanket state, $y_{t+1}$ is independent of the member state, so the conditional mutual information that defines the closure defect is zero. The innovation $y_{t+1} - \lambda y_t - Fk^s\bar{s}_S(t) = \bar{\zeta}_t$ is the average read noise of round $t$, independent of $y_{0:t}$ and of the whole input path, because the sensors read the autonomous world and no member. Hence $I(y_{t+1}; y_{0:t-1} \mid y_t, \bar{s}_{S,0:t}) = 0$ for every $t$, and the argument uses no property of the initial law of the members. When that law is Gaussian and independent of the world and of the sensor noise, the input path is independent of the initial state and of the read noise, the proviso of Definition 5.1(i) holds, and the supremum over $t$, which is zero, is the Markov defect of that definition in its input-conditional form. In the stationary state the input path is correlated with the initial member state through the world, so the proviso is not met and the claim is the vanishing of the conditional mutual information only; the defect of the law of $y$ without conditioning on the input is not computed and is not claimed.

(ii) The sensors are memoryless readers of $\eta$ with independent noises and read no interior variable, which is (S); the world is autonomous, so the exterior reads no member and (a2) holds with $\mathsf{L}_\mathrm{E} = 0$, and the abstract physics is the world's own with $\bar{F} = F$ and $\rho(F) < 1$, so the pair $(\bar{F}, \Lambda^s)$ is detectable, as (H-phys) requires. The members fuse with the level-0 gains of Definition 7.2(v) and not with the consensus filter at the steady-state covariances of the pooled model $(F, Q, R^s/(nM))$: the protocol of (H-prot) is absent by construction. Its conclusion fails as well at the base point, where the coarse recursion weighs the pooled observation with $k^s = 0.026$ against the pooled filter's

0.379 (Check S47.179). With (H-prot) false the theorem is not applicable, and no conclusion of it is used; the closure of the coarse state is the direct statement (i).

(iii) The model of Definition S19.1 has no acts and no decision states.

(iv) Under uniform partner draws redrawn every round, $\mathsf{p}_c = \kappa_2/(2M-1)$ on every edge at every state, and the argument of Theorem S19.9(b) gives one globally attracting fixed point with equal bias on all parent edges. At $\gamma = 0$ the attention weights of Definition S19.1 are uniform, which is the same draw. With partners drawn once and kept, the attendance of an edge is 0 or 1 for all time and the map is a different one, which this argument does not treat.

(v) By Proposition S19.11(b), and by (c) for stationary laws with the symmetry of the split, each superunit has contrast 1 under the translation-aware null, below every admissible level $r_0 > 1$; the partition into superunits does not persist with blocks at $r \geq r_0$, which is what a null of type S requires in Table 6. The grade of Definition 5.19 requires survival of every null of type S, which gives the final statement. □

## S41 Proofs for Section S20

Throughout, $E_i \in \mathbb{R}^{d_u \times d_i}$ is the inclusion of member $i$'s coordinates, so that $u_i = E_i^\top u$ and $\sum_i E_i K_{ii}^{-1} E_i^\top = K_{\mathrm{d}}^{-1}$, and $n \geq 2$. In the setting of Definition S20.1, $G_i^{\mathrm{act}}(u) = \frac{1}{2}\|\mathsf{z}_i + \sum_j \mathsf{M}_j^{(i)} u_j\|^2_{\Lambda_i^\star}$, so as a function of $u_i$ at fixed $u_{-i}$,

$$G_i^{\mathrm{act}}(u) = \tfrac{1}{2} u_i^\top K_{ii} u_i + u_i^\top \Big( \sum_{j \neq i} K_{ij} u_j + \mathsf{h}_i \Big) + \mathsf{c}_i(u_{-i}), \tag{S136}$$

which gives (S70) and the Gaussian conditional $p_i(\cdot \mid u_{-i}) \propto e^{-\gamma G_i^{\mathrm{act}}}$ of Definition S20.1; $K_{ii} \succ 0$ because $\mathsf{M}_i^{(i)}$ has full column rank. Write $T_i$ for the kernel that redraws $u_i$ from $p_i(\cdot \mid u_{-i})$ and $T = \frac{1}{n}\sum_i T_i$ for the heat-bath kernel. With $\xi \sim \mathcal{N}(0, (\gamma K_{ii})^{-1})$ independent of $u$, the update of member $i$ is

$$u^+ = \mathsf{A}_i u - E_i K_{ii}^{-1} \mathsf{h}_i + E_i \xi, \qquad \mathsf{A}_i = I - E_i K_{ii}^{-1} E_i^\top K. \tag{S137}$$

*Proof of Theorem S20.2.* (a) The mixed second difference of a quadratic function over displacements $\Delta_i$ and $\Delta_j$ of two different blocks of coordinates is $\Delta_i^\top H_{ij} \Delta_j$, with $H$ its Hessian. The Hessian block of $G_i^{\mathrm{act}}$ is $\mathsf{M}_i^{(i)\top} \Lambda_i^\star \mathsf{M}_j^{(i)} = K_{ij}$ and that of $G_j^{\mathrm{act}}$ is $\mathsf{M}_i^{(j)\top} \Lambda_j^\star \mathsf{M}_j^{(j)} = K_{ji}^\top$. Hence $\mathcal{L}_{ij} = \partial^2_{ij} G_j^{\mathrm{act}} - \partial^2_{ij} G_i^{\mathrm{act}} = \Delta_i^\top (K_{ji}^\top - K_{ij}) \Delta_j$, which is (S71) because $(K_{\mathrm{a}})_{ij} = (K_{ij} - K_{ji}^\top)/2$. The diagonal blocks of $K_{\mathrm{a}}$ vanish since each $K_{ii}$ is symmetric. If $K_{\mathrm{a}} = 0$, the criterion (S25) holds on every finite restriction, which is then an exact potential game and reversible by Theorem S5.17(a) and Theorem 5.10(b). If $(K_{\mathrm{a}})_{ij} \neq 0$, choose $\Delta_i, \Delta_j$ with $\Delta_i^\top (K_{\mathrm{a}})_{ij} \Delta_j \neq 0$; the restriction to $\mathcal{U}_i = \{0, \Delta_i\}$, $\mathcal{U}_j = \{0, \Delta_j\}$ and $\mathcal{U}_k = \{0\}$ otherwise has a four-cycle with ratio $e^{-\gamma \mathcal{L}_{ij}} \neq 1$ and is not reversible.

(b) Sufficiency. Let $K = K^\top \succ 0$, $\Phi(u) = \frac{1}{2} u^\top K u + \mathsf{h}^\top u$ and $\mu \propto e^{-\gamma \Phi}$, that is $\mu = \mathcal{N}(u^{\mathrm{N}}, (\gamma K)^{-1})$. By symmetry of $K$ and (S136), $\Phi - G_i^{\mathrm{act}}$ does not depend on $u_i$, so the conditional law of $u_i$ given $u_{-i}$ under $\mu$ is $p_i(\cdot \mid u_{-i})$. The measure $\mu(\mathrm{d}u) T_i(u, \mathrm{d}v)$ then has, on $\{u_{-i} = v_{-i}\}$, the density $\mu_{-i}(u_{-i}) p_i(u_i \mid u_{-i}) p_i(v_i \mid u_{-i})$, which is symmetric in $(u_i, v_i)$. Each $T_i$, and so $T$, is reversible with respect to $\mu$. For uniqueness, run two copies with the same selected member and the same $\xi$. By (S137) their difference obeys $e^+ = \mathsf{A}_i e$, and $\mathsf{A}_i$ is the projector along range $E_i$ that is orthogonal in the inner product of $K$, so $\|\mathsf{A}_i e\|_K^2 = \|e\|_K^2 - (E_i^\top K e)^\top K_{ii}^{-1} (E_i^\top K e)$ and

$$\mathbb{E}[\|e^+\|_K^2 \mid e] = \|e\|_K^2 - \frac{1}{n} (Ke)^\top K_{\mathrm{d}}^{-1} (Ke) \leq \Big(1 - \frac{\kappa}{n}\Big) \|e\|_K^2, \tag{S138}$$

because $(Ke)^\top K_{\mathrm{d}}^{-1} Ke = w^\top K^{1/2} K_{\mathrm{d}}^{-1} K^{1/2} w$ with $w = K^{1/2} e$, and $K^{1/2} K_{\mathrm{d}}^{-1} K^{1/2}$ has the spectrum of $K_{\mathrm{d}}^{-1/2} K K_{\mathrm{d}}^{-1/2}$, with smallest eigenvalue $\kappa > 0$. If $\nu$ is stationary, start one copy in $\nu$ and an independent one in $\mu$; then $e_t \to 0$ in probability while the two marginal laws stay $\nu$ and $\mu$, so $\nu = \mu$.

Necessity. Let $\mu$ be a probability law with $\mathfrak{M} = \mu \otimes T$ invariant under the swap $(u, v) \mapsto (v, u)$, and put $\mathfrak{M}_i = \mu \otimes T_i$, $S_i = \{(u, v) : u_{-i} = v_{-i}, u_i \neq v_i\}$. The sets $S_i$ are swap invariant and pairwise disjoint; $\mathfrak{M}_i$ is carried by $S_i$, since $p_i$ has a density, and gives no mass to $S_j$, $j \neq i$. Hence $\frac{1}{n}\mathfrak{M}_i$ is the restriction of $\mathfrak{M}$ to $S_i$ and is swap invariant, so its two marginals agree: $\mu T_i = \mu$ for every $i$. It follows that under $\mu$ the conditional law of $u_i$ given $u_{-i}$ is $p_i(\cdot \mid u_{-i})$, and that $\mu = \mu T_1 \cdots T_n$ has the density $m(v) = \int \mu(\mathrm{d}u) \prod_i p_i(v_i \mid v_1, \ldots, v_{i-1}, u_{i+1}, \ldots, u_n)$, which is positive everywhere. For each $i$ there is therefore a measurable $c_i$ with

$$\ln m(u) = -\gamma G_i^{\mathrm{act}}(u) + c_i(u_{-i}) \qquad \text{for almost every } u. \tag{S139}$$

Fix $i \neq j$ and displacements $\Delta_i, \Delta_j$, and let $\mathsf{D} f(u) = f(u + \Delta_i + \Delta_j) - f(u + \Delta_i) - f(u + \Delta_j) + f(u)$. Null sets are translation invariant, so for almost every $u$ the identity (S139) holds for $i$ and for $j$ at all four points. A function that does not depend on $u_i$, or does not depend on $u_j$, is annihilated by $\mathsf{D}$. Therefore $-\gamma \Delta_i^\top K_{ij} \Delta_j = \mathsf{D} \ln m(u) = -\gamma \Delta_i^\top K_{ji}^\top \Delta_j$ for all displacements, and $K_{ij} = K_{ji}^\top$: $K$ is symmetric. Then $\Phi - G_i^{\mathrm{act}}$ does not depend on $u_i$, and $h = \ln m + \gamma \Phi$ agrees almost everywhere with a function of $u_{-i}$, for every $i$. Let $g = \arctan h$ and let $\mathsf{P}_i$ integrate out $u_i$ against a standard Gaussian law; by Fubini's theorem $\mathsf{P}_i$ respects equality almost everywhere, and $\mathsf{P}_i g = g$ almost everywhere. Hence $g = \mathsf{P}_n \cdots \mathsf{P}_1 g$ almost everywhere, and the right side is a constant. So $m$ is proportional to $e^{-\gamma\Phi}$. If $w^\top K w \leq 0$ for some $w \neq 0$, then $s \mapsto \Phi(u + sw)$ is concave for every $u$, $\int e^{-\gamma\Phi(u+sw)} \mathrm{d}s = \infty$, and $e^{-\gamma\Phi}$ is not integrable. Hence $K \succ 0$ and $\mu = \mathcal{N}(u^{\mathrm{N}}, (\gamma K)^{-1})$.

(c) The solution is $u_t = e^{-Kt} u_0 - \int_0^t e^{-Kr} \mathsf{h} \mathrm{d}r + \sqrt{2/\gamma} \int_0^t e^{-K(t-r)} \mathrm{d}\mathsf{W}_r$. If every eigenvalue of $K$ has positive real part, $e^{-Kt} \to 0$ exponentially, $\Sigma = (2/\gamma) \int_0^\infty e^{-Kr} e^{-K^\top r} \mathrm{d}r$ converges, is positive definite, and satisfies $K\Sigma + \Sigma K^\top = -(2/\gamma) \int_0^\infty \frac{\mathrm{d}}{\mathrm{d}r}(e^{-Kr} e^{-K^\top r}) \mathrm{d}r = (2/\gamma) I$. The solution of this equation is unique: a solution $X$ of $KX + XK^\top = 0$ has $e^{-Kr} X e^{-K^\top r}$ constant in $r$ and tending to zero. Given $u_0$, the law of $u_t$ is Gaussian with mean tending to $u^{\mathrm{N}} = -K^{-1}\mathsf{h}$ and covariance tending to $\Sigma$, so $\mathcal{N}(u^{\mathrm{N}}, \Sigma)$ is the unique stationary law. Conversely, let $\lambda$ be an eigenvalue with $\operatorname{Re}\lambda \leq 0$ and $w \neq 0$ a left eigenvector, $w^* K = \lambda w^*$. Then $y_t = w^* u_t = a_t(u_0) + \mathsf{N}_t$, where $a_t$ is a function of $u_0$ and $\mathsf{N}_t = \sqrt{2/\gamma} \int_0^t e^{-\lambda(t-r)} w^* \mathrm{d}\mathsf{W}_r$ is a centered complex Gaussian variable independent of $u_0$ with $\mathbb{E}|\mathsf{N}_t|^2 = (2/\gamma)|w|^2 \int_0^t e^{-2r \operatorname{Re}\lambda} \mathrm{d}r \geq (2/\gamma)|w|^2 t$. One of its real and imaginary parts has variance at least $|w|^2 t/\gamma$, and a real variable $a + X$ with $X \sim \mathcal{N}(0, \sigma^2)$ independent of $a$ has $\mathbb{P}(|a + X| \leq R) \leq 2R/(\sigma\sqrt{2\pi})$. Under a stationary law the numbers $\mathbb{P}(|\operatorname{Re} y_t| \leq R)$ and $\mathbb{P}(|\operatorname{Im} y_t| \leq R)$ do not depend on $t$, and the smaller of them would be at most $2R\sqrt{\gamma}/(|w|\sqrt{2\pi t})$ for every $t$, hence zero for every $R$, which is impossible. So there is no stationary law.

Let now $\mathsf{Q} = K\Sigma - \gamma^{-1} I$. Then $\mathsf{Q} + \mathsf{Q}^\top = K\Sigma + \Sigma K^\top - (2/\gamma) I = 0$ and $K = (\gamma^{-1} I + \mathsf{Q})\Sigma^{-1}$. The Fokker–Planck equation is $\partial_t p = -\nabla \cdot (p\mathsf{v})$ with current velocity $\mathsf{v} = -K(u - u^{\mathrm{N}}) - \gamma^{-1} \nabla \ln p$; at the stationary density $\nabla \ln p = -\Sigma^{-1}(u - u^{\mathrm{N}})$ and $\mathsf{v} = -(K - \gamma^{-1}\Sigma^{-1})(u - u^{\mathrm{N}}) = -\mathsf{Q}\Sigma^{-1}(u - u^{\mathrm{N}})$. Its mean square is $\operatorname{tr}(\mathsf{Q}\Sigma^{-1}\mathsf{Q}^\top) \geq 0$, which vanishes only for $\mathsf{Q} = 0$. Expanding $K\Sigma K^\top = (\gamma^{-1} I + \mathsf{Q})\Sigma^{-1}(\gamma^{-1} I + \mathsf{Q})^\top$ and using $\operatorname{tr}(\mathsf{Q}\Sigma^{-1}) = 0$ gives $\operatorname{tr}(K\Sigma K^\top) = \gamma^{-2} \operatorname{tr}\Sigma^{-1} + \operatorname{tr}(\mathsf{Q}\Sigma^{-1}\mathsf{Q}^\top)$ and $\operatorname{tr} K = \gamma^{-1} \operatorname{tr}\Sigma^{-1}$, which is (S72). The quantity $\gamma\mathbb{E}|\mathsf{v}|^2$ is what is usually called the entropy production rate of the diffusion; only its vanishing is used here. The stationary process is Gaussian with $\operatorname{Cov}(u_\tau, u_0) = C(\tau) = e^{-K\tau}\Sigma$ for $\tau \geq 0$, and time reversal replaces $C(\tau)$ by $C(\tau)^\top$, so the process is reversible if and only if $C(\tau)$ is symmetric for all $\tau$. Differentiating at $\tau = 0$, this forces $K\Sigma$ symmetric, that is $\mathsf{Q} = 0$; conversely $\mathsf{Q} = 0$ gives $K\Sigma = \Sigma K^\top$, hence $K^k \Sigma = \Sigma (K^\top)^k$ for all $k$ and $C(\tau)$ symmetric. Finally $\mathsf{Q} = 0$ means $K = (\gamma\Sigma)^{-1}$, which is symmetric, and if $K$ is symmetric

then $(\gamma K)^{-1}$ solves the Lyapunov equation, so $\Sigma = (\gamma K)^{-1}$ and $\mathsf{Q} = 0$. □

**Lemma S41.1** (Moments of heat-bath deliberation)**.** *In the setting of Definition S20.1 let $K$ be invertible, $\bar{\mathsf{A}} = \frac{1}{n}\sum_i \mathsf{A}_i = I - \frac{1}{n}K_{\mathrm{d}}^{-1}K$ and $\mathcal{T}(S) = \frac{1}{n}\sum_i \mathsf{A}_i S \mathsf{A}_i^\top$ on symmetric matrices. Under heat-bath deliberation the mean $\mathsf{m}_t$ and the second moment $S_t = \mathbb{E}(u_t - u^{\mathrm{N}})(u_t - u^{\mathrm{N}})^\top$ obey*

$$\mathsf{m}_{t+1} - u^{\mathrm{N}} = \bar{\mathsf{A}}(\mathsf{m}_t - u^{\mathrm{N}}), \qquad S_{t+1} = \mathcal{T}(S_t) + \frac{1}{n\gamma}K_{\mathrm{d}}^{-1}. \tag{S140}$$

*Every stationary law with a finite first moment has mean $u^{\mathrm{N}}$. A stationary law with finite second moments exists if and only if the spectral radius of $\mathcal{T}$ is below one, and its second moment is then the unique fixed point of (S140).*

*Proof.* Since $E_i^\top K u^{\mathrm{N}} = -\mathsf{h}_i$, (S137) reads $u^+ - u^{\mathrm{N}} = \mathsf{A}_i(u - u^{\mathrm{N}}) + E_i\xi$, and averaging over the selected member and over $\xi$ gives (S140), with $\frac{1}{n}\sum_i E_i(\gamma K_{ii})^{-1}E_i^\top = (n\gamma)^{-1}K_{\mathrm{d}}^{-1}$. A stationary mean satisfies $K_{\mathrm{d}}^{-1}K(\mathsf{m} - u^{\mathrm{N}}) = 0$, so $\mathsf{m} = u^{\mathrm{N}}$. The map $\mathcal{T}$ and its adjoint $\mathcal{T}^*(X) = \frac{1}{n}\sum_i \mathsf{A}_i^\top X \mathsf{A}_i$ for the trace inner product preserve the cone of positive semidefinite matrices, which is closed, convex, pointed, self-dual and has nonempty interior. A linear map of a finite-dimensional space that preserves such a cone has its spectral radius $\mathsf{r}$ as an eigenvalue with an eigenvector in the cone (the finite-dimensional Krein–Rutman theorem). Let $X \succeq 0$, $X \neq 0$, with $\mathcal{T}^*X = \mathsf{r}X$. If a stationary second moment $S$ exists, pairing $S = \mathcal{T}(S) + (n\gamma)^{-1}K_{\mathrm{d}}^{-1}$ with $X$ gives $(1 - \mathsf{r})\operatorname{tr}(XS) = (n\gamma)^{-1}\operatorname{tr}(XK_{\mathrm{d}}^{-1}) > 0$, so $\mathsf{r} < 1$. If $\mathsf{r} < 1$, the fixed point $S_\infty = (n\gamma)^{-1}\sum_{k\geq 0}\mathcal{T}^k(K_{\mathrm{d}}^{-1})$ exists and is unique, and from the point mass at $u^{\mathrm{N}}$ the second moments increase to $S_\infty$, so the laws $\nu_t$ are tight. The kernel $T$ maps bounded continuous functions to bounded continuous functions by (S137), so a weak limit point of the time averages of $\nu_t$ is stationary, and its second moment is finite by lower semicontinuity and equals $S_\infty$ by uniqueness. □

*Proof of Proposition S20.3.* Langevin deliberation. A real $2 \times 2$ matrix has both eigenvalues in the open right half plane if and only if its trace and determinant are positive. Here $\operatorname{tr} K = K_{11} + K_{22} > 0$ and $\det K = K_{11}K_{22}(1 - \chi)$, so by Theorem S20.2(c) a stationary law exists if and only if $\chi < 1$.

Heat-bath deliberation. Write the two updates as $u_1^+ = a_1u_2 + \beta_1 + \xi_1$ and $u_2^+ = a_2u_1 + \beta_2 + \xi_2$, with $a_1 = -K_{12}/K_{11}$, $a_2 = -K_{21}/K_{22}$, $\beta_i = -\mathsf{h}_i/K_{ii}$ and $\xi_i \sim \mathcal{N}(0, (\gamma K_{ii})^{-1})$, so that $a_1a_2 = \chi$. Let $\mu$ be a stationary law, with no moment assumption, and $\nu_1, \nu_2$ its marginals. The first marginal of $\mu T_2$ is $\nu_1$ and that of $\mu T_1$ is the law of $a_1U_2 + \beta_1 + \xi_1$ with $U_2 \sim \nu_2$, so $\mu = \frac{1}{2}\mu T_1 + \frac{1}{2}\mu T_2$ gives $\nu_1 = \operatorname{law}(a_1U_2 + \beta_1 + \xi_1)$, and in the same way $\nu_2 = \operatorname{law}(a_2U_1 + \beta_2 + \xi_2)$. For the characteristic functions $\varphi_i$ of $\nu_i$ this reads $\varphi_1(t) = e^{\mathrm{i}t\beta_1}\varphi_2(a_1t)e^{-t^2/(2\gamma K_{11})}$ and $\varphi_2(t) = e^{\mathrm{i}t\beta_2}\varphi_1(a_2t)e^{-t^2/(2\gamma K_{22})}$, hence

$$\varphi_1(t) = e^{\mathrm{i}t(\beta_1 + a_1\beta_2)}\varphi_1(\chi t)e^{-\sigma_1^2t^2/2}, \qquad \sigma_1^2 = \frac{1}{\gamma K_{11}} + \frac{a_1^2}{\gamma K_{22}} > 0. \tag{S141}$$

If $|\chi| \geq 1$, iterating (S141) $k$ times gives $|\varphi_1(t)| \leq \exp(-\sigma_1^2t^2\sum_{j<k}\chi^{2j}/2) \leq e^{-k\sigma_1^2t^2/2}$ for every $k$, so $\varphi_1(t) = 0$ for $t \neq 0$, which contradicts continuity at $\varphi_1(0) = 1$. There is no stationary law. If $|\chi| < 1$, the same iteration and $\varphi_1(\chi^k t) \to 1$ give $\varphi_1(t) = \exp(\mathrm{i}t\mathsf{m}_1 - s_1t^2/2)$ with $\mathsf{m}_1 = (\beta_1 + a_1\beta_2)/(1 - \chi)$ and $s_1 = \sigma_1^2/(1 - \chi^2)$: the marginals are $\nu_i = \mathcal{N}(\mathsf{m}_i, s_i)$, uniquely. The law $\mu T_1$ depends on $\mu$ only through $\nu_2$ and is the Gaussian law $\mathsf{G}_1$ of $(a_1U_2 + \beta_1 + \xi_1, U_2)$, with mean $(\mathsf{m}_1, \mathsf{m}_2)$ and covariance $\left(\begin{smallmatrix} s_1 & a_1s_2 \\ a_1s_2 & s_2 \end{smallmatrix}\right)$; likewise $\mu T_2 = \mathsf{G}_2$ with the same mean and covariance $\left(\begin{smallmatrix} s_1 & a_2s_1 \\ a_2s_1 & s_2 \end{smallmatrix}\right)$. So $\mu = \frac{1}{2}\mathsf{G}_1 + \frac{1}{2}\mathsf{G}_2$ is the only candidate, and it is stationary because its marginals are $\nu_1$ and $\nu_2$, whence $\mu T_i = \mathsf{G}_i$. The mean satisfies $\mathsf{m}_1 = a_1\mathsf{m}_2 + \beta_1$ and $\mathsf{m}_2 = a_2\mathsf{m}_1 + \beta_2$, that is $K\mathsf{m} = -\mathsf{h}$, so $\mathsf{m} = u^{\mathrm{N}}$. An equal mixture of two Gaussian laws with one mean and different covariances is not Gaussian: along a

direction in which the variances $v_1 \neq v_2$ differ, the fourth central moment $3(v_1^2+v_2^2)/2$ exceeds the Gaussian value $3(v_1+v_2)^2/4$. The covariances agree if and only if $a_1 s_2 = a_2 s_1$, and

$$\gamma(1-\chi^2)(a_1 s_2 - a_2 s_1) = \frac{a_1}{K_{22}} + \frac{a_1 a_2^2}{K_{11}} - \frac{a_2}{K_{11}} - \frac{a_2 a_1^2}{K_{22}} = (1-\chi)\frac{K_{21}-K_{12}}{K_{11}K_{22}},$$

which vanishes if and only if $K_{12} = K_{21}$.

(a) The matrix is realized by $\mathsf{M}_1 = e_1$, $\mathsf{M}_2 = e_2$ in $\mathbb{R}^2$, shared by both members, with $\Lambda_1^\star = \left(\begin{smallmatrix} 1 & c \\ c & \lambda \end{smallmatrix}\right)$, $\Lambda_2^\star = \left(\begin{smallmatrix} \lambda & -c \\ -c & 1 \end{smallmatrix}\right)$, $\lambda > c^2$, and $\mathsf{z}_i = 0$: the members agree on the effects and disagree on the sign of the correlation they prefer between the two outcomes. Here $\chi = -c^2$. The eigenvalues of $K$ are $1 \pm \mathrm{i}c$, $K + K^\top = 2I$ shows that $\Sigma = \gamma^{-1} I$ solves the Lyapunov equation, $\mathsf{Q} = \gamma^{-1}\left(\begin{smallmatrix} 0 & c \\ -c & 0 \end{smallmatrix}\right)$, and $\gamma \operatorname{tr}(\mathsf{Q}\Sigma^{-1}\mathsf{Q}^\top) = \gamma^2 \operatorname{tr}(\mathsf{Q}\mathsf{Q}^\top) = 2c^2$. For the heat bath, $|\chi| < 1$ means $|c| < 1$, $a_1 = -c$, $a_2 = c$, $\sigma_1^2 = \sigma_2^2 = (1+c^2)/\gamma$ and $s_1 = s_2 = (1+c^2)/(\gamma(1-c^4)) = s$, which gives the two components. Their average covariance is $sI$: the acts are uncorrelated and dependent, with $\mathbb{E}u_1^2 u_2^2 = s^2(1+2c^2)$ against $s^2$ for independent acts. For the moments, $\mathsf{A}_1 = \left(\begin{smallmatrix} 0 & -c \\ 0 & 1 \end{smallmatrix}\right)$ and $\mathsf{A}_2 = \left(\begin{smallmatrix} 1 & 0 \\ c & 0 \end{smallmatrix}\right)$, so $\bar{\mathsf{A}} = \frac{1}{2}\left(\begin{smallmatrix} 1 & -c \\ c & 1 \end{smallmatrix}\right)$ has eigenvalues $(1 \pm \mathrm{i}c)/2$ of modulus $\sqrt{1+c^2}/2$, below one exactly for $c^2 < 3$, and on $S = \left(\begin{smallmatrix} s_1 & r \\ r & s_2 \end{smallmatrix}\right)$ the map $\mathcal{T}$ of Lemma S41.1 acts as $(s_1, s_2, r) \mapsto ((s_1 + c^2 s_2)/2, (c^2 s_1 + s_2)/2, c(s_1 - s_2)/2)$, with eigenvalues $(1+c^2)/2$, $(1-c^2)/2$ and 0. In particular $\operatorname{tr} S_{t+1} = \frac{1+c^2}{2} \operatorname{tr} S_t + \gamma^{-1}$ by (S140), so for $c^2 \geq 1$ the variance grows at least as $t/\gamma$ from every initial law with finite second moments, while for $c^2 < 3$ the mean converges to $u^{\mathrm{N}} = 0$.

(b) With the columns $\mathsf{M}_1 = (1, \alpha)^\top$ and $\mathsf{M}_2 = (\alpha, 1)^\top$, $K_{11} = \mathsf{M}_1^\top \Lambda_1^\star \mathsf{M}_1 = \epsilon + \alpha^2 = K_{22}$ and $K_{12} = \mathsf{M}_1^\top \Lambda_1^\star \mathsf{M}_2 = \alpha(1+\epsilon) = \mathsf{M}_2^\top \Lambda_2^\star \mathsf{M}_1 = K_{21}$. So $K$ is symmetric, the game is an exact potential game on every finite restriction by Theorem S20.2(a), and $\chi = \alpha^2(1+\epsilon)^2/(\epsilon+\alpha^2)^2$. For $\alpha > 0$, $\chi > 1$ if and only if $\alpha(1+\epsilon) > \epsilon + \alpha^2$, that is $(1-\alpha)(\alpha - \epsilon) > 0$, that is $\epsilon < \alpha < 1$. At $\alpha = 0.5$, $\epsilon = 0.1$, $K = \left(\begin{smallmatrix} 0.35 & 0.55 \\ 0.55 & 0.35 \end{smallmatrix}\right)$ has eigenvalues $-0.2$ and $0.9$ and $\chi = 0.3025/0.1225 > 1$; by the first part neither protocol has a stationary law. The potential $\frac{1}{2}u^\top K u + \mathsf{h}^\top u$ is unbounded below along $u_1 = -u_2$: member 1 weights the second outcome, which member 2's act moves most, and answers each move of member 2 by a larger move of its own. With shared precision and effect models $K = \mathsf{M}^\top \Lambda^\star \mathsf{M}$ is a Gram matrix, so $0 \leq \chi \leq 1$ by the Cauchy–Schwarz inequality, with $\chi = 1$ only for collinear effect directions; for $\mathsf{M}$ of full column rank $K \succ 0$ and both protocols have the stationary law $\mathcal{N}(u^{\mathrm{N}}, (\gamma K)^{-1})$. □

*Proof of Proposition S20.4.* Here $K_{ij} = \mathsf{M}_i^\top \Lambda^\star \mathsf{M}_j$, so $K = \mathsf{M}^\top \Lambda^\star \mathsf{M}$ is symmetric and positive definite. By Theorem S20.2(b) the heat bath has the unique stationary law $\mathcal{N}(u^{\mathrm{N}}, (\gamma K)^{-1})$, and by Theorem S20.2(c) so has Langevin deliberation, since $\Sigma = (\gamma K)^{-1}$ solves the Lyapunov equation. With $G_\Sigma = \sum_i G_i^{\mathrm{act}}$ and $G_i^{\mathrm{act}}(u) = \frac{1}{2}\|\mathsf{z}_i + \mathsf{M}u\|_{\Lambda^\star}^2$, the identity $\sum_i \|\mathsf{z}_i + v\|^2 = n\|\bar{\mathsf{z}} + v\|^2 + \sum_i \|\mathsf{z}_i - \bar{\mathsf{z}}\|^2$ gives $G_\Sigma(u) = \frac{n}{2}\|\bar{\mathsf{z}} + \mathsf{M}u\|_{\Lambda^\star}^2 + \text{const}$, a quadratic function with Hessian $nK$ and minimizer $u^{\mathrm{C}} = -K^{-1}\mathsf{M}^\top \Lambda^\star \bar{\mathsf{z}}$. Hence $G_\Sigma(u) - \min G_\Sigma = \frac{n}{2}(u - u^{\mathrm{C}})^\top K (u - u^{\mathrm{C}})$ and

$$\mathbb{E}_\mu G_\Sigma - \min G_\Sigma = \frac{n}{2}(u^{\mathrm{N}} - u^{\mathrm{C}})^\top K (u^{\mathrm{N}} - u^{\mathrm{C}}) + \frac{n}{2}\operatorname{tr}(K(\gamma K)^{-1}).$$

The last term is $n d_u/(2\gamma)$. Block $i$ of $K(u^{\mathrm{N}} - u^{\mathrm{C}})$ is $-\mathsf{h}_i + \mathsf{M}_i^\top \Lambda^\star \bar{\mathsf{z}} = -\mathsf{w}_i$, so the first term is $\frac{n}{2}\mathsf{w}^\top K^{-1}\mathsf{w}$, which is (S73). A single agent that samples $u \propto e^{-\gamma G_\Sigma}$ has the law $\mathcal{N}(u^{\mathrm{C}}, (\gamma n K)^{-1})$ and pays $\frac{n}{2}\operatorname{tr}(K(\gamma n K)^{-1}) = d_u/(2\gamma)$.

For the group model, $\mathsf{z}_{\mathrm{N}} = -\mathsf{M}u^{\mathrm{N}}$ satisfies $\mathsf{M}_j^\top \Lambda^\star \mathsf{z}_{\mathrm{N}} = -(K u^{\mathrm{N}})_j = \mathsf{h}_j = \mathsf{M}_j^\top \Lambda^\star \mathsf{z}_j$. In (S26) with $\mathsf{z}_C = \mathsf{z}_{\mathrm{N}}$ the own-act terms $\langle \mathsf{z}_j - \mathsf{z}_{\mathrm{N}}, \mathsf{M}_j u_j \rangle_{\Lambda^\star}$ therefore vanish, the potential is the group act part $\frac{1}{2}\|\mathsf{z}_{\mathrm{N}} + \mathsf{M}u\|_{\Lambda^\star}^2 = \frac{1}{2}(u - u^{\mathrm{N}})^\top K (u - u^{\mathrm{N}})$, and $\mu \propto e^{-\gamma G_C^{\mathrm{act}}}$ is the group posterior with flat habits. A displacement $\mathsf{z}$ gives the same group posterior if and only if $\mathsf{M}^\top \Lambda^\star \mathsf{z} = \mathsf{h}$, and $\mathsf{z}_{\mathrm{N}}$ is the only such displacement in the range of $\mathsf{M}$.

For two scalar acts write $\mathsf{M}_i = \mathsf{m}_i e_i$ with $\mathsf{m}_i > 0$ and $\|e_i\|_{\Lambda^\star} = 1$, and $\mathsf{d} = (\mathsf{z}_1 - \mathsf{z}_2)/2 = \mathsf{z}_1 - \bar{\mathsf{z}} = -(\mathsf{z}_2 - \bar{\mathsf{z}})$. Then $K = \mathsf{DRD}$ with $\mathsf{D} = \operatorname{diag}(\mathsf{m}_1, \mathsf{m}_2)$ and $\mathsf{R} = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{smallmatrix}\right)$, and $\mathsf{w} = \mathsf{D}(\eta_1, -\eta_2)^\top$. Hence $\mathsf{w}^\top K^{-1}\mathsf{w} = (\eta_1, -\eta_2)\mathsf{R}^{-1}(\eta_1, -\eta_2)^\top = (\eta_1^2 + 2\cos\theta\eta_1\eta_2 + \eta_2^2)/\sin^2\theta$, which is (S74) since $n/2 = 1$. As $\theta \to 0$ the numerator tends to $4\eta_1^2$, so the bias diverges as $\sin^{-2}\theta$ unless the members agree along the common direction. □

*Proof of Theorem S20.5.* Let $\mathsf{R} = K_{\mathrm{d}}^{-1/2} K K_{\mathrm{d}}^{-1/2} \succ 0$, with identity diagonal blocks. Its eigenvalues $\lambda_k$ average to one, so $\kappa = \min_k \lambda_k \in (0, 1]$, and $x^\top \mathsf{R} x = \|\sum_i \mathsf{R}^{1/2} E_i x_i\|^2 \le n \sum_i \|\mathsf{R}^{1/2} E_i x_i\|^2 = n\|x\|^2$, so $\lambda_k \in [\kappa, n]$. The matrix $K^{1/2} K_{\mathrm{d}}^{-1} K^{1/2}$ has the same spectrum.

(a) If $W_{2,K}(\nu_0, \mu) = \infty$ there is nothing to prove. Otherwise take an optimal coupling $(u_0, u_0')$ of $\nu_0$ and $\mu$ and run both copies with the same selected members and the same $\xi$. Then $u_t' \sim \mu$ for all $t$ and, by (S138), $W_{2,K}(\nu_t, \mu)^2 \le \mathbb{E}\|u_t - u_t'\|_K^2 \le (1 - \kappa/n)^t W_{2,K}(\nu_0, \mu)^2$. By (S140) the mean error is multiplied at each event by $\bar{\mathsf{A}} = I - \frac{1}{n} K_{\mathrm{d}}^{-1} K$. Since $K^{1/2} \bar{\mathsf{A}} K^{-1/2} = I - \frac{1}{n} K^{1/2} K_{\mathrm{d}}^{-1} K^{1/2}$ is symmetric with eigenvalues $1 - \lambda_k/n \in [0, 1 - \kappa/n]$, $\bar{\mathsf{A}}$ is self-adjoint in the inner product of $K$ with norm $1 - \kappa/n$, attained on the eigenvector of the smallest $\lambda_k$.

(b) Put $x = (\gamma K)^{1/2}(u - u^{\mathrm{N}})$, so that $x \sim \mathcal{N}(0, I)$ under $\mu$. By (S137), an update of member $i$ reads $x^+ = \mathsf{Q}_i x + (\gamma K)^{1/2} E_i \xi$ with $\mathsf{Q}_i = K^{1/2} \mathsf{A}_i K^{-1/2} = I - K^{1/2} E_i K_{ii}^{-1} E_i^\top K^{1/2}$, the orthogonal projector onto the complement of $\operatorname{range}(K^{1/2} E_i)$, and the noise has covariance $I - \mathsf{Q}_i$. Hence $T_i f(x) = \mathbb{E} f(\mathsf{Q}_i x + (I - \mathsf{Q}_i)\zeta)$ with $\zeta \sim \mathcal{N}(0, I)$. For $\varepsilon_a(x) = \exp(a^\top x - \|a\|^2/2)$ this gives $T_i \varepsilon_a = \exp(a^\top \mathsf{Q}_i x - \|a\|^2/2 + \|(I - \mathsf{Q}_i)a\|^2/2) = \varepsilon_{\mathsf{Q}_i a}$. Decompose $L^2(\mathcal{N}(0, I)) = \bigoplus_{k \ge 0} \mathcal{H}_k$ into Wiener chaoses and let $\mathsf{I}_k$ be the map from symmetric $k$-tensors onto $\mathcal{H}_k$ with $\mathsf{I}_k(a^{\otimes k}) = \|a\|^k \mathrm{He}_k(a^\top x/\|a\|)$, for which $\langle \mathsf{I}_k(a^{\otimes k}), \mathsf{I}_k(b^{\otimes k}) \rangle = k! \langle a, b \rangle^k$ and $\varepsilon_{sa} = \sum_k s^k \mathsf{I}_k(a^{\otimes k})/k!$ in $L^2$. Comparing the coefficients of $s^k$ in $T_i \varepsilon_{sa} = \varepsilon_{s\mathsf{Q}_i a}$ gives $T_i \mathsf{I}_k(a^{\otimes k}) = \mathsf{I}_k((\mathsf{Q}_i a)^{\otimes k})$: every $T_i$ preserves each chaos, and $T$ is unitarily equivalent on $\mathcal{H}_k$ to the restriction of $\frac{1}{n}\sum_i \mathsf{Q}_i^{\otimes k}$ to symmetric tensors. On $\mathcal{H}_0$, the constants, it is the identity. On $\mathcal{H}_1$ it is $\frac{1}{n}\sum_i \mathsf{Q}_i = I - \frac{1}{n} K^{1/2} K_{\mathrm{d}}^{-1} K^{1/2}$, with largest eigenvalue $1 - \kappa/n$. For $k \ge 2$, $\mathsf{Q}_i^{\otimes k}$ is a product of the commuting projectors $\mathsf{Q}_i \otimes I \otimes \cdots \otimes I$, …, $I \otimes \cdots \otimes I \otimes \mathsf{Q}_i$, hence a projector below each of them, and $0 \preceq \frac{1}{n}\sum_i \mathsf{Q}_i^{\otimes k} \preceq (\frac{1}{n}\sum_i \mathsf{Q}_i) \otimes I \otimes \cdots \otimes I \preceq (1 - \kappa/n) I$. The spectrum of the self-adjoint operator $T$ on the complement of the constants therefore lies in $[0, 1 - \kappa/n]$ and contains $1 - \kappa/n$.

(c) For two members $\mathsf{R} = \left(\begin{smallmatrix} I & \mathsf{C} \\ \mathsf{C}^\top & I \end{smallmatrix}\right)$ with $\mathsf{C} = K_{11}^{-1/2} K_{12} K_{22}^{-1/2}$, whose eigenvalues are $1 \pm \sigma_k(\mathsf{C})$ and 1, so $\kappa = 1 - \sigma_{\max}(\mathsf{C})$. Under shared $\Lambda^\star$ and effect models, $\mathsf{U}_i = (\Lambda^\star)^{1/2} \mathsf{M}_i K_{ii}^{-1/2}$ has orthonormal columns spanning $\operatorname{range}((\Lambda^\star)^{1/2} \mathsf{M}_i)$ and $\mathsf{C} = \mathsf{U}_1^\top \mathsf{U}_2$, whose singular values are by definition the cosines of the principal angles between the two ranges.

(d) The convention is that at event $t$ the selected member responds to the loads $\mathsf{h}_t$, after which the loads move; $u_t^{\mathrm{N}} = -K^{-1}\mathsf{h}_t$ and $\mu_t = \mathcal{N}(u_t^{\mathrm{N}}, (\gamma K)^{-1})$. With $\mathsf{e}_t = \mathbb{E}_{\nu_t} u - u_t^{\mathrm{N}}$, (S140) gives $\mathsf{e}_{t+1} = \bar{\mathsf{A}} \mathsf{e}_t - (u_{t+1}^{\mathrm{N}} - u_t^{\mathrm{N}})$, so $\|\mathsf{e}_{t+1}\|_K \le (1 - \kappa/n)\|\mathsf{e}_t\|_K + \delta$ and $\|\mathsf{e}_t\|_K \le (1 - \kappa/n)^t \|\mathsf{e}_0\|_K + \delta \sum_{j<t} (1 - \kappa/n)^j$, which is the first bound. For a constant drift $\mathsf{v}$ per event the error converges to $-(I - \bar{\mathsf{A}})^{-1}\mathsf{v} = -nK^{-1}K_{\mathrm{d}}\mathsf{v}$; if $\mathsf{v}$ is an eigenvector of $\bar{\mathsf{A}}$ for $1 - \kappa/n$ with $\|\mathsf{v}\|_K = \delta$, this limit is $-(n/\kappa)\mathsf{v}$, of norm $n\delta/\kappa$, so the constant cannot be improved. For the laws, $\mu_t$ is stationary for the kernel $T^{(t)}$ with loads $\mathsf{h}_t$, the contraction of (a) applies to any pair of laws, and two Gaussian laws with equal covariances are at transport distance equal to the distance of their means. Hence

$$W_{2,K}(\nu_{t+1}, \mu_{t+1}) \le W_{2,K}(\nu_t T^{(t)}, \mu_t T^{(t)}) + W_{2,K}(\mu_t, \mu_{t+1}) \le (1 - \kappa/n)^{1/2} W_{2,K}(\nu_t, \mu_t) + \delta,$$

and the second bound follows from $1/(1 - \sqrt{1 - \kappa/n}) \le 2n/\kappa$. □

**Proposition S41.2** (Rotation does not slow Langevin deliberation)**.** *Let every eigenvalue of $K$ have positive real part, let $\Sigma$ solve $K\Sigma + \Sigma K^\top = (2/\gamma) I$ and $K_0 = (\gamma\Sigma)^{-1}$. Every eigenvalue $\lambda$ of*

*$K$ satisfies $\lambda_{\min}(K_0) \le \operatorname{Re}\lambda \le \lambda_{\max}(K_0)$. Conversely, for every $\Sigma \succ 0$ and antisymmetric $\mathsf{Q}$ the matrix $(\gamma^{-1}I + \mathsf{Q})\Sigma^{-1}$ has all eigenvalues in the open right half plane.*

*Proof.* Let $K = (\gamma^{-1}I + \mathsf{Q})\Sigma^{-1}$ with $\mathsf{Q}$ real antisymmetric, as in Theorem S20.2(c) or as given, $Kx = \lambda x$ with $x \neq 0$ complex, and $y = \Sigma^{-1}x$. Then $(\gamma^{-1}I+\mathsf{Q})y = \lambda\Sigma y$, and multiplying by $y^*$ gives $\gamma^{-1}\|y\|^2 + y^*\mathsf{Q}y = \lambda y^*\Sigma y$. The number $y^*\mathsf{Q}y$ is imaginary and $y^*\Sigma y > 0$, so $\operatorname{Re}\lambda = \gamma^{-1}\|y\|^2/(y^*\Sigma y)$, which lies between $1/(\gamma\lambda_{\max}(\Sigma))$ and $1/(\gamma\lambda_{\min}(\Sigma))$. □

Among the interaction matrices with one stationary law the potential one therefore has the slowest asymptotic decay of means and lagged covariances; the heat bath behaves in the opposite way in Proposition S20.3(a), where rotation raises the modulus $\sqrt{1+c^2}/2$ of the mean map.

*Proof of Proposition S20.6.* Write $p_i(\cdot \mid \pi_{-i}) \propto E_i e^{-\gamma G_i(\cdot,\pi_{-i})}$ for the conditionals and $d_{\mathrm{H}}$ for the Hamming distance. If $\pi_{-i}$ and $\pi'_{-i}$ differ only in member $j$'s policy, $p_i(\cdot \mid \pi'_{-i}) \propto p_i(\cdot \mid \pi_{-i})e^{g}$ with $g = -\gamma(G_i(\cdot,\pi'_{-i}) - G_i(\cdot,\pi_{-i}))$, of oscillation at most $\gamma\omega_{ij}$, and Lemma S29.2 gives $\mathfrak{J}_{ij} \le \tanh(\gamma\omega_{ij}/4)$.

One event from adjacent profiles. Let $\pi$ and $\pi'$ differ only in member $j$'s policy, select the same member $i$ in both copies, and draw the two new policies from a maximal coupling of $p_i(\cdot \mid \pi_{-i})$ and $p_i(\cdot \mid \pi'_{-i})$. If $i = j$ the two conditionals coincide and the copies coalesce. If $i \neq j$ the copies still differ at $j$ and differ at $i$ with probability at most $\mathfrak{J}_{ij}$. Hence the expected Hamming distance after the event is at most $\frac{1}{n}\sum_{i\neq j}(1+\mathfrak{J}_{ij}) \le 1 - \frac{1}{n} + \frac{\mathfrak{D}}{n} = \rho$.

One event from arbitrary laws. Two profiles at Hamming distance $k$ are joined by a path of $k$ adjacent steps, so by the triangle inequality $W_1(\delta_\pi T, \delta_{\pi'}T) \le \rho d_{\mathrm{H}}(\pi,\pi')$. For laws $\nu,\nu'$ with optimal coupling $\vartheta$, mixing optimal couplings of $\delta_\pi T$ and $\delta_{\pi'}T$ over $\vartheta$ gives a coupling of $\nu T$ and $\nu' T$, so $W_1(\nu T, \nu' T) \le \rho W_1(\nu,\nu')$. If $\mathfrak{D} < 1$ the stationary law is unique and $W_1(\nu_t, \mathbb{P}^\infty) \le \rho^t W_1(\nu_0,\mathbb{P}^\infty) \le n\rho^t$, the diameter being $n$; and TV $\le W_1$ because $d_{\mathrm{H}}(\pi,\pi') \ge \mathbf{1}[\pi \neq \pi']$. This is (S76).

Two profiles of expected free energies. Let $T'$ be the kernel of $G'$ and $\mathbb{P}'^\infty$ a stationary law of it. From one profile $\pi$, select the same member $i$ and couple $p_i(\cdot \mid \pi_{-i})$ and $p'_i(\cdot \mid \pi_{-i}) \propto p_i(\cdot \mid \pi_{-i})e^{-\gamma(G'_i - G_i)}$ maximally; by Lemma S29.2 the copies differ, in one coordinate, with probability at most $\tanh(\gamma\omega_i/4)$. So $W_1(\delta_\pi T, \delta_\pi T') \le \frac{1}{n}\sum_i \tanh(\gamma\omega_i/4) = \tau$ for every $\pi$, and

$$W_1(\mathbb{P}^\infty, \mathbb{P}'^\infty) \le W_1(\mathbb{P}^\infty T, \mathbb{P}'^\infty T) + W_1(\mathbb{P}'^\infty T, \mathbb{P}'^\infty T') \le \rho W_1(\mathbb{P}^\infty, \mathbb{P}'^\infty) + \tau,$$

which gives (S77) since $1 - \rho = (1-\mathfrak{D})/n$. Only the influence of one of the two chains enters.

Moving expected free energies. Let $T_t$ be the kernel of the profile $G_{\cdot,t}$ in force at event $t$, $\nu_{t+1} = \nu_t T_t$, $\mathfrak{D}_t \le \bar{\mathfrak{D}} < 1$ its influence constant and $\mathbb{P}^\infty_t$ its stationary law, which is unique because $T_t$ contracts $W_1$, with $\omega_{i,t} = \max_{\pi_{-i}} \operatorname{osc}_{\pi_i}(G_{i,t+1} - G_{i,t})$. The bound (S77), applied to the profiles at events $t$ and $t+1$ with the influence constant $\mathfrak{D}_t$ of the first, gives $W_1(\mathbb{P}^\infty_t, \mathbb{P}^\infty_{t+1}) \le \sum_i \tanh(\gamma\omega_{i,t}/4)/(1-\mathfrak{D}_t) \le n\bar\tau/(1-\bar{\mathfrak{D}})$, the kernel $T_t$ contracts $W_1$ by $1-(1-\mathfrak{D}_t)/n \le \bar\rho$, and $\mathbb{P}^\infty_t T_t = \mathbb{P}^\infty_t$, so

$$W_1(\nu_{t+1}, \mathbb{P}^\infty_{t+1}) \le W_1(\nu_t T_t, \mathbb{P}^\infty_t T_t) + W_1(\mathbb{P}^\infty_t, \mathbb{P}^\infty_{t+1}) \le \bar\rho W_1(\nu_t, \mathbb{P}^\infty_t) + \frac{n\bar\tau}{1-\bar{\mathfrak{D}}},$$

and summing the geometric series gives (S78). If each member's expected free energy moves by at most $\dot\omega/n$ in oscillation per event, that is $\dot\omega$ per round of $n$ events, then $\tanh(\gamma\omega_{i,t}/4) \le \gamma\dot\omega/(4n)$, so $\bar\tau$ can be taken to be $\gamma\dot\omega/(4n)$, and $W_1/n$, the expected fraction of members whose policy differs from that of a coupled draw of the current stationary law, is at most $\gamma\dot\omega/(4(1-\bar{\mathfrak{D}})^2)$ once the transient has decayed. Since $n\rho^t \le ne^{-t(1-\mathfrak{D})/n}$, the law is within $\epsilon$ of stationarity after $\ln(n/\epsilon)/(1-\mathfrak{D})$ rounds. □

*Proof of Proposition S20.7.* (a) If Langevin deliberation with matrix $K$ has the stationary law $\mathcal{N}(u^{\mathrm{N}}, \Sigma)$, then $K = (\gamma^{-1} I + \mathsf{Q})\Sigma^{-1}$ with $\mathsf{Q}$ antisymmetric by Theorem S20.2(c). Conversely, for antisymmetric $\mathsf{Q}$ the matrix $K = (\gamma^{-1} I + \mathsf{Q})\Sigma^{-1}$ satisfies $K\Sigma + \Sigma K^{\top} = (2/\gamma) I$ and has its spectrum in the open right half plane by Proposition S41.2, so with loads $\mathsf{h} = -K u^{\mathrm{N}}$ its stationary law is $\mathcal{N}(u^{\mathrm{N}}, \Sigma)$. The matrix $K$ is symmetric if and only if $\mathsf{Q}\Sigma^{-1} + \Sigma^{-1}\mathsf{Q} = 0$, that is $\Sigma\mathsf{Q} + \mathsf{Q}\Sigma = 0$, which forces $\mathsf{Q} = 0$ because the spectra of $\Sigma$ and $-\Sigma$ are disjoint. For realizability, a matrix $K$ whose diagonal blocks are symmetric and positive definite is the interaction matrix of a quadratic deliberation: with outcome dimension at least $\max_i d_i$ and isometries $\mathsf{V}_i$ of $\mathbb{R}^{d_i}$ into the outcome space, take $\Lambda_i^{\star} = I$, $\mathsf{M}_i^{(i)} = \mathsf{V}_i K_{ii}^{1/2}$ and $\mathsf{M}_j^{(i)} = \mathsf{V}_i K_{ii}^{-1/2} K_{ij}$, and $\mathsf{z}_i = \mathsf{V}_i K_{ii}^{-1/2} \mathsf{h}_i$. The diagonal blocks of $(\gamma^{-1} I + \mathsf{Q})\Sigma^{-1}$ are $\gamma^{-1}(\Sigma^{-1})_{ii} + (\mathsf{Q}\Sigma^{-1})_{ii}$; for scalar acts they are positive for all small $\mathsf{Q}$, and for vector acts the realizable small $\mathsf{Q}$ are those for which every $(\mathsf{Q}\Sigma^{-1})_{ii}$ is symmetric.

(b) The change of variable $u \mapsto u - u^{\mathrm{N}}$ maps the deliberation with loads $\mathsf{h}$ to the one with loads zero under either protocol, by (S137), so the existence of a stationary law does not depend on $\mathsf{h}$. A Langevin stationary law is $\mathcal{N}(u^{\mathrm{N}}, \Sigma)$ by Theorem S20.2(c), and a heat-bath stationary law with a finite first moment has mean $u^{\mathrm{N}}$ by Lemma S41.1; in both cases the mean is $u^{\mathrm{N}} = -K^{-1}\mathsf{h}$. Hence $-\partial \mathbb{E} u / \partial \mathsf{h} = K^{-1}$, which is symmetric if and only if $K$ is.

(c) For $\tau \geq 0$, $\mathrm{Cov}(u_\tau, u_0) = e^{-K\tau}\Sigma$, whose derivative at $\tau = 0$ is $-K\Sigma$. The symmetry statement is in the proof of Theorem S20.2(c), and $K^{-1} = \Sigma(\gamma^{-1} I + \mathsf{Q})^{-1} = \gamma\Sigma(I + \gamma\mathsf{Q})^{-1}$, where $I + \gamma\mathsf{Q}$ is invertible because $\mathsf{Q}$ has imaginary spectrum. □

The proofs for Section S20.2 rest on one lemma about Gaussian pairs, which is used at a stationary horizon in Theorem S20.8 and between two arbitrary times in Proposition S20.11. Recall $\iota(s) = -\frac{1}{2}\ln(1 - e^{-2s})$ on $(0, \infty)$, with $\iota(\infty) = 0$. Since $e^{-2s} + e^{-2\iota(s)} = 1$ is symmetric in $s$ and $\iota(s)$, $\iota$ is a decreasing involution of $(0, \infty)$, and $\iota(\theta) \geq c$ if and only if $\theta \leq \iota(c) = \varsigma(c)$.

**Lemma S41.3** (Information retained by a frame)**.** *Let $(\mathsf{x}, \mathsf{x}')$ be jointly Gaussian with $\mathsf{x} \in \mathbb{R}^N$, $S = \mathrm{Cov}(\mathsf{x}) \succ 0$ and $\mathrm{Cov}(\mathsf{x} \mid \mathsf{x}') = S^{1/2}(I - \mathsf{H})S^{1/2}$, where $\mathsf{H}$ is symmetric with orthonormal eigenvectors $\mathsf{v}_1, \dots, \mathsf{v}_N$ and eigenvalues $e^{-2\theta_1} \geq \cdots \geq e^{-2\theta_N}$, $0 < \theta_1 \leq \cdots \leq \theta_N \leq \infty$. Put $\theta_0 = 0$ and $\theta_{N+1} = \infty$.*

*(a) $I(\mathsf{x}; \mathsf{x}') = \sum_k \iota(\theta_k)$. For every $\Phi \in \mathbb{R}^{N \times M}$ with orthonormal columns, $I(\Phi^{\top}\mathsf{x}; \mathsf{x}') \leq \sum_{k \leq M} \iota(\theta_k)$, with equality if $S^{1/2} \operatorname{range} \Phi$ is spanned by $\mathsf{v}_1, \dots, \mathsf{v}_M$ and, if $\theta_M < \theta_{M+1}$, only then.*

*(b) For $c > 0$ let $\mathsf{j}(M) = \sum_{k > M} \iota(\theta_k) + cM$, which by (a) is the minimum of $I(\mathsf{x}; \mathsf{x}') - I(\Phi^{\top}\mathsf{x}; \mathsf{x}') + c \operatorname{rank} \Phi$ over the frames of rank $M$. A rank $M \in \{0, \dots, N\}$ minimizes $\mathsf{j}$ if and only if $\theta_M \leq \varsigma(c) \leq \theta_{M+1}$, it is the only minimizing rank if and only if both inequalities are strict, and $M^* = \#\{k : \theta_k < \varsigma(c)\}$ is always a minimizing rank.*

*Proof.* (a) Let $\Psi = S^{1/2}\Phi$, which has rank $M$, and write $\Psi = W\mathsf{R}_0$ with $W \in \mathbb{R}^{N \times M}$ having orthonormal columns and $\mathsf{R}_0$ invertible, so that $\operatorname{range} W = S^{1/2} \operatorname{range} \Phi$. Then $\mathrm{Cov}(\Phi^{\top}\mathsf{x}) = \Psi^{\top}\Psi$ and $\mathrm{Cov}(\Phi^{\top}\mathsf{x} \mid \mathsf{x}') = \Psi^{\top}(I - \mathsf{H})\Psi$, and the Gaussian formula $I = \frac{1}{2}\ln\det \mathrm{Cov}(\Phi^{\top}\mathsf{x}) - \frac{1}{2}\ln\det \mathrm{Cov}(\Phi^{\top}\mathsf{x} \mid \mathsf{x}')$ gives

$$I(\Phi^{\top}\mathsf{x}; \mathsf{x}') = -\tfrac{1}{2}\ln\det\bigl(W^{\top}(I - \mathsf{H})W\bigr).$$

The eigenvalues of $I - \mathsf{H}$ are $\mu_k = 1 - e^{-2\theta_k}$, with $0 < \mu_1 \leq \cdots \leq \mu_N \leq 1$ and $-\frac{1}{2}\ln\mu_k = \iota(\theta_k)$; the case $\Phi = I$ is the first claim. Let $\beta_1 \leq \cdots \leq \beta_M$ be the eigenvalues of $W^{\top}(I - \mathsf{H})W$. By the Courant–Fischer principle, $\beta_k$ is the minimum over the $k$-dimensional subspaces of $\operatorname{range} W$ of the largest value of the quadratic form of $I - \mathsf{H}$ on unit vectors of the subspace, and $\mu_k$ is the same minimum over all $k$-dimensional subspaces of $\mathbb{R}^N$, so $\beta_k \geq \mu_k$ and $\det(W^{\top}(I - \mathsf{H})W) \geq \prod_{k \leq M} \mu_k$,

which is the inequality. If range $W$ is spanned by $\mathsf{v}_1, \dots, \mathsf{v}_M$, then $W^\top(I - \mathsf{H})W$ is orthogonally similar to $\operatorname{diag}(\mu_1, \dots, \mu_M)$ and equality holds. Conversely, equality forces $\beta_k = \mu_k$ for every $k \le M$, hence $\operatorname{tr}(W^\top(I - \mathsf{H})W) = \sum_{k \le M} \mu_k$. With $p_j = \|W^\top \mathsf{v}_j\|^2 \in [0, 1]$ and $\sum_j p_j = M$,

$$\operatorname{tr}(W^\top(I - \mathsf{H})W) - \sum_{k \le M} \mu_k = \sum_{j > M} \mu_j p_j - \sum_{k \le M} \mu_k (1 - p_k) \ge (\mu_{M+1} - \mu_M) \sum_{j > M} p_j,$$

because $\sum_{k \le M}(1 - p_k) = \sum_{j > M} p_j$. If $\theta_M < \theta_{M+1}$ the factor $\mu_{M+1} - \mu_M$ is positive, so $p_j = 0$ for $j > M$: the range of $W$ is orthogonal to $\mathsf{v}_{M+1}, \dots, \mathsf{v}_N$ and, having dimension $M$, equals the span of $\mathsf{v}_1, \dots, \mathsf{v}_M$.

(b) $\mathsf{j}(M) - \mathsf{j}(M-1) = c - \iota(\theta_M)$ is nondecreasing in $M$ because $\iota$ decreases, so $\mathsf{j}$ is convex on $\{0, \dots, N\}$ and $M$ minimizes it if and only if $\mathsf{j}(M) - \mathsf{j}(M-1) \le 0 \le \mathsf{j}(M+1) - \mathsf{j}(M)$, each inequality being required when the difference is defined. This reads $\iota(\theta_M) \ge c \ge \iota(\theta_{M+1})$, that is $\theta_M \le \varsigma(c) \le \theta_{M+1}$. If one of the two is an equality, the neighboring rank has the same value; if both are strict, convexity makes $M$ the only minimizer. By its definition $M^*$ satisfies $\theta_{M^*} < \varsigma(c) \le \theta_{M^*+1}$. □

*Proof of Theorem S20.8.* In the stationary law $\operatorname{Cov}(x_0) = \operatorname{Cov}(x_\tau) = \Pi^{-1}$ and $x_\tau = e^{-\Pi\tau}x_0 + \zeta$ with $\zeta$ Gaussian and independent of $x_0$, so $\operatorname{Cov}(x_0, x_\tau) = \Pi^{-1}e^{-\Pi\tau}$ and

$$\operatorname{Cov}(x_0 \mid x_\tau) = \Pi^{-1} - \Pi^{-1}e^{-\Pi\tau}\Pi e^{-\Pi\tau}\Pi^{-1} = \Pi^{-1/2}(I - e^{-2\Pi\tau})\Pi^{-1/2}.$$

Lemma S41.3 applies with $S = \Pi^{-1}$, $\mathsf{H} = e^{-2\Pi\tau}$, $\theta_k = \varpi_k\tau$ and $\mathsf{v}_k$ the eigenvectors of $\Pi$.

(a) This is part (a) of the lemma, since $S^{1/2} \operatorname{range}\Phi = \Pi^{-1/2} \operatorname{range}\Phi$ equals $\operatorname{range}\mathsf{V}_\mathrm{s}$ if and only if $\operatorname{range}\Phi$ does, the slow eigenspace being invariant under $\Pi^{1/2}$.

(b) By (a), the minimum of $J_{\tau,c}$ over the frames of rank $M$ is $\mathsf{j}(M)$, attained on the frames with range $\operatorname{range}\mathsf{V}_\mathrm{s}$, so the minimum over all frames is $\min_M \mathsf{j}(M)$. Part (b) of the lemma gives (S80) and its uniqueness when no $\varpi_k\tau$ equals $\varsigma(c)$. If $\varsigma(c)/\varpi_{M+1} < \tau < \varsigma(c)/\varpi_M$, then $\varpi_M < \varpi_{M+1}$, $M$ is the only minimizing rank, and by (a) a frame of rank $M$ attains $\mathsf{j}(M)$ only if its range is the slow eigenspace. Conversely, if the minimizers are exactly the frames of rank $M$ on the slow eigenspace, then $M$ is the only minimizing rank, and both inequalities $\varpi_M\tau < \varsigma(c) < \varpi_{M+1}\tau$ are strict. The interval (S81) is nonempty exactly when $\varpi_M < \varpi_{M+1}$, and for $1 \le M < N$ the logarithms of its endpoints differ by $\ln(\varpi_{M+1}/\varpi_M)$, in which $c$ does not appear.

(c) If $\operatorname{range}\Phi$ is invariant, $\Pi$ is block diagonal in the decomposition $\operatorname{range}\Phi \oplus (\operatorname{range}\Phi)^\perp$, with blocks $\Pi_\Phi$ and $\Pi_{\perp\perp}$, so its spectrum is the union of theirs. With $\varkappa_\perp > \varkappa_\Phi$ the $M$ smallest eigenvalues are those of $\Pi_\Phi$: $\operatorname{range}\Phi = \operatorname{range}\mathsf{V}_\mathrm{s}$, $\varpi_M = \varkappa_\Phi$ and $\varpi_{M+1} = \varkappa_\perp$. By (a), $J_{\tau,c}(\Phi) = \mathsf{j}(M)$, so $\Phi$ minimizes over all frames if and only if $M$ is a minimizing rank, that is $\varkappa_\Phi\tau \le \varsigma(c) \le \varkappa_\perp\tau$. For an equitable partition with uniform leak the span of the block indicators is invariant (Lemma 13.3(a)), and, with $\varkappa_\mathrm{in}^{\min}$ the minimum over the blocks with at least two members (one exists because $\operatorname{rank}\Phi < N$), Theorem 13.8(d) gives $\varkappa_\perp \ge \varkappa_\mathrm{in}^{\min}$ and $0 < \varkappa_\Phi \le 2\varkappa_\mathrm{out}^{\max} - \varepsilon^\Pi$. Under $\varkappa_\mathrm{in}^{\min} > 2\varkappa_\mathrm{out}^{\max} - \varepsilon^\Pi$ the interval $[\varsigma(c)/\varkappa_\mathrm{in}^{\min}, \varsigma(c)/(2\varkappa_\mathrm{out}^{\max} - \varepsilon^\Pi)]$ is therefore contained in $[\varsigma(c)/\varkappa_\perp, \varsigma(c)/\varkappa_\Phi]$. □

For $\Pi = \varepsilon^\Pi I + L$ the eigenvalues are $\varpi_k = \varepsilon^\Pi + \lambda_k(L)$, so the condition $\varpi_k < \varsigma(c)/\tau$ of (S80) is $\lambda_k(L) < \varsigma(c)/\tau - \varepsilon^\Pi$, the strict mode rule of Section S13.6 at $1/\tau_\mathrm{L} = \varsigma(c)/\tau - \varepsilon^\Pi$. A mode with $\varpi_k\tau = \varsigma(c)$ can be kept or discarded at no change of the criterion (Lemma S41.3(b)), so the convention at equality does not affect optimality. Since $\varsigma(\iota(1)) = 1$, the price $c = \iota(1) = -\frac{1}{2}\ln(1 - e^{-2}) \approx 0.0727$ makes $\tau_\mathrm{L} = \tau$ at zero leak, and for $\tau \ge \varsigma(c)/\varepsilon^\Pi$ the right-hand side is not positive and nothing is retained. The reading of the diffusion time as a prediction horizon and the optimality statement are ours and concern the diffusion of Theorem 13.8; we do not attribute them to Villegas et al. (2023).

*Remark* S41.4 (The closure defect does not select a scale). Let a criterion have the form $\mathsf{f}(\mathcal{D})+c\,\operatorname{rank}\Phi$ with $\mathsf{f}\geq 0$ and $\mathsf{f}(0)=0$, or the same with the memory kernel $\Gamma$ of Theorem 13.8(b) in place of $\mathcal{D}$. Both $\mathcal{D}$ and $\Gamma$ vanish on every frame with invariant range (Lemma 13.3(a); $\Pi_{\perp\Phi}=0$). The empty frame is one, so it attains the least possible value 0 and the criterion selects nothing at every price. Restricted to a fixed rank $M\geq 1$ the criterion is minimized by every frame spanned by $M$ eigenvectors of $\Pi$, the $M$ fastest included, although by Theorem S20.8(a) these retain the least predictive information among invariant frames. In the nested medium every level has $\mathcal{D}=0$ (Theorem 13.6(a)), so the defect is silent about which level to use. The closure defect measures whether a description is self-contained, which is a different property from being worth keeping, and this is why (S79) uses predictive information. The statement is exact; the reading of the two properties is ours.

*Proof of Theorem S20.9.* Let $\mathsf{P}_l$, $0\leq l\leq n_{\mathrm{H}}$, be the orthogonal projection of $\mathbb{R}^N$ onto the vectors that are constant on the level-$l$ blocks, $(\mathsf{P}_l)_{ij}=b^{-l}$ if $i$ and $j$ lie in one level-$l$ block and 0 otherwise, so that $\mathsf{P}_0=I$, $\operatorname{rank}\mathsf{P}_l=b^{n_{\mathrm{H}}-l}$ and $\mathsf{P}_l\mathsf{P}_{l'}=\mathsf{P}_{\max(l,l')}$. The indicator of the pairs whose smallest common block has level $l\geq 1$ is $b^l\mathsf{P}_l-b^{l-1}\mathsf{P}_{l-1}$, and every agent has $b^l-b^{l-1}$ such partners, so

$$L=\sum_{l=1}^{n_{\mathrm{H}}}a_l^{\mathrm{H}}\big[(b^l-b^{l-1})I-b^l\mathsf{P}_l+b^{l-1}\mathsf{P}_{l-1}\big].$$

Put $\mathsf{E}_m=\mathsf{P}_{m-1}-\mathsf{P}_m$ for $1\leq m\leq n_{\mathrm{H}}$ and $\mathsf{E}_{n_{\mathrm{H}}+1}=\mathsf{P}_{n_{\mathrm{H}}}$. These are orthogonal projections with $\mathsf{E}_m\mathsf{E}_{m'}=0$ for $m\neq m'$, they sum to $I$, $\operatorname{rank}\mathsf{E}_m=(b-1)b^{n_{\mathrm{H}}-m}$ for $m\leq n_{\mathrm{H}}$, and $\mathsf{P}_l\mathsf{E}_m=\mathsf{E}_m$ if $l<m$ and 0 otherwise. On range $\mathsf{E}_m$ the bracket therefore equals 0 for $l<m$, $b^m$ for $l=m$ and $b^l-b^{l-1}$ for $l>m$, so

$$L=\sum_{m=1}^{n_{\mathrm{H}}}\mathsf{g}_m\mathsf{E}_m,\qquad \mathsf{g}_m=b^m a_m^{\mathrm{H}}+\sum_{l>m}(b^l-b^{l-1})a_l^{\mathrm{H}},\qquad \mathsf{g}_m-\mathsf{g}_{m+1}=b^m(a_m^{\mathrm{H}}-a_{m+1}^{\mathrm{H}}),\tag{S142}$$

with $a_{n_{\mathrm{H}}+1}^{\mathrm{H}}=0$ and $\mathsf{g}_{n_{\mathrm{H}}+1}=0$. These identities hold for arbitrary level couplings. For $a_l^{\mathrm{H}}=A_{\mathrm{h}}\mathfrak{q}^{l-1}/((b-1)b^{l-1})$ one has $(b^l-b^{l-1})a_l^{\mathrm{H}}=A_{\mathrm{h}}\mathfrak{q}^{l-1}$ and $b^m a_m^{\mathrm{H}}=A_{\mathrm{h}}\mathfrak{q}^{m-1}+A_{\mathrm{h}}\mathfrak{q}^{m-1}/(b-1)$, so $\mathsf{g}_m$ is the quantity of Proposition 10.7, and $\mathsf{g}_m-\mathsf{g}_{m+1}=A_{\mathrm{h}}\mathfrak{q}^{m-1}(b-\mathfrak{q})/(b-1)$ for $m<n_{\mathrm{H}}$, positive exactly when $\mathfrak{q}<b$.

(a) By (S142), $\Pi=\varepsilon^{\Pi}\mathsf{E}_{n_{\mathrm{H}}+1}+\sum_m(\varepsilon^{\Pi}+\mathsf{g}_m)\mathsf{E}_m$ with $\mathsf{g}_1>\cdots>\mathsf{g}_{n_{\mathrm{H}}}>0$, which is the spectrum claimed. The number of eigenvalues not exceeding $\varepsilon^{\Pi}+\mathsf{g}_{l+1}$ is $1+\sum_{m>l}(b-1)b^{n_{\mathrm{H}}-m}=b^{n_{\mathrm{H}}-l}$, and the corresponding eigenspace is $\bigoplus_{m>l}\operatorname{range}\mathsf{E}_m=\operatorname{range}\mathsf{P}_l$, the span of the level-$l$ block indicators. Hence for $M=b^{n_{\mathrm{H}}-l}$, $0\leq l\leq n_{\mathrm{H}}$, one has $\varpi_M=\varepsilon^{\Pi}+\mathsf{g}_{l+1}=1/\tau_{l+1}$ and $\varpi_{M+1}=\varepsilon^{\Pi}+\mathsf{g}_l=1/\tau_l$ (with $\varpi_{N+1}=\infty$ and $\tau_0=0$ for $l=0$), and Theorem S20.8(b) gives the windows $\varsigma(c)\tau_l<\tau<\varsigma(c)\tau_{l+1}$. For $M=0$ the window is $\tau>\varsigma(c)/\varpi_1=\varsigma(c)/\varepsilon^{\Pi}$. Every other rank has $\varpi_M=\varpi_{M+1}$ and by Lemma S41.3(b) minimizes only at the single horizon $\tau=\varsigma(c)/\varpi_M$.

(b) $\sum_{l'=l}^{n_{\mathrm{H}}}\mathfrak{q}^{l'-1}=(\mathfrak{q}^{l-1}-\mathfrak{q}^{n_{\mathrm{H}}})/(1-\mathfrak{q})$, so $\mathsf{g}_l=\mathsf{g}_1^{\infty}\mathfrak{q}^{l-1}-A_{\mathrm{h}}\mathfrak{q}^{n_{\mathrm{H}}}/(1-\mathfrak{q})$ with $\mathsf{g}_1^{\infty}=A_{\mathrm{h}}(1/(1-\mathfrak{q})+1/(b-1))$, which converges to $\mathsf{g}_1^{\infty}\mathfrak{q}^{l-1}$ as $n_{\mathrm{H}}\to\infty$. Since $|C_l|^{z_{\mathrm{H}}}=b^{l\ln(1/\mathfrak{q})/\ln b}=\mathfrak{q}^{-l}$, the limit of $\tau_l$ is $1/(\varepsilon^{\Pi}+\mathsf{g}_1^{\infty}/(\mathfrak{q}|C_l|^{z_{\mathrm{H}}}))$, which is (S82). If level-$l$ blocks have linear size proportional to $b^{l/D_e}$, then $|C_l|^{z_{\mathrm{H}}}$ is that size to the power $D_e z_{\mathrm{H}}=y_{\mathrm{RG}}$ (Theorem 13.6(b)). The second term of the denominator of (S82) relative to the first is $\varepsilon^{\Pi}\mathfrak{q}^{1-l}/\mathsf{g}_1^{\infty}=\mathfrak{x}_l A_{\mathrm{h}}\mathfrak{q}/\mathsf{g}_1^{\infty}$, so the saturation at $1/\varepsilon^{\Pi}$ sets in where $\mathfrak{x}_l$ is of order one.

(c) By (S142), $e^{-\tau_{\mathrm{L}}L}=\sum_{m=1}^{n_{\mathrm{H}}}e^{-\tau_{\mathrm{L}}\mathsf{g}_m}(\mathsf{P}_{m-1}-\mathsf{P}_m)+\mathsf{P}_{n_{\mathrm{H}}}$. For a pair whose smallest common block has level $m_0$ ($m_0=0$ on the diagonal), $(\mathsf{P}_l)_{ij}=b^{-l}$ for $l\geq m_0$ and 0 for $l<m_0$, so the entry is

$$\mathsf{k}(m_0)=b^{-n_{\mathrm{H}}}+\sum_{m>m_0}e^{-\tau_{\mathrm{L}}\mathsf{g}_m}\big(b^{-(m-1)}-b^{-m}\big)-\mathbf{1}\{m_0\geq 1\}e^{-\tau_{\mathrm{L}}\mathsf{g}_{m_0}}b^{-m_0},$$

a function of the level alone. For $0 \le m < n_{\mathrm{H}}$, $\mathsf{k}(m) - \mathsf{k}(m+1) = b^{-m}(e^{-\tau_{\mathrm{L}}\mathsf{g}_{m+1}} - e^{-\tau_{\mathrm{L}}\mathsf{g}_m})$ with $e^{-\tau_{\mathrm{L}}\mathsf{g}_0} = 0$, which is positive because $\mathsf{g}_{m+1} < \mathsf{g}_m$, and $\mathsf{k}(n_{\mathrm{H}}) = b^{-n_{\mathrm{H}}}(1 - e^{-\tau_{\mathrm{L}}\mathsf{g}_{n_{\mathrm{H}}}}) > 0$. The density matrix is this kernel divided by its trace. The mode rule of Section S13.6 keeps the eigenvalues strictly below $1/\tau_{\mathrm{L}}$ and discards a mode at equality; under the opposite convention at equality each window would lose its left endpoint and gain its right one. For $1/\mathsf{g}_l \le \tau_{\mathrm{L}} < 1/\mathsf{g}_{l+1}$ the kept eigenvalues are 0 and the $\mathsf{g}_m$ with $m > l$, in number $b^{n_{\mathrm{H}}-l}$. Pairs are joined in the order of increasing level, in an arbitrary order within a level, until $b^{n_{\mathrm{H}}-l}$ clusters remain. When all pairs of level at most $m$ have been joined, the clusters are the level-$m$ blocks, $b^{n_{\mathrm{H}}-m}$ in number. While pairs of level $m$ are being joined, each join merges two clusters inside one level-$m$ block or does nothing, so the clusters refine the level-$m$ blocks, and their number is at least $b^{n_{\mathrm{H}}-m}$ with equality only when they are the level-$m$ blocks. Since the number of clusters at the end of level $l-1$ is $b^{n_{\mathrm{H}}-l+1} > b^{n_{\mathrm{H}}-l}$, it first equals $b^{n_{\mathrm{H}}-l}$ during level $l$, at which moment the clusters are the level-$l$ blocks, whatever the order within the level. For $l = 0$ no pair is joined.

Let $C$ be a level-$l$ block, $1 \le l < n_{\mathrm{H}}$. Its induced medium is the nested medium of depth $l$ with the same couplings, so by (S142) at depth $l$, $\lambda_2(L_C) = b^l a_l^{\mathrm{H}} = bA_{\mathrm{h}}\mathsf{q}^{l-1}/(b-1)$, and $\mathrm{cut}(C)/|C| = \sum_{l'>l}(b^{l'} - b^{l'-1})a_{l'}^{\mathrm{H}} = \sum_{l'>l} A_{\mathrm{h}}\mathsf{q}^{l'-1}$. Hence

$$\mathsf{g}_l = \lambda_2(L_C) + \frac{\mathrm{cut}(C)}{|C|}, \qquad \mathsf{g}_{l+1} = \frac{\mathrm{cut}(C)}{|C|} + \frac{\mathsf{q}}{b}\lambda_2(L_C).$$

The block $C$ is a macronode exactly for $\tau_{\mathrm{L}} \in [1/\mathsf{g}_l, 1/\mathsf{g}_{l+1})$. Since $1/\mathsf{g}_l < 1/\lambda_2(L_C)$ and $1/\mathsf{g}_{l+1} < |C|/\mathrm{cut}(C)$, this window meets the interval $(1/\lambda_2(L_C), |C|/\mathrm{cut}(C))$ of Open problem S13.21 if and only if $\mathsf{g}_{l+1} < \lambda_2(L_C)$, that is $\mathrm{cut}(C)/|C| < (1 - \mathsf{q}/b)\lambda_2(L_C)$, that is $r_\star(C) > b/(b-\mathsf{q})$. □

*Proof of Proposition S20.10.* Let $b \ge 3$. The induced graph on $C$ is complete on $b-1$ vertices, so $\lambda_2(L_C) = b-1$; each vertex of $C$ has one edge to the remaining vertex, so $\mathrm{cut}(C) = |C| = b-1$ and $r_\star(C) = b-1$. The Laplacian of the complete graph has eigenvalues 0 and $b$, the latter of multiplicity $b-1$. For $\tau_{\mathrm{L}} < 1/b$ all $b$ modes are kept and the macronodes are the singletons; for $\tau_{\mathrm{L}} \ge 1/b$ one mode is kept and the single macronode is the whole graph (the strict rule of Section S13.6 discards the mode at $\tau_{\mathrm{L}} = 1/b$; keeping it would give the singletons there). The region $C$ is neither. The interval $(1/\lambda_2(L_C), |C|/\mathrm{cut}(C))$ of Open problem S13.21 is $(1/(b-1), 1)$, which lies in $\tau_{\mathrm{L}} \ge 1/b$. Here $N = b$, and for every constant $\mathsf{c}_0$ there is $b$ with $b - 1 > \mathsf{c}_0 \ln b$, so no constant makes the threshold conjecture true. The whole graph has zero cut, so adding the missing vertex to $C$ raises $r_\star$ without bound: $C$ is not a local maximizer. □

The same failure for a level-one block of the nested medium with one site removed is a computation at one parameter value and is reported as such in Check S47.188.

**Proposition S41.5** (Mean specific heat of the infinite hierarchy)**.** *Let* $\mathsf{q} < 1$, $\mathsf{g}_1^\infty = A_{\mathrm{h}}(1/(1-\mathsf{q}) + 1/(b-1))$ *and* $\mathsf{z}(\tau) = \sum_{l\ge1}(b-1)b^{-l}e^{-\tau\mathsf{g}_1^\infty\mathsf{q}^{l-1}}$. *As* $n_{\mathrm{H}} \to \infty$, $N^{-1}\operatorname{tr} e^{-\tau L}$ *converges to* $\mathsf{z}(\tau)$ *locally uniformly on* $(0,\infty)$ *together with its first two derivatives, so the entropy of* $\varrho_\tau$ *minus* $\ln N$ *converges to* $\mathsf{s}(\tau) = \ln\mathsf{z} - \tau\mathsf{z}'/\mathsf{z}$ *and its specific heat to* $\mathsf{C}(\tau) = -\mathrm{d}\mathsf{s}/\mathrm{d}\ln\tau$. *The limit satisfies*

$$\lim_{\tau\to\infty}\big[\mathsf{s}(\tau) - \mathsf{s}(\tau/\mathsf{q})\big] = \ln b, \qquad \textit{so} \qquad \frac{1}{\ln(1/\mathsf{q})}\int_{\ln\tau}^{\ln\tau+\ln(1/\mathsf{q})} \mathsf{C}\,\mathrm{d}\ln\tau' \longrightarrow \frac{\ln b}{\ln(1/\mathsf{q})} = \frac{1}{z_{\mathrm{H}}}.$$

*Proof.* By Theorem S20.9(a) and the proof of its part (b), $N^{-1}\operatorname{tr} e^{-\tau L} = b^{-n_{\mathrm{H}}} + \sum_{l=1}^{n_{\mathrm{H}}}(b-1)b^{-l}e^{-\tau\mathsf{g}_l}$ with $\mathsf{g}_l = \mathsf{g}_1^\infty\mathsf{q}^{l-1} - A_{\mathrm{h}}\mathsf{q}^{n_{\mathrm{H}}}/(1-\mathsf{q}) \le \mathsf{g}_1^\infty$. The $k$th derivative of the $l$th term is bounded by $(b-1)b^{-l}(\mathsf{g}_1^\infty)^k$ uniformly in $\tau$ and $n_{\mathrm{H}}$, and each term converges locally uniformly with its derivatives, so the series converges as stated. With $Z = \operatorname{tr} e^{-\tau L}$, $-\operatorname{tr}\varrho_\tau\ln\varrho_\tau = \tau\operatorname{tr}(L\varrho_\tau) + \ln Z = \ln Z - \tau Z'/Z$,

which gives the entropy; $\mathsf{z} > 0$ gives the convergence of the entropy and of the specific heat, which involves $\mathsf{z}''$.

Separating the first term of $\mathsf{z}(\tau/\mathfrak{q})$ gives $\mathsf{z}(\tau/\mathfrak{q}) = \mathsf{z}(\tau)/b + \mathsf{R}(\tau)$ with $\mathsf{R}(\tau) = (b-1)b^{-1}e^{-\tau \mathsf{g}_1^\infty/\mathfrak{q}}$. Since $(\tau/\mathfrak{q})\mathsf{z}'(\tau/\mathfrak{q}) = \tau \frac{\mathrm{d}}{\mathrm{d}\tau}\mathsf{z}(\tau/\mathfrak{q})$,

$$\mathsf{s}(\tau/\mathfrak{q}) - \mathsf{s}(\tau) = -\ln b + \ln\Big(1 + \frac{b\mathsf{R}}{\mathsf{z}}\Big) - \frac{\tau b(\mathsf{R}'\mathsf{z} - \mathsf{z}'\mathsf{R})}{(\mathsf{z} + b\mathsf{R})\mathsf{z}}.$$

Three bounds finish the proof. First, $\tau|\mathsf{R}'| = (\tau \mathsf{g}_1^\infty/\mathfrak{q})\mathsf{R}$. Second,

$$\tau|\mathsf{z}'| = \sum_l (b-1)b^{-l}(\tau \mathsf{g}_1^\infty \mathfrak{q}^{l-1})e^{-\tau \mathsf{g}_1^\infty \mathfrak{q}^{l-1}} \le e^{-1},$$

because $xe^{-x} \le e^{-1}$ and the weights sum to one. Third, for $\tau \mathsf{g}_1^\infty \ge 1$ let $l$ be the least integer with $\tau \mathsf{g}_1^\infty \mathfrak{q}^{l-1} \le 1$; then $l - 1 \le \ln(\tau \mathsf{g}_1^\infty)/\ln(1/\mathfrak{q}) + 1$, and keeping only that term,

$$\mathsf{z}(\tau) \ge (b-1)b^{-l}e^{-1} \ge (b-1)b^{-2}e^{-1}(\tau \mathsf{g}_1^\infty)^{-1/z_{\mathrm{H}}}.$$

Hence $1/\mathsf{z}$ grows at most polynomially while $\mathsf{R}$ and $\tau|\mathsf{R}'|$ decay exponentially, so $b\mathsf{R}/\mathsf{z} \to 0$ and

$$\Big|\frac{\tau b(\mathsf{R}'\mathsf{z} - \mathsf{z}'\mathsf{R})}{(\mathsf{z} + b\mathsf{R})\mathsf{z}}\Big| \le \frac{b\tau|\mathsf{R}'|}{\mathsf{z}} + \frac{b\mathsf{R}}{e\mathsf{z}^2} \longrightarrow 0.$$

The integral of $\mathsf{C}$ over one period of $\ln\tau$ is $\mathsf{s}(\tau) - \mathsf{s}(\tau/\mathfrak{q})$. □

The proposition fixes the mean of the specific heat over a period and says nothing about its variation within one. That the variation does not die out is observed in Check S47.189 and is not proved.

*Proof of Proposition S20.11.* The identities (S142) hold at each time, so $\Pi(t) = \sum_{m=1}^{n_{\mathrm{H}}+1}(\varepsilon^\Pi(t) + \mathsf{g}_m(t))\mathsf{E}_m$ with $\mathsf{g}_m(t)$ as in (S83): all $\Pi(t)$ lie in the commutative algebra spanned by the fixed projections $\mathsf{E}_m$, and the equation has a unique strong solution for locally integrable coefficients.

(a) Let $\Phi$ hold an orthonormal basis of range $\mathsf{P}_l$, so $\Phi\Phi^\top = \mathsf{P}_l$. Since $\Pi(t)$ commutes with $\mathsf{P}_l$, $\Pi(t)\Phi = \mathsf{P}_l\Pi(t)\Phi = \Phi\Pi_\Phi(t)$, hence $\Phi^\top\Pi(t) = \Pi_\Phi(t)\Phi^\top$ and $\mathrm{d}y = -\Pi_\Phi(t)y\mathrm{d}t + \sqrt{2}\mathrm{d}\mathsf{B}'_t$ with $\mathsf{B}' = \Phi^\top\mathsf{B}$ a standard Brownian motion. This is a closed equation for $y$ with deterministic coefficients, whatever the law of $x_0$.

(b) Applying $\mathsf{E}_l$ to the equation, $\mathrm{d}x^{(l)} = -(\varepsilon^\Pi(t) + \mathsf{g}_l(t))x^{(l)}\mathrm{d}t + \sqrt{2}\mathsf{E}_l\mathrm{d}\mathsf{B}_t$, so

$$x_t^{(l)} = e^{-\Theta_l(s,t)}x_s^{(l)} + \sqrt{2}\int_s^t e^{-\Theta_l(r,t)}\mathsf{E}_l\mathrm{d}\mathsf{B}_r,$$

where the integral is a centered Gaussian vector with covariance $2\int_s^t e^{-2\Theta_l(r,t)}\mathrm{d}r\mathsf{E}_l$, independent of $(x_r)_{r\le s}$. This is the claim; when $x_s$ has no first moment the conditional expectation is understood in this sense. Each clock is absolutely continuous and additive, $\Theta_l(s,r) + \Theta_l(r,t) = \Theta_l(s,t)$, and strictly increasing in $t$ because its speed $\varepsilon^\Pi + \mathsf{g}_l$ is positive. If the leak were allowed to vanish on an interval, the clock of the population mean would stall there, and that of level $l$ wherever in addition the couplings of levels $l$ and above vanish.

(c) By (b), $x_t = \mathsf{U}x_s + \zeta$ with $\mathsf{U} = \sum_l e^{-\Theta_l(s,t)}\mathsf{E}_l$ and $\zeta$ Gaussian, independent of $x_s$, with covariance $\sum_l 2\int_s^t e^{-2\Theta_l(r,t)}\mathrm{d}r\mathsf{E}_l$. If $x_s$ is Gaussian with covariance $S = \sum_l v_l(s)\mathsf{E}_l$, the pair is jointly Gaussian, $\mathrm{Cov}(x_t) = \sum_l v_l(t)\mathsf{E}_l$ with $v_l(t)$ as stated, and

$$\mathrm{Cov}(x_s \mid x_t) = S - S\mathsf{U}\,\mathrm{Cov}(x_t)^{-1}\mathsf{U}S = S^{1/2}(I - \mathsf{H})S^{1/2}, \qquad \mathsf{H} = \sum_l \frac{e^{-2\Theta_l(s,t)}v_l(s)}{v_l(t)}\mathsf{E}_l = \sum_l e^{-2\tilde{\Theta}_l}\mathsf{E}_l.$$

The canonical correlations are the square roots of the eigenvalues of $\mathsf{H}$, that is $e^{-\tilde{\Theta}_l}$ with multiplicity $\operatorname{rank}\mathsf{E}_l$, and $\tilde{\Theta}_l > 0$ because $v_l(t) > e^{-2\Theta_l(s,t)}v_l(s)$. Additivity of the clock gives

$$e^{2\tilde{\Theta}_l} = 1 + \frac{2}{v_l(s)}\int_s^t e^{2\Theta_l(s,r)}\mathrm{d}r.$$

If $a_l^{\mathrm{H}} > a_{l+1}^{\mathrm{H}}$ almost everywhere on $[s,t]$, then $\mathsf{g}_l - \mathsf{g}_{l+1} = b^l(a_l^{\mathrm{H}} - a_{l+1}^{\mathrm{H}}) > 0$ there by (S142), so $\Theta_l(s,r) > \Theta_{l+1}(s,r)$ for $s < r \le t$, and with $v_l(s) \le v_{l+1}(s)$ the right-hand side is strictly larger for $l$ than for $l+1$: $\tilde{\Theta}_1 > \cdots > \tilde{\Theta}_{n_{\mathrm{H}}+1}$. Apply Lemma S41.3 to $(x_s, x_t)$. For $M = b^{n_{\mathrm{H}}-l}$ the ordered exponents satisfy $\theta_M = \tilde{\Theta}_{l+1} < \theta_{M+1} = \tilde{\Theta}_l$, the span of the first $M$ eigenvectors of $\mathsf{H}$ is $\bigoplus_{m>l}\operatorname{range}\mathsf{E}_m = \operatorname{range}\mathsf{P}_l$, and $S^{1/2}$ maps this space onto itself. The argument of Theorem S20.8(b) then shows that the frames with range range $\mathsf{P}_l$ are the only minimizers exactly when $\tilde{\Theta}_{l+1} < \varsigma(c) < \tilde{\Theta}_l$. The hypothesis $v_l(s) \le v_{l+1}(s)$ holds for an isotropic law and for the stationary law of any nested medium with decreasing level couplings, where $v_l = 1/(\varepsilon^{\Pi} + \mathsf{g}_l)$; in a constant medium started there, $\tilde{\Theta}_l = (\varepsilon^{\Pi} + \mathsf{g}_l)(t-s)$ and the statement is Theorem S20.9(a). □

*Remark* S41.6 (What blocks a general non-stationary statement). Lemma S41.3 holds for any jointly Gaussian pair, so for any linear Gaussian dynamics and any two times $s < t$ the frames that minimize the criterion are those whose image under $\operatorname{Cov}(x_s)^{1/2}$ spans the leading eigenvectors of $\mathsf{H}(s,t)$, and the selected rank is the number of canonical correlations above $e^{-\varsigma(c)}$. What Proposition S20.11 adds on a fixed tree, and what fails in general, is that these frames are the same for all pairs of times and are closed. Two exact computations locate the failure. For a fixed frame $\Phi$ with complement $\Phi_\perp$ and continuous symmetric $\Pi(t)$, eliminating $\Phi_\perp^\top x$ gives

$$\dot{y}(t) = -\Pi_\Phi(t)y(t) + \int_s^t \Gamma(t,r)y(r)\mathrm{d}r + \mathfrak{f}(t), \qquad \Gamma(t,r) = \Pi_{\perp\Phi}(t)^\top \mathsf{U}_\perp(t,r)\Pi_{\perp\Phi}(r),$$

with $\mathsf{U}_\perp$ the propagator of $\dot{w} = -\Pi_{\perp\perp}(t)w$ and $\mathfrak{f}$ a forcing that does not depend on $y$; since $\Gamma(t,t) = \Pi_{\perp\Phi}(t)^\top\Pi_{\perp\Phi}(t)$, the kernel vanishes identically if and only if range $\Phi$ is invariant under every $\Pi(t)$. For a differentiable frame $\Phi(t)$ whose range is invariant under $\Pi(t)$ at each time, $y = \Phi(t)^\top x$ obeys $\mathrm{d}y = (\dot{\Phi}^\top\Phi - \Pi_\Phi)y\mathrm{d}t + \dot{\Phi}^\top\Phi_\perp\Phi_\perp^\top x\mathrm{d}t + \sqrt{2}\Phi^\top\mathrm{d}\mathsf{B}_t$, and $\dot{\Phi}^\top\Phi_\perp = 0$ for all $t$ if and only if the projection $\Phi\Phi^\top$ is constant. Exact closure at all times therefore requires a common invariant frame, which is the situation of Proposition S20.11. For a nonreversible stationary drift $K$ with covariance $\Sigma$, $\mathsf{H}(\tau) = e^{-\tilde{\mathsf{K}}^\top\tau}e^{-\tilde{\mathsf{K}}\tau}$ with $\tilde{\mathsf{K}} = \Sigma^{-1/2}K\Sigma^{1/2}$; its eigenvectors are those of $\tilde{\mathsf{K}} + \tilde{\mathsf{K}}^\top$ at every horizon when $\tilde{\mathsf{K}}$ is normal, and we have no horizon-independent description of them otherwise. The computations are exact. That they are the only obstruction is not claimed, and a scale-time law for these cases is part of Open problem S20.13.

## S42 Proofs for Section S21

Throughout, a pair $(\Lambda, h)$ stands for the density proportional to $\exp(h^\top x - \frac{1}{2}x^\top\Lambda x)$, pairs are added and subtracted componentwise, $\pi_0 = (P_0^{-1}, P_0^{-1}m_0)$ is the prior pair, and for $\Lambda \succ 0$ the prediction map is

$$\Pi(\Lambda, h) = (\Lambda^-, \Lambda^- A\Lambda^{-1}h), \qquad \Lambda^- = (A\Lambda^{-1}A^\top + Q_x)^{-1}, \tag{S143}$$

which is defined whenever $A$ is invertible or $Q_x \succ 0$. If $x_t$ has pair $\pi$ given some data, then $x_{t+1}$ has pair $\Pi(\pi)$ given the same data, and conditioning on a further observation $o = H_i x_{t+1} + v$ with $v \sim \mathcal{N}(0, R_i)$ independent of everything else adds the item $(\Lambda_i^{\mathrm{o}}, H_i^\top R_i^{-1}o)$; this is the recursive filter of Kalman (1960) written in information form, with the sensing noise kept as a separate term, and it follows from Bayes' rule because the likelihood of $o$ is proportional to $\exp(o^\top R_i^{-1}H_i x - \frac{1}{2}x^\top\Lambda_i^{\mathrm{o}}x)$.

**Lemma S42.1** (Fusion identity)**.** *Let $x, a, b, c$ have a joint density with $a$ and $b$ conditionally independent given $(x, c)$, and let $p(x \mid c) > 0$. Then*

$$p(x \mid a, b, c) \propto \frac{p(x \mid a, c)p(x \mid b, c)}{p(x \mid c)} \tag{S144}$$

*as a function of $x$. For Gaussian laws the pair of the left side is the pair given $(a, c)$ plus the pair given $(b, c)$ minus the pair given $c$.*

*Proof.* By the conditional independence, $p(x \mid a, b, c) \propto p(x \mid c)p(a \mid x, c)p(b \mid x, c)$. Bayes' rule gives $p(a \mid x, c) = p(x \mid a, c)p(a \mid c)/p(x \mid c)$ and the same for $b$; the factors $p(a \mid c)$ and $p(b \mid c)$ do not depend on $x$. The Gaussian statement is the identity of exponents. □

With $a$ and $b$ the observations in $D_u \setminus D_v$ and $D_v \setminus D_u$ and $c$ those in $D_u \cap D_v$, the lemma says that exact fusion over a link needs the posterior given the common information, and nothing else, provided the two private parts are conditionally independent given the state and the common part. The results below are about when that proviso holds and about what it costs to store the common part.

*Proof of Theorem S21.2.* (a) Since $Q_x = 0$, $x_t = A^t x_0$ and the observation $o_{v,t} = H_v A^t x_0 + v_{v,t}$ has, as a likelihood for $x_0$, the pair $e_{v,t} = ((A^t)^\top \Lambda^{\mathrm{o}}_v A^t, (A^t)^\top h^{\mathrm{o}}_{v,t})$. Observations are independent given $x_0$, so for any finite set $D$ of items the posterior of $x_0$ given the observations in $D$ has pair $\pi_0 + \sum_{e \in D} e$, and the posterior of $x_t$ is its image under $A^t$. The image has a pair when $A$ is invertible; for a singular $A$ the law of $x_t$ can be degenerate, and exactness is then the statement about $x_0$. It therefore suffices to prove the two displayed identities.

For a link $uv$ let $T_u(uv)$ and $T_v(uv)$ be the node sets of the two components of the tree with that link removed, containing $u$ and $v$ respectively. Call the member that made an observation the source of its item. We prove by induction on the number of events that after every history the following hold for every node $v$ and every link $uv$:

(i) $s_v = \pi_0 + \sum_{e \in D_v} e$;

(ii) $c_{uv} = \pi_0 + \sum_{e \in D_u \cap D_v} e$;

(iii) an item with source in $T_u(uv)$ that lies in $D_v$ lies in $D_u$, and an item with source in $T_v(uv)$ that lies in $D_u$ lies in $D_v$.

Before any event all information sets are empty and all pairs equal $\pi_0$.

An observation event at node $w$ adds a new item $e$, which lies in no information set, to $D_w$ and to $s_w$. Then (i) is preserved; no intersection $D_u \cap D_v$ changes, because $e$ lies in one set only, so (ii) is preserved; and (iii) is preserved because its two hypotheses concern an item in the information set of the endpoint on the far side from the source, and $e$ lies only in $D_w$ with $w$ on the side of its own source.

A report $u \to v$ replaces $D_v$ by $D'_v = D_u \cup D_v$ and leaves every other set unchanged. By (i) and (ii) before the event,

$$s_v + s_u - c_{uv} = \pi_0 + \sum_{D_v} e + \sum_{D_u} e - \sum_{D_u \cap D_v} e = \pi_0 + \sum_{D_u \cup D_v} e,$$

which is (i) for $v$ afterwards, and $D_u \cap D'_v = D_u$, so the new channel pair $s_u$ is (ii) for the link $uv$. For another link $vw$ at $v$, (ii) is preserved if $D'_v \cap D_w = D_v \cap D_w$, that is, if no item $e$ lies in $D_u \cap D_w$ and outside $D_v$. Suppose there were one. If its source lies in $T_u(uv)$, then it lies in $T_v(vw)$,

because the branch of $u$ at $v$ is on the side of $v$ of the link $vw$, and (iii) for $vw$ gives $e \in D_v$ from $e \in D_w$. If its source lies in $T_v(uv)$, (iii) for $uv$ gives $e \in D_v$ from $e \in D_u$. Both contradict $e \notin D_v$. Links not incident to $v$ are untouched.

It remains to check (iii) after the report. For the link $uv$: an item with source in $T_u(uv)$ lying in $D'_v$ lies in $D_v$ or in $D_u$, and in the first case in $D_u$ by (iii) before; an item with source in $T_v(uv)$ lying in $D_u$ lies in $D_v \subseteq D'_v$. For a link $vw$ with $w \neq u$: an item with source in $T_v(vw)$ lying in $D_w$ lies in $D_v \subseteq D'_v$; an item with source in $T_w(vw)$ lying in $D'_v$ lies in $D_v$, and then in $D_w$ by (iii) before, or in $D_u \setminus D_v$, which is impossible because $T_w(vw) \subseteq T_v(uv)$ and (iii) for $uv$ would put it in $D_v$. For links not incident to $v$ nothing changed. This completes the induction. The stored statistic of a link is one pair, a symmetric $d \times d$ matrix and a $d$-vector, $d(d+3)/2$ numbers. A two-way exchange over $uv$ is the report $u \to v$ followed by the report $v \to u$ and leaves $s_u = s_v = c_{uv}$ equal to the old $s_u + s_v - c_{uv}$.

(b) By (S143) and the information-form update, the private filter of member $i$ satisfies $p_{i,t} = \Pi(p_{i,t-1}) + e_{i,t}$ with $e_{i,t} = (\Lambda^o_i, h^o_{i,t})$ and $p_{i,0} = \pi_0$, and every $p_{i,t}$ has positive definite precision. Since $c_{i,t-1} = p_{i,t-1}$ for $t \geq 2$ and $c_{i,0} = \pi_0 = p_{i,0}$, the hub's increment from member $i$ is $p_{i,t} - \Pi(c_{i,t-1}) = e_{i,t}$ exactly, so $s_t = \Pi(s_{t-1}) + \sum_i e_{i,t}$ with $s_0 = \pi_0$, which is the centralized Kalman filter in information form. In the variant with feedback every member holds $s_{t-1}$ after round $t-1$ and $c_{i,t-1} = s_{t-1}$, so $p_{i,t} = \Pi(s_{t-1}) + e_{i,t}$ and again $p_{i,t} - \Pi(c_{i,t-1}) = e_{i,t}$. □

*Remark* S42.2 (The stored pair must be predicted). If the hub of Theorem S21.2(b) subtracts the last report $c_{i,t-1}$ and not its prediction, its recursion becomes $s_t = \Pi(s_{t-1}) + \sum_i e_{i,t} + \sum_i (\Pi(p_{i,t-1}) - p_{i,t-1})$. In a scalar world the precision component of $\Pi(\lambda) - \lambda$ is $\lambda(1 - F^2 - Q_x\lambda)/(F^2 + Q_x\lambda)$, which for $\lambda > 0$ vanishes only at $\lambda = (1 - F^2)/Q_x$, so with identical scalar members the increment is exact only when their common precision equals that value. In Check S47.192 the unpredicted rule deviates from the centralized filter by a median of $1.4 \times 10^3$ in the pair.

Part (b) of the theorem needs a report at every step. The next proposition is the smallest exact instance of the failure when reports are less frequent.

**Proposition S42.3** (One pair per link fails under intermittent reports)**.** *Let $d = 1$, $A = F = 1$, $Q_x = 1$ and $x_0 \sim \mathcal{N}(0, 1)$. Two members observe $o_{i,t} = x_t + v_{i,t}$ with $R = 1$ at $t = 1, 2$, run private filters and report once, after the second observation, to a hub that holds the prior and applies the rule of Theorem S21.2(b) with the stored pair predicted over the two steps, $s_2 = \Pi^2(\pi_0) + \sum_i (p_{i,2} - \Pi^2(\pi_0))$. Then the hub claims precision $43/15$, the exact posterior precision of $x_2$ given the four observations is $19/7 < 43/15$, and the mean-square error of the hub's mean is $693/1849$, with $693/1849 > 7/19 > 15/43$: the hub's mean is worse than the exact posterior mean, and its error is $231/215$ times its claimed variance.*

*Proof.* With $F = Q_x = R = 1$ and $x_0 \sim \mathcal{N}(0, 1)$, the state $x_1$ has prior variance 2. A member's filter has precision $\frac{1}{2} + 1 = \frac{3}{2}$ after its first observation, predicted variance $\frac{2}{3} + 1 = \frac{5}{3}$, and precision $\frac{3}{5} + 1 = \frac{8}{5}$ after its second. The two-step prediction of the common prior has variance 3 and precision $\frac{1}{3}$. The hub's rule therefore claims $\frac{1}{3} + 2(\frac{8}{5} - \frac{1}{3}) = \frac{43}{15}$. For the exact posterior, the prior precision of $(x_1, x_2)$ is $\left(\begin{smallmatrix} 3/2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, since $x_1 \sim \mathcal{N}(0, 2)$ and $x_2 = x_1 + w$ with $w \sim \mathcal{N}(0, 1)$; two unit observations of each state add 2 to each diagonal entry, giving $J = \left(\begin{smallmatrix} 7/2 & -1 \\ -1 & 3 \end{smallmatrix}\right)$ with $\det J = \frac{19}{2}$ and posterior variance of $x_2$ equal to $J_{11}/\det J = \frac{7}{19}$. Since $43 \cdot 7 = 301 > 285 = 19 \cdot 15$, the claimed precision exceeds the exact one. The posterior mean minimizes the mean-square error among functions of the four observations and attains $\frac{7}{19}$, so the hub's mean has mean-square error at least $\frac{7}{19} > \frac{15}{43}$.

The error of the hub's mean is computed directly. The information vector of a member is $o_{i,1}$ after the first observation, $\frac{3}{5} \cdot \frac{2}{3} o_{i,1} = \frac{2}{5} o_{i,1}$ after the prediction, and $\frac{2}{5} o_{i,1} + o_{i,2}$ after the second

observation. The two-step prediction of the prior pair has information vector zero, so the hub's mean is

$$\hat{x} = \tfrac{15}{43}\sum_{i=1}^{2}(\tfrac{2}{5}o_{i,1} + o_{i,2}) = \tfrac{6}{43}(o_{1,1} + o_{2,1}) + \tfrac{15}{43}(o_{1,2} + o_{2,2}).$$

With $o_{i,1} = x_1 + v_{i,1}$, $o_{i,2} = x_1 + w + v_{i,2}$ and $x_2 = x_1 + w$, where $w = w_1^x$, its error is

$$\hat{x} - x_2 = -\tfrac{1}{43}x_1 - \tfrac{13}{43}w + \tfrac{6}{43}(v_{1,1} + v_{2,1}) + \tfrac{15}{43}(v_{1,2} + v_{2,2}),$$

a sum of independent terms with $\operatorname{Var} x_1 = 2$ and unit variance otherwise. Its mean-square error is $(2 + 169 + 72 + 450)/43^2 = 693/1849$, and $693 \cdot 19 = 13167 > 12943 = 7 \cdot 1849$. The ratio to the claimed variance $\frac{15}{43}$ is $693/(43 \cdot 15) = 231/215$.

The hypothesis of Lemma S42.1 fails here: given $x_2$, the first observations of the two members are both informative about $x_1$, which is not determined by $x_2$ when $Q_x \succ 0$, so the two windows are dependent given $x_2$ and the common prior. □

For the window results, index the steps of the window by $\tau = 1, \dots, k$, let $x_0 \sim \mathcal{N}(m, P)$ with $P \succ 0$ be the law of the state at the last contact, and stack the window trajectory as $X = (x_1, \dots, x_k) \in \mathbb{R}^{kd}$. Let $E_\tau$ be the $kd \times d$ matrix that has the identity in block $\tau$ and zeros elsewhere.

**Lemma S42.4** (Window trajectory)**.** *Let $Q_x \succ 0$.*

*(a) The covariance of $X$ is positive definite. Write $J_0$ for its inverse and $j_0 = J_0 \mathbb{E}X$.*

*(b) If evidence about $X$ enters through likelihoods $\exp(g_\tau^\top x_\tau - \frac{1}{2}x_\tau^\top D_\tau x_\tau)$ with $D_\tau \succeq 0$, one for each step, then the posterior of $X$ is Gaussian with precision $J_0 + \operatorname{diag}(D_1, \dots, D_k)$ and mean $(J_0 + \operatorname{diag}(D_\tau))^{-1}(j_0 + g)$, $g = (g_1, \dots, g_k)$, and under it*

$$\operatorname{Cov}(x_\tau, x_k) = G_\tau G_{\tau+1} \cdots G_{k-1} \operatorname{Cov}(x_k), \qquad G_\tau = P_{\tau|\tau} A^\top (A P_{\tau|\tau} A^\top + Q_x)^{-1}, \tag{S145}$$

*for $\tau < k$, where $P_{\tau|\tau} \succ 0$ is the covariance of $x_\tau$ given the evidence of steps $1, \dots, \tau$. If $A$ is invertible, every $\operatorname{Cov}(x_\tau, x_k)$ is invertible.*

*Proof.* (a) Let $a = (a_1, \dots, a_k) \neq 0$ and let $\tau^*$ be the largest index with $a_{\tau^*} \neq 0$. Since $x_{\tau^*} = A x_{\tau^*-1} + w_{\tau^*-1}^x$, the variable $\sum_\tau a_\tau^\top x_\tau$ is the sum of $a_{\tau^*}^\top w_{\tau^*-1}^x$ and a variable independent of it, so its variance is at least $a_{\tau^*}^\top Q_x a_{\tau^*} > 0$.

(b) The posterior density is the prior density of $X$ times the likelihoods, which gives the precision and the mean. The joint density of $X$ and the evidence factorizes along the chain, $p(x_1) \prod_\tau p(x_{\tau+1} \mid x_\tau) \prod_\tau \psi_\tau(x_\tau)$ with $\psi_\tau$ the likelihood of step $\tau$, so under the posterior $X$ is a Markov chain, and the conditional density of $x_\tau$ given $x_{\tau+1}, \dots, x_k$ is proportional in $x_\tau$ to $\alpha_\tau(x_\tau) p(x_{\tau+1} \mid x_\tau)$, where $\alpha_\tau$ is the density of $x_\tau$ given the evidence of steps $1, \dots, \tau$, the Gaussian $\mathcal{N}(m_{\tau|\tau}, P_{\tau|\tau})$. Its precision is the inverse of a predicted covariance $A P' A^\top + Q_x \succ 0$ plus $D_\tau \succeq 0$, so $P_{\tau|\tau} \succ 0$. Under $\alpha_\tau(x_\tau) p(x_{\tau+1} \mid x_\tau)$ the pair $(x_\tau, x_{\tau+1})$ is Gaussian with $\operatorname{Cov}(x_\tau, x_{\tau+1}) = P_{\tau|\tau} A^\top$ and $\operatorname{Cov}(x_{\tau+1}) = A P_{\tau|\tau} A^\top + Q_x$, hence

$$\mathbb{E}[x_\tau \mid x_{\tau+1}, \dots, x_k] = m_{\tau|\tau} + G_\tau (x_{\tau+1} - A m_{\tau|\tau})$$

under the posterior. Therefore $\operatorname{Cov}(x_\tau, x_k) = \operatorname{Cov}(\mathbb{E}[x_\tau \mid x_{\tau+1}, \dots, x_k], x_k) = G_\tau \operatorname{Cov}(x_{\tau+1}, x_k)$, and (S145) follows by descending induction from $\tau = k - 1$. When $A$ is invertible each $G_\tau$ is a product of invertible matrices, and $\operatorname{Cov}(x_k) \succ 0$ as a diagonal block of a positive definite matrix. □

*Proof of Theorem S21.3.* Member $i$'s observations enter the posterior of $X$ through the likelihoods $\exp(h_{i,\tau}^{\mathrm{o}\top} x_\tau - \frac{1}{2} x_\tau^\top \Lambda_i^{\mathrm{o}} x_\tau)$. Write $\ell = (h_{i,1}^{\mathrm{o}}, \dots, h_{i,k}^{\mathrm{o}})$ for the window ledger, $D = \operatorname{diag}(D_1, \dots, D_k)$ and $g = (g_1, \dots, g_k)$ for the hub-side evidence, and

$$J_D = J_0 + I_k \otimes \Lambda_i^{\mathrm{o}} + D.$$

By Lemma S42.4(b) the posterior mean of $x_k$ given both bodies of evidence is $E_k^\top J_D^{-1}(j_0 + \ell + g)$. Every $D_\tau \succeq 0$ is the information matrix of some observation, for instance $H = D_\tau^{1/2}$ with unit noise, so every block-diagonal $D \succeq 0$ is an admissible hub-side evidence.

Let $o$ and $o'$ be two values of member $i$'s window data with $\varphi_i(o) = \varphi_i(o')$, with ledgers $\ell$ and $\ell'$, and put $z = \ell - \ell'$. The encoder is blind, so the same message is sent whatever $D$ and $g$ are, and the hub's output, a function of the message, of $(D, g)$ and of the model, is the same for $o$ and $o'$. By exactness it equals the posterior mean in both cases, hence

$$E_k^\top J_D^{-1} z = 0 \qquad \text{for every block-diagonal } D \succeq 0. \tag{S146}$$

Take $D = I_{kd}$, put $C = J_I^{-1} \succ 0$, fix $\tau$ and a symmetric $d \times d$ matrix $S$. For $|\epsilon|$ small, $I_{kd} + \epsilon E_\tau S E_\tau^\top$ is block diagonal and positive semidefinite, so (S146) holds along this line, and differentiating at $\epsilon = 0$ with $\frac{\mathrm{d}}{\mathrm{d}\epsilon} J^{-1} = -J^{-1} \dot{J} J^{-1}$ gives

$$E_k^\top C E_\tau S E_\tau^\top C z = 0.$$

The matrix $E_k^\top C E_\tau$ is the posterior covariance $\operatorname{Cov}(x_k, x_\tau)$ under the evidence with precision $J_I$, which has the form of Lemma S42.4(b) with step information $\Lambda_i^{\mathrm{o}} + I_d$; it is invertible because $A$ is. With $S = I_d$ this yields $E_\tau^\top C z = 0$. This holds for every $\tau$, so $Cz = 0$ and $z = 0$. Thus $\varphi_i(o) = \varphi_i(o')$ implies $\ell = \ell'$: the ledger is a function of the message.

If $\varphi_i(o) = \Phi o$ is linear, the linear map $L = I_k \otimes H_i^\top R_i^{-1}$ that sends the data to the ledger satisfies $\ker \Phi \subseteq \ker L$, hence $\operatorname{rank} \Phi \geq \operatorname{rank} L = k \operatorname{rank} H_i$. Conversely the ledger is sufficient: a hub that receives $\ell$ and knows $\Lambda_i^{\mathrm{o}}$ has all the likelihoods of member $i$ and computes the exact posterior of $X$, by running the Kalman filter through the window from the pair it held at the last contact. The ledger is computed by member $i$ from its own data and its own $(H_i, R_i)$, so the encoder is blind, and its coordinates in a basis of the range of $H_i^\top$ are $k \operatorname{rank} H_i$ numbers. □

*Remark* S42.5 (Scope of the bound). The proof uses invertibility of $A$ only to make every $\operatorname{Cov}(x_k, x_\tau)$ invertible. It is enough that these blocks are nonzero for one positive definite block-diagonal $D$: if $Y = \operatorname{Cov}(x_k, x_\tau) \neq 0$ and $u = E_\tau^\top C z \neq 0$, the choice $S = uu^\top$ gives $Yu = 0$ and then $S = yu^\top + uy^\top$ with $Yy \neq 0$ gives $Yy|u|^2 = 0$, a contradiction. With a singular $A$ the bound was still attained in Check S47.194, and with a nonzero nilpotent $A$ in Check S47.200. Some condition on $A$ is needed, however. For $A = 0$ the states of the window are independent, every $\operatorname{Cov}(x_k, x_\tau)$ with $\tau < k$ vanishes under every block-diagonal evidence, and the posterior of $x_k$ depends on the window of member $i$ only through $h_{i,k}^{\mathrm{o}}$; that vector alone, $\operatorname{rank} H_i$ numbers, is then an exact blind message, and the conclusion of Theorem S21.3 is false for $k \geq 2$ and $H_i \neq 0$. The bound also fails for $Q_x = 0$, where Theorem S21.2(a) applies, and it is a statement about encoders fixed without knowledge of the other evidence; Proposition S42.6 shows what that knowledge buys.

**Proposition S42.6** (Model-aware summaries of dimension $d$)**.** *Let $A$ and $Q_x \succeq 0$ be arbitrary, and let all $n$ members report to a hub at the same times, every $k$ steps, the hub holding no evidence other than the prior and the reports. For every contact there are matrices $G_1, \dots, G_n$ with $d$ rows, which every member can compute from $(A, Q_x, (H_j, R_j)_{j \leq n}, P_0, k)$ and the number of the contact, such that the messages $z_i = G_i O_i \in \mathbb{R}^d$, with $O_i$ the observations of member $i$ during the window, let the hub compute the centralized Kalman posterior at every contact. No pair is stored per link.*

*Proof.* Suppose the hub's law at a contact is $\mathcal{N}(m, P)$ and equals the posterior of the state given all observations so far. By the Markov property the data $O = (O_1, \dots, O_n)$ of the next window, $O_i = (o_{i,1}, \dots, o_{i,k})$, and the state $x_k$ at the next contact are jointly Gaussian given the earlier observations, with the law obtained from $x_0 \sim \mathcal{N}(m, P)$. The covariance of $O$ is positive definite because the sensing noises are, so

$$\mathbb{E}[x_k \mid O] = A^k m + G(O - \mathbb{E}O), \qquad G = \operatorname{Cov}(x_k, O)\operatorname{Cov}(O)^{-1},$$

and the conditional covariance $\operatorname{Cov}(x_k) - G\operatorname{Cov}(O, x_k)$ does not depend on the data. Covariances of jointly Gaussian variables do not depend on means, so $G$ is a function of $(A, Q_x, (H_j, R_j)_j, P, k)$ alone. Split $G = (G_1, \dots, G_n)$ by member. Member $i$ sends $z_i = G_i O_i \in \mathbb{R}^d$; the hub, which knows $m$, forms $A^k m - G\mathbb{E}O + \sum_i z_i$ and the conditional covariance, and so holds the posterior given all observations up to the new contact. The new covariance is again a deterministic function of the model, so every member can compute the next gain without hearing from the hub. Induction over contacts, starting from $\mathcal{N}(m_0, P_0)$, proves the claim. □

**Proposition S42.7** (The static triangle)**.** *Let $Q_x = 0$ and $A = I$. Three members on a triangle hold the common prior pair $\pi_0$ and one item each, $s_j = \pi_0 + e_j$, and exchange with both neighbors simultaneously in rounds $r = 1, 2, \dots$.*

*(a) Under the product rule $s_i \leftarrow \sum_{j \in N[i]} s_j$ on the closed neighborhood, the analog of rule (P) of Proposition S12.7, every node holds $3^r \pi_0 + 3^{r-1} \sum_j e_j$ after $r$ rounds.*

*(b) Under the channel rule $s_i \leftarrow s_i + \sum_{j \in N(i)} (s_j - c_{ij})$ and $c_{ij} \leftarrow s_i + s_j - c_{ij}$, with old values on the right and $c_{ij} = \pi_0$ initially, the prior coefficient of every node stays one, the node pairs are exact after the first round, and the evidence coefficients of a node sum to $2r + 1$ after $r$ rounds, against the exact 3.*

*Proof.* Both rules are linear with integer coefficients, so every pair they produce is a formal combination $\kappa\pi_0 + \sum_j m_j e_j$ of the prior pair and the three items; call $\kappa$ its prior coefficient and $\sum_j m_j$ its weight. Both functionals are linear and invariant under relabeling of the members, and both rules and the initial configuration are invariant under relabeling, so in every round all nodes share one prior coefficient and one weight, and so do all links.

(a) The product rule triples the common prior coefficient and the common weight of the nodes in every round, starting from 1 and 1. After the first round $s_i = 3\pi_0 + \sum_j e_j$ for every $i$, the three nodes hold the same pair, and each later round multiplies it by 3.

(b) Let $(\alpha_r, S_r)$ be the prior coefficient and weight of a node after $r$ rounds and $(\gamma_r, C_r)$ those of a link, with $\alpha_0 = \gamma_0 = 1$, $S_0 = 1$ and $C_0 = 0$. The rule gives $\alpha_{r+1} = 3\alpha_r - 2\gamma_r$ and $\gamma_{r+1} = 2\alpha_r - \gamma_r$, so $\alpha_r = \gamma_r = 1$ for all $r$, and $S_{r+1} = 3S_r - 2C_r$, $C_{r+1} = 2S_r - C_r$, so $S_{r+1} - C_{r+1} = S_r - C_r = 1$ and $S_{r+1} = S_r + 2$, that is, $S_r = 2r + 1$. In the first round $s_j - c_{ij} = e_j$ for both neighbors, so $s_i = \pi_0 + \sum_j e_j$, the exact pair. With identical sensing the precision component of the weight statement is $\Lambda_0 + (2r + 1)\Lambda^{\mathrm{o}}$ against the exact $\Lambda_0 + 3\Lambda^{\mathrm{o}}$. □

The channel rule fails on the triangle from the second round because the hypothesis that carried Theorem S21.2(a) fails: after the first round the item of the third member lies in $D_i \cap D_j$ without having crossed the link $ij$, so the link's pair no longer describes the common information of its endpoints.

*Proof of Theorem S21.4.* An anonymous protocol consists of an initial state $\sigma^0$ and three maps: a message map $\mu(\sigma, p)$ giving the message sent on port $p \in \{1, 2\}$, an update map $U(\sigma, o, m_1, m_2)$

giving the next state from the current state, the datum of the step if there is one, and the messages received on the two ports, and an output map from states to pairs. The maps and $\sigma^0$ are the same at every node and do not depend on the graph. A port labeling assigns to every node a bijection between $\{1, 2\}$ and its two links.

Let $n \geq 3$, write the nodes of $C_n$ as the residues modulo $n$ and those of $C_{2n}$ as the residues modulo $2n$, with $v$ adjacent to $v \pm 1$, and let $q(v) = v \bmod n$ map the second onto the first. Since $n \geq 3$, the three nodes $v-1, v, v+1$ of $C_{2n}$ have distinct images, so $q$ maps the closed neighborhood of $v$ bijectively onto that of $q(v)$. Fix a port labeling of $C_n$ and lift it: port $p$ of $v$ in $C_{2n}$ is the link to the neighbor $w$ for which $q(w)$ is the neighbor of $q(v)$ at port $p$. Fix data $(o_s)_{s<n}$ on $C_n$ and give node $v$ of $C_{2n}$ the datum $o_{q(v)}$.

We claim that after every round the state of node $v$ in $C_{2n}$ equals the state of node $q(v)$ in $C_n$. Initially all states equal $\sigma^0$. Suppose the claim holds before a round. Node $v$ receives on port $p$ the message $\mu(\sigma_w, p_w)$, where $w$ is its neighbor at port $p$ and $p_w$ the port of $w$ toward $v$. By the lifted labeling, $q(w)$ is the neighbor of $q(v)$ at port $p$ and its port toward $q(v)$ is $p_w$; by the induction hypothesis $\sigma_w = \sigma_{q(w)}$. So $v$ receives on each port the message that $q(v)$ receives on the same port, the data agree, and the update map returns the same state.

Hence the outputs agree: for every round, the pair that node $v$ outputs in $C_{2n}$ under the lifted data is the pair that $q(v)$ outputs in $C_n$. If the protocol were exact on $C_n$ from round $T_n$ on and on $C_{2n}$ from round $T_{2n}$ on, then for every later round the first pair would have precision $\Lambda_0 + 2n\Lambda^{\mathrm{o}}$, since each of the $2n$ members of $C_{2n}$ made one observation with information $\Lambda^{\mathrm{o}}$, and the second would have precision $\Lambda_0 + n\Lambda^{\mathrm{o}}$. These differ because $\Lambda^{\mathrm{o}} \neq 0$. □

The pair of sizes $n$ and $2n$ is not special. With constant data the same comparison works for any two sizes.

**Corollary S42.8** (Any two ring sizes)**.** *In the setting of Theorem S21.4 let $3 \leq n < m$. No anonymous protocol outputs the exact pair at every node from some round on, for every value of the data and every labeling of the ports, on both $C_n$ and $C_m$.*

*Proof.* Label the ports of both rings so that port 1 of node $v$ leads to $v+1$ and port 2 to $v-1$, and give every node of both rings the same datum. All nodes of both rings are then in one common state after every round. This holds initially. If all nodes are in the state $\sigma$ before a round, every node receives $\mu(\sigma, 2)$ on port 1 and $\mu(\sigma, 1)$ on port 2 and has the same datum, so the update map returns one state for all of them. The outputs on the two rings therefore agree in every round, while the exact precisions $\Lambda_0 + n\Lambda^{\mathrm{o}}$ and $\Lambda_0 + m\Lambda^{\mathrm{o}}$ differ because $\Lambda^{\mathrm{o}} \neq 0$. □

The theorem and the corollary exclude one deterministic rule that is exact for every value of the data on rings of two sizes. The next proposition shows that none of these three requirements can be dropped.

**Proposition S42.9** (What the obstruction leaves open)**.** *Let $Q_x = 0$ and $A = I$, and let identical members with common $(H, R)$ and $\Lambda^{\mathrm{o}} \neq 0$ observe once, $o_s = Hx + v_s$ with item $e_s = (\Lambda^{\mathrm{o}}, H^\top R^{-1} o_s)$, and then exchange messages in synchronous rounds, without identifiers.*

*(a) Fix $N \geq 3$ and consider the following rule for nodes with two ports. In round 1 a node sends its item with counter 1 on both ports. In every later round a node that received $(e, c)$ on port $p$ in the previous round sends $(e, c+1)$ on its other port if $c + 1 \leq N - 1$, and sends nothing for that message otherwise. A node keeps one pair, initially $\pi_0$ plus its own item, and adds to it every item that arrives on its port 1. This rule is an anonymous protocol. On $C_N$, for every value of the data and every labeling of the ports, every node holds $\pi_0 + \sum_s e_s$ from round*

$N-1$ *on, and it stores one pair and at most two items with their counters. On* $C_{2N}$ *every node ends with precision* $\Lambda_0 + N\Lambda^{\mathrm{o}}$*, against the exact* $\Lambda_0 + 2N\Lambda^{\mathrm{o}}$*.*

*(b) On a connected graph* $G$ *with* $n$ *nodes, let every node keep a finite set of observed values, initially its own, send the set to every neighbor in every round, replace its set by the union of that set with the sets received, and output* $\pi_0 + \sum_o (\Lambda^{\mathrm{o}}, H^\top R^{-1} o)$*, the sum running over the values* $o$ *in its set. This rule is an anonymous protocol. If the* $n$ *observed values are pairwise distinct, an event of probability one, the output of* $v$ *after* $r$ *rounds is* $\pi_0 + \sum_{s:\mathrm{dist}(v,s)\le r} e_s$*, the exact pair for its information set, and it equals* $\pi_0 + \sum_s e_s$ *from round* $\mathrm{ecc}(v)$ *on. If only* $m < n$ *of the values are distinct, the output has precision* $\Lambda_0 + m\Lambda^{\mathrm{o}}$ *from round* $\mathrm{ecc}(v)$ *on and is not exact.*

*(c) Let every member draw a key independently of everything else from a law on* $\mathbb{R}$ *without atoms, and run the rule of (b) on the pairs of key and observed value, with the output summed over the pairs held. For every value of the data the outputs are those of (b) for distinct values, with probability one over the keys.*

*Proof.* (a) The rule consists of an initial state, a message map, an update map and an output map that are the same at every node and contain the constant $N$ and no reference to the graph, so it is an anonymous protocol in the sense of Definition S21.1. A node absorbs its datum into its state before the first round; the proofs of Theorem S21.4 and Corollary S42.8 cover this, since nodes with equal data are then in equal states before the first round. Let the ring have $M \in \{N, 2N\}$ nodes $w_j$, with $j$ a residue modulo $M$ and $w_j$ adjacent to $w_{j\pm1}$. Whatever the labeling, the other port of the port of $w_{j+1}$ toward $w_{j+2}$ is its port toward $w_j$, so a message keeps its direction of travel. We claim that for $1 \le r \le N-1$ the only message sent in round $r$ from $w_{j+1}$ to $w_j$ is $(e_{w_{j+r}}, r)$, and that no message is sent in a round $r \ge N$. For $r = 1$ this is the rule. If it holds for some $r \le N-2$, then in round $r$ the node $w_{j+1}$ received $(e_{w_{j+1+r}}, r)$ from $w_{j+2}$ and one message from $w_j$; in round $r+1$ it forwards the first to $w_j$ with counter $r+1 \le N-1$ and the second to $w_{j+2}$. The messages received in round $N-1$ carry the counter $N-1$ and are not forwarded. The same holds in the other direction with $w_{j-r}$ in place of $w_{j+r}$. Port 1 of $w_j$ leads to $w_{j+1}$ or to $w_{j-1}$, and so receives in rounds $1, \dots, N-1$ the items of $w_{j\pm1}, \dots, w_{j\pm(N-1)}$ with one fixed sign. For $M = N$ these are the $N-1$ other nodes, each once, and the pair of $w_j$ is $\pi_0 + \sum_s e_s$ after round $N-1$ and unchanged afterwards. For $M = 2N$ they are $N-1$ distinct other nodes, and the pair has precision $\Lambda_0 + N\Lambda^{\mathrm{o}}$. Between rounds a node holds its pair and the at most two messages it received in the last round.

(b) The rule uses neither identifiers nor the graph. Let $V_v^r$ be the set of $v$ after $r$ rounds. By induction on $r$, $V_v^r = \bigcup_{u \in N[v]} V_u^{r-1} = \{o_s : \mathrm{dist}(v,s) \le r\}$, because a node is within distance $r$ of $v$ exactly when it is within distance $r-1$ of a member of the closed neighborhood of $v$. The same induction, with every node reporting to every neighbor in every round, gives the information set $D_v = \{s : \mathrm{dist}(v,s) \le r\}$. If the values are pairwise distinct, $V_v^r$ contains one value for each source in $D_v$ and the output is $\pi_0 + \sum_{s \in D_v} e_s$, which is the posterior pair given $o_{D_v}$ because the observations are independent given $x$. For $s \ne s'$ the difference $o_s - o_{s'} = v_s - v_{s'}$ has the law $\mathcal{N}(0, 2R)$ with $R \succ 0$ and is nonzero with probability one, and there are finitely many pairs of sources. From round $\mathrm{ecc}(v)$ on, $V_v^r$ is the set of all observed values; if it has $m < n$ elements the output has precision $\Lambda_0 + m\Lambda^{\mathrm{o}}$, while the posterior precision is $\Lambda_0 + n\Lambda^{\mathrm{o}}$ whatever the values, and these differ because $\Lambda^{\mathrm{o}} \ne 0$.

(c) Keys drawn independently from a law without atoms are pairwise distinct with probability one. On that event the pairs of key and value are pairwise distinct whatever the data, and the induction of (b) applies to them. □

Part (a) shows that Theorem S21.4 cannot be strengthened to a single ring size: a rule may contain the size of the ring it was built for, and by Corollary S42.8 what no rule can do is serve two sizes. Part (b) shows that exactness for every value of the data cannot be weakened to exactness with probability one. The lifted data in the proof of Theorem S21.4 repeat every value twice, and the data in the proof of Corollary S42.8 are constant, so both lie in the null set on which the rule of (b) fails. Part (c) shows that the theorem does not extend to randomized rules. The rules of (b) and (c) store up to $n$ values or keys per node, so they are no cheaper than the cumulative pairs of Proposition S42.11. The three rules and the lifting are run in Check S47.201.

*Remark* S42.10 (What the obstruction is about). No bound on the dimension of states or messages was used in the proof of Theorem S21.4, so the obstruction on a cycle is not a shortage of storage. Nor is it the absence of identifiers as such, by Proposition S42.9, and randomized rules without identifiers can be exact on cycles with probability one by Proposition S42.9(c). For continuous rules the gap between probability one and every value of the data closes. Suppose that states and messages lie in Euclidean spaces, that the message, update and output maps are continuous, and that on a given ring with a given labeling of the ports the rule is exact with probability one from a round that does not depend on the data. The output at such a round is a continuous function of the data, and so is the exact pair $\pi_0 + \sum_s e_s$. The data $(o_s)_s$ have a Gaussian law with covariance $\mathbf{1}\mathbf{1}^\top \otimes HP_0H^\top + I \otimes R \succ 0$, so an event of probability one has a complement of Lebesgue measure zero and is dense, and two continuous functions that agree on a dense set agree everywhere. Such a rule is exact for every value of the data, and Theorem S21.4 applies to it. The rule of Proposition S42.9(b) is discontinuous where two observed values coincide. When the round from which exactness holds may depend on the data the density argument does not apply, and that case is not settled here. The claimed variances of the two local rules in Check S47.195 are the same for every ring size while the exact variance is not, which is in line with the lifting argument and is not an instance of the theorem, since that check has process noise and repeated observations.

The lifting argument has a precedent for finite alphabets. Hendrickx et al. (2010) study identical deterministic nodes without identifiers on graphs with labeled ports, with inputs from a finite set, and show that a family of functions computed by such nodes with one algorithm for all network sizes, even with infinite memory, depends only on the frequencies of the input values (their Theorem 3.1). Their proof compares a ring with the ring obtained by repeating it $m$ times, on which every node passes through the same states as its image (their Lemma 3.3). A function of the frequencies does not change when the ring is repeated, which is the property used in Theorem S21.4. That theorem is the counterpart for real-valued data and states of any dimension; it is proved above and is not derived from their result. Only the part of their paper that contains these statements was read for this section.

*Proof of Proposition S21.5.* (a) Fix an item $(s, \tau)$ and argue by induction on $\ell = \operatorname{dist}(v, s)$ that $v$ first acquires it in round $\tau + \max(\ell - 1, 0)$. For $\ell = 0$ the item is created at step $\tau$, before round $\tau$. For $\ell = 1$ the source sends its fresh item in round $\tau$ and not before. For $\ell \geq 2$ every neighbor $u$ of $v$ has $\operatorname{dist}(u, s) \geq \ell - 1 \geq 1$, first acquires the item in round $\tau + \operatorname{dist}(u, s) - 1 \geq \tau + \ell - 2$ by the induction hypothesis, and sends it in the following round and in no other; one neighbor has distance exactly $\ell - 1$. So $v$ first receives the item in round $\tau + \ell - 1$. The items of one source thus first reach $v$ in the order of their times, one per round, and a copy that arrives in a later round carries a time that does not exceed the last time $v$ holds for that source. A node that accepts an arriving item exactly when its time exceeds the last time it holds for the source, taking one copy if several arrive together, therefore holds each item once and acquires it at its first arrival; this is how duplicates are discarded by key, and it uses one stored time per source. The resulting sets are those obtained when, in every round, every node receives the whole information set that each neighbor

held before the round together with the neighbor's fresh item: by the same induction, the least delay over neighbors is $\min_u(\delta(u,s)+1) = \delta(v,s)$ for $\operatorname{dist}(v,s) \geq 2$.

(b) In the state-space model the observations of step $\tau$ are independent of all other variables given $x_\tau$. For a set $D$ of items, the pair $\pi(\tau)$ of $x_\tau$ given the items of $D$ with time at most $\tau$ therefore satisfies $\pi(\tau) = \Pi(\pi(\tau-1)) + \sum_{(s,\tau)\in D} e_{s,\tau}$ with $\pi(0) = \pi_0$, the information-form filter with whatever observations are present. Put $b = t - \operatorname{ecc}(v) + 1$ for $t \geq \operatorname{ecc}(v) - 1$. Since $\delta(v,s) \leq \operatorname{ecc}(v) - 1$ for every $s$, by (a) the set $D_v(t)$ contains every item with time at most $b$, so $\pi(b)$ is the pair of the centralized filter at time $b$; the node keeps it as its base pair and advances it by one step per round, as the slice $b+1$ becomes complete. From the base pair and the stored items with times in $(b,t]$ the recursion returns the pair of $p(x_t \mid o_{D_v(t)})$. The items of source $s$ that the filter needs are those with $b < \tau \leq t - \delta(v,s)$, in number $\operatorname{ecc}(v) - 1 - \delta(v,s)$. The node holds more than these. The item of a source $s \neq v$ that it acquired in round $t$ has time $t - \delta(v,s)$ and must be sent in round $t+1$; when $\delta(v,s) = \operatorname{ecc}(v) - 1$ this time is $b$, the item has been absorbed into the base pair, and it is kept for one more round only to be forwarded. After round $t$ the node therefore holds one item of each source $s \neq v$ with $\delta(v,s) = \operatorname{ecc}(v) - 1$ and $\operatorname{ecc}(v) - 1 - \delta(v,s)$ items of every other source, at most $\max(\operatorname{ecc}(v) - 1, 1)$ per source and $n \max(\operatorname{ecc}(v) - 1, 1)$ in all, besides the $n$ times used to discard duplicates. For $\operatorname{ecc}(v) = 1$ the filter needs no stored item and the node holds the $n-1$ items it has yet to forward, so the count $\sum_s(\operatorname{ecc}(v) - 1 - \delta(v,s))$ alone understates the storage. In a round a node sends on each link its fresh item and, for each other source, at most the one item it first acquired in the previous round, at most $n$ items.

(c) The recursion of (b) uses the items of a time slice only through their sum. By (a) the item $(s,\tau)$ reaches $v$ for the first time in round $\tau + \delta(v,s)$, so a node that knows the delays accepts an arriving item with key $(s,\tau)$ exactly when the round is $\tau + \delta(v,s)$, taking one copy if several neighbors deliver it in that round, and adds it to the pair of slice $\tau$. The open slices are $b+1, \ldots, t$, of which the last is always complete only for the sources at delay zero, so $\operatorname{ecc}(v) - 1$ slice pairs are kept besides the base pair. The items acquired in a round are still sent singly in the next round, so the node also holds up to $n-1$ keyed items for one round, and no key outlives the round in which its item is forwarded. The storage of this variant is $\operatorname{ecc}(v)$ pairs, a table of $n$ distances and that buffer; it is free of the window length per source and not of $n$. □

The window ledger of Proposition S21.5, run numerically in Checks S47.198 and S47.200, is needed because of process noise. In a static world the items of one source may be merged, and a counter per source replaces the keys.

**Proposition S42.11** (Per-source cumulative pairs)**.** *Let $Q_x = 0$, let $G$ be any connected graph whose $n$ members carry identifiers, and number the items of member $s$ in order of observation, $e_{s,1}, e_{s,2}, \ldots$, as pairs about $x_0$. Every node $v$ stores for each source $s$ a counter $c_v(s)$ and a pair $E_v(s)$, initially zero. An observation by $v$ increases $c_v(v)$ by one and adds the new item to $E_v(v)$. A report $u \to v$ replaces $(c_v(s), E_v(s))$ by $(c_u(s), E_u(s))$ for every source $s$ with $c_u(s) > c_v(s)$. Then after any finite sequence of observations and reports*

$$D_v = \{(s,m) : m \leq c_v(s)\} \qquad \text{and} \qquad \pi_0 + \sum_s E_v(s) = \pi_0 + \sum_{e\in D_v} e,$$

*so the protocol is exact with $n$ pairs and $n$ counters per node and per message, whatever the number of rounds and whatever cycles $G$ has.*

*Proof.* We show by induction on the number of events that for all $v$ and $s$ the items of $s$ in $D_v$ are the first $c_v(s)$ and $E_v(s)$ is their sum. Initially all sets are empty. Items of $s$ are created at $s$ and enter $D_s$ at once, so $D_s$ contains every item of $s$ created so far and $c_u(s) \leq c_s(s)$ for every $u$; in particular

a report never replaces a node's entry for itself, and the item created by an observation of $v$ has number $c_v(v)+1$. An observation therefore preserves the claim. A report $u \to v$ replaces $D_v$ by $D_u \cup D_v$; for each source the union of two initial segments is the longer one, whose counter and sum are those of the entry with the larger counter. Exactness follows as in the proof of Theorem S21.2(a), because for $Q_x = 0$ the posterior pair of $x_0$ is the prior pair plus the sum of the items held. □

The storage grows with the number of members and not with time (Check S47.197). Whether growth with $n$ is necessary on graphs with cycles is Open problem S21.10. For $Q_x \succ 0$ and invertible $A$ a blind member that reports intermittently cannot merge the items of a window into one cumulative pair, by Theorem S21.3.

**Lemma S42.12** (Convex combinations of pairs are consistent)**.** *Let $m_1, \dots, m_J$ be estimators of a random vector $x$ whose errors $\tilde{e}_j = m_j - x$ have finite second moments, and let $P_j \succ 0$ satisfy $\mathbb{E}\tilde{e}_j\tilde{e}_j^\top \preceq P_j$. Let $\omega_j \geq 0$ be deterministic weights with $\sum_j \omega_j = 1$, and put*

$$P^{-1} = \sum_j \omega_j P_j^{-1}, \qquad m = P\sum_j \omega_j P_j^{-1} m_j.$$

*Then $\mathbb{E}(m-x)(m-x)^\top \preceq P$, whatever the joint law of the errors.*

*Proof.* Since $P\sum_j \omega_j P_j^{-1} = I$, the error is $m - x = P\sum_j \omega_j P_j^{-1}\tilde{e}_j$. Fix $u \in \mathbb{R}^d$ and put $\alpha_j = P_j^{-1}Pu$, so that $u^\top(m-x) = \sum_j \omega_j \alpha_j^\top \tilde{e}_j$. By convexity of the square, $(\sum_j \omega_j a_j)^2 \leq \sum_j \omega_j a_j^2$, hence

$$\mathbb{E}(u^\top(m-x))^2 \leq \sum_j \omega_j \alpha_j^\top (\mathbb{E}\tilde{e}_j\tilde{e}_j^\top)\alpha_j \leq \sum_j \omega_j \alpha_j^\top P_j \alpha_j = u^\top P\Big(\sum_j \omega_j P_j^{-1}\Big)Pu = u^\top Pu.$$

□

The lemma is the consistency property associated with covariance intersection (Section S3.5); the proof above is self-contained, and the original source was not consulted for it.

*Proof of Proposition S21.6.* Write $P_i$ for the claimed covariance of member $i$ and $\tilde{P}_i = \mathbb{E}(m_i - x_t)(m_i - x_t)^\top$ for the second-moment matrix of its error, which the statement calls its error covariance, at the stage of the step under discussion. The claimed covariances are deterministic and positive definite at every stage: the prediction $AP_iA^\top + Q_x$ is positive definite because $A$ is invertible or $Q_x \succ 0$, the update adds $\Lambda_i^{\mathrm{o}} \succeq 0$ to a positive definite precision, and a convex combination of positive definite precisions is positive definite. The initial means are deterministic, so the mean of a member at any stage is a function of the observations made so far, and its error is a function of $x_0$, of the process noises and of the sensing noises up to that stage.

(a) We show $\tilde{P}_i \preceq P_i$ after every stage by induction; it holds initially by hypothesis. Prediction: $m_i^- = Am_i$ and $x_t = Ax_{t-1} + w_{t-1}^x$, so the error becomes $A\tilde{e}_i - w_{t-1}^x$. The error $\tilde{e}_i$ is a function of $x_0$, of the process noises before $t-1$ and of the sensing noises up to $t-1$, so it is independent of $w_{t-1}^x$, which has mean zero, and $\tilde{P}_i^- = A\tilde{P}_iA^\top + Q_x \preceq AP_iA^\top + Q_x = P_i^-$. Observation: with the gain $K = P_i^- H_i^\top (H_iP_i^-H_i^\top + R_i)^{-1}$ computed from the claimed covariance, $m_i = m_i^- + K(o_{i,t} - H_im_i^-)$ has error $(I-KH_i)\tilde{e}_i^- + Kv_{i,t}$, and $v_{i,t}$ has mean zero and is independent of $\tilde{e}_i^-$, so

$$\tilde{P}_i = (I-KH_i)\tilde{P}_i^-(I-KH_i)^\top + KR_iK^\top \preceq (I-KH_i)P_i^-(I-KH_i)^\top + KR_iK^\top = P_i,$$

the last equality being the Joseph form of the Kalman update, valid for this gain. Fusion: the weights are deterministic, because the claimed covariances are deterministic sequences, and Lemma S42.12

applies to the members of the closed neighborhood whatever the dependence among their errors. No statistic about past exchanges entered.

(b) If the members are identical and start from a common prior, their claimed covariances are equal after every stage by induction: prediction and observation act identically on equal covariances, and a convex combination of equal precisions is that precision. The fusion stage therefore leaves the claimed covariance unchanged, and the claimed covariance follows the Riccati recursion of a member that never communicates. For the last claim of (b) the members need not be identical. If all start from $\pi_0$, each has precision $\Lambda_1^- + \Lambda_j^{\mathrm{o}}$ after the first observation, with $\Lambda_1^- = (AP_0A^\top + Q_x)^{-1}$, and member $i$ has $\Lambda_1^- + \sum_{j\in N[i]} \omega_{ij}\Lambda_j^{\mathrm{o}}$ after the first fusion. Its information set is then $\{(j,1) : j \in N[i]\}$, and the exact posterior precision of $x_1$ given these observations is $\Lambda_1^- + \sum_{j\in N[i]} \Lambda_j^{\mathrm{o}}$. The difference is $\sum_{j\in N[i]}(1-\omega_{ij})\Lambda_j^{\mathrm{o}} \succeq 0$. If two members $j_1, j_2$ of $N[i]$ have nonzero $\Lambda_j^{\mathrm{o}}$, then $\omega_{ij_1} + \omega_{ij_2} \le 1$, one of the two weights is at most $\frac{1}{2}$, and the difference is nonzero, so the output of member $i$ is not the exact pair.

(c) After the fusion of step $t$ the mean of $i$ is a function of the means of its closed neighborhood after the observation stage, hence of the observations of step $t$ made at distance at most one and of the neighborhood's means after the fusion of step $t-1$. By induction on $t$ it is a function of the observations $(s,\tau)$ with $\tau \le t - \max(\mathrm{dist}(i,s) - 1, 0)$, the set $D_i(t)$ of Proposition S21.5(a). Here the deterministic initial means are used. For a function $m$ of those observations, $\mathbb{E}(m - x_t)(m - x_t)^\top$ is the second-moment matrix of $m - \mathbb{E}[x_t \mid o_{D_i(t)}]$ plus the posterior covariance, which does not depend on the data in a Gaussian model, so $\tilde{P}_i \succeq \mathrm{Cov}(x_t \mid o_{D_i(t)})$.

(d) Here $d = 1$, $H_i = 1$, $R_i = R$ and $A = F$, and all members start from $\pi_0$. On the complete graph with equal weights $1/n$, every closed neighborhood is the whole group and by (b) the claimed precisions are equal, so after each fusion all members hold the same mean $\bar{m}_t$, the average of the updated means. With the common gain $K_t = P_t^-/(P_t^- + R)$ of the solitary filter and $\bar{o}_t = \frac{1}{n}\sum_i o_{i,t}$,

$$\bar{m}_t = F\bar{m}_{t-1} + K_t(\bar{o}_t - F\bar{m}_{t-1}),$$

whose error obeys $\tilde{e}_t = (1-K_t)(F\tilde{e}_{t-1} - w_{t-1}^x) + K_t\bar{v}_t$ with $\mathrm{Var}\,\bar{v}_t = R/n$ and the three terms independent. Hence the mean-square error $M_t = \mathbb{E}\tilde{e}_t^2$ satisfies $M_t = (1-K_t)^2(F^2M_{t-1}+Q)+K_t^2R/n$. The claimed variance follows $P_t = f(P_{t-1})$ with $f(P) = R(F^2P+Q)/(F^2P+Q+R)$. For $Q > 0$ the map $f$ is nondecreasing and concave on $P \ge 0$, with $f(0) > 0$ and $f < R$, so $f(P) - P$ is concave, positive at 0 and negative for $P \ge R$, and has exactly one zero $P_{\mathrm{loc}} > 0$. If $P < P_{\mathrm{loc}}$ then $P < f(P) \le f(P_{\mathrm{loc}}) = P_{\mathrm{loc}}$, and if $P > P_{\mathrm{loc}}$ then $P_{\mathrm{loc}} \le f(P) < P$, so the iterates are monotone and bounded and converge to the only fixed point. Therefore $P_t \to P_{\mathrm{loc}}$, $P_t^- \to P^- = F^2P_{\mathrm{loc}} + Q$ and $K_t \to K = P^-/(P^- + R) = P_{\mathrm{loc}}/R$, the last equality because $P_{\mathrm{loc}} = P^-R/(P^- + R)$. At the fixed point $P_{\mathrm{loc}} = (1-K)P^-$, so $P^- = F^2(1-K)P^- + Q$, hence $1 - F^2(1-K) = Q/P^- > 0$ and, since $0 < 1-K < 1$, $a = F^2(1-K)^2 < 1$. The recursion for $M_t$ is affine with coefficients that converge to $a < 1$ and to $b = (1-K)^2Q + K^2R/n$, so $M_t \to b/(1-a)$, the stated value. □

*Proof of Proposition S21.8.* Since $Q_x = 0$ and $A = I$ the state is one vector $x$, all observations are independent given $x$, and the posterior pair given any set of observations and summaries of distinct observations is the prior pair plus the sum of their items. The summary $y = S_io$ of an observation $o = H_ix + v$ is an observation with sensing matrix $S_iH_i$ and noise covariance $S_iR_iS_i^\top \succ 0$, so its item has precision $\Lambda_i^S$. Also

$$\Lambda_i^{\mathrm{o}} - \Lambda_i^S = H_i^\top R_i^{-1/2}(I - \mathsf{P})R_i^{-1/2}H_i \succeq 0, \qquad \mathsf{P} = R_i^{1/2}S_i^\top(S_iR_iS_i^\top)^{-1}S_iR_i^{1/2},$$

because $\mathsf{P}$ is symmetric with $\mathsf{P}^2 = \mathsf{P}$, an orthogonal projector. Equality holds when the range of $R_i^{-1/2}H_i$ lies in that of $R_i^{1/2}S_i^\top$, for instance when the rows of $S_i$ span the row space of $H_i^\top R_i^{-1}$, which takes rank $H_i$ numbers per observation.

(a) Run the protocol of Theorem S21.2(a) on the star, with $D_z$ the information set of the record and $C_i = D_i \cap D_z$ that of the link to position $i$. When the occupant leaves, the items of $D_i \setminus D_z$ leave every information set, and the newcomer starts with the empty set. On a star an item of position $i$ reaches another position only through the record, so the items of $D_i \setminus D_z$ lie in no other set and their removal leaves (i) and (ii) intact elsewhere. The predecessor's items that remain in $D_z$ are from then on counted as items whose source is the record. Treated as a new node with $s = c_i = \pi_0$, the newcomer then satisfies the invariants (i) to (iii) of that proof: (i) and (ii) hold because its set is empty, (iii) holds on the link to position $i$ because no item has the newcomer as its source and its set is empty, and (iii) is unchanged on every other link $jz$ because the record and position $i$ lie on the same side of that link. Without the change of source (iii) would fail on the link to position $i$, where the record holds items of the predecessor that the newcomer does not. The read is the report $z \to i$, after which $s_i = s_z$ and $c_i = s_z$; this is the reset, and the induction of Theorem S21.2(a) continues from it. Suppose instead that the newcomer copies $s_z = \pi_0 + \sum_{D^{\mathrm{r}}} e$ at the read, with $D^{\mathrm{r}}$ the record's set at that moment, while the link keeps $c_i = \pi_0 + \sum_{C_i} e$ from the predecessor; $C_i \subseteq D^{\mathrm{r}}$ because the record's set only grows. Let the newcomer then make observations with item set $N$ and report, the record's set having grown to $D'_z \supseteq D^{\mathrm{r}}$ meanwhile. The rule gives

$$s_z \leftarrow \Big(\pi_0 + \sum_{D'_z} e\Big) + \Big(\pi_0 + \sum_{D^{\mathrm{r}}} e + \sum_{N} e\Big) - \Big(\pi_0 + \sum_{C_i} e\Big) = \pi_0 + \sum_{D'_z \cup N} e + \sum_{D^{\mathrm{r}} \setminus C_i} e,$$

the exact pair plus a second copy of every item of $D^{\mathrm{r}} \setminus C_i$. If the newcomer does not read and starts from $\pi_0$ while the old link pair is kept, the same computation gives the exact pair minus $\sum_{C_i} e$, an undercount of what the predecessor shared.

(b) The record holds the prior pair and the items of the written summaries, each once, so its pair is the exact posterior pair given what was written, with precision $\Lambda_0 + \sum_i N_i^W(t)\Lambda_i^S$, where $N_i^W(t)$ counts the observations of position $i$ up to $t$ that were written. An observation made at step $j$ of a window of $k$ steps is written at the end of the window exactly when its observer is not replaced at the ends of steps $j, \dots, k-1$, an event of probability $(1-g)^{k-j}$. Summing over $j$ and over the $t/k$ windows, $\mathbb{E}N_i^W(t) = \frac{t}{k}\sum_{j=1}^{k}(1-g)^{k-j} = t\rho(g,k)$. As the average of $1, (1-g), \dots, (1-g)^{k-1}$, $\rho$ is nonincreasing in $k$ and in $g$, equals one for $k = 1$ and tends to one as $g \to 0$.

(c) At a write time $t$, before the replacements of that step, the occupant of position $i$ has made the last $a_i$ observations of the position, holds them raw, and has written them all; the other $N_i^W - a_i$ written observations of the position are known only through their summaries, and the remaining $t - N_i^W$ are known to nobody present. The summaries of the present occupant are functions of its raw observations, so the collective posterior has precision

$$\Lambda_{\mathrm{coll}}(t) = \Lambda_0 + \sum_i \big[(N_i^W - a_i)\Lambda_i^S + a_i\Lambda_i^{\mathrm{o}}\big] = \Lambda_0 + \sum_i N_i^W\Lambda_i^S + \sum_i a_i(\Lambda_i^{\mathrm{o}} - \Lambda_i^S).$$

The tenure satisfies $a_i(t) = 1$ if the position was refilled at the end of step $t-1$ and $a_i(t) = a_i(t-1)+1$ otherwise, so $\mathbb{E}a_i(t) = 1 + (1-g)\mathbb{E}a_i(t-1)$ with $\mathbb{E}a_i(1) = 1$, that is, $\mathbb{E}a_i(t) = (1-(1-g)^t)/g$. Taking expectations gives the formula for $\mathbb{E}\Lambda_{\mathrm{coll}}$. Subtracting from $\Lambda_0 + t\sum_i \Lambda_i^{\mathrm{o}}$ gives the deficit, a sum of positive semidefinite terms since $t \geq N_i^W \geq a_i$. For $\Lambda_i^{\mathrm{o}} \neq 0$ the term of position $i$ vanishes exactly when $N_i^W = t$ and either $a_i = t$ or $\Lambda_i^S = \Lambda_i^{\mathrm{o}}$. If every leaver writes its unwritten observations on exit, then $N_i^W = t$ and $\rho$ is replaced by one; if in addition summaries are lossless, the deficit is zero for every $g$ and $k$. □

*Remark* S42.13 (Turnover with process noise). For $Q_x \succ 0$ and full-rate reports of full pairs, turnover is free under Theorem S21.2(b) provided the link pair is reset: a newcomer that reads $s_t$ and whose

link pair is set to $s_t$ produces the increment $\Pi(s_t) + e_{i,t+1} - \Pi(s_t) = e_{i,t+1}$ at the next step. For $k > 1$ a leaver's unwritten window is lost, and by Theorem S21.3 what a blind exit write must contain to avoid the loss is the window ledger, up to $k \operatorname{rank} H_i$ numbers, and not a pair. The sizes of the loss in Check S47.199 are numerical.

*Remark* S42.14 (Memory accounting). The entries of Table S13 follow from the statements. A pair is a symmetric $d \times d$ matrix and a $d$-vector, $d(d+3)/2$ numbers. Channel pairs take one pair for each of the $n-1$ links of a tree (Theorem S21.2), whatever the number of rounds $T$. An explicit ledger keyed by observation holds up to $nT$ items of rank $H_i$ numbers each, with their keys, at every node that has heard from everyone; it removes duplicates on any graph and under any schedule, and for $Q_x \succ 0$ it is exact because the keys carry the times. Per-source cumulative pairs take $n$ pairs and $n$ counters per node (Proposition S42.11), and the ledger keyed by member and time at most $n \max(\operatorname{ecc}(v) - 1, 1)$ items and $n$ times, or $\operatorname{ecc}(v)$ pairs with a distance table and a buffer of $n-1$ items that wait one round to be forwarded (Proposition S21.5). A blind member that reports every $k$ steps sends $k \operatorname{rank} H_i$ numbers per contact (Theorem S21.3), a model-aware one $d$ numbers under a synchronous schedule (Proposition S42.6). Covariance intersection stores nothing beyond the node pairs (Proposition S21.6). Covariance intersection is consistent and, when two members of a closed neighborhood are informative, not exact. Among the schemes of the table the channel pairs are the only one whose storage per link depends on neither $n$ nor $T$, and they are exact only on trees. Every exact scheme for graphs with cycles in this section stores an amount that grows with $n$ or with the diameter, with one exception that is not a scheme for general graphs: the hop-counter rule of Proposition S42.9(a) stores one pair and two items, and it is exact on the one ring size it was built for, with one observation per member.

## S43 Proofs for Section S22

Throughout, $\|\cdot\|$ is the Euclidean norm of a vector and the operator norm of a matrix, $\|\cdot\|_F$ is the Frobenius norm, $\kappa(P) = \lambda_{\max}(P)/\lambda_{\min}(P)$, $s_{\min}(H)$ is the smallest singular value of $H$, and $\mathrm{O}(k)$ is the orthogonal group. On a fixed edge we write $P = P_i$, $H = H_j$, $\hat{H} = \hat{H}_{ij}$, $O = O_{ij}$, and so on. Three lemmas are used repeatedly. They are classical facts about Gaussian vectors and matrices, proved here so that the section does not rest on sources we did not consult.

**Lemma S43.1** (Inverse moments of a Gaussian norm)**.** *Let $X$ be a Gaussian vector in a Euclidean space $V$ of dimension $d \geq 1$ with mean $\theta$ and covariance $\Sigma \succeq \lambda I_V$, $\lambda > 0$.*

*(a) For $0 < p < d$, $\mathbb{E}\|X\|^{-p} \leq 1 + 2\big[(2\lambda)^{d/2}\Gamma(d/2)(d-p)\big]^{-1}$.*

*(b) For $d \geq 3$, $\mathbb{E}\|X\|^{-2} \leq [\lambda(d-2)]^{-1}$, with equality when $\theta = 0$ and $\Sigma = \lambda I_V$.*

*(c) For $p \geq d$ and every continuous $g \geq 0$ on the unit sphere of $V$ that is not identically zero, $\mathbb{E}[g(X/\|X\|)\|X\|^{-p}] = \infty$.*

*Proof.* The density $f$ of $X$ with respect to Lebesgue measure on $V$ is continuous, positive and bounded by $(2\pi\lambda)^{-d/2}$. (a) Split at $\|x\| = 1$: the outer part contributes at most one, and the inner part at most $(2\pi\lambda)^{-d/2} \int_{\|x\|<1} \|x\|^{-p} dx = (2\pi\lambda)^{-d/2} \cdot 2\pi^{d/2}\Gamma(d/2)^{-1}(d-p)^{-1}$. (c) Let $c_0 > 0$ be the minimum of $f$ on the closed unit ball. In polar coordinates the expectation is at least $c_0 \big(\int g(\omega) d\omega\big)\big(\int_0^1 r^{d-1-p} dr\big)$, where the first integral is positive (for $d = 1$ it is the sum of $g$ over the two unit vectors) and the second is infinite for $p \geq d$. (b) Write $X = X_1 + X_2$ with $X_1 \sim \mathcal{N}(\theta, \lambda I_V)$ and $X_2 \sim \mathcal{N}(0, \Sigma - \lambda I_V)$ independent, so that $X \sim \mathcal{N}(\theta + X_2, \lambda I_V)$ given $X_2$. For $Y \sim \mathcal{N}(\vartheta, \lambda I_V)$

the identities $s^{-1} = \int_0^\infty e^{-ts}dt$ and $\mathbb{E}e^{-t\|Y\|^2} = (1+2t\lambda)^{-d/2}\exp(-t\|\vartheta\|^2/(1+2t\lambda))$ give, by Tonelli's theorem,

$$\mathbb{E}\|Y\|^{-2} = \int_0^\infty (1+2t\lambda)^{-d/2}\exp\Big(-\frac{t\|\vartheta\|^2}{1+2t\lambda}\Big)dt \le \int_0^\infty (1+2t\lambda)^{-d/2}dt = \frac{1}{\lambda(d-2)}, \tag{S147}$$

with equality at $\vartheta = 0$. Taking the expectation over $X_2$ proves (b). □

**Lemma S43.2** (Reduction of a lift to one vector)**.** *Let $G \in \mathbb{R}^{r\times K}$ have full row rank $r \le K$, let $y \in \mathbb{R}^r$ be nonzero, $\hat{y} = y/\|y\|$, and let $[\hat{y}, Y_\perp]$ be an orthogonal $r \times r$ matrix. Put $a = G^\top\hat{y}$, $B = G^\top Y_\perp$, let $\Pi_B$ be the orthogonal projector onto $\mathrm{col}(B)$ and $c = (I-\Pi_B)a$. Then $c \neq 0$ and*

$$G^\top(GG^\top)^{-1}y = \|y\|\frac{c}{\|c\|^2}, \qquad y^\top(GG^\top)^{-1}y = \frac{\|y\|^2}{\|c\|^2}. \tag{S148}$$

*Proof.* The matrix $G^\top[\hat{y}, Y_\perp] = [a, B]$ has full column rank, so $a \notin \mathrm{col}(B)$ and $c \neq 0$. The vector $x = G^\top(GG^\top)^{-1}y$ is the unique element of $\mathrm{col}(G^\top)$ with $Gx = y$. The vector $x' = \|y\|c/\|c\|^2$ lies in $\mathrm{col}([a, B]) = \mathrm{col}(G^\top)$, and $[\hat{y}, Y_\perp]^\top Gx' = [a, B]^\top x' = \|y\|(1, 0, \dots, 0)^\top$, because $a^\top c = \|c\|^2$ and $B^\top c = 0$. Hence $Gx' = \|y\|\hat{y} = y$ and $x' = x$. The second identity is $\|x\|^2$. □

**Lemma S43.3** (Second moments of a matrix-normal error)**.** *Let $W \in \mathbb{R}^{r\times K}$ have independent standard normal entries and $\Delta = R^{1/2}W\Pi^{1/2}$ with $R \succeq 0$ and $\Pi \succeq 0$, so that $\mathbb{E}[\Delta_{ak}\Delta_{bl}] = R_{ab}\Pi_{kl}$. For $A \in \mathbb{R}^{K\times K}$, $B \in \mathbb{R}^{r\times r}$ and $C \in \mathbb{R}^{K\times r}$,*

$$\mathbb{E}[\Delta A\Delta^\top] = \mathrm{tr}(A\Pi)R, \qquad \mathbb{E}[\Delta^\top B\Delta] = \mathrm{tr}(BR)\Pi, \qquad \mathbb{E}[\Delta C\Delta] = RC^\top\Pi. \tag{S149}$$

*Proof.* Entrywise, $\mathbb{E}[\Delta A\Delta^\top]_{ab} = \sum_{kl} R_{ab}\Pi_{kl}A_{kl}$, $\mathbb{E}[\Delta^\top B\Delta]_{kl} = \sum_{ab} R_{ab}\Pi_{kl}B_{ab}$ and $\mathbb{E}[\Delta C\Delta]_{al} = \sum_{kb} R_{ab}\Pi_{kl}C_{kb}$. □

## S43.1 Moments of the understanding error

*Proof of Theorem S22.1.* Write $\sigma = \sigma_H$, $r = r_c$, $\hat{H} = H + \sigma W$ with $W$ a matrix of independent standard normal entries, and $G = \hat{H}P^{1/2}$, which has full row rank almost surely. Since $O = P^{1/2}G^\top(GG^\top)^{-1}$, the reading is $\hat{m} = (I - O\hat{H})m_i + Oy$ and

$$\psi = H(I - O\hat{H})m_i + HOy - u, \qquad I - O\hat{H} = P^{1/2}(I - G^\top(GG^\top)^{-1}G)P^{-1/2}. \tag{S150}$$

The middle factor of the second expression is an orthogonal projector, so $\|(I - O\hat{H})m_i\| \le \kappa(P)^{1/2}\|m_i\|$ for every realization of the code model. All dependence on small singular values of $\hat{H}$ therefore sits in $Oy$.

Condition on $(m_i, u, \nu)$ with $y \neq 0$ and apply Lemma S43.2 to $G$. Because $W^\top[\hat{y}, Y_\perp]$ again has independent standard normal entries, $W^\top\hat{y}$ and $W^\top Y_\perp$ are independent; hence $a = P^{1/2}H^\top\hat{y} + \sigma P^{1/2}W^\top\hat{y} \sim \mathcal{N}(P^{1/2}H^\top\hat{y}, \sigma^2 P)$ is independent of $B = P^{1/2}\hat{H}^\top Y_\perp$, which has rank $r-1$ almost surely. Given $B$, the vector $c = (I-\Pi_B)a$ is Gaussian on $V_B = \mathrm{col}(B)^\perp$, of dimension $d = K - r + 1$, with covariance $\sigma^2(I-\Pi_B)P(I-\Pi_B) \succeq \sigma^2\lambda_{\min}(P)I_{V_B}$ on $V_B$. By Eq. (S148),

$$Oy = \|y\|\frac{P^{1/2}c}{\|c\|^2}, \qquad \|HOy\| \le \|H\|\lambda_{\max}(P)^{1/2}\frac{\|y\|}{\|c\|}. \tag{S151}$$

Finite moments. Let $0 < p < d$. From Eqs. (S150) and (S151), $\|\psi\|^p \le 3^p(\|H\|^p\kappa(P)^{p/2}\|m_i\|^p + \|H\|^p\lambda_{\max}(P)^{p/2}\|y\|^p\|c\|^{-p} + \|u\|^p)$, and Lemma S43.1(a) with $\lambda = \sigma^2\lambda_{\min}(P)$ bounds $\mathbb{E}[\|c\|^{-p}$ |

$m_i, u, \nu, B]$ by a constant, so $\mathbb{E}\|\psi\|^p < \infty$. For $p = 2$ and $d \geq 3$, Lemma S43.1(b) gives $\mathbb{E}[\|c\|^{-2} \mid m_i, u, \nu, B] \leq [\sigma^2 \lambda_{\min}(P)(K - r - 1)]^{-1}$, and $(x_1 + x_2 + x_3)^2 \leq 3(x_1^2 + x_2^2 + x_3^2)$ with $\mathbb{E}\|y\|^2 = \mathbb{E}\|u\|^2 + \operatorname{tr} R_c$ gives the bound Eq. (S85).

Infinite moments. Let $p \geq d$. The remainder $\varrho = \psi - HOy = H(I - O\hat{H})m_i - u$ has $\mathbb{E}\|\varrho\|^p < \infty$, and $\|HOy\|^p \leq 2^p(\|\psi\|^p + \|\varrho\|^p)$, so it suffices to show $\mathbb{E}\|HOy\|^p = \infty$. By Eq. (S151), $\|HOy\|^p = \|y\|^p g_B(c/\|c\|)\|c\|^{-p}$ with $g_B(\omega) = \|HP^{1/2}\omega\|^p$. The subspace $P^{1/2}V_B$ has dimension $d = K - r + 1$, which exceeds $\dim\ker H = K - r$, so $g_B$ does not vanish identically on the unit sphere of $V_B$. By Lemma S43.1(c) the conditional expectation given $(m_i, u, \nu, B)$ is infinite whenever $y \neq 0$, an event of positive probability.

Tail bound. Given $B$, the density of $c$ on $V_B$ is at most $(2\pi\sigma^2\lambda_{\min}(P))^{-d/2}$, and the ball of radius $s$ in $V_B$ has volume $\pi^{d/2}s^d/\Gamma(d/2+1)$, so $\Pr(\|c\| < s \mid m_i, u, \nu, B) \leq s^d\big[(2\sigma^2\lambda_{\min}(P))^{d/2}\Gamma(d/2+1)\big]^{-1}$. By Eqs. (S150) and (S151), $\|\psi\| > t$ with $t > t_0$ forces $\|c\| < \|H\|\lambda_{\max}(P)^{1/2}\|y\|/(t - t_0)$, which is Eq. (S86). □

*Proof of Proposition S22.2.* Write $\sigma = \sigma_H$, $\hat{H} = H + \sigma W$, $\phi_0 = y - Hm_i$ and $\mathcal{A}_\rho = \{\sigma\|W\|_F \leq \rho\}$. On $\mathcal{A}_\rho$ the smallest singular value of $\hat{H}$ is at least $s_{\min}(H) - \rho > 0$, so the receiver map $O(\hat{H}) = P\hat{H}^\top(\hat{H}P\hat{H}^\top)^{-1}$ and the reading $\hat{m} = m_i + O(\hat{H})(y - \hat{H}m_i)$ are real-analytic functions of $\sigma W$ with derivatives of every order bounded on $\mathcal{A}_\rho$, and $\hat{m}$ is affine in $\nu$.

Differentiating $O$ at $H$ in the direction $W$ and using $O_0HP = PH^\top O_0^\top$ gives

$$DO[W] = PW^\top S_0^{-1} - O_0(WPH^\top + HPW^\top)S_0^{-1} = P_0^+W^\top S_0^{-1} - O_0WO_0, \tag{S152}$$

so that $\hat{m} = \hat{m}_0 + \sigma\hat{m}_1 + \varrho_2$ with

$$\hat{m}_1 = DO[W]\phi_0 - O_0Wm_i = P_0^+W^\top S_0^{-1}\phi_0 - O_0W\hat{m}_0, \tag{S153}$$

and $\|\varrho_2\| \leq C(1 + \|\nu\|)\sigma^2\|W\|_F^2$ on $\mathcal{A}_\rho$. Since $\psi = \nu - \sigma W\hat{m}$ by Eq. (S4),

$$\psi\psi^\top = \nu\nu^\top - \sigma\big[\nu(W\hat{m}_0)^\top + W\hat{m}_0\nu^\top\big] + \sigma^2\big[W\hat{m}_0\hat{m}_0^\top W^\top - \nu(W\hat{m}_1)^\top - W\hat{m}_1\nu^\top\big] + \sigma^3 T_3(W, \nu) + \varrho_4, \tag{S154}$$

where $T_3$ is a homogeneous cubic in $W$ and $\|\varrho_4\| \leq C(1 + \|\nu\|^2)\sigma^4\|W\|_F^4$ on $\mathcal{A}_\rho$.

The set $\mathcal{A}_\rho$ and the law of $W$ are invariant under $W \mapsto -W$, so the terms of odd order in $W$ have zero mean on $\mathcal{A}_\rho$. For the terms of order zero and two, restricting to $\mathcal{A}_\rho$ changes the mean by at most a constant times $\mathbb{E}[(1 + \|W\|_F^2); \mathcal{A}_\rho^{\mathrm{c}}]$, and the Chernoff bound for a chi-square variable with $r_cK$ degrees of freedom gives

$$\Pr(\mathcal{A}_\rho^{\mathrm{c}}) = \Pr(\chi^2_{r_cK} > \rho^2/\sigma^2) \leq 2^{r_cK/2}\exp(-\rho^2/(4\sigma^2)), \tag{S155}$$

which is smaller than every power of $\sigma$. It remains to evaluate the unrestricted mean of the second-order bracket. By Lemma S43.3 with $R = I$ and $\Pi = I$, $\mathbb{E}_W[W\hat{m}_0\hat{m}_0^\top W^\top] = \|\hat{m}_0\|^2 I$, and by Eq. (S153),

$$\mathbb{E}_W\big[(W\hat{m}_1)^\top\big] = \phi_0^\top S_0^{-1}\mathbb{E}_W[WP_0^+W^\top] - \hat{m}_0^\top\mathbb{E}_W[WO_0W]^\top = \operatorname{tr}(P_0^+)\phi_0^\top S_0^{-1} - \hat{m}_0^\top O_0. \tag{S156}$$

With $\phi_0 = u - Hm_i + \nu$ and $\hat{m}_0 = \bar{m}_0 + O_0\nu$ we have $\mathbb{E}[\nu\phi_0^\top] = R_c$, $\mathbb{E}[\nu\hat{m}_0^\top] = R_cO_0^\top$ and $\mathbb{E}\|\hat{m}_0\|^2 = \|\bar{m}_0\|^2 + \operatorname{tr}(O_0R_cO_0^\top)$. Hence $\mathbb{E}[\nu(W\hat{m}_1)^\top] = \operatorname{tr}(P_0^+)R_cS_0^{-1} - R_cO_0^\top O_0$, and collecting terms in Eq. (S154) gives Eq. (S87) for the truncated moment, with a remainder bounded by $C\sigma^4\mathbb{E}[(1 + \|\nu\|^2)\|W\|_F^4]$.

Let $K \geq r_c+2$, so that $d = K-r_c+1 \geq 3$, and fix $1 < p < d/2$. The proof of Theorem S22.1 and Lemma S43.1(a) with $\lambda = \sigma^2\lambda_{\min}(P)$ give $\mathbb{E}\|\psi\|^{2p} \leq C(1+\sigma^{-d})$ for $\sigma \leq 1$. By Hölder's inequality and Eq. (S155),

$$\mathbb{E}\big[\|\psi\|^2; \mathcal{A}_\rho^{\mathrm{c}}\big] \leq \big(\mathbb{E}\|\psi\|^{2p}\big)^{1/p} \Pr(\mathcal{A}_\rho^{\mathrm{c}})^{1-1/p} \leq C\sigma^{-d/p} \exp\Big(-\frac{(p-1)\rho^2}{4p\sigma^2}\Big), \tag{S157}$$

which is $O(\sigma^4)$, so the expansion holds for the full second moment. □

*Remark* S43.4 (Sign of the correction and the share of the first term). The first term of Eq. (S87) is the term $\sigma_H^2\mathbb{E}\|\hat{m}_{ij}\|^2 I$ named in Section S2.4. The remaining terms have the same order, are proportional to the channel noise, and come from the correlation between the noise and the response of the reading to the code error, which is the dependence that Open problem S2.9 asks about. Their sign is not fixed. For a square code, $P_0^+ = 0$ and $O_0 = H^{-1}$, so the correction is $\sigma_H^2(R_cH^{-\top}H^{-1} + H^{-\top}H^{-1}R_c)$, with positive trace. In general the three noise terms have trace $\sigma_H^2(-2\operatorname{tr}(P_0^+)\operatorname{tr}(R_cS_0^{-1}) + 2\operatorname{tr}(R_cO_0^\top O_0))$. Since $O_0^\top O_0 = S_0^{-1}HP^2H^\top S_0^{-1} \preceq \lambda_{\max}(P)S_0^{-1}$ and $\operatorname{tr}(P_0^+) \geq (K-r_c)\lambda_{\min}(P)$, this trace is negative whenever $R_c \neq 0$ and $K - r_c > \lambda_{\max}(P)/\lambda_{\min}(P)$, and the first term alone then overstates the trace of the excess at second order. For $P = pI$ the trace equals $-2\sigma_H^2 p(K-r_c-1)\operatorname{tr}(R_cS_0^{-1})$, which is positive for a square code, zero for $K = r_c+1$ and negative for $K \geq r_c+2$. The share of the excess carried by the first term is therefore instance dependent (Check S47.203).

### S43.2 Online code learning

The first statement collects classical properties of least squares in the notation of Section 2.5. It is not new; it is proved here because the later statements use its three parts and no source for it was consulted.

**Proposition S43.5** (Recursive least squares for a fixed code)**.** *Run Eq. (S2) with $R_w = R_c$ on grounded rounds $y_t = H\phi_t + \nu_t$ with regressors $\hat{\phi}_t$, where the $\nu_t \sim \mathcal{N}(0, R_c)$ are independent over rounds.*

*(a) If $P_0^H = \Pi_0 \otimes R_c$ with $\Pi_0 \succ 0$, then $P_t^H = \Pi_t \otimes R_c$ for all $t$, with $\Pi_{t+1}^{-1} = \Pi_t^{-1} + \hat{\phi}_t\hat{\phi}_t^\top$, and the update is $\hat{H}_{t+1} = \hat{H}_t + (y_t - \hat{H}_t\hat{\phi}_t)g_t^\top$ with $g_t = \Pi_t\hat{\phi}_t/(1+\hat{\phi}_t^\top\Pi_t\hat{\phi}_t)$, which does not involve $R_c$.*

*(b) Let $\mathcal{N} = \{w \in \mathbb{R}^K : \sum_t (w^\top\hat{\phi}_t)^2 < \infty\}$. Then $v^\top\Pi_T v \to 0$ as $T \to \infty$ if and only if $v \perp \mathcal{N}$. If the regressors are stationary and ergodic with second moment $M_2$, then $\mathcal{N} = \ker M_2$ almost surely.*

*(c) Let $\hat{\phi}_t = \phi_t$ be independent draws from $\mathcal{N}(0, M_2)$ with $M_2 \succ 0$, independent of the noise, and let $\hat{H}_T$ be the least-squares estimate from $T \geq K$ rounds. Then*

$$\mathbb{E}\|\hat{H}_T - H\|_F^2 = \frac{\operatorname{tr}(R_c)\operatorname{tr}(M_2^{-1})}{T-K-1} \qquad \text{for } T \geq K+2, \tag{S158}$$

*and the left side is infinite for $T \in \{K, K+1\}$.*

Part (b) is persistent excitation in the terms of Proposition 2.7: the code is learned exactly on the directions of content that keep occurring. Part (c) gives the rate $1/T$ with a constant set by the least exercised directions, and the infinite risk at $T \in \{K, K+1\}$ has the same origin as Theorem S22.1, the inversion of a Gaussian matrix that is nearly square.

*Proof.* (a) Suppose $P_t^H = \Pi_t \otimes R_c$ and write $\hat\phi = \hat\phi_t$, $X = \hat\phi^\top \otimes I_{r_c}$. Then $XP_t^H X^\top + R_c = (1+\hat\phi^\top\Pi_t\hat\phi)R_c$ and $P_t^H X^\top = (\Pi_t\hat\phi)\otimes R_c$, so the gain of Eq. (S2) is $g_t \otimes I_{r_c}$ with $g_t = \Pi_t\hat\phi/(1+\hat\phi^\top\Pi_t\hat\phi)$, free of $R_c$, and $P_{t+1}^H = P_t^H - (g_t\otimes I)XP_t^H = (\Pi_t - g_t\hat\phi^\top\Pi_t)\otimes R_c$. The mean update is $\mathrm{vec}\hat H_{t+1} = \mathrm{vec}\hat H_t + (g_t\otimes I)(y_t - \hat H_t\hat\phi)$, which is $\hat H_{t+1} = \hat H_t + (y_t - \hat H_t\hat\phi)g_t^\top$ because $\mathrm{vec}(ab^\top) = b\otimes a$. The Sherman–Morrison identity turns the recursion for $\Pi_t$ into $\Pi_{t+1}^{-1} = \Pi_t^{-1} + \hat\phi_t\hat\phi_t^\top$.

(b) The set $\mathcal{N}$ is a linear subspace because $(a+b)^2 \le 2a^2+2b^2$. The sequence $\Pi_T$ is nonincreasing in the Loewner order and bounded below by zero, so $\Pi_\infty = \lim_T \Pi_T$ exists. Let $w \in \mathcal{N}$ and $v \in \mathbb{R}^K$. By the Cauchy–Schwarz inequality, $(v^\top w)^2 \le (v^\top\Pi_T v)(w^\top\Pi_T^{-1}w)$, and $w^\top\Pi_T^{-1}w \le w^\top\Pi_0^{-1}w + \sum_t(w^\top\hat\phi_t)^2 < \infty$ uniformly in $T$. Hence $v^\top\Pi_\infty v > 0$ whenever $v$ is not orthogonal to $\mathcal{N}$. Conversely, put $w = \Pi_\infty v$. From $\Pi_\infty \preceq \Pi_T$ we get $\|\Pi_T^{-1/2}\Pi_\infty^{1/2}\| \le 1$, hence $w^\top\Pi_T^{-1}w \le v^\top\Pi_\infty v$ for every $T$, so $\sum_t(w^\top\hat\phi_t)^2 \le v^\top\Pi_\infty v$ and $w \in \mathcal{N}$. If $v \perp \mathcal{N}$ then $v^\top\Pi_\infty v = v^\top w = 0$. For the ergodic case, $T^{-1}\sum_{t<T}\hat\phi_t\hat\phi_t^\top \to M_2$ almost surely. On that event $\sum_t(w^\top\hat\phi_t)^2 = \infty$ for every $w$ with $w^\top M_2 w > 0$, while $w^\top M_2 w = 0$ gives $w^\top\hat\phi_t = 0$ for all $t$ almost surely, and it suffices to check this on a basis of $\ker M_2$. Hence $\mathcal{N} = \ker M_2$ almost surely.

(c) With $\Phi$ the $T\times K$ matrix of regressors and $W_T = \Phi^\top\Phi$, which is invertible almost surely for $T \ge K$, the least-squares estimate satisfies $\hat H_T - H = (\sum_t \nu_t\phi_t^\top)W_T^{-1}$, so $\mathbb{E}[\|\hat H_T - H\|_F^2 \mid \Phi] = \mathrm{tr}(R_c)\,\mathrm{tr}(W_T^{-1})$. Write $\Phi = \tilde\Phi M_2^{1/2}$ with $\tilde\Phi$ a matrix of independent standard normal entries and fix a unit vector $v$. Applying Lemma S43.2 to $G = \tilde\Phi^\top \in \mathbb{R}^{K\times T}$ and $x = M_2^{-1/2}v$ gives $v^\top W_T^{-1}v = x^\top(GG^\top)^{-1}x = \|x\|^2/\|c\|^2$, where, as in the proof of Theorem S22.1, $c$ is a standard Gaussian vector on a subspace of dimension $T-K+1$ given $B$. Thus $v^\top W_T^{-1}v$ is distributed as $v^\top M_2^{-1}v$ divided by a chi-square variable with $T-K+1$ degrees of freedom. By Lemma S43.1(b,c) the inverse chi-square has mean $1/(T-K-1)$ for $T \ge K+2$ and infinite mean for $T \in \{K, K+1\}$. Summing over an orthonormal basis gives Eq. (S158). □

**Lemma S43.6** (Operator Jensen inequalities)**.** *Let $X$ be a random symmetric positive definite matrix with $\mathbb{E}X$ and $\mathbb{E}X^{-1}$ finite. Then $\mathbb{E}[X^{-1}] \succeq (\mathbb{E}X)^{-1}$. For $q > 0$ the map $f(A) = (A^{-1} + qI)^{-1} = q^{-1}(I - (I+qA)^{-1})$ on positive definite matrices is monotone in the Loewner order, and $\mathbb{E}f(A) \preceq f(\mathbb{E}A)$ for every random positive definite $A$ with finite mean.*

*Proof.* The block matrix with rows $(X, I)$ and $(I, X^{-1})$ is positive semidefinite, hence so is its expectation, and the Schur complement of the block $\mathbb{E}X$ is $\mathbb{E}[X^{-1}] - (\mathbb{E}X)^{-1} \succeq 0$. The second expression for $f$ follows from $(A^{-1}+qI)^{-1} = A(I+qA)^{-1}$. Monotonicity holds because $A \preceq B$ implies $(I+qB)^{-1} \preceq (I+qA)^{-1}$, and the Jensen inequality follows from the first claim applied to $X = I + qA$, for which $X^{-1} \preceq I$. □

*Proof of Theorem S22.3.* Let $\mathcal{H}_t$ be generated by $(\phi_s, y_s)_{s<t}$. By induction the conditional law of $\ell_t$ given $\mathcal{H}_t$ is $\mathcal{N}(\hat\ell_t^-, P_t^-)$: the regressor $\phi_t$ is conditionally independent of $\ell_t$ given $\mathcal{H}_t$, so conditioning on it leaves this law unchanged; $y_t = \phi_t^\top\ell_t + \nu_t$ is then a Gaussian observation with known regressor, which gives $P_t = P_t^- - P_t^-\phi_t\phi_t^\top P_t^-/(\rho + \phi_t^\top P_t^-\phi_t)$; and the drift adds $qI$. In particular $\mathbb{E}\|\ell_t - \hat\ell_t^-\|^2 = \mathbb{E}\,\mathrm{tr}\,P_t^-$, and $qI \preceq P_t^- \preceq P_0^- + qtI$ for $t \ge 1$.

(a) Put $J_t = (P_t^-)^{-1}$. The update reads $P_t^{-1} = J_t + \phi_t\phi_t^\top/\rho$ and $J_{t+1} = f(J_t + \phi_t\phi_t^\top/\rho)$ with $f$ as in Lemma S43.6. Define $\bar J_0 = J_0$ and $\bar J_{t+1} = f(\bar J_t + \tilde M/\rho)$. If $\mathbb{E}J_t \preceq \bar J_t$, then by Lemma S43.6 and $\mathbb{E}[\phi_t\phi_t^\top] \preceq \tilde M$,

$$\mathbb{E}J_{t+1} \preceq f\big(\mathbb{E}J_t + \mathbb{E}[\phi_t\phi_t^\top]/\rho\big) \preceq f\big(\bar J_t + \tilde M/\rho\big) = \bar J_{t+1}, \tag{S159}$$

so $\mathbb{E}J_t \preceq \bar J_t$ for all $t$. No independence between $\phi_t$ and $J_t$ was used. A second application of Lemma S43.6 gives $\mathbb{E}P_t^- = \mathbb{E}[J_t^{-1}] \succeq (\mathbb{E}J_t)^{-1} \succeq \bar J_t^{-1} = \bar P_t$, and $\bar P_t$ obeys Eq. (S88). If $\bar P_0 = p_0 I$, then

$\bar{P}_t$ is diagonal in an eigenbasis of $\tilde{M}$ with entries iterating the scalar maps $\tau_k(p) = p\rho/(\rho + \tilde{\mu}_k p) + q$. For $\tilde{\mu}_k > 0$ the map $\tau_k$ is increasing and concave with $\tau_k(0) = q > 0$, so its iterates converge monotonically to its positive fixed point, the positive root of $\tilde{\mu}_k p^2 - q\tilde{\mu}_k p - q\rho = 0$, which is $p_k^\star = q/2 + (q^2/4 + q\rho/\tilde{\mu}_k)^{1/2}$. For $\tilde{\mu}_k = 0$ the iterates are $p_0 + qt$. A general $P_0^-$ lies between two multiples of the identity, the map $P \mapsto (P^{-1} + \tilde{M}/\rho)^{-1} + qI$ is monotone, and both isotropic sequences have the same limit, so $\operatorname{tr} \bar{P}_t \to \sum_k p_k^\star$, which gives Eq. (S89).

(b) Because the regressors are now independent of the code and the noise, $\hat{\ell}_t^- = \mathbb{E}[\ell_t \mid \mathcal{H}_t]$ has the least mean-square error among $\mathcal{H}_t$-measurable estimators. Compare it with the constant-gain recursion $\tilde{\ell}_{t+1} = \tilde{\ell}_t + \Gamma\phi_t(y_t - \phi_t^\top \tilde{\ell}_t)$, $\tilde{\ell}_0 = \hat{\ell}_0^-$, where $\Gamma = V \operatorname{diag}(g_k) V^\top$ and $g_k = (q/(\rho\tilde{\mu}_k))^{1/2}$. Its error $e_t = \ell_t - \tilde{\ell}_t$ obeys $e_{t+1} = (I - \Gamma\phi_t\phi_t^\top)e_t - \Gamma\phi_t\nu_t + w_t$ with $\phi_t$ independent of $e_t$. With $\Sigma_t = \mathbb{E}[e_t e_t^\top]$ and the Gaussian identity $\mathbb{E}[\phi\phi^\top\Sigma\phi\phi^\top] = p_{\mathrm{a}}(2M_2\Sigma M_2 + \operatorname{tr}(\Sigma M_2)M_2)$,

$$\Sigma_{t+1} = \Sigma_t - \Gamma\tilde{M}\Sigma_t - \Sigma_t\tilde{M}\Gamma + p_{\mathrm{a}}\Gamma(2M_2\Sigma_t M_2 + \operatorname{tr}(\Sigma_t M_2)M_2)\Gamma + \rho\Gamma\tilde{M}\Gamma + qI. \tag{S160}$$

The diagonal entries $s_k = (V^\top\Sigma_t V)_{kk}$ evolve on their own. With $a_k = g_k\tilde{\mu}_k = (q\tilde{\mu}_k/\rho)^{1/2}$ and $a_k g_k = q/\rho$,

$$s_k' = \Big(1 - 2a_k + \frac{2a_k^2}{p_{\mathrm{a}}}\Big)s_k + \frac{q}{\rho}\sum_l \mu_l s_l + 2q. \tag{S161}$$

This is an affine map $s \mapsto As + b$ with $b = 2q\mathbf{1}$ and $A$ entrywise nonnegative, since $1 - 2a + 2a^2/p_{\mathrm{a}} \geq (1-a)^2 + a^2$. Let $\bar{s}_k = (1+\epsilon)q/a_k = (1+\epsilon)(q\rho/\tilde{\mu}_k)^{1/2}$. Substituting, $(A\bar{s} + b)_k \leq \bar{s}_k$ is equivalent to $1 + (1+\epsilon)\xi \leq (1+\epsilon)(1-\gamma_k)$, where $\gamma_k = a_k/p_{\mathrm{a}} = (q\mu_k/(\rho p_{\mathrm{a}}))^{1/2}$ and $\xi = \frac{1}{2}\sum_l (q\mu_l/(\rho p_{\mathrm{a}}))^{1/2}$. With $\gamma = \max_k \gamma_k$, $\gamma + \xi < 1$ and $1 + \epsilon = (1 - \gamma - \xi)^{-1}$ this holds for every $k$. Then $A\bar{s} \leq \bar{s} - b \leq \theta\bar{s}$ with $\theta = \max_k(1 - b_k/\bar{s}_k) < 1$, so $A^n\bar{s} \leq \theta^n\bar{s}$; since $\bar{s}$ is positive and $A$ nonnegative, $A^n \to 0$, the iterates converge from every start to the fixed point $s^\star$, and the iterates started at $\bar{s}$ decrease, so $s^\star \leq \bar{s}$. Therefore $\limsup_t \mathbb{E} \operatorname{tr} P_t^- \leq \lim_t \mathbb{E}\|e_t\|^2 = \sum_k s_k^\star \leq (1 + \epsilon_q)\sum_k (q\rho/\tilde{\mu}_k)^{1/2}$.

(c) If $P_t^- v = \lambda_t v$ and $v^\top\phi_t = 0$, then $\phi_t^\top P_t^- v = 0$, so $P_t v = \lambda_t v$ and $P_{t+1}^- v = (\lambda_t + q)v$. A round with $\phi_t = 0$ leaves $P_t = P_t^-$, so $\tau$ consecutive unattended rounds add $q\tau I$. □

*Remark* S43.7 (Noisy grounding and several rows). If the receiver sees a noisy regressor $\hat{\phi}_t$ in place of $\phi_t$, with grounding noise independent of the code and of the channel noise given the regressors, the floor of Theorem S22.3(a) still holds: a receiver that also saw $\phi_t$ would have the conditional law computed above, and its information contains that of the actual receiver. For a code with $r_c$ rows, channel covariance $R_c = \operatorname{diag}(\rho_a)$, independent rows in the prior and independent row drifts, the rows decouple and the floor for $\mathbb{E}\|H_t - \hat{H}_t^-\|_F^2$ is the sum of the row floors, $\sum_a \sum_k [q/2 + (q^2/4 + q\rho_a/\tilde{\mu}_k)^{1/2}]$. The scaling $(q\rho/\tilde{\mu})^{1/2}$ per direction is the classical balance between lag and noise in tracking a random walk; the content of the theorem here is that the floor needs only a bound on the second moment of what the receiver is allowed to attend, and so applies to attention rules that depend on the learned calibration.

### S43.3 Joint learning of code and content

**Lemma S43.8** (Equal Gram matrices). *Let $A, A' \in \mathbb{R}^{r\times k}$ with $AA^\top = A'A'^\top$. Then $A' = AO$ for some $O \in \mathrm{O}(k)$.*

*Proof.* For every $x \in \mathbb{R}^r$, $\|A^\top x\| = \|A'^\top x\|$, so $A^\top x \mapsto A'^\top x$ is a well-defined linear isometry from $\operatorname{col}(A^\top)$ onto $\operatorname{col}(A'^\top)$. The two subspaces of $\mathbb{R}^k$ have equal dimension, so the isometry extends to some $\tilde{O} \in \mathrm{O}(k)$ with $\tilde{O}A^\top = A'^\top$. Take $O = \tilde{O}^\top$. □

*Proof of Theorem S22.4.* The stationary covariance of the referent is $\Sigma_\phi = \sum_a s_a \Pi_a$ with $s_a = q_a/(1-f_a^2) > 0$, because $\Sigma_\phi = F\Sigma_\phi F + Q$ has a unique solution and this one satisfies it. The message process is centered, stationary and Gaussian, so its law is determined by the autocovariances

$$\Lambda_k(H) = \mathbb{E}[y_{t+k} y_t^\top] = HF^k \Sigma_\phi H^\top + \delta_{k0} R_c = \sum_a f_a^k s_a H \Pi_a H^\top + \delta_{k0} R_c, \qquad k \geq 0. \tag{S162}$$

Suppose $\Lambda_k(H') = \Lambda_k(H)$ for $0 \leq k \leq m-1$, where $m$ is the number of distinct eigenvalues, and put $D_a = H'\Pi_a H'^\top - H\Pi_a H^\top$. Then $\sum_a f_a^k s_a D_a = 0$ for these $k$. The Vandermonde matrix $(f_a^k)$ is invertible because the $f_a$ are distinct (with $0^0 = 1$ if one of them vanishes), so $D_a = 0$ for every $a$. With $\Pi_a = V_a V_a^\top$ this says that $HV_a$ and $H'V_a$ have equal Gram matrices, and Lemma S43.8 gives $O_a \in \mathrm{O}(k_a)$ with $H'V_a = HV_aO_a$. Then $H' = \sum_a H'V_aV_a^\top = HT$ with $T = \sum_a V_aO_aV_a^\top$, which is orthogonal and commutes with $F$. Conversely, an orthogonal $T$ that commutes with $F$ preserves each eigenspace, hence commutes with each $\Pi_a$, and $HT\Pi_aT^\top H^\top = H\Pi_aH^\top$, so $\Lambda_k(HT) = \Lambda_k(H)$ for all $k$.

Dimension. The stabilizer of $H$ in $\mathcal{G}_F$ consists of the $T = \sum_a V_aO_aV_a^\top$ with $HV_aO_a = HV_a$ for every $a$, that is, with $O_a$ fixing the row space of $HV_a$ pointwise. It is isomorphic to $\prod_a \mathrm{O}(k_a - \varrho_a)$, and the orbit has the dimension Eq. (S91).

Symmetry. If $T \in \mathrm{GL}(K)$ and $(T\phi_t)$ has the law of $(\phi_t)$, then $TFT^{-1} = F$ and $TQT^\top = Q$. The first condition makes $T$ block diagonal, $T = \sum_a V_aT_aV_a^\top$, and the second then reads $q_aT_aT_a^\top = q_aI$, so $T \in \mathcal{G}_F$. The converse is immediate. Hence $\mathcal{G}_F$ is the full linear symmetry group of the referent law, and $(H, \phi) \mapsto (HT^{-1}, T\phi)$ leaves the messages unchanged. □

*Proof of Corollary S22.5.* The ungrounded stream identifies the orbit $H\mathcal{G}_F$ by Theorem S22.4, and the grounded pairs identify $H\mathcal{P}_M$ and nothing more by Proposition 2.7(b). The two data sets are independent, so $H'$ gives the same joint law if and only if $H' = HT$ with $T \in \mathcal{G}_F$ and $HT\mathcal{P}_M = H\mathcal{P}_M$. If $H$ is injective the second condition is $T\mathcal{P}_M = \mathcal{P}_M$: $T$ fixes $\mathrm{col}(M_2)$ pointwise. Since $T$ commutes with every $\Pi_a$, $Tx = x$ holds if and only if $T\Pi_a x = \Pi_a x$ for every $a$. So the condition is that $O_a$ fixes the $m_a$-dimensional subspace $V_a^\top \mathrm{col}(M_2)$ pointwise for every $a$, and such $O_a$ form a group isomorphic to $\mathrm{O}(k_a - m_a)$. The group $\prod_a \mathrm{O}(k_a - m_a)$ is trivial if and only if $m_a = k_a$ for all $a$, and finite if and only if $k_a - m_a \leq 1$ for all $a$. Because $H$ is injective, distinct $T$ give distinct codes. The two special cases follow by taking one block with $k_1 = K$ and $m_1 = \mathrm{rank}\, M_2$, and $K$ blocks with $k_a = 1$ and $m_a = 1$ exactly when $\mathcal{P}_M v_a \neq 0$. □

*Remark* S43.9 (Grounding inside the same stream). Corollary S22.5 treats the grounded pairs as a separate sample. The conclusion is the same when the receiver observes, along the stream itself, a component $z_t = C\phi_t$ of the referent, with $\mathcal{P}$ the projector onto the row space of $C$. The joint process $(y_t, z_t)$ is centered, stationary and Gaussian; the autocovariances of $z$ do not involve $H$, and the cross-covariances are $\mathbb{E}[y_{t+k}z_t^\top] = \sum_a f_a^k s_a H\Pi_a C^\top$ and their transposes. The Vandermonde argument identifies $H\Pi_a\mathcal{P}$ for each $a$, so the identified set is $\{HT : T \in \mathcal{G}_F,\ HT\Pi_a\mathcal{P} = H\Pi_a\mathcal{P} \text{ for all } a\}$. For injective $H$ the condition is that $T$ fixes each $\Pi_a \mathrm{col}(\mathcal{P})$ pointwise, which is the condition found in the proof above.

**Proposition S43.10** (Local identifiability on the gauge quotient)**.** *In the setting of Theorem S22.4 let $\varrho_a = \mathrm{rank}(HV_a)$. The kernel of the differential of $H \mapsto (\Lambda_k(H))_{k\geq 0}$ at $H$ contains the tangent space $\{H\Xi : \Xi^\top = -\Xi,\ \Xi F = F\Xi\}$ of the orbit, with codimension $\sum_a (r_c - \varrho_a)(k_a - \varrho_a)$ in the kernel. The two coincide if and only if $\varrho_a = \min(r_c, k_a)$ for every $a$, which holds for all $H$ outside a proper algebraic subset.*

*Proof.* By Eq. (S162) and the Vandermonde argument, $X$ lies in the kernel if and only if $X_aA_a^\top + A_aX_a^\top = 0$ for every $a$, where $A_a = HV_a$ and $X_a = XV_a$ are $r_c \times k_a$ and the blocks $X_a$ are free.

Fix $a$, drop the index, and take a singular value decomposition $A = U\tilde{A}N^\top$ in which $\tilde{A}$ has an invertible diagonal block $D$ of size $\varrho$ in its upper left corner and zeros elsewhere. In the blocks of $\tilde{X} = U^\top X N$, the condition is that $\tilde{X}_{11}D$ is skew-symmetric and $\tilde{X}_{21} = 0$, with $\tilde{X}_{12}$ and $\tilde{X}_{22}$ free. The solution space has dimension $\varrho(\varrho-1)/2 + \varrho(k-\varrho) + (r_c-\varrho)(k-\varrho)$. The orbit directions $A\Xi$ with $\Xi$ skew-symmetric are those with $D^{-1}\tilde{X}_{11}$ skew-symmetric, $\tilde{X}_{12}$ free and $\tilde{X}_{21} = 0$, $\tilde{X}_{22} = 0$; they satisfy the condition and span dimension $\varrho(\varrho-1)/2 + \varrho(k-\varrho)$. The difference $(r_c-\varrho)(k-\varrho)$ vanishes exactly when $\varrho = \min(r_c, k)$. □

Let $R_c \succ 0$. The vector of $n \geq m$ consecutive messages is centered Gaussian with a nonsingular covariance $\Sigma_n(H)$ whose blocks are the $\Lambda_k(H)$ with $k < n$ and their transposes, and its Fisher information form at $H$ in the direction $X$ is $\frac{1}{2}\operatorname{tr}\big(\Sigma_n^{-1} d\Sigma_n[X] \Sigma_n^{-1} d\Sigma_n[X]\big)$. This form vanishes if and only if $d\Sigma_n[X] = 0$, that is, by the Vandermonde argument, if and only if $X$ lies in the kernel of Proposition S43.10. At a point of full block rank the Fisher information of the message stream is therefore nondegenerate on the quotient by the gauge group, which is the local form of identifiability up to gauge. At rank-deficient blocks the Fisher information has additional null directions although, by Theorem S22.4, the set of observationally equivalent codes is still the orbit; such points are singular for the likelihood, and the standard root-$T$ asymptotics do not apply there. Gaussianity is used throughout: the statements concern the information in second moments, and non-Gaussian referents can carry more.

### S43.4 Decomposition of the receiver's risk

*Proof of Theorem S22.6.* Write $P = P_i$ and $\iota = H(\eta - m_i) + \nu \sim \mathcal{N}(0, S^*)$. Since $y - \hat{H}m_i = \iota + h - \Delta m_i$,

$$\hat{m} - \eta = \underbrace{\mathcal{K}^*\iota - (\eta - m_i)}_{e_{\mathrm{ch}}} + \underbrace{(\hat{\mathcal{K}} - \mathcal{K}^*)\iota}_{e_{\mathrm{gain}}} + \underbrace{\hat{\mathcal{K}}(h - \bar{h})}_{e_{\mathrm{dev}}} + \underbrace{\hat{\mathcal{K}}(\bar{h} - \Delta m_i)}_{e_{\mathrm{off}}}. \tag{S163}$$

The pair $(e_{\mathrm{ch}}, \iota)$ is jointly Gaussian and centered with $\operatorname{Cov}(e_{\mathrm{ch}}, \iota) = \mathcal{K}^* S^* - P H^\top = 0$, so $e_{\mathrm{ch}}$ is independent of $\iota$, and $\operatorname{Cov}(e_{\mathrm{ch}}) = P - \mathcal{K}^* S^* \mathcal{K}^{*\top} = P^+$. Both are functions of $(\eta, \nu)$, hence independent of $(h, \hat{H})$. The four vectors are square integrable: the first is Gaussian, $\mathbb{E}\|e_{\mathrm{gain}}\|^2 = \mathbb{E}\operatorname{tr}\big((\hat{\mathcal{K}} - \mathcal{K}^*) S^* (\hat{\mathcal{K}} - \mathcal{K}^*)^\top\big)$ and $\mathbb{E}\|e_{\mathrm{dev}}\|^2 = \mathbb{E}\operatorname{tr}(\hat{\mathcal{K}} \Sigma_h \hat{\mathcal{K}}^\top)$ are finite because $\mathbb{E}\|\hat{\mathcal{K}}\|_F^2 < \infty$, and $\|e_{\mathrm{off}}\| \leq \|\hat{\mathcal{K}}\|\|\bar{h}\| + \|\hat{\mathcal{K}}\Delta\|_F \|m_i\|$.

Orthogonality. The vector $e_{\mathrm{ch}}$ is centered and independent of $(\iota, h, \hat{H})$, of which the other three vectors are functions, so $\mathbb{E}\langle e_{\mathrm{ch}}, e\rangle_M = 0$ for each of them. Given $\hat{H}$, the vectors $\iota$ and $h - \bar{h}$ are independent and centered and $e_{\mathrm{off}}$ is fixed, so the conditional expectations of $\langle e_{\mathrm{gain}}, e_{\mathrm{dev}}\rangle_M$, $\langle e_{\mathrm{gain}}, e_{\mathrm{off}}\rangle_M$ and $\langle e_{\mathrm{dev}}, e_{\mathrm{off}}\rangle_M$ vanish. This is (b). Summing the four squared norms and expanding $\|\hat{\mathcal{K}}\bar{h} - \hat{\mathcal{K}}\Delta m_i\|_M^2$ gives Eq. (S92), which is (a).

(c) For a given gain the offset $e_{\mathrm{off}} = \hat{\mathcal{K}}(\bar{h} - \Delta m_i)$ depends on $(\bar{h}, \Delta)$ only through $\bar{h} - \Delta m_i$. The gain is a function of $\hat{H} = H + \Delta$, so two code errors with the same $\Delta m_i$ can give different offsets, and the first claim cannot be made for the offset as a whole. For example, let $K = 2$, $r_c = 1$, $P = I$, $H = (1, 0)$, $m_i = (1, 0)^\top$, and take the regularized gain of (d) with $\hat{R} = 1$: the code errors $\Delta = (0, 0)$ and $\Delta = (0, 1)$ both have $\Delta m_i = 0$, and their gains are $(1/2, 0)^\top$ and $(1/3, 1/3)^\top$, so the offsets $\hat{\mathcal{K}}\bar{h}$ differ whenever $\bar{h} \neq 0$ (a random instance is in Check S47.208). The gain does not depend on $\bar{h}$ or on $m_i$, so $C_{\mathrm{int}} = -2\mathbb{E}\langle \hat{\mathcal{K}}\bar{h}, \hat{\mathcal{K}}\Delta m_i\rangle_M$ is bilinear in $(\bar{h}, m_i)$ and vanishes when $\bar{h} = 0$ or $m_i = 0$. For Eq. (S93), split $C_{\mathrm{obj}} = \mathbb{E}\operatorname{tr}(M\hat{\mathcal{K}}\Sigma_h\hat{\mathcal{K}}^\top) + \mathbb{E}\|\hat{\mathcal{K}}\bar{h}\|_M^2$ and complete the square, $\mathbb{E}\|\hat{\mathcal{K}}\bar{h}\|_M^2 + \mathbb{E}\|\hat{\mathcal{K}}\Delta m_i\|_M^2 + C_{\mathrm{int}} = \mathbb{E}\|e_{\mathrm{off}}\|_M^2$. The right side of Eq. (S93) is a sum of two nonnegative terms, and $\mathbb{E}\|e_{\mathrm{off}}\|_M^2 = 0$ if and only if $M^{1/2} e_{\mathrm{off}} = 0$ almost surely, which gives the two cancellation statements. For the example let $K = r_c = 1$, $H = 1$, $M = 1$ and $\Sigma_h = 0$, and let the code error

$\Delta = \varepsilon$ and the gain $\hat{\mathcal{K}} = k$ be constants. The squared offset is $k^2(\bar{h} - \varepsilon m_i)^2$. With $\bar{h} = \varepsilon m_i$ it vanishes, while the separate terms $k^2\bar{h}^2$ and $k^2\varepsilon^2 m_i^2$ are equal and positive; with $\bar{h} = -\varepsilon m_i$ it is $4k^2\varepsilon^2 m_i^2$, twice their sum. The mean of the innovation $y - \hat{H}m_i$ is $\bar{h} - \Delta m_i$ for fixed $\Delta$, an affine function of $m_i$, which is determined by its values at $K+1$ affinely independent prior means and by no fewer.

(d) For the regularized gain put $G = \hat{R}^{-1/2}\hat{H}P^{1/2}$, so that $\hat{\mathcal{K}} = P^{1/2}G^\top(GG^\top + I)^{-1}\hat{R}^{-1/2}$. The singular values of $G^\top(GG^\top + I)^{-1}$ are $s/(s^2+1) \le \frac{1}{2}$ for the singular values $s$ of $G$. For the receiver map, $\|Oy\| \le \|O\|_F\|y\|$ and the proof of Theorem S22.1 shows $\mathbb{E}\|Oy\|^2 = \infty$ for $y \neq 0$ when $K \le r_c + 1$.

Finally, $P^+ - P_0^+ = PH^\top((HPH^\top)^{-1} - S^{*-1})HP \succeq 0$ because $S^* \succeq HPH^\top$, which gives the split of the channel term. □

**Corollary S43.11** (Calibration cost of a learned code)**.** *In the setting of Theorem S22.6 let $h = 0$, let $R \succ 0$, let the gain be $\hat{\mathcal{K}} = P\hat{H}^\top(\hat{H}P\hat{H}^\top + R)^{-1}$, and let $\Delta = R^{1/2}W\Pi^{1/2}$ with $W$ a matrix of independent standard normal entries, so that the covariance of* $\mathrm{vec}\Delta$ *is $\Pi \otimes R$ as in Proposition S43.5(a). As $\Pi \to 0$,*

$$\begin{aligned} C_{\mathrm{cal}} = {} & \operatorname{tr}(RS^{*-1})\operatorname{tr}(MP^+\Pi P^+) - 2\operatorname{tr}(MP^+\Pi\mathcal{K}^*R\mathcal{K}^{*\top}) \\ & + \big(\operatorname{tr}((P-P^+)\Pi) + m_i^\top\Pi m_i\big)\operatorname{tr}(M\mathcal{K}^*R\mathcal{K}^{*\top}) + O(\|\Pi\|^2). \end{aligned} \tag{S164}$$

*Proof.* Write $\mathcal{K}(\hat{H}) = P^{1/2}G^\top N^{-1}$ with $G = \hat{H}P^{1/2}$ and $N = GG^\top + R$. A derivative of order $n$ of $G \mapsto G^\top N^{-1}$ in a direction $X$ is a sum of products in which factors $N^{-1}$ alternate with factors $XG^\top$, $GX^\top$ or $XX^\top$, preceded by $G^\top$ or $X^\top$. Splitting each $N^{-1}$ as $N^{-1/2}N^{-1/2}$, every factor $G$ or $G^\top$ is adjacent to its own $N^{-1/2}$, and $\|N^{-1/2}G\| \le 1$, $\|N^{-1/2}\| \le \|R^{-1/2}\|$. Hence the derivatives of every order of $\mathcal{K}$ are bounded uniformly in $\hat{H}$, by constants times powers of $\|X\|$. The first derivative at $H$ is, as in Eq. (S152) with $S^*$ in place of $S_0$,

$$D\mathcal{K}[\Delta] = P^+\Delta^\top S^{*-1} - \mathcal{K}^*\Delta\mathcal{K}^*, \tag{S165}$$

using $P - \mathcal{K}^*HP = P^+$. By Taylor's theorem $\hat{\mathcal{K}} - \mathcal{K}^* = D\mathcal{K}[\Delta] + \frac{1}{2}D^2\mathcal{K}[\Delta,\Delta] + \varrho_3$ with $\|\varrho_3\| \le C\|\Delta\|^3$ for all $\Delta$. Inserting this into $C_{\mathrm{cal}}$, the terms of order three in $\Delta$ have zero mean because the law of $\Delta$ is symmetric, and the remaining terms beyond order two are bounded by $C\mathbb{E}(\|\Delta\|_F^4 + \|\Delta\|_F^6) = O(\|\Pi\|^2)$. The terms of order two are $\mathbb{E}\operatorname{tr}(MD\mathcal{K}[\Delta]S^*D\mathcal{K}[\Delta]^\top) + \mathbb{E}\|\mathcal{K}^*\Delta m_i\|_M^2$. Expanding with Eq. (S165) and applying Lemma S43.3,

$$\begin{aligned} \mathbb{E}[P^+\Delta^\top S^{*-1}\Delta P^+] &= \operatorname{tr}(RS^{*-1})P^+\Pi P^+, \\ \mathbb{E}[\mathcal{K}^*\Delta\mathcal{K}^*\Delta P^+] &= \mathcal{K}^*R\mathcal{K}^{*\top}\Pi P^+, \\ \mathbb{E}[\mathcal{K}^*\Delta\mathcal{K}^*S^*\mathcal{K}^{*\top}\Delta^\top\mathcal{K}^{*\top}] &= \operatorname{tr}((P-P^+)\Pi)\mathcal{K}^*R\mathcal{K}^{*\top}, \\ \mathbb{E}[\mathcal{K}^*\Delta m_i m_i^\top\Delta^\top\mathcal{K}^{*\top}] &= (m_i^\top\Pi m_i)\mathcal{K}^*R\mathcal{K}^{*\top}, \end{aligned}$$

where $\mathcal{K}^*S^*\mathcal{K}^{*\top} = P - P^+$. Taking the trace against $M$, with the cross term counted twice, gives Eq. (S164). □

*Remark* S43.12 (Learning curve and plateau). For a fixed code learned from $T$ grounded rounds with exactly observed content, Proposition S43.5 gives $\Pi_T = (\Pi_0^{-1} + \sum_{t<T}\phi_t\phi_t^\top)^{-1}$, which is $M_2^{-1}/T$ to leading order when $M_2 \succ 0$, so the calibration cost Eq. (S164) decays like $1/T$ with a constant read off by setting $\Pi = M_2^{-1}$. For a drifting code the error covariance does not fall below the floor of Theorem S22.3, of order $(q\rho/\tilde{\mu}_k)^{1/2}$ per direction. Equation (S164) assumes the Kronecker form

$\Pi \otimes R$, which a drifting code keeps when the drift covariance has the form $Q_\Pi \otimes R$, for instance for a single row, where $\Pi_t = P_t^-/\rho$, or for independent row drifts of equal variance with $R = \rho I$. The terms of second order in Eq. (S164) are the expectation of a nonnegative quadratic form in $\Delta$, hence a linear and nonnegative, and so monotone, function of $\Pi \succeq 0$; since the function is linear, its mean over the regressor history is its value at $\mathbb{E}\Pi_t$, and the floor $\mathbb{E}P_t^- \succeq \bar{P}_t$ bounds it from below by its value at the floor. That value is of order $q^{1/2}$ when $M \succ 0$ and $m_i \neq 0$, because the leading term is at least $(m_i^\top \Pi m_i)\operatorname{tr}(M\mathcal{K}^* R \mathcal{K}^{*\top})$ and $\mathcal{K}^* \neq 0$. In the attention model of Theorem S22.3(b) with $q$ small enough that $\gamma + \xi < 1$, Eq. (S90) bounds $\operatorname{tr}\mathbb{E}P_t^-$ in the limit at the same order, and a linear nonnegative function of $\Pi$ is at most its value at the identity times $\operatorname{tr}\Pi$; there the leading term of the calibration cost levels off at a plateau of order $q^{1/2}$. This is a statement about the leading term only: the remainder $O(\|\Pi\|^2)$ has not been bounded along the tracking recursion, where $\Pi$ is random through the regressors. In addition Corollary S43.11 holds the true code fixed and the code error random, whereas given a tracking history the code model $\hat{H}$ is fixed and the true code $H = \hat{H} - \Delta$ is random, so the constants $\mathcal{K}^*$, $S^*$ and $P^+$ of the expansion are themselves random there. In that conditional form no expansion is needed. If, given the history, $\Delta$ is centered Gaussian with covariance $\Pi \otimes R$ and independent of $(\eta, \nu)$, and $h = 0$, then $\hat{m} - \eta = (\hat{\mathcal{K}}\hat{H} - I)(\eta - m_i) + \hat{\mathcal{K}}\nu - \hat{\mathcal{K}}\Delta\eta$, the last vector is centered given $(\eta, \nu)$, and $\mathbb{E}[\Delta\eta\eta^\top\Delta^\top \mid \eta] = (\eta^\top\Pi\eta)R$, so the conditional risk is exactly

$$\operatorname{tr}(M\hat{P}^+) + (\operatorname{tr}(\Pi P) + m_i^\top \Pi m_i)\operatorname{tr}(M\hat{\mathcal{K}}R\hat{\mathcal{K}}^\top), \qquad \hat{P}^+ = P - \hat{\mathcal{K}}(\hat{H}P\hat{H}^\top + R)\hat{\mathcal{K}}^\top, \tag{S166}$$

with $\hat{\mathcal{K}} = P\hat{H}^\top(\hat{H}P\hat{H}^\top + R)^{-1}$ (Check S47.208). The second term is linear and nondecreasing in $\Pi$ and does not vanish while $\Pi \neq 0$, $M \succ 0$, $R \succ 0$ and $\hat{H} \neq 0$. The risk of a receiver that learns its interface is therefore the channel loss plus a calibration term that learning removes at rate $1/T$ for a fixed code and cannot remove for a drifting one.

**Proposition S43.13** (A learned code absorbs linear deviations)**.** *On grounded rounds let the receiver observe the content $\phi_t$ exactly, and let $y_t = H\phi_t + h_t + \nu_t$ with deviation $h_t = D\phi_t + d + \zeta_t$, where $D \in \mathbb{R}^{r_c \times K}$ and $d \in \mathbb{R}^{r_c}$ are fixed. Let the triples $(\phi_t, \zeta_t, \nu_t)$ be independent and identically distributed over rounds with mutually independent components, where $\phi_t$ has mean $\bar{m}$, second moment $M_2$ and covariance $C_\phi = M_2 - \bar{m}\bar{m}^\top$, $\zeta_t$ is centered with covariance $\Sigma_\zeta$, and $\nu_t$ is centered with covariance $R$.*

*(a) If $C_\phi \succ 0$, least squares with an intercept converges almost surely to $(H + D, d)$.*

*(b) If $M_2 \succ 0$, least squares without an intercept converges almost surely to $H + D + d\bar{m}^\top M_2^{-1}$.*

*(c) If $\zeta_t$ and $\nu_t$ are Gaussian, the law of the sample depends on $(H, D, d, \Sigma_\zeta, R)$ only through $(H + D, d, R + \Sigma_\zeta)$. For $\eta \sim \mathcal{N}(m_i, P)$ and a message $y = H\eta + D\eta + d + \zeta + \nu$ with $R + \Sigma_\zeta \succ 0$, the reading with code $H + D$, intercept $d$ and noise covariance $R + \Sigma_\zeta$ is the conditional mean of $\eta$, with risk $\operatorname{tr}(MP_D^+)$, where*

$$P_D^+ = P - P(H + D)^\top\big((H + D)P(H + D)^\top + R + \Sigma_\zeta\big)^{-1}(H + D)P. \tag{S167}$$

*For $D = -H$ this is $P_D^+ = P$, and for $D = (a - 1)H$ with $a > 1$ and $\Sigma_\zeta = 0$ it satisfies $P_D^+ \preceq P^+$.*

*Proof.* Write $y_t = (H + D)\phi_t + d + \varepsilon_t$ with $\varepsilon_t = \zeta_t + \nu_t$, which is centered, independent of $\phi_t$ and has covariance $R + \Sigma_\zeta$. (a) The second-moment matrix of the augmented regressor $(\phi_t, 1)$ is positive definite if and only if $C_\phi \succ 0$, since its Schur complement with respect to the entry 1 is

$M_2 - \bar{m}\bar{m}^\top = C_\phi$. By the strong law of large numbers the empirical normal equations converge almost surely to the population ones, whose unique solution is $(H + D, d)$ because $\mathbb{E}[\varepsilon_t(\phi_t^\top, 1)] = 0$. (b) Without an intercept the limit is $\mathbb{E}[y_t\phi_t^\top]M_2^{-1} = ((H + D)M_2 + d\bar{m}^\top)M_2^{-1}$. (c) Given $\phi_t$, the message is Gaussian with mean $(H + D)\phi_t + d$ and covariance $R + \Sigma_\zeta$, the rounds are independent, and the law of the regressors does not involve the parameters, so the law of the sample depends on $(H, D, d, \Sigma_\zeta, R)$ only through $(H + D, d, R + \Sigma_\zeta)$. For the reading, $y - d = (H + D)\eta + \varepsilon$ is a linear Gaussian observation of $\eta \sim \mathcal{N}(m_i, P)$ with code $H + D$ and noise covariance $R + \Sigma_\zeta$, so the conditional mean is the stated reading and its risk is $\mathrm{tr}(MP_D^+)$. With $D = -H$ the message is independent of $\eta$ and $P_D^+ = P$. With $D = (a-1)H$, $a > 1$ and $\Sigma_\zeta = 0$, the observation is equivalent to $H\eta + \nu/a$, whose noise covariance $R/a^2 \preceq R$ gives $P_D^+ \preceq P^+$. □

*Remark* S43.14 (What the decomposition is relative to). Proposition S43.13 limits the reading of Theorem S22.6. The decomposition takes a declared code $H$ and a deviation independent of the content. A deviation that is linear in the content is, for a receiver that learns from grounded pairs, a different code: it moves risk between the channel term and the calibration term and leaves no objective term. A constant deviation is absorbed by an intercept, or confounded with the code through the mean of the grounded content when no intercept is fitted. Only the idiosyncratic part $\zeta$ survives learning, and its law is that of additional channel noise. Whether a sender is biased or merely uses another code is therefore not a question that grounded message data can answer in the linear-Gaussian setting; it needs a reference outside the data, such as a declared convention or knowledge of the sender's objective. The relation to strategic communication, where the sender chooses its deviation in response to the receiver's reading, is interpretive here; no equilibrium is computed (Open problem S22.11).

### S43.5 Transfer along paths and through a hub

*Proof of Theorem S22.7.* (a) Insert $U_{i_u}U_{i_u}^{-1}$ between consecutive factors of $\Theta_{\mathfrak{p}} = \Theta_{i_L i_{L-1}} \cdots \Theta_{i_1 i_0}$: then $U_{i_L}^{-1}\Theta_{\mathfrak{p}}U_{i_0}$ becomes the product of the factors $U_{i_u}^{-1}\Theta_{i_u i_{u-1}}U_{i_{u-1}} = \Upsilon_{i_u i_{u-1}}$ in the same order, which is $\Upsilon_{\mathfrak{p}}$. For a directed closed walk this is $\mathrm{Hol}(\mathfrak{c}) = U_{i_0}\Upsilon_{\mathfrak{c}}U_{i_0}^{-1}$ in the convention of Section 6.3.

(b) Let $\Upsilon = I + \Delta$ with $\mathbb{E}\Delta = 0$ and $\mathbb{E}[\Delta^\top\Delta] = \delta^2 I$, and let $z$ be a square-integrable random vector independent of $\Upsilon$. Then $\mathbb{E}\|\Upsilon z\|^2 = \mathbb{E}\|z\|^2 + 2\mathbb{E}[z^\top\mathbb{E}[\Delta]z] + \mathbb{E}[z^\top\mathbb{E}[\Delta^\top\Delta]z] = (1 + \delta^2)\mathbb{E}\|z\|^2$. Applying this to the factors of $\Upsilon_{\mathfrak{p}}x$ from the outermost inward gives $\mathbb{E}\|\Upsilon_{\mathfrak{p}}x\|^2 = \prod_e(1 + \delta_e^2)\|x\|^2$. Independence gives $\mathbb{E}\Upsilon_{\mathfrak{p}} = I$, so

$$\mathbb{E}\|(\Upsilon_{\mathfrak{p}} - I)x\|^2 = \mathbb{E}\|\Upsilon_{\mathfrak{p}}x\|^2 - 2x^\top\mathbb{E}[\Upsilon_{\mathfrak{p}}]x + \|x\|^2 = \Big(\prod_e(1 + \delta_e^2) - 1\Big)\|x\|^2. \tag{S168}$$

A matrix of independent $\mathcal{N}(0, \delta^2/K)$ entries has $\mathbb{E}[\Delta^\top\Delta] = \delta^2 I$. Without isotropy the same steps give $\mathbb{E}[\Upsilon_{\mathfrak{p}}^\top\Upsilon_{\mathfrak{p}}] = \mathcal{T}_1(\mathcal{T}_2(\cdots\mathcal{T}_L(I)))$ with $\mathcal{T}_u(A) = A + \mathbb{E}[\Delta_u^\top A\Delta_u] \succeq A$ for $A \succeq 0$, where $\Delta_u$ is the error of the $u$th edge; each $\mathcal{T}_u$ is monotone, so the error $x^\top(\mathbb{E}[\Upsilon_{\mathfrak{p}}^\top\Upsilon_{\mathfrak{p}}] - I)x$ does not decrease when the walk is extended.

(c) With $\Theta_{ij} = g_i g_j^{-1}$ the factors of every holonomy cancel in pairs, also along steps that run against an edge, where the factor is $(g_{i_{u-1}}g_{i_u}^{-1})^{-1}$. So $\mathrm{Hol}(\mathfrak{c}) = I$ for every closed walk and every value of the maps $g_i$, and $\dim\mathrm{Cons}(\Theta) = K$ by Theorem 6.5(c). The mistranslation is $\Upsilon_{ij} = U_i^{-1}g_iU_h(U_j^{-1}g_jU_h)^{-1} = \Upsilon_i\Upsilon_j^{-1}$, and $\Theta_{ij} = \tilde{U}_i\tilde{U}_j^{-1}$ with $\tilde{U}_i = g_iU_h = U_i\Upsilon_i$. If $\|\Upsilon_k - I\| \le \epsilon_k < 1$, the Neumann series gives $\|\Upsilon_k^{-1}\| \le (1 - \epsilon_k)^{-1}$, and $\Upsilon_i\Upsilon_k^{-1} - I = (\Upsilon_i - \Upsilon_k)\Upsilon_k^{-1}$ with $\|\Upsilon_i - \Upsilon_k\| \le \epsilon_i + \epsilon_k$ gives Eq. (S95).

(d) A shortest directed closed walk of positive length at $i_0$ exists because the graph is strongly connected and $n \ge 2$, and it visits no agent twice, since otherwise a shorter one could be cut out

of it. Its edges are therefore distinct, and $\mathrm{Hol}(\mathfrak{c}) = \Theta^{(L)}A$ with $A = \Theta^{(L-1)}\cdots\Theta^{(1)}$ independent of $\Theta^{(L)}$. All factors are invertible almost surely, because singular matrices form a Lebesgue-null set. Now $\mathrm{Hol}(\mathfrak{c}) - I = (\Theta^{(L)} - A^{-1})A$ is singular if and only if $\det(\Theta^{(L)} - A^{-1}) = 0$. Given $A$, this is the zero set of a nonzero polynomial in the entries of $\Theta^{(L)}$, a Lebesgue-null set, so it has conditional probability zero. Hence $\mathrm{Fix}(\mathrm{Hol}(\mathfrak{c})) = \{0\}$ almost surely, and $\mathrm{Cons}(\Theta) = \{0\}$ by Theorem 6.5(a). The moment formula is (b) applied to $\Upsilon_{\mathfrak{c}}$ together with (a). □

*Remark* S43.15 (No mean-square zero-shot bound for Gaussian spokes). The bound Eq. (S95) is deterministic because a moment bound of the kind in Theorem S22.7(b) is false for the hub design with Gaussian spoke errors. Let $\Upsilon_k = I + \Delta_k$ with independent $\mathcal{N}(0, \tau^2)$ entries and $x \neq 0$. The first row $w^\top$ of $\Upsilon_k^{-1}$ is orthogonal to the columns $c_2, \dots, c_K$ of $\Upsilon_k$ and has $w^\top c_1 = 1$, so $w^\top x = (n^\top x)/(n^\top c_1)$ for a unit normal $n$ of their span. Given $c_2, \dots, c_K$, the scalar $n^\top c_1$ is Gaussian with variance $\tau^2$, and $n^\top x \neq 0$ almost surely, so Lemma S43.1(c) with $d = p = 1$ gives $\mathbb{E}\|\Upsilon_k^{-1}x\| \geq \mathbb{E}|w^\top x| = \infty$. For an independent $\Upsilon_i$ with mean $I$, Jensen's inequality gives $\mathbb{E}\|\Upsilon_i\Upsilon_k^{-1}x\| \geq \mathbb{E}\|\Upsilon_k^{-1}x\|$, so the hub mistranslation $\Upsilon_{ik} - I$ has no finite first moment. This is the same mechanism as in Theorem S22.1: inverting a learned square map. The first-order statement survives: $\Upsilon_i\Upsilon_k^{-1} - I = (\Delta_i - \Delta_k) - (\Delta_i - \Delta_k)\Delta_k(I + \Delta_k)^{-1}$, and the last term is at most $2\|\Delta_k\|(\|\Delta_i\| + \|\Delta_k\|)$ in norm when $\|\Delta_k\| \leq \frac{1}{2}$, so to first order the error between two agents that never met is the difference of their two spoke errors.

*Remark* S43.16 (Separate speaking and listening maps). A hub route can avoid inverses: agent $i$ trains a listening map $a_i$, an estimate of $U_iU_h^{-1}$, sender $k$ uses a speaking map $b_k$, an estimate of $U_hU_k^{-1}$, and $\Theta_{ik} = a_ib_k$. The mistranslation is a product of two factors, $U_i^{-1}a_iU_h$ and $U_h^{-1}b_kU_k$, and when these are independent Theorem S22.7(b) gives the exact mean-square error $(1+\delta_i^2)(1+\delta_k^2) - 1$ for every pair. The price is flatness. On the complete graph with $n \geq 3$ agents these translations are flat if and only if $b_ka_k = I$ for every $k$. Indeed the holonomy of $i_0 \to i_1 \to \cdots \to i_L = i_0$ is $a_{i_0}c_{i_{L-1}}\cdots c_{i_1}b_{i_0}$ with $c_k = b_ka_k$, which is the identity when every $c_k$ is; conversely the two-cycles give $c_k = c_i^{-1}$ for $i \neq k$, so with three agents $c_1 = c_3 = c_2^{-1}$ and $c_1 = c_1^{-1}$, and the three-cycle $1 \to 2 \to 3 \to 1$ gives $c_3c_2 = c_1^{-1}$, that is $c_1 = I$. A star in which the hub agent and the spokes train their translations independently realizes this design along its relays, and when the laws of the translations are absolutely continuous, Theorem S22.7(d) shows that almost surely it has no consensus configuration except zero.

**Corollary S43.17** (Budget comparison). *Suppose a translation trained on $T'$ grounded rounds has a mistranslation that satisfies the hypotheses of Theorem S22.7(b) with $\delta^2 = c/T'$, independently across maps, and let every agent spend $T$ rounds.*

*(a) In the pairwise design each agent trains its $n-1$ incoming translations on $T/(n-1)$ rounds each. A direct edge has mean-square mistranslation $c(n-1)/T$ and a relay over $L$ edges $(1 + c(n-1)/T)^L - 1$; the collective trains $n(n-1)$ maps.*

*(b) In a hub design with two independently trained legs per route, each on $T_h$ rounds, every pair of agents, including pairs that never interacted, has mean-square mistranslation $(1 + c/T_h)^2 - 1$. This is smaller than the direct pairwise value $c(n-1)/T$ of (a) if and only if $n > 3 + c/T$ when $T_h = T$, and if and only if $n > 5 + 4c/T$ when $T_h = T/2$.*

*Proof.* (a) is Eq. (S94) with $\delta_e^2 = c(n-1)/T$. (b) By Eq. (S94) with two edges the error is $2c/T_h + c^2/T_h^2$. For $T_h = T$ it is below $c(n-1)/T$ if and only if $2 + c/T < n - 1$, and for $T_h = T/2$ if and only if $4 + 4c/T < n - 1$. □

The case $T_h = T$ describes a dedicated hub agent that trains the legs into the hub with $T$ rounds per spoke, so that the hub spends $(n-1)T$ rounds in all; the case $T_h = T/2$ describes a shared code for which each agent trains both a speaking and a listening map from its own budget. The assumption $\delta^2 = c/T'$ is the leading order of Eq. (S158); it is an assumption about the learner, and the corollary is arithmetic given it. With one invertible map per agent the leading-order error is the sum of two spoke errors, $2c/T$, by Remark S43.15, and the comparison with the pairwise design at that order turns at $n = 3$.

## S44 Proofs for Section S23

The proofs follow the order of Section S23. Assumption S23.1 is in force throughout, and in this appendix we abbreviate $\mathsf{Q} = Q^{\Xi}$ and $\widetilde{\mathsf{Q}} = \widetilde{Q}^{\Xi}$, keeping $Q$ for the innovation variance of the scalar world of Definition S23.11. The transition density of Eq. (23) is $k(x' \mid x) = \mathcal{N}(x'; Ax, \mathsf{Q})$, the stationary density is $\pi = \mathcal{N}(0, \Sigma)$, and $\mathrm{Cov}(X_{t+1}, X_t) = A\Sigma$. Beyond the two lemmas below we use only the nonnegativity of relative entropy, its monotonicity under a measurable map applied to both arguments, and the chain rule for mutual information.

**Lemma S44.1** (Relative entropy of centered Gaussian laws)**.** *For positive definite $k \times k$ matrices $S_0$ and $S_1$,*

$$D_{\mathrm{KL}}(\mathcal{N}(0, S_0) \| \mathcal{N}(0, S_1)) = \tfrac{1}{2}\big[\mathrm{tr}(S_1^{-1} S_0) - k + \ln\det S_1 - \ln\det S_0\big],$$

*and the value is zero if and only if $S_0 = S_1$.*

*Proof.* With $U \sim \mathcal{N}(0, S_0)$, the logarithm of the ratio of the two densities at $U$ is $-\frac{1}{2}U^\top S_0^{-1} U + \frac{1}{2}U^\top S_1^{-1} U - \frac{1}{2}\ln\det S_0 + \frac{1}{2}\ln\det S_1$, and $\mathbb{E}U^\top S^{-1} U = \mathrm{tr}(S^{-1} S_0)$. Relative entropy vanishes only when the two laws agree, and a centered Gaussian law determines its covariance. □

**Lemma S44.2** (Stationary covariances)**.** *(i) $\Sigma \succeq \mathsf{Q} \succ 0$, and the covariance of $(X_0, \dots, X_n)$ is positive definite for every $n \geq 0$.*

*(ii) $\widetilde{\mathsf{Q}} = \mathrm{Cov}(X_0 \mid X_1) \succ 0$ and $\det \widetilde{\mathsf{Q}} = \det \mathsf{Q}$.*

*(iii) $\mathsf{J} - A\mathsf{J}A^\top = A\mathsf{Q} - \mathsf{Q}A^\top$, and $\mathsf{J} = \sum_{k \geq 0} A^k (A\mathsf{Q} - \mathsf{Q}A^\top)(A^\top)^k$. In particular $\mathsf{J} = 0$ if and only if $A\mathsf{Q} = \mathsf{Q}A^\top$.*

*Proof.* (i) Since $\rho(A) < 1$, $\Sigma = \sum_{k \geq 0} A^k \mathsf{Q} (A^\top)^k \succeq \mathsf{Q}$. The vector $(X_0, \dots, X_n)$ is the image of $(X_0, \Xi_0, \dots, \Xi_{n-1})$, whose covariance $\mathrm{diag}(\Sigma, \mathsf{Q}, \dots, \mathsf{Q})$ is positive definite, under a linear map that is block lower triangular with identity diagonal blocks, hence invertible.

(ii) By (i) the covariance of $(X_0, X_1)$, with blocks $\Sigma$, $\Sigma A^\top$, $A\Sigma$, $\Sigma$, is positive definite. Its Schur complement with respect to the first diagonal block is $\Sigma - A\Sigma\Sigma^{-1}\Sigma A^\top = \mathsf{Q}$, and with respect to the second it is $\Sigma - \Sigma A^\top \Sigma^{-1} A\Sigma = \widetilde{\mathsf{Q}}$, the conditional covariance of $X_0$ given $X_1$. Both are positive definite, and the determinant of the joint covariance equals $\det\Sigma \det\mathsf{Q}$ and $\det\Sigma \det\widetilde{\mathsf{Q}}$.

(iii) Using $\mathsf{Q} = \Sigma - A\Sigma A^\top$,

$$A\mathsf{Q} - \mathsf{Q}A^\top = A\Sigma - A^2\Sigma A^\top - \Sigma A^\top + A\Sigma(A^\top)^2 = (A\Sigma - \Sigma A^\top) - A(A\Sigma - \Sigma A^\top)A^\top.$$

If $Z = AZA^\top$ then $Z = A^k Z (A^\top)^k \to 0$, so the equation $Z - AZA^\top = B$ has at most one solution, and the convergent series $\sum_k A^k B (A^\top)^k$ is one. □

*Proof of Theorem S23.3.* We first show that the window divergence is additive. At the point $(x_0, \dots, x_n)$ the law of $(X_0, \dots, X_n)$ and the law of the reversed window $(X_n, \dots, X_0)$ have the densities

$$\pi(x_0) \prod_{t<n} k(x_{t+1} \mid x_t) \qquad \text{and} \qquad \pi(x_n) \prod_{t<n} k(x_t \mid x_{t+1}).$$

Both are nondegenerate Gaussian densities by Lemma S44.2(i), and the logarithm of their ratio is a quadratic form, so its expectation is finite and

$$\mathsf{D}_n = \mathbb{E}[\ln \pi(X_0) - \ln \pi(X_n)] + \sum_{t<n} \mathbb{E} \ln \frac{k(X_{t+1} \mid X_t)}{k(X_t \mid X_{t+1})} = n\mathbb{E} \ln \frac{k(X_1 \mid X_0)}{k(X_0 \mid X_1)},$$

because $X_0$ and $X_n$ have the same law and the pairs $(X_t, X_{t+1})$ have the same law for all $t$. Hence $\mathsf{D}_n = n\mathsf{D}_1$ and $\mathsf{ep} = \mathsf{D}_1$.

Next we compute the one-step value. Since $\ln k(x' \mid x) = -\frac{1}{2}(x' - Ax)^\top \mathsf{Q}^{-1}(x' - Ax) - \frac{1}{2} \ln \det(2\pi\mathsf{Q})$ and $\mathbb{E}(X_1 - AX_0)(X_1 - AX_0)^\top = \mathsf{Q}$,

$$\mathsf{ep} = \tfrac{1}{2} \operatorname{tr}(\mathsf{Q}^{-1}(\mathsf{M} - \mathsf{Q})), \qquad \mathsf{M} = \mathbb{E}(X_0 - AX_1)(X_0 - AX_1)^\top = \Sigma - \Sigma(A^\top)^2 - A^2\Sigma + A\Sigma A^\top.$$

With $\mathsf{Q} = \Sigma - A\Sigma A^\top$ this gives $\mathsf{M} - \mathsf{Q} = 2A\Sigma A^\top - A^2\Sigma - \Sigma(A^\top)^2 = \mathsf{J}A^\top - A\mathsf{J}$. Since $\mathsf{J}^\top = -\mathsf{J}$ and $\mathsf{Q}^{-1}$ is symmetric, $\operatorname{tr}(\mathsf{Q}^{-1}A\mathsf{J}) = \operatorname{tr}(\mathsf{J}^\top A^\top \mathsf{Q}^{-1}) = -\operatorname{tr}(\mathsf{Q}^{-1}\mathsf{J}A^\top)$, so $\mathsf{ep} = \operatorname{tr}(\mathsf{Q}^{-1}\mathsf{J}A^\top)$, the first expression in Eq. (S97).

For the decomposition into two nonnegative terms, note that

$$\mathsf{J}\Sigma^{-1}\mathsf{J}^\top = (A\Sigma - \Sigma A^\top)\Sigma^{-1}(\Sigma A^\top - A\Sigma) = A\Sigma A^\top - A^2\Sigma - \Sigma(A^\top)^2 + \Sigma A^\top \Sigma^{-1} A\Sigma,$$

so $\mathsf{M} = \widetilde{\mathsf{Q}} + \mathsf{J}\Sigma^{-1}\mathsf{J}^\top$ and the second expression in Eq. (S97) follows. Its second term is $\frac{1}{2}\|\mathsf{Q}^{-1/2}\mathsf{J}\Sigma^{-1/2}\|_F^2 \geq 0$, with equality exactly when $\mathsf{J} = 0$. For the first term, the eigenvalues of $\mathsf{Q}^{-1/2}\widetilde{\mathsf{Q}}\mathsf{Q}^{-1/2}$ are positive with product $\det \widetilde{\mathsf{Q}} / \det \mathsf{Q} = 1$ by Lemma S44.2(ii), so their arithmetic mean is at least one, that is, $\operatorname{tr}(\mathsf{Q}^{-1}\widetilde{\mathsf{Q}}) \geq d$ with $d = \dim X$, and $\frac{1}{2}\operatorname{tr}(\mathsf{Q}^{-1}(\widetilde{\mathsf{Q}} - \mathsf{Q})) \geq 0$.

For the equivalences, if $\mathsf{ep} = 0$ both terms vanish, so $\mathsf{J} = 0$; if $\mathsf{J} = 0$ the first expression gives $\mathsf{ep} = 0$. By Lemma S44.2(iii), $\mathsf{J} = 0$ if and only if $A\mathsf{Q} = \mathsf{Q}A^\top$. The chain is time reversible when $(X_0, \dots, X_n)$ and $(X_n, \dots, X_0)$ have the same law for every $n$, that is, when $\mathsf{D}_n = 0$ for every $n$, which by additivity is $\mathsf{ep} = 0$.

For normal drift, let $AA^\top = A^\top A$ and $\mathsf{Q} = qI$. Then $\Sigma = q\sum_k (AA^\top)^k = q(I - AA^\top)^{-1}$ commutes with $A$ and $A^\top$, and $\mathsf{ep} = q^{-1}\operatorname{tr}(\mathsf{J}A^\top) = \operatorname{tr}((A - A^\top)(I - AA^\top)^{-1}A^\top)$. A real normal matrix is unitarily diagonalizable, $A = U\Lambda U^*$ and $A^\top = U\overline{\Lambda}U^*$, so

$$\mathsf{ep} = \sum_k \frac{(\lambda_k - \overline{\lambda_k})\overline{\lambda_k}}{1 - |\lambda_k|^2} = \sum_k \frac{|\lambda_k|^2 - \operatorname{Re}(\lambda_k^2)}{1 - |\lambda_k|^2} = \sum_k \frac{2(\operatorname{Im} \lambda_k)^2}{1 - |\lambda_k|^2},$$

where the second equality holds because the sum is real and the last because $|\lambda|^2 - \operatorname{Re}(\lambda^2) = 2(\operatorname{Im}\lambda)^2$. □

*Proof of Corollary S23.4.* Let $\mathsf{ep} = 0$, so that $A\mathsf{Q} = \mathsf{Q}A^\top$ by Theorem S23.3, and let $\mathsf{Q}$ be block diagonal over $(\mathsf{I}, \mathsf{S}, \mathsf{A}, \mathsf{E})$. The block $(\mathsf{E}, \mathsf{A})$ of this identity reads $A_{\mathsf{EA}}\mathsf{Q}_{\mathsf{AA}} = \mathsf{Q}_{\mathsf{EE}}A_{\mathsf{AE}}^\top$. No active variable reads an exterior variable (Definition 4.1), so $A_{\mathsf{AE}} = 0$ and, $\mathsf{Q}_{\mathsf{AA}}$ being invertible, $A_{\mathsf{EA}} = 0$. An active variable drives at least one variable that is not collective, so every column of $A_{\mathsf{EA}}$ is nonzero; hence $\mathsf{A} = \varnothing$. The block $(\mathsf{E}, \mathsf{S})$ reads $A_{\mathsf{ES}}\mathsf{Q}_{\mathsf{SS}} = \mathsf{Q}_{\mathsf{EE}}A_{\mathsf{SE}}^\top$, which is the stated reciprocity. For the last sentence, a collective with an active state violates the first conclusion. If $\mathsf{S} \neq \varnothing$ and $A_{\mathsf{ES}} = 0$, reciprocity would force $A_{\mathsf{SE}} = 0$, whereas every sensory variable reads a variable that is not collective, so every row of $A_{\mathsf{SE}}$ is nonzero. In both cases $\mathsf{ep} \neq 0$, and $\mathsf{ep} \geq 0$. □

*Remark* S44.3 (Scope of the reversible case of Proposition S4.4). For the canonical partition with noise independent across the four classes $(\mathsf{I},\mathsf{S},\mathsf{A},\mathsf{E})$, in particular for per-channel noise, the hypotheses of Proposition S4.4(c) contain those of Corollary S23.4 with $\mathsf{ep}=0$. In that case the proposition describes collectives without active states whose sensory states drive the exterior with the weights $\mathsf{Q}_{\mathsf{EE}}A_{\mathsf{SE}}^{\top}\mathsf{Q}_{\mathsf{SS}}^{-1}$. This restricts its range of application and does not affect its proof; the sentence on agents that sense an exogenous world, in the paragraph that follows the proposition, is the case $A_{\mathsf{ES}}=0$ with $\mathsf{S}\neq\varnothing$. If the noise of sensory and active states is correlated, the block $(\mathsf{E},\mathsf{B})$ of $A\mathsf{Q}=\mathsf{Q}A^{\top}$ gives $A_{\mathsf{EB}}=\mathsf{Q}_{\mathsf{EE}}A_{\mathsf{BE}}^{\top}\mathsf{Q}_{\mathsf{BB}}^{-1}$, whose active columns $\mathsf{Q}_{\mathsf{EE}}A_{\mathsf{SE}}^{\top}(\mathsf{Q}_{\mathsf{BB}}^{-1})_{\mathsf{SA}}$ need not vanish, so the corollary does not extend to noise that is independent only across $(\mathsf{I},\mathsf{B},\mathsf{E})$. A witness is $A=\kappa\mathsf{Q}M$ with $\kappa>0$, $M$ symmetric, $M_{\mathsf{IE}}=0$ and $M_{\mathsf{AE}}=-\mathsf{Q}_{\mathsf{AA}}^{-1}\mathsf{Q}_{\mathsf{AS}}M_{\mathsf{SE}}$: then $A\mathsf{Q}=\kappa\mathsf{Q}M\mathsf{Q}=\mathsf{Q}A^{\top}$, $A_{\mathsf{IE}}=0$, $A_{\mathsf{EI}}=0$ and $A_{\mathsf{AE}}=0$, while $A_{\mathsf{EA}}=\kappa\mathsf{Q}_{\mathsf{EE}}M_{\mathsf{AE}}^{\top}$ and $A_{\mathsf{SE}}=\kappa(\mathsf{Q}_{\mathsf{SS}}-\mathsf{Q}_{\mathsf{SA}}\mathsf{Q}_{\mathsf{AA}}^{-1}\mathsf{Q}_{\mathsf{AS}})M_{\mathsf{SE}}$. With one variable of each class, ordered $(\mathsf{I},\mathsf{S},\mathsf{A},\mathsf{E})$, take $\kappa=\frac{1}{2}$,

$$\mathsf{Q}=\begin{pmatrix}0.5&0&0&0\\0&1&0.45&0\\0&0.45&0.8&0\\0&0&0&1.2\end{pmatrix},\qquad M=\begin{pmatrix}0.7&0.3&-0.2&0\\0.3&-0.4&0.25&0.6\\-0.2&0.25&0.5&-0.3375\\0&0.6&-0.3375&0.3\end{pmatrix}.$$

Then $\mathsf{Q}\succ 0$, $\rho(A)=0.356$, $A_{\mathsf{SE}}=717/3200$, $A_{\mathsf{ES}}=9/25$, $A_{\mathsf{EA}}=-81/400$ and $A_{\mathsf{AE}}=0$, so the partition is canonical with one sensory and one active variable, and $\mathsf{ep}=0$ by Theorem S23.3. With the same $A$ and the entry $\mathsf{Q}_{\mathsf{SA}}$ set to zero, $\mathsf{ep}=0.0722$ (Check S47.214).

*Proof of Theorem S23.5.* Write $\mathbf{Y}_n=(Y_0,\dots,Y_n)$ and $\mathbf{Y}_n^{\mathrm{r}}=(Y_n,\dots,Y_0)=\mathsf{P}_n\mathbf{Y}_n$.

(a) Nonnegativity is that of relative entropy. The map $(x_0,\dots,x_n)\mapsto(Cx_0,\dots,Cx_n)$ sends the law of $(X_0,\dots,X_n)$ to that of $\mathbf{Y}_n$ and the law of $(X_n,\dots,X_0)$ to that of $\mathbf{Y}_n^{\mathrm{r}}$, so $\mathsf{D}_n(C)\le\mathsf{D}_n=n\mathsf{ep}$ by monotonicity and Theorem S23.3. Dropping the last coordinate sends the law of $\mathbf{Y}_{n+1}$ to that of $\mathbf{Y}_n$ and the law of $\mathbf{Y}_{n+1}^{\mathrm{r}}=(Y_{n+1},\dots,Y_0)$ to that of $(Y_{n+1},\dots,Y_1)$, which equals the law of $\mathbf{Y}_n^{\mathrm{r}}$ by stationarity; hence $\mathsf{D}_n(C)\le\mathsf{D}_{n+1}(C)$. Dividing $0\le\mathsf{D}_n(C)\le n\mathsf{ep}$ by $n$ places the hidden dissipation in $[0,\mathsf{ep}]$.

(b) For $s\ge t$, $\mathrm{Cov}(Y_s,Y_t)=CA^{s-t}\Sigma C^{\top}=\Gamma_{s-t}$. The covariance $\mathsf{T}_n$ of $\mathbf{Y}_n$ is the image of the covariance of $(X_0,\dots,X_n)$, positive definite by Lemma S44.2(i), under the map $I\otimes C$ of full row rank, so $\mathsf{T}_n\succ 0$. The covariance of $\mathbf{Y}_n^{\mathrm{r}}$ is $\mathsf{P}_n\mathsf{T}_n\mathsf{P}_n^{\top}$, with the same determinant, and Lemma S44.1 with $k=(n+1)m$ gives the formula.

(c) By Lemma S44.1, $\mathsf{D}_n(C)=0$ if and only if $\mathsf{T}_n=\mathsf{P}_n\mathsf{T}_n\mathsf{P}_n^{\top}$. The block $(s,t)$ of the right side is $\mathrm{Cov}(Y_{n-s},Y_{n-t})$, which is $\Gamma_{s-t}^{\top}$ for $s\ge t$. Hence $\mathsf{D}_n(C)=0$ if and only if $\Gamma_k=\Gamma_k^{\top}$ for $k=1,\dots,n$, and $\mathsf{D}_n(C)=0$ for all $n$ if and only if every $\Gamma_k$ is symmetric. By the Cayley–Hamilton theorem each $A^k$ with $k\ge d$ is a real linear combination of $I,A,\dots,A^{d-1}$, so each $\Gamma_k$ is a real linear combination of $\Gamma_0,\dots,\Gamma_{d-1}$, and $\Gamma_0=C\Sigma C^{\top}$ is symmetric. Symmetry for $k=1,\dots,d-1$ therefore implies symmetry for all $k$.

(d) For $m=1$ each $\Gamma_k$ is a number, so (c) applies, $\mathsf{D}_n(C)=0$ for all $n$, and the hidden dissipation is $\mathsf{ep}-0$. □

*Details for Example S23.6.* The cyclic shift $P$ is a permutation matrix, hence orthogonal, and $A=cI+aP$ is normal with eigenvalues $\lambda_j=c+ae^{2\pi ij/3}$, $j=0,1,2$, and $\rho(A)\le|c|+|a|<1$. The eigenvalue $\lambda_0$ is real, and $\lambda_1,\lambda_2$ have imaginary parts $\pm\frac{\sqrt{3}}{2}a$ and squared modulus $c^2-ac+a^2$. The last formula of Theorem S23.3 gives $\mathsf{ep}=2\cdot 2\cdot\frac{3}{4}a^2/(1-c^2+ac-a^2)=3a^2/(1-c^2+ac-a^2)$, which is $0.75/0.81=25/27$ at $c=0.3$, $a=0.5$. The mean $\frac{1}{3}(1,1,1)X$ and a single coordinate are scalar descriptions, reversible by Theorem S23.5(d). The value for a pair of agents is numerical:

$\mathsf{D}_n(C)/n = 0.34596$ at $n = 128$ and the integral Eq. (S98) equals 0.34665 (Check S47.210); Theorem S23.5(a) guarantees only that it lies in $[0, 25/27]$. □

### Information flow

In the setting of Theorem S23.8, $\mathsf{Q} = \operatorname{diag}(\mathsf{Q}_{\mathrm{uu}}, \mathsf{Q}_{\mathrm{vv}})$, so $k(x' \mid x) = k_{\mathrm{u}}(x'_{\mathrm{u}} \mid x)k_{\mathrm{v}}(x'_{\mathrm{v}} \mid x)$ with $k_{\mathrm{u}}(x'_{\mathrm{u}} \mid x_{\mathrm{u}}, x_{\mathrm{v}}) = \mathcal{N}(x'_{\mathrm{u}}; A_{\mathrm{uu}}x_{\mathrm{u}} + A_{\mathrm{uv}}x_{\mathrm{v}}, \mathsf{Q}_{\mathrm{uu}})$. In Eq. (S99) the symbol $k_{\mathrm{u}}(x_{\mathrm{u}} \mid x'_{\mathrm{u}}, x_{\mathrm{v}})$ is this function with its first two arguments exchanged, and $k_{\mathrm{u}}(X_{\mathrm{u}} \mid X') = k_{\mathrm{u}}(X_{\mathrm{u}} \mid X'_{\mathrm{u}}, X'_{\mathrm{v}})$. The covariance of $(X, X')$ is positive definite by Lemma S44.2(i), so every subvector of $(X, X')$ has a nondegenerate Gaussian law, all entropies and mutual informations below are finite, and so are the expectations in Eq. (S99), which are expectations of quadratic forms:

$$\begin{aligned} \mathsf{s}_{\mathrm{u}} &= \tfrac{1}{2}\operatorname{tr}\big(\mathsf{Q}_{\mathrm{uu}}^{-1}(\mathbb{E}\mathsf{l}\mathsf{l}^{\top} - \mathsf{Q}_{\mathrm{uu}})\big), \qquad \mathsf{l} = X_{\mathrm{u}} - A_{\mathrm{uu}}X'_{\mathrm{u}} - A_{\mathrm{uv}}X_{\mathrm{v}}, \\ \mathsf{ep}_{\mathrm{u}} &= \tfrac{1}{2}\operatorname{tr}\big(\mathsf{Q}_{\mathrm{uu}}^{-1}(\mathbb{E}\mathsf{l}'\mathsf{l}'^{\top} - \mathsf{Q}_{\mathrm{uu}})\big), \quad \mathsf{l}' = X_{\mathrm{u}} - A_{\mathrm{uu}}X'_{\mathrm{u}} - A_{\mathrm{uv}}X'_{\mathrm{v}}. \end{aligned} \tag{S169}$$

The two differ only in whether the partner enters the reversed step with its old or its new value.

*Proof of Theorem S23.8.* (a) By the proof of Theorem S23.3, $\mathsf{ep} = \mathbb{E}\ln\big(k(X' \mid X)/k(X \mid X')\big)$, and the factorization of $k$ splits the logarithm into the two terms whose expectations are $\mathsf{ep}_{\mathrm{u}}$ and $\mathsf{ep}_{\mathrm{v}}$. For the lower bound let $\mathsf{R} = k_{\mathrm{u}}(X_{\mathrm{u}} \mid X')/k_{\mathrm{u}}(X'_{\mathrm{u}} \mid X)$, so that $\mathsf{ep}_{\mathrm{u}} = -\mathbb{E}\ln\mathsf{R} \geq -\ln\mathbb{E}\mathsf{R}$ by Jensen's inequality. The joint density of $(X, X')$ is $\pi(x)k_{\mathrm{u}}(x'_{\mathrm{u}} \mid x)k_{\mathrm{v}}(x'_{\mathrm{v}} \mid x)$, hence

$$\mathbb{E}\mathsf{R} = \int \pi(x)k_{\mathrm{v}}(x'_{\mathrm{v}} \mid x)\Big[\int k_{\mathrm{u}}(x_{\mathrm{u}} \mid x'_{\mathrm{u}}, x'_{\mathrm{v}})dx'_{\mathrm{u}}\Big]dxdx'_{\mathrm{v}}.$$

When $A_{\mathrm{uu}}$ is invertible the substitution $y = A_{\mathrm{uu}}x'_{\mathrm{u}}$ shows that the inner integral of $\mathcal{N}(x_{\mathrm{u}}; A_{\mathrm{uu}}x'_{\mathrm{u}} + A_{\mathrm{uv}}x'_{\mathrm{v}}, \mathsf{Q}_{\mathrm{uu}})$ over $x'_{\mathrm{u}}$ is $|\det A_{\mathrm{uu}}|^{-1}$, and the remaining integral is one. Hence $\mathbb{E}\mathsf{R} = |\det A_{\mathrm{uu}}|^{-1}$ and $\mathsf{ep}_{\mathrm{u}} \geq \ln|\det A_{\mathrm{uu}}|$.

(b) The function $x_{\mathrm{u}} \mapsto k_{\mathrm{u}}(x_{\mathrm{u}} \mid x'_{\mathrm{u}}, x_{\mathrm{v}})$ is a normal density, so $\pi^{+}(x'_{\mathrm{u}}, x_{\mathrm{v}})k_{\mathrm{u}}(x_{\mathrm{u}} \mid x'_{\mathrm{u}}, x_{\mathrm{v}})$ is a probability density, that of the nondegenerate Gaussian law $P^{\mathrm{fr}}$. The law $P$ of $(X_{\mathrm{u}}, X_{\mathrm{v}}, X'_{\mathrm{u}})$ has density $\pi(x_{\mathrm{u}}, x_{\mathrm{v}})k_{\mathrm{u}}(x'_{\mathrm{u}} \mid x_{\mathrm{u}}, x_{\mathrm{v}})$. Therefore

$$D_{\mathrm{KL}}(P\|P^{\mathrm{fr}}) = \mathbb{E}\ln\frac{k_{\mathrm{u}}(X'_{\mathrm{u}} \mid X_{\mathrm{u}}, X_{\mathrm{v}})}{k_{\mathrm{u}}(X_{\mathrm{u}} \mid X'_{\mathrm{u}}, X_{\mathrm{v}})} + \mathbb{E}\ln\frac{\pi(X_{\mathrm{u}}, X_{\mathrm{v}})}{\pi^{+}(X'_{\mathrm{u}}, X_{\mathrm{v}})} = \mathsf{s}_{\mathrm{u}} - h(X_{\mathrm{u}}, X_{\mathrm{v}}) + h(X'_{\mathrm{u}}, X_{\mathrm{v}}),$$

with $h$ the differential entropy. Since $X'_{\mathrm{u}}$ and $X_{\mathrm{u}}$ have the same law,

$$\mathsf{i}_{\mathrm{u}} = \big[h(X'_{\mathrm{u}}) + h(X_{\mathrm{v}}) - h(X'_{\mathrm{u}}, X_{\mathrm{v}})\big] - \big[h(X_{\mathrm{u}}) + h(X_{\mathrm{v}}) - h(X_{\mathrm{u}}, X_{\mathrm{v}})\big] = h(X_{\mathrm{u}}, X_{\mathrm{v}}) - h(X'_{\mathrm{u}}, X_{\mathrm{v}}),$$

and the two displays give $D_{\mathrm{KL}}(P\|P^{\mathrm{fr}}) = \mathsf{s}_{\mathrm{u}} - \mathsf{i}_{\mathrm{u}}$.

(c) The chain rule applied in two orders gives

$$I(X_{\mathrm{u}}; X_{\mathrm{v}}) + I(X'_{\mathrm{u}}; X_{\mathrm{v}} \mid X_{\mathrm{u}}) = I(X'_{\mathrm{u}}, X_{\mathrm{u}}; X_{\mathrm{v}}) = I(X'_{\mathrm{u}}; X_{\mathrm{v}}) + I(X_{\mathrm{u}}; X_{\mathrm{v}} \mid X'_{\mathrm{u}}) \geq I(X'_{\mathrm{u}}; X_{\mathrm{v}}),$$

which is $\mathsf{i}_{\mathrm{u}} \leq \mathsf{te}_{\mathrm{v}\to\mathrm{u}}$. Since $X'_{\mathrm{u}} = A_{\mathrm{uu}}X_{\mathrm{u}} + A_{\mathrm{uv}}X_{\mathrm{v}} + \Xi_{\mathrm{u}}$ with $\Xi_{\mathrm{u}} \sim \mathcal{N}(0, \mathsf{Q}_{\mathrm{uu}})$ independent of $X$, the conditional covariance of $X'_{\mathrm{u}}$ given $X_{\mathrm{u}}$ is $\mathsf{Q}_{\mathrm{uu}} + A_{\mathrm{uv}}\Sigma_{\mathrm{v}|\mathrm{u}}A_{\mathrm{uv}}^{\top}$ and given $(X_{\mathrm{u}}, X_{\mathrm{v}})$ it is $\mathsf{Q}_{\mathrm{uu}}$, so

$$\mathsf{te}_{\mathrm{v}\to\mathrm{u}} = \tfrac{1}{2}\ln\det\big(\mathsf{Q}_{\mathrm{uu}} + A_{\mathrm{uv}}\Sigma_{\mathrm{v}|\mathrm{u}}A_{\mathrm{uv}}^{\top}\big) - \tfrac{1}{2}\ln\det\mathsf{Q}_{\mathrm{uu}},$$

which is the stated formula.

(d) By stationarity $I(X'_{\mathrm{u}}; X'_{\mathrm{v}}) = I(X_{\mathrm{u}}; X_{\mathrm{v}})$, so $\mathsf{i}_{\mathrm{u}} + \mathsf{i}_{\mathrm{v}} = I(X'_{\mathrm{u}}; X_{\mathrm{v}}) + I(X_{\mathrm{u}}; X'_{\mathrm{v}}) - 2I(X_{\mathrm{u}}; X_{\mathrm{v}})$, which is the negative of the bracket in the statement. □

*Proof of Corollary S23.9.* A matrix $\mathsf{Q}$ that is block diagonal over $(\mathsf{I},\mathsf{S},\mathsf{A},\mathsf{E})$ is block diagonal over $(\mathrm{u},\mathrm{v}) = (\mathsf{I}\cup\mathsf{B},\mathsf{E})$, so Theorem S23.8 applies in both directions. By Definition 4.1 no internal variable reads or drives an exterior variable and no active variable reads one: $A_{\mathsf{IE}} = 0$, $A_{\mathsf{EI}} = 0$ and $A_{\mathsf{AE}} = 0$. The only nonzero rows of $A_{\mathrm{uv}}$ are therefore the sensory rows, $A_{\mathrm{uv}}\Sigma_{\mathsf{E}|\mathrm{u}}A_{\mathrm{uv}}^{\top}$ vanishes outside its block $(\mathsf{S},\mathsf{S})$, where it equals $A_{\mathsf{SE}}\Sigma_{\mathsf{E}|\mathrm{u}}A_{\mathsf{SE}}^{\top}$, and since $\mathsf{Q}_{\mathrm{uu}} = \mathrm{diag}(\mathsf{Q}_{\mathsf{II}},\mathsf{Q}_{\mathsf{SS}},\mathsf{Q}_{\mathsf{AA}})$ the determinant in Theorem S23.8(c) reduces to the first line of Eq. (S101). In the other direction the only nonzero columns of $A_{\mathrm{vu}}$ are the blanket columns, so $A_{\mathrm{vu}}\Sigma_{\mathrm{u}|\mathsf{E}}A_{\mathrm{vu}}^{\top} = A_{\mathsf{EB}}\Sigma_{\mathsf{B}|\mathsf{E}}A_{\mathsf{EB}}^{\top}$, where $\Sigma_{\mathsf{B}|\mathsf{E}} = \Sigma_{\mathsf{BB}} - \Sigma_{\mathsf{BE}}\Sigma_{\mathsf{EE}}^{-1}\Sigma_{\mathsf{EB}}$ is the blanket block of $\Sigma_{\mathrm{u}|\mathsf{E}}$; this is the second line. The inequalities are Theorem S23.8(b) and (c) for each side. □

**Example S44.4** (A four-variable blanket)**.** Take one variable of each class, ordered $(\mathsf{I},\mathsf{S},\mathsf{A},\mathsf{E})$, with

$$A = \begin{pmatrix} 0.5 & 0.4 & 0 & 0 \\ 0 & 0.3 & 0 & 0.6 \\ 0.5 & 0 & 0.2 & 0 \\ 0 & 0 & 0.4 & 0.7 \end{pmatrix}, \qquad \mathsf{Q} = \mathrm{diag}(0.2, 0.1, 0.1, 1).$$

The sensory variable reads the exterior, the internal variable reads the sensory one, the active variable reads the internal one and drives the exterior, and $\rho(A) = 0.936$. By Corollary S23.4 the process is irreversible, and Theorem S23.3 gives $\mathsf{ep} = 6.0995$ nats per round. For the collective $\mathrm{u} = \mathsf{I}\cup\mathsf{B}$ the computation of Check S47.211 gives $\mathsf{s}_{\mathrm{u}} = 5.1336$, $\mathsf{i}_{\mathrm{u}} = 0.7848$, $\mathsf{te}_{\mathsf{E}\to\mathrm{u}} = 0.8225$ and $\mathsf{ep}_{\mathrm{u}} = 6.0995$; for the exterior, $\mathsf{s}_{\mathsf{E}} = 0.0206$, $\mathsf{i}_{\mathsf{E}} = -0.1505$, $\mathsf{te}_{\mathrm{u}\to\mathsf{E}} = 0.0336$ and $\mathsf{ep}_{\mathsf{E}} = -5.1\times10^{-5}$. All inequalities of Theorem S23.8 and Corollary S23.9 hold, the collective gains about 0.78 nats about its exterior per round through one sensory channel whose transfer entropy is 0.82, and three statements familiar from continuous time fail: $\mathsf{i}_{\mathrm{u}} + \mathsf{i}_{\mathsf{E}} = 0.634 \neq 0$, $\mathsf{s}_{\mathrm{u}} + \mathsf{s}_{\mathsf{E}} - \mathsf{ep} = -0.945 \neq 0$, and $\mathsf{ep}_{\mathsf{E}} < 0$.

**Example S44.5** (A negative partial entropy production)**.** For two scalar blocks with $A = \left(\begin{smallmatrix} 0.1 & 0.05 \\ 0.9 & 0.5 \end{smallmatrix}\right)$ and $\mathsf{Q} = \mathrm{diag}(1, 0.05)$, Eq. (S169) gives $\mathsf{ep}_{\mathrm{u}} = -0.0451$ and $\mathsf{ep}_{\mathrm{v}} = 16.2362$, with sum $\mathsf{ep} = 16.1911$ and with $\mathsf{ep}_{\mathrm{u}} \geq \ln|A_{\mathrm{uu}}| = -2.30$ as Theorem S23.8(a) requires (Check S47.211). The partial terms of the additive decomposition are therefore not entropy productions of the parts, and the Jensen bound is the only general lower bound we have for them.

*Remark* S44.6 (What does not transfer from continuous time). For a bipartite Markov jump process, in which the two parts never change in the same instant, Horowitz and Esposito (2014) show that the entropy production rate is the sum of two partial rates, each nonnegative and, in a steady state, equal to the entropy flow of one part minus the information flow into it, and that the two information flows then cancel. Their derivation is for jump processes; we cite no corresponding result for diffusions. In discrete time both blocks are updated in one round, and the reversed step of block u can be taken with the partner at its old value, which gives $\mathsf{s}_{\mathrm{u}}$, or at its new value, which gives $\mathsf{ep}_{\mathrm{u}}$. The first choice keeps the second-law form $\mathsf{s}_{\mathrm{u}} \geq \mathsf{i}_{\mathrm{u}}$ of Theorem S23.8(b) and loses additivity; the second keeps additivity, Theorem S23.8(a), and loses the sign. The sum of the information flows is the mixed second difference of Theorem S23.8(d), which is negative in 24 of 400 random instances and positive in the others (Check S47.211). Under the scaling $A = I + hG$, $\mathsf{Q} = hD$ the two defects $\mathsf{s}_{\mathrm{u}} + \mathsf{s}_{\mathrm{v}} - \mathsf{ep}$ and $\mathsf{i}_{\mathrm{u}} + \mathsf{i}_{\mathrm{v}}$ halve, per unit time, each time $h$ is halved in the instance of Check S47.211, consistent with defects of order $h^2$ per round. This scaling is a numerical observation on one instance; a proof is part of Open problem S23.17, and we neither cite nor prove that the identities of a bipartite diffusion are recovered as $h \to 0$.

### The bridge and the scale theorem

*Proof of Proposition S23.10.* (a) Since $R \succ 0$ and $\Sigma_x \succeq 0$, $y$ is Gaussian with positive definite covariance $\Sigma_x + R$ and $y$ given $x$ has covariance $R$, so

$$I(y;x) = h(y) - h(y \mid x) = \tfrac{1}{2}\ln\det(\Sigma_x + R) - \tfrac{1}{2}\ln\det R = \tfrac{1}{2}\ln\det\bigl(I + R^{-1/2}\Sigma_x R^{-1/2}\bigr),$$

and Assumption S2 gives the cost. Let $\lambda_1, \dots, \lambda_m \geq 0$ be the eigenvalues of $R^{-1/2}\Sigma_x R^{-1/2}$. Under the thermal clause $E_{\mathrm{sig}} = \tfrac{1}{2}k_{\mathrm{B}}\mathsf{T}\mathbb{E}x^{\top}R^{-1}x = \tfrac{1}{2}k_{\mathrm{B}}\mathsf{T}\,\mathrm{tr}(R^{-1}\Sigma_x) = \tfrac{1}{2}k_{\mathrm{B}}\mathsf{T}\sum_k \lambda_k$, while $I(y;x) = \tfrac{1}{2}\sum_k \ln(1+\lambda_k) \leq \tfrac{1}{2}\sum_k \lambda_k$, so $E_{\mathrm{sig}} \geq k_{\mathrm{B}}\mathsf{T} I(y;x)$. For a scalar channel $\lambda_1 = e^{2I(y;x)} - 1$, which gives the last formula.

(b) As in the statement, $q_e$ is the variance of $w_t$ and $P_e = q_e/(1-\mathsf{f}^2)$ is the stationary variance of the source. The pair $X = (z, e)$ obeys Eq. (23) with

$$A = \begin{pmatrix} \mathsf{r} & \omega \\ 0 & \mathsf{f} \end{pmatrix}, \qquad \mathsf{Q} = \mathrm{diag}(q_z, q_e) \succ 0, \qquad \rho(A) = \max\{|\mathsf{r}|, |\mathsf{f}|\} < 1,$$

so Assumption S23.1 holds with blocks $\mathrm{u} = z$ and $\mathrm{v} = e$ and independent noise, and Theorem S23.8(b) gives $\mathsf{s}_z \geq \mathsf{i}_z$. For jointly Gaussian scalars with correlation $\varrho$ the mutual information is $-\tfrac{1}{2}\ln(1-\varrho^2)$, so it remains to find the correlation of $z_{t+1}$ and $e_t$. The noise $w_t$ is independent of $(e_t, z_t, \zeta_t^z)$ and hence of $z_{t+1}$, so $\mathrm{Cov}(z_{t+1}, e_{t+1}) = \mathrm{Cov}(z_{t+1}, \mathsf{f}e_t + w_t) = \mathsf{f}\,\mathrm{Cov}(z_{t+1}, e_t)$. By stationarity the left side is $\varrho_{ze}\sqrt{V_z P_e}$ with $V_z$ the stationary variance of the record, and $e_t$ has variance $P_e$, so the correlation of $z_{t+1}$ and $e_t$ is $\varrho_{ze}/\mathsf{f}$. Its square is below one because the covariance of $(X_t, X_{t+1})$ is positive definite by Lemma S44.2(i). Therefore

$$\mathsf{i}_z = I(z_{t+1}; e_t) - I(z_t; e_t) = \tfrac{1}{2}\ln\frac{1-\varrho_{ze}^2}{1-\varrho_{ze}^2/\mathsf{f}^2},$$

which is nonnegative because $|\mathsf{f}| < 1$. In closed form, stationarity of $\mathsf{c} = \mathrm{Cov}(z_t, e_t)$ and of $V_z$ gives $\mathsf{c} = \mathsf{f}(\omega P_e + \mathsf{r}\mathsf{c})$ and $V_z = \omega^2 P_e + \mathsf{r}^2 V_z + 2\omega\mathsf{r}\mathsf{c} + q_z$, that is,

$$\mathsf{c} = \frac{\mathsf{f}\omega P_e}{1-\mathsf{f}\mathsf{r}}, \qquad V_z = \frac{\omega^2 P_e + 2\omega\mathsf{r}\mathsf{c} + q_z}{1-\mathsf{r}^2}, \qquad \varrho_{ze}^2 = \frac{\mathsf{c}^2}{V_z P_e}.$$

For the limit, hold $P_e$, $\omega$, $\mathsf{r}$ and $q_z$ fixed. The quantity $1-\varrho_{ze}^2/\mathsf{f}^2$ is the ratio of the conditional variance of $z_{t+1}$ given $e_t$ to $V_z$, and that conditional variance is at least $q_z$ because $\zeta_t^z$ is independent of $(e_t, z_t)$. The triangle inequality in $L^2$ applied to $z_{t+1} = \mathsf{r}z_t + \omega e_t + \zeta_t^z$ gives $\sqrt{V_z} \leq |\mathsf{r}|\sqrt{V_z} + |\omega|\sqrt{P_e} + \sqrt{q_z}$, hence $V_z \leq \bar{V} = \bigl(|\omega|\sqrt{P_e} + \sqrt{q_z}\bigr)^2/(1-|\mathsf{r}|)^2$ for every $\mathsf{f}$. With $\ln(1+a) \leq a$ and $\varrho_{ze}^2/\mathsf{f}^2 < 1$,

$$\mathsf{i}_z = \tfrac{1}{2}\ln\Bigl(1 + \frac{(\varrho_{ze}^2/\mathsf{f}^2)(1-\mathsf{f}^2)}{1-\varrho_{ze}^2/\mathsf{f}^2}\Bigr) \leq \frac{1-\mathsf{f}^2}{2(1-\varrho_{ze}^2/\mathsf{f}^2)} \leq \frac{(1-\mathsf{f}^2)\bar{V}}{2q_z},$$

which tends to zero as $|\mathsf{f}| \to 1$. The bound uses that $\bar{V}$ does not depend on $\mathsf{f}$, which is where the fixed variance $P_e$ enters. At fixed $q_e$ the variance $P_e$ diverges as $|\mathsf{f}| \to 1$ and the limit fails. Let $\mathsf{u}$ be the conditional variance of $z_t$ given $e_t$. Since $z_t$ and $z_{t+1}$ have the same law, $\mathsf{i}_z = h(z_t \mid e_t) - h(z_{t+1} \mid e_t) = \tfrac{1}{2}\ln(\mathsf{u}/(\mathsf{r}^2\mathsf{u} + q_z))$. The stationary source has the backward representation $e_{s-1} = \mathsf{f}e_s + b_s$ with $b_s \sim \mathcal{N}(0, q_e)$ independent of $(e_s, e_{s+1}, \dots)$, so that $e_{t-1-k} - \mathsf{f}^{k+1}e_t = \sum_{j=0}^{k}\mathsf{f}^{k-j}b_{t-j}$, and inserting this in $z_t = \sum_{k\geq 0}\mathsf{r}^k(\omega e_{t-1-k} + \zeta_{t-1-k}^z)$ gives

$$\mathsf{u} = \frac{1}{1-\mathsf{r}^2}\Bigl(q_z + \frac{\omega^2 q_e}{(1-\mathsf{f}\mathsf{r})^2}\Bigr).$$

At fixed $q_e$ this has a finite limit as $\mathsf{f} \to 1$, and likewise as $\mathsf{f} \to -1$, with $(1-\mathsf{r}^2)\mathsf{u} > q_z$ because $\omega \neq 0$, so $\mathsf{i}_z$ tends to a positive number (Check S47.214). □

*Remark* S44.7 (The accounting behind the margins). In the complete graph agent $i$ operates one sensor register, whose input is $\eta_t$ and which acquires $I_s = \frac{1}{2}\ln(1 + P_0/R^s)$ nats, and $N-1$ message registers, each with input $s_j$ of variance $P_0 + R^s$, acquiring $I_m = \frac{1}{2}\ln(1 + (P_0 + R^s)/R_c)$ nats. In the record architecture each agent operates its sensor register, one write register on the record with input $s_j$, and one read register with input $z$ of variance $P_0 + (R^s + R_c)/N$, acquiring

$$I_r(N) = \tfrac{1}{2}\ln\Big(1 + \frac{P_0 + (R^s + R_c)/N}{R_c}\Big),$$

which decreases in $N$ to $\frac{1}{2}\ln(1 + P_0/R_c)$. By Assumption S2 the work per agent and round is at least $k_{\mathrm{B}}\mathsf{T}\mathsf{c}_\bullet(N)$, by Assumption S3 the harvest is at most $\mathsf{x}k_{\mathrm{B}}\mathsf{T}I_\bullet^{\mathrm{pred}}(N)$, and the surplus, harvest minus work, is at most $k_{\mathrm{B}}\mathsf{T}\mathsf{m}_\bullet(N)$. Averaging at the record is counted as free, which can only favor the record; its effect on the comparison is discussed in Remark S44.8.

*Proof of Theorem S23.12.* Throughout, $w_t$ is independent of $\eta_t$ and of everything stored in round $t$, so for any stored Gaussian data $\mathsf{d}$ with $p = \mathrm{Var}(\eta_t \mid \mathsf{d})$ one has $\mathrm{Var}(\eta_{t+1} \mid \mathsf{d}) = F^2 p + Q$ and

$$I(\mathsf{d}; \eta_{t+1}) = \tfrac{1}{2}\ln\frac{P_0}{F^2 p + Q}, \qquad I(\mathsf{d}; \eta_t) = \tfrac{1}{2}\ln\frac{P_0}{p}. \tag{S170}$$

Since $Q = (1 - F^2)P_0$, the first quantity is smaller than the second exactly when $p < P_0$, and it is smaller than $\frac{1}{2}\ln(P_0/Q) = I_\eta$ exactly when $p > 0$.

(a) Given $\eta_t$, the data $(s_i, (y_{ij})_{j\neq i})$ of agent $i$ in $K_N$ are independent observations $s_i = \eta_t + v_i$ and $y_{ij} = \eta_t + v_j + \nu_{ij}$ of $\eta_t$ with noise variances $R^s$ and $R^s + R_c$. Precisions add, so $p = p_K(N)$ as stated, and Eq. (S170) is the formula for $I_K^{\mathrm{pred}}(N)$. It increases with $N$ because $p_K(N)$ decreases, and it is below $I_\eta$ because $p_K(N) > 0$ and $F \neq 0$.

(b) Both architectures store $s_i$, so $p \leq (1/P_0 + 1/R^s)^{-1} < P_0$ and, by the remark after Eq. (S170), $I_\bullet^{\mathrm{pred}}(N) < I(\mathsf{d}; \eta_t)$. It remains to bound $I(\mathsf{d}; \eta_t)$ by the cost. In $K_N$ write $y = (y_{ij})_{j\neq i}$. The chain rule, the fact that conditioning does not increase entropy, and subadditivity of entropy give

$$I(s_i, y; \eta_t) = I_s + h(y \mid s_i) - h(y \mid s_i, \eta_t) \leq I_s + \sum_{j\neq i} h(y_{ij}) - h(y \mid \eta_t, (s_j)_j).$$

Given all sensor values the entries of $y$ are the independent variables $s_j + \nu_{ij}$, so the last entropy is $\sum_{j\neq i} h(\nu_{ij}) = \sum_{j\neq i} h(y_{ij} \mid s_j)$, and the right side is $I_s + \sum_{j\neq i} I(y_{ij}; s_j) = I_s + (N-1)I_m = \mathsf{c}_K(N)$. In the record architecture $\nu_i^r$ is independent of $(z, \eta_t, s_i)$, so

$$\begin{aligned} I(s_i, r_i; \eta_t) &= I_s + h(r_i \mid s_i) - h(r_i \mid s_i, \eta_t) \leq I_s + h(r_i) - h(r_i \mid z, s_i, \eta_t) \\ &= I_s + h(r_i) - h(r_i \mid z) = I_s + I_r(N), \end{aligned}$$

which is less than $\mathsf{c}_R(N)$ because $I_m > 0$. Hence $I_\bullet^{\mathrm{pred}}(N) < \mathsf{c}_\bullet(N)$ and $\mathsf{m}_\bullet(N) < (\mathsf{x}-1)I_\bullet^{\mathrm{pred}}(N) \leq 0$ when $\mathsf{x} \leq 1$.

(c) If $\mathsf{x} = 0$ then $\mathsf{m}_K(N) = -\mathsf{c}_K(N) < 0$. If $\mathsf{x} > 0$, part (a) gives $\mathsf{m}_K(N) < \mathsf{x}I_\eta - I_s - (N-1)I_m$, and the right side is at most zero exactly when $N \geq 1 + (\mathsf{x}I_\eta - I_s)/I_m = N_K$.

(d) Agent $i$ stores $s_i = \eta_t + v_i$ and $r_i = \eta_t + \mathsf{n}_i$ with $\mathsf{n}_i = N^{-1}\sum_j (v_j + \nu_j^w) + \nu_i^r$. The noise pair $(v_i, \mathsf{n}_i)$ is independent of $\eta_t$ and has variances $\mathsf{a} = R^s$ and $\mathsf{b} = (R^s + R_c)/N + R_c$ and covariance $\mathsf{c} = R^s/N$, with $\mathsf{ab} - \mathsf{c}^2 > 0$. Two observations with unit loading and this noise covariance carry precision $(\mathsf{a} + \mathsf{b} - 2\mathsf{c})/(\mathsf{ab} - \mathsf{c}^2)$ about $\eta_t$, so

$$I_R^{\mathrm{pred}}(N) = \tfrac{1}{2}\ln\frac{P_0}{F^2 p_R(N) + Q}, \qquad p_R(N) = \Big(\frac{1}{P_0} + \frac{\mathsf{a} + \mathsf{b} - 2\mathsf{c}}{\mathsf{ab} - \mathsf{c}^2}\Big)^{-1}.$$

As $N \to \infty$, $\mathsf{b} \to R_c$ and $\mathsf{c} \to 0$, the precision tends to $1/R^s + 1/R_c$, $p_R(N) \to p_\infty$, and $I_r(N) \to \frac{1}{2}\ln(1 + P_0/R_c)$ by Remark S44.7, which gives the limit $\mathsf{m}_R^\infty$. For the comparison,

$$\mathsf{m}_R(N) - \mathsf{m}_K(N) = \mathsf{x}(I_R^{\mathrm{pred}}(N) - I_K^{\mathrm{pred}}(N)) + (N-2)I_m - I_r(N) \geq -\mathsf{x}I_\eta + (N-2)I_m - I_r(1),$$

using $I_R^{\mathrm{pred}}(N) \geq 0$, $I_K^{\mathrm{pred}}(N) < I_\eta$ from (a), and $I_r(N) \leq I_r(1)$, where $I_r(1) = \frac{1}{2}\ln(1 + (P_0 + R^s + R_c)/R_c)$ is the value of the formula of Remark S44.7 at $N = 1$. The right side is positive when $N > 2 + (\mathsf{x}I_\eta + I_r(1))/I_m$. □

*Remark* S44.8 (What the premises supply and what they do not). Every statement of Sections S23.4 and S23.5 that mentions $k_{\mathrm{B}}\mathsf{T}$ is conditional on the premise it cites or, for the signal energy of Proposition S23.10(a), on the thermal clause stated there, and neither the premises nor that clause are derived from the model. Assumption S2 is taken over from Sagawa and Ueda (2009), whose bound is stated for a memory with finitely many outcomes that is measured against a system and then erased in contact with a bath; we have not checked that a register holding a real-valued Gaussian reading falls under those hypotheses, and for that reason the premise is stated directly for the mutual information acquired. Assumption S3 is a declared linear cap on harvest, and the exchange ratio $\mathsf{x}$ is not computed from a model of the exterior. The margins are one-sided for three further reasons: the lower bounds on work need not be attainable, the upper bound on harvest need not be attainable, and costs of the agents' own computation, of averaging at the record, and of maintaining the record against the flow $\mathsf{i}_z$ of Proposition S23.10(b) are omitted. Adding the omitted costs can only lower the margins, so Theorem S23.12(b) and (c) are unaffected; the comparison in Theorem S23.12(d) is a statement about the declared costs $\mathsf{c}_K$ and $\mathsf{c}_R$ only, and a cost of the record that grew with $N$ per agent would move its crossover. The conclusion that survives all of this is the one that uses the bounds in their valid direction: a negative margin excludes an architecture under the premises, and the linear growth of $\mathsf{c}_K$ against the cap $\mathsf{x}I_\eta$ makes the complete graph excluded beyond $N_K$.

## S45 Proofs for Section S24

The statements of Section S24 are proved in their order of appearance, after two lemmas on $2 \times 2$ matrices. Three secondary results (Propositions S45.3, S45.6 and S45.7), a worked example and an extended comparison with Bourrat et al. (2022) follow. Throughout, $\tilde{\varepsilon} = \varepsilon + g(1 - \mathsf{f})$ and $\tilde{\beta} = \beta + g\mathsf{f}$ as in Corollary 12.5, and $\mathsf{p}_1 = \varepsilon + \beta + g = \tilde{\varepsilon} + \tilde{\beta}$, $\mathsf{p}_2 = \omega + \delta$, $\mathsf{E} = \omega\tilde{\beta}$ and $\varpi_\pm$ are as in the proof of Proposition 5.14, so that $\varpi_- \leq \varpi_+$ are the roots of

$$\chi(s) = (\mathsf{p}_1 - s)(\mathsf{p}_2 - s) - \mathsf{E}, \tag{S171}$$

and $\tau_{\mathrm{id}}^{-1} = \varpi_-$. The standing hypotheses are those of Section S24: the hypotheses of Proposition 5.14, among them $g > 0$, $\omega > 0$ and $\det \mathsf{A} = \mathsf{p}_1\mathsf{p}_2 - \mathsf{E} = \omega\tilde{\varepsilon} + \delta\mathsf{p}_1 > 0$, together with $\tilde{\beta} > 0$ and $\mathsf{T} > 0$. Of these, Lemma S45.2 uses only $\omega > 0$, $\tilde{\beta} > 0$ and $\det \mathsf{A} > 0$. Inequalities between vectors or between matrices are entrywise, $\mathbf{1} = (1, 1)^\top$, and $\mathsf{G}(t) = e^{t\mathsf{A}}$ with entries $\mathsf{G}_{ij}(t)$.

**Lemma S45.1** (Nonnegative $2 \times 2$ matrices). *Let $\mathsf{Q} = (q_{ij})$ be a real $2 \times 2$ matrix with nonnegative entries and put $\lambda_\pm = \frac{1}{2}[q_{11} + q_{22} \pm ((q_{11} - q_{22})^2 + 4q_{12}q_{21})^{1/2}]$.*

*(i) The eigenvalues of $\mathsf{Q}$ are $\lambda_\pm$; they are real, $\lambda_+ \geq \max\{q_{11}, q_{22}\}$ and $\lambda_+ \geq |\lambda_-|$. Hence the spectral radius is $\rho(\mathsf{Q}) = \lambda_+$, a continuous function of the entries.*

*(ii) There are nonzero vectors $w \geq 0$ and $v \geq 0$ with $\mathsf{Q}w = \lambda_+ w$ and $v^\top\mathsf{Q} = \lambda_+ v^\top$.*

*(iii)* *If $\vartheta > 0$ and $a\vartheta^\top \le \vartheta^\top \mathsf{Q} \le b\vartheta^\top$ for real numbers $a \le b$, then $a \le \lambda_+ \le b$. If $\mathsf{Q}\mathbf{1} \le \mathbf{1}$ then $\lambda_+ \le 1$, and if $\mathsf{Q}\mathbf{1} < \mathbf{1}$ then $\lambda_+ < 1$.*

*(iv)* *If all entries are positive, then $\lambda_+ > |\lambda_-|$, the vectors $w$ and $v$ of (ii) are positive and unique up to scale, and $\lambda_+^{-k}\mathsf{Q}^k \to wv^\top/(v^\top w)$ as $k \to \infty$.*

*Proof.* (i) The characteristic polynomial is $\lambda^2 - (q_{11}+q_{22})\lambda + q_{11}q_{22} - q_{12}q_{21}$, with discriminant $D = (q_{11}-q_{22})^2 + 4q_{12}q_{21} \ge (q_{11}-q_{22})^2 \ge 0$. Its roots are $\lambda_\pm$, they are real, and $\lambda_+ \ge \frac{1}{2}[q_{11}+q_{22}+|q_{11}-q_{22}|] = \max\{q_{11},q_{22}\}$. Since $\lambda_+ + \lambda_- = q_{11}+q_{22} \ge 0$ and $\lambda_+ \ge \lambda_-$, also $\lambda_+ \ge -\lambda_-$, so $\lambda_+ \ge |\lambda_-|$. Continuity is read off the formula.

(ii) The characteristic equation at $\lambda_+$ reads $(\lambda_+ - q_{11})(\lambda_+ - q_{22}) = q_{12}q_{21}$. If $q_{12} > 0$, the vector $w = (q_{12}, \lambda_+ - q_{11})^\top$ is nonzero and nonnegative by (i), the first row of $\mathsf{Q}w$ is $q_{11}q_{12} + q_{12}(\lambda_+ - q_{11}) = \lambda_+ q_{12}$, and the second row is $q_{21}q_{12} + q_{22}(\lambda_+ - q_{11}) = (\lambda_+ - q_{11})(\lambda_+ - q_{22}) + q_{22}(\lambda_+ - q_{11}) = \lambda_+(\lambda_+ - q_{11})$. If $q_{12} = 0$ and $q_{22} \ge q_{11}$, then $\lambda_+ = q_{22}$ and $w = (0,1)^\top$ works. If $q_{12} = 0$ and $q_{11} > q_{22}$, then $\lambda_+ = q_{11}$ and $w = (q_{11}-q_{22}, q_{21})^\top$ works, because the second row of $\mathsf{Q}w$ is $q_{21}(q_{11}-q_{22}) + q_{22}q_{21} = q_{11}q_{21}$. The vector $v$ is obtained by applying this to $\mathsf{Q}^\top$, which has the same $\lambda_+$.

(iii) With $w$ from (ii), $\vartheta^\top w > 0$, and multiplying the hypothesis on the right by $w \ge 0$ gives $a\vartheta^\top w \le \vartheta^\top \mathsf{Q} w = \lambda_+ \vartheta^\top w \le b\vartheta^\top w$. With $v$ from (ii), $v^\top \mathbf{1} > 0$ and $\lambda_+ v^\top \mathbf{1} = v^\top \mathsf{Q}\mathbf{1} \le v^\top \mathbf{1}$, with strict inequality when $\mathsf{Q}\mathbf{1} < \mathbf{1}$ because $v \ge 0$ is nonzero.

(iv) Now $D > 0$, so $\lambda_+ > \lambda_-$, and $\lambda_+ + \lambda_- > 0$, so $\lambda_+ > -\lambda_-$. In the characteristic equation the right side is positive and both factors on the left are nonnegative by (i), so both are positive; hence $w = (q_{12}, \lambda_+ - q_{11})^\top$ and $v = (q_{21}, \lambda_+ - q_{11})^\top$ are positive. The eigenvalues are distinct, so each eigenvector is unique up to scale and $\mathsf{Q}$ is diagonalizable. If $w_-$ is a right eigenvector of $\lambda_-$, then $\lambda_+ v^\top w_- = v^\top \mathsf{Q} w_- = \lambda_- v^\top w_-$ gives $v^\top w_- = 0$, so $\Pi = wv^\top/(v^\top w)$ fixes $w$ and annihilates $w_-$, and $\mathsf{Q}^k = \lambda_+^k \Pi + \lambda_-^k (I - \Pi)$. Dividing by $\lambda_+^k$ and using $|\lambda_-/\lambda_+| < 1$ gives the limit. □

**Lemma S45.2** (The relaxation semigroup)**.** *Under the standing hypotheses the following hold.*

*(i)* *$0 < \varpi_- < \min\{\mathsf{p}_1, \mathsf{p}_2\} \le \max\{\mathsf{p}_1, \mathsf{p}_2\} < \varpi_+$, and*

$$\varpi_+ - \mathsf{p}_2 = \mathsf{p}_1 - \varpi_-, \qquad \varpi_+ - \mathsf{p}_1 = \mathsf{p}_2 - \varpi_-, \qquad (\mathsf{p}_1 - \varpi_-)(\mathsf{p}_2 - \varpi_-) = \mathsf{E}. \tag{S172}$$

*The number $\mathsf{u} = (\mathsf{p}_1 - \varpi_-)/(\varpi_+ - \varpi_-) = \frac{1}{2}[1 + (\mathsf{p}_1 - \mathsf{p}_2)((\mathsf{p}_1 - \mathsf{p}_2)^2 + 4\mathsf{E})^{-1/2}]$ lies in $(0,1)$, and $1 - \mathsf{u} = (\mathsf{p}_2 - \varpi_-)/(\varpi_+ - \varpi_-)$.*

*(ii)* *For every $t \ge 0$,*

$$\begin{aligned} \mathsf{G}(t) &= \frac{e^{-\varpi_- t}(\mathsf{A} + \varpi_+ I) - e^{-\varpi_+ t}(\mathsf{A} + \varpi_- I)}{\varpi_+ - \varpi_-} \\ &= \begin{pmatrix} (1-\mathsf{u})e^{-\varpi_- t} + \mathsf{u}e^{-\varpi_+ t} & \tilde{\beta}\dfrac{e^{-\varpi_- t} - e^{-\varpi_+ t}}{\varpi_+ - \varpi_-} \\ \omega\dfrac{e^{-\varpi_- t} - e^{-\varpi_+ t}}{\varpi_+ - \varpi_-} & \mathsf{u}e^{-\varpi_- t} + (1-\mathsf{u})e^{-\varpi_+ t} \end{pmatrix}, \end{aligned} \tag{S173}$$

*and all entries of $\mathsf{G}(t)$ are positive for $t > 0$.*

*(iii)* *The vectors $\ell = (\omega, \mathsf{p}_1 - \varpi_-)^\top$ and $r = (\tilde{\beta}, \mathsf{p}_1 - \varpi_-)^\top$ are positive, $\ell^\top \mathsf{A} = -\varpi_- \ell^\top$, $\mathsf{A}r = -\varpi_- r$, $\ell^\top r = (\mathsf{p}_1 - \varpi_-)(\varpi_+ - \varpi_-)$, and $e^{\varpi_- t}\mathsf{G}(t) \to r\ell^\top/(\ell^\top r)$ as $t \to \infty$.*

*(iv)* *$\mathsf{G}(t)\mathbf{1} < \mathbf{1}$ for every $t > 0$.*

*(v)* $\chi(\tilde\varepsilon) = \tilde\beta(\delta-\tilde\varepsilon)$ *and* $\chi(\delta) = \omega(\tilde\varepsilon-\delta)$. *Consequently* $\min\{\delta,\tilde\varepsilon\} \le \varpi_- \le \max\{\delta,\tilde\varepsilon\}$, *both inequalities are strict when* $\delta \neq \tilde\varepsilon$, *and* $\varpi_- = \delta = \tilde\varepsilon$ *when* $\delta = \tilde\varepsilon$.

*Proof.* (i) The polynomial $\chi$ of (S171) is a monic quadratic with real roots $\varpi_- \le \varpi_+$ and is negative exactly on $(\varpi_-,\varpi_+)$. Since $\chi(\mathsf{p}_1) = \chi(\mathsf{p}_2) = -\mathsf{E} < 0$, both $\mathsf{p}_1$ and $\mathsf{p}_2$ lie strictly between the roots. The product of the roots is $\det \mathsf{A} > 0$ and $\varpi_+ > 0$, so $\varpi_- > 0$. The sum of the roots is $\mathsf{p}_1+\mathsf{p}_2$, which gives the first two identities, and $\chi(\varpi_-) = 0$ is the third. Then $\mathsf{u} \in (0,1)$ because $\varpi_- < \mathsf{p}_1 < \varpi_+$, the expression for $1-\mathsf{u}$ follows from the second identity, and the closed form of $\mathsf{u}$ from $\varpi_+ - \varpi_- = ((\mathsf{p}_1-\mathsf{p}_2)^2 + 4\mathsf{E})^{1/2}$ and $\mathsf{p}_1 - \varpi_- = \frac12(\mathsf{p}_1-\mathsf{p}_2) + \frac12(\varpi_+-\varpi_-)$.

(ii) Let $\mathsf{X}(t)$ be the middle expression of (S173). Then $\mathsf{X}(0) = I$. By the Cayley–Hamilton theorem $(\mathsf{A}+\varpi_- I)(\mathsf{A}+\varpi_+ I) = 0$, so $\mathsf{A}(\mathsf{A}+\varpi_+ I) = -\varpi_-(\mathsf{A}+\varpi_+ I)$ and $\mathsf{A}(\mathsf{A}+\varpi_- I) = -\varpi_+(\mathsf{A}+\varpi_- I)$, whence $\mathsf{A}\mathsf{X}(t) = \mathrm{d}\mathsf{X}(t)/\mathrm{d}t$. The linear equation $\mathrm{d}\mathsf{X}/\mathrm{d}t = \mathsf{A}\mathsf{X}$ with $\mathsf{X}(0) = I$ has the unique solution $e^{t\mathsf{A}}$. The entries follow from

$$\mathsf{A}+\varpi_+ I = \begin{pmatrix} \mathsf{p}_2-\varpi_- & \tilde\beta \\ \omega & \mathsf{p}_1-\varpi_- \end{pmatrix}, \qquad \mathsf{A}+\varpi_- I = \begin{pmatrix} -(\mathsf{p}_1-\varpi_-) & \tilde\beta \\ \omega & -(\mathsf{p}_2-\varpi_-) \end{pmatrix},$$

where (S172) was used in the first matrix. For $t > 0$ one has $e^{-\varpi_- t} > e^{-\varpi_+ t} > 0$, and $\omega$, $\tilde\beta$, $\mathsf{u}$ and $1-\mathsf{u}$ are positive, so every entry is positive.

(iii) Positivity is (i). The first component of $\ell^\top \mathsf{A}$ is $-\mathsf{p}_1\omega + \omega(\mathsf{p}_1-\varpi_-) = -\varpi_-\omega$, and the second is $\tilde\beta\omega - \mathsf{p}_2(\mathsf{p}_1-\varpi_-) = (\mathsf{p}_1-\varpi_-)(\mathsf{p}_2-\varpi_-) - \mathsf{p}_2(\mathsf{p}_1-\varpi_-) = -\varpi_-(\mathsf{p}_1-\varpi_-)$ by (S172); the computation for $r$ is the same with $\omega$ and $\tilde\beta$ exchanged. Next, $\ell^\top r = \mathsf{E} + (\mathsf{p}_1-\varpi_-)^2 = (\mathsf{p}_1-\varpi_-)(\mathsf{p}_1+\mathsf{p}_2-2\varpi_-) = (\mathsf{p}_1-\varpi_-)(\varpi_+-\varpi_-)$. Finally, using $\omega\tilde\beta = (\mathsf{p}_1-\varpi_-)(\mathsf{p}_2-\varpi_-)$, one checks entry by entry that $r\ell^\top = (\mathsf{p}_1-\varpi_-)(\mathsf{A}+\varpi_+ I)$, so by (S173) $e^{\varpi_- t}\mathsf{G}(t) = r\ell^\top/(\ell^\top r) - e^{-(\varpi_+-\varpi_-)t}(\mathsf{A}+\varpi_- I)/(\varpi_+-\varpi_-)$, and the second term vanishes as $t \to \infty$.

(iv) $\mathsf{A}\mathbf{1} = (-\mathsf{p}_1+\tilde\beta, \omega-\mathsf{p}_2)^\top = (-\tilde\varepsilon, -\delta)^\top \le 0$, and this vector is nonzero because $\det \mathsf{A} = \omega\tilde\varepsilon + \delta\mathsf{p}_1 > 0$. Since $\mathsf{G}(s) > 0$ for $s > 0$ by (ii), $\mathrm{d}(\mathsf{G}(s)\mathbf{1})/\mathrm{d}s = \mathsf{G}(s)\mathsf{A}\mathbf{1} < 0$ for $s > 0$, and integrating from 0 to $t$ gives $\mathsf{G}(t)\mathbf{1} < \mathbf{1}$.

(v) $\chi(\tilde\varepsilon) = \tilde\beta(\omega+\delta-\tilde\varepsilon) - \omega\tilde\beta = \tilde\beta(\delta-\tilde\varepsilon)$ and $\chi(\delta) = (\tilde\varepsilon+\tilde\beta-\delta)\omega - \omega\tilde\beta = \omega(\tilde\varepsilon-\delta)$. By (i), $\tilde\varepsilon < \mathsf{p}_1 < \varpi_+$ and $\delta < \mathsf{p}_2 < \varpi_+$, so for $s \in \{\tilde\varepsilon, \delta\}$ the sign of $\chi(s)$ decides the position of $s$ relative to $\varpi_-$: $s < \varpi_-$ when $\chi(s) > 0$, $s > \varpi_-$ when $\chi(s) < 0$, and $s = \varpi_-$ when $\chi(s) = 0$. If $\delta < \tilde\varepsilon$ then $\chi(\tilde\varepsilon) < 0 < \chi(\delta)$, so $\delta < \varpi_- < \tilde\varepsilon$. If $\delta > \tilde\varepsilon$ the signs are reversed and $\tilde\varepsilon < \varpi_- < \delta$. If $\delta = \tilde\varepsilon$ both values vanish. □

*Proof of Proposition S24.2.* Let $\mathcal{H}$ be the parental information of Definition S24.1, and for one daughter let $\xi_i \in \{0,1\}$ indicate that position $i$ is a founder.

(a) The daughter's mean at birth is $\bar\mu_0' = \frac1n \sum_i [\xi_i\mu_i + (1-\xi_i)(\mathsf{f}z_0' + \nu_i)]$ and its record is $z_0' = \mathsf{c}z + \zeta$. The founder set is independent of $\mathcal{H}$ with $\mathbb{E}\xi_i = \bar{\mathsf{b}}$, and $\zeta$ and the $\nu_i$ have zero mean and are independent of $\mathcal{H}$ and of the founder set, so $\mathbb{E}[z_0' \mid \mathcal{H}] = \mathsf{c}z$ and $\mathbb{E}[\bar\mu_0' \mid \mathcal{H}] = \bar{\mathsf{b}}\bar\mu + (1-\bar{\mathsf{b}})\mathsf{f}\mathsf{c}z$; that is, $\mathbb{E}[\mathsf{x}_0' \mid \mathcal{H}] = \mathsf{F}\mathsf{x}$. By step (iv) of Definition S24.1 the daughter's noise increments after birth have zero conditional mean given $\mathcal{H}$ and the daughter's own past, so the argument of Proposition 5.14(a) applies to the daughter with $\mathcal{H}$ adjoined to the conditioning set and gives $\mathbb{E}[\mathsf{x}'(\mathsf{T}) \mid \mathcal{H}, \mathsf{x}_0'] = \mathsf{G}(\mathsf{T})\mathsf{x}_0'$. The tower property gives $\mathbb{E}[\mathsf{x}'(\mathsf{T}) \mid \mathcal{H}] = \mathsf{G}(\mathsf{T})\mathsf{F}\mathsf{x} = \mathsf{M}\mathsf{x}$. For a parent drawn from a population with finite second moments and covariance $\Sigma$ of $\mathsf{x}$, the parent's state is $\mathcal{H}$-measurable, so $\mathrm{Cov}(\mathsf{x}'(\mathsf{T}), \mathsf{x}) = \mathrm{Cov}(\mathbb{E}[\mathsf{x}'(\mathsf{T}) \mid \mathcal{H}], \mathsf{x}) = \mathsf{M}\Sigma$, and the regression of $\mathsf{a}^\top\mathsf{x}'(\mathsf{T})$ on $\mathsf{a}^\top\mathsf{x}$ has slope $\mathsf{a}^\top\mathsf{M}\Sigma\mathsf{a}/(\mathsf{a}^\top\Sigma\mathsf{a})$ whenever the denominator is positive.

(b) By Lemma S45.2(ii) $\mathsf{G}(\mathsf{T}) > 0$, and $\mathsf{F} \ge 0$, so $\mathsf{M} \ge 0$, with entries

$$\mathsf{M}_{11} = \bar{\mathsf{b}}\mathsf{G}_{11}, \quad \mathsf{M}_{21} = \bar{\mathsf{b}}\mathsf{G}_{21}, \quad \mathsf{M}_{12} = \mathsf{c}[(1-\bar{\mathsf{b}})\mathsf{f}\mathsf{G}_{11} + \mathsf{G}_{12}], \quad \mathsf{M}_{22} = \mathsf{c}[(1-\bar{\mathsf{b}})\mathsf{f}\mathsf{G}_{21} + \mathsf{G}_{22}], \tag{S174}$$

all evaluated at $\mathsf{T}$. Lemma S45.1(i) and (ii) give the formula (S104) for $\mathsf{h} = \rho(\mathsf{M})$ and a nonzero $\mathsf{l} \geq 0$ with $\mathsf{l}^\top \mathsf{M} = \mathsf{h}\mathsf{l}^\top$. Then $\mathbb{E}[y' \mid \mathcal{H}] = \mathsf{l}^\top \mathsf{M}\mathsf{x} = \mathsf{h}y$ by (a), and the slope of (a) with $\mathsf{a} = \mathsf{l}$ is $\mathsf{h}\mathsf{l}^\top \Sigma \mathsf{l}/(\mathsf{l}^\top \Sigma \mathsf{l}) = \mathsf{h}$. Since $\mathsf{F}\mathbf{1} = (\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f}\mathsf{c}, \mathsf{c})^\top \leq \mathbf{1}$ and $\mathsf{G}(\mathsf{T}) \geq 0$, Lemma S45.2(iv) gives $\mathsf{M}\mathbf{1} \leq \mathsf{G}(\mathsf{T})\mathbf{1} < \mathbf{1}$, and Lemma S45.1(iii) gives $\mathsf{h} < 1$. When $\mathsf{b} = 0$ the first column of $\mathsf{M}$ vanishes, $\mathsf{h} = \mathsf{M}_{22}$ and $\mathsf{l} = (0,1)^\top$; when $\mathsf{c} = 0$ the second column vanishes, $\mathsf{h} = \mathsf{M}_{11} = \bar{\mathsf{b}}\mathsf{G}_{11}(\mathsf{T})$ and $\mathsf{l} = (1,0)^\top$ if $\mathsf{b} \geq 1$.

(c) If $\mathsf{b} \geq 1$ and $\mathsf{c} > 0$, every entry in (S174) is positive, so Lemma S45.1(iv) applies: $\mathsf{l} > 0$ is unique up to scale and $\mathsf{h}^{-k}\mathsf{M}^k \to r_{\mathsf{M}}\mathsf{l}^\top/(\mathsf{l}^\top r_{\mathsf{M}})$, where $r_{\mathsf{M}} > 0$ is the right eigenvector of $\mathsf{h}$. Consider a line of descent in which every collective founds at least one daughter, so that the line exists for every $k$, and in which the daughter followed is selected by a rule independent of the states and noises of all collectives on the line. Let $\mathcal{H}_k$ be the information of generation $k$ on the line, with $\mathcal{H}_0 = \mathcal{H}$ and $\mathcal{H}_k \subseteq \mathcal{H}_{k+1}$. Because the selection rule is independent of the daughters' states and noises, conditioning on which daughter is followed does not change the conditional law of that daughter's state, and (a) applied to generation $k$ gives $\mathbb{E}[\mathsf{x}^{(k+1)} \mid \mathcal{H}_k] = \mathsf{M}\mathsf{x}^{(k)}$. The tower property and induction on $k$ give $\mathbb{E}[\mathsf{x}^{(k)} \mid \mathcal{H}] = \mathsf{M}^k\mathsf{x}$, so $\mathsf{h}^{-k}\mathbb{E}[\mathsf{a}^\top \mathsf{x}^{(k)} \mid \mathcal{H}] \to (\mathsf{a}^\top r_{\mathsf{M}})(\mathsf{l}^\top \mathsf{x})/(\mathsf{l}^\top r_{\mathsf{M}})$, which is the stated limit; it is nonzero exactly when $\mathsf{a}^\top r_{\mathsf{M}} \neq 0$ and $y \neq 0$, since $\mathsf{l}^\top r_{\mathsf{M}} > 0$. A rule that selects the daughter by its state, for example the daughter with the largest trait, is excluded: the selected daughter's conditional mean is then not $\mathsf{M}\mathsf{x}^{(k)}$ in general. If $\mathsf{a}^\top r_{\mathsf{M}} = 0$ with $\mathsf{a} \neq 0$, then $\mathsf{a}$ is a left eigenvector of $\mathsf{M}$ for its second eigenvalue $\lambda_- = \operatorname{tr}\mathsf{M} - \mathsf{h}$, because the eigenvalues are distinct and the left eigenvector of one annihilates the right eigenvector of the other, and $\mathbb{E}[\mathsf{a}^\top \mathsf{x}^{(k)} \mid \mathcal{H}] = \lambda_-^k \mathsf{a}^\top \mathsf{x}$ with $|\lambda_-| < \mathsf{h}$.

(d) In the variant without a record, $\bar{\mu}_0' = \frac{1}{n}\sum_i [\xi_i \mu_i + (1-\xi_i)(\mathsf{f} m_S + \nu_i)] = (\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f}) m_S + \frac{1}{n}\sum_i (1-\xi_i)\nu_i$, where $m_S = \frac{1}{\mathsf{b}}\sum_i \xi_i \mu_i$ has conditional mean $\bar{\mu}$ given $\mathcal{H}$. By the last sentence of Proposition 5.14(a) the conditional mean of the members' mean then relaxes at the rate $\tilde{\varepsilon}$, so $\mathbb{E}[\bar{\mu}'(\mathsf{T}) \mid \mathcal{H}] = (\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f}) e^{-\tilde{\varepsilon}\mathsf{T}} \bar{\mu} = \mathsf{h}_0 \bar{\mu}$. □

*Proof of Theorem S24.3.* (a) With $\ell > 0$ from Lemma S45.2(iii), $\ell^\top \mathsf{M} = \ell^\top \mathsf{G}(\mathsf{T})\mathsf{F} = e^{-\varpi_- \mathsf{T}} \ell^\top \mathsf{F}$, and

$$\ell^\top \mathsf{F} = (\bar{\mathsf{b}}\omega,\ \omega(1-\bar{\mathsf{b}})\mathsf{f}\mathsf{c} + \mathsf{c}(\mathsf{p}_1 - \varpi_-)) = (\bar{\mathsf{b}}\ell_1, \mathsf{c}\mathsf{k}\ell_2), \qquad \mathsf{k} = 1 + \frac{(1-\bar{\mathsf{b}})\mathsf{f}\omega}{\mathsf{p}_1 - \varpi_-}. \tag{S175}$$

Hence $e^{-\varpi_- \mathsf{T}} \min\{\bar{\mathsf{b}}, \mathsf{c}\mathsf{k}\} \ell^\top \leq \ell^\top \mathsf{M} \leq e^{-\varpi_- \mathsf{T}} \max\{\bar{\mathsf{b}}, \mathsf{c}\mathsf{k}\} \ell^\top$, and Lemma S45.1(iii) gives the two-sided bound of (S105). By Lemma S45.1(i), $\mathsf{h} \geq \max\{\mathsf{M}_{11}, \mathsf{M}_{22}\}$, and by (S174) and (S173), $\mathsf{M}_{11} \geq \bar{\mathsf{b}}(1-\mathsf{u}) e^{-\varpi_- \mathsf{T}}$ and $\mathsf{M}_{22} \geq \mathsf{c}\mathsf{G}_{22}(\mathsf{T}) \geq \mathsf{c}\mathsf{u} e^{-\varpi_- \mathsf{T}}$, which is the third bound. For the limits, $e^{\varpi_- \mathsf{T}} \mathsf{M} \to r \ell^\top \mathsf{F}/(\ell^\top r)$ as $\mathsf{T} \to \infty$ by Lemma S45.2(iii). The limit is a nonnegative matrix of rank at most one, whose spectral radius is its trace $\ell^\top \mathsf{F} r/(\ell^\top r)$, and the spectral radius is continuous (Lemma S45.1(i)), so by (S175) and (S172)

$$\lim_{\mathsf{T}\to\infty} e^{\varpi_- \mathsf{T}} \mathsf{h} = \frac{\bar{\mathsf{b}}\omega\tilde{\beta} + \mathsf{c}\mathsf{k}(\mathsf{p}_1 - \varpi_-)^2}{(\mathsf{p}_1 - \varpi_-)(\varpi_+ - \varpi_-)} = \bar{\mathsf{b}} \frac{\mathsf{p}_2 - \varpi_-}{\varpi_+ - \varpi_-} + \mathsf{c}\mathsf{k} \frac{\mathsf{p}_1 - \varpi_-}{\varpi_+ - \varpi_-} = (1-\mathsf{u})\bar{\mathsf{b}} + \mathsf{u}\mathsf{c}\mathsf{k}.$$

As $\mathsf{T} \to 0$, $\mathsf{M} \to \mathsf{F}$, which is triangular with spectral radius $\max\{\bar{\mathsf{b}}, \mathsf{c}\}$.

(b) This is Lemma S45.2(v), with $\tau_{\mathrm{id}}^{-1} = \varpi_-$ and the identity time $1/\tilde{\varepsilon}$ that Proposition 5.14(a) gives without a record.

(c) By (a), $\mathsf{h} \geq \mathsf{c}\mathsf{u} e^{-\varpi_- \mathsf{T}} > 0$ for every $\mathsf{b}$, while $\mathsf{h}_0 = (\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f}) e^{-\tilde{\varepsilon}\mathsf{T}}$ for $\mathsf{b} \geq 1$ and $\mathsf{h}_0 = 0$ for $\mathsf{b} = 0$. Dividing gives the bound, and $\tilde{\varepsilon} - \varpi_- > 0$ by (b) when $\delta < \tilde{\varepsilon}$, so the bound diverges with $\mathsf{T}$; the weaker form uses $\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f} \leq 1$.

(d) By (a), $\mathsf{h}/\mathsf{h}_0 \leq \max\{\bar{\mathsf{b}}, \mathsf{c}\mathsf{k}\} e^{-(\varpi_- - \tilde{\varepsilon})\mathsf{T}}/(\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f})$, and $\varpi_- \geq \tilde{\varepsilon}$ by (b) when $\delta \geq \tilde{\varepsilon}$, with strict inequality when $\delta > \tilde{\varepsilon}$. The prefactor does not depend on $\mathsf{T}$. For the rate, when $\mathsf{b} \geq 1$ the

limit in (a) is at least $(1-\mathsf{u})\bar{\mathsf{b}} > 0$, so $\frac{1}{\mathsf{T}}\log \mathsf{h} \to -\varpi_-$, while $\frac{1}{\mathsf{T}}\log \mathsf{h}_0 \to -\tilde{\varepsilon}$. If $\mathsf{c} = 0$, the second column of $\mathsf{M}$ vanishes and $\mathsf{h} = \mathsf{M}_{11} = \bar{\mathsf{b}}\mathsf{G}_{11}(\mathsf{T}) \le \bar{\mathsf{b}}$, because $\mathsf{G}_{11}(\mathsf{T})$ is a convex combination of $e^{-\varpi_-\mathsf{T}}$ and $e^{-\varpi_+\mathsf{T}}$.

(e) Let the record be written only by non-members whose writes do not depend on the members' states, with locally integrable path $z_{\mathrm{ext}}(\cdot)$. As in Proposition 5.14(b), the path is then a function of the exterior's exogenous input, which is the conditioning set of Definition 5.13; the argument below uses this only through the three conditional hypotheses of the statement. The hypotheses are stated conditionally on the pair $(\mathcal{H}, z_{\mathrm{ext}})$, since independence of the path from the founder set and from the noises separately would not suffice once the path is added to the conditioning set: given $\mathcal{H}$ and $z_{\mathrm{ext}}$ the founder set is uniform, so $\mathbb{E}[\xi_i \mid \mathcal{H}, z_{\mathrm{ext}}] = \bar{\mathsf{b}}$, the entry noises have $\mathbb{E}[\nu_i \mid \mathcal{H}, z_{\mathrm{ext}}, \xi] = 0$, and the relaxation noise of the daughter has zero conditional mean increments given $\mathcal{H}$, $z_{\mathrm{ext}}$ and the daughter's past. Recruits at fission start at $\mathsf{f} z_{\mathrm{ext}}(0) + \nu_i$, so $\mathbb{E}[\bar{\mu}'_0 \mid \mathcal{H}, z_{\mathrm{ext}}] = \bar{\mathsf{b}}\bar{\mu} + (1-\bar{\mathsf{b}})\mathsf{f} z_{\mathrm{ext}}(0)$. As in the proof of Proposition 5.14(b), the first row of (62) holds with the record path given: the member-level drifts sum to $-\mathsf{p}_1\bar{\mu}' + \tilde{\beta} z_{\mathrm{ext}}$ because the communication term has zero sum over members, and the noise increments drop out of the conditional mean by the third hypothesis. Hence $m(t) = \mathbb{E}[\bar{\mu}'(t) \mid \mathcal{H}, z_{\mathrm{ext}}]$ solves $\mathrm{d}m/\mathrm{d}t = -\mathsf{p}_1 m + \tilde{\beta} z_{\mathrm{ext}}(t)$, and

$$\mathbb{E}[\bar{\mu}'(\mathsf{T}) \mid \mathcal{H}, z_{\mathrm{ext}}] = \bar{\mathsf{b}} e^{-\mathsf{p}_1\mathsf{T}}\bar{\mu} + e^{-\mathsf{p}_1\mathsf{T}}(1-\bar{\mathsf{b}})\mathsf{f} z_{\mathrm{ext}}(0) + \tilde{\beta}\int_0^{\mathsf{T}} e^{-\mathsf{p}_1(\mathsf{T}-s)} z_{\mathrm{ext}}(s)\mathrm{d}s.$$

The last two terms are functions of the record path alone, so the coefficient of the parent's mean given the record path is $\bar{\mathsf{b}} e^{-\mathsf{p}_1\mathsf{T}}$. Since $\mathsf{p}_1 = \varepsilon + \beta + g \ge g$ and $\mathsf{p}_1 = \tilde{\varepsilon} + \tilde{\beta} \ge \tilde{\varepsilon}$, it is at most $\bar{\mathsf{b}} e^{-g\mathsf{T}}$ and at most $\bar{\mathsf{b}} e^{-\tilde{\varepsilon}\mathsf{T}} \le \mathsf{h}_0$. □

*Proof of Corollary S24.4.* By Theorem S24.3(a), $\mathsf{h} \ge \mathsf{c}\mathsf{u} e^{-\mathsf{T}/\tau_{\mathrm{id}}}$, and the right side is at least $\mathsf{h}_*$ exactly when $\mathsf{T} \le \tau_{\mathrm{id}} \log(\mathsf{c}\mathsf{u}/\mathsf{h}_*)$. Without a record, $\mathsf{h}_0 = (\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f}) e^{-\tilde{\varepsilon}\mathsf{T}} \ge \mathsf{h}_*$ is the stated condition on $\mathsf{T}$ after taking logarithms. □

*Proof of Proposition S24.5.* (a) With the indicators $\xi_i$ of the proof of Proposition S24.2, the deviations of the daughter's birth state from its conditional mean are

$$\bar{\mu}'_0 - \mathbb{E}[\bar{\mu}'_0 \mid \mathcal{H}] = \frac{1}{n}\sum_i (\xi_i - \bar{\mathsf{b}})\mu_i + (1-\bar{\mathsf{b}})\mathsf{f}\zeta + \frac{1}{n}\sum_i (1-\xi_i)\nu_i, \qquad z'_0 - \mathbb{E}[z'_0 \mid \mathcal{H}] = \zeta.$$

The founder set, $\zeta$ and the $\nu_i$ are independent of each other and of $\mathcal{H}$, and $\zeta$ and the $\nu_i$ have zero mean, so the three terms of the first deviation are uncorrelated given $\mathcal{H}$. For a uniformly drawn set of $\mathsf{b}$ positions out of $n \ge 2$, $\operatorname{Var}\xi_i = \bar{\mathsf{b}}(1-\bar{\mathsf{b}})$ and, for $i \ne j$, $\operatorname{Cov}(\xi_i, \xi_j) = \frac{\mathsf{b}(\mathsf{b}-1)}{n(n-1)} - \bar{\mathsf{b}}^2 = -\bar{\mathsf{b}}(1-\bar{\mathsf{b}})/(n-1)$. Hence

$$\begin{aligned}\operatorname{Var}\Big(\frac{1}{n}\sum_i \xi_i\mu_i \,\Big|\, \mathcal{H}\Big) &= \frac{\bar{\mathsf{b}}(1-\bar{\mathsf{b}})}{n^2}\Big[\sum_i \mu_i^2 - \frac{(\sum_i \mu_i)^2 - \sum_i \mu_i^2}{n-1}\Big] \\ &= \frac{\bar{\mathsf{b}}(1-\bar{\mathsf{b}})}{n^2}\frac{n}{n-1}\sum_i (\mu_i - \bar{\mu})^2 = \frac{\bar{\mathsf{b}}(1-\bar{\mathsf{b}})}{n-1}\mathsf{v}_{\mathrm{w}}.\end{aligned}$$

The third term has variance $\frac{1}{n^2}\mathbb{E}[\sum_i (1-\xi_i)]\mathsf{s}_\nu^2 = (1-\bar{\mathsf{b}})\mathsf{s}_\nu^2/n$ and the second $(1-\bar{\mathsf{b}})^2\mathsf{f}^2\mathsf{s}_z^2$; the covariance of the two deviations is $(1-\bar{\mathsf{b}})\mathsf{f}\mathsf{s}_z^2$, and the variance of the second deviation is $\mathsf{s}_z^2$. This is (S106).

(b) Write $\mathsf{x}'(\mathsf{T}) - \mathsf{M}\mathsf{x} = \mathsf{G}(\mathsf{T})(\mathsf{x}'_0 - \mathsf{F}\mathsf{x}) + (\mathsf{x}'(\mathsf{T}) - \mathsf{G}(\mathsf{T})\mathsf{x}'_0)$. The second term has zero conditional mean given $\mathcal{H}$ and the daughter's birth state (proof of Proposition S24.2(a)), and the first is a function of these, so the two are uncorrelated given $\mathcal{H}$, and $\operatorname{Cov}(\mathsf{x}'(\mathsf{T}) \mid \mathcal{H}) = \mathsf{G}(\mathsf{T})\mathsf{N}_{\mathrm{fis}}\mathsf{G}(\mathsf{T})^\top + \mathsf{N}_{\mathrm{rel}}$

with $\mathsf{N}_{\mathrm{rel}}$ the conditional covariance of the second term. Contracting with $\mathsf{l}$ gives the conditional variance of $y'$. Since $\mathbb{E}[y' \mid \mathcal{H}] = \mathsf{h}y$, the law of total variance, applied to a parent drawn from the parental distribution and its one daughter, gives $\mathrm{Var}(y') = \mathrm{Var}(\mathbb{E}[y' \mid \mathcal{H}]) + \mathbb{E}[\mathrm{Var}(y' \mid \mathcal{H})] = \mathsf{h}^2\mathsf{V} + \mathsf{s}^2$. Here Var is taken over the draw of the parent and over the noises of fission and relaxation; for a finite population of $N \geq 2$ parents whose daughters' deviations $y'_j - \mathsf{h}y_j$ are uncorrelated across parents given $\mathcal{H}$, the sample variance of the daughters' traits with divisor $N-1$ has conditional expectation $\mathsf{h}^2$ times the parents' sample variance plus $\frac{1}{N}\sum_j \mathrm{Var}(y'_j \mid \mathcal{H})$, and its realized value fluctuates about that expectation. The statement claims nothing more for a finite population. If $\mathsf{s}^2$ is the same in every generation, the recursion $\mathsf{V}_{k+1} = \mathsf{h}^2\mathsf{V}_k + \mathsf{s}^2$ with $\mathsf{h} < 1$ converges to $\mathsf{s}^2/(1-\mathsf{h}^2)$. At that value with $\mathsf{s}^2 > 0$ one has $\mathrm{Var}(y') = \mathrm{Var}(y) = \mathsf{V} > 0$ and $\mathrm{Cov}(y', y) = \mathsf{h}\mathsf{V}$ by Proposition S24.2(a), so $\mathrm{Corr}(y', y) = \mathsf{h}\mathsf{V}/(\mathsf{V}\mathsf{V})^{1/2} = \mathsf{h}$; for $\mathsf{s}^2 = 0$ the limit is 0 and the correlation is undefined.

(c) In the variant without a record, by the proof of Proposition S24.2(d), $\bar{\mu}'_0 - \mathbb{E}[\bar{\mu}'_0 \mid \mathcal{H}] = (\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f})(m_S - \bar{\mu}) + \frac{1}{n}\sum_i (1-\xi_i)\nu_i$. The computation of (a) with $\frac{1}{\mathsf{b}}$ in place of $\frac{1}{n}$ gives $\mathrm{Var}(m_S \mid \mathcal{H}) = \frac{\bar{\mathsf{b}}(1-\bar{\mathsf{b}})}{\mathsf{b}^2}\frac{n^2}{n-1}\mathsf{v}_{\mathrm{w}} = \frac{\mathsf{v}_{\mathrm{w}}}{\mathsf{b}}\frac{n-\mathsf{b}}{n-1}$, and the two terms are uncorrelated, which gives the stated variance. Finally $\mathsf{h}_0 = (\mathsf{f} + (1-\mathsf{f})\mathsf{b}/n)e^{-\tilde{\varepsilon}\mathsf{T}}$ is nondecreasing in $\mathsf{b}$, strictly increasing when $\mathsf{f} < 1$. □

**Proposition S45.3** (Founders against recruits at long generation times)**.** *Let* $\mathsf{P}(\bar{\mathsf{b}}) = (1-\mathsf{u})\bar{\mathsf{b}} + \mathsf{u}\mathsf{c}\mathsf{k}$ *be the limit of* $\mathsf{h}e^{\mathsf{T}/\tau_{\mathrm{id}}}$ *in Theorem S24.3(a), regarded as a function of the founder fraction. It is affine in* $\bar{\mathsf{b}}$ *with slope*

$$\frac{\mathrm{d}\mathsf{P}}{\mathrm{d}\bar{\mathsf{b}}} = \frac{\omega(1-\mathsf{c}\mathsf{f}) + \delta - \varpi_-}{\varpi_+ - \varpi_-}. \tag{S176}$$

*Consequently there is a* $\mathsf{T}_0$ *such that for all* $\mathsf{T} \geq \mathsf{T}_0$ *the heritability* $\mathsf{h}$ *is strictly increasing in* $\mathsf{b} \in \{0, \dots, n\}$ *when* $\varpi_- < \delta + \omega(1-\mathsf{c}\mathsf{f})$ *and strictly decreasing when* $\varpi_- > \delta + \omega(1-\mathsf{c}\mathsf{f})$*. When* $\mathsf{c} = \mathsf{f} = 1$ *the second case holds exactly when* $\delta < \tilde{\varepsilon}$*.*

*Proof.* By (S175), $\mathsf{k}$ is affine in $\bar{\mathsf{b}}$ with slope $-\mathsf{f}\omega/(\mathsf{p}_1 - \varpi_-)$, so $\mathrm{d}\mathsf{P}/\mathrm{d}\bar{\mathsf{b}} = (1-\mathsf{u}) - \mathsf{u}\mathsf{c}\mathsf{f}\omega/(\mathsf{p}_1 - \varpi_-) = [(\mathsf{p}_2 - \varpi_-) - \mathsf{c}\mathsf{f}\omega]/(\varpi_+ - \varpi_-)$ by Lemma S45.2(i), and $\mathsf{p}_2 = \omega + \delta$. For each of the finitely many values $\mathsf{b} \in \{0, \dots, n\}$, $e^{\varpi_- \mathsf{T}}\mathsf{h} \to \mathsf{P}(\mathsf{b}/n)$ as $\mathsf{T} \to \infty$ by Theorem S24.3(a). When the slope is nonzero the limits are strictly ordered in $\mathsf{b}$, so there is a common $\mathsf{T}_0$ beyond which the values of $\mathsf{h}$ are ordered in the same way. When $\mathsf{c} = \mathsf{f} = 1$ the slope has the sign of $\delta - \varpi_-$, which is negative exactly when $\delta < \tilde{\varepsilon}$ by Lemma S45.2(v). □

The mechanism is visible in the slow mode $r$ of Lemma S45.2(iii). Replacing one recruit by one founder changes $\ell^\top \mathsf{F}\mathsf{x}$ by $\omega(\bar{\mu} - \mathsf{f}\mathsf{c}z)/n$, and on the slow mode $\bar{\mu}/z = \tilde{\beta}/(\mathsf{p}_1 - \varpi_-) = (\mathsf{p}_2 - \varpi_-)/\omega$. When the record decays more slowly than the members forget, the record coordinate of the slow mode exceeds the members' mean, and a recruit who reads a faithful copy of the record carries more of the heritable trait than a founder does. The comparison of Check S47.217 shows the inversion at short generation times as well in one instance, which the proposition does not cover.

**Lemma S45.4** (Covariance identity)**.** *Let* $y_1, \dots, y_N$ *be real numbers and* $\mathsf{W}_1, \dots, \mathsf{W}_N$ *nonnegative numbers with* $\bar{\mathsf{W}} = \frac{1}{N}\sum_j \mathsf{W}_j > 0$*. For each* $j$ *with* $\mathsf{W}_j > 0$ *let* $\bar{y}'_j$ *be a real number and* $\Delta y_j = \bar{y}'_j - y_j$*, and put* $\Delta y_j = 0$ *when* $\mathsf{W}_j = 0$*. With* $\bar{y} = \frac{1}{N}\sum_j y_j$*,* $\bar{y}' = \sum_j \mathsf{W}_j \bar{y}'_j / \sum_j \mathsf{W}_j$ *and* $\mathrm{Cov}_j(\mathsf{W}_j, y_j) = \frac{1}{N}\sum_j \mathsf{W}_j y_j - \bar{\mathsf{W}}\bar{y}$*,*

$$\bar{\mathsf{W}}(\bar{y}' - \bar{y}) = \mathrm{Cov}_j(\mathsf{W}_j, y_j) + \frac{1}{N}\sum_j \mathsf{W}_j \Delta y_j. \tag{S177}$$

*Proof.* $N\bar{\mathsf{W}}\bar{y}' = \sum_j \mathsf{W}_j \bar{y}'_j = \sum_j \mathsf{W}_j y_j + \sum_j \mathsf{W}_j \Delta y_j$. Dividing by $N$ and subtracting $\bar{\mathsf{W}}\bar{y}$ gives (S177). □

*Proof of Proposition S24.6.* Apply Lemma S45.4 with $y_j = \mathsf{a}^\top \mathsf{x}_j$ and $\bar{y}'_j$ the mean of $\mathsf{a}^\top \mathsf{x}'(\mathsf{T})$ over the $\mathsf{W}_j$ daughters of parent $j$, and divide by $\bar{\mathsf{W}}$: $\bar{y}' - \bar{y} = \mathrm{Cov}_j(\mathsf{w}_j, y_j) + \frac{1}{N}\sum_j \mathsf{w}_j \Delta y_j$. The numbers $\mathsf{W}_j$ and $y_j$ are $\mathcal{H}$-measurable, and each daughter of parent $j$ is produced by a fission of Definition S24.1, so by Proposition S24.2(a) $\mathbb{E}[\bar{y}'_j \mid \mathcal{H}] = \mathsf{a}^\top \mathsf{M} \mathsf{x}_j$ and $\mathbb{E}[\Delta y_j \mid \mathcal{H}] = \mathsf{a}^\top(\mathsf{M} - I)\mathsf{x}_j$ when $\mathsf{W}_j > 0$. Since $\frac{1}{N}\sum_j \mathsf{w}_j = 1$, $\frac{1}{N}\sum_j \mathsf{w}_j \mathsf{a}^\top(\mathsf{M}-I)\mathsf{x}_j = \mathrm{Cov}_j(\mathsf{w}_j, \mathsf{a}^\top(\mathsf{M}-I)\mathsf{x}_j) + \mathsf{a}^\top(\mathsf{M}-I)\bar{\mathsf{x}}$, and adding $\mathrm{Cov}_j(\mathsf{w}_j, \mathsf{a}^\top \mathsf{x}_j)$ gives (S107). For $\mathsf{a} = \mathsf{l}$, $\mathsf{l}^\top \mathsf{M} = \mathsf{h}\mathsf{l}^\top$, so the transmission term is $\frac{1}{N}\sum_j \mathsf{w}_j(\mathsf{h}-1)y_j = -(1-\mathsf{h})(\bar{y} + \mathsf{d}_{\mathrm{sel}})$, and the sum of the two terms is $\mathsf{h}\mathsf{d}_{\mathrm{sel}} - (1-\mathsf{h})\bar{y}$, which is positive exactly when $\mathsf{h}\mathsf{d}_{\mathrm{sel}} > (1-\mathsf{h})\bar{y}$. If all $\mathsf{W}_j$ are equal then $\mathsf{d}_{\mathrm{sel}} = 0$ and $\mathbb{E}[\bar{y}' \mid \mathcal{H}] = \mathsf{h}\bar{y}$; if this holds in every generation, the tower property gives $\mathbb{E}[\bar{y}_k] = \mathsf{h}^k \mathbb{E}[\bar{y}_0]$. □

*Proof of Proposition S24.8.* Write $\mathcal{N}(\cdot; m, \mathsf{V})$ for the normal density.

*Recursion.* For $\mathsf{V} > 0$ the trait density among the parents of the daughters is proportional to $\mathsf{W}(y)\mathcal{N}(y; \bar{y}, \mathsf{V})$, the exponential of a quadratic with second-order coefficient $-\frac{1}{2}(1/\mathsf{V} + 1/\mathsf{v}_{\mathrm{s}})$. It is therefore normal with variance $\mathsf{V}\mathsf{v}_{\mathrm{s}}/(\mathsf{V}+\mathsf{v}_{\mathrm{s}})$ and mean $(\bar{y}/\mathsf{V} + \mathsf{y}_*/\mathsf{v}_{\mathrm{s}})\mathsf{V}\mathsf{v}_{\mathrm{s}}/(\mathsf{V}+\mathsf{v}_{\mathrm{s}}) = \bar{y} + \frac{\mathsf{V}}{\mathsf{V}+\mathsf{v}_{\mathrm{s}}}(\mathsf{y}_* - \bar{y})$; for $\mathsf{V} = 0$ the weighting leaves the point mass at $\bar{y}$ unchanged and the same formulas hold. A daughter's trait is $\mathsf{h}y$ plus an independent normal variable of variance $\mathsf{s}^2$, so the daughters' traits are normal with the mean and variance of (S109). The selection differential is $\mathsf{d}_{\mathrm{sel}} = \frac{\mathsf{V}}{\mathsf{V}+\mathsf{v}_{\mathrm{s}}}(\mathsf{y}_* - \bar{y})$, and the recursion for the mean is $\bar{y}' = \mathsf{h}(\bar{y} + \mathsf{d}_{\mathrm{sel}})$, in agreement with (S108).

*Variance.* Let $\Phi(\mathsf{V}) = \mathsf{h}^2\mathsf{v}_{\mathrm{s}}\mathsf{V}/(\mathsf{V}+\mathsf{v}_{\mathrm{s}}) + \mathsf{s}^2$ on $[0,\infty)$. Then

$$\Phi(\mathsf{V}) - \mathsf{V} = -\frac{\mathsf{q}(\mathsf{V})}{\mathsf{V}+\mathsf{v}_{\mathrm{s}}}, \qquad \mathsf{q}(\mathsf{V}) = \mathsf{V}^2 + \mathsf{V}(\mathsf{v}_{\mathrm{s}}(1-\mathsf{h}^2) - \mathsf{s}^2) - \mathsf{s}^2\mathsf{v}_{\mathrm{s}}. \tag{S178}$$

The quadratic $\mathsf{q}$ has roots of product $-\mathsf{s}^2\mathsf{v}_{\mathrm{s}} < 0$, hence exactly one positive root, which is $\hat{\mathsf{V}}$ of (S110); $\mathsf{q} < 0$ on $[0, \hat{\mathsf{V}})$ and $\mathsf{q} > 0$ on $(\hat{\mathsf{V}}, \infty)$. Since $\Phi \geq \mathsf{s}^2$, $\mathsf{V}_k \geq \mathsf{s}^2$ for $k \geq 1$ and $\hat{\mathsf{V}} = \Phi(\hat{\mathsf{V}}) \geq \mathsf{s}^2$. On $[\mathsf{s}^2, \infty)$, $0 \leq \Phi'(\mathsf{V}) = \mathsf{h}^2\mathsf{v}_{\mathrm{s}}^2/(\mathsf{V}+\mathsf{v}_{\mathrm{s}})^2 \leq \theta^2$ with $\theta = \mathsf{v}_{\mathrm{s}}/(\mathsf{s}^2+\mathsf{v}_{\mathrm{s}}) < 1$, so $|\mathsf{V}_{k+1} - \hat{\mathsf{V}}| \leq \theta^2 |\mathsf{V}_k - \hat{\mathsf{V}}|$ for $k \geq 1$ and $\mathsf{V}_k \to \hat{\mathsf{V}}$.

*Mean.* Put $\mathsf{i}_k = \mathsf{V}_k/(\mathsf{V}_k + \mathsf{v}_{\mathrm{s}})$, so that $\mathsf{i}_k \to \mathsf{i} = \hat{\mathsf{V}}/(\hat{\mathsf{V}} + \mathsf{v}_{\mathrm{s}}) > 0$ and $\bar{y}_{k+1} = \mathsf{h}(1-\mathsf{i}_k)\bar{y}_k + \mathsf{h}\mathsf{i}_k \mathsf{y}_*$. The number $1 - \mathsf{h} + \mathsf{h}\mathsf{i}$ is positive, because $1 - \mathsf{h} > 0$ or $\mathsf{h} = 1$ and $\mathsf{i} > 0$, and $\hat{y}$ of (S110) is the unique solution of $\hat{y} = \mathsf{h}(1-\mathsf{i})\hat{y} + \mathsf{h}\mathsf{i}\mathsf{y}_*$. Subtracting, $e_k = \bar{y}_k - \hat{y}$ satisfies $e_{k+1} = \mathsf{h}(1-\mathsf{i}_k)e_k + \mathsf{h}(\mathsf{i}_k - \mathsf{i})(\mathsf{y}_* - \hat{y})$. For $k \geq 1$, $\mathsf{i}_k \geq \mathsf{s}^2/(\mathsf{s}^2+\mathsf{v}_{\mathrm{s}})$, so $0 \leq \mathsf{h}(1-\mathsf{i}_k) \leq \theta$ and $|e_{k+1}| \leq \theta|e_k| + \eta_k$ with $\eta_k = |\mathsf{i}_k - \mathsf{i}||\mathsf{y}_* - \hat{y}| \to 0$. The sequence $|e_k|$ is bounded, and its upper limit $L$ satisfies $L \leq \theta L$, so $L = 0$.

*Dependence on* $\mathsf{h}$ *and* $\mathsf{s}^2$. For $\mathsf{V} > 0$, $\Phi(\mathsf{V})$ is strictly increasing in $\mathsf{h} \in [0,1]$ and in $\mathsf{s}^2$. If $\Phi_1 \leq \Phi_2$ are the maps for two parameter values, with strict inequality on $(0,\infty)$, and $\hat{\mathsf{V}}_1$, $\hat{\mathsf{V}}_2$ their fixed points, then $\Phi_2(\hat{\mathsf{V}}_1) - \hat{\mathsf{V}}_1 > \Phi_1(\hat{\mathsf{V}}_1) - \hat{\mathsf{V}}_1 = 0$, and by (S178) the function $\Phi_2(\mathsf{V}) - \mathsf{V}$ is positive only below $\hat{\mathsf{V}}_2$, so $\hat{\mathsf{V}}_1 < \hat{\mathsf{V}}_2$. Hence $\hat{\mathsf{V}}$ and $\mathsf{i}$ are strictly increasing in $\mathsf{h}$ and in $\mathsf{s}^2$; they are continuous in both by (S110), and $\hat{\mathsf{V}} = \mathsf{s}^2$ at $\mathsf{h} = 0$, where $\mathsf{q}(\mathsf{V}) = (\mathsf{V} - \mathsf{s}^2)(\mathsf{V} + \mathsf{v}_{\mathrm{s}})$. The retained fraction is $\mathsf{R} = \hat{y}/\mathsf{y}_* = \mathsf{h}\mathsf{i}/(1 - \mathsf{h} + \mathsf{h}\mathsf{i})$, continuous on $[0,1]$, with $\mathsf{R} = 0$ at $\mathsf{h} = 0$, $\mathsf{R} = 1$ at $\mathsf{h} = 1$ and $\mathsf{R} > 0$ for $\mathsf{h} > 0$. For $0 < \mathsf{h} \leq 1$, $\mathsf{R} = [1 + (1-\mathsf{h})/(\mathsf{h}\mathsf{i})]^{-1}$, and $(1-\mathsf{h})/\mathsf{h}$ is nonnegative and strictly decreasing while $1/\mathsf{i}$ is positive and strictly decreasing, so their product is strictly decreasing on $(0,1]$ and $\mathsf{R}$ is strictly increasing there, hence on $[0,1]$. A threshold would be a value $\mathsf{h}_{\mathrm{c}} > 0$ with $\mathsf{R} = 0$ for $\mathsf{h} \leq \mathsf{h}_{\mathrm{c}}$, or a discontinuity of $\mathsf{R}$; neither exists. For fixed $0 < \mathsf{h} < 1$, $\mathsf{R}$ is strictly increasing in $\mathsf{i}$, hence in $\mathsf{s}^2$. From $\hat{\mathsf{V}} = \Phi(\hat{\mathsf{V}}) \leq \mathsf{h}^2\hat{\mathsf{V}} + \mathsf{s}^2$ one gets $\hat{\mathsf{V}} \leq \mathsf{s}^2/(1-\mathsf{h}^2)$, so $\hat{\mathsf{V}} \to 0$, $\mathsf{i} \to 0$ and $\mathsf{R} \leq \mathsf{h}\mathsf{i}/(1-\mathsf{h}) \to 0$ as $\mathsf{s}^2 \to 0$. Finally, for $\mathsf{h} > 0$, $\mathsf{R} \geq \frac{1}{2}$ exactly when $(1-\mathsf{h})/(\mathsf{h}\mathsf{i}) \leq 1$, that is, when $\mathsf{h}\mathsf{i} \geq 1 - \mathsf{h}$; at $\mathsf{h} = 0$ both $\mathsf{R} \geq \frac{1}{2}$ and $\mathsf{h}\mathsf{i} \geq 1 - \mathsf{h}$ fail. □

The price of the variance that sustains the response is a load in mean fitness. At the fixed point

the mean fitness is

$$\int \mathsf{W}(y)\mathcal{N}(y;\hat{y},\hat{\mathsf{V}})\mathrm{d}y = \Big(\frac{\mathsf{v_s}}{\mathsf{v_s}+\hat{\mathsf{V}}}\Big)^{1/2}\exp\Big(-\frac{(\hat{y}-\mathsf{y}_*)^2}{2(\mathsf{v_s}+\hat{\mathsf{V}})}\Big), \tag{S179}$$

because $\mathsf{W}(y) = (2\pi\mathsf{v_s})^{1/2}\mathcal{N}(\mathsf{y}_*; y, \mathsf{v_s})$, so the integral is $(2\pi\mathsf{v_s})^{1/2}$ times the density at $\mathsf{y}_*$ of the sum of independent normal variables with laws $\mathcal{N}(\hat{y},\hat{\mathsf{V}})$ and $\mathcal{N}(0,\mathsf{v_s})$. The two bounds $\mathsf{s}^2 \le \hat{\mathsf{V}} \le \mathsf{s}^2/(1-\mathsf{h}^2)$ of the proof, the second for $\mathsf{h}<1$, also give the weak-selection limit: as $\mathsf{v_s}\to\infty$ with $\mathsf{h}<1$ fixed, $\mathsf{v_s}/(\hat{\mathsf{V}}+\mathsf{v_s})\to 1$ and $\hat{\mathsf{V}}\to\mathsf{s}^2/(1-\mathsf{h}^2)$, the neutral value of Proposition S24.5(b). At $\mathsf{h}=1$ the fixed point is $\hat{\mathsf{V}} = \frac{1}{2}[\mathsf{s}^2 + (\mathsf{s}^4+4\mathsf{s}^2\mathsf{v_s})^{1/2}]$ and $\hat{y}=\mathsf{y}_*$.

**Lemma S45.5** (Monotonicity in the entries)**.** *If* $\mathsf{Q}$ *and* $\mathsf{Q}'$ *are* $2\times 2$ *matrices with* $0\le\mathsf{Q}\le\mathsf{Q}'$*, then* $\rho(\mathsf{Q})\le\rho(\mathsf{Q}')$*. Consequently* $\mathsf{h}$ *is nondecreasing in the copy fidelity* $\mathsf{c}$*, and under Assumption S24.7 with* $\mathsf{s}^2$ *held fixed the retained fraction is nondecreasing in* $\mathsf{c}$*.*

*Proof.* Let $\chi_{\mathsf{Q}}(\lambda) = (\lambda-q_{11})(\lambda-q_{22})-q_{12}q_{21}$ and $\lambda' = \rho(\mathsf{Q}')$. By Lemma S45.1(i), $\lambda' \ge \max\{q'_{11},q'_{22}\} \ge \max\{q_{11},q_{22}\}$, so $0\le\lambda'-q'_{ii}\le\lambda'-q_{ii}$, and $q'_{12}q'_{21}\ge q_{12}q_{21}$; hence $\chi_{\mathsf{Q}}(\lambda')\ge\chi_{\mathsf{Q}'}(\lambda')=0$. The parabola $\chi_{\mathsf{Q}}$ is negative strictly between its roots and its larger root $\rho(\mathsf{Q})$ is at least its vertex $\frac{1}{2}(q_{11}+q_{22})$; since $\lambda'$ is at least the vertex and $\chi_{\mathsf{Q}}(\lambda')\ge 0$, $\lambda'\ge\rho(\mathsf{Q})$. By (S174) the entries $\mathsf{M}_{12}$ and $\mathsf{M}_{22}$ are proportional to $\mathsf{c}$ and the others do not depend on it, which gives the monotonicity of $\mathsf{h}$; the last claim follows from Proposition S24.8. □

The founder fraction has no such monotonicity: $\mathsf{M}_{11}$ and $\mathsf{M}_{21}$ increase with $\bar{\mathsf{b}}$ while $\mathsf{M}_{12}$ and $\mathsf{M}_{22}$ decrease when $\mathsf{fc}>0$, and Proposition S45.3 shows that either direction occurs.

**Proposition S45.6** (Directional selection)**.** *In Assumption S24.7 replace the fitness by* $\mathsf{W}(y)=e^{\mathsf{r}y}$ *with* $\mathsf{r}\in\mathbb{R}$*. Then* $\bar{y}' = \mathsf{h}(\bar{y}+\mathsf{rV})$ *and* $\mathsf{V}' = \mathsf{h}^2\mathsf{V}+\mathsf{s}^2$*. For* $\mathsf{h}<1$ *the recursion converges from every initial condition to*

$$\hat{\mathsf{V}} = \frac{\mathsf{s}^2}{1-\mathsf{h}^2}, \qquad \hat{y} = \frac{\mathsf{hrs}^2}{(1-\mathsf{h})^2(1+\mathsf{h})}. \tag{S180}$$

*For* $\mathsf{h}=1$ *and* $\mathsf{r}\ne 0$ *the variance and the absolute value of the mean grow without bound.*

*Proof.* The density proportional to $e^{\mathsf{r}y}\mathcal{N}(y;\bar{y},\mathsf{V})$ is $\mathcal{N}(y;\bar{y}+\mathsf{rV},\mathsf{V})$, by completing the square, and transmission multiplies the mean by $\mathsf{h}$ and maps the variance to $\mathsf{h}^2\mathsf{V}+\mathsf{s}^2$. For $\mathsf{h}<1$ the variance recursion is an affine contraction with fixed point $\mathsf{s}^2/(1-\mathsf{h}^2)$, and $\bar{y}_{k+1}=\mathsf{h}\bar{y}_k+\mathsf{hr}\mathsf{V}_k$ is an affine contraction with convergent forcing, so by the argument used for the mean in the proof of Proposition S24.8 it converges to $\mathsf{hr}\hat{\mathsf{V}}/(1-\mathsf{h})$. For $\mathsf{h}=1$, $\mathsf{V}_k=\mathsf{V}_0+k\mathsf{s}^2$ and $\bar{y}_k=\bar{y}_0+\mathsf{r}\sum_{j<k}\mathsf{V}_j$. □

With a fitness that is unbounded in the trait the stationary displacement again has the sign of $\mathsf{r}$ for every $\mathsf{h}>0$ and diverges as $\mathsf{h}\to 1$; a threshold is absent here too.

**Proposition S45.7** (A two-type model with a threshold)**.** *Let an infinite population of collectives consist of carriers of a convention and blanks. A carrier founds* $\mathsf{m}>1$ *daughters for each daughter of a blank, a daughter of a carrier is a carrier with probability* $\mathsf{q}\in(0,1]$ *and a blank otherwise, and every daughter of a blank is a blank. The frequency* $p$ *of carriers obeys*

$$p' = \frac{\mathsf{mq}p}{1+(\mathsf{m}-1)p}. \tag{S181}$$

*If* $\mathsf{mq}\le 1$ *then* $p_k\to 0$ *from every* $p_0\in[0,1]$*. If* $\mathsf{mq}>1$ *and* $p_0>0$ *then* $p_k\to\hat{p}=(\mathsf{mq}-1)/(\mathsf{m}-1)$*.*

| $\mathsf{T}$ | $\mathsf{h}$ | $\mathsf{h}e^{\mathsf{T}/\tau_{\mathrm{id}}}$ | $\mathsf{h}_0$ (no record) | blank record ($\mathsf{c}=0$) | external record |
|---|---|---|---|---|---|
| 1 | 0.9255 | 0.9410 | $3.6\times10^{-2}$ | $5.8\times10^{-3}$ | $4.9\times10^{-3}$ |
| 5 | 0.8657 | 0.9406 | $6.4\times10^{-4}$ | $1.0\times10^{-3}$ | $2.9\times10^{-8}$ |
| 20 | 0.6750 | 0.9406 | $1.7\times10^{-10}$ | $7.9\times10^{-4}$ | $7.2\times10^{-28}$ |
| 60 | 0.3476 | 0.9406 | $4.8\times10^{-28}$ | $4.1\times10^{-4}$ | $3.7\times10^{-80}$ |

Table S15: Heritability of the convention in Example S45.8 as a function of the generation time $\mathsf{T}$, in units of the residence time. The third column converges to 0.9406; the last three columns are the transmission coefficients without a record, with a record that is blank at birth, and with a record written only by non-members.

*Proof.* The daughters of carriers are a fraction $\mathsf{m}p/(\mathsf{m}p+1-p)$ of all daughters, and a fraction $\mathsf{q}$ of them are carriers, which is (S181). The map $\Psi(p)=\mathsf{mq}p/(1+(\mathsf{m}-1)p)$ is increasing on $[0,1]$ with values in $[0,\mathsf{q}]$, and $\Psi(p)/p=\mathsf{mq}/(1+(\mathsf{m}-1)p)$ is strictly decreasing in $p$. Its fixed points are 0 and the solution $\hat{p}$ of $1+(\mathsf{m}-1)p=\mathsf{mq}$, which lies in $(0,1]$ exactly when $\mathsf{mq}>1$. If $\mathsf{mq}\le1$, then $\Psi(p)<p$ for every $p>0$, so $p_k$ decreases to a fixed point in $[0,p_0]$, which must be 0. If $\mathsf{mq}>1$ and $0<p<\hat{p}$, then $\Psi(p)/p>\Psi(\hat{p})/\hat{p}=1$ and $\Psi(p)<\Psi(\hat{p})=\hat{p}$, so $p_k$ increases to a positive fixed point, which is $\hat{p}$; if $p>\hat{p}$ the inequalities are reversed and $p_k$ decreases to $\hat{p}$. □

This model lies outside the linear-Gaussian setting and is included for contrast with Proposition S24.8. The threshold $\mathsf{mq}=1$ exists because the advantage of a carrier is bounded and the blank state is absorbing. In the Gaussian model the loss $(1-\mathsf{h})y$ and the selection differential are both linear in the displacement and the transmission noise restores variance in every generation, so selection always retains a positive fraction. A retention probability $\mathsf{q}$ would have to be derived from a bistable convention dynamics of the kind conjectured in Conjecture S5.22, which is Open problem S24.13.

**Example S45.8** (A durable record with naive recruits)**.** Take the residence time as the unit of time, $g=1$, and let $\varepsilon=0.01$, $\beta=2$, $\mathsf{f}=0$, $\omega=0.05$, $\delta=0$, $\mathsf{c}=0.95$, $n=20$ and $\mathsf{b}=2$. Then $\tilde{\varepsilon}=1.01$, $\tilde{\beta}=2$, $\tau_{\mathrm{id}}=60.27$ against $1/\tilde{\varepsilon}=0.990$ without a record, $\mathsf{u}=0.9890$, $\mathsf{k}=1$ and the limit of Theorem S24.3(a) is $(1-\mathsf{u})\bar{\mathsf{b}}+\mathsf{uc}=0.9406$. Table S15 lists the heritability at four generation times, together with the value $\mathsf{h}_0$ without a record, the value $\bar{\mathsf{b}}\mathsf{G}_{11}(\mathsf{T})$ when the record is blank at birth, and the coefficient $\bar{\mathsf{b}}e^{-\mathsf{p}_1\mathsf{T}}$ of Theorem S24.3(e) for a record written by non-members. By Corollary S24.4, $\mathsf{h}\ge\frac{1}{2}$ for every generation time up to $\tau_{\mathrm{id}}\log(2\mathsf{cu})=38.0$ residence times, whereas $\mathsf{h}_0\le\bar{\mathsf{b}}=0.1$ at every generation time. Naive recruits and two founders out of twenty leave almost nothing to inherit through the members, and the copied record carries the convention across the fission.

*Remark* S45.9 (Comparison with tradeoff breaking). Bourrat et al. (2022) examine the description of an evolutionary transition in individuality as a decoupling of the fitness of collectives from the fitness of their particles. Their result is that when both fitnesses are measured over commensurable time intervals, in a population whose distribution of collective sizes has stabilized, the two coincide, so the fitnesses cannot decouple. They propose to describe transitions through traits and the tradeoffs among them, with the breaking of a tradeoff as a marker of a transition when it is observed in the context of one; they state that tradeoff breaking can occur outside transitions and that such an event taken alone does not identify a transition. Three points of contact with the fission model are exact. First, the model has no particle-level fitness: members are recruited and replaced and are not born of members, so the comparison of the two fitnesses cannot be made inside it; if one counts

occupied positions instead, there are $n$ of them in every collective at every census, their number grows at the rate of the number of collectives, and the agreement of the two rates holds trivially and tests nothing. Second, the construction is trait-based in the sense those authors recommend: the fitness $\mathsf{W}$ is a declared function of a collective trait, and Propositions S24.6 and S24.8 follow the trait. Third, the model has a single trait, so it contains no tradeoff and cannot express the breaking of one. The following suggestion is interpretive and is not a result. The manuscript does contain two collective quantities governed by the same parameter, the read rate $\beta$, one of which can improve while the other cannot. Differentiating $\chi(\varpi_-) = 0$ in (S171) with respect to $\beta$, with $\partial\chi/\partial\beta = \delta - s$ and $\chi'(\varpi_-) = -(\varpi_+ - \varpi_-)$, gives $\partial\varpi_-/\partial\beta = (\delta - \varpi_-)/(\varpi_+ - \varpi_-)$, which is negative exactly when $\delta < \tilde{\varepsilon}$ by Lemma S45.2(v). When the record decays more slowly than the members forget, reading a member-written record more strongly therefore lengthens the identity time of a convention (compare Corollary 12.5) and lowers the exponential rate at which its heritability decays with the generation time in (S105). The heritability at a fixed generation time need not increase with $\beta$: for a record that is blank at birth, $\mathsf{h} = \bar{\mathsf{b}}\mathsf{G}_{11}(\mathsf{T}) = \bar{\mathsf{b}}(1 - \mathsf{p}_1\mathsf{T}) + O(\mathsf{T}^2)$ decreases with $\beta$ at short generation times (Check S47.216). For a grounded coordinate the common-mode factor that bounds every member's stationary error from below is nondecreasing in $\beta$ (Theorem 12.7(c), where $\beta$ is the per-round read gain of the discrete-time record of Lemma 12.6 and $\beta, \omega > 0$), so stronger reading never lowers that floor. Whether fitness functions of these two quantities define a tradeoff in the sense of those authors, and whether a change of organization can break it, is Open problem S24.14.

## S46 Proofs for Section S25

Throughout, $e_i$ is the $i$th coordinate vector, $z = e^{\mathrm{i}\theta}$ is a point of the unit circle, $M^*$ is the conjugate transpose, and for a stationary process $\Gamma_h = \mathrm{Cov}(x_{t+h}, x_t)$. All processes are zero-mean Gaussian, so that independence is the vanishing of covariances and every conditional mutual information is one half of a log ratio of conditional variance determinants.

*Proof of Proposition S25.2.* Given $\mathcal{X}_{\mathrm{sh}} = (x_0, \mathsf{d})$ the recursion Eq. (S111) reads $x_{t+1} = Ax_t + c_t + \zeta_t$ with $c_t = \mathsf{L}\mathsf{d}_t$ known, so the conditional law of the path is Gaussian with a known mean and with the covariance of the process $\tilde{x}_{t+1} = A\tilde{x}_t + \zeta_t$, $\tilde{x}_0 = 0$. If $A$ is diagonal, $\tilde{x}_i$ is a function of $(\zeta_i(s))_s$ alone; these families are independent across $i$ because $D$ is diagonal, so every pairwise conditional mutual information in Eq. (63) vanishes. Conversely, $\tilde{x}_1 = \zeta_0$ and $\tilde{x}_2 = A\zeta_0 + \zeta_1$ give

$$\mathrm{Cov}(x_2(i), x_1(j) \mid \mathcal{X}_{\mathrm{sh}}) = A_{ij}D_{jj}. \tag{S182}$$

If $A_{ij} \neq 0$ for some $i \neq j$ this is nonzero, the paths of $i$ and $j$ over $[0{:}\bar{t}]$ with $\bar{t} \geq 2$ are conditionally dependent, and the corresponding term of Eq. (63) is positive. For $\bar{t} = 1$ the conditional covariance of $x_1$ is $D$, which is diagonal, so the restriction $\bar{t} \geq 2$ cannot be dropped. □

*Proof of Theorem S25.3.* The stationary covariance of Eq. (S111) with a white driver of unit covariance solves $\Sigma = A\Sigma A^\top + \mathsf{L}\mathsf{L}^\top + D$, and the solution is unique when $\rho(A) < 1$.

(a) With $A = aI$ and $D = \delta I$ the equation is $(1 - a^2)\Sigma = \mathsf{L}\mathsf{L}^\top + \delta I$. The matrix $(1 - a^2)\Sigma - \delta I$ is positive semidefinite exactly when $\delta \leq (1 - a^2)\lambda_{\min}(\Sigma)$, and any $\mathsf{L}$ with $\mathsf{L}\mathsf{L}^\top$ equal to it gives the claim. At $\delta = (1 - a^2)\lambda_{\min}(\Sigma)$ the matrix $\Sigma - \lambda_{\min}(\Sigma)I$ is singular, so the rank of $\mathsf{L}\mathsf{L}^\top$ is at most $n - 1$.

(b) A driver-free model has stationary covariance $\Sigma$ exactly when $\Sigma = A\Sigma A^\top + D$. Put $N = \Sigma^{-1/2}A\Sigma^{1/2}$. The equation is $NN^\top = I - \Sigma^{-1/2}D\Sigma^{-1/2}$. A solution requires the right side to be positive semidefinite, that is $D \preceq \Sigma$, and then it has a positive semidefinite square root $R$, with $\|R\|_2 < 1$ because $D \succ 0$. By the polar decomposition, $NN^\top = R^2$ holds exactly when $N = RU$ for

some orthogonal $U$: every square $N$ factors as $(NN^\top)^{1/2}U$, and conversely $RU(RU)^\top = R^2$. This is Eq. (S112). For stability, $A$ is similar to $N = RU$ and $\|RU\|_2 = \|R\|_2 < 1$, so $\rho(A) < 1$ for every admissible $D$ and $U$. If $A$ and $D$ are both diagonal, the off-diagonal entries of $\Sigma = A\Sigma A^\top + D$ satisfy $\Sigma_{ij}(1 - A_{ii}A_{jj}) = 0$ with $|A_{ii}A_{jj}| < 1$, hence $\Sigma$ is diagonal; so $A$ is not diagonal when $\Sigma$ is not.

(c) With $\Sigma = I$ and $D = \delta I$, Eq. (S112) gives $A = \sqrt{1-\delta}U$, and $AA^\top + \delta I = I$ directly. □

*Remark* S46.1 (Bounded driver rank). For an interaction-free model with diagonal $A = \mathrm{diag}(a_i)$ and a white driver of rank $k$ and unit covariance, the stationary covariance has $\Sigma_{ij} = (\mathsf{L}\mathsf{L}^\top)_{ij}/(1 - a_i a_j)$ for $i \neq j$ and $\Sigma_{ii} = ((\mathsf{L}\mathsf{L}^\top)_{ii} + D_{ii})/(1 - a_i^2)$. With equal $a_i$ and driver rank at most $k$ the realizable covariances are the factor-analytic set $\{D' + L' : D' \succ 0 \text{ diagonal}, L' \succeq 0, \mathrm{rank}\, L' \leq k\}$; for $k = 1$ and $n \geq 4$ its members satisfy $\Sigma_{ij}\Sigma_{lm} = \Sigma_{il}\Sigma_{jm}$ for distinct indices. With equal memories a rank bound therefore makes the interaction-free hypothesis refutable from the covariance, while a covariance inside the set remains compatible with interaction by Theorem S25.3(b).

With unequal memories the relations among four entries fail in general: for $k = 1$ and distinct indices,

$$\frac{\Sigma_{ij}\Sigma_{lm}}{\Sigma_{il}\Sigma_{jm}} = \frac{(1 - a_i a_l)(1 - a_j a_m)}{(1 - a_i a_j)(1 - a_l a_m)}, \tag{S183}$$

which equals one only for special memories. The hypothesis of an interaction-free model with a white driver of rank one is still refutable from the covariance when $n \geq 5$. For $|\tau| < 1$ the substitution $a_i \mapsto a_i' = (a_i + \tau)/(1 + \tau a_i)$, $\mathsf{L}_i \mapsto \mathsf{L}_i' = \mathsf{L}_i\sqrt{1-\tau^2}/(1 + \tau a_i)$ maps $(-1, 1)$ to itself and leaves every $\mathsf{L}_i\mathsf{L}_j/(1 - a_i a_j)$ unchanged, the case $i = j$ included, because $1 - a_i' a_j' = (1 - \tau^2)(1 - a_i a_j)/((1 + \tau a_i)(1 + \tau a_j))$. Taking $\tau = -a_1$ sets one memory to zero, so the off-diagonal entries of the realizable covariances form the image of a continuously differentiable map of the $2n - 1$ parameters $(a_2, \ldots, a_n, \mathsf{L}_1, \ldots, \mathsf{L}_n)$ into $\mathbb{R}^{n(n-1)/2}$. For $n \geq 5$ one has $2n - 1 < n(n-1)/2$, and the image of a continuously differentiable map from a space of smaller dimension has Lebesgue measure zero, so almost every covariance refutes the hypothesis. For $n = 4$ there are six off-diagonal entries and seven parameters, and a numerical Jacobian of the map has rank six at a random point (Check S47.229), so no equality constraint is indicated there; the case $n = 4$ and the case of rank $k \geq 2$ with unequal memories are not settled here.

*Proof of Proposition S25.4.* Since $|\phi| < 1$ the driver has the stationary solution $\mathsf{d}_t = \sum_{j \geq 0} \phi^j \omega_{t-1-j}$, in which $\mathsf{d}_0$ is a function of the $\omega_s$ with $s < 0$ and is independent of $\zeta$ and of the $\omega_s$ with $s \geq 0$. The pair $(x, \mathsf{d})$ obeys $x_{t+1} = Ax_t + \mathsf{L}\mathsf{d}_t + \zeta_t$, $\mathsf{d}_{t+1} = \phi\mathsf{d}_t + \omega_t$. This is the recursion Eq. (S111) for the $n + 1$ states $(x, \mathsf{d})$ with coupling matrix $\left(\begin{smallmatrix} A & \mathsf{L} \\ 0 & \phi \end{smallmatrix}\right)$, no driver, and independent inputs $(\zeta, \omega)$: the last agent has a zero row outside its diagonal, so it hears nobody, and its column $\mathsf{L}$ is its weight in the members' updates. The joint law is the same under both descriptions because the equations and the input laws are the same. Under the first classification $\omega$, and with it the path of $\mathsf{d}$, belongs to $\mathcal{X}_{\mathrm{sh}}$, and Proposition S25.2 gives $\Omega_{\mathrm{int}}(C) = 0$ since $A$ is diagonal. Under the second, $\mathcal{X}_{\mathrm{sh}}$ contains the initial states $(x_0, \mathsf{d}_0)$ and not $\omega$. Then $\mathsf{d}_1 = \phi\mathsf{d}_0 + \omega_0$ and $x_i(2) = A_{ii}x_i(1) + \mathsf{L}_i\mathsf{d}_1 + \zeta_i(1)$, so

$$\mathrm{Cov}(x_1(2), x_2(2) \mid \mathcal{X}_{\mathrm{sh}}) = \mathsf{L}_1\mathsf{L}_2 \,\mathrm{Var}\,\omega_0 \neq 0, \tag{S184}$$

because $x_1(1)$, $x_2(1)$, $\zeta_1(1)$, $\zeta_2(1)$ are conditionally independent of each other and of $\omega_0$. The paths of members 1 and 2 are jointly Gaussian and conditionally correlated for $\bar{t} \geq 2$, so their conditional mutual information is positive, and $\Omega_{\mathrm{int}}(C)$, an average of nonnegative pairwise terms (Eq. (63)), is positive. The member-restricted order parameter under the second classification adds $\omega$, the private input of the only non-member, to the conditioning set; with $\mathsf{d}_0$ it fixes the driver path, and the

argument of Proposition S25.2 gives zero. Under the first classification the population is $C$ and there are no non-members. □

For the path law we write $\mathcal{H}_t(x)$ for the closed linear span of the coordinates of $x_s$, $s \le t$, and use the two-sided stationary solutions. The spectral density of a stationary process with summable autocovariances is $f(z) = \sum_h \Gamma_h z^h$ on $|z| = 1$; for the driver-free model it is $f(z) = (I - Az)^{-1} D (I - Az)^{-*}$.

**Lemma S46.2** (Innovations of the recorded process under a driver)**.** *In an interaction-free or interacting model Eq. (S111) with coupling $A_0$, private covariance $D_0$ and a regular driver, the one-step prediction error covariance of $x_{t+1}$ given $\mathcal{H}_t(x)$ is $D_0 + \mathsf{L} R \mathsf{L}^\top$ with $R \succeq Q_{\mathsf{d}}$.*

*Proof.* Put $y_t = x_{t+1} - A_0 x_t = \mathsf{L}\mathsf{d}_t + \zeta_t$. Each $y_s$ with $s \le t-1$ lies in $\mathcal{H}_t(x)$, and $x_t = \sum_{j \ge 0} A_0^j y_{t-1-j}$ converges in mean square because $\rho(A_0) < 1$, so $\mathcal{H}_t(x) = \mathcal{H}_{t-1}(y)$. Since $A_0 x_t \in \mathcal{H}_t(x)$, the prediction error of $x_{t+1}$ is that of $y_t$ given $\mathcal{H}_{t-1}(y)$. The space $\mathcal{H}_{t-1}(y)$ is contained in $\mathcal{K} = \mathcal{H}_{t-1}(\mathsf{d}) \oplus \mathcal{H}_{t-1}(\zeta)$, the sum being orthogonal because $\mathsf{d}$ and $\zeta$ are independent. The variable $\zeta_t$ is orthogonal to $\mathcal{K}$ and to $\mathsf{d}_t$, so the error equals $\zeta_t + \mathsf{L}(\mathsf{d}_t - \mathbb{E}[\mathsf{d}_t \mid \mathcal{H}_{t-1}(y)])$ with orthogonal summands, and its covariance is $D_0 + \mathsf{L} R \mathsf{L}^\top$ with $R$ the covariance of the second bracket. Projecting on the smaller space gives the larger error, and $\mathsf{d}_t$ is orthogonal to $\mathcal{H}_{t-1}(\zeta)$, hence $R \succeq \mathrm{Cov}(\mathsf{d}_t - \mathbb{E}[\mathsf{d}_t \mid \mathcal{K}]) = \mathrm{Cov}(\mathsf{d}_t - \mathbb{E}[\mathsf{d}_t \mid \mathcal{H}_{t-1}(\mathsf{d})]) = Q_{\mathsf{d}}$. □

*Proof of Theorem S25.5.* (a) Coordinate range. In the driver-free model $x_{t+1} - A x_t = \zeta_t$ is independent of $\mathcal{H}_t(x)$, so the prediction error covariance is $D$. The prediction error covariance is a functional of the path law, so Lemma S46.2 gives $D - D_0 = \mathsf{L} R \mathsf{L}^\top$ with $R \succeq Q_{\mathsf{d}} \succ 0$. Because $\mathsf{L}$ has full column rank $k$, the right side is positive semidefinite of rank $k$ with range equal to the range of $\mathsf{L}$. The left side is diagonal, and the range of a diagonal matrix is spanned by the $e_i$ with nonzero diagonal entry. Hence the range of $\mathsf{L}$ is $\mathbb{R}^T = \mathrm{span}\{e_i : i \in T\}$ with $|T| = k$.

Support. In the interaction-free model $y_t = x_{t+1} - A_0 x_t$ is a filter of $x$ with transfer function $z^{-1}(I - A_0 z)$, so its spectral density is $(I - A_0 z) f(z) (I - A_0 z)^* = G D G^*$ with

$$G(z) = (I - A_0 z)(I - Az)^{-1}. \tag{S185}$$

Since $y = \mathsf{L}\mathsf{d} + \zeta$ with independent summands, the spectral measure of $\mathsf{L}\mathsf{d}$ is that of $y$ minus that of $\zeta$; it is therefore absolutely continuous with density

$$f_{\mathsf{L}\mathsf{d}}(z) = G(z) D G(z)^* - D_0 \succeq 0, \qquad |z| = 1, \tag{S186}$$

and, because $\mathsf{L}\mathsf{d}_t \in \mathbb{R}^T$, its rows and columns outside $T$ vanish. Let $P$ be the coordinate projector on $T^c$ and $\bar{P} = I - P$. From $P(GDG^* - D_0) = 0$,

$$PG(z)D = PD_0 \big(G(z)^{-1}\big)^* = PD_0 (I - A_0 z^{-1})^{-1} (I - A^\top z^{-1}), \qquad |z| = 1, \tag{S187}$$

where we used $\bar{z} = z^{-1}$ on the circle and that the matrices are real. The left side is analytic on a neighborhood of $\{|z| \le 1\}$ because $\rho(A) < 1$; the right side is analytic on a neighborhood of $\{|z| \ge 1\}$ because the entries of $A_0$ have modulus below one, and it tends to $PD_0$ as $z \to \infty$. The two agree on the circle, so by the identity theorem they are restrictions of one entire function, which is bounded and therefore constant by Liouville's theorem. Its value at $z = 0$ is $PD$ and its value at infinity is $PD_0$. Hence $PD = PD_0$ and $PG(z) = P$ identically, that is $P(I - A_0 z) = P(I - Az)$ and $PA = PA_0$: the rows of $A$ outside $T$ are diagonal and agree with $A_0$, and $D = D_0$ outside $T$. Next, $\bar{P}(GDG^* - D_0)P = 0$ and $\bar{P} D_0 P = 0$ give $\bar{P} G D G^* P = 0$; with $G^* P = (PG)^* = P$ and $D$ diagonal and invertible this is $\bar{P} G(z) P = 0$. So $G$ is block diagonal for the splitting $(T, T^c)$, with identity

block on $T^c$, and so is $I - Az = G(z)^{-1}(I - A_0 z)$. Hence $A_{T,T^c} = 0$, every off-diagonal entry of $A$ lies in $T \times T$, and $T \supseteq \mathcal{E}(A)$.

(b) Order the agents so that $A = \operatorname{diag}(A_T, \Lambda)$ with $\Lambda$ diagonal, which is possible because $T \supseteq \mathcal{E}(A)$, and let $A_0 = \operatorname{diag}(A_{0,T}, \Lambda)$ with $A_{0,T}$ diagonal and stable. Then $G = \operatorname{diag}(G_T, I)$ with $G_T(z) = (I - A_{0,T}z)(I - A_T z)^{-1}$, which is continuous and invertible on the circle. Put $c = \min_{|z|=1} \lambda_{\min}(G_T D_T G_T^*) > 0$, take $D_0 = \operatorname{diag}(D_{0,T}, D_{T^c})$ with $0 \prec D_{0,T} \prec cI$, and define $f_{\mathsf{d}} = G_T D_T G_T^* - D_{0,T}$. It is continuous, Hermitian, satisfies $f_{\mathsf{d}}(\bar{z}) = \overline{f_{\mathsf{d}}(z)}$, and $f_{\mathsf{d}} \succeq \varepsilon I$ with $\varepsilon = c - \|D_{0,T}\|_2 > 0$. Its Fourier coefficients $\Gamma_h^{\mathsf{d}}$ are real with $\Gamma_{-h}^{\mathsf{d}} = \Gamma_h^{\mathsf{d}\top}$, and for finitely many vectors $v_t$ the quadratic form $\sum_{s,t} v_s^\top \Gamma_{s-t}^{\mathsf{d}} v_t$ equals $(2\pi)^{-1} \int v(\theta)^* f_{\mathsf{d}} v(\theta) d\theta \geq 0$ for the trigonometric polynomial $v(\theta)$ built from the $v_t$. By Kolmogorov's extension theorem there is a stationary Gaussian process $\mathsf{d}$ in $\mathbb{R}^{|T|}$ with these autocovariances, which we take independent of $\zeta$. It is regular: $f_{\mathsf{d}} - \varepsilon I$ is again a spectral density, so $\mathsf{d}$ has the law of $\mathsf{d}' + \epsilon$ with $\epsilon$ white of covariance $\varepsilon I$ and independent of $\mathsf{d}'$, and the argument of Lemma S46.2 applied to this sum shows $Q_{\mathsf{d}} \succeq \varepsilon I$. With $\mathsf{L}$ the inclusion of $\mathbb{R}^T$, the process $y = \mathsf{L}\mathsf{d} + \zeta$ has spectral density $\operatorname{diag}(f_{\mathsf{d}} + D_{0,T}, D_{T^c}) = GDG^*$, and $x_t = \sum_{j \geq 0} A_0^j y_{t-1-j}$ has spectral density $(I - A_0 z)^{-1} GDG^* (I - A_0 z)^{-*} = (I - Az)^{-1} D (I - Az)^{-*}$. Two zero-mean stationary Gaussian processes with the same spectral density have the same law.

(c) Both claims are contrapositives of (a). A vector with two nonzero entries does not span a coordinate subspace. □

*Remark* S46.3 (Drivers of full rank). With $T$ the whole collective, Theorem S25.5(b) says that every driver-free law is the law of an interaction-free model with a regular driver of rank $n$. The same holds beyond first order: if a stationary Gaussian process has a continuous spectral density $f \succeq cI$, then for $0 < \delta < c$ it has the law of $\zeta_t + \mathsf{d}_t$ with $\zeta$ white of covariance $\delta I$ and an independent regular driver of density $f - \delta I$, which is the interaction-free model with $A = 0$ after a shift of one round; the driver is built from the density $f - \delta I \succeq (c - \delta) I$ by the Kolmogorov extension and regularity argument in the proof of Theorem S25.5(b), which uses only continuity, the symmetry $f(\bar{z}) = \overline{f(z)}$ of a real process and the lower bound. Without a restriction on the driver, a recorded path law whose spectral density is continuous and bounded below never refutes the absence of interaction, even when every member is recorded.

*Proof of Proposition S25.6.* (a) With a white driver of unit covariance, $u_t = \mathsf{L}\mathsf{d}_t + \zeta_t$ is independent across rounds with covariance $Q = D + \mathsf{L}\mathsf{L}^\top$, and $u_t$ is independent of $x_s$ for $s \leq t$. Hence $\Gamma_1 = \operatorname{Cov}(Ax_t + u_t, x_t) = A\Gamma_0$ and $\Gamma_0 = A\Gamma_0 A^\top + Q$. Since $\Gamma_0 \succeq Q \succeq D \succ 0$, $\Gamma_0$ is invertible, $A = \Gamma_1 \Gamma_0^{-1}$ and $Q = \Gamma_0 - A\Gamma_0 A^\top$. By Proposition S25.2 the event $\Omega_{\text{int}}(C) > 0$ is the event that $A$ is not diagonal.

(b) For $k = 1$ the off-diagonal entries of $Q$ are $Q_{ij} = \mathsf{L}_i \mathsf{L}_j$. With at least three nonzero loadings, $\mathsf{L}_i \neq 0$ exactly when $Q_{ij} \neq 0$ for some $j \neq i$, so the set of loaded agents is identified. For a loaded agent $i$ and two further loaded agents $j \neq l$, $Q_{ij} Q_{il} / Q_{jl} = \mathsf{L}_i^2$; for an agent that is not loaded $\mathsf{L}_i^2 = 0$. Then $D_{ii} = Q_{ii} - \mathsf{L}_i^2$. By the proof of Proposition S25.2, the conditional law of the path given $\mathcal{X}_{\text{sh}}$ has a covariance that depends on $(A, D)$ and $\bar{t}$ only, and $\Omega_{\text{int}}(C)$ is a function of that covariance.

(c) Conditionally on $\mathcal{X}_{\text{sh}}$, the centered path satisfies $\tilde{x}_1 = \zeta_0$ and $\tilde{x}_2 = A\zeta_0 + \zeta_1$, so for $i \neq j$

$$\operatorname{Corr}(x_2(i), x_1(j) \mid \mathcal{X}_{\text{sh}})^2 = \frac{A_{ij}^2 D_{jj}}{A_{ij}^2 D_{jj} + \sum_{l \neq j} A_{il}^2 D_{ll} + D_{ii}}. \tag{S188}$$

The right side is nondecreasing in $D_{jj}$ and nonincreasing in the other diagonal entries of $D$. Since $Q - D = \mathsf{L}\mathsf{L}^\top$ has a nonnegative diagonal, $D_{ll} \leq Q_{ll}$, and $D_{jj} \geq d_j^-$ by hypothesis, so Eq. (S188) is at least $\varrho_{ij}^2$. The conditional covariance does not depend on the value of $\mathcal{X}_{\text{sh}}$, so the conditional mutual information of the two scalars is $-\frac{1}{2}\log(1 - \text{Corr}^2)$. For $\bar{t} \geq 2$ the scalar $x_2(i)$ is a coordinate

of the path of $i$ and $x_1(j)$ a coordinate of the path of $j$, and mutual information does not increase under coordinate projections, so

$$I(\mu_i[0{:}\bar{t}]; \mu_j[0{:}\bar{t}] \mid \mathcal{X}_{\mathrm{sh}}) \geq -\tfrac{1}{2}\log(1-\varrho_{ij}^2). \tag{S189}$$

The same holds with $i$ and $j$ exchanged, which gives the maximum, and averaging over pairs gives Eq. (S113). Under $(\mathsf{L}\mathsf{L}^\top)_{ii} \leq v_i$ one has $D_{ii} = Q_{ii} - (\mathsf{L}\mathsf{L}^\top)_{ii} \geq Q_{ii} - v_i$, and $A$, $Q$ are identified by (a), so the bound is a function of the recorded law and of the $v_i$. □

*Remark* S46.4 (The bound is not sharp). The estimate Eq. (S113) uses one pair of scalars per pair of agents and the worst case of each variance separately. In Check S47.226 it is 0.0055 against a minimum of 0.018 over the admissible box. The exact range of $\Omega_{\mathrm{int}}(C)$ under rank and variance bounds is Open problem S25.15.

*Proof of Proposition S25.7.* Let $\rho(\Phi) < 1$ and let $\omega$ be white with covariance $Q_\omega$. The stationary driver has spectral density $(I-\Phi z)^{-1}Q_\omega(I-\Phi z)^{-*}$, the input $\mathsf{L}\mathsf{d}+\zeta$ has density $D + \mathsf{L}(I-\Phi z)^{-1}Q_\omega(I-\Phi z)^{-*}\mathsf{L}^\top$ by independence, and

$$f(z) = (I-Az)^{-1}\Big[D + \mathsf{L}(I-\Phi z)^{-1}Q_\omega(I-\Phi z)^{-*}\mathsf{L}^\top\Big](I-Az)^{-*}. \tag{S190}$$

On the circle $(I-Mz)^{-*} = (I-M^\top z^{-1})^{-1}$ for a real matrix $M$, so $f$ extends to a rational function whose poles lie where one of $I-Az$, $I-\Phi z$, $zI-A^\top$, $zI-\Phi^\top$ is singular, that is in the reciprocals of the nonzero eigenvalues of $A$ and $\Phi$ and in the eigenvalues themselves. The reflection of $1/\lambda$ in the unit circle is $\bar{\lambda}$, and the spectrum of a real matrix is closed under conjugation.

For $n = k = 1$, Eq. (S190) is

$$f(z) = \frac{D|1-\phi z|^2 + \beta^2}{|1-az|^2|1-\phi z|^2} = \frac{D(1+\phi^2)+\beta^2 - 2D\phi\cos\theta}{|1-az|^2|1-\phi z|^2}. \tag{S191}$$

The denominator is symmetric in $(a,\phi)$. The model with memory $\phi$, driver pole $a$, private variance $D'$ and driver innovation variance $\beta'^2$ has numerator $D'(1+a^2)+\beta'^2 - 2D'a\cos\theta$. The two numerators agree for all $\theta$ exactly when $D'a = D\phi$ and $D'(1+a^2)+\beta'^2 = D(1+\phi^2)+\beta^2$, that is

$$D' = \frac{D\phi}{a}, \qquad \beta'^2 = \beta^2 + \frac{D}{a}(a(1+\phi^2) - \phi(1+a^2)) = \beta^2 + \frac{D(a-\phi)(1-a\phi)}{a}. \tag{S192}$$

For $0 < \phi < a < 1$ these satisfy $0 < D' < D$ and $\beta'^2 > \beta^2 \geq 0$, so the second parameter point is admissible and differs from the first. Both models are stable, and zero-mean stationary Gaussian processes with the same spectral density have the same law. □

*Proof of Proposition S25.8.* Within an epoch the driver equals a constant vector $\mathsf{d}$, and given $\mathsf{d}$ the stationary law of Eq. (S111) has mean $(I-A)^{-1}\mathsf{L}\mathsf{d} = M\mathsf{d}$, where $I-A$ is invertible because $\rho(A) < 1$, and covariance $\Sigma^{\mathrm{loc}}$. A snapshot is therefore $x = M\mathsf{d} + x^{\mathrm{loc}}$ with $x^{\mathrm{loc}} \sim \mathcal{N}(0, \Sigma^{\mathrm{loc}})$ independent of $\mathsf{d} \sim \mathcal{N}(0, I_k)$. The joint density of $(x, \mathsf{d})$ is proportional to $\exp(-\frac{1}{2}(x-M\mathsf{d})^\top \Pi^{\mathrm{loc}}(x-M\mathsf{d}) - \frac{1}{2}\mathsf{d}^\top\mathsf{d})$, so the joint concentration matrix has blocks

$$K_O = \Pi^{\mathrm{loc}}, \qquad K_{O,H} = -\Pi^{\mathrm{loc}}M, \qquad K_H = I + M^\top\Pi^{\mathrm{loc}}M, \tag{S193}$$

for the observed block $O$ and the hidden block $H$. The concentration of the marginal law of $x$ is the Schur complement $K_O - K_{O,H}K_H^{-1}K_{H,O} = \Pi^{\mathrm{loc}} - \mathsf{K}$, which is also the Woodbury form of

$(\Sigma^{\mathrm{loc}} + MM^\top)^{-1}$. The matrix $\mathsf{K}$ is positive semidefinite because $K_H \succ 0$, and its rank is that of $\Pi^{\mathrm{loc}} M$, which is $k$ because $\Pi^{\mathrm{loc}}$ and $I - A$ are invertible and $\mathsf{L}$ has full column rank.

If $A$ is diagonal, the off-diagonal entries of $\Sigma^{\mathrm{loc}} = A\Sigma^{\mathrm{loc}}A^\top + D$ satisfy $\Sigma^{\mathrm{loc}}_{ij}(1 - A_{ii}A_{jj}) = 0$, so $\Sigma^{\mathrm{loc}}$ and $\Pi^{\mathrm{loc}}$ are diagonal. Hence a nondiagonal $\Pi^{\mathrm{loc}}$ implies a nondiagonal $A$, and $\Omega_{\mathrm{int}}(C) > 0$ by Proposition S25.2, which holds for any driver path contained in $\mathcal{X}_{\mathrm{sh}}$. For the converse take $A = \sqrt{1-\delta}U$ with $U$ orthogonal and not diagonal and $D = \delta I$: by Theorem S25.3(c), $\Sigma^{\mathrm{loc}} = \Pi^{\mathrm{loc}} = I$ while $A$ is not diagonal. □

*Remark* S46.5 (Scope of the sparse and low-rank result). Equation (S193) is the setting of Chandrasekaran et al. (2012): a jointly Gaussian vector of observed and hidden variables whose observed marginal concentration is a matrix $S^* = K_O$ minus a positive semidefinite matrix $L^* = K_{O,H}K_H^{-1}K_{H,O}$ of rank equal to the number of hidden variables. In their notation, $\Omega$ is the tangent space at $S^*$ to the matrices with its support, $T$ the tangent space at $L^*$ to the matrices of its rank, $\mu(\Omega)$ is bounded by the largest number of nonzeros in a row of $S^*$, and $\xi(T)$ is small when the row space of $L^*$ is incoherent with the coordinate axes. Transversality of $\Omega$ and $T$, meaning that they meet only in the origin, is necessary and sufficient for identifiability with respect to the tangent spaces and sufficient for local identifiability with respect to the two varieties, and a sufficiently small product $\mu(\Omega)\xi(T)$ guarantees it. Their Theorem 4.1 assumes the bound on $\mu(\Omega)\xi(T)$ of their Proposition 3.3, which is phrased through the Fisher information at the true model, and a trade-off parameter in the range given there. In the present notation, with $n$ observed variables and $N$ independent samples, it requires $N$ at least of order $n/\xi(T)^4$, a regularization parameter of order $\xi(T)^{-1}\sqrt{n/N}$, a smallest nonzero singular value of $L^*$ at least of order $\xi(T)^{-3}\sqrt{n/N}$ and a smallest nonzero entry of $S^*$ at least of order $(\xi(T)\mu(\Omega))^{-1}\sqrt{n/N}$. Its conclusion is that a regularized maximum-likelihood convex program returns the sign pattern of $S^*$ and the rank of $L^*$ with probability greater than $1 - 2e^{-n}$. The constants of these conditions are not evaluated for the present model. Three limits bear on its use here. The theorem concerns independent Gaussian samples, so it applies to snapshots across epochs and not to a path within one. Its incoherence condition asks that the driver load diffusely, the regime in which Theorem S25.5(c) also separates the driver from interaction. Its sparse term is $\Pi^{\mathrm{loc}}$, whose support is not the support of $A$: for a normal network with $D = qI$ one has $\Pi^{\mathrm{loc}} = (I - AA^\top)/q$ as in Theorem 4.9, supported on pairs with a common source, and for $A = \sqrt{1-\delta}U$ it is the identity. Recovery of $S^*$ therefore certifies interaction when it returns an off-diagonal entry and is silent otherwise.

*Proof of Theorem S25.9.* In either regime the stationary solution is $x_t = x_t^{\mathrm{com}} + x_t^{\mathrm{loc}}$ with

$$x_t^{\mathrm{com}} = \sum_{j\geq 0} A^j \mathsf{L}\mathsf{d}_{t-1-j}, \qquad x_t^{\mathrm{loc}} = \sum_{j\geq 0} A^j \zeta_{t-1-j}, \tag{S194}$$

and the two parts are independent because the driver is independent of the private inputs, which under the intervention include the injected symbols by (I1).

(a) Let $A$ be diagonal. By (I3) the intervention leaves $(A, D)$ unchanged, by (I2) it leaves $\mathsf{L}$ unchanged, and by (I1) the driver has the same law and remains independent of the private inputs. The law of $x$ is a function of $(A, D, \mathsf{L})$ and of the driver law through Eq. (S194), so it is the same in both regimes, and so is the law of every recorded subset. By contraposition, a difference in the law of some recorded subset implies that $A$ is not diagonal, and Proposition S25.2 gives $\Omega_{\mathrm{int}}(C) > 0$.

(b) Write $\Gamma_h = \Gamma_h^{\mathrm{com}} + \Gamma_h^{\mathrm{loc}}$ with $\Gamma_h^{\mathrm{loc}} = A^h\Sigma^{\mathrm{loc}}$ for $h \geq 0$. The identity $(I - Az)^{-1}\mathsf{L} = (I - A^{\mathfrak{N}}z)^{-1}\mathsf{L}$ between power series says that $A^j\mathsf{L} = (A^{\mathfrak{N}})^j\mathsf{L}$ for all $j \geq 0$; by induction on $j$ it is equivalent to $(A - A^{\mathfrak{N}})(A^{\mathfrak{N}})^j\mathsf{L} = 0$ for all $j \geq 0$. Then $x^{\mathrm{com}}$ is the same function of the driver path in both

regimes, and by (I1) $\Gamma_h^{\mathrm{com}}$ is the same. Hence

$$\Gamma_h - \Gamma_h^{\mathfrak{N}} = A^h \Sigma^{\mathrm{loc}} - (A^{\mathfrak{N}})^h \Sigma^{\mathrm{loc},\mathfrak{N}}, \qquad h \geq 0. \tag{S195}$$

Because $A^{\mathfrak{N}}$ and $D^{\mathfrak{N}}$ are diagonal, so are $\Sigma^{\mathrm{loc},\mathfrak{N}}$ and the subtracted term, which proves the first claim. If $A$ is diagonal then $\Sigma^{\mathrm{loc}}$ and $A\Sigma^{\mathrm{loc}}$ are diagonal. Conversely, if the differences at $h = 0$ and $h = 1$ are diagonal, then $\Sigma^{\mathrm{loc}}$ is diagonal, with positive diagonal since $\Sigma^{\mathrm{loc}} \succeq D \succ 0$, and $A\Sigma^{\mathrm{loc}}$ is diagonal, so $A = (A\Sigma^{\mathrm{loc}})(\Sigma^{\mathrm{loc}})^{-1}$ is diagonal. Proposition S25.2 turns this into the statement about $\Omega_{\mathrm{int}}(C)$; the equivalence uses the records of all members, and for a recorded subset Eq. (S195) gives the corresponding entries.

For the level-0 network with scalar exact codes, $W_{ii} = F(1 - k_i^s - \sum_j k_{ij})$ and $W_{ij} = Fk_{ij}$, so every row of $W$ sums to $F(1-k_i^s)$. With equal sensing gains $k_i^s = k^s$ this is $W\mathbf{1} = \lambda\mathbf{1}$, $\lambda = F(1-k^s)$, and $\mathsf{L} = Fk^s\mathbf{1}$. Removing the message slots and returning their weight to the prior prediction leaves the total precision of each receiver, and with it the sensing gain, unchanged, in line with the convention of Definition 5.17 that a null keeps the agents' sensing; it gives $W^{\mathfrak{N}} = \lambda I$. The operation raises the weight of the prior prediction; if the priors that the convention keeps include this weight, that is a second reason why slot removal is not one of the tabulated nulls. Then $W^j\mathsf{L} = \lambda^j\mathsf{L} = (W^{\mathfrak{N}})^j\mathsf{L}$ for all $j$.

(c) Under the instrumented intervention the update of member $i$ reads

$$x_i(t+1) = A_{ii}^{\mathfrak{N}} x_i(t) + \mathsf{L}_i \mathsf{d}_t + \sum_{l \neq i} A_{il}\xi_{il}(t) + \zeta_i(t), \tag{S196}$$

because the receiver keeps its gain $A_{il}$ on each slot. The state $x_i(t)$ is a function of the driver path and of the symbols and private inputs before round $t$. The symbol $\xi_{ij}(t)$ is independent of the driver by (I1), of the symbols of earlier rounds because the symbols are independent across rounds, and of all other inputs by hypothesis; it is therefore independent of $x_i(t)$, of $\mathsf{L}_i\mathsf{d}_t$, of the other symbols of round $t$ and of $\zeta_i(t)$. Hence $\mathrm{Cov}^{\mathfrak{N}}(x_i(t+1), \xi_{ij}(t)) = A_{ij}s_{ij}$. Independence across rounds cannot be dropped: for a stationary symbol with $\mathrm{Cov}(\xi_{ij}(t+h), \xi_{ij}(t)) = s_{ij}\rho^{|h|}$ one finds $\mathrm{Cov}^{\mathfrak{N}}(x_i(t), \xi_{ij}(t)) = A_{ij}s_{ij}\rho/(1 - A_{ii}^{\mathfrak{N}}\rho)$, and the ratio in (c) is $A_{ij}/(1 - A_{ii}^{\mathfrak{N}}\rho)$. □

*Remark* S46.6 (Where the difference formula fails). The common-mode condition of Theorem S25.9(b) is restrictive. Under $\mathfrak{N}_{\mathrm{lp}}$, which keeps slots and assigned precisions, $W^{\mathfrak{N}} = \mathrm{diag}(W)$, and with equal sensing gains $(W^{\mathfrak{N}})^j\mathsf{L}$ has entries $Fk^sW_{ii}^j$, which equal $Fk^s\lambda^j$ for all $j$ only if $\sum_j k_{ij} = 0$: the condition fails for every agent that listens. Under $\mathfrak{N}_{\mathrm{rn}}$ the deleted capacity is reassigned to the remaining sources; whenever part of it reaches the sensing slot, $\mathsf{L}$ changes and (I2) fails. With slot removal and unequal sensing gains, $W^{\mathfrak{N}} = \mathrm{diag}(F(1 - k_i^s))$ and

$$(W\mathsf{L} - W^{\mathfrak{N}}\mathsf{L})_i = F^2 \sum_j k_{ij}(k_j^s - k_i^s), \tag{S197}$$

which vanishes for every $i$ only in special cases, so the common-mode condition fails already at first order. In these cases $\Gamma_h - \Gamma_h^{\mathfrak{N}}$ contains the term $\Gamma_h^{\mathrm{com}} - \Gamma_h^{\mathrm{com},\mathfrak{N}}$, which is not diagonal and depends on the unrecorded driver (Check S47.225); parts (a) and (c) remain valid, since neither uses the condition. The proof of (a) uses (I2) and (I3) only for a diagonal $A$; there no message capacity is in use, so a reassignment rule that moves nothing when nothing is deleted keeps $(A, D, \mathsf{L})$, and the conclusion of (a) then holds for $\mathfrak{N}_{\mathrm{rn}}$ as well, given (I1). Definition 5.17 does not fix the reassignment rule, so this is stated conditionally.

*Proof of Proposition S25.11.* (a) Relative entropy between two laws is zero only if they agree, so $\mathfrak{c}_{\mathfrak{N}} > 0$ means that the recorded laws differ, and Theorem S25.9(a) gives $\Omega_{\mathrm{int}}(C) > 0$. The recorded

law is a marginal of the full law, so the full laws differ as well. By Definition 5.17 a null whose intervention leaves the model unchanged is not applicable and imposes no requirement, and the type D clause of Definition 5.19 asks that the structure differ from every applicable null of type D. For an applicable null, a selector that is a function of the law returns the same partition for equal laws, so a difference of the full laws is necessary for the type D clause at that null; it is not sufficient, because that clause concerns the selected partition on an open set of the admitted class (Definition 5.17). The argument does not make $\mathfrak{c}_{\mathfrak{N}} > 0$ necessary. In the driver-free model with two members, $A_{12} = 0 \neq A_{21}$, and only member 1 recorded, the row of member 1 in $(A, D)$ is the same in both regimes, so its recorded law is unchanged and $\mathfrak{c}_{\mathfrak{N}} = 0$, while $\operatorname{Cov}(x_2(t+1), x_1(t)) = A_{21} \operatorname{Var}(x_1)/(1 - A_{11}A_{22}) \neq 0$ before the intervention and zero after it.

(b) All variables are jointly Gaussian and stationary with $|a| < 1$. Put $\sigma_1^2 = \operatorname{Var}(y_{t+1} \mid y_t)$ and $\sigma_\infty^2 = \operatorname{Var}(y_{t+1} \mid y_{-\infty:t})$, so that $\hat{\mathfrak{d}}_{\mathrm{M}} = \frac{1}{2}\log(\sigma_1^2/\sigma_\infty^2) \geq 0$. The noise $\mathfrak{n}_t$ is independent of $u$ and of $y_s$, $s \leq t$, hence $\sigma_\infty^2 = q + \operatorname{Var}(\beta u_t \mid y_{-\infty:t}) \geq q$. Also $\sigma_1^2 = \operatorname{Var}(\beta u_t + \mathfrak{n}_t \mid y_t) \leq \beta^2 \operatorname{Var}(u_t) + q$. The quotient gives the bound. If $u$ is white, $\beta u_t + \mathfrak{n}_t$ is independent of $y_s$, $s \leq t$, the process $y$ is first-order autoregressive, and $\sigma_1^2 = \sigma_\infty^2$.

(c) Take $A = 0$, $\mathsf{L} = \mathbf{1}$, $D = \delta I$ and the scalar driver $\mathsf{d}_{t+1} = \phi \mathsf{d}_t + \omega_t$ with $\operatorname{Var} \omega_t = \sigma^2$ and $0 < \phi < 1$. The model is interaction-free, so $\Omega_{\mathrm{int}}(C) = 0$ by Proposition S25.2, and by the proof of Theorem S25.9(a) a channel null satisfying (I1) to (I3) leaves its law unchanged. The collective average is $\bar{y}_{t+1} = \mathsf{d}_t + \bar{\zeta}_t$ with $\operatorname{Var} \bar{\zeta}_t = r = \delta/n$. With $v = \sigma^2/(1 - \phi^2)$ the variance of the driver, $\operatorname{Var} \bar{y} = v + r$ and $\operatorname{Cov}(\bar{y}_{t+h}, \bar{y}_t) = \phi^{|h|} v$ for $h \neq 0$, so

$$\tau_y = 1 + \frac{2v}{v+r} \cdot \frac{\phi}{1-\phi}, \qquad \varrho_1 = \frac{\phi v}{v+r}, \qquad \sigma_1^2 = (v+r)(1-\varrho_1^2) = \frac{\sigma^2 v + 2vr + r^2}{v+r} \leq \sigma^2 + 2r. \tag{S198}$$

Since $\bar{y}_{t+1} = \phi \mathsf{d}_{t-1} + \omega_{t-1} + \bar{\zeta}_t$ and $\omega_{t-1}$, $\bar{\zeta}_t$ are independent of $\bar{y}_s$, $s \leq t$, we have $\sigma_\infty^2 \geq \sigma^2 + r$. Therefore

$$\hat{\mathfrak{d}}_{\mathrm{M}} \leq \tfrac{1}{2} \log \frac{\sigma^2 + 2r}{\sigma^2 + r} \leq \frac{\delta}{2n\sigma^2}, \qquad R^2 = 1 - \frac{\sigma_\infty^2}{v+r} \geq 1 - \frac{\sigma_1^2}{v+r} = \varrho_1^2. \tag{S199}$$

Given $\varepsilon$ and $\tau_0$, choose $\phi$ with $\phi^2 > 1 - \varepsilon$ and $1 + \phi/(1-\phi) \geq \tau_0$, and then $n$ so large that $\delta/(2n\sigma^2) \leq \varepsilon$, $\varrho_1^2 \geq 1 - \varepsilon$ and $v/(v+r) \geq \frac{1}{2}$; the last two are possible because $\varrho_1 \to \phi$ and $v/(v+r) \to 1$ as $n \to \infty$. □

*Remark* S46.7 (Relation to the grade). The grade $(r_0, \alpha_{\mathrm{M}}, \alpha_{\mathrm{I}}, \alpha_{\mathrm{pr}}, \mathfrak{A}_0)$ of Definition 5.19 has four kinds of clause: the collective clause at level $r_0$, the closure clauses, the clauses on nulls, and the agency index. In the stationary state without input, the Markov defect of Definition 5.1 is the limit of $I(y_{t+1}; y_{0:t-1} \mid y_t)$ over the history length, and the memory clause is $\mathfrak{d}_{\mathrm{M}} \leq \alpha_{\mathrm{M}}$. For a stationary Gaussian scalar $y$ with $\operatorname{Var}(y_1 \mid y_{-\infty:0}) > 0$, as in Proposition S25.11(b) and (c), this limit is the recorded memory defect $\hat{\mathfrak{d}}_{\mathrm{M}}$ of Definition S25.10: by stationarity the term with history length $t$ is $\frac{1}{2}\log(\operatorname{Var}(y_1 \mid y_0)/\operatorname{Var}(y_1 \mid y_{-t:0}))$, and $\operatorname{Var}(y_1 \mid y_{-t:0})$ decreases to $\operatorname{Var}(y_1 \mid y_{-\infty:0})$ as $t \to \infty$, the projections on an increasing family of closed subspaces converging to the projection on the closure of their union. When the blanket carries input, Definition 5.1 takes the defect for the law of $y$ given the input path, which presupposes that the input is recorded; a variable that is exactly closed given its blanket input has input-conditional defect zero. With the input unrecorded, records give only $\hat{\mathfrak{d}}_{\mathrm{M}}$. Proposition S25.11(c) shows that a collection of non-interacting members has $\hat{\mathfrak{d}}_{\mathrm{M}} \leq \varepsilon$ at every tolerance, and its input-conditional defect is zero as well, the average being independent across rounds given the driver path, while an exactly closed variable can have $\hat{\mathfrak{d}}_{\mathrm{M}} > 0$ when its input is colored and unrecorded, as in the instance computed in Check S47.227, and (b) bounds it. The grade also allows a tolerance $\alpha_{\Xi}$ on the Markov error in place of $\alpha_{\mathrm{M}}$, a continuous-time measure that is not treated here. The collective clause refers to the contrast of Definition 4.10, a

functional of the world-conditional precision of the population (Definition 4.12). That precision is conditional on the world path (Definition 4.7), and with the driver in the place of the world it is not a functional of the recorded law: the two models of Theorem S25.5(b) share a recorded law, and their covariances given the driver path differ, because the driver contributes positive variance on $T$ in one model and is absent from the other. For an interaction-free model that precision is diagonal, so that $L_C = 0$ and $\varkappa_{\mathrm{in}}(C)$, the least eigenvalue of the compression of $\mathrm{diag}(\varepsilon^{\Pi})$ on $C$ to the complement of $\mathbf{1}_C$, is at most the mean $\bar{\varepsilon}_C$ of the $n-1$ eigenvalues of that compression, whose trace is $(n-1)\bar{\varepsilon}_C$; hence $r(C) \le 1$ and no region is a collective at an admissible level, which exceeds $\varepsilon_C^{\max}/\bar{\varepsilon}_C \ge 1$. The model of Proposition S25.11(c) thus fails the collective clause, which refers to a precision that the recorded law does not determine. The clauses on nulls, namely the type D clause for the applicable nulls, the partner-randomization test at level $\alpha_{\mathrm{pr}}$ and survival under the nulls of type S, refer to interventions on the full model (Definition 5.17). A channel null that satisfies (I1) to (I3) leaves an interaction-free model unchanged, so by the applicability clause of that definition it is not applicable to such a model and imposes no requirement on it. The agency index refers to the habit law, which the policy law under one regime does not determine (Proposition S25.12(c)). From records under one regime only necessary conditions of these clauses can be examined. The recorded quantity that carries certifying weight is the control score, and only in the direction of Proposition S25.11(a). This reading of the clauses is an interpretation of Definitions 5.1 and 5.19, and the proposition is its mathematical content.

*Proof of Proposition S25.12.* With gains $k_i^s$, $k_{ij}$, sensing noise $v_i$ of variance $r_i$ and channel noise $\nu_{ij}$ of variance $\sigma_{ij}^2$, the level-0 update with scalar exact codes is $m_i(t+1) = F[(1 - k_i^s - \sum_j k_{ij})m_i(t) + k_i^s(\eta_t + v_i(t)) + \sum_j k_{ij}(m_j(t) + \nu_{ij}(t))]$, that is $m(t+1) = Wm(t) + b\eta_t + \zeta(t)$ with $b_i = Fk_i^s$ and independent $\zeta_i(t)$ of variance $q_i$. The joint law of beliefs and world is a function of $(W, b, q)$ and of the world law. The map from assigned precisions to gains, $k_i^s = \pi_i^s/\Pi_i$, $k_{ij} = \pi_{ij}/\Pi_i$, is a hypothesis of the proposition, consistent with the convex gains of Remark S2.14; it is homogeneous of degree zero in $(p_i, \pi_i^s, \pi_{i\cdot})$, which gives (a). The variances $(r_i, \sigma_{i\cdot}^2)$ enter through $q_i$ only, which gives (b). For (c), $Ee^{-\gamma g}\exp(-\gamma' \cdot \gamma(G-g)/\gamma') = E\exp(-\gamma G)$, so the unnormalized weights of the policies agree at every $X$. Adding a function of $X$ to $G$ multiplies all weights at $X$ by one factor, which the normalization removes.

Separation (a). Let the gains be known before and after an intervention that sets $\pi_{ij}$ to a known $\pi^\star > 0$ and leaves the other assigned precisions of $i$ unchanged, and let these others be not all zero, so that $\Pi_i - \pi_{ij} > 0$ and $k_{ij} < 1$. The new gain is $k'_{ij} = \pi^\star/(\Pi_i - \pi_{ij} + \pi^\star) \in (0,1)$, so $\Pi_i - \pi_{ij} = \pi^\star(1-k'_{ij})/k'_{ij}$, and $\Pi_i - \pi_{ij} = \Pi_i(1-k_{ij})$ gives $\Pi_i$. Then $\pi_i^s = k_i^s\Pi_i$, $\pi_{il} = k_{il}\Pi_i$ and $p_i$ follow. The scale $c_i$ is separate for each member, so one such intervention is needed for each. Separation (b). If slot $j$ of receiver $i$ delivers noise of known variance $s$ in place of $m_j + \nu_{ij}$ and the gain is kept, the private variance changes from $D_{ii} = q_i$ to $q_i + F^2k_{ij}^2(s - \sigma_{ij}^2) = D_{ii} + W_{ij}^2(s - \sigma_{ij}^2)$, which gives $\sigma_{ij}^2$ when $W_{ij} \neq 0$; all channels of $i$ give $\sigma_{i\cdot}^2$ and then $r_i$ from $q_i$ when $k_i^s \neq 0$. Separation (c). The log odds of two policies at $X$ are $\log(E(\pi)/E(\pi')) - \gamma(G(\pi, X) - G(\pi', X))$. A known shift $\Delta G$ at fixed $E$ and $\gamma$ changes them by $-\gamma(\Delta G(\pi, X) - \Delta G(\pi', X))$, which gives $\gamma$ when the shift separates two policies. The habit null sets $\gamma = 0$, so the policy law is then $E$. With $E$ and $\gamma$ known, $G$ is determined up to a function of $X$. □

**Corollary S46.8** (The classes are the whole fibre)**.** *(i) Let the world state be recorded, and let $0 < |F| < 1$, $\rho(W) < 1$, the world noise variance be positive and $q_i > 0$ for every $i$ at both points. Two parameter points of Proposition S25.12 with $F$ and the world noise fixed give the same stationary joint law of beliefs and world if and only if they are related by the scalings (a) and the reallocations (b). (ii) On a finite policy set, two triples $(E, \gamma, G)$ and $(E', \gamma', G')$ with $E, E' > 0$ and $\gamma, \gamma' > 0$*

*give the same policy law at every $X$ if and only if $G' = \gamma(G - g)/\gamma' + c(X)$ with $g = \gamma^{-1}\log(E/E')$ and some function $c$ of $X$ alone.*

*Proof.* (i) The augmented state $(m_t, \eta_t)$ follows a stable first-order recursion with input covariance $\mathrm{diag}(q, \sigma_w^2) \succ 0$, so its stationary covariance is positive definite and the regression of $m(t+1)$ on $(m(t), \eta_t)$ returns $(W, b)$ and the residual variances $q$. Thus the law determines $(W, b, q)$, and conversely. The gains are $k_i^s = b_i/F$ and $k_{ij} = W_{ij}/F$; the vector $(p_i, \pi_i^s, \pi_{i\cdot})/\Pi_i = (1 - k_i^s - \sum_j k_{ij}, k_i^s, k_{i\cdot})$ determines the assigned precisions up to the factor $\Pi_i$, and $q_i$ is one linear equation in $(r_i, \sigma_{i\cdot}^2)$. (ii) Equality of the two laws at $X$ is $\log E - \gamma G = \log E' - \gamma' G' + c(X)$ for a normalizing term $c(X)$, which is the stated relation after division by $\gamma'$ and renaming $c$. □

*Remark* S46.9 (Inference is not in the linear law). The recursion $m(t+1) = Wm(t) + b\eta_t + \zeta(t)$ is also the law of members that hold no generative model and average their inputs with fixed weights. Under the precision map of Proposition S25.12, an inferring member changes its gain when an assigned precision is changed, by the formula in separation (a), and a fixed-weight averager does not. The response of the gains to a declared change of reliability therefore separates the two readings, and no functional of the law under a single regime does. That the first reading deserves the name active inference is an interpretation; the equality of laws is exact.

*Remark* S46.10 (The recovery example). Check S47.228 fits three model classes to the exact law of a window of six rounds of a three-agent level-0 network with unrecorded world, by minimizing the relative entropy from the truth, and then compares their predictions for a jamming intervention that was not used in the fit. The collective class contains the truth and recovers it from the start that was tried, and the Jacobian of the window covariance with respect to its sixteen parameters has full rank at the truth, which is local identifiability at that point and nothing more. Recovery by local search depends on the start: of 7 random starts, 2 reached the truth and 5 reached a second strict local minimum, with relative entropy $6.5 \times 10^{-3}$, driver pole 0.581 in place of 0.9, and largest coupling eigenvalue 0.909 in place of 0.644. This point assigns the slow mode to the coupling and a faster one to the driver, an exchange that resembles the pole swap of Proposition S25.7; its relative entropy is positive, so it is a different law, and its prediction of the jammed law is off by 1.33 nats.

The other two classes do not contain the truth. Theorem S25.5(c) does not show this, because the truth has both interaction and a driver and the autoregressive class has a full innovation covariance; two direct certificates do. A stationary first-order autoregression has $\Gamma_h = A^h\Gamma_0$ and hence $\Gamma_2 = \Gamma_1\Gamma_0^{-1}\Gamma_1$, while at the truth the largest entry of $\Gamma_2 - \Gamma_1\Gamma_0^{-1}\Gamma_1$ is 0.068 in modulus. For independent agents with memories $a_i$ and a scalar stationary driver, the cross-covariance of $x_i(t+1) - a_i x_i(t) = \mathsf{L}_i\mathsf{d}_t + \zeta_i(t)$ and $x_j(t+1) - a_j x_j(t)$ at lag $h$ is $\mathsf{L}_i\mathsf{L}_j\,\mathrm{Cov}(\mathsf{d}_{t+h}, \mathsf{d}_t)$, which is even in $h$; with $g_h = \mathrm{Cov}(x_i(t+h), x_j(t))$ it equals $(1 + a_i a_j)g_h - a_j g_{h+1} - a_i g_{h-1}$, so the $4 \times 3$ matrix with rows $(g_h - g_{-h}, g_{h-1} - g_{-h-1}, g_{h+1} - g_{1-h})$, $h = 1, \ldots, 4$, annihilates $(1 + a_i a_j, -a_i, -a_j)$ and has rank at most two. At the truth its smallest singular values for the three pairs are $8 \times 10^{-4}$, $3 \times 10^{-4}$ and $3.3 \times 10^{-3}$, against about $10^{-16}$ for members of the class computed the same way. The autoregressive minimum 0.310 is the global minimum of its class: the relative entropy over the window is the sum of a term for the first round, which vanishes when the stationary covariance of the fitted model is $\Gamma_0$, and transition terms, which are minimized by the regression coefficient $\Gamma_1\Gamma_0^{-1}$ and its residual covariance, and the fitted model with these coefficients has stationary covariance $\Gamma_0$. The value 0.408 for independent agents was reached from the start of the check and from 10 of 10 random starts and is not proven to be global. The misfit of the two classes is therefore expected, and the content of the example is the size of the held-out error and the fact that the effective linear law, which fits the recorded window better than the independent class, predicts the intervention worse under either reading of its coefficients. The fits are at the population level; no finite-sample

claim is made.

# Part IV
# Numerical checks

## S47 Numerical checks

The checks below confirm proofs, closed forms and counterexamples on small instances. Each was run once, single threaded, in double precision, with the sizes and parameters stated. A check does not replace a proof; where a statement rests on a check alone it is labeled a numerical observation.

### S47.1 Checks for Section 2

**Check S47.1** (Value of one message under different retained information)**.** In a scalar static world with prior precision one, one common source and one private source per agent, each of precision one, and a noiseless message carrying the sender's posterior mean, the mutual information between the state and the message given the receiver's retained information was computed by exact Gaussian conditioning on the full joint law. A receiver that keeps its raw data gains 0.1438 nats, equal to $\frac{1}{2}\ln(4/3)$; one that keeps only its posterior mean gains 0.1003 nats, equal to $\frac{1}{2}\ln(11/9)$; a level-1 receiver believes it gains 0.3466 nats, equal to $\frac{1}{2}\ln 2$ (Proposition 2.5).

**Check S47.2** (Update subspace and prior completion)**.** For Proposition 2.2, 200 random instances were drawn with state dimension $K$ between 3 and 6, code rank between 1 and $K-1$, random positive-definite priors and noises, and every third instance given a rank-deficient code by repeating a scaled row. The largest component of $m_i^+ - m_i$ outside $\mathrm{col}(P_i\hat{H}^\top)$, together with the largest component of the columns of $P_i\hat{H}^\top$ outside $\mathrm{col}(P_i - P_i^+)$, was $1.4\times10^{-14}$; the rank of $P_i - P_i^+$ equaled $\operatorname{rank}\hat{H}$ in every instance. For a functional $f$ with $f^\top P_i\hat{H}^\top = 0$ ($K=5$, rank two) the changes of the mean and the variance of $f^\top\eta$ were $2\times10^{-16}$ and $6\times10^{-17}$. The factorization (6) held to $2.2\times10^{-17}$ for the gain and $2.5\times10^{-16}$ for the mean. For Proposition 2.3(a), a two-component Gaussian mixture prior on $\mathbb{R}^2$ was discretized on a $401\times401$ grid and updated with a likelihood that depends on the first coordinate only: the conditional law of the second coordinate given the first was unchanged to $7\times10^{-18}$ on every row with mass, while the marginal of the first coordinate moved by total variation 0.63.

**Check S47.3** (Two receivers and the disagreement example)**.** For Proposition S2.1, 200 random pairs of receivers (state dimension 2 to 5, code rank 1 to $K$) gave a largest error of $1.1\times10^{-15}$ in the decomposition (a) and a largest relative error of $2.7\times10^{-15}$ in the identity (d), with the expectation computed in closed form. When the two prior covariances differed by a matrix of norm 1.31 whose product with $\hat{H}^\top$ vanishes, the gains differed by $2.2\times10^{-16}$; when the priors differed along a sent direction they differed by 0.22, as part (b) requires. The equal-covariance identity (c) held to $1.7\times10^{-14}$, and in the noiseless limit (e) both receivers reproduced the message exactly in the coded coordinates while their posterior means differed by 0.107 in $\ker\hat{H}$. For Example S2.2 at $\varrho=0.8$, $\sigma^2=0.5$, the prior divergence was 3.5556, equal to $2\varrho^2/(1-\varrho^2)$; at $y=0,1,2,3$ the gaps of the posterior means were 0, 1.0667, 2.1333 and 3.2000, equal to $2\varrho y/(1+\sigma^2)$, and the posterior divergences were 1.1852, 2.7654, 7.5062 and 15.4074, equal to (S1); the coefficient of $y^2$ was 1.5802 and the predictive variance at which the expected divergence equals the prior one was 1.5000.

**Check S47.4** (Learning codes)**.** (i) Exact joint attention with $K = 4$, $r_c = 2$, a content of rank three and 60,000 rounds: the relative error of $\hat{H}M_\phi$ was $5.0 \times 10^{-4}$, and along the direction in which the content is identically zero the minimum-norm estimate was $2 \times 10^{-15}$ while the true code there was $(-0.30, -0.25)$ (Proposition 2.7(a,b)). (ii) For the recursive learner with prior covariance $2I$: the posterior mean along that direction equaled the prior mean exactly, the posterior variance there equaled the prior variance to a relative $10^{-10}$, and recursion and batch formula agreed to $8.5 \times 10^{-13}$ after 300 steps (Proposition S2.3). (iii) Noisy grounding with $K = 3$, $r_c = 2$ and 200,000 rounds: raw least squares matched the attenuated limit within 0.002 in every entry, and regression on the calibrated statistic matched the true code within 0.004 (Proposition 2.7(c)). (iv) Ungrounded learning of a rank-one code on a two-dimensional state with 300,000 rounds matched the limit of Proposition S2.4(b) within 0.005 for independent and for correlated sensing noises; in four scalar configurations with 400,000 rounds least squares matched $Hk_j^s(1 + R_{ji}^s/P_0)$ within 0.0012, including zero bias for identical data. (v) Partial grounding at $p_g = 0$, 0.1, 0.5, 0.9 and 1 ($P_0 = 1$, sender noise 1, receiver noise 0.5, code 2, referent variance 1.5): pooled least squares matched Proposition S2.4(c) within 0.003, and least squares on the grounded rounds alone matched the true code within 0.001 whenever $p_g > 0$. (vi) For Proposition S2.3(d), 3000 replicates of 400 independent rounds with $K = 3$, $r_c = 2$, a content of rank two and Gaussian grounding noise gave $T$ times the covariance of the error on the identified block within 0.033 of $M_2^+ \otimes R_{\text{res}}$, whose largest entry is 1.73, for the identity and for a correlated working noise.

**Check S47.5** (Transfer kernel and records)**.** The impulse response of a linear joint process with a world, a channel and a record, simulated for a direct, a stigmergic and a performative act over 14 lags, equaled the mode sum (9) to $5.6 \times 10^{-17}$. For records with retention $1 - \omega - \delta$ equal to 0, 0.5, 0.9 and 0.99 the normalized mean delays were 1, 2, 10 and 100, as Proposition S2.5(a) gives. The closed form (S3) matched the stationary Lyapunov solution to $4.4 \times 10^{-16}$ over 500 random draws. At six pairs $(F, \text{snr})$ the grid maximizer of (S3), the closed form for $1 - \omega^*$ and $F(1 - k^*)$ agreed to five digits, and the maximum equaled $1 - P^-/P_0$: $(0.5, 1)$ gave $1 - \omega^* = 0.26795$ and squared correlation 0.13397; $(0.9, 1)$ gave 0.62679 and 0.56411; $(0.9, 10)$ gave 0.25884 and 0.75230; $(0.99, 0.5)$ gave 0.90419 and 0.81020; $(0.999, 5)$ gave 0.90485 and 0.97919; $(0.7, 100)$ gave 0.01334 and 0.48519.

**Check S47.6** (Being understood and being predicted)**.** With $K = 3$ and $r_c = 2$: (i) the second moment of the prediction error matched Proposition S2.10(a) within Monte Carlo error over 300,000 draws at $\sigma_H^2 = 0$ and 0.05 (largest deviations 0.12 and 0.045 on entries of size 34); (ii) at known codes the second moment of the understanding error equaled $R_c$ (Proposition S2.8) within Monte Carlo error over 100,000 draws for honest content and for content with a deviation $h_j = (1, -1)$; at $\sigma_H^2 = 0.02$, with $\hat{H}_{ij} = H_j + \sigma_H W$ and $W$ a standard Gaussian matrix, the second moment of the understanding error of this instance is infinite, because $\mathbb{E}\|\psi_{ij}\|^p$ is finite only for $p < K - r_c + 1 = 2$ (Theorem S22.1), so no untruncated average is reported. The script `p2_understanding.py` (instance seed 17, Monte Carlo seed 1702, 400,000 draws per case) computed the truncated moment $\mathbb{E}[\psi_{ij}\psi_{ij}^\top; \sigma_H\|W\|_F \le \rho]$ at $\rho = 0.749$, nine tenths of the smallest singular value 0.832 of $H_j$, an event of frequency 0.99989 on which the lift is bounded. Its diagonal exceeded that of $R_c$ times the frequency by 0.0947 and 0.0744 for honest content and by 0.0942 and 0.0760 for $h_j = (1, -1)$, with standard errors at most 0.0011, and the term $\sigma_H^2\mathbb{E}\|\hat{m}_{ij}\|^2$ with the reading computed at the true code, 0.0658 and 0.0681 in the two cases, was 69 and 88 percent of the excess in the first and 72 and 90 percent in the second. For a second instance with $K = 6$ and $r_c = 2$ (instance seed 23), where the second moment and the variance of its estimator are finite, the untruncated diagonal excess over $R_c$ was 0.329 and 0.370 for honest content and 0.383 and 0.453 for $h_j = (1, -1)$, with standard errors at most 0.0020, and the same term, 0.323 and 0.377, was 98 and 87 percent of it in the first case and 98 and 83 percent in the second. The same script evaluated the

second-order excess of Proposition S22.2 in closed form on both instances, with the same average over the receiver's mean: the term is 0.816 and 0.973 of it for honest content and 0.821 and 0.974 for $h_j = (1, -1)$ in the first instance, and 1.017 and 1.046, then 1.015 and 1.039, in the second. The Monte Carlo excess is 3 to 25 percent above the second-order value on every entry, by at least five standard errors, a difference that the proposition assigns to its $O(\sigma_H^4)$ remainder at $\sigma_H^2 = 0.02$. These shares belong to the computed instances (Remark S43.4); in them the further terms from the code dependence of the lift are not negligible; (iii) at $\sigma_H^2 = 0$ the expected-free-energy decision, evaluated as risk plus ambiguity from the Gaussian laws, agreed with the threshold rule of (b) in 2000 of 2000 random instances; for scalar beliefs with $H_j = 1$ the gap $\frac{1}{2}\ln(2\pi e S_{ji}) - \mathbb{H}[\varphi_{ij}]$, computed by quadrature, was 0 at $(\bar{m}_{ji}, \bar{P}_{ji}, \sigma_H^2, R_c) = (1, 1, 0, 0.5)$, 0.0025 at $(1, 1, 0.05, 0.5)$, 0.0473 at $(1, 1, 0.5, 0.5)$, 0.1118 at $(0, 1, 1, 0.1)$ and 0.1508 at $(0, 4, 1, 0.05)$; in the last instance with $\Psi_{\mathrm{p}} = 10$, $p^{\varnothing} = \frac{1}{2}$ and $g_{\mathrm{use}} = 0$, $S_{ji} = 8.05$, $\Xi_{ji} = 0.0680$ and $\Xi_{ji}^{\mathrm{ex}} = -0.0829$, so at disagreement 0.7746, where the left side of (b) is 0.0300, the inequality with $\Xi_{ji}$ holds while addressing costs 0.1129 more expected free energy than silence; (iv) numerical maximization of the content posterior matched $u_j^*$ of (c) to $10^{-6}$ at $\gamma = 0.5$, 2 and 50, with $\|u_j^* - H_j \bar{m}_{ji}\|$ falling from 2.20 to 0.92 and 0.050; (v) the identity (d) with the sender's own belief held fixed gave a being-predicted risk of 17.094 in closed form against a Monte Carlo value of 17.136 over 200,000 draws, with standard error 0.031; with the own belief updated from the same message, the matrix identity of (d) agreed with a Monte Carlo estimate over 400,000 draws to 0.005 on entries up to 4.4 for a random gain $L_j$, and in the scalar instance $m_j = 2$, $\bar{m}_{ji} = 0$, $\bar{P}_{ji} = 1$, $R_c = 0.5$ the second moment 5 without listening became 2.2778, 0.7778 and 0.4568 at own gains 1/3, 2/3 and 8/9, while at $d_{ji} = 0$, $\bar{P}_{ji} = 0.1$, $R_c = 10$ and $L_j = 0.5$ it rose from 0.1 to 2.525; for a random instance with $K = 3$, $r_c = 2$ and $\sigma_H^2 = 0.05$ the expected risk change $\frac{1}{2}\operatorname{tr}(\Psi_{\mathrm{p}}^{-1} H_j \Delta_{ji} H_j^{\top})$ of (d) was 0.5727 in closed form against a Monte Carlo value of 0.5703 over 400,000 draws, with standard error 0.0039, it was zero at $L_j = 0$ and at a gain with $H_j L_j = 0$ to $2 \times 10^{-16}$, and on the ray through the same gain it vanished to $10^{-16}$ at the one nonzero multiple $\ell/q$ of the proof, where the Monte Carlo value was 0.0019 with standard error 0.0027.

**Check S47.7** (Frames)**.** (i) A Kalman filter run in internal coordinates with $K = 3$ for 50 steps returned decoded beliefs equal to the world-coordinate filter to $1.8 \times 10^{-15}$ (Proposition S2.12(a)). (ii) With $K = 4$ and $r_c = 2$, perturbing an encoding by a matrix in the stabilizer changed the effective code by $1.2 \times 10^{-15}$, and the stabilizer had dimension $8 = K(K - r_c)$ (Proposition S2.13). (iii) For a misidentified code with $K = 4$, $r_c = 2$, the excess risk of Proposition S26.1(a) was 0.1691 against a Monte Carlo value of 0.1694; the criterion (b) agreed with direct computation in 3000 random cases with $K$ between 2 and 4; the rank-one noiseless formula held to $2.8 \times 10^{-14}$ with no threshold mismatch in 3000 cases. For an isotropic two-dimensional prior with $p = 1$ the mean-square errors at rotations of 0, 30, 59, 60, 61 and 90 degrees were 1.0000, 1.2679, 1.9699, 2.0000, 2.0304 and 3.0000, equal to $3 - 2\cos\vartheta$, against 2 in silence.

**Check S47.8** (Common mode with mistranslated codes)**.** A scalar level-0 network of six agents with $F = 0.9$, world innovation variance 1, sensing gains drawn in $[0.1, 0.4]$, message gains in $[0, 0.15]$ on about 60 percent of the pairs, and true and modeled code gains drawn independently in $[0.7, 1.3]$ had $\rho(W) = 0.774$ and mistranslation coefficients $B_i^{\mathrm{mis}}$ up to 0.049 in magnitude. The errors after 60 rounds and the world path were written as explicit linear maps of independent inputs, and the joint Gaussian law was conditioned on the world path by a Schur complement. The conditional covariance equaled $\sum_t W^t Q^{\zeta} (W^{\top})^t$ to $2.4 \times 10^{-15}$ and the stationary Lyapunov solution to $3.5 \times 10^{-15}$, while the world added correlations up to 0.978 to the unconditional covariance (Lemma 2.13(b,c)). An independent computation with four agents, mistranslation coefficients up to 0.05 and a window of twelve rounds reproduced the local-noise covariance to $5.0 \times 10^{-16}$, with the world adding up to 0.59 to the unconditional covariance.

### S47.2 Checks for Section 3

The value checks compare closed forms with exact Gaussian conditioning of the full joint law of the world, the sources, the channel noise and the messages, built as explicit linear maps of independent inputs. Two ensembles of random static common-source worlds are used. Write $W_d = GG^\top/d + 0.3I_d$ for a random $d \times d$ matrix in which $G$ has independent standard normal entries, drawn afresh at each use. In the source ensemble the world dimension $K$ is uniform on $\{1, \dots, 4\}$ (on $\{2, 3, 4\}$ in the search of Check S47.10), the number of sources is uniform on $\{2, \dots, 5\}$ and each source dimension is uniform on $\{1, \dots, K\}$; the source maps $H_u^s$ have independent standard normal entries, $\Lambda_0 = 0.5W_K$ and $R_u^s = 0.7W_d$; the sender's source set is a uniformly drawn subset of a size uniform on $\{1, \dots, |\mathcal{U}|\}$, and the receiver's is drawn in the same way with a size uniform on $\{0, \dots, |\mathcal{U}|\}$. The precision ensemble draws the precisions of Definition 3.2 directly: $K$ is uniform on $\{2, 3, 4\}$, $\Lambda_0 = 0.5W_K$, and the shared source precision $\sum_{u \in \mathcal{U}_i \cap \mathcal{U}_j} \Gamma_u$, the sender's private precision $\Lambda_{j \setminus i}$ and the receiver's private precision each have the form $BB^\top/\rho$ with $B$ a $K \times \rho$ matrix of independent standard normal entries, where $\rho$ is uniform on $\{0, \dots, K\}$ for the shared and the receiver's precision (the zero matrix at $\rho = 0$) and on $\{1, \dots, K\}$ for the sender's. In both ensembles codes have independent standard normal entries, and the channel noise of a code with $r$ rows is $R_c = sW_r$ with the law of $s$ stated in each check. Counts of sign events belong to these ensembles and to one seed of the generator; a fresh sample reproduces their frequencies, not the exact counts.

**Check S47.9** (Believed and true values)**.** Submodularity of the level-0 believed value (Proposition S3.1) was tested in 2000 random instances without a violation. For Example S3.2 the computed values are 0.8047 for each sender and 3.3429 for the pair. Over 150 random worlds of the source ensemble (code rank uniform on $\{1, \dots, K\}$, $s = 0.3$) the closed form (16) agrees with exact conditioning to $2.6 \times 10^{-14}$; with invertible noiseless codes the form of Lemma 3.3(c) agrees to $7.3 \times 10^{-11}$ and no instance has $\mathcal{V}_i^1 < \mathcal{V}_i^*$. In one world with $K = 3$ and four sources, the error covariances of Lemma 3.3(a) agree with the explicit linear maps to $1.0 \times 10^{-16}$, and over the 150 worlds the matrix (S7) agrees with the direct expansion of the level-1 error to $3.7 \times 10^{-15}$. An independent recomputation, conditioning the joint law of $(\eta, o_{\mathrm{com}}, o_i, o_j, \nu, y_{ij})$ in the symmetric scalar configuration, reproduces (17) to $3.0 \times 10^{-15}$ over 300 configurations, and the ratio of actual to claimed level-1 variance equals $1 + \phi$ at $(\Lambda_0, \Gamma_{\mathrm{com}}, \Gamma_{\mathrm{pr}}) = (1, 1, 1)$ (claimed 0.1667, actual 0.2778, ratio 1.6667) and $(0.2, 2, 0.5)$ (claimed 0.1852, actual 0.3361, ratio 1.8148).

**Check S47.10** (Scope of overvaluation and undervaluation)**.** The band (S6) matches the sign of $\mathcal{V}_i^* - \mathcal{V}_i^1$ in 50,000 random draws of the scalar track model. Part (b) of Proposition 3.4 holds in 20,000 random scalar configurations (prior, shared and private precisions in $(0, 3)$, code gains of modulus in $(0.05, 5)$, channel noise in $(0, 5)$), part (d) in 5000 random aligned worlds, where the matrix value also equals the sum of the scalar values (for example 0.13398440543537 both ways), and part (e) in 5000 random instances with $\mathcal{U}_i \subseteq \mathcal{U}_j$ (precision ensemble with the receiver's private precision set to zero, code rank uniform on $\{1, \dots, K\}$, $s$ uniform on $(0.01, 2)$), among which $\mathcal{V}_i^* > \mathcal{V}_i^1$ in 386. Random searches give Numerical observation S3.10: among 4000 instances of the source ensemble with code rank uniform on $\{1, \dots, K-1\}$ and $s$ uniform on $(0.001, 2)$, $\mathcal{V}_i^* > \mathcal{V}_i^1$ in 454 (largest excess 0.947) and $\mathcal{V}_i^* > \mathcal{V}_i^0$ in 17 (largest excess 0.211); among 5000 instances of the precision ensemble with $K \times K$ codes, invertible with probability one, and $s$ uniform on $(0.01, 3)$, $\mathcal{V}_i^* > \mathcal{V}_i^1$ in 221 (largest excess 0.696), and in a second sample of 5000 from the same ensemble $\mathcal{V}_i^* > \mathcal{V}_i^0$ in 2 (largest excess 0.143). The largest excesses are sample maxima and vary with the seed: the rank-limited search under another seed gives 473 and 19 of 4000 with largest excesses 1.48 and 0.53. Example S3.8: at $\alpha_1 = \Gamma_{\mathrm{pr}} = 1$, $\alpha_2 = 0.01$, $\mathcal{V}_i^* = 2.3076$ and $\mathcal{V}_i^1 = 1.9732$ at $R_c = 0$, and $(\mathcal{V}_i^*, \mathcal{V}_i^1, \mathcal{V}_i^0) = (2.2157, 1.9263, 3.1133)$ at $R_c = 0.1$; at $\alpha_1 = 10$ the gap shrinks (2.3076

against 2.2654), as the limit $\frac{1}{2}\ln((\alpha_1+\Gamma_{\rm pr})/\alpha_1)$ predicts. Example S3.9: at $\psi=0.8$, $R_c=0.01$, $\mathcal{V}_i^*=1.5868$ at precisions $10^4$ and 1.5898 at $10^6$ against the limit 1.5899, with $\mathcal{V}_i^1=1.35\times10^{-4}$ and $\mathcal{V}_i^0=4.97\times10^{-3}$ at $10^4$; at $\psi=0.5$, $R_c=0.1$ and precisions 10, $10^2$, $10^4$, $\mathcal{V}_i^*=0.421$, 0.515, 0.528 against the limit 0.528 while $\mathcal{V}_i^0=0.323$, 0.047, 0.0005. For the full-rank code at $\psi=0.8$, $\sigma_c^2=0.1$ and source precisions $\Gamma_1=\Gamma_2=10^5$, the computed values $\mathcal{V}_i^*=0.96598$ and $\mathcal{V}_i^0=0.76309$ approach the limits 0.96605 and 0.76303.

**Check S47.11** (Closed-form ratios)**.** Over 300 random symmetric configurations, (17) agrees with exact conditioning to $8.9\times10^{-15}$. At $\varrho_c=0.5$:

| $\phi$ | $\mathcal{V}_i^*$ | $\mathcal{V}_i^1/\mathcal{V}_i^*$ | $\mathcal{V}_i^0/\mathcal{V}_i^*$ |
|---|---|---|---|
| 0 | 0.255413 | 1 | 2.151 |
| 0.5 | 0.111572 | 2.289 | 4.923 |
| 0.9 | 0.00826465 | 30.90 | 66.46 |
| 0.99 | $9.80\times10^{-5}$ | 2605.5 | 5603.5 |
| 0.999 | $9.98\times10^{-7}$ | $2.56\times10^5$ | $5.50\times10^5$ |

At $\varrho_c=0$ the ratio $\mathcal{V}_i^1/\mathcal{V}_i^*$ is 1, 1.7095, 7.2725 and 69.66 at $\phi=0$, 0.5, 0.9, 0.99. Part (b) agrees with a belief-only receiver under prior precision $10^{-6}$ to $7.9\times10^{-8}$. For part (c), with $\frac{1}{2}\ln(1+P_i\Gamma_{\rm rel})=0.3466$, the true value is 0.1341, 0.1928, 0.2623, 0.3206, 0.3410, 0.3460 and the level-1 value 0.2350, 0.3942, 0.6931, 1.2825, 2.0554, 3.1993 at $n_{\rm r}=1$, 2, 5, 20, 100, 1000; at $n_{\rm r}=3$ exact conditioning on all three messages gives 0.2259926, equal to the formula to machine precision.

**Check S47.12** (Dynamic cross-covariance)**.** For $(F,Q)$ and sensing noise variances of $i$ and $j$ equal to $(0.9,0.19;1.0,0.5)$, $(0.5,0.75;2.0,2.0)$ and $(0.99,0.0199;0.3,3.0)$, with steady-state Kalman gains, iterating the covariance recursion of the filtered errors gives 0.114603, 0.403005 and 0.050006, equal to (S8) to six digits, with error correlations 0.4549, 0.6464 and 0.4254.

**Check S47.13** (Echo-chamber bounds)**.** On 400 random stable level-0 networks of 3 to 11 agents with random directed attention graphs, heterogeneous gains and some pure listeners, the stationary variances computed from the Lyapunov equation never fall below (20); the smallest ratio of variance to bound is 1.0057. An independent recomputation on 300 random networks with common world noise gives the smallest ratio 1.0003. On 300 random closed sets of 3 to 9 agents with a common sensing gain and symmetric weights (reversible $M$), the disagreement never exceeds (21); the largest ratio is 0.2423, and 0.2141 in the independent recomputation on 300 networks.

**Check S47.14** (Closed pure-listening sets)**.** Static world with prior variance 1, five agents, channel noise $R_c=1$, independent initial errors of unit variance (initial pooled information $\frac{1}{2}\ln 6=0.896$ nats), level-0 gains from the claimed precision as in Proposition S3.12(c). After $2\times10^4$ rounds: on the complete graph the mean-square error is 0.2599, the disagreement $1.67\times10^{-5}$, the claimed variance $1.25\times10^{-5}$, the believed information 5.645 nats and the group's true information 0.789 nats; on the directed cycle 0.3290, $1.01\times10^{-4}$, $5.00\times10^{-5}$, 4.952 and 0.698. A star in which the hub listens to all and the leaves to the hub, which (c) does not cover, also reaches consensus, on a hub-weighted average: 0.3445, $4.58\times10^{-5}$, $1.25\times10^{-5}$, 5.645 and 0.681. In a stationary world with $F=0.98$, $Q=1$ ($P_0=25.2525$) and channel noise $R_c=0.2$, closed complete groups of pure listeners of sizes 4, 10 and 30 with fixed per-edge message gains 0.2, 0.09 and 0.03, not derived from a claimed precision, have mean-square errors 25.4160, 25.3006 and 25.2616, all above $P_0$, disagreements 0.0479, 0.0283 and 0.0101, and echo ratios (mean-square error over disagreement, with $\Psi=1$) 530, 895 and 2495.

**Check S47.15** (Complete group with law headers)**.** Stationary values for Example S3.13 ($F = 0.995$, $Q = 0.01$, $R_i^s = 1$, $R_c = 0.2$, $P_0 = 1.0025$, isolated predicted error 0.1000), from the claimed Riccati fixed point and the Lyapunov equation of the actual errors; the shared part is the variance of the members' mean error $n^{-1}\sum_i e_i$, which differs from the common-mode floor quoted below the table, and the centralized optimum is the error of a correlation-aware fuser of all members' data.

| $n$ | claimed $\hat{P}^-$ | actual error | shared part | disagreement | centralized | believed/true |
|---|---|---|---|---|---|---|
| 1 | 0.1000 | 0.1000 | 0.1000 | 0 | 0.1000 | 1.000 |
| 2 | 0.0488 | 0.1590 | 0.1535 | 0.0219 | 0.0733 | 1.626 |
| 4 | 0.0318 | 0.2225 | 0.2157 | 0.0181 | 0.0539 | 2.196 |
| 8 | 0.0231 | 0.3016 | 0.2956 | 0.0136 | 0.0400 | 2.807 |
| 16 | 0.0179 | 0.4041 | 0.3996 | 0.0096 | 0.0301 | 3.514 |
| 32 | 0.0146 | 0.5283 | 0.5252 | 0.0064 | 0.0232 | 4.322 |
| 64 | 0.0126 | 0.6607 | 0.6587 | 0.0041 | 0.0184 | 5.179 |

The last column is the ratio of believed to true information about the current state accumulated by a member: believed information is $\frac{1}{2}\ln(P_0/\hat{P}^-)$ and true information is $I(\eta_t; m_i(t))$ under the actual stationary joint law. Per round, believed information is the claimed one-round gain $\frac{1}{2}\ln(\hat{P}^-/\hat{P}^+)$, with $\hat{P}^+$ the claimed filtered variance, and available information is $I(\eta_t; s_i(t), (y_{ij}(t))_{j\neq i} \mid m_i(t))$ under the actual law; their ratio rises from 1.475 at $n = 2$ to 4.237 at $n = 64$. At the claimed fixed point the floor $Q/(1 - F^2(1 - k^s)^2)$ of Proposition 3.8 is 0.1161 at $n = 2$ ($k^s = 0.0392$), 0.2781 at $n = 8$ and 0.6567 at $n = 64$: below the actual errors and above the isolated error 0.1000.

## S47.3 Checks for Section 4

All computations below are exact Gaussian linear algebra in double precision (Lyapunov and Stein solves, Schur complements, linear solves, spectra, Fourier inversion of translation-invariant symbols, and exhaustive enumeration of small regions); no stochastic simulation enters any of them. Where two independent computations were made, both are reported.

**Check S47.16** (Path blankets, the collider and the initial slice)**.** For Theorem 4.3: in a system with two-dimensional interior, active, sensory and exterior blocks, sensory channels that do not read the interior, active channels that do not read the exterior, independent noise and spectral radius 0.8, the window leak from a conditionally independent start, computed from the path precision, is at most $1.8 \times 10^{-15}$ nats for windows up to 20 rounds, and the stationary leak rate computed from the spectral density is $3.6 \times 10^{-19}$. With correlated sensory and active noise, so that the collider term has norm 0.468, the window leaks are 0.060, 0.184 and 0.391 at $T = 2$, 5 and 10 and the leak rate is 0.0413 nats per round. An independent computation from window covariances found leaks below $8.9 \times 10^{-15}$ in 20 random systems with the split at $T = 2$, 5, 9, and positive leaks in all 20 after one collider entry was added. For Example S4.2: the window leaks from a conditionally independent start grow by exactly 0.143841, 0.087177 and 0.003420 nats per round at $(\alpha_1, \alpha_2) = (1, 1)$, $(0.5, 2)$ and $(0.3, 0.3)$, equal to (S13); a collider with persistent variables (autoregression coefficients 0.6, 0.3, $-0.5$ and weights 0.8, 0.7) has a stationary leak rate of 0.059151, equal to the slope of the window leak to six digits. For part (c): on a ring of ten agents with random nearest-neighbor weights and largest row sum 0.8, a partition with two interior agents and a shell two agents thick has window leak equal to the initial-slice term, $1.085 \times 10^{-4}$ nats for $T = 2$ to 5 from the stationary start, and below $10^{-15}$ from a product start; with a shell one agent thick whose agents read both sides the leak grows by about $2.1 \times 10^{-3}$ nats per round while the initial-slice term stays near $8.8 \times 10^{-3}$. An independent instance gave equality with the initial-slice term to seven digits at

$T = 1$, 2, 3, 5 and a one-layer increment of $1.19 \times 10^{-3}$ per round. For Proposition S4.3: on a ring of 30 agents with random local weights, an arc of 6 agents has window leak ($T = 4$, conditionally independent start) $1.06 \times 10^{-2}$ with a shell one agent thick and $7 \times 10^{-15}$ with shells two or three agents thick. For Proposition S4.4 and Theorem 4.23(d): in the scalar feed-forward chain with unit noise variances, the stationary instantaneous leak is 0.03892524, 0.00228861 and 0.01314003 nats at (exterior persistence, coupling) $= (0.9, 1)$, $(0.5, 0.7)$ and $(0.95, 2)$, computed both numerically and from the closed form of the stationary moments, while the path leak from a conditionally independent start is zero; at $t = 8$ the instantaneous leak 0.0341 lies below the memory bound 0.631, and the leaks given the blanket history and given the augmented blanket are below $1.5 \times 10^{-15}$ in magnitude. For a reversible three-variable system with noise variances 0.7, 1.4 and 0.9, the interior-exterior entry of the stationary precision equals minus the collider term, 0.1322875656, to ten digits.

**Check S47.17** (Records in the partition)**.** For Proposition 4.4, at agent level: a chain of four agents in the roles interior, shell, shell, exterior, with a record, has leak rate 0 without the record; $8.37 \times 10^{-4}$ when the record is written by the interior agent, read by the exterior agent and marginalized; 0 when the same record is kept and placed in the blanket; 0.0279 when it is placed in the exterior; 0 when it is written by the outer shell agent and read by the exterior agent; and 0 for an internal archive written by the interior agent and read by the inner shell agent. A shared archive written by the interior and exterior agents leaks 0.0273, 0.0022 and $2.7 \times 10^{-4}$ when placed in the exterior, placed in the blanket or marginalized, and a shared archive written by the inner shell agent and the exterior agent leaks 0 in the exterior or in the blanket.

**Check S47.18** (Thick shells and precision decay)**.** For Theorem S4.5: on a ring of 80 agents with random local weights and row sums at most 0.9, $\lambda_{\min}(\Sigma) = 0.584$, and the covariance bound with $m_\Sigma = 0.499$ and $B_\Sigma = 3.625$ guarantees the rate $m_\Pi = 0.010$ with $\mathcal{C}(m_\Pi) = 0.292$; the precision bound holds for every pair, the largest ratio of $|\Pi_{ij}|$ to the bound being 0.40, while the fitted decay rates are 0.939 per hop for the precision and 0.934 for the covariance. For an arc of 10 agents with shells of width $h$ on both sides the instantaneous leak is $1.14 \times 10^{-2}$, $3.65 \times 10^{-4}$, $3.34 \times 10^{-4}$, $1.30 \times 10^{-5}$, $1.93 \times 10^{-7}$, $3.05 \times 10^{-12}$ and $1.15 \times 10^{-14}$ at $h = 1$, 2, 3, 5, 8, 12 and 16, decaying at 1.99 per unit width against twice the fitted precision rate, 1.88. On a $14 \times 14$ torus with random local weights the fitted rates are 0.73 for the covariance and 0.81 for the precision. A ring with translation-invariant directed passing, self weight 0.3 and neighbor weights 0.45 and 0.15, which is normal but not symmetric, has stationary precision equal to $I - WW^\top$ to $1.9 \times 10^{-15}$ and entries at most $1.6 \times 10^{-15}$ at distance three or more.

**Check S47.19** (Markov length and correlation length)**.** For Proposition 4.6(a): on a path of 401 agents with $a = 0.45$ (so $\lambda = 0.9$) and $q_{\text{src}} = 1$, Lyapunov solves of the full network give covariances exactly zero at hop distance three or more, precision decay per listener step 1.76275, 0.96242, 0.31492 and 0.09996 against $\operatorname{arccosh}(1 + \psi/2)$ at $\psi = 4$, 1, 0.1 and 0.01, and Markov lengths fitted from conditional mutual information over widths up to 30 of 0.567, 1.036, 3.154 and 9.36 hops against 0.567, 1.039, 3.175 and 10.00; at $\psi = 0.001$ the window is pre-asymptotic (21.7 fitted against 31.6). The condition number of $\Sigma$ from these solves is 2.000 and 5.000 at $\psi = 4$ and 1, and 49.38, 493.8 and 4938 at $\psi = 0.1$, 0.01 and 0.001. The closed form $\operatorname{cond}(\Sigma) = 1/(a^2\psi) = 4.938/\psi$ of the proposition needs $q_{\text{lis}} \le (1 - 4a^2)q_{\text{src}}$, that is $\psi \le (1 - 4a^2)/a^2 = 0.938$ at $a = 0.45$, and it gives the last three values; at $\psi = 4$ and 1 that condition fails and the values are $1 + 4/\psi$, as the proof of the proposition shows. An independent computation on 241 positions found the precision entry between adjacent listeners equal to the closed form (0.84356 at $\psi = 1$) and Markov lengths 0.567, 1.039, 3.175 and 9.993 hops, fitted over the last ten even widths up to 40. For part (b): the partial-correlation

formula matches direct computation to ten digits at $\alpha \in \{0.5, 0.9, 1\}$ and $h \in \{1, 4, 10\}$; at $\alpha = \frac{1}{2}$ the conditional mutual information has log-slope $-1.3863 = 2\ln\alpha$, and at $\alpha = 1$ it is $2.56 \times 10^{-3}$ at $h = 12$, equal to $-\frac{1}{2}\ln(1 - 1/196)$. Fitted slopes over $h \le 12$ at $\alpha = 0.9$ and 0.99 ($-0.2688$ and $-0.1979$ against $2\ln\alpha = -0.2107$ and $-0.0201$) are pre-asymptotic, while the exact formula reproduces every computed value. For part (c): on a ring of 120 agents with nearest-neighbor passing at leak $\varepsilon = 10^{-3}$, the conditional mutual information between an agent and the far region given a shell of width $h$ is $5.4 \times 10^{-2}$ at $h = 1$ for $c = 0.3$ and below $3 \times 10^{-14}$ in magnitude at $h = 2$, 3 and 6 for $c = 0.3$ and $c = 0$, with $\xi = 18.7$ hops at $c = 0.3$ and 22.4 hops at $c = 0$. On a non-normal ring with directed drift and heterogeneous noise, $\xi = 3.26$ hops while the conditional mutual information decays at 1.45 per unit width, $\xi_M \approx 0.69$ hops.

**Check S47.20** (The massless limit)**.** For Numerical observation S4.9(i): symbolic algebra confirms that the closed-form precision symbol of the Jordan-block agents satisfies the Stein equation, inverts the covariance symbol and has the stated determinant, each residual being exactly zero. Fourier inversion on 8192 sites with $a = 0.3$ reproduces the predicted precision decay rate to three or four digits in ten parameter sets, for example 0.57533 against 0.57504 at $(\varepsilon, \tilde{a}) = (0.1, 0.05)$, 0.21011 against 0.21016 at $(0.001, 0.05)$ and 0.10276 against 0.10289 at $(0.003, 0.003)$. The covariance symbol has a triple pole, so its envelope carries a polynomial prefactor; fits without and with a quadratic prefactor bracket the predicted covariance rate in every case, for example $0.0497 < 0.0577 < 0.0597$ at $\varepsilon = 0.001$. The predicted ratio $\xi_M/\xi$ is 0.137 at $(\varepsilon, \tilde{a}) = (0.001, 0.05)$, outside the row-sum class (row sum 1.025), 0.4858 for $\tilde{a} = \varepsilon$ inside the class, and 0.5000 for $\tilde{a} \ll \varepsilon$. Direct conditional mutual information on rings of 128 two-component agents decays at 1.1821, 0.6810 and 0.4988 per hop of width against twice the precision rate, 1.1803, 0.6791 and 0.4994. For (ii): three families of 200 scalar agents with leaks $\varepsilon \in \{0.1, 0.03, 0.01, 0.003\}$, with heterogeneous random weights and uniform noise, with directed drift weights $(0.3, 0.5, 0.2)$ and noise variances drawn in $[1, 2]$, and with both, give $\xi$ from about 1.7 to between 8.2 and 15.7 hops; $\xi_M/\xi$ lies in $[0.25, 0.63]$ on open chains and on rings decreases monotonically in all three families, from 0.44, 0.36 and 0.44 to 0.26, 0.18 and 0.18. For (iii): in 150 random translation-invariant two-component systems in the row-sum class with isotropic noise, the ratio of the precision decay length to the correlation length had median 0.92 and maximum 1.93, attained where $\xi = 0.42$ hops and the precision decays at 1.24 per hop; with noise covariances $\mathrm{diag}(1, 10)$ and $\mathrm{diag}(1, 100)$ the medians were 1.16 and 2.22 and the maxima 5.3 and 11.4.

**Check S47.21** (The world-conditional law)**.** For Proposition 4.8: for a six-agent level-0 network with mistranslation coefficients up to 0.049 in magnitude and $\rho(W) = 0.774$, the error at round $T$ and the world path were built as explicit linear maps of independent inputs and the joint Gaussian was conditioned on the path by a Schur complement; the conditional covariance equals $\sum_t W^t Q^\zeta (W^\top)^t$ to $2.4 \times 10^{-15}$ and the stationary Lyapunov solution to $3.5 \times 10^{-15}$, while the world adds correlations up to 0.978 to the unconditional covariance. An independent computation with four agents, mistranslation up to 0.05 and twelve rounds reproduces the local-noise covariance to $5.0 \times 10^{-16}$, the world adding up to 0.59.

**Check S47.22** (Diffusive world-conditional precisions)**.** For Theorem 4.9: in 300 random reversible instances with 4 to 11 agents, random noise variances and row sums at most $\lambda \in [0.5, 0.99]$, (29) matches the inverse of the Lyapunov solution to relative error $2.8 \times 10^{-15}$, off-diagonal entries are at most $1.1 \times 10^{-15}$, and $q_i \varepsilon_i^{\Pi} - (1 - \lambda^2) \ge 0.0127$. In 200 normal instances (circulants with drift, and combinations of a random permutation matrix, its transpose and the identity with nonnegative coefficients) (30) matches to $3.8 \times 10^{-15}$. On a $16 \times 16$ torus with $(c, a, q) = (0.2, 0.15, 0.7)$ the couplings are 0.08571, 0.03214 and 0.06429 and the leak 0.51429, equal to $2ca/q$, $a^2/q$, $2a^2/q$

and $(1-\lambda^2)/q$, and the entry at distance three is $9\times10^{-16}$. An independent computation with 300 reversible instances built from symmetric matrices gives agreement to $3.4\times10^{-15}$ and $\min_i(q_i\varepsilon_i^{\Pi}-(1-\lambda^2))=0.0151$, and with 200 normal instances to $1.2\times10^{-14}$.

**Check S47.23** (Non-diffusive precisions)**.** For Example S4.12: among 2000 random directed $W\geq0$ on four agents with row sums below one and $Q^{\zeta}=I$, 674 have a positive off-diagonal entry of $\Pi^{c}$, the largest 0.043 in an instance of common-listener type; an independent sampling scheme gives 365 of 2000, the largest 0.189. A three-agent common-listener instance gives $\Pi^{c}_{12}=+0.0124$. In six-agent systems in which the two silent agents have a common listener and also feed separate listeners that share a third silent source, $\Pi^{c}_{12}<0$ in 1328 of 2000 draws. The signed instance has $\rho(W)=0.7$, $\Pi^{c}_{13}=+0.12$ and $\Pi^{c}_{12}=-0.16$. A directed three-agent chain with rows $(0.5,0,0)$, $(0.45,0.4,0)$ and $(0.3,0.3,0.3)$ has off-diagonal entries $-0.149$, $-0.096$ and $-0.226$, and a directed relay line of level-0 agents has none positive.

**Check S47.24** (Marginal precisions)**.** For Lemma S4.30: in 200 random diffusive media of 10 to 39 agents with random regions, the members' marginal precision has no positive off-diagonal entry, its leaks exceed the original ones by between $-1.0\times10^{-15}$ and the cut coupling plus $1.3\times10^{-15}$, the leak formula holds to $2.3\times10^{-15}$, and its inverse equals $\Sigma_{CC}$ to $3.6\times10^{-15}$. For Remark S4.13: on a ring of 2000 agents with message weight 0.2, leak 0.002 and one record per site, the agents' marginal precision has no positive off-diagonal entry with a durable local record and with a perishable record read at a distance when records are noiseless; with the durable record and record noise variance 1 it has 11 positive entries, the largest $7.6\times10^{-3}$, so it is not diffusive.

**Check S47.25** (Declared geometry)**.** For Remark S4.16: two disjoint cliques of $n$ agents with $(n,a/\varepsilon^{\Pi})=(4,10)$, $(3,5)$ and $(4,0.4)$ have clique contrasts 41.000, 16.000 and 2.600. Enumeration of every region of the exchangeable comparison population gives the baselines 35.286, 13.000 and 2.371, attained by the whole population and equal to the closed form, with significances 1.162, 1.231 and 1.096; under the graph clause the baseline equals the clique contrast and no region is a collective. For two cliques of six agents with coupling 0.2 joined by a perfect matching of coupling 0.05 at leak 0.1, the block contrast is 8.6667; the declared graph has 36 edges with mean coupling 0.175, and enumeration of its 4083 regions gives the baseline 4.5000, attained by the whole comparison population, so the significance is 1.926; the exchangeable baseline is 12.4545. For the sentence after Definition 4.11: in the two-group model, whose groups at the split are isolated cliques, the exchangeable population with the split's mean coupling and leak has baseline $1+\frac{2(n-1)}{2n-1}(r(A)-1)$, which at $n=4$ and $r(A)=8.7329$ is 7.6282, giving significance 1.1448; a first computation gave the same values, and the model's own exchangeable state, solved at full precision from the closed forms of Lemma 7.7, has baseline 7.9543. An independent computation reproduced the numbers for the cliques, for the matched cliques and the value 7.6282. Over the 162 points of Numerical observation S7.6, with both calibration conditions solved from the same closed forms (one root each at every point, the split's mean-square error at the base point agreeing with a direct Lyapunov solve), the baseline of the model's exchangeable state exceeds that of the exchangeable population with the split's mean coupling and leak at every point, by a factor between 1.0049 and 1.1124, and the split exceeds both, with significance between 1.0095 and 1.4148 against the first and between 1.0303 and 1.4762 against the second.

**Check S47.26** (Whole populations)**.** For Proposition S4.17(b), with heterogeneous leaks drawn in $[0.01,1]$: in 300 rings of ten agents with independent couplings at displacements one and two, 300 tori of $4\times5$ sites with independent couplings along both axes and one diagonal, 300 sparse populations of ten agents compared with the exchangeable population, and 300 weighted hypercubes of eight

agents compared under the graph clause, $\varkappa_{\mathrm{in}}(V)$ never exceeded $\varkappa_{\mathrm{in}}(V_{\mathrm{hom}})$, the smallest margins being 0.086, 0.14, 0.81 and 0.25. A first computation with 200 rings, 200 tori and 200 exchangeable populations found no exception either. For (c) and (d): the path of (d) has $\lambda_2(L^{\Pi}) = 0.600000$ against 0.585786 and $R^{\star}_{\mathrm{hom}} = 4$, attained by an end pair, so (c) guarantees a collective for $\varepsilon^{\Pi} \le 0.2$; enumeration of all its regions gives $r(V) = 61.000$ against the baseline 59.579 at $\varepsilon^{\Pi} = 0.01$, 7.000 against 6.858 at 0.1, equality at 2.800 when $\varepsilon^{\Pi} = \frac{1}{3}$, and 1.600 against 2.000 at 1, and bisection places the threshold at $\frac{1}{3}$ to ten digits. In 100 boxes of $2 \times 4$ sites with uniform leak, random vertical couplings and horizontal couplings stronger in the middle, 41 had $\lambda_2(L^{\Pi}) > \lambda_2(L_{\mathrm{hom}})$, and all 41 were collectives at some of nine leaks between $10^{-4}$ and 10; in these boxes and in 60 boxes of $3 \times 3$ sites with independent couplings, none of which had the inequality, there was no violation of (a), of the sufficient bound of (c) or of the equivalence at a given leak. In exact rational arithmetic the path of (d) has spectrum 0, $\frac{3}{5}$, $\frac{9}{5}$ and $\frac{18}{5}$, the comparison path has $\lambda_2 = 2 - \sqrt{2}$, and $1 + \frac{3}{5}/\varepsilon^{\Pi}$ equals the contrast of an end pair, of the middle pair and of a triple of the comparison path exactly at $\varepsilon^{\Pi} = \frac{1}{3}$, $\frac{3}{2}$ and 3. The transitivity on agents that (b) requires with heterogeneous leaks cannot be dropped: a star with three leaves and unit couplings, leak 0.1 on each leaf and none at the center, declared as its graph, which is edge-transitive but not vertex-transitive, has $\varkappa_{\mathrm{in}}(V) = 1.1$ against $\bar{\varepsilon} + \lambda_2(L_{\mathrm{hom}}) = 1.075$, and $r(V) = 14.667$ exceeds both its baseline 14.333 and $\varepsilon_V^{\max}/\bar{\varepsilon}_V = \frac{4}{3}$, so it is a collective of its comparison medium.

**Check S47.27** (Homogeneous lattices and block media)**.** For Theorem 4.13 with nearest-neighbor coupling on $\mathbb{Z}^2$: over 79 regions (squares of side 2 to 14 and random connected regions of 2 to 400 sites) every inequality holds; the largest $r_\star$ is 1.000, at the $2 \times 2$ square, squares of side 3, 6, 10 and 14 give 0.75, 0.40, 0.24 and 0.18, random regions with at least 8 sites give 0.02 to 0.12, and the largest contrast is 1.000 at $\varepsilon^{\Pi} \in \{0, 0.01, 0.1\}$; the supremum over $n$ of the bound is 6.63. An independent computation over 57 regions at $\varepsilon^{\Pi} \in \{10^{-12}, 0.01, 0.1\}$ found no violation. For Proposition S4.19: for $6 \times 6$ blocks with $a_{\mathrm{in}} = 1$, $r_\star = 0.40$, 4.02, 40.2 and 402 at $a_{\mathrm{out}} = 1$, 0.1, 0.01 and 0.001, and the contrast is 0.40, 3.98, 35.1 and 161 at $\varepsilon^{\Pi} = 10^{-3}$ and 0.48, 2.21, 3.45 and 3.66 at $\varepsilon^{\Pi} = 0.1$. In a $3 \times 3$ arrangement of $6 \times 6$ blocks at $\varepsilon^{\Pi} = 10^{-3}$ the gap $\lambda_{10}(\Pi)/\lambda_9(\Pi)$ is 1.11, 3.27, 24.9 and 134.6 for the same couplings, and (S17) guarantees 18.8 at $a_{\mathrm{out}} = 0.01$.

**Check S47.28** (Observed baselines)**.** For Numerical observation S4.20: in one dimension every region connected by the kernel with up to nine sites was enumerated (for a kernel of range two such a region lies in a window of $2n - 1$ sites), and in two dimensions every subset of a $4 \times 4$ window containing a fixed corner; for each region $r$ was computed at the medium's own leak and $r_\star$ from the couplings. For nearest-neighbor coupling on $\mathbb{Z}$ at leak $10^{-3}$ the largest contrast by size is 1.999, 1.499, 1.171, 0.955 and 0.804 for $n = 2$ to 6. For nearest-neighbor passing at $(c, a) = (0.3, 0.3)$ it is 1.000, 1.279, 1.186, 1.108, 1.040, 0.993, 0.960 and 0.937 for $n = 2$ to 9. For the two-dimensional kernel of Table 4 the largest value over $n$ of the bound (35) is 17.523, attained at $n = 17$.

**Check S47.29** (The size bound)**.** For Theorem 4.14: on $\mathbb{Z}^2$, for the precision kernel of nearest-neighbor passing at $(c, a, q) = (0.2, 0.15, 0.7)$, with couplings at axis distances one and two and on diagonals and $\varsigma_1^2 = \varsigma_2^2 = 0.6857$, and for a random anisotropic kernel of range three with $\varsigma_1^2 = 6.47$ and $\varsigma_2^2 = 8.37$, 39 regions (squares, two-row strips, Eden clusters of 3 to 250 sites and sparse random sets) at three leaks each gave no violation; the largest ratio $\lambda_2(L_C)/(\varsigma_d^2/(2\,\mathrm{Var}_d(C)))$ is 0.77. In a heterogeneous medium on a $30 \times 30$ window, with random couplings of range two and random leaks in $[0.01, 0.05]$, 150 random regions gave no violation of either form of (37), with the Euclidean metric; an independent computation on a $24 \times 24$ window with 150 regions found largest ratios of $r$ to the bound of 0.022 for the coordinate form and 0.010 for the metric form. Isolated boxes with nearest-neighbor coupling give $\mathrm{sd}_d(C)(r(C) - 1)^{1/2}/\xi_{2,d} = 0.8557$, 0.9037, 0.9067 and 0.9069 at

sides 4, 16, 64 and 256 in one dimension and 0.8557, 0.8940 and 0.9037 at sides 4, 8 and 16 in two, against $\pi/\sqrt{12} = 0.9069$.

**Check S47.30** (The bridge theorem)**.** For Theorem 4.18: the two $6 \times 6$ grids of Table 5, with closed-form $\lambda_2$, linear solves for the block averages and a minimum spanning tree for the single-linkage ultrametric, give the entries of the table, and an independent computation reproduces every printed digit; the row at $\varepsilon^{\Pi} = 0.1$, $r(C_1) = 3.6192$, $r(C_1 \cup C_2) = 1.0323$ in $[1.0312, 1.0333]$, $\mathrm{Corr}(\bar{e}_1, \bar{e}_2) = 0.0160$ in $[0.0157, 0.0164]$ and $\mathcal{U}(C_1) = 3.474$, was computed twice, once as the other rows and once with the ultrametric from minimax paths over all 72 agents; $\lambda_2(L^{\Pi}) = 0.003234$ against $2\bar{a}^{\partial} = 0.003333$ and $2\bar{a}^{\partial} - \delta_{\cup} = 0.003123$, so $r(C_1 \cup C_2) = 1 + \lambda_2(L^{\Pi})/\varepsilon^{\Pi}$ equals 2 at $\varepsilon^{\Pi} = 0.003234$. The block-average correlation, computed by linear solves at 1401 leaks between $10^{-6}$ and 10, decreases in the leak and equals $\frac{1}{2}$ at $\varepsilon^{\Pi} = 0.001617$ (by root finding), inside the interval $[0.001564, 0.001667]$ that the two sides of (42) give at $\theta_0 = \frac{1}{2}$. Over $10^{-5} \le \varepsilon^{\Pi} \le 10$ the relative width of the bracket (41) is at most 0.0647 and the width of the bracket (42) at most 0.0160. The effective-resistance contrast of one grid is $\mathcal{U}^R(C_1) = 16.64$, above the bound 9.20 of (iii)(a), and $\mathcal{U}(C_1)/(\frac{N\varepsilon^{\Pi}}{2}\mathcal{U}^R(C_1)) = 0.778$, 0.970, 0.997 and 0.9997 at $\varepsilon^{\Pi} = 10^{-3}$, $10^{-4}$, $10^{-5}$ and $10^{-6}$. At strong grounding $\mathcal{U}(C_1) = 4.071$, 4.459, 4.586 and 4.603 at $\varepsilon^{\Pi} = 1$, 10, 100 and 1000, against $\ln 100 = 4.605$, while $r(C_1)$ falls to 1.0003. For an equitable partition into three blocks of seven agents (a cycle inside each block, one partner in each other block), $\Phi^{\top}\Sigma\Phi = \Pi_{\Phi}^{-1}$ to $2 \times 10^{-11}$ and $4 \times 10^{-15}$ at $\varepsilon^{\Pi} = 10^{-3}$ and 0.1, and one added cross edge produces errors of $2.4 \times 10^{-2}$ and $6.5 \times 10^{-3}$. For rings of $M = 2$, 4, 8 and 16 path blocks of four agents with internal coupling 1, one edge of weight 0.05 between consecutive blocks and $\varepsilon^{\Pi} = 0.01$, $\|\mathcal{D}\| = 1.21 \times 10^{-3}$, $1.98 \times 10^{-3}$, $1.98 \times 10^{-3}$ and $2.04 \times 10^{-3}$, the spectral bound of (ii) is $4.20 \times 10^{-3}$ in all four cases, and the same bound with $\sigma_{\mathrm{cut}}^2$ in place of the squared spectral norm is $4.20 \times 10^{-3}$, $1.05 \times 10^{-2}$, $2.10 \times 10^{-2}$ and $4.20 \times 10^{-2}$. Over 246 random closed two-block media with heterogeneous cut couplings and $\varepsilon^{\Pi} \in [10^{-4}, 1]$, and over 300 more in an independent computation, (41) and (42) were never violated; the median width of the correlation bracket is $3.1 \times 10^{-3}$ and $5.9 \times 10^{-3}$ in the two sets.

**Check S47.31** (The baseline of the two-grid population)**.** For Numerical observation S4.22: the comparison box has 66 horizontal pairs with coupling sum 60.06 and 60 vertical pairs with coupling sum 60, so $\bar{a}_{\mathrm{h}} = 0.91$ and $\bar{a}_{\mathrm{v}} = 1$ under the lattice clause and the edge mean is $120.06/126 = 0.9528571$ under the graph clause; $\lambda_2(L_{\mathrm{hom}}) = 0.0620149962$ and 0.0649356396 from the spectra, equal to $\bar{a}_{\mathrm{h}}(2 - 2\cos(\pi/12))$ and $0.9528571(2 - 2\cos(\pi/12))$. Connected regions were listed by growing connected sets one adjacent site at a time, which lists every connected set because a connected set of $k \ge 2$ sites contains a connected set of $k - 1$ sites, obtained by removing a leaf of a spanning tree. There are 72, 126, 328, 897, 2572, 7604, 22936 and 69892 connected regions of 1 to 8 sites, the same sets as those of a second extension scheme, which was itself checked against all subsets of a box of $3 \times 4$ sites. The bound (U1) and the cut bound of Lemma S28.6 hold on every enumerated region, and all its bounds hold on every enumerated complement. For each competitor the two sets of leaks of the observation were computed from the linear inequalities, with the analytic bounds padded by a relative $10^{-12}$ in the safe direction; their union was $(0, \infty)$ for all 104944 competitors in both clauses. On 1801 leaks between $10^{-6}$ and $10^{3}$ the right side of (S18) exceeds every other competitor by a relative margin of at least $2.7 \times 10^{-4}$ under the lattice clause and $4.8 \times 10^{-4}$ under the graph clause; the margin shrinks only because every contrast tends to one as the leak grows. An independent computation, with its own enumeration and its own implementation of the bounds, which it checked again on every enumerated region and complement, found no failure in either clause and reproduced the interval ends $5.059949 \times 10^{-4}$ and 0.422333 (lattice clause) and $5.375105 \times 10^{-4}$ and 0.367245 (graph clause), unchanged when the intersection is taken over every competitor; the admissibility thresholds 0.0620150 and 0.0649356; the largest significances 3.3211 at $\varepsilon^{\Pi} = 0.01216$ and 3.1956 at

0.01255; the baselines 621.15, 7.2015, 1.8199 and 1.4425 (lattice clause) and 650.36, 7.4936, 1.9050 and 1.4879 (graph clause) at the four leaks of Table 5, and 19.60 at $\varepsilon^{\Pi} = 1/300$, the lower end of the window; and the thresholds 0.0661151 and 0.0694451 for the $2 \times 2$ squares at the corners of the box, with largest ratios to the baseline 1.04916 at $\varepsilon^{\Pi} = 0.0738$ and 1.00832 at $\varepsilon^{\Pi} = 1.380$. A third computation, with an enumeration of its own validated against all subsets of the box of $3 \times 4$ sites, reproduced the certificate and these numbers, found every bound of Lemma S28.6 satisfied on every enumerated region and complement, and found that among the connected regions of the population itself with 2 to 8 sites only these four squares and the four $2 \times 2$ squares at the ends of the seam, with cut 1.005 per member, thresholds 0.0668018 and 0.2261017 (under the graph clause these squares fall below the corner-square term of (S18) up to 0.2261017) and largest ratios 1.04429 and 1.00638, exceed the baseline at some leak between $10^{-6}$ and $10^{3}$. On a box of $12 \times 3$ sites with unit couplings the method does not close: with regions of up to 8 sites enumerated the widest bracket is $[1.930, 2.764]$ at $\varepsilon^{\Pi} = 0.075$, where the block of $3 \times 3$ sites at an end has contrast 2.633, and with regions of up to 12 sites it is $[2.787, 2.920]$ at $\varepsilon^{\Pi} = 0.040$. Of the 1565 rectangles of 2 to 71 sites of the two-grid population itself, for each of which the two comparisons with the terms of (S18) are linear inequalities in the leak and were solved exactly for all $\varepsilon^{\Pi} > 0$, only the two grids and the eight $2 \times 2$ squares above exceed the baseline at some leak, under both clauses, with the thresholds and largest ratios already stated. Regions of the population with more than eight sites that are neither rectangles nor the union were not examined.

**Check S47.32** (Contrast and coherence)**.** For Proposition S4.24: over 300 random regions in random diffusive media with uniform leak, no violation of (a) or (b) occurred, and the bound (S19) reaches at most 0.758 of the actual value. For (c) and Numerical observation S4.25, isolated regions were set at contrast $r = 5$ by choosing $\varepsilon^{\Pi} = \lambda_2(L_C)/(r-1)$, and $\varepsilon^{\Pi} T_C$ was computed from the spectrum of $L_C$: for intervals of 8, 32, 128 and 512 members it is 0.3440, 0.3556, 0.3563 and 0.3563, below $\pi^2/(6(r-1)) = 0.4112$, and a direct computation of the mean of $\mathrm{Corr}(e_i, \bar{e}_C)^{-2} - 1$ for 8 members gives 0.343985, equal to $\varepsilon^{\Pi} T_C$. The values for square and cubic boxes are those of Numerical observation S4.25, against the logarithmic estimate $(\pi/(2(r-1)))\ln \ell_b$, which gives 0.82, 1.09, 1.36 and 1.63, and the linear estimate $0.2527\pi^2 \ell_b/(r-1)$, which gives 2.49, 4.99, 9.98 and 19.95.

**Check S47.33** (Coarse media)**.** For Lemma 4.20: in 200 random diffusive media of 8 to 14 agents with heterogeneous leaks, partitioned at random into 2 to 5 blocks of unequal sizes, the off-diagonal entries of $\mathsf{P}\Pi\mathsf{P}^{\top}$ equal minus the cut couplings exactly, its row sums equal the block sums of the leaks to $8.9 \times 10^{-15}$, and its smallest eigenvalue over all instances is 0.10. The precision of the block averages given zero disagreement inside every block, computed by conditioning the joint Gaussian, equals $\mathsf{P}\Pi\mathsf{P}^{\top}$ to a relative $4.2 \times 10^{-15}$; the precision of the block averages without conditioning equals $\mathsf{P}\Pi\mathsf{P}^{\top} - \mathrm{diag}(n_\alpha)^{1/2}\mathcal{D}\,\mathrm{diag}(n_\alpha)^{1/2}$ to a relative $1.8 \times 10^{-15}$, with smallest eigenvalue of $\mathcal{D}$ equal to $-2.3 \times 10^{-16}$; and a second random coarsening composes with the first to $1.4 \times 10^{-14}$. For three blocks of seven agents with a cycle of unit couplings inside each block and one partner of coupling 0.05 in each other block, an equitable partition, $\mathcal{D}$ vanishes to $10^{-31}$ at the leaks $10^{-3}$ and 0.1, and one added cross edge makes its largest entry $3.5 \times 10^{-4}$ and $3.2 \times 10^{-4}$. For blocks of equal size three the block-sum and orthonormal conventions agree to $3.6 \times 10^{-15}$ and give equal contrasts, while for blocks of sizes 5, 2, 3 and 2 the coarse contrast of the first two blocks is 1.274 in the block-sum convention and the rates (31) evaluated on the orthonormal block precision give 1.295, although that precision is not diffusive there, its first row sum being $-0.88$. A second computation, over 300 random media of 6 to 15 agents with leaks that may vanish, found the same identities to $2.3 \times 10^{-14}$, including the conditional precision obtained by conditioning instead on the deviations of members from their block averages, and found that with leaks 0.05, 0.1 and 0.3 inside each of

two equitably coupled blocks of three agents the largest entry of $\mathcal{D}$ is $8.5 \times 10^{-3}$, while it vanishes to $10^{-31}$ when the leak is constant on each block.

**Check S47.34** (Nesting in one medium and across levels)**.** For Numerical observation S4.27: the couplings of the nested medium are 1, 0.1, 0.01 and 0.001 between agents whose smallest common block has level 1 to 4. The coarse media were built as $\mathsf{P}_l \Pi \mathsf{P}_l^{\top}$, their comparison media by giving every displacement class of the boxes of 16, 8 and 4 units its mean coupling and every unit the mean leak, and their baselines by enumerating all 65519, 247 and 11 regions. Couplings and leak are precision-side quantities, $a_{ij}^{\Pi}$ and $\varepsilon^{\Pi}$. At $\varepsilon^{\Pi} = 0.1$ the baselines are 2.4239, 1.7893 and 1.3006, the blocks of 2, 4 and 8 agents have contrasts 6.0345, 3.3784 and 1.6667 in the level-0 medium, and the same numbers are their coarse contrasts in the media of levels 0, 1 and 2. Exact leak intervals, from the linear inequalities against every competitor, are the following: in the level-0 medium the pairs are collectives for $\varepsilon^{\Pi} > 0.0107$, the blocks of four for $0.0124 < \varepsilon^{\Pi} < 0.2945$ and the blocks of eight at no leak, while level by level the three block sizes are collectives for $\varepsilon^{\Pi} > 0.0107$, $\varepsilon^{\Pi} > 0.00777$ and $\varepsilon^{\Pi} > 0.00573$. A first computation gave the same baselines, contrasts and intervals, and a third, with its own construction of the coarse and comparison media, reproduced them.

**Check S47.35** (Second-order precision of a weakly coupled linear diffusion)**.** For the expansion of the stationary precision in Section 17.2: the drift is $-I + gA$ on six variables with noise covariance $2\sigma^2 I$ and $\sigma = 0.7$, the stationary covariance is the solution of the Lyapunov equation from a direct solver, and the precision is its inverse. For a Gaussian random $A$ scaled to spectral norm one, the largest entry of the difference between the exact precision and the second-order form is $1.58 \times 10^{-3}$, $2.00 \times 10^{-4}$, $2.51 \times 10^{-5}$, $3.13 \times 10^{-6}$ and $3.92 \times 10^{-7}$ at $g = 0.2$, 0.1, 0.05, 0.025 and 0.0125, with successive ratios 7.91, 7.98, 7.995 and 7.999. In a second instance with two interior, two blanket and two exterior variables, no direct coupling between interior and exterior, and each blanket variable reading one side only while driving both, the interior–exterior blocks of $A + A^{\top}$ and of $A^{\top}A$ vanish identically, and the interior–exterior block of the exact precision differs from $-(g^2/4\sigma^2) \sum_b A_{\mathsf{I}b} A_{\mathsf{E}b}^{\top}$ by $4.3 \times 10^{-4}$ at $g = 0.2$ and by $1.1 \times 10^{-7}$ at $g = 0.0125$, with successive ratios 7.84, 7.93, 7.97 and 7.99. For a normal $A$ whose skew part has spectral norm 0.17, built as an orthogonal conjugate of rotation-scaling blocks, the exact precision equals $\sigma^{-2}[I - \frac{g}{2}(A + A^{\top})]$ to $6 \times 10^{-15}$ at $g = 0.2$, 0.5 and 0.9.

## S47.4 Checks for Section 5

Each check below evaluates exact Gaussian linear algebra, exact stationary laws of finite Markov chains or small Monte Carlo experiments on the instances stated. Where a second, independent implementation of the same statement was run, its results are reported in the same check.

**Check S47.36** (Exact reduction and closure)**.** A random stable linear system with six interior variables (update matrix scaled to spectral radius 0.9, random positive definite noise covariance) and a two-dimensional coarse variable was simulated along one sample path, and the right side of (46) was evaluated term by term. With the Mori complement the identity held to $1.8 \times 10^{-14}$ over 24 steps, $\rho(A_{\perp\perp}) = 0.641$, $|\mathrm{Cov}(\mathfrak{f}_t, y_0)| \le 8.8 \times 10^{-14}$, and $\|\Gamma_l\| = 0.43$, 0.097, 0.027, 0.0041 at $l = 1, 2, 4, 8$; over 200 random stable systems the largest $\rho(A_{\perp\perp})$ under the Mori complement was 0.923. With a generic complement $\rho(A_{\perp\perp}) = 5.12$, $\|\Gamma_8\| = 1.9 \times 10^{6}$, and the identity held to $9.6 \times 10^{-8}$ against memory terms of size $3.5 \times 10^{8}$ over 10 steps. The closure defect (47) for a random $\mathsf{P}$ was 0.48335, equal to the directly computed conditional mutual information. The conditional mutual information $I(y_{t+1}; y_{0:t-1} \mid y_t)$ was 0.1662, 0.1968, 0.2185 and 0.21907 at history lengths $t = 1, 2, 5, 12$, nondecreasing and below the closure defect throughout, and the Markov

defect $\mathfrak{d}_{\mathrm{M}}$, its limit, computed from the stationary Riccati equation of the filter, was 0.21907. For the two-dimensional system $A = \left(\begin{smallmatrix} 0.6 & 0.9 \\ 0 & 0 \end{smallmatrix}\right)$, $Q^{\mathfrak{n}} = I$, $\mathsf{P} = \left(\begin{smallmatrix} 1 & 0 \end{smallmatrix}\right)$ discussed after Theorem 5.3, the closure defect was $0.29666 = \frac{1}{2}\ln 1.81$ and $I(y_{t+1}; y_{0:t-1} \mid y_t)$ vanished to round-off (below $10^{-15}$) at history lengths 1, 2, 5 and 10. For $\mathsf{P}$ spanning a slow left eigenspace of the first system, the closure defect by both computations and the conditional mutual information at history length one were below $3 \times 10^{-15}$. For arbitrary real matrices with two retained and three hidden coordinates the representation was confirmed on 50 random instances over 25 rounds (largest deviation $5.6 \times 10^{-9}$, round-off on values that grow), and the filter of Theorem 5.3(d) agreed with brute-force Gaussian conditioning of the joint path law on 20 instances with one retained and two hidden coordinates over six rounds (largest deviations $3.6 \times 10^{-14}$ in the mean and $9.7 \times 10^{-14}$ in the covariance). For a random linear network recursion with two retained boundary coordinates and five eliminated interior coordinates the identity held to $2 \times 10^{-15}$ over 30 steps, the memory decayed by the factor 0.6694 per step against the eliminated block's spectral radius 0.6690, and the rank of the block Hankel matrix was 5 for a generic five-agent eliminated block and 4 when one of its modes was made unobservable.

**Check S47.37** (Markov closure of the collective average)**.** For symmetric drift, a $4 \times 4$ region of the square lattice with unit couplings ($\lambda_2 = 0.586$ for its Laplacian), uniform leak $10^{-3}$, and its left column of four members leaking with cut coupling $\bar{a}^{\partial} \in \{1, 0.3, 0.1, 0.01\}$ gave: the tangent of the angle between $\phi_C$ and the slow eigenvector 0.345, 0.148, 0.0546, 0.0057, against the bound $\sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}})$ of Theorem 5.4(a), which is 1.29, 0.254, 0.0772, 0.0074; the memory weight $1.2 \times 10^{-2}$, $1.7 \times 10^{-3}$, $1.1 \times 10^{-4}$, $1.8 \times 10^{-7}$, each below $\tan^2 \angle \cdot \varpi_1/\varpi_2$ ($2.1 \times 10^{-2}$, $2.0 \times 10^{-3}$, $1.2 \times 10^{-4}$, $1.9 \times 10^{-7}$); the Markov error $8.3 \times 10^{-3}$, $1.2 \times 10^{-3}$, $9.2 \times 10^{-5}$, $1.8 \times 10^{-7}$, against the bound (49), which is 0.71, $8.4 \times 10^{-3}$, $2.6 \times 10^{-4}$ and $3.3 \times 10^{-7}$; and agreement of the reduced equation with the full system through its Laplace transform to $10^{-13}$. In the first case, with $\varkappa_{\mathrm{out}} = 0.251$, $\varkappa_{\mathrm{in}} = 0.587$ and $\sigma_C = 0.433$, the dispersion exceeds the gap $\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}} = 0.336$ and both bounds still hold, although they are loose; the bounds of Lemma 11.10(b), which use the rate $\lambda_{\min}(ZMZ^{\top}) = 0.707$ of $(M, \phi_C)$ in place of $\varkappa_{\mathrm{in}}$, are 0.949 on the tangent of the angle and 0.32 on the Markov error. The same held at leak 0.05, where the bound (49) is 0.78, $1.3 \times 10^{-2}$, $7.0 \times 10^{-4}$ and $4.6 \times 10^{-6}$ against Markov errors $1.0 \times 10^{-2}$, $1.8 \times 10^{-3}$, $2.1 \times 10^{-4}$ and $1.9 \times 10^{-6}$. An independent implementation on 200 random weighted graphs with heterogeneous leaks and boundary couplings found the Laplace-domain identity to $4.7 \times 10^{-15}$, the kernel bound attained at $u = 0$, the sine of the angle at most 81 percent of $\sigma_C/(\varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}})$, the memory weight at most 0.97 of $\tan^2 \angle \cdot \varpi_1/\varpi_2$, and the Markov error at most 0.88 times the memory weight.

For general drift, over 300 random stable directed regions of 6 to 12 members the Lyapunov residual was below $1.7 \times 10^{-14}$ and the Laplace-domain form of the reduced equation held to $2.7 \times 10^{-15}$; the kernel bound of part (b)(i) was used up to 0.969, the eigenvalue bound of (b)(ii) up to 0.395, with $\varpi_1$ real in every case and equal to the fixed-point value to $6.7 \times 10^{-15}$, and the intermediate bound $2|\mathsf{k}||\mathbf{X} + \mathbf{V}\mathsf{k}|/\operatorname{Var}(y)$ of the proof up to 0.35 in the 88 regions where it applies, with no violation. In 300 high-contrast regions (symmetric internal coupling scaled by 5 to 50 plus a random directed part) the intermediate bound was used up to 0.48, and the explicit bound (b)(iii) applied in 293 regions with the Markov error at most 0.012 of it. An independent implementation on 400 stable random directed regions, half of them of high contrast, found the kernel bound used up to 0.971, part (b)(ii) applicable in 369 regions with every claim holding (largest eigenvalue ratio 0.310), and the explicit bound applicable in 201 regions with the Markov error at most $8.4 \times 10^{-3}$ of it.

**Check S47.38** (The symmetric-dispersion counterexample)**.** For random directed attention patterns with the leaks (S20) on 6, 8 and 10 members, $\sigma_{\mathrm{sym}}$ was below $5 \times 10^{-16}$ (zero by construction),

$\sigma_{\mathrm{row}} = \sigma_{\mathrm{col}} = 0.83$, 1.02, 0.76, the drift-based contrast 2.49, 2.63, 1.59, and the Markov error $1.93 \times 10^{-2}$, $1.46 \times 10^{-2}$, $4.85 \times 10^{-2}$. In the eight-member family with symmetric ring coupling scaled by 1, 4, 16 and 64 against a fixed directed pattern, $\sigma_{\mathrm{row}} = \sigma_{\mathrm{col}} = 0.2165$ throughout; the drift-based contrast was 7.18, 13.3, 36.8, 130.5; $\sigma_{\mathrm{col}}/(\varkappa_{\mathrm{in}}^{M} - \varkappa_{\mathrm{out}}^{M}) = 0.117$, 0.059, 0.020, 0.0056; $\|M - M^{\top}\|/\varkappa_{\mathrm{in}}^{M} = 1.07$, 0.58, 0.21, 0.059; and the Markov error $6.6 \times 10^{-4}$, $1.0 \times 10^{-4}$, $5.7 \times 10^{-6}$, $1.4 \times 10^{-7}$, below the intermediate bound of the proof ($2.5 \times 10^{-3}$, $3.1 \times 10^{-4}$, $1.4 \times 10^{-5}$, $3.1 \times 10^{-7}$). An independent implementation reproduced the structure exactly, with $\sigma_{\mathrm{sym}} \leq 3.4 \times 10^{-16}$ and $\Gamma(0) = -\sigma_{\mathrm{circ}}^{2}$ (for instance $-0.4113$ with $\sigma_{\mathrm{circ}} = 0.6413$), and found Markov errors $3.3 \times 10^{-2}$, $5.1 \times 10^{-3}$ and $1.3 \times 10^{-2}$ on 6, 8 and 10 members.

**Check S47.39** (Drift-based and precision-based rates)**.** In the 300 random stable directed regions of Check S47.37 the identity $M = (I + J)\Pi$ held to $7.5 \times 10^{-14}$, and the relative difference between the outward rates computed from $M_{\mathrm{sym}}$ and from $\Pi$ had median 0.18 and maximum 0.62. In the 400 regions of the independent implementation the identity held to $3.0 \times 10^{-14}$, the relative difference had median 0.026 and maximum 0.148, and the second bound of (51) held in every instance. In 4000 further random stable drifts on two to six coordinates, each with a random unit vector $f$, the first bound of (51) held without violation, with equality on two coordinates, and the left side was at most 0.93 of the second bound. For reversible drift, $J = 0$, the contrasts still differ through the cut couplings: on a path of three members with unit couplings, uniform leak 0.05 and a cut coupling 1 on one end member, $\varkappa_{\mathrm{out}}^{M} = \varkappa_{\mathrm{out}} = 0.3833$, $\varkappa_{\mathrm{in}} = 1.05$ and $r(C) = 2.739$, whereas $\varkappa_{\mathrm{in}}^{M} = \lambda_{\min}(Z\Pi_{CC}^{c}Z^{\top}) = 1.5014$ and $r^{M} = 3.917$; with the cut coupling on the middle member instead, $\varkappa_{\mathrm{in}}^{M} = \varkappa_{\mathrm{in}}$ and the two contrasts coincide. The two-sided bound $\varkappa_{\mathrm{in}} \leq \varkappa_{\mathrm{in}}^{M} \leq \varkappa_{\mathrm{in}} + \max_{i \in C} a_{i}^{\partial}$ for $M = \Pi_{CC}^{c}$ with the exterior held fixed, which follows from $0 \preceq Z\,\mathrm{diag}(a^{\partial})Z^{\top} \preceq \max_{i} a_{i}^{\partial} I$ and Weyl's inequality, held in 2000 random reversible regions on two to seven members; the lower bound is attained when a minimizing mode vanishes on the members that carry cut couplings, as in the middle-member example, and the upper bound when the cut couplings are uniform.

**Check S47.40** (Pooling and consensus)**.** For members' Gaussian posteriors about a two-dimensional external state, the prior-corrected pool reproduced the centralized precision and mean to $2 \times 10^{-15}$; the logarithmic pool with weights $1/n$ carried 0.41 of the centralized information, measured by the ratio of log determinants, and the product of posteriors 1.25.

For consensus on information contributions on a ring of eight members with one chord and Metropolis weights ($\rho_{\perp} = 0.848$), the bound (52) held in 20 sensor configurations, with equality at $n_{\mathrm{c}} = 0$; the largest relative entropy of a member's belief to the union posterior fell from 45.6 nats at $n_{\mathrm{c}} = 0$ to 3.9 at $n_{\mathrm{c}} = 4$, 0.52 at $n_{\mathrm{c}} = 8$ and $1.1 \times 10^{-4}$ at $n_{\mathrm{c}} = 32$. In the dynamic world, with closed-loop spectral radius 0.519 and bound constants $M_{\mathrm{cl}} = 1$, $\alpha_{\mathrm{cl}} = 0.760$ and $\|\bar{P}\| = 0.143$, an independent sample path of 400 world steps was drawn for each $n_{\mathrm{c}} \in \{1, 2, 4, 8, 16\}$, and both sides of (53) were evaluated on that path, the bound with the supremum over the 400 steps of the dispersion $(\sum_{i} |h_{i}^{s}(t) - \bar{h}^{s}(t)|^{2})^{1/2}$ of the same path. The largest deviation from the centralized estimate was 12.1, 7.7, 5.3, 2.5 and 0.95, each below the pathwise bound of its own path (76.8, 56.6, 37.0, 19.0, 7.1, with dispersion suprema 19.0, 16.5, 15.0, 14.9 and 20.7 and the common constant $M_{\mathrm{cl}} n \|\bar{P}\|/(1 - \alpha_{\mathrm{cl}}) = 4.77$), which is conservative by a factor between six and eight.

For literal consensus at level 1 (Proposition S5.5), the limits equaled $\mathsf{v}^{\top} h(0)/\mathsf{v}^{\top}\Lambda(0)$ to five digits on a star of five, a ring of six and a path of five members. With a proper prior the limit missed the union mean ($-0.139$ against $-0.222$ on the star, 1.061 against 1.579 on the ring); the prior weight was 0.397 against 0.124 on the star and 0.394 against 0.098 on the ring, where it equals $n\Lambda_{\circ}/(n\Lambda_{\circ} + \sum_{j} \Lambda_{j}^{s})$; and the mean-squared error of the limit was 0.215 against 0.124 on the star. With a nearly flat prior the limit equaled the union mean on the ring and missed it on the

path (1.265 against 1.184). Believed variances after 100 rounds were below $10^{-43}$, and averaging of posterior means converged to their plain average, which also missed the union mean.

**Check S47.41** (Agency under the protocol)**.** For two data-sharing members whose input directions are separated by an angle of 45, 22.5 or 0 degrees, with unit slot noise, costs 0.3 and 0.2 and $\gamma = 2$, the stationary law of asynchronous deliberation equaled the group posterior to $3 \times 10^{-16}$, with detailed-balance residual $10^{-16}$, and its log odds ratio was $-\gamma I(o_1; o_2) = -0.1335$, $-0.2400$ and $-0.2877$ at the three angles. Three members with random input directions agreed to $3 \times 10^{-16}$. With an interaction term added to the members' expected free energies with unequal weights (0.5 and $-0.3$) the four-cycle ratio was 4.953 and the detailed-balance residual 0.022; with equal weights, a potential game, the ratio was 1.000000 and the residual $10^{-17}$. An independent implementation with member-specific pragmatic terms on co-members' slots, non-uniform product habits, three-dimensional hidden states and two or three members with two or three policies found the asynchronous law equal to the group posterior to $1.2 \times 10^{-15}$ and the potential identity to $1.3 \times 10^{-15}$.

Under synchronous deliberation of two data-sharing members the stationary law had log odds ratio 0.0000 at every angle, against $-0.1335$ to $-0.2877$ for the group posterior, at total-variation distance up to 0.036. The independent implementation found the two-member synchronous law to be a product law to $2 \times 10^{-16}$ at distance 0.015 from the group posterior and, with three members, a law that is not a product law (deviation $5.4 \times 10^{-4}$) at distance 0.033 from the group posterior.

Without data sharing, for identical inputs, $I(o_1; o_2) = 0.1438$, the group log odds ratio was $-0.2877$, the deliberation law's 0, and their total-variation distance 0.057; the independent implementation found distances up to 0.157. For the unrepresented driver at the angle of 45 degrees and $\gamma = 2$, the component log odds ratios were $-0.1335$ and the mixture log odds ratios $+1.534$, $+3.942$ and $+0.202$ for cost shifts $\{-0.5, 1.0\}$, $\{-1.0, 2.0\}$ and $\{0, 0.6\}$.

**Check S47.42** (Deliberation with heterogeneous beliefs)**.** Sixteen instances had two or three members and two or three policies each, each policy other than the null opening one or two Gaussian slots of dimension one or two on a three-dimensional hidden state, Gaussian preferences, member beliefs with different means and covariance dispersion 0.1 or 0.5, and $\gamma \in \{1, 4\}$; one more instance had three members, three policies and $\gamma = 12$. The decomposition of Lemma 5.9 held to $2.7 \times 10^{-15}$ and the cycle formula of Theorem 5.10(a) to $1.3 \times 10^{-14}$. The distance $\mathrm{TV}(\mathbb{P}^\infty, \mathbb{P}_\Phi)$ ranged from $3.0 \times 10^{-5}$ to 0.121, and no bound of Theorem 5.11 was violated. For instance, with three members, three policies, $\gamma = 1$ and dispersion 0.1, the distance was $4.7 \times 10^{-3}$ against the contraction bound $2.2 \times 10^{-2}$ ($\mathfrak{D}_\Phi = 0.32$) and the pointwise tree bound 0.343; the spectral bound gave $1.4 \times 10^{-2}$ for $\|\mathbb{P}^\infty/\mathbb{P}_\Phi - 1\|$, whose value was $1.07 \times 10^{-2}$; and the tree bound on $\max|\ln(\mathbb{P}^\infty/\mathbb{P}_\Phi)|$ was 1.43 against the value 0.026. At $\gamma = 12$ the distance was 0.040, while $\mathfrak{D}_\Phi = 0.554$ gave the vacuous bound 1.33, $e^{\gamma \sigma_{\max}} - 1 = 2.29$ exceeded $\mathrm{gap}_\Phi = 0.264$, and the pointwise tree bound was 1.000. With one covariance and different means (six instances) $\mathbb{P}^\infty$ equaled the tilted Gibbs law to $1.3 \times 10^{-15}$, at total-variation distance 0.063 to 0.29 from the untilted group posterior. Over 60 random instances the largest ratio $|\mathcal{R}_i|/(\dim(o_A)\mathrm{Th}_i)$ was 0.39. With a common belief and member-specific precisions, the cycle formula of Remark S5.8 held to $1.9 \times 10^{-15}$ over 20 instances. Over 20000 random tilts the ratio of the distance to $\tanh(\operatorname{osc} g/4)$ was at most 1.000000, attained by the two-point law (0.314021 at $\operatorname{osc} g = 1.3$). An independent implementation with three members, three policies opening 0, 1 or 2 scalar Gaussian slots on a two-dimensional hidden state, and 16 instances with dispersion 0.02 to 0.3 and $\gamma \in \{0.5, 1, 2, 4\}$ confirmed the decomposition to $5.3 \times 10^{-15}$ and the cycle formulas to $1.9 \times 10^{-14}$ and $1.3 \times 10^{-14}$, found the dispersion ratio at most 0.319 and distances between $1.3 \times 10^{-11}$ and $4.6 \times 10^{-3}$ with no violation of any bound, and confirmed the translated case to $1.9 \times 10^{-15}$.

**Check S47.43** (Agency index)**.** At one decision state, with the product law of two members without data sharing as the actual law and the class of group models whose sector laws equal the members' with a free precision and uniform habits held fixed, the index was 1.000, 0.977, 0.871, 0.632, 0.434 and 0.311 as the overlap of the two inputs increased ($I(o_1;o_2) = 0$, 0.032, 0.067, 0.104, 0.128, 0.144), with optimal precision between 1.57 and 2.44. In the 17 instances of Check S47.42, $1 - \mathfrak{A}$ computed by convex fitting over free product habits and a free precision lay between 0 and $3.8 \times 10^{-3}$, always below $D_{\mathrm{KL}}(\mathbb{P}^{\infty} \| \mathbb{P}_{\Phi}) / D_{\mathrm{KL}}(\mathbb{P}^{\infty} \| E)$, which was at most $4.8 \times 10^{-2}$; the fitted precisions lay between 0.88 and 12.7. In an instance with two members, two policies each, uniform habits, no preferences, $\gamma = 2$, beliefs $\mathcal{N}(0, I)$ and $\mathcal{N}(0, \mathrm{diag}(3, 0.5))$ on a two-dimensional hidden state and scalar slots with unit noise along directions at angles 0 and 0.4, the members' redundancies were 0.119 and 0.375, the cycle log ratio was $-0.5106$ and the detailed-balance residual $5.9 \times 10^{-3}$, so the chain was not reversible, while $\mathbb{P}^{\infty}$ equaled to $1.3 \times 10^{-14}$ the group posterior with belief $\mathcal{N}(0, 2.070 I)$, precision 2 and product habits $(0.563, 0.437)$ and $(0.504, 0.496)$.

**Check S47.44** (Joint performative acts)**.** In 30 instances with two or three members, a shared preference precision, differing targets and beliefs, and additive effects, the oscillation of $G_i^{\mathrm{act}} - \Phi^{\mathrm{act}}$ in $u_i$ was at most $1.2 \times 10^{-14}$ and the stationary law equaled the Gibbs law of $\Phi^{\mathrm{act}}$ to $4.5 \times 10^{-15}$. With member-specific preference precisions the cycle formula of Theorem S5.17(a) held to $2.8 \times 10^{-14}$, and with member-specific effect models to $6.2 \times 10^{-15}$. With a complementary joint act $u_1 u_2 \mathsf{v}_{12}$ and a shared precision, the cycle log ratio matched $\gamma \langle \mathsf{z}_1 - \mathsf{z}_2, \mathsf{H}_\star \mathsf{v}_{12} \rangle_{\Lambda^\star}$ to $2.1 \times 10^{-14}$ in 30 instances (for example $-0.675436$, $+2.393957$ and $-19.142035$; detailed-balance residual $2.3 \times 10^{-3}$ in the first), while games with common displacements $\mathsf{z}_i$ equaled the Gibbs law of $G_C^{\mathrm{act}}$ to $6.7 \times 10^{-16}$. In a further instance with common targets $o_i^\star$ and different belief means the cycle log ratio was $-0.2968$, equal to the same formula, and replacing the two displacements by a common one gave the Gibbs law of $G_C^{\mathrm{act}}$ to $2.2 \times 10^{-16}$. Under the group posterior the log odds ratio of two members acting was $-2.0000$ for aligned additive effects, $+2.0000$ for opposed effects and $+2.7800$ for a complementary joint act, each equal to $-\gamma \partial_{ij}^2 G_C$. With three acts per member, changing the members' selection probabilities from $(0.5, 0.5)$ to $(0.85, 0.15)$ moved the stationary law by total variation 0.3009, 0.0045 and 0.0002 with divergent displacements, and by at most $1.4 \times 10^{-14}$ with common displacements.

**Check S47.45** (Identity under turnover)**.** With $g = 1$, $\varepsilon = 0.01$ and $\delta = 0.001$, the closed form of $\tau_{\mathrm{id}}^{-1}$ in Proposition 5.14 agreed with numerically computed eigenvalues to six digits on the grid $\beta \in \{0, 1, 10, 100\}$, $\omega \in \{0.01, 0.1, 1\}$, $\mathsf{f} \in \{0, 0.9\}$; the approximation was within 1 percent at $\omega = 0.01$ and within 9 percent at $\omega = 0.1$. Over about $10^5$ random draws of the six parameters, log-uniform rates in $[e^{-6}, e^4]$ and uniform $\mathsf{f}$, with $\omega + \delta \le \frac{1}{2}(\varepsilon + \beta + g)$ the rate never exceeded the comparison value $\delta + \omega(\varepsilon + g(1 - \mathsf{f}))/(\varepsilon + \beta + g)$ and fell below it by at most 0.986 times the bound $2\omega(\omega + \delta)/(\varepsilon + \beta + g)$. The ratio of the identity time of the pair $(\bar{\mu}, z)$ to the residence time ranged from 1.00 ($\beta = 0$, $\mathsf{f} = 0$, $\omega = 1$) to 989 ($\beta = 100$, $\omega = 0.01$, $\mathsf{f} = 0.9$). At the three grid points with $\beta = 0$ and $\mathsf{f} = 0$ the record is unread, the slow eigenvector has no component on $\bar{\mu}$, and the pair's ratios of 90.9, 9.90 and 1.00 at $\omega = 0.01$, 0.1 and 1 are the record's own, the members' mean decorrelating in 0.99 residence times; at every other grid point the component of the slow eigenvector on $\bar{\mu}$ was at least 0.44 in a unit vector. The stationary covariance of the pair under noise in $\bar{\mu}$ alone was positive definite at all 24 grid points. Without a record the ratio was 0.99, 1.96, 9.09 and 50.0 at $\mathsf{f} = 0$, 0.5, 0.9 and 0.99. For the conditioning set of Definition 5.13, a stationary scalar statistic $Y_{t+1} = 0.9 Y_t + e_t$ with unit innovation variance, heard by an exterior variable $x_{t+1} = 0.5 x_t + Y_t + \nu_t$, had $I(Y_0; Y_5) = 0.2144$ nats without conditioning and given the exterior's exogenous inputs, computed exactly from linear maps of independent inputs; given the exterior path on rounds 0 to 5 it was 0.214, $8.0 \times 10^{-5}$ and $4.7 \times 10^{-11}$ nats at noise variances $10^6$, 1 and 0.1.

**Check S47.46** (Writer-activated leak)**.** The closed form (S28) matched a direct spectral computation to eight digits in six instances, including record retention $1 - \delta_z = 0.999$ and nonzero record noise (for example 0.02556593 and 0.00004984). On a layered chain of four agents (an interior agent, two shell agents and an exterior agent) the leak was 0 without a record; with a record written by the interior agent and read by the exterior agent it was $8.37 \times 10^{-4}$ with the record marginalized, 0 with the record kept and assigned to the blanket, and 0.0279 with the record placed in the exterior; with the record written by a shell agent, 0; for an internal archive written by the interior agent and read by a shell agent, 0; for an archive written by the interior and the exterior agents, 0.0273, 0.0022 and $2.7 \times 10^{-4}$ as exterior, blanket and marginalized; and for an archive written by a shell agent and the exterior agent, 0 in both placements. With $\omega = 0.01$ fixed, the local exponent of $\iota_{\text{path}}$ in $\beta$ between $\beta = 0.01$ and $\beta = 0.1$ fell from 2.00 at retention 0.5 to 1.34 at retention 0.9999, where $\iota_{\text{path}} = 4.525 \times 10^{-4}$ at $\beta = 0.1$, which the leading term of the second form of (S29) reproduces to the digits shown; with $\omega = \delta_z$ the exponent stayed at 2.00.

**Check S47.47** (The order parameter and the nulls)**.** A linear-Gaussian model had three scalar agents, a scalar world with coefficient 0.9, a scalar record and optional acts, and a window of six rounds, with the order parameter for one pair computed exactly from linear maps of independent inputs. The full model with messages, record and acts gave 0.687 given the shared drive and 0.600 given the world path; with messages and record deleted and acts kept, $4.0 \times 10^{-2}$ and $2.3 \times 10^{-5}$; deletion and the frozen control, both with acts decoupled, 0 to $2 \times 10^{-15}$; the record as the only channel, 0.248, and 0 to $10^{-14}$ under its deletion or freezing; frozen broadcast messages, 0 when the broadcast noise is in the conditioning set and $5.8 \times 10^{-2}$ when it is not. The collider of Example S5.24, with world coefficient 0.9, agent states $\mu_i(t+1) = 0.4\mu_i(t) + 0.5p_i(t)$ with standard normal private noises, act gains 0.30 and 0.25 on the world, no sensing, messages or record, unit variances of the world innovations and initial states, and agent paths over rounds 1 to 6, gave $1.8 \times 10^{-15}$ given the drive and $1.39 \times 10^{-3}$ given the world path, both 0 with acts decoupled. With both act gains multiplied by a common factor of 0.125, 0.25, 0.5, 1, 2 and 4 the value given the world path was $3.62 \times 10^{-7}$, $5.78 \times 10^{-6}$, $9.13 \times 10^{-5}$, $1.39 \times 10^{-3}$, $1.87 \times 10^{-2}$ and 0.179, with local exponents 4.00, 3.98, 3.93, 3.75 and 3.26 between successive factors. Over 2000 draws of signed coefficients with a message path from one agent to the other and acts decoupled, the smallest order parameter was $8.1 \times 10^{-5}$ and none fell below $10^{-10}$; along a single message gain in $[-1, 1]$ with no other path it vanished only at zero. Over all regions of 200 random diagonal precisions on six members the largest contrast was 1.000000. An independent implementation in different six-round models, with world coefficient 0.8, agent retentions 0.6, 0.5 and 0.4, and unit variances of the private noises, the world innovations and the initial states, found a collider with act gains 0.7 and $-0.42$ at 0 given the drive and 0.2017 given the world path, a larger value than the first instance because its act gains and private noise variances are larger, and the masking instance of Example S5.25, in which agent 1 acts with gain 0.7 and agent 3 senses the world with gain 0.5, at 0.614 and $4 \times 10^{-16}$, a smallest value of $1.46 \times 10^{-3}$ over 1000 signed draws with a path, and again a largest contrast of 1.000000. In a three-agent level-0 model with world coefficient 0.9, sensing gain 0.3 and message gain 0.2 over eight rounds, the conditional mutual information of two agents' paths given the world path and the initial state was 0.287 nats in the full model, below $3 \times 10^{-14}$ under the deletions and the frozen control, and 0.123 under the frozen control with broadcast noise outside the conditioning set, while the dependence given the initial state alone, with the common drive not conditioned on, stayed large (3.80 nats under the deletions and 3.00 under the frozen control) and the final belief correlation lay between 0.86 and 0.98. For the member-restricted order parameter of Definition 5.15, two scalar level-0 members with sensing gain 0.3 that read one outsider with gain 0.4 through receiver-specific unit-variance read noise, with no channel between them, no acts, world coefficient 0.9 and unit

innovation and sensing noise variances, had conditional mutual information 0.0290 nats between their paths over rounds 1 to 6 given the shared drive, and 0 to round-off once the outsider's private inputs joined the conditioning set or the read gain was set to zero.

**Check S47.48** (Operational tests)**.** The likelihood-ratio test was run under $\mathfrak{N}_{\mathrm{lp}}$ with acts decoupled, three-dimensional path blocks $U_i$ and $U_j$ with nonsingular conditional covariance (the initial states, which are functions of the shared drive, are not among their coordinates), a shared drive of dimension 20 and nine degrees of freedom, with 300 replications per sample size. At $n_{\mathrm{rep}} = 200$ the mean of $2n_{\mathrm{rep}}\hat{I}$ was 9.98 against the chi-squared mean 9, and the rejection rate at the 0.95 quantile was 0.097 raw and 0.043 with the finite-sample multiplier; at $n_{\mathrm{rep}} = 800$ the values were 9.48, 0.063 and 0.060 (standard error about 0.013). Because the null law of the statistic does not depend on the parameters, an independent simulation with standard normal $U_i$, $U_j$ and drive of the same dimensions tests the same law: over 3000 replications at $n_{\mathrm{rep}} = 200$ the mean was 10.22 and the rejection rates 0.091 raw and 0.049 with the multiplier, and over 1500 replications at $n_{\mathrm{rep}} = 800$ they were 9.23, 0.059 and 0.050; the multiplier predicts the means $9n_{\mathrm{rep}}/(n_{\mathrm{rep}} - 24.5) = 10.26$ and 9.28. With one coordinate that is a function of the drive kept in $U_i$, the smallest eigenvalue of the residual covariance was $1.5 \times 10^{-30}$ and the statistic was undefined. The power at $n_{\mathrm{rep}} = 400$ was 0.075, 0.135 and 0.425 for message gains 0.02, 0.05 and 0.1, whose exact order parameters are $4.3 \times 10^{-4}$, $2.7 \times 10^{-3}$ and $1.0 \times 10^{-2}$. The structure test used eight agents in two groups, attention capacity $\kappa = 2$, level-0 updates, and the selector equal to the largest ratio of mean within-block to mean between-block world-conditional mutual information over balanced bipartitions; the observed value 4.651 compared with a null median 1.882, 95 percent quantile 3.016 and maximum 11.256 over 999 quenched partner randomizations at matched rates, $p = 0.007$.

## S47.5 Checks for Section 6

The checks below were each run once, single threaded and in double precision. Each confirms a closed form, a bound or a counterexample of Section 6 on the instances it evaluates and on no others.

**Check S47.49** (One message and Gaussian contraction)**.** In four random instances of the setting of Lemma S30.1 the closed form (S115) agreed with a direct computation from the joint covariance of $(\eta, y)$ to ten digits (values 0.9258843256, 0.4456534130, 0.7426156703, 1.5119045934), and each value lay below $\sigma_1(M_{XY})^2 I(\eta; m)$. For the contraction coefficient of Proposition S6.1, the ratio of output to input relative entropy divided by $\sigma_1(M_{XY})^2$ was at most 0.999956 over 300 random Gaussian input laws; mean shifts along the top canonical direction gave ratios equal to $\sigma_1(M_{XY})^2$ to twelve digits in three instances (0.941107656591, 0.938151358084, 0.752322620659); for 200 one-dimensional Gaussian mixtures, by quadrature, the largest ratio divided by the squared correlation was 0.9999996; and for 25 two-dimensional mixtures through channels with canonical correlations $(\sigma_1, \sigma_2)$ the largest ratio divided by $\sigma_1^2$ was 0.935. No ratio exceeded the coefficient.

**Check S47.50** (Relay paths)**.** For four random Gaussian Markov chains $X_0, X_1, X_2, X_3$ of dimensions 3, 4, 2, 3, the top canonical correlation of $(X_0, X_3)$ equaled the top singular value of $M_3 M_2 M_1$ to ten digits (0.7088578942, 0.8770328208, 0.8179696032, 0.8290112856) and lay strictly below the products of per-hop top singular values (0.866837, 0.969954, 0.852493, 0.920113). In a scalar chain of six hops the endpoint correlation equaled the product of per-hop correlations, 0.0014709787. For the relay conventions of Section S30.1 ($P_0 = 1$, source noise 0.25, channel noise 0.5), the information carried after 1, 2, 3, 4, 6 and 12 hops was 0.3389, 0.1400, 0.04178, 0.005568, $5.756 \times 10^{-8}$ and zero to double precision for posterior-mean forwarding; 0.3811, 0.2197, 0.1353, 0.0860, 0.03641

and 0.003092 for unit-variance forwarding; and 0.4236, 0.2939, 0.2260, 0.1839, 0.1341 and 0.07421 for unbiased forwarding. For the relay chain in a persistent scalar world, the ratio of consecutive stationary cross-covariances with the current state equaled the factor $\psi = F^2 g/(1 - F^2(1 - g))$ to ten digits at $(F, g) = (0.95, 0.5)$, $(0.99, 0.8)$ and $(0.9, 0.2)$, where $\psi = 0.8223234624$, $0.9752481405$ and $0.4602272727$.

**Check S47.51** (Information about a distant source on the ring)**.** On a ring of 400 agents with $W_{ii} = c$ and $W_{ij} = a$ for neighbors, $G(d) = \sum_{n \geq 0} ((W^n)_{0d})^2$ was computed exactly by Fourier sums. The fitted decay rate of $\ln G(d) + \frac{1}{2} \ln d$ was 0.75118, 0.32803 and 0.40306 at $(c, \varepsilon) = (0.25, 0.05)$, $(0.25, 0.01)$ and $(0, 0.02)$, against $2m = 0.75150$, 0.32843 and 0.40338; without the correction $\frac{1}{2} \ln d$ the fitted rates were 0.77774, 0.34433 and 0.41999. The rate agrees with Proposition S6.3(b); the correction $\frac{1}{2} \ln d$, which the saddle-point estimate suggests, improves the fit but is not proved.

**Check S47.52** (Calibration and withdrawal)**.** In 500 random instances of the static two-group model, with $K$ from 2 to 5 and proper and improper orthogonal mistranslations, the calibrated noises of Theorem 6.4(a) agreed with exact second moments to $1.8 \times 10^{-15}$ under both C1g and C1i. The top-$\kappa$ rule of part (b) agreed with brute-force maximization of believed information in 2000 of 2000 random heterogeneous instances, in two independent computations, and the withdrawal inequality (65) agreed with direct comparison of calibrated precisions in 5000 of 5000 instances. For two homogeneous groups of four agents ($\sigma_\eta^2 = 1$, $R^s = 1$, $R_c = 0.1$, $K = 2$, $\kappa = 2$) with a cross-group mistranslation rotating by $\phi$:

| $\phi$ | 0 | 0.2 | 0.5 | 1.0 | 2.0 | $\pi$ |
|---|---|---|---|---|---|---|
| $\varpi$ cross-group, C1g | 0.7143 | 0.6945 | 0.6080 | 0.4311 | 0.2363 | 0.1852 |
| $\varpi$ cross-group, C1i | 0.7143 | 0.7043 | 0.6569 | 0.5377 | 0.3551 | 0.2941 |
| error, C0 with one cross-group sender | 0.2917 | 0.2958 | 0.3172 | 0.3874 | 0.5867 | 0.7083 |

The in-group calibrated precision was 0.7143 throughout. The C1 receiver attended two in-group senders for every $\phi > 0$ and kept the error 0.2917, the error of exact inference; its own sensing alone gave 0.5000. In the setting of Example S6.4 the believed value of the in-group sender was 0.1527 and that of the cross-group sender 0.6399, 0.3703 and 0.2405 at rotations 0.3, 1.0 and $\pi/2$, with sensing noise 1 in $A$ and 0.5 in $B$; the cross edge was attended in each case. With sensing noise 1 in both groups the cross-group values were 0.4016, 0.2768 and 0.1975, again above the in-group value.

**Check S47.53** (Holonomy and consensus)**.** On random strongly connected digraphs with $N = 6$ and $K = 3$, three instances each with consistent translations, with one common fixed vector and with generic translations, the dimension of the kernel of the incidence operator, the common fixed dimension of the fundamental-cycle holonomies and the common fixed dimension of about 400 sampled directed closed walks agreed: 3, 1 and 0 respectively. On the acyclic digraph with edges $0 \to 1$, $1 \to 2$, $0 \to 2$ and generic translations the consensus space was trivial. On a four-cycle with $K = 3$, flat holonomy, a rotation about one axis and a generic holonomy gave consensus spaces of dimension 3, 1 and 0, each equal to the dimension of the fixed space of the holonomy. For the learned holonomy of Section S30.3, on a directed four-cycle with $K = 3$ whose edges carry random codes of ranks 1, 2, 1 and 2, encodings $U_i = e^{a_i}$ with random $a_i$ of entrywise scale $s$, true generators $\log(U_i U_j^{-1})$ and learned generators $(I - \Pi_e) \log(U_i U_j^{-1})$, the holonomy of the learned translations differed from $I - \sum_e \Pi_e (a_i - a_j)$ by $5.2 \times 10^{-2}$, $4.5 \times 10^{-4}$ and $4.9 \times 10^{-6}$ in the largest entry at $s = 0.1$, 0.01 and 0.001, against holonomy defects 0.23, $2.0 \times 10^{-2}$ and $1.8 \times 10^{-3}$, so the error falls as $s^2$; the coded differences $T_e$ times learned minus true generators vanished to $7 \times 10^{-17}$; and with one code on every edge the learned holonomy differed from $I$ by $1.5 \times 10^{-6}$ at $s = 0.001$, a second-order amount.

**Check S47.54** (Frustration)**.** On cycles with O(2) translations of random edge angles and holonomy angle $\phi$, the spectrum of Proposition 6.6(b) matched the computed one to at most $5.3 \times 10^{-15}$ at $(N, \phi) = (5, 0.4)$, $(12, 2.0)$, $(40, \pi)$ and $(7, 0)$, with least eigenvalues 0.006397, 0.027714, 0.006165 and 0 against single-cycle lower bounds 0.006315, 0.019669, 0.002500 and 0. Over 60 random undirected graphs with O(2) and O(3) translations and unit precisions, the ratio of the least eigenvalue to the lower bound of part (c) was at least 1.007 and its ratio to the upper bound at most 0.669; in one instance with $N = 6$ the lower bound was $1.1 \times 10^{-7}$ against a least eigenvalue of $7.9 \times 10^{-4}$. For Proposition S6.7, the directed three-cycle example gave a trivial consensus space and $\dim H^0(\mathfrak{S}) = 3$, and over 200 random coded instances $\dim H^0(\mathfrak{S}) \geq NK - |\mathcal{E}_\mathrm{a}| r_c$ held in all, with equality to $\max\{NK - |\mathcal{E}_\mathrm{a}| r_c, 0\}$ in 196. On a $24 \times 24$ torus with SO(2) translations and uniform flux 0.0109, 0.0218, 0.0436 and 0.0873 per plaquette, total fluxes $2\pi$ times 1, 2, 4 and 8, and each lattice edge attended in one direction, the least eigenvalue divided by the flux per plaquette was 0.999, 0.997, 0.995 and 0.989, and the lowest level had multiplicity 2, 4, 8 and 16 as a real eigenspace of the $2N \times 2N$ operator $\mathcal{L}$, that is, complex multiplicity 1, 2, 4 and 8 for the associated magnetic Laplacian, since $\mathcal{L}$ commutes with the quarter turn applied in every fiber. With both directions of every edge attended the least eigenvalue doubles (ratios 1.997, 1.995, 1.989 and 1.978) and the multiplicities are unchanged.

**Check S47.55** (Cycle amplification)**.** Among 3000 random three-cycles in two dimensions with noiseless codes of rank one, the largest spectral radius of the cycle product was at most 0.9987 with a common prior covariance and 2.598 with heterogeneous priors; a second sample of 3000 gave 0.9996 and 3.052. For the two receivers of Example S6.11(i), the spectral radius equaled $\alpha_a \alpha_b x/(1 + R_c)^2$ to four digits at $(\alpha_a, \alpha_b, x) = (0.9, 0.9, 4)$, $(0.5, 0.5, 4)$ and $(0.9, 0.9, 1)$ with $R_c = 0.1$ (values 2.6777, 0.8264, 0.6694), and in the first case the norm of the tenth power was $2.2 \times 10^4$. Over 20000 random cycles of dimensions 2 to 4, lengths 2 to 5 and code ranks 1 to $K$, noisy or noiseless, 304 amplified and the bound $\prod_e \mathfrak{t}_e^{1/2}$ of Theorem 6.8 was never violated (largest ratio of spectral radius to bound 0.9998); over 20000 noisy cycles 164 amplified, the sharpened bound (66) was never violated and was attained to four digits, and none of the 1760 draws meeting the spread condition of Proposition S6.10(b) amplified. An independent sample of 20000 cycles, 30 percent of them noiseless, gave 1595 amplifying cycles and no violation of either bound. With priors within Thompson distance $\delta_\mathrm{T}$ of a common prior (three agents, two dimensions, codes of rank one, 4000 draws per value), the largest spectral radius at edge signal-to-noise ratio near 4 was 0.51, 0.52, 0.53, 0.57, 0.66, 0.83, 1.07 and 1.91 at $\delta_\mathrm{T} = 0$, 0.05, 0.1, 0.25, 0.5, 0.75, 1 and 1.5, and no draw meeting the stricter condition $\sum_{(i \to j) \in \mathfrak{c}} \mathrm{sp}(P_i \| P_j) \leq \sum_{e \in \mathfrak{c}} \ln(1 + \mathrm{snr}_e^{-1})$ amplified. Over 20000 random cycles the largest spectral radius was 2.445 with channel noise only and 0.250 with assigned noise $H_i P_i H_i^\top + R_c$. A common code with common noise gave at most 0.999985 (5000 draws), a common code of rank one with sender-specific scalar noises at most 0.998525 (5000 draws), and a common code of rank at least two with sender-specific noise covariances amplified: with common code $I_2$, the priors of Example S6.11(i) at $\alpha_a = \alpha_b = 0.9$ and $x = 4$, and assigned noise covariances $\mathrm{diag}(0.1, M)$ for sender $b$ and $\mathrm{diag}(M, 0.1)$ for sender $a$, the spectral radius was 1.3568, 2.3586, 2.6410 and 2.6773 at $M = 1$, 10, 100 and $10^4$, against the limit $\alpha^2 x/(1 + r)^2 = 2.6777$; at $M = 10^6$ it was 10.7107, 42.842 and 171.35 at $x = 16$, 64 and 256, against limits 10.7107, 42.843 and 171.37; with the two noise covariances exchanged it stayed below 0.41, and with common noise $0.1 I_2$ it was 0.8615. The rank-one formula (S117) and its factorization held to $2.8 \times 10^{-11}$ and $1.3 \times 10^{-15}$ over 3000 cycles. The round-trip map of Theorem 6.8(b) satisfied $|\rho(\bar{A}_\mathfrak{c}) - \max\{\rho(A_\mathfrak{c}), 1\}| \leq 8.9 \times 10^{-14}$ over 2000 noiseless cycles. The family of Example S6.11(ii) gave $\rho(A_\mathfrak{c}) = 1.254400$, 1.092025, 1.019696 and 1.003988 at $u = 0.2$, 0.05, 0.01 and 0.002, equal to $(1 - u)^2(1 + 2u)^2$, at Thompson distances 0.8927, 0.3995, 0.1601 and 0.0671, in two independent computations. For Numerical observation S6.12,

priors drawn as exact posteriors of one common prior given private linear-Gaussian data gave largest spectral radii 6.26 without channel noise and 5.48 with noise 0.05 over 20000 draws.

**Check S47.56** (Code games)**.** The exact 16-state logit chain on a ring of four agents with random pulls at $\gamma = 1.7$ had a stationary law equal to the Gibbs law of Proposition 6.10(a) to $4.4 \times 10^{-16}$ for symmetric couplings $J = 1$, with detailed-balance defect $5.9 \times 10^{-17}$; with couplings 1 and 0.3 in the two directions the stationary law differed from the Gibbs law by 0.179 and detailed balance failed by $2.0 \times 10^{-3}$. On a ring of six agents, exact chains for three count-based values (the C1 value with both partners attended, the C1 value with capacity one, and a linear value) equaled the Gibbs law of the pairwise game with $J = \frac{1}{2}(\mathcal{F}(2) - \mathcal{F}(0))$ to $9.4 \times 10^{-16}$. On the triangular prism, whose agents have three partners, the two C1 values gave chains that violate detailed balance by $1.5 \times 10^{-4}$ and $1.6 \times 10^{-3}$ and differ from the Gibbs law of the pairwise game with $J = \frac{1}{3}(\mathcal{F}(3) - \mathcal{F}(0))$ by $1.4 \times 10^{-3}$ and $1.2 \times 10^{-2}$, while the linear value gave agreement to $1.5 \times 10^{-15}$. On the complete graphs on four and five agents, with the C1 value at $K = 2$, $\varpi^{=} = 1$, $\varpi^{\neq} = 0.2$ and $\Lambda^{\text{own}} = 1$, without capacity and with capacity one, random pulls and $\gamma = 1.7$, the potential $\Phi$ of Proposition 6.10(c) reproduced every unilateral cost change to $1.3 \times 10^{-15}$, and the exact logit chain had the Gibbs law of $\Phi$ to $1.9 \times 10^{-15}$ with detailed-balance defect at most $1.4 \times 10^{-16}$, although $\nabla^{\pm}\mathcal{F}(u)$ took the values 0.6286, 0.5754 and 0.6286 at $u = 0, 1, 2$ on four agents. Over twelve graphs (chains, rings, complete graphs, the complete graph on four agents with an edge removed and with a pendant agent, the prism, the complete bipartite graph on three and three agents and a star) and four values (the two C1 values, an affine and a quadratic one), the summed cost changes over the whole configuration space were path independent to $5.4 \times 10^{-15}$ in the cases that meet the edge condition of Proposition 6.10(c) and path dependent by at least 0.053 in all the others.

**Check S47.57** (One-dimensional code domains)**.** Exact minimizers of the pairwise potential on open chains were computed by dynamic programming with $h = 1$:

| $J/h$ | agents | mean domain length | ratio to $(J/h)^2$ |
|---|---|---|---|
| 1 | $4 \times 10^5$ | 4.71 | 4.71 |
| 2 | $4 \times 10^5$ | 9.98 | 2.50 |
| 4 | $4 \times 10^5$ | 26.75 | 1.67 |
| 8 | $4 \times 10^5$ | 83.40 | 1.30 |
| 16 | $4 \times 10^5$ | 306.5 | 1.20 |
| 16 | $1.5 \times 10^6$ | 289.5 | 1.13 |
| 32 | $1.5 \times 10^6$ | 1085.4 | 1.06 |

The local exponent of the mean domain length in $J/h$ was 1.88 between 8 and 16 and 1.91 between 16 and 32, approaching 2 slowly because of the overshoot of the discrete walk. The structural lemma held in every domain with walls on both sides, both inequalities of the counting lemma held, and the number of domains equaled the number of greedy switches (84843, 14951, 4796, 1305) or exceeded it by one (40084 against 40083) in the runs with $4 \times 10^5$ agents. Without pulls, logit dynamics sampled on an open chain at $\gamma J = 2$ gave $\Pr(\sigma_0 \neq \sigma_d) = 0.1199$, 0.3725 and 0.4983 at $d = 1$, 5 and 20, against $\frac{1}{2}(1 - \tanh^d(\gamma J/2)) = 0.1192$, 0.3719 and 0.4978, and at $\gamma J = 4$ gave 0.0177, 0.0828 and 0.2596 against 0.0180, 0.0837 and 0.2597.

**Check S47.58** (Bundling of walls)**.** Exact joint minimizers for two binary features on a chain with $J = 3$ and independent standard Gaussian pulls were computed by dynamic programming. The probability that a wall of the second feature sits on the same edge as a wall of the first was 0.057 when two mismatches cost $2J$ (independence predicts 0.0574), 0.366 when they cost $1.5J$

(independence 0.0658) and 0.662 when they cost $1.1J$ (independence 0.0904); the probabilities of a wall of the second feature within two edges were 0.279, 0.470 and 0.724.

**Check S47.59** (Grounding tension)**.** On the ring with $F = 0.99$, $Q = 0.02$, $R^s = 1$ and $R_c = 0.1$ ($P_0 = 1.005$, isolated Kalman error 0.1415), the level-0 family in the assigned message precision gave the values below, with $\varpi_s = 1$; the column headed floor is $Q/(\varepsilon(2-\varepsilon))$, and the last two columns are the fidelity of a message given the receiver's prior mean alone, $I(\eta_t; y_{ij} \mid m_i)$, and its believed fidelity, in nats.

| $\varpi_y/\varpi_s$ | $k^s$ | $k^y$ | $\xi$ | $\mathbb{E}e_i^2$ | $\hat{P}^-$ | floor | given $m_i$ | believed |
|---|---|---|---|---|---|---|---|---|
| 0.01 | 0.1227 | 0.0012 | 0.213 | 0.1405 | 0.1403 | 0.0814 | 0.0612 | 0.001 |
| 0.1 | 0.1131 | 0.0113 | 0.392 | 0.1347 | 0.1308 | 0.0873 | 0.0423 | 0.006 |
| 0.3 | 0.0976 | 0.0293 | 0.587 | 0.1325 | 0.1156 | 0.0991 | 0.0217 | 0.017 |
| 1 | 0.0700 | 0.0700 | 0.976 | 0.1501 | 0.0886 | 0.1313 | 0.0056 | 0.042 |
| 3 | 0.0436 | 0.1307 | 1.587 | 0.2098 | 0.0627 | 0.1933 | 0.0024 | 0.086 |
| 10 | 0.0223 | 0.2227 | 2.639 | 0.3505 | 0.0418 | 0.3172 | 0.0041 | 0.175 |
| 30 | 0.0107 | 0.3197 | 3.935 | 0.5621 | 0.0304 | 0.4917 | 0.0076 | 0.324 |
| 100 | 0.0041 | 0.4122 | 5.391 | 0.8445 | 0.0240 | 0.7152 | 0.0064 | 0.612 |
| 1000 | 0.0005 | 0.4878 | 6.794 | 1.1913 | 0.0205 | 0.9590 | 0.0003 | 1.534 |

The marginal fidelity of a message peaked near 0.743 at $\varpi_y/\varpi_s = 0.1$ and fell to 0.001. The receiver fidelity $I(\eta_t; y_{ij} \mid \mathcal{I}_i)$ of Definition 6.1, with $\mathcal{I}_i$ the receiver's signals through round $t$ and both neighbors' messages of all earlier rounds, was computed from the steady-state filter for the joint law of the world and the prior errors on a ring of 60 agents, with values unchanged to six decimals on 30 agents: 0.0169, 0.0165, 0.0155, 0.0119, 0.0062, 0.0016, 0.0003, $3 \times 10^{-5}$ and less than $10^{-6}$ along the nine rows of the table. Conditioning on earlier rounds only gave 0.0188, 0.0134 and 0.0070 at $\varpi_y/\varpi_s = 0.01$, 1 and 3, and conditioning in addition on the other neighbor's current message gave 0.0159, 0.0113 and 0.0060. The receiver fidelity therefore exceeds the fidelity given the prior mean at $\varpi_y/\varpi_s = 1$ and 3 and lies below it at the other grid values. Over the grid $\xi$ increased with $\varpi_y$, which is checked and not proved, and the smallest error on the grid was 0.1325 at $\varpi_y/\varpi_s = 0.3$. On a logarithmic grid of 20001 message precisions in $[10^{-3}, 10^3]\varpi_s$, evaluated with (68), the error was smallest, 0.1323, at $\varpi_y/\varpi_s = 0.247$, where $\xi = 0.544$; $\xi$ increased along this grid as well, to 6.794, below the bound $1/\operatorname{arccosh}(1/F) = 7.042$. An independent Lyapunov solution on a ring of 400 agents gave the errors 0.150071, 0.562109 and 0.128875 at $(F, \varpi_y/\varpi_s) = (0.99, 1)$, $(0.99, 30)$ and $(0.95, 3)$, and 0.255072 at $(0.95, 3)$ with $Q = 0.05$, $R^s = 0.5$, $R_c = 0.2$ and $\varpi_s = 2$, all equal to (68) to six digits. Over 300 random level-0 networks with common world noise the smallest ratio of error to the common-mode floor $Q/(1 - F^2(1 - k^s_{\max})^2)$ was 1.0003. For Numerical observation S6.20, the social optimum over common gains was $k^s = 0.1494$, $k^y = 0.1354$, error 0.1154, $\xi = 0.963$; the correlation-aware best response reached a symmetric fixed point after 42 iterations at $k^s = 0.1054$, $k^y = 0.1185$, error 0.1233, $\xi = 1.051$; the level-0 receiver with message precision moment-matched to innovations reached its fixed point at the noise floor, $\varpi_y = 10$, with $k^s = 0.0223$, $k^y = 0.2227$, error 0.3505, $\xi = 2.639$ and claimed $\hat{P}^- = 0.0418$; with sensing precision also moment-matched it assigned $\varpi_s = 0.7232$ against the true 1.0, with $k^s = 0.0162$, $k^y = 0.2246$, error 0.4248 and $\xi = 2.934$.

## S47.6 Checks for Section 7

Unless stated otherwise the checks use the base point of Section 7.2: $F = 0.99$, $Q = 1 - F^2$, $R^s = 1$, $R_c = 0.05$, $n = 4$, $\kappa = 3$, $\vartheta = \pi/2$, $\alpha = 0.1$ and $h = 0.995$, so $\mathsf{p}^* = 0.0503$. Linear solves had at most 324 unknowns, and stochastic simulations used at most eight agents, or 200 replicas of eight agents sharing one world path.

**Check S47.60** (Single-edge bistability)**.** Single-edge model of Proposition S7.2 with $P_0 = 1$, receiver sensing noise variance 1, reading noise variance $R_c = 0.1$, drift variance $\sigma_{\mathrm{dr}}^2 = 0.002$ and angle-observation noise $\sigma_\alpha^2 = 0.05$. With in-group sender noise $R_a^s = 1$ the end values $(\mathcal{A}(0), \mathcal{A}(1))$ are $(-0.1441, 0.1508)$, $(-0.1210, 0.4105)$, $(-0.1073, 0.6940)$ and $(-0.0972, 1.0693)$ for cross-sender noise $R_b^s = 0.6$, 0.3, 0.15 and 0.05: bistable at $\gamma = \infty$ in all four. On the grid $R_a^s \in \{0.3, 0.6, 1, 2, 4, 8\}$, $R_b^s \in \{0.02, 0.1, 0.3, 1, 3\}$ all three classes of the proposition occur, split only when the cross sender is worse even when learned and merged only when the in-group sender is poor. Fixed points of $\mathsf{p} = 1/(1 + e^{-\gamma \mathcal{A}(\mathsf{p})})$ at $R_a^s = 1$, $R_b^s = 0.3$, found by root finding in logit coordinates: one at $\gamma = 5$, 20 and 50 ($\mathsf{p} = 0.886$, 0.9997, 1.000); three at each of $\gamma = 80$, 100, 120, 150, 200 and 400, for example $6.3 \times 10^{-5}$, $5.9 \times 10^{-4}$ and 1.000 at $\gamma = 80$. The estimate $\log(10^3)/|\mathcal{A}(0)| \approx 57$ lies between 50 and 80.

**Check S47.61** (Calibration constants)**.** The calibration fixed point at the split was solved with direct Lyapunov solves of the joint world-and-group system, not with the closed forms of Lemma 7.7: $\hat{\sigma}_{\mathrm{in}}^2 = 0.29872$, MSE $= 0.24872$, $\Lambda_0 = 20.0646$, $\varpi_{\mathrm{in}} = 3.3477$, gains $k^0 = 0.63322$, $k^s = 0.03321$, $\bar{k} = 0.11119$. The direct value $2\,\mathrm{Cov}_1(\eta_t, m_j(t)) = 1.24123$ equals the closed form of Lemma 7.3, and the calibrated noise of an unlearned cross edge computed from explicit two-dimensional second moments is $1.53995 = \hat{\sigma}_{\mathrm{in}}^2 + D_{\mathrm{cal}}$. An independent evaluation of the closed forms gives the same values to all printed digits.

**Check S47.62** (Attention bounds)**.** Exact set softmax over the 35 sets of three senders among a receiver's seven others, at the frozen base-point calibration ($\varpi_\times(\pi/2) = 0.6494$). Over 3000 random states $x \in [0,1]^4$ and five values of $\gamma$, the largest ratio $\mathsf{p}_j/\bar{\mathsf{p}}(x_j)$ was 1.000000, so Lemma 7.5(a) is sharp; an independent implementation found the same. The supremum is approached only as $x_j \to 0$ with $\gamma \geq 160$. With $x_j$ fixed and the other three cross edges on a grid of six values each, the largest ratio at $x_j = \frac{1}{2}$ is 0.59, 0.67, 0.79, 0.74, 0.55 and 0.30 at $\gamma = 5$, 20, 40, 80, 160 and 320, and at $x_j = 1$ it is at most 0.65: $\Delta\mathcal{V}$ is a lower bound on the value gap $\frac{K}{2}\log((\Lambda_0 + \kappa\varpi_{\mathrm{in}})/(\Lambda_0 + (\kappa-1)\varpi_{\mathrm{in}} + \varpi_\times(x\vartheta)))$ of a set otherwise filled with in-group edges, which is 0.0630 at $x = \frac{1}{2}$ and 0.0939 at $x = 1$. For part (b) the smallest ratio to the lower bound was 1.141 and the largest ratio to the upper bound 0.867. Constants: $\Delta\mathcal{V}(\frac{1}{2}) = 0.0592$, $\Delta\mathcal{V}(1) = 0.0858$, $\Delta\mathcal{V}_{\max} = 0.3132$, $\mathcal{L} = 1.0890$, $\mathsf{c}_\mathcal{V} = 0.0186$, $\gamma_S = 68.2$, $\gamma_M = 6.84$.

**Check S47.63** (Fixed points and thresholds)**.** Symmetric fixed points of $\Phi$ with the exact set softmax at $\vartheta \in \{\pi/4, \pi/2, \pi\}$ and $\gamma$ from 5 to 1600, with Jacobian multipliers; on the grids used for $\vartheta \in \{\pi/4, \pi/2, \pi\}$ and $\gamma \in \{20, 200\}$ every increment of $\mathsf{P}$ was negative, as Theorem 7.6(a) requires. At $\vartheta = \pi/2$ the fixed points are those of Table S4; at $\vartheta = \pi/4$ the symmetric map is merged only up to $\gamma = 80$ and bistable from 120 to 240, and at $\vartheta = \pi$ merged only up to $\gamma = 40$. An independent recomputation reproduced $x = 0.1129$ ($\gamma = 40$), 0.1187 (60), 0.1274, 0.2999, 0.9659 (80, multipliers 0.9762 and 0.9937, 1.0142 and 1.0266, 0.9953 and 0.9953), 0.9992 (120) and 1.0000 (160); iteration from the declared start reached 0.9659 at $\gamma = 80$ and the merged point at 40 and 60. The computed cross derivative $\mathsf{p}'_{\mathrm{cross}}$ was nonnegative at every fixed point computed, for example 0.346 at the merged point and $6.8 \times 10^{-6}$ on the split branch at $\gamma = 80$. Conditions (i) and (ii) of Theorem 7.6(b): the smallest fixed point of $\bar{g}$ is 0.0879 at $\gamma = 240$, so (i) fails, and 1.0000 at 244, 245, 250, 320 and beyond, where both conditions hold, with $2\gamma\mathcal{L}n\bar{\mathsf{p}}(\underline{x})$ below $10^{-8}$ from 320 on. On a geometric grid of ratio 1.15 the first $\gamma$ at which (i) and (ii) hold equals the first at which a single learned bridge collapses at $\vartheta = \pi/4$ (995.7) and $\vartheta = \pi/2$ (246.1), and is 140.7 against 70.0 at $\vartheta = \pi$. By bisection on (i) and (ii), both of which hold at every point above the bisection bracket of a 300-point geometric grid in $\gamma$ reaching $4\mathsf{m}_\kappa^2/(\mathsf{c}_\mathcal{V}^+\mathsf{p}^{*2})$, $\gamma_T = 242.11$ at the base point ($\mathsf{p}^* = 0.05025$), and with $\mathsf{p}^* = 0.05$, 0.025, 0.0125, 0.00625 (varied through $h$ at $\alpha = 0.1$): $\gamma_S = 68.2$,

80.4, 92.3, 104.1, $\gamma_T = 244.22$, 861.13, 3255.56, 12684.3, and $\mathsf{p}^{*2}\gamma_T = 0.6106$, 0.5382, 0.5087, 0.4955; at $\mathsf{p}^* = 3.125\times10^{-3}$, $1.5625\times10^{-3}$, $7.8125\times10^{-4}$ and $2\times10^{-4}$ the product is 0.4893, 0.4862, 0.4848 and 0.4837. The constants of the lower bound of Theorem 7.6(b) are $\mathsf{m}_\kappa = 0.52485$ at $\mathsf{t}_\kappa = 1.0167$ and $\mathsf{c}_\mathcal{V}^+ = 0.56999$, so $\mathsf{m}_\kappa^2/\mathsf{c}_\mathcal{V}^+ = 0.4833$ and $\mathsf{m}_\kappa/(2\mathsf{t}_\kappa) = 0.258$; condition (i) fails at $\gamma = \mathsf{m}_\kappa^2/(\mathsf{c}_\mathcal{V}^+\mathsf{p}^{*2})$ at each of these $\mathsf{p}^*$, and $\Delta\mathcal{V}(y)/(\mathsf{c}_\mathcal{V}^+y^2) \le 1$ on a grid of $[0,1]$. The sufficient bound $2\kappa^2/(e\mathsf{c}_\mathcal{V}\mathsf{p}^{*2})$ is $1.41\times10^5$ at the base point, about 580 times $\gamma_T$. With calibration solved self-consistently from exact second moments, the cross attention and fixed point move from $1.7752\times10^{-3}$ and 0.9659 to $1.7934\times10^{-3}$ and 0.9656 on the split branch at $\gamma = 80$, are unchanged to four digits at $\gamma = 120$, and move from 0.3442 and 0.1275 to 0.4172 and 0.1076 on the merged branch at $\gamma = 80$ and from 0.1130 to 0.1060 at $\gamma = 40$. Solving $x\mathsf{P}(x) = \mathsf{p}^*(1-x)$ for $\gamma$ at each $x$ continues the symmetric fixed points into one S-shaped curve with folds at $\gamma = 105.89$ ($x = 0.1691$) and 60.053 ($x = 0.6318$). Along the merged branch the symmetry-breaking multiplier equals one at $\gamma = 87.92$ ($x = 0.1324$) and is 1.00007 at $\gamma = 88$, 1.0067 at 95, 1.0125 at 100 and 1.0260 at the fold, where the symmetric multiplier reaches one; both multipliers of the split branch are below one from $\gamma = 60.11$ ($x = 0.6551$). Two computations of this continuation, one of them made for Fig. S1, agree to the digits shown.

**Check S47.64** (Bridged states)**.** Per-receiver fixed points with $n_\mathrm{b}$ learned edges, iterated from $x = 0.02$ on those edges and $x = 1$ on the others. At $\gamma = 80$: $n_\mathrm{b} = 1, 2, 3, 4$ give learned edges at 0.066, 0.085, 0.105, 0.127, attended with probability 0.711, 0.542, 0.428, 0.344, with spectral radii 0.9951, 0.9951, 0.9951, 0.9937 (the split, $n_\mathrm{b} = 0$, has 0.9953). At $\gamma = 120$, $n_\mathrm{b} = 1$ and 2 survive (0.068 and 0.091), a start with three learned edges ends with one, and a start with four ends at the split; at $\gamma = 160$ and 240 only $n_\mathrm{b} = 1$ survives ($x = 0.071$, attention 0.657; $x = 0.089$, attention 0.513), and starts with more learned edges end with one or none; from $\gamma = 400$ to 3200 every start ends at the split. A single edge started at $x = 0.02$ or 0.1 stays learned at $\gamma = 160$ and one started at $x \ge 0.2$ collapses; at $\gamma = 400$ and 1600 every start collapses. An independent implementation found the same survival at 160 and 240 and collapse at 400. Continuing the two-class fixed points ($n_\mathrm{b}$ learned edges at one bias fraction, the others at another) in $\gamma$ from the states at $\gamma = 80$ by Newton's method, with stability from the eigenvalues of the full $4\times4$ Jacobian computed by central differences, gives the stable ranges: $n_\mathrm{b} = 3$ from its fold at $\gamma = 43.41$, where the unlearned edge has fallen to $x = 0.56$, to 111.3, where a double multiplier of symmetry breaking among the learned edges crosses one; $n_\mathrm{b} = 2$ from 46.69, just above its fold at 46.67, to 151.2; and $n_\mathrm{b} = 1$ from 51.54, just above its fold at 51.51, to its fold at 240.98 ($x = 0.0927$), located by maximizing $\gamma$ along the branch parametrized by the learned edge's bias fraction. Their union is $43.4 < \gamma < 241.0$, the shaded range of Fig. S1; the continuation made for that figure gives the same folds and stability ends within 0.4 of these (111.0, 46.69, 151.0 and 51.55). The network with one reciprocal bridge per receiver at $\gamma = 160$ (receiver $i$ of $A$ keeps sender $i+n$, and symmetrically) was solved exactly in the frozen-slow model: group contrast 4.507 and union contrast 1.757 from the isotropic part (half-trace of each $2\times2$ block) of the world-conditional covariance, whose relative departure from the form $\Sigma^{\mathrm{sc}}\otimes I_2$ after rotating $B$ by the mean cross error is 0.44; with two bridges per receiver the state does not exist at $\gamma = 160$.

**Check S47.65** (The world-conditional law at the split)**.** For $\kappa = n-1$ the closed forms of Lemma 7.7 and Theorem 7.8 were compared with doubling Lyapunov solves at ten parameter points ($F$ from 0.9 to 0.999, $R^s \in \{0.25, 1, 4\}$, $R_c \in \{0.05, 0.5\}$, $n \in \{2, 4, 8\}$): largest error $1.1\times10^{-16}$ in the mean-square error and $5.7\times10^{-14}$ in the contrast. For $\kappa < n-1$ the closed forms were compared with exact random-coefficient second moments at $(F, R^s, \kappa) = (0.98, 1, 2)$, $(0.99, 1, 1)$ and $(0.99, 4, 2)$ with $n = 4$: contrasts 5.255805, 4.868779 and 11.708162 and the mean-square errors agree to six digits, and the precisions are diffusive. At the $\gamma = \infty$ split the world-conditional covariance between the groups is $3.4\times10^{-33}$, the off-diagonal precision entries within a group all equal $-55.9701$, and

$r(A) = 8.7329 = (1 - \lambda_\perp^2)/(1 - \lambda^2)$. Subsets of $A$ with two, three and four members have contrasts 1.00000, 2.31816 and 8.73289, equal to $(\varepsilon^\Pi + n_S a^\Pi)/(\varepsilon^\Pi + (n - n_S)a^\Pi)$, and a three-member region meeting both groups has 0.18146.

**Check S47.66** (The baseline and the grid)**.** The comparison medium at the base point (eight agents, each attending three of the seven others uniformly at random, exact translations, its own calibration root $\hat{\sigma}^2 = 0.29632$) was solved by brute force, with $\mathbb{E}[W \otimes W]$ assembled from the 35 attended sets of each row (64 unknowns); it matches the exchangeable closed form, $\mathsf{s}_0^h = 0.004504$ and $\mathsf{s}_1^h = 0.003915$, to all printed digits. Its largest contrasts by region size are 0.441, 0.675, 1.000, 1.482, 2.270, 3.790 and 7.954 for two to eight members, so $\bar{r}_{\text{hom}} = 7.954$ and the admissible window is $(7.954, 8.733]$. On the grid of Numerical observation S7.6 one calibration root was found at every point, with $\mathrm{dMSE}/\mathrm{d}\hat{\sigma}^2_{\text{in}} \in [-0.4908, -0.0099]$; the split contrast lies in $[1.242, 60.483]$; the split's group contrast exceeds the comparison medium's whole-population contrast at all 162 points, with ratio in $[1.009, 1.415]$. At $R^s = 1$, $R_c = 0.05$ the pairs (split, whole comparison population) are $(2.207, 1.974)$, $(2.750, 2.635)$ and $(3.313, 3.257)$ at $F = 0.9$ and $n = 2$, 4, 8, and $(6.195, 4.869)$, $(8.733, 7.954)$ and $(12.100, 11.619)$ at $F = 0.99$. An independent evaluation of the closed forms on the same grid, scanning the calibration condition for sign changes on 4001 logarithmically spaced values of the assigned noise between $R_c$ and 50, found a single root for the split and a single root for the comparison medium at every point, the latter with $\mathrm{dMSE}/\mathrm{d}\hat{\sigma}^2 \in [-0.4912, -0.0083]$, so both are nondegenerate; it found $r_{\text{sub}} < \bar{r}_{\text{hom}}$ at every point (largest ratio 0.871), reproduced the split contrasts and the significance range $[1.0095, 1.4148]$, and located the narrowest window at $F = 0.9$, $R^s = 0.25$, $R_c = 0.01$, $n = 8$, where $r(A) = 2.2685$ and $\bar{r}_{\text{hom}} = 2.2472$.

**Check S47.67** (Finite policy precision)**.** Exact random-coefficient second moments of the frozen-slow model at the mean-field fixed points, with calibration iterated to self-consistency (24 to 27 outer iterations). Split branch at $\gamma = 120$, 160 and 80: $x = 0.9992$, 1.0000 and 0.9659; $r(A) = 8.7317$, 8.7329 and 8.6763; $r(A \cup B) = 1.0002$, 1.0000 and 1.0080; all diffusive, with departure from $\Sigma^{\text{sc}} \otimes I_2$ after the rotation of group $B$ between $1.0 \times 10^{-14}$ and $3.1 \times 10^{-14}$. Merged branch at $\gamma = 80$, 40 and 10: $x = 0.1274$, 0.1129 and 0.1066; $r(A) = 1.0431$, 1.0187 and 1.0070; $r(A \cup B) = 7.6880$, 7.8005 and 7.8531; all diffusive. Enumerating all 247 regions of at least two members: on the split branch at $\gamma = 80$ exactly $A$ and $B$ exceed 7.954 (at 8.6763), the largest contrasts by size being 0.999, 2.314, 8.676, 0.465, 0.440, 0.551 and 1.008 for two to eight members; on the merged branch at $\gamma = 80$ and 10 no region does, the largest being the whole population at 7.688 and 7.853. The annealed estimate at $\gamma = 80$ with $\mathsf{p}_\times = 1.793 \times 10^{-3}$ and $k_\times = 0.0249$ gives 8.6979 against the exact 8.6763.

**Check S47.68** (Nulls and the order parameter)**.** Renormalized deletion (isolated level-0 receivers) gives $r(A) = 1.000000$ and a diagonal precision. Partner randomization at $\kappa = 3$ partners per round, with idealized calibration solved at the null's own state (in-group assigned noise 0.48299 with unlearned cross translations, 0.29916 with learned ones), gives $r(A) = 1.3505$, $r(A \cup B) = 3.9528$ and cross-group world-conditional covariance $2.844 \times 10^{-3}$ with unlearned cross translations, and $r(A) = 1.0038$, $r(A \cup B) = 7.8676$ and $3.879 \times 10^{-3}$ with translations learned at the null's rate, $x = \mathsf{p}^*/(\mathsf{p}^* + 3/7)$; the comparison medium gives $r(A) = 1.0000$, $r(A \cup B) = 7.9543$. Enumerating all 247 regions of at least two members, the largest contrasts by size for two to eight members are 0.595, 0.901, 1.351, 1.272, 1.707, 2.451, 3.953 under partner randomization with unlearned translations and 0.442, 0.677, 1.004, 1.479, 2.262, 3.768, 7.868 with learned ones, so no region exceeds 7.954 in either. The order parameter at the $\gamma = \infty$ split, by exact Gaussian path information given the world path and the initial state, is 0, 0.02408, 0.14036 and 0.50994 nats for a within-group pair over windows of one, two, four and eight rounds, and below $1.5 \times 10^{-14}$ in absolute value for a cross-group pair. At level 3, which is not admissible and is used only for Remark S7.5: the split selects exactly $A$

and $B$; the merged state at $\gamma = 40$ and the comparison medium select the whole population and its eight seven-member regions; partner randomization with unlearned translations selects only the whole population. The largest contrasts by size are 1.000, 2.318, 8.733, 0.463, 0.437, 0.547, 1.000 at the split and 0.446, 0.685, 1.019, 1.474, 2.248, 3.739, 7.800 in the merged state, for two to eight members.

**Check S47.69** (The full model with idealized calibration)**.** Full stochastic model with $\sigma_\alpha^2 = 0.01$, eight independent replicates per cell, 8000 rounds, statistics over the last 3000: the values of Table 7, with minima and maxima over replicates of the cross attention $3.9 \times 10^{-1}$ to $4.0 \times 10^{-1}$ at $\gamma = 40$, $1.6 \times 10^{-3}$ to $2.1 \times 10^{-3}$ at $\gamma = 80$ from the declared start, 0 to $1.0 \times 10^{-5}$ at $\gamma = 160$ from the declared start (one attended cross edge-round among the $3000 \times 32 = 96{,}000$ of one replicate and none in the other seven, hence the average $1.3 \times 10^{-6}$), and $4.6 \times 10^{-3}$ to $5.7 \times 10^{-2}$ at $\gamma = 160$ from the merged start, where no replicate kept cross attention above 0.1. A single learned bridge attended with probability at least 0.34 would contribute at least $0.34/32 \approx 0.011$ to the cross attention per edge of the 32 directed cross edges. Replica estimates of the world-conditional covariance (200 replicas sharing one world path, 1500 rounds of burn-in, 2500 samples every two rounds; isotropic part after rotating $B$ by the mean cross error) at $\gamma = 160$ from the declared start gave $(r(A), r(B), r(A \cup B)) = (8.934, 8.997, 0.997)$, $(8.900, 8.874, 1.009)$, $(8.863, 8.874, 1.004)$ and $(8.916, 8.866, 1.008)$ in four runs, and $(8.741, 8.699, 1.007)$ with $\sigma_\alpha^2 = 0$, against the values $(8.7329, 8.7329, 1.0000)$ of Lemma 7.7, which assumes exact in-group translations. The eight group contrasts at $\sigma_\alpha^2 = 0.01$ lie 1.5 to 3.0 percent above 8.7329, with mean 8.903, and the two at $\sigma_\alpha^2 = 0$ lie within 0.4 percent of it. A simulation of group $A$ alone at the $\gamma = \infty$ split with the frozen split gains and in-group angles following the learning rule of Definition 7.2 (16 world paths with 64 replicas each, 400 rounds of burn-in, 3600 samples, one seed, the same random numbers at both noise levels) gave $r(A) = 8.712$ at $\sigma_\alpha^2 = 0$ and 8.947 at $\sigma_\alpha^2 = 0.01$, a shift of 2.7 percent, with $\mathsf{s}_0$ higher by 0.8 percent, $\mathsf{s}_1$ higher by 3.9 percent and angle standard deviation 0.022 radians. At $\gamma = 40$ from the merged start $(1.164, 1.167, 7.271)$ against $(1.0187, 1.0187, 7.8005)$. An independent simulation of the slow recursion alone, with frozen idealized calibration and attended sets drawn every round (eight replicates of four receivers, 8000 rounds), gave cross attention and mean bias fraction 0.392 and 0.116 ($\gamma = 40$), $1.77 \times 10^{-3}$ and 0.967 with all eight replicates free of learned cross edges ($\gamma = 80$, declared start), 0.247 and 0.587 ($\gamma = 80$, merged start), $5 \times 10^{-6}$ and 1.000 ($\gamma = 160$, declared start), and 0.0239 and 0.963 ($\gamma = 160$, merged start).

**Check S47.70** (Online calibration)**.** Full model with online calibration from the symmetric start, $\sigma_\alpha^2 = 0.01$, eight replicates per cell, 6000 rounds; the counts are those of Table 8, and the mean number of cross edges per receiver ending below $0.3\vartheta$ ranges from 0.016 ($\gamma = 160$, $\alpha_{\mathrm{cal}} = 0.2$) to 2.078 ($\gamma = 80$, $\alpha_{\mathrm{cal}} = 0.5$). A rerun with other seeds and 8000 rounds at $\alpha_{\mathrm{cal}} = 0.05$ gave 6, 1 and 2 replicates of 8 in the split's attention pattern at $\gamma = 160$, 400, 1600, with 0.355, 0.536 and 0.621 stale in-group estimates per receiver (in-group edges dropped after an unlucky high estimate and never sampled again). Two variants test the lock-in: relaxing unattended estimates to their start value removes the stale estimates but re-explores every cross edge and gave 0 of 8 at all three values of $\gamma$; shrinking estimates toward the start value gave 6, 2 and 3 of 8. For independent residuals with two degrees of freedom the calibration estimate has relative standard deviation $(\alpha_{\mathrm{cal}}/(2 - \alpha_{\mathrm{cal}}))^{1/2} \approx 0.16$ at $\alpha_{\mathrm{cal}} = 0.05$, and more for the time-correlated residuals of the model. Per-replicate contrasts in a separate run of eight replicates at $\gamma = 400$, $\alpha_{\mathrm{cal}} = 0.05$ were estimated from the time-averaged covariance of the local part of the error over the last 5000 of 8000 rounds, given the realized attention and translations, by the isotropic part of each $2 \times 2$ block, whose departures from the form $\Sigma^{\mathrm{sc}} \otimes I_2$ were between 0.06 and 0.16: replicates with two or three cross edges below $0.3\vartheta$ have $(r(A), r(B), r(A \cup B)) = (7.585, 4.795, 1.944)$, $(3.331, 3.112, 3.054)$,

$(3.397, 4.035, 2.454)$, $(3.010, 3.076, 3.155)$ and $(2.441, 5.007, 3.665)$; replicates in the split's attention pattern have $(5.471, 7.620, 2.021)$, $(9.890, 8.858, 1.098)$ and $(3.687, 4.060, 3.582)$. The same estimator at the split with idealized calibration ($\gamma = 160$) gave group contrasts from 8.018 to 9.891. The cutoff $0.3\vartheta$ used throughout this check lies below the bias fraction $\frac{1}{2}$ that separates learned from unlearned edges for the map, so the pattern counts are upper bounds on those at the cutoff $\frac{1}{2}$. Sampling error of the counts: the cell $\gamma = 400$, $\alpha_{\mathrm{cal}} = 0.05$ gave 0 of 8 in Table 8, 1 of 8 in the rerun and 3 of 8 in the contrast run, and the cell $\gamma = 160$, $\alpha_{\mathrm{cal}} = 0.05$ gave 3 and 6 of 8; a count out of eight has binomial standard deviation up to 1.41, and two-sided Fisher exact tests give $p = 0.20$ for 0 against 3 and $p = 0.31$ for 3 against 6, so the reruns are mutually consistent. In the table only the differences of the cell with 7 of 8 from the cells with 0, 1 and 2 of 8 ($p = 0.001$, 0.010, 0.041) and of the cell with 5 of 8 from the cells with 0 of 8 ($p = 0.026$) fall below 0.05, without correction for the number of comparisons; 7 against 3 gives $p = 0.12$ and 7 against 5 gives $p = 0.57$.

**Check S47.71** (Abundant capacity)**.** At $\kappa = n = 4$ and $\gamma = \infty$ the bias fraction of an edge attended every round is $\mathsf{p}^*/(\mathsf{p}^* + 1) = 0.0478$. With reciprocal learned bridges the frozen-slow second moments give $r(A) = 2.462$ and $r(A \cup B) = 5.137$, with departure $4.8 \times 10^{-17}$ from $\Sigma^{\mathrm{sc}} \otimes I_2$ after the rotation of group $B$, against 8.733 and 1.000 at $\kappa = n - 1$.

**Check S47.72** (The grounding tension)**.** Calibration fixed points at $F = 0.99$, $R_c = 0.05$, $n = 4$, $\kappa = 3$: at $R^s = 0.25$, 1 and 4, $(\hat{\sigma}^2_{\mathrm{in}}, D_{\mathrm{cal}}, r(A)) = (0.16861, 1.62085, 4.6230)$, $(0.29872, 1.24123, 8.7329)$ and $(0.52407, 0.73941, 14.3314)$, so $D_{\mathrm{cal}}/\hat{\sigma}^2_{\mathrm{in}} = 9.61$, 4.16 and 1.41 while the split contrast rises. An independent evaluation of the frozen constants at the three points gives $\Delta\mathcal{V}(\frac{1}{2}) = 0.0907$, 0.0592 and 0.0262, hence $\gamma_S = 44.5$, 68.2 and 153.9 at $\vartheta = \pi/2$ and $\mathsf{p}^* = 0.0503$.

## S47.7 Checks for Section 8

These computations confirm results of Section 8 on small instances. Where a second, independent computation was made, both are reported.

**Check S47.73** (A moment bound at spectral radius one)**.** For the lazy walk of Lemma 8.2(c) on $\mathbb{Z}^3$, quadrature of $G(E) = \int_0^\infty e^{-Es} I_0(s)^3 \mathrm{d}s$ gave $G(3) = 0.505462$ and $G(5) = 0.214294$, hence the variance $2(G(3) + G(5))q = 1.439512q$. The mean of $(1 - \hat{W}(p)^2)^{-1}$ over midpoint Fourier grids with 40 and 80 points per axis was 1.4256 and 1.4326, and the partial sums of $\sum_m (2\pi)^{-3} \int \hat{W}^{2m} \mathrm{d}p$ on a grid with 48 points per axis were 1.329, 1.404 and 1.427 when summed through $m = 10$, 100 and 1000, increasing and bounded. For the lazy nearest-neighbor walks on $\mathbb{Z}$ ($W_{xx} = 1/2$, neighbor weights 1/4) and on $\mathbb{Z}^2$ ($W_{xx} = 1/3$, neighbor weights 1/6) the grid means grew without bound as the grid was refined, 25.4, 100.4 and 400.4 on $\mathbb{Z}$ and 2.75, 3.41 and 4.07 on $\mathbb{Z}^2$ at 50, 200 and 800 points per axis.

**Check S47.74** (Light cone of a nonlinear rule)**.** The synchronous update $\mu_i \leftarrow \tanh(\sum_k B_{ik}\mu_k) + h_i$ on a ring of 30 agents, with $B_{ii}$ uniform on $[0.2, 0.6]$, the two neighbor entries of each row uniform on $[-0.5, 0.5]$, offsets $h_i$ normal with standard deviation 0.1, a standard normal initial state, and the influence bound $\bar{J}_{ik} = |B_{ik}|$. The Jacobian $J_{ij}(t, 0)$ vanished outside the cone $d(i, j) \le t$ to machine precision at $t = 3$, 6 and 12, and no entry exceeded $(\bar{J}^t)_{ij}$. The largest row sums of $\bar{J}^t$ were 1.51, 1.93 and 3.03, against $g^t_{\mathrm{tot}} = 2.10$, 4.39 and 19.3.

**Check S47.75** (Poisson clocks)**.** A ring of 40 agents with clock rate one and linear scalar updates with coefficient matrix $A$. For random convex weights with largest diagonal entry $\bar{J}_{\mathrm{diag}} = 0.726$ and largest off-diagonal row sum $g = 0.762$, the exact mean $e^{\nu t(A-I)}$ never exceeded the bound of Proposition S8.3 at $t = 2$, 5 and 10, the largest ratio falling from 0.887 to 0.453; with constant

diagonal 0.5 the ratio was 1.000000 at all three times, the predicted equality. For signed weights a dominating process with weights $|A|$ was never exceeded pathwise, and the Monte Carlo excess over the bound stayed within one standard error. For the directed copy chain the expected influence $\mathbb{E}J_{ij}(t) = e^{-\nu t}(\nu t)^d/d!$ equals the middle member of (S31), 0.17547 at $t = 5$, $d = 5$ and 0.12511 at $t = 10$, $d = 10$, from the matrix exponential. The probability that a signal held at the source has crossed $d$ hops by time $t$ is a different quantity and equals the last member $e^{-\nu t}\sum_{n\ge d}(\nu t)^n/n!$, 0.55951 and 0.54207 at the same points, with Monte Carlo values 0.555 and 0.542. The velocity $\inf_{\varphi>0}(ge^{\varphi} + \bar{J}_{\mathrm{diag}} - 1)/\varphi$ evaluated to 2.7183 at $g = \bar{J}_{\mathrm{diag}} = 1$ and to 1.0001 at $g = 1$, $\bar{J}_{\mathrm{diag}} = 0$.

**Check S47.76** (Belief-dependent routing and the switch chain)**.** Three hundred random routing instances with five policies, linear updates and linear expected free energy: the largest ratio of the exact Jacobian of the averaged map to $\bar{J}(\gamma)$ was 0.572, with no violation. In the chain of Example S8.4, the response 20 hops away to a perturbation of $10^{-6}$ at the head equaled $(\gamma\alpha_0/2)^{20}\cdot 10^{-6}$ for $\gamma\alpha_0 = 1$, 1.8, 2 and 2.5 (for example $9.537 \times 10^{-13}$ at $\gamma\alpha_0 = 1$), was 0.734 for $\gamma\alpha_0 = 4$, where the linear value 1.049 lies outside the range of the map and the response approaches the fixed point $\mu^* = 0.9575$ over further hops (0.9575 at 25 hops), and was 1.0 at $\gamma = \infty$. With sampled routing and message noise of standard deviation 0.5, quadrature gave the flip probabilities 0.15546, 0.06767, 0.03430, 0.02345 and $\Pr(Z < -2) = 0.02275$, and the correlation lengths of the routing bit 2.69, 6.88, 14.07, 20.82 and 21.47 hops, for $\gamma\alpha_0 = 2$, 4, 8, 32 and $\infty$; Monte Carlo estimates from $4 \times 10^5$ draws agreed within two standard errors. Simulated correlations of the routing bit agreed with $(1 - 2p)^n$, for example 0.478 against 0.483 at $n = 5$ and $\gamma\alpha_0 = 4$.

**Check S47.77** (One common code)**.** This check collects the evidence for Conjecture S8.2. The instances were drawn as follows. The two-agent loops live in a two-dimensional world with $F = fI$, $Q = I$ and unit message and sensing noise variances, with $f = 0.5 + 0.49/(1 + e^{-x})$ for $x$ normal with standard deviation 2, and with code and sensing rows of independent normal entries with standard deviation 3. The three-dimensional instances have $f$ uniform on $[0.5, 0.97]$, $Q = I$ and code and sensing matrices of independent normal entries with standard deviation 2, 120 draws for each of the rank pairs $(1, 1)$, $(2, 1)$ and $(1, 2)$. The switching instances have $f$ uniform on $[0.8, 0.995]$, $Q = qI$ with $q$ uniform on $[0.05, 1]$, and rank-one code and sensing rows of normal entries with standard deviations 3 and 6. In random two-agent reciprocal loops that sense and listen, the largest ratio $\rho(W)/f$ was 1.0000000000 over 300 draws and after Nelder–Mead refinement with a common code, against 100 of 300 draws above one with independent codes (largest 1.1673, and 1.2071 after refinement). In three-dimensional worlds with code and sensing ranks one and two, none of 360 common-code instances exceeded one, against 85 of 360 with independent codes (largest 1.1772). For period-two switching between listening and sensing, none of 300 common-code instances exceeded $f$ per round, against 112 of 300 with independent codes (largest 1.0941). The common-code instances with symmetric listening that confirm Proposition 8.3(b) and the comparison of Corollary 8.19(b), and the one-round reception maps on random directed listening graphs, none of which exceeded spectral radius one, are those of Check S47.99; they are reported only there.

**Check S47.78** (Clustering without contraction)**.** Rings of 40 scalar agents over 18 rounds from a diagonal initial covariance, in four cases: a time-invariant signed $W$ with $g_0 = 1.52$ and spectral radius 0.77; a time-varying signed schedule with time-varying noise, $g_0 = 1.49$; a nonnegative $W$ with every row sum above one, $g_0 = 1.60$ and spectral radius 1.18; and a signed $W$ with spectral radius 1.33, $g_0 = 2.22$. In each case the self weight and the two neighbor weights of an agent have moduli uniform on $[0.2, 1]$ times a common scale (0.55, 0.5, 0.62 and 0.8 in the four cases), with signs flipped independently with probability one half in the signed cases; the noise is diagonal with variances uniform on $[0.5, 1.5]$, redrawn every round in the time-varying case, and the initial

variances are uniform on $[0,2]$. Every covariance entry with $d(i,j) > 2t$ was exactly zero. Over all pairs and rounds, the largest ratio of $\|\Sigma_{ij}(t)\|$ to the bound of Theorem 8.7(a) was 0.42 at $\varphi = 0$, 0.23 at $\varphi = 0.5$, 0.20 at $\varphi = 1$ and 0.23 at $\varphi = 2$, and the normalized ratio of (b) was at most 0.06. An independent computation on a ring of 40 agents with a signed time-varying schedule ($g_0 = 2.67$; self and nearest-neighbor entries uniform on $[-0.9, 0.9]$ and noise variances uniform on $[0.2, 1]$, both redrawn every round, initial variances uniform on $[0.5, 1.5]$) over 16 rounds found no nonzero entry beyond $2t$ and largest ratios 0.207, 0.106 and 0.105 at $\varphi = 0$, 0.5 and 1. The lag bound (78) was tested separately on rings of 24 scalar agents over 12 rounds with lags $h = 1, \dots, 4$, for a time-invariant signed $W$ ($g_0 = 2.41$, spectral radius 1.42), a time-varying signed schedule with time-varying noise ($g_0 = 3.79$) and a nonnegative $W$ with every row sum above one ($g_0 = 1.63$, spectral radius 1.43); the self and nearest-neighbor entries are normal with standard deviation 0.5 in the signed cases and uniform on $[0.3, 0.6]$ in the nonnegative case, the initial variances are uniform on $[0.2, 1]$, and the noise covariance is the identity except in the time-varying case, where its diagonal is uniform on $[0.3, 1]$ and redrawn every round. Every entry of $\mathrm{Cov}(e(t+h), e(t))$ with $d(i,j) > 2t + h$ was exactly zero, entries with $2t < d(i,j) \le 2t + h$ were nonzero, and the largest ratio of an entry to the bound was 0.25 at $\varphi = 0$, 0.15 at $\varphi = 0.5$ and 0.08 at $\varphi = 1$. On the scalar chain with $W_{xx} = 0.2$, $W_{x,x\pm1} = 0.3$, $q = 1$ and zero initial covariance, the lagged covariance at $t = 2$, $h = 1$, $d = 3$ is 0.027, against the bound 0.512 at $\varphi = 0$.

**Check S47.79** (Rank-one dominance in the amplifying profile)**.** The profile of Example S8.1 rebuilt from its specification has dominant eigenvalue 1.0613266750560013, simple and real, second modulus 0.412032, and block row sums 1.0999 for both agents. From the covariance recursion of its mean map, the per-round growth $\frac{1}{2}\ln(\|\Sigma(t)\|/\|\Sigma(t-1)\|)$ was 0.06294, 0.05969, 0.05952013 and 0.05951971 at $t = 25$, 50, 100 and 150, against $\ln\rho = 0.05951971$. One minus the top canonical correlation of the two agents was $1.6 \times 10^{-2}$, $8.0 \times 10^{-4}$, $2.1 \times 10^{-6}$, $5.4 \times 10^{-9}$ and $1.4 \times 10^{-11}$ at $t = 25$, 50, 100, 150 and 200, evaluated in 60-digit arithmetic because double precision loses the last value to cancellation; the last three values fall by the factor $\rho^{-100} = 2.6007 \times 10^{-3}$ per 50 rounds, the rate $\rho^{-2t}$ of (S32).

**Check S47.80** (An amplifying site)**.** An open chain of 801 agents with $c = a = 0.3$ ($\lambda = 0.9$) and an amplifying center. For $\rho = 1.1$ and 1.2 the symmetric eigensolver reproduced $\rho$ to ten digits, the second eigenvalue was $0.899982 \le 0.9$, and the fitted eigenvector decay was 0.79536555 against $m_\rho = 0.79536546$ and 0.96242368 against 0.96242365. From the direct recursion $\Sigma(t+1) = W\Sigma(t)W^\top + qI$, the fronts at $\theta = 0.5$ and 0.9, resolved to whole hops, stood at 8, 19, 29 and 6, 17, 28 at $t = 60$, 150, 240 for $\rho = 1.1$, with fitted speeds over $t \ge 150$ of 0.1152 and 0.1182 against $v_\mathrm{f} = 0.1198$; for $\rho = 1.2$ the speeds were 0.1946 and 0.1855 against 0.1894. The lower bound of Proposition S8.7(c) held at all 340 and 382 pairs $(t, x)$ checked with $\mathfrak{s}_x > 1$, and the decomposition of (b) agreed with the direct recursion at $t = 40$ to $2.5 \times 10^{-14}$. An independent computation on a chain of 401 agents found the top eigenvalue equal to $c + (4a^2 + b_0^2)^{1/2}$ to ten digits, the second eigenvalue 0.899927, the eigenvector decay 0.75848615 against $\operatorname{arcsinh}(b_0/(2a)) = 0.75848614$, $\|B(t)P_v\| = 1.3 \times 10^{-11}$, and the spectrum of $B(t)$ on $v^\perp$ in $[1.0000, 5.2584] \subset [q, q/(1-\lambda^2)] = [1, 5.2632]$.

**Check S47.81** (Spectral form with signed weights)**.** Three symmetric rings of 80 agents with signed weights, spectral radius 0.95 and largest row sums 1.227, 1.276 and 1.271, so that the row-sum bound is unusable. Each ring has diagonal entries uniform on $[-0.3, 0.3]$ and symmetric bond weights uniform on $[-0.5, 0.5]$, rescaled to spectral radius 0.95, with noise covariance $qI$. The result: no violation of the bound of Proposition S8.10, with largest ratios 0.508, 0.478 and 0.539.

**Check S47.82** (Diffusive clustering)**.** A $20 \times 20$ torus with symmetric bond conductances uniform on $[0.5, 1.5]$ and self-conductances uniform on $[0.5, 2]$, with $P$ the conductances divided by their row

sums, $W = (1-\varepsilon)P$ and noise covariance $qI$, at $\varepsilon = 0.05$ and 0.01: no violation of (S34), with the bound minimized over $\vartheta \in \{0.05, 0.10, \dots, 0.95\}$. The fitted decay rates of the largest covariance over hop distances 4 to 17 were 0.390 and 0.153, against the guaranteed rates 0.320 and 0.142 and the row-sum rates 0.051 and 0.010. On a $15 \times 15$ torus with bond conductances uniform on $[0.2, 1]$, self-conductances uniform on $[0.5, 2]$, $\lambda = 0.97$ and $q = 1$, $|\Sigma_{ij}|$ used at most 2.3 percent of the bound.

**Check S47.83** (Heterogeneous media)**.** Rings of 300 agents with $F = 0.97$ and nearest-neighbor attention, the ring form of the upper bound being computed by image sums. Homogeneous control: $m_{\mathrm{up}} = m_{\mathrm{low}} =$ fitted rate $= 0.48879$. Mild heterogeneity (leaks in $[0.0495, 0.0591]$, couplings in $[0.233, 0.252]$): the guaranteed rate, computed here and in the cases below as the largest tilt $\varphi$ with $\Psi(\varphi) < 1$ in Proposition S8.12(a), was 0.4424, slightly above the closed form $m_{\mathrm{up}} = \operatorname{arccosh}(1+\varepsilon_{\min}/(2a_{\max})) = 0.4399$ and below the fitted rates 0.4667 and 0.4699 of the largest and smallest correlation at each distance, which are below $m_{\mathrm{low}} = 0.6296$, so $\xi \in [1.59, 2.26]$ hops. Strong symmetric heterogeneity: largest tilt 0.3591 (closed form 0.3405) $\le 0.4795, 0.5057 \le 1.5920$. Strong directed heterogeneity, where the two neighbor weights of an agent differ and the closed form does not apply: largest tilt $0.1599 \le 0.4906, 0.5649 \le 2.1048$. The upper bound was used at most to 0.21 on all resolved entries, $\Sigma - q_{\min}(I - W_{\mathrm{low}}^2)^{-1} \ge -1.9 \times 10^{-15}$ (round-off), and $\min_i(1-\varepsilon_i) \le \rho(W) \le \max_i(1-\varepsilon_i)$ in every case. On an open chain the rows of the two one-sided end agents lower the largest admissible tilt to 0.2644, against 0.3591 on the ring drawn from the same law, and the bound with $\bar{\psi}$ taken over all rows holds with largest ratio 0.11; on an open $18 \times 18$ grid both bounds hold along the axis and the diagonal (largest ratio 0.045, smallest gap $4.1 \times 10^{-7}$); on a ring with a period-three schedule of heterogeneous directed gains, after 600 rounds the largest ratio was 0.071.

**Check S47.84** (The ring formula)**.** The closed form (82) against a Lyapunov solution: relative difference at most $2.1 \times 10^{-15}$ at $(c, \varepsilon) = (0.25, 0.2)$, $(0.25, 0.02)$, $(0.6, 0.01)$ and $(0.05, 0.05)$, and, with ring images, absolute difference at most $1.9 \times 10^{-13}$ on rings of 50 to 100 agents at four further parameter sets, one with $c = 0$. At $(c, \varepsilon) = (0.25, 0.02)$ the rate is $m = 0.23355 = 2\operatorname{arcsinh}\sqrt{\varepsilon/(4a)}$, against $\sqrt{\varepsilon/a} = 0.23408$.

**Check S47.85** (Ornstein–Zernike decay in two dimensions)**.** On a $512 \times 512$ torus, for the resolvent $(I-W)^{-1}$ at $c = 0.2$: at $\varepsilon = 0.05$ the ratio to the form (S35) rose to 0.9963, 0.9965 and 0.9958 along the axis, the diagonal and the direction $(2, 1)$ at $r \approx 30\xi$, and similarly at $\varepsilon = 0.01$; the relative anisotropy of the support function was 0.00537 against the small-$\varepsilon$ value $\varepsilon/(48a) = 0.00556$ at $\varepsilon = 0.05$, and 0.00105 against 0.00105 at $\varepsilon = 0.01$; at $\varepsilon = 0.3$ the support function gives 1.4254 and 1.4806 along the axis and the diagonal, and the local rates corrected for the prefactor approach 1.4260 and 1.4822. By Fourier inversion on a $1024 \times 1024$ torus at three parameter sets, the fitted pointwise axis rates were 0.51023, 0.22421 and 1.23420 against $\operatorname{arccosh}(1+\varepsilon/(2a)) = 0.51083$, 0.22455 and 1.23024, and the diagonal rates 0.51260, 0.22442 and 1.26222 against $\sqrt{2}\operatorname{arccosh}(1+\varepsilon/(4a)) = 0.51357$, 0.22478 and 1.26655. At $(c, a) = (0.1, 0.2)$ in two dimensions the hyperplane-summed decay rate was $0.6931 = \operatorname{arccosh}(1+\varepsilon/(2a))$, and (S41) with $\cosh m = 1+\varepsilon/(2a)$ gave $\xi_2 = 1.3765$, the computed value, whereas the one-dimensional relation $\cosh m = (1-c)/(2a)$ would give 0.6156.

**Check S47.86** (Second-moment lengths)**.** (a) By direct summation, $\xi_2 = 2.70327$ and 6.26909 in one dimension and 1.91150 and 4.43292 in two at $(c, \varepsilon) = (0.2, 0.05)$ and $(0.2, 0.01)$, equal to (83) to all printed digits; an independent summation gave $\xi_2^2 = 19.30303$ and 9.65152 in one and two dimensions, equal to the formula, with $a/\varepsilon = 19.5$ and 9.75. (b) On a ring of 4096 sites with four random kernels of range one to four, the covariance-side and precision-side lengths agreed to

all printed digits (3.058728, 23.469588, 7.390093, 2.808015), the root $m_d$ of Proposition S8.27(b) equaled the fitted decay rate to six digits, and (S40) held, for example $23.4696 \le 23.4892 \le 23.6398$. For the medium of Theorem 8.10 in one dimension, $\xi_{\mathrm{OZ}}/\xi_2 = 1.041$, 1.003 and 1.0004 at $\xi_2 = 1.69$, 7.02 and 15.79. (c) In two dimensions the on-site variance grew with slope 0.19867 in $\ln(1/\varepsilon)$ between $\varepsilon = 10^{-3}$ and $10^{-4}$, against $q/(8\pi a) = 0.19897$.

**Check S47.87** (Persistence)**.** Three strongly non-normal $W$ with $\rho(W) = 0.9$: $\|\tilde{W}\|$ equaled $(1 - \lambda_{\min}(\Sigma^{-1/2} Q^{\zeta} \Sigma^{-1/2}))^{1/2}$ to five digits (0.98319, 0.98812, 0.99996), and $\|\tilde{W}^h\| \le \|\tilde{W}\|^h$ at $h = 1$, 5 and 20, while $\|\tilde{W}^h\|$ tracked $\rho^h$ for large $h$; in the third case $\|\tilde{W}\| = 0.99996$ while $\|\tilde{W}^{20}\| = 0.47$.

**Check S47.88** (Autocovariance and diffusive scaling)**.** In the medium of Theorem 8.10 with $c > 0$, the local autocovariance was compared with the continuum approximation $\frac{q}{2} \int_h^\infty e^{-\varepsilon_\lambda u} (4\pi a u)^{-D/2} \mathrm{d}u$, $\varepsilon_\lambda = \ln(1/\lambda)$: at $\varepsilon = 0.002$ the ratio lay in $[0.994, 1.005]$ for $h$ from 10 to 3000 in one dimension and in $[0.993, 1.017]$ in two. That range has $\varepsilon h \le 6$, where the integral form applies and the asymptote (S38), its limit for $\varepsilon h \gg 1$ at small $\varepsilon$, is not yet accurate: at $\varepsilon = 0.002$ and $h = 3000$ the ratio of (S37) to (S38) was 0.932 in one dimension and 0.872 in two. The asymptote itself was tested at $c = 0.2$ and $\varepsilon = 0.05$, where the same ratio was 0.8715, 0.9583, 0.9886 and 0.9969 at $h = 50$, 200, 800 and 3000 in one dimension and 0.7703, 0.9201, 0.9775 and 0.9938 in two; the shortfalls at $h = 3000$, 0.0031 and 0.0062, equal $D\lambda^2/((1-\lambda^2)h)$ to the digits shown, about $D/(2\varepsilon h)$. Local decay rates approached $\varepsilon_\lambda + D/(2h)$, for example 0.01021 and 0.01037 at $h = 3000$ and $\varepsilon = 0.01$ in one and two dimensions. The ratio $\xi_2^2/\tau$ was 0.39499 against $a = 0.395$ at $\varepsilon = 0.01$ in one dimension, and 0.19750 against 0.1975 in two.

**Check S47.89** (The record model)**.** (a) Over 3000 random reduced models with one to four records, the root of $g_z$ equaled $\lambda_{\min}(\mathsf{R})$ to a relative error of $8.5 \times 10^{-13}$, and $\rho(I - \mathsf{R}) = 1 - \varepsilon_{\mathrm{rec}}$ to $2.9 \times 10^{-15}$ whenever $I - \mathsf{R} \ge 0$; an independent computation found $1.3 \times 10^{-12}$ and $3.1 \times 10^{-15}$ in 2993 nonnegative cases. (b) No sign mismatch between $\varepsilon - \varepsilon_{\mathrm{rec}}$ and the criterion of Proposition 8.12(c) occurred in 15962 stable one-record draws with $\upsilon$ zero, small or arbitrary in $[0, 1)$, and no mismatch of (86) in 20000 several-record draws; the brackets of Proposition 8.12(a) and (b) held in 20000 draws. (c) Mixed records at $\varepsilon = 0.002$ and $\upsilon = 0$: the durable record $(\beta, \omega, \delta) = (0.05, 0.01, 10^{-4})$ alone gave $\varepsilon_{\mathrm{rec}} = 0.000408$ and multiplied $\tau$ by 4.900; with the perishable record $(0.2, 0.1, 0.01)$ added, $\varepsilon_{\mathrm{rec}} = 0.002291$ and $\tau$ fell to 0.873 of its value without records. (d) The exact reduction of Proposition 8.11(d) held to $4.1 \times 10^{-15}$ on random networks of 4 to 12 agents with one to three records and heterogeneous write weights. (e) Special cases: one record with $(\varepsilon, \beta, \omega, \delta) = (0.01, 0.2, 0.1, 0)$ gave $\varepsilon_{\mathrm{rec}} = 0.003260$ and $\tau = 306.24$; in two turnover instances the secular roots equaled the exact identity rates 0.0012379 and 0.0021905. (f) For level-0 reading with fixed prior precision the displacement fraction equaled the sensing gain before reading in three instances (0.166667, 0.142857, 0.019608). (g) Thirty agents on a ring and one archive with retention $1 - \delta$: $\rho(W) = 0.9000$, 0.9862 and 0.9951 for $\delta = 0.1$, 0.01 and 0.001, at least the archive's self-weights 0.894, 0.984 and 0.993, and an agent's autocorrelation at lag 50 rose from $1.1 \times 10^{-4}$ to $5.2 \times 10^{-2}$.

**Check S47.90** (The length with records)**.** A ring of 2000 sites with $a = 0.2$ and $\varepsilon = 0.002$, so that $\xi_2 = 9.995$ hops and $\tau = 499.5$ rounds without records. For no record; one record $(\beta, \omega, \delta, \ell_r) = (0.1, 0.01, 10^{-4}, 0)$; $(0.1, 0.1, 5 \times 10^{-3}, 6)$; $(0.1, 0.01, 5 \times 10^{-3}, 6)$; $(0.1, 0.01, 10^{-4}, 6)$; and two records $(0.05, 0.01, 10^{-4}, 0)$ and $(0.05, 0.1, 5 \times 10^{-3}, 6)$, the real-space second moment gave $\xi_2 = 9.995$, 7.265, 16.308, 3.592, 22.527 and 12.957 hops, equal to formula (a) of Proposition 8.13 in every case. The slow-branch value $(2(1 - x_0)\mathcal{D}_{\mathrm{eff}}/(x_0(2 - x_0)))^{1/2}$, the leading term in the proof of (b), was 9.995, 8.078, 16.927, 6.278, 25.742 and 14.333, and the first-order form of (90) 10.000, 8.165, 16.903, 6.202, 25.820 and 14.832; the correlation times were 499.5, 3702.5, 286.1, 211.3, 3702.5 and

1381.1. In every configuration (89) agreed with the first-order perturbation of the slow eigenvalue to $4.2 \times 10^{-16}$. For the asymptotics of (b), the leaks of the second, fourth and sixth configurations, with $q_z = 0.5$, were divided by 10, 100 and 1000: the ratio of $\xi_2^2$ to $\mathcal{D}_{\mathrm{eff}}/\varepsilon_{\mathrm{rec}}$ tended to one, its relative error divided by $\varepsilon_{\mathrm{rec}}$ settling at 167.3, 181.4 and 123.2, as the error term of (b) requires. The fitted decay lengths of the covariance in the second to fifth configurations were 8.18, 17.06, 6.86 and 25.90 hops. On the same ring, the agents' marginal world-conditional precision had no positive off-diagonal entry with no record, with a durable local record and no record noise, and with a perishable record read at range and no record noise; with the durable record and record noise of variance one it had eleven positive off-diagonal entries, the largest $7.6 \times 10^{-3}$.

**Check S47.91** (Hub component of a global record)**.** One record read by all agents of a ring with weight $\beta$ and written uniformly with total weight $\omega$, no record noise, in the setting of Remark S8.17. On a ring of 12 agents with $a = 0.1$, $\varepsilon' = 10^{-3}$, $\beta = 0.2$, $\omega = 0.05$ and $\delta = 10^{-4}$, the agents' covariance from the discrete Lyapunov equation equaled the branch $p \neq 0$ continued to all wavevectors plus the constant $[\hat{\Sigma}_{00}(0) - q/(1-(1-\varepsilon'-\beta)^2)]/N$ to within $2 \times 10^{-12}$. With $s = (\varepsilon' + \mathcal{R}\delta)(\beta+\omega)/(\beta\omega)$, the constant was the following fraction of $\hat{\Sigma}_{00}(0)/N$ at $(\varepsilon', \beta, \omega, \delta)$: 0.996 at $(10^{-4}, 0.1, 0.1, 10^{-4})$, where $s = 0.004$; 0.962 at $(10^{-3}, 0.2, 0.05, 10^{-4})$, where $s = 0.035$; 0.484 at $(10^{-5}, 0.1, 10^{-3}, 10^{-5})$, where $s = 1.02$; 0.093 at $(10^{-5}, 0.1, 10^{-3}, 10^{-4})$ and 0.078 at $(10^{-2}, 0.1, 10^{-3}, 0)$, where $s = 10.1$. In ten instances the fraction stayed within 0.015 of $1/(1+s)$. The last instance has $\varepsilon_{\mathrm{rec}} \approx 10^{-4} \ll \beta$, so smallness of the effective leak beside the read weight does not make the hub component dominant.

**Check S47.92** (Algebraic decay)**.** A ring of $2^{17}$ sites with $c = 0.3$ and $\lambda = 0.95$. The ratio of $\Sigma(x)$ to $2q\lambda a \mathsf{J}(x)/(1-\lambda^2)^2$ fell from 1.3034 at $x = 25$ to 1.0006 at $x = 3200$ for $\sigma = 1.5$, and from 5.1427 to 1.0000 for $\sigma = 3$, with fitted tail exponents 2.503 and 4.000 between $x = 800$ and 3200 (against $1 + \sigma = 2.5$ and 4); for $\sigma = 0.5$ it rose from 0.11 to 0.76 over the same range, with corrections decaying roughly as $x^{-0.48}$.

**Check S47.93** (Fronts)**.** The front $r_{0.1}(t)$ of $\Pr(|Y_t| \geq r)$ in one dimension, on rings of $2^{20}$ sites for $\sigma = 0.5$ and 0.75 and of $2^{17}$ sites for $\sigma = 1$, 1.5 and 3, with the fronts at most 3.2 percent of the half-length. Least-squares fits of $\ln r_{0.1}$ against $\ln t$ over doubling times, $t = 1$ to 8 for $\sigma = 0.5$, $t = 2$ to 32 for $\sigma = 0.75$ and $t = 16$ to 512 for the others, gave exponents 1.897, 1.305, 0.988, 0.671 and 0.469 for $\sigma = 0.5$, 0.75, 1, 1.5 and 3, against 2, 1.333, 1, 0.667 and 0.5. The fits depend on the window: at $\sigma = 3$, where the fronts are only 8 to 39 sites, the window $t = 32$ to 256 gave 0.497, and at $\sigma = 0.5$ the local slopes between successive times were 1.879, 1.906 and 1.901, so the shortfall of five percent there did not close within the window. The single-big-jump ratio $\Pr(Y_t = x)/(t \Pr(X_1 = x))$ at $\sigma = 1.5$ and $t = 16$ was 1.0133, 1.0017, 1.0002 and 1.0000 at $x = 200$, 800, 3200 and 12800.

**Check S47.94** (Gaussian broadcasting on trees)**.** The recursion of Theorem S8.19(a) agreed with direct Gaussian conditioning to ten digits on binary trees of depth 7 (255 vertices), ternary trees of depth 5 (364 vertices) and Galton–Watson trees, with heterogeneous edge correlations and leaf noise 0 or 0.3. On $b$-ary trees: at $b\alpha^2 = 0.8$, $I_{40} = 2.216 \times 10^{-5}$ against the asymptotic $2.215 \times 10^{-5}$; at $b\alpha^2 = 1$, $I_{40} = 2.440 \times 10^{-2}$ against $2.500 \times 10^{-2}$; at $b\alpha^2 = 1.25$ and 1.8 the limits 0.2554 and 1.0986. The closed form of the union bound in Theorem 8.16 agreed with truncated sums at three parameter sets, for example 0.02000000 at $\Delta s_e = 0.1$, $s_m = 0.5$, $d = 3$.

**Check S47.95** (Gaussian information percolation)**.** Four hundred random trees with edge correlations in $[0.05, 0.99]$, three leaf-noise levels and five source signal-to-noise ratios: the largest ratio of the left side of (S39) to the right side was 1.0000, attained in the small-signal limit of a single edge, with no violation.

**Check S47.96** (Coherence volume and susceptibilities)**.** (a) Disks of radius 2, 4, 8 and 12 (13 to 441 agents) in the two-dimensional medium of Theorem 8.10 on a $128 \times 128$ torus, at $\varepsilon = 0.05$ ($\xi_{\mathrm{OZ}} = 1.96$, $\chi = 7.0$) and $\varepsilon = 0.01$ ($\xi_{\mathrm{OZ}} = 4.45$, $\chi = 28.4$): the best correlational coherence from the quadratic program equaled the smallest member correlation with $p^{*\top} X$ to four digits, and $|C|\theta^{*2} \le \lambda_{\max}(\mathrm{Corr}_C) \le \chi$ in every case; at $\varepsilon = 0.01$ the disk of 441 agents has $\theta^* = 0.19$. (b) For 180 random subsets of 60 random nonnegative networks the minimax value matched $\min_i \mathrm{Corr}(X_i, Z^*)$ to $1.5 \times 10^{-7}$, the chain of Theorem 8.18(b) held, and the largest susceptibility was 0.19 of the bound of Corollary 8.19(a). (c) On the complete graph at $\lambda = 0.9$, $N\theta^{*2} = 3.690$, 5.048 and 5.241 at $N = 10$, 100 and 1000, against $1/(1-\lambda^2) = 5.263$. (d) Lattice susceptibilities: $\chi/(2\xi_{\mathrm{OZ}}) = 0.81$, 0.88 and 0.94 in one dimension as $\varepsilon$ falls from 0.04 to 0.0025; in three dimensions ($c = 0.2$, a $48^3$ torus) $\chi$ equaled $q\xi_2^2/(2a\lambda\Sigma(0))$ to rounding, as it must, since $\chi = q/((1-\lambda^2)\Sigma(0))$ and Eq. (83) make this an identity in every dimension; the form $q\xi_2^2/(2a\Sigma(0))$ of Corollary 8.19(d) is therefore $\lambda\chi$ and lies below $\chi$ by the relative amount $\varepsilon$ exactly, which is 4, 1, 0.6 and 0.25 percent at $\varepsilon = 0.04$, 0.01, 0.006 and 0.0025; in two dimensions the leading logarithmic form is far from the exact $\chi$ at these $\varepsilon$, so the exact $\Sigma(0)$ must be used. (e) A directed copy chain with weights 0.95 has $\rho(W) = 0$, so $\tau = 0$, while $1/(1-\lambda_r\lambda_c) = 10.26$ and $\chi = 1$: the bound of Corollary 8.19(a) holds but is not a memory time.

**Check S47.97** (Products of projections)**.** (a) Two codes in six dimensions, with three-dimensional transmitted subspaces and a one-dimensional common subspace, in a common prior metric: $\|(P_{V_2}P_{V_1})^n - P_\cap\| = c^{2n-1}$ to seven digits for $n = 1$ to 20, with $c = 0.683035$. (b) Three codes in cyclic order, four-dimensional subspaces with a common vector: $c_{\mathrm{cyc}} = 0.963301$, and $\|A_{\mathrm{cyc}}^n - P_\cap\| \le c_{\mathrm{cyc}}^n$ at $n = 1$, 5, 10, 20 and 40 (at $n = 40$, 0.0755 against 0.224). (c) Quasi-periodic orders whose blocks are random permutations of the three codes: $c_* = 0.96930$, and the bound of Theorem S8.24(b) held at every block.

**Check S47.98** (Lyapunov spectra of random relays)**.** A six-dimensional world, rank-two codes $H = \mathrm{diag}(1, 0.6)U^\top$ with $U$ a Haar-random $6 \times 2$ frame, message noise $0.1I$ and identity prior, so that the hop map is $A = UGU^\top$ with $G = \mathrm{diag}(1/1.1, 0.36/0.46)$. The law of the hop map is invariant under conjugation by orthogonal matrices, so the uniform measure on $k$-planes is stationary and $\Gamma_1 + \cdots + \Gamma_k$ is the expectation of the logarithm of the $k$-volume of $AE_k$ for a fixed orthonormal $k$-frame $E_k$; the exponents are averages of this one-step quantity over $2 \times 10^6$ independent draws, and path products with repeated orthogonalization (400 paths of 300 hops) agreed with them within their standard errors. Pure relays: exponents $-0.918$ and $-1.506$ per hop, gap 0.588 with standard error 0.001, in agreement with the values $\Gamma_1 = -3/4 + \ln((d_1+d_2)/2) = -0.9174$ and $\Gamma_1 + \Gamma_2 = \ln(d_1 d_2) - 25/12 = -2.4238$ that the same one-step expectation gives in closed form for $G = \mathrm{diag}(d_1, d_2)$, whose gap is 0.5889; abstraction length 1.70 hops, so the ratio $\sigma_2/\sigma_1$ of a typical product falls below 0.01 after $\ln 100/0.588 \approx 8$ hops. The product of the $2 \times 2$ factors $GU_m^\top U_{m-1}$ of Section S8.8 gave the same two exponents within the standard errors, with sums $-2.424$ and $-2.427$, and the smallest $|\det U_m^\top U_{m-1}|$ among $1.2 \times 10^5$ draws was $5 \times 10^{-6}$, so every factor was invertible. The crossing hop scatters widely: over 2000 independent realizations the second singular value was still above 0.01 of the largest in 74, 44, 24 and 4 percent of them at $n = 5$, 8, 11 and 20, with median ratio 0.041 at $n = 5$ and 0.007 at $n = 8$. Receivers that keep part of their own belief, with hop map $(1-\omega)I + \omega A$ and $\omega = 1/2$, so that the retention weight is $1/2$: six exponents from $-0.423$ to $-0.545$, gap 0.0296 with standard error 0.0002, abstraction length 33.8 hops, and in one realization 6, 5 and 2 singular values above the threshold at $n = 10$, 20 and 60. A single path of 4000 hops determines this gap only to about $\pm 0.004$, hence the length only to within several hops. With identical frames and isotropic gains the rank stayed at the code rank, two.

### S47.8 Checks for Section 9

**Check S47.99** (One common rank-one code)**.** For Proposition 9.1(a) the reception map (93) was built for 400 random symmetric listening graphs on $N = 4$ agents in $K = 3$ dimensions, with random positive-definite covariances (not fixed points), one random rank-one code with equal scales, and sensing maps of random rank 0 to 2. The largest eigenvalue modulus was 1.000000000000, the largest imaginary part of an eigenvalue $3.2 \times 10^{-15}$, and the eigenvalue 1 occurred in all 400 instances, as part (a) predicts, since every sensing map had a nonzero kernel. For 400 random directed listening graphs the largest modulus was again 1.000000000000, with imaginary parts up to 0.377, so the real spectrum needs symmetric listening. For part (b), 400 random instances with $N = 3$ to 8 agents, $K = 2$ or 3, sensing ranks 0 to $K - 1$, random covariances and symmetric listening gave no violation of $\rho(W) \le f$, a largest ratio of $\|W^t\|_2$ to $K_{\tilde{M}} f^t$ of 0.4413 over $t \le 60$, a largest ratio of $\lambda_{\max}(\Sigma^c)$ to $K_{\tilde{M}}^2 q/(1 - f^2)$ of 0.1296, and, for random unit readouts, a largest ratio of the top eigenvalue of the readouts' correlation matrix to $\lambda_{\max}(\Sigma^c)/q_{\min}$ of 0.7248, the comparison used in Corollary 8.19(b); the largest block row sum of $W$ exceeded 1 in 377 of the 400 instances, so the row-sum contraction of the clustering theorem did not apply there. These instances also confirm Proposition 8.3(b). For two agents listening to each other without sensing, with a common code $H_1 = H_2 = T_c$, the eigenvalue moduli of the one-round disagreement map of Lemma S9.4 agreed with 1 and $|1 - T_c(\mathcal{K}_{12} + \mathcal{K}_{21})|$, with the two gains of the lemma, to within $4.4 \times 10^{-16}$ over 2000 random draws.

**Check S47.100** (The disagreement criterion)**.** For Lemma S9.4, 20000 random instances were drawn with $K$ uniform in $\{2, 3, 4\}$, covariances $AA^\top + 0.05I$ with standard normal $A$, standard normal codes $H_1, H_2$ and channel noise variance exponential with mean one. With the level-1 gains of the lemma, the one-round disagreement map had spectral radius above one in 2983 instances, and in every instance exactly when the stated inequality held. With level-0 gains, in which the assigned noise omits the sender's posted variance, the map expanded in 3109 instances, 126 of them without the inequality and always through a complex pair of eigenvalues, so the criterion is specific to level 1.

**Check S47.101** (The amplifying counterexample and its capacity-one alternations)**.** For the two-agent counterexample ($K = 2$, $F = 0.9I_2$, $Q = I_2$, codes $(3, 0)$ and $(0, 3)$, sensing rows $(3, 6)$ and $(6, 3)$, unit sensing and channel noise variances, both agents sensing and listening to each other at level 1), the covariance fixed point was computed by Riccati iteration, with last change $7.0 \times 10^{-13}$. The frozen mean map has eigenvalue moduli 0.0171, 0.0244, 0.4120 and 1.0613266751, the last a simple real eigenvalue, and its block row sums are 1.0999 for both agents. The spectral radius agrees with Dennis (2026a). In the same instance at capacity one, where each agent either senses or listens in a round, each of the four frozen profiles has a mean map of spectral radius 0.9, and along the periodic covariance orbit every period-two alternation of two such profiles contracts at level 1, the largest rate per round being 0.9752822296, attained when the agents exchange the roles of sensing and listening every round. The growth by 1.0703173057 per round that the report states for the alternation of two individually stable capacity-one profiles belongs to a second instance, its CEX-1, whose parameters the report's summary does not give; they are taken from the working record behind the report: $F = 0.99I_2$, $Q = 0.1I_2$, the same codes $(3, 0)$ and $(0, 3)$, sensing rows $(-5, 10)$ and $(10, -5)$, unit sensing and channel noise variances, capacity one, and the period-two schedule in which both agents listen and then both sense. With these parameters the rate was recomputed at level 1 along the periodic covariance orbit (`cex1_recheck.py`, deterministic, last change $8.2 \times 10^{-14}$): the two-round mean map has eigenvalue moduli 0.0344, 0.0367, 0.4824 and 1.1455791349, so the growth per round is 1.0703173057, and each of the two frozen profiles, both

listening or both sensing, has spectral radius 0.99.

**Check S47.102** (Grounding rate in the counterexample)**.** In the two-agent counterexample $P_0 = I_2/0.19$ and $R_c = 1$, so $\varpi_Q = 0.81$, $\mathrm{snr}_j = 9/0.19 = 47.368$ for both codes, $\varpi_H = \mathrm{snr}_j/(1+\mathrm{snr}_j) = 0.979325$, and $\bar{J} = 1$. The condition $\varpi_g < 1$ of Proposition 9.2 reads $1 - p_g < 1/(0.81 \times 1.979325) = 0.623732$, that is $p_g > 0.376268$; this was evaluated in double precision.

**Check S47.103** (Echo defect with and without frames)**.** A symbolic computation confirmed that the mean square of $\phi_{kj} + \phi_{jk}$ in Remark S9.3 is (S42); that under the frame-consistent prior the difference between a reciprocal and a nonreciprocal edge is $-\sigma_0^2 g[g - h(1-g)]$ with frames and $h\sigma_0^2 g(1-g)$ at identity encodings; that the derivative with respect to $g_{jk}$ at equal gains is $h\sigma_0^2(1-2g)$ in both cases; and, for $h = 1$, that $(1-g)^2\sigma_0^2 + g(1-g)\sigma_0^2 = (1-g)\sigma_0^2$, so that the learning noise makes a calibrated learner's error variance equal to its posterior variance.

## S47.9 Checks for Section 10

The checks below confirm the closed forms of Section 10 on instances and supply the numbers its interpretations quote. All are exact linear algebra (Lyapunov and linear solves, Fourier sums one mode at a time, spectral formulas), root finding by bisection, exact minimizers by dynamic programming, or Monte Carlo sampling where stated.

**Check S47.104** (Crossover distance)**.** Two modalities on a ring with $(a_1, \varepsilon_1) = (0.3, 0.1)$ and $(a_2, \varepsilon_2) = (0.3, 0.01)$, so $c_1 = 0.3$, $c_2 = 0.39$, $m_1^0 = 0.56962$ and $m_2^0 = 0.18232$, with noise $qI$ (Proposition 10.3). The closed form of part (b) was compared with a direct Fourier sum on a ring of 4000 sites at distances up to 120, above a numerical floor of $10^{-11}$: the largest relative error is $4.5 \times 10^{-7}$, $5.5 \times 10^{-7}$, $8.0 \times 10^{-7}$ and $1.1 \times 10^{-8}$ at $a_\times = 0$, $10^{-3}$, $5 \times 10^{-3}$ and 0.02. The Fourier sum agrees with a Lyapunov solve on a ring of 160 sites to $4 \times 10^{-16}$ relative to $\Sigma_{11}(0)$. At $a_\times = 10^{-3}$, $m_+ = 0.56965$, $m_- = 0.18222$, $\mathcal{A}_+ = 2.7730$ and $\mathcal{A}_- = 1.12 \times 10^{-3}$; the crossover distance is 20.16 from the exact amplitudes and 20.17 from the small-coupling formula of part (c), and the local decay rate $-\mathrm{d}\ln\Sigma_{11}/\mathrm{d}d$, evaluated on the two non-alternating terms of part (b) and equal to the central difference $\frac{1}{2}\ln(\Sigma_{11}(d-1)/\Sigma_{11}(d+1))$ of the Fourier sum to the digits shown, is 0.562 at $d = 10$, 0.382 at $d = 20$, 0.1824 at $d = 40$ and 0.1822 from $d = 60$ on. Near the crossover the rate falls by about 0.037 per site, so the forward difference $\ln(\Sigma_{11}(d)/\Sigma_{11}(d+1))$, which estimates the rate at $d + \frac{1}{2}$, is 0.363 at $d = 20$. At $a_\times = 5 \times 10^{-3}$ the two crossover values are 11.74 and 11.86, and at $a_\times = 0.02$ they are 3.64 and 4.70, where the small-coupling formula is no longer accurate. The alternating rates are $m'_+ = 1.41$ and $m'_- = 1.48$, both above $m_-$.

**Check S47.105** (Coupled contents share one light cone)**.** Three agents, world $(\eta^{(1)}, \eta^{(2)})$ with $F_{11} = 0.9$, $F_{22} = 0.8$, off-diagonal entries $f_\times$ and $Q = I$ (Example S10.1). Codes read $\eta^{(1)}$ with gain 2 and $\eta^{(2)}$ with gain 1.5; agent 2 reads agent 1 in modality 1 and agent 3 reads agent 2 in modality 2; agents 1 and 3 sense $\eta^{(1)}$ with gains 1.3 and 0.9 and agent 2 senses $\eta^{(2)}$ with gain 1.1, all with unit noise. Receivers read at level 1, adding the sender's coded variance to a unit channel noise. Steady-state covariances were obtained by iterating the coupled Riccati maps, and the sensitivity of $e_3^{(1)}(t + n_\mathrm{r})$ to $e_1^{(1)}(t)$ was read from the $n_\mathrm{r}$-th power of the linearized round map. At $n_\mathrm{r} = 1$ it is zero for every $f_\times$; at $f_\times = 0$ it is zero for $n_\mathrm{r} = 1, 2, 3$. The ratio of the sensitivity at $n_\mathrm{r} = 2$ to $f_\times^2$ is 0.4071, 0.4118, 0.4123, 0.4123 and 0.4123 at $f_\times = 0.1$, 0.03, 0.01, 0.003 and 0.001.

**Check S47.106** (Barrier transmission and homogeneous rings)**.** Chains of 801 sites with $a = 0.3$ and leak $\varepsilon = 0.02$ or 0.003 ($\xi = 3.88$ or 10.00 hops), with the barrier weight $a/n_b$ for $n_b \in \{1, 2, 10, 100, 1000\}$ (Proposition 10.4). In all ten cases the ratio $\mathsf{G}(x, y)/\mathsf{G}^0(x, y)$ at three pairs

straddling the barrier equals $\mathcal{T}$ of part (a) to six digits (for example 0.464612 at $\xi = 3.88$, $n_b = 10$, and 0.019651 at $\xi = 10.00$, $n_b = 1000$), and the ratio of the correlation coefficients of the full stationary covariance for a far pair agrees with $\mathcal{T}$ to within $10^{-6}$. The equivalent distances at $n_b = 1000$ are 18.87 and 39.31 hops. A second, independent computation on a chain of 4001 sites reproduced $\mathcal{T}$ to six digits for $n_b \in \{2, 10, 100, 1000\}$ at both leaks, and on the 801-site chain the bound of part (c) holds at four pairs with the left side at most 0.21 of the right side. On a ring of 120 agents with $a = 0.3$, $\varepsilon = 0.01$ and update $W = (1-\varepsilon)I - L$, the single-linkage dendrogram of the correlations of $q(I - W^2)^{-1}$ has the single merge height 0.29060 without barriers (Proposition S10.4); with two barriers with $n_b = 100$ on opposite bonds it returns exactly the two arcs of 60 agents, with ultrametric contrast 2.7935 against $\ln(1/\mathcal{T}) = 2.3026$. The positive contrast of an arc between barriers is specific to strong barriers or large self weight. On a ring of 60 agents with two opposite barriers and $\mathcal{U}$ of one arc of 30 computed from single linkage on the full ring, the values at $a = 0.45$, $\varepsilon = 0.01$ (self weight 0.09) are $-0.073$, $-0.038$, $-0.012$, 0.013, 0.276 and 2.488 at $n_b = 1.001$, 1.05, 1.5, 2, 5 and 100, the negative values arising because next-nearest neighbors are more correlated than nearest neighbors; at $(a, \varepsilon) = (0.3, 0.01)$, $(0.2, 0.05)$ and $(0.1, 0.001)$ the contrast is positive at all six values of $n_b$ and exceeds $\ln(1/\mathcal{T})$.

**Check S47.107** (Bypass through a coupled modality)**.** Open chain of 300 agents with the parameters of Check S47.104 and a cut in modality 1 only (Proposition S10.6). At 20 hops across the cut, $\mathsf{G}_{11}$ equals the leading bypass term to within 0.2, 2.0 and 25 percent for $a_\times = 10^{-3}$, $3 \times 10^{-3}$ and $10^{-2}$, and the entrywise lower bound holds in every case. The ratio of same-side to cross-cut values at equal distance is $1.87 \times 10^3$, 725, 300, 57.5, 2.09 and 1.0 at $d = 2$, 4, 6, 10, 20 and 40 for $a_\times = 10^{-3}$; 208, 81.1, 34.1, 7.28, 1.13 and 1.0 for $a_\times = 3 \times 10^{-3}$; and 18.5, 7.78, 3.82, 1.56, 1.02 and 1.0 for $a_\times = 10^{-2}$. An independent computation reproduced the agreement with the leading term to within 0.2, 1.9 and 24 percent, and on a random ten-agent instance with a cut in modality 1 the series of the proposition matches $\mathsf{G}_{11}$ across the cut to $3 \times 10^{-11}$.

**Check S47.108** (Records as media)**.** Ring of 2000 sites in the record model with normalized reading, $a = 0.2$ and $\varepsilon = 0.002$, computed one Fourier mode at a time; without records $\xi_2 = 9.995$ hops and $\tau = 499.5$ rounds (Remark S10.9). One record per site with parameters $(\beta, \omega, \delta, \ell_r)$: $(0.1, 0.01, 10^{-4}, 0)$ gives $\xi_2 = 7.265$ and $\tau = 3702.5$; $(0.1, 0.1, 5 \times 10^{-3}, 6)$ gives 16.308 and 286.1; $(0.1, 0.01, 5 \times 10^{-3}, 6)$ gives 3.592 and 211.3; and $(0.1, 0.01, 10^{-4}, 6)$ gives 22.527 and 3702.5. In each case $\xi_2$ from the exact second-moment formula of Proposition 8.13(a) agrees with the real-space second moment of the agents' covariance to all printed digits. In all four cases the sign of $\xi_2 - 9.995$ is that of $\omega \ell_r^2 \varepsilon - 2a\delta$, the leading-order criterion of Remark S10.9.

**Check S47.109** (Séguy curves)**.** (i) The arcsine law of Proposition S10.13(a) was compared with $4 \times 10^5$ Monte Carlo samples at $\Gamma \in \{0, 0.2, 0.5, 0.8, 0.95, 0.99\}$; the largest difference is 0.0018. (ii) With $\xi = 20$, the local log-slope $\ln(\mathcal{S}(2d)/\mathcal{S}(d))/\ln 2$ on the doubling grid $d = 1, 2, 4, \ldots, 128$ is, between $d = 1$ and $d = 2$, 0.49 for $\Gamma = e^{-d/\xi}$, 0.29 for the correlation of a two-dimensional lattice field with nearest-neighbor coupling and leak, and 1.00 for the Gaussian correlator $\Gamma = e^{-(d/\xi)^2}$, of width $\sigma_{\mathrm{g}} = \xi/\sqrt{2}$; it is also 1.00 at the width $\sigma_{\mathrm{g}} = \xi$ drawn in Fig. S2. Between $d = 64$ and $d = 128$, the first step beyond $3\xi$, all of these slopes are below 0.04. (iii) Exact minimizers by dynamic programming of the pairwise potential on an open chain with Gaussian pulls and $J/h = 4$, the stochastically stable states of the pairwise code game, for 64 independent features on 40000 sites: wall probability 0.03754 (mean domain length 26.6, inside the interval $[8, 423]$ of Theorem 6.11, whose proof covers the pairwise minimizer on open chains), $\mathcal{S}(1), \mathcal{S}(2), \mathcal{S}(3) = 0.03754, 0.07507, 0.11245$ against $dp_{\mathrm{w}} = 0.03754, 0.07508, 0.11262$, and a local log-slope of 0.995 between $d = 1$ and $d = 4$, against 0.473 for the arcsine curve with the same large-distance rate. The curve reaches 0.501 by

$d = 50$, faster than the form $\frac{1}{2}(1 - e^{-2d/\bar{\ell}})$ of independently placed walls with mean domain length $\bar{\ell}$ (0.488 at $d = 50$, and 0.297 against the observed 0.367 at $d = 12$), because domain lengths are more regular than exponential: their standard deviation is 21.8 against a mean of 26.8 over eight features. In a rerun of these minimizers on the same pull sample, which reproduced the ground-state values above, the local decay rate of $\frac{1}{2} - \mathcal{S}$ over windows of four hops rose steadily from 0.114 between $d = 4$ and $d = 8$ to 0.157 between $d = 24$ and $d = 28$, where the standard error over features reaches eight percent of $\frac{1}{2} - \mathcal{S}$, against the rate $2/\bar{\ell} = 0.075$ of independently placed walls; these samples do not establish an exponential tail. (iv) Sampling $2 \times 10^5$ bonds of the open chain at $h_i = 0$ reproduces part (c) to three digits at $\gamma J = 2$ and 4, for example $\mathcal{S}(5) = 0.3725$ against 0.3719 at $\gamma J = 2$.

**Check S47.110** (Holonomy of exchange rates)**.** Complete graph on 12 nodes in two zones of six, with true potentials plus independent Gaussian edge errors of standard deviation 0.02 inside zones and 0.30 across (Proposition S10.15). With exact rates the largest triangle holonomy is $4.4 \times 10^{-16}$, and the least-squares residual is orthogonal to gradients to $1.6 \times 10^{-14}$. With noisy rates, the holonomy equals that of the residual to $5.6 \times 10^{-16}$; the mean squared holonomy is 0.00121 over triangles inside a zone (predicted 0.00120) and 0.182 over triangles crossing the border between the zones (predicted 0.180).

**Check S47.111** (The Walrasian price)**.** A symbolic computation for $N = 3$ confirms the clearing price of Proposition 10.5(a) and that the pooled posterior mean depends on the signals only through $\sum_i s_i/R_i^s$. With $P_0 = R_i^s = 1$, $\bar{\eta}_0 = 0$ and $\eta = 2$, the naive clearing price is 1.217, 1.059 and 0.994 at $N = 10$, $10^2$ and $10^4$ (limit 1), while the pooled posterior mean is 2.21, 2.10 and 1.987.

**Check S47.112** (The Hayek dimension)**.** $\eta \sim \mathcal{N}(0, I_4)$, a private signal $s_i = \eta + v_i$ with noise variance 4 in every component, and a public message of rank two that reads the first two components (Proposition S10.16). The conditional variance $\mathrm{Var}(\psi^\top \eta \mid s_i, M\eta)$ is 0 for $\psi = (1, 0, 0, 0)$ and $(0.6, -0.8, 0, 0)$, 0.80 for $(0, 0, 1, 0)$ and 0.40 for $(0.5, 0.5, 0.5, 0.5)$, as the row-space criterion predicts.

**Check S47.113** (Polity size)**.** (i) On the line with $\xi = g = 1$ (Proposition 10.6(c),(d)): under rule C, for $\bar{f}_{\mathrm{C}} \in \{0.01, 0.1, 0.5, 1, 1.5, 1.9, 1.99\}$ the numerical minimizer of the per-capita cost equals $2\xi\varrho_{\mathrm{C}}$ to five digits, for example 0.20699 at $\bar{f}_{\mathrm{C}} = 0.01$, where $2(f\xi/g)^{1/2} = 0.2$, and 14.860 at 1.99; for $\bar{f}_{\mathrm{C}} \in \{2, 2.5\}$ the cost is nonincreasing in size. Under rule M, for $\bar{f}_{\mathrm{M}} \in \{0.01, 0.1, 0.5, 1, 1.5, 1.9, 1.99\}$ the numerical minimizer equals $\xi\varrho_{\mathrm{M}}$ to five digits, for example 0.18116 at $\bar{f}_{\mathrm{M}} = 0.01$, where $(6f\xi/g)^{1/2} = 0.17321$, and 40.000 at 1.9, where $4\xi/(2 - \bar{f}_{\mathrm{M}}) = 40$; for $\bar{f}_{\mathrm{M}} \in \{2, 2.5\}$ no finite polity beats $g/2$, and $\psi_{\mathrm{M}}$ is decreasing on a grid from $10^{-4}$ to $10^4$. For the 999 equally spaced values $f/(g\xi) \in \{0.001, \dots, 0.999\}$, solving both root equations by bisection gives a ratio $d^*_{\mathrm{M}}/d^*_{\mathrm{C}}$ that increases along the grid from 1.2359, at 0.001, near the small-cost limit $\sqrt{6}/2 = 1.2247$, to 596, at 0.999, near the rule-M threshold, with 1.9368 at 0.5, as part (d) requires; a symbolic expansion of the series in its proof gives vanishing coefficients of $\varrho^0$, $\varrho^1$ and $\varrho^2$, reproduces the closed form of $c_{\mathsf{n}}$ for $2 \le \mathsf{n} \le 13$, and gives positive $c_3, \dots, c_{13}$. (ii) A two-dimensional field whose correlation is that of $(\varepsilon I + L)^{-1}$ for nearest-neighbor weight 0.25 and $\varepsilon = 0.0025$ ($\xi \approx 10$), on a periodic lattice of $512^2$ sites. Its correlation volume is $\chi = 155.87 = 1/(\varepsilon \mathsf{G}(0))$, an exact lattice identity. Under rule C the best disc beats $g$ at $f/(g\chi) = 0.05$, 0.2, 0.5 and 0.9 (at 0.9, with 6224 members and cost $0.99833g$) and not at 1.1. Under rule M the best disc beats $g/2$ at $f/(g\chi) = 0.1$, 0.3 and 0.45 (at 0.45, radius 153 and cost $0.499963g$) and not at 0.55.

**Check S47.114** (Distance elasticity)**.** The function $\mathrm{el}_{K_0}(x) = xK_1(x)/K_0(x)$ of Proposition S10.24(b) takes the values 0.3197, 0.4060, 0.6680, 1.4296, 2.4561, 5.4789 and 10.4886 at $x = 0.05$, 0.1, 0.3, 1, 2, 5 and 10, increasing and approaching $x + \frac{1}{2}$; on a grid of $3 \times 10^5$ points in $[0.01, 30]$ it is increasing and satisfies its Riccati equation to $7 \times 10^{-5}$, the accuracy of the finite

differences. At a distance of 1000 km the elasticity is 4.475, 2.456, 1.430 and 0.896 for $\xi = 250$, 500, 1000 and 2000 km. For the integral of part (c) with $\alpha_{\mathrm{el}} = 0.9$, numerical integration matched $d^{-0.9}\int_0^\infty \varrho^{-0.1}e^{-\varrho}\mathrm{d}\varrho$ to four digits at $d = 10$ and 100 and to 0.2 percent at $d = 1000$.

**Check S47.115** (Nested coupling)**.** The spectral formula of Proposition 10.7(a) agrees with a direct inverse to $2 \times 10^{-16}$ for $b = 4$, $n_{\mathrm{H}} = 5$, and to $10^{-15}$ for $b = 3$, $n_{\mathrm{H}} = 4$, $A_{\mathrm{h}} = 0.7$, $\mathfrak{q} = 0.6$ and precision leak $\varepsilon^{\Pi} = 0.05$. For $b = 2$, $n_{\mathrm{H}} = 40$, $A_{\mathrm{h}} = 1$, $\mathfrak{q} = 0.7$ and $\varepsilon^{\Pi} = 10^{-7}$, the ratio $\mathsf{G}(l+1)/\mathsf{G}(l)$ is 0.7143, 0.7143 and 0.7144 at $l = 1$, 3 and 5 against $1/(b\mathfrak{q}) = 0.7143$, and drifts to 0.717 at $l = 16$ as the constant mode enters; with $n_{\mathrm{H}} = 60$ and $\varepsilon^{\Pi} = 10^{-12}$ the correlations $\mathsf{G}(l)/\mathsf{G}(0)$ at $l = 1, \dots, 5$ equal the limit of part (b) to four digits. For $b = 4$, $n_{\mathrm{H}} = 20$, $A_{\mathrm{h}} = 1$, $\mathfrak{q} = 0.3$ and $\varepsilon^{\Pi} = 10^{-9}$, the ratio is 0.825 at $l = 1$ against 0.833, and falls to 0.799 at $l = 8$ and 0.475 at $l = 16$ as the leak cutoff $l^* \approx 18.7$ approaches. For $b = 4$, $n_{\mathrm{H}} = 5$, $A_{\mathrm{h}} = 0.5$, $\mathfrak{q} = 0.5$ and $\varepsilon^{\Pi} = 0.01$, with agents on a line, the per-level ratios are 0.58 to 0.65, a power-law fit over distances 4 to 256 gives the exponent 0.31 against 0.34 from the fitted ratio, and the local elasticity oscillates between 0.15 and 0.57 with the period of the hierarchy.

**Check S47.116** (The cap on unconditional contrast)**.** Ring of 60 agents with $a = 0.3$, barrier weights $a/100$ on two opposite bonds, update $W = (1-\varepsilon)I - L$, local noise variance 1 and world innovation variance $Q = 1$, so that $\lambda = 1 - \varepsilon$ and $\Sigma_{\mathrm{cm}} = Q/(1-\lambda^2)$ (Proposition S10.27). For one arc of 30 agents, the ultrametric contrast from world-conditional correlations is 3.365, 2.796 and 2.120 at $\varepsilon = 0.05$, 0.01 and 0.002, and from unconditional correlations 0.1138, 0.0716 and 0.0408, below the caps 0.2999, 0.1522 and 0.0704. The common-mode variances are 10.3, 50.3 and 250.3, and the largest world-conditional variances 3.59, 8.26 and 18.25. The cap does not vanish at weak grounding: it is 0.0223 at $\varepsilon = 10^{-4}$ and 0.0166 at $\varepsilon = 10^{-6}$, against the limit $\ln(1+q/(NQ)) = \ln(1+1/60) = 0.01653$; here $W$ is symmetric, $\Sigma^c = q(I-W^2)^{-1}$, and the uniform mode contributes $q/(N(1-\lambda^2))$ to every diagonal entry while the other modes stay bounded. The world-conditional contrast is unchanged by a rescaling of the local noise, while the unconditional one depends on $q/Q$: with local noise variance $q = 100$ and $Q = 1$ the unconditional contrasts at the three leaks are 2.470, 2.040 and 1.445 under caps 3.583, 2.859 and 2.116, and with $q = 1000$ they are 3.222, 2.680 and 2.016 against the world-conditional 3.365, 2.796 and 2.120. For nearly isolated blocks the cap is set by the block size: with two blocks of three agents, weight 0.2 on every bond within a block, one bond of weight $a_\times$ between the blocks, $W = \lambda(I-L)$ and $q = Q = 1$, the values at $(\lambda, a_\times) = (0.9999, 10^{-6})$ are $(1-\lambda^2)\max_i \Sigma^c_{ii} = 0.3324$, cap 0.2870, unconditional contrast of one block 0.2857 and world-conditional contrast 5.7064, and at $(0.999, 10^{-5})$ they are 0.3338, 0.2880, 0.2851 and 5.6996, against $\ln(1+q/(nQ)) = \ln(4/3) = 0.2877$. At the fixed weight $a_\times = 0.05$ and $\lambda = 0.9999$ the cap is 0.1548, near $\ln(7/6) = 0.1542$, and both contrasts are small, 0.0007 unconditional and 0.0050 world-conditional.

## S47.10 Checks for Section 11

Each check below evaluates exact Gaussian linear algebra (path covariances and precisions, discrete Lyapunov equations, conditional covariances) or exact stationary laws of finite Markov chains on the instances stated. Where a second, independent implementation of the same statement was run, its results are reported in the same check.

**Check S47.117** (Coarse path blankets)**.** The code was validated on Example S4.2 with $\alpha_1 = \alpha_2 = 1$, whose leak rate 0.143841 equals $\frac{1}{2}\ln\frac{4}{3}$. The test system has three members whose internal states read their own sensory channels, each other and the coarse act $\alpha_1 + \alpha_2$ (noise variance 0.3), three sensory channels that read a scalar exterior state with noise variances 0.5, 1 and 2, two active

channels driven by the internal states, one of which also reads a sensory channel (noise variance 0.2), and an exterior of two variables, an autoregressive state with coefficient 0.9 driven by the coarse act and an outside agent that reads the coarse act and that state (noise variance 0.25). With internal states $\mu_1, \mu_2, \mu_3$, sensory channels $X_1, X_2, X_3$, active channels $\alpha_1, \alpha_2$, exterior state $\eta$ and outside agent $x_{\mathrm{e}}$, in this order, the recursion $X_{t+1} = AX_t + \Xi_t$ has independent noises and the blocks

$$A_{\mathsf{II}} = \begin{pmatrix} 0.3276 & 0.0657 & 0.0264 \\ -0.1043 & 0.3724 & 0.0357 \\ -0.0430 & 0.0465 & 0.3292 \end{pmatrix}, \quad A_{\mathsf{IS}} = \begin{pmatrix} 1.2147 & 0.0014 & 0.0273 \\ -0.0368 & 0.5919 & -0.0241 \\ 0.0299 & 0.0020 & 0.2854 \end{pmatrix},$$
$$A_{\mathsf{AI}} = \begin{pmatrix} -0.1102 & 0.5176 & 0.4027 \\ -1.0845 & -0.7556 & -0.0699 \end{pmatrix},$$

each of the two columns of $A_{\mathsf{IA}}$ equal to $(-0.0782, -0.0257, 0.0008)^{\top}$, $X_k(t+1) = \eta(t) + \Xi_{X_k}$, coefficient 0.2 of $\alpha_1$ on $X_1$, $\eta(t+1) = 0.9\eta + 0.15(\alpha_1 + \alpha_2) + \Xi_\eta$ and $x_{\mathrm{e}}(t+1) = 0.5x_{\mathrm{e}} + 0.3(\alpha_1 + \alpha_2) + 0.2\eta + \Xi_{x_{\mathrm{e}}}$, all other entries zero and $\rho(A) = 0.968$; the entries are $0.3I + 0.08G_1$, $\mathrm{diag}(0.6/R^s_k) + 0.05G_2$, $0.1g_3$ and $0.4G_4$ for standard normal draws $G_1, G_2, g_3, G_4$, printed to four decimals, and with the printed entries the leaks of the next two sentences are reproduced to three significant digits, the round-off values staying below $3 \times 10^{-14}$ in magnitude. With the information-form $s_C$ and the residuals placed by Proposition 11.4(a), the leak between the interior with residuals and the exterior given $b_C$, from an independent start, was $1.8 \times 10^{-15}$ on four rounds and $2.8 \times 10^{-14}$ on ten, $I(X_{\mathsf{I}}; X_{\mathsf{E}} \mid b_C)$ was below $4 \times 10^{-15}$ in magnitude, and the weak leak was $1.8 \times 10^{-15}$; from the stationary start the window leak was $3.6374 \times 10^{-3}$ on both six and twelve rounds, the initial-slice term, with rate $-7.6 \times 10^{-9}$. With the equal-weight summary $s_C = X_1 + X_2 + X_3$ the leak rate with the residuals on the interior side was 0.0309 nats per round, $I(X_{\mathsf{I}}; X_{\mathsf{E}} \mid b_C) = 0.118$ and the weak leak 0.0279 on ten rounds. The closed form (S43) agreed with window differences of path computations to six digits, 0.135600 at $R^s = (0.5, 2)$ and 0.619053 at $R^s = (0.2, 5)$ with $q = 0.1$ and $g_k = 1/R^s_k$, and both vanish at $R^s_1 = R^s_2$. With channel memories in the same system the values of Example S11.3 were obtained; in addition, the weak leaks on ten rounds were $1.1 \times 10^{-14}$, 0.0096 and 0.0037, and the leak rates with the residual on the interior side $-2.8 \times 10^{-8}$, 0.0108 and 0.0124, for memories $(0.5, 0.5, 0.5)$, $(0.2, 0.5, 0.8)$ and $(0, 0, 0.6)$. For Example S11.4 (interior coefficient 0.5 on itself and 0.6 on $s_C$, active coefficients 0.8 and $-0.5$, exterior coefficient 0.8 on itself and 0.3 on each act, noise variances 0.2, 0.5, 0.3, 0.3, 0.4), the leak on ten rounds was 0.1133, and the leak rate with the residual $\alpha_1 - \alpha_2$ on the interior side 0.0197, when the sensor reads $\alpha_1$, and these were 0 and $-1.1 \times 10^{-7}$ when it reads $\alpha_1 + \alpha_2$. An independent implementation on eight random systems built to satisfy (a1) to (a3), with members reading their own channels individually and active channels also reading sensory channels, found the conditional mutual information of interior and exterior paths given the coarse blanket path over five rounds at most $8.9 \times 10^{-16}$ nats, against $5.7 \times 10^{-3}$ to 0.69 nats with an equal-weight summary, and reproduced both closed-form rates.

**Check S47.118** (The leak bound)**.** In the system of Check S47.117 the information-form weights were perturbed by $\mathsf{h}(1, -1, 0)$ with a noise-orthogonal residual assignment (off-block noise covariance at most $4 \times 10^{-16}$). The leak rates were $8.15 \times 10^{-4}$, $7.16 \times 10^{-5}$, $7.90 \times 10^{-6}$ and $7.09 \times 10^{-7}$ at $\mathsf{h} = 0.1$, 0.03, 0.01 and 0.003, so that rate over $\mathsf{h}^2$ lay between 0.0788 and 0.0815. The constants were $\mathfrak{s}_{\mathsf{I}'} = 3.27$ to 3.29, $\mathfrak{s}_{\mathsf{E}'} = 10.89$, $\mathfrak{v}_{\mathsf{I}'} = 1.58$ to 1.61, $\mathfrak{v}_{\mathsf{E}'} = 1.00$ and $\mathfrak{c}_C = 45.1$; the hypothesis $\mathfrak{c}_C \mathsf{k} < 1$ held for $\mathsf{h} \le 0.01$, where the rate bound was 0.341 and 0.019 against the computed $7.9 \times 10^{-6}$ and $7.1 \times 10^{-7}$. With the exact conditional-variance constant $(\|J^{-1}_{\mathsf{UU}}\| \|J_{ZZ}\|)^{1/2} = 2.15$ in place of $\mathfrak{c}_C$ the same argument gives $4.6 \times 10^{-4}$ and $4.1 \times 10^{-5}$.

**Check S47.119** (Exact abstract physics)**.** An exterior of four variables in two blocks was built to be lumpable, with coarse weights $\left(\begin{smallmatrix}0.7 & 0.15\\ 0.1 & 0.6\end{smallmatrix}\right)$, read by two sensors and driven by two active channels through their sum ($\rho(A) = 0.868$). Each coarse weight $\bar{F}_{ab}$ is split over the two rows of block $a$ as $\bar{F}_{ab}\mathsf{w}$ and $\bar{F}_{ab}(1-\mathsf{w})$ with a weight $\mathsf{w}$ uniform on $[0.2, 0.8]$ drawn for each column of block $b$, and each active column of $A_{\mathsf{EA}}$ is split in the same way from the coarse input weights $(0.3, -0.2)$; the sensors read the first exterior variable and half the sum of the second block, $A_{\mathsf{II}} = \left(\begin{smallmatrix}0.5 & 0.1\\ 0.2 & 0.4\end{smallmatrix}\right)$, $A_{\mathsf{IS}} = \left(\begin{smallmatrix}0.3 & 0.1\\ 0.1 & 0.3\end{smallmatrix}\right)$, $A_{\mathsf{AI}} = \left(\begin{smallmatrix}0.5 & -0.3\\ 0.2 & 0.4\end{smallmatrix}\right)$, and the noise is diagonal with variances uniform on $[0.2, 0.6]$. The draws of this check, in the order in which they are listed, come from NumPy's default generator with seed 7, so the values that follow belong to that instance. The algebraic residuals of Theorem 11.8(a) were 0 and $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}} = 2.2 \times 10^{-16}$; adding 0.05 to the coefficient of the first exterior variable on the third gave $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}} = 1.8 \times 10^{-4}$, and adding 0.05 to its coefficient on the first active channel alone gave $4.2 \times 10^{-4}$. For part (f), the exterior was $A_{\mathsf{EE}} = \mathsf{S}\left(\begin{smallmatrix}\mathsf{A}_{11} & 0\\ \mathsf{A}_{21} & \mathsf{A}_{22}\end{smallmatrix}\right)\mathsf{S}^{-1}$ with $\mathsf{S}$ a $5 \times 5$ standard normal matrix, $\mathsf{A}_{11} = \left(\begin{smallmatrix}0.8 & 0.2\\ -0.1 & 0.5\end{smallmatrix}\right)$, $\mathsf{A}_{21}$ equal to 0.3 times a $3 \times 2$ standard normal matrix and $\mathsf{A}_{22} = \mathrm{diag}(0.6, -0.4, 0.3)$, with $\mathsf{P}_{\mathrm{E}} = (1, 0.4, 0, 0, 0)\mathsf{S}^{-1}$, one sensor reading $\eta_C$, one active channel entering the exterior through $0.2\mathsf{S}(1, 0.5, 0, 0, 0)^{\top}$, two internal states with $A_{\mathsf{II}} = \left(\begin{smallmatrix}0.5 & 0.1\\ 0 & 0.4\end{smallmatrix}\right)$, $A_{\mathsf{IS}} = (0.3, 0.2)^{\top}$ and $A_{\mathsf{AI}} = (0.4, -0.3)$, and diagonal noise with variances uniform on $[0.2, 0.5]$. This five-dimensional exterior, with a three-dimensional invariant subspace in $\ker \mathsf{P}_{\mathrm{E}}$ and a one-dimensional $\eta_C$, had $\operatorname{rank} \mathcal{O} = 2$; the extension spanned by $\mathcal{O}$ satisfied the kernel condition to $3.7 \times 10^{-15}$ and had $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}} = 1.5 \times 10^{-13}$, against $1.5 \times 10^{-3}$ for $\eta_C$ alone ($\rho(A) = 0.835$). An independent implementation found the lumpable residual $1.1 \times 10^{-16}$ with $\mathfrak{d}_{\mathrm{cl}}^{\mathrm{E}} = 0$, $5.0 \times 10^{-4}$ after a perturbation of 0.05 and $2.3 \times 10^{-3}$ when the exterior reads one active channel; in five random exteriors with a three-dimensional invariant subspace in $\ker \mathsf{P}_{\mathrm{E}}$, $\operatorname{rank} \mathcal{O} = 3$ equaled $\operatorname{rank} \mathsf{P}_{\mathrm{E}}$ plus the observability rank of the memory system, and $\operatorname{row} \mathcal{O}$ was invariant to $5.2 \times 10^{-14}$.

**Check S47.120** (Minimal order and closure order)**.** For $\eta_{t+1} = \frac{1}{4}\eta_t + \sqrt{7/8}\, z_t + \mathfrak{n}_t$ and $z_{t+1} = \frac{1}{2} z_t + \mathfrak{n}'_t$ with independent noises of unit variance, $\operatorname{rank} \mathcal{O} - \operatorname{rank} \mathsf{P}_{\mathrm{E}} = 1$, the stationary variance of $\eta$ was 2.666667, its autocorrelations at lags 1 to 5 equaled $2^{-k}$ to eight decimals, and the one-step predictive variance was 2.00000000 both given $\eta_t$ alone and given thirteen past values, so that the Markov defect was 0 to machine precision. With the coefficient $\sqrt{1/2}$ in place of $\sqrt{7/8}$ the two predictive variances were 1.59340659 and 1.59307033 and the Markov defect $1.06 \times 10^{-4}$ nats. The exact statement follows from the autocovariances $\frac{17}{8}$ and $-\frac{1}{2}$ of $\sqrt{7/8}\,\mathfrak{n}'_{t-1} + \mathfrak{n}_t - \frac{1}{2}\mathfrak{n}_{t-1}$, which are those of $\varepsilon_t - \frac{1}{4}\varepsilon_{t-1}$ for a white sequence $\varepsilon$ of variance 2.

**Check S47.121** (Closure lemmas)**.** For Lemma 11.11, 500 random reversible recursions were drawn in dimension 3 to 8, with $\tilde{A}$ of nonnegative spectrum and top eigenvalue in $[0.9, 0.995]$, a random positive definite $Q^{\mathfrak{n}}$ and coordinates $\phi$ obtained from the slow eigenvector by adding random vectors of norm up to about 1.2 and normalizing; 403 had $\varkappa_{\mathrm{in}}^{\mathrm{d}} > \varkappa_{\mathrm{out}}^{\mathrm{d}}$, and 142 of these had $\sigma^{\mathrm{d}} \geq \varkappa_{\mathrm{in}}^{\mathrm{d}} - \varkappa_{\mathrm{out}}^{\mathrm{d}}$. The largest ratios were 0.9958 for the tangent bound, 0.9986 for the memory weight against the first bound of (103), 0.9849 for the first against the second, and 0.9832 for $\Xi^{\mathrm{d}}$ against the memory weight; the bracket on $\varpi_1$ held in every instance. With the precision rates of (b), all 403 instances had $\varkappa_{\mathrm{in}}^{\Pi} > \varkappa_{\mathrm{out}}^{\Pi}$, and the largest ratios were 0.9977 for the tangent bound and 0.9898 for the memory weight against the bound. The autocorrelation computed from $(A, Q^{\mathfrak{n}})$ by a discrete Lyapunov solve agreed with the whitened spectral formula to $4.2 \times 10^{-15}$. For mixed spectra, in 215 instances with $\varkappa_{\mathrm{in}}^{\mathrm{d}} > \varkappa_{\mathrm{out}}^{\mathrm{d}}$, the largest ratios were 0.979 for the tangent bound and 0.844 for $\Xi^{\mathrm{d}}$ against the bound of (c). In earlier computations, 394 random instances meeting the hypothesis $\sigma^{\mathrm{d}} < \varkappa_{\mathrm{in}}^{\mathrm{d}} - \varkappa_{\mathrm{out}}^{\mathrm{d}}$ gave ratios of $\Xi^{\mathrm{d}}$ to the bound $(\sigma^{\mathrm{d}})^2/((\varkappa_{\mathrm{in}}^{\mathrm{d}} - \varkappa_{\mathrm{out}}^{\mathrm{d}})^2 - (\sigma^{\mathrm{d}})^2) \cdot \varkappa_{\mathrm{out}}^{\mathrm{d}}(2 - \varkappa_{\mathrm{out}}^{\mathrm{d}})/(\varkappa_{\mathrm{in}}^{\mathrm{d}}(2 - \varkappa_{\mathrm{in}}^{\mathrm{d}}))$ at most 0.917, and 300 symmetric $W_{CC} \succeq 0$ on 6 to 15 members with $Q_{CC}^{\zeta} = qI$ gave ratios at most 0.115 to the last bound of (103).

For Lemma 11.10(b), 400 symmetric drifts in dimension 3 to 8 with a slow eigenvalue in $[0.02, 0.4]$ and the others in $[0.5, 5]$ gave 267 instances with $\varkappa_{\mathrm{in}}(M,\phi) > \varkappa_{\mathrm{out}}(M,\phi)$, of which 91 had $\sigma(M,\phi) \geq \varkappa_{\mathrm{in}}(M,\phi) - \varkappa_{\mathrm{out}}(M,\phi)$. The largest ratios were 0.993 for the tangent bound, 0.998 for the memory weight against $\tan^2 \angle(\phi, \mathsf{u}_1)\varpi_1/\varpi_2$, 0.984 for the latter against (102), and 0.962 for the Markov error, sampled on 20000 lags, against the memory weight. For part (a), 96 symmetric drifts in dimension 3 to 6 with a slow eigenvalue in $[0.05, 0.3]$, the others in $[1, 4]$ and coordinates near the slow eigenvector, 21 of them with $\sigma(M,\phi) \geq \varkappa_{\mathrm{in}}(M,\phi) - \varkappa_{\mathrm{out}}(M,\phi)$, gave ratios of the Markov error to the bound (49) of Theorem 5.4(a), evaluated at the rates of $(M,\phi)$, at most 0.586, and 13 non-normal drifts meeting the hypothesis of (b)(iii) gave ratios at most 0.0073.

**Check S47.122** (Closure notions)**.** For three random stable six-dimensional systems with two-dimensional $y$ and no input, the finite-history dependence $I(y_{t+1}; x_t \mid y_{t-L:t})$ at $L = 0$ equaled the closure defect of Theorem 5.3 (1.00556, 0.73105 and 0.27781) and decreased with $L$ (at $L = 8$: 0.58140, 0.42157 and 0.24497), as Proposition S11.8(b) requires. For a system with $A_{y\perp} = 0$ it was zero to $1.1 \times 10^{-16}$ at every $L$.

**Check S47.123** (A price and many traders)**.** For Example S11.5 with $\mathsf{f}_p = 0.9$, $\mathsf{b}_p = 0.3$, $\mathsf{b}_x = -0.5$, $q_p = 0.05$, $q_x = 0.1$, no act input and persistences evenly spaced on $[0.3, 0.7]$: $\Gamma_1 = 0$ and $\Gamma_l/\mathsf{b}_x \geq 0$; the memory weight $\sum_{l\geq 2}|\Gamma_l|$ was 0.0370, 0.0313, 0.0300 and 0.0297 at $N = 10$, 40, 160 and 640, against the bound 0.0905, 0.0779, 0.0750 and 0.0743; the relative predictive defect of $\bar{x}$ was $2.5\times10^{-3}$, $7.4 \times 10^{-4}$, $3.2 \times 10^{-4}$ and $2.1 \times 10^{-4}$, that of the price below $2 \times 10^{-16}$, and $\mathfrak{d}^{\mathrm{E}}_{\mathrm{cl}}$ was 0.0126, 0.0131, 0.0217 and 0.0558. With all persistences equal to 0.5, $\mathfrak{d}^{\mathrm{E}}_{\mathrm{cl}}$ was below $5 \times 10^{-13}$ at every $N$. An independent implementation reproduced the kernel signs, the memory weights and the bounds.

**Check S47.124** (Another collective as exterior)**.** For the ring of eight in Example S11.7, the smallest eigenvalue of $W_{C'C'}$ was 0.941, 0.864, 0.585, 0.185 and 0.025 at $a_{\mathrm{in}} = 0.01$, 0.03, 0.1, 0.2 and 0.24, so the spectrum is nonnegative. The dispersion was $\sigma^{\mathrm{d}} = 0.00866$ and the gaps $\varkappa^{\mathrm{d}}_{\mathrm{in}} - \varkappa^{\mathrm{d}}_{\mathrm{out}}$ were 0.00086, 0.0126, 0.0536, 0.112 and 0.136. The Markov errors were $7.756 \times 10^{-3}$, $1.184 \times 10^{-3}$, $7.245 \times 10^{-5}$, $1.272 \times 10^{-5}$ and $8.120 \times 10^{-6}$, computed from the spectral formula of the proof of Lemma 11.11 over 6000 lags. The bound of Lemma 11.11(a) was 96.4, 0.260, $5.87\times10^{-3}$, $7.45\times10^{-4}$ and $4.36\times10^{-4}$, and that of (b) 85.0, 0.238, $4.58\times10^{-3}$, $4.46\times10^{-4}$ and $2.32\times10^{-4}$; the weaker drift bound with $(\sigma^{\mathrm{d}})^2/((\varkappa^{\mathrm{d}}_{\mathrm{in}} - \varkappa^{\mathrm{d}}_{\mathrm{out}})^2 - (\sigma^{\mathrm{d}})^2)$ in place of $(\sigma^{\mathrm{d}}/(\varkappa^{\mathrm{d}}_{\mathrm{in}} - \varkappa^{\mathrm{d}}_{\mathrm{out}}))^2$, which needs $\sigma^{\mathrm{d}} < \varkappa^{\mathrm{d}}_{\mathrm{in}} - \varkappa^{\mathrm{d}}_{\mathrm{out}}$, was inapplicable at the first value and gave 0.494, $6.0 \times 10^{-3}$, $7.5 \times 10^{-4}$ and $4.4 \times 10^{-4}$ at the others. The precision-based contrasts of $I - W^2_{C'C'}$ were 1.06, 1.83, 4.46, 8.02 and 9.38. With $C$'s act as a white input of unit variance the closure defect of the average was $3.9 \times 10^{-4}$, $2.0 \times 10^{-4}$, $8.2 \times 10^{-5}$, $5.0 \times 10^{-5}$ and $4.6 \times 10^{-5}$.

**Check S47.125** (Coarse inference)**.** For Proposition 11.12, a two-dimensional coarse exterior with five members, random sensing rows and noises and a known random act was simulated over 400 steps. The steady state was $\bar{P} = \left(\begin{smallmatrix} 0.18146 & 0.05227 \\ 0.05227 & 0.09334 \end{smallmatrix}\right)$ with closed-loop spectral radius 0.658. With a random column-stochastic $\mathcal{W}$ that is not symmetric, the member average differed from the centralized filter by at most $2.7 \times 10^{-15}$ at $n_{\mathrm{c}} = 0$, 1 and 3. With the lazy ring ($\rho_\perp = 0.655$) the largest deviation norm was 19.1, 7.86, 5.06, 2.17 and 0.398 at $n_{\mathrm{c}} = 0$, 1, 2, 4 and 8, below the bound of the form of Proposition 5.7(b), 60.9, 39.8, 26.1, 11.2 and 2.05. For Proposition S11.9(b) in the same model, the literal belief had variances $(0.023, 0.032)$, the exact one $(0.181, 0.093)$ and the literal estimator's true errors $(0.690, 0.210)$, at an expected relative entropy of 15.3 nats; the exact filter's computed error covariance equaled $\bar{P}$ to $10^{-17}$. An independent implementation on ten systems found the member average equal to the centralized steady-state filter to $1.1 \times 10^{-14}$ over 300 steps at $n_{\mathrm{c}} = 0$, 1 and 3.

**Check S47.126** (Level-0 averages)**.** For five scalar members with random internal attention gains in $[0, 0.08]$, sensing gains in $[0.05, 0.2]$ and outside gains in $[0, 0.1]$, the decomposition (S45), checked with $F = 1$ and with $F = 0.9$ on 200 random states, inputs and noises per instance, agreed with the direct average to $9 \times 10^{-16}$. At $F = 1$ the closure defect of the average, computed from the stationary law of the update of the means with unit noise, was 0.0149 for generic gains (spread of the column sums 0.365); $-5.6 \times 10^{-17}$, zero to round-off, for unequal leaks (spread 0.279) chosen to cancel an attention imbalance of the same spread, so that the column sums were equal to $2.2 \times 10^{-16}$; 0 for equal leaks and symmetric attention; and 0.0112 for equal leaks and the directed attention of the first case. At $F = 0.9$, on separately drawn instances of the same four kinds, it was $8.3 \times 10^{-4}$ (spread of the column sums 0.092), zero to $2 \times 10^{-16}$ in the two cases with equal column sums, and $2.4 \times 10^{-3}$ (spread 0.140).

**Check S47.127** (Two firms and a price)**.** The values of Table S12 and of the text after it were computed from the stationary law and the path laws of the fine joint process of Example S11.11, of dimension 49 or 33; the predictive law was computed by conditioning the stationary path law on thirty rounds of the firm's blanket, and the deliberation law as the exact stationary law of the logit chain on the 81 profiles. The leaks in the first two rows on (i) were computed with the sensory and active residuals placed with the interior. An independent implementation reproduced them and computed the leaks of Definition 11.2 directly: the strong coarse leak was below $3 \times 10^{-12}$ in magnitude from the independent start and 0.1742, 0.1740 (exact model) and 0.0909, 0.0909 (bounded model) on six and twelve rounds from the stationary start, and with the equal-weight summary the strong and weak leak rates were those of the table, against 0.0440 and 0.0389 with the residuals placed with the interior. With $C_1$ running its exact model and noiseless members, the members' dispersion in the price coordinate, $\mathbb{E} \sum_i \mathrm{dev}_i^2$, was 0.477, 0.0829, 0.0207 and 0.0013 at $n_c = 0$, 1, 2 and 4, ratios 0.174, 0.043 and 0.0027 to the value at $n_c = 0$, below $\rho_\perp^{2n_c} = 0.25$, 0.0625 and 0.0039. At the decision state the coarse predictive mean was $m_C = 0.3$, the members' price beliefs were $m_i = m_C + \mathrm{dev}_i$ with deviations drawn from the stationary dispersion and a common variance, and each act moved the next price by $\mathsf{b}_p u_i$. The total-variation distance between the stationary law of the chain and the group posterior with tilted habits was $9.3 \times 10^{-16}$, $1.1 \times 10^{-15}$ and $7.6 \times 10^{-16}$ at $n_c = 0$, 1 and 4, detailed balance held to $2 \times 10^{-17}$, the largest tilt was 0.249, 0.136 and 0.018, and the expected coarse act was $-0.605$, $-0.641$ and $-0.650$: the firm sells when it believes the price is above target. With preference precisions $(2, 3, 5, 6)$, detailed balance failed by 6 to $7 \times 10^{-4}$ and the distance to the shared-precision group posterior was 0.127, 0.060 and 0.062. An independent implementation found firm $C_2$'s filtered variance 0.12571, also from the quadratic $2.43x^2 + 0.49x - 0.1 = 0$, firm $C_1$'s exact predictive and filtered variances of the price from the five-dimensional abstract physics, 0.11772 and 0.08167, and the bounded model's claimed filtered variance 0.11089, whose prediction $0.81 \cdot 0.11089 + 0.1 = 0.18982$ is the claimed variance of the table. For Lemma S35.1 with a singular information matrix, firm $C_1$'s five-dimensional abstract physics has $\Lambda^s = 3.75 e_1 e_1^\top$ and $\rho(\bar{F}) = 0.618$; the iteration of $\Psi$ reached the same fixed point from three starts to $10^{-16}$, with closed-loop spectral radius 0.636. On 200 random four-dimensional models with information matrices of rank two and $\bar{F} = \mathsf{K} + \mathsf{L}\Lambda^s$, $\rho(\mathsf{K}) = 0.7$, the bound in the proof of the lemma held as an identity to $10^{-14}$, the iterates from two starts agreed to $10^{-13}$ and the closed-loop spectral radius was at most 0.875; for $\bar{F} = \mathrm{diag}(0.5, 1.2)$, $\bar{Q} = I$ and $\Lambda^s = \mathrm{diag}(1, 0)$, which is not detectable, the iterates diverged.

## S47.11 Checks for Section 12

Each check below evaluates exact Gaussian linear algebra (continuous and discrete Lyapunov equations, block Toeplitz covariances of sampled paths, joint precisions and their conditional and

marginal blocks, eigendecompositions) on the instances stated, and symbolic simplification where stated. Where a second, independent implementation of the same statement was run, its results are reported in the same check.

**Check S47.128** (The member's model of its collective)**.** For Lemma S12.4, 60 random stable systems with $\dim x$ from 3 to 5 and $\dim y$ equal to 1 or 2 were drawn. The relative entropy rate to the moment-matched first-order autoregression, computed as the difference of the relative entropies of windows of lengths $T$ and $T-1$ at large $T$, equaled $\frac{1}{2}\ln\det(S_1 S_\infty^{-1})$ to $1.4\times10^{-14}$. The inequality $\mathfrak{d}_{\mathrm{M}}^{\infty} \le \mathfrak{d}_{\mathrm{cl}}$ held in every instance, with ratios from 0 to 0.684 and median 0.138; changing the autoregression coefficient by up to 0.05 or the noise variance by the factors 0.9 and 1.1 never lowered the rate; and a system with $A\ker\mathsf{P}\subseteq\ker\mathsf{P}$ had $\mathfrak{d}_{\mathrm{cl}}=0$ and $\mathfrak{d}_{\mathrm{M}}^{\infty}=1.1\times10^{-16}$.

For Proposition S12.5, 40 random diffusive populations with a denser region (contrast 2.09 to 29.1) satisfied the sine bound of Theorem 5.4(a) with largest ratio 0.822. On 401 frequencies from 0 to $10^4$ the ratio $\mathcal{S}_y/\mathcal{S}_{\mathrm{OU}}$ stayed in $[\cos^2\alpha_C, 1]$, exceeding $\cos^2\alpha_C$ by at least $3.5\times10^{-8}$; the generalized eigenvalues of the covariances at eight random times lay in $[\cos^2\alpha_C, 1]$; and the relative entropy at those times was at most the first bound of (S49), with largest ratio 0.888. A second set of 300 regions with $\varkappa_{\mathrm{in}} > \varkappa_{\mathrm{out}}$ (contrast 1.03 to 80.7), 35 of them with $\sigma_C \ge \varkappa_{\mathrm{in}} - \varkappa_{\mathrm{out}}$, gave largest ratios 0.929 for $\tan^2\alpha_C$ against $t^2$, with $t = s/(r(C)-1)$, and 0.759 for the relative entropy at eight random times against $\frac{m}{2}(-x-\ln(1-x))$ at $x=t^2/(1+t^2)$; the last inequality of (c) held in every instance. For twelve members with pair coupling $1/12$, three of them with cut coupling 0.3, at leak 0.02: $r(C)=10.737$, $s=1.3674$, $t=0.14044$ and $\sin^2\alpha_C = 1.229\times10^{-2}$ against $t^2/(1+t^2)=1.934\times10^{-2}$. The bound per sampled time is $3.81\times10^{-5}$ nats at the true angle, $9.54\times10^{-5}$ in the form of (c) and $9.92\times10^{-5}$ from the sine bound.

**Check S47.129** (Messages and predictive coding)**.** For Theorem 12.2, 200 random Gaussian trees with 3 to 11 nodes of dimension 1 or 2 were drawn, with random link gains, link noises, sensing matrices and data. The posterior from the two passes agreed with the posterior from the joint precision to $1.2\times10^{-13}$, and every downward message agreed with the law of its node given the data outside its subtree, obtained by conditioning the joint law, to $2.5\times10^{-12}$; an independent implementation on 150 random trees found both deviations below $8.9\times10^{-14}$. For Proposition S12.6, in 200 random trees of the same kind with a proper root prior, the gradient of $\mathcal{F}$ at a random point agreed with (S50) to $5.3\times10^{-16}$; the block Gauss–Seidel iteration had spectral radius at most 0.9952, and 3000 sweeps reached the exact posterior mean to $3.7\times10^{-7}$; and at every node the mean-field covariance lay below the exact marginal posterior covariance.

**Check S47.130** (Literal reading, records and depth)**.** For Proposition 12.3, in 500 random collectives the mean-square error of the literal child computed from the joint law agreed with (108) to a relative error $8.7\times10^{-16}$ and the overconfidence factor with (109) to $8.9\times10^{-16}$, and in every instance the claimed variance lay below the exact posterior variance, which lay below the actual mean-square error. Symbolic simplification reproduced (108) and (109), and an independent implementation confirmed them symbolically, found the supremum of $\mathrm{oc}_1$ at $\psi=0$ to be 2, approached as $\mathsf{w}\to1$, and the largest ratio of the mean-square error to the exact variance at $\psi=0$ to be 1.125 at $\mathsf{w}=1/3$. With unit information per member, $\Lambda_\circ=0.25$ and $\psi=0$, $\mathrm{oc}_1$ is 1.3810, 1.1778 and 1.0198 for 4, 10 and 100 members.

For Proposition S12.7 with three members of unit information and $\Lambda_\circ=1$, the iterated rules agreed with Table 13 to $4\times10^{-16}$. After eight rounds the precisions were 4, 25, 9 and 16401, the mean-square errors 0.2500, 0.3088, 0.2757 and 0.2800, and the overconfidence factors 1, 7.72, 2.48 and $4.59\times10^3$ under (E), (U), (A) and (P), in both implementations. Symbolic computation reproduced the table; for $n=3$, $\Lambda_\circ=1$ and $\Lambda^{\mathrm{o}}=2$ the limit error under (P) is $3/8$ against the union value $1/3$.

For Proposition S12.8(a), in 20 random systems over 60 steps the record under rule (U) equaled the centralized Kalman filter to $4.1 \times 10^{-14}$, and an independent implementation found $8.9 \times 10^{-15}$. For (b), the fixed point $\Lambda_z = 207.18$ gives the claimed variance 0.00483; the mean-square error from the closed form of the proof and from a Lyapunov solve is 0.2876, 59.6 times the claim; and the centralized filter's stationary variance is 0.0528.

For Proposition S12.9, in four groups of five members with unit information, $\Lambda_\circ = 0.2$ and $\Psi = 0$, exact cavities give claimed, actual and optimal variance 0.04950. Literal reading at the member level only puts weights in the ratio $2:1:1$ on the member's own datum, a group-mate's and another group's, with actual variance 0.0516 against the claimed 0.0472; literal reading at both levels gives $3:2:1$, with actual variance 0.0586 against the claimed 0.0382. With link noise variances 0.3 at the groups and 0.2 at the members the double cavity is exact (0.209409 claimed and actual). At $\Lambda_\circ = 0.5$ the shortcut's believed precision over the exact one is 1.846, 1.980 and 1.998 for two groups of three, five of ten and twenty of twenty members, its actual over claimed variance 1.917, 1.990 and 1.999, and at 50 members its actual error is 0.01990 against the exact 0.01980. An independent implementation at $\Lambda_\circ = 0.2$ found the ratios 1.839, 1.980 and 1.998 and reproduced the variances of the literal readings.

**Check S47.131** (Records: drift, shared error and nesting)**.** For Proposition 12.4, 3000 random rings were drawn with 4 to 15 members, coupling, leak, read weight and write weight log-uniform on $[10^{-2}, 10^{1/2}]$, $[10^{-3}, 10^{-1}]$, $[10^{-2}, 1]$ and $[10^{-3}, 1]$, and $\delta = 0$ in about half of them and log-uniform on $[10^{-4}, 10^{-1}]$ otherwise. The gain (111) agreed with the spectrum of the full drift to a relative deviation $8.0 \times 10^{-11}$. In the 540 draws meeting the hypothesis of (c) both bounds held, with extreme ratios 1.0000; in the 1926 draws meeting the hypothesis of (d), $\mathsf{g}_{\mathrm{sep}}/\mathsf{g}_\times$ ranged from $2.0 \times 10^{-5}$ to 4.78, and its largest ratio to the bound of (112) was 0.9998. An independent computation on 400 random rings found the same agreement of the gain with the spectrum. On the ring of ten with neighbor coupling 1, $\varepsilon = 0.01$, $\beta = 0.2$ and $\omega = 0.1$: $\lambda_2(L_C) = 0.3820$, slowest rate 0.003260 against $\varepsilon/(1+\mathcal{R}) = 0.003333$, second rate 0.3067, and gain 2.400 against the product 4.531; with $\omega = 0.01$ the gain is 12.30 against 31.72; with coupling 0.05 and $\omega = 0.1$, where (c) applies, it is 24.15 against 23.62.

For the condition stated after Proposition 12.4, on the same ring with $\omega = 0.1$ and $\delta = 0$, $\mathsf{r}_\circ = 0.197981$ and $(\mathsf{r}_\circ - \varepsilon)(\mathsf{r}_\circ - \omega)/\mathsf{r}_\circ = 0.093032$, where the gain equals 1 to $10^{-12}$. On 401 values of $\beta$ in $[10^{-4}, 1]$ the gain from the spectrum of the $11 \times 11$ drift agreed with (111) to $2.7 \times 10^{-12}$ and was below one exactly below that value, and at $\beta = 10^{-9}$ it was $0.255124 = \omega/(\varepsilon + \lambda_2(L_C))$. Over 200000 random draws with $\delta = 0$ and $(\varepsilon, \omega, \lambda_2(L_C), \beta)$ log-uniform on $[10^{-3}, 1]$, $[10^{-3}, 1]$, $[10^{-2}, 10]$ and $[10^{-5}, 10]$, the gain was below one in 70696 draws; among the draws with $|\mathsf{g}_{\mathrm{sep}} - 1| > 10^{-9}$ it was below one exactly when $\beta$ was below the threshold, and $\mathsf{g}_{\mathrm{sep}} = \mathsf{r}_+^2/(\omega(\varepsilon + \lambda_2(L_C)))$ held to $7.6 \times 10^{-9}$ whenever $\mathsf{r}_+ < \varepsilon + \beta + \lambda_2(L_C)$. For Corollary 12.5, in 20000 random draws of $(\varepsilon, \beta, g, \mathsf{f}, \omega, \delta)$, with $\mathsf{f}$ the newcomer fidelity and $\delta = 0$ in about half of them, $-\mathsf{A}$ of Proposition 5.14(a) agreed with $\mathsf{R}$ at $(\tilde{\varepsilon}, \tilde{\beta})$, and the closed form of $\tau_{\mathrm{id}}^{-1}$ there with the smaller eigenvalue of that matrix, both to $2.5 \times 10^{-9}$, the largest deviation coming from cancellation in the closed form at small rates, and neither the bracket for $\delta = 0$ nor the lower bound of the corollary was violated.

For Lemma 12.6(b) and Theorem 12.7, the closed form (113) agreed with the stationary variance of the common mode from a Lyapunov solve to a relative error $6.6 \times 10^{-15}$ over 2000 draws with $\beta + \omega \le 1$ and $\beta + k^s \le 1$, none of them with $\mathsf{c}_2 < 0$, and the series $\sum_s \mathsf{f}_s^2$ agreed with the Lyapunov solve to $8.1 \times 10^{-16}$ on 50 draws. In 300 further draws there was no violation of the monotonicity of (c) or of the bound of (b). At $F = 0.99$, $k^s = 0.05$, $\beta = 0.2$ and $\omega = 0.1$ $(1 + \mathcal{R} = 3)$ the factor $\mathsf{F}_{\mathrm{com}}$ is 16.77 against 8.661 without the record, a ratio 1.936; with $F = 0.999$ the ratio is 2.553 and with $F = 0.9$ it is 1.113; at $F = 0.9999$, $k^s = 0.02$, $\beta = 0.4$ and $\omega = 0.05$ $(1 + \mathcal{R} = 9)$ the

factor is 207.1 against 25.13, a ratio 8.240. The symbolic common-mode factor reduces at $\beta = 0$ to $1/(1-F^2(1-k^s)^2)$. For (e), symbolic computation confirmed that the closed form of the proof equals the solution of the Stein equation of the common mode and the solution of the three linear equations of the proof, that $\mathsf{K}(x) = x\mathsf{K}_1(x)$ and $\mathsf{K}_1(\mathsf{p}_\mathrm{A}) = (1 - \mathsf{p}_\mathrm{m}\mathsf{p}_\mathrm{z} + \mathsf{p}_\mathrm{A})\mathsf{T}_-(\mathsf{p}_\mathrm{A})\mathsf{T}_+(\mathsf{p}_\mathrm{A})$ identically, and that the limit is $(1+\mathsf{y})(\beta+\omega)/(\omega+\mathsf{y}(\beta+\omega))$; at 25 random rational points the closed form agreed with (113), evaluated with exact radicals, to 40 digits. Over 400000 random points with $F = 1 - 10^{-u}$, $u$ uniform on $[0, 9]$, $\beta + \omega \le 1$ and $\beta + k^s \le 1$, the largest value of $\mathsf{F}_\mathrm{com}(1 - F^2(1 - k^s)^2)/(1 + \mathcal{R})$ was 0.99999997 and the vertex of $\mathsf{K}_1$ was at least 1.007; on the 2670 of them with $F < 1 - 10^{-6}$ among the first 4000, the closed form agreed with a direct solve of the Stein equation to $7.9 \times 10^{-12}$. At $\beta = 0.4$, $\omega = 0.05$ and $k^s = 10^{-5}$, with $1 - F = \mathsf{y}k^s$, the ratio $\mathsf{F}_\mathrm{com}(1 - F^2(1 - k^s)^2)$ is 8.99978, 8.33923, 5.21035 and 1.79992 for $\mathsf{y} = 0$, 0.01, 0.1 and 1, against the limits 9, 8.33945, 5.21053 and 1.8 (at $\mathsf{y} = 0$ the closed form is taken at $F = 1$, where it is continuous).

For Proposition S12.12, 1500 random chains of 2 to 5 levels and 1500 random stars with 1 to 4 records produced no violation of (S51), with $\varepsilon_\mathrm{rec}/\bar{\mathsf{e}}_\mathsf{u}$ between 0.060 and 1. At $\varepsilon = 0.01$, $\beta = 0.2$, $\omega = 0.1$ and $\delta = 0$ the slowest rate 0.003260 lies in the bracket $[0.003258, 0.003333]$, and the correlation time $-1/\ln(1 - \varepsilon_\mathrm{rec}) = 306.24$ equals the one from the secular equation of Proposition 8.11.

**Check S47.132** (Records as nodes of the precision)**.** For Proposition S12.13, a ring of eight members (neighbor coupling 1, leak 0.02, $q = 1$) joined to a ring of eight outsiders by cross couplings 0.05 between matched members has $r(C) = 8.6541$, the value of (S52). At the reversible record noise the joint precision is diffusive (largest positive off-diagonal entry below $4 \times 10^{-15}$ of the scale), and $r(C')$ is 12.9501 for $\beta = 0.2$ at both $\omega = 0.1$ and $\omega = 0.01$, 25.8073 for $\beta = 1$ and $\omega = 0.1$, and 11.3314 for $\beta = 0.2$, $\omega = 0.1$ and $\delta = 0.005$, each equal to (S52); the members' marginal contrasts are 11.5112, 11.5112, 22.9398 and 10.1326, as (c) gives. With a noiseless record or with $q_z = q$ the joint precision has positive off-diagonal entries of $1.7 \times 10^{-4}$ to $6.4 \times 10^{-3}$ of the scale. In discrete time with coupling 0.2, cross coupling 0.01 and leak 0.02, $r(C) = 4.2908$, and the reversible record gives diffusive joint and marginal precisions with $r(C') = 10.6073$ at $\beta = 0.2$ for $\omega = 0.1$ and $\omega = 0.01$ and $r(C') = 14.8621$ at $\beta = 0.4$; the noiseless record and $q_z = q$ again give positive entries.

For Example S12.15, the closed form (S53) agreed with the inverse of the Lyapunov solution in 300 random closed collectives to a relative error $7.2 \times 10^{-13}$, and the members' marginal precision was diffusive in all of them. For the closed ring of eight with $\beta = 0.2$ and $\omega = 0.1$, the noiseless record gives every uncoupled pair of members the entry $\omega/(nq) = 0.0125$ and the record row sum 0.512; with outsiders the largest positive entry is $3.2 \times 10^{-3}$ of the largest entry in magnitude for the joint precision and $2.5 \times 10^{-4}$, between two outsiders, for the agents' marginal precision. With the outsiders joined at two matched pairs only, these entries are $3.8 \times 10^{-3}$ and $4.6 \times 10^{-5}$.

For Numerical observation S12.16, the ratio $q_z/q_\mathrm{rev}$ took the values 0, 0.5, 0.9, 0.99, 1, 1.01, 1.1, 2 and 10 for $\delta = 0$, and, in an independent implementation in which the outsiders were joined at two matched pairs only, 0, 0.25, 0.5, 0.9, 0.99, 1, 1.01, 1.1, 2 and 4 for $\delta = 0$ and $\delta = 0.01$; entries are relative to the largest entry of the precision in magnitude. For $\delta = 0$, below $q_\mathrm{rev}$ the largest positive member-member entry falls from $4.1 \times 10^{-3}$ at 0 to $3.5 \times 10^{-5}$ at 0.99 for the closed collective, and from $3.2 \times 10^{-3}$ to $3.0 \times 10^{-5}$ with outsiders; above it the closed collective has no positive off-diagonal entry but a negative record row sum, $-4.5 \times 10^{-4}$ at 1.01 and $-2.1 \times 10^{-2}$ at 2, and with outsiders the largest positive record-outsider entry grows from $4.7 \times 10^{-5}$ at 1.01 to $6.1 \times 10^{-3}$ at 10. The independent implementation found the same pattern for $\delta = 0$. For $\delta = 0.01$ the closed collective is diffusive at 1, 1.01, 1.1 and 2, has positive member-member entries at 0.99 and below and a negative record row sum at 4, while with outsiders positive entries appear at every value except 1, both for two matched pairs and, in a recomputation over the same ratios and 10, for all

eight.

For Proposition S12.17, on a closed ring of 200 members with coupling 1, $\varepsilon = 0.01$, $\beta = 0.2$, $\omega = 0.1$ and $q = 1$ at the reversible record noise, the covariance of antipodal members is 0.4762, the value $q\beta/(N\varepsilon(\varepsilon+\beta))$, against $4.55 \times 10^{-4}$ without the record, and the variance of the members' mean is 0.5 with and without the record. With a noiseless record the antipodal covariance is 0.1536 and the variance of the mean 0.1774, the value of the closed form.

**Check S47.133** (Top-down precision)**.** For Numerical observation S12.19(a), the stationary law of eight level-0 members on the complete graph and their record was computed from a discrete Lyapunov equation, with the members' sensing and record gains set from the claimed variances at the fixed point. As the record precision $\mathsf{r}_z$ took the values 0, 0.1, 0.3, 1, 3, 10 and 30, the record gain rose from 0 to 0.3985 and the sensing gain fell from 0.0276 to 0.0133; the mean-square errors were 0.2815, 0.2875, 0.2998, 0.3426, 0.4495, 0.6631 and 0.8579, above the shared floor $Q\mathsf{F}_{\mathrm{com}}$ (0.2732 to 0.8374) by at most 0.0205; the disagreement was between 0.0073 and 0.0376; and the echo ratios were 35.95, 37.06, 39.29, 46.74, 60.29, 49.02 and 22.79. The smallest error over $\mathsf{r}_z \in [0, 10]$ was at $\mathsf{r}_z = 0$ in this instance and with sensing noise variance 4, with message precision 0.2, and with $\omega = 0.5$. On rings the finite-difference derivative of the error at $\mathsf{r}_z = 0$ was $6.3 \times 10^{-2}$, $4.7 \times 10^{-2}$, $1.5 \times 10^{-2}$, $7.0 \times 10^{-2}$ and $1.9 \times 10^{-2}$ for $(n, \varpi_{\mathrm{y}}, \omega)$ equal to $(40, 2, 0.05)$, $(40, 10, 0.05)$, $(100, 10, 0.2)$, $(100, 10, 0.02)$ and $(40, 2, 0.5)$.

For (b), one member fuses its prior mean, its sensing, its neighbors' messages and the record with the minimum-variance linear weights under the population's stationary law. On the complete graph of eight with population record precision 1 and 0, its posterior variance is 0.19024 and 0.17980 with the record against 0.19704 and 0.18814 without, with record weights $-0.2464$ and $-0.2663$; on the ring of 40 with population record precision 0.3 and 0 it is 0.14079 and 0.13622 against 0.14638 and 0.14119, with record weights $-0.2067$ and $-0.1902$. The relative reductions, computed from the unrounded variances, are 3.446, 4.432, 3.816 and 3.519 percent.

For (c) and Proposition S12.20, the stationary covariance was split by linearity into the part driven by the world innovations, which equals $Q\mathsf{F}_{\mathrm{com}}$ of (113) to nine digits, and the part driven by the local noises. With sensing noise variance 4, message precision 10 and channel and record read noise variances 0.1, as $\mathsf{r}_z$ took the values 0, 1, 3, 10 and 30: on the complete graph of eight with $F = 0.9$, $Q = 0.02$ and $\omega = 0.05$ the errors were 0.111056, 0.110796, 0.110419, 0.110038 and 0.112207, with common-mode part rising from 0.103200 to 0.104352 and local-noise part 0.007857, 0.007513, 0.006988, 0.006226 and 0.007855; on the complete graph of eight with $F = 0.99$, $Q = 0.001$ and $\omega = 0.05$ the errors were 0.052056, 0.051819, 0.051493, 0.051185 and 0.052275, with common-mode part rising from 0.046569 to 0.048758; and on the ring of 40 with $F = 0.99$, $Q = 0.001$ and $\omega = 0.2$ they were 0.050778, 0.049747, 0.048964, 0.049237 and 0.052700, with common-mode part rising from 0.043638 to 0.047307. The forward-difference derivatives of the error at $\mathsf{r}_z = 0$ were $-2.9 \times 10^{-4}$, $-2.7 \times 10^{-4}$ and $-1.5 \times 10^{-3}$, against $6.0 \times 10^{-2}$ in the instance of (a), where the local-noise part is 0.008261 of 0.281458; changing only $F$ to 0.9 and the message precision to 10 in that instance gives $-4.5 \times 10^{-5}$. In 1799 random instances with 3 to 8 members on rings and complete graphs and $\beta + \omega \le 1$, the sensing gain fell and the common-mode part rose above its value at $\mathsf{r}_z = 0$ in every case.

**Check S47.134** (Conformity and code domains)**.** For Proposition 12.8(a), 200 random unilateral moves changed $G_i$ and $\mathcal{V}$ by equal amounts to $3.6 \times 10^{-14}$, and an independent implementation found $1.6 \times 10^{-14}$. For (c), two blocks of six with pair coupling 0.2 inside the blocks, a perfect matching of coupling 0.05 between them and $\mathsf{a}_{\mathrm{p}} = 0.1$ have block contrast 8.6667 without records and 24.6667 with $\beta_{\mathrm{r}} = 2$, as (c) gives, in both implementations. For (d), homogeneous attention on the complete graph of twelve with pair coupling $1/11$, $\mathsf{a}_{\mathrm{p}} = 0.1$ and no records gives no set

of six a contrast above 1, and the whole population has contrast 11.909. Two disjoint cliques of four with $J = 1$ and $\mathsf{a}_\mathrm{p} = 0.1$ have contrast $41.000 = 1 + 4J/\mathsf{a}_\mathrm{p}$ against the exchangeable baseline $35.286 = 1 + 2m(m-1)J/((2m-1)\mathsf{a}_\mathrm{p})$ with $m = 4$, a significance of 1.162, while under the graph clause the comparison medium is the population and the baseline is 41.000; cliques of three at $\mathsf{a}_\mathrm{p} = 0.2$ give 16 against 13, and cliques of six at $\mathsf{a}_\mathrm{p} = 0.1$ give 61 against 55.545. For the two blocks of six above the graph clause gives the baseline 4.500, with mean coupling 0.175 on the declared edges, so each block is a collective with significance 1.926.

For Numerical observation S12.23, the calibrated in-group noise at the base point is 0.29872, with mean-square error 0.24872. Conditioning on the joint law of the literal split gives the values 0.00443, 0.01379, 0.01932 and 0.02288 nats, over both coordinates, for zero to three cross senders among three, against the believed value 0.40582 nats of any three senders under calibrated level-0 reading, and the threshold 0.00936 nats; both implementations agree. Under the translation-aware null every sender has calibrated noise 0.29872, so all sets of three tie.

For Example S12.24, the closed form of the ring baseline, the larger of the whole ring's contrast and an adjacent pair's, agreed with the supremum over all arcs and the whole ring for $N = 5$, 8, 12 and 60, $\bar{a} = 0.8$ and 1, at 26 leaks each, to $1.1 \times 10^{-11}$, and for $N = 12$ and $\bar{a} = 1$ with the supremum over all 4083 subsets of at least two sites at four leaks to $8.9 \times 10^{-16}$. At $N = 60$, $a = 1$ and $\varepsilon^{\Pi} = 0.002$ the contrasts, baselines and significances of the table were computed from the precisions, and a scan over 601 leaks in $[10^{-5}, 10]$ found the domains of 6, 10, 20, 30 and 29 sites to qualify up to 0.389, 0.1096, 0.02512, 0.01096 and 0.01175, the grid points just below the thresholds of the table. The domain of 31 sites has $\lambda_{31} = 0.010261$ below $2\bar{a}(1 - \cos(2\pi/N)) = 0.010591$. The exchangeable baselines are 848.5, 916.3, 967.1 and 984.1 for 10, 6, 3 and 2 walls. The exclusion of domains of three or more sites under the exchangeable clause needs at most $N/2$ walls: with 58 walls on the ring of 60, 57 single sites and one domain of three, $N\bar{a}_\mathrm{ex} = 4/59 = 0.0678$ is below $\lambda_3 = 1$. Equal domains of $n$ sites, $n$ dividing 60, are collectives at some leak exactly for $n = 2$, 3, 4, 5, 6, 10, 12, 15, 20 and 30. When the declared graph is the ring with the walls removed, for domains of ten the baseline equals the domain contrast 49.94. An independent implementation reproduced the contrasts 134.97, 49.94, 13.31, 6.86 and 6.48 and the baselines 5.57, 5.93, 6.20 and 6.30.

**Check S47.135** (Three levels)**.** For Theorem 12.9, in 200 random three-level populations the spectrum of the drift equaled the union of the three families to a relative deviation $3.6 \times 10^{-15}$, and every eigenvalue was real and positive. For five groups of ten, each a complete graph with unit coupling, with $\varepsilon = 0.01$, $\beta_1 = 0.2$, $\omega_1 = 0.05$, $\beta_2 = 0.1$ and $\omega_2 = 0.02$ ($\mathcal{R}_1 = 4$, $\mathcal{R}_2 = 5$), the society rates are $3.765 \times 10^{-4}$, $9.249 \times 10^{-2}$ and 0.2871, against the first-order value $\varepsilon/(1 + \mathcal{R}_1 + \mathcal{R}_1\mathcal{R}_2) = 4.000 \times 10^{-4}$ and $\varepsilon = 0.01$ without records; the group-level rates are $7.560 \times 10^{-2}$ and 0.2844, the slower inside the bracket $[0.0597, 0.1194]$ of (c); and the member-deviation rate is 10.210. The separations, ratios of the slowest rates of consecutive families, are 200.8 between the group level and the society and 135.1 between the members and the group level. With $\delta_1 = 10$ and $\omega_2 = 0.5$, the other parameters unchanged, the slowest society rate 0.20898 and the slower group-level rate 0.20899 coincide to four digits although $\varepsilon/(1 + \mathcal{R}_1 + \mathcal{R}_1\mathcal{R}_2) = 0.0017$ lies far below $\det \mathsf{R}_\mathrm{grp}/\operatorname{tr} \mathsf{R}_\mathrm{grp} = 0.2048$; the bound of (c) is 6.90 there, so the condition of (c) fails, as it must. For random chains of the agents and one to three records, with random read and write weights and leaks at every node scaled by $10^{-2}$, $10^{-3}$ and $10^{-4}$, the relative error of the first-order rate $\bar{\mathsf{e}}_\mathsf{u}$ of Proposition S12.12 was $5.5 \times 10^{-3}$, $5.5 \times 10^{-4}$ and $5.5 \times 10^{-5}$ with one record, $3.5 \times 10^{-3}$, $3.5 \times 10^{-4}$ and $3.5 \times 10^{-5}$ with two and $3.3 \times 10^{-3}$, $3.4 \times 10^{-4}$ and $3.4 \times 10^{-5}$ with three, linear in the leak scale.

For the hypotheses of (b), 300 further populations of 2 to 5 groups of 2 to 6 members, each group a complete graph with unit coupling, with leak log-uniform on $[10^{-3}, 1]$, read and write weights log-uniform on $[10^{-3}, 10]$ and decays zero or log-uniform on $[10^{-3}, 1]$, gave eigenvalues whose

imaginary parts were at most $2.1 \times 10^{-16}$ of the largest modulus and whose real parts were all positive. For the five groups of ten above, $\omega_2 = \delta_2 = 0$ makes 0 an eigenvalue, while $\omega_2 = 0$ with $\delta_2 = 0.01$ gives the smallest eigenvalue 0.01, and $\beta_2 = 0$ with $\omega_2 = 0.02$ gives 0.00194.

### S47.12 Checks for Section 13

Each check below evaluates exact Gaussian linear algebra (Schur complements, Lyapunov solutions, Fourier inversion on tori, spectral decompositions) on the instances stated. Where a second, independent implementation of the same statement was run, its results are reported in the same check.

**Check S47.136** (Decimation on $\mathbb{Z}$)**.** Rings of $60b$ sites with $(\varepsilon^{\Pi}, a^{\Pi}) \in \{(0.01, 1), (0.3, 0.7), (2, 0.5)\}$ and $b \in \{2, 3, 5\}$ were decimated by Schur complements. The coarse couplings and leaks agreed with (116) to 13 digits, for instance $a^{\Pi,(1)} = 0.4975124378$ and $\varepsilon^{\Pi,(1)} = 0.01995024876$ at $(0.01, 1)$ and $b = 2$; every coupling beyond range one vanished exactly, and $m^{(1)}/m = b$ to 12 digits. The map $\mathfrak{x} \mapsto \mathfrak{x}(\mathfrak{x} + 4)$ reproduced $\mathfrak{x}^{(1)} = 0.0401$ at $\mathfrak{x} = 0.01$. The squared second-moment length of the decimated field agreed with $1/(4\sinh^2(bm/2))$ to $10^{-8}$ relative at $\varepsilon^{\Pi} \in \{10^{-3}, 10^{-2}, 0.1\}$ and $b \in \{2, 4\}$. An independent implementation found the largest relative deviation $3.2 \times 10^{-13}$ over the three media.

**Check S47.137** (Nearest-neighbor passing on $\mathbb{Z}$)**.** For $(c, a, q) = (0.25, 0.3, 1)$ and $(0.1, 0.44, 0.7)$ and $b \in \{2, 3, 4\}$ on a ring of 400 sites, the two-pole formula of Proposition S13.1(a) matched the decimated covariance to $1.7 \times 10^{-14}$ relative, the largest off-diagonal entry of every decimated precision was negative, and the ratios of consecutive tail couplings, 0.120538, 0.024868, 0.021248 for the first medium and 0.363636, 0.053705, 0.152641 for the second, equaled $e^{-\operatorname{arccosh} u_*}$ to six digits, the odd factor $b = 3$ included.

**Check S47.138** (Decimation on $\mathbb{Z}^2$)**.** On a $16 \times 16$ torus with $\varepsilon^{\Pi} = 0.05$ and $a^{\Pi} = 1$ the checkerboard decimation gave the couplings 0.493827 and 0.246914 at the rotated nearest and diagonal displacements and the leak 0.0993827, as Example S13.2 predicts. The further decimation onto $2\mathbb{Z} \times 2\mathbb{Z}$ gave all 63 couplings per site positive, decaying from 0.4575 at the coarse displacement $(1, 0)$ to 0.1154 at $(1, 1)$, 0.0189 at $(2, 0)$ and 0.0017 at $(3, 0)$. For Numerical observation S13.3 the covariance of the nearest-neighbor field with $a^{\Pi} = 1$ on a $256 \times 256$ torus was computed from its symbol by Fourier inversion, subsampled on $2\mathbb{Z} \times 2\mathbb{Z}$ and on $4\mathbb{Z} \times 4\mathbb{Z}$, and inverted by Fourier transform. At $\varepsilon^{\Pi} = 0.5$, 0.05, 0.005 and 0.0005 the coarse couplings at $(1, 0)$ to $(5, 0)$ after the first decimation were $(0.3565, 8.92 \times 10^{-3}, 5.95 \times 10^{-4}, 3.98 \times 10^{-5}, 2.67 \times 10^{-6})$, $(0.4575, 1.89 \times 10^{-2}, 1.71 \times 10^{-3}, 1.55 \times 10^{-4}, 1.41 \times 10^{-5})$, $(0.4712, 2.06 \times 10^{-2}, 1.92 \times 10^{-3}, 1.81 \times 10^{-4}, 1.71 \times 10^{-5})$ and $(0.4726, 2.07 \times 10^{-2}, 1.95 \times 10^{-3}, 1.84 \times 10^{-4}, 1.74 \times 10^{-5})$, with consecutive ratios between 0.0250 and 0.0948; after the second they were $(0.0698, 1.34 \times 10^{-3}, 3.71 \times 10^{-5}, 1.08 \times 10^{-6}, 3.21 \times 10^{-8})$, $(0.2349, 1.87 \times 10^{-2}, 1.85 \times 10^{-3}, 1.91 \times 10^{-4}, 2.01 \times 10^{-5})$, $(0.2844, 2.76 \times 10^{-2}, 3.28 \times 10^{-3}, 4.06 \times 10^{-4}, 5.13 \times 10^{-5})$ and $(0.2903, 2.88 \times 10^{-2}, 3.49 \times 10^{-3}, 4.39 \times 10^{-4}, 5.66 \times 10^{-5})$, with ratios between 0.0192 and 0.1288. The ratios increase with the displacement: at $\varepsilon^{\Pi} = 0.0005$ the ratios of the couplings at $(x + 1, 0)$ and $(x, 0)$ for $x = 1, \dots, 8$ were 0.0439, 0.0938, 0.0943, 0.0948, 0.0953, 0.0957, 0.0961, 0.0964 after the first decimation and 0.0993, 0.1210, 0.1260, 0.1288, 0.1309, 0.1324, 0.1336, 0.1346 after the second, the same to the printed digits on a $512 \times 512$ torus. No coarse precision had a positive off-diagonal entry.

**Check S47.139** (Flows of the second-moment length and of the axis rate)**.** In one dimension, for the nearest-neighbor precision at $\varepsilon^{\Pi} \in \{10^{-3}, 10^{-2}, 0.1\}$ and $b \in \{2, 4\}$, both formulas of (119) agreed with the covariance-side lengths of the coarse fields to $10^{-8}$ relative; at $\varepsilon^{\Pi} = 10^{-2}$ and $b = 2$

the squared lengths were 24.93766 (decimation) and 25.06234 (averaging), against $\xi_2^2/b^2 = 25$. In two dimensions, on a $480 \times 480$ torus at $\varepsilon^{\Pi} \in \{0.01, 0.1\}$ and $b \in \{2, 3, 4\}$, they agreed to all eight printed digits; at $\varepsilon^{\Pi} = 0.01$ and $b = 4$ the lengths were 2.44809 and 2.51551 against $\xi_2/b = 2.5$, and at $\varepsilon^{\Pi} = 0.1$ and $b = 4$ they were 0.66218 and 0.83652 against 0.79057. The decimated length lay below $\xi_2/b$ in all six two-dimensional cases. On a $510 \times 510$ torus, for passing with $(c, a) = (0.2, 0.19)$ and $(0.3, 0.17)$, the axis rates of the decimated and of the block-averaged fields equaled $bm_1$ to five digits or better for $b = 2, 3$, with relative deviation at most $1.6 \times 10^{-6}$; for instance $m_1 = 0.45489907$, and at $b = 2$ the decimated field gave 0.90979956 and the averaged one 0.90979814, against $2m_1 = 0.90979814$. An independent implementation reproduced the one-dimensional formulas to $1.7 \times 10^{-9}$ on a ring of 24000 sites and the two-dimensional values to every printed digit.

**Check S47.140** (Block averages)**.** On a ring of 1200 sites with $\varepsilon^{\Pi} \in \{10^{-3}, 10^{-2}, 0.2\}$ and $b \in \{2, 3, 4, 8\}$, the ratio of the precision entries of the block averages at distances two and one equaled $-\mathsf{h}_b$ of Example S13.4 to six digits in all twelve cases, with a positive entry at distance two; at $\varepsilon^{\Pi} = 10^{-3}$ and $b = 2$ it was $-0.171512$, against the limit $-0.171573$. An independent implementation found the largest relative error $3.8 \times 10^{-14}$ and the limits 0.1716 and $2 - \sqrt{3}$. On a $240 \times 240$ torus with the nearest-neighbor kernel, $b \in \{2, 3\}$ and $\varepsilon^{\Pi} \in \{10^{-3}, 0.05\}$, the block-average precision had positive off-diagonal entries: counted per site, as the number of nonzero coarse displacements at which the entry of the precision kernel exceeds $10^{-12}$, there were 316 and 308 for $b = 2$ and 376 and 372 for $b = 3$, at the smaller and the larger leak, and the same four counts on a $480 \times 480$ torus. The entries decay with the displacement, so the count grows as the threshold is lowered, from between 144 and 184 at the threshold $10^{-8}$ to between 216 and 268 at $10^{-10}$; below about $10^{-14}$ rounding dominates. At $\varepsilon^{\Pi} = 10^{-3}$ its coupling at the coarse displacement $(2, 0)$ was $-0.272$ for $b = 2$ and $-0.451$ for $b = 3$, against 2.031 and 2.590 at $(1, 0)$.

**Check S47.141** (Long-range couplings)**.** The kernel $a^{\Pi}(x) = 1/x^2$ on $\mathbb{Z}$ has the symbol $\varepsilon^{\Pi} + \pi|p| - p^2/2$ on $[-\pi, \pi]$, so $\sigma = 1$, $\sigma' = 2$ and $\mathsf{a}_\sigma = \pi$. The coarse symbols (118) were evaluated in closed form at $p' = 0$, $h$ and $2h$ with $h = 10^{-5}\varepsilon^{\Pi}$, and $\varepsilon^{\Pi,(1)}$ and $\mathsf{a}_\sigma^{(1)}$ were extracted from the first two orders. At $\varepsilon^{\Pi} \in \{10^{-2}, 10^{-3}\}$ and $b \in \{2, 3, 5\}$, $\varepsilon^{\Pi,(1)}$ agreed with Proposition S13.5(b) to ten digits for both maps and $\mathsf{a}_\sigma^{(1)}$ to $2 \times 10^{-7}$ relative; the factor by which $\varepsilon^{\Pi}/\mathsf{a}_\sigma$ was multiplied was 2.00404, 3.01365 and 5.05263 for decimation at $\varepsilon^{\Pi} = 10^{-2}$, against $b(1 + \varepsilon^{\Pi}\mathsf{S}_1)$, and 2.00000, 3.00000 and 5.00000 for averaging.

**Check S47.142** (The nested medium)**.** For $(b, n_{\mathrm{H}}, A_{\mathrm{h}}, \mathfrak{q}, \varepsilon^{\Pi}) = (3, 5, 1, 0.5, 10^{-3})$, $(2, 8, 1, 0.7, 10^{-4})$ and $(4, 4, 2, 0.3, 0.05)$, with 243, 256 and 256 agents, $\|\Pi_{\perp\Phi}\|_F \le 1.6 \times 10^{-14}$, the coarse precision equaled the nested precision with amplitude $\mathfrak{q}A_{\mathrm{h}}$ to $1.4 \times 10^{-15}$, the inverse covariance of the block coordinates equaled $\Pi_\Phi$ to $1.1 \times 10^{-12}$, and the contrast of the blocks of every level agreed to ten digits in the fine medium, by (120) and in the coarse medium; the third medium has $r = 3.073152$, 2.992958 and 2.788462 for blocks of levels one to three. For this medium and $r_0 = 2$, the formula of Theorem 13.6(d) gives $l^* = 2$, and blocks of levels one to three reach $r_0$. In finite hierarchies the top levels have small cuts: in the second medium $r$ rises from 0.934 at level one to 2.855 at level seven. An independent implementation on the media $(3, 4, 1, 0.5, 10^{-3})$, $(2, 7, 1, 0.7, 10^{-4})$ and $(4, 3, 2, 0.3, 0.05)$ found the equitability residual at most $7.3 \times 10^{-16}$, the error of the flow at most $8.9 \times 10^{-16}$ and equal contrasts in the fine medium, by the formula and in the coarse medium.

**Check S47.143** (Depth)**.** For decimation of the nearest-neighbor precision on $\mathbb{Z}$ with $a^{\Pi} = 1$ and $b = 2$, the largest block lengths $2^l$ at which the necessary condition of Proposition 13.7(a) holds, at $(r_0, \varepsilon^{\Pi}) = (2, 10^{-4})$, $(2, 10^{-3})$, $(5, 10^{-4})$ and $(5, 10^{-3})$, were 128, 32, 64 and 16, below the bounds $(2/m)\operatorname{arcsinh}((r_0 - 1)^{-1/2}) = 176.3$, 55.7, 96.2 and 30.4 and $2\xi_{\mathrm{OZ}}/(r_0 - 1)^{1/2} = 200.0$, 63.2,

100.0 and 31.6, and at the next level the best adjacent pair had contrast 1.083, 1.149, 1.348 and 1.473, below $r_0$. The condition is only necessary: in these homogeneous media every region has contrast below 2, an adjacent pair having $(\mathfrak{r}^{(l)}+2)/(\mathfrak{r}^{(l)}+1)$, so no set reaches either level. For Proposition 13.7(d), 60 random symmetric translation-invariant kernels on rings, of range one to six, with leaks in $[10^{-3},1]$ and blocks of two to five sites, gave coarse-medium lengths that agreed with the formula to $4.5\times10^{-13}$ relative and never exceeded $\xi_2^2/B$; nearest-neighbor kernels at $\varepsilon^{\Pi}=10^{-3}$ gave 500, 333.33, 250 and 125 for $B=2$, 3, 4 and 8, equal to $\xi_2^2/B$. Six random kernels of range two on tori of $12\times12$ square blocks of side two or three agreed with the formula to $6.3\times10^{-14}$, with ratios to $\xi_2^2/B$ at most 0.715.

**Check S47.144** (Nested collectives level by level)**.** In the coarse media built by iterating $\mathsf{Q}\Pi^{(l)}\mathsf{Q}^{\top}$, with $\mathsf{Q}$ the indicator matrix of the next partition, for $(b,n_{\mathrm{H}},A_{\mathrm{h}},\mathfrak{q},\varepsilon^{\Pi})$ equal to $(3,5,1,0.5,10^{-3})$, $(2,8,1,0.7,10^{-4})$, $(4,4,2,0.3,0.05)$ and $(2,8,1,0.2,0.7)$, the contrast of the blocks of the next level and the exchangeable baseline agreed with the closed forms at every level, the baseline to $5\times10^{-7}$ at values up to $3.2\times10^{4}$. The first three media have $\mathfrak{q}\geq1/b$ and $\Delta_{\mathrm{H}}<0$ at every level, and no level qualified at any of 121 leaks in $[10^{-4},10^{2}]$. In the fourth the first qualifying leak of the grid at each level $l=0,\dots,6$ was the first grid point above the threshold $A_{\mathrm{h}}\mathfrak{q}^{l}T_{\mathrm{H}}$, which is 0.6336, 0.1285, 0.02643, 0.005593, 0.001255, $3.174\times10^{-4}$ and $1.024\times10^{-4}$; at $\varepsilon^{\Pi}=1$ the windows of admissible levels, $(2.2549,2.4000]$, $(1.2520,1.3333]$, $(1.0508,1.0693]$ and so on down to $(1.00010,1.00012]$, are disjoint, as (e) requires. The sign of $\Delta_{\mathrm{H}}$ on 260739 triples ($b=2,\dots,10$, 999 values of $\mathfrak{q}$, $m=2,\dots,30$) showed no exception to (c) or (d). For the embedding, the sixteen agents of the medium with $b=2$, $n_{\mathrm{H}}=4$ and $A_{\mathrm{h}}=1$ were placed at the sites $0,\dots,15$ of $\mathbb{Z}$, each level was given the lattice clause of Definition 4.11 on its own box, and every subset of every comparison box was enumerated. Judged level by level, the blocks of all three levels are collectives for $\varepsilon^{\Pi}$ above 0.00220, 0.0107, 0.0399, 0.780 and 3.44 at $\mathfrak{q}=0.1$, 0.2, 0.3, 0.5 and 0.55; at $\mathfrak{q}=0.6$ the third level never qualifies, and at $\mathfrak{q}=0.7$ no level does.

For (f), $\Delta_{\mathrm{H}}(b,1/(2b-1),m)$ was computed in exact rational arithmetic for $b=2,\dots,12$ and $m=2,\dots,60$: every value is negative and equals $(2b-1)^{1-m}/(b-1)-S_m/(b^m-1)$, with $-1/((2b-1)(b+1))$ at $m=2$, for instance $-0.1111$ for $b=2$. The strict increase in $m$, with the numerator identity of the proof, held at all 8120 exact triples with $b=2,\dots,8$, $\mathfrak{q}=j/(40(2b-1))$ for $j=1,\dots,40$ and $m=2,\dots,30$. In double precision, 3580209 triples ($b=2,\dots,10$, $\mathfrak{q}$ from 0.0005 to 0.9995 in steps of 0.0005, $m=2,\dots,200$) gave no positive $\Delta_{\mathrm{H}}$ for $\mathfrak{q}\geq1/(2b-1)$, an upward-closed set of $m$ with $\Delta_{\mathrm{H}}>0$ for every smaller $\mathfrak{q}$, and $m^{*}=2$ exactly below $b/(2b^2-1)$; for $b=2$ the zeros of $\Delta_{\mathrm{H}}(2,\cdot,m)$ are 0.2857, 0.3062, 0.3182, 0.3321 and 0.3333 at $m=2$, 3, 4, 8 and 16. In explicit media with $(b,n_{\mathrm{H}},A_{\mathrm{h}})=(2,8,1)$ and $\mathfrak{q}\in\{0.3,1/3,0.36,0.45\}$, and with $(3,5,1)$ and $\mathfrak{q}\in\{0.19,0.2,0.25\}$, the contrast of the next-level blocks and the exchangeable baseline, both computed from the iterated coarse matrices at 121 leaks in $[10^{-4},10^{2}]$, agreed with the criterion of (b) at every leak and level: at $\mathfrak{q}=0.3$ and 0.19, where $m^{*}=3$, the blocks of levels 1 to $n_{\mathrm{H}}-2$ qualify above their thresholds and those of level $n_{\mathrm{H}}-1$ never do, and at the other five values no block of any level qualifies under either judgment.

**Check S47.145** (Memory of partitioned populations)**.** Square blocks on tori of $4\times4$ blocks, internal nearest-neighbor coupling 1, coupling $a_{\mathrm{out}}$ across blocks and uniform leak $\varepsilon^{\Pi}$, were computed for (side, $a_{\mathrm{out}}$, $\varepsilon^{\Pi}$) $=(6,0.01,10^{-3})$, $(6,0.001,10^{-3})$, $(4,0.01,10^{-2})$, $(8,0.002,10^{-3})$ and $(6,0.01,0.1)$. The identity of Theorem 13.8(a) held to $2.2\times10^{-16}$ and every bound of (a) and (c) held. At $(6,0.01,10^{-3})$, contrast 35.1, $\|\mathcal{D}\|=1.95\times10^{-4}$ against the bound $6.6\times10^{-4}$, $\sin\Theta=0.0244$ against 0.0524, the relative fast weight $1.5\times10^{-5}$ against $1.5\times10^{-4}$, and the drift error $2.0\times10^{-4}$ against $7.0\times10^{-4}$; the nearest-neighbor coupling of the exact coarse precision was 0.9892 times the Galerkin coupling, and 0.9989 times at $a_{\mathrm{out}}=0.001$, contrast 161. The single-block closure

bound of Theorem 5.20, on one block with its exterior held fixed, gave $\Xi \le 1.9 \times 10^{-5}$ against the computed $2.4 \times 10^{-7}$. An independent implementation on 200 random partitioned populations with two to four blocks, uniform leak and weak cuts found no violation of any bound of (a), (c) or (d), the largest ratio of $\sin\Theta$ to its bound 0.83, and $\int_0^\infty \Gamma(u)\mathrm{d}u = \mathcal{D}$ to the quadrature error $2.4 \times 10^{-8}$ against $|\mathcal{D}| = 2.8 \times 10^{-4}$. A further run on 4000 random populations of two to five blocks with one to six members each, with a leak drawn separately for every agent over several orders of magnitude, found no violation of the three bounds of Theorem 13.8(d) in their general form, nor of the bounds on $\|\mathcal{D}\|$ and $|\mathcal{D}_{XY}|$ of Theorem 13.10(a), beyond relative rounding errors of $3 \times 10^{-15}$; in the same populations $|\mathcal{D}\mathsf{n}|$, with $\mathsf{n} = (|X|^{1/2})_X$, was as large as $\|\mathcal{D}\||\mathsf{n}|$ to three digits, so Lemma 13.3(c) does not extend to nonuniform leaks. The Frobenius quantity $\sigma^2_{\mathrm{cut}}$ is extensive in the number of blocks while the spectral quantities are not: at $(6, 0.01, 10^{-3})$ on tori of $4 \times 4$ and of $6 \times 6$ blocks, $\sigma^2_{\mathrm{cut}}/(\varkappa^{\min}_{\mathrm{in}}\varkappa_{\mathrm{out}}) = 0.776$ and 1.746, $\|\Pi_{\perp\Phi}\|^2/(\varkappa^{\min}_{\mathrm{in}}\varkappa_{\mathrm{out}}) = 0.0862$ on both, and $\|\mathcal{D}\|/\varkappa_{\mathrm{out}} = 0.0255$ and 0.0262; every block satisfied the bound of Lemma 13.3(b), with $\sigma^2_X = 1.00 \times 10^{-4}$ against $2.67 \times 10^{-4}$. For two cliques of $n$ agents with internal coupling 1, joined by one edge of weight $n/2$, at leak 0.01 and $n = 20$, 50, 100 and 200, the contrast was 39.2, 98.1, 196.1 and 392.2, the exact coarse coupling was 0.513, 0.505, 0.503 and 0.501 times the Galerkin coupling, $\|\mathcal{D}\|/\varkappa_{\mathrm{out}}$ ranged from 0.955 to 0.978 and $\|\Pi_{\perp\Phi}\|^2/(\varkappa_{\mathrm{in}}\varkappa_{\mathrm{out}})$ from 1.86 to 1.95, and $\sigma^2_X = 9.5$, 24.5, 49.5 and 99.5 against the bounds 10, 25, 50 and 100. At $n = 20$ the relative fast weight was $1.7 \times 10^{-4}$ against the bound $2.7 \times 10^{-3}$ of Theorem 13.8(c), and the drift of the slow part differed from $\Pi_\Phi$ by 0.49 of its norm.

**Check S47.146** (Per-particle slow modes)**.** In the block media of Check S47.145, with the frame of per-block Perron vectors, at contrast 35.1: $\|\mathcal{D}^{\mathsf{P}}\| = 1.59 \times 10^{-4}$ against the bound $2.45 \times 10^{-4}$, $\sin\Theta = 0.0237$ against 0.0319, the Galerkin coupling within $2 \times 10^{-4}$ relative of the exact one, and the slowest retained rate $1.0368 \times 10^{-3}$ against the exact $10^{-3}$, with the shift bound $2.5 \times 10^{-4}$. AR0 and AR1, the latter adding white noise of intensity $2\Pi^{\top}_{\perp\Phi}\Pi^{-2}_{\perp\perp}\Pi_{\perp\Phi}$, had stationary relative entropies $1.40 \times 10^{-3}$ and $1.36 \times 10^{-3}$ to the exact law, with largest relative variance errors of 2.3 percent; at contrast 161 they were $1.9 \times 10^{-6}$. The frame of block averages at contrast 35.1 retained the slowest rate $10^{-3}$ exactly, with relative entropy $1.52 \times 10^{-3}$ for AR0. An independent implementation found no violation of the bound of Proposition 13.9(c) in 200 instances, with relative shifts of the slowest retained rate up to 0.52, and the mean coarse rate at most the mean fine rate in 100 of 100 instances.

**Check S47.147** (Without contrast)**.** For the homogeneous nearest-neighbor precision on $\mathbb{Z}^2$ with coupling 1 and square blocks of side $b$, the coarse medium has coupling $b$ between adjacent blocks and leak $b^2\varepsilon^{\Pi}$, hence the squared second-moment length $1/(b\varepsilon^{\Pi})$ per axis in block units. It was compared with the squared length of the block averages, computed from Eq. (119) and, independently, from the real-space second moment of the covariance of the block averages on a torus of $960 \times 960$ sites, about 30 correlation lengths at $\varepsilon^{\Pi} = 10^{-3}$; the two computations gave the same ratios to four decimals. The squared length of the coarse medium was 1.9995, 3.9950 and 5.9826 times that of the block averages for $b = 2$, 4 and 6 at $\varepsilon^{\Pi} = 10^{-3}$, and 3.5726 times for $b = 4$ at $\varepsilon^{\Pi} = 0.1$. On a torus of $240 \times 240$ sites, under four correlation lengths in half-width at $\varepsilon^{\Pi} = 10^{-3}$, the real-space moment is truncated and the ratios read 2.41, 4.82 and 7.21. On a ring with blocks of $b$ sites at $\varepsilon^{\Pi} = 10^{-3}$, the squared lengths of the coarse medium, 500, 250 and 125, were 2.000, 3.995 and 7.958 times the exact values 250.06, 62.58 and 15.71 for $b = 2$, 4 and 8.

**Check S47.148** (Elimination of internal states)**.** On three random partitioned drifts the identities of Proposition S13.10(a) and (b) held to $4.4 \times 10^{-16}$, $|\mathrm{Cov}(x_\perp, x_{\mathrm{ret}})| \le 5.6 \times 10^{-17}$, and $\|M_{\perp\mathrm{r}}\|$ lay between 0.96 and 1.68, so the kernel does not vanish. For Numerical observation S13.11, in the first

instance the noise intensity of the memoryless reduction was positive semidefinite at every speed factor, and the difference between its drift $-M_{\mathrm{ri}}\mathsf{C}_{\mathrm{ir}}$ and the quasi-static drift $M_{\mathrm{ri}}M_{\mathrm{ii}}^{-1}M_{\mathrm{ir}}$ fell from 0.378 to 0.0100, from $\mathtt{s} = 10$ on in the same proportion as the error. An independent implementation found the identity residual at most $9.8 \times 10^{-15}$, $|\mathrm{Cov}(x_{\perp}, x_{\mathrm{ret}})| \le 2.2 \times 10^{-16}$, for a second instance drawn independently the deviations $4.4 \times 10^{-2}$, $2.0 \times 10^{-2}$, $8.3 \times 10^{-3}$, $3.1 \times 10^{-3}$ and $9.5 \times 10^{-4}$ at the same speed factors, and, for (c), a nonzero mean $(-0.0139, 0.0429)$ of $x_{\mathrm{ret}}(0.05)$ from $x_{\mathrm{ret}}(0) = 0$ and $x_{\mathrm{int}}(0) = z$ with $M_{\mathrm{ri}} z \neq 0$.

**Check S47.149** (Aggregated codes)**.** With $K = 3$, a rank-two code reading $(\mathbf{e}_1, \mathbf{e}_2)$, branching four, five levels and Gaussian angles of spread 1.5, so that $\mathbb{E}\cos\vartheta = e^{-1.125} = 0.325$: rotations about $\mathbf{e}_1$ gave the second singular value 0.171, 0.287, 0.575, 0.385 and 0.323 at levels one to five, the first staying 1; rotations about $\mathbf{e}_3$ gave equal singular values 0.603, 0.320, 0.284, 0.321 and 0.326; frames drawn from the Haar law on SO(3) gave $(0.382, 0.106)$, $(0.272, 0.096)$, $(0.138, 0.052)$, $(0.074, 0.024)$ and $(0.046, 0.036)$; and aggregation after true translation kept $(1, 1)$.

**Check S47.150** (Meta-level relays)**.** Twelve rank-two codes in $\mathbb{R}^4$ that share one direction gave the averaged hop $\bar{P}$ with eigenvalues $(1, 0.4263, 0.3206, 0.2531)$, hence rank four. The effective rank of $\bar{P}^n$ at the relative threshold 0.1 was $4, 3, 1, 1, \dots$ for $n = 1, 2, 3, 4, \dots$ up to $n = 30$, nonincreasing with the limit $1 = \dim V_{\cap}$, and a relay through one member at a time in cyclic order gave $2, 2, 2, 2, 1, \dots$.

**Check S47.151** (Literal reading across levels)**.** With literal consensus on cycles at both levels, four groups of five and six groups of three gave hierarchical limit means 0.2993052160 and 0.2535794074, equal to the prediction with the prior counted $N$ times to ten digits, against union posterior means 0.765 and 0.680. With one complete group of five among cycles, its level-two weight was 0.357, 0.607, 0.877, 0.982 and 0.9999 at $n_{\mathrm{c},1} = 1$, 3, 6, 10 and 20 rounds at level one, its weight ratio to another group was $(5/3)^{n_{\mathrm{c},1}}$, and the top-level mean tended to that group's own literal mean $2/3$. An independent implementation with other data gave 0.1905046646 and $-0.0787871578$, equal to the prediction, and the information form gave the union posterior means 0.184390 and $-0.216733$ exactly.

**Check S47.152** (Laplacian renormalization)**.** Two $6 \times 6$ grids with coupling 1 joined by six edges of coupling 0.01 have $\lambda_2(L^{\Pi}) = 0.0032336$ and $\lambda_3(L^{\Pi}) = 0.2679$, and the specific heat has maxima at $\tau_{\mathrm{L}} = 1.24$, 8.32 and 742; here and below, maxima of the specific heat were located on a grid of 400 points per decade over $\tau_{\mathrm{L}} \in [10^{-3}, 10^{6}]$ and refined by bounded scalar maximization. The procedure returned the two grids at $\tau_{\mathrm{L}} = 11.2$ and 103 and one macronode at 928. At $\varepsilon^{\Pi} \in \{10^{-4}, 10^{-2}, 1\}$ the macronodes for $\varepsilon^{\Pi} I + L^{\Pi}$ with the cutoff applied to the shifted spectrum equaled those for $L^{\Pi}$, while the unshifted cutoff kept 2, 0 and 0 modes; the criterion $r \ge 2$ selected the grids and the union, the grids only, and nothing. In Example S13.20, where $\lambda_2(L^{\Pi}) = 2.2205 \times 10^{-4}$ and $\lambda_3(L^{\Pi}) = 0.26795$, the specific heat has maxima at $\tau_{\mathrm{L}} = 0.0382$, 1.21 and $1.08 \times 10^{4}$. A scan of the procedure over $\tau_{\mathrm{L}} \in [10^{-3}, 10^{5}]$ at 400 points per decade, with endpoints refined on linear grids of spacing below $10^{-4}$, returned $A$ and $A'$ as macronodes for $\tau_{\mathrm{L}}$ in $[0.0167, 0.2830)$, $[0.3060, 0.3871)$ and $[0.7887, 0.9998]$ and for no other value, $A \cup A'$ for $1 \le \tau_{\mathrm{L}} < 1/\lambda_2 = 4503.4$, and $B$ for $1/\lambda_3 = 3.732 \le \tau_{\mathrm{L}} < 4503.4$; no scanned value returned $A$, $A'$ and $B$ together. Every computation of this check keeps the modes with eigenvalue strictly below $1/\tau_{\mathrm{L}}$. The two endpoints at which a threshold is attained were computed separately with the number of kept modes set by index, so that no floating-point comparison at equality enters (script `ren_p2_lrg_equality.py`, deterministic): at $\tau_{\mathrm{L}} = 1/\lambda_3$ the strict rule keeps two modes and returns $A \cup A'$ and $B$, while keeping the mode at equality gives three macronodes of sizes 1, 12 and 35 and does not return $B$; at $\tau_{\mathrm{L}} = 1/\lambda_2$ the strict rule keeps one mode and returns the whole population. On a homogeneous $16 \times 16$ torus the specific heat

lay between 0.191 and 0.245 for $\tau_{\mathrm{L}} \in [1, 10]$, without a peak; the macronodes at $\tau_{\mathrm{L}} = 0.5$, 1 and 4 included one cluster of 154, 235 and 252 sites, and the largest macronode contrast at $\varepsilon^{\Pi} = 0.01$ was 2.01, below the contrast 16.2 of the whole torus. In the nested medium with $b = 4$, $n_{\mathrm{H}} = 3$, $\mathfrak{q} = 0.2$ and $A_{\mathrm{h}} = 1$, with distinct Laplacian eigenvalues 0, 0.0533, 0.3067 and 1.5733, the specific heat has maxima at $\tau_{\mathrm{L}} = 2.21$, 11.05 and 53.3, the procedure returned exactly the level-one blocks at $\tau_{\mathrm{L}} = 2$ and the level-two blocks at 10, and the contrasts of these blocks were 5.54 and 6.53 at $\varepsilon^{\Pi} = 10^{-3}$ and 1.88 and 1.22 at $\varepsilon^{\Pi} = 1$. For Open problem S13.21, a clique $C$ of $n$ agents with unit couplings joined by one unit edge to a path of $n_{\mathrm{p}}$ sites has $\lambda_2(L_C) = n$, $\mathrm{cut}(C) = 1$ and $r_\star(C) = n^2$, and the regions obtained from $C$ by removing one member or adding the adjacent site of the path had ratio at most $n + 1$ (a site that is not adjacent gives a disconnected region, of ratio 0), so $C$ is a strict local maximizer of $r_\star$. For $(n, n_{\mathrm{p}}) = (8, 64)$, $(16, 256)$ and $(24, 400)$, with $r_\star(C)/\ln N = 15.0$, 45.7 and 95.2, the procedure did not return $C$ at any of 3058, 439 and 60 values of $\tau_{\mathrm{L}}$ in the interval $(1/n, n)$ of Open problem S13.21 (logarithmic grids of 3000, 200 and 60 points, and in the first two cases the geometric midpoints of consecutive thresholds $1/\lambda_k$ inside the interval); at least 8, 21 and 27 modes were kept there, and for $n = 16$ and 24 no two members of $C$ were joined at any scanned value. On logarithmic grids over $[10^{-3}, 10^{6}]$ the procedure returned $C$ at 31 of 2000 values for $n = 8$, all in $[9.75, 13.31]$, at one of 200 values for $n = 16$, $\tau_{\mathrm{L}} = 45.5$, and at none of 120 for $n = 24$, so that every returned value lies above the interval (script `ren_p2_lrg_restricted.py`, ties broken at random with seed 7). A scan on a grid does not exclude windows narrower than its spacing.

**Check S47.153** (Coarse contrast)**.** In 200 random populations with four to seven equal blocks of two to five agents, couplings drawn within blocks with probability 0.9 and weights in $[0.5, 2]$ and across blocks with probability 0.25 and weights in $[0, 0.2]$, so that the partitions are not equitable, leaks in $[10^{-3}, 1]$ and a random set $S$ of at least two blocks, $r^{(1)}(S) \geq r(\bigcup S)$ held in every instance, with equality in 13, and the contrasts computed from $\Pi_\Phi$ and from the coarse medium $\mathsf{P}\Pi\mathsf{P}^{\top}$ agreed to $10^{-9}$ relative.

## S47.13 Checks for Section 14

The checks below confirm the closed forms of Section 14 on small instances and supply the numbers of its examples. They are exact linear algebra (Gaussian conditioning, Lyapunov solves, eigen- and singular value decompositions), root finding by bisection, Nelder–Mead searches over raw sensor matrices with the budget imposed, and one-dimensional quadrature. Where an independent recomputation was made, its instance and outcome are reported as well.

**Check S47.154** (Information and the optimal sensors)**.** A random pair $(\mathsf{X}, \mathsf{Y})$ of dimensions five and four with canonical correlations $(0.928, 0.725, 0.573, 0.169)$ and $\kappa = 2$ (Lemma 14.2 and Theorem 14.3). Over 4000 random two-channel sensors the formula of Lemma 14.2(ii) and the direct formula from the joint covariance of $(s, \mathsf{Y})$ agree to $6.1 \times 10^{-15}$, the ratio of $I(s; \mathsf{Y})$ to the bound of Theorem 14.3(a) is at most 0.918, and the canonical sensor attains the bound to twelve digits (0.584985927655). The water-filling values 0.16689, 0.58077 and 1.05531 nats at $I_{\max} = 0.2$, 0.8 and 2.0 (water levels 0.806, 0.555, 0.261; one, one and two variates read) agree with the best of twelve Nelder–Mead searches over all sensors to $10^{-10}$. With a random weight, the error bound of (c) holds on 4000 sensors, and the reverse water-filling errors 142.30322369 and 99.20984032 at $I_{\max} = 0.3$ and 1.5 agree with numerical optimization to $4 \times 10^{-10}$; for $\Upsilon = \Sigma_{\mathsf{Y}}^{-1}$, $\|\mathsf{J}_\Upsilon - \mathsf{C}\mathsf{C}^{\top}\|_{\max} = 4.4 \times 10^{-16}$. The noiseless rank-two optimum 1.35835 nats of (e) is attained by the canonical subspace, and the best of 4000 random subspaces reaches 1.25814. An independent recomputation on another random pair, with canonical correlations $(0.964, 0.872, 0.498, 0.304)$, found the ratio to the bound of (a) at most 0.890 over 3000 sensors, water-filling values 0.18303, 0.67827 and 1.41330 nats equal to

a numerical maximum over raw sensor matrices at the same three budgets, and the active set of the information criterion contained in that of the error criterion in 500 of 500 random cases, as (d) states.

**Check S47.155** (Summaries of the past)**.** A random non-normal three-dimensional exterior and $h = 2$ (Lemma S14.1). The canonical correlations of the pair formed by a two-step window of the past and $\eta_{t+2}$ and those of $(\eta_t, \eta_{t+2})$ agree to ten digits, $(0.72923, 0.33603, 0.06270)$, and so do the budget-limited optima for $I_{\max} \in \{0.3, 1.0\}$ and $\kappa \in \{1, 2\}$ (0.137174 and 0.307917 nats).

**Check S47.156** (Horizon dependence)**.** (a) Four modes with $\lambda(F) = (0.98, 0.9, 0.7, 0.4)$ in a random orthonormal basis and $Q = 0.5I$, so that detailed balance holds to $4.4 \times 10^{-16}$, with $I_{\max} = 1.5$ (Proposition 14.4(a)). The computed canonical correlations equal $|\lambda(F)|^h$ at $h = 1, 2, 5, 10, 20, 40, 80$; $\mathcal{I}_2(h) = 0.869$, 0.915, 1.061, 1.338, 1.991, 3.517 and 6.833 there, so two modes are read for $h \le 10$ and one for $h \ge 20$, and the sufficient horizon of (a) is 17.61. (b) Four modes with $\lambda(F) = (0.95, 0.8, 0.5, 0.3)$ in a random non-orthogonal basis, $\|FP_0 - P_0F^\top\|_{\max} = 1.59$. For $\kappa = 1$, $\sin\vartheta_h$ falls from 0.674 at $h = 1$ through 0.274 at $h = 6$ to 0.0213 at $h = 20$, against $(0.8/0.95)^h = 0.842$, 0.357 and 0.032; for $\kappa = 2$ it falls from 0.288 to $1.39 \times 10^{-4}$, against $(0.5/0.8)^{20} = 8.3 \times 10^{-5}$ at $h = 20$. In an independent non-normal instance the explicit bound of (b) was never violated, but its condition $\mathfrak{a}_h < \mathfrak{b}_h$ failed, so that it bounded nothing, at every computed $h \le 20$ for $\kappa = 1$ and $h \le 10$ for $\kappa = 2$ (it is 0.19 at $h = 20$). (c) Two modes, $\lambda_{\mathrm{s}}(F) = 0.97$ and $\lambda_{\mathrm{f}}(F) = 0.6$ with innovation variances 0.02 and 1, so $\mathsf{p}_{\mathrm{s}} = 0.3384$ and $\mathsf{p}_{\mathrm{f}} = 1.5625$: $h^* = 1.592$, the rank-one error optimum reads the fast mode at $h = 1$ and the slow mode from $h = 2$ on, and the information criterion reads the slow mode at every $h \ge 1$. With a third mode of eigenvalue 0.9 and stationary variance 5 added, the entries $\lambda_{\mathsf{j}}(F)^{2h}\mathsf{p}_{\mathsf{j}}$ at $h = 1$ are $(0.318, 4.05, 0.5625)$ and the rank-one error optimum reads the added mode at every $h$ from 1 to 5, while the fast mode still ranks above the slow mode at $h = 1$ and below it from $h = 2$ on.

**Check S47.157** (Control and expected free energy)**.** (a) A random four-dimensional exterior, $h = 3$, two controls and three outcomes (Theorem 14.6(a)). Over six random sensors, the optimal expected cost minus $\frac{1}{2}\operatorname{tr}(\Upsilon_{\mathrm{ctl}}\operatorname{Cov}(\eta_{t+h} \mid s))$ is the same constant 5.5301278619, with spread $4.4 \times 10^{-15}$; perturbing the optimal act rule in 200 random directions raises the cost every time (least increase $2.8 \times 10^{-5}$), and $\operatorname{rank}\Upsilon_{\mathrm{ctl}} = 2$. (b) In a separate random five-dimensional exterior with $h = 2$, three outcomes and one control, the eigenvalues of $\mathsf{J}_{\mathrm{ctl}}$ are $(1.587, 0, 0, 0, 0)$. (c) Four reversible modes, $h = 2$, with $\sigma^2_{\mathsf{j}} = (0.885, 0.522, 0.130, 0.008)$ and $\lambda_{\mathsf{j}}(\mathsf{J}_{\mathrm{ctl}}) = (8.99, 1.88, 0.101, 8.9 \times 10^{-4})$, both non-increasing along the common basis: the minimum of $G^{\mathrm{att}}_h$ over two-channel sensors with $I_{\max} = 1.2$, found by sixteen Nelder–Mead searches, equals the separable closed form up to the sensor-independent constant $\mathfrak{c}_0 + \frac{1}{2}\operatorname{tr}(\Upsilon_{\mathrm{ctl}}P_0)$, the sum over the two channels being $-5.0154728020$, with capacities 1.007 and 0.193 nats on the two slowest modes. For Remark S14.3, two independent modes with $\lambda(F) = (0.95, 0.5)$ and innovation variances $(0.01, 1)$ have stationary variances $(0.10256, 1.33333)$; the gain-budget sensor of the present state reads the fast mode, and the canonical sensor reads the slow mode at every horizon. For Example S14.5, a search over the sensor's angle at $h = 1, 2, 3, 4, 6, 8$ put the optimum at a pure mode in every case, and a comparison of the two pure modes at every $h$ from 1 to 8 gives the slow mode throughout with $w_{\mathrm{p}} = 4$, and the fast mode at $h = 1, 2$ and the slow mode from $h = 3$ on with $w_{\mathrm{p}} = 40$.

**Check S47.158** (Variables constant at the agent's scale)**.** Scalar exteriors with $F_{\mathrm{g}} \in \{0, 0.5, 0.9, 0.99, 1\}$ and $Q_{\mathrm{g}}/R^s \in \{1, 10^{-2}, 10^{-4}, 10^{-6}\}$ (Lemma S14.7). In all twenty cases the fixed point solves the quadratic to $3 \times 10^{-17}$ and the steady-state information per reading obeys the bound; at $F_{\mathrm{g}} = 1$ and $Q_{\mathrm{g}}/R^s = 10^{-6}$ it is $5.0 \times 10^{-4}$ nats per reading against the bound $5.002 \times 10^{-4}$, and at $F_{\mathrm{g}} = 0$ and $Q_{\mathrm{g}}/R^s = 10^{-6}$ it is $5.0 \times 10^{-7}$ nats.

**Check S47.159** (Volume-time average)**.** Units $r_{\mathrm{cell}} = D_{\mathrm{m}} = \bar{c}_{\mathrm{L}} = 1$ (Proposition S14.8). The lens-volume integral $\int_0^2 4\pi z^2 V_{\cap}(z)\mathrm{d}z$ equals $|\mathbb{B}|^2$ and $\int_0^{\infty} \mathfrak{K}$ equals $8\pi/15$, both to twelve digits. The variance computed by quadrature from the exact covariance, divided by the leading term, is 0.7425, 0.9102, 0.9707 and 0.9906 at $T_{\mathrm{int}} = 10$, $10^2$, $10^3$ and $10^4$; the remainder $\mathfrak{q}_T$ is $4.0 \times 10^{-2}$, $4.2 \times 10^{-3}$, $4.3 \times 10^{-4}$ and $4.3 \times 10^{-5}$, between 0.40 and 0.46 of its bound; and $(1 - \mathrm{ratio})/(\pi T_{\mathrm{int}})^{-1/2}$ is 1.443, 1.592, 1.643 and 1.659, tending to 5/3.

**Check S47.160** (The cell model)**.** The parameters of Example S14.9, with $\bar{c}_{\mathrm{L}} = 1$ nM $= 0.6022\ \mu\mathrm{m}^{-3}$. The relative error of the chemical channel is $1.1263 \times 10^{-2}$ in the leading term, the two-term value of its finite-time factor is

$$1 - \frac{5}{3}\Big(\frac{r_{\mathrm{cell}}^2}{\pi D_{\mathrm{m}} T_{\mathrm{int}}}\Big)^{1/2} = 0.7897,$$

and $r_{\mathrm{cell}}^2/D_{\mathrm{m}} = 0.050$ s. The variance computed by quadrature from the exact covariance at $T_{\mathrm{int}} = 20 r_{\mathrm{cell}}^2/D_{\mathrm{m}}$ is 0.8101 times the leading term, so that $\mathfrak{q}_T = 0.0204$, against the bound 0.0488, and the exact relative error is $1.0138 \times 10^{-2}$. For signals of 5 and 1 percent, with the leading-term variance: signal-to-noise ratios 19.707 and 0.788, captured fractions 0.9517 and 0.4408, predictive information 0.7552 and 0.2238 nats at 1 s and 0.0689 and 0.0308 nats at 10 s. With the exact variance the ratios are 24.326 and 0.973, the captured fractions 0.9605 and 0.4932, and the predictive information 0.7718 and 0.2586 nats at 1 s and 0.0696 and 0.0345 nats at 10 s. These values take the reading to be $\eta^{\mathrm{L}}$ at the end of the window plus the thermal noise. For the reading that averages the Ornstein–Uhlenbeck signal over the window, with $a = T_{\mathrm{int}}/\tau_{\mathrm{L}} = 0.1$ and horizon $h$ from the end of the window, the squared correlation with $\eta^{\mathrm{L}}_{t+h}$ is $e^{-2h/\tau_{\mathrm{L}}}$ times $(a^{-1}(1 - e^{-a}))^2 = 0.9056$, divided by the sum of $2a^{-2}(a - 1 + e^{-a}) = 0.9675$ and the reciprocal of the signal-to-noise ratio of the point-sample model (the value 0.9675 agrees with a quadrature of the double integral of the autocorrelation to seven digits), and the predictive information is 0.6513 and 0.2014 nats at 1 s and 0.0641 and 0.0282 nats at 10 s with the leading-term variance, and 0.6642 and 0.2323 nats at 1 s and 0.0648 and 0.0317 nats at 10 s with the exact variance. The autocorrelation of the volume average of the thermal fluctuation is $7.10 \times 10^{-2}$, $2.89 \times 10^{-3}$ and $9.38 \times 10^{-5}$ at lags 1, 10 and 100 in units of $r_{\mathrm{cell}}^2/D_{\mathrm{m}}$, against the large-lag form $9.40 \times 10^{-2}$, $2.97 \times 10^{-3}$ and $9.40 \times 10^{-5}$. For windows of length $T_{\mathrm{int}} = 20 r_{\mathrm{cell}}^2/D_{\mathrm{m}}$ (one second) starting $T_{\mathrm{int}}$, $5T_{\mathrm{int}}$ and $10T_{\mathrm{int}}$ apart the correlations, normalized by the exact variance, are $6.36 \times 10^{-2}$, $2.93 \times 10^{-3}$ and $1.03 \times 10^{-3}$ (leading-order approximations $6.16 \times 10^{-2}$, $2.35 \times 10^{-3}$ and $8.3 \times 10^{-4}$), carrying, in the Gaussian approximation, $2.0 \times 10^{-3}$, $4.3 \times 10^{-6}$ and $5.3 \times 10^{-7}$ nats. A protein of 50 kDa with buoyancy factor 0.27 has buoyant mass $2.242 \times 10^{-23}$ kg, $x_{\mathrm{g}} = 5.14 \times 10^{-7}$ at 310 K over 10 $\mu$m, signal-to-noise ratio $2.20 \times 10^{-14}$ per reading and $4.5 \times 10^{13}$ readings for a ratio of one, 13.66 orders of magnitude below a ratio of one and 13.55 below the leading-term ratio of the 1-percent ligand signal. The chemical reading pools $0.6022 \times 523.6 = 315.3$ molecules over 30.9 effective snapshots (snapshot variance $1/315.3$ against the exact relative variance $1.028 \times 10^{-4}$ of the reading), $9.73 \times 10^3$ molecule-samples; a gravity reading that pools as many independent protein heights has ratio $2.14 \times 10^{-10}$, 9.67 orders of magnitude below one. In the leading term the inverse relative variance is $5\pi D_{\mathrm{m}} r_{\mathrm{cell}} \bar{c}_{\mathrm{L}} T_{\mathrm{int}}/3 = 7.88 \times 10^3$ molecule counts, 25.0 snapshots, at which the pooled gravity ratio is $1.73 \times 10^{-10}$, 9.76 orders of magnitude below one. A particle of radius 1 $\mu$m and density excess 500 kg/m$^3$ has $x_{\mathrm{g}} = 48.0$. The exact ratio $1 - (x_{\mathrm{g}}/2)^2/\sinh^2(x_{\mathrm{g}}/2)$ of item (4), which agrees with $x_{\mathrm{g}}^2 \operatorname{Var}(\mathsf{z})/Z_{\mathrm{box}}^2$ computed by quadrature of the truncated Boltzmann law to eight digits, is 0.0793 at $x_{\mathrm{g}} = 1$, 0.600 at $x_{\mathrm{g}} = \sqrt{12}$, where the small-$x_{\mathrm{g}}$ form gives one, 0.79 at $x_{\mathrm{g}} = 4.656$ (radius 508 nm at the density of protein), 0.9955 at $x_{\mathrm{g}} = 10$ and $1 - 3.3 \times 10^{-18}$ at $x_{\mathrm{g}} = 48$; it increases on a grid of step 0.1 up to $x_{\mathrm{g}} = 60$ and coincides with $x_{\mathrm{g}}^2/12$ to five digits for the protein. The probability that the particle lies in the upper half of the compartment is $1/(1 + e^{x_{\mathrm{g}}/2})$, confirmed by quadrature, and equals 0.3775 at

$x_{\mathrm{g}} = 1$ and $3.78 \times 10^{-11}$ at $x_{\mathrm{g}} = 48$.

**Check S47.161** (Aggregates)**.** Types (Proposition S14.12): nineteen scalar agents in two types of sizes 7 and 12 with a scalar world input of persistence 0.9 and unit innovation variance; the own persistences are 0.6 and 0.3, the couplings to the two type averages are $(0.1, 0.15)$ for the first type and $(0.2, 0.25)$ for the second, the world loadings are 0.5 and $-0.4$, and the idiosyncratic innovation variance is 0.7. The covariance of the type averages from the full twenty-dimensional Lyapunov equation equals that of the three-dimensional closed model to $2.7 \times 10^{-14}$, the invariance residual is $1.1 \times 10^{-16}$, and the finite-minus-infinite covariance equals $\Sigma^{\mathrm{fl}}$ to $1.0 \times 10^{-15}$ (diagonal 0.2311 and 0.1196); changing one agent's persistence by 0.2 makes the invariance defect $2.9 \times 10^{-2}$. Memory (Proposition S14.13): twenty agents with persistences 0.9 and 0.5 in equal numbers and $q_{\mathrm{com}} = q_{\mathrm{id}} = 1$ have $\mathrm{Corr}(\bar{x}_t, \bar{x}_{t+u}) = (1, 0.779, 0.641, 0.547, 0.477, \dots)$ against $0.779^u = (1, 0.779, 0.607, 0.473, 0.369, \dots)$, a largest $\Delta_u^{\mathrm{agg}}$ of 0.162 over $u \le 7$, and the moment formula of (b) agrees with the Lyapunov solution at every lag; equal persistences 0.8 give a geometric autocorrelation to $2.2 \times 10^{-16}$. Over 200 random narrow persistence laws the largest $\Delta_u^{\mathrm{agg}}$ is at most 0.49 of the bound of (d) (an independent recomputation on 200 other aggregates gave at most 0.785). Long memory, for $\mathbb{Q}(\mathrm{d}\alpha) = \mathfrak{p}(1-\alpha)^{\mathfrak{p}-1}\mathrm{d}\alpha$, so $\mathsf{f}_0(1)\Gamma(\mathfrak{p}) = \Gamma(\mathfrak{p}+1)$: $l^{\mathfrak{p}}\mathfrak{m}_l$ is 0.91900 at $l = 10^4$ against $\Gamma(1.75) = 0.91906$ for $\mathfrak{p} = 0.75$, and 0.89347 against 0.89352 for $\mathfrak{p} = 0.6$; $\mathsf{R}(u)u^{2\mathfrak{p}-1}$, computed in closed form, approaches the predicted constant with relative errors 0.35, 0.20, 0.11 and 0.062 at $u = 10$, $10^2$, $10^3$ and $10^4$ for $\mathfrak{p} = 0.75$, and 0.097, 0.039, 0.015 and 0.0062 for $\mathfrak{p} = 0.6$, ratios per decade close to $10^{-(1-\mathfrak{p})}$. For (e), in exact rational arithmetic the observability matrix of $(N^{-1}\mathbf{1}^\top, \mathrm{diag}(\alpha_i))$ has rank 3 for six agents with three distinct persistences, rank 4 for five agents with the four distinct values 0, $\pm\frac{1}{3}$ and $\frac{9}{10}$, rank 5 for five distinct values and rank 1 for equal persistences, so that $d^* = \mathfrak{T} - 1$ in each case, and the Hankel matrix of the autocovariances $\mathsf{R}(u)$ with $q_{\mathrm{com}} = 1$ and $q_{\mathrm{id}} = \frac{1}{2}$ has rank $\mathfrak{T}$.

**Check S47.162** (The Hayek dimension at a horizon)**.** Six reversible modes with eigenvalues 0.99, 0.95, 0.9, 0.75, 0.5 and 0.3 and ten random decision functionals (Theorem 14.7(b)). The instance is drawn with the NumPy generator `default_rng(311109)` in this order: a $6 \times 6$ standard normal matrix whose orthogonal QR factor $\Omega$ gives $F = \Omega\,\mathrm{diag}(\lambda_1, \dots, \lambda_6)\Omega^\top$ with the eigenvalues in the order listed; six uniform draws $q_j$ on $[0.5, 1.5]$, here $(0.6669, 1.0690, 0.5760, 1.4686, 1.3877, 0.6739)$, which are the eigenvalues of the process-noise covariance in the same basis, so that $P_0 = \Omega\,\mathrm{diag}(q_j/(1-\lambda_j^2))\Omega^\top$; and a $6 \times 10$ standard normal matrix whose columns are the $\psi_i$. The effective Hayek dimension at $\mathfrak{e} = 0.05$ is $4, 3, 2, 2, 2, 2, 1, 1, 1$ at $h = 0, 1, 2, 5, 10, 20, 50, 100, 200$, never above the count bound $6, 5, 4, 3, 2, 1, 1, 1$ at the horizons from $h = 1$ on (the count is the trivial 6 at $h = 0$); the relative excess of the best rank-one message falls from 0.342 at $h = 0$ to $2.3 \times 10^{-8}$ at $h = 200$, and that of the message reading the slowest mode from 0.392 to $2.4 \times 10^{-8}$. The exact rank of $\mathsf{D}_h$ is six at every $h$ because $F$ is invertible; a numerical rank with a relative tolerance falls at large $h$ by underflow.

**Check S47.163** (A firm with a production lag)**.** $F = \mathrm{diag}(0.95, 0.6)$ and $P_0 = \mathrm{diag}(1, 2)$ (Example S14.14). The excess losses computed by Gaussian conditioning equal the formulas at $h = 0, \dots, 8$ to all printed digits, for example 0.08167 and 0.72000 for the spot price and the slow index at $h = 1$ and 0.31273 and 0.03359 at $h = 4$; the futures loss is zero to $10^{-16}$, as it is by construction, the futures functional being the decision functional. The spot price wins for $h \le 2$ and the index for $h \ge 3$, as $h^\times = 2.187$ requires. With price noise of variance 0.1 the spot loss at $h = 0$ is 0.097. At $h = 4$ the information-optimal rank-one functional is $(1, 0)$ and the decision-optimal one $(1, 0.159)$. An independent recomputation reproduced the losses at $h = 0, \dots, 6$, $h^\times$ and both functionals.

**Check S47.164** (A person sensing the economy)**.** The model of Example S14.16, with stationary variances $(1.0252, 0.9042, 0.7910, 1.0187)$. The eigenvalues of $\mathsf{J}$ are $(1.11774, 0.26304, 0.01642, 0)$, and

the three channel functionals on (trend, cycle, local prices, noise) are $(1, 0.4614, 0.3102, 0.0409)$, $(0.3074, -0.154, -1, -0.1344)$ and $(0.3331, -1, 0.3013, 0.0427)$. Reverse water-filling at budgets 0.05, 0.3, 1 and 3 nats gives water levels 1.0114, 0.6134, 0.1995 and 0.0270 and capacities (0.05), (0.3), $(0.8617, 0.1383)$ and $(1.8617, 1.1383)$; the second channel enters at 0.7234 nats and the third at 3.4971 nats.

**Check S47.165** (Composition across levels)**.** A random Gaussian Markov chain of dimensions $(4, 3, 2)$ (Lemma S14.17): $\|\mathsf{C}_{13} - \mathsf{C}_{12}\mathsf{C}_{23}\|_{\max} = 4.4 \times 10^{-16}$, and the canonical correlations $(0.936, 0.593)$ of the chain's ends lie below the product bounds $(0.958, 0.912)$.

**Check S47.166** (Physical and abstract channels)**.** (c) Five reversible modes with eigenvalues 0.98, 0.9, 0.7, 0.5 and 0.2, noiseless aggregates that read the two slowest, a three-channel physical sensor of a random mixture with unit noise, and a random scalar decision (Proposition S14.18). The process-noise variances of the five modes are 0.1, 0.3, 0.6, 1 and 1, so that $P_0 = \mathrm{diag}(q_j/(1-\lambda_j^2))$; the rows of the physical sensing matrix are $(0.1608, 1.0953, 1.2422, 1.5681, 0.7805)$, $(0.2032, -0.1399, 0.1708, -1.9159, 0.8412)$ and $(-0.0265, -2.7307, 0.1142, -0.5602, 0.2954)$, and the decision functional is $\psi = (0.8242, -0.5926, 0.1046, -0.1101, -1.2728)$, values that reproduce the figures below to the digits shown. The information that the physical channel adds is 0.181, $3.2\times10^{-3}$, $5.2 \times 10^{-5}$, $2.7 \times 10^{-5}$, $7.1 \times 10^{-7}$ and $5.0 \times 10^{-10}$ nats at $h = 0, 1, 2, 5, 10, 20$, with the squared partial correlation below $\mathfrak{r}_h^2$ at every $h$, where $\mathfrak{r}_0^2 = 1$ and the displayed bound on $\mathfrak{r}_h^2$ applies from $h = 1$ on (for example $6.4 \times 10^{-3}$ against $4.1 \times 10^{-2}$ at $h = 1$ and $1.4 \times 10^{-6}$ against $3.7 \times 10^{-6}$ at $h = 10$); for a decision in the row space of the aggregates the conditional covariance of (b) is zero. (d) Decision eigenvalues $(2.0, 0.4)$ for a firm and $(1.2, 0.9)$ for a polity: with 0.1 nats only the firm's leading channel is read, with 0.3 nats also the polity's leading channel, and with 0.6 and 1.5 nats also the polity's second channel; the polity's channels enter at 0.2554 and 0.5431 nats and the firm's second channel at 1.7595 nats.

## S47.14 Checks for Section S18

**Check S47.167** (Path identity, uniform bound and time step)**.** The identity Eq. (S58) was compared with the relative entropy of the stacked Gaussian path vectors under random affine feedback with randomization. For $n = 5$, $k = 2$, $p = 2$, $T = 6$ (`verify_apx_a.py`) the direct value, the identity and its second form agree to $9.6 \times 10^{-12}$, and the conditional covariance of $x_t$ given $(y_{0:t}, \omega)$ agrees with the policy-free $P_t$ of Lemma S39.1 to $4.0 \times 10^{-12}$. For $n = 6$, $k = 2$, $p = 2$, $T = 12$ (`check_path.py`) five trials agree to $1.8 \times 10^{-14}$.

For Corollary S18.3, with $n = 5$, $k = 2$, $p = 1$, $T = 25$ and 300 random policies in $\mathcal{P}(\kappa, \bar{c})$, the largest observed $(\mathbb{E}|x_t|^2)^{1/2}$ is 3.01 against $\bar{X} = 34.8$, the largest $(\mathbb{E}|x_{\perp,t}|^2)^{1/2}$ is 6.13 against $\bar{X}_\perp = 103.6$, the largest $(\mathbb{E}|\delta_t|^2)^{1/2}$ is 7.11 against $\bar{\mathfrak{d}} = 189.4$, and the largest relative entropy beyond the initial term is 332.0 against a bound of $4.8 \times 10^5$. The bound holds and is loose by three orders of magnitude on this instance. In the scalar family of Remark S39.2 with $\mathsf{a} = 0.6$, the gain $\kappa = 0.3$ ($g_\kappa = 0.75$) gives $\mathbb{E}\delta_t^2 = 0.107, 0.117, 0.118$ at $t = 10, 20, 39$, and $\kappa = 0.5$ ($g_\kappa = 1.25$) gives $2.13, 16.0, 609$, with ratio $1.21 = (\mathsf{a} + \kappa)^2$ per step.

For Proposition S18.4, with $n = 4$, $k = 2$, $p = 1$, $\tau = 1$ and $\mathsf{h} = 0.1, 0.05, 0.025, 0.0125$: with $\bar{\mathsf{q}} = \mathsf{q}^y$ the relative entropy beyond the initial term is $0.2601, 0.2574, 0.2561, 0.2555$, below the drift bounds $0.3025, 0.3003, 0.2993, 0.2987$; with $\bar{\mathsf{q}} = 1.3\mathsf{q}^y$ it is $0.506, 0.820, 1.451, 2.714$, and $\mathsf{h}$ times it is $0.0506, 0.0410, 0.0363, 0.0339$, decreasing toward $\tau h(\bar{\mathsf{q}}^{-1}\mathsf{q}^y) = 0.0316$.

**Check S47.168** (Kernel floor and decay in the order)**.** For the two-variable system of Proposition S18.5 with $\mathsf{b} = 0.5$, $\mathsf{g} = 1$, the rate Eq. (S60) was computed in two ways: as the increment of the

relative entropy of the stacked path vectors between horizons 39 and 40 (`verify_apx_a.py`), and from the stationary innovation variance (`check_memory.py`). The two agree to six digits at $\mathtt{a} = 0.9$ and $\sigma^2 = 1, 2.4839, 6.2632$, the values that both scripts evaluate; the other values come from the first script. With $\mathtt{a} = 0.9$: $V = 5.2632$, $P_\infty = 1.4839$, and the rate is $2.1767, 0.7608, 0.4624, 0.5590$ at $\sigma^2 = 1, 2.4839, 6.2632, 12.5263$, with minimum $0.4624 = \frac{1}{2}\ln((1+V)/(1+P_\infty))$ at $\sigma^2 = 1 + V$. With $\mathtt{a} = 0.5$ the minimum is 0.0449. With $\mathtt{a} = 0$: $V = P_\infty = 1$ and the rate at $\sigma^2 = 2$ is 0 to machine precision, which is the exception recorded in the proposition. The memory kernels $\Gamma_l$ for the complement $x_2$ are zero for $l = 1, \dots, 5$. The windowed filter of order $d = 0, 1, 2, 4, 8$ has per-step relative entropy $0.761, 0.0999, 0.0131, 2.3 \times 10^{-4}, 6.7 \times 10^{-8}$ on the same system at $\mathtt{a} = 0.9$ (`check_memory.py`).

For Theorem S18.6, with $n = 5$, $k = 2$, $p = 1$, $T = 14$ (`verify_apx_a.py`): the recursion Eq. (S125) agrees with Gaussian conditioning to $1.6 \times 10^{-13}$ and the Joseph form Eq. (S126) to $4.4 \times 10^{-15}$; over $d \le 8$ and $d + 1 \le t \le 12$ the largest ratio of the left side of Eq. (S61) to its right side is 0.040 for a generic initial law and 0.034 for a sharp one. At $t = d$ with the sharp initial law the ratio is 726, so the restriction $t \ge d+1$ is needed. With $n = 6$, $k = 2$, $p = 1$ (`check_memory.py`): $\rho(A_{\perp\perp}) = 0.856$, $\rho(\Phi) = 0.855$, $\theta = 0.867$, and the per-step relative entropy against the bound is $4.5 \times 10^{-2}/4.34$ at $d = 0$, $2.1 \times 10^{-3}/2.12$ at $d = 5$ and $2.2 \times 10^{-5}/0.44$ at $d = 16$. The bound holds with a constant that is loose by two to four orders of magnitude here.

With $A_{\perp\perp} = 0$, $k = 1$, one unresolved coordinate and unit noise variances, the filter transition is $\Phi = 0$ with $P^\perp = 1$ when $Q^{\perp y} = 0$, and $\rho(\Phi) = 0.5$ with $P^\perp = 0.6$ when $Q^{\perp y} = 0.8$: correlated noise gives the predictor memory although the complement has none (`check_memory.py`).

**Check S47.169** (Excess relative entropy of a Markov filter)**.** The exact excess $\Delta\mathcal{F}_t$ was computed at $t = 28$ from the joint Gaussian law of the fine state, the record and the declared filter, for 30 random instances with $n = 5$, $k = 2$, $p = 1$, a two-dimensional record and affine feedback on the record with independent randomization (`verify_apx_b.py`). The fixed-point identity Eq. (S127) holds to $5 \times 10^{-14}$, the sandwich $c_- P_{\mathrm{b}}^- \preceq P_{\mathrm{e}} \preceq c_+ P_{\mathrm{b}}^-$ holds with margin $2.3 \times 10^{-4}$, and with the exact $\Delta$ the ratio of the exact excess to the right side of Eq. (S62) is at most 0.37; for the filtered law and Eq. (S130) it is at most 0.053. The a priori constants built from $\vartheta$ gave $c_- \le 0$, and were therefore inadmissible, on all 30 instances.

On a scaling family without input in which $A_{y\perp}$ is multiplied by $\varepsilon = 0.4, 0.2, 0.1, 0.05, 0.025, 0.0125$, with $\bar{A} = A_{yy}$ and $\bar{Q} = Q^y$ (same script), the exact predictive excess is $4.65 \times 10^{-2}$, $1.09 \times 10^{-2}$, $2.58 \times 10^{-3}$, $6.24 \times 10^{-4}$, $1.53 \times 10^{-4}$, $3.80 \times 10^{-5}$, a factor between 4.0 and 4.3 per halving, so it is of second order in the defect on this family. The bound with the exact $\Delta$ exceeds it by a factor of 2.3 to 2.4. With $\vartheta$ the constant $c_-$ is $-0.21, 0.44, 0.73, 0.87, 0.93, 0.97$, so the bound is void at $\varepsilon = 0.4$ and exceeds the exact value by a factor of 80 down to 57 afterwards. For the filtered law the factors are between 2.8 and 7.1 with the exact $\Delta$ and between 1400 and 2000 with $\vartheta$. On a second family of the same kind with $n = 6$, $k = 2$ and constants built from $\|\Delta\|$ (`check_posterior.py`) the predictive excess is 0.572, 0.106, $2.19 \times 10^{-2}$, $5.06 \times 10^{-3}$, $1.23 \times 10^{-3}$, $3.05 \times 10^{-4}$, the bound with the exact $\Delta$ exceeds it by a factor of 3.5 to 3.7, and with $\vartheta$ it is void at $\varepsilon = 0.4, 0.2$ and exceeds the exact value by a factor of 15 down to 12 afterwards.

For Corollary S18.8, on 40 random instances without input, with weak unresolved coupling and the moment-matched model: $|D\Sigma_x C^\top| \le 6.1 \times 10^{-15}$, the inequality $\sigma_D^2 \le \|Q^y\|(e^{2\mathfrak{d}_{\mathrm{cl}}} - 1)$ holds with margin at least $1.4 \times 10^{-4}$, and on the 23 instances with $\mathsf{x} < 1$ the chain from the exact excess through the bound of the theorem with $\vartheta$ to the bound of the corollary holds, the last exceeding the exact excess by a factor between 64 and $9 \times 10^7$. For Proposition S39.6, at horizon 9 on 20 instances with a generic initial law: equality at $t = 0$ to $9 \times 10^{-16}$ and a largest ratio of the left side of Eq. (S131) to its right side of 0.69 for $t \ge 1$.

**Check S47.170** (Two firms and a price, approximately)**.** The two-firm model of Eq. (S46) and Check S47.127 was rebuilt twice: at the coarse level with eight variables, the five exterior variables of firm $C_1$, its act, the error of its filter and its estimate one round earlier (`check_firms.py`); and in the protocol clock with nine variables, with the exact predictive law obtained by conditioning the stationary path law on a window of 40 rounds of the record of firm $C_1$ (`verify_apx_c.py`). The noise variances of the coarse variables are not printed in Example S11.11; they were reconstructed from its member-level description (act noise $4 \times 0.01$ for the sum of four acts of variance 0.01, averaged computation noise $10^{-3}/4$), so the agreement below is agreement with the example and not an independent derivation of its inputs. Both rebuilds reproduce the values of Table S12 and Check S47.127: claimed filtered and predictive variances 0.11089 and 0.18982, exact predictive and filtered variances of the price 0.11772 and 0.08167, error variance of the bounded predictor 0.12734, and excess relative entropy 0.07432 nats, which splits as 0.04897 from the covariance and 0.02535 from the mean.

For Theorem S18.7 on this instance, with $C$ selecting the price, $\bar{A} = 0.9$, $\bar{Q} = 0.1$ and $Q^y = 0.05$: $\Delta = 0.00157$, $c_- = 0.5157$, $c_+ = 1$, the sandwich reads $0.09788 \le 0.11772 \le 0.18982$, $\mathbb{E}|\delta_t|^2 = 0.00349$ in the closed loop, and the right side of Eq. (S62) is 0.279 against the exact 0.0743. The covariance term $\frac{1}{2}g(c_-) = 0.0890$ accounts for about a third of the bound, because the firm declares twice the true price noise. With the a priori constants, $\sigma_D^2 = 0.00351$, $\varsigma_y^2 = 0.2075$ and $\vartheta = 0.0521$ give $c_- = -0.021$, so the a priori form of the theorem does not apply to this firm. The split of the excess into calibration and accuracy parts printed by `check_posterior.py` ($0.0350 + 0.0393$) uses a different decomposition of the same total.

**Check S47.171** (Scores, policy posteriors and levels)**.** For Proposition S18.9, with $n = 5$, $k = 2$, $p = 1$ (`verify_apx_c.py`): under affine feedback, on 20 instances and $t \le 10$, the closed-form Gaussian $W_2$ distance is at most 0.77 of $w_t$ and $|G_t - \hat{G}_t|$ is at most 0.56 of $a_t(2\hat{G}_t^{1/2} + a_t)$; under the Lipschitz feedback $u = 0.5\tanh(\mathtt{d}^\top y)$, with 20000 synchronously coupled samples and $t \le 8$, the coupled $L^2$ distance, an upper bound on $W_2$ up to sampling error, is at most 0.82 of $w_t$.

For Proposition S18.10, on 3000 random instances with 2 to 7 policies the largest ratios of the three left sides to their bounds are 1.000000 for the log-ratio, 0.999426 for the relative entropy and 1.000000 for the total variation, none above 1. For two policies with $\gamma \operatorname{osc} = 1.3$ and $p = (0.342990, 0.657010)$, the total variation distance is $0.3140209253 = \tanh(1.3/4)$. A second script, `check_levels.py`, finds ratios 0.9999996, 0.99993, 0.99996 on 2000 instances.

For Theorem S18.11, with dimensions $8 \to 4 \to 2 \to 1$, $T = 12$ and 30 random instances (`verify_apx_c.py`): the cocycle (a) holds to $4.4 \times 10^{-16}$, the largest ratio $\mathfrak{e}_{l+1}/(\|C_{l+1}\|\mathfrak{e}_l + \mathfrak{d}_{l+1})$ is 0.874, and the largest ratio $\varepsilon_{l+1}/(\|C_{l+1}\|\varepsilon_l + \delta_{l+1})$ is 0.985. In a perturbed nested medium with relative perturbation 0.04 (`check_levels.py`), the totals $\mathfrak{e}_l$ are $6.43 \times 10^{-3}$, $5.85 \times 10^{-3}$, $6.97 \times 10^{-3}$ against the recursion values $6.43 \times 10^{-3}$, $9.35 \times 10^{-3}$, $1.11 \times 10^{-2}$; the path relative entropies are $3.7 \times 10^{-4}$, $6.2 \times 10^{-5}$, $6.7 \times 10^{-5}$ against the bounds $2.1 \times 10^{-3}$, $1.7 \times 10^{-3}$, $2.4 \times 10^{-3}$; and the Wasserstein errors $\varepsilon_l$ are $1.63 \times 10^{-2}$, $6.66 \times 10^{-3}$, $8.15 \times 10^{-3}$ against $1.63 \times 10^{-2}$, $2.36 \times 10^{-2}$, $2.03 \times 10^{-2}$. The total error is not monotone in the level.

## S47.15 Checks for Section S19

**Check S47.172** (Inherited law and nested split law)**.** For Proposition S19.2 and Lemma S19.3: at the base point the self-consistency equations were solved by damped iteration for unit sizes $(4)$, $(4, 4)$, $(4, 4, 4)$, $(3, 2)$ and $(2, 3, 2)$, that is, for 4 to 64 agents, and the closed forms (S64) were compared with discrete Lyapunov solves of the full member recursion and of the joint recursion with the world. World-conditional variances and mean-square errors of all unit averages agree to

$2 \times 10^{-14}$, the spectrum of the member matrix is $\{\lambda, \mu_1, \dots, \mu_L\}$, and $2\,\mathrm{Cov}_1(\eta, \cdot)$ computed for a member and for a parent average both equal $D_L$ to six digits. For sizes (4) the solve returns $\hat{\Lambda}^+ = 30.1075$, $k^s = 0.033214$, $k^0 = 0.633216$, $\bar{k} = 0.11119$, $\hat{\sigma}^2_{\mathrm{in}} = 0.298717$, $\Lambda_0 = 20.0646$ and $D_{\mathrm{cal}} = 1.241230$, the constants of the formation section, with $\lambda = 0.957118$, $\lambda_\perp = 0.516806$ and $q = 2.8988 \times 10^{-3}$. The parent has lag variance $Q/(1-\lambda^2) = 0.237115$, fluctuation variance $q/(n(1-\lambda^2)) = 0.008635 = \mathsf{s}_0/n + \mathsf{s}_1$ with $\mathsf{s}_0 = 0.003955$ and $\mathsf{s}_1 = 0.007646$, and mean-square error 0.245750 against 0.248717 for a member, whose deviation from the parent has variance 0.002966; the lag carries 95.3% of the member error. The exact filter for the coarse model $(F, Q, R^s/n)$ has predicted variance 0.078390 and gain 0.238710, whereas the parent inherits the gain $k^s = 0.033214$ and the claimed predicted variance 0.052453; an isolated Kalman member has predicted variance 0.141067. For sizes $(4, 4)$ the solve gives $\hat{\Lambda}^+ = 42.0915$, $k^s = 0.023758$, $\varpi = (2.8161, 3.1623)$, $\mu = (0.478403, 0.668965)$ and mean-square errors $(0.30510, 0.30372, 0.30324)$ for member, parent and superunit; for $(4, 4, 4)$ it gives $\hat{\Lambda}^+ = 51.2202$ and member error 0.34570. Uniqueness of the solution was not tested beyond convergence of the iteration from one start.

**Check S47.173** (Thresholds by level)**.** For Theorem S19.4, Corollary S19.5, and Numerical observation S19.6: with $n_\ell = 4$, $\kappa_\ell = 3$, $M = 4$ and the geometry $\vartheta_{\mathrm{n}} = 0$, $\vartheta_{\mathrm{f}} = \vartheta_{\max} = \pi/2$ at every level, the nested solutions of depth 0 to 6 have $\hat{\Lambda}^+ = 8.089$, 30.108, 42.092, 51.220, 58.962, 65.876, 72.232, $k^s = 0.1236$, 0.0332, 0.0238, 0.0195, 0.0170, 0.0152, 0.0138, member mean-square error 0.1411, 0.2487, 0.3051, 0.3457, 0.3773, 0.4032, 0.4252 and $D_L = 1.718$, 1.241, 1.078, 0.980, 0.910, 0.856, 0.811. Freezing the constants of the depth-$\ell$ solution, with base precision $\hat{\Lambda}^+ - \kappa_\ell \varpi_\ell$, gives for levels 1 to 6 the separation thresholds 68.15, 109.56, 161.04, 218.95, 281.37, 347.55 and the merging thresholds 6.845, 11.193, 16.146, 21.392, 26.798, 32.327; level 1 reproduces (73). Freezing the inherited constants of depth $\ell - 1$ gives 27.29, 76.42, 128.44, 186.00, 247.63, 312.78 and 2.378, 8.182, 13.483, 18.895, 24.378, 29.938, with $(\Lambda, \hat{\sigma}^2, D) = (30.1075, 0.25825, 1.24123)$ at level 2. Successive ratios of the separation threshold in the first convention are 1.608, 1.470, 1.360, 1.285, 1.235 against squared ratios of base precision 2.64, 1.71, 1.43, 1.31, 1.24. The proved lower bound on the merging threshold with the sharper constant $2K\kappa F^2(1-\lambda^2)^2/(QR^s(1-F^2)(1-\lambda F)\Lambda^2)$ in place of $\kappa C/\Lambda^2$ evaluates to 0.216, 0.763, 1.530, 2.454, 3.507, 4.674, a factor 7 to 32 below the computed thresholds, and the bound with $C$ is weaker by a further factor of 374 at level 1, so (S66) fixes the direction and the asymptotic order of the growth and not its size at these depths. For inherited frames with $n = 4$, $M = 3$, $\kappa_2 = 2$, $\vartheta_{\mathrm{n}} = \pi/8$, $\vartheta_{\mathrm{f}} = \pi/2$, $\vartheta_{\max} = 3\pi/4$ and $R_c^{(2)} = R_c/n$, the inherited constants give $\gamma_S^{(2)} = 113.88$, $\gamma_M^{(2)} = 10.303$ and the near-edge condition of part (b) from $\gamma = 45.86$; the constants of the depth-2 solution with sizes $(4, 3)$ give 141.27, 12.808 and 56.89. On the individual channel the level-1 separation threshold at cross mismatch $\pi/2$, $\pi/8$, $\pi/16$ is 68.2, 493.2, 1853.3.

**Check S47.174** (Two-level mean-field iteration)**.** For Theorem S19.4 and Numerical observation S19.7: population of 24 agents in two superunits of $M = 3$ parents of $n = 4$, parent frames $\{0, \pi/16, \pi/8\}$ and $5\pi/8 + \{0, \pi/16, \pi/8\}$. The map (72) was iterated 4000 steps from $x = 1$ for a receiver in the parent at angle 0, with exact softmax attendance, on the individual channel ($\kappa = 3$ among 23 senders, 1771 sets, constants $\Lambda_0 = 20.0646$, $\hat{\sigma}^2 = 0.29872$, $D = 1.24123$, in-parent translations exact) and on the aggregate channel ($\kappa_2 = 2$ among 5 parents, 10 sets, inherited constants $\Lambda = 30.1075$, $\hat{\sigma}^2 = 0.25825$, $D = 1.24123$). Aggregate channel: at $\gamma = 8 < \gamma_M^{(2)} = 10.30$ all five edges are learned, with bias fractions 0.107 to 0.116 below the bound 0.386 of part (c); at $\gamma = 120 > \gamma_S^{(2)} = 113.88$ the far edges end at 0.9994 or above and the near edges at 0.04785, within the bounds 0.5464 and 0.0543 of parts (a) and (b), with $\bar{\mathsf{p}}_2 = 0.0417$; at $\gamma = 2000$ the far edges are 1 and the near edges $0.04785 = \mathsf{p}^*/(\mathsf{p}^* + 1)$. Between the thresholds, all edges are learned for $\gamma = 10, 12, 15, 20, 30$, the state is mixed at $\gamma = 45$ with far edges $(0.077, 0.718, 0.742)$, and the

superunits are separate at $\gamma = 60$ with far edges 0.849 to 0.878 and far attendance below 0.009, so the sufficient thresholds are not sharp. Individual channel, cross edges by sending parent in the order $\pi/16, \pi/8$, far: $\gamma = 60$ gives $(0.155, 0.159, 0.983, 0.985, 0.986)$ and $\gamma = 120$ gives $(0.153, 0.162, 1, 1, 1)$, so the first event at these temperatures produces the superunit directly; for $\gamma = 250, 400, 600, 900$ the parent at $\pi/16$ remains bridged, at about 0.106, while the parent at $\pi/8$ is separated; at $\gamma = 1300$ all cross edges exceed 0.998, and at $\gamma = 2000$ in-parent attendance is 1 to six digits and total cross attendance is $8.1 \times 10^{-7}$.

**Check S47.175** (Residual translation inside a level-2 unit)**.** For Remark S19.17 and Proposition S19.2: one superunit of $M = 3$ parents of $n = 4$ with frames $\{0, \pi/16, \pi/8\}$, gains from the depth-2 solution with sizes $(4, 3)$, and the near-edge residual fractions $y = 0.04785$ of the aggregate-channel map at $\gamma = 120$ on all six ordered parent pairs. The joint Lyapunov solve of the 12 members and the world, in dimension 26, gives member mean-square error 0.288600 with exact translations, equal to the closed form (S64), and $(0.288645, 0.288607, 0.288645)$ by parent with the residual rotations, a relative change of at most $1.6 \times 10^{-4}$; with untranslated aggregates ($y = 1$) the errors are $(0.308166, 0.291751, 0.308166)$. The raw average of the three parent broadcasts contracts the common component by $\rho = 0.987190$, so $1 - \rho^2 = 0.025456$.

**Check S47.176** (Aggregate of a frame-incoherent parent)**.** For Lemma S19.8: script `rfm_obstruction.py`, part (A). The joint second moments of $(\eta, m_1, \dots, m_n)$ at the level-1 split are obtained from the discrete Lyapunov equation, and $R_c^{(2)} + \mathbb{E}\|X - \eta\|^2/K$ is evaluated from them at the five angles $\phi \in \{0, 0.4, 1.1, \pi/2, 2.5\}$ and compared with (S67). With $n = 4$ the largest absolute difference is $5.8 \times 10^{-15}$ for aligned frames ($\rho = 1$, $\hat\sigma_\rho^2 = 0.25825$), $5.3 \times 10^{-15}$ for frames drawn in $[-0.6, 0.6]$ ($\rho = 0.9545$, $\hat\sigma_\rho^2 = 0.27150$) and $3.1 \times 10^{-15}$ for frames drawn in $[-\pi, \pi]$ ($\rho = 0.4823$, $\hat\sigma_\rho^2 = 0.52792$). With $n = 5$ and frames at the vertices of a regular pentagon ($\rho = 0$) it is $4.4 \times 10^{-16}$, and the calibrated noise is 1.01067 at every angle.

**Check S47.177** (Obstruction bound and thresholds)**.** For Theorem S19.9: script `rfm_obstruction.py`, parts (B) and (C), with $n = 4$, $M = 3$, $\kappa_2 = 2$, near mismatch 0, far mismatch $\vartheta_{\max} = \pi/2$, and $\ell_2 = 2.0744$. For $\rho = 1, 0.75, 0.5, 0.25, 0.1$ the value span $\Delta\mathcal{V}_{\max}^{(2)}$ is 0.18556, 0.12079, 0.06458, 0.02499, 0.00846, below $B_2 = 0.21292$, 0.13501, 0.07062, 0.02692, 0.00905; the thresholds are $\gamma_S^{(2)} = 62.53$, 108.29, 245.06, 795.41, 2690.98 and $\gamma_M^{(2)} = 11.179$, 17.175, 32.121, 82.997, 245.086, above $\ell_2/B_2 = 9.743$, 15.365, 29.376, 77.060, 229.105. At $\rho = 0$ the span and $B_2$ are zero. At $\rho = 1$ and $\vartheta_{\max} = \pi/2, \pi/4, \pi/8, \pi/16$ the thresholds are $\gamma_S^{(2)} = 62.53$, 134.37, 421.79, 1571.51 and $\gamma_M^{(2)} = 11.179$, 16.282, 36.805, 118.886. The smallest group size with $\gamma < \gamma_M^{(2)}$ is $n = 4$, 8, 21, 39 at $\gamma = 10$, 20, 50, 100. The closed-form sufficient size of part (c), with $C = 4.97 \times 10^5$ and $\varpi_{\min} = 0.01895$, is $8.2 \times 10^4$, $1.2 \times 10^5$, $1.8 \times 10^5$, $2.6 \times 10^5$ at the same values of $\gamma$: the bound is correct and far from sharp at this base point, where $C$ carries the factor $(1 - F^2)^{-2}$.

**Check S47.178** (Contrast at the split and under the null)**.** For Propositions S19.10 and S19.11: script `rfm_null.py`. The stationary world-conditional covariance is obtained by iterating the second-moment recursion (4000 steps at the split; under the null, 600 steps inside a damped fixed-point iteration of the calibration, which stops at a change below $10^{-12}$ after 55 sweeps for $n = 4$, $M = 3$, $\kappa_2 = 2$), with the exact first and second moments of uniform draws of $\kappa_2$ parents out of $2M - 1$. Split, $n = 4$, $M = 3$: $\lambda = 0.96437$, $\mu_1 = 0.49041$, $\mu_2 = 0.70758$; the contrast of the superunit is 7.1343 from the member precision, from the precision of the parent aggregates and from (S69); a parent inside the superunit has 1.3286, against 8.7329 for the same parent at the level-1 split. For $(n, M) = (4, 2)$ the two values are 5.6329 and 2.2829, for $(2, 4)$ they are 8.8577 and 0.5974. Null, $n = 4$, $M = 3$, $\kappa_2 = 2$: the row sums of the mean member matrix equal $\lambda = 0.96440$ to machine

precision; the precision couplings are 57.749 inside a parent and 8.641 across parents; unions of $1, \dots, 6$ parents have contrast 1.2721, 0.6142, 1.0000, 1.6280, 2.8311, 6.0655, and all 20 unions of three parents have 1.000000. For $(4, 2)$ with $\kappa_2 = 1$ the six balanced unions have 1.000000 and the whole population 4.2184; for $(2, 4)$ with $\kappa_2 = 3$ the 70 balanced unions have 1.000000 and the whole population 7.9887; for $(4, 3)$ with $\kappa_2 = 1$ the balanced unions have 1.0000 and the whole population 3.9208. The windows between the whole-population value under the null and the split value are $(6.07, 7.13]$, $(4.22, 5.63]$ and $(7.99, 8.86]$. The baseline of Definition 4.11 is a supremum over regions of every size, and the unions of parents are not all of them. For $n = 4$, $M = 3$, $\kappa_2 = 2$ the script `rfm_baseline_regions.py` takes the printed couplings 57.749 and 8.641, the uniform leak 40.94 that reproduces the whole-population value, and enumerates the 208 types of region with at least two members, a type being the list of member counts by parent: the largest contrast is 6.0655, at the whole population, and the next are 4.2811, 3.4406, 3.2261, 3.0104, at the whole population less one member, less two members of one parent, less two members of different parents, and less three members of one parent. The lower end 6.07 of the first window is therefore the baseline over all regions, at the precision of the printed couplings. For $(4, 2)$ and $(2, 4)$ the supremum was taken over unions of parents only, and the lower ends of those two windows are not checked against other regions.

**Check S47.179** (Turnover and the pooled filter)**.** For Propositions S19.14 and S19.15, parts (D) and (E) of the script `rfm_obstruction.py`, with $n = 4$, $M = 3$, $\kappa_2 = 2$, $\vartheta_{\max} = \pi/2$. Reset translations ($f_\alpha = 0$): $g_c = 0.08634$; at $g = 0$, 0.002, 0.005, 0.01, 0.02 the constant $\mathsf{p}^*(g)$ is 0.0503, 0.0704, 0.1008, 0.1518, 0.2554, with $\gamma_S^{(2)} = 62.53$, 56.36, 49.61, 41.55, 30.36 and $\gamma_M^{(2)} = 11.179$, 9.363, 7.430, 5.223, 2.419; at $g = 0.05$, $\mathsf{p}^* = 0.5792$ exceeds $\kappa_2/(2M - 1) = 0.4$ and $\gamma_S^{(2)} = 6.43$; at $g = 0.08$, $\mathsf{p}^* = 0.9242$ and $\gamma_S^{(2)} < 0$. Coarse memory at the depth-2 split, $\lambda = 0.96437$: it outlasts the membership for $g > 0.03563$ at $f = 1$ and for $g > 0.06881$ at $f = 0.5$, and for no $g \le 1$ at $f = 0$. Faithful replacement ($f = 1$, copy noise $R_c/n$, gains frozen at the split, 4000 steps of the second-moment recursion with the replacement variance on the diagonal): the contrast of the superunit from the precision of the parent aggregates is 7.1343, 7.3797, 8.1928, 9.9202, 11.2391 at $g = 0$, 0.01, 0.05, 0.2, 0.5, while the conditional variance of the superunit mean grows from 0.002023 to 0.009687. Pooled filter: the superunit mean has MSE = 0.28635 and gain $k^s = 0.02589$, against the steady-state predicted variance 0.05085 and gain 0.37897 of the Kalman filter of $(F, Q, R^s/12)$; the relative entropy between the two centered Gaussians is 1.451 nats per coordinate.

**Check S47.180** (Recomputation with separate code)**.** For Lemma S19.3, Theorems S19.4 and S19.9, and Propositions S19.10 to S19.12 and S19.14: scripts `check/ic_1_laws.py` to `check/ic_5_turnover.py`, which share no code with the scripts above. The nested mean-square errors agree with direct Lyapunov solutions to $4 \times 10^{-15}$ for sizes $(2, 3)$, $(3, 2, 2)$, $(4, 3)$; the attention bounds of Theorems S19.4 and S19.9 hold on 8000 and 3000 random states by enumeration of all attended sets; the contrasts 7.1343, 1.3286, 8.7329 and the null values of Check S47.178 are reproduced. Signs of the offsets (Proposition S19.12(a)): a reader with gains $k^0 = 0.3$, $k^s = 0.2$ fusing two sensing-only senders at gain 0.25 each has mean-square error 0.4364 with offsets $(0.8, 0.8)$ and 0.1952 with $(0.8, -0.8)$. Null from the split (Proposition S19.11(c)), $n = 4$, $M = 3$, $\kappa_2 = 2$, calibration on the current law: the entrywise distance of the member covariance from its symmetrized form is $6.0 \times 10^{-4}$, $6.6 \times 10^{-5}$, $1.7 \times 10^{-8}$ after 1, 5, 20 rounds, the largest one-round ratio is $0.576 < F^2$, and the contrast of a superunit falls from 7.134 to 4.974, 1.543, 1.0001. Turnover with $f = 1$: the contrast of the parent aggregates recorded after fusion is 7.1343, 7.3797, 8.1928, 9.9202, 11.2391 at $g = 0$, 0.01, 0.05, 0.2, 0.5, and recorded after the replacement step it is 7.1343 at each of these $g$. Not recomputed with separate code: the mean-field orbits of Check S47.174 behind Numerical

observation S19.7, the two numbers quoted in Remark S19.17 (the size condition of Check S47.177 and the residual-translation change of Check S47.175), and, in Check S47.179, the thresholds under turnover and the relative entropy 1.451.

### S47.16 Checks for Section S20

All checks run single-threaded in double precision with fixed seeds; each script finishes in under two seconds. The scripts are `obj_check_potential.py`, `obj_check_time.py`, `obj_check_tracking.py`, `obj_check_scale.py` and `obj_check_clocks.py`. Three further scripts written separately, `obj_verify_a.py`, `obj_verify_b.py` and `obj_verify_c.py`, repeat the checks on other instances and seeds without reusing the first five; no statement of Section S20 failed in them.

**Check S47.181** (Potential and rotational parts of the interaction matrix)**.** Script `obj_check_potential.py`, seed 7, $\gamma = 2$, three members with act dimensions $(2, 1, 2)$, outcome dimension 6. With shared precision and effect models the interaction matrix is symmetric to $7.1 \times 10^{-15}$, and the heat-bath moment recursion of Lemma S41.1 has the fixed point $(u^{\mathrm{N}}, (\gamma K)^{-1})$ to $2.0 \times 10^{-15}$ in the mean and $2.5 \times 10^{-16}$ in the covariance. With member-specific precisions and effect models the asymmetry $\|K - K^{\top}\|$ is 12.58 and the smallest real part of the spectrum of $K$ is 2.146. The matrix $\mathsf{Q} = K\Sigma - \gamma^{-1} I$ of Theorem S20.2(c) is antisymmetric to $8.0 \times 10^{-15}$ with norm 0.0518, and the two expressions for the entropy production in (S72) both give 1.46455. The heat-bath mean and second-moment recursions have spectral radii 0.9016 and 0.8253, the heat-bath stationary mean equals $u^{\mathrm{N}}$ to $6.1 \times 10^{-16}$, and the heat-bath and Langevin stationary covariances differ by 0.0659 in relative Frobenius norm. For random displacements of two members the four-cycle sum computed from the expected free energies and from (S71) agree at 11.0668476113.

**Check S47.182** (Protocol dependence and escalation)**.** Script `obj_check_potential.py`, rotational pair $K = \left(\begin{smallmatrix} 1 & c \\ -c & 1 \end{smallmatrix}\right)$, $\mathsf{h} = 0$, $\gamma = 2$. The spectral radii of the heat-bath mean and second-moment recursions are 0.5590 and 0.6250 at $c = 0.5$, 0.6727 and 0.9050 at $c = 0.9$, 0.7810 and 1.2200 at $c = 1.2$, and 1.1180 and 2.5000 at $c = 2$, equal to $\sqrt{1 + c^2}/2$ and $(1 + c^2)/2$. The heat-bath stationary variance is 0.6667 at $c = 0.5$ and 2.6316 at $c = 0.9$, equal to $1/(\gamma(1 - c^2))$, against the Langevin variance 0.5 at every $c$. A simulation of 400000 heat-bath events at $c = 0.5$ gives variance 0.66707, covariance 0.00030 and $\mathbb{E} u_1^2 u_2^2 = 0.67435$; the mixture of Proposition S20.3(a) predicts 0.66667, 0 and 0.66667, and a Gaussian law with the same covariance would give 0.44444. For the escalation pair of Proposition S20.3(b), $K = \left(\begin{smallmatrix} 0.35 & 0.55 \\ 0.55 & 0.35 \end{smallmatrix}\right)$ has determinant $-0.18$ and eigenvalues $-0.2$ and 0.9, and the heat-bath mean and second-moment recursions have spectral radii 1.2857 and 1.7347.

**Check S47.183** (Cost of disagreement)**.** Script `obj_check_time.py`, seed 11, $\gamma = 2$, three members with act dimensions $(2, 1, 2)$, outcome dimension 7, shared precision and effect models, random displacements. The expected collective objective under the stationary law minus the cooperative minimum is 372.23225016729907 by direct evaluation and 372.2322501672994 by (S73), with bias term 368.4822, fluctuation term $3.75 = n d_u/(2\gamma)$ and team fluctuation $1.25 = d_u/(2\gamma)$. The negotiated displacement satisfies $\max_j \|\mathsf{M}_j^{\top} \Lambda^{\star} (\mathsf{z}_j - \mathsf{z}_{\mathrm{N}})\| = 5.7 \times 10^{-14}$. For two scalar acts with $\Lambda^{\star} = I$, $\mathsf{M}_1 = e_1$, $\mathsf{M}_2 = 2(\cos\theta, \sin\theta, 0)^{\top}$ and half-difference of displacements $(0.3, 0.4, 0.5)$, the bias term and the rate $\kappa = 1 - \cos\theta$ are 0.491028 and 0.637642 at $\theta = 1.2$, 1.720774 and 0.174664 at $\theta = 0.6$, 11.378880 and 0.019933 at $\theta = 0.2$, and 153.602059 and 0.001250 at $\theta = 0.05$, matching (S74); the relaxation times $2/\kappa$ are 3.14, 11.45, 100.33 and 1600.33 events.

**Check S47.184** (Deliberation time and tracking for quadratic preferences)**.** Script `obj_check_time.py`, same instance. The spectrum of $K_{\mathrm{d}}^{-1/2} K K_{\mathrm{d}}^{-1/2}$ lies in $[0.11480, 2.05559]$, so $\kappa = 0.11480$

with $n = 3$. The largest eigenvalue of the mean map $\bar{\mathsf{A}}$ is 0.9617348142605592 against $1 - \kappa/n = 0.9617348142605584$. The matrices $\mathsf{Q}_i = K^{1/2}\mathsf{A}_i K^{-1/2}$ are symmetric and idempotent to $2.6 \times 10^{-15}$ and $3.3 \times 10^{-16}$, and the largest eigenvalues of the averaged second and third tensor powers on symmetric tensors are 0.92669 and 0.89473, below the degree-one value as Theorem S20.5(b) requires. For two members with act dimensions $(2, 3)$, $\kappa = 0.0925030155$ equals one minus the cosine of the smallest principal angle to ten digits. With an optimum drifting at constant velocity, after 4000 events the lag equals $-nK^{-1}K_{\mathrm{d}}\mathsf{v}$, with $\mathsf{v}$ the drift of $u^{\mathrm{N}}$ per event, to $4.3 \times 10^{-18}$, its $K$-norm is 0.02599 against the bound $n\delta/\kappa = 0.3115$ of (S75), and for a drift along the slowest eigenvector the limiting lag norm 0.0261334155493 equals the bound to all printed digits; at finite $t$ and zero initial lag the lag norm is $\delta(1 - (1 - \kappa/n)^t)n/\kappa$, strictly below the bound.

**Check S47.185** (Identification of conflict)**.** Script `obj_check_time.py`, a rotational $K$ with smallest real part of the spectrum 9.8334 and the potential matrix $K_0 = (\gamma\Sigma)^{-1}$ with the same Langevin stationary covariance (difference $6.9 \times 10^{-17}$). The lagged covariances of the two dynamics differ by 0.00194 at lag 0.2; the asymmetry $\|C(\tau) - C(\tau)^{\top}\|$ is 0.00368 for $K$ and $3.5 \times 10^{-18}$ for $K_0$. The response matrix $K^{-1}$ has asymmetry 0.0254, equals $\gamma\Sigma(I + \gamma\mathsf{Q})^{-1}$ to $2.8 \times 10^{-17}$, and differs from the potential response $\gamma\Sigma$ by 0.0162. The smallest real part 9.833 of the spectrum of $K$ exceeds the smallest eigenvalue 8.207 of $K_0$, as Proposition S41.2 requires.

**Check S47.186** (Deliberation time and tracking for finite policy sets)**.** Script `obj_check_tracking.py`, seed 5, three members with three policies each (27 profiles), $\gamma = 1$, flat habits, expected free energies drawn independently as 1.2 times uniform variables on $[0, 1]$, so that no potential exists (the four-cycle sum for the first two members at a base profile is 0.1503). Transport distances on the Hamming metric are computed exactly by linear programming. The influence matrix of Proposition S20.6 has off-diagonal entries $(0.3725, 0.2698; 0.2893, 0.3045; 0.2714, 0.2324)$ by rows, $\mathfrak{D} = 0.60495$ and $\rho = 0.86832$; the largest ratio of the transport distance after one event to the Hamming distance before it, over 60 random pairs of profiles and 40 adjacent pairs, is 0.84982, attained on adjacent pairs. Perturbing every member's expected free energy by $\delta$ times a standard normal table moves the stationary law by 0.00820, 0.04130 and 0.12455 in transport distance for $\delta = 0.02$, 0.1 and 0.3, against the bounds 0.10412, 0.51977 and 1.53838 of (S77). For expected free energies moving as $G_{i,t} = G_{i,0} + 0.1\sin(2\pi t/300)\mathsf{H}_i$ over 300 events from a point mass, $\sup_t \mathfrak{D}_t = 0.67015$ and the largest per-event drift is 0.0014358; the distance of the deliberation law to the current stationary law is 0.00149, 0.00122, 0.00265, 0.00141, 0.00115 and 0.00264 at $t = 50, 100, \ldots, 300$, against the bound of (S78), which falls from 0.12451 to 0.11877. The bound holds and is conservative by a factor of about 45 in this instance. On the separately written instance of `obj_verify_b.py`, with $\sup_t \mathfrak{D}_t = 0.71409$ and largest per-event drift 0.006917, the limiting bound is 0.76163 against distances between 0.00122 and 0.00670 at $t = 20, 40, \ldots, 140$, a factor above 110; the bound is not shown to be tight (Open problem S20.12(e)).

**Check S47.187** (Scale selection in the nested medium)**.** Script `obj_check_scale.py`, nested medium with $b = 3$, $n_{\mathrm{H}} = 4$ $(N = 81)$, $A_{\mathrm{h}} = 1$, $\mathfrak{q} = 0.5$, $\varepsilon^{\Pi} = 10^{-3}$. The level rates are $\mathsf{g}_l = (2.375, 1.125, 0.5, 0.1875)$, equal to the closed form of Proposition 10.7, and the spectrum of $\Pi$ consists of $\varepsilon^{\Pi}$ and $\varepsilon^{\Pi} + \mathsf{g}_l$ with multiplicities $(b-1)b^{n_{\mathrm{H}}-l}$, to $5.8 \times 10^{-15}$. At resolution price $c = 0.05$ the predicted window edges $\varsigma(c)\tau_l$ are 0.49498, 1.04448, 2.34747 and 6.23917 and the leak cutoff is $\varsigma(c)/\varepsilon^{\Pi} = 1176.08$. Over 400 logarithmically spaced horizons in $[0.02, 2000]$ the minimizing rank of (S79), the count of (S80) and the level predicted by Theorem S20.9(a) agree at every horizon. The logarithmic widths of the windows are 0.7467, 0.8098, 0.9775 and 5.2391, against $\ln(1/\mathfrak{q}) = 0.6931$ for an infinitely deep hierarchy. At horizon 1 the level-two block frame retains predictive information 5.638438106083655, equal to the optimum 5.638438106083537 over rank-nine frames, and the best of 200 random rank-nine frames retains 3.7323.

**Check S47.188** (The Laplacian procedure in the nested medium and outside it)**.** Script `obj_check_scale.py`, same medium, the procedure of Section S13.6 with ties broken at random. At each of 60 diffusion times logarithmically spaced in $[0.05, 50]$ the macronodes are the level blocks predicted by Theorem S20.9(c), with level changes at $1/\mathtt{g}_l = (0.4211, 0.8889, 2.0, 5.3333)$. The level blocks have grounding-free ratios $r_\star = (1.714, 2.0, 3.0)$ for levels one to three, all above $b/(b-\mathtt{q}) = 1.2$, and the interval of Open problem S13.21 meets the returning window at each level. For the complete graph on $b$ vertices with one vertex removed from the region, $r_\star(C) = 5$, 11 and 23 against $\ln N = 1.792$, 2.485 and 3.178 at $b = 6$, 12 and 24; over 200 diffusion times in $[10^{-3}, 10]$ the procedure returns only $b$ singletons or one macronode, and never $C$. In the nested medium with $b = 8$, $n_{\mathrm{H}} = 2$, $\mathtt{q} = 0.01$, a level-one block with one site removed has $r_\star = 6.542$, equal to $(b-1)/(1+(b-1)\mathtt{q})$, against $\ln N = 4.159$, and is never returned over 240 diffusion times in $[10^{-2}, 10^4]$.

**Check S47.189** (Specific heat of the nested medium)**.** Script `obj_check_scale.py`, infinite-depth level rates $\mathtt{g}_l = \mathtt{q}^{l-1}(1/(1-\mathtt{q}) + 1/(b-1))$ truncated at 120 levels, diffusion times spanning two logarithmic periods starting at $1/\mathtt{g}_{60}$. The average of $-\mathrm{d}S/\mathrm{d}\ln\tau_{\mathrm{L}}$ over a period is 1.584963 for $(b, \mathtt{q}) = (3, 0.5)$, 0.500000 for $(2, 0.25)$ and 10.427173 for $(3, 0.9)$, equal in each case to $\ln b/\ln(1/\mathtt{q})$ to the printed digits. The specific heat ranges over $[1.5816, 1.5883]$, $[0.4518, 0.5481]$ and $[10.4272, 10.4272]$ within a period: it oscillates, with an amplitude that grows as $\mathtt{q}$ decreases.

**Check S47.190** (Level clocks in a time-dependent nested medium)**.** Script `obj_check_scale.py`, $b = 3$, $n_{\mathrm{H}} = 3$ ($N = 27$), $A_{\mathrm{h}} = 1$, $\mathtt{q} = 0.5$, level-one block frame $\Phi$, couplings $a_l^{\mathrm{H}}(t) = a_l^{\mathrm{H}}(1 + 0.6\sin(2t + l))$ and leak $\varepsilon^{\Pi}(t) = 0.05(1 + 0.5\cos 3t)$, held constant on 40 steps of length 0.05, arbitrary initial covariance. The clocks of (S83) over the window are 0.09767 for the uniform mode and 4.37583, 1.65702 and 0.65140 for levels one to three. The propagator restricted to each level subspace equals $e^{-\Theta_l}$ times the identity to $1.5 \times 10^{-16}$; the block coordinates do not receive from their complement, the largest entry of $\Phi^{\top}\mathsf{U}(I - \Phi\Phi^{\top})$ being $1.3 \times 10^{-16}$; and the block covariance computed from the closed block equation equals the block covariance of the full system to $2.7 \times 10^{-15}$. Adding $\sin(5t)$ times a fixed symmetric Gaussian matrix of scale 0.05, which does not respect the tree, raises that entry to 0.0124, so the closure is a property of the nested structure. Script `obj_check_clocks.py` tests Proposition S20.11(c) with $b = 2$, $n_{\mathrm{H}} = 4$ ($N = 16$), $\mathtt{q} = 0.5$, couplings $a_l^{\mathrm{H}}(t) = a_l^{\mathrm{H}}(1 + 0.3\sin(1.7t + 0.4l))$, the same leak, the window $[0, 1.2]$ in 240 steps and initial level variances $(0.15, 0.3, 0.5, 0.9)$ for levels one to four and 4 for the mean. The clocks are $(4.32533, 2.07639, 0.94483, 0.39377)$ and 0.05631 for the mean, the corrected clocks $(4.65498, 2.43403, 1.38981, 0.82496)$ and 0.24828, and the sixteen canonical correlations of the state at the two ends of the window equal $e^{-\tilde{\Theta}_l}$ with the level multiplicities to $3.4 \times 10^{-15}$. For a price chosen in each of the five windows $\tilde{\Theta}_{l+1} < \varsigma(c) < \tilde{\Theta}_l$ the minimizing rank is $b^{n_{\mathrm{H}}-l}$ and the level-$l$ block frame attains the minimum of the criterion to $1.4 \times 10^{-15}$; the best of 320 random frames exceeds it by 0.0030, 0.0766, 0.1069 and 0.1123 at levels one to four.

## S47.17 Checks for Section S21

**Check S47.191** (Reproduction of the record filter)**.** Script `prv01_reproduce.py`. For Proposition S12.8(b) with $F = 0.99$, $Q = 0.02$, $n = 5$ and unit sensing noise, the fixed point of the claimed precision under rule (P) is 207.18, a claimed variance of 0.004827. The mean-square error of the record's mean, computed once from the stationary error recursion and once from the joint covariance of state and record mean, is 0.28757 by both routes, an overconfidence of 59.58, and the variance of the centralized Kalman filter is 0.05281. The three numbers of Proposition S12.8(b) are reproduced. For part (a), on 20 random systems of state dimension at most 3 run for 60 steps, rule (U) relative to the record's prediction agreed with the centralized filter to $3.8 \times 10^{-8}$ in absolute terms; these

systems were not normalized for conditioning, and Check S47.130 reports agreement to $4.1 \times 10^{-14}$ for that part.

**Check S47.192** (Channel pairs on trees and stars)**.** Script `prv02_tree_star.py`, parts (a) and (b). For Theorem S21.2(a), 20 random trees with 6 nodes, state dimension 2, a random matrix $A$ scaled to spectral radius one, $Q_x = 0$ and 12 steps were run; at every step each node observed with probability 0.7 and each link exchanged with probability 0.5, in random order. Every node pair agreed with the prior pair plus the sum of the evidence items in its information set, tracked separately as a set, to $1.7 \times 10^{-13}$; at the end a node held on average 41.8 of the 51.6 items generated. For Theorem S21.2(b), 20 random stars with 4 members, state dimension 3, two-dimensional observations, $Q_x \succ 0$ and 40 steps were run with a report from every member at every step. The hub with the predicted channel pair agreed with the centralized Kalman filter to $3.5 \times 10^{-10}$. With the channel pair stored as last reported and not predicted, the median deviation was $1.4 \times 10^{3}$.

**Check S47.193** (Intermittent reports)**.** Script `prv02_tree_star.py`, part (c). In exact rational arithmetic with $F = Q = R = 1$, $x_0 \sim \mathcal{N}(0, 1)$, two members and one report after two steps, the member precision is $8/5$, the hub's claimed precision $43/15$ and the exact posterior precision $19/7$, as in Proposition S42.3. For $F = 0.99$, $Q = 0.02$, $R = 1$ and $n = 5$ with a report every $k$ steps, the stationary claimed variance of the hub and the actual mean-square error of its mean, obtained by exact propagation of noise coefficients over 720 steps, are

| $k$ | 1 | 2 | 3 | 5 | 10 |
|---|---|---|---|---|---|
| claimed variance | 0.05281 | 0.04915 | 0.04625 | 0.04205 | 0.03642 |
| mean-square error | 0.05281 | 0.05308 | 0.05356 | 0.05493 | 0.05913 |
| error over claim | 1.000 | 1.080 | 1.158 | 1.306 | 1.624 |
| error over centralized | 1.000 | 1.005 | 1.014 | 1.040 | 1.120 |

against the centralized variance 0.05281.

**Check S47.194** (Window ledgers and model-aware summaries)**.** Script `prv03_window.py`. For Theorem S21.3, the rows $E_k^\top J_D^{-1}$ restricted to the range of $\operatorname{diag}(H_i^\top, \dots, H_i^\top)$ were stacked over $4kd$ random block-diagonal positive semidefinite matrices $D$, for ten random systems in each of six configurations $(d, k, \operatorname{rank} H_i)$: $(1, 2, 1)$, $(1, 5, 1)$, $(2, 3, 1)$, $(2, 3, 2)$, $(3, 4, 2)$ and $(3, 4, 1)$. The rank of the stack, with relative threshold $10^{-9}$, was $k \operatorname{rank} H_i$ in every instance: 2, 5, 3, 6, 8 and 4. Ten systems with $d = 2$, $k = 3$, $\operatorname{rank} H_i = 2$ and a singular $A$ also gave rank 6, so invertibility of $A$ is sufficient for the bound and not necessary. Some condition on $A$ is needed all the same: for $A = 0$ the stack has rank $d$ and the bound fails (Remark S42.5 and Check S47.200). For Proposition S42.6, 20 random systems with $d = 2$, $n = 3$, gap $k = 4$ and 8 contacts were run; the hub that receives one $d$-vector per member and contact agreed with the centralized Kalman filter to $8.9 \times 10^{-15}$.

**Check S47.195** (Rings without provenance)**.** Script `prv04_ring.py`. For Proposition S42.7, the recursions on the static triangle gave claimed evidence $3, 9, 27, 81, 243, 729$ for the naive product and $3, 5, 7, 9, 11, 13$ for the channel rule over six rounds, against the exact 3. For the dynamic ring with $F = 0.99$, $Q = 0.02$, $R = 1$, one observation per member and one simultaneous exchange with both neighbors per step, stationary values after 700 steps by exact propagation of noise coefficients are as follows; the exact variance is that of the posterior given every observation that can have arrived,

with relay delay $\max(\text{dist}-1, 0)$.

| | | | naive product | | | channel rule | | |
|---|---|---|---|---|---|---|---|---|
| $n$ | centralized | exact | claim | error | error/claim | claim | error | error/claim |
| 3 | 0.07000 | 0.07000 | 0.00949 | 0.25563 | 26.94 | 0.05098 | 0.07763 | 1.523 |
| 4 | 0.05981 | 0.06362 | 0.00949 | 0.25480 | 26.85 | 0.05098 | 0.06891 | 1.352 |
| 5 | 0.05281 | 0.05905 | 0.00949 | 0.25433 | 26.80 | 0.05098 | 0.06274 | 1.231 |
| 6 | 0.04762 | 0.05717 | 0.00949 | 0.25405 | 26.77 | 0.05098 | 0.06017 | 1.180 |
| 8 | 0.04032 | 0.05503 | 0.00949 | 0.25374 | 26.74 | 0.05098 | 0.05690 | 1.116 |

The claimed variances of both rules do not depend on $n$, while the exact variance does. This is in line with the lifting argument behind Theorem S21.4 and is not an instance of that theorem, which concerns a static world with one observation per member. The two rules contain no constant tied to a ring size; a rule built for one size can register it (Proposition S42.9(a)).

**Check S47.196** (Covariance intersection on the ring)**.** Script `prv04_ring.py`. In the setting of Check S47.195, covariance intersection with equal weights 1/3 on the closed neighborhood has stationary claimed variance 0.12394 for every $n$, equal to the stationary variance of a solitary filter obtained from the scalar Riccati fixed point, as Proposition S21.6 states. The actual mean-square errors are 0.08261, 0.07884, 0.07690, 0.07601 and 0.07532 for $n = 3, 4, 5, 6, 8$, so the ratio of error to claim is 0.667, 0.636, 0.620, 0.613 and 0.608, below one in every case, and the ratio of error to the exact variance is 1.180, 1.239, 1.302, 1.329 and 1.369. A solitary member has error 0.12394, a ratio to the exact variance of 1.771, 1.948, 2.099, 2.168 and 2.252. Script `prv08_ci_closed.py` compares the closed form of Proposition S21.6(d) on the complete graph with a direct propagation of the $n \times n$ error covariance over 3000 steps, for three parameter triples $(F, Q, R)$ and $n = 3, 5, 8$; the two agree to the five digits printed in all nine cases. At $F = 0.99$, $Q = 0.02$, $R = 1$ the error is 0.08261, 0.07435 and 0.06970, a ratio to the centralized variance of 1.180, 1.408 and 1.729, and the triangle value coincides with the ring value above because the triangle is complete.

**Check S47.197** (Per-source cumulative pairs)**.** Script `prv07_clock.py`. For Proposition S42.11, the protocol was run with $Q_x = 0$, state dimension 2, a random $A$ scaled to spectral radius one and 12 steps on rings of 3 and 5 members, a ring of 6 with two chords and the complete graph on 4 nodes, ten random systems each; at every step each member observed with probability 0.7 and each link carried a one-way report in a random direction with probability 0.5, in random order. After every event the pair $\pi_0 + \sum_s E_v(s)$ of every node agreed with the prior pair plus the sum over its information set, tracked separately as a set of keys, to $8.5 \times 10^{-14}$, $1.8 \times 10^{-12}$, $1.7 \times 10^{-13}$ and $1.1 \times 10^{-13}$ respectively; at the end a node held on average between 72 and 89 percent of the items generated, depending on the graph.

**Check S47.198** (Window ledger on graphs with cycles)**.** Script `prv05_ledger.py`. For Proposition S21.5, the ledger protocol was run on rings of 3, 5 and 6 members and on a graph of six nodes with two independent cycles, with state dimension 2, $Q_x \succ 0$, scalar observations, 9 steps and 5 random systems each. Every node posterior agreed with Gaussian conditioning of the joint law on the set $\{(s, \tau) : \tau \le t - \delta(v, s)\}$ to $3.7 \times 10^{-14}$, $2.6 \times 10^{-13}$, $3.4 \times 10^{-13}$ and $5.3 \times 10^{-13}$ respectively. In every run each item $(s, \tau)$ first arrived at node $v$ in round $\tau + \delta(v, s)$, and the number of stored items equaled $\sum_s (\text{ecc}(v) - 1 - \delta(v, s))$ once $t \ge \text{ecc}(v) - 1$. That number counts the items the filter uses. The items a node holds for one more round in order to forward them are counted in Check S47.200.

**Check S47.199** (Turnover)**.** Script `prv06_turnover.py`. For Proposition S21.8(b), the fraction of observations that reach the record was estimated from $4 \times 10^5$ simulated position-windows and compared with $\rho(g,k)$ of (S84): for $g = 0.02$ the pairs (simulated, formula) are $(0.9901, 0.9900)$, $(0.9606, 0.9608)$ and $(0.9148, 0.9146)$ at $k = 2, 5, 10$; for $g = 0.10$ they are $(0.9499, 0.9500)$, $(0.8190, 0.8190)$ and $(0.6513, 0.6513)$; for $g = 0.30$ they are $(0.8497, 0.8500)$, $(0.5542, 0.5546)$ and $(0.3240, 0.3239)$; at $k = 1$ both equal one. For part (c), the mean number of own observations held by the present occupant of a position after 400 steps was 20.09 at $g = 0.05$ and 4.997 at $g = 0.20$, against $(1 - (1 - g)^t)/g$, which is 20.000 and 5.000 at $t = 400$; the estimates use $2 \times 10^5$ positions and have standard errors near 0.04 and 0.01. For the dynamic scalar world with $F = 0.99$, $Q = 0.02$, $R = 1$, $n = 5$ and a record that fuses exactly what it receives, the mean record variance at write times over $2 \times 10^4$ windows, divided by the centralized variance 0.05281, was 1.000 at $g = 0$ for every $k$ and at $k = 1$ for every $g$; 1.010, 1.029 and 1.039 at $g = 0.05$ and $k = 2, 5, 10$; and 1.044, 1.128 and 1.187 at $g = 0.20$.

**Check S47.200** (Independent reruns)**.** Scripts `prvchk1_star.py` to `prvchk5_turnover.py`, written for the independent check of this section without reference to the scripts above and by other routes: posteriors by batch Gaussian conditioning in covariance form, mean-square errors by a recursion of joint second moments, and expectations under turnover by enumeration in rational arithmetic.

For Theorem S21.2(a), on 15 random trees with 5 nodes, $d = 2$, $Q_x = 0$ and 10 steps, every node pair agreed with conditioning on its information set to $8.7 \times 10^{-14}$, and the same rule on a triangle deviated by 0.15 in the median. For Theorem S21.2(b), 15 random stars at full rate agreed with a Kalman filter in covariance form to $8.0 \times 10^{-15}$ in both variants, and the channel pair stored without prediction gave a median deviation of $9.3 \times 10^2$. For Proposition S42.3, rational arithmetic returned the member precision 8/5, the claimed precision 43/15, the exact precision 19/7 and the mean-square error 693/1849 of the hub's mean, and the table of Check S47.193 was reproduced to the digits printed. For rule (P) of Check S47.191 the claimed precision 207.18 was reproduced. A simulation of $4 \times 10^5$ steps gave a mean-square error of 0.2828 with a standard error near 0.0035, which is consistent with 0.28757; that value was not recomputed by a second exact route.

For Theorem S21.3, the rows $E_k^\top J_D^{-1}$ were stacked without restriction to the range of $H_i^\top$, a test of the stronger statement that the rank is $kd$. Eight random systems with invertible $A$ in each of the configurations $(d, k, \operatorname{rank} H_i) = (1, 2, 1)$, $(1, 4, 1)$, $(2, 3, 1)$, $(2, 3, 2)$ and $(3, 3, 2)$ gave rank $kd$, and so did eight matrices $A$ of rank one and the nilpotent matrix with rows $(0, 1)$ and $(0, 0)$, for $d = 2$ and $k = 3$. For $A = 0$, $d = 2$, $k = 3$ and $H_i = I$ the rank was 2 and not 6, the failure described in Remark S42.5. For Proposition S42.6, ten systems with $d = 2$, $n = 3$, $k = 3$ and three contacts agreed with conditioning on the whole history to $8.0 \times 10^{-15}$.

For Proposition S21.5, rings of 4 and 7 members, a path of 5 and a graph of six nodes with two cycles were run with $d = 2$, 8 steps and three systems each. The node posteriors of the ledger and of the variant (c) that sums time slices agreed with conditioning on $\{(s, \tau) : \tau \le t - \delta(v, s)\}$ to $1.7 \times 10^{-14}$, and every item first arrived in round $\tau + \delta(v, s)$. The largest number of items sent over a link in one round was $n$ on all four graphs. The items that a node held in order to forward them numbered up to $n - 1$, and up to 1, 2, 2 and 2 of them on the four graphs lay outside the window that the filter uses; these are the items that the count $\sum_s (\operatorname{ecc}(v) - 1 - \delta(v, s))$ omits. For Proposition S42.11, on a ring of 4, a ring of 6 with two chords and the complete graph on 4 nodes, six systems each, the information sets were initial segments after every event and the pairs agreed with conditioning to $3.6 \times 10^{-14}$.

The tables of Checks S47.195 and S47.196 were reproduced to the digits printed, with the exact variance obtained by conditioning on a finite window of the stationary process. The closed form

of Proposition S21.6(d) agreed with the second-moment recursion to five digits for $(F, Q, R) = (0.99, 0.02, 1)$, $(0.9, 0.5, 2)$ and $(1, 0.1, 0.5)$ with $n = 3, 5, 8$. For parts (a) and (c), ten systems with $d = 2$, four distinct members on a triangle with a pendant node, random convex weights and 8 steps were run: the claimed covariance minus the error covariance and the error covariance minus the exact posterior covariance were positive semidefinite at every node and step, with smallest eigenvalues $1.3 \times 10^{-5}$ and $1.4 \times 10^{-6}$. For part (b), identical scalar members on that graph with $F = 0.99$, $Q = 0.02$, $R = 1$ and unit prior variance claimed precision 1.99990 after one step, the value of a solitary member, against the exact 2.99990 and 4.99990 at the nodes of degree one and three; after 5 and 50 steps the claimed variance equaled the solitary 0.187781 and 0.123938.

For Proposition S21.8(b) and (c), all replacement patterns of one position were enumerated in rational arithmetic for $g \in \{1/10, 3/10, 1/2\}$, $k = 1, \dots, 4$ and $t = k, 2k, 3k$: in all 36 cases the expected number of written observations equaled $t\rho(g, k)$ and the expected tenure equaled $(1 - (1 - g)^t)/g$. On 20 sample paths with $d = 2$, $n = 3$, $k = 3$, $t = 6$ and summaries of rank one, the collective precision of part (c) agreed with conditioning on the stacked data to a relative $4.4 \times 10^{-13}$. A symbolic star with integer coefficients, 200 runs of 40 events with 1919 replacements, confirmed the two statements of part (a) and the corresponding count when the newcomer does not read. For the dynamic record of Numerical observation S21.9, $2 \times 10^4$ windows gave ratios 1.010, 1.029 and 1.038 at $g = 0.05$ and 1.043, 1.129 and 1.186 at $g = 0.20$ for $k = 2, 5, 10$, with standard errors between 0.0001 and 0.0009. Four of the six differ from Check S47.199 by one unit in the third decimal, which is the resolution of both runs.

**Check S47.201** (Anonymous rules on rings)**.** Scripts `prvchk3_cycles.py` and `prvchk6_sizes.py`. For the lifting in the proof of Theorem S21.4, 20 random anonymous rules with states in $\mathbb{R}^4$ and a random labeling of the ports were run for 12 rounds on rings of 3 to 5 nodes and on their lifts with 6 to 10 nodes; after the last round the state of every node of the lift was bitwise equal to that of its image. For Corollary S42.8, 20 random anonymous rules were run for 15 rounds on rings of 3, 4, 5, 7 and 10 nodes with oriented ports and one common datum; all nodes of all five rings were in one common state after every round, bitwise. For Proposition S42.9(a), the rule with hop counters was run on rings of $N = 3, \dots, 8$ nodes with random labelings of the ports, 30 runs: every node held each item once from round $N - 1$ on and did not yet hold all of them after round $N - 2$, at most two items were in transit at a node, and the rule built for $N = 5$ counted 5 of 10 items on a ring of 10. For part (b), the rule with observed values as keys was exact for the information set after every round and complete from round $\operatorname{diam}(G)$ on 20 random connected graphs with 4 to 8 nodes; on a ring of 5 with two equal values it counted 4 items, and on a ring of 6 with constant data it counted one. For part (c), with keys drawn uniformly from $[0, 1]$ the rule counted every item in 200 draws on both of these rings.

### S47.18 Checks for Section S22

**Check S47.202** (Tail of the understanding error)**.** For Theorem S22.1, instances were drawn with a code $H$ whose singular values are evenly spaced in $[1, 1.6]$, prior covariance $P = 0.5I + 0.1AA^\top$ for a standard Gaussian matrix $A$, channel covariance $R_c = 0.05I + 0.02\mathbf{1}\mathbf{1}^\top$ and code error scale $\sigma_H = 0.7$. For each pair $(K, r_c)$, $4 \times 10^5$ draws of the understanding error $\psi$ were generated and the tail index of $\|\psi\|^2$ was estimated by the Hill estimator on the 2000 largest values; the moment threshold of the theorem corresponds to the index $d/2$ with $d = K - r_c + 1$. The estimates were 0.469 and 0.490 against 0.5 for $(K, r_c) = (2, 2)$ and $(3, 3)$, 1.037 and 1.037 against 1 for $(3, 2)$ and $(4, 3)$, 1.448 and 1.488 against 1.5 for $(4, 2)$ and $(5, 3)$, and 2.556 against 2 for $(5, 2)$. The last estimate misses the predicted value by 28 percent; at this sample fraction the estimator is not reliable for the

lighter tail, and the case $d = 4$ rests on the proof alone. The running mean of $\|\psi\|^2$ over the first $10^3$, $10^4$, $10^5$ and $4 \times 10^5$ draws was 665, 834, $1.9 \times 10^5$, $3.9 \times 10^5$ at $(2, 2)$ and $1.2 \times 10^4$, $6.6 \times 10^4$, $2.0 \times 10^7$, $5.6 \times 10^7$ at $(3, 3)$, where $d = 1$; it was 3.70, 3.67, 4.12, 28.4 at $(3, 2)$ and 9.98, 9.62, 10.9, 19.8 at $(4, 3)$, where $d = 2$ and the second moment diverges logarithmically; and over the last three sample sizes it was 1.27, 1.25, 1.25 at $(4, 2)$, 19.7, 19.0, 18.9 at $(5, 3)$ and 0.446, 0.448, 0.450 at $(5, 2)$, where $d \geq 3$. At small $\sigma_H$ the divergence is carried by rare draws in which the code error is comparable to the smallest singular value of $H$: at $K = 3$, $r_c = 2$ and $\sigma_H^2 = 0.02$, one draw of $4 \times 10^5$ had $\sigma_H \|W\|_F > 0.9$ (Check S47.203). A sample of $10^5$ draws at that scale usually contains no such draw and returns a stable average although the moment is infinite. For its instance with these dimensions Check S47.6 therefore reports no untruncated average and computes the moment truncated to an event $\sigma_H \|W\|_F \leq \rho$ with $\rho$ below the smallest singular value of the code.

**Check S47.203** (Second-order expansion of the understanding error)**.** For Proposition S22.2, with the instance family of Check S47.202 and truncation radius 0.9, below the smallest singular value of $H$, the truncated moment $\mathbb{E}[\psi\psi^\top; \sigma_H \|W\|_F \leq 0.9] - R_c$ was estimated with the channel noise integrated in closed form, from $4 \times 10^5$ draws ($3 \times 10^5$ in the last two cases). The ratio of the first term $\sigma_H^2 \mathbb{E}\|\hat{m}_0\|^2$ to the diagonal entries of the estimate was 0.99 and 0.95 at $(K, r_c, \sigma_H^2) = (3, 2, 0.02)$, 1.00 and 0.98 at $(3, 2, 0.005)$, 1.08 and 1.06 at $(5, 2, 0.02)$, 1.08 and 1.09 at $(5, 2, 0.005)$, between 0.95 and 0.97 at $(4, 4, 0.005)$, and between 1.06 and 1.15 at $(6, 3, 0.01)$. The first term thus understates the excess in some instances and overstates it in others, and the overstatement at $(5, 2)$ does not shrink with $\sigma_H$. Adding the three remaining terms of Eq. (S87) reduced the largest entrywise gap in each case, in the same order, from $1.40 \times 10^{-3}$ to $1.08 \times 10^{-3}$, from $1.57 \times 10^{-4}$ to $7.8 \times 10^{-5}$, from $3.62 \times 10^{-3}$ to $1.75 \times 10^{-3}$, from $9.06 \times 10^{-4}$ to $1.28 \times 10^{-4}$, from $2.10 \times 10^{-3}$ to $1.55 \times 10^{-3}$, and from $4.41 \times 10^{-3}$ to $2.64 \times 10^{-3}$. The order of the remainder was tested separately with $10^5$ antithetic pairs $(W, -W)$, which cancel the odd orders. The mean of the second-order Taylor term agreed with the closed form within 1.8 standard errors in all four instances, and the ratio of the largest entry of the mean remainder at $\sigma_H^2 = 0.004$ to that at $\sigma_H^2 = 0.001$ was 16.2, 15.3, 16.4 and 15.7 for $(K, r_c) = (3, 2)$, $(5, 2)$, $(4, 4)$ and $(6, 3)$, against 16 for a remainder of order $\sigma_H^4$. The remainder is largest for the square code, about $50\sigma_H^4$ per diagonal entry at $K = r_c = 4$.

**Check S47.204** (Least squares for a fixed code and tracking of a drifting one)**.** For Proposition S43.5(a), the recursion Eq. (S2) on the vectorized code and the recursion for $\Pi_t$ were run side by side for 12 rounds at $K = 3$ and $r_c = 2$: the covariances $P_t^H$ and $\Pi_t \otimes R_c$ differed by at most $1.0 \times 10^{-17}$ and the estimates by $1.4 \times 10^{-16}$. For part (c), with $M_2 = \mathrm{diag}(1, 0.5, 0.25)$ and $4 \times 10^4$ replications, the mean of $\|\hat{H}_T - H\|_F^2$ at $T = 5, 6, 8, 20$ was 2.711, 1.315, 0.6633, 0.1665 against 2.664, 1.332, 0.6659, 0.1665 from Eq. (S158). The standard errors at $T = 8$ and $T = 20$ were 0.0027 and 0.0002; for $T \leq 6$ the squared error has infinite variance and no standard error is quoted, so the averages at $T = 5$ and $T = 6$ are weaker evidence than those at $T = 8$ and $T = 20$. For Theorem S22.3, one row with $K = 3$, $\rho = 0.05$, $M_2 = \mathrm{diag}(1, 0.5, 0.25)$ and $P_0^- = 0.1I$ was tracked by the exact recursion for $6 \times 10^4$ rounds, and averages were taken over the last four fifths. With $p_a = 1$ and $q = 10^{-2}$, $10^{-3}$, $10^{-4}$, the first expression of Eq. (S89) was 0.1149, 0.0328, 0.01002; the average of $\mathrm{tr}\, P_t^-$ was 0.1611, 0.0373, 0.01046 and the realized squared error 0.1618, 0.0378, 0.01067; the stationary error of the constant-gain rule used in the proof was 0.655, 0.0420, 0.01073; and the right side of Eq. (S90) was 1.666, 0.0444, 0.0109. With $p_a = 0.3$ and $q = 10^{-3}$, $10^{-4}$, the floor was 0.0585 and 0.01817, the realized error 0.0789 and 0.02072, and the upper bound 0.1247 and 0.02176. The source script did not evaluate the stationary error of the constant-gain rule for $p_a = 0.3$; an independent recomputation gave 0.1084 and 0.0211 (Check S47.208), between the floor and the upper bound in both cases. The ratio of the average of $\mathrm{tr}\, P_t^-$ to the floor was 1.40, 1.14, 1.04 at $p_a = 1$ and 1.31, 1.10 at $p_a = 0.3$, decreasing toward one with $q$ as parts (a) and (b) together require.

With the third component of the regressor set to zero and $q = 10^{-3}$, the entry $P_{33}^{-}$ after 2000 rounds was 2.1000, equal to $p_0 + qt$.

**Check S47.205** (Gauge of ungrounded learning)**.** For Theorem S22.4, with $K = 3$, a random orthogonal eigenbasis and the stationary content covariance $\Sigma = \sum_a q_a(1 - f_a^2)^{-1}\Pi_a$, the Jacobian with respect to the code of the autocovariances $HF^k\Sigma H^\top$, $k = 0, \dots, 4$, was computed by central differences at a random code. Its nullity at $r_c = 3$ and at $r_c = 2$ was 3 and 3 for $F = 0.9I$, 1 and 1 for eigenvalues $(0.9, 0.9, 0.5)$, and 0 and 0 for eigenvalues $(0.9, 0.6, 0.3)$, equal in each case to Eq. (S91) and consistent with Proposition S43.10. For the simple spectrum the eight sign changes $T = V\,\mathrm{diag}(\pm 1)V^\top$ altered the autocovariances by at most $1.2 \times 10^{-14}$, a generic rotation altered them by 4.99, and at $r_c = 3$ exactly one of the eight codes $HT$ agreed with $H$ on a grounded direction in general position, as Corollary S22.5 states.

**Check S47.206** (Composition along a path and hub holonomy)**.** For Theorem S22.7(b), with $K = 3$, errors $\Delta_e$ with independent $\mathcal{N}(0, \delta^2/K)$ entries, $\delta^2 = 0.05$, a unit vector $x$ and $10^5$ replications, the mean of $\|(\Upsilon_{\mathfrak{p}} - I)x\|^2$ for $L = 1, \dots, 6$ was 0.0500, 0.1027, 0.1577, 0.2150, 0.2753, 0.3384, against $(1.05)^L - 1 = 0.0500$, 0.1025, 0.1576, 0.2155, 0.2763, 0.3401 from Eq. (S94). The standard errors grow from $10^{-4}$ at $L = 1$ to $1.1 \times 10^{-3}$ at $L = 6$, and the largest deviation is 1.6 standard errors; the six averages share their draws and are not independent. For part (c), three hub maps with spoke errors of entry scale 0.3 gave a triangle holonomy within $1.1 \times 10^{-14}$ of the identity.

**Check S47.207** (Risk decomposition, calibration cost and absorption)**.** For Theorem S22.6(a), with $K = 3$, two message rows, $M = \mathrm{diag}(1, 2, 0.5)$, a fixed code error with entries of scale 0.15, $\bar{h} = (0.3, -0.2)^\top$, $\Sigma_h = 0.02I$, the gain $\hat{\mathcal{K}} = P\hat{H}^\top(\hat{H}P\hat{H}^\top + R)^{-1}$ and $4 \times 10^5$ draws, the risk was 1.3544 with standard error 0.0023, against $0.8239 + 0.2589 + 0.1381 + 0.1340 = 1.3549$ for the channel, calibration, objective and interaction terms of Eq. (S92). The interaction term is positive in this instance and nearly as large as the objective term. For Corollary S43.11, with $\Pi = s\Pi_0$ for a fixed $\Pi_0 \succ 0$ and $2 \times 10^5$ antithetic pairs, the mean calibration cost was 0.006470 (standard error $1.5 \times 10^{-5}$) at $s = 0.02$ and 0.001618 ($4 \times 10^{-6}$) at $s = 0.005$, against 0.006468 and 0.001617 from the leading terms of Eq. (S164). For Proposition S43.13, $2 \times 10^5$ grounded rounds with content mean $0.7\mathbf{1}$, content covariance $\mathrm{diag}(1, 0.64, 0.36)$, a deviation matrix $D$ with entries of scale 0.3, $d = (0.5, -0.4)^\top$ and $\Sigma_\zeta = 0.04I$ gave a least-squares fit with intercept within 0.0019 of $H + D$ and 0.0010 of $d$, and a fit without intercept within $1.3 \times 10^{-3}$ of $H + D + d\bar{m}^\top M_2^{-1}$ evaluated at the sample moments.

**Check S47.208** (Independent recomputation)**.** The statements of this section were recomputed with separately written scripts, on separately drawn instances except for the tracking run, which uses the parameters of Check S47.204 with independently drawn regressors. For Theorem S22.1, at $\sigma_H = 0.7$ with $(m_i, u, \nu)$ fixed and $4 \times 10^5$ draws of $W$, the slope of the logarithm of the empirical survival function of $\|\psi\|$ against $\log t$ was $-0.96$, $-2.39$, $-3.11$ at $(K, r_c) = (2, 2)$, $(3, 2)$, $(4, 2)$ and $-1.01$, $-1.97$, $-3.16$ at $(3, 3)$, $(4, 3)$, $(5, 3)$, against $-d$; the fit at $(3, 2)$ is 20 percent steeper than predicted. The empirical tail was at most 0.21 times the bound Eq. (S86). The Monte Carlo second moment was 2.65, 1.72, 1.95 against 345, 302, 2069 from Eq. (S85) at $(K, r_c, \sigma_H) = (4, 2, 0.7)$, $(6, 2, 0.7)$, $(7, 3, 0.3)$, so that bound holds with a wide margin and is far from tight. For Proposition S22.2, with the channel noise integrated in closed form and antithetic pairs, the coefficient of $\sigma_H^2$ extrapolated from $\sigma_H = 0.01$, 0.02, 0.04 agreed with the bracket of Eq. (S87) within relative error $7 \times 10^{-3}$ at $(K, r_c) = (2, 2)$, $(3, 2)$, $(4, 2)$, $(6, 2)$, $(5, 3)$, and the ratio of successive differences was between 3.97 and 4.21, against 4 for a remainder of order $\sigma_H^4$. Over 2000 random instances at $K = 3$ and $r_c = 2$, the ratio of the first term of the bracket to its diagonal entries, computed from the closed form, had

minimum 0.605, fifth percentile 0.955, median 0.999, ninety-fifth percentile 1.047 and maximum 25.1; the extremes occur where a diagonal entry of the bracket is small. For Remark S43.4, the trace of the three noise terms was negative in all 1657 random instances with $K - r_c > \kappa(P)$, and for $P = pI$ it equaled $-2\sigma_H^2 p(K - r_c - 1)\operatorname{tr}(R_c S_0^{-1})$ to $1.1 \times 10^{-14}$.

For Proposition S43.5(c), with $K = 3$, $M_2 = \operatorname{diag}(1, 0.5, 0.25)$ and $\operatorname{tr} R_c = 0.16$, the mean of $\|\hat{H}_T - H\|_F^2$ at $T = 5, 6, 8, 20$ was 1.147, 0.558, 0.279, 0.0700 against 1.120, 0.560, 0.280, 0.0700, and at $T = 4$ the two halves of the sample gave 4.69 and 6.61, the instability expected of an infinite mean. For Theorem S22.3, the first expression of Eq. (S89), the stationary error of the constant-gain rule and the right side of Eq. (S90) agreed with Check S47.204 to the digits quoted there; the average of $\operatorname{tr} P_t^-$ was 0.1615, 0.0374, 0.01050 at $p_a = 1$ and 0.0770, 0.0200 at $p_a = 0.3$; and the stationary error of the constant-gain rule at $p_a = 0.3$ was 0.1084 and 0.0211 from the fixed point and 0.1082 and 0.0211 in simulation. With an attention rule that depends on the receiver's history, 4000 replicas and 150 rounds, the smallest eigenvalue of $\mathbb{E}P_t^- - \bar{P}_t$ over all rounds was $5.6 \times 10^{-15}$, nonnegative as part (a) requires.

For Theorem S22.4 and Proposition S43.10, the nullity of the differential of the autocovariances, computed from its closed form at $K = 3$, was 3, 3, 2 at $r_c = 3, 2, 1$ for $F = 0.9I$, was 1 for eigenvalues $(0.9, 0.9, 0.5)$ and 0 for $(0.9, 0.6, 0.3)$, equal to Eq. (S91); at a rank-deficient block ($K = 4$, multiplicities $(3, 1)$, $r_c = 2$, $\operatorname{rank}(HV_1) = 1$) it was 4, the orbit dimension 2 plus the excess 2 of Proposition S43.10. At simple spectrum and $K = r_c = 3$, a Gauss–Newton search from 60 random starts for codes with the autocovariances of a given code converged every time to one of the eight codes of the orbit and reached all eight. For Corollary S22.5, at $K = r_c = 4$ with multiplicities $(3, 1)$, grounding of rank 0, 1, 2, 3 in general position left nullity 3, 1, 0, 0, the dimension of $\mathrm{O}(3 - m_1)$, and at rank 2 a reflection gave a second code that differs from $H$ by 3.06 in its largest entry, agrees with $H$ on $\operatorname{col}(M_2)$ to $1.4 \times 10^{-15}$ and has the same autocovariances to $9.8 \times 10^{-15}$. With isotropic dynamics, $K = r_c = 3$, $F = 0.7I$ and $\operatorname{rank} M_2 = 2$, the grounded pairs alone leave a family of dimension 3; with the stream the nullity was 0, and the reflection across $\operatorname{col}(M_2)$ gave the one other code, at distance 2.36 in the largest entry and equal to $H$ on $\operatorname{col}(M_2)$ to $2.3 \times 10^{-16}$. The stream therefore reduces the unidentified set under isotropic dynamics although it does not change the condition for point identification.

For Theorem S22.6, with $K = 3$, two message rows, a random code error with independent entries of standard deviation 0.35, a regularized gain with $\hat{R} \neq R$ and $4 \times 10^5$ draws, the risk computed from the definitions was 3.5695 with standard error 0.0068, against $2.5326 + 0.7397 + 0.5243 - 0.2288 = 3.5678$ for the four terms of Eq. (S92) computed from a separate sample; the interaction term is negative in this instance. The six weighted inner products of the four error vectors were within 1.3 standard errors of zero. Two code errors with the same value of $\bar{h} - \Delta m_i$ gave offsets that differ by 0.177 in norm, so the offset is not a function of $\bar{h} - \Delta m_i$ alone. Over $2 \times 10^4$ code models with scales spread over five orders of magnitude, the largest norm of the regularized gain was 2.2217 against the bound 2.2384 of part (d). For Corollary S43.11, the mean calibration cost was $2.819 \times 10^{-2}$, $6.972 \times 10^{-3}$, $1.738 \times 10^{-3}$ at $\Pi = s\Pi_1$ with $s = 0.02, 0.005, 0.00125$, against $2.783 \times 10^{-2}$, $6.956 \times 10^{-3}$, $1.739 \times 10^{-3}$ from the leading terms. For Eq. (S166), with the code model fixed and the true code random, the risk was 3.5753 with standard error 0.0078 against $3.5030 + 0.0680 = 3.5710$. For Proposition S43.13, least squares with an intercept on $4 \times 10^5$ rounds was within 0.0027 of $H + D$ and 0.0041 of $d$, the fit without intercept was within 0.0018 of $H + D + d\bar{m}^\top M_2^{-1}$, and the risk of the reading at the learned code was 1.8942 with standard error 0.0039 against $\operatorname{tr}(MP_D^+) = 1.8944$, compared with 3.7804 at the declared code.

For Theorem S22.7, the conjugation identity of part (a) held to $3.8 \times 10^{-14}$ over 200 random walks; the mean of $\|(\Upsilon_{\mathfrak{p}} - I)x\|^2$ at $\delta^2 = 0.05$ and $L = 1, \dots, 6$ was 0.0499, 0.1022, 0.1572, 0.2152, 0.2759, 0.3393, against 0.0500, 0.1025, 0.1576, 0.2155, 0.2763, 0.3401 from Eq. (S94), with standard

errors between 0.0001 and 0.0008 (the six averages come from nested walks and are not independent); hub holonomies over random closed walks were within $6.8 \times 10^{-15}$ of the identity with a consensus space of dimension $K = 3$; the ratio of the two sides of Eq. (S95) was at most 0.967 over $2 \times 10^4$ random spoke pairs and equal to 1 for scalar spokes $(1+\epsilon_i)I$ and $(1-\epsilon_k)I$, so the bound is attained; and independent pairwise translations with errors of scale $10^{-3}$ had a trivial consensus space in all 200 draws on a directed triangle, on the complete digraph with three agents and on a bidirected star with four agents. For Remark S43.15 the tail slope of the zero-shot mistranslation with Gaussian spokes was $-1.006$ against $-1$. For Remark S43.16 the mean-square mistranslation was 0.1336 with standard error 0.0002 against 0.1336. For Corollary S43.17 the two thresholds agreed with a direct comparison of the two expressions for every $n$ from 2 to 40 on a grid of $c/T$ in $(0, 3]$.

### S47.19 Checks for Section S23

All checks are deterministic linear-algebra computations in double precision with fixed seeds; none is a simulation. Stationary covariances are obtained by solving the discrete Lyapunov equation as a linear system, and window divergences are evaluated from the full covariance of the window, of size $(n+1)m$, with the Gaussian formula of Theorem S23.5(b). The scripts are `nrg_check_ep.py`, `nrg_check_spectral.py`, `nrg_check_flow.py`, `nrg_check_viability.py` and `nrg_check_scale.py`; each runs in a few seconds on one thread. Check S47.214 reports a rerun by three scripts written separately from these, `nrg_indep_a.py`, `nrg_indep_b.py` and `nrg_indep_c.py`. Random sweeps are finite tests: they can refute a statement and they do not prove one.

**Check S47.209** (Entropy production formula)**.** Script `nrg_check_ep.py`, parts 1 to 3. For $d \in \{2, 3, 5\}$ a Gaussian random $A$ is rescaled to $\rho(A) = 0.8$ and $Q^{\Xi} = GG^{\top} + 0.5I$ with Gaussian $G$ (seed 20261109). The two expressions in Eq. (S97) agree to ten digits, with $\mathsf{ep} = 30.1116$, 1.01655 and 11.6960, and $|\mathsf{D}_n - n\mathsf{ep}| \le 9.1 \times 10^{-13}$ for $n \in \{1, 2, 5, 8\}$, where $\mathsf{D}_n$ is computed from the window covariance without use of the Markov property. For $d = 4$ and $A = Q^{\Xi}M$ with $M$ symmetric, rescaled to $\rho(A) = 0.7$, the entries of $AQ^{\Xi} - Q^{\Xi}A^{\top}$ and of $\mathsf{J}$ are below $1.2 \times 10^{-16}$ and $\mathsf{ep}$ evaluates to $-7.7 \times 10^{-34}$; after a Gaussian perturbation of $A$ of size 0.05 the identity $\mathsf{J} - A\mathsf{J}A^{\top} = AQ^{\Xi} - Q^{\Xi}A^{\top}$ of Lemma S44.2 holds with residual $7.8 \times 10^{-16}$ and $\mathsf{ep} = 0.077874$. For the directed ring of Example S23.6 at $c = 0.3$, $a = 0.5$, $q = 1$, the closed form and the normal-matrix formula of Theorem S23.3 both give $\mathsf{ep} = 0.9259259259 = 25/27$.

**Check S47.210** (Coarse descriptions)**.** Scripts `nrg_check_ep.py`, parts 3 and 4, and `nrg_check_spectral.py`. In the directed ring the mean and the single-agent description have $\mathsf{D}_n(C)/n = 0$ to machine precision for $n \in \{1, 2, 4, 8, 16, 32\}$, as Theorem S23.5(d) requires. The description by agents 1 and 2 has $\mathsf{D}_n(C)/n = 0.257767$, 0.301990, 0.324550, 0.335599, 0.341126, 0.343890, 0.345272, 0.345963 at $n = 1, 2, 4, 8, 16, 32, 64, 128$. The integral Eq. (S98), by the midpoint rule with 4096 nodes, is $\bar{\mathsf{e}} = 0.346653$, which is 37.4% of $\mathsf{ep}$, and $n(\bar{\mathsf{e}} - \mathsf{D}_n(C)/n) = 0.08844$ at $n = 32$, 64 and 128; the same integral with $C = I$ returns $0.925926 = \mathsf{ep}$. A second instance has $d = 4$, Gaussian $A$ rescaled to $\rho(A) = 0.7$, diagonal $Q^{\Xi}$ with entries uniform in $[0.5, 2]$ and a Gaussian $2 \times 4$ matrix $C$ (seed 5), with $\mathsf{ep} = 6.148810$. There $\mathsf{D}_n(C)/n = 0.899610$, 0.799223, 0.781282, 0.772260, 0.766787, 0.763230 at $n = 1, \dots, 6$ and 0.758769, 0.752084, 0.748741, 0.747070, 0.746234 at $n = 8, 16, 32, 64, 128$, which is decreasing, while $\bar{\mathsf{e}} = 0.745398$ and $n(\bar{\mathsf{e}} - \mathsf{D}_n(C)/n) = -0.10697$ at $n = 32$, 64 and 128. The increments $\mathsf{D}_{n+1}(C) - \mathsf{D}_n(C)$ for $n = 1, \dots, 6$ are 0.698835, 0.745400, 0.745195, 0.744893, 0.745444, 0.745379, positive as Theorem S23.5(a) requires and not monotone. A third instance with $d = 5$, $\rho(A) = 0.85$ and diagonal $Q^{\Xi}$ has $\mathsf{ep} = 16.237119$; a Gaussian scalar description has $\mathsf{D}_n(C)$ equal to zero in double precision for $n \in \{1, 5, 12\}$, and a Gaussian two-dimensional description has $\max_k \|\Gamma_k - \Gamma_k^{\top}\|_{\max} = 16.57$ and $\mathsf{D}_n(C)/n = 1.656752$, 2.006642,

2.359673, 2.531009, 2.613740, 2.655110 at $n = 1, 2, 4, 8, 16, 32$, the last of which is 16% of $\mathsf{ep}$. These numbers support Numerical observation S23.7 and do not prove it.

**Check S47.211** (Information flow in discrete time)**.** Scripts `nrg_check_flow.py` and `nrg_check_viability.py`, part 2. Over 200 random bipartitions with $d \in \{2, \dots, 5\}$, $\rho(A)$ uniform in $[0.3, 0.95]$ and block-diagonal $Q^{\Xi}$ (seed 7), the identities $\mathsf{s}_\mathrm{u} - \mathsf{i}_\mathrm{u} = D_{\mathrm{KL}}(P \| P^{\mathrm{fr}})$ and $\mathsf{ep}_\mathrm{u} + \mathsf{ep}_\mathrm{v} = \mathsf{ep}$ hold with relative error at most $9.0 \times 10^{-14}$, the smallest value of $\mathsf{s}_\mathrm{u} - \mathsf{i}_\mathrm{u}$ is $2.7 \times 10^{-7}$, the smallest value of $\mathsf{te}_{\mathrm{v}\to\mathrm{u}} - \mathsf{i}_\mathrm{u}$ is $9.0 \times 10^{-8}$, and the smallest partial entropy production is $-0.1204$. Over 400 further bipartitions with $d \in \{2, 3, 4\}$ and diagonal $Q^{\Xi}$ (seed 11), $\mathsf{i}_\mathrm{u} + \mathsf{i}_\mathrm{v}$ ranges from $-1.3156$ to 1.5631 and is negative in 24 cases. The instance of Example S44.4 has $\rho(A) = 0.9356$, $\mathsf{ep} = 6.099484$ and $I(X_\mathrm{u}; X_\mathsf{E}) = 0.461376$; on the side of the collective $\mathsf{s}_\mathrm{u} = 5.133610$, $\mathsf{i}_\mathrm{u} = 0.784838$, $\mathsf{ep}_\mathrm{u} = 6.099535$ and $\mathsf{te}_{\mathsf{E}\to\mathrm{u}} = 0.822459$ from both the definition and the first line of Eq. (S101); on the side of the exterior $\mathsf{s}_\mathsf{E} = 0.020580$, $\mathsf{i}_\mathsf{E} = -0.150468$, $\mathsf{ep}_\mathsf{E} = -0.000051$ and $\mathsf{te}_{\mathrm{u}\to\mathsf{E}} = 0.033553$ from both the definition and the second line. Hence $\mathsf{i}_\mathrm{u} + \mathsf{i}_\mathsf{E} = 0.634370$ and $\mathsf{s}_\mathrm{u} + \mathsf{s}_\mathsf{E} - \mathsf{ep} = -0.945295$. The instance of Example S44.5 has $\mathsf{ep} = 16.191080$, $\mathsf{ep}_\mathrm{u} = -0.045101$, $\mathsf{ep}_\mathrm{v} = 16.236181$ and $\ln|A_{\mathrm{uu}}| = -2.3026$; the rotation-like instance $A = \left(\begin{smallmatrix} 0.2 & 0.9 \\ -0.9 & 0.2 \end{smallmatrix}\right)$, $Q^{\Xi} = I$ has $\mathsf{ep} = 21.6$, $\mathsf{ep}_\mathrm{u} = \mathsf{ep}_\mathrm{v} = 10.8$, $\mathsf{s}_\mathrm{u} = 6.48$ and $\mathsf{i}_\mathrm{u} = 0.830366$. For the diffusive scaling $A = I + hG$, $Q^{\Xi} = hD$ with $G = \left(\begin{smallmatrix} -1 & 0.8 \\ -0.5 & -1.2 \end{smallmatrix}\right)$ and $D = \mathrm{diag}(1, 0.5)$, at $h = 0.2$, 0.1, 0.05, 0.025, 0.0125 the rate $\mathsf{ep}/h$ is 0.861702, 0.794118, 0.764151, 0.750000, 0.743119, the defect $(\mathsf{s}_\mathrm{u} + \mathsf{s}_\mathrm{v} - \mathsf{ep})/h$ is $-0.095702$, $-0.044058$, $-0.021189$, $-0.010396$, $-0.005150$, and $(\mathsf{i}_\mathrm{u} + \mathsf{i}_\mathrm{v})/h$ is 0.081845, 0.040816, 0.020411, 0.010210, 0.005107. Both defects halve with $h$ per unit time, that is, they are of order $h^2$ per round in this instance.

**Check S47.212** (Maintenance of a record)**.** Script `nrg_check_viability.py`, part 1. Over 300 random instances of Proposition S23.10(b), with $\mathsf{f}$ uniform in $[-0.95, 0.95]$, $\omega$ in $[0.05, 0.9]$, $\mathsf{r}$ in $[-0.9, 0.9]$, source noise variance in $[0.2, 2]$ and $q_z$ in $[0.05, 2]$ (seed 3), the value of $\mathsf{i}_z$ computed from the two mutual informations agrees with Eq. (S102) to $2.2 \times 10^{-15}$, the smallest value of $\mathsf{s}_z - \mathsf{i}_z$ is $4.7 \times 10^{-6}$ and the smallest value of $\mathsf{i}_z$ is $3.2 \times 10^{-4}$. At $\omega = 0.5$, $\mathsf{r} = 0.4$, $q_z = 0.1$ and source variance one, the triples $(I(z_t; e_t), \mathsf{i}_z, \mathsf{s}_z)$ are $(0.0948, 0.49207, 2.18750)$ at $\mathsf{f} = 0.5$, $(0.3432, 0.40473, 1.02941)$ at 0.8, $(0.6953, 0.19723, 0.28226)$ at 0.95, $(0.8929, 0.05313, 0.05795)$ at 0.99 and $(0.9536, 0.00578, 0.00583)$ at 0.999. The information held by the record rises while the flow needed to keep it falls to zero, and in this family the entropy flow $\mathsf{s}_z$ falls with it; the proposition proves the limit only for $\mathsf{i}_z$.

**Check S47.213** (Complete graph against a record)**.** Scripts `nrg_check_viability.py`, part 3, and `nrg_check_scale.py`. At $F = 0.9$, $Q = 0.19$, $R^s = 1$, $R_c = 0.5$ and $\mathsf{x} = 20$: $P_0 = 1$, $I_\eta = 0.83037$, $I_s = 0.34657$, $I_m = 0.80472$, $I_r(1) = 0.89588$ and $N_K = 21.207$. The triples $(I^{\mathrm{pred}}_\bullet, \mathsf{c}_\bullet, \mathsf{m}_\bullet)$ for the complete graph and for the record are $(0.3529, 1.151, 5.906)$ and $(0.2989, 1.903, 4.075)$ at $N = 2$; $(0.5059, 3.565, 6.552)$ and $(0.3722, 1.792, 5.653)$ at $N = 5$; $(0.5832, 5.980, 5.684)$ and $(0.4015, 1.759, 6.270)$ at $N = 8$; $(0.6784, 12.417, 1.150)$ and $(0.4313, 1.731, 6.895)$ at $N = 16$; $(0.7028, 15.636, -1.581)$ and $(0.4380, 1.725, 7.034)$ at $N = 20$; $(0.7813, 47.825, -32.199)$ and $(0.4572, 1.709, 7.434)$ at $N = 60$. The complete-graph margin is largest at $N = 4$, where it is 6.59, it is 0.484 at $N = 17$ and $-0.194$ at $N = 18$, and it is negative for every $N$ from 18 to 200, in agreement with and somewhat before the bound $N_K$ of Theorem S23.12(c). The record margin exceeds the complete-graph margin for every $N$ from 7 to 200 and for no smaller $N$, well before the bound 23.75 of Theorem S23.12(d), and it increases to $\mathsf{m}^{\infty}_R = 7.6516$ with $I^{\mathrm{pred}}_{R,\infty} = 0.46761$. The closed forms for $I^{\mathrm{pred}}_K$ and $I^{\mathrm{pred}}_R$ were compared at $N = 4$ with mutual informations computed from the joint covariance of $\eta_{t+1}$, $\eta_t$, the sensors, the messages and the record; both agree to six digits, 0.467609 and 0.355801. Over 500 random draws of $(F, Q, R^s, R_c, \mathsf{x})$ with $|F| \in [0.05, 0.98]$, $Q \in [0.05, 2]$, $R^s, R_c \in [0.05, 3]$ and $\mathsf{x} \in [0, 40]$ (seed 17) and all $N$ from 2 to 80, none of the inequalities of Theorem S23.12(a) to (d) is violated.

**Check S47.214** (Independent rerun)**.** Scripts `nrg_indep_a.py`, `nrg_indep_b.py` and `nrg_indep_c.py`, written separately from the five scripts above, with their own generators and seeds; they run in 0.6, 1.4 and 0.3 seconds on one thread. The five scripts above were rerun as well and print the numbers quoted in Checks S47.209 to S47.213.

For Theorem S23.3, over 60 instances with $d \in \{2,\dots,5\}$ and full $Q^{\Xi}$, $\mathsf{D}_n = n\mathsf{ep}$ holds with relative error at most $1.1\times10^{-14}$ for $n \le 6$, the two expressions of Eq. (S97) agree to $9.4\times10^{-16}$, and the two terms of the second expression are positive, with smallest values $6.3\times10^{-5}$ and $1.1\times10^{-4}$. For 20 normal matrices $A$ with $Q^{\Xi} = qI$ the eigenvalue formula agrees to $6.2\times10^{-15}$. For the matrix $A = \left(\begin{smallmatrix} 0.5 & 0.8 \\ 0 & 0.3 \end{smallmatrix}\right)$ with $Q^{\Xi} = I$, which is not normal and has real spectrum, $\mathsf{ep} = 0.752941$, so the eigenvalue formula does not extend beyond normal drift. For Corollary S23.4, the smallest $\mathsf{ep}$ over 400 canonical instances with an active state is 0.106, and over 400 instances without active states and with $A_{\mathsf{ES}} = 0$ it is $5.2\times10^{-3}$. The witness of Remark S44.3 has $\rho(A) = 0.355893$, entries of $AQ^{\Xi} - Q^{\Xi}A^{\top}$ below $1.5\times10^{-17}$ and $\mathsf{ep}$ equal to zero in double precision, and the same $A$ with $Q^{\Xi}_{\mathsf{SA}}$ set to zero has $\mathsf{ep} = 0.072179$.

For Theorem S23.5, over 60 instances and $n = 1,\dots,6$ no inequality of part (a) is violated, and scalar descriptions have $|\mathsf{D}_n(C)| \le 3.6\times10^{-15}$. An instance built to have $\Gamma_1$ symmetric and $\Gamma_2$ not symmetric has $\mathsf{D}_1(C) = 0$ in double precision and $\mathsf{D}_2(C) = 0.014990$, so the symmetry of the first lag alone does not give reversibility in part (c). The series $\mathsf{D}_n(C)/n$, the integrals $\bar{\mathsf{e}} = 0.346653$ and 0.745398 and the increments of Check S47.210 are reproduced, with $n\bar{\mathsf{e}} - \mathsf{D}_n(C) = 0.088436$ and $-0.106968$ at $n = 16$, 32, 64 and 128. In four further random instances the difference $n\bar{\mathsf{e}} - \mathsf{D}_n(C)$ takes the same value to six digits at $n = 24$, 48 and 96: 0.049988 ($d = 5$, $m = 2$), 0.000000 ($d = m = 3$, where $\bar{\mathsf{e}} = \mathsf{ep}$), $-0.163726$ ($d = 4$, $m = 2$) and 0.096358 ($d = 5$, $m = 2$). For the nonlinear scalar summary $Y = X_1 + X_2^2$, Isserlis' formula gives

$$\mathbb{E}[Y_t^2 Y_{t+1}] - \mathbb{E}[Y_t Y_{t+1}^2] = (\Gamma^X_{12} - \Gamma^X_{21})(4\Sigma_{12} - 2(\Gamma^X_{12} + \Gamma^X_{21})), \qquad \Gamma^X = A\Sigma,$$

and the left side vanishes for a reversible stationary series because $(Y_t, Y_{t+1})$ and $(Y_{t+1}, Y_t)$ then have the same law. In the directed ring at $c = 0.3$, $a = 0.5$, $q = 1$, with the orientation of the shift used in the script, $\Gamma^X_{12} = 0.411523$, $\Gamma^X_{21} = 1.028807$ and $\Sigma_{12} = 0.514403$, and the difference is 0.5080526 both from this formula and from a tensor Gauss–Hermite rule with ten nodes per coordinate, which is exact for the polynomial degrees involved; a reversible control gives $4\times10^{-15}$. The opposite orientation changes the sign.

For Theorem S23.8, over 300 random bipartitions with $d \in \{2,\dots,6\}$ and full blocks $Q^{\Xi}_{\mathrm{uu}}$ and $Q^{\Xi}_{\mathrm{vv}}$: the additivity in (a) holds to $8.9\times10^{-16}$, the smallest value of $\mathsf{ep}_{\mathrm{u}} - \ln|\det A_{\mathrm{uu}}|$ is 0.084, and $\mathsf{ep}_{\mathrm{u}} < 0$ in 46 of 600 blocks with minimum $-0.2324$; the identity in (b) holds to $5.9\times10^{-14}$ with smallest value $3.6\times10^{-6}$ of $\mathsf{s}_{\mathrm{u}} - \mathsf{i}_{\mathrm{u}}$; the formula in (c) holds to $2.2\times10^{-14}$ with smallest value $7.8\times10^{-8}$ of $\mathsf{te}_{\mathrm{v}\to\mathrm{u}} - \mathsf{i}_{\mathrm{u}}$; the identity in (d) holds to $5.3\times10^{-15}$, and $\mathsf{i}_{\mathrm{u}} + \mathsf{i}_{\mathrm{v}}$ is negative in 16 instances. The numbers of Example S44.4, of Example S44.5, of the rotation-like instance and of the diffusive scaling in Check S47.211 are reproduced to the digits printed there, and the entropy production rate of the diffusion $dX = GX\,dt + D^{1/2}\,dW$ is 0.736364, the value that $\mathsf{ep}/h$ approaches. For Corollary S23.9, over 200 random canonical instances with classes of one to three variables the two lines of Eq. (S101) agree with the definition to $1.4\times10^{-14}$ and neither bound is violated.

For Proposition S23.10(a), over 100 channels the formula for $I(y;x)$ holds to $9.3\times10^{-15}$ and the smallest value of $E_{\mathrm{sig}}/(k_{\mathrm{B}}\mathsf{T}) - I(y;x)$ is $1.7\times10^{-6}$. For part (b), over 400 records with both signs of $\mathsf{f}$, $\omega$ and $\mathsf{r}$, Eq. (S102) holds to $7.2\times10^{-15}$, the smallest values of $\mathsf{s}_z - \mathsf{i}_z$ and of $\mathsf{i}_z$ are $9.2\times10^{-5}$ and $4.1\times10^{-4}$, and the bound $(1-\mathsf{f}^2)\bar{V}/(2q_z)$ of the proof exceeds $\mathsf{i}_z$ by at least 0.108. The table of Check S47.212 is reproduced. At $\omega = 0.5$, $\mathsf{r} = 0.4$, $q_z = 0.1$ and $\mathsf{f} = 0.9$, 0.99, 0.999, 0.9999, the flow $\mathsf{i}_z$ is 0.299875, 0.053127, 0.005776, 0.000583 at fixed source variance $P_e = 1$, and 0.299875,

0.322278, 0.324632, 0.324868 at fixed source noise variance $q_e = 0.19$, where the closed form in the proof of Proposition S23.10 gives the limit 0.324895; that closed form agrees with the definition to $1.4 \times 10^{-15}$ over 500 further draws. The limit $\mathsf{i}_z \to 0$ therefore requires the source variance to be held fixed.

For Theorem S23.12, all mutual informations were computed from the joint covariance of $\eta_{t+1}$, $\eta_t$, the sensors, the messages and the record, without the closed forms. Every number of Check S47.213 at the parameters quoted in Section S23.5 is reproduced, and over 200 random parameter draws with both signs of $F$ and $N \le 150$ none of the inequalities of parts (a) to (d) is violated, part (c) being tested in 4142 cases and the comparison of part (d) in 3567; $\mathsf{m}_R(10^7)$ differs from $\mathsf{m}_R^{\infty}$ by at most $4.0 \times 10^{-7}$ in relative terms. No sweep produced a counterexample to a theorem, proposition, lemma or corollary of Section S23.

## S47.20 Checks for Section S24

The checks below evaluate the closed forms of Section S24 on random parameter draws, compare the transmission matrix with a simulation of the fission life cycle at the level of individual members, and iterate the selection recursions against their fixed points and against a simulated population. Every statement of Sections S24 and S45 has a proof; the checks guard the algebra and the modeling of the life cycle, and each one ends by naming what it does not test. Deviations are the largest found over the stated draws.

**Check S47.215** (Identity rate of the manuscript re-derived)**.** The section builds on Proposition 5.14, so its formulas were re-derived and tested independently of Check S47.45. On 20000 draws with $\varepsilon$, $\beta$, $g$, $\omega$ and $\delta$ uniform on $[0,3]$, $\omega$ floored at $10^{-3}$, and $\mathsf{f}$ uniform on $[0,1]$, all of which had $\det \mathsf{A} > 10^{-9}$, the closed form of $\varpi_{\pm}$ agreed with the numerically computed eigenvalues of $-\mathsf{A}$ to $3.6 \times 10^{-15}$. The rate $\varpi_-$ never exceeded the comparison value $\delta + \omega\tilde{\varepsilon}/\mathsf{p}_1$, the smallest gap being $6.3 \times 10^{-7}$, and on the 6163 draws with $\mathsf{p}_2 \le \frac{1}{2}\mathsf{p}_1$ it fell below that value by at most 0.794 times $2\omega\mathsf{p}_2/\mathsf{p}_1$. With $g = 1$ and $\varepsilon = 0.01$ the identity time without a record was 0.99, 1.96, 9.09 and 50.0 residence times at $\mathsf{f} = 0$, 0.5, 0.9 and 0.99, and with a record ($\beta = 100$, $\omega = 0.01$, $\mathsf{f} = 0.9$, $\delta = 0.001$) it was 989.3. These agree with the values reported in Check S47.45, and no discrepancy with Proposition 5.14(a) was found. Part (b) of that proposition was not tested numerically.

**Check S47.216** (Heritability closed forms and the record channel)**.** On 40000 draws with $\varepsilon$, $\beta$, $g$, $\omega$ and $\delta$ uniform on $[0,2]$, $\omega$ floored at $10^{-3}$, $\mathsf{f}$, $\mathsf{c}$ and $\bar{\mathsf{b}}$ uniform on $[0,1]$ and $\mathsf{T}$ uniform on $[0.05,6]$, none of which had $\tilde{\beta} < 10^{-3}$, the closed form (S173) agreed with $e^{\mathsf{TA}}$ computed by eigendecomposition to $1.9 \times 10^{-15}$, and (S104) agreed with the numerically computed spectral radius of $\mathsf{M}$ to $2.2 \times 10^{-16}$. Every draw satisfied the three inequalities of (S105): the smallest values of $\mathsf{h} - e^{-\varpi_-\mathsf{T}} \min\{\bar{\mathsf{b}}, \mathsf{ck}\}$, of $e^{-\varpi_-\mathsf{T}} \max\{\bar{\mathsf{b}}, \mathsf{ck}\} - \mathsf{h}$ and of $\mathsf{h} - e^{-\varpi_-\mathsf{T}} \max\{\mathsf{cu}, \bar{\mathsf{b}}(1-\mathsf{u})\}$ were $5.6 \times 10^{-10}$, $7.9 \times 10^{-10}$ and $4.8 \times 10^{-11}$, all nonnegative, and $1 - \mathsf{h} \ge 0.048$ throughout. The equivalence of $\varpi_- < \tilde{\varepsilon}$ and $\delta < \tilde{\varepsilon}$ in Theorem S24.3(b) held on every draw. On the draws with $\delta \ge \tilde{\varepsilon}$ the ratio of $\mathsf{h}/\mathsf{h}_0$ to $\max\{\bar{\mathsf{b}}, \mathsf{ck}\}/(\bar{\mathsf{b}} + (1-\bar{\mathsf{b}})\mathsf{f})$ was at most 0.9991, a consequence of part (d); the bound of part (d) itself is the upper inequality of (S105) divided by $\mathsf{h}_0$. For Example S45.8 the script returned $\tau_{\mathrm{id}} = 60.27$, $1/\tilde{\varepsilon} = 0.990$, $\mathsf{u} = 0.9890$, the limit 0.9406 and the entries of Table S15, with $\mathsf{h}e^{\mathsf{T}/\tau_{\mathrm{id}}} = 0.9410$, 0.9406, 0.9406 and 0.9406 at $\mathsf{T} = 1$, 5, 20 and 60.

For the dependence on the read rate discussed in Remark S45.9, on 20000 further draws from the same law with $\beta$ floored at $10^{-2}$, none of which had $\varpi_+ - \varpi_- < 0.05$, the formula $\partial\varpi_-/\partial\beta = (\delta - \varpi_-)/(\varpi_+ - \varpi_-)$ agreed with a central difference of step $10^{-5}$ to $4.5 \times 10^{-8}$, and its sign was negative exactly on the draws with $\delta < \tilde{\varepsilon}$. On 2094 of the 13853 draws with $\delta < \tilde{\varepsilon}$ the heritability at the drawn generation time was nevertheless lower at $\beta + 10^{-3}$ than at $\beta$. One

instance with a record that is blank at birth, $\varepsilon = 0.1$, $g = 1$, $\mathsf{f} = 0.5$, $\omega = 0.2$, $\delta = 0$, $\mathsf{c} = 0$, $\bar{\mathsf{b}} = 0.5$ and $\mathsf{T} = 0.2$: at $\beta = 0.5$, 1 and 2 the identity time is 14.42, 18.72 and 27.19 while $\mathsf{h} = 0.3647$, 0.3308 and 0.2723. Heritability at a fixed generation time is therefore not increasing in the read rate, although the identity time is whenever $\delta < \tilde{\varepsilon}$.

Not tested: Theorem S24.3(e) was not simulated by this script, and only its coefficient $\bar{\mathsf{b}}e^{-\mathsf{p}_1\mathsf{T}}$ is tabulated (its conditional mean is computed exactly in Check S47.221); the limits $\mathsf{T} \to \infty$ and $\mathsf{T} \to 0$ of part (a) are covered by Check S47.217.

**Check S47.217** (Founders against recruits)**.** On 20000 draws with rates uniform on $[0, 2]$, $\omega$ floored at $10^{-3}$, and $\mathsf{f}$ and $\mathsf{c}$ uniform on $[0, 1]$, the bracket $\min\{\delta, \tilde{\varepsilon}\} \le \varpi_- \le \max\{\delta, \tilde{\varepsilon}\}$ of Lemma S45.2(v) held on every draw with tolerance $10^{-12}$. The long-generation prefactor was computed as the spectral radius of $e^{\varpi_-\mathsf{T}}\mathsf{G}(\mathsf{T})\mathsf{F}$ at $(\varpi_+ - \varpi_-)\mathsf{T} = 60$ for $\bar{\mathsf{b}} \in \{0, \frac{1}{2}, 1\}$; it agreed with $\mathsf{P}(\bar{\mathsf{b}}) = (1 - \mathsf{u})\bar{\mathsf{b}} + \mathsf{uck}$ to $6.7 \times 10^{-16}$, the difference $\mathsf{P}(1) - \mathsf{P}(0)$ agreed with the slope (S176) to $1.7 \times 10^{-15}$, and the sign of the difference agreed with the sign of $\omega(1 - \mathsf{cf}) + \delta - \varpi_-$ on every draw. The spectral radius of $\mathsf{F}$ equaled $\max\{\bar{\mathsf{b}}, \mathsf{c}\}$ to $1.1 \times 10^{-16}$ at one uniformly drawn $\bar{\mathsf{b}}$ per draw, which is the limit $\mathsf{T} \to 0$ of Theorem S24.3(a). An instance of the inversion, with faithful copy and faithful recruits and a record that does not decay, $\varepsilon = 0.2$, $\beta = 1$, $g = 1$, $\mathsf{f} = 1$, $\omega = 0.5$, $\delta = 0$, $\mathsf{c} = 1$ and $n = 10$, has $\varpi_- = 0.0376 > \delta + \omega(1 - \mathsf{cf}) = 0$, and the heritability at $\mathsf{b} = 0$, 1, 5 and 10 is 0.99166, 0.99089, 0.98732 and 0.98140 at $\mathsf{T} = 0.5$, 0.84064, 0.83946, 0.83471 and 0.82878 at $\mathsf{T} = 5$, and 0.47856, 0.47788, 0.47518 and 0.47180 at $\mathsf{T} = 20$. In this instance every founder lowers the heritability at all three generation times, including the short one, which Proposition S45.3 does not cover; the threshold $\mathsf{T}_0$ of that proposition was not computed, and monotonicity in $\mathsf{b}$ at intermediate generation times was not tested on the random draws.

**Check S47.218** (Fission life cycle at the level of members)**.** One generation of Definition S24.1 was simulated member by member for 20000 parents with one daughter each, with $n = 8$, $\mathsf{b} = 2$, $\varepsilon = 0.05$, $\beta = 0.8$, $g = 0.5$, $\mathsf{f} = 0.6$, $\omega = 0.4$, $\delta = 0.02$, $\mathsf{c} = 0.9$, $\mathsf{T} = 2$, $\mathsf{s}_\nu = 0.3$ and $\mathsf{s}_z = 0.2$. Parents' member states were a common standard normal component plus independent deviations of standard deviation 0.5, and the parent's record was 0.7 times the common component plus noise of standard deviation 0.6. The relaxation used an Euler step of 0.01 (200 steps) with mean-conserving consensus among members at rate 1, private diffusion of intensity 0.1, and replacement of each member in each step with probability $1 - e^{-0.005}$. The conditional mean of this discrete scheme is exactly linear with a known $2 \times 2$ one-step matrix, so the Monte Carlo error and the discretization error can be separated. The regression of the daughter's state at birth on the parent's state returned $(0.2473, 0.4061; -0.0019, 0.8988)$, rows separated by a semicolon, against $\mathsf{F} = (0.25, 0.405; 0, 0.9)$. The regression of the daughter's final state returned $(0.0502, 0.5910; 0.0496, 0.6980)$ against $(0.0525, 0.5901; 0.0513, 0.6984)$ for the discrete scheme, a largest deviation of $2.3 \times 10^{-3}$, and the discrete matrix differs from $\mathsf{M} = e^{\mathsf{TA}}\mathsf{F} = (0.0525, 0.5904; 0.0511, 0.6989)$ by at most $4.6 \times 10^{-4}$. The cross-covariance of final and parental states differed from the product of the discrete matrix with the parental covariance by at most $2.0 \times 10^{-3}$, for entries up to 0.625. The parent-offspring regression slope of the heritable trait was 0.7405 with standard error 0.0012, against the spectral radius 0.7423 of the discrete matrix (1.5 standard errors) and $\mathsf{h} = 0.7426$. The empirical covariance of the birth state about $\mathsf{Fx}$ was $(0.02267, 0.01822; 0.01822, 0.04001)$ against $(0.02239, 0.01800; 0.01800, 0.04000)$ from (S106) with $\mathsf{v}_\mathrm{w}$ replaced by its mean over parents; the nominal relative sampling error of a variance estimated from 20000 normal draws is 1 percent. In the variant without a record the slope was 0.4272 with standard error 0.0013, against 0.4248 for the discrete scheme (1.8 standard errors) and $\mathsf{h}_0 = 0.4246$, and the variance of the members' mean at birth about its conditional mean was 0.05342 against 0.05422 from Proposition S24.5(c). Not tested: the covariance $\mathsf{N}_\mathrm{rel}$ added during relaxation, which

is an input and was not compared with any formula, the convergence along lines of descent in Proposition S24.2(c), and the variance recursion of Proposition S24.5(b) over several generations.

**Check S47.219** (Price decomposition)**.** For 500 parents with traits drawn from $\mathcal{N}(0.5, 1)$, numbers of daughters Poisson with mean $e^{0.5(y_j-\bar{y})}$, and daughters' traits $\mathsf{h}y_j$ plus independent normal noise with $\mathsf{h} = 0.8$ and standard deviation 0.3, the two sides of (S177) differed by at most $4.4 \times 10^{-16}$ over 400 independent draws of the offspring generation. With the parents and their fitnesses held fixed, the mean realized change of the trait mean over those 400 draws was 0.2731 with standard error 0.0006, against $\mathsf{hd}_{\mathrm{sel}} - (1-\mathsf{h})\bar{y} = 0.2741$ from (S108) (1.7 standard errors); over 20000 further draws it was 0.27419 with standard error 0.00009, against 0.27407 (1.3 standard errors). In the member-level simulation of Check S47.218, with numbers of daughters Poisson with mean $e^{0.4(y_j-\bar{y})}$ and the one simulated daughter of each parent weighted by $\mathsf{W}_j$, the realized change of the mean heritable trait was 0.2780 against 0.2792 predicted with the spectral radius of the discrete matrix, with a Monte Carlo standard error of 0.0017, selection differential $\mathsf{d}_{\mathrm{sel}} = 0.3749$ and $\bar{y} = -0.0036$. The weighting reproduces the conditional expectation in (S108) and not the sampling variance of a population in which each parent founds $\mathsf{W}_j$ distinct daughters. The general form (S107) for a trait other than the heritable one was not tested separately.

**Check S47.220** (Selection-transmission balance and the absence of a threshold)**.** With $\mathsf{h} = 0.85$, $\mathsf{s}^2 = 0.04$, $\mathsf{v}_{\mathrm{s}} = 1$ and $\mathsf{y}_* = 2$, 400 iterations of (S109) from $(\mathsf{V}, \bar{y}) = (0, -3)$ and from $(5, 4)$ both gave $(0.113847, 0.733532)$, equal to (S110) to the six digits printed. A simulated population of 200000 collectives, with parents sampled in proportion to $\mathsf{W}$ and daughters' traits drawn as in Assumption S24.7, run for 120 generations from a standard normal start, had variance 0.11386 (standard deviation across generations 0.00056) and mean 0.73255 (0.00167) over generations 61 to 120, and mean fitness 0.46069 (0.00090) over generations 61 to 119 against 0.46121 from (S179). At $\mathsf{s}^2 = 0.04$ and $\mathsf{v}_{\mathrm{s}} = 1$ the retained fraction $\hat{y}/\mathsf{y}_*$ was 0.0020, 0.0099, 0.0292, 0.0789, 0.2597, 0.5228, 0.7373, 0.9451 and 1.0000 at $\mathsf{h} = 0.05$, 0.2, 0.4, 0.6, 0.8, 0.9, 0.95, 0.99 and 1; on a grid of 20001 values of $\mathsf{h}$ in $[10^{-4}, 1]$ it was strictly increasing, with smallest value $3.9 \times 10^{-6}$. For Proposition S45.6 with $\mathsf{r} = 0.3$ and the same $\mathsf{h}$ and $\mathsf{s}^2$, 2000 iterations gave $(0.144144, 0.245045)$, equal to (S180). For the two-type map (S181) with $\mathsf{m} = 3$ and $p_0 = \frac{1}{2}$, the frequency after 20000 generations was 0 to six decimals at $\mathsf{q} = 0.2$, $2.5 \times 10^{-5}$ at $\mathsf{q} = \frac{1}{3}$, 0.250000 at $\mathsf{q} = 0.5$ and 0.850000 at $\mathsf{q} = 0.9$, against the limits 0, 0, 0.25 and 0.85 of Proposition S45.7; at the threshold $\mathsf{mq} = 1$ the decay is algebraic, $p_k = 1/(2 + 2k)$, which is the value found. The simulated population draws daughters from the Gaussian kernel of Assumption S24.7 and therefore tests the recursion under the closure, not the closure itself; the monotonicity of the retained fraction in $\mathsf{s}^2$ is proved and was not checked numerically by this script; Check S47.221 checks it on a grid.

**Check S47.221** (Independent recomputation by enumeration)**.** This check was written separately from the scripts behind the six checks above. It uses its own matrix exponential, a Taylor series with scaling and squaring, and it replaces Monte Carlo sampling of the life cycle by exact computation of conditional means and by enumeration of founder sets. On 6000 draws with $\varepsilon$, $\beta$ and $\delta$ equal to 0 with probability 0.15 and otherwise uniform on $[0, 2]$, $g$ uniform on $[0.05, 2]$, $\omega$ uniform on $[10^{-3}, 2]$, $\mathsf{f}$ and $\mathsf{c}$ equal to 0 and to 1 with probability 0.15 each and otherwise uniform on $[0, 1]$, $n$ uniform on $\{2, \ldots, 12\}$, $\mathsf{b}$ uniform on $\{0, \ldots, n\}$ and $\mathsf{T}$ log-uniform on $[0.01, 30]$, retained when $\tilde{\beta} > 10^{-6}$ and $\det \mathsf{A} > 10^{-9}$, the closed form (S173) agreed with the exponential to $1.8 \times 10^{-13}$ and (S104) with the numerically computed spectral radius to $2.2 \times 10^{-16}$. No draw violated the ordering of the roots, $0 < \mathsf{u} < 1$, the positivity and row-sum bound of $\mathsf{G}(\mathsf{T})$, the bracket of Lemma S45.2(v), $\mathsf{h} < 1$, any of the four inequalities of (S105), or the ratio bounds of Theorem S24.3(c) and (d), at relative tolerance $10^{-12}$; the largest heritability was 0.999946. The limit of part (a) as $\mathsf{T} \to \infty$ was reproduced to

$4.2 \times 10^{-13}$ and the limit as $\mathsf{T} \to 0$ to $9.5 \times 10^{-8}$ at $\mathsf{T} = 10^{-9}$. On 20000 random nonnegative $2 \times 2$ matrices with each entry zero with probability $\frac{1}{2}$, Lemma S45.1(iii) and Lemma S45.5 had no violation, and on 500 draws with $\mathsf{b} \geq 1$ and $\mathsf{c} > 0$ the matrix limit of Proposition S24.2(c) held to $5.1 \times 10^{-13}$. In Example S45.8 at $\mathsf{T} = 20$ the row vector orthogonal to $r_{\mathsf{M}}$ is a left eigenvector of $\mathsf{M}$ with eigenvalue zero to rounding, so the conditional mean of that trait is lost in one generation while the heritable trait retains 0.675; this is the exception that the statement of Proposition S24.2(c) records. The script reproduced every entry of Table S15.

For the life cycle, 300 draws with $n \in \{2, \dots, 6\}$ and $\mathsf{T} \leq 8$ were used with a random mean-conserving communication matrix (a symmetric weighted Laplacian plus a directed cycle, with zero column sums). The $(n+1)$-dimensional linear system for the conditional means of the members and the record was exponentiated, and the birth map was averaged over every founder set of size $\mathsf{b}$. The resulting map from the parent's member states and record to the daughter's mean and record equaled $\mathsf{M}$ applied to the parent's mean and record to $5.2 \times 10^{-14}$, which confirms Proposition S24.2(a) at the level of members without sampling. Without a record every member's coefficient equaled $\mathsf{h}_0/n$ to $3.5 \times 10^{-15}$, and with an externally written record every member's coefficient equaled $\bar{\mathsf{b}} e^{-\mathsf{p}_1 \mathsf{T}}/n$ to $1.3 \times 10^{-15}$, while on 40 of the draws the forced solution computed by a fourth-order Runge-Kutta scheme for the record path $z_{\mathrm{ext}}(t) = 0.3 + \sin 2t + 0.2t$ agreed with the formula in the proof of Theorem S24.3(e) to $1.4 \times 10^{-11}$. For all 25 pairs $(n, \mathsf{b})$ with $2 \leq n \leq 6$ and two-point noises, enumeration of founder sets and noise values gave the conditional mean $\mathsf{Fx}$ to $2.1 \times 10^{-15}$, the covariance (S106) to $1.7 \times 10^{-16}$, and the variance of Proposition S24.5(c) to $4.4 \times 10^{-16}$. The difference between 0.05342 and 0.05422 reported in Check S47.218 is therefore sampling error and not an error of the formula.

For selection, exact expectations over 200 random populations of 2 to 29 parents, with random nonnegative $2 \times 2$ transmission matrices and Poisson numbers of daughters, gave (S107) for a random trait vector to $4.4 \times 10^{-16}$ and (S108) to $3.3 \times 10^{-16}$. On 3000 draws of $(\mathsf{h}, \mathsf{s}^2, \mathsf{v}_{\mathrm{s}})$ the iterated recursion (S109) reached (S110) to $4.4 \times 10^{-16}$ in the variance and $2.7 \times 10^{-15}$ in the mean, the selection step agreed with numerical quadrature to $6.7 \times 10^{-16}$ and (S179) to $5.8 \times 10^{-12}$, and the equivalence of $\hat{y}/\mathsf{y}_* \geq \frac{1}{2}$ with $\mathsf{hi} \geq 1 - \mathsf{h}$ held on every draw. For 200 pairs $(\mathsf{s}^2, \mathsf{v}_{\mathrm{s}})$ the retained fraction was strictly increasing in $\mathsf{h}$ on a grid of 2001 points of $[0, 1]$, and for 200 values of $\mathsf{h}$ in $[0.02, 0.98]$ it was strictly increasing in $\mathsf{s}^2$ on a logarithmic grid of 1501 points of $[10^{-6}, 10^{3}]$, with value at most $3.4 \times 10^{-9}$ at $\mathsf{s}^2 = 10^{-12}$. The fixed point (S180) was reached to $2.7 \times 10^{-14}$, the limits of Proposition S45.7 to $1.3 \times 10^{-15}$ on 971 draws with $|\mathsf{mq} - 1| > 0.05$, and at $\mathsf{mq} = 1$ with $\mathsf{m} = 3$ and $p_0 = \frac{1}{2}$ the frequency after 1000 generations was $1/2002$.

For Proposition S45.3, on 2000 draws with $n = 8$ the ordering of $\mathsf{h}$ in $\mathsf{b}$ at $(\varpi_+ - \varpi_-)\mathsf{T} = 45$ agreed with the sign of (S176) on all 1981 draws with slope larger than $10^{-3}$ in absolute value. On a grid of 60 generation times log-spaced in $[0.02, 60]$ no draw had a heritability that was neither nondecreasing nor nonincreasing in $\mathsf{b}$; this is a numerical observation on these draws and not a proved statement. Lowering the copy fidelity of the inversion instance of Check S47.217 from $\mathsf{c} = 1$ to $\mathsf{c} = 0.9$ gives $\varpi_- = 0.0376 < \delta + \omega(1 - \mathsf{cf}) = 0.05$, and the heritability at $\mathsf{b} = 0$, 1, 5 and 10 is then 0.89249, 0.89312, 0.89608 and 0.90113 at $\mathsf{T} = 0.5$ and 0.43070, 0.43092, 0.43182 and 0.43294 at $\mathsf{T} = 20$, increasing in $\mathsf{b}$ as the proposition predicts for long generations. With $\mathsf{c} = 1$ the heritability of that instance was strictly decreasing in $\mathsf{b}$ at each of 400 generation times log-spaced in $[10^{-3}, 200]$.

Not tested: the covariance $\mathsf{N}_{\mathrm{rel}}$, which remains an unspecified input; any property of the stochastic paths beyond conditional means and the birth covariance; the sampling variance of a finite population in Proposition S24.5(b) and Proposition S24.6; the closure Assumption S24.7 as a description of the fission model; and the threshold $\mathsf{T}_0$ of Proposition S45.3, for which no bound is given.

## S47.21 Checks for Section S25

**Check S47.222** (Three realizations of one covariance)**.** For Theorem S25.3, a covariance $\Sigma = GG^\top/n + 0.5I$ with $n = 5$ and a standard normal $G$ (seed 0) was realized three ways, each stationary covariance being computed from the discrete Lyapunov equation. (a) With $A = 0.6I$ and $D = \delta I$, $\delta = 0.5(1-a^2)\lambda_{\min}(\Sigma)$, the matrix $(1-a^2)\Sigma - \delta I$ has smallest eigenvalue 0.1672, so it is a white driver covariance, and the stationary covariance agrees with $\Sigma$ to $1.1 \times 10^{-16}$; at $\delta = (1-a^2)\lambda_{\min}(\Sigma)$ the driver covariance has rank 4. (b) With a random diagonal $D \preceq \Sigma$ and a random orthogonal $U$, the matrix of Eq. (S112) has spectral radius 0.899, its stationary covariance agrees with $\Sigma$ to $1.1 \times 10^{-14}$, its largest off-diagonal entry is 1.29 in modulus, and the largest off-diagonal entry of $AD$, the conditional covariance of Eq. (S182), is 0.192. (c) With $A = \sqrt{0.7}U$ and $D = 0.3I$ the stationary covariance agrees with the identity to $4.4 \times 10^{-16}$ while the largest off-diagonal entry of $A$ is 0.613.

**Check S47.223** (Innovations under a driver and the full-rank construction)**.** For Lemma S46.2, the interaction-free model with $n = 3$, $A_0 = \mathrm{diag}(0.5, 0.3, 0.6)$, $D_0 = \mathrm{diag}(0.4, 0.5, 0.3)$, loading $\mathsf{L} = (0.7, 0.5, 0.9)^\top$ and a scalar first-order driver with pole 0.8 and innovation variance $Q_{\mathsf{d}} = 1$ was put in state-space form, and the one-step prediction error covariance of the recorded process was computed from the stationary Riccati equation. It equals $D_0 + R\mathsf{L}\mathsf{L}^\top$ to $2.5 \times 10^{-13}$ with $R = 1.1204 \geq Q_{\mathsf{d}}$, and its largest off-diagonal entry is 0.706, so it is not the innovation covariance of any driver-free model. For Theorem S25.5(b), with $A$ having the two off-diagonal entries $A_{12} = 0.3$, $A_{21} = 0.2$, diagonal $(0.5, 0.4, 0.6)$, $D = \mathrm{diag}(0.5, 0.4, 0.3)$, $T = \{1, 2\}$ and $A_{0,T} = \mathrm{diag}(0.5, 0.4)$, the smallest eigenvalue of $G_T D_T G_T^* - D_{0,T}$ over 2001 frequencies in $[0, \pi]$ is 0.1232 for $D_{0,T} = 0.2D_T$, $-0.0100$ for $0.5D_T$ and $-0.2328$ for $D_T$: the driver density exists for small $D_{0,T}$ and fails to exist for large, as the constant $c$ in the statement requires. For part (a), a broadcaster with $A_{21} = A_{31} = 0.3$ and $T = \{2, 3\}$, which omits the sender, gives entries of modulus 0.178 in the block $(GDG^*)_{T,T^c}$ at $\theta = 1$, so the residual is not supported on $T$.

**Check S47.224** (Pole swap and the static form)**.** For Proposition S25.7, the scalar model with $(a, \phi, D, \beta^2) = (0.7, 0.4, 0.5, 0.3)$ and the swapped model $(0.4, 0.7, 0.2857, 0.4543)$ were put in joint state-space form and their autocovariances computed from the Lyapunov equation; they agree to $4.4 \times 10^{-16}$ for lags up to 12, with $\Gamma_0 = 2.2253$ and $\Gamma_1 = 1.7561$. For Proposition S25.8, with $n = 4$, $k = 2$, a random coupling scaled to spectral radius 0.6, random diagonal $D$ with entries in $[0.3, 0.8]$ and a random loading (seed 3), the inverse of $\Sigma^{\mathrm{loc}} + MM^\top$ agrees with $\Pi^{\mathrm{loc}} - \mathsf{K}$ to $2.8 \times 10^{-16}$, and $\mathsf{K}$ has eigenvalues 0, 0, 1.154, 1.611, hence rank 2.

**Check S47.225** (The difference formula and the instrument)**.** For Theorem S25.9, a level-0 network with $n = 3$, $F = 0.9$, world noise variance one, message gains $k_{12} = 0.20$, $k_{13} = 0.10$, $k_{21} = 0.15$, $k_{23} = 0.25$, $k_{31} = 0.05$, $k_{32} = 0.30$ and private variances $(0.05, 0.08, 0.06)$ was compared with its intervened versions through exact lagged covariances of the joint recursion of beliefs and world. (a) With equal sensing gains 0.3 and the message slots removed, $W^{\mathfrak{N}} = F(1-k^s)I$ with private variances $(0.03, 0.05, 0.04)$, the off-diagonal entries of $\Gamma_h - \Gamma_h^{\mathfrak{N}}$ agree with those of $W^h\Sigma^{\mathrm{loc}}$ to $5.7 \times 10^{-16}$ for $h \leq 4$; the off-diagonal entries of $\Sigma^{\mathrm{loc}}$ are 0.0132, 0.0121, 0.0166. (b) With the same equal sensing gains, the slots jammed and the gains kept, $W^{\mathfrak{N}} = \mathrm{diag}(W)$, the off-diagonal entries of $\Gamma_0 - \Gamma_0^{\mathfrak{N}}$ exceed those of $\Sigma^{\mathrm{loc}}$ by 1.54, 1.49, 1.58, a driver term two orders of magnitude larger than the interaction term. With slots removed and unequal sensing gains $(0.3, 0.2, 0.4)$ the residual driver term is 0.205, $-0.177$, 0.120, and $W\mathsf{L} - W^{\mathfrak{N}}\mathsf{L} = (-0.0081, 0.0527, -0.0527)$, the value of Eq. (S197). The difference formula therefore fails outside the common-mode condition. (c) In a simulation of $2 \times 10^5$ rounds of the network with unequal sensing gains, with every slot jammed by recorded symbols of variance 0.5 and gains kept (seed 1), the instrument estimate of the six cross gains has

largest error 0.0033 against a largest standard error of 0.0034. The simulation keeps the private variances under jamming, as when the channels carry no noise of their own or the symbols pass through the same noisy channels; the population value of the estimate does not depend on this choice. The agreement of the two numbers is particular to the seed: in an independent simulation of the same design with seeds 5, 6 and 7 the largest errors were 0.0061, 0.0056 and 0.0030 against the same standard error, and over 16 seeds of $6 \times 10^4$ rounds the root mean square of the error divided by its standard error was 0.90, as an unbiased estimate requires. With a first-order symbol of lag-one correlation 0.6 in place of a white one, the population ratio for the gain $A_{12} = 0.18$ is 0.2296, the value $A_{12}/(1 - 0.36 \times 0.6)$ given in the proof of Theorem S25.9(c).

**Check S47.226** (White driver: identification and the lower bound)**.** For Proposition S25.6, a model with $n = 4$, a coupling matrix with six nonzero off-diagonal entries between 0.1 and 0.3, diagonal $(0.5, 0.4, 0.5, 0.3)$, $D = \mathrm{diag}(0.5, 0.4, 0.3, 0.6)$ and a white rank-one driver with loading $(0.7, 0.5, 0.9, 0.4)$ was used. From the exact $\Gamma_0$ and $\Gamma_1$, part (a) returns $A$ to $2.2 \times 10^{-16}$ and the triad formula of part (b) returns $D$ to $4.4 \times 10^{-16}$. The order parameter over the window $\bar{t} = 4$, computed from the conditional path covariance, is 0.0740 with the identified $D$ and 0.0820 if the innovation variances $Q_{ii}$ are used in its place, so the correction for the driver matters at the second digit. For part (c) with $\bar{t} = 2$ and the variance bound $v_i = 1.2\mathsf{L}_i^2$, the bound Eq. (S113) is 0.0055, while $\Omega_{\mathrm{int}}(C)$ over 2000 random points and the 16 corners of the admissible box $D_{ii} \in [Q_{ii} - v_i, Q_{ii}]$ ranges from 0.0180 to 0.0306. The bound holds and is loose by a factor of about three.

**Check S47.227** (Recorded grade of non-interacting members)**.** For Proposition S25.11(c), the average of $n$ memoryless members with private variance $\delta = 1$ that sense a first-order driver with pole 0.98 and innovation variance one was evaluated from its exact autocovariances, the infinite-past prediction variance being computed in closed form from the spectral factorization of the density $\sigma^2/|1-\phi z|^2 + \delta/n$, which is that of a first-order autoregressive moving-average process. The recorded memory defect, predictability and persistence time are $1.3 \times 10^{-2}$, 0.943, 98.0 for $n = 4$; $4.4 \times 10^{-5}$, 0.960, 99.0 for $n = 100$; and $4.6 \times 10^{-9}$, 0.960, 99.0 for $n = 10^4$. In the last case the script prints $4.76 \times 10^{-9}$, an artifact of cancellation in double precision in the root that gives the moving-average coefficient; with $r = \delta/n$ and $B = \sigma^2 + r(1 + \phi^2)$ one has $\sigma_1^2 = B - r^2\phi^2/(v + r)$ exactly and $\sigma_\infty^2 = B - r^2\phi^2/B$ up to a relative error of order $r^4$, which gives $\hat{\mathfrak{d}}_{\mathrm{M}} = 4.61 \times 10^{-9}$, the value also obtained with 50-digit arithmetic. The bound $\delta/(2n\sigma^2)$ of Eq. (S199) is 0.125, 0.005 and $5 \times 10^{-5}$ in the three cases. For part (b), a scalar closed variable with $a = 0.6$, $\beta = 0.8$, $q = 0.5$ and a first-order input of unit variance has recorded memory defect 0 for a white input, $7.4 \times 10^{-3}$ for input pole 0.5 and $2 \times 10^{-5}$ for input pole 0.9, against the bound 0.412; in the last case the moving-average root of the recorded process, 0.592, nearly cancels the pole 0.6.

**Check S47.228** (Model recovery and a held-out intervention)**.** The data-generating model is the level-0 network of Check S47.225 with sensing gains $(0.3, 0.2, 0.4)$, so that the coupling has diagonal $(0.36, 0.36, 0.225)$ and loading $(0.27, 0.18, 0.36)$, and with the world unrecorded. Each model class was fitted to the exact law of a window of six rounds (an 18-dimensional Gaussian) by minimizing the relative entropy from the truth with a quasi-Newton method from the start $W = 0.3I$, loadings 0.3, variances 0.1, pole 0.5, and then asked to predict the window law when every channel delivers noise of variance 0.5 with gains kept, an intervention not used in the fit. The collective class (coupling, loading, private variances and driver pole; 16 parameters) reaches a relative entropy of $1.3 \times 10^{-12}$, recovers the coupling to $5.2 \times 10^{-7}$, the loading to $1.3 \times 10^{-7}$, the variances to $4.0 \times 10^{-8}$ and the pole as 0.9000003, and predicts the intervened law with relative entropy $5.9 \times 10^{-12}$; the Jacobian of the window covariance at the truth has singular values between 195 and 0.076, so its rank is 16. Independent agents with a rank-one first-order driver (10 parameters) fit with relative entropy 0.408,

with driver pole 0.959, and, predicting no change under jamming, miss the intervened law by 2.02. A first-order autoregression with full innovation covariance (15 parameters) fits with relative entropy 0.310; its fitted coupling has off-diagonal entries up to 0.448 where the true entries are at most 0.27, and it misses the intervened law by 4.64 when its off-diagonal coefficients are read as channels and jammed and by 3.15 when it predicts no change. All quantities are at the population level, in nats. The numbers above are from the stated starting point. From random starts (independent rerun) the collective class reaches the truth in 2 of 7 and otherwise a second strict local minimum, with relative entropy $6.5 \times 10^{-3}$, gradient norm $4 \times 10^{-6}$, Hessian eigenvalues between 0.27 and 859, driver pole 0.581, coupling eigenvalues $(0.054, 0.270, 0.909)$ against $(0.032, 0.270, 0.644)$ at the truth, and a held-out error of 1.33. The independent class reaches 0.408 from 10 of 10 random starts, and the autoregression reaches 0.310 from 7 of 7, which is the value at the regression coefficient $\Gamma_1\Gamma_0^{-1}$ and the global minimum of that class (Remark S46.10). Neither of these two classes contains the truth: the largest entry of $\Gamma_2 - \Gamma_1\Gamma_0^{-1}\Gamma_1$ at the truth is 0.068 in modulus, and the matrix of lag differences of Remark S46.10 has smallest singular values $8 \times 10^{-4}$, $3 \times 10^{-4}$ and $3.3 \times 10^{-3}$ for the three pairs, with largest singular values between 0.74 and 0.90.

**Check S47.229** (Rank-one white driver with unequal memories)**.** For Remark S46.1, the off-diagonal entries $\mathsf{L}_i\mathsf{L}_j/(1 - a_i a_j)$ of an interaction-free model with a white driver of rank one were evaluated for 200 random draws of $n$ between 3 and 7, memories in $(-0.9, 0.9)$, standard normal loadings and $\tau \in (-0.95, 0.95)$ (seed 11); the substitution of the remark changes them by at most $8.4 \times 10^{-15}$ in relative terms. With memories $(0.5, -0.3, 0.7, 0.2)$, loading $(0.7, 0.5, 0.9, 0.4)$ and $D = \mathrm{diag}(0.5, 0.4, 0.3, 0.6)$ the formula agrees with the solution of the Lyapunov equation to machine precision, and the ratio $\Sigma_{12}\Sigma_{34}/(\Sigma_{13}\Sigma_{24})$ is 0.6967, the value of Eq. (S183), where equal memories give one. The Jacobian of the map from $(a, \mathsf{L})$ to the off-diagonal entries, by central differences at one random point for each $n$, has rank 6, 9, 11, 13 for $n = 4, 5, 6, 7$, against 6, 10, 15, 21 off-diagonal entries and $2n - 1 = 7, 9, 11, 13$ free parameters.

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